TECHNICAL SUPPLEMENT TO COLLISION RECORDS, ARITHMETIC, AND RAW LOCAL INVERSION IN A TRIANGULAR LORENTZ GAS

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ABSTRACT. We study exact displacement, collision count and flight time for actual returns in a triangular periodic Lorentz gas. We prove that the complete physical boundary source has a finite distributional derivative at each collision count, uniformly in the radius, the original exact labels and the guard width at that count. The proof writes the unchanged smooth guard as a signed threshold integral and combines definable finite-word geometry with the complete density bound. A two-sided averaging inequality then bounds the unrecovered height by paired incoming and outgoing flux, without charging an isolated jump. Finally the ordered scalar pointwise error equals the bounded-Lipschitz path-numerator error on the same roofs. The collision-uniform decay of the paired flux remains the quantitative estimate needed for the unrestricted arithmetic pointwise law.

The physical record and the pointwise problem. Let $R \in I = [9/20, 47/100]$ and let $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ be the actual return record under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, where $c_* = 91/(10000\pi)$. The occupation convention counts collisions $0, \dots, m-1$ and tests terminal membership at collision m. Write

$$Z_{m,R}(k,t,n) = \frac{(k_1, k_2, t - m\bar{\tau}_R, n - mc_*)}{\sqrt{m}}.$$

The original pointwise objective is, for every $M < \infty$,

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t:\, |Z_{m,R}(k,t,n)|\le M} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{0.1}$$

Here $\mathcal{L}_{m,R}$ retains the full finite arithmetic transition kernel, including zero classes. The new results concern this exact record; they do not replace it by a window or a pooled discrete label. Let $\mathcal{E}_M$ be the collision limsup of the left side of (0.1). For a family of densities set

$$\mathcal{H}(f) = \limsup_{m\to\infty} \sup_{R,n,k} m^2 \|f_{n,k,m,R}\|_{L^\infty(\mathbb{R})}. \tag{0.2}$$

Every density norm is an essential norm. Widths and reserves are fixed before this collision limit. The complete source. Let $b = b^{\varepsilon}_{n,k,m,R}$ be the density of $(L_{m,R})_\#((1 - H^{\varepsilon}_{m,R})\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*)$. Its original positive decomposition is

$$b = s_{71,1} + d_{71,1} + b^{\varepsilon,\mathrm{inc}} + e_{70}.$$

The term e_{70} contains first clearance outside the selected disks; first incidence is separate. We keep all these states. The canonical current is Db, constructed directly from this full source. Our first result strengthens its topology without changing the source.

Theorem A (Finite variation of the complete current)**.** *At each fixed collision count $m \ge 2$, the full density b belongs to $BV(\mathbb{R})$ and its derivative has total mass zero. There is a finite constant*

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C_m, uniform in R, ε, n, k at that count, such that $\|Db\|_{\mathrm{TV}} \le C_m$. A possible bound is

$$C_m = 4(J_m + 1)(CA^m/c_*)\|d\Theta\|_{\mathrm{TV}}^{2m+1},$$

where J_m is the uniform monotonicity number of the finite threshold family. No collision-uniform bound on $m^2 C_m$ is part of this assertion.

Proof. Lemma 74.1 represents the original smooth guard exactly. Lemma 74.2 combines finite physical geometry, parameterized integration and definable monotonicity. Theorem 74.3 integrates the resulting variation bound to obtain the original full current. □

Theorem B (Paired flux and raw height)**.** *For a nonnegative compactly supported BV density,*

$$[b - 2A_\delta b]_+(t) \le \min\{(Db)^+([t-\delta, t]), (Db)^-([t, t+\delta])\} \quad \text{for a.e. } t.$$

Let $\mathcal{P}_\varepsilon(\delta)$ be the ordered normalized essential supremum of the right side, defined in (75.2)*. For every sufficiently large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$ and any fixed $\delta(B) > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathcal{P}_{\varepsilon(B)}(\delta(B)). \tag{0.3}$$

The choice of B and its width precedes $m \to \infty$; only then may $B \to \infty$. The reserve is the constant two.

Proof. Lemma 75.1 compares both one-sided averages. Theorem 75.2 applies the resulting bound to the original source and the fixed-width local-variation estimate. The source-preserving allocation with reserve two still retains $s_{71,1}$; no original residual is discarded. □

Theorem C (Equality of scalar and path errors)**.** *On the inherited common positive path-remainder theorem, the ordered bounded-Lipschitz path-numerator error equals $\mathcal{E}_M$ on every original central target. Thus the scalar pointwise endpoint and the same-roof collision-path numerator have the same vanishing condition. Positive-reference normalization gives the conditional collision bridge and forward essential arithmetic likelihood when that endpoint holds.*

Proof. Theorem 76.1 proves the equality by positivity of a common path remainder. Corollary 76.2 gives the normalization and the exact central excess comparison. Zero arithmetic classes are never normalized. □

The missing quantitative assertion is decay of $\mathcal{P}_{\varepsilon(B)}(\delta(B))$ for a legal choice of widths, or directly of the equivalent central excess. Finite variation at each count does not imply this assertion. In particular the present results do not yet establish (0.1). They prove the full finite-measure property, remove the charge for isolated physical jumps, and identify the scalar estimate that also controls the path numerator. The complete inherited proof chain follows the new argument. Its preceding opening is retained in Appendix Q.

Input and proof-page guide for revision 85

This guide points to the complete proofs actually retained in this supplement. Page numbers are resolved by the current native build, not copied from an earlier PDF. The new article's complete-segment parametrization and residence-band estimates are proved in its Sections 6–8; the table identifies their imported inequalities. An entry is a dependency citation, not independent proof certification.

Input used	Retained theorem and exact display	Proof page
Endpoint field and radial growth	Theorem 55.2; (55.7)	358
Full angular-density variation	Lemma 8.1; (8.4)	234
Common physical coarea kernels	Proposition A.1; (A.2)	238
True-exit interval mass	Theorem 5.1; (5.3)	246
Surface local mass	Theorem 5.3; (5.8)	247
Two-sided surface payment	Theorem 5.2; (5.7)	246
Critical optical volume	Theorem 26.1; (26.2)	302
Fractional and truncated gradients	Theorem 26.2; (26.7) and (26.9)	302
Complete depth-weighted source	Corollary 36.2; (36.4)	322
Scalar positive remainder	Theorem 181.4	678
Scalar-to-path equality	Theorem 76.1	395

The common-space norms and spectral inputs underlying the volume estimates remain (117.1)–(117.3) and Theorem 157.3. The table makes no claim that an external Lorentz-gas local limit theorem establishes this paper's complete four-record endpoint. Historical module 189 remains explicitly superseded for its ambiguous dyadic notation.

Input and proof-page guide for revision 86

All page numbers below refer to the actual compiled supplement. These entries identify inherited mathematical inputs, not independent audits. The focused article proves its mixed-moment, microscopic-envelope and reachable-fiber theorems in full.

Input	Retained theorem and display	Proof page
Capped joint endpoint/incidence bound	Theorem 25.3; (25.8)	301
Unconditioned capped incidence bound	Theorem 35.3; (35.8)	320
Actual fractional source	Theorem 26.2; (26.5)	302
Borel maximal-segment map	Theorem 7.1; (7.2)	214
Full angular-density variation	Lemma 8.1; (8.4)	234
Surface charge sandwich	Theorem 9.1; (9.2)	219
Original short-segment source	Theorem 9.2; (9.4)	219
Active offset correction	(6.1); historical module 189 remains explicitly superseded	248
Positive scalar remainder	Theorem 181.4	678
Scalar-to-path equality	Theorem 76.1	395

No fixed-cutoff theorem in this table is being applied at cutoff zero. The new finite-count optical moment and selected-sector envelope are proved directly in the article. The unproved zero-residence and reachable moderate-sensitivity conditions remain separate.

Revision 87: active input and proof guide

The active article proves its positive-density trace inequality and its shrinking-cutoff envelope in full. Its inherited inputs are fixed here by their theorem labels in this complete archive. Page numbers refer to this supplement, not to an earlier PDF.

The actual mixed moment and its finite-count remainder are Theorem 4.2, page 198. The Borel segment bijection and exact density are Theorem 7.1, page 214. The physical angular-density variation is Lemma 8.1, page 234.

The complete surface theorem with a fixed positive residence cutoff is Theorem 8.2, page 218. Its leading constant is cutoff independent; its assertion does not set the cutoff to zero. The common physical kernel is Proposition A.1, page 203.

The original positive scalar remainder is Theorem 181.4, page 678. The active offset-variable correction is equation (6.1), page 248; the unaltered historical module 189 is explicitly marked superseded at its inclusion.

The original weighted source on initial criticality or zero total residence remains a positive contribution in the active article. No input listed here asserts that its concentration charge is zero.

Revision 88: active inputs and proof locations

The active article proves the physical residence cap, its capped negative moment, the whole-segment weighted trace, the interval-pulse inequality and the raw local-norm consequence in full. The following inherited statements are the precise interfaces used in those proofs.

The physical layer geometry is Lemma 178.1, page 670; the finite-count exit bound is Theorem 5.1, page 246. The actual fractional volume source is in Theorem 26.2, page 302.

The maximal-segment Borel bijection is Theorem 7.1, page 214; the full angular variation is Lemma 8.1, page 234. The actual two-sided first-exit comparison is Theorem 5.2, page 246. Long-residence essential height is Theorem 7.2, page 233.

The original scalar remainder is Theorem 181.4, page 678. Its remaining original-source pointwise criterion is Theorem 34.2, page 189. The active offset formula is (6.1); the notice at historical module 189 remains in place. None of the new finite-norm conclusions promotes these remaining essential-height charges to zero.

1. The critical-distribution dependency map

The two revision-88 variants have different modules 205–207. This archive retains the weak-local-norm variant. The complete parallel optical-moment variant is separately preserved with its exact source tree; no theorem is silently substituted between the two variants.

The indispensable chain for the new critical bound is

$$\begin{gathered}\text{physical cap and finite-count short-exit estimate}\\ \Downarrow\\ \text{capped residence distribution} \quad + \quad \text{whole-segment optical trace}\\ \Downarrow \quad \text{retain } \min\{|E|,\ell\} \text{ before summation}\\ \text{critical weak bound for the complete first-exit surface}\\ \Downarrow \quad \text{original source shift at } W^1=1\\ \text{critical weak bound for the original raw density}\\ \Downarrow \quad \text{local mass convergence}\\ \text{critical logarithmic convergence and diffusive-window limits.}\end{gathered}$$

The first two inputs are retained below as Theorems 25.2 and 26.2; the capped distribution itself is equation (25.2). The full Borel segment map and its physical density are stated in the retained article and traced to Theorem 7.1. The fractional source is the actual physical layer-cake input in equations (24.7) and (24.8). The source shift and scalar reconstruction appear in equations (24.3) and (24.9).

The new journal proves the dyadic endpoint sum, its finite-count capped remainder, the logarithmic mass interpolation, and the exact rescaling factors directly. None of these inputs includes a pointwise bound for the original exceptional source. The old module-189 notation remains explicitly superseded at its own inclusion, without any alteration of its archived bytes.

Part 1. Positive angular intervals and paid critical arms

Retained revision-104 abstract. We study the exact-label pointwise raw local limit problem for actual returns in a triangular Lorentz gas. The remaining upper source is the positive minimum of two weighted endpoint traces on short regular segments. We prove an angular-interval comparison for this same positive measure. The comparison uses endpoint order, integrated trace balance and physical overlap, but no reciprocal endpoint velocity. On regular backward-critical arms it gives an essential-height bound through stationary outgoing roofs, uniformly in the relative position inside the segment. An integral comparison first isolates any shortage of outgoing trace. On these critical arms its entire shortage is then paid by the actual uncut optical volume source, without an outgoing trace-fraction lower bound. The recovered source has height $O(B^{-1/192})$ for uniformly balanced arms, or $O(B^{-1/384})$ under a growing balance budget. An exact positive decomposition retains uncovered physical pairs; no complete coverage theorem is inferred. The original pointwise assertion remains unchanged; decay of that complete deficit is not asserted.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge a I_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Comparison at a stationary endpoint.** The previous endpoint-distance flows compare nearby trajectories of the same exact event. Their paired coarea densities are $c^\varepsilon/|a'|$ and $c^\varepsilon/|b'|$, where c^ε is the positive minimum of the weighted traces at the two ends of a maximal regular segment. These formulas remove the relative-position degeneracy but still require nonzero endpoint speed along a useful comparison path.

The argument below uses intervals of actual angular trajectories instead of a single comparison path. If all trajectories in a comparison interval contain a one-sided roof translate of every source trajectory, the integral of their positive trace controls the source at its original roof. The ratio of the two integrated traces and the physical overlap are the only numerical costs. Stationary endpoint velocity and rapid local variation of the minimum do not enter that ratio. Theorem 5.2 verifies it on critical arms with an explicit level-width condition and the original uncut optical variation. Theorem efthm:v104-paid-arms also pays the entire comparison-trace shortage. Its

whole-family height bound includes all relative positions, without a lower original residence cutoff.

The pointwise problem is still (2.5). The new theorems estimate specified positive source families at that topology, not a replacement finite-norm problem. Theorem 6.1 retains a nonnegative remainder with a precise comparison-trace deficit. Its vanishing on the complete physical family remains the last requirement in the resulting scalar reduction.

3. The inherited physical source in the required form

Fix the radius, collision count and exact displacement and occupation labels. On the disjoint original space of regular pairing sheets, write $a(\theta) < u < b(\theta)$ for a maximal roof segment. On the regular set away from $g = 0$, put

$$q = (\alpha_0, \alpha_m), \quad F = L, \quad g = \nabla_q F = R(-p_0, p_m), \quad z = |g|, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i.$$

Here $\omega = |\partial_{p_0}\alpha_m|^{-1}$ and $\chi = |\det D_q(F, \arg g)|$. On the regular set away from $g = 0$, put

$$q = (\alpha_0, \alpha_m), \quad F = L, \quad g = \nabla_q F = R(-p_0, p_m), \quad z = |g|, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i.$$

Here $\omega = |\partial_{p_0}\alpha_m|^{-1}$ and $\chi = |\det D_q(F, \arg g)|$. The physical angular density is

$$A(u,\theta) = \frac{\omega(u,\theta)}{4\pi c_* \chi(u,\theta)}, \qquad f^\varepsilon = AW^\varepsilon, \qquad c^\varepsilon(\theta) = \min\{f^\varepsilon(a+,\theta), f^\varepsilon(b-,\theta)\}.$$

The initial section visit is counted and the terminal section visit is tested separately, as in (2.1). Every angular sheet carries this same exact event; its incident physical component is part of its identity.

Input statement 3.1 (Positive pairing and local concentration)**.** The original paired measure is $c^\varepsilon(\theta)\, d\theta\, du$ on $a < u < b$. For $0 < \Delta$ let $\mathsf{T}^\varepsilon_\Delta$ be its roof density restricted to $0 < \ell = b - a < \Delta$, without an additional selector. At every previously fixed $r, \Delta, \varepsilon > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r \mathsf{T}^\varepsilon_\Delta\|_\infty \le C\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\, du. \tag{3.1}$$

The leading constant is independent of the previously fixed widths. For fixed m, R, n, k, all nonnegative angular restrictions use the same coarea kernels outside a roof-null set. Countable selections use common versions. The uniform order is $\sup_R \operatorname{ess\,sup}_t$; one universal null set for all radii is not asserted.

This is the positive barycentric source and complete paired-average estimate of Supplement Theorem 4.2 and equation (5.1). The paired-average estimate follows there from the original weighted source and the actual uncut inverse-gradient volume source. It is not an assumed essential-height estimate for the paired source. The same inherited analytic input gives the actual uncut source

$$v^0(t)\, dt = F_\#(I_m z^{-1} \mathbf{1}_\Pi \nu^*_R), \qquad \limsup_m \sup_{R,n,k} m^2 \|A_\rho v^0\|_\infty \le C \tag{3.2}$$

for every previously fixed $\rho > 0$, with a width-independent leading constant. It is finite at fixed count. On each regular segment,

$$|\partial_u f^\varepsilon(u,\theta)| \le C(1+\varepsilon^{-1}) A(u,\theta) I_m(u,\theta)/z(u,\theta). \tag{3.3}$$

These are the actual uncut volume and weighted optical variation statements in Supplement Input statement 3.2 and its physical input. No unweighted pointwise height bound for v^0 is assumed. The bound is used only on a previously fixed positive window after the physical sum.

The Borel physical sheet summation retains multiplicity one; its historical proof and both endpoint trace conventions are unchanged.

Input statement 3.2 (Regular critical entrance and scalar reduction)**.** At a regular nongrazing backward-critical extension q_*, the incoming roof is constant and

$$f^{\varepsilon}(a+,\theta)=\frac{N_*}{e_\perp(\theta)^{\mathsf T}K_*e_\perp(\theta)},\quad N_*=\frac{\omega(q_*)W^{\varepsilon}(q_*)}{4\pi c_*},\quad \lambda_\circ I\le K_*\le \Lambda_\circ I, \tag{3.4}$$

where $\lambda_\circ=9\sqrt{19}/200$ and $\Lambda_\circ=47/6$. The incoming trace can be positive. A singular stationary extension is not included in this formula merely because its incoming roof is critical.

Under the original outer schedule

$$\varepsilon(B)=A_0B^{-1/12},\qquad q_0^*=1/8,\qquad r_0=c_0\varepsilon(B)^4,\qquad \delta=6q_0^*r_0, \tag{3.5}$$

the preceding positive remainder $\mathsf U_B^{\mathrm{rem}}$ is the original short paired source on failure of the relative clock and both endpoint-distance clocks. With

$$\mathcal H_M(T)=\limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal C_{m,R,n,k}(M)} T(t),$$

one has

$$\mathcal E_M\le \mathcal H_M(\mathsf U_B^{\mathrm{rem}})+C_MB^{-1/384}. \tag{3.6}$$

Along this same schedule, the original pointwise assertion is equivalent to $\mathcal H_M(\mathsf U_B^{\mathrm{rem}})\to 0$.

The trace formula is Supplement Proposition 7.2; the scalar statement is Supplement Theorem 8.2. The earlier relative and endpoint-distance comparisons remain available and unchanged. The new comparison is applied specifically to their positive failure source below.

4. Positive comparison without an endpoint velocity

We work on the original disjoint space of physical pairing sheets. An angular point on that space is denoted by ξ; its measure is $d\theta$, its endpoint roofs are $a(\xi)<b(\xi)$, and

$$c(\xi)=\min\{f^{\varepsilon}(a+,\xi),f^{\varepsilon}(b-,\xi)\},\qquad \ell(\xi)=b(\xi)-a(\xi).$$

The suppressed superscript on c is always the original ε. In these coordinates the positive paired measure is $c(\xi)\,d\theta\,du$ on $a(\xi)<u<b(\xi)$. In particular

$$\mathsf T_\Delta^{\varepsilon}(t)=\int c(\xi)\mathbf 1_{\{0<\ell(\xi)<\Delta\}}\mathbf 1_{\{a(\xi)<t<b(\xi)\}}\,d\theta. \tag{4.1}$$

All sheets carry the same fixed exact event. No occupation class is pooled, and coincident physical faces retain their incident component.

Fix $0<\delta<\Delta$ and $r>0$. A comparison pair consists of Borel angular sets I_j,J_j on an actual physical sheet, and a sign $\tau_j\in\{-1,1\}$. Its source has $0<\ell\le\delta$ on I_j; its comparison set has $0<\ell<\Delta$ on J_j. For almost every $(\xi,\zeta)\in I_j\times J_j$ assume the corresponding inclusion

$$\begin{aligned}\tau_j=1:&\qquad a(\zeta)\le a(\xi),\quad b(\zeta)\ge b(\xi)+r,\\ \tau_j=-1:&\qquad a(\zeta)\le a(\xi)-r,\quad b(\zeta)\ge b(\xi).\end{aligned} \tag{4.2}$$

The sets I_j are disjoint in the physical angular space. Suppose also that the comparison sets have overlap at most N:

$$\sum_j \mathbf 1_{J_j}(\xi)\le N\quad\text{for almost every }\xi. \tag{4.3}$$

This is a condition on physical angular sets, not on the number of words. For disjoint angular arms it holds with $N = 1$. Write $C_j = \int_{I_j} c\,d\theta$ and $D_j = \int_{J_j} c\,d\theta$, and assume these numbers are finite. Zero values are allowed.

Theorem 4.1 (A positive angular-interval comparison)**.** *Let $H(\xi, t)$ be any original Borel selector with $0 \le H \le 1$, and let $\kappa > 0$. Define*

$$\alpha_j = \begin{cases} \min\{1, \kappa D_j / C_j\}, & C_j > 0, \\ 0, & C_j = 0, \end{cases} \qquad T^{\varepsilon}_{H,j}(t) = \int_{I_j} c(\xi) H(\xi, t) \mathbf{1}_{\{a(\xi) < t < b(\xi)\}}\, d\theta.$$

At every finite collision count and for almost every target roof,

$$\sum_j \alpha_j T^{\varepsilon}_{H,j}(t) \le \frac{N\kappa}{r} \int_{t-r}^{t+r} \mathsf{T}^{\varepsilon}_{\Delta}(u)\, du. \tag{4.4}$$

The remaining source is exactly $\sum_j (1 - \alpha_j) T^{\varepsilon}_{H,j}$, and satisfies

$$0 \le \sum_j (1 - \alpha_j) T^{\varepsilon}_{H,j}(t) \le \sum_j (C_j - \kappa D_j)_+ \mathbf{1}_{\{T^{\varepsilon}_{1,j}(t) > 0\}}. \tag{4.5}$$

If $C_j \le \kappa D_j$ whenever $C_j > 0$, then the left side of (4.4) *is the complete selected source on the union of the I_j.*

Proof. Fix j and a roof t for which $T^{\varepsilon}_{H,j}(t) > 0$. The set of source angles with $a(\xi) < t < b(\xi)$ and positive selected weight then has positive angular measure. By Fubini, (4.2) holds between some such source angle and almost every comparison angle. If $\tau_j = 1$, every $u \in (t, t + r)$ belongs to the corresponding comparison segment; if $\tau_j = -1$, the same assertion holds for $u \in (t - r, t)$. Writing $T_{J_j}(u) = \int_{J_j} c \mathbf{1}_{\{a<u<b\}}\, d\theta$, we obtain

$$r D_j \le \int_{t + \tau_j (0,r)} T_{J_j}(u)\, du, \qquad \alpha_j T^{\varepsilon}_{H,j}(t) \le \alpha_j C_j \le \kappa D_j.$$

When the selected source is zero the desired comparison is immediate. Thus each summand is bounded by the corresponding one-sided integral multiplied by κ/r. All such intervals lie in $(t - r, t + r)$. Tonelli and (4.3) give $\sum_j T_{J_j}(u) \le N \mathsf{T}^{\varepsilon}_{\Delta}(u)$ and prove (4.4). The remaining source is positive because $0 \le \alpha_j \le 1$. Since $(1 - \alpha_j) C_j = (C_j - \kappa D_j)_+$ for $C_j > 0$, its displayed upper bound follows. The case $C_j = 0$ contributes nothing. Countable sums follow from monotone convergence of nonnegative finite sums. No collision limsup has been taken. □

This proof compares integrals of the original positive endpoint minimum. It neither differentiates that minimum nor divides by a', b' or $(1 - s)a' + sb'$. There is no flow through a stationary point and no assertion that two prescribed trajectories are joined by an admissible clock. The selector remains at the starting roof; only its positive comparison source is enlarged.

Corollary 4.2 (The physically summed essential height)**.** *Suppose N and κ are uniform in collision count, radius and exact labels. Under* (3.1)*,*

$$\mathcal{H}_M \left(\sum_j \alpha_j T^{\varepsilon}_{H,j} \right) \le CN\kappa \{ \varepsilon^{1/16} + (1 + \varepsilon^{-1}) \Delta \}. \tag{4.6}$$

All widths and constants in this statement are fixed before the collision limit. The estimate permits any number of source intervals and comparison intervals, subject to the stated overlap.

Proof. The right side of (4.4) is $2N\kappa A_r \mathsf{T}^{\varepsilon}_{\Delta}(t)$. Apply the complete paired-average estimate only after the finite-count sum and the pointwise comparison have been formed. □

5. Stationary endpoint roofs at a positive critical entrance

The preceding comparison can be verified on a class for which the endpoint speed has no positive lower bound. The condition on the outgoing trace is an integral condition. Neither a derivative of that trace nor a pointwise lower bound for the positive minimum is used.

Consider an oriented angular arm $x = \sigma(\theta - \theta_*) > 0$, $\sigma \in \{-1, 1\}$, on one actual pairing sheet with a regular nongrazing critical entrance. The incoming roof is a_* and, by (3.4), its weighted trace is

$$f_-^\varepsilon(x) = \frac{N_*}{Q_*(x)}, \qquad \lambda_\circ \le Q_*(x) \le \Lambda_\circ, \qquad N_* \ge 0. \tag{5.1}$$

Suppose the outgoing roof on this arm is

$$b(x) = a_* + d + \psi(x), \qquad 0 \le d \le \delta/2, \tag{5.2}$$

where ψ is continuous, strictly increasing, positive for $x > 0$, and tends to zero at the left endpoint. Assume that its levels $\delta/2, 2\delta, 3\delta$ are reached at points $x_1 < x_2 < x_3$ inside this same physical arm. Set

$$I = (0, x_1), \qquad J = (x_2, x_3).$$

The existence of x_3 is an actual angular-reach condition. It is not inferred from an algebraic continuation beyond a true endpoint.

If $N_* > 0$, put

$$w(x) = \min\{1, f_+^\varepsilon(x)/f_-^\varepsilon(x)\}, \qquad \mu_J = \frac{1}{|J|}\int_J w(x)\,dx. \tag{5.3}$$

Thus $c^\varepsilon = f_-^\varepsilon w$ exactly. The fraction may vanish on subsets of positive angular measure. If $N_* = 0$, the whole paired source on the arm is zero and is treated separately.

Lemma 5.1 (Width and positive-trace balance)**.** *For $N_* > 0$ and $\mu_J > 0$,*

$$\frac{\int_I c^\varepsilon dx}{\int_J c^\varepsilon dx} \le \frac{\Lambda_\circ}{\lambda_\circ \mu_J}\frac{x_1}{x_3 - x_2}. \tag{5.4}$$

If, in addition, ψ is locally absolutely continuous and

$$\frac{x\psi'(x)}{\psi(x)} \le K \quad \text{for almost every } x \in (x_2, x_3), \qquad K > 0, \tag{5.5}$$

then

$$\frac{\int_I c^\varepsilon dx}{\int_J c^\varepsilon dx} \le \frac{\Lambda_\circ K}{\lambda_\circ \mu_J \log(3/2)}. \tag{5.6}$$

No lower bound on ψ' or bound on $\partial_x \log \psi'$ is part of either assertion.

Proof. On I one has $c^\varepsilon \le N_*/\lambda_\circ$; on J, $c^\varepsilon \ge N_* w/\Lambda_\circ$. Integration proves (5.4). Since $\psi \ge 2\delta$ on $[x_2, x_3]$, its logarithm is absolutely continuous there. Hence

$$\log(3/2) = \int_{x_2}^{x_3} \frac{\psi'}{\psi}\,dx \le K \log(x_3/x_2).$$

It follows that

$$x_3 - x_2 \ge x_2\big((3/2)^{1/K} - 1\big) \ge x_1 \log(3/2)/K.$$

Substitution gives (5.6). The numerator N_* cancels; no positive lower bound for its size is required. □

Theorem 5.2 (A stationary-arm essential-height estimate)**.** *Take any disjoint family of actual angular arms with* (5.1)–(5.2)*. Suppose their comparison intervals satisfy either*

$$\frac{\Lambda_\circ}{\lambda_\circ \mu_J}\frac{x_1}{x_3-x_2} \le \kappa,$$

or the sufficient uniform bounds $K \le K_0$ and $\mu_J \ge \mu_0 > 0$ in Lemma 5.1. Let $T^{\varepsilon}_{H,\mathrm{arm}}$ be any original selected paired source on the union of their source intervals I. At finite collision count,

$$T^{\varepsilon}_{H,\mathrm{arm}}(t) \le \frac{\kappa}{\delta}\int_t^{t+\delta} \mathsf{T}^{\varepsilon}_{4\delta}(u)\,du \quad \text{for almost every } t. \tag{5.7}$$

Consequently

$$\mathcal{H}_M(T^{\varepsilon}_{H,\mathrm{arm}}) \le C\kappa\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\delta\}, \qquad \kappa = \frac{\Lambda_\circ K_0}{\lambda_\circ \mu_0 \log(3/2)} \tag{5.8}$$

in the sufficient case. Every original relative position $0 < s < 1$ is included. The family may have arbitrarily many physical words.

Proof. For a source angle $x \in I$ the residence is at most $d + \delta/2 \le \delta$. For a comparison angle $y \in J$, it is less than $d + 3\delta \le 7\delta/2 < 4\delta$. Their incoming roofs are equal, and

$$b(y) - b(x) > 2\delta - \delta/2 = 3\delta/2 > \delta.$$

Thus the positive inclusion (4.2) holds with $r = \delta$. Lemma 5.1 proves the mass balance. Different arms are disjoint in the original physical angular space; therefore their comparison intervals have overlap one, even if geometrical boundary points coincide on different incident sheets. The proof of Theorem 4.1, with every sign positive, gives (5.7). Its right side is at most $2\kappa A_\delta \mathsf{T}^{\varepsilon}_{4\delta}$. The inherited paired-average estimate proves (5.8). Arms with $N_* = 0$ have no positive minimum and add zero. □

The arm assumption, including the weighted fraction in (5.3), is imposed on actual endpoint traces. It is weaker than a positive pointwise lower bound or logarithmic variation bound on their minimum. Failure of this average condition is quantified by the positive deficit in (4.5), not assigned zero mass.

5.1. **Paying a deficient comparison trace by the original volume.** The average fraction in (5.3) need not be bounded below when the original weighted trace variation is available. Its possible loss has a direct pointwise payment. Retain the actual uncut source v^0 from (3.2).

Theorem 5.3 (Stationary-arm height without a trace-fraction hypothesis)**.** *On a disjoint family of the actual critical arms above, assume only* (5.1)–(5.2) *and a uniform geometric width ratio*

$$\frac{\Lambda_\circ}{\lambda_\circ}\frac{x_1}{x_3-x_2} \le \kappa.$$

There is no lower bound on μ_J. For every original selector $0 \le H \le 1$, at finite count and almost every target roof,

$$\begin{aligned} T^{\varepsilon}_{H,\mathrm{arm}}(t) \le \frac{\kappa}{\delta}\int_t^{t+\delta} \mathsf{T}^{\varepsilon}_{4\delta}(u)\,du \\ + C\kappa(1+\varepsilon^{-1})\int_{t-\delta}^{t+4\delta} v^0(u)\,du. \end{aligned} \tag{5.9}$$

The positive comparison deficit itself obeys

$$\sum_j (1-\alpha_j) T^{\varepsilon}_{H,j}(t) \le C\kappa(1+\varepsilon^{-1})\int_{t-\delta}^{t+4\delta} v^0(u)\,du. \tag{5.10}$$

Consequently the entire selected arm source satisfies

$$\mathcal{H}_M(T^{\varepsilon}_{H,\mathrm{arm}}) \le C\kappa\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\delta\}. \tag{5.11}$$

The sufficient profile condition (5.5) *gives* $\kappa = \Lambda_\circ K_0/(\lambda_\circ \log(3/2))$ *for* $K \le K_0$. *No source endpoint-speed, incidence, residence or relative-position lower cutoff is imposed.*

Proof. For one arm set

$$P_j = \int_{J_j} f^{\varepsilon}_{-}(x)\,dx, \qquad E_j = \int_{J_j} [f^{\varepsilon}_{-}(x) - f^{\varepsilon}_{+}(x)]_+\,dx.$$

The positive minimum gives the exact identity $D_j = P_j - E_j$. The critical incoming formula and the geometric width ratio imply $C_j \le \kappa P_j$. Therefore

$$(C_j - \kappa D_j)_+ \le \kappa E_j. \tag{5.12}$$

This is valid even when $D_j = 0$. Arms with $N_* = 0$ again contribute zero; no division by a vanishing incoming trace is used.

The retained derivative bound for the original weighted density, integrated on the same comparison segment, yields

$$[f^{\varepsilon}_{-}(x) - f^{\varepsilon}_{+}(x)]_+ \le C(1+\varepsilon^{-1}) \int_{a_j}^{b(x)} A(u,x) I_m(u,x)/z(u,x)\,du.$$

One may first integrate on compact interior subintervals and pass to the one-sided traces. The uncut volume input and Tonelli justify the positive limiting integral. This pays a possible decrease of the trace; it does not assume that the two traces are comparable.

At a roof with $T^{\varepsilon}_{H,j}(t) > 0$, some original source segment on I_j contains t. Since every such segment has length at most δ and has the same incoming roof a_j, one has $t - \delta < a_j < t$. Each comparison segment on J_j has length less than 4δ. Its entire interior consequently lies in $(t-\delta, t+4\delta)$. The comparison angular sets are disjoint on the original physical sheet space. After their interiors are summed, the Borel segment identity bounds their combined variation by the integral of the same complete v^0 over this one roof window, with multiplicity one. Equation (5.12) and (4.5) prove (5.10). Adding the paid comparison in Theorem 4.1 gives (5.9). The first term uses only positive roof translates, since the incoming roof is constant on each arm.

Finally take the collision limsup with $\delta, \varepsilon, \kappa$ fixed. The paired-average input estimates the first integral. The uncut volume input estimates the second by $C\kappa(1+\varepsilon^{-1})\delta$. Auxiliary spectral bands are enlarged only afterwards. This proves (5.11). □

Thus poor outgoing trace balance on these arms is not assigned to an uncontrolled positive remainder: its entire shortage is estimated at the original target roof by (5.10). What still requires physical information is the level reach and width geometry, and coverage of the complete remaining paired family by such arms or other comparisons.

Corollary 5.4 (Finite-order stationary profiles)**.** *Suppose on an actual arm*

$$\psi(x) = \beta x^k(1+O(x^\eta)), \qquad \psi'(x) = k\beta x^{k-1}(1+O(x^\eta)), \qquad \beta, \eta > 0, \quad k \ge 1.$$

For all sufficiently small widths on that arm, (5.5) *holds with* $K = k+1$. *Provided the required physical reach holds, Theorem 5.3 applies also for* $k > 1$, *when the endpoint speed vanishes at the arm's initial point. No uniform bound on physical orders* k, *on the size of the asymptotic neighborhood or on comparison reach is inferred from finite-count definability.*

Proof. The displayed asymptotics give $x\psi'/\psi \to k$. The assertion then follows from Lemma 5.1 and Theorem 5.3. □

5.2. **Recovery can occur when every preceding clock is small.** This elementary pairing-sheet calculation distinguishes the new comparison from admission by any of the three previous flows. It is not asserted to be another physical Lorentz realization. Set $a = 0$, $b(x) = x^k$, $c = 1$, $k \geq 2$, and restrict the comparison collar to $0 < \ell < 4\delta$. The incoming-distance clock is zero. For the outgoing-distance flow at positive roof time h,

$$x_h = (x^k + h)^{1/k}, \qquad J_h^+ = \left(\frac{x}{x_h}\right)^{k-1}.$$

Its forward contribution for $0 < h < \delta$ is at most

$$\frac{kx^{k-1}}{\delta}\big((x^k + \delta)^{1/k} - x\big). \tag{5.13}$$

The negative-time contribution is at most x^k/δ, because the starting gap must remain smaller than the new residence. At fixed relative position s, the relative clock has total roof reach at most $4s\delta$, so its capped value is at most $4s$.

For any fixed $q > 0$, take $s < q/8$ and then take $x/\delta^{1/k}$ sufficiently small. Both terms in (5.13) and its negative-time bound tend to zero; all three old clocks are below q on a set of positive paired measure. Nevertheless the intervals

$$I = (0, (\delta/2)^{1/k}), \qquad J = ((2\delta)^{1/k}, (3\delta)^{1/k})$$

satisfy the positive comparison, with $C_I/D_J \leq k/\log(3/2)$. The new inequality controls their entire source, including that positive-measure clock-failure set. Its proof uses neither a velocity bound on I nor passage of a flow through $x = 0$.

6. The recovered source and the original pointwise theorem

Retain the original schedule in (3.5). Let H_B be exactly the original selector for the source which fails all three preceding clocks, so that $\mathsf{U}_B^{\mathrm{rem}} = T_{H_B}^{\varepsilon(B)}$. This selector includes the original short-source test and is supported on $\ell \leq \delta(B)$. It is not redefined by the present comparison.

Choose comparison pairs as in Theorem 4.1, with $r = \delta$, $\Delta = 4\delta$ and constants N_B, κ_B fixed before the collision limit. They may, for example, be the disjoint critical arms of Theorem 5.2. The exact split is

$$\begin{aligned}
\mathsf{V}_B^{\mathrm{int}}(t) &= \sum_j \alpha_j T_{H_B,j}^{\varepsilon(B)}(t), \\
\mathsf{R}_B^{\mathrm{int}}(t) &= T_{H_B \mathbf{1}_{(\cup_j I_j)^c}}^{\varepsilon(B)}(t) + \sum_j (1 - \alpha_j) T_{H_B,j}^{\varepsilon(B)}(t).
\end{aligned} \tag{6.1}$$

Hence $\mathsf{U}_B^{\mathrm{rem}} = \mathsf{V}_B^{\mathrm{int}} + \mathsf{R}_B^{\mathrm{int}}$ at finite count, with both terms nonnegative. The second sum is an explicitly weighted shortage of comparison trace, with upper bound (4.5). The uncovered source in the first term is retained in full.

Theorem 6.1 (Positive recovery at the original outer schedule)**.** *Suppose $N_B\kappa_B \leq C\varepsilon(B)^{-1/32}$ uniformly in collision count, radius and exact labels. Then*

$$\mathcal{H}_M(\mathsf{V}_B^{\mathrm{int}}) \leq C_M B^{-1/384}, \qquad \mathcal{E}_M \leq \mathcal{H}_M(\mathsf{R}_B^{\mathrm{int}}) + C_M B^{-1/384}. \tag{6.2}$$

If $N_B\kappa_B$ is bounded independently of B, the first bound improves to $C_M B^{-1/192}$. Along the same schedule the original pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{int}}) \to 0$.

Proof. Apply Corollary 4.2 to the exact starting selector H_B. Since $\delta = O(\varepsilon^4)$,

$$\mathcal{H}_M(\mathsf{V}_B^{\mathrm{int}}) \leq CN_B\kappa_B\{\varepsilon^{1/16} + (1 + \varepsilon^{-1})4\delta\} \leq C(\varepsilon^{1/32} + \varepsilon^{95/32}).$$

This is $O(B^{-1/384})$; for bounded $N_B\kappa_B$ the leading power is $\varepsilon^{1/16} = O(B^{-1/192})$. Insert the exact positive split (6.1) into the inherited scalar bound (3.6). This proves sufficiency. Conversely, the retained necessity statement gives $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{rem}}) \to 0$ under scalar closure. Since $0 \le \mathsf{R}_B^{\mathrm{int}} \le \mathsf{U}_B^{\mathrm{rem}}$, necessity follows. The complete physical sums precede the height functional in every step. □

For the critical-arm theorem a uniform K_0 and $\mu_0 > 0$ give the stronger rate for the entire selected family. More generally, any outer choice with $K_B/\mu_B \le C\varepsilon(B)^{-1/32}$ gives the first rate. This allows an expanding family of finite-order profiles, but is not a claim that their order or trace fraction is uniformly controlled on all physical pairs. Every such threshold is chosen from B, never from the collision count.

Corollary 6.2 (Removing the whole stationary-critical arm contribution)**.** *Suppose the comparison pairs are the disjoint actual critical arms of Theorem 5.3, with $\kappa_B \le C\varepsilon(B)^{-1/32}$. Then*

$$\mathcal{H}_M\left(\sum_j (1-\alpha_j) T_{H_B,j}^{\varepsilon(B)}\right) \le C\varepsilon(B)^{95/32}.$$

The complete original source on these arms, including its entire three-clock-failure part, has height $O(B^{-1/384})$. With a bounded geometric ratio the latter rate is $O(B^{-1/192})$. In particular, setting $\mathsf{U}_B^{\mathrm{outside}} = T_{H_B \mathbf{1}_{(\cup_j I_j)^c}}^{\varepsilon(B)}$, one has

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B^{\mathrm{outside}}) + C_M B^{-1/384}. \tag{6.3}$$

Along this schedule the original pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{outside}}) \to 0$.

Proof. Apply (5.10) on the original starting selector. The complete uncut volume estimate gives $C\kappa_B(1+\varepsilon^{-1})\delta = O(\varepsilon^{95/32})$. Adding this paid shortage to (6.2) removes the entire angular-arm source, not only a fraction admitted by trace balance. The exact decomposition into that source and its positive outside complement proves (6.3). Necessity follows from domination by the original $\mathsf{U}_B^{\mathrm{rem}}$, and sufficiency from the new estimate. No count-dependent width is chosen. □

6.1. **The remaining physical estimate.** The new comparison has an actual essential-height conclusion on the positive recovered source. It does not turn an average, a signed boundary current or a finite local norm into the target theorem. Stationary endpoint speed and rapid pointwise variation of the minimum are no longer obstructions on the interval families just estimated. The model calculation shows that this recovered family need not have been admitted by any earlier clock.

Two separate matters are still not proved on the complete Lorentz family: coverage by comparison intervals with a schedule-compatible physical overlap and mass balance, and vanishing of the uncovered source. On the critical arms of Theorem 5.3, the entire comparison-trace shortage is already paid by the original uncut volume. In particular a flat endpoint pulse on an entire isolated chamber need not have any comparison interval satisfying (4.2); its positive mass is not removed. Loss of physical continuation before the comparison levels, and singular critical extensions remain in the outside source. General comparison families without the critical incoming normalization may still have a trace shortage in $\mathsf{R}_B^{\mathrm{int}}$.

Thus (6.2) is not a proof that the full remaining source has vanishing height. The exact-label pointwise theorem remains the governing assertion (2.5); its full closure and the corresponding unrestricted point-conditioned consequences are not promoted. The original arithmetic zero classes remain in the unnormalized theorem, and scalar closure still precedes any denominator, same-roof ratio or unrestricted likelihood statement.

Appendix A. Proof dependencies and the complete remaining source

The original event and target are (2.1) and (2.5). The inherited inputs include the complete positive paired measure, its fixed-width concentration, actual uncut optical volume, weighted variation, regular critical trace and the original scalar schedule. Their historical proofs remain in the archive; no new spectral or pressure theorem is assumed.

Endpoint inclusion and integrated positive-trace balance give Theorem 4.1 at finite collision count. The critical incoming normalization and the width of one actual comparison interval give Lemma 5.1. If the outgoing trace is too small, Theorem 5.3 pays its entire shortage by the original uncut volume in one common roof window. The resulting height estimate needs no outgoing trace-fraction lower bound. Corollary 6.2 then removes the entire selected critical-arm contribution from the old three-clock failure. Physical overlap is counted on the disjoint angular sheet space, not by counting words or strata.

The finite-order condition is local to an actual arm. Its comparison levels must be reached before any true boundary, and its geometric ratio must meet the stated uniform or outer-budget bound. No uniform order, reach or complete physical coverage is inferred from finite-count definability. The stationary model checks the distinction from the old clocks; it is not an additional Lorentz realization.

The complete remaining source is positive and lies outside the recovered physical families. Vanishing of that source in the original essential-height norm is still required for the organizing theorem. The complete preceding article and all earlier mathematical cores are reproduced in the supplement, with the two revision-88 variants separately identified and preserved.

Part 2. Endpoint-distance coarea and positive recovery

Retained revision-103 abstract. We study the exact-label pointwise raw local limit problem for actual returns in a triangular Lorentz gas. The remaining upper source is the positive minimum of the two weighted endpoint traces on each short regular segment. For this same measure we construct two comparisons preserving distance in roof time from one endpoint, rather than relative position between the endpoints. Their coarea densities contain neither a residence factor nor a relative-position factor. An expanding-endpoint criterion therefore admits all relative positions, including approach to a positive backward-critical trace and to a fold of the relative-position map. After physical summation, the certified source has essential height $O(B^{-1/192})$ under the original outer schedule. The two clocks also recover part of the previous failure source with height $O(B^{-1/384})$. The residual which fails all three comparisons remains positive and is not proved to vanish. The full pointwise target, exact labels, original guard, arithmetic classes and preceding proofs are retained.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U=\{|\alpha-\pi/6|<3/50,\ |p|<3/20\},\qquad Y_R^*=Y_R\cup U,\qquad c_*=\frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*)=c_*$. We always use $\nu_R^*=c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L=L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R}=\sum_{j=0}^{m-1}\mathbf{1}_{Y_R^*}\circ T_R^j$. The event is

$$\Pi_{n,k,m,R}=Y_R^*\cap\{K^{\mathrm{c}}_{m,R}=k,\ A_{m,R}=n\}\cap T_R^{-m}Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j=\sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0\le\Theta\le1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0=47/53$, $d_j=\min(j,m-j)$ and $d_j'=\min(j,m-1-j)$, set

$$H^\varepsilon=\prod_{j=0}^{m}\Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right)\prod_{j=0}^{m-1}\Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right),\qquad W^\varepsilon=1-H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t=L_{\#}(W^\varepsilon\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^m B_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Two additional directions on the positive paired measure.** Write $a(\theta) < u < b(\theta)$ for an original maximal regular segment, $\ell = b - a$, and $c^\varepsilon = \min\{f^\varepsilon(a+), f^\varepsilon(b-)\}$. The earlier transverse comparison fixes $s = (u-a)/\ell$ and has speed $(1-s)a' + sb'$. At a backward critical entrance with $a' = 0$ this speed degenerates as $s \downarrow 0$; it may also vanish when the two endpoint derivatives have opposite signs.

We instead fix either $u - a$ or $b - u$. The exact paired densities in the two resulting roof coordinates are $c^\varepsilon/|a'|$ and $c^\varepsilon/|b'|$. The absence of ℓ and s is an identity of measures, not an asymptotic cancellation. A one-sided expansion of ℓ keeps every original interior position inside its segment, even when the starting residence is arbitrarily short. The comparison still moves between actual trajectories of the same exact event; it does not continue the radial orbit beyond its first exit or change any section decision.

Theorem 8.1 gives a finite-count pointwise comparison after physical sheets have been summed. Theorem 9.1 and Corollary 9.2 prove its uniformity in relative position, with the precise reach and trace-variation hypotheses. Theorem 8.2 pays the recovered positive source and retains the exact remainder. In particular a conditional endpoint certificate is not identified with admission of every physical pair.

The original barycentric comparison is stated first with its proof. It remains available unchanged and is combined with, not replaced by, the two endpoint-distance comparisons. The additional critical-trace calculation distinguishes a positive critical entrance from a genuinely grazing end. All clocks below use pullback Jacobians for the paired measure; their integral covariance is printed explicitly.

3. The original sources and the inputs used here

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar\nu = \mathbf{1}_\Pi \nu_R^*$ and let μ_t be its original roof kernels. The Borel maximal-segment parametrization gives $\Psi(\theta, u)$ for $a(\theta) < u < b(\theta)$, with

$$A(u,\theta) = \frac{\omega(\Psi(\theta,u))}{4\pi c_* \chi(\Psi(\theta,u))}, \quad \omega = |\partial_{p_0}\alpha_m|^{-1}, \quad \chi = |\det D_q(F, \arg g)|, \quad g = \nabla_q F.$$

Thus $d\bar\nu = A\,du\,d\theta$ after disjoint physical summation. Let $z = |g|$ and $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$. We use the following assertions with precisely their original scope.

Input statement 3.1 (Physical geometry and trace comparison)**.** The finite-count physical graph and section decisions are definable. The endpoint Hessian has uniform positive lower and upper bounds; for a fixed lifted itinerary the endpoint-momentum map is globally injective. The maximal-segment Borel parametrization is one-to-one, and its kernel identities retain incident components at coincident physical faces. For $f^\varepsilon = AW^\varepsilon$ the one-sided traces exist, and

$$|\partial_u f^\varepsilon| \le C(1 + \varepsilon^{-1}) A I_m / z.$$

The trace is zero at a genuinely grazing end, not necessarily at a backward critical end. If a Borel target selector H is supported on $0 < \ell = b - a \le \delta$, let

$$R_H^\varepsilon(t) = \int \mathbf{1}_H W^\varepsilon \, d\mu_t, \quad T_H^\varepsilon(t) = \sum \int \mathbf{1}_{\{a<t<b\}} \mathbf{1}_H(\Psi(\theta,t)) \min(f^\varepsilon(a+), f^\varepsilon(b-))\, d\theta.$$

Then, on the original kernels and at every finite count,

$$|R_H^\varepsilon(t) - T_H^\varepsilon(t)| \le C(1+\varepsilon^{-1}) \int_{t-\delta}^{t+\delta} v^0(u)\, du, \qquad v^0\, dt = F_\#(I_m z^{-1} \bar\nu). \tag{3.1}$$

The selector is held at the target; no trace of an artificial selector boundary is taken.

The Borel map is Supplement Theorem 7.1. The physical momentum uniqueness and trace comparison are Supplement Theorems 7.1 and 6.1. The first theorem's label is recorded also in the dependency table below. No independent recertification of the inherited continuum chain is implicit in its restatement here.

Input statement 3.2 (Uniform volume concentration and scalar reduction)**.** For every previously fixed $r, \varepsilon > 0$, with $A_r h(t) = (2r)^{-1} \int_{t-r}^{t+r} h(u)\, du$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad \limsup_m \sup_{R,n,k} m^2 \|A_r v^0\|_\infty \le C. \tag{3.2}$$

The constants are independent of those fixed widths. The actual uniform weighted momentum capacity and its geometric-pressure proof supply the second assertion without an additional capacity premise. The original positive scalar remainder is

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \varepsilon(B) = A_0 B^{-1/12}, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}.$$

Use the fixed radial clock threshold $q = 1/8$, radius $r = c\varepsilon(B)^4$, gradient threshold $a = \sqrt{qr}$, and short-residence threshold $\delta = 6qr$. The retained low-clock failure source has a disjoint short part S supported on $\ell \le \delta$. With $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t\in\mathcal{C}(M)} h(t)$,

$$\mathcal{E}_M \le \mathcal{H}_M(T_{\mathsf{S}}^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1+\varepsilon(B)^{-1})qr. \tag{3.3}$$

The original pointwise law is equivalent along this schedule to $\mathcal{H}_M(T_{\mathsf{S}}^{\varepsilon(B)}) \to 0$.

The uncut volume estimate is Supplement Theorem 11.3; the scalar bound is Supplement Theorem 12.2 together with its retained necessity argument. The original low-clock selector, including the comparison collar and the original exact-event lifetimes, is reproduced in that article; it is not differentiated or modified in the new argument.

Every auxiliary band is fixed before the collision limit and enlarged only afterwards, with B outermost. At fixed m, R, n, k, common roof kernels and joint roof/time kernels are used. Uniformity is $\sup_R \operatorname{ess\,sup}_t$, not a claim of one null set for all radii or all uncountable selector families. The new finite-count comparisons are measure identities before their uniform norms are taken.

4. A transverse coordinate for the positive paired source

Fix the collision count, radius and exact label. A maximal segment of that original event has roof interval $(a(\theta), b(\theta))$, length $\ell = b - a$, and physical density $A(u, \theta)$. Its paired weight is

$$c^\varepsilon(\theta) = \min\{f^\varepsilon(a+, \theta), f^\varepsilon(b-, \theta)\}, \qquad f^\varepsilon = AW^\varepsilon.$$

The minimum is taken on the same physical segment, before summation. A zero trace contributes no paired mass. In particular genuinely grazing ends contribute zero; a backward critical trace is retained.

Lemma 4.1 (Regular pairing chambers)**.** *At each fixed count and radius, the original paired measure admits angular chambers on which a, b are C^2 and c^ε is locally Lipschitz. These chambers represent all paired mass except for transverse-null strata. Their interiors are counted once. There is no assertion of a collision-uniform bound on their number or on their derivatives. At a backward critical entrance, a may be constant while c^ε is nonzero.*

Proof. Use the finite physical graph and its exact section decisions, not a partition of symbolic extensions. On each lifted itinerary the map from endpoint angles to g has positive Jacobian and is globally injective by the retained momentum uniqueness theorem. On finitely many polar charts with $z > 0$, its image is a definable set. Along one ray its connected components are the

original maximal segments: $\partial_z F = ze^{\mathsf{T}}K^{-1}e > 0$. Definable cell decomposition and one-variable piecewise differentiability therefore give finitely many angular intervals on which the component endpoints, their limiting contact data and their roof values are C^2, after further subdivision if necessary [9]. No estimate of the number of pieces is used.

At a nongrazing endpoint the stationary word has a smooth local extension; this is not physical continuation through a first-hit boundary. The clearance is the minimum of finitely many smooth functions locally. Thus the original smooth guard has locally Lipschitz endpoint values. The two angular density factors have the same property away from $g = 0$. At a critical incoming end the positive endpoint Hessian gives a local inverse in g at zero. The limiting angular factor is $e_\perp^{\mathsf{T}}K(q_*)e_\perp$, bounded below by $\lambda_\circ$, and the optical factor has its ordinary value at q_*. The resulting trace is locally Lipschitz in the direction e and need not vanish. This also gives the assertion in that case. Taking a minimum preserves local Lipschitz regularity.

The disjoint physical-segment parametrization assigns each component once. The finitely many angular subdivision points have zero transverse measure; their roof saturations have zero paired measure by Tonelli. Original Borel seams are only assignments. Smoothly compatible chamber charts can be continued across such seams. Where pairing regularity actually ends, the transverse comparison below is stopped, and its failure source is retained. No part of the original event is removed by that stopping convention. □

On a chamber put $0 < s < 1$ and set

$$\mathscr{F}(\theta, s) = a(\theta) + s\ell(\theta), \qquad v(\theta, s) = \partial_\theta \mathscr{F} = (1 - s)a'(\theta) + sb'(\theta). \tag{4.1}$$

Thus s is the relative position between the two original ends. Define a positive measure on the chamber by

$$d\pi^\varepsilon = c^\varepsilon(\theta)\ell(\theta)\, d\theta\, ds. \tag{4.2}$$

It is a representation of the paired source, not a new collision probability or a redefinition of the guard.

Theorem 4.2 (Exact paired coarea and transverse Jacobian)**.** *For any original Borel target selector H, the roof pushforward of $\mathbf{1}_H\pi^\varepsilon$ has density T_H^ε. Here H is evaluated at $\Psi(\theta, \mathscr{F}(\theta, s))$. On the open regular region $c^\varepsilon > 0$, $v \neq 0$, define*

$$V^\perp = v^{-1}\partial_\theta, \qquad \varrho^\varepsilon = \frac{c^\varepsilon \ell}{|v|}. \tag{4.3}$$

The partial flow $\Theta_h^\perp$ obeys $\mathscr{F} \circ \Theta_h^\perp = \mathscr{F} + h$ and preserves s. Its Jacobian relative to (4.2) *is*

$$J_h^\perp(y) = \frac{\varrho^\varepsilon(\Theta_h^\perp y)}{\varrho^\varepsilon(y)}. \tag{4.4}$$

Consequently its target-direction coarea covariance holds jointly for almost every (t, h) and all nonnegative Borel integrands, on the same stopped survivor graph. Physical sheets are summed with multiplicity one.

Proof. For any nonnegative test ϕ the change of variable $u = a + s\ell$ gives

$$\sum \int \int_0^1 \phi(\mathscr{F}) c^\varepsilon \ell \mathbf{1}_H \, ds\, d\theta = \sum \int \int_a^b \phi(u) c^\varepsilon \mathbf{1}_H(\Psi(\theta, u))\, du\, d\theta.$$

This is precisely the defining paired source, including the target selector. It proves the first assertion also across $v = 0$.

On $v \neq 0$ use $(t, s) = (\mathscr{F}, s)$ as local coordinates. The measure is $\varrho^\varepsilon(t, s)\, dt\, ds$ and the new flow is translation in t. This proves (4.4), including negative times. Equivalently, at fixed s the

angular derivative of the flow is $v(y)/v(\Theta_h^\perp y)$; multiplying by the ratio of the weights $c^\varepsilon \ell$ gives the same positive Jacobian. The absolute value on v cannot be omitted.

Uniqueness for the C^1 field makes the stopped graph one-to-one. Local changes of variables, a countable determining class and Tonelli give joint coarea kernels and the stated covariance. The same argument is valid for graph selectors depending on (y, h). Compatible coordinate seams do not create multiple copies of a physical segment. The construction never moves a state to another occupation or displacement class. It is a transverse comparison between actual trajectories, not the original dynamical roof flow. The set $v = 0$ is assigned to comparison failure whenever it carries paired mass; no null-set assertion about that entire set is needed. □

5. A pointwise estimate of the positive paired part

Write $\mathsf{T}^\varepsilon_\Delta$ for the complete paired source on $0 < \ell < \Delta$, with no target selection beyond the original exact event. All incoming endpoint types are included. The preceding uncut trace comparison and source estimates imply

$$\limsup_m \sup_{R,n,k} m^2 \|A_r \mathsf{T}^\varepsilon_\Delta\|_\infty \le C\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\} \quad (r, \Delta, \varepsilon > 0 \text{ fixed}). \tag{5.1}$$

The constant is independent of these previously fixed widths. Indeed $\mathsf{T}^\varepsilon_\Delta \le b^\varepsilon + C(1+\varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0(u)\, du$. Apply the fixed-width bounds to $A_r b^\varepsilon$ and $A_r v^0$; translation invariance of the latter supremum pays the convolution with the factor 2Δ. This argument uses neither a pointwise bound on v^0 nor a boundary spectral insertion.

Fix $0 < \delta < \Delta$ and use $\Theta^\perp$ on

$$\mathcal{D}^\varepsilon_\Delta = \{0 < \ell < \Delta,\ c^\varepsilon > 0,\ v \ne 0\}$$

inside the regular pairing chambers. Stop at the boundary of this comparison domain or at an actual loss of chamber regularity. The original trace outside that domain is retained, not discarded. For an interior starting pair y define the capped transverse clock

$$k^\perp_{r,\Delta}(y) = \frac{1}{r} \int_{-r}^{r} \mathbf{1}_{\{\Theta_h^\perp y \text{ exists}\}} \min\{1, J_h^\perp(y)\}\, dh. \tag{5.2}$$

It is zero on all other starting pairs. Its range is $[0, 2]$ and it is Borel. The trace weight already contains the original guard at both endpoints; there is no extra guard mismatch in this clock.

For a Borel selector H supported on $\ell \le \delta$ and $0 < q_\perp \le 1$, let $T^\varepsilon_{H,\mathrm{adm}}$ and $T^\varepsilon_{H,\mathrm{fail}}$ be its paired source restricted, respectively, to $k^\perp_{r,\Delta} \ge q_\perp$ and $k^\perp_{r,\Delta} < q_\perp$. Their nonnegative sum is exactly T^ε_H, including every nonregular or zero-speed pair.

Theorem 5.1 (Complete transverse-clock trace estimate)**.** *At every finite count and almost every target roof,*

$$T^\varepsilon_{H,\mathrm{adm}}(t) \le \frac{1}{q_\perp r} \int_{t-r}^{t+r} \mathsf{T}^\varepsilon_\Delta(u)\, du. \tag{5.3}$$

Consequently, uniformly in the original radius and exact labels,

$$\mathcal{H}_M(T^\varepsilon_{H,\mathrm{adm}}) \le \frac{C}{q_\perp}\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{5.4}$$

This is an essential-height estimate for the positive common trace itself. It imposes no positive lower original residence, gradient or internal incidence cutoff on its admitted part. No estimate of an outgoing-minus-incoming difference is substituted for it.

Proof. Let π_t^{ε} be the paired kernels supplied by Theorem 4.2. On admitted starting pairs, $q_{\perp} \le k_{r,\Delta}^{\perp}$. Integrate this inequality with the target selector held at t. For each fixed comparison time,

$$\int \mathbf{1}_H \mathbf{1}_{\mathrm{adm}} \mathbf{1}_{\text{survives } h} \min(1, J_h^{\perp})\, d\pi_t^{\varepsilon} \le \int \mathbf{1}_H \mathbf{1}_{\mathrm{adm}} \mathbf{1}_{\text{survives } h} J_h^{\perp}\, d\pi_t^{\varepsilon} \le \mathsf{T}_{\Delta}^{\varepsilon}(t+h).$$

The last inequality is target-direction coarea covariance on the one-to-one graph. It is valid after all physical sheets are summed. The transported target need not still satisfy H; it does belong to the identical exact event and has total residence below Δ. Thus only the selector is enlarged in a positive upper comparison, not the displacement, occupation, collision count or section tests. Integrate over $-r < h < r$ and divide by $q_{\perp} r$. Joint almost-everywhere covariance and Tonelli prove (5.3) without selecting a fiber version separately at every time. Equation (5.1) then proves (5.4), with all parameters fixed before the collision limit and auxiliary bands enlarged only after it. □

Remark 5.2 (What has and has not been averaged). The left side of (5.3) is at the original target roof; only its upper comparison is averaged. The clock changes angular direction while preserving relative position between the same two physical ends. It is therefore different from the original radial clock, whose trajectory can terminate inside a short segment. The Jacobian is that of the positive paired measure, not that of the original volume with the minimum suppressed. No word, chamber-count or transverse-overlap constant is inserted. The unadmitted part is a genuine positive remainder and is not asserted small by this theorem.

6. Two-ended certificates and the remaining scalar source

The clock has an explicit geometric sufficient test. In particular, short original residence is not by itself a reason for transverse failure. The test also states which additional control is still required on the full paired family.

Theorem 6.1 (A relative two-ended transverse certificate)**.** *Fix a starting pair (θ_0, s) with $0 < \ell(\theta_0) \le \delta$ and $c^{\varepsilon}(\theta_0) > 0$. Suppose that in at least one angular direction there is an interval of length $L_\theta > 0$ in the same regular pairing chamber on which, at this fixed relative coordinate,*

$$|v| \ge \nu > 0, \qquad |\partial_\theta \log \ell| \le A, \qquad |\partial_\theta \log c^{\varepsilon}| \le A, \qquad |\partial_\theta \log |v|| \le A \tag{6.1}$$

almost everywhere. Positive ℓ, c^{ε} and the sign of v persist on this interval. If

$$0 < r \le \nu \min\{L_\theta, \log 2/(3A)\}, \tag{6.2}$$

with the second term interpreted as infinity when $A = 0$, then $k_{r,2\delta}^{\perp}(\theta_0, s) \ge 1/2$. The conclusion has no lower bound on $\ell(\theta_0)$.

A sufficient condition for the bounds on v, simultaneously for all $s \in (0,1)$, is that a', b' have the same sign, $|a'|, |b'| \ge \nu$, and $|a''| \le A|a'|$, $|b''| \le A|b'|$ on that interval.

Proof. Choose the roof direction corresponding to the supplied angular side and the sign of v. Since $d\mathscr{F}/d\theta = v$, every roof displacement of size at most r is reached before an angular displacement of r/ν. Positivity and the logarithmic bounds give

$$\ell(\theta) \le e^{Ar/\nu} \ell(\theta_0) \le 2^{1/3}\delta < 2\delta, \qquad \frac{\varrho^{\varepsilon}(\theta, s)}{\varrho^{\varepsilon}(\theta_0, s)} \ge e^{-3Ar/\nu} \ge \frac{1}{2}.$$

Thus the entire one-sided comparison interval exists in $\mathcal{D}_{2\delta}^{\varepsilon}$, and its contribution to (5.2) is at least $1/2$. A statement at an interval endpoint is obtained from smaller times and monotone convergence. The last assertion follows because $v = (1-s)a' + sb'$ is a convex combination of two derivatives of the same sign and $|\partial_\theta v| \le A((1-s)|a'| + s|b'|) = A|v|$. □

For example, on an actual chamber satisfying $a = a_0 + v_0\theta$, $b = a + \ell_0$ and $c^\varepsilon = c_0 e^{\beta\theta}$, the Jacobian is exactly $\exp(\beta h/v_0)$ and the admissible radius is independent of $\ell_0 > 0$. This calculation is a test of the physical-chamber hypotheses, not an alternative realization of the Lorentz problem. A lower residence cutoff would miss precisely this uniformity. Conversely, at a backward critical entrance $a' = 0$ one has $v = sb'$; the theorem can apply for interior s, but gives no uniform conclusion as s tends to zero. That possible positive critical trace remains in the failure source.

Keep the original schedule from the preceding scalar theorem:

$$\varepsilon = \varepsilon(B) = A_0 B^{-1/12}, \quad q_0^* = 1/8, \quad r_0 = c\varepsilon^4, \quad a_0 = \sqrt{q_0^* r_0}, \quad \delta = 6q_0^* r_0.$$

The starred symbol denotes only the fixed clock threshold, not the optical contraction constant q_0. Let S be the original low-clock short residual for this schedule. Independently choose

$$\Delta = 2\delta, \qquad r_\perp = \delta, \qquad q_\perp = \varepsilon^{1/32}. \tag{6.3}$$

Define the actual nonnegative remaining paired source

$$\mathsf{U}_B(t) = T^{\varepsilon(B)}_{\mathsf{S},\mathrm{fail}}(t) \quad \text{for } (r_\perp, \Delta, q_\perp) \text{ in (6.3)}. \tag{6.4}$$

It includes every positive paired trace not admitted by the transverse comparison, including nonregular pairing strata of positive paired mass and zero transverse speed. The target selector is still the original S.

Theorem 6.2 (Schedule-compatible positive-trace reduction)**.** *Under the retained physical and four-record analytic inputs and the proved uniform-capacity theorem, for all sufficiently large B,*

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B) + C_M B^{-1/384}. \tag{6.5}$$

The admitted part of the positive paired trace satisfies

$$\mathcal{H}_M(T^{\varepsilon(B)}_{\mathsf{S},\mathrm{adm}}) \le C(\varepsilon^{1/32} + \varepsilon^{95/32}). \tag{6.6}$$

Along this schedule the original pointwise theorem is equivalent to $\mathcal{H}_M(\mathsf{U}_B) \to 0$ for every fixed M. No vanishing assertion about U_B is made here.

Proof. The paired trace splits positively and exactly into the admitted and failure sources at finite collision count. Insert (6.3) in Theorem 5.1. Since $\Delta = O(\varepsilon^4)$, its two costs are $\varepsilon^{1/16-1/32}$ and $O(\varepsilon^{4-1-1/32})$, proving (6.6). The previously established scalar bound has the additional terms $C_M B^{-1/192}$, $O(\varepsilon^{257/64})$ and $O(\varepsilon^3)$. Each is dominated by a constant times $B^{-1/384}$, proving (6.5). Every new comparison width and threshold is fixed as a function of B before the collision limit; no band or cutoff depends on that count.

Sufficiency follows from (6.5). For necessity, scalar closure and the positive scalar remainder make the original bad source small in the prescribed ordered height. The selected short volume source is below it. The established uniform short trace-comparison error is $O((1+\varepsilon^{-1})\delta)$, so the whole positive $T^\varepsilon_{\mathsf{S}}$ tends to zero. Since $0 \le \mathsf{U}_B \le T^\varepsilon_{\mathsf{S}}$, its height also tends to zero. This argument does not require transversality of a critical entrance or a bound on the number of physical sheets. □

Theorem 6.1 discharges transverse admission on every physical chamber family satisfying its numerical tests. It does not establish uniform tests for all actual pairs. Controlling the remaining mass at short angular reach, nearly stationary boundary roofs or rapid relative trace variation is the outstanding geometric step. This is a positive boundary source, not a signed cancellation, an averaged target, or a replacement exact event. The uniform pressure theorem alone does

not provide these transverse tests. The original upper pointwise half and unrestricted point-conditioned consequences are therefore not promoted. Their target and arithmetic restrictions are unchanged.

7. Endpoint-distance coordinates for the same positive trace

The relative coordinate in (4.1) need not be preserved by a comparison. On a regular pairing chamber let $e_- = a$, $e_+ = b$, and put

$$x_- = u - a(\theta), \qquad x_+ = b(\theta) - u, \qquad 0 < x_\pm < \ell(\theta).$$

These are distances in the roof variable, not spatial distances on the billiard table. In either coordinate the original paired measure is

$$d\pi^\varepsilon = c^\varepsilon(\theta)\, d\theta\, dx_\pm. \tag{7.1}$$

Indeed $x_- = s\ell$ and $x_+ = (1-s)\ell$, so the factor ℓ in (4.2) is exactly the coordinate Jacobian. No trace minimum, original weight or source selector is changed.

Theorem 7.1 (Two endpoint-distance flows and their covariance)**.** *Fix $\Delta > 0$. On the regular set*

$$c^\varepsilon > 0, \quad 0 < x_\pm < \ell < \Delta, \quad e'_\pm \neq 0,$$

define $V_\pm = (e'_\pm)^{-1}\partial_\theta$ at fixed $x_\pm$. The corresponding stopped flows $\Theta_h^\pm$ have unit roof speed. In (θ, s) coordinates their fields are

$$\begin{aligned} V_- &= \frac{1}{a'}\left(\partial_\theta - \frac{s\ell'}{\ell}\partial_s\right), \\ V_+ &= \frac{1}{b'}\left(\partial_\theta + \frac{(1-s)\ell'}{\ell}\partial_s\right). \end{aligned} \tag{7.2}$$

Their densities in $(t, x_\pm)$ and pullback Jacobians are

$$\rho_\pm^\varepsilon = \frac{c^\varepsilon}{|e'_\pm|}, \qquad J_h^\pm(y) = \frac{\rho_\pm^\varepsilon(\Theta_h^\pm y)}{\rho_\pm^\varepsilon(y)}. \tag{7.3}$$

Writing $S_h^\pm(t)$ for the stopped survivor fiber, the convention is

$$\int_{S_h^\pm(t)} \varphi(\Theta_h^\pm y) J_h^\pm(y)\, d\pi_t^\varepsilon(y) = \int_{\Theta_h^\pm S_h^\pm(t)} \varphi(z)\, d\pi_{t+h}^\varepsilon(z). \tag{7.4}$$

For fixed collision count, radius and exact label this holds jointly for almost every (t, h) and all nonnegative Borel integrands. Each flow counts physical interiors with multiplicity one. The two flows are distinct comparisons, not two copies of the source.

Proof. At fixed x_- the roof is $a(\theta) + x_-$, and at fixed x_+ it is $b(\theta) - x_+$. In both cases its angular derivative is $e'_\pm$. Thus $V_\pm \mathscr{F} = 1$. Differentiating the two identities for $x_\pm$ proves (7.2). The triangular coordinate change from $(\theta, x_\pm)$ to $(t, x_\pm)$ has absolute determinant $|e'_\pm|$. Applying it to (7.1) gives (7.3). In these coordinates the flow translates t and fixes $x_\pm$; its pullback density is therefore the displayed target-to-source ratio, not its reciprocal.

On the open comparison domain the field is C^1. Its partial flow and inverse are unique until a true comparison boundary is reached. Apply change of variables on this one-to-one graph, then disintegrate in (t, h). The original countable physical charts and a countable determining algebra identify one joint kernel version. Monotone approximation proves (7.4) for all nonnegative integrands, also with Borel graph restrictions. Tonelli sums the disjoint physical pieces. Smoothly compatible chart seams are crossed by coordinate change; actual loss of pairing regularity remains a stopping event. Neither $e'_\pm = 0$ nor nonregular positive source is discarded: its corresponding clock will be zero. □

Proposition 7.2 (The nonvanishing trace at a regular critical entrance)**.** *Suppose a chamber has a backward critical entrance with a regular nongrazing stationary extension at q_*. Put $K_* = D_q^2 F(q_*)$, $e(\theta) = (\cos\theta, \sin\theta)$, and let $e_\perp$ be its positive quarter-turn. Its incoming roof is $a = F(q_*)$, independent of θ, and its original weighted incoming trace is*

$$f^\varepsilon(a+, \theta) = \frac{\omega(q_*) W^\varepsilon(q_*)}{4\pi c_* \, e_\perp(\theta)^\mathsf{T} K_* e_\perp(\theta)}. \tag{7.5}$$

If its numerator is positive, then

$$|\partial_\theta \log f^\varepsilon(a+, \theta)| \le 2\Lambda_\circ / \lambda_\circ.$$

Nevertheless c^ε is still the minimum with the outgoing trace. This bound alone gives no relative bound for that minimum.

Proof. The positive Hessian gives a local inverse $q(g)$ near $g = 0$. All rays in this critical chamber approach the same q_*, so their incoming roofs agree. At $g = ze$ direct differentiation gives

$$\det D_q(F, \arg g) = e_\perp^\mathsf{T} K(q) e_\perp.$$

The optical factor and original guard have continuous values at the regular nongrazing extension. Taking $z \downarrow 0$ proves (7.5); no physical continuation beyond a first-hit or section boundary is asserted. The derivative of the quadratic form has magnitude at most $2\Lambda_\circ$, while the form is at least $\lambda_\circ$. Division proves the bound. A coincident singular extension not covered by the hypothesis remains in the original source, not in the scope of this trace calculation. □

Lemma 7.3 (Which zero-speed set can have paired mass)**.** *Within the regular chambers, the set $v = 0$ with $|a'| + |b'| > 0$ is π^ε-null. Apart from transverse-null sets, the remaining zero-speed source consists of angular intervals on which both a and b are constant. On such an interval I, its unselected roof density is exactly*

$$\left(\int_I c^\varepsilon(\theta) \, d\theta \right) \mathbf{1}_{(a_I, b_I)}(t).$$

No smallness of the sum of these flat-interval pulses is asserted.

Proof. For a fixed θ with $(a', b') \neq (0, 0)$, the affine function $(1 - s)a' + sb'$ has at most one zero. The measure in s is absolutely continuous, proving the first assertion by Tonelli. The endpoint maps are definable by Lemma 4.1; their common derivative-zero set is a finite union of points and intervals at fixed count. The maps are constant on each interval. Integration in the original u coordinate gives the displayed density. Points have zero transverse mass. The conclusion concerns exact zero speed, not an estimate for neighborhoods of folds or stationary roofs. □

8. Recovering positive trace missed by the relative clock

For each of the two flows define, on its comparison domain,

$$k^\pm_{r,\Delta}(y) = \frac{1}{r} \int_{-r}^{r} \mathbf{1}_{\{\Theta_h^\pm y \text{ exists}\}} \min\{1, J_h^\pm(y)\} \, dh, \tag{8.1}$$

and set it to zero elsewhere. It is Borel and lies in $[0, 2]$. The roof kernels are those of the same original positive paired measure as in Section 4.

Theorem 8.1 (Pointwise height from the two endpoint-distance clocks)**.** *Let $0 < \delta < \Delta$, $r > 0$ and $0 < q \le 1$. For an original Borel selector H supported on $\ell \le \delta$, let*

$$G = \{\max(k^-_{r,\Delta}, k^+_{r,\Delta}) \ge q\}.$$

At every finite collision count and almost every original target roof,

$$T^{\varepsilon}_{H\cap G}(t) \le \frac{2}{qr}\int_{t-r}^{t+r} \mathsf{T}^{\varepsilon}_{\Delta}(u)\,du. \tag{8.2}$$

Consequently

$$\mathcal{H}_M(T^{\varepsilon}_{H\cap G}) \le \frac{C}{q}\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{8.3}$$

The estimate acts on the positive endpoint minimum itself. There is no lower residence, initial-gradient or internal-incidence cutoff. The numerical factor two counts comparison directions, not words.

Proof. Disjointize G into $G_- = \{k^- \ge q\}$ and $G_+ = \{k^- < q,\ k^+ \ge q\}$. On either set, integrate $q \le k^{\pm}$ against $\mathbf{1}_H\, d\pi^{\varepsilon}_t$. For each time h, use $\min(1, J^{\pm}_h) \le J^{\pm}_h$ and (7.4). The transported restriction is bounded above by the complete paired target with $\ell < \Delta$, after all physical sheets have been summed. The target need not satisfy the original selector H, which is held at the starting roof; it has the same exact event and labels. This gives

$$qT^{\varepsilon}_{H\cap G_\pm}(t) \le \frac{1}{r}\int_{-r}^{r} \mathsf{T}^{\varepsilon}_{\Delta}(t+h)\,dh.$$

Adding the two estimates proves (8.2). Apply (5.1) to obtain (8.3). All widths and thresholds are fixed before the collision limit. No separate estimate of sheet count, no differentiation of the selector, and no signed cancellation enters. □

Keep the original schedule and original selector S in (6.3) and (6.4). In addition to its unchanged relative clock with collar 2δ, use the two new clocks with

$$r_g = \delta, \qquad \Delta_g = 4\delta, \qquad q_g = \varepsilon(B)^{1/32}. \tag{8.4}$$

Let F_B be the original relative-clock failure set, so that $\mathsf{U}_B = T^{\varepsilon(B)}_{\mathsf{S}\cap F_B}$. With G_B the gap-admitted set from Theorem 8.1, define the positive sources

$$\mathsf{V}_B = T^{\varepsilon(B)}_{\mathsf{S}\cap F_B\cap G_B}, \qquad \mathsf{U}^{\mathrm{rem}}_B = T^{\varepsilon(B)}_{\mathsf{S}\cap F_B\cap G_B^c}. \tag{8.5}$$

They satisfy $\mathsf{U}_B = \mathsf{V}_B + \mathsf{U}^{\mathrm{rem}}_B$ on the original kernels. In particular $0 \le \mathsf{U}^{\mathrm{rem}}_B \le \mathsf{U}_B$ at finite count.

Theorem 8.2 (A smaller positive failure source with paid recovery)**.** *For all sufficiently large outer B,*

$$\mathcal{H}_M(\mathsf{V}_B) \le C_M B^{-1/384}. \tag{8.6}$$

Moreover the original scalar error satisfies

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}^{\mathrm{rem}}_B) + C_M B^{-1/384}. \tag{8.7}$$

Along this same schedule the original pointwise theorem is equivalent to $\mathcal{H}_M(\mathsf{U}^{\mathrm{rem}}_B) \to 0$ for every fixed M. The first assertion estimates the newly recovered part; the last limit is not asserted here.

Proof. Apply (8.3) with $H = \mathsf{S}\cap F_B$. Since $\delta = O(\varepsilon^4)$, the upper bound is

$$C\varepsilon^{-1/32}\{\varepsilon^{1/16} + (1+\varepsilon^{-1})4\delta\} \le C(\varepsilon^{1/32} + \varepsilon^{95/32}).$$

This is $O(B^{-1/384})$. Substitute the exact positive split (8.5) in Theorem 6.2. This proves (8.7) and sufficiency. Conversely, scalar closure implies $\mathcal{H}_M(\mathsf{U}_B) \to 0$ by the retained necessity argument; domination then implies the stated limit for $\mathsf{U}^{\mathrm{rem}}_B$. No limit is exchanged with a collision-dependent collar or a changing Fourier band. □

The next section proves a geometric admission theorem which is uniform in the relative coordinate, including its approach to a backward critical entrance and to a relative-coordinate fold. Thus (8.5) is accompanied by a new pointwise estimate and a new comparison direction, not merely by new names for the old positive remainder. Failure of all three clocks still retains its full positive source, including short angular reach and uncontrolled variation. It is not inferred small from its definition.

9. An endpoint certificate uniform up to critical corners

Write $\theta = \theta_0 + \sigma x$, $0 \le x \le L_\theta$, for one angular side, where $\sigma \in \{-1, 1\}$. Derivatives below are with respect to x. A certificate is imposed on the actual regular pairing chamber, not on a symbolic continuation of the word.

Theorem 9.1 (An expanding endpoint-distance certificate)**.** *Choose either $e = a$ or $e = b$. Suppose the chamber continues over this angular interval, $c^\varepsilon > 0$, and for constants $\nu > 0$, $A, K \ge 0$ one has*

$$|\dot e| \ge \nu, \qquad 0 \le \dot\ell \le K|\dot e|, \qquad |(\log c^\varepsilon)^\cdot| \le A, \qquad |(\log|\dot e|)^\cdot| \le A \tag{9.1}$$

almost everywhere. Let $0 < \ell(\theta_0) \le \delta$ and assume

$$r \le \nu L_\theta, \qquad 2Ar/\nu \le \log 2, \qquad \delta + Kr < \Delta. \tag{9.2}$$

Then the clock for the endpoint e satisfies

$$k^e_{r,\Delta}(\theta_0, s) \ge \frac{1}{2} \quad \text{for every } 0 < s < 1.$$

Here $k^a = k^-$ and $k^b = k^+$. The conclusion uses no lower bound on $\ell(\theta_0)$ or on $s, 1-s$, and no bound on $\partial_\theta \log \ell$. The sign of $\dot e$ may be either positive or negative.

Proof. The nonzero continuous derivative has a constant sign τ. For every $0 \le h \le r$, the equation $e(\theta_h) - e(\theta_0) = \tau h$ has a unique solution on the chosen side, with angular displacement at most h/ν. Choose the gap to that endpoint at the starting state and keep it fixed. Initially it lies strictly between zero and $\ell(\theta_0)$. The residence is nondecreasing on the chosen side and

$$\ell(\theta_0) \le \ell(\theta_h) \le \ell(\theta_0) + K|e(\theta_h) - e(\theta_0)| \le \delta + Kr < \Delta.$$

The fixed gap therefore stays strictly inside the same segment. There is no loss at its other endpoint, however close the starting relative coordinate was to that endpoint. The regularity assumption and positive weight keep the rest of the comparison domain intact. This curve is exactly the partial flow at roof time τh. By (7.3) and the two logarithmic bounds,

$$J^e_{\tau h} \ge \exp(-2Ah/\nu) \ge \tfrac{1}{2}.$$

One full one-sided time interval of length r now contributes at least $1/2$ to (8.1). Endpoint values of the angular interval are handled by smaller times and monotone convergence. □

Corollary 9.2 (Critical entrances and opposite endpoint slopes)**.** *On a regular backward-critical chamber with a constant, choose an angular side on which b increases. Then $\dot\ell = \dot b$ and Theorem 9.1 applies with $e = b$, $K = 1$, provided its speed, weight and reach hypotheses hold. The conclusion is uniform as $s \downarrow 0$, although the relative-clock speed is $v = sb'$.*

On a side with $\dot a \le 0 \le \dot b$, if one endpoint derivative has magnitude at least the other throughout the comparison interval, use that endpoint as e. Then $0 \le \dot\ell \le 2|\dot e|$. With $\Delta = 4\delta$ and $r = \delta$ the residence constraint holds with $K = 2$. Subject to the same remaining hypotheses, every s is admitted, including a value at which $(1-s)a' + sb' = 0$.

Proof. For the first assertion differentiate $b - a$. For the second, $\dot{\ell} = \dot{b} + |\dot{a}|$ is at most twice the larger endpoint speed. The certificate and $\delta + 2\delta < 4\delta$ finish the proof. No sign is assigned to the positive incoming critical trace, which is explicitly given in Proposition 7.2. □

For any Borel physical subfamily on which this certificate holds with $\Delta = 4\delta, r = \delta$, Theorem 8.1 can be used with threshold $1/2$. Its complete positive trace has ordered height

$$C\{\varepsilon^{1/16} + (1 + \varepsilon^{-1})\delta\} = O(B^{-1/192}). \tag{9.3}$$

The estimate is independent of how many physical chambers satisfy the tests. This is a height assertion for the certified physical source, not just a sheetwise derivative bound.

9.1. **Two checks of the newly recovered geometry.** The following elementary chamber models test the comparison formulas. They are not asserted to be additional Lorentz realizations. Take $c^{\varepsilon} = 1$, $r = \delta$, and a collar $0 < \ell < 4\delta$. If $a = a_*$ and $b = a_* + \theta$, then the outgoing-gap flow is

$$\theta_h = \theta + h, \qquad u_h = u + h, \qquad J_h^+ = 1.$$

For $\ell(\theta) \le \delta$ its whole forward interval $0 < h < r$ exists, at every starting relative position. In contrast, at fixed s the relative flow has total roof reach at most $4s\delta$, so its capped clock is at most $4s$. Thus for any fixed positive admission threshold there is a positive-measure critical-corner strip missed by the relative clock and recovered by the gap clock.

For $a = -\theta$, $b = \theta$, $\theta > 0$, the outgoing-gap flow again has $\theta_h = \theta + h$, $u_h = u + h$ and $J_h^+ = 1$. Its residence increases by $2h$, so all initial $\ell \le \delta$ survive the full forward comparison. The relative speed is $2s - 1$; its total roof reach within the collar is at most $2\delta|2s - 1|$. It is zero at the fold and arbitrarily small on a positive-measure neighborhood, while the gap clock remains at least one. The tests also show why neither the s factor nor the relative residence derivative belongs in the new Jacobian.

9.2. **The remaining full-source statement.** The recovered height bounds do not verify the certificate on every physical pair. In particular simultaneous stationary endpoint roofs, flat-interval pulses from Lemma 7.3, insufficient one-sided comparison reach and rapid variation of the positive minimum may remain in $\mathsf{U}_B^{\mathrm{rem}}$. The proof does not set a nonregular or backward-critical trace to zero. Establishing $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{rem}}) \to 0$ on the full Lorentz family remains the required upper pointwise step. The original pointwise objective and all scalar-first arithmetic restrictions are unchanged.

Appendix A. The dependency chain of the current comparison

The original event and scalar target are (2.1) and (2.5). Input statements 3.1 and 3.2 give the exact physical kernels, positive short-trace comparison, uncut volume concentration and scalar reduction. Their proof references remain those displayed beside the inputs, including the uniform-capacity theorem in the complete archive. No new spectral insertion theorem is assumed.

The relative-position measure, Jacobian and admitted-source bound are Theorems 4.2 and 5.1. The new chain is

$$\text{endpoint-distance coordinates} \longrightarrow (7.4) \longrightarrow (8.2) \longrightarrow (8.6).$$

The expanding-endpoint hypotheses in (9.1) are sufficient conditions on actual chambers. Under them (9.3) is an essential-height theorem for the entire selected physical family, independent of its chamber count. The two affine chamber calculations are checks of this new coordinate mechanism, not new realizations of the physical model.

The full scalar theorem still requires $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{rem}}) \to 0$. The zero-speed decomposition in Lemma 7.3 does not control near-stationary pairs or the flat-interval pulse sum. The critical-trace

formula controls its incoming angular factor, not the relative variation of an arbitrary outgoing trace. These distinct limitations are retained in the source-level endpoint.

After scalar closure, the archived positive path-remainder and collision-to-return statements keep their original hypotheses. No arithmetic zero class is normalized, and no same-roof ratio, pointwise denominator or unrestricted likelihood assertion follows from the admitted-source estimate alone. The complete preceding article is reproduced in the archive; historical revisions are not substituted for independent verification of the inherited inputs.

Part 3. Positive paired coarea and the relative transverse clock

Retained revision-102 abstract. We study the exact-label pointwise raw local limit problem for actual returns in a triangular Lorentz gas. The remaining upper source is the positive minimum of two physical endpoint traces on each short regular segment. We represent this minimum by a barycentric coarea measure and construct a transverse comparison which preserves relative position between the two ends. Its Jacobian is computed before physical sheets are summed. A capped transverse clock then gives an essential-height estimate for the admitted positive paired source itself, rather than for its outgoing-minus-incoming difference. A relative endpoint certificate applies without a lower residence cutoff. Under the original reconstruction schedule the admitted contribution is $O(B^{-1/384})$. The unadmitted transverse source is retained explicitly; no assertion that it vanishes on the full Lorentz family is made. The original pointwise objective, exact labels, physical guard, arithmetic classes and all historical proofs are unchanged.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on

$[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_\#(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Positive paired mass and a transverse comparison.** The uniform momentum-capacity theorem has removed the singular volume-source premise from the preceding scalar reduction. Its remaining boundary quantity is positive: two large endpoint traces can have a small difference. We therefore put the positive minimum itself into a coarea measure. The new comparison moves in angular direction, keeping fixed the relative position between the same two physical ends. It does not require a radial trajectory to survive beyond its original short residence.

Theorem 4.2 constructs this measure and its Jacobian. Theorem 5.1 estimates its whole admitted part in essential height after physical summation, with no word-count or chart-overlap factor. Theorem 6.1 gives a test in terms of the two endpoint roof maps and their positive trace density. Its constants do not deteriorate merely because the original residence tends to zero. Theorem 6.2 inserts the estimate into the unchanged scalar target.

Admission on the complete nongrazing paired family is not established. The remaining positive source includes short transverse comparison lifetimes, folds of the endpoint roof maps, and

insufficient control of relative trace variation. A backward critical trace is not set to zero. These distinctions are part of the theorem statements, not changes of the governing objective.

3. The original sources and the inputs used here

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar{\nu} = \mathbf{1}_{\Pi}\nu_R^*$ and let μ_t be its original roof kernels. The Borel maximal-segment parametrization gives $\Psi(\theta, u)$ for $a(\theta) < u < b(\theta)$, with

$$A(u,\theta) = \frac{\omega(\Psi(\theta,u))}{4\pi c_* \chi(\Psi(\theta,u))}, \quad \omega = |\partial_{p_0}\alpha_m|^{-1}, \quad \chi = |\det D_q(F, \arg g)|, \quad g = \nabla_q F.$$

Thus $d\bar{\nu} = A\,du\,d\theta$ after disjoint physical summation. Let $z = |g|$ and $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$. We use the following assertions with precisely their original scope.

Input statement 3.1 (Physical geometry and trace comparison)**.** The finite-count physical graph and section decisions are definable. The endpoint Hessian has uniform positive lower and upper bounds; for a fixed lifted itinerary the endpoint-momentum map is globally injective. The maximal-segment Borel parametrization is one-to-one, and its kernel identities retain incident components at coincident physical faces. For $f^\varepsilon = AW^\varepsilon$ the one-sided traces exist, and

$$|\partial_u f^\varepsilon| \le C(1+\varepsilon^{-1})AI_m/z.$$

The trace is zero at a genuinely grazing end, not necessarily at a backward critical end. If a Borel target selector H is supported on $0 < \ell = b - a \le \delta$, let

$$R_H^\varepsilon(t) = \int \mathbf{1}_H W^\varepsilon \, d\mu_t, \quad T_H^\varepsilon(t) = \sum \int \mathbf{1}_{\{a<t<b\}} \mathbf{1}_H(\Psi(\theta,t)) \min(f^\varepsilon(a+), f^\varepsilon(b-))\, d\theta.$$

Then, on the original kernels and at every finite count,

$$|R_H^\varepsilon(t) - T_H^\varepsilon(t)| \le C(1+\varepsilon^{-1}) \int_{t-\delta}^{t+\delta} v^0(u)\,du, \qquad v^0\,dt = F_{\#}(I_m z^{-1}\bar{\nu}). \tag{3.1}$$

The selector is held at the target; no trace of an artificial selector boundary is taken.

The Borel map is Supplement Theorem 7.1. The physical momentum uniqueness and trace comparison are Supplement Theorems 7.1 and 6.1. The first theorem's label is recorded also in the dependency table below. No independent recertification of the inherited continuum chain is implicit in its restatement here.

Input statement 3.2 (Uniform volume concentration and scalar reduction)**.** For every previously fixed $r, \varepsilon > 0$, with $A_r h(t) = (2r)^{-1}\int_{t-r}^{t+r} h(u)\,du$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad \limsup_m \sup_{R,n,k} m^2 \|A_r v^0\|_\infty \le C. \tag{3.2}$$

The constants are independent of those fixed widths. The actual uniform weighted momentum capacity and its geometric-pressure proof supply the second assertion without an additional capacity premise. The original positive scalar remainder is

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \varepsilon(B) = A_0 B^{-1/12}, \qquad \limsup_m \sup_{R,n,k} \operatorname{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}.$$

Use the fixed radial clock threshold $q = 1/8$, radius $r = c\varepsilon(B)^4$, gradient threshold $a = \sqrt{qr}$, and short-residence threshold $\delta = 6qr$. The retained low-clock failure source has a disjoint short part S supported on $\ell \le \delta$. With $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t\in\mathcal{C}(M)} h(t)$,

$$\mathcal{E}_M \le \mathcal{H}_M(T_{\mathsf{S}}^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1+\varepsilon(B)^{-1})qr. \tag{3.3}$$

The original pointwise law is equivalent along this schedule to $\mathcal{H}_M(T_{\mathsf{S}}^{\varepsilon(B)}) \to 0$.

The uncut volume estimate is Supplement Theorem 11.3; the scalar bound is Supplement Theorem 12.2 together with its retained necessity argument. The original low-clock selector, including the comparison collar and the original exact-event lifetimes, is reproduced in that article; it is not differentiated or modified in the new argument.

Every auxiliary band is fixed before the collision limit and enlarged only afterwards, with B outermost. At fixed m, R, n, k, common roof kernels and joint roof/time kernels are used. Uniformity is $\sup_R \operatorname{ess\,sup}_t$, not a claim of one null set for all radii or all uncountable selector families. The new finite-count comparisons are measure identities before their uniform norms are taken.

4. A transverse coordinate for the positive paired source

Fix the collision count, radius and exact label. A maximal segment of that original event has roof interval $(a(\theta), b(\theta))$, length $\ell = b - a$, and physical density $A(u, \theta)$. Its paired weight is

$$c^{\varepsilon}(\theta) = \min\{f^{\varepsilon}(a+, \theta), f^{\varepsilon}(b-, \theta)\}, \qquad f^{\varepsilon} = AW^{\varepsilon}.$$

The minimum is taken on the same physical segment, before summation. A zero trace contributes no paired mass. In particular genuinely grazing ends contribute zero; a backward critical trace is retained.

Lemma 4.1 (Regular pairing chambers)**.** *At each fixed count and radius, the original paired measure admits angular chambers on which a, b are C^2 and c^{ε} is locally Lipschitz. These chambers represent all paired mass except for transverse-null strata. Their interiors are counted once. There is no assertion of a collision-uniform bound on their number or on their derivatives. At a backward critical entrance, a may be constant while c^{ε} is nonzero.*

Proof. Use the finite physical graph and its exact section decisions, not a partition of symbolic extensions. On each lifted itinerary the map from endpoint angles to g has positive Jacobian and is globally injective by the retained momentum uniqueness theorem. On finitely many polar charts with $z > 0$, its image is a definable set. Along one ray its connected components are the original maximal segments: $\partial_z F = z e^{\mathsf{T}} K^{-1} e > 0$. Definable cell decomposition and one-variable piecewise differentiability therefore give finitely many angular intervals on which the component endpoints, their limiting contact data and their roof values are C^2, after further subdivision if necessary [9]. No estimate of the number of pieces is used.

At a nongrazing endpoint the stationary word has a smooth local extension; this is not physical continuation through a first-hit boundary. The clearance is the minimum of finitely many smooth functions locally. Thus the original smooth guard has locally Lipschitz endpoint values. The two angular density factors have the same property away from $g = 0$. At a critical incoming end the positive endpoint Hessian gives a local inverse in g at zero. The limiting angular factor is $e_{\perp}^{\mathsf{T}} K(q_*) e_{\perp}$, bounded below by $\lambda_\circ$, and the optical factor has its ordinary value at q_*. The resulting trace is locally Lipschitz in the direction e and need not vanish. This also gives the assertion in that case. Taking a minimum preserves local Lipschitz regularity.

The disjoint physical-segment parametrization assigns each component once. The finitely many angular subdivision points have zero transverse measure; their roof saturations have zero paired measure by Tonelli. Original Borel seams are only assignments. Smoothly compatible chamber charts can be continued across such seams. Where pairing regularity actually ends, the transverse comparison below is stopped, and its failure source is retained. No part of the original event is removed by that stopping convention. □

On a chamber put $0 < s < 1$ and set

$$\mathscr{F}(\theta, s) = a(\theta) + s\ell(\theta), \qquad v(\theta, s) = \partial_\theta \mathscr{F} = (1 - s)a'(\theta) + sb'(\theta). \tag{4.1}$$

Thus s is the relative position between the two original ends. Define a positive measure on the chamber by

$$d\pi^{\varepsilon} = c^{\varepsilon}(\theta)\ell(\theta)\, d\theta\, ds. \tag{4.2}$$

It is a representation of the paired source, not a new collision probability or a redefinition of the guard.

Theorem 4.2 (Exact paired coarea and transverse Jacobian)**.** *For any original Borel target selector H, the roof pushforward of $\mathbf{1}_H\pi^{\varepsilon}$ has density T^{ε}_H. Here H is evaluated at $\Psi(\theta, \mathscr{F}(\theta, s))$. On the open regular region $c^{\varepsilon} > 0$, $v \neq 0$, define*

$$V^{\perp} = v^{-1}\partial_{\theta}, \qquad \varrho^{\varepsilon} = \frac{c^{\varepsilon}\ell}{|v|}. \tag{4.3}$$

The partial flow $\Theta^{\perp}_h$ obeys $\mathscr{F} \circ \Theta^{\perp}_h = \mathscr{F} + h$ and preserves s. Its Jacobian relative to (4.2) *is*

$$J^{\perp}_h(y) = \frac{\varrho^{\varepsilon}(\Theta^{\perp}_h y)}{\varrho^{\varepsilon}(y)}. \tag{4.4}$$

Consequently its target-direction coarea covariance holds jointly for almost every (t, h) and all nonnegative Borel integrands, on the same stopped survivor graph. Physical sheets are summed with multiplicity one.

Proof. For any nonnegative test ϕ the change of variable $u = a + s\ell$ gives

$$\sum \int \int_0^1 \phi(\mathscr{F}) c^{\varepsilon}\ell \mathbf{1}_H\, ds\, d\theta = \sum \int \int_a^b \phi(u) c^{\varepsilon} \mathbf{1}_H(\Psi(\theta, u))\, du\, d\theta.$$

This is precisely the defining paired source, including the target selector. It proves the first assertion also across $v = 0$.

On $v \neq 0$ use $(t, s) = (\mathscr{F}, s)$ as local coordinates. The measure is $\varrho^{\varepsilon}(t, s)\, dt\, ds$ and the new flow is translation in t. This proves (4.4), including negative times. Equivalently, at fixed s the angular derivative of the flow is $v(y)/v(\Theta^{\perp}_h y)$; multiplying by the ratio of the weights $c^{\varepsilon}\ell$ gives the same positive Jacobian. The absolute value on v cannot be omitted.

Uniqueness for the C^1 field makes the stopped graph one-to-one. Local changes of variables, a countable determining class and Tonelli give joint coarea kernels and the stated covariance. The same argument is valid for graph selectors depending on (y, h). Compatible coordinate seams do not create multiple copies of a physical segment. The construction never moves a state to another occupation or displacement class. It is a transverse comparison between actual trajectories, not the original dynamical roof flow. The set $v = 0$ is assigned to comparison failure whenever it carries paired mass; no null-set assertion about that entire set is needed. □

5. A pointwise estimate of the positive paired part

Write $\mathsf{T}^{\varepsilon}_{\Delta}$ for the complete paired source on $0 < \ell < \Delta$, with no target selection beyond the original exact event. All incoming endpoint types are included. The preceding uncut trace comparison and source estimates imply

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_r \mathsf{T}^{\varepsilon}_{\Delta}\|_{\infty} \leq C\{\varepsilon^{1/16} + (1 + \varepsilon^{-1})\Delta\} \quad (r, \Delta, \varepsilon > 0 \text{ fixed}). \tag{5.1}$$

The constant is independent of these previously fixed widths. Indeed $\mathsf{T}^{\varepsilon}_{\Delta} \leq b^{\varepsilon} + C(1 + \varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0(u)\, du$. Apply the fixed-width bounds to $A_r b^{\varepsilon}$ and $A_r v^0$; translation invariance of the latter supremum pays the convolution with the factor 2Δ. This argument uses neither a pointwise bound on v^0 nor a boundary spectral insertion.

Fix $0 < \delta < \Delta$ and use $\Theta^{\perp}$ on

$$\mathcal{D}^{\varepsilon}_{\Delta} = \{0 < \ell < \Delta,\ c^{\varepsilon} > 0,\ v \neq 0\}$$

inside the regular pairing chambers. Stop at the boundary of this comparison domain or at an actual loss of chamber regularity. The original trace outside that domain is retained, not discarded. For an interior starting pair y define the capped transverse clock

$$k^{\perp}_{r,\Delta}(y) = \frac{1}{r}\int_{-r}^{r} \mathbf{1}_{\{\Theta^{\perp}_{h} y \text{ exists}\}} \min\{1, J^{\perp}_{h}(y)\}\, dh. \tag{5.2}$$

It is zero on all other starting pairs. Its range is $[0, 2]$ and it is Borel. The trace weight already contains the original guard at both endpoints; there is no extra guard mismatch in this clock.

For a Borel selector H supported on $\ell \le \delta$ and $0 < q_{\perp} \le 1$, let $T^{\varepsilon}_{H,\mathrm{adm}}$ and $T^{\varepsilon}_{H,\mathrm{fail}}$ be its paired source restricted, respectively, to $k^{\perp}_{r,\Delta} \ge q_{\perp}$ and $k^{\perp}_{r,\Delta} < q_{\perp}$. Their nonnegative sum is exactly T^{ε}_{H}, including every nonregular or zero-speed pair.

Theorem 5.1 (Complete transverse-clock trace estimate)**.** *At every finite count and almost every target roof,*

$$T^{\varepsilon}_{H,\mathrm{adm}}(t) \le \frac{1}{q_{\perp} r}\int_{t-r}^{t+r} \mathsf{T}^{\varepsilon}_{\Delta}(u)\, du. \tag{5.3}$$

Consequently, uniformly in the original radius and exact labels,

$$\mathcal{H}_{M}(T^{\varepsilon}_{H,\mathrm{adm}}) \le \frac{C}{q_{\perp}}\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{5.4}$$

This is an essential-height estimate for the positive common trace itself. It imposes no positive lower original residence, gradient or internal incidence cutoff on its admitted part. No estimate of an outgoing-minus-incoming difference is substituted for it.

Proof. Let π^{ε}_{t} be the paired kernels supplied by Theorem 4.2. On admitted starting pairs, $q_{\perp} \le k^{\perp}_{r,\Delta}$. Integrate this inequality with the target selector held at t. For each fixed comparison time,

$$\int \mathbf{1}_{H}\mathbf{1}_{\mathrm{adm}}\mathbf{1}_{\text{survives } h} \min(1, J^{\perp}_{h})\, d\pi^{\varepsilon}_{t} \le \int \mathbf{1}_{H}\mathbf{1}_{\mathrm{adm}}\mathbf{1}_{\text{survives } h} J^{\perp}_{h}\, d\pi^{\varepsilon}_{t} \le \mathsf{T}^{\varepsilon}_{\Delta}(t+h).$$

The last inequality is target-direction coarea covariance on the one-to-one graph. It is valid after all physical sheets are summed. The transported target need not still satisfy H; it does belong to the identical exact event and has total residence below Δ. Thus only the selector is enlarged in a positive upper comparison, not the displacement, occupation, collision count or section tests. Integrate over $-r < h < r$ and divide by $q_{\perp} r$. Joint almost-everywhere covariance and Tonelli prove (5.3) without selecting a fiber version separately at every time. Equation (5.1) then proves (5.4), with all parameters fixed before the collision limit and auxiliary bands enlarged only after it. □

Remark 5.2 (What has and has not been averaged)*.* The left side of (5.3) is at the original target roof; only its upper comparison is averaged. The clock changes angular direction while preserving relative position between the same two physical ends. It is therefore different from the original radial clock, whose trajectory can terminate inside a short segment. The Jacobian is that of the positive paired measure, not that of the original volume with the minimum suppressed. No word, chamber-count or transverse-overlap constant is inserted. The unadmitted part is a genuine positive remainder and is not asserted small by this theorem.

6. Two-ended certificates and the remaining scalar source

The clock has an explicit geometric sufficient test. In particular, short original residence is not by itself a reason for transverse failure. The test also states which additional control is still required on the full paired family.

Theorem 6.1 (A relative two-ended transverse certificate)**.** *Fix a starting pair (θ_0, s) with $0 < \ell(\theta_0) \le \delta$ and $c^\varepsilon(\theta_0) > 0$. Suppose that in at least one angular direction there is an interval of length $L_\theta > 0$ in the same regular pairing chamber on which, at this fixed relative coordinate,*

$$|v| \ge \nu > 0, \qquad |\partial_\theta \log \ell| \le A, \qquad |\partial_\theta \log c^\varepsilon| \le A, \qquad |\partial_\theta \log |v|| \le A \tag{6.1}$$

almost everywhere. Positive ℓ, c^ε and the sign of v persist on this interval. If

$$0 < r \le \nu \min\{L_\theta, \log 2/(3A)\}, \tag{6.2}$$

with the second term interpreted as infinity when $A = 0$, then $k^{\perp}_{r,2\delta}(\theta_0, s) \ge 1/2$. The conclusion has no lower bound on $\ell(\theta_0)$.

A sufficient condition for the bounds on v, simultaneously for all $s \in (0,1)$, is that a', b' have the same sign, $|a'|, |b'| \ge \nu$, and $|a''| \le A|a'|$, $|b''| \le A|b'|$ on that interval.

Proof. Choose the roof direction corresponding to the supplied angular side and the sign of v. Since $d\mathscr{F}/d\theta = v$, every roof displacement of size at most r is reached before an angular displacement of r/ν. Positivity and the logarithmic bounds give

$$\ell(\theta) \le e^{Ar/\nu}\ell(\theta_0) \le 2^{1/3}\delta < 2\delta, \qquad \frac{\varrho^\varepsilon(\theta, s)}{\varrho^\varepsilon(\theta_0, s)} \ge e^{-3Ar/\nu} \ge \frac{1}{2}.$$

Thus the entire one-sided comparison interval exists in $\mathcal{D}^\varepsilon_{2\delta}$, and its contribution to (5.2) is at least $1/2$. A statement at an interval endpoint is obtained from smaller times and monotone convergence. The last assertion follows because $v = (1-s)a' + sb'$ is a convex combination of two derivatives of the same sign and $|\partial_\theta v| \le A((1-s)|a'| + s|b'|) = A|v|$. □

For example, on an actual chamber satisfying $a = a_0 + v_0\theta$, $b = a + \ell_0$ and $c^\varepsilon = c_0 e^{\beta\theta}$, the Jacobian is exactly $\exp(\beta h/v_0)$ and the admissible radius is independent of $\ell_0 > 0$. This calculation is a test of the physical-chamber hypotheses, not an alternative realization of the Lorentz problem. A lower residence cutoff would miss precisely this uniformity. Conversely, at a backward critical entrance $a' = 0$ one has $v = sb'$; the theorem can apply for interior s, but gives no uniform conclusion as s tends to zero. That possible positive critical trace remains in the failure source.

Keep the original schedule from the preceding scalar theorem:

$$\varepsilon = \varepsilon(B) = A_0 B^{-1/12}, \quad q_0^* = 1/8, \quad r_0 = c\varepsilon^4, \quad a_0 = \sqrt{q_0^* r_0}, \quad \delta = 6q_0^* r_0.$$

The starred symbol denotes only the fixed clock threshold, not the optical contraction constant q_0. Let S be the original low-clock short residual for this schedule. Independently choose

$$\Delta = 2\delta, \qquad r_\perp = \delta, \qquad q_\perp = \varepsilon^{1/32}. \tag{6.3}$$

Define the actual nonnegative remaining paired source

$$\mathsf{U}_B(t) = T^{\varepsilon(B)}_{\mathsf{S},\mathrm{fail}}(t) \quad \text{for } (r_\perp, \Delta, q_\perp) \text{ in (6.3)}. \tag{6.4}$$

It includes every positive paired trace not admitted by the transverse comparison, including nonregular pairing strata of positive paired mass and zero transverse speed. The target selector is still the original S.

Theorem 6.2 (Schedule-compatible positive-trace reduction)**.** *Under the retained physical and four-record analytic inputs and the proved uniform-capacity theorem, for all sufficiently large B,*

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B) + C_M B^{-1/384}. \tag{6.5}$$

The admitted part of the positive paired trace satisfies

$$\mathcal{H}_M(T^{\varepsilon(B)}_{\mathsf{S},\mathrm{adm}}) \le C(\varepsilon^{1/32} + \varepsilon^{95/32}). \tag{6.6}$$

Along this schedule the original pointwise theorem is equivalent to $\mathcal{H}_M(\mathsf{U}_B) \to 0$ *for every fixed* M*. No vanishing assertion about* U_B *is made here.*

Proof. The paired trace splits positively and exactly into the admitted and failure sources at finite collision count. Insert (6.3) in Theorem 5.1. Since $\Delta = O(\varepsilon^4)$, its two costs are $\varepsilon^{1/16-1/32}$ and $O(\varepsilon^{4-1-1/32})$, proving (6.6). The previously established scalar bound has the additional terms $C_M B^{-1/192}$, $O(\varepsilon^{257/64})$ and $O(\varepsilon^3)$. Each is dominated by a constant times $B^{-1/384}$, proving (6.5). Every new comparison width and threshold is fixed as a function of B before the collision limit; no band or cutoff depends on that count.

Sufficiency follows from (6.5). For necessity, scalar closure and the positive scalar remainder make the original bad source small in the prescribed ordered height. The selected short volume source is below it. The established uniform short trace-comparison error is $O((1+\varepsilon^{-1})\delta)$, so the whole positive $T^{\varepsilon}_{\mathsf{S}}$ tends to zero. Since $0 \le \mathsf{U}_B \le T^{\varepsilon}_{\mathsf{S}}$, its height also tends to zero. This argument does not require transversality of a critical entrance or a bound on the number of physical sheets. □

Theorem 6.1 discharges transverse admission on every physical chamber family satisfying its numerical tests. It does not establish uniform tests for all actual pairs. Controlling the remaining mass at short angular reach, nearly stationary boundary roofs or rapid relative trace variation is the outstanding geometric step. This is a positive boundary source, not a signed cancellation, an averaged target, or a replacement exact event. The uniform pressure theorem alone does not provide these transverse tests. The original upper pointwise half and unrestricted point-conditioned consequences are therefore not promoted. Their target and arithmetic restrictions are unchanged.

Appendix A. The active dependency chain

The new scalar estimate depends on three inherited interfaces and three new arguments. The original reconstruction and paired-source comparison are restated in Section 3. Uniform uncut concentration for v^0 uses the geometric-pressure and momentum-capacity theorem in Supplement Sections 11–12. The Borel maximal-segment map and the finite physical graph supply only source identity and fixed-count regular pairing chambers.

The new path is

$$\text{paired coarea} \longrightarrow \text{capped transverse clock} \longrightarrow \text{admitted paired height} \longrightarrow \text{original scalar bound.}$$

The numerical two-ended certificate is a sufficient physical test for admission. It is not used as an assumption about all chambers. The outstanding theorem would bound the unadmitted paired source in (6.4) on the whole Lorentz family. No critical entrance, arithmetic zero class or target selector is removed on the way to that statement.

All 243 inherited modules and the complete revision-100 article are compiled in the archive, with their source bytes unchanged. The revision-101 submission-state report contained no new author paper; there is no missing mathematical article inserted under that label. This appendix identifies the active proof dependencies, not an independent certification of every historical theorem.

Part 4. Uniform geometric capacity and the positive paired trace

Retained revision-100 abstract. We study the exact-label pointwise raw local limit problem for actual returns in a triangular Lorentz gas. A finite transverse testing family, together with uniform growth and whole-cell distortion estimates, bounds the geometric stable-Jacobian

pressure uniformly over the prescribed interval of radii and every collision count. The construction uses no parameter-uniform mixing time. The previously established marked comparison then bounds the actual weighted endpoint-momentum capacity uniformly, discharging its spectral-capacity condition at every fixed auxiliary band. A positive two-cap layer cake gives uncut first inverse-gradient concentration and nearby negative-gradient moments on the original exact-label source. The long critical residual and the full short-trace comparison error consequently have uniform essential-height bounds without an additional capacity premise. The original pointwise objective is retained. The positive paired minimum on nongrazing short segments remains a distinct unproved height estimate; no unrestricted scalar or point-conditioned law is inferred from capacity alone. All preceding mathematics and both historical source variants are preserved.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^m B_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right].\tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow 0.\tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Uniform pressure on the original radius family.** The principal target remains (2.5). The complete finite-count gradient cap, critical target certificate and paired physical trace comparison are retained below with their proofs. The global momentum uniqueness and marked pressure comparison are also retained: they are the precise bridge from dynamical pressure to the original optical fiber, not an analogy with a symbolic count.

The new geometric argument is Theorem 10.3. It compares every long backward continuity cell to one of finitely many short stable test curves and pays their homogeneous inverse branches by a uniform growth estimate. Dropping only the final stable factor leaves a uniformly finite physical split. This removes the need to infer uniform constants from a fixed-table mixing argument. The resulting bound is simultaneous in radius and collision count.

Theorem 11.1 discharges the full-family spectral capacity condition. Theorem 11.3 gives the actual uncut source with weight I_mz^{-p} for $1\le p<129/128$; this is not a higher power of I_m/z. Theorem 12.1 then bounds the complete critical residual and every selected short-source comparison error in essential height, uniformly in the radius. These are new family-wide conclusions, not an interchange of a fixed-radius limit with a supremum.

The remaining term in Theorem 12.2 is the positive minimum of the two original weighted traces on each nongrazing short segment. Its decay is not proved by the pressure theorem. The original uniform pointwise objective is not replaced by the fixed-radius or finite-norm consequences already in the archive.

3. Physical source, signed measures, and retained estimates

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar\nu = \mathbf{1}_\Pi \nu_R^*$ for the original exact-event source. The retained Borel parametrization is a bijection between its regular part and maximal segments (σ, u), $a_\sigma < u < b_\sigma$. Its transverse measure $d\sigma$ is the disjoint angular measure and its density is

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad d\bar\nu = A_\sigma(u)\, du\, d\sigma.$$

Here $F = u$, $z = |\nabla_q F|$, $q = (\alpha_0, \alpha_m)$, $\omega = |\partial_{p_0}\alpha_m|^{-1}$, and $\chi = |\det D_q(F, \arg\nabla_q F)|$. All fibers and signed measures below use these same kernels. Let $z_\sigma^- = z(\Psi_\sigma(a_\sigma+))$ and $\ell_\sigma = b_\sigma - a_\sigma$.

Input statement 3.1 (Complete segment geometry and signed balance)**.** Put $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. On every regular segment,

$$2\lambda_\circ(v-u) \le z(v)^2 - z(u)^2 \le 2\Lambda_\circ(v-u), \tag{3.1}$$

$$|A_\sigma'| \le C A_\sigma I_m / z, \qquad |(W^\varepsilon)'| \le C\varepsilon^{-1}/z, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i. \tag{3.2}$$

The complete density $f_\sigma^\varepsilon = A_\sigma W^\varepsilon$ has one-sided traces and is absolutely continuous on the closed segment. The sum of its endpoint traces and absolute derivative integrals is finite at each fixed count. A grazing endpoint has zero trace; a critical backward endpoint need not. The geometric domain is a finite definable family at each fixed count. Its component count is used for existence of measures only, not for normalized estimates.

For every Borel prefix endpoint $c_\sigma \in [a_\sigma, b_\sigma]$ put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma.$$

The original union $z < \zeta$ or $\ell < \delta$ is exactly such a prefix, and $c_\sigma = b_\sigma$ gives b^ε. Define

$$\mathsf{I}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma,$$

$$\mathsf{D}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma,$$

$$\mathsf{G}_c^{\varepsilon,\pm}(H) = \int\int_{(a_\sigma, c_\sigma)\cap H} [\pm (f_\sigma^\varepsilon)'(u)]_+\, du\, d\sigma.$$

Their complete finite-count signed identity is

$$DR_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{3.3}$$

Every artificial prefix cut is negative. All true entrance types and one-sided incident physical components are retained.

This is the endpoint-field, optical, Borel-segment and signed-prefix chain, culminating in retained modules 220–222. In particular, A_σ includes both angular factors. No reversal of the billiard record or change of the section is used at a backward end.

Input statement 3.2 (Backward-end and critical volume estimates)**.** Set $\alpha = 129/128$ and $\gamma = 1/32$. Let

$$e_s^-(t) = \int \mathbf{1}_{\{t - a_\sigma \le s\}}\, d\mu_t, \qquad u_q(t)\, dt = F_\#(I_m \mathbf{1}_{\{z<q\}}\bar\nu), \qquad p_q(t)\, dt = F_\#(\mathbf{1}_{\{z<q\}}\bar\nu).$$

For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\theta_{B_*} < 1$ such that, at finite count and small $s, q > 0$,

$$m^2 \int_J e_s^- \le Cs^\gamma(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\theta_{B_*}^m, \sqrt{s}\}, \tag{3.4}$$

$$m^2 \int_J u_q \le Cq^\alpha(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 q^{1/2}\theta_{B_*}^m. \tag{3.5}$$

The constants are uniform in radius, exact labels, marks where present, and interval location. Since $I_m \ge 1$, the latter also bounds p_q. Consequently, at every fixed $s, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_s^- \le CLs^\gamma.$$

For the actual positive source $v^\rho(t)\,dt = F_\#(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar\nu)$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\rho \le CL \quad (\rho, L > 0 \text{ fixed}). \tag{3.6}$$

The leading constant is independent of the fixed positive cutoff; its finite-count remainder is not.

Equation (3.4) is the same-event backward-end estimate of module 211, including critical entrances. For (3.5), multiply the actual fractional-source bound of module 187, at exponent $1/4$, by $q^{1/4}$: pointwise, $I_m\mathbf{1}_{z<q} \le q^{1/4} I_m z^{-1/4}\mathbf{1}_{z<q}$. The leading power is q^α and the remainder power is $q^{1/2}$. The audited cutoff estimate is retained in modules 196 and 222. These are volume-source theorems, not assumptions on entrance flux.

Input statement 3.3 (Original scalar reconstruction)**.** For every fixed $\varepsilon, r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\,du. \tag{3.7}$$

For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards. The reconstruction band B is outermost.

We use $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} h(t)$ for a nonnegative source family. Unqualified $\|\cdot\|_\infty$ denotes essential supremum on the full roof line. Identities hold for common kernels at fixed m, R, n, k; the uniform order is $\sup_R \operatorname{ess\,sup}_t$. Countably many chosen radii share versions. No simultaneous Lebesgue-point assertion for every uncountable selector family is required.

3.1. **The residual retained from the preceding revision.** The physical comparison flow $\widehat\Phi$ crosses section-decision walls on the regular endpoint collar $|p_0|, |p_m| < 19/20$, and stops at true physical or backward critical ends. It has the same roof and transverse coordinates as the original flow while it stays in the original exact event. Its complete angular density is $\widehat A$. Only comparison times returning to the identical exact event are counted in the capped clock

$$c_r(x) = \frac{1}{r}\int_{(-r,r)\cap(-\widehat\tau^-,\widehat\tau^+)} \mathbf{1}_\Pi(\widehat\Phi_s x)\min\{1, \widehat A(\widehat\Phi_s x)/\widehat A(x)\}\,ds.$$

For $0 < q \le 1/2$, $d = 2qr$ and $a > 0$, set

$$P = \{\tau^- > 2d\} \cup \{\tau^- \le 2d,\ z \ge a,\ \tau^+ > d\}, \qquad Y = \Pi \cap \{c_r < q\} \cap P^c.$$

The lifetimes in P belong to the original stopped exact event, not the enlarged comparison domain. Write $Y^{\varepsilon}(t) = \int_{F=t} W^{\varepsilon} \mathbf{1}_Y \, d\mu_t$.

Input statement 3.4 (Recovered failure source and scalar reduction)**.** The capped-clock and running-compression theorems of retained modules 232–234 give, with every parameter fixed before the collision limit,

$$\mathcal{H}_M(b^{\varepsilon} - Y^{\varepsilon}) \le Cq^{-1}\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\} + Cqr.$$

The support is $Y \subset \{\tau^- \le 4qr, z < a\} \cup \{\ell \le 6qr\}$. The collar has no exit at widths $r \le r_{\mathrm{col}} = R_-^2/(800\Lambda_\circ)$. For $q = 1/8$, $r = c\varepsilon(B)^4 \le r_{\mathrm{col}}$ and $a = \sqrt{qr}$, the unchanged scalar objective satisfies

$$\mathcal{E}_M \le \mathcal{H}_M(Y^{\varepsilon(B)}) + C_M B^{-1/192}. \tag{3.9}$$

No residual-height decay is part of this input.

The support and the low-clock condition will both be used below. Dropping the latter would lose the density-decrease mechanism on the long critical component.

4. A quadratic physical cap at the critical gradient

The finite-count spectral error in (3.5) has only a square-root gain in the gradient cutoff. Integrating that error against the first inverse gradient is not legitimate. We obtain a second, quadratic bound from the two-dimensional physical map. Its constant is an explicitly weighted multiplicity, retained throughout rather than replaced by an unestimated word count.

Fix $0 < b_0 < 1/8$ within the range of (3.5). On the regular exact event put $G(x) = R(-p_0(x), p_m(x))$ and $z = |G|$. The retained endpoint Hessian satisfies $\lambda_\circ I \le K = D_q^2 F \le \Lambda_\circ I$ in endpoint angles $q = (\alpha_0, \alpha_m)$. The optical normalization is $\omega = |\partial_{p_0}\alpha_m|^{-1}$. Define, with physical states counted once,

$$D_{m,R,n,k}(v) = \sum_{x \in \Pi^{\mathrm{reg}} : G(x) = v} \frac{I_m(x)\omega(x)}{4\pi c_* \det K(x)}, \tag{4.1}$$

$$\mathcal{K}_m = 1 + \pi \sup_{R,n,k} \operatorname*{ess\,sup}_{|v| < b_0} D_{m,R,n,k}(v). \tag{4.2}$$

These definitions involve neither the roof target nor an inverse power of z. The factor I_m is the full optical insertion from (3.2).

Lemma 4.1 (Physical area formula and a finite weighted cap)**.** *For every finite m, $\mathcal{K}_m < \infty$, uniformly over the permitted radius and exact labels. For $0 < b < b_0$ and every interval J,*

$$\int_J u_b(t)\, dt \le \mathcal{K}_m b^2. \tag{4.3}$$

More precisely, if N_m bounds the cardinalities of the regular physical fibers of G at count m, and $v_* = 2\tau_-/R_+$, then

$$\mathcal{K}_m \le 1 + CN_m(1 + v_*)^{-(m-1)}. \tag{4.4}$$

Equation (4.4) *establishes finiteness only; no collision-uniform estimate of its right side is assumed.*

Proof. First variation gives $F_{\alpha_0} = -Rp_0$. Change of variables from (α_0, p_0) to endpoint angles has density ω. The Jacobian of $q \mapsto G(q)$ is $\det K > 0$. Thus the area formula, applied on injective local pieces and summed without multiplicity of charts, gives

$$G_{\#}(I_m \bar{\nu}) = D_{m,R,n,k}(v)\, dv.$$

Equivalently $|\det D_{(\alpha_0,p_0)}G| = \det K/\omega$. This identity uses the original collision normalization and counts physical preimages, not representatives in overlapping charts. Truncation of I_m, followed by monotone convergence, justifies the formula before its boundedness has been established.

Here is a bound for each summand uniform through internal grazing. For flight lengths $L_i \geq \tau_-$ the retained positive optical recursion is

$$y_1 = L_0, \quad f_0 = 1, \quad f_i = f_{i-1} + \frac{2y_i}{Rc_i}, \quad y_{i+1} = y_i + L_i f_i, \quad \omega = \frac{Rc_0c_m}{y_m}.$$

Consequently

$$y_m \geq \tau_- \prod_{i=1}^{m-1} (1 + v_*/c_i).$$

For a selected internal incidence,

$$\frac{1}{c_i} \prod_j (1 + v_*/c_j)^{-1} \leq v_*^{-1}(1 + v_*)^{-(m-2)}.$$

Since $\sum_i q_0^{d_i} \leq 2q_0/(1 - q_0)$, it follows that $I_m\omega \leq C(1 + v_*)^{-(m-1)}$. For one flight the empty product gives the same conclusion after enlarging C. Division by $\det K \geq \lambda_\circ^2$ bounds each summand.

At a fixed count, the finite-horizon restriction permits only finitely many lifted scatterer itineraries. The retained finite definable physical graph, expressed in finitely many circle charts, includes the first-hit tests, radius parameter and section decisions. The nonzero Jacobian makes every regular G-fiber discrete. Uniform finiteness for definable families therefore gives a finite N_m [9]. Displacement and occupation take finitely many values at this count. This is used only for fixed-count finiteness and proves (4.4). Finally discard the roof test on the upper side, and integrate (4.1) over the disk $|v| < b$. Its area is πb^2, proving (4.3). No change is made to the source being estimated. □

Lemma 4.2 (Integration of two different positive caps)**.** *For $E, H, b > 0$,*

$$b^{-1} \min(Eb^{1/2}, Hb^2) + \int_0^b v^{-2} \min(Ev^{1/2}, Hv^2)\, dv \leq 3 \min\{Hb, E^{2/3}H^{1/3}\}. \tag{4.5}$$

Proof. The crossing is $v_* = (E/H)^{2/3}$. For $b \leq v_*$ the left side is $2Hb$. For $b > v_*$ it is $3E^{2/3}H^{1/3} - Eb^{-1/2}$. These formulas prove the claim. The minimum is a cap on the positive source estimate, not on a signed spectral remainder considered by itself. □

Define the actual untruncated source

$$w_b(t)\, dt = F_\#(I_m z^{-1} \mathbf{1}_{\{0<z<b\}} \bar{\nu}).$$

It has a finite, absolutely continuous roof measure at every count.

Theorem 4.3 (First inverse gradient with a complete finite-count error)**.** *For one fixed auxiliary band B_*, all $m \geq 2$, $0 < b < b_0$ and intervals J of length L,*

$$m^2 \int_J w_b \leq Cb^{\alpha-1}(L + B_*^{-1}) + C \min\left\{m^2\mathcal{K}_m b,\ C_{B_*}^{2/3}(1 + B_*L)^{2/3}m^2(\mathcal{K}_m\theta_{B_*}^{2m})^{1/3}\right\}. \tag{4.6}$$

The integration goes to gradient zero at finite count. In particular there is no error proportional to a negative power of a lower cutoff. The weighted multiplicity $\mathcal{K}_m$ is not asserted bounded or subexponential in m.

Proof. Let $M(v) = m^2 \int_J u_v$. The two positive bounds give

$$M(v) \le C(L + B_*^{-1})v^\alpha + \min\{Ev^{1/2}, Hv^2\}, \quad E = C_{B_*}(1 + B_*L)m^2\theta_{B_*}^m, \quad H = m^2\mathcal{K}_m.$$

Indeed a nonnegative quantity bounded by both $A + D$ and V is bounded by $A + \min(D, V)$. Positive layer cake gives exactly

$$m^2 \int_J w_b = b^{-1}M(b) + \int_0^b v^{-2}M(v)\,dv.$$

Apply it first above a positive lower endpoint, and then use monotone convergence. The leading term is integrable because $\alpha > 1$; its multiplier is $1 + (\alpha - 1)^{-1}$. Lemma 4.2 integrates the full error. This proves the inequality and finiteness. Absolute continuity follows from coarea on compact regular roof charts and exhaustion; the finite critical set has zero source area. □

We isolate the additional estimate which would make this error negligible. For an unbounded sequence of previously fixed auxiliary bands, consider the condition

$$m^6\mathcal{K}_m\theta_{B_*}^{2m} \longrightarrow 0. \tag{SC}$$

Subexponential growth of $\mathcal{K}_m$ is sufficient, but is not proved here. Under (SC), first taking the collision limit and only then the indicated band sequence gives

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J w_b \le CLb^{1/128}. \tag{4.7}$$

Adding $z \ge b_0$ by (3.6) gives the bound CL for $v^0 = F_\#(I_m z^{-1}\bar{\nu})/dt$ at each fixed L. These are implications of the explicit condition, not unconditional zero-cutoff conclusions. Geometric-pressure results for stable Jacobian partition sums [25] concern a different weighted sum; no identification with (4.2) or uniform parameter bound is imported from them.

5. Separate critical and short-residence residuals

Keep the original residual $Y = Y_{a,q,r}$ and write $d = 2qr$. We assign the overlap to the short-residence component:

$$S = Y \cap \{\ell \le 3d\}, \qquad C = Y \cap \{\ell > 3d\}, \qquad Y^\varepsilon = S^\varepsilon + C^\varepsilon. \tag{5.1}$$

Here and below a superscript ε denotes the original weight W^ε on the indicated target restriction. The letters C, S are target sets, not new probability measures.

Lemma 5.1 (A full forward interval on the critical component)**.** *Every target in C satisfies*

$$\tau^- \le 2d, \qquad 0 < z < a, \qquad \tau^+ > d. \tag{5.2}$$

For $0 \le s \le d$, the trajectory remains in the original exact event and

$$0 < z(\Phi_s x) < b, \qquad b = (a^2 + 2\Lambda_\circ d)^{1/2}.$$

Moreover some $s \in [0, d]$ has $A(\Phi_s x) < A(x)/2$.

Proof. The complement of the old protected set is $\{\tau^- \le 2d\} \cap (\{z < a\} \cup \{\tau^+ \le d\})$. Since $\ell > 3d$, its second alternative is impossible and $\tau^+ > d$. Radial growth proves the upper bound on z. If $A(\Phi_s x)/A(x) \ge 1/2$ on this interval, the same-event times alone contribute $d/(2r) = q$ to the capped clock. This contradicts $c_r < q$. The direction choice introduces no new comparison-domain availability assumption. □

Theorem 5.2 (Uncut critical loss paid by the first inverse gradient)**.** *Assume $b < b_0$. At every finite collision count, for almost every target roof,*

$$C^{\varepsilon}(t) \le C_0 \int_t^{t+d} w_b(u)\, du. \tag{5.3}$$

Thus Theorem 4.3, with $L = d$, is an explicit bound for the complete critical residual, with no positive lower gradient cutoff. Under (SC)*,*

$$\limsup_m \sup_{R,n,k} m^2 \|C^{\varepsilon}\|_{\infty} \le C d b^{1/128}. \tag{5.4}$$

Without that additional condition, the same height bound holds for $C \cap \{z \ge c\}$ at each previously fixed $c > 0$. That last statement does not imply the uncut assertion by exchanging limits.

Proof. For a target selected in Lemma 5.1,

$$A(x)/2 \le \int_0^d |\partial_u A(\Phi_s x)|\, ds \le C \int_0^d A(\Phi_s x) I_m(\Phi_s x)/z(\Phi_s x)\, ds.$$

Multiply by $W^{\varepsilon}(x) \le 1$. At each fixed s the same-event transport is injective in the physical transversal; its image is inside the original exact fiber and has $z < b$. Therefore summing the right side gives a submeasure of $w_b(t+s)$. The untruncated measure has already been proved finite at fixed count. Tonelli and the common kernels prove (5.3). This is a target-height comparison, not an integral in starting roofs, and no target weight is replaced by an unweighted terminal flux.

Under (SC), apply (4.7) to the fixed interval of length d. For the assertion with $z(x) \ge c > 0$, use radial monotonicity and integrate (3.5) only above c. The leading layer cake is at most $C b^{\alpha-1}(d + B_*^{-1})$, and the finite-count error is $C_{B_*}(1 + B_* d) m^2 c^{-1/2} \theta_{B_*}^m$. Fixing c, d, B_* before the collision limit and enlarging B_* afterwards gives the same bound. This proof neither establishes (SC) nor removes the fixed c inside that limit. □

For the retained choice $q = 1/8$, $a = \sqrt{qr}$, we have $a^2 = d/2$. Consequently (5.4), whenever its stated hypothesis is verified, becomes

$$\mathcal{H}_M(C^{\varepsilon}) \le C(qr)^{257/256}. \tag{5.5}$$

The exponent is $1 + (\alpha - 1)/2$. It comes from integrating the first inverse gradient against the critical-volume power, not from the small mass of C itself. All internal incidences and all initial positive gradients are present in the finite-count comparison.

5.1. **A finite-count certificate rather than a hidden hypothesis.** The exact cost of the uncut comparison is visible in (4.6): at each fixed band it contains

$$m^2 (\mathcal{K}_m \theta_{B_*}^{2m})^{1/3}.$$

The theorem estimates the full source with that cost retained. A source-specific bound on the weighted momentum-fiber multiplicity would turn it into the vanishing critical-height theorem (5.5). Finiteness of N_m, a pressure bound for another partition, or a fixed positive-gradient theorem does not supply that bound. The condition remains an explicit proof obligation.

6. Two-ended physical traces on the short component

For a maximal original segment write $f_\sigma^{\varepsilon}(u) = A_\sigma(u) W^{\varepsilon}(\Psi_\sigma(u))$. Its nonnegative one-sided traces are denoted by f_σ^- and f_σ^+. These are the complete physical traces in (3.3); a critical entrance trace need not be zero. The comparison below neither pools the occupation classes nor identifies coincident faces from different incident components.

Let H be any Borel target restriction supported on $0 < \ell \le \delta$. Define its paired trace source at the original roof by

$$T_H^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < b_\sigma\}} \mathbf{1}_H(\Psi_\sigma(t)) \min\{f_\sigma^-, f_\sigma^+\}\, d\sigma. \tag{6.1}$$

The selector is evaluated at the target in both quantities being compared. In particular it can be the old low-clock residual, which need not be a smooth or monotone selector along a segment.

Theorem 6.1 (Complete short-source and paired-trace comparison)**.** *At every finite count and almost every roof,*

$$|R_H^\varepsilon(t) - T_H^\varepsilon(t)| \le C(1+\varepsilon^{-1}) \int_{t-\delta}^{t+\delta} v^0(u)\, du, \qquad v^0\, dt = F_\#(I_m z^{-1} \bar\nu). \tag{6.2}$$

This untruncated positive source is finite at each count by Theorem 4.3 and the bounded-gradient part. Under (SC)*,*

$$\limsup_m \sup_{R,n,k} m^2 \|R_H^\varepsilon - T_H^\varepsilon\|_\infty \le C(1+\varepsilon^{-1})\delta. \tag{6.3}$$

The constants are independent of the Borel target selector. In particular, the entire selected source on short segments with at least one grazing endpoint satisfies the same height bound.

Proof. Absolute continuity on the closed segment gives, for every interior u where the representative is defined,

$$|f_\sigma^\varepsilon(u) - \min(f_\sigma^-, f_\sigma^+)| \le \int_{a_\sigma}^{b_\sigma} |(f_\sigma^\varepsilon)'(v)|\, dv.$$

Both angular factors and the original guard give

$$|(f_\sigma^\varepsilon)'| \le C A_\sigma I_m / z + C\varepsilon^{-1} A_\sigma / z \le C(1+\varepsilon^{-1}) A_\sigma I_m / z.$$

Multiply the first inequality by the target selector. All segments of length at most δ containing t lie in $(t-\delta, t+\delta)$. At each intermediate roof the Borel segment bijection counts their interiors once, so the sum of these positive integrals is bounded by the displayed integral of v^0. The absolute value of the projected difference is bounded by this sum of absolute differences. Truncate and exhaust the original regular source, then use Tonelli. This proves (6.2) without taking a derivative of the Borel selector or a trace on its artificial boundary.

Under (SC), the complete first inverse gradient has fixed-width bound CL, as proved after (4.7). Apply it at $L = 2\delta$. If an endpoint is grazing, its original weighted trace is zero, so the minimum in (6.1) is zero. This proves the final assertion. An incoming critical endpoint with nonzero trace is not included in that zero-trace argument. □

This comparison does not claim that a minimum commutes with projection, or that (6.1) is the Jordan positive part of a signed current. It controls the error of a specific positive paired-trace comparison by the full physical derivative. The inherited signed forcing remains formed after projection.

Theorem 6.2 (A scalar reduction with distinct geometric obligations)**.** *Choose $q = 1/8$, $r = c\varepsilon(B)^4 \le r_{\mathrm{col}}$, $a = \sqrt{qr}$ and $\delta = 6qr$, with $c > 0$ fixed and small. Use $H = S$ from* (5.1) *in* (6.1)*. If* (SC) *holds, then*

$$\mathcal{E}_M \le \mathcal{H}_M(T_S^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1+\varepsilon(B)^{-1})qr. \tag{6.4}$$

Under that same condition, the original pointwise theorem is equivalent along this schedule to $\mathcal{H}_M(T_S^{\varepsilon(B)}) \to 0$. Neither (SC) *nor this paired-trace height decay is asserted established for the full Lorentz family.*

Proof. The retained finite-count scalar reduction bounds b^ε by Y^ε plus the already paid clock and protected compression terms. Split $Y = C \sqcup S$ before projection. Theorem 5.2 pays C, and Theorem 6.1 compares S to T_S. The retained positive remainder proves (6.4). Every scale and auxiliary band is fixed before the collision limit; (SC) is invoked only at each such fixed band. The band sequence is enlarged afterwards, and B is last.

For the schedule, the additional powers are $(qr)^{257/256} = O(\varepsilon^{257/64})$ and $(1+\varepsilon^{-1})qr = O(\varepsilon^3)$; both vanish. Sufficiency follows. Conversely $S^\varepsilon \le b^\varepsilon$, so scalar closure and the positive remainder imply $\mathcal{H}_M(S^{\varepsilon(B)}) = O(B^{-1/192})$. Equation (6.3) then implies decay of T_S. This necessity statement is conditional on the same spectral-capacity estimate as its sufficiency statement. □

Without (SC), the finite-count comparisons (5.3) and (6.2) remain valid with the explicit error in (4.6). Thus the new reduction identifies two separate tasks: control a weighted two-dimensional momentum-fiber multiplicity against the spectral error, and control the original paired positive boundary traces on genuinely short segments. The first is not a roof-height assumption; the second retains the exact target selector, physical endpoints and original guard. Finite-count finiteness does not prove either asymptotic task. Scalar upper closure, not a finite-norm consequence, still precedes every unrestricted point-conditioned downstream assertion.

7. Global momentum uniqueness and the stable optical weight

The multiplicity in (4.2) can be estimated dynamically. Two separate facts are needed: a global uniqueness assertion for each lifted itinerary, and a comparison of the optical weight with a genuine stable Jacobian. Neither follows merely from the nonzero local endpoint Hessian.

Theorem 7.1 (One physical orbit per itinerary and momentum pair)**.** *Fix a finite lifted sequence of circular scatterers and two endpoint tangential momenta in $(-1,1)$. There is at most one regular physical trajectory with that itinerary and those endpoint momenta. Internal incidences need not be bounded below uniformly. The assertion concerns physical contact points, not their overlapping angular coordinates.*

Proof. Write $Q = (Q_0, \dots, Q_m)$ for the contact positions and

$$S(Q) = \sum_{j=0}^{m-1} |Q_{j+1} - Q_j|.$$

This is a convex function of the Cartesian contact variables. It is differentiable at every configuration with positive flight lengths. For two such configurations $Q, \widehat{Q}$, monotonicity of its subgradient gives

$$\sum_{j=0}^{m} (\nabla_j S(Q) - \nabla_j S(\widehat{Q})) \cdot (Q_j - \widehat{Q}_j) \ge 0. \tag{7.1}$$

Let n_j be the outward unit normal of the jth circular scatterer and let t_j be its fixed oriented tangent. Reflection and first variation give

$$\nabla_0 S = -c_0 n_0 - p_0 t_0, \quad \nabla_j S = -2c_j n_j \ (0 < j < m), \quad \nabla_m S = -c_m n_m + p_m t_m.$$

At the two trajectories $c_0 = \widehat{c}_0 > 0$ and $c_m = \widehat{c}_m > 0$, since their endpoint momenta agree. The scatterer centers agree at every index, so $Q_j - \widehat{Q}_j = R(n_j - \widehat{n}_j)$. Rotation by a right angle is skew symmetric; consequently $(t_j - \widehat{t}_j) \cdot (n_j - \widehat{n}_j) = 0$. The left side of (7.1) therefore equals

$$\begin{aligned} &- Rc_0 |n_0 - \widehat{n}_0|^2 - Rc_m |n_m - \widehat{n}_m|^2 \\ &- R \sum_{j=1}^{m-1} (c_j + \widehat{c}_j) |n_j - \widehat{n}_j|^2. \end{aligned}$$

Every coefficient is strictly positive. Thus all contact positions coincide. The flight directions and physical state then coincide as well. Repeated visits to a scatterer cause no change: the contact variables at different times are distinct Cartesian variables. No convexity of a physical word domain, existence of an interpolating physical trajectory, or global angular lift is needed. □

For the Jacobian comparison use standard billiard coordinates (s, φ), where s is arclength and $|p| = |\sin\varphi|$. Signs are fixed consistently on each oriented boundary component. Let M' be the full-measure set regular in both infinite time directions. Write $J_R^u T_R^j$ and $J_R^s T_R^j$ for the positive arclength Jacobians on its invariant unstable and stable lines. The invariant cones have slopes in fixed compact positive and negative intervals, respectively, with a uniform angle between them [25, Section 1.1].

Lemma 7.2 (Optical weight and removal of one incidence)**.** *There are uniform geometric constants C_0, C_1 such that on M', for every $m \geq 1$,*

$$C_0^{-1} c_m J_R^s T_R^m(x) \leq \omega(x) \leq C_0 c_m J_R^s T_R^m(x), \qquad \frac{J_R^s T_R(x)}{c(x)} \leq C_1. \tag{7.2}$$

In particular, for an internal mark $1 \leq i < m$,

$$\frac{\omega(x)}{c_i(x)} \leq C_0 C_1 J_R^s T_R^i(x) J_R^s T_R^{m-i-1}(T_R^{i+1} x). \tag{7.3}$$

The stable factor at the marked time is removed, not evaluated with a singular inverse-incidence multiplier left over.

Proof. Up to consistent orientation signs, one flight has derivative

$$\frac{1}{c'} \begin{pmatrix} \tau\varkappa + c & \tau \\ \tau\varkappa\varkappa' + \varkappa c' + \varkappa' c & \tau\varkappa' + c' \end{pmatrix}, \qquad \varkappa = \varkappa' = R^{-1}.$$

Its determinant is c/c'. Every entry is positive. The ratio of its first column to its second, in either row, lies between $\varkappa_{\min}$ and $\varkappa_{\max} + \tau_-^{-1}$. Left multiplication by a positive matrix preserves these componentwise inequalities. If b_m is the upper-right entry of the m-flight product, the image of the vertical vector has slope in the unstable cone. Comparing it with a unit invariant unstable vector at the source consequently gives

$$C^{-1} b_m \leq J_R^u T_R^m(x) \leq C b_m.$$

Here the constants do not accumulate with m: both the initial column comparison and the final slope comparison use fixed cones. Changing from φ_0 to p_0 and from s_m to α_m gives $\omega = R c_0 / b_m$.

Put $e(x) = \sin\angle(E^s(x), E^u(x))$. Conservation of collision area in these coordinates is the exact identity

$$J_R^u T_R^m(x) J_R^s T_R^m(x) \frac{e(T_R^m x)}{e(x)} = \frac{c_0}{c_m}.$$

The function e is bounded above and below by positive cone constants. Combining the last three formulas proves the two-sided comparison in (7.2). For one flight, if v is the initial unstable slope, then

$$J_R^u T_R(x) = \frac{\tau\varkappa + c + \tau v}{c'} \frac{\sqrt{1 + (v')^2}}{\sqrt{1 + v^2}}.$$

The same area identity shows that $J_R^s T_R / c$ is bounded by a fixed constant divided by $\tau\varkappa + c + \tau v$. The latter is bounded below by a positive geometric constant. This proves the one-step bound. Factor the stable cocycle at time i and use $c_m \leq 1$ to obtain (7.3). □

The area formula in module 235 can be restricted to M' without altering its almost-everywhere momentum density. The omitted set has zero collision area. On countably many regular endpoint charts G is a local diffeomorphism, so its image has zero two-dimensional Lebesgue measure. Thus the infinite-time invariant lines used above do not add a restriction to the original finite-record source.

8. A marked pressure comparison for the actual momentum fibers

Normalize the initial lifted scatterer to the origin. Let $\mathcal{W}_j(R)$ be the finite set of realizable physical itineraries of j flights, retaining the lifted displacement of each step. If $[w]_R$ is the regular domain of w, define

$$P_j(R) = \sum_{w\in\mathcal{W}_j(R)} \sup_{x\in[w]_R\cap M'} J_R^s T_R^j(x), \qquad P_0(R) = 1. \tag{8.1}$$

These are geometric upper sums, not replacements for the exact event. They impose no occupation or section test; those tests are dropped only on the upper side of the comparison below. There is one fixed finite step alphabet of size D: successive lifted centers are at distance at most $\tau_+ + 2R_+$. It follows in particular that every $P_j(R)$ is finite and that each fixed-j supremum over R is finite.

For comparison with the established pressure theory let $\mathcal{Q}_j(R)$ be the stable-Jacobian sum at potential parameter one over the homogeneous continuity partition $\mathcal{M}_0^{j,H}$ of [25, equations (2.4), (2.11)]. Its cells retain the true singularities through the terminal time and homogeneity cuts through time $j-1$. Each cell determines one lifted physical itinerary. Splitting into those cells can only increase a sum of suprema of a positive function. Hence

$$P_j(R) \le \mathcal{Q}_j(R). \tag{8.2}$$

No identification of the two sums is asserted.

Let $\mathcal{K}_m(R)$ denote (4.2) with the radius held fixed. Thus $\mathcal{K}_m = \sup_R \mathcal{K}_m(R)$.

Theorem 8.1 (Complete momentum-fiber pressure bound)**.** *With a constant depending only on the uniform physical geometry,*

$$\mathcal{K}_m(R) \le 1 + C\left[P_m(R) + D\sum_{i=1}^{m-1} q_0^{d_i} P_i(R) P_{m-i-1}(R)\right]. \tag{8.3}$$

All physical preimages, internal incidences and original exact labels are included. There is no factor counting homogeneity strips at the marked collision.

Proof. Fix an almost-everywhere momentum target and one exact event. By Theorem 7.1, each itinerary supplies at most one preimage. The denominator in (4.1) is at least $4\pi c_*\lambda_\circ^2$. The term without a mark is therefore bounded by $CP_m(R)$ using Lemma 7.2.

For the mark at i, use (7.3). A full itinerary is uniquely determined by its first i flights, its single step at time i, and its remaining $m-i-1$ flights, with the last block translated back to the initial fundamental scatterer. The middle step has at most D choices. The weighted sum of its possible preimages is consequently at most $CDP_i(R)P_{m-i-1}(R)$. This argument uses physical itineraries, not the infinitely many homogeneity strips whose inverse-incidence weight would no longer be summable after removing the marked stable factor. Multiplying by $q_0^{d_i}$ and summing proves the bound after taking the required essential and exact-label suprema. The finite horizon alphabet and global uniqueness supply every multiplicity used in this argument. □

Corollary 8.2 (Bounded capacity at each fixed radius)**.** *For every fixed $R \in I$,*

$$\sup_{m\ge1} \mathcal{K}_m(R) < \infty. \tag{8.4}$$

In particular $m^6\mathcal{K}_m(R)\theta^{2m} \to 0$ *for every fixed* $0 < \theta < 1$.

Proof. For each fixed finite-horizon dispersing table, the pressure theorem [25, Lemma 3.8, proof] gives $\sup_j \mathcal{Q}_j(R) < \infty$. Apply (8.2) and (8.3), and use $\sum_i q_0^{d_i} \le 2q_0/(1-q_0)$. This use of the pressure theorem is now justified by an explicit comparison of the same optical weight and momentum preimages; it is not a comparison of cardinalities. □

The constant in the cited fixed-table pressure theorem has not been shown here to be uniform as R varies. The quantifiers in (8.4) must not be interchanged. The following finite-block theorem gives an explicit family-uniform alternative.

Theorem 8.3 (Uniform finite-block capacity certificate)**.** *Put* $\overline{P}_j = \sup_{R\in I} P_j(R)$. *For every fixed integer* $L \ge 1$ *define*

$$\rho_L = \max\{1, \overline{P}_L^{1/L}\}, \qquad C_L = \max_{0\le j<L} \overline{P}_j \rho_L^{-j}.$$

There is an explicit finite geometric multiple A_L *of* $1 + C_L + DC_L^2 \sum_{i\ge1} 2q_0^i$ *such that*

$$\mathcal{K}_m \le A_L \rho_L^m \qquad (m \ge 1). \tag{8.5}$$

Consequently (SC) holds at a fixed auxiliary band whenever, for some fixed L,

$$\rho_L \theta_{B_*}^2 < 1. \tag{BC}$$

If this certificate holds at an unbounded sequence of previously fixed bands, all radius-uniform consequences of (SC) in the retained article follow. No numerical verification of (BC) over I *is claimed in this revision.*

Proof. Splitting a physical itinerary into two consecutive blocks is injective into the pair of block itineraries. The stable cocycle gives $P_{j+k}(R) \le P_j(R)P_k(R)$, so also $\overline{P}_{j+k} \le \overline{P}_j\overline{P}_k$. Writing $m = vL + j$, $0 \le j < L$, proves $\overline{P}_m \le C_L\rho_L^m$. Insert this and the corresponding bound for the two factors in (8.3). Since $\rho_L \ge 1$ and the depth weights are summable, it gives (8.5). Under (BC),

$$m^6\mathcal{K}_m\theta_{B_*}^{2m} \le A_L m^6(\rho_L\theta_{B_*}^2)^m \longrightarrow 0.$$

All suprema over R, block lengths and bands are in the displayed order. The block is fixed before m and is allowed to depend on the previously chosen auxiliary band, never on collision count. □

In particular

$$\limsup_m m^{-1}\log\mathcal{K}_m \le \inf_{L\ge1} \max\{0, L^{-1}\log\overline{P}_L\}.$$

This is an actual uniform growth estimate expressed through geometric partition sums. It replaces the unquantified fiber cardinality of module 235. Establishing that its right side is zero, or certifying the strict inequalities (BC), remains a family-uniform question; pointwise-in-R pressure bounds alone do not settle it.

9. Unconditional fixed-radius recovery and the uniform endpoint

The preceding comparison discharges the spectral-capacity hypothesis for each fixed physical radius. We state the resulting source estimates with that quantifier visible. For a nonnegative family let

$$\mathcal{H}_{M,R}(f) = \limsup_m \sup_{n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} f(t).$$

It differs from $\mathcal{H}_M$ by placing the radius outside the collision limsup.

Theorem 9.1 (Uncut recovery with no capacity hypothesis at fixed radius)**.** *For every fixed $R \in I$, every fixed interval length $L > 0$, and $0 < b < b_0$,*

$$\limsup_m \sup_{n,k} \sup_{|J|=L} m^2 \int_J w_b \le CLb^{1/128}, \qquad \limsup_m \sup_{n,k} \sup_{|J|=L} m^2 \int_J v^0 \le CL. \tag{9.1}$$

The leading constant is independent of R; the collision count needed for a prescribed error is not asserted uniform in R. For the disjoint residuals of module 236 and for every original Borel short-segment selector H of total residence at most δ,

$$\mathcal{H}_{M,R}(\mathsf{C}^\varepsilon) \le C(qr)^{257/256}, \tag{9.2}$$

$$\limsup_m \sup_{n,k} m^2 \|R_H^\varepsilon - T_H^\varepsilon\|_\infty \le C(1+\varepsilon^{-1})\delta. \tag{9.3}$$

In particular (9.3) *bounds the entire selected short source with a grazing physical endpoint, without any positive initial-gradient or internal-incidence cutoff.*

Proof. Let $K_R = \sup_m \mathcal{K}_m(R) < \infty$. The finite-count estimate (4.6), before taking any limit, has remainder at most

$$CC_{B_*}^{2/3}(1+B_*L)^{2/3}m^2K_R^{1/3}\theta_{B_*}^{2m/3}.$$

It vanishes for each fixed R, L, B_*. Enlarge B_* only after the collision limit. The leading constants in the inherited local source estimate and the new geometric comparison are uniform, proving (9.1). Adding $z \ge b_0$ uses the retained positive-cutoff estimate, not a new zero-cutoff exchange.

The finite-count critical certificate (5.3) is at the target roof, and so (9.1) on its interval of length $d = 2qr$ proves (9.2). The two-ended finite-count comparison (6.2), with $L = 2\delta$, proves (9.3). Neither proof differentiates a Borel target selector. At a grazing physical endpoint one original weighted trace is zero, so its same-segment minimum is zero. A backward critical trace is not deleted by that argument. □

For one fixed R let $\mathcal{E}_{M,R}$ denote the collision limsup of the original scalar error. With the retained schedule $q = 1/8$, $r = c\varepsilon(B)^4$, $a = \sqrt{qr}$ and $\delta = 6qr$, the preceding theorem gives, without (SC),

$$\mathcal{E}_{M,R} \le \mathcal{H}_{M,R}(T_{\mathsf{S}}^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1+\varepsilon(B)^{-1})qr. \tag{9.4}$$

This follows by the original positive scalar decomposition and the disjoint residual partition before their suprema are taken. The positive paired trace on short nongrazing segments has not been proved small, even at a fixed radius; (9.4) is not a claim of pointwise scalar closure.

For the original radius-uniform target, (BC) on an unbounded sequence of fixed auxiliary bands implies the same displayed bound with $\mathcal{E}_M$ and $\mathcal{H}_M$. This is Theorem 6.2 with its capacity condition supplied by Theorem 8.3. There are still two separate family-uniform tasks: certify this pressure-growth margin, and prove decay of the positive same-segment trace. The new comparison proves the first at every fixed radius but does not turn that assertion into $\limsup_m \sup_R$ by compactness alone. Nor does it identify the paired minimum with the Jordan positive part of a projected current.

The original exact labels, guard, section and arithmetic reference remain those of (2.5). All original unconditional finite-roof-norm and lower pointwise conclusions are retained; the unrestricted upper pointwise half, and the downstream conclusions requiring it, are not promoted. No zero arithmetic class is normalized.

10. Uniform geometric pressure by finitely many transverse tests

The fixed-table conclusion in Corollary 8.2 is not interchanged with a supremum over tables. We prove a uniform bound directly. The argument uses expansion and distortion, but no mixing time, invariant Cantor rectangle or parameter-dependent spectral gap.

We use standard collision coordinates (s,φ) in this section; s is physical arclength and $p=\sin\varphi$. Jacobians along curves refer to Euclidean arclength in these coordinates. The index of a homogeneity strip is denoted by k; its width is $O(|k|^{-3})$. Let $\mathcal{M}_0^{m,H}(R)$ be the homogeneous continuity partition of [25, (2.4)], and set

$$Q_m(R)=\sum_{A\in\mathcal{M}_0^{m,H}(R)}\sup_{x\in A\cap M_R'}J^sT_R^m(x),\qquad Q_0(R)=1. \tag{10.1}$$

These are geometric partition sums, without the section, guard or four-record twist. They are used only as positive upper bounds for the original exact-label fibers through Theorem 8.1.

Lemma 10.1 (Uniform geometric constants)**.** *The given circular family admits one choice of cone bounds, homogeneity threshold, short-curve length and constants with the following properties.*

For every homogeneous short cone-stable curve W, let $\mathcal{G}_j(W)$ be its homogeneous inverse-image components, subdivided at the chosen length scale. Then

$$\sum_{V\in\mathcal{G}_j(W)}\sup_V J_VT_R^j\le C_g,\qquad j\ge 0. \tag{10.2}$$

On one homogeneous backward continuity cell, the ratio of the inverse Jacobians on any two short cone-stable curves is between C_d^{-1} and C_d, independently of the iterate.

For $A\in\mathcal{M}_0^{m,H}(R)$, let $B_{m-1}(A)$ be the cell of $\mathcal{M}_{-m+1}^{0,H}(R)\vee H$ containing $T_R^{m-1}A$. There are $\delta_*>0$ and $c_{\mathrm{long}}>0$, independent of R,m, such that

$$c_{\mathrm{long}}Q_m(R)\le\sum_{\substack{A\in\mathcal{M}_0^{m,H}(R)\\ \operatorname{diam}^u B_{m-1}(A)\ge\delta_*}}\sup_A J^sT_R^m. \tag{10.3}$$

The geometric constant c_{long} is unrelated to the unchanged section normalization c_*.

Proof. The common geometric bounds are a positive minimum free flight, a finite maximum free flight, positive lower and upper curvatures, and bounded third derivatives of the circular boundaries. The number of possible one-flight lattice displacements is uniformly finite. Physical arclengths can be identified with fixed collision circles by uniformly bounded linear changes of coordinate. Thus [26, Theorem 2.5 and Remark 2.7] applies with one set of geometric constants. Its uniform growth estimate, Lemma 3.2(b), is (10.2). The whole-cell distortion statement is [25, Lemma 2.2]; its proof sums the common hyperbolic contraction and homogeneous distortion bounds and therefore gives one C_d on this family. It compares different stable curves, not merely two points on the same curve.

For clarity, the uniformity needed in the long-cell lemma is a finite geometric choice, rather than a compactness assertion about $\sup_m Q_m(R)$. Use [25, Lemmas 3.3, 3.5 and 3.6] only at potential parameter one and additional potential zero. In their proof choose the one-step small-curve contraction $\vartheta<1$ and its prefactor C_0 from the common growth bounds. Choose an integer n_1 so that the resulting geometric loss series $C_0\sum_{j\ge1}\vartheta^{jn_1}$ is smaller than the fixed loss fraction used in Lemma 3.5. This choice is independent of R. The short-curve threshold for the first $2n_1$ iterates is then chosen using the same one-step expansion and homogeneous tail. Only finitely many iterates have been specified at this stage. Their branchwise Hölder moduli and complexity bounds are uniform: the free-flight formula has the same square-root contact singularity, bounded

geometric coefficients and a finite common displacement alphabet. The identical construction applies to T_R^{-1}.

In the proof of Lemma 3.6 choose the smaller of the forward and backward length thresholds, and then choose n_2 with $C\Lambda^{-n_2}$ below that threshold, where $\Lambda > 1$ is the common cone expansion. Partition a backward cell by its most recent ancestor having the prescribed unstable length. The branches which have not yet reached that length are charged to the preceding small-curve geometric series. Applying distortion and summing that series leaves a fixed positive fraction of the full pressure on cells with a long unstable curve. This is precisely the inequality of that lemma, with its $\delta_2/3$ renamed δ_*. Its remaining finitely many initial iterates are handled by reducing the common length threshold: the homogeneous tails have uniformly summable weights, and the remaining finite physical branches have the same uniform geometric bounds. This gives (10.3). No step of this choice uses Proposition 3.14 of that paper, its mixing argument, or its table-dependent return time. The section normalization is not used in this geometric lemma. □

Lemma 10.2 (Finite testing of long backward cells)**.** *There are integers $N_*, D_* < \infty$ and, for each R, homogeneous short cone-stable curves $W_1(R), \dots, W_{N_*}(R)$ such that every cell B in* (10.3) *meets at least one test curve. The same N_* works for all R and all m. At most D_* different forward cells A have a given $B_{m-1}(A) = B$.*

Distinct backward cells assigned to one test can be represented by distinct members of $\mathcal{G}_{m-1}(W_j)$. The representatives are chosen for summation only and introduce no new physical stopping rule.

Proof. A homogeneous cone-unstable arc of length δ_* has a coordinate projection of length at least $c\delta_*$. Since its slope is bounded above and away from zero, it cannot lie in a homogeneity strip of width smaller than $c'\delta_*$. Hence only finitely many strip indices can occur. Work in finitely many overlapping lifted collision charts. Choose a slope strictly inside the common stable cone. The affine coordinate transverse to lines of that slope increases along every cone-unstable arc at speed bounded below by a positive constant. A grid of transverse levels with mesh less than $c''\delta_*$ must therefore meet every such arc in its interior. Split these grid lines into overlapping short pieces and intersect with the finitely many relevant homogeneity strips. This produces at most N_* tests. The overlap removes chart-end ambiguities. A generic translation of the grids avoids coincidence with true singular arcs; an arbitrarily small such translation preserves the strict mesh inequality. Tests are chosen separately for each R, with the same numerical bounds; no radius-independent exceptional null set is required.

The boundary of a backward continuity cell consists of cone-unstable arcs and horizontal homogeneity boundaries, in lifted collision coordinates. Such cells are convex along a cone-stable test: successive crossings of their unstable boundaries separate different cells, and an arc cannot end freely in a regular interior. This remains true under inverse iteration, by invariance of the unstable cone and the fact that new arc ends lie on previous singular or strip boundaries. Consequently each connected test intersection has one homogeneous inverse branch. Extra length subdivisions can only refine it; choose one of them. Different B's give disjoint inverse branches, so no member is charged twice.

Finally $T_R^{m-1}A$ is obtained from B by the last physical one-step singularity cuts. There is no homogeneity cut at the final image in the definition of $\mathcal{M}_0^{m,H}$. The one-step true singularity set has a uniformly bounded number of lifted cone-stable arcs, because only a fixed finite set of disks can be the next hit. Each such arc crosses a cone-stable-convex backward cell in at most one connected arc. Their ordered graph cuts make at most $D_*' + 1$ components; take $D_* = D_*' + 1$. This last bound counts physical one-step cuts, not the countably many homogeneous strips. □

Theorem 10.3 (Uniform bounded geometric pressure)**.** *There is $Q_* < \infty$ such that*

$$\sup_{R \in [R_-, R_+]} \sup_{m \geq 0} Q_m(R) \leq Q_*. \tag{10.4}$$

In particular the actual geometric itinerary sums $P_m(R)$ in (8.1) satisfy the same bound.

Proof. There is a uniform one-step bound $J^s T_R \le C_1$. For every forward cell A counted in (10.3), put $B = B_{m-1}(A)$. Dropping the last stable factor gives

$$\sup_A J^s T_R^m \le C_1 \sup_{y \in B \cap M'_R} J^s T_R^{m-1}(T_R^{-m+1} y).$$

Assign B to its first meeting test W_j. Compare a true stable curve through y to that test inside the same homogeneous backward cell, using Lemma 10.1. Inverting the two backward Jacobians bounds the last supremum by $C_d \sup_V J_V T_R^{m-1}$ for a representing homogeneous inverse component V of W_j. The distortion bound is uniform over the whole cell; the test need not meet the particular final one-step subcell $T_R^{m-1} A$.

There are at most D_* such subcells per B. The injective assignment in Lemma 10.2 and the curve growth bound therefore give, for $m \ge 2$,

$$c_{\mathrm{long}} Q_m(R) \le C_1 D_* C_d \sum_{j=1}^{N_*} \sum_{V \in \mathcal{G}_{m-1}(W_j)} \sup_V J_V T_R^{m-1} \le C_1 D_* C_d N_* C_g.$$

The same geometric one-step estimates bound Q_1 uniformly, and $Q_0 = 1$. Taking the larger of these constants proves the assertion. Finally the homogeneous continuity partition refines each physical itinerary; positivity gives $P_m(R) \le Q_m(R)$. Every sum is taken at finite collision count before either supremum. In particular (10.4) is not obtained by interchanging $\sup_R$ with a fixed-table collision limit. □

11. Uniform momentum capacity and uncut gradient moments

The preceding pressure theorem applies to the exact optical weight through the already proved marked comparison. No unweighted word count or unspecified finite fiber multiplicity enters the conclusion.

Theorem 11.1 (Uniform weighted momentum capacity)**.** *For the original radius family and every finite collision count,*

$$\mathcal{K}_m \le K_* := 1 + C\left(Q_* + D Q_*^2 \frac{2q_0}{1-q_0}\right) < \infty. \tag{11.1}$$

Consequently, for every fixed admissible auxiliary band,

$$m^6 \mathcal{K}_m \theta_{B_*}^{2m} \longrightarrow 0. \tag{11.2}$$

The strict finite-block criterion of Theorem 8.3 is also satisfied analytically for a sufficiently large fixed block at each such band.

Proof. Insert $P_j(R) \le Q_*$ in Theorem 8.1. The complete depth sum is at most $2q_0/(1-q_0)$; its coefficient D counts the single omitted physical step. Global fixed-itinerary momentum uniqueness and the optical/stable comparison have already been proved before this insertion. Thus this bounds the actual $I_m \omega / \det K$ sum on each momentum fiber, including all internal incidence scales and all exact labels.

Since $\theta_{B_*} < 1$, the right side of $m^6 \mathcal{K}_m \theta_{B_*}^{2m} \le K_* m^6 \theta_{B_*}^{2m}$ tends to zero with B_* fixed. Also $\sup_R P_L(R) \le Q_*$, so that

$$\max\{1, (\sup_R P_L(R))^{1/L}\} \theta_{B_*}^2 < 1 \quad \text{if} \quad L > \frac{\log \max(1, Q_*)}{-2 \log \theta_{B_*}}.$$

This is an existence certificate with finite constants, not a numerical interval computation of an optimal block or mixing rate. The block may depend on the already fixed auxiliary band, never on m. Equation (11.2) itself does not require a block choice at all. □

Put $\alpha = 129/128$. For $1 \le p < \alpha$ and $0 < b < b_0$, define an actual nonnegative measure on the same exact event by

$$w_{p,b}(t)\,dt = F_{\#}\Big(I_m z^{-p}\mathbf{1}_{\{0<z<b\}}\bar{\nu}\Big). \tag{11.3}$$

Here the exponent acts on z, not on I_m; no assertion about $(I_m/z)^p$ is made. Values on the original area-null critical set are immaterial. The source is untruncated at $z = 0$.

Lemma 11.2 (The two-cap moment integral)**.** *If $E, H, b > 0$ and $1/2 < p < 2$, then*

$$\begin{aligned} &b^{-p}\min\{Eb^{1/2}, Hb^2\} + p\int_0^b v^{-p-1}\min\{Ev^{1/2}, Hv^2\}\,dv \\ &\qquad \le C_p \min\{Hb^{2-p}, E^{(4-2p)/3}H^{(2p-1)/3}\}. \end{aligned} \tag{11.4}$$

Proof. The quadratic cap alone gives $2Hb^{2-p}/(2-p)$. The caps agree at $v_0 = (E/H)^{2/3}$. Below v_0 use Hv^2; above it use $Ev^{1/2}$. The two resulting integrals are bounded by multiples of Hv_0^{2-p} and $Ev_0^{1/2-p}$, respectively, and both equal $E^{(4-2p)/3}H^{(2p-1)/3}$. The boundary term has the same bound. Finiteness at zero uses $p < 2$, and at the extended upper limit it uses $p > 1/2$. Both inequalities hold together, proving their minimum after increasing C_p. □

Theorem 11.3 (Uniform uncut first and nearby gradient moments)**.** *For every $1 \le p < \alpha$, every fixed auxiliary band B_* and $|J| = L$, the source in* (11.3) *satisfies*

$$\begin{aligned} m^2\int_J w_{p,b} &\le C_p b^{\alpha-p}(L + B_*^{-1}) \\ &\quad + C_p \min\Big\{m^2K_*b^{2-p}, [C_{B_*}(1+B_*L)]^{(4-2p)/3}m^2K_*^{(2p-1)/3}\theta_{B_*}^{m(4-2p)/3}\Big\}. \end{aligned} \tag{11.5}$$

Constants are uniform in radius, exact labels, collision count and interval location. In particular, for each fixed $b, L > 0$,

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} m^2\int_J w_{p,b} \le C_p L b^{\alpha-p}. \tag{11.6}$$

The full source $v^0 = F_{\#}(I_m/z\,\bar{\nu})/dt$ obeys

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} m^2\int_J v^0 \le CL. \tag{11.7}$$

No spectral-capacity hypothesis remains in these conclusions.

Proof. The endpoint-momentum area formula and Theorem 11.1 give $\int_J u_v \le K_* v^2$, at every count and every $v > 0$ in range. The inherited critical-volume estimate gives

$$m^2\int_J u_v \le Cv^\alpha(L + B_*^{-1}) + Ev^{1/2}, \qquad E = C_{B_*}(1+B_*L)m^2\theta_{B_*}^m.$$

The source on the left is nonnegative. Combining its two upper bounds therefore retains the leading term and caps the error by $\min(Ev^{1/2}, Hv^2)$, with $H = m^2K_*$.

Apply the positive identity

$$z^{-p}\mathbf{1}_{\{z<b\}} = b^{-p}\mathbf{1}_{\{z<b\}} + p\int_0^b v^{-p-1}\mathbf{1}_{\{z<v\}}\,dv.$$

Integrate first from a positive lower endpoint and then use monotone convergence. The leading integral is finite because $p < \alpha$; Lemma 11.2 pays the entire error down to zero. The exponents of m^2 in the two factors sum to one, giving (11.5). The physical quadratic bound also proves finiteness of the measure at each count. Local coarea away from its area-null critical set, followed by monotone exhaustion, gives its density.

Fix b, L, B_* before the collision limit. The second expression in the minimum tends to zero uniformly in R, n, k. Enlarge B_* only afterwards to obtain (11.6). For $p = 1$, add the region $z \geq b_0/2$ using the inherited complete I_m source bound. This proves (11.7). Neither a collision-dependent cutoff nor a band growing with m is used. □

12. Uniform critical recovery and the positive short trace

The capacity obstruction and the positive short-trace obstruction are different. The former has now been discharged uniformly in the radius. We record exactly which essential-height conclusions this supplies, and keep the latter source explicit.

Use the original residual partition from Section 5: $d = 2qr$, $\mathsf{S} = Y \cap \{\ell \leq 3d\}$ and $\mathsf{C} = Y \cap \{\ell > 3d\}$. For a Borel target selector H supported on segments of total length at most δ, write R_H^ε for its original weighted source and T_H^ε for the same-segment minimum of the two weighted physical endpoint traces. No selector trace is introduced.

Theorem 12.1 (Radius-uniform recovery without a capacity premise)**.** *For the entire prescribed radius interval, the critical target source satisfies*

$$\limsup_m \sup_{R,n,k} m^2 \|\mathsf{C}^\varepsilon\|_\infty \leq C d (a^2 + 2\Lambda_\circ d)^{1/256}. \tag{12.1}$$

Every previously fixed Borel selection on short segments satisfies

$$\limsup_m \sup_{R,n,k} m^2 \|R_H^\varepsilon - T_H^\varepsilon\|_\infty \leq C(1 + \varepsilon^{-1})\delta. \tag{12.2}$$

In particular this bounds the complete selected source on all segments with an actual grazing endpoint, since their paired minimum is zero. Neither estimate requires a positive initial-gradient or internal-incidence cutoff. The supremum is over the original exact labels, not a neighboring-label enlargement.

Proof. The finite-count target certificate in Theorem 5.2 reads

$$\mathsf{C}^\varepsilon(t) \leq C_0 \int_t^{t+d} w_{1,b}(u)\, du, \qquad b = (a^2 + 2\Lambda_\circ d)^{1/2}.$$

Its low-clock argument forces a loss of at least one half of the initial angular density inside a full surviving forward interval. The target-direction map is injective at each intermediate roof. Apply (11.6) with interval length d; $b^{1/128} = (a^2 + 2\Lambda_\circ d)^{1/256}$ gives (12.1).

For the other assertion the inherited same-segment trace comparison is already a finite-count inequality on common kernels:

$$|R_H^\varepsilon(t) - T_H^\varepsilon(t)| \leq C(1 + \varepsilon^{-1}) \int_{t-\delta}^{t+\delta} v^0(u)\, du.$$

The target selector is held at t and not differentiated. The Borel segment bijection supplies multiplicity one for all joining intervals. Equation (11.7) now pays the right side uniformly in the radius, proving (12.2). A grazing endpoint has zero original optical trace. A backward critical endpoint need not, and is not discarded by this statement. □

Theorem 12.2 (The original uniform pointwise reduction)**.** *Keep $q = 1/8$, $\varepsilon(B) = A_0 B^{-1/12}$, $r = c\varepsilon(B)^4$, $a = \sqrt{qr}$ and $\delta = 6qr$. For all sufficiently large fixed B,*

$$\mathcal{E}_M \leq \mathcal{H}_M(T_{\mathsf{S}}^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1 + \varepsilon(B)^{-1})qr. \tag{12.3}$$

Unlike the earlier family-wide reduction, this assertion has no additional condition (SC) or finite-block margin as a hypothesis. It uses the inherited physical and four-record analytic inputs and the uniform pressure theorem proved above.

Proof. The running-compression reduction is an identity and upper comparison on the original source before its roof supremum. Its residual is the disjoint sum $\mathsf{C}^{\varepsilon} + \mathsf{S}^{\varepsilon}$. Apply Theorem 12.1 to each part and the retained positive scalar remainder. With $a^2 = qr$ and $d = 2qr$, the critical cost in (12.1) is $C(qr)^{257/256}$. The paired comparison cost is $C(1 + \varepsilon^{-1})qr$. All auxiliary bands are fixed before the collision limit and enlarged afterwards. The common uniform constants permit the radius supremum inside that limit; no fixed-radius conclusion is used in its place. This gives (12.3).

Under the retained outer schedule the explicit costs have sizes $B^{-1/192}$, $O(\varepsilon^{257/64})$ and $O(\varepsilon^3)$. They tend to zero with B last. Thus decay of the positive paired height along this schedule suffices to finish the original uniform upper half, and hence the full pointwise theorem with the retained lower half. That paired-height decay is not proved here. □

Remark 12.3 (The remaining positive source). The minimum in $T^{\varepsilon}_{\mathsf{S}}$ is taken before the physical sheets are summed. It is not the Jordan positive part of an already projected signed current. A small difference between the two traces cannot bound their positive minimum. On nongrazing short segments both traces may be positive, including at a backward critical entrance. Theorems 11.1 and 12.1 remove the separate family-uniform capacity premise; they do not supply a boundary deconcentration theorem for this positive minimum. Accordingly the unrestricted upper pointwise law, full raw local limit and unrestricted point-conditioned consequences are not marked proved. The original target, all zero arithmetic classes and the scalar-first normalization order remain unchanged.

Appendix A. Dependencies of the uniform conclusion

The new proof separates the following four steps. The first is purely geometric and has no exact-label spectral input. Uniform short-curve growth is taken in the geometric family of Demers–Zhang, with the common arclength identification explicitly allowed there. The whole-cell distortion and long-cell reduction of Baladi–Demers are used at potential parameter one. Lemma 10.1 records the finite choices required for their uniformity. The finite transverse testing and final-step multiplicity are proved in Lemma 10.2. No uniform mixing rate is invoked.

Second, the exact optical-to-pressure bridge is Theorem 8.1, based on the global fixed-itinerary uniqueness and the nonaccumulating stable-Jacobian comparison in the preceding section. Thus the pressure theorem controls the actual weighted momentum fiber, not just a partition count. Third, the existing endpoint-momentum area formula and critical-volume input are combined by the positive layer cake in Theorem 11.3; all cutoffs reach zero before the collision limit. The finite-count error is displayed there.

Fourth, the existing target loss certificate and same-segment comparison are applied before the roof supremum. Their common coarea versions retain the one-sided incident components, both physical endpoints and arbitrary Borel target selection. Theorem 12.1 changes the available uniformity, not these finite-count identities. The final positive paired trace has not been bounded. The conditional historical formulas in the archive remain verbatim; the active uniform capacity authority is Theorem 11.1.

Part 5. Global momentum uniqueness and marked physical pressure

Retained revision-99 abstract. We study the exact-label pointwise raw local limit problem for actual returns in a triangular Lorentz gas. Convexity of the Cartesian flight action proves global uniqueness for a prescribed lifted itinerary and two endpoint momenta. A comparison of the inverse optical twist with the invariant stable Jacobian then controls the actual weighted momentum-fiber multiplicity by a marked pressure sum. The mark removes one stable factor; a

finite physical step alphabet pays the resulting multiplicity without an infinite homogeneity-strip count. Established fixed-table pressure bounds imply bounded optical capacity at every fixed radius. Thus the first inverse-gradient insertion, long critical residual height and grazing-ended short-source estimates no longer require a capacity hypothesis at fixed radius. A submultiplicative finite-block theorem separately gives a radius-uniform exponential growth bound and a sufficient spectral margin. The original uniform pointwise target is unchanged: uniform certification of that margin and decay of the positive nongrazing paired short trace remain to be proved. All earlier proofs and both historical source variants are retained.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. The arithmetic reference and the target.

For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$

be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R,a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R,a_j), \quad H_j = -\nabla^2 \log \lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \geq aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4}\int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\leq n\leq m,\,k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The weighted multiplicity and the pressure comparison.** The principal target is (2.5), with the original source, normalization and complete arithmetic reference. The finite-count quadratic cap and both physical residual mechanisms are retained below with their proofs. Their capacity condition concerns a weighted sum over *endpoint-momentum fibers*, not an ordinary symbolic word count. Theorem 7.1 removes the possible multiplicity inside one physical itinerary by a global Cartesian convexity argument. Lemma 7.2 then compares its precise optical weight with the stable Jacobian and removes each singular incidence at its marked time.

The resulting theorem compares the complete momentum-fiber capacity with geometric partition sums. For each fixed radius it is bounded uniformly in collision count by the established fixed-table pressure theory; see Corollary 8.2. For the radius-uniform problem, Theorem 8.3 gives a finite-block exponential growth bound and its explicit strict spectral margin. These are different quantifiers. The former is not silently promoted to the latter by compactness of the parameter interval.

At fixed radius the new estimate makes the critical recovery and the complete short-trace comparison unconditional relative to the retained analytic and geometric inputs. The positive same-segment minimum on short nongrazing segments remains an additional source-height question, even at one radius. The original uniform theorem is not replaced by the fixed-radius consequence, nor is the paired minimum identified with a projected signed cancellation.

3. Physical source, signed measures, and retained estimates

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar\nu = \mathbf{1}_\Pi \nu_R^*$ for the original exact-event source. The retained Borel parametrization is a bijection between its regular part and maximal segments (σ,u), $a_\sigma < u < b_\sigma$. Its transverse measure $d\sigma$ is the disjoint angular measure and its density is

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_*\chi(\Psi_\sigma(u))}, \qquad d\bar\nu = A_\sigma(u)\,du\,d\sigma.$$

Here $F = u$, $z = |\nabla_q F|$, $q = (\alpha_0, \alpha_m)$, $\omega = |\partial_{p_0}\alpha_m|^{-1}$, and $\chi = |\det D_q(F, \arg\nabla_q F)|$. All fibers and signed measures below use these same kernels. Let $z_\sigma^- = z(\Psi_\sigma(a_\sigma+))$ and $\ell_\sigma = b_\sigma - a_\sigma$.

Input statement 3.1 (Complete segment geometry and signed balance)**.** Put $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. On every regular segment,

$$2\lambda_\circ(v-u) \le z(v)^2 - z(u)^2 \le 2\Lambda_\circ(v-u), \tag{3.1}$$

$$|A_\sigma'| \le CA_\sigma I_m/z, \qquad |(W^\varepsilon)'| \le C\varepsilon^{-1}/z, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i. \tag{3.2}$$

The complete density $f_\sigma^\varepsilon = A_\sigma W^\varepsilon$ has one-sided traces and is absolutely continuous on the closed segment. The sum of its endpoint traces and absolute derivative integrals is finite at each fixed count. A grazing endpoint has zero trace; a critical backward endpoint need not. The geometric domain is a finite definable family at each fixed count. Its component count is used for existence of measures only, not for normalized estimates.

For every Borel prefix endpoint $c_\sigma \in [a_\sigma, b_\sigma]$ put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma.$$

The original union $z < \zeta$ or $\ell < \delta$ is exactly such a prefix, and $c_\sigma = b_\sigma$ gives b^ε. Define

$$\mathsf{I}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma,$$

$$\mathsf{D}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma,$$

$$\mathsf{G}_c^{\varepsilon,\pm}(H) = \int\int_{(a_\sigma, c_\sigma)\cap H} [\pm(f_\sigma^\varepsilon)'(u)]_+\, du\, d\sigma.$$

Their complete finite-count signed identity is

$$DR_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{3.3}$$

Every artificial prefix cut is negative. All true entrance types and one-sided incident physical components are retained.

This is the endpoint-field, optical, Borel-segment and signed-prefix chain, culminating in retained modules 220–222. In particular, A_σ includes both angular factors. No reversal of the billiard record or change of the section is used at a backward end.

Input statement 3.2 (Backward-end and critical volume estimates)**.** Set $\alpha = 129/128$ and $\gamma = 1/32$. Let

$$e_s^-(t) = \int \mathbf{1}_{\{t - a_\sigma \le s\}}\, d\mu_t, \qquad u_q(t)\, dt = F_\#(I_m \mathbf{1}_{\{z<q\}}\bar\nu), \qquad p_q(t)\, dt = F_\#(\mathbf{1}_{\{z<q\}}\bar\nu).$$

For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\theta_{B_*} < 1$ such that, at finite count and small $s, q > 0$,

$$m^2 \int_J e_s^- \le Cs^\gamma(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\theta_{B_*}^m, \sqrt{s}\}, \tag{3.4}$$

$$m^2 \int_J u_q \le Cq^\alpha(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 q^{1/2}\theta_{B_*}^m. \tag{3.5}$$

The constants are uniform in radius, exact labels, marks where present, and interval location. Since $I_m \ge 1$, the latter also bounds p_q. Consequently, at every fixed $s, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_s^- \le CLs^\gamma.$$

For the actual positive source $v^\rho(t)\,dt = F_\#(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar\nu)$,

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} m^2\int_J v^\rho \le CL \quad (\rho, L>0 \text{ fixed}). \tag{3.6}$$

The leading constant is independent of the fixed positive cutoff; its finite-count remainder is not.

Equation (3.4) is the same-event backward-end estimate of module 211, including critical entrances. For (3.5), multiply the actual fractional-source bound of module 187, at exponent $1/4$, by $q^{1/4}$: pointwise, $I_m\mathbf{1}_{z<q} \le q^{1/4} I_m z^{-1/4}\mathbf{1}_{z<q}$. The leading power is q^α and the remainder power is $q^{1/2}$. The audited cutoff estimate is retained in modules 196 and 222. These are volume-source theorems, not assumptions on entrance flux.

Input statement 3.3 (Original scalar reconstruction)**.** For every fixed $\varepsilon, r>0$,

$$\limsup_m \sup_{R,n,k} m^2\|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\,du. \tag{3.7}$$

For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards. The reconstruction band B is outermost.

We use $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} h(t)$ for a nonnegative source family. Unqualified $\|\cdot\|_\infty$ denotes essential supremum on the full roof line. Identities hold for common kernels at fixed m, R, n, k; the uniform order is $\sup_R \operatorname{ess\,sup}_t$. Countably many chosen radii share versions. No simultaneous Lebesgue-point assertion for every uncountable selector family is required.

3.1. **The residual retained from the preceding revision.** The physical comparison flow $\widehat\Phi$ crosses section-decision walls on the regular endpoint collar $|p_0|, |p_m| < 19/20$, and stops at true physical or backward critical ends. It has the same roof and transverse coordinates as the original flow while it stays in the original exact event. Its complete angular density is $\widehat A$. Only comparison times returning to the identical exact event are counted in the capped clock

$$c_r(x) = \frac{1}{r}\int_{(-r,r)\cap(-\widehat\tau^-,\widehat\tau^+)} \mathbf{1}_\Pi(\widehat\Phi_s x)\min\{1, \widehat A(\widehat\Phi_s x)/\widehat A(x)\}\,ds.$$

For $0<q\le 1/2$, $d = 2qr$ and $a>0$, set

$$P = \{\tau^- > 2d\}\cup\{\tau^-\le 2d,\ z\ge a,\ \tau^+ > d\}, \qquad Y = \Pi\cap\{c_r<q\}\cap P^c.$$

The lifetimes in P belong to the original stopped exact event, not the enlarged comparison domain. Write $Y^\varepsilon(t) = \int_{F=t} W^\varepsilon \mathbf{1}_Y\,d\mu_t$.

Input statement 3.4 (Recovered failure source and scalar reduction)**.** The capped-clock and running-compression theorems of retained modules 232–234 give, with every parameter fixed before the collision limit,

$$\mathcal{H}_M(b^\varepsilon - Y^\varepsilon) \le Cq^{-1}\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\} + Cqr.$$

The support is $Y\subset\{\tau^-\le 4qr, z<a\}\cup\{\ell\le 6qr\}$. The collar has no exit at widths $r\le r_{\mathrm{col}} = R_-^2/(800\Lambda_\circ)$. For $q=1/8$, $r = c\varepsilon(B)^4 \le r_{\mathrm{col}}$ and $a=\sqrt{qr}$, the unchanged scalar objective satisfies

$$\mathcal{E}_M \le \mathcal{H}_M(Y^{\varepsilon(B)}) + C_M B^{-1/192}. \tag{3.9}$$

No residual-height decay is part of this input.

The support and the low-clock condition will both be used below. Dropping the latter would lose the density-decrease mechanism on the long critical component.

4. A quadratic physical cap at the critical gradient

The finite-count spectral error in (3.5) has only a square-root gain in the gradient cutoff. Integrating that error against the first inverse gradient is not legitimate. We obtain a second, quadratic bound from the two-dimensional physical map. Its constant is an explicitly weighted multiplicity, retained throughout rather than replaced by an unestimated word count.

Fix $0 < b_0 < 1/8$ within the range of (3.5). On the regular exact event put $G(x) = R(-p_0(x), p_m(x))$ and $z = |G|$. The retained endpoint Hessian satisfies $\lambda_\circ I \le K = D_q^2 F \le \Lambda_\circ I$ in endpoint angles $q = (\alpha_0, \alpha_m)$. The optical normalization is $\omega = |\partial_{p_0}\alpha_m|^{-1}$. Define, with physical states counted once,

$$D_{m,R,n,k}(v) = \sum_{x\in\Pi^{\mathrm{reg}}:G(x)=v} \frac{I_m(x)\omega(x)}{4\pi c_* \det K(x)}, \tag{4.1}$$

$$\mathcal{K}_m = 1 + \pi \sup_{R,n,k} \operatorname*{ess\,sup}_{|v|<b_0} D_{m,R,n,k}(v). \tag{4.2}$$

These definitions involve neither the roof target nor an inverse power of z. The factor I_m is the full optical insertion from (3.2).

Lemma 4.1 (Physical area formula and a finite weighted cap)**.** *For every finite m, $\mathcal{K}_m < \infty$, uniformly over the permitted radius and exact labels. For $0 < b < b_0$ and every interval J,*

$$\int_J u_b(t)\,dt \le \mathcal{K}_m b^2. \tag{4.3}$$

More precisely, if N_m bounds the cardinalities of the regular physical fibers of G at count m, and $v_* = 2\tau_-/R_+$, then

$$\mathcal{K}_m \le 1 + CN_m(1+v_*)^{-(m-1)}. \tag{4.4}$$

Equation (4.4) *establishes finiteness only; no collision-uniform estimate of its right side is assumed.*

Proof. First variation gives $F_{\alpha_0} = -Rp_0$. Change of variables from (α_0, p_0) to endpoint angles has density ω. The Jacobian of $q \mapsto G(q)$ is $\det K > 0$. Thus the area formula, applied on injective local pieces and summed without multiplicity of charts, gives

$$G_\#(I_m\bar\nu) = D_{m,R,n,k}(v)\,dv.$$

Equivalently $|\det D_{(\alpha_0,p_0)}G| = \det K/\omega$. This identity uses the original collision normalization and counts physical preimages, not representatives in overlapping charts. Truncation of I_m, followed by monotone convergence, justifies the formula before its boundedness has been established.

Here is a bound for each summand uniform through internal grazing. For flight lengths $L_i \ge \tau_-$ the retained positive optical recursion is

$$y_1 = L_0, \quad f_0 = 1, \quad f_i = f_{i-1} + \frac{2y_i}{Rc_i}, \quad y_{i+1} = y_i + L_i f_i, \quad \omega = \frac{Rc_0c_m}{y_m}.$$

Consequently

$$y_m \ge \tau_- \prod_{i=1}^{m-1}(1 + v_*/c_i).$$

For a selected internal incidence,

$$\frac{1}{c_i}\prod_j (1+v_*/c_j)^{-1} \le v_*^{-1}(1+v_*)^{-(m-2)}.$$

Since $\sum_i q_0^{d_i} \le 2q_0/(1-q_0)$, it follows that $I_m\omega \le C(1+v_*)^{-(m-1)}$. For one flight the empty product gives the same conclusion after enlarging C. Division by $\det K \ge \lambda_\circ^2$ bounds each summand.

At a fixed count, the finite-horizon restriction permits only finitely many lifted scatterer itineraries. The retained finite definable physical graph, expressed in finitely many circle charts, includes the first-hit tests, radius parameter and section decisions. The nonzero Jacobian makes every regular G-fiber discrete. Uniform finiteness for definable families therefore gives a finite N_m [9]. Displacement and occupation take finitely many values at this count. This is used only for fixed-count finiteness and proves (4.4). Finally discard the roof test on the upper side, and integrate (4.1) over the disk $|v| < b$. Its area is πb^2, proving (4.3). No change is made to the source being estimated. □

Lemma 4.2 (Integration of two different positive caps)**.** *For $E, H, b > 0$,*

$$b^{-1}\min(Eb^{1/2}, Hb^2) + \int_0^b v^{-2}\min(Ev^{1/2}, Hv^2)\,dv \le 3\min\{Hb, E^{2/3}H^{1/3}\}. \tag{4.5}$$

Proof. The crossing is $v_* = (E/H)^{2/3}$. For $b \le v_*$ the left side is $2Hb$. For $b > v_*$ it is $3E^{2/3}H^{1/3} - Eb^{-1/2}$. These formulas prove the claim. The minimum is a cap on the positive source estimate, not on a signed spectral remainder considered by itself. □

Define the actual untruncated source

$$w_b(t)\,dt = F_\#(I_m z^{-1}\mathbf{1}_{\{0<z<b\}}\bar\nu).$$

It has a finite, absolutely continuous roof measure at every count.

Theorem 4.3 (First inverse gradient with a complete finite-count error)**.** *For one fixed auxiliary band B_*, all $m \ge 2$, $0 < b < b_0$ and intervals J of length L,*

$$m^2\int_J w_b \le Cb^{\alpha-1}(L + B_*^{-1}) + C\min\left\{m^2\mathcal{K}_m b,\ C_{B_*}^{2/3}(1+B_*L)^{2/3}m^2(\mathcal{K}_m\theta_{B_*}^{2m})^{1/3}\right\}. \tag{4.6}$$

The integration goes to gradient zero at finite count. In particular there is no error proportional to a negative power of a lower cutoff. The weighted multiplicity $\mathcal{K}_m$ is not asserted bounded or subexponential in m.

Proof. Let $M(v) = m^2\int_J u_v$. The two positive bounds give

$$M(v) \le C(L + B_*^{-1})v^\alpha + \min\{Ev^{1/2}, Hv^2\}, \quad E = C_{B_*}(1+B_*L)m^2\theta_{B_*}^m, \quad H = m^2\mathcal{K}_m.$$

Indeed a nonnegative quantity bounded by both $A + D$ and V is bounded by $A + \min(D, V)$. Positive layer cake gives exactly

$$m^2\int_J w_b = b^{-1}M(b) + \int_0^b v^{-2}M(v)\,dv.$$

Apply it first above a positive lower endpoint, and then use monotone convergence. The leading term is integrable because $\alpha > 1$; its multiplier is $1 + (\alpha-1)^{-1}$. Lemma 4.2 integrates the full error. This proves the inequality and finiteness. Absolute continuity follows from coarea on compact regular roof charts and exhaustion; the finite critical set has zero source area. □

We isolate the additional estimate which would make this error negligible. For an unbounded sequence of previously fixed auxiliary bands, consider the condition

$$m^6 \mathcal{K}_m \theta_{B_*}^{2m} \longrightarrow 0. \tag{SC}$$

Subexponential growth of $\mathcal{K}_m$ is sufficient, but is not proved here. Under (SC), first taking the collision limit and only then the indicated band sequence gives

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J w_b \le CLb^{1/128}. \tag{4.7}$$

Adding $z \ge b_0$ by (3.6) gives the bound CL for $v^0 = F_\#(I_m z^{-1}\bar{\nu})/dt$ at each fixed L. These are implications of the explicit condition, not unconditional zero-cutoff conclusions. Geometric-pressure results for stable Jacobian partition sums [25] concern a different weighted sum; no identification with (4.2) or uniform parameter bound is imported from them.

5. Separate critical and short-residence residuals

Keep the original residual $Y = Y_{a,q,r}$ and write $d = 2qr$. We assign the overlap to the short-residence component:

$$S = Y \cap \{\ell \le 3d\}, \qquad C = Y \cap \{\ell > 3d\}, \qquad Y^\varepsilon = S^\varepsilon + C^\varepsilon. \tag{5.1}$$

Here and below a superscript ε denotes the original weight W^ε on the indicated target restriction. The letters C, S are target sets, not new probability measures.

Lemma 5.1 (A full forward interval on the critical component)**.** *Every target in C satisfies*

$$\tau^- \le 2d, \qquad 0 < z < a, \qquad \tau^+ > d. \tag{5.2}$$

For $0 \le s \le d$, the trajectory remains in the original exact event and

$$0 < z(\Phi_s x) < b, \qquad b = (a^2 + 2\Lambda_\circ d)^{1/2}.$$

Moreover some $s \in [0, d]$ has $A(\Phi_s x) < A(x)/2$.

Proof. The complement of the old protected set is $\{\tau^- \le 2d\} \cap (\{z < a\} \cup \{\tau^+ \le d\})$. Since $\ell > 3d$, its second alternative is impossible and $\tau^+ > d$. Radial growth proves the upper bound on z. If $A(\Phi_s x)/A(x) \ge 1/2$ on this interval, the same-event times alone contribute $d/(2r) = q$ to the capped clock. This contradicts $c_r < q$. The direction choice introduces no new comparison-domain availability assumption. □

Theorem 5.2 (Uncut critical loss paid by the first inverse gradient)**.** *Assume $b < b_0$. At every finite collision count, for almost every target roof,*

$$C^\varepsilon(t) \le C_0 \int_t^{t+d} w_b(u)\, du. \tag{5.3}$$

Thus Theorem 4.3, with $L = d$, is an explicit bound for the complete critical residual, with no positive lower gradient cutoff. Under (SC)*,*

$$\limsup_m \sup_{R,n,k} m^2 \|C^\varepsilon\|_\infty \le Cdb^{1/128}. \tag{5.4}$$

Without that additional condition, the same height bound holds for $C \cap \{z \ge c\}$ at each previously fixed $c > 0$. That last statement does not imply the uncut assertion by exchanging limits.

Proof. For a target selected in Lemma 5.1,

$$A(x)/2 \le \int_0^d |\partial_u A(\Phi_s x)|\, ds \le C \int_0^d A(\Phi_s x) I_m(\Phi_s x)/z(\Phi_s x)\, ds.$$

Multiply by $W^\varepsilon(x) \le 1$. At each fixed s the same-event transport is injective in the physical transversal; its image is inside the original exact fiber and has $z < b$. Therefore summing the right side gives a submeasure of $w_b(t+s)$. The untruncated measure has already been proved finite at fixed count. Tonelli and the common kernels prove (5.3). This is a target-height comparison, not an integral in starting roofs, and no target weight is replaced by an unweighted terminal flux.

Under (SC), apply (4.7) to the fixed interval of length d. For the assertion with $z(x) \ge c > 0$, use radial monotonicity and integrate (3.5) only above c. The leading layer cake is at most $Cb^{\alpha-1}(d + B_*^{-1})$, and the finite-count error is $C_{B_*}(1 + B_* d)m^2 c^{-1/2}\theta_{B_*}^m$. Fixing c, d, B_* before the collision limit and enlarging B_* afterwards gives the same bound. This proof neither establishes (SC) nor removes the fixed c inside that limit. □

For the retained choice $q = 1/8$, $a = \sqrt{qr}$, we have $a^2 = d/2$. Consequently (5.4), whenever its stated hypothesis is verified, becomes

$$\mathcal{H}_M(C^\varepsilon) \le C(qr)^{257/256}. \tag{5.5}$$

The exponent is $1 + (\alpha - 1)/2$. It comes from integrating the first inverse gradient against the critical-volume power, not from the small mass of C itself. All internal incidences and all initial positive gradients are present in the finite-count comparison.

5.1. **A finite-count certificate rather than a hidden hypothesis.** The exact cost of the uncut comparison is visible in (4.6): at each fixed band it contains

$$m^2(\mathcal{K}_m \theta_{B_*}^{2m})^{1/3}.$$

The theorem estimates the full source with that cost retained. A source-specific bound on the weighted momentum-fiber multiplicity would turn it into the vanishing critical-height theorem (5.5). Finiteness of N_m, a pressure bound for another partition, or a fixed positive-gradient theorem does not supply that bound. The condition remains an explicit proof obligation.

6. Two-ended physical traces on the short component

For a maximal original segment write $f_\sigma^\varepsilon(u) = A_\sigma(u) W^\varepsilon(\Psi_\sigma(u))$. Its nonnegative one-sided traces are denoted by f_σ^- and f_σ^+. These are the complete physical traces in (3.3); a critical entrance trace need not be zero. The comparison below neither pools the occupation classes nor identifies coincident faces from different incident components.

Let H be any Borel target restriction supported on $0 < \ell \le \delta$. Define its paired trace source at the original roof by

$$T_H^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < b_\sigma\}} \mathbf{1}_H(\Psi_\sigma(t)) \min\{f_\sigma^-, f_\sigma^+\}\, d\sigma. \tag{6.1}$$

The selector is evaluated at the target in both quantities being compared. In particular it can be the old low-clock residual, which need not be a smooth or monotone selector along a segment.

Theorem 6.1 (Complete short-source and paired-trace comparison)**.** *At every finite count and almost every roof,*

$$|R_H^\varepsilon(t) - T_H^\varepsilon(t)| \le C(1 + \varepsilon^{-1}) \int_{t-\delta}^{t+\delta} v^0(u)\, du, \qquad v^0\, dt = F_\#(I_m z^{-1} \bar\nu). \tag{6.2}$$

This untruncated positive source is finite at each count by Theorem 4.3 and the bounded-gradient part. Under (SC)*,*

$$\limsup_{m} \sup_{R,n,k} m^2 \|R_H^\varepsilon - T_H^\varepsilon\|_\infty \le C(1+\varepsilon^{-1})\delta. \tag{6.3}$$

The constants are independent of the Borel target selector. In particular, the entire selected source on short segments with at least one grazing endpoint satisfies the same height bound.

Proof. Absolute continuity on the closed segment gives, for every interior u where the representative is defined,

$$|f_\sigma^\varepsilon(u) - \min(f_\sigma^-, f_\sigma^+)| \le \int_{a_\sigma}^{b_\sigma} |(f_\sigma^\varepsilon)'(v)|\, dv.$$

Both angular factors and the original guard give

$$|(f_\sigma^\varepsilon)'| \le C A_\sigma I_m/z + C\varepsilon^{-1} A_\sigma/z \le C(1+\varepsilon^{-1}) A_\sigma I_m/z.$$

Multiply the first inequality by the target selector. All segments of length at most δ containing t lie in $(t-\delta, t+\delta)$. At each intermediate roof the Borel segment bijection counts their interiors once, so the sum of these positive integrals is bounded by the displayed integral of v^0. The absolute value of the projected difference is bounded by this sum of absolute differences. Truncate and exhaust the original regular source, then use Tonelli. This proves (6.2) without taking a derivative of the Borel selector or a trace on its artificial boundary.

Under (SC), the complete first inverse gradient has fixed-width bound CL, as proved after (4.7). Apply it at $L = 2\delta$. If an endpoint is grazing, its original weighted trace is zero, so the minimum in (6.1) is zero. This proves the final assertion. An incoming critical endpoint with nonzero trace is not included in that zero-trace argument. □

This comparison does not claim that a minimum commutes with projection, or that (6.1) is the Jordan positive part of a signed current. It controls the error of a specific positive paired-trace comparison by the full physical derivative. The inherited signed forcing remains formed after projection.

Theorem 6.2 (A scalar reduction with distinct geometric obligations)**.** *Choose $q = 1/8$, $r = c\varepsilon(B)^4 \le r_{\mathrm{col}}$, $a = \sqrt{qr}$ and $\delta = 6qr$, with $c > 0$ fixed and small. Use $H = S$ from* (5.1) *in* (6.1)*. If* (SC) *holds, then*

$$\mathcal{E}_M \le \mathcal{H}_M(T_S^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1+\varepsilon(B)^{-1})qr. \tag{6.4}$$

Under that same condition, the original pointwise theorem is equivalent along this schedule to $\mathcal{H}_M(T_S^{\varepsilon(B)}) \to 0$. Neither (SC) *nor this paired-trace height decay is asserted established for the full Lorentz family.*

Proof. The retained finite-count scalar reduction bounds b^ε by Y^ε plus the already paid clock and protected compression terms. Split $Y = C \sqcup S$ before projection. Theorem 5.2 pays C, and Theorem 6.1 compares S to T_S. The retained positive remainder proves (6.4). Every scale and auxiliary band is fixed before the collision limit; (SC) is invoked only at each such fixed band. The band sequence is enlarged afterwards, and B is last.

For the schedule, the additional powers are $(qr)^{257/256} = O(\varepsilon^{257/64})$ and $(1+\varepsilon^{-1})qr = O(\varepsilon^3)$; both vanish. Sufficiency follows. Conversely $S^\varepsilon \le b^\varepsilon$, so scalar closure and the positive remainder imply $\mathcal{H}_M(S^{\varepsilon(B)}) = O(B^{-1/192})$. Equation (6.3) then implies decay of T_S. This necessity statement is conditional on the same spectral-capacity estimate as its sufficiency statement. □

Without (SC), the finite-count comparisons (5.3) and (6.2) remain valid with the explicit error in (4.6). Thus the new reduction identifies two separate tasks: control a weighted two-dimensional momentum-fiber multiplicity against the spectral error, and control the original paired positive

boundary traces on genuinely short segments. The first is not a roof-height assumption; the second retains the exact target selector, physical endpoints and original guard. Finite-count finiteness does not prove either asymptotic task. Scalar upper closure, not a finite-norm consequence, still precedes every unrestricted point-conditioned downstream assertion.

7. Global momentum uniqueness and the stable optical weight

The multiplicity in (4.2) can be estimated dynamically. Two separate facts are needed: a global uniqueness assertion for each lifted itinerary, and a comparison of the optical weight with a genuine stable Jacobian. Neither follows merely from the nonzero local endpoint Hessian.

Theorem 7.1 (One physical orbit per itinerary and momentum pair)**.** *Fix a finite lifted sequence of circular scatterers and two endpoint tangential momenta in* $(-1,1)$*. There is at most one regular physical trajectory with that itinerary and those endpoint momenta. Internal incidences need not be bounded below uniformly. The assertion concerns physical contact points, not their overlapping angular coordinates.*

Proof. Write $Q=(Q_0,\dots,Q_m)$ for the contact positions and

$$S(Q)=\sum_{j=0}^{m-1}|Q_{j+1}-Q_j|.$$

This is a convex function of the Cartesian contact variables. It is differentiable at every configuration with positive flight lengths. For two such configurations $Q,\widehat{Q}$, monotonicity of its subgradient gives

$$\sum_{j=0}^{m}(\nabla_j S(Q)-\nabla_j S(\widehat{Q}))\cdot(Q_j-\widehat{Q}_j)\geq 0. \tag{7.1}$$

Let n_j be the outward unit normal of the jth circular scatterer and let t_j be its fixed oriented tangent. Reflection and first variation give

$$\nabla_0 S=-c_0n_0-p_0t_0,\quad \nabla_j S=-2c_jn_j\ (0<j<m),\quad \nabla_m S=-c_mn_m+p_mt_m.$$

At the two trajectories $c_0=\widehat{c}_0>0$ and $c_m=\widehat{c}_m>0$, since their endpoint momenta agree. The scatterer centers agree at every index, so $Q_j-\widehat{Q}_j=R(n_j-\widehat{n}_j)$. Rotation by a right angle is skew symmetric; consequently $(t_j-\widehat{t}_j)\cdot(n_j-\widehat{n}_j)=0$. The left side of (7.1) therefore equals

$$\begin{aligned}&-Rc_0|n_0-\widehat{n}_0|^2-Rc_m|n_m-\widehat{n}_m|^2\\&\qquad-R\sum_{j=1}^{m-1}(c_j+\widehat{c}_j)|n_j-\widehat{n}_j|^2.\end{aligned}$$

Every coefficient is strictly positive. Thus all contact positions coincide. The flight directions and physical state then coincide as well. Repeated visits to a scatterer cause no change: the contact variables at different times are distinct Cartesian variables. No convexity of a physical word domain, existence of an interpolating physical trajectory, or global angular lift is needed. □

For the Jacobian comparison use standard billiard coordinates (s,φ), where s is arclength and $|p|=|\sin\varphi|$. Signs are fixed consistently on each oriented boundary component. Let M' be the full-measure set regular in both infinite time directions. Write $J_R^uT_R^j$ and $J_R^sT_R^j$ for the positive arclength Jacobians on its invariant unstable and stable lines. The invariant cones have slopes in fixed compact positive and negative intervals, respectively, with a uniform angle between them [25, Section 1.1].

Lemma 7.2 (Optical weight and removal of one incidence)**.** *There are uniform geometric constants C_0, C_1 such that on M', for every $m \ge 1$,*

$$C_0^{-1} c_m J_R^s T_R^m(x) \le \omega(x) \le C_0 c_m J_R^s T_R^m(x), \qquad \frac{J_R^s T_R(x)}{c(x)} \le C_1. \tag{7.2}$$

In particular, for an internal mark $1 \le i < m$,

$$\frac{\omega(x)}{c_i(x)} \le C_0 C_1 J_R^s T_R^i(x) J_R^s T_R^{m-i-1}(T_R^{i+1} x). \tag{7.3}$$

The stable factor at the marked time is removed, not evaluated with a singular inverse-incidence multiplier left over.

Proof. Up to consistent orientation signs, one flight has derivative

$$\frac{1}{c'} \begin{pmatrix} \tau\varkappa + c & \tau \\ \tau\varkappa\varkappa' + \varkappa c' + \varkappa' c & \tau\varkappa' + c' \end{pmatrix}, \qquad \varkappa = \varkappa' = R^{-1}.$$

Its determinant is c/c'. Every entry is positive. The ratio of its first column to its second, in either row, lies between $\varkappa_{\min}$ and $\varkappa_{\max} + \tau_-^{-1}$. Left multiplication by a positive matrix preserves these componentwise inequalities. If b_m is the upper-right entry of the m-flight product, the image of the vertical vector has slope in the unstable cone. Comparing it with a unit invariant unstable vector at the source consequently gives

$$C^{-1} b_m \le J_R^u T_R^m(x) \le C b_m.$$

Here the constants do not accumulate with m: both the initial column comparison and the final slope comparison use fixed cones. Changing from φ_0 to p_0 and from s_m to α_m gives $\omega = R c_0 / b_m$.

Put $e(x) = \sin \angle(E^s(x), E^u(x))$. Conservation of collision area in these coordinates is the exact identity

$$J_R^u T_R^m(x) J_R^s T_R^m(x) \frac{e(T_R^m x)}{e(x)} = \frac{c_0}{c_m}.$$

The function e is bounded above and below by positive cone constants. Combining the last three formulas proves the two-sided comparison in (7.2). For one flight, if v is the initial unstable slope, then

$$J_R^u T_R(x) = \frac{\tau\varkappa + c + \tau v}{c'} \frac{\sqrt{1 + (v')^2}}{\sqrt{1 + v^2}}.$$

The same area identity shows that $J_R^s T_R / c$ is bounded by a fixed constant divided by $\tau\varkappa + c + \tau v$. The latter is bounded below by a positive geometric constant. This proves the one-step bound. Factor the stable cocycle at time i and use $c_m \le 1$ to obtain (7.3). □

The area formula in module 235 can be restricted to M' without altering its almost-everywhere momentum density. The omitted set has zero collision area. On countably many regular endpoint charts G is a local diffeomorphism, so its image has zero two-dimensional Lebesgue measure. Thus the infinite-time invariant lines used above do not add a restriction to the original finite-record source.

8. A marked pressure comparison for the actual momentum fibers

Normalize the initial lifted scatterer to the origin. Let $\mathcal{W}_j(R)$ be the finite set of realizable physical itineraries of j flights, retaining the lifted displacement of each step. If $[w]_R$ is the regular domain of w, define

$$P_j(R) = \sum_{w \in \mathcal{W}_j(R)} \sup_{x \in [w]_R \cap M'} J_R^s T_R^j(x), \qquad P_0(R) = 1. \tag{8.1}$$

These are geometric upper sums, not replacements for the exact event. They impose no occupation or section test; those tests are dropped only on the upper side of the comparison below. There is one fixed finite step alphabet of size D: successive lifted centers are at distance at most $\tau_+ + 2R_+$. It follows in particular that every $P_j(R)$ is finite and that each fixed-j supremum over R is finite.

For comparison with the established pressure theory let $\mathcal{Q}_j(R)$ be the stable-Jacobian sum at potential parameter one over the homogeneous continuity partition $\mathcal{M}_0^{j,H}$ of [25, equations (2.4), (2.11)]. Its cells retain the true singularities through the terminal time and homogeneity cuts through time $j-1$. Each cell determines one lifted physical itinerary. Splitting into those cells can only increase a sum of suprema of a positive function. Hence

$$P_j(R) \le \mathcal{Q}_j(R). \tag{8.2}$$

No identification of the two sums is asserted.

Let $\mathcal{K}_m(R)$ denote (4.2) with the radius held fixed. Thus $\mathcal{K}_m = \sup_R \mathcal{K}_m(R)$.

Theorem 8.1 (Complete momentum-fiber pressure bound)**.** *With a constant depending only on the uniform physical geometry,*

$$\mathcal{K}_m(R) \le 1 + C\left[P_m(R) + D\sum_{i=1}^{m-1} q_0^{d_i} P_i(R) P_{m-i-1}(R)\right]. \tag{8.3}$$

All physical preimages, internal incidences and original exact labels are included. There is no factor counting homogeneity strips at the marked collision.

Proof. Fix an almost-everywhere momentum target and one exact event. By Theorem 7.1, each itinerary supplies at most one preimage. The denominator in (4.1) is at least $4\pi c_* \lambda_\circ^2$. The term without a mark is therefore bounded by $CP_m(R)$ using Lemma 7.2.

For the mark at i, use (7.3). A full itinerary is uniquely determined by its first i flights, its single step at time i, and its remaining $m-i-1$ flights, with the last block translated back to the initial fundamental scatterer. The middle step has at most D choices. The weighted sum of its possible preimages is consequently at most $CDP_i(R)P_{m-i-1}(R)$. This argument uses physical itineraries, not the infinitely many homogeneity strips whose inverse-incidence weight would no longer be summable after removing the marked stable factor. Multiplying by $q_0^{d_i}$ and summing proves the bound after taking the required essential and exact-label suprema. The finite horizon alphabet and global uniqueness supply every multiplicity used in this argument. □

Corollary 8.2 (Bounded capacity at each fixed radius)**.** *For every fixed $R \in I$,*

$$\sup_{m\ge 1} \mathcal{K}_m(R) < \infty. \tag{8.4}$$

In particular $m^6 \mathcal{K}_m(R)\theta^{2m} \to 0$ for every fixed $0 < \theta < 1$.

Proof. For each fixed finite-horizon dispersing table, the pressure theorem [25, Lemma 3.8, proof] gives $\sup_j \mathcal{Q}_j(R) < \infty$. Apply (8.2) and (8.3), and use $\sum_i q_0^{d_i} \le 2q_0/(1-q_0)$. This use of the pressure theorem is now justified by an explicit comparison of the same optical weight and momentum preimages; it is not a comparison of cardinalities. □

The constant in the cited fixed-table pressure theorem has not been shown here to be uniform as R varies. The quantifiers in (8.4) must not be interchanged. The following finite-block theorem gives an explicit family-uniform alternative.

Theorem 8.3 (Uniform finite-block capacity certificate)**.** *Put $\overline{P}_j = \sup_{R\in I} P_j(R)$. For every fixed integer $L \ge 1$ define*

$$\rho_L = \max\{1, \overline{P}_L^{1/L}\}, \qquad C_L = \max_{0\le j<L} \overline{P}_j \rho_L^{-j}.$$

There is an explicit finite geometric multiple A_L of $1 + C_L + DC_L^2 \sum_{i\ge1} 2q_0^i$ such that

$$\mathcal{K}_m \le A_L \rho_L^m \qquad (m \ge 1). \tag{8.5}$$

Consequently (SC) holds at a fixed auxiliary band whenever, for some fixed L,

$$\rho_L \theta_{B_*}^2 < 1. \tag{BC}$$

If this certificate holds at an unbounded sequence of previously fixed bands, all radius-uniform consequences of (SC) in the retained article follow. No numerical verification of (BC) over I is claimed in this revision.

Proof. Splitting a physical itinerary into two consecutive blocks is injective into the pair of block itineraries. The stable cocycle gives $P_{j+k}(R) \le P_j(R)P_k(R)$, so also $\overline{P}_{j+k} \le \overline{P}_j \overline{P}_k$. Writing $m = vL + j$, $0 \le j < L$, proves $\overline{P}_m \le C_L \rho_L^m$. Insert this and the corresponding bound for the two factors in (8.3). Since $\rho_L \ge 1$ and the depth weights are summable, it gives (8.5). Under (BC),

$$m^6 \mathcal{K}_m \theta_{B_*}^{2m} \le A_L m^6 (\rho_L \theta_{B_*}^2)^m \longrightarrow 0.$$

All suprema over R, block lengths and bands are in the displayed order. The block is fixed before m and is allowed to depend on the previously chosen auxiliary band, never on collision count. □

In particular

$$\limsup_m m^{-1} \log \mathcal{K}_m \le \inf_{L\ge1} \max\{0, L^{-1} \log \overline{P}_L\}.$$

This is an actual uniform growth estimate expressed through geometric partition sums. It replaces the unquantified fiber cardinality of module 235. Establishing that its right side is zero, or certifying the strict inequalities (BC), remains a family-uniform question; pointwise-in-R pressure bounds alone do not settle it.

9. Unconditional fixed-radius recovery and the uniform endpoint

The preceding comparison discharges the spectral-capacity hypothesis for each fixed physical radius. We state the resulting source estimates with that quantifier visible. For a nonnegative family let

$$\mathcal{H}_{M,R}(f) = \limsup_m \sup_{n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} f(t).$$

It differs from $\mathcal{H}_M$ by placing the radius outside the collision limsup.

Theorem 9.1 (Uncut recovery with no capacity hypothesis at fixed radius)**.** *For every fixed $R \in I$, every fixed interval length $L > 0$, and $0 < b < b_0$,*

$$\limsup_m \sup_{n,k} \sup_{|J|=L} m^2 \int_J w_b \le CLb^{1/128}, \qquad \limsup_m \sup_{n,k} \sup_{|J|=L} m^2 \int_J v^0 \le CL. \tag{9.1}$$

The leading constant is independent of R; the collision count needed for a prescribed error is not asserted uniform in R. For the disjoint residuals of module 236 and for every original Borel short-segment selector H of total residence at most δ,

$$\mathcal{H}_{M,R}(\mathsf{C}^\varepsilon) \le C(qr)^{257/256}, \tag{9.2}$$

$$\limsup_m \sup_{n,k} m^2 \|R_H^\varepsilon - T_H^\varepsilon\|_\infty \le C(1 + \varepsilon^{-1})\delta. \tag{9.3}$$

In particular (9.3) *bounds the entire selected short source with a grazing physical endpoint, without any positive initial-gradient or internal-incidence cutoff.*

Proof. Let $K_R = \sup_m \mathcal{K}_m(R) < \infty$. The finite-count estimate (4.6), before taking any limit, has remainder at most

$$CC_{B_*}^{2/3}(1 + B_*L)^{2/3}m^2K_R^{1/3}\theta_{B_*}^{2m/3}.$$

It vanishes for each fixed R, L, B_*. Enlarge B_* only after the collision limit. The leading constants in the inherited local source estimate and the new geometric comparison are uniform, proving (9.1). Adding $z \geq b_0$ uses the retained positive-cutoff estimate, not a new zero-cutoff exchange.

The finite-count critical certificate (5.3) is at the target roof, and so (9.1) on its interval of length $d = 2qr$ proves (9.2). The two-ended finite-count comparison (6.2), with $L = 2\delta$, proves (9.3). Neither proof differentiates a Borel target selector. At a grazing physical endpoint one original weighted trace is zero, so its same-segment minimum is zero. A backward critical trace is not deleted by that argument. □

For one fixed R let $\mathcal{E}_{M,R}$ denote the collision limsup of the original scalar error. With the retained schedule $q = 1/8$, $r = c\varepsilon(B)^4$, $a = \sqrt{qr}$ and $\delta = 6qr$, the preceding theorem gives, without (SC),

$$\mathcal{E}_{M,R} \leq \mathcal{H}_{M,R}(T_{\mathsf{S}}^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1 + \varepsilon(B)^{-1})qr. \tag{9.4}$$

This follows by the original positive scalar decomposition and the disjoint residual partition before their suprema are taken. The positive paired trace on short nongrazing segments has not been proved small, even at a fixed radius; (9.4) is not a claim of pointwise scalar closure.

For the original radius-uniform target, (BC) on an unbounded sequence of fixed auxiliary bands implies the same displayed bound with $\mathcal{E}_M$ and $\mathcal{H}_M$. This is Theorem 6.2 with its capacity condition supplied by Theorem 8.3. There are still two separate family-uniform tasks: certify this pressure-growth margin, and prove decay of the positive same-segment trace. The new comparison proves the first at every fixed radius but does not turn that assertion into $\limsup_m \sup_R$ by compactness alone. Nor does it identify the paired minimum with the Jordan positive part of a projected current.

The original exact labels, guard, section and arithmetic reference remain those of (2.5). All original unconditional finite-roof-norm and lower pointwise conclusions are retained; the unrestricted upper pointwise half, and the downstream conclusions requiring it, are not promoted. No zero arithmetic class is normalized.

Appendix A. Dependencies and quantifiers of the active proof

The original problem and physical conventions precede the new proofs. The inherited analytic interfaces are stated in the article before module 235: the finite-count critical-volume estimate, original positive scalar remainder, full angular density, Borel segment map and running-compression residual. They retain their exact parameter and null-set conventions.

The new global uniqueness theorem uses only the physical reflection law and convexity of the Cartesian sum of flight lengths. The optical comparison uses the standard invariant cones and collision-area Jacobian, with every change of coordinates displayed. The marked pressure comparison uses both of these results, the lower endpoint Hessian bound and the finite lifted-step alphabet. The external input is the bounded fixed-table stable-Jacobian partition sum at parameter one in [25, Lemma 3.8, proof]; its partition is specified in (8.2). It is not cited as an endpoint momentum-fiber theorem: that comparison is proved here.

The finite-block estimate takes the parameter supremum first. Its strict condition (BC) must be verified at each of an unbounded sequence of fixed auxiliary bands to imply the original uniform (SC). The fixed-radius theorem instead takes the collision limsup with the radius held fixed. It has no additional capacity hypothesis but does not assert uniform convergence in radius. In either case the positive paired short trace is a separate unresolved height estimate.

The complete preceding article, all 237 inherited cores, both v88 variants and their original source files remain in the repository and archive. Their historical conditional statements are preserved rather than rewritten as if they had used the present comparison. The active claim authority is the new source manifest and this theorem sequence, not a superseded historical status flag. No source receipt, finite fixture or successful build is a continuum proof certificate.

Part 6. Quadratic momentum caps and paired physical traces

Retained revision-98 abstract. We retain the exact-label pointwise raw local limit problem for actual returns in a triangular Lorentz gas. A physical area formula in the two endpoint momenta gives a quadratic small-gradient cap with an explicit weighted optical multiplicity. Combining this cap with the inherited spectral estimate makes the first inverse-gradient layer cake integrable to zero at every finite collision count, with its entire error retained. We separate the preceding residual into long near-critical targets and short physical segments. The first is paid by the new untruncated insertion; the second is compared to the minimum of its two actual weighted endpoint traces. The error of that comparison is a complete physical volume term. An explicit spectral-capacity condition would make these errors negligible and give a critical-residual height bound of order $(qr)^{257/256}$. That condition is not verified, and the remaining paired short-trace height is not proved to vanish for the full family. All original sources, labels, arithmetic classes and earlier proofs remain unchanged; the pointwise objective is not replaced by a finite-norm law.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K_{m,R}^{\mathrm{c}}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K_{m,R}^{\mathrm{c}} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on

$[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_\#(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The finite-count estimate and the two physical mechanisms.** The new estimate acts on the actual insertion $I_m/|g|$, without a positive lower gradient cutoff. Its leading local term is $Cb^{1/128}(|J| + B_*^{-1})$. Its complete finite-count error is bounded by an explicit function of the weighted two-dimensional momentum-fiber multiplicity $\mathcal{K}_m$ and the fixed-band decay factor $\theta_{B_*}^m$; see Theorem 4.3. The weighted multiplicity is finite at each count, but its growth has not been bounded at the rate needed for scalar closure.

The residual is split disjointly by total physical residence, with the overlap assigned to the short component. On the longer component a low capped clock forces a definite decrease of the original angular density on a full forward interval. On the short component we compare the target density to its two physical endpoint traces using the same Borel segment, with no trace on an artificial target-selector graph. Both comparisons hold at finite count before any asymptotic condition. The conditional height consequences and their exact additional hypothesis are stated separately, in Theorems 5.2 and 6.1.

3. Physical source, signed measures, and retained estimates

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar\nu = \mathbf{1}_\Pi \nu_R^*$ for the original exact-event source. The retained Borel parametrization is a bijection between its regular part and maximal segments (σ, u), $a_\sigma < u < b_\sigma$. Its transverse measure $d\sigma$ is the disjoint angular measure and its density is

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad d\bar\nu = A_\sigma(u)\, du\, d\sigma.$$

Here $F = u$, $z = |\nabla_q F|$, $q = (\alpha_0, \alpha_m)$, $\omega = |\partial_{p_0}\alpha_m|^{-1}$, and $\chi = |\det D_q(F, \arg \nabla_q F)|$. All fibers and signed measures below use these same kernels. Let $z_\sigma^- = z(\Psi_\sigma(a_\sigma+))$ and $\ell_\sigma = b_\sigma - a_\sigma$.

Input statement 3.1 (Complete segment geometry and signed balance)**.** Put $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. On every regular segment,

$$2\lambda_\circ(v-u) \le z(v)^2 - z(u)^2 \le 2\Lambda_\circ(v-u), \tag{3.1}$$

$$|A_\sigma'| \le C A_\sigma I_m / z, \qquad |(W^\varepsilon)'| \le C\varepsilon^{-1}/z, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i. \tag{3.2}$$

The complete density $f_\sigma^\varepsilon = A_\sigma W^\varepsilon$ has one-sided traces and is absolutely continuous on the closed segment. The sum of its endpoint traces and absolute derivative integrals is finite at each fixed count. A grazing endpoint has zero trace; a critical backward endpoint need not. The geometric domain is a finite definable family at each fixed count. Its component count is used for existence of measures only, not for normalized estimates.

For every Borel prefix endpoint $c_\sigma \in [a_\sigma, b_\sigma]$ put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma.$$

The original union $z < \zeta$ or $\ell < \delta$ is exactly such a prefix, and $c_\sigma = b_\sigma$ gives b^ε. Define

$$\mathsf{I}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma,$$

$$\mathsf{D}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma,$$

$$\mathsf{G}_c^{\varepsilon,\pm}(H) = \int\int_{(a_\sigma, c_\sigma)\cap H} [\pm (f_\sigma^\varepsilon)'(u)]_+\, du\, d\sigma.$$

Their complete finite-count signed identity is

$$DR_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{3.3}$$

Every artificial prefix cut is negative. All true entrance types and one-sided incident physical components are retained.

This is the endpoint-field, optical, Borel-segment and signed-prefix chain, culminating in retained modules 220–222. In particular, A_σ includes both angular factors. No reversal of the billiard record or change of the section is used at a backward end.

Input statement 3.2 (Backward-end and critical volume estimates)**.** Set $\alpha = 129/128$ and $\gamma = 1/32$. Let

$$e_s^-(t) = \int \mathbf{1}_{\{t - a_\sigma \le s\}}\, d\mu_t, \qquad u_q(t)\, dt = F_\#(I_m \mathbf{1}_{\{z<q\}}\bar\nu), \qquad p_q(t)\, dt = F_\#(\mathbf{1}_{\{z<q\}}\bar\nu).$$

For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\theta_{B_*} < 1$ such that, at finite count and small $s, q > 0$,

$$m^2 \int_J e_s^- \le Cs^\gamma(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\theta_{B_*}^m, \sqrt{s}\}, \tag{3.4}$$

$$m^2 \int_J u_q \le Cq^\alpha(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 q^{1/2}\theta_{B_*}^m. \tag{3.5}$$

The constants are uniform in radius, exact labels, marks where present, and interval location. Since $I_m \ge 1$, the latter also bounds p_q. Consequently, at every fixed $s, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_s^- \le CLs^\gamma.$$

For the actual positive source $v^\rho(t)\,dt = F_\#(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar\nu)$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\rho \le CL \quad (\rho, L > 0 \text{ fixed}). \tag{3.6}$$

The leading constant is independent of the fixed positive cutoff; its finite-count remainder is not.

Equation (3.4) is the same-event backward-end estimate of module 211, including critical entrances. For (3.5), multiply the actual fractional-source bound of module 187, at exponent $1/4$, by $q^{1/4}$: pointwise, $I_m\mathbf{1}_{z<q} \le q^{1/4} I_m z^{-1/4}\mathbf{1}_{z<q}$. The leading power is q^α and the remainder power is $q^{1/2}$. The audited cutoff estimate is retained in modules 196 and 222. These are volume-source theorems, not assumptions on entrance flux.

Input statement 3.3 (Original scalar reconstruction)**.** For every fixed $\varepsilon, r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\,du. \tag{3.7}$$

For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards. The reconstruction band B is outermost.

We use $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} h(t)$ for a nonnegative source family. Unqualified $\|\cdot\|_\infty$ denotes essential supremum on the full roof line. Identities hold for common kernels at fixed m, R, n, k; the uniform order is $\sup_R \operatorname{ess\,sup}_t$. Countably many chosen radii share versions. No simultaneous Lebesgue-point assertion for every uncountable selector family is required.

3.1. **The residual retained from the preceding revision.** The physical comparison flow $\widehat\Phi$ crosses section-decision walls on the regular endpoint collar $|p_0|, |p_m| < 19/20$, and stops at true physical or backward critical ends. It has the same roof and transverse coordinates as the original flow while it stays in the original exact event. Its complete angular density is $\widehat A$. Only comparison times returning to the identical exact event are counted in the capped clock

$$c_r(x) = \frac{1}{r}\int_{(-r,r)\cap(-\widehat\tau^-,\widehat\tau^+)} \mathbf{1}_\Pi(\widehat\Phi_s x)\min\{1, \widehat A(\widehat\Phi_s x)/\widehat A(x)\}\,ds.$$

For $0 < q \le 1/2$, $d = 2qr$ and $a > 0$, set

$$P = \{\tau^- > 2d\} \cup \{\tau^- \le 2d,\ z \ge a,\ \tau^+ > d\}, \qquad Y = \Pi \cap \{c_r < q\} \cap P^c.$$

The lifetimes in P belong to the original stopped exact event, not the enlarged comparison domain. Write $Y^{\varepsilon}(t) = \int_{F=t} W^{\varepsilon} \mathbf{1}_Y \, d\mu_t$.

Input statement 3.4 (Recovered failure source and scalar reduction)**.** The capped-clock and running-compression theorems of retained modules 232–234 give, with every parameter fixed before the collision limit,

$$\mathcal{H}_M(b^{\varepsilon} - Y^{\varepsilon}) \le Cq^{-1}\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\} + Cqr.$$

The support is $Y \subset \{\tau^- \le 4qr, z < a\} \cup \{\ell \le 6qr\}$. The collar has no exit at widths $r \le r_{\mathrm{col}} = R_-^2/(800\Lambda_\circ)$. For $q = 1/8$, $r = c\varepsilon(B)^4 \le r_{\mathrm{col}}$ and $a = \sqrt{qr}$, the unchanged scalar objective satisfies

$$\mathcal{E}_M \le \mathcal{H}_M(Y^{\varepsilon(B)}) + C_M B^{-1/192}. \tag{3.9}$$

No residual-height decay is part of this input.

The support and the low-clock condition will both be used below. Dropping the latter would lose the density-decrease mechanism on the long critical component.

4. A quadratic physical cap at the critical gradient

The finite-count spectral error in (3.5) has only a square-root gain in the gradient cutoff. Integrating that error against the first inverse gradient is not legitimate. We obtain a second, quadratic bound from the two-dimensional physical map. Its constant is an explicitly weighted multiplicity, retained throughout rather than replaced by an unestimated word count.

Fix $0 < b_0 < 1/8$ within the range of (3.5). On the regular exact event put $G(x) = R(-p_0(x), p_m(x))$ and $z = |G|$. The retained endpoint Hessian satisfies $\lambda_\circ I \le K = D_q^2 F \le \Lambda_\circ I$ in endpoint angles $q = (\alpha_0, \alpha_m)$. The optical normalization is $\omega = |\partial_{p_0}\alpha_m|^{-1}$. Define, with physical states counted once,

$$D_{m,R,n,k}(v) = \sum_{x \in \Pi^{\mathrm{reg}} : G(x) = v} \frac{I_m(x)\omega(x)}{4\pi c_* \det K(x)}, \tag{4.1}$$

$$\mathcal{K}_m = 1 + \pi \sup_{R,n,k} \operatorname*{ess\,sup}_{|v| < b_0} D_{m,R,n,k}(v). \tag{4.2}$$

These definitions involve neither the roof target nor an inverse power of z. The factor I_m is the full optical insertion from (3.2).

Lemma 4.1 (Physical area formula and a finite weighted cap)**.** *For every finite m, $\mathcal{K}_m < \infty$, uniformly over the permitted radius and exact labels. For $0 < b < b_0$ and every interval J,*

$$\int_J u_b(t)\, dt \le \mathcal{K}_m b^2. \tag{4.3}$$

More precisely, if N_m bounds the cardinalities of the regular physical fibers of G at count m, and $v_* = 2\tau_-/R_+$, then

$$\mathcal{K}_m \le 1 + CN_m(1 + v_*)^{-(m-1)}. \tag{4.4}$$

Equation (4.4) *establishes finiteness only; no collision-uniform estimate of its right side is assumed.*

Proof. First variation gives $F_{\alpha_0} = -Rp_0$. Change of variables from (α_0, p_0) to endpoint angles has density ω. The Jacobian of $q \mapsto G(q)$ is $\det K > 0$. Thus the area formula, applied on injective local pieces and summed without multiplicity of charts, gives

$$G_{\#}(I_m \bar{\nu}) = D_{m,R,n,k}(v)\, dv.$$

Equivalently $|\det D_{(\alpha_0,p_0)}G| = \det K/\omega$. This identity uses the original collision normalization and counts physical preimages, not representatives in overlapping charts. Truncation of I_m, followed by monotone convergence, justifies the formula before its boundedness has been established.

Here is a bound for each summand uniform through internal grazing. For flight lengths $L_i \geq \tau_-$ the retained positive optical recursion is

$$y_1 = L_0, \quad f_0 = 1, \quad f_i = f_{i-1} + \frac{2y_i}{Rc_i}, \quad y_{i+1} = y_i + L_i f_i, \quad \omega = \frac{Rc_0c_m}{y_m}.$$

Consequently

$$y_m \geq \tau_- \prod_{i=1}^{m-1} (1 + v_*/c_i).$$

For a selected internal incidence,

$$\frac{1}{c_i} \prod_j (1 + v_*/c_j)^{-1} \leq v_*^{-1}(1 + v_*)^{-(m-2)}.$$

Since $\sum_i q_0^{d_i} \leq 2q_0/(1 - q_0)$, it follows that $I_m\omega \leq C(1 + v_*)^{-(m-1)}$. For one flight the empty product gives the same conclusion after enlarging C. Division by $\det K \geq \lambda_\circ^2$ bounds each summand.

At a fixed count, the finite-horizon restriction permits only finitely many lifted scatterer itineraries. The retained finite definable physical graph, expressed in finitely many circle charts, includes the first-hit tests, radius parameter and section decisions. The nonzero Jacobian makes every regular G-fiber discrete. Uniform finiteness for definable families therefore gives a finite N_m [9]. Displacement and occupation take finitely many values at this count. This is used only for fixed-count finiteness and proves (4.4). Finally discard the roof test on the upper side, and integrate (4.1) over the disk $|v| < b$. Its area is πb^2, proving (4.3). No change is made to the source being estimated. □

Lemma 4.2 (Integration of two different positive caps)**.** *For $E, H, b > 0$,*

$$b^{-1} \min(Eb^{1/2}, Hb^2) + \int_0^b v^{-2} \min(Ev^{1/2}, Hv^2)\, dv \leq 3 \min\{Hb, E^{2/3}H^{1/3}\}. \tag{4.5}$$

Proof. The crossing is $v_* = (E/H)^{2/3}$. For $b \leq v_*$ the left side is $2Hb$. For $b > v_*$ it is $3E^{2/3}H^{1/3} - Eb^{-1/2}$. These formulas prove the claim. The minimum is a cap on the positive source estimate, not on a signed spectral remainder considered by itself. □

Define the actual untruncated source

$$w_b(t)\, dt = F_\#(I_m z^{-1} \mathbf{1}_{\{0<z<b\}} \bar{\nu}).$$

It has a finite, absolutely continuous roof measure at every count.

Theorem 4.3 (First inverse gradient with a complete finite-count error)**.** *For one fixed auxiliary band B_*, all $m \geq 2$, $0 < b < b_0$ and intervals J of length L,*

$$m^2 \int_J w_b \leq Cb^{\alpha-1}(L + B_*^{-1}) + C \min\left\{m^2\mathcal{K}_m b,\ C_{B_*}^{2/3}(1 + B_*L)^{2/3}m^2(\mathcal{K}_m\theta_{B_*}^{2m})^{1/3}\right\}. \tag{4.6}$$

The integration goes to gradient zero at finite count. In particular there is no error proportional to a negative power of a lower cutoff. The weighted multiplicity $\mathcal{K}_m$ is not asserted bounded or subexponential in m.

Proof. Let $M(v) = m^2 \int_J u_v$. The two positive bounds give

$$M(v) \le C(L + B_*^{-1})v^\alpha + \min\{Ev^{1/2}, Hv^2\}, \quad E = C_{B_*}(1 + B_*L)m^2\theta_{B_*}^m, \quad H = m^2\mathcal{K}_m.$$

Indeed a nonnegative quantity bounded by both $A + D$ and V is bounded by $A + \min(D, V)$. Positive layer cake gives exactly

$$m^2 \int_J w_b = b^{-1}M(b) + \int_0^b v^{-2}M(v)\,dv.$$

Apply it first above a positive lower endpoint, and then use monotone convergence. The leading term is integrable because $\alpha > 1$; its multiplier is $1 + (\alpha - 1)^{-1}$. Lemma 4.2 integrates the full error. This proves the inequality and finiteness. Absolute continuity follows from coarea on compact regular roof charts and exhaustion; the finite critical set has zero source area. □

We isolate the additional estimate which would make this error negligible. For an unbounded sequence of previously fixed auxiliary bands, consider the condition

$$m^6\mathcal{K}_m\theta_{B_*}^{2m} \longrightarrow 0. \tag{SC}$$

Subexponential growth of $\mathcal{K}_m$ is sufficient, but is not proved here. Under (SC), first taking the collision limit and only then the indicated band sequence gives

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J w_b \le CLb^{1/128}. \tag{4.7}$$

Adding $z \ge b_0$ by (3.6) gives the bound CL for $v^0 = F_\#(I_m z^{-1}\bar{\nu})/dt$ at each fixed L. These are implications of the explicit condition, not unconditional zero-cutoff conclusions. Geometric-pressure results for stable Jacobian partition sums [25] concern a different weighted sum; no identification with (4.2) or uniform parameter bound is imported from them.

5. Separate critical and short-residence residuals

Keep the original residual $Y = Y_{a,q,r}$ and write $d = 2qr$. We assign the overlap to the short-residence component:

$$S = Y \cap \{\ell \le 3d\}, \qquad C = Y \cap \{\ell > 3d\}, \qquad Y^\varepsilon = S^\varepsilon + C^\varepsilon. \tag{5.1}$$

Here and below a superscript ε denotes the original weight W^ε on the indicated target restriction. The letters C, S are target sets, not new probability measures.

Lemma 5.1 (A full forward interval on the critical component)**.** *Every target in C satisfies*

$$\tau^- \le 2d, \qquad 0 < z < a, \qquad \tau^+ > d. \tag{5.2}$$

For $0 \le s \le d$, the trajectory remains in the original exact event and

$$0 < z(\Phi_s x) < b, \qquad b = (a^2 + 2\Lambda_\circ d)^{1/2}.$$

Moreover some $s \in [0, d]$ has $A(\Phi_s x) < A(x)/2$.

Proof. The complement of the old protected set is $\{\tau^- \le 2d\} \cap (\{z < a\} \cup \{\tau^+ \le d\})$. Since $\ell > 3d$, its second alternative is impossible and $\tau^+ > d$. Radial growth proves the upper bound on z. If $A(\Phi_s x)/A(x) \ge 1/2$ on this interval, the same-event times alone contribute $d/(2r) = q$ to the capped clock. This contradicts $c_r < q$. The direction choice introduces no new comparison-domain availability assumption. □

Theorem 5.2 (Uncut critical loss paid by the first inverse gradient)**.** *Assume $b < b_0$. At every finite collision count, for almost every target roof,*

$$C^{\varepsilon}(t) \le C_0 \int_t^{t+d} w_b(u)\, du. \tag{5.3}$$

Thus Theorem 4.3, with $L = d$, is an explicit bound for the complete critical residual, with no positive lower gradient cutoff. Under (SC)*,*

$$\limsup_m \sup_{R,n,k} m^2 \|C^{\varepsilon}\|_{\infty} \le C d b^{1/128}. \tag{5.4}$$

Without that additional condition, the same height bound holds for $C \cap \{z \ge c\}$ at each previously fixed $c > 0$. That last statement does not imply the uncut assertion by exchanging limits.

Proof. For a target selected in Lemma 5.1,

$$A(x)/2 \le \int_0^d |\partial_u A(\Phi_s x)|\, ds \le C \int_0^d A(\Phi_s x) I_m(\Phi_s x)/z(\Phi_s x)\, ds.$$

Multiply by $W^{\varepsilon}(x) \le 1$. At each fixed s the same-event transport is injective in the physical transversal; its image is inside the original exact fiber and has $z < b$. Therefore summing the right side gives a submeasure of $w_b(t+s)$. The untruncated measure has already been proved finite at fixed count. Tonelli and the common kernels prove (5.3). This is a target-height comparison, not an integral in starting roofs, and no target weight is replaced by an unweighted terminal flux.

Under (SC), apply (4.7) to the fixed interval of length d. For the assertion with $z(x) \ge c > 0$, use radial monotonicity and integrate (3.5) only above c. The leading layer cake is at most $C b^{\alpha-1}(d + B_*^{-1})$, and the finite-count error is $C_{B_*}(1 + B_* d) m^2 c^{-1/2} \theta_{B_*}^m$. Fixing c, d, B_* before the collision limit and enlarging B_* afterwards gives the same bound. This proof neither establishes (SC) nor removes the fixed c inside that limit. □

For the retained choice $q = 1/8$, $a = \sqrt{qr}$, we have $a^2 = d/2$. Consequently (5.4), whenever its stated hypothesis is verified, becomes

$$\mathcal{H}_M(C^{\varepsilon}) \le C(qr)^{257/256}. \tag{5.5}$$

The exponent is $1 + (\alpha - 1)/2$. It comes from integrating the first inverse gradient against the critical-volume power, not from the small mass of C itself. All internal incidences and all initial positive gradients are present in the finite-count comparison.

5.1. **A finite-count certificate rather than a hidden hypothesis.** The exact cost of the uncut comparison is visible in (4.6): at each fixed band it contains

$$m^2 (\mathcal{K}_m \theta_{B_*}^{2m})^{1/3}.$$

The theorem estimates the full source with that cost retained. A source-specific bound on the weighted momentum-fiber multiplicity would turn it into the vanishing critical-height theorem (5.5). Finiteness of N_m, a pressure bound for another partition, or a fixed positive-gradient theorem does not supply that bound. The condition remains an explicit proof obligation.

6. Two-ended physical traces on the short component

For a maximal original segment write $f_\sigma^{\varepsilon}(u) = A_\sigma(u) W^{\varepsilon}(\Psi_\sigma(u))$. Its nonnegative one-sided traces are denoted by f_σ^- and f_σ^+. These are the complete physical traces in (3.3); a critical entrance trace need not be zero. The comparison below neither pools the occupation classes nor identifies coincident faces from different incident components.

Let H be any Borel target restriction supported on $0 < \ell \le \delta$. Define its paired trace source at the original roof by

$$T_H^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < b_\sigma\}} \mathbf{1}_H(\Psi_\sigma(t)) \min\{f_\sigma^-, f_\sigma^+\}\, d\sigma. \tag{6.1}$$

The selector is evaluated at the target in both quantities being compared. In particular it can be the old low-clock residual, which need not be a smooth or monotone selector along a segment.

Theorem 6.1 (Complete short-source and paired-trace comparison)**.** *At every finite count and almost every roof,*

$$|R_H^\varepsilon(t) - T_H^\varepsilon(t)| \le C(1+\varepsilon^{-1}) \int_{t-\delta}^{t+\delta} v^0(u)\, du, \qquad v^0\, dt = F_\#(I_m z^{-1} \bar\nu). \tag{6.2}$$

This untruncated positive source is finite at each count by Theorem 4.3 and the bounded-gradient part. Under (SC)*,*

$$\limsup_m \sup_{R,n,k} m^2 \|R_H^\varepsilon - T_H^\varepsilon\|_\infty \le C(1+\varepsilon^{-1})\delta. \tag{6.3}$$

The constants are independent of the Borel target selector. In particular, the entire selected source on short segments with at least one grazing endpoint satisfies the same height bound.

Proof. Absolute continuity on the closed segment gives, for every interior u where the representative is defined,

$$|f_\sigma^\varepsilon(u) - \min(f_\sigma^-, f_\sigma^+)| \le \int_{a_\sigma}^{b_\sigma} |(f_\sigma^\varepsilon)'(v)|\, dv.$$

Both angular factors and the original guard give

$$|(f_\sigma^\varepsilon)'| \le C A_\sigma I_m / z + C\varepsilon^{-1} A_\sigma / z \le C(1+\varepsilon^{-1}) A_\sigma I_m / z.$$

Multiply the first inequality by the target selector. All segments of length at most δ containing t lie in $(t-\delta, t+\delta)$. At each intermediate roof the Borel segment bijection counts their interiors once, so the sum of these positive integrals is bounded by the displayed integral of v^0. The absolute value of the projected difference is bounded by this sum of absolute differences. Truncate and exhaust the original regular source, then use Tonelli. This proves (6.2) without taking a derivative of the Borel selector or a trace on its artificial boundary.

Under (SC), the complete first inverse gradient has fixed-width bound CL, as proved after (4.7). Apply it at $L = 2\delta$. If an endpoint is grazing, its original weighted trace is zero, so the minimum in (6.1) is zero. This proves the final assertion. An incoming critical endpoint with nonzero trace is not included in that zero-trace argument. □

This comparison does not claim that a minimum commutes with projection, or that (6.1) is the Jordan positive part of a signed current. It controls the error of a specific positive paired-trace comparison by the full physical derivative. The inherited signed forcing remains formed after projection.

Theorem 6.2 (A scalar reduction with distinct geometric obligations)**.** *Choose $q = 1/8$, $r = c\varepsilon(B)^4 \le r_{\mathrm{col}}$, $a = \sqrt{qr}$ and $\delta = 6qr$, with $c > 0$ fixed and small. Use $H = S$ from* (5.1) *in* (6.1)*. If* (SC) *holds, then*

$$\mathcal{E}_M \le \mathcal{H}_M(T_S^{\varepsilon(B)}) + C_M B^{-1/192} + C(qr)^{257/256} + C(1+\varepsilon(B)^{-1})qr. \tag{6.4}$$

Under that same condition, the original pointwise theorem is equivalent along this schedule to $\mathcal{H}_M(T_S^{\varepsilon(B)}) \to 0$. Neither (SC) *nor this paired-trace height decay is asserted established for the full Lorentz family.*

Proof. The retained finite-count scalar reduction bounds b^{ε} by Y^{ε} plus the already paid clock and protected compression terms. Split $Y = C \sqcup S$ before projection. Theorem 5.2 pays C, and Theorem 6.1 compares S to T_S. The retained positive remainder proves (6.4). Every scale and auxiliary band is fixed before the collision limit; (SC) is invoked only at each such fixed band. The band sequence is enlarged afterwards, and B is last.

For the schedule, the additional powers are $(qr)^{257/256} = O(\varepsilon^{257/64})$ and $(1+\varepsilon^{-1})qr = O(\varepsilon^{3})$; both vanish. Sufficiency follows. Conversely $S^{\varepsilon} \le b^{\varepsilon}$, so scalar closure and the positive remainder imply $\mathcal{H}_M(S^{\varepsilon(B)}) = O(B^{-1/192})$. Equation (6.3) then implies decay of T_S. This necessity statement is conditional on the same spectral-capacity estimate as its sufficiency statement. □

Without (SC), the finite-count comparisons (5.3) and (6.2) remain valid with the explicit error in (4.6). Thus the new reduction identifies two separate tasks: control a weighted two-dimensional momentum-fiber multiplicity against the spectral error, and control the original paired positive boundary traces on genuinely short segments. The first is not a roof-height assumption; the second retains the exact target selector, physical endpoints and original guard. Finite-count finiteness does not prove either asymptotic task. Scalar upper closure, not a finite-norm consequence, still precedes every unrestricted point-conditioned downstream assertion.

Appendix A. Dependencies and the exact new hypothesis

The physical map and uniform finite-count definability used in Lemma 4.1 are in archived module 164. The endpoint-angle Hessian and first variation are in module 174; the positive continuant and optical normalization are in modules 182 and 184. These give the area density and cancellation of each marked inverse incidence without a word-count estimate at collision scale.

The spectral input to Theorem 4.3 is precisely (3.5), from the actual fractional-source layer cake in module 187. The second cap is proved in the present article; its constant $\mathcal{K}_m$ is not silently replaced by a uniform number. Condition (SC) is a new, unverified sufficient estimate on that explicitly defined sequence. The finite-count insertion theorem does not require it.

The critical-target comparison uses the capped clock of module 232 and the protected-set calculation of modules 233–234. The short-trace comparison uses the Borel maximal-segment bijection of module 196, complete endpoint traces and signed balance of modules 220–222, and the original guard derivative. Both signs of endpoint failure and a possibly nonzero backward critical trace are retained. The scalar reconnection uses only (3.9) and the original positive remainder. No source theorem is assumed outside the identical exact event.

All historical statements are preserved in the archive. The active claims do not infer a zero-cutoff collision limit from fixed-count finiteness, identify stable-Jacobian pressure with a momentum-fiber sum, or promote the paired short-trace height to a proved vanishing quantity. Independent specialist verification of these continuum interfaces remains separate from source and build qualification.

Part 7. Capped coarea occupation and failure-source recovery

Retained revision-97 abstract. We study the upper pointwise local limit problem for the exact record of actual returns in a triangular Lorentz gas. Capping the physical coarea ratio before time integration gives an occupation anchor using both available directions and only returns to the identical selected event. This recovers targets excluded by a whole-interval or worst-time test. Independently, we bound the running compression of the complete physical angular density at each target roof on prescribed same-event intervals. The positive-cutoff inverse-gradient input pays this maximum without counting words or visits. Together these estimates quantitatively

reduce the preceding three-way failure source to a subsource near an original entrance with small gradient, or on a short original segment. The artificial endpoint collar has no contribution at a uniform small radius. Every recovered contribution has an explicit vanishing height bound along the reconstruction schedule. The residual essential height is not proved zero; the original pointwise theorem and all exact physical and arithmetic data remain unchanged.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. The arithmetic reference and the target.

For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R, a_j), \quad H_j = -\nabla^2 \log\lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \geq aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} Hv/2}\, \mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\leq n\leq m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Quantitative recovery of failed targets.** The new results concern the original failed-test source, not only a newly selected family of regular targets. With $Z^{\varepsilon}_{r,D,\beta}$ denoting the preceding disjoint interruption, compression and occupation failures, Theorem 6.1 constructs an original subsource $Y^{\varepsilon} \leq Z^{\varepsilon}$ and proves

$$\mathcal{H}_M(Z^{\varepsilon} - Y^{\varepsilon}) \leq Cq^{-1}\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\} + Cqr, \qquad q = \beta/D \leq 1/2.$$

This is essential height at the target roof. It is not an integrated translation or exceptional-set estimate. The residual support is specified by the original lifetimes in (6.3). The running maximum responsible for the Cqr term is estimated in Theorem 5.1 before taking the collision limit.

All new theorems use the following explicitly retained inputs. The comparison-domain anchor uses no source estimate outside the exact event. The singular-volume estimate for running compression is applied only while the trajectory remains in that event. This distinction prevents label changes from entering a fixed-label spectral estimate.

3. Physical source, signed measures, and retained estimates

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar{\nu} = \mathbf{1}_{\Pi}\nu_R^*$ for the original exact-event source. The retained Borel parametrization is a bijection between its regular part and maximal segments (σ, u), $a_\sigma < u < b_\sigma$. Its transverse measure $d\sigma$ is the disjoint angular measure and its density is

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad d\bar{\nu} = A_\sigma(u)\, du\, d\sigma.$$

Here $F = u$, $z = |\nabla_q F|$, $q = (\alpha_0, \alpha_m)$, $\omega = |\partial_{p_0}\alpha_m|^{-1}$, and $\chi = |\det D_q(F, \arg\nabla_q F)|$. All fibers and signed measures below use these same kernels. Let $z_\sigma^- = z(\Psi_\sigma(a_\sigma+))$ and $\ell_\sigma = b_\sigma - a_\sigma$.

Input statement 3.1 (Complete segment geometry and signed balance)**.** Put $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. On every regular segment,

$$2\lambda_\circ(v - u) \leq z(v)^2 - z(u)^2 \leq 2\Lambda_\circ(v - u), \tag{3.1}$$

$$|A'_\sigma| \leq CA_\sigma I_m/z, \qquad |(W^{\varepsilon})'| \leq C\varepsilon^{-1}/z, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i. \tag{3.2}$$

The complete density $f_\sigma^\varepsilon = A_\sigma W^\varepsilon$ has one-sided traces and is absolutely continuous on the closed segment. The sum of its endpoint traces and absolute derivative integrals is finite at each fixed count. A grazing endpoint has zero trace; a critical backward endpoint need not. The geometric domain is a finite definable family at each fixed count. Its component count is used for existence of measures only, not for normalized estimates.

For every Borel prefix endpoint $c_\sigma \in [a_\sigma, b_\sigma]$ put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma.$$

The original union $z < \zeta$ or $\ell < \delta$ is exactly such a prefix, and $c_\sigma = b_\sigma$ gives b^ε. Define

$$\mathsf{I}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma,$$

$$\mathsf{D}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma,$$

$$\mathsf{G}_c^{\varepsilon,\pm}(H) = \int \int_{(a_\sigma, c_\sigma)\cap H} [\pm (f_\sigma^\varepsilon)'(u)]_+\, du\, d\sigma.$$

Their complete finite-count signed identity is

$$DR_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{3.3}$$

Every artificial prefix cut is negative. All true entrance types and one-sided incident physical components are retained.

This is the endpoint-field, optical, Borel-segment and signed-prefix chain, culminating in retained modules 220–222. In particular, A_σ includes both angular factors. No reversal of the billiard record or change of the section is used at a backward end.

Input statement 3.2 (Backward-end and critical volume estimates)**.** Set $\alpha = 129/128$ and $\gamma = 1/32$. Let

$$e_s^-(t) = \int \mathbf{1}_{\{t - a_\sigma \le s\}}\, d\mu_t, \qquad u_q(t)\, dt = F_\#(I_m \mathbf{1}_{\{z<q\}}\bar\nu), \qquad p_q(t)\, dt = F_\#(\mathbf{1}_{\{z<q\}}\bar\nu).$$

For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\theta_{B_*} < 1$ such that, at finite count and small $s, q > 0$,

$$m^2 \int_J e_s^- \le Cs^\gamma(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\theta_{B_*}^m, \sqrt{s}\}, \tag{3.4}$$

$$m^2 \int_J u_q \le Cq^\alpha(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 q^{1/2}\theta_{B_*}^m. \tag{3.5}$$

The constants are uniform in radius, exact labels, marks where present, and interval location. Since $I_m \ge 1$, the latter also bounds p_q. Consequently, at every fixed $s, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_s^- \le CLs^\gamma.$$

For the actual positive source $v^\rho(t)\, dt = F_\#(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar\nu)$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\rho \le CL \quad (\rho, L > 0 \text{ fixed}). \tag{3.6}$$

The leading constant is independent of the fixed positive cutoff; its finite-count remainder is not.

Equation (3.4) is the same-event backward-end estimate of module 211, including critical entrances. For (3.5), multiply the actual fractional-source bound of module 187, at exponent 1/4, by $q^{1/4}$: pointwise, $I_m \mathbf{1}_{z<q} \le q^{1/4} I_m z^{-1/4} \mathbf{1}_{z<q}$. The leading power is q^{α} and the remainder power is $q^{1/2}$. The audited cutoff estimate is retained in modules 196 and 222. These are volume-source theorems, not assumptions on entrance flux.

Input statement 3.3 (Original scalar reconstruction)**.** For every fixed $\varepsilon, r > 0$,

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_r b^{\varepsilon}\|_{\infty} \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r} \int_{t-r}^{t+r} f(u)\, du. \tag{3.7}$$

For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards. The reconstruction band B is outermost.

We use $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} h(t)$ for a nonnegative source family. Unqualified $\|\cdot\|_{\infty}$ denotes essential supremum on the full roof line. Identities hold for common kernels at fixed m, R, n, k; the uniform order is $\sup_R \operatorname{ess\,sup}_t$. Countably many chosen radii share versions. No simultaneous Lebesgue-point assertion for every uncountable selector family is required.

3.1. **The retained physical comparison and the old failure source.**

Input statement 3.4 (Physical comparison at a fixed exact label)**.** Before imposing section decisions, the regular physical collar $|p_0|, |p_m| < 19/20$, $z > 0$, carries the partial comparison flow

$$\widehat{X} = K^{-1} g / (g^{\mathsf{T}} K^{-1} g), \qquad \widehat{X} F = 1, \quad \widehat{X} \vartheta = 0.$$

It stops at true physical, collar and backward critical ends, but crosses section-decision walls. On that collar $\lambda_* I \le K \le \Lambda_\circ I$, where $\lambda_* = 9\sqrt{39}/400$. The complete physical segment density is $\widehat{A} = \omega/(4\pi c_* \chi)$, and its restriction to Π is the original density A. The signed-time survivor graph is Borel, and each fixed-time flow is injective with common coarea kernels. Along an available interval of roof length at most r,

$$|W^{\varepsilon}(\widehat{\Phi}_s x) - W^{\varepsilon}(x)| \le \kappa_{\varepsilon}(r) = \min\{1, C_\Theta \varepsilon^{-1} \sqrt{r}\}.$$

Squared radial speed lies between $2\lambda_*$ and $2\Lambda_\circ$. No spectral estimate outside the original exact event is assumed.

This is the retained comparison-flow lemma, labeled `lem:v96-flow` in module 229; its proof uses the general endpoint Hessian, the contact inverse, the cubic action estimate and the Borel transversal from modules 174, 177, 180 and 196. The original stopped lifetimes $\tau^{\pm}$ and total residence $\ell = \tau^- + \tau^+$ are not the comparison lifetimes $\widehat{\tau}^{\pm}$.

For $H \subset \Pi$, set $R^{\varepsilon}_H(t) = \int_{F=t} \mathbf{1}_H W^{\varepsilon}\, d\mu_t$ and $U_H(t) = \int_{F=t} \mathbf{1}_H\, d\mu_t$. An old comparison direction $e \in \{-1, 1\}$ is available if it survives for all $0 \le s \le r$. On an available direction put

$$d_{e,r}(x) = \sup_{0 \le s \le r} \frac{\widehat{A}(x)}{\widehat{A}(\widehat{\Phi}_{es} x)}, \qquad o_{H,e,r}(x) = r^{-1} \int_0^r \mathbf{1}_H(\widehat{\Phi}_{es} x)\, ds.$$

Old admission means that some direction has $d_{e,r} \le D$ and $o_{H,e,r} \ge \beta$. For $H = \Pi$, N_T means neither direction is available; N_D means availability but no available direction has $d \le D$; N_O means a bounded-ratio direction exists but none of them has $o \ge \beta$. Their disjoint sum is N, and

$$Z^{\varepsilon}_{r,D,\beta}(t) = \int_{F=t} \mathbf{1}_N W^{\varepsilon}\, d\mu_t.$$

The new proofs estimate a portion of this very source, not a source obtained by changing its label or its stopping convention.

4. Capped coarea occupation on the original event

A worst-time density ratio can fail at a time which supplies no same-label anchor. Requiring a whole one-sided interval can also reject two shorter intervals which together supply sufficient mass. We retain all available times and cap their coarea contribution before integrating. No state of a different label supplies mass.

Lemma 4.1 (The collar is not a microscopic obstruction)**.** *Put* $r_{\mathrm{col}} = R_-^2/(800\Lambda_\circ)$*. A comparison trajectory starting in* Π *cannot reach the artificial collar boundary in either direction within roof distance* r_{col}*. This statement does not assert continuation past a true physical or critical end.*

Proof. The original section gives $|p_0|, |p_m| < 17/20$. Indeed its largest positive momentum is at most $2(53/100)/(1 + (53/100)^2) + 1/200 < 17/20$; all other section rectangles have smaller absolute momenta. Along the comparison flow, g has constant direction and its squared norm has derivative between $2\lambda_*$ and $2\Lambda_\circ$. Therefore, for either sign of a surviving time s,

$$|p_i(\widehat{\Phi}_s x) - p_i(x)| \le R_-^{-1}|g(\widehat{\Phi}_s x) - g(x)| \le R_-^{-1}\sqrt{2\Lambda_\circ|s|}, \qquad i = 0, m.$$

At the stated radius this is at most $1/20$. Hence the momenta remain below $9/10$, strictly inside the collar $19/20$. The same estimate holds at a limiting first boundary time, so a collar end cannot be the first obstruction, including a simultaneous failure. A backward critical end is not excluded by this argument. □

For $x \in H \subset \Pi$, let $T_r(x) = (-r, r) \cap (-\widehat{\tau}^-(x), \widehat{\tau}^+(x))$. Define the dimensionless capped occupation

$$c_{H,r}(x) = \frac{1}{r}\int_{T_r(x)} \mathbf{1}_H(\widehat{\Phi}_s x) \min\left\{1, \frac{\widehat{A}(\widehat{\Phi}_s x)}{\widehat{A}(x)}\right\} ds. \tag{4.1}$$

It belongs to $[0, 2]$. It is a Borel function by the measurable survivor graph and Tonelli. The density ratio is taken along the same physical transverse coordinate; it is not a ratio of projected roof densities. For $0 < q \le 1/2$ put $G_{H,r,q} = H \cap \{c_{H,r} \ge q\}$.

Theorem 4.2 (An anchor using all available coarea mass)**.** *At finite collision count and almost every target roof,*

$$R^{\varepsilon}_{G_{H,r,q}}(t) \le \frac{1}{qr}\int_{t-r}^{t+r} \{R^{\varepsilon}_H(u) + \kappa_\varepsilon(r) U_H(u)\}\, du, \tag{4.2}$$

$$U_{G_{H,r,q}}(t) \le \frac{1}{qr}\int_{t-r}^{t+r} U_H(u)\, du. \tag{4.3}$$

If $q = \beta/D \le 1/2$*, the previously admitted set* $H^{\mathrm{adm}}_{r,D,\beta}$ *is contained in* $G_{H,r,q}$*. No lower gradient or residence cutoff is used in this theorem.*

Proof. Write $A_0 = \widehat{A}(x)$, $A_s = \widehat{A}(\widehat{\Phi}_s x)$ and $W_s = W^{\varepsilon}(\widehat{\Phi}_s x)$. On a sampled return to H, the unchanged guard gives

$$\min(A_0, A_s) W_0 \le A_s\{W_s + \kappa_\varepsilon(r)\}.$$

Integrating this inequality and using $c_{H,r}(x) \ge q$ bounds $A_0 W_0$ by the right side divided by qr. Integrate over target transverse coordinates. At each fixed signed time the comparison flow is injective, preserves the transverse coordinate, and maps the selected targets into the same exact H fiber. Thus their A_s mass is a submeasure of that original fiber. Tonelli proves (4.2); taking weight one proves (4.3). The two signs are already the two halves of the displayed roof integral. Multiple returns introduce no multiplicity at a fixed sampled roof.

On a previously admitted one-sided interval, at least βr times return to H, and $A_s/A_0 \ge D^{-1}$ there. Their capped contribution is at least $\beta/D = q$. This proves containment. All identities use common kernels at each fixed choice of parameters; no universal null set for uncountably many selectors is asserted. □

For $H = \Pi$, denote $c_{\Pi,r}$ by c_r. Let $N = N_T \cup N_D \cup N_O$ be the preceding disjoint failed-test set. The theorem already estimates an actual part of that old failure source:

$$\mathcal{H}_M\left(R^{\varepsilon}_{N\cap\{c_r\ge q\}}\right) \le \frac{C}{q}\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\}, \qquad q = \beta/D \le 1/2. \tag{4.4}$$

This follows from the two original fixed-width source estimates. Unavailable times and times of a different label contribute zero to (4.1); a very large inverse ratio at one such time does not by itself reject the target. This is a containment and an estimated recovery, not a claim that every physical failure is recovered.

5. Running compression at a prescribed target roof

The next estimate controls a maximum over the sampled time before projection and before the collision limit. Its norm is essential height in the target roof, not an integral over starting roofs. The sampled interval is deliberately kept in the original exact event: no singular-volume theorem on the enlarged comparison domain is assumed.

Fix $a, r > 0$, $0 < q \le 1/2$, and put $d = 2qr$. Using the original stopped lifetimes, define disjoint target classes

$$\begin{aligned} P_- &= \{\tau^- > 2d\},\\ P_+ &= \{\tau^- \le 2d,\ z \ge a,\ \tau^+ > d\}, \qquad P = P_- \cup P_+. \end{aligned} \tag{5.1}$$

Choose sign $e = -1$ on P_- and $e = 1$ on P_+. The chosen interval of length d lies entirely in the same original exact-event segment. Set

$$\rho = \min\{a, \sqrt{2\lambda_\circ d}\}, \qquad v^\rho(u)\,du = F_{\#}(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar\nu).$$

On P_+ radial growth gives $z(\Phi_s x) \ge a$; on P_- it gives $z(\Phi_{-s}x)^2 \ge 2\lambda_\circ(\tau^-(x) - s) > 2\lambda_\circ d$ for $0 \le s \le d$. Thus the same positive cutoff ρ works on both classes, including every small internal incidence.

For $D_0 > 1$ define the original weighted running-compression source

$$C^{\varepsilon}_{a,q,r,D_0}(t) = \int_{F=t} W^{\varepsilon}\mathbf{1}_P \mathbf{1}_{\{\sup_{0\le s\le d} A(x)/A(\Phi_{es}x) > D_0\}}\, d\mu_t(x). \tag{5.2}$$

The supremum can be taken over rational times and endpoints.

Theorem 5.1 (A complete target-height bound for running loss)**.** *At every finite count and almost every roof,*

$$C^{\varepsilon}_{a,q,r,D_0}(t) \le \frac{C}{1 - D_0^{-1}} \int_{t-d}^{t+d} v^\rho(u)\,du. \tag{5.3}$$

Consequently, for all previously fixed a, q, r, D_0,

$$\limsup_m \sup_{R,n,k} m^2 \|C^{\varepsilon}_{a,q,r,D_0}\|_\infty \le \frac{Cqr}{1 - D_0^{-1}}. \tag{5.4}$$

The leading constant is independent of a and ε. Neither $a = 0$ nor a collision-dependent cutoff is asserted.

Proof. For a target counted in (5.2), continuity provides a sampled time at which $A_s < A_0/D_0$. Hence

$$(1 - D_0^{-1})A_0 \le \int_0^d |\partial_u A(\Phi_{es}x)|\, ds.$$

The factor on the left pays the original target density, not the terminal density. Multiply by $W^\varepsilon(x) \le 1$ and use $|\partial_u A| \le CAI_m/z$. At every sampled roof the Borel segment bijection counts the selected states once, inside the same exact event. The lower bound for z therefore identifies their sum as a submeasure of v^ρ at that roof. The negative and positive target-sign classes use disjoint halves of $(t-d, t+d)$. This proves (5.3) by Tonelli, without removing the density ratio or taking a positive part after a false change of source. Apply the positive-cutoff volume estimate (3.6) to the fixed interval of length $2d$ to obtain (5.4). Its finite-count remainder may depend on ρ; the leading constant after the stated limit does not. □

Corollary 5.2 (Low capped occupation on the whole protected class)**.** *For $H = \Pi$,*

$$R^\varepsilon_{P\cap\{c_r<q\}}(t) \le C\int_{t-d}^{t+d} v^\rho(u)\, du, \tag{5.5}$$

$$\mathcal{H}_M(R^\varepsilon_{P\cap\{c_r<q\}}) \le Cqr. \tag{5.6}$$

Proof. If $A_s/A_0 \ge 1/2$ throughout the chosen original interval, that interval alone contributes at least $d/(2r) = q$ to c_r. Therefore $c_r < q$ implies that the running ratio exceeds two. Apply Theorem 5.1 with $D_0 = 2$. □

In particular the running loss is paid even if a later part of the physical comparison encounters many section decisions or fails the old worst-time test. The proof samples only the displayed initial same-event interval; it does not estimate optical variation accumulated at states carrying a different exact occupation label.

6. Quantitative reduction of the old failure source

Keep $q = \beta/D \le 1/2$, use $H = \Pi$ in the capped clock, and use P from (5.1). Define a residual target set and its original weighted source by

$$Y_{a,q,r} = \Pi \cap \{c_r < q\} \cap P^c,$$

$$Y^\varepsilon_{a,q,r}(t) = \int_{F=t} W^\varepsilon \mathbf{1}_{Y_{a,q,r}}\, d\mu_t. \tag{6.1}$$

No new measure is placed on the boundary or on a sampled orbit. Every target is in exactly one of $\{c_r \ge q\}$, $\{c_r < q\} \cap P$, and $Y_{a,q,r}$.

Theorem 6.1 (An estimated part of every previous failure class)**.** *At finite count the residual is a subsource of the old three-way failure source and has the geometric support*

$$0 \le Y^\varepsilon_{a,q,r} \le Z^\varepsilon_{r,D,\beta}, \tag{6.2}$$

$$Y_{a,q,r} \subset \{\tau^- \le 4qr,\ z < a\} \ \cup\ \{\ell \le 6qr\}. \tag{6.3}$$

At almost every target roof,

$$b^\varepsilon(t) \le Y^\varepsilon_{a,q,r}(t) + \frac{1}{qr}\int_{t-r}^{t+r} \{b^\varepsilon(u) + \kappa_\varepsilon(r)p(u)\}\, du + C\int_{t-2qr}^{t+2qr} v^\rho(u)\, du. \tag{6.4}$$

The recovered part of the old *failure source obeys the actual height estimate*

$$\mathcal{H}_M(Z^\varepsilon_{r,D,\beta} - Y^\varepsilon_{a,q,r}) \le \frac{C}{q}\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\} + Cqr. \tag{6.5}$$

All parameters here are fixed before the collision limit. The constant in (6.5) *is independent of the positive* a.

Proof. Old admission implies $c_r \geq q$ by Theorem 4.2. Thus $\{c_r < q\} \subset N$, which proves (6.2). The complement of P is exactly

$$\{\tau^- \leq 4qr\} \cap (\{z < a\} \cup \{\tau^+ \leq 2qr\}).$$

In the second alternative $\ell = \tau^- + \tau^+ \leq 6qr$, proving (6.3). Split the full source by the three disjoint sets preceding the theorem. The capped anchor estimates the first; Corollary 5.2 estimates the second. This proves (6.4) at finite count.

The same split restricted to N gives a nonnegative recovered source $Z - Y$, bounded by those same two paid terms. Apply the original fixed-width estimates for b^ε and p, and the audited positive-cutoff estimate for v^ρ. This proves (6.5). The sum of the recovered parts of N_T, N_D, N_O is precisely $Z - Y$: it is not estimated by counting three overlapping events or by subtracting signed boundary fluxes. □

The capped anchor admits some targets rejected by the old one-sided or worst-time tests, and the running-loss theorem pays the remaining low-clock targets on P. The theorem proves containment and a bound on the recovered source. It does not assert a positive lower bound for the amount recovered in every Lorentz fiber.

Corollary 6.2 (A sharper original-source support at one legal schedule)**.** *Choose* $D = 2$, $\beta = 1/4$, $q = 1/8$, $r(B) = c\varepsilon(B)^4 \leq r_{\mathrm{col}}$, *and* $a(B) = \sqrt{qr(B)}$, *with fixed sufficiently small* $c > 0$. *Then*

$$\mathcal{E}_M \leq \mathcal{H}_M(Y^{\varepsilon(B)}_{a(B),q,r(B)}) + C_M B^{-1/192}. \tag{6.6}$$

Moreover the recovered old failure in (6.5) *is* $O(B^{-1/192})$. *The residual contains only targets with* $z < \sqrt{qr}$ *near an original entrance or total original residence at most* $6qr$. *Its fixed-window mass satisfies*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J Y^\varepsilon_{a,q,r}(t)\, dt \leq CL\{(qr)^{129/256} + (qr)^{1/32}\}. \tag{6.7}$$

The last assertion is a mass estimate, not an essential-height estimate.

Proof. Insert (6.4) into the retained positive scalar remainder. Its explicit errors have orders $\varepsilon^{1/16}$, $\varepsilon^{-1}\sqrt{r} = O(\varepsilon)$, and $qr = O(\varepsilon^4)$. With $\varepsilon = A_0 B^{-1/12}$ the slowest is $B^{-1/192}$. All widths, including $\rho = \min(a, \sqrt{4\lambda_\circ qr})$, are fixed before the collision limit; auxiliary bands are enlarged afterwards. For (6.7), use (6.3), $W \leq 1$, the critical-volume bound at $a = \sqrt{qr}$, and $\{\ell \leq 6qr\} \subset \{\tau^- \leq 6qr\}$. The two original volume estimates give the displayed powers after the same legal limits. □

For completeness, the original pointwise theorem along this schedule is equivalent to $\mathcal{H}_M(Y^{\varepsilon(B)}_{a(B),q,r(B)}) \to 0$. Sufficiency follows from (6.6); necessity follows from $Y \leq b^\varepsilon$ and the positive remainder. This equivalence is secondary to the estimated reduction (6.5). The remaining residual height is not proved to vanish.

The retained bidirectional signed forcing is formed after projection and satisfies $0 \leq J^\varepsilon_r \leq b^\varepsilon$. Thus (6.4) bounds it without moving a minimum or a positive part into a sheet integral. Since the schedule also has $(1 + \varepsilon^{-1})r \to 0$, its retained scalar equivalence applies. No joint-forcing-zero, unrestricted pointwise denominator, same-roof ratio, essential-likelihood or arithmetic-zero normalization assertion is promoted by the present reduction.

Appendix A. Dependency and source conventions

The unchanged physical formulation is (2.1), with the reference (2.4) and pointwise goal (2.5). The complete preceding article is included in the technical archive. All 231 earlier core files retain their bytes.

The new collar lemma uses only the section momentum bound and the radial comparison-flow identity. The capped anchor uses the comparison coarea density, fixed-time injectivity, and the original guard modulus. The running-compression theorem uses the original stopped lifetimes, radial growth, the derivative of both angular-density factors, and the positive-cutoff source estimate (3.6). That last input is proved in retained modules 196 and 222 from the finite-count critical-volume layer cake; its leading constant is cutoff independent, but its finite-count remainder is not. The cutoff in the new proof is positive and fixed before every collision limit.

The final recovery theorem combines these two distinct comparisons and uses the original scalar average and positive remainder. The support mass estimate additionally uses the original backward-end and critical-volume estimates. No new theorem on a spectral insertion outside the exact event, no uniform untruncated first inverse gradient, and no interchange of a zero-gradient limit with a collision limsup is required or asserted.

The Borel and signed-measure foundations remain those of the retained geometric chain, with the standard bounded-variation and definable integration references [13, 15, 9]. Source preservation and finite diagnostics do not supply an independent reconstruction of that chain. The new proofs provide an estimated recovered source; they do not establish vanishing of the remaining critical/short-source essential height. The two distinct revision-88 variants retain their separate source identities throughout the archive.

Part 8. Exact-label physical continuation and occupation

Retained revision-96 abstract. We study the remaining upper pointwise local limit problem for the exact record of actual returns in a triangular Lorentz gas. A section-decision end need not terminate the underlying regular physical word. We construct a comparison flow across such decisions and prove an exact-label occupation inequality at every finite collision count. Only sampled states returning to the original selected event supply anchor mass; no occupation labels are pooled. This estimates the positive two-nongrazing short-segment packet itself on the bounded-compression recurrent class, and estimates the entire recurrent target-critical core without a lower gradient cutoff or a finite-annulus truncation. A prescribed-radius pointwise inequality localizes the remaining source to physical interruption, compression, and insufficient same-label occupation. The original source, guard, arithmetic classes and pointwise objective are unchanged. Exhaustion of all possible spikes by the controlled class is not proved, and the unrestricted pointwise theorem is not claimed complete.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\,\mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U=\{|\alpha-\pi/6|<3/50,\ |p|<3/20\},\qquad Y_R^*=Y_R\cup U,\qquad c_*=\frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*)=c_*$. We always use $\nu_R^*=c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L=L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R}=\sum_{j=0}^{m-1}\mathbf{1}_{Y_R^*}\circ T_R^j$. The event is

$$\Pi_{n,k,m,R}=Y_R^*\cap\{K^{\mathrm{c}}_{m,R}=k,\ A_{m,R}=n\}\cap T_R^{-m}Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j=\sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0\le\Theta\le1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0=47/53$, $d_j=\min(j,m-j)$ and $d_j'=\min(j,m-1-j)$, set

$$H^\varepsilon=\prod_{j=0}^{m}\Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right)\prod_{j=0}^{m-1}\Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right),\qquad W^\varepsilon=1-H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t=L_{\#}(W^\varepsilon\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^m B_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The positive packet and its returning anchors.** The principal target remains (2.5). The new question is whether the positive two-ended packet can be paid by the original source at other roofs, even when its own exact-event segment is short. Theorem 4.2 answers this on a class defined by physical continuation, a bound on the inverse physical Jacobian, and the time spent in the same exact selected event. Its bound has no word-count or return-count factor. Theorem 5.3 applies it to the previously residual two-nongrazing minimum, not only to the error between that minimum and the target source.

Theorem 5.1 also gives a target-critical estimate with all gradients below the upper threshold present at finite count. This closes the disappearing-annulus issue on the admitted class; it does not claim that every critical target is admitted. The full original source is partitioned exactly in Theorem 6.1. The three failed tests there, rather than a silently removed physical subsource, remain in the endpoint budget. All prior results and their hypotheses are retained.

3. Physical source, signed measures, and retained estimates

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar{\nu} = \mathbf{1}_{\Pi}\nu_R^*$ for the original exact-event source. The retained Borel parametrization is a bijection between its regular part and maximal segments (σ, u), $a_\sigma < u < b_\sigma$. Its transverse measure $d\sigma$ is the disjoint angular measure and its density is

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad d\bar{\nu} = A_\sigma(u)\, du\, d\sigma.$$

Here $F = u$, $z = |\nabla_q F|$, $q = (\alpha_0, \alpha_m)$, $\omega = |\partial_{p_0}\alpha_m|^{-1}$, and $\chi = |\det D_q(F, \arg \nabla_q F)|$. All fibers and signed measures below use these same kernels. Let $z_\sigma^- = z(\Psi_\sigma(a_\sigma+))$ and $\ell_\sigma = b_\sigma - a_\sigma$.

Input statement 3.1 (Complete segment geometry and signed balance)**.** Put $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. On every regular segment,

$$2\lambda_\circ(v-u) \le z(v)^2 - z(u)^2 \le 2\Lambda_\circ(v-u), \tag{3.1}$$

$$|A_\sigma'| \le C A_\sigma I_m / z, \qquad |(W^\varepsilon)'| \le C\varepsilon^{-1}/z, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i. \tag{3.2}$$

The complete density $f_\sigma^\varepsilon = A_\sigma W^\varepsilon$ has one-sided traces and is absolutely continuous on the closed segment. The sum of its endpoint traces and absolute derivative integrals is finite at each fixed count. A grazing endpoint has zero trace; a critical backward endpoint need not. The geometric domain is a finite definable family at each fixed count. Its component count is used for existence of measures only, not for normalized estimates.

For every Borel prefix endpoint $c_\sigma \in [a_\sigma, b_\sigma]$ put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma.$$

The original union $z < \zeta$ or $\ell < \delta$ is exactly such a prefix, and $c_\sigma = b_\sigma$ gives b^ε. Define

$$\mathsf{I}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma,$$

$$\mathsf{D}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma,$$

$$\mathsf{G}_c^{\varepsilon,\pm}(H) = \int \int_{(a_\sigma, c_\sigma)\cap H} [\pm (f_\sigma^\varepsilon)'(u)]_+\, du\, d\sigma.$$

Their complete finite-count signed identity is

$$DR_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{3.3}$$

Every artificial prefix cut is negative. All true entrance types and one-sided incident physical components are retained.

This is the endpoint-field, optical, Borel-segment and signed-prefix chain, culminating in retained modules 220–222. In particular, A_σ includes both angular factors. No reversal of the billiard record or change of the section is used at a backward end.

Input statement 3.2 (Backward-end and critical volume estimates)**.** Set $\alpha = 129/128$ and $\gamma = 1/32$. Let

$$e_s^-(t) = \int \mathbf{1}_{\{t-a_\sigma \le s\}}\, d\mu_t, \qquad u_q(t)\,dt = F_\#(I_m \mathbf{1}_{\{z<q\}}\bar\nu), \qquad p_q(t)\,dt = F_\#(\mathbf{1}_{\{z<q\}}\bar\nu).$$

For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\theta_{B_*} < 1$ such that, at finite count and small $s, q > 0$,

$$m^2 \int_J e_s^- \le Cs^\gamma(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\theta_{B_*}^m, \sqrt{s}\}, \tag{3.4}$$

$$m^2 \int_J u_q \le Cq^\alpha(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 q^{1/2}\theta_{B_*}^m. \tag{3.5}$$

The constants are uniform in radius, exact labels, marks where present, and interval location. Since $I_m \ge 1$, the latter also bounds p_q. Consequently, at every fixed $s, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_s^- \le CLs^\gamma.$$

For the actual positive source $v^\rho(t)\,dt = F_\#(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar\nu)$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\rho \le CL \quad (\rho, L > 0 \text{ fixed}). \tag{3.6}$$

The leading constant is independent of the fixed positive cutoff; its finite-count remainder is not.

Equation (3.4) is the same-event backward-end estimate of module 211, including critical entrances. For (3.5), multiply the actual fractional-source bound of module 187, at exponent $1/4$, by $q^{1/4}$: pointwise, $I_m \mathbf{1}_{z<q} \le q^{1/4} I_m z^{-1/4}\mathbf{1}_{z<q}$. The leading power is q^α and the remainder power is $q^{1/2}$. The audited cutoff estimate is retained in modules 196 and 222. These are volume-source theorems, not assumptions on entrance flux.

Input statement 3.3 (Original scalar reconstruction)**.** For every fixed $\varepsilon, r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\,du. \tag{3.7}$$

For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards. The reconstruction band B is outermost.

We use $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} h(t)$ for a nonnegative source family. Unqualified $\|\cdot\|_\infty$ denotes essential supremum on the full roof line. Identities hold for common kernels at fixed m, R, n, k; the uniform order is $\sup_R \operatorname{ess\,sup}_t$. Countably many chosen radii share versions. No simultaneous Lebesgue-point assertion for every uncountable selector family is required.

4. Physical continuation and exact-label occupation

A maximal segment of an exact event can be short because of section *decisions*, while its physical word remains regular for a longer roof interval. We use this distinction only for comparison. The original stopped segments, endpoint traces, occupation index and source measure are not changed.

Fix $p_* = 19/20$ and put $\lambda_* = 9\sqrt{39}/400$. Let $\widehat{\Omega}_m$ be the actual regular m-flight collision states with $|p_0|, |p_m| < p_*$ and $z > 0$, before imposing any section decisions. The physical word and displacement are fixed on a component. A comparison trajectory stops at a true physical boundary, at the collar boundary, or at a backward critical end. It does not stop at a section-decision wall. Write $\widehat{\Phi}_s$ for this partial flow. The original event $\Pi = \Pi_{n,k,m,R}$ is always tested at the sampled state, including its initial and terminal tests and the exact value n.

Lemma 4.1 (A comparison flow with the same physical density)**.** *On $\widehat{\Omega}_m$ the field*

$$\widehat{X} = \frac{K^{-1}g}{g^{\mathsf{T}}K^{-1}g}$$

is smooth on each physical component, with $\lambda_* I \le K \le \Lambda_\circ I$, $\widehat{X}F = 1$ and $\widehat{X}\vartheta = 0$. Its complete-physical-segment coordinates (ξ, u) have density

$$\widehat{A}_\xi(u) = \frac{\omega}{4\pi c_* \chi}, \qquad d(c_*^{-1}\nu) = \widehat{A}_\xi(u)\, du\, d\xi.$$

They restrict to the original coarea measure wherever Π holds. For a surviving comparison interval of roof length at most r,

$$|W^\varepsilon(\widehat{\Phi}_s x) - W^\varepsilon(x)| \le \kappa_\varepsilon(r) := \min\{1, C_\Theta \varepsilon^{-1}\sqrt{r}\}, \tag{4.1}$$

$$|\partial_u \log \widehat{A}_\xi| \le C_* I_m / z. \tag{4.2}$$

All constants are independent of the collision count. The comparison flow is not a continuation through a grazing or competing-hit boundary.

Proof. The general endpoint Hessian inequality, before the section restriction, is

$$R\,\mathrm{diag}(c_0, c_m) \le K \le (2R_+^2/\tau_- + R_+)I.$$

Its proof in retained module 174 uses the supporting polygonal action and the positive internal Schur complement, not section membership. On the collar, $c_0, c_m > \sqrt{39}/20$, giving the stated constants. The identities for F, ϑ and χ follow by differentiating $g/|g|$, exactly as for the original field. They give the displayed area density, including both optical factors. The rational-roof Borel transversal construction of module 196 applies to this open physical domain. At any fixed roof the comparison map is injective on its domain; two different starting states cannot meet under a smooth flow. This remains true when the same orbit leaves and reenters Π.

The contact inverse of module 177 is independent of section decisions. It bounds contact motion by $Cq_0^{d_j}$ times endpoint-curve length. Radial growth with λ_* bounds that length by $C\sqrt{r}$, in either direction along a surviving interval. The original product guard then gives (4.1); no monotonicity of Θ is used. The optical log-sensitivity computation of module 182 requires only nonzero internal incidences and a positive lower endpoint cosine. On this fixed collar it gives $|\widehat{X}\log\omega| \le CI_m/z$. The stationary cubic estimate of module 180 and $\chi \ge \lambda_*$ give $|\widehat{X}\log\chi| \le C/z$. Subtracting the two logarithmic derivatives proves (4.2). Thus the extension uses the same local geometric formulas, not a spectral theorem outside Π or an assumed source bound on the comparison domain. □

Let H be any Borel subset of the original exact event. The choices $H = \Pi$, $H = \Pi \cap \{z < a\}$, $H = \Pi \cap \{\ell < \delta\}$, and their original union are important. Here ℓ is always the length of the *original stopped exact-event segment*, recomputed at each sampled state. Define

$$R_H^{\varepsilon}(t) = \int_{F=t} \mathbf{1}_H W^{\varepsilon}\, d\mu_t, \qquad U_H(t) = \int_{F=t} \mathbf{1}_H\, d\mu_t.$$

For a sign $e \in \{-1, 1\}$, a comparison interval is available when $\widehat{\Phi}_{es}x$ exists for $0 \le s \le r$. On such an interval put

$$d_{e,r}(x) = \sup_{0\le s\le r} \frac{\widehat{A}(x)}{\widehat{A}(\widehat{\Phi}_{es}x)},$$
$$o_{H,e,r}(x) = \frac{1}{r}\int_0^r \mathbf{1}_H(\widehat{\Phi}_{es}x)\, ds. \tag{4.3}$$

Set $d_{e,r} = +\infty$ if the interval is unavailable. The ratio is intrinsic: it is the inverse physical coarea Jacobian in the same (F, ϑ) coordinates. Fix $D \ge 1$ and $0 < \beta \le 1$. A target in H is *admitted* if, for at least one sign, $d_{e,r} \le D$ and $o_{H,e,r} \ge \beta$. Denote this Borel target set by $H^{\mathrm{adm}}_{r,D,\beta}$. The supremum in d can be taken on rational times by continuity; the occupation integral is Borel by Tonelli. Choose the positive sign when both signs are admissible.

Theorem 4.2 (Exact-label occupation anchoring)**.** *At every finite collision count and almost every target roof,*

$$R^{\varepsilon}_{H^{\mathrm{adm}}}(t) \le \frac{D}{\beta r}\int_{t-r}^{t+r} \big(R_H^{\varepsilon}(u) + \kappa_{\varepsilon}(r) U_H(u)\big)\, du, \tag{4.4}$$

$$R^{\varepsilon}_{H^{\mathrm{adm}}}(t) \le U_{H^{\mathrm{adm}}}(t) \le \frac{D}{\beta r}\int_{t-r}^{t+r} U_H(u)\, du. \tag{4.5}$$

There is no lower gradient, incidence or original-residence cutoff. There is no factor counting visits, words, or exact-event components.

Proof. Fix an admitted target and its chosen sign. At every sampled time with $\widehat{\Phi}_{es}x \in H$, the density ratio and guard modulus give

$$\widehat{A}(x) W^{\varepsilon}(x) \le D\widehat{A}(\widehat{\Phi}_{es}x)\{W^{\varepsilon}(\widehat{\Phi}_{es}x) + \kappa_{\varepsilon}(r)\}.$$

Integrate over these sampled times, whose total length is at least βr. Then integrate over the target transverse coordinates. For each fixed s and each sign, flow injectivity and preservation of ϑ identify the transported targets with a subfamily of the *same exact-label* fiber at $t + es$. Its density is exactly $\widehat{A}(\widehat{\Phi}_{es}x)$; the inverse Jacobian has already been paid by D. Adding the two disjoint target-sign classes yields (4.4). Apply the identical argument with weight one to obtain (4.5).

Intermediate states outside H supply no anchor mass. In particular states with a different occupation value are never substituted. The flow may reenter H many times, but at any fixed sampled roof each transported target is counted only once. Tonelli and the physical Borel coordinates justify the integrals. As usual each fixed parameter choice has its common almost-everywhere version; no common Lebesgue set for uncountably many Borel selectors is asserted. □

A directly checkable sufficient distortion test is

$$C_* \int_0^r \frac{I_m(\widehat{\Phi}_{es}x)}{z(\widehat{\Phi}_{es}x)}\, ds \le \log D.$$

It follows from (4.2). This is an integrated physical test, not an incidence corridor. Its satisfaction on all potential spike trajectories is not assumed.

5. Recurrent two-ended packets and the full radial core

The occupation theorem estimates a positive packet itself, rather than only a difference between its two traces. The estimated family can contain arbitrarily short exact-event segments: the anchors need not lie in the original segment, but must return to the same selected event on the same regular physical word.

Set $\alpha = 129/128$ and $\gamma = 1/32$, and write

$$H_{a,\delta} = \Pi \cap \{z < a \text{ or } \ell < \delta\}.$$

The intersection is counted once. Its unweighted density satisfies $U_{H_{a,\delta}} \le p_a + e_\delta^-$, so the retained critical-volume and backward-end estimates give, at fixed auxiliary band,

$$\begin{aligned} m^2 \int_J U_{H_{a,\delta}} &\le C(a^\alpha + \delta^\gamma)(|J| + B_*^{-1}) \\ &\quad + C_{B_*}(1 + B_*|J|)\{m^2 a^{1/2}\theta_{B_*}^m + m^3 \min(\theta_{B_*}^m, \sqrt{\delta})\}. \end{aligned} \tag{5.1}$$

This is an estimate on the original exact event, not on the enlarged comparison domain.

Theorem 5.1 (Untruncated recurrent-source height)**.** *For fixed $r, D, \beta, a, \delta > 0$ in their stated ranges,*

$$\limsup_m \sup_{R,n,k} m^2 \|R^\varepsilon_{H^{\mathrm{adm}}_{a,\delta}}\|_\infty \le C\frac{D}{\beta}(a^\alpha + \delta^\gamma). \tag{5.2}$$

The constant is independent of ε and of the previously fixed comparison radius. At finite count the right side before taking limits is $D/(\beta r)$ times the right side of (5.1) *with $|J| = 2r$.*

The same conclusions separately hold for $H = \Pi \cap \{z < a\}$ with only a^α, and for $H = \Pi \cap \{\ell < \delta\}$ with only δ^γ. In particular, for fixed D, β, the original choice $a = h^{1/8}$, $\delta = h^{1/4}$ gives

$$\mathcal{H}_M(R^\varepsilon_{H^{\mathrm{adm}}_{a,\delta}}) \le C(D/\beta)h^{1/128}. \tag{5.3}$$

The radius may be a prescribed function of h, fixed before each collision limit.

Proof. Apply (4.5) and then (5.1) on $[t-r, t+r]$. The estimate is uniform in the interval location. Fix all displayed parameters and B_*, take the collision limsup, and only then enlarge B_*. The floor B_*^{-1}/r disappears in this order, and both spectral remainders vanish. The leading interval length cancels the r^{-1} factor. The two single restrictions use the corresponding single input. Finally $a^\alpha = h^{129/1024}$ and $\delta^\gamma = h^{1/128}$, proving (5.3). □

This radial conclusion contains every annulus below a at finite count, including annuli whose locations depend on m. It is not a sum of fixed-annulus limsups. It does not assert that every such annulus is admitted: comparison survival, physical compression and same-label occupation remain explicit tests.

Corollary 5.2 (No additional radial-return test in the inner core)**.** *Let $0 < r \le a^2/(8\Lambda_\circ)$. Targets with $z(t) < a/2$ which pass the continuation tests for $H = \Pi$ also pass them for $H = \Pi \cap \{z < a\}$. Their complete weighted source has ordered height at most $C(D/\beta)a^{129/128}$, with no lower gradient cutoff.*

Proof. On any available backward interval z decreases. On an available forward interval, $z(t+s)^2 \le a^2/4 + 2\Lambda_\circ r \le a^2/2$. Thus every sample returning to Π also lies in $z < a$. Both occupation fractions coincide, and the distortion test is unchanged. Apply the single-critical-restriction part of Theorem 5.1. □

For the packet application put $H_\delta = \Pi \cap \{\ell < \delta\}$. At a target roof select original segments with $z_\sigma^- \ge \rho$, $0 < \ell_\sigma < \delta$, two nongrazing endpoints, and admitted target in H_δ for the tests of (4.3). Define

$$\mathcal{M}_{\rho,\delta}^{\varepsilon,\mathrm{adm}}(t) = \int_{\text{this target family}} \min\{f_\sigma^\varepsilon(a_\sigma+), f_\sigma^\varepsilon(b_\sigma-)\}\, d\sigma.$$

The minimum is still on the two ends of the same original segment. The continuation anchors do not replace either endpoint.

Theorem 5.3 (Decay of the recurrent two-nongrazing packet)**.** *At each previously fixed positive entrance-gradient cutoff ρ,*

$$\mathcal{H}_M(\mathcal{M}_{\rho,\delta}^{\varepsilon,\mathrm{adm}}) \le C(D/\beta)\delta^{1/32} + C(1+\varepsilon^{-1})\delta. \tag{5.4}$$

Consequently this complete admitted packet has zero ordered zero-residence charge for fixed ε, D, β and a fixed comparison radius, without any lower residence cutoff. The statement also holds for target cutoff $z(t) \ge \rho$ when $\delta \le \rho^2/(8\Lambda_\circ)$.

If admission is tested for $H = \Pi$ instead of H_δ, the corresponding two-nongrazing packet obeys

$$\mathcal{H}_M(\mathcal{M}_{\rho,\delta}^{\varepsilon,\Pi\text{-adm}}) \le C(D/\beta)\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\} + C(1+\varepsilon^{-1})\delta. \tag{5.5}$$

This requires only returns to the same exact event, not returns to short exact-event segments.

Proof. The two-ended variation proof of retained module 226 applies to any Borel subfamily at the target roof. It gives

$$\mathcal{M}_{\rho,\delta}^{\varepsilon,\mathrm{adm}}(t) \le R_{H_\delta^{\mathrm{adm}}}^\varepsilon(t) + C(1+\varepsilon^{-1})\int_{t-\delta}^{t+\delta} v^\rho(u)\, du.$$

Indeed the absolute difference between a target value and its same-segment endpoint minimum is bounded by the whole derivative integral. The physical Borel parametrization counts that integral once. Theorem 5.1 estimates the first term; the retained positive-cutoff theorem estimates the second. This proves (5.4). For the target-gradient version, radial growth bounds the gradient on the entire short segment below by $\rho/2$, as in the retained packet corollary. For (5.5), use the same variation comparison with admission to Π, apply (4.4), and use the original fixed-width bounds for b^ε and p. □

The new assertion concerns the positive minimum packet, including segments with two nonzero traces. It is not the previous grazing-ended zero theorem. It can apply when a short exact-label segment is followed by arbitrarily many section-decision crossings, provided the same label and same selector occupy the required total roof fraction. Neither a bound on the number of crossings nor a pooling of their labels is used. The packet carried by targets failing these tests is not estimated by (5.4).

6. Localization of the original pointwise forcing

Take $H = \Pi$ in the occupation tests. Partition its nonadmitted targets into the following disjoint Borel classes. The first, N_T, has no available comparison interval of length r in either direction. The second, N_D, has at least one available interval but no available sign with $d_{e,r} \le D$. The third, N_O, has such a bounded-compression sign but no such sign with $o_{\Pi,e,r} \ge \beta$. Thus physical interruption, compression and insufficient same-label occupation are charged separately and only once. Define their original weighted source sum by

$$Z_{r,D,\beta}^\varepsilon(t) = \int_{F=t} W^\varepsilon(\mathbf{1}_{N_T} + \mathbf{1}_{N_D} + \mathbf{1}_{N_O})\, d\mu_t. \tag{6.1}$$

In particular $0 \le Z_{r,D,\beta}^\varepsilon \le b^\varepsilon$. No new source is placed on a comparison-domain boundary.

Theorem 6.1 (Prescribed-radius localization of the full source)**.** *At every finite count and almost every roof,*

$$b^{\varepsilon}(t) \le Z^{\varepsilon}_{r,D,\beta}(t) + \frac{D}{\beta r}\int_{t-r}^{t+r} \{b^{\varepsilon}(u) + \kappa_{\varepsilon}(r)p(u)\}\,du. \tag{6.2}$$

The same bound dominates the bidirectional forcing of module 228, because that forcing is bounded by the original target source. Consequently, with $\varepsilon = \varepsilon(B)$,

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathcal{H}_M(Z^{\varepsilon}_{r,D,\beta}) + C\frac{D}{\beta}\{\varepsilon^{1/16} + \varepsilon^{-1}\sqrt{r}\}. \tag{6.3}$$

The first and last terms are estimated. The middle term retains the complete weighted mass of the three failed geometric tests.

Proof. The exact partition of b^{ε} into admitted and nonadmitted parts, followed by Theorem 4.2, proves (6.2). It is a statement at the target roof, not an integral in the target variable. The retained bidirectional forcing obeys $0 \le J \le b^{\varepsilon}$ at finite count, so no minimum is moved inside the physical-sheet integral.

For fixed B, r, D, β the complete source average is at most $C\varepsilon^{1/16}$ in the normalized limit. The unweighted average has a common bound by the same theorem at auxiliary guard parameter one, where $W^1 = 1$. Apply these two estimates to (6.2), then insert the original positive scalar remainder. This proves (6.3), without any spectral estimate outside the exact event. □

Corollary 6.2 (A legal outer schedule and its exact obligation)**.** *Suppose $D(B)/\beta(B) \le C\varepsilon(B)^{-\tau}$, where $0 \le \tau < 1/16$, and set $r(B) = c\varepsilon(B)^N$ with $N > 2(1+\tau)$ and sufficiently small fixed $c > 0$. The original pointwise theorem is equivalent to*

$$\mathcal{H}_M(Z^{\varepsilon(B)}_{r(B),D(B),\beta(B)}) \longrightarrow 0 \quad (B \to \infty) \tag{6.4}$$

for every fixed M. The same schedule preserves the equivalence with vanishing of the retained bidirectional forcing.

Proof. The explicit errors have powers $\varepsilon^{1/16-\tau}$ and $\varepsilon^{N/2-1-\tau}$, both positive. This proves sufficiency from (6.3). Conversely the original pointwise law and the positive scalar remainder give $\mathcal{H}_M(b^{\varepsilon(B)}) \le C_M B^{-1/192}$. The domination $Z \le b^{\varepsilon}$ proves necessity. The retained bidirectional equivalence applies to this same prescribed radius, since $(1+\varepsilon^{-1})r \to 0$. One simple schedule is $D = 2$, $\beta = 1/4$, $r = c\varepsilon^4$. All these parameters, including each sampling radius, are fixed before their collision limit; auxiliary bands are enlarged afterwards and B is last. □

For an original exceptional selector $H_{a,\delta}$ the identical three-way partition is made using returns to $H_{a,\delta}$ itself. Theorem 5.1 then localizes its essential-height charge to failed continuation, compression or selected-event occupation, with the explicit admitted error $C(D/\beta)(a^{\alpha} + \delta^{\gamma})$. Together with Theorem 5.3, this controls the recurrent part of the two-nongrazing packet and the complete recurrent radial core, not just a finite annulus head.

No theorem here proves (6.4) for the full Lorentz source. A short two-nongrazing pulse may have too little same-label occupation, and a critical target may fail physical continuation or the compression test. These possibilities remain positive original-source terms. In the elementary comparison model $\widehat{A} \equiv 1$ and $\Pi = (0, \delta)$, occupation over a fixed one-sided radius is at most δ/r; the theorem does not erase that pulse. This records the boundary between a proved persistence mechanism and the still required exhaustion of all spike mechanisms. The original scalar, arithmetic and unrestricted downstream claims are therefore not promoted.

Appendix A. Proof dependencies and comparison scope

The new occupation argument uses the general, pre-section Hessian inequality of module 174, the contact inverse and original product-guard motion of module 177, the stationary cubic formula of module 180, and the optical sensitivity formula of module 182. Their proofs require only a fixed positive endpoint cosine on the comparison collar, not section membership. Module 196 supplies the rational-roof Borel parametrization; its construction is applied to the regular physical domain before restricting the sampled event. The new comparison never continues through a physical singularity.

The source estimates used after sampling are exclusively those of the original exact event: the backward-end estimate of module 211, the critical-volume layer cake of module 187 in the explicit form restated above, and the original bad-source average and positive scalar remainder. The packet comparison is module 226 on the same Borel target subfamily. The joint forcing is the post-projection minimum of module 228, not a sheetwise minimum. The two objects are never identified.

The full prior article is reproduced in the supplement under its own reference namespace. Every inherited theorem source is preserved. Preservation does not independently certify the four-record spectral input or the common-kernel construction. The present proofs are the comparison-domain construction, exact-label occupation transfer, its positive packet and radial-source consequences, and the three-way pointwise localization. The full failed-test source is not proved small.

Part 9. Two-ended packets and intrinsic radial residence

Retained revision-95 abstract. We continue the upper pointwise local limit problem for the exact record of actual returns in a triangular Lorentz gas. On short physical segments, the original weighted source is approximated in essential height by the smaller of its two endpoint traces. At every fixed positive entrance gradient this eliminates the zero-residence contribution of all segments with a grazing endpoint. A two-sided anchor on the intrinsic radial scale also estimates complete critical annuli without requiring a prescribed shell crossing. The residual thin segments cannot originate at a critical point. Finally we retain signed projection in both roof directions; the minimum of the two resulting forcings is no larger than the previous backward forcing and gives an equivalent pointwise criterion. These results preserve the original source and every exact label. The remaining two-nongrazing bridge and zero-gradient concentrations are not asserted to vanish; the unrestricted pointwise theorem remains the objective.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k, m, t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R, \vartheta)$ and $\mathsf{P}_j(R, \vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Two physical ends rather than one positive inlet.** The target remains (2.5). The new proof addresses the short-segment and critical-core sources directly. It does not interpolate a finite norm into an essential supremum or normalize the upper excess. Theorem 4.1 proves a complete, finite-count comparison with a two-ended packet. One of its consequences is a vanishing zero-residence height for every grazing-ended subfamily at fixed positive entrance gradient. No positive lower residence cutoff is inserted.

Theorem 5.1 treats targets which remain inside a previously prescribed critical shell but have enough residence on the scale of their own gradient. It chooses an anchor on either physical side and sums the physical energies with multiplicity one. The leading bound in one annulus is $Cq^{129/128} + C(1+\varepsilon^{-1})q^{257/128}$. Its finite-count remainder is displayed, so the difference between a finite annulus sum and a zero-gradient limit is explicit.

Theorem 6.1 retains entrance and terminal signs through projection in both directions. A critical entrance which was positive in the backward balance is negative in the forward balance. Their minimum is no larger than either individual forcing; it is not a sum of two positive estimates. Its vanishing is equivalent to the same original scalar law (Theorem 6.2); verification of that condition still requires control of the remaining concentrations.

3. Physical source, signed measures, and retained estimates

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar{\nu} = \mathbf{1}_{\Pi}\nu_R^*$ for the original exact-event source. The retained Borel parametrization is a bijection between its regular part and maximal segments (σ, u), $a_\sigma < u < b_\sigma$. Its transverse measure $d\sigma$ is the disjoint angular measure and its density is

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad d\bar{\nu} = A_\sigma(u)\, du\, d\sigma.$$

Here $F = u$, $z = |\nabla_q F|$, $q = (\alpha_0, \alpha_m)$, $\omega = |\partial_{p_0}\alpha_m|^{-1}$, and $\chi = |\det D_q(F, \arg \nabla_q F)|$. All fibers and signed measures below use these same kernels. Let $z_\sigma^- = z(\Psi_\sigma(a_\sigma+))$ and $\ell_\sigma = b_\sigma - a_\sigma$.

Input statement 3.1 (Complete segment geometry and signed balance)**.** Put $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. On every regular segment,

$$2\lambda_\circ(v-u) \le z(v)^2 - z(u)^2 \le 2\Lambda_\circ(v-u), \tag{3.1}$$

$$|A_\sigma'| \le C A_\sigma I_m / z, \qquad |(W^\varepsilon)'| \le C\varepsilon^{-1}/z, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i. \tag{3.2}$$

The complete density $f_\sigma^\varepsilon = A_\sigma W^\varepsilon$ has one-sided traces and is absolutely continuous on the closed segment. The sum of its endpoint traces and absolute derivative integrals is finite at each fixed count. A grazing endpoint has zero trace; a critical backward endpoint need not. The geometric domain is a finite definable family at each fixed count. Its component count is used for existence of measures only, not for normalized estimates.

For every Borel prefix endpoint $c_\sigma \in [a_\sigma, b_\sigma]$ put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma.$$

The original union $z < \zeta$ or $\ell < \delta$ is exactly such a prefix, and $c_\sigma = b_\sigma$ gives b^ε. Define

$$\mathsf{I}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma,$$
$$\mathsf{D}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma,$$
$$\mathsf{G}_c^{\varepsilon,\pm}(H) = \int\int_{(a_\sigma,c_\sigma)\cap H} [\pm(f_\sigma^\varepsilon)'(u)]_+\, du\, d\sigma.$$

Their complete finite-count signed identity is

$$DR_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{3.3}$$

Every artificial prefix cut is negative. All true entrance types and one-sided incident physical components are retained.

This is the endpoint-field, optical, Borel-segment and signed-prefix chain, culminating in retained modules 220–222. In particular, A_σ includes both angular factors. No reversal of the billiard record or change of the section is used at a backward end.

Input statement 3.2 (Backward-end and critical volume estimates)**.** Set $\alpha = 129/128$ and $\gamma = 1/32$. Let

$$e_s^-(t) = \int \mathbf{1}_{\{t-a_\sigma \le s\}}\, d\mu_t, \qquad u_q(t)\, dt = F_\#(I_m \mathbf{1}_{\{z<q\}}\bar\nu), \qquad p_q(t)\, dt = F_\#(\mathbf{1}_{\{z<q\}}\bar\nu).$$

For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\theta_{B_*} < 1$ such that, at finite count and small $s, q > 0$,

$$m^2 \int_J e_s^- \le Cs^\gamma(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\theta_{B_*}^m, \sqrt{s}\}, \tag{3.4}$$
$$m^2 \int_J u_q \le Cq^\alpha(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 q^{1/2}\theta_{B_*}^m. \tag{3.5}$$

The constants are uniform in radius, exact labels, marks where present, and interval location. Since $I_m \ge 1$, the latter also bounds p_q. Consequently, at every fixed $s, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_s^- \le CLs^\gamma.$$

For the actual positive source $v^\rho(t)\, dt = F_\#(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar\nu)$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\rho \le CL \quad (\rho, L > 0 \text{ fixed}). \tag{3.6}$$

The leading constant is independent of the fixed positive cutoff; its finite-count remainder is not.

Equation (3.4) is the same-event backward-end estimate of module 211, including critical entrances. For (3.5), multiply the actual fractional-source bound of module 187, at exponent 1/4, by $q^{1/4}$: pointwise, $I_m\mathbf{1}_{z<q} \le q^{1/4} I_m z^{-1/4}\mathbf{1}_{z<q}$. The leading power is q^α and the remainder power is $q^{1/2}$. The audited cutoff estimate is retained in modules 196 and 222. These are volume-source theorems, not assumptions on entrance flux.

Input statement 3.3 (Original scalar reconstruction)**.** For every fixed $\varepsilon, r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\, du. \tag{3.7}$$

For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards. The reconstruction band B is outermost.

We use $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} h(t)$ for a nonnegative source family. Unqualified $\|\cdot\|_\infty$ denotes essential supremum on the full roof line. Identities hold for common kernels at fixed m, R, n, k; the uniform order is $\sup_R \operatorname{ess\,sup}_t$. Countably many chosen radii share versions. No simultaneous Lebesgue-point assertion for every uncountable selector family is required.

Fix $q_* > 0$ within the small-gradient range of Input 3.2. Every radial scale below is chosen so that its stated upper gradient threshold is less than q_*.

4. Two-ended packets on the actual short segments

An inlet trace alone can overestimate a short pulse. Its other physical end gives a second trace on the same incident component. We retain both before summing the original source. In this section the endpoints are physical endpoints a_σ, b_σ, not artificial prefix cuts. Put $E_\varepsilon = 1 + \varepsilon^{-1}$ and

$$F_\sigma^- = f_\sigma^\varepsilon(a_\sigma+), \qquad F_\sigma^+ = f_\sigma^\varepsilon(b_\sigma-), \qquad M_\sigma = \min\{F_\sigma^-, F_\sigma^+\}.$$

For fixed $\rho, \delta > 0$ let

$$\mathcal{D}_{\rho,\delta}(t) = \{\sigma : a_\sigma < t < b_\sigma,\ z_\sigma^- \ge \rho,\ 0 < \ell_\sigma \le \delta\}.$$

Integrating over this same family defines four nonnegative functions:

$$\begin{aligned} Q^\varepsilon_{\rho,\delta}(t) &= \int_{\mathcal{D}_{\rho,\delta}(t)} f_\sigma^\varepsilon(t)\, d\sigma, \\ I^\varepsilon_{\rho,\delta}(t) = \int_{\mathcal{D}_{\rho,\delta}(t)} F_\sigma^-\, d\sigma, &\quad O^\varepsilon_{\rho,\delta}(t) = \int_{\mathcal{D}_{\rho,\delta}(t)} F_\sigma^+\, d\sigma, \quad M^\varepsilon_{\rho,\delta}(t) = \int_{\mathcal{D}_{\rho,\delta}(t)} M_\sigma\, d\sigma. \end{aligned} \tag{4.1}$$

In particular Q is a restriction of the original weighted collision source. The minimum in M is taken on each physical segment, not between two projected totals, and no sign cancellation is assumed.

Theorem 4.1 (A complete two-ended replacement)**.** *At every finite count and almost every roof,*

$$\begin{aligned} |Q^\varepsilon_{\rho,\delta} - M^\varepsilon_{\rho,\delta}|(t) &\le V^\varepsilon_{\rho,\delta}(t), \\ |I^\varepsilon_{\rho,\delta} - M^\varepsilon_{\rho,\delta}|(t) + |O^\varepsilon_{\rho,\delta} - M^\varepsilon_{\rho,\delta}|(t) &\le V^\varepsilon_{\rho,\delta}(t), \\ V^\varepsilon_{\rho,\delta}(t) &:= C E_\varepsilon \int_{t-\delta}^{t+\delta} v^\rho(u)\, du. \end{aligned} \tag{4.2}$$

Consequently every error in (4.2) *has ordered height at most $CE_\varepsilon\delta$. At each fixed positive ρ the four zero-residence charges agree, as extended nonnegative numbers, when δ decreases after the collision limit. The common charge is carried entirely by segments having two nongrazing physical endpoints.*

Proof. On a selected segment radial monotonicity gives $z(u) \ge \rho$ throughout. The product rule and the retained full-density estimates give

$$|(f_\sigma^\varepsilon)'(u)| \le C E_\varepsilon A_\sigma(u) I_m(u)/z(u).$$

Absolute continuity, including the one-sided physical traces, implies

$$|f_\sigma^\varepsilon(t) - M_\sigma| \le \int_{a_\sigma}^{b_\sigma} |(f_\sigma^\varepsilon)'|, \qquad |F_\sigma^- - F_\sigma^+| \le \int_{a_\sigma}^{b_\sigma} |(f_\sigma^\varepsilon)'|.$$

Moreover $(F_\sigma^- - M_\sigma) + (F_\sigma^+ - M_\sigma) = |F_\sigma^- - F_\sigma^+|$. All quantities on the latter left side are nonnegative. Every selected segment lies in $(t-\delta, t+\delta)$, and its intermediate physical states are counted once by the original Borel parametrization. Summing the variation integrals is therefore bounded by the actual v^ρ source on that interval. This proves the finite-count assertions. Apply (3.6) at the previously fixed width 2δ. The common leading constant is independent of the positive ρ, but its finite-count remainder is not being removed at $\rho = 0$.

Each packet increases with δ, so its ordered zero-residence limit exists. The pairwise differences have height tending to zero; the triangle inequality in both directions proves equality, also for infinite limits. If either endpoint is grazing, its weighted trace vanishes, hence $M_\sigma = 0$. Restricting the same proof to that Borel family gives height at most $CE_\varepsilon\delta$ for its complete original source and for either endpoint packet. Thus that family's zero-residence charge is zero. No positive lower residence cutoff was imposed in the finite-count formula. □

Corollary 4.2 (A target-gradient version)**.** *Replace $z_\sigma^- \ge \rho$ in the selected family by $z_\sigma(t) \ge \rho$. If $0 < \delta \le \rho^2/(8\Lambda_\circ)$, all conclusions of* (4.2) *hold with $v^{\rho/2}$ on the right. In particular the original target-gradient-restricted short source with a grazing endpoint has zero ordered zero-residence charge at each fixed positive ρ.*

Proof. For every earlier point of that segment, $z(u)^2 \ge z(t)^2 - 2\Lambda_\circ\delta \ge 3\rho^2/4$. At later points z only increases. The complete interval is therefore above $\rho/2$, and the proof of the theorem applies unchanged. □

The result eliminates all grazing-ended contributions to the noncritical zero-residence inlet, including segments with a nongrazing entrance and a grazing exit. It does not estimate the common positive packet of two nongrazing ends. Nor does equality of the four charges permit an exchange of the zero-gradient and collision limits.

5. Critical targets on their intrinsic residence scale

A history can remain in the target core of a prescribed shell and still have enough residence to provide its own anchor. We now use an anchor of roof length proportional to the square of the target gradient, on whichever physical side is longer. This does not require a crossing of the shell prescribed in the preceding revision.

We first record the shell input with the finite-count remainder visible. For $0 < b < D < q_*$ set

$$v^{b,D}(u)\,du = F_\#(I_m z^{-1}\mathbf{1}_{\{b\le z<D\}}\bar\nu).$$

The retained critical-volume estimate implies

$$m^2 \int_J v^{b,D} \le CD^{\alpha-1}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 b^{-1/2}\theta_{B_*}^m. \tag{5.1}$$

Indeed multiply the positive identity for $z^{-1}\mathbf{1}_{b\le z<D}$ by I_m and bound it by $D^{-1}I_m\mathbf{1}_{z<D} + \int_b^D s^{-2}I_m\mathbf{1}_{z<s}\,ds$. The two integrals are bounded by $CD^{\alpha-1}$ and $2b^{-1/2}$, respectively. This is precisely the finite positive layer cake of retained module 224, not an untruncated first-inverse-gradient estimate.

Put

$$\eta_0 = \frac{1}{16\Lambda_\circ}, \qquad d_q = \frac{q^2}{256\Lambda_\circ}, \qquad J_{t,q} = (t - 4d_q, t + 4d_q).$$

For any original prefix c, let $H^{\varepsilon}_{c,q}$ be its actual target source on

$$q \le z(t) < 2q, \qquad \ell_\sigma \ge \eta_0 q^2.$$

Only targets are selected: subsequent anchor values may lie outside the selected prefix, but remain in its same physical segment.

Theorem 5.1 (Two-sided intrinsic anchor at a critical target)**.** *For $3q < q_*$, every finite count and almost every target roof,*

$$H^{\varepsilon}_{c,q}(t) \le d_q^{-1} \int_{J_{t,q}} p_{3q}(u)\,du + CE_\varepsilon \int_{J_{t,q}} v^{q/2,3q}(u)\,du. \tag{5.2}$$

In particular, at each previously fixed $q > 0$,

$$\limsup_m \sup_{R,n,k,c} m^2 \|H^{\varepsilon}_{c,q}\|_\infty \le Cq^{129/128} + CE_\varepsilon q^{257/128}. \tag{5.3}$$

There is no incidence cutoff and no required shell crossing. One fixed-band finite-count upper bound for its normalized height is

$$C\{q^{\alpha-2} + E_\varepsilon q^{\alpha-1}\}(q^2 + B_*^{-1}) + C_{B_*}(1 + B_* q^2) m^2 \theta^m_{B_*}(q^{-3/2} + E_\varepsilon q^{-1/2}). \tag{5.4}$$

Proof. The selected segment has length at least $16d_q$. One side of the target therefore has length at least $8d_q$. Choose the interval $(t-3d_q, t-2d_q)$ on the backward side if that side is long enough; otherwise choose $(t+2d_q, t+3d_q)$. This is a Borel choice on the retained segment space. Each joining interval lies in $J_{t,q}$ and in the same physical segment. Radial growth gives throughout

$$z(u)^2 \ge q^2 - 8\Lambda_\circ d_q > q^2/4, \qquad z(u)^2 \le 4q^2 + 8\Lambda_\circ d_q < 9q^2.$$

Average the elementary inequality $f^{\varepsilon}_\sigma(t) \le f^{\varepsilon}_\sigma(v) + \int_{[v,t]} |(f^{\varepsilon}_\sigma)'|$ over the anchor. Here $[v,t]$ means the unoriented joining interval. Use $f^\varepsilon \le A$ on the first term and $|(f^\varepsilon)'| \le CE_\varepsilon A I_m/z$ on the second. At each intermediate roof, all joining intervals form a subfamily of the actual physical fiber. The Borel bijection gives multiplicity one, so their complete sum is bounded by the two original source integrals in (5.2). The anchor is not a new normalized source and does not change the target event.

Apply (3.5) and (5.1), using $|J_{t,q}| = 8d_q$. Since d_q is a fixed physical multiple of q^2, this gives (5.4). Fix q, ε, B_* before taking the collision limsup; only afterwards enlarge B_*. The remaining terms are $Cq^\alpha + CE_\varepsilon q^{\alpha+1}$. This proves (5.3). □

There is a complementary geometric distinction which is useful at residence zero. If

$$\ell_\sigma < \eta_0 z(t)^2, \tag{5.5}$$

then

$$(z^-_\sigma)^2 \ge (1 - 2\Lambda_\circ \eta_0) z(t)^2 = \tfrac{7}{8} z(t)^2. \tag{5.6}$$

Such a segment cannot have a critical backward endpoint. Conversely, every segment born at a critical endpoint satisfies $\ell_\sigma > t - a_\sigma \ge z(t)^2/(2\Lambda_\circ)$, and hence belongs to the relatively long class rather than (5.5).

Proposition 5.2 (All thin grazing-ended targets in one radial annulus)**.** *Let $G^{\varepsilon}_{c,q}$ be the original target source satisfying $q \le z(t) < 2q$,* (5.5)*, and having at least one grazing physical endpoint. Then*

$$\limsup_m \sup_{R,n,k,c} m^2 \|G^{\varepsilon}_{c,q}\|_\infty \le CE_\varepsilon q^{257/128}. \tag{5.7}$$

Proof. A selected segment has length at most $4\eta_0 q^2$ and throughout it $q/2 < z < 3q$, by the same radial inequalities. At least one physical endpoint has zero weighted trace. The fundamental theorem therefore bounds its target value by the absolute derivative integral over that segment. Sum through the Borel bijection and apply (5.1) on $(t - 4\eta_0 q^2, t + 4\eta_0 q^2)$. After the fixed-$q$ collision limit and auxiliary-band enlargement the bound is $CE_\varepsilon q^2 q^{\alpha-1}$, as asserted. □

For fixed $0 < a < q$ the annuli $(2^{-j-1}q, 2^{-j}q]$ meeting $[a, q)$ are a finite family. Applying the last two results and summing their geometric leading bounds proves

$$\mathcal{H}_M\Big(R_c^\varepsilon \mathbf{1}_{\{a\le z<q\}} \mathbf{1}_{\{\ell\ge\eta_0 z^2\}\cup\{\text{a grazing endpoint}\}}\Big) \le Cq^\alpha + CE_\varepsilon q^{\alpha+1}. \tag{5.8}$$

The notation means restriction on the physical target before projection. The constant is independent of the previously fixed positive a. The physical annulus sum remains finite before the collision limsup. Formula (5.4) explains why sending a to zero inside that limit is not licensed: its fixed-band floor and its finite-count remainder are not summable there. The unresolved part can still concentrate in those disappearing annuli, or on thin segments with two nongrazing endpoints.

6. Bidirectional signed forcing at the original roof

A positive entrance is a creation term in the backward comparison but a loss term in the forward comparison. We retain both comparisons, including all opposing physical and artificial cuts, and take their minimum only after projection. This does not reverse or reindex the billiard record.

Write $R = R_c^\varepsilon$ and, for a fixed $\rho > 0$, put

$$\begin{aligned} \mathsf{H}_\rho^+ &= \mathsf{G}_c^{\varepsilon,+,\ge\rho}, & \mathsf{H}_\rho^- &= \mathsf{G}_c^{\varepsilon,-,\ge\rho}, \\ \mathsf{K}_\rho^- &= DR - \mathsf{H}_\rho^+, & \mathsf{K}_\rho^+ &= -DR - \mathsf{H}_\rho^-. \end{aligned} \tag{6.1}$$

Superscripts on K indicate the side of the target; those on H indicate the sign of the density derivative. Thus

$$\mathsf{K}_\rho^+ = \mathsf{D}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,-,<\rho} - \mathsf{I}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,+}.$$

In this forward measure a critical entrance is negative. An upper prefix cut is positive and is not discarded.

Let $A_r^- R = r^{-1}\int_{t-r}^t R$ and $A_r^+ R = r^{-1}\int_t^{t+r} R$. For finite signed measures define

$$\mathcal{T}_r^-\mu(t) = \int_{(t-r,t]} \frac{u-t+r}{r}\, d\mu(u), \qquad \mathcal{T}_r^+\mu(t) = \int_{[t,t+r)} \frac{t+r-u}{r}\, d\mu(u).$$

At Lebesgue roofs of R and away from atoms, set

$$\begin{aligned} S_{r,\rho}^- &= [\mathcal{T}_r^- \mathsf{K}_\rho^-]_+, & S_{r,\rho}^+ &= [\mathcal{T}_r^+ \mathsf{K}_\rho^+]_+, \\ J_{c;r,\rho}^\varepsilon &= \min\{S_{r,\rho}^-, S_{r,\rho}^+\}. \end{aligned} \tag{6.2}$$

Define $S^{-,\sharp}$ and $S^{+,\sharp}$ by taking the essential supremum of $[\mathsf{K}^-((s,t])]_+$ over $t - r < s < t$ and $[\mathsf{K}^+([t,s))]_+$ over $t < s < t + r$, respectively; put $J^\sharp = \min\{S^{-,\sharp}, S^{+,\sharp}\}$. Finite-count BV representatives give Borel versions. No common Lebesgue-point set for an uncountable collection of parameters is asserted.

Theorem 6.1 (A smaller two-sided pointwise budget)**.** *For every previously fixed $r, \rho > 0$, at finite count and almost every original roof,*

$$0 \le J^{\varepsilon}_{c;r,\rho} \le J^{\varepsilon,\sharp}_{c;r,\rho} \le R^{\varepsilon}_{c}, \tag{6.3}$$

$$R^{\varepsilon}_{c}(t) \le \max\{A^{-}_{r} R^{\varepsilon}_{c}(t), A^{+}_{r} R^{\varepsilon}_{c}(t)\} + J^{\varepsilon}_{c;r,\rho}(t) + CE_{\varepsilon} \int_{t-r}^{t+r} v^{\rho}(u)\, du. \tag{6.4}$$

The joint forcing is bounded by the backward signed forcing of module 225. If its sharp version plus the last error is at most $R^{\varepsilon}_{c}(t) - \lambda$, then the source is at least λ almost everywhere on at least one of the two full prescribed adjacent intervals. All sources, losses and positive parts are formed at finite count, before any uniform height is taken.

Proof. Stieltjes integration of the two identities in (6.1) gives

$$R - A^{-}_{r} R = \mathcal{T}^{-}_{r} \mathsf{K}^{-}_{\rho} + \mathcal{T}^{-}_{r} \mathsf{H}^{+}_{\rho},$$
$$R - A^{+}_{r} R = \mathcal{T}^{+}_{r} \mathsf{K}^{+}_{\rho} + \mathcal{T}^{+}_{r} \mathsf{H}^{-}_{\rho}.$$

The signs can also be checked by integrating the fundamental theorem between the target and the earlier or later roof. Each last term is nonnegative. Each signed interval increment, after subtraction of its H term, is at most $R(t)$ because the other endpoint source is nonnegative. This proves the domination for both sharp forcings. Their ramped expressions are averages of the corresponding interval increments, so $S^{\pm} \le S^{\pm,\sharp}$. Taking minima proves (6.3).

The derivative bound, with its physical angular density retained, gives

$$\mathsf{H}^{+}_{\rho}((t-r,t]) + \mathsf{H}^{-}_{\rho}([t,t+r)) \le CE_{\varepsilon} \int_{t-r}^{t+r} v^{\rho}(u)\, du.$$

This follows from multiplicity-one coarea summation, not from a component count. Either ramped H term is at most this common error. Take the smaller of the two upper comparisons and use $\min\{x+s, y+t\} \le \max\{x,y\} + \min\{s,t\}$. This proves (6.4). The two interval identities likewise give lower bounds for $R(s)$ on the backward and forward interval. Choose the side attaining the smaller sharp forcing to obtain the asserted persistence statement.

All physical cuts and all artificial prefix cuts are already present in DR. The positive parts in (6.2) are therefore taken after their signed projection, not sheet by sheet. In particular a critical entrance trace alone cannot be positive forcing in both directions. Forward terminations and negative critical growth can still make the forward forcing large. □

The minimum of two projected forcings is not the integral of the minimum of two sheetwise forcings. No such interchange is used. The packet M of Section 4 is a separate positive majorant with its own proved error bound; it is not identified with J or with a Jordan part of DR.

6.1. **The original scalar endpoint and legal rates.** For the full bad source take $c_{\sigma} = b_{\sigma}$. Define

$$\mathfrak{J}_{\varepsilon,M}(r) = \lim_{\rho \downarrow 0} \mathcal{H}_{M}(J^{\varepsilon}_{r,\rho}), \qquad \mathfrak{J}^{\sharp}_{\varepsilon,M}(r) = \lim_{\rho \downarrow 0} \mathcal{H}_{M}(J^{\varepsilon,\sharp}_{r,\rho}).$$

Both limits are monotone extended nonnegative limits: lowering ρ increases the removed positive measures $\mathsf{H}^{\pm}_{\rho}$.

Theorem 6.2 (Bidirectional criterion for the unchanged pointwise law)**.** *For all sufficiently large fixed B and every fixed $r > 0$,*

$$\mathcal{E}_{M} \le C_{M} B^{-1/192} + \mathfrak{J}_{\varepsilon(B),M}(r) + CE_{\varepsilon(B)} r. \tag{6.5}$$

For any prescribed $r(B) > 0$ with $E_{\varepsilon(B)} r(B) \to 0$, the original pointwise assertion is equivalent to either $\mathfrak{J}_{\varepsilon(B),M}(r(B)) \to 0$ or $\mathfrak{J}^{\sharp}_{\varepsilon(B),M}(r(B)) \to 0$, for every fixed M.

Proof. For a nonnegative source both one-sided averages are at most $2A_r b^{\varepsilon}$. Fix B, r, ρ and the auxiliary bands; apply the original fixed-width bad-source estimate, the cutoff volume estimate on length $2r$, and the scalar positive remainder. Only after the collision limsup and allowed band enlargement do we let ρ decrease. Since $\varepsilon(B)^{1/16} = O(B^{-1/192})$, the result is (6.5). This proves sufficiency of either criterion. Conversely the original pointwise law and its positive remainder give $\mathcal{H}_M(b^{\varepsilon(B)}) \le C_M B^{-1/192}$. Domination (6.3) proves necessity for every prescribed radius, without exchanging limits. □

The explicit new errors admit a common legal schedule. With $r(B) = c\varepsilon(B)^N$, $q(B) = \sqrt{r(B)}$ and $N > 1$,

$$E_{\varepsilon(B)} r(B) \to 0, \qquad q(B)^{\alpha} + E_{\varepsilon(B)} q(B)^{\alpha+1} \to 0.$$

The same schedule pays the grazing-ended short-packet error when $\delta = r(B)$ at each previously fixed positive entrance gradient. These rates do not prove that the remaining two-nongrazing packet charge or the zero-gradient target concentration vanishes. Nor do they justify an infinite radial sum inside a collision limsup. The theorem is an improved pointwise-controlling criterion, not its verification. The full scalar law and unrestricted point-conditioned consequences remain the governing objectives, with their original arithmetic restrictions intact.

7. The remaining upper-source estimate

The short noncritical source is now represented, up to a vanishing height error, by a packet requiring two nongrazing physical traces. At fixed positive entrance gradient its grazing-ended part has zero ordered residence charge. Critical annuli with sufficient relative residence have an additional pointwise estimate even when they do not cross the previously prescribed shell. The complementary relatively thin segments have strictly positive entrance gradient.

The pointwise law is not concluded from these statements. A family of positive annular estimates with a common leading constant does not authorize removal of a lower gradient cutoff inside a collision limsup. Nor does the minimum of two unknown signed forcings have to vanish. The complete two-nongrazing short packet and a possible concentration in disappearing critical annuli remain positive source-level obligations. All finite-norm, one-sided lower and conditioning theorems retain their historical scope; unrestricted point-conditioned laws and arithmetic-zero normalization are not promoted here. The full preceding article and every inherited proof remain in the technical archive.

Appendix A. Proof dependencies and the physical source

The original problem and the three input statements preceding the new proofs fix every source used in this article. The collision source is $\mathbf{1}_{\Pi}\nu_R^*$ and its angular density is $\omega/(4\pi c_* \chi)$. No independent boundary spectral theorem is assumed.

The two-ended packet theorem uses the Borel segment disintegration (module 196), the full angular-density derivative (modules 180–185), the original guard derivative (177), the one-sided trace theorem (220), and the positive-cutoff volume estimate audited in 196 and 222. Its endpoint minimum is new; the historical inlet-volume and outlet-volume comparisons remain separately credited to 198 and 223.

The intrinsic anchor uses the same disintegration and radial growth, and the critical-volume estimate from 187 in the explicit form (3.5). Its shell layer cake is restated and proved in (5.1); that calculation is inherited from 224. The target-adapted two-sided anchor and relatively thin classification are the new use of these inputs.

The bidirectional comparison uses the fixed-count signed-prefix identity of 221 and the projected Stieltjes calculation of 225. The forward calculation is written in full and includes its positive terminal and prefix cuts. Standard BV terminology follows [13]. The scalar conclusion uses only

the original fixed-width source bound and positive scalar remainder stated in Input 3.3. No path-remainder theorem enters the new upper estimate.

The main article directly loads only modules 226–228. The complete archive retains modules 1–225 and the full preceding article, with historical reference namespaces. Preservation is not an independent reconstruction of the inherited four-record spectral or coarea proofs.

Part 10. Radial entrance packets and signed projected forcing

Retained revision-94 abstract. We study the upper pointwise local limit problem for the exact collision record of actual returns in a triangular Lorentz gas. The complete noncritical entrance source has an essential-height estimate at every fixed positive residence cutoff, with a leading constant independent of both cutoffs. Its zero-residence contribution is identified with the original weighted short-segment source. We also estimate all critical histories which traverse a prescribed radial shell before the target roof. Their entrance and growth are retained as a signed packet; its actual target height is bounded without a residence or incidence cutoff. Finally, projecting the complete signed physical balance before taking positive parts gives a forcing term smaller than the separate positive entrance and growth budgets. Its vanishing is equivalent to the original pointwise assertion, in the prescribed order of limits. The quantitative entrance and radial-transit estimates do not establish vanishing of the remaining critical-target and zero-residence concentration. The original pointwise theorem and all exact physical data remain unchanged.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on

$[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_\#(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Entrance geometry and signed upper forcing.** The positive scalar remainder makes the upper sign the outstanding part of (2.5). The previous signed prefix balance identified two sufficient positive mechanisms, entrance traces and critical positive growth. Those positive pieces may overestimate the projected source, since physical and prefix losses can cancel them. The present argument retains that cancellation while estimating two complete geometric families in essential height.

Theorem 4.1 controls every noncritical inlet at each fixed positive residence cutoff by $C(r^{1/32}+r)$, uniformly in that cutoff and the original guard. Proposition 4.2 identifies the precise uncut remainder as an original collision-source restriction. Theorem 5.2 controls the actual target source of every history crossing the shell $\sqrt{r} \le z \le 2\sqrt{r}$ within the backward window by

$$Cr^{129/256} + C(1+\varepsilon^{-1})r^{257/256}.$$

This includes genuine critical directional births; zero area at a critical point is never used to erase its trace.

Theorems 6.1 and 6.3 then give a smaller, signed pointwise forcing and an equivalent upper criterion. These are not finite-norm embeddings. The forcing remains inside the essential supremum and collision limsup before its gradient cutoff is removed. Its zero value is not proved by the two partial height estimates.

3. Physical source, signed measures, and retained estimates

Fix m, R, n, k until a uniform norm is explicitly taken. Write $\bar{\nu} = \mathbf{1}_{\Pi}\nu_R^*$ for the original exact-event source. The retained Borel parametrization is a bijection between its regular part and maximal segments (σ, u), $a_\sigma < u < b_\sigma$. Its transverse measure $d\sigma$ is the disjoint angular measure and its density is

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad d\bar{\nu} = A_\sigma(u)\, du\, d\sigma.$$

Here $F = u$, $z = |\nabla_q F|$, $q = (\alpha_0, \alpha_m)$, $\omega = |\partial_{p_0}\alpha_m|^{-1}$, and $\chi = |\det D_q(F, \arg \nabla_q F)|$. All fibers and signed measures below use these same kernels. Let $z_\sigma^- = z(\Psi_\sigma(a_\sigma+))$ and $\ell_\sigma = b_\sigma - a_\sigma$.

Input statement 3.1 (Complete segment geometry and signed balance)**.** Put $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. On every regular segment,

$$2\lambda_\circ(v-u) \le z(v)^2 - z(u)^2 \le 2\Lambda_\circ(v-u), \tag{3.1}$$

$$|A_\sigma'| \le C A_\sigma I_m / z, \qquad |(W^\varepsilon)'| \le C\varepsilon^{-1}/z, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i. \tag{3.2}$$

The complete density $f_\sigma^\varepsilon = A_\sigma W^\varepsilon$ has one-sided traces and is absolutely continuous on the closed segment. The sum of its endpoint traces and absolute derivative integrals is finite at each fixed count. A grazing endpoint has zero trace; a critical backward endpoint need not. The geometric domain is a finite definable family at each fixed count. Its component count is used for existence of measures only, not for normalized estimates.

For every Borel prefix endpoint $c_\sigma \in [a_\sigma, b_\sigma]$ put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma.$$

The original union $z < \zeta$ or $\ell < \delta$ is exactly such a prefix, and $c_\sigma = b_\sigma$ gives b^ε. Define

$$\mathsf{I}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma,$$

$$\mathsf{D}_c^\varepsilon = \int_{a_\sigma < c_\sigma} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma,$$

$$\mathsf{G}_c^{\varepsilon,\pm}(H) = \int\int_{(a_\sigma, c_\sigma)\cap H} [\pm (f_\sigma^\varepsilon)'(u)]_+\, du\, d\sigma.$$

Their complete finite-count signed identity is

$$DR_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{3.3}$$

Every artificial prefix cut is negative. All true entrance types and one-sided incident physical components are retained.

This is the endpoint-field, optical, Borel-segment and signed-prefix chain, culminating in retained modules 220–222. In particular, A_σ includes both angular factors. No reversal of the billiard record or change of the section is used at a backward end.

Input statement 3.2 (Backward-end and critical volume estimates)**.** Set $\alpha = 129/128$ and $\gamma = 1/32$. Let

$$e_s^-(t) = \int \mathbf{1}_{\{t-a_\sigma \le s\}}\, d\mu_t, \qquad u_q(t)\,dt = F_\#(I_m \mathbf{1}_{\{z<q\}}\bar{\nu}), \qquad p_q(t)\,dt = F_\#(\mathbf{1}_{\{z<q\}}\bar{\nu}).$$

For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\theta_{B_*} < 1$ such that, at finite count and small $s, q > 0$,

$$m^2 \int_J e_s^- \le Cs^\gamma(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\theta_{B_*}^m, \sqrt{s}\}, \tag{3.4}$$

$$m^2 \int_J u_q \le Cq^\alpha(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 q^{1/2}\theta_{B_*}^m. \tag{3.5}$$

The constants are uniform in radius, exact labels, marks where present, and interval location. Since $I_m \ge 1$, the latter also bounds p_q. Consequently, at every fixed $s, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_s^- \le CLs^\gamma.$$

For the actual positive source $v^\rho(t)\,dt = F_\#(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar{\nu})$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\rho \le CL \quad (\rho, L > 0 \text{ fixed}). \tag{3.6}$$

The leading constant is independent of the fixed positive cutoff; its finite-count remainder is not.

Equation (3.4) is the same-event backward-end estimate of module 211, including critical entrances. For (3.5), multiply the actual fractional-source bound of module 187, at exponent $1/4$, by $q^{1/4}$: pointwise, $I_m\mathbf{1}_{z<q} \le q^{1/4} I_m z^{-1/4}\mathbf{1}_{z<q}$. The leading power is q^α and the remainder power is $q^{1/2}$. The audited cutoff estimate is retained in modules 196 and 222. These are volume-source theorems, not assumptions on entrance flux.

Input statement 3.3 (Original scalar reconstruction)**.** For every fixed $\varepsilon, r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\,du. \tag{3.7}$$

For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards. The reconstruction band B is outermost.

We use $\mathcal{H}_M(h) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} h(t)$ for a nonnegative source family. Unqualified $\|\cdot\|_\infty$ denotes essential supremum on the full roof line. Identities hold for common kernels at fixed m, R, n, k; the uniform order is $\sup_R \operatorname{ess\,sup}_t$. Countably many chosen radii share versions. No simultaneous Lebesgue-point assertion for every uncountable selector family is required.

4. Actual entrances at every positive residence scale

An incoming trace is attached to the incident maximal segment, not to a boundary point without its physical component. We first separate its types. If an endpoint or internal incidence vanishes, assign the end to the grazing class. Among the remaining ends assign those with $z_\sigma^- = 0$ to the critical class. Every other end is a nongrazing physical or section-decision entrance with $z_\sigma^- > 0$. Simultaneous section and clearance failures can be assigned to the first failing test in a

fixed finite ordering. These assignments are Borel: the limiting contact data and lifetimes are Borel, and all tests are closed conditions on those data. They form a disjoint partition. If all limiting physical margins and z_σ^- were strictly positive, regular continuation would contradict maximality of the same exact-event segment. This exhausts the entrance types without counting chart seams. The grazing trace vanishes; the critical directional trace is retained.

Put $k_{r,t}(u) = (u - t + r)/r$ for $t - r < u \le t$. For fixed positive ρ, δ, define the noncritical, positive-residence entrance assignment

$$B_{c,r}^{\varepsilon;\rho,\delta}(t) = \int_{a_\sigma < t < c_\sigma} \mathbf{1}_{\{t-r<a_\sigma,\ z_\sigma^- \ge \rho,\ \ell_\sigma \ge \delta\}} k_{r,t}(a_\sigma) f_\sigma^\varepsilon(a_\sigma+)\, d\sigma. \tag{4.1}$$

Thus only entrances whose same selected segment reaches the target are counted. A superscript $\sharp$ removes the ramp. In particular no later re-entry is included.

Theorem 4.1 (Complete noncritical entrance height)**.** *For all sufficiently small fixed $r > 0$ and every fixed $\rho, \delta > 0$,*

$$\limsup_m \sup_{R,n,k,c} m^2 \|B_{c,r}^{\varepsilon;\rho,\delta,\sharp}\|_\infty \le C(r^{1/32} + r). \tag{4.2}$$

The leading constant is independent of ε, ρ, δ and the Borel prefix. All noncritical nongrazing entrance types are included. Each cutoff is positive and fixed before the collision limit; the theorem does not assert the zero-cutoff conclusion.

Proof. It is enough to bound the larger source with entrance trace $A_\sigma(a_\sigma+)$ and without the prefix restriction, since $0 \le W^\varepsilon \le 1$. On a selected segment, radial monotonicity keeps $z \ge \rho$. If $\ell_\sigma > 2r$ and $a_\sigma \in (t - r, t)$, absolute continuity gives

$$A_\sigma(a_\sigma+) \le \frac{1}{r} \int_{a_\sigma}^{a_\sigma + r} A_\sigma(u)\, du + C \int_{a_\sigma}^{a_\sigma + r} A_\sigma(u) I_m(u)/z(u)\, du.$$

All these anchors lie in $(t - r, t + r)$ and have backward lifetime at most r. The Borel segment map counts their interiors once. Summing therefore bounds the long part by

$$\frac{1}{r} \int_{t-r}^{t+r} e_r^-(u)\, du + C \int_{t-r}^{t+r} v^\rho(u)\, du. \tag{4.3}$$

The ordered bound is $C(r^{1/32} + r)$ by the retained volume inputs. This is a trace comparison with the original density, not an identification of entrance flux and unweighted starting mass.

For a residence band $s < \ell_\sigma \le 2s$, integrate the same fundamental theorem over the complete segment:

$$A_\sigma(a_\sigma+) \le \frac{1}{s} \int_{a_\sigma}^{b_\sigma} A_\sigma(u)\, du + C \int_{a_\sigma}^{b_\sigma} A_\sigma(u) I_m(u)/z(u)\, du.$$

If the segment contains t, its interior lies in $(t - 2s, t + 2s)$ and has backward lifetime at most $2s$. Complete summation gives

$$\int_{a_\sigma < t < b_\sigma} \mathbf{1}_{\{s<\ell_\sigma \le 2s,\ z_\sigma^- \ge \rho\}} A_\sigma(a_\sigma+)\, d\sigma \le \frac{1}{s} \int_{t-2s}^{t+2s} e_{2s}^- + C \int_{t-2s}^{t+2s} v^\rho. \tag{4.4}$$

Its ordered height is $C(s^{1/32} + s)$. For a previously fixed $\delta > 0$, only finitely many bands $s_j = 2^{-j} r$ are needed to cover $\delta \le \ell \le 2r$. Sum them before the collision limsup. Their leading bounds form convergent geometric series independent of the number of bands. Combine with (4.3). No physical-word count or uniform estimate at $\rho = 0$ was used. □

4.1. **The zero-residence inlet is an original source, not a new loss.** Keep $\rho > 0$ fixed. For the full original source set

$$Z_\delta^{\varepsilon,\rho}(t) = \int_{a_\sigma<t<b_\sigma} \mathbf{1}_{\{\ell_\sigma\le\delta,\ z_\sigma^-\ge\rho\}} f_\sigma^\varepsilon(a_\sigma+)\, d\sigma,$$
$$Q_\delta^{\varepsilon,\rho}(t) = \int_{a_\sigma<t<b_\sigma} \mathbf{1}_{\{\ell_\sigma\le\delta,\ z_\sigma^-\ge\rho\}} f_\sigma^\varepsilon(t)\, d\sigma. \tag{4.5}$$

The second density is a restriction of the original weighted collision source to the same actual segments, with no transported normalization.

Proposition 4.2 (Exact identification of the noncritical inlet charge)**.** *For every fixed $\rho, \delta > 0$,*

$$|Z_\delta^{\varepsilon,\rho}(t) - Q_\delta^{\varepsilon,\rho}(t)| \le C(1+\varepsilon^{-1}) \int_{t-\delta}^t v^\rho(u)\, du. \tag{4.6}$$

Consequently their monotone zero-residence ordered heights agree:

$$\mathfrak{E}_{\varepsilon,M}^\rho := \lim_{\delta\downarrow 0} \mathcal{H}_M(Z_\delta^{\varepsilon,\rho}) = \lim_{\delta\downarrow 0} \mathcal{H}_M(Q_\delta^{\varepsilon,\rho}).$$

If $B_r^{\varepsilon,\rho}$ is the full ramped entrance assignment with $z_\sigma^- \ge \rho$ and no residence cutoff, then

$$\mathfrak{E}_{\varepsilon,M}^\rho \le \mathcal{H}_M(B_r^{\varepsilon,\rho}) \le \mathfrak{E}_{\varepsilon,M}^\rho + C(r^{1/32} + r). \tag{4.7}$$

All assertions hold for extended nonnegative heights; their zero value is not inferred from cutoff-independent leading constants.

Proof. On the selected segment $a_\sigma > t-\delta$ and $z \ge \rho$. Integrate $|(f_\sigma^\varepsilon)'| \le C(1+\varepsilon^{-1})A_\sigma I_m/z$ from a_σ to t and sum by the common Borel bijection. This proves (4.6). Its ordered norm is at most $C(1+\varepsilon^{-1})\delta$, proving equality of the two monotone limits by inequalities, without subtracting infinities. For $0 < \delta < r$ the ramp on every selected short segment lies between $1-\delta/r$ and one. Hence

$$(1-\delta/r)Z_\delta^{\varepsilon,\rho} \le B_r^{\varepsilon,\rho} \le Z_\delta^{\varepsilon,\rho} + B_r^{\varepsilon;\rho,\delta,\sharp}.$$

Apply Theorem 4.1, take the ordered height, and only then decrease δ. The first comparison remains valid when the limiting height is infinite. □

5. Critical histories which cross a radial shell

The entrance and subsequent growth of one history should not always be estimated separately. We now bound the actual target source of all histories which traverse a prescribed radial shell in the backward window. This includes births at critical directional entrances and arbitrarily small internal incidences.

Let $q_* > 0$ be a fixed small-gradient range for (3.5). For $0 < b < D$ define the actual shell source

$$v^{b,D}(u)\, du = F_\#(I_m z^{-1} \mathbf{1}_{\{b\le z<D\}} \bar\nu).$$

Lemma 5.1 (The upper-gradient gain with its finite-count remainder)**.** *For every fixed auxiliary band and $0 < b < D < q_*$ sufficiently small,*

$$m^2 \int_J v^{b,D} \le CD^{\alpha-1}(|J| + B_*^{-1}) + C_{B_*}(1+B_*|J|)m^2 b^{-1/2}\theta_{B_*}^m. \tag{5.1}$$

In particular its ordered fixed-window mass is at most $CD^{\alpha-1}|J|$, with leading constant independent of the previously fixed positive b.

Proof. The positive majorant

$$I_m z^{-1}\mathbf{1}_{\{b\le z<D\}} \le D^{-1}I_m\mathbf{1}_{\{z<D\}} + \int_b^D q^{-2}I_m\mathbf{1}_{\{z<q\}}\,dq$$

permits direct use of (3.5). The leading integral is bounded by $D^{\alpha-1}(1+1/(\alpha-1))$. Its finite-count error is bounded by $D^{-1/2}+\int_b^D q^{-3/2}\,dq \le 2b^{-1/2}$. Tonelli applies to this finite positive layer cake. Fix b, D, J, B_*, take the collision limsup and then enlarge the auxiliary band. No integration to $b=0$ is made. □

For a live target ancestor $a_\sigma < t < c_\sigma$, let $u_\sigma^0 = \max\{a_\sigma, t-r\}$ and use the right trace at that roof. Write $\mathcal{T}_{c;r,b}(t)$ for all such ancestors satisfying

$$z_\sigma(u_\sigma^0+) < b, \qquad z_\sigma(t) \ge 2b. \tag{5.2}$$

Define their actual target contribution by

$$N^\varepsilon_{c;r,b}(t) = \int_{\mathcal{T}_{c;r,b}(t)} f^\varepsilon_\sigma(t)\,d\sigma.$$

The selector is Borel and contains no positive residence or incidence cutoff. A crossing itself supplies the geometric length used below.

Theorem 5.2 (Complete radial-transit target height)**.** *Put $D = (b^2+2\Lambda_\circ r)^{1/2}$ and suppose $D < q_*$. At every finite count and almost every target roof,*

$$N^\varepsilon_{c;r,b}(t) \le \frac{2\Lambda_\circ}{3b^2}\int_{t-r}^t p_{2b}(u)\,du + C(1+\varepsilon^{-1})\int_{t-r}^t v^{b,D}(u)\,du. \tag{5.3}$$

Thus, for fixed $r, b > 0$ before the collision limit,

$$\limsup_m \sup_{R,n,k,c} m^2\|N^\varepsilon_{c;r,b}\|_\infty \le Crb^{\alpha-2} + C(1+\varepsilon^{-1})rD^{\alpha-1}. \tag{5.4}$$

In particular $b=\sqrt{r}$ gives

$$\limsup_m \sup_{R,n,k,c} m^2\|N^\varepsilon_{c;r,\sqrt{r}}\|_\infty \le Cr^{129/256} + C(1+\varepsilon^{-1})r^{257/256}. \tag{5.5}$$

This is an essential-height estimate for the complete selected target source, not an integrated starting-roof estimate.

Proof. If $D \le 2b$, the selected family is empty. Otherwise $2b < D < q_*$, so the small-gradient source bounds apply throughout the shell. For a selected ancestor, strict radial monotonicity gives unique roofs $u_b < u_{2b} \le t$ with $z(u_b) = b$, $z(u_{2b}) = 2b$. Both lie after u_σ^0 and hence in $(t-r, t]$. Equation (3.1) gives

$$u_{2b} - u_b \ge \frac{3b^2}{2\Lambda_\circ}, \qquad z(t) < (b^2+2\Lambda_\circ r)^{1/2} = D.$$

For $v \in (u_b, u_{2b})$, the same selected prefix contains the whole interval (v,t). The fundamental theorem gives

$$f^\varepsilon_\sigma(t) \le f^\varepsilon_\sigma(v) + \int_v^t [(f^\varepsilon_\sigma)'(u)]_+\,du.$$

Average v over the shell. Bound its reciprocal roof length by $2\Lambda_\circ/(3b^2)$, and the derivative integral by the integral from u_b to t. In this latter interval $b \le z < D$ and $[(f^\varepsilon)']_+ \le C(1+\varepsilon^{-1})AI_m/z$. Complete summation of the shell interiors is a submeasure of the original source restricted to $z < 2b$; complete summation of the later intervals is a submeasure of its $v^{b,D}$ insertion. This uses the Borel bijection with multiplicity one at each intermediate roof, not a bound for the

number of ancestors. Since $W^\varepsilon \le 1$, the first sum is bounded by p_{2b}. This proves (5.3) without any inverse-Jacobian deletion.

The critical volume bound and Lemma 5.1, at interval length r, imply (5.4) after fixing all widths and the auxiliary band before the collision limit. The finite-count error contains $b^{-1/2}$ and is not declared uniform at $b = 0$. The choice $b = \sqrt{r}$ is made before that limit; it yields $\alpha/2 = 129/256$ and $(\alpha+1)/2 = 257/256$. □

5.1. **What is estimated in a signed critical birth packet.** For an ancestor born within the window, integration by parts gives

$$k_{r,t}(a_\sigma) f^\varepsilon_\sigma(a_\sigma+) + \int_{a_\sigma}^t k_{r,t}(u)(f^\varepsilon_\sigma)'(u)\,du = f^\varepsilon_\sigma(t) - \frac{1}{r}\int_{a_\sigma}^t f^\varepsilon_\sigma(u)\,du. \tag{5.6}$$

The positive part of the left side is at most the actual target value. Hence Theorem 5.2 estimates the complete signed entrance-and-growth packet of every critical or near-critical newborn ancestor satisfying (5.2). It does not estimate its two positive terms separately. A grazing birth has zero trace but may have subsequent growth; it is included. A critical birth may have a nonzero trace; it is included as well. Histories which stay below $2b$ at the target are not covered by this crossing theorem and remain in the exact pointwise problem.

6. Signed projected forcing and the original upper endpoint

Taking positive parts on every sheet can overestimate the forcing of the projected source. We retain all physical and prefix losses through projection. Only then is a positive part taken. The resulting forcing is smaller than the earlier sufficient positive budgets and gives an equivalent criterion for the original scalar endpoint.

Restrict $\mathsf{G}^{\varepsilon,+}_c$ at its physical intermediate state to $z < \rho$ or $z \ge \rho$ and set

$$\mathsf{K}^\varepsilon_{c,\rho} = \mathsf{I}^\varepsilon_c + \mathsf{G}^{\varepsilon,+,<\rho}_c - \mathsf{D}^\varepsilon_c - \mathsf{G}^{\varepsilon,-}_c. \tag{6.1}$$

This is a finite signed measure at fixed count. For such a measure put

$$\mathcal{T}_r\mu(t) = \int_{(t-r,t]} k_{r,t}(u)\,d\mu(u), \qquad A^-_r f(t) = \frac{1}{r}\int_{t-r}^t f(u)\,du.$$

Define two nonnegative functions

$$\begin{aligned} S^\varepsilon_{c;r,\rho}(t) &= [\mathcal{T}_r \mathsf{K}^\varepsilon_{c,\rho}(t)]_+, \\ S^{\varepsilon,\sharp}_{c;r,\rho}(t) &= \operatorname*{ess\,sup}_{s\in(t-r,t)} [\mathsf{K}^\varepsilon_{c,\rho}((s,t])]_+. \end{aligned} \tag{6.2}$$

The supremum is in the earlier roof only. BV representatives and Fubini give Borel versions of both functions; pointwise assertions below are at Lebesgue roofs. They are not Jordan decompositions of DR^ε_c.

Theorem 6.1 (A complete signed pointwise comparison)**.** *For every fixed $r, \rho > 0$ and almost every roof,*

$$0 \le S^\varepsilon_{c;r,\rho} \le S^{\varepsilon,\sharp}_{c;r,\rho} \le R^\varepsilon_c, \tag{6.3}$$

$$R^\varepsilon_c(t) \le A^-_r R^\varepsilon_c(t) + S^\varepsilon_{c;r,\rho}(t) + C(1+\varepsilon^{-1})\int_{t-r}^t v^\rho(u)\,du. \tag{6.4}$$

For almost every $s \in (t-r, t)$ one also has

$$R^\varepsilon_c(s) \ge R^\varepsilon_c(t) - S^{\varepsilon,\sharp}_{c;r,\rho}(t) - C(1+\varepsilon^{-1})\int_{t-r}^t v^\rho(u)\,du. \tag{6.5}$$

Completed prefixes contribute nonpositively before the positive part in (6.2). *Moreover*

$$S^{\varepsilon}_{c;r,\rho}(t) \le B^{\varepsilon}_{c,r}(t) + G^{\varepsilon,+,<\rho}_{c,r}(t), \tag{6.6}$$

where the right side is the retained target-ancestral ramped budget. The analogous inequality holds with all ramps removed.

Proof. By the exact signed balance, $DR^{\varepsilon}_c = \mathsf{K}^{\varepsilon}_{c,\rho} + \mathsf{G}^{\varepsilon,+,\ge\rho}_c$. Stieltjes integration gives

$$R^{\varepsilon}_c(t) - A^-_r R^{\varepsilon}_c(t) = \mathcal{T}_r \mathsf{K}^{\varepsilon}_{c,\rho}(t) + \mathcal{T}_r \mathsf{G}^{\varepsilon,+,\ge\rho}_c(t). \tag{6.7}$$

The last term is nonnegative. Since $A^-_r R^{\varepsilon}_c \ge 0$, its subtraction proves $S^{\varepsilon}_{c;r,\rho} \le R^{\varepsilon}_c$. For almost every earlier roof, the interval form of the same identity is

$$\mathsf{K}^{\varepsilon}_{c,\rho}((s,t]) = R^{\varepsilon}_c(t) - R^{\varepsilon}_c(s) - \mathsf{G}^{\varepsilon,+,\ge\rho}_c((s,t]) \le R^{\varepsilon}_c(t).$$

It proves the sharp domination and the persistence bound. Averaging this interval identity in s gives (6.7); hence the ramped positive part is bounded by the sharp one. Finally, the full physical product derivative and multiplicity-one source sum give

$$\mathsf{G}^{\varepsilon,+,\ge\rho}_c((t-r,t]) \le C(1+\varepsilon^{-1}) \int_{t-r}^{t} v^{\rho}(u)\, du.$$

This proves (6.4) and (6.5).

For the finer comparison, do the same calculation on one zero-extended prefix. If $c_\sigma \le t$, its ramped derivative is $-A^-_r(\mathbf{1}_{(a_\sigma,c_\sigma)} f^{\varepsilon}_\sigma)(t) \le 0$; subtracting its noncritical positive growth only decreases it. A future prefix contributes zero. For a live target ancestor there is no upper cut in the window, and its remaining signed contribution is at most its entrance trace plus critical positive growth. Sum all signed contributions before taking the positive part. Fixed-count finite variation justifies the interchange. This proves (6.6); the interval form proves the sharp version. In particular no unobserved completed history is charged as positive forcing. □

6.1. **Where the signed forcing can still be positive.** Write $R^{\varepsilon,<2b}_c$ for the original prefix source restricted at the target to $z(t) < 2b$, and let $B^{\varepsilon,b,\sharp}_{c,r}$ be its entrance budget restricted to $z^-_\sigma \ge b$, without a residence cutoff.

Proposition 6.2 (Critical-target and inlet localization)**.** *For $0 < \rho \le b$ and the same r, b as in Theorem 5.2,*

$$S^{\varepsilon,\sharp}_{c;r,\rho}(t) \le R^{\varepsilon,<2b}_c(t) + B^{\varepsilon,b,\sharp}_{c,r}(t) + N^{\varepsilon}_{c;r,b}(t). \tag{6.8}$$

Thus radial-transiting critical packets are already estimated, and all noncritical inlet residence bands above a fixed positive cutoff are already estimated. The two remaining terms are restrictions of the original critical target source and original zero-residence inlet source, not separate absolute critical-growth charges.

Proof. Completed or future prefixes contribute nonpositively to every earlier interval comparison. For live prefixes, first separate targets with $z(t) < 2b$. Each sheet's signed interval increment is at most its current value, so their total positive contribution is at most $R^{\varepsilon,<2b}_c(t)$. For the remaining targets use $u^0_\sigma = \max(a_\sigma, t-r)$. If $z(u^0_\sigma+) < b$, the sheet belongs to the radial-transit family and its signed increment is at most $f^{\varepsilon}_\sigma(t)$. If $z(u^0_\sigma+) \ge b$, no critical positive growth occurs in the window. An old sheet then contributes nonpositively; a newborn sheet contributes at most its entrance trace. These bounds hold for every earlier roof outside the common null set, so they survive the earlier-roof essential supremum. All source restrictions are Borel and all sheet interiors are counted once. □

6.2. **An equivalent upper-endpoint criterion.** For the full source $c_\sigma = b_\sigma$ omit c and define

$$\mathfrak{S}_{\varepsilon,M}(r) = \lim_{\rho\downarrow 0} \mathcal{H}_M(S^{\varepsilon}_{r,\rho}), \qquad \mathfrak{S}^{\sharp}_{\varepsilon,M}(r) = \lim_{\rho\downarrow 0} \mathcal{H}_M(S^{\varepsilon,\sharp}_{r,\rho}).$$

Both limits exist monotonically: decreasing ρ removes a positive part from (6.1). The complete finite-count signed sum, its positive part and its essential height all precede this limit.

Theorem 6.3 (Signed forcing at the original pointwise topology)**.** *For all large fixed B and fixed $r > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{S}_{\varepsilon(B),M}(r) + C(1+\varepsilon(B)^{-1})r. \tag{6.9}$$

For any prescribed positive $r(B)$ with $(1+\varepsilon(B)^{-1})r(B) \to 0$, the original pointwise assertion is equivalent to each of

$$\mathfrak{S}_{\varepsilon(B),M}(r(B)) \longrightarrow 0, \qquad \mathfrak{S}^{\sharp}_{\varepsilon(B),M}(r(B)) \longrightarrow 0 \quad \textit{for every fixed } M. \tag{6.10}$$

The forcing in this criterion is formed after signed projection; it is not the sum of the earlier two sufficient positive charges. No vanishing assertion in (6.10) *is made here.*

Proof. Apply (6.4) to b^{ε}. Its backward average is at most $2A_r b^{\varepsilon}$. Fix B, r, ρ and all auxiliary bands before the collision limsup. Inputs 3.2 and 3.3 then give

$$\mathcal{H}_M(b^{\varepsilon}) \le C\varepsilon^{1/16} + \mathcal{H}_M(S^{\varepsilon}_{r,\rho}) + C(1+\varepsilon^{-1})r.$$

Only now decrease ρ. Insert the original positive remainder and $\varepsilon(B)^{1/16} = O(B^{-1/192})$ to obtain (6.9). This proves sufficiency of either criterion, since the ramped forcing is bounded by the sharp one.

Conversely, if the original pointwise assertion holds, the positive remainder gives $\mathcal{H}_M(b^{\varepsilon(B)}) \le C_M B^{-1/192}$. Both signed forcings are bounded by $b^{\varepsilon(B)}$ at every finite count by (6.3). Their ordered heights therefore satisfy the same bound for every r, ρ. This proves necessity without exchanging any cutoff and collision limit. The legal example $r(B) = c\varepsilon(B)^2$ makes the explicit last term vanish, but does not prove either signed charge zero. □

7. The remaining upper-source estimate

The entry theorem uses a complete physical-volume anchor for each selected segment and only a finite residence-band sum before the collision limit. The radial theorem obtains its anchor length from actual passage through a gradient shell. Neither argument sums a single-sheet modulus by an uncontrolled transverse multiplicity. The signed theorem removes completed prefixes and preserves all negative physical and artificial-cut contributions through projection.

The pointwise target nevertheless still requires vanishing of the signed forcing in (6.10). Equation (6.8) records what is not estimated: a target which remains in the small-gradient core, and the original weighted zero-residence inlet concentration. Small fixed-window mass of these sources does not bound their essential height. The equality of the inlet and original-source charges is not a proof that either is zero. Nor does a signed cancellation identity alone supply a bound for its remaining positive part.

All original labels, section decisions, first-hit tests, weights and arithmetic classes remain in (2.5). The earlier lower pointwise half, finite-norm laws and their conditioned consequences retain their exact hypotheses. No point-conditioned likelihood, kernel, unrestricted ratio or normalization of arithmetic zero classes is promoted. The complete preceding article and every inherited proof remain in the separate technical archive.

Appendix A. Dependencies and normalization

The present article directly loads the three new modules 223–225. The complete technical archive contains all 222 inherited core files and the full preceding article. The principal dependencies are as follows; module numbers refer to their unchanged source files.

The exact event, guard and arithmetic reference are the retained physical problem. Modules 174, 177, 180–182 and 196 provide the endpoint field, contact motion, optical recurrence, full angular density and Borel segment parametrization. Module 220 gives the absolute optical cancellation, complete-sheet traces and fixed-count variation. Module 221 gives the exact signed prefix identity used in (3.3). These ingredients retain the same mixed endpoint-angle and internal-arclength convention.

The backward-end estimate is the two-sided physical-end theorem of module 211; the precise display used here is restated as (3.4). The small-gradient volume estimate is obtained from the actual fractional insertion in module 187, with its finite-count remainder as restated in the inherited journal input `eq:v83-fractional-finite-input`. The positive-cutoff layer cake was audited in module 196 and used for complete noncritical growth in module 222. Its lower cutoff is fixed before the collision limit in every application here. These dependencies are explicitly restated in Inputs 3.2 and 3.3; no surface spectral insertion is assumed.

The first new theorem uses the incoming, rather than outgoing, trace and the actual backward layer. The second theorem uses the common source on a target-dependent radial shell. The third uses the BV Stieltjes identity for finite signed roof measures [13]; it is proved directly to fix the ramp direction and the sign of all cuts. No Jordan decomposition is interchanged with projection. Historical fixed-count definability and threshold integration remain credited to [9, 15]; neither gives a collision-normalized component count.

All interval lengths and cutoffs in the quantitative estimates are fixed before their collision limits. The illustrative choice $r(B) = c\varepsilon(B)^2$ also fixes the shell width $\sqrt{r(B)}$ first. It estimates the displayed radial-transit terms but does not remove either remaining concentration. The two revision-88 source variants remain separately preserved; their incompatible module histories are not merged.

Part 11. Absolute optical traces and signed ancestral upper forcing

Retained revision-93 abstract. We return to the upper pointwise local limit problem for the exact collision record of actual returns in a triangular Lorentz gas. The positive optical denominator cancels each individual incidence singularity in the absolute angular density, giving a collision-uniform one-half modulus on every physical sheet. Its complete fixed-count signed balance shows that the original critical-or-short-residence selector creates only negative interior cuts. Positive increments come from actual entrance traces and positive physical density growth. We prove a prescribed-scale pointwise comparison retaining only the ancestors of the selected target fiber, together with an upper-spike persistence dichotomy on that same source. The complete noncritical growth term has ordered height of order the roof width, with a leading constant independent of every fixed positive gradient cutoff. Actual ancestral entrances and critical positive growth remain explicit in the scalar upper bound. No vanishing of their concentration is inferred, and the original unrestricted pointwise theorem remains to be completed. All earlier finite-norm and conditioning results are retained.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1, 1)$

for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\,\mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that

analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The upper sign and its physical origins.** The organizing assertion is still (2.5). The historical one-sided lower law and the finite-norm conclusions remain available, but do not answer the upper-spike question. The new argument studies the original positive weighted density before taking its absolute roof increments. It uses the monotonicity of the actual exceptional selector along each maximal physical segment, not an embedding of a finite norm into an essential supremum.

Theorems 5.1 and 5.2 give its signed balance and a pointwise inequality at a prescribed roof radius. Theorem 6.1 estimates the full noncritical positive variation after the complete physical summation. Thus a high value either persists on the prescribed backward interval or is paid by a quantified entrance or critical-growth contribution of the same target's ancestors. The final upper bound in Theorem 6.2 keeps those contributions rather than declaring them small.

3. Retained inputs and conventions

The next statements specify the prior analytic and geometric results used here. They are part of the frozen proof archive, not new assumptions silently substituted for the original event. Exact proof locations are recorded in the accompanying dependency map. The new argument below requires no path-remainder input.

Input statement 3.1 (Physical optical geometry and common fibers)**.** At fixed m,R,n,k, the regular exact event, with critical points removed, has a Borel parametrization by its complete maximal segments (σ,u), $a_\sigma<u<b_\sigma$. The transverse measure $d\sigma$ is the disjoint angular measure. In endpoint angles,

$$F=L,\quad g=\nabla F=R(-p_0,p_m),\quad z=|g|,\quad K=D^2F,\quad \vartheta=\arg g,\quad X=\frac{K^{-1}g}{g^{\mathsf T}K^{-1}g}.$$

Here $XF=1$, $X\vartheta=0$, $\lambda_\circ I\le K\le\Lambda_\circ I$, and $\lambda_\circ\le\chi:=|\det D(F,\vartheta)|\le\Lambda_\circ$, where $\lambda_\circ=9\sqrt{19}/200$ and $\Lambda_\circ=47/6$. The physical source and its common roof kernels are

$$d\bar\nu=A_\sigma(u)\,du\,d\sigma,\qquad A_\sigma=\omega/(4\pi c_*\chi),\qquad \omega=|\partial_{p_0}\alpha_m|^{-1}. \tag{3.1}$$

For every surviving segment and $s>0$,

$$2\lambda_\circ s\le z(u+s)^2-z(u)^2\le 2\Lambda_\circ s,\qquad |\partial_u A_\sigma|\le CA_\sigma I_m/z,\quad I_m=1+\sum_{i=1}^{m-1}q_0^{d_i}/c_i.$$

The complete original guard is locally Lipschitz on the physical graph and satisfies $|\partial_u W^\varepsilon|\le C\varepsilon^{-1}/z$ almost everywhere. All constants depend only on the fixed physical family and the original smooth Θ.

For $V_i=2/(Rc_i)$ the positive optical recursion is

$$y_1=\ell_0,\quad F_0=1,\quad F_i=F_{i-1}+V_iy_i,\quad y_{i+1}=y_i+\ell_iF_i,\qquad \omega=Rc_0c_m/y_m. \tag{3.2}$$

The F_i in this recurrence are optical variables, not the roof F. The physical flights satisfy $\ell_i\ge\tau_-$.

The flow stops at actual physical and section-decision boundaries; Borel chart seams are not ends. All contact data have one-sided limits. A finite backward end is critical or is a true

incidence, clearance or section-decision end; there is no forward interior critical termination. At a fixed count the unweighted physical-word and endpoint graph is a finite definable family in rational circle coordinates. No definability of Θ is assumed.

This is the endpoint-field, optical-recursion, all-contact-motion, full-angular-variation and Borel-segment chain in the retained archive. The formula for the angular density includes both the inverse twist and angular Jacobian; it is not Lebesgue density in endpoint angles.

Input statement 3.2 (The positive scalar and cutoff-source estimates)**.** For every fixed $\varepsilon, r > 0$,

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_r b^{\varepsilon}\|_{\infty} \le C\varepsilon^{1/16}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\,ds. \tag{3.3}$$

For $\rho > 0$ define the actual positive source

$$v^{\rho}(t)\,dt = F_{\#}(I_m z^{-1}\mathbf{1}_{\{z\ge\rho\}}\bar{\nu}).$$

For every previously fixed $\rho, L > 0$,

$$\limsup_{m} \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\rho} \le C_0 L, \qquad C_0 \text{ independent of } \rho, L. \tag{3.4}$$

This estimate is not stated at $\rho = 0$. Its inherited finite-count remainder contains a factor $\rho^{-1/2}$.

Writing $P = m^2 p$, $G = \mathcal{L}$, the scalar decomposition is

$$P - G = m^2 b^{\varepsilon(B)} + e_B, \quad \varepsilon(B) = A_0 B^{-1/12}, \quad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.5}$$

The original union density $R^{\varepsilon}_{\zeta,\delta}$ has the ordered local mass estimate

$$\limsup_{m} \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J R^{\varepsilon}_{\zeta,\delta} \le C_L(\zeta^{129/128} + \delta^{1/32}).$$

The complete unweakened weak-$L^{32/31}$ raw limit and original-union estimate are retained; they are not required to justify the signed balance. Every auxiliary band is fixed before its collision limit and enlarged only afterwards. The reconstruction band B is last.

The fixed-width bound is on the complete original source. The cutoff estimate is the audited positive layer-cake theorem, not a theorem for a boundary insertion. The one-sided lower law follows from (3.5) and remains historically credited. The present revision returns to the opposite, upper sign.

All original densities are common coarea versions for fixed m, R, n, k. Signed identities are identities of finite measures at fixed count. Pointwise conclusions are understood at their Lebesgue points, with the permitted uniform order $\sup_R \operatorname{ess\,sup}_t$. The Borel segment formulas define common versions for all positive selectors; we do not require one set of Lebesgue points for all radii and every uncountable family of cutoffs simultaneously.

4. Absolute optical control and physical traces

We begin with an estimate before taking the sum over physical sheets. Unlike a logarithmic derivative, its constant does not contain an inverse incidence. This estimate will justify the complete signed balance, not an estimate obtained by counting words.

Lemma 4.1 (Cancellation in the positive optical denominator)**.** *For every regular physical word and every internal contact* $1 \le i < m$,

$$\omega \le C_0 := R_+/\tau_-, \qquad \frac{\omega}{c_i} \le C_1 := \frac{R_+^2}{2\tau_-^2}, \qquad \omega I_m \le C_0 + \frac{2q_0}{1-q_0}C_1. \tag{4.1}$$

The constants are independent of the collision count, the word, the radius and every positive incidence. The internal assertion is empty when $m = 1$.

Proof. All entries of the positive recursion in Input 3.1 are nonnegative and y_j is increasing. Thus $y_m \geq y_1 = \ell_0$. For an internal contact,

$$y_m \geq y_{i+1} \geq \ell_i V_i y_i \geq \frac{2\ell_i \ell_0}{Rc_i} \geq \frac{2\tau_-^2}{Rc_i}.$$

Insert these two lower bounds in $\omega = Rc_0 c_m / y_m$ and use $c_0 c_m \leq 1$. Finally, each positive depth occurs at most twice, so $\sum_{i=1}^{m-1} q_0^{d_i} \leq 2q_0/(1-q_0)$. The estimate is for the absolute density; it does not assert that $\nabla \log \omega$ is bounded near grazing. □

On a maximal physical segment σ, write

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad f_\sigma^\varepsilon(u) = A_\sigma(u) W^\varepsilon(\Psi_\sigma(u)), \qquad a_\sigma < u < b_\sigma.$$

The transverse measure $d\sigma$ denotes the countable disjoint sum of the angular measures in the retained Borel parametrization.

Theorem 4.2 (Uniform traces on each complete physical sheet)**.** *There is a physical constant C such that*

$$0 \leq A_\sigma \leq C, \qquad |\partial_u f_\sigma^\varepsilon(u)| \leq C(1+\varepsilon^{-1}) z(\Psi_\sigma(u))^{-1}, \tag{4.2}$$

$$|f_\sigma^\varepsilon(v) - f_\sigma^\varepsilon(u)| \leq C(1+\varepsilon^{-1})\sqrt{|v-u|} \qquad (u, v \in [a_\sigma, b_\sigma]). \tag{4.3}$$

In the last assertion the endpoint values are the unique one-sided traces. The trace is zero at any endpoint where an internal incidence vanishes. A backward critical endpoint is not required to have zero trace.

At each fixed collision count all the following quantities are finite:

$$\int \left(f_\sigma^\varepsilon(a_\sigma+) + f_\sigma^\varepsilon(b_\sigma-) + \int_{a_\sigma}^{b_\sigma} |\partial_u f_\sigma^\varepsilon(u)|\, du \right) d\sigma. \tag{4.4}$$

No collision-uniform bound for (4.4) *is asserted.*

Proof. The retained full-density estimate is $|A_\sigma'| \leq C A_\sigma I_m / z$. The original guard is locally Lipschitz on the physical graph and $|(W^\varepsilon)'| \leq C\varepsilon^{-1}/z$ almost everywhere. The product rule, $0 \leq W^\varepsilon \leq 1$, the lower bound for χ and Lemma 4.1 prove (4.2). A change of the minimizing third disk in a clearance does not affect this almost-everywhere argument.

Radial growth gives, by taking the earlier roof down to a_σ,

$$z(\Psi_\sigma(s))^2 \geq 2\lambda_\circ (s - a_\sigma).$$

For $a_\sigma < u < v < b_\sigma$, integration yields

$$\int_u^v \frac{ds}{z(\Psi_\sigma(s))} \leq \sqrt{\frac{2}{\lambda_\circ}} (\sqrt{v-a_\sigma} - \sqrt{u-a_\sigma}) \leq \sqrt{\frac{2}{\lambda_\circ}} \sqrt{v-u}.$$

This proves the extension and its modulus. If $c_i \to 0$, then $\omega \leq C_1 c_i \to 0$; the other factors are bounded.

For completeness, the transverse parameter space has finite measure at fixed count. There are finitely many candidate physical words, because the flight horizon is finite. In finitely many rational circle-coordinate charts their reflection, first-hit and section decisions are definable. On the regular graph the direction $g/|g|$ is also definable. Its one-dimensional fibers have finitely many connected components, with a bound for this fixed definable family. Each component is

exactly one maximal roof segment: its tangent is X and $XF = 1$. Definable finiteness, equally expressed in finitely many bounded angular charts, therefore gives $\int d\sigma \le CN_m < \infty$ for some N_m. The Borel rational-roof selection counts each component once and preserves its angle. This argument concerns the geometric domain, not definability of the smooth guard. See [9] for the finite-family principle.

The roof lies in an interval of length at most $m\tau_+$. The derivative integral on each sheet is consequently at most $C(1+\varepsilon^{-1})\sqrt{m\tau_+}$ and its two traces are bounded. This proves (4.4). The number N_m is used only for existence of these fixed-count measures. Every collision-normalized estimate below instead uses the exact physical coarea identity. □

Remark 4.3 (A critical trace is not point mass in the original area)*.* The original area of a critical point is zero. Its directional trace may nevertheless give an atom in the derivative of the roof density. For example, the pushforward of planar area by $F(q) = |q|^2/2$ near its minimum has density $2\pi\mathbf{1}_{\{t>0\}}$ and derivative $2\pi\delta_0$ there. We retain this distinction, already present in the historical angular-residue calculation. Critical entrances are not replaced by artificial positive area and are not deleted.

5. The signed balance of the original source

The original exceptional selector has a useful sign. Along a maximal segment the total residence is constant and z is increasing. Thus selection by small initial gradient can end along the forward flow, but cannot begin at an interior roof.

Choose any Borel function $c_\sigma \in [a_\sigma, b_\sigma]$ and put

$$R_c^\varepsilon(t) = \int \mathbf{1}_{\{a_\sigma < t < c_\sigma\}} f_\sigma^\varepsilon(t)\, d\sigma. \tag{5.1}$$

For the original union $z < \zeta$ or $\ell < \delta$, take $c_\sigma = b_\sigma$ when $b_\sigma - a_\sigma < \delta$, and otherwise let c_σ be the endpoint of the initial interval $z < \zeta$. This is Borel by strict radial monotonicity and the Borel segment map. The empty and full intervals are allowed. Formula (5.1) is exactly the original weighted union, not the sum of its two overlapping restrictions. Taking every $c_\sigma = b_\sigma$ gives the original bad-source density b^ε.

Define nonnegative roof measures, with the integrands zero when $c_\sigma = a_\sigma$, by

$$\begin{aligned} \mathsf{I}_c^\varepsilon &= \int \mathbf{1}_{\{a_\sigma < c_\sigma\}} f_\sigma^\varepsilon(a_\sigma+)\delta_{a_\sigma}\, d\sigma, \\ \mathsf{D}_c^\varepsilon &= \int \mathbf{1}_{\{a_\sigma < c_\sigma\}} f_\sigma^\varepsilon(c_\sigma-)\delta_{c_\sigma}\, d\sigma, \\ \mathsf{G}_c^{\varepsilon,\pm}(H) &= \int \int_{(a_\sigma, c_\sigma)\cap H} [\pm\partial_u f_\sigma^\varepsilon(u)]_+\, du\, d\sigma. \end{aligned} \tag{5.2}$$

Here H is any Borel roof set and δ_v is a unit point mass. All four measures are finite at fixed count by Theorem 4.2.

Theorem 5.1 (Exact balance with nonpositive artificial cuts)**.** *The distributional derivative of the original prefix source is*

$$D_t R_c^\varepsilon = \mathsf{I}_c^\varepsilon + \mathsf{G}_c^{\varepsilon,+} - \mathsf{D}_c^\varepsilon - \mathsf{G}_c^{\varepsilon,-}. \tag{5.3}$$

All interior prefix boundaries occur with the negative sign. The positive entrance measure is supported on true backward physical or section-decision ends and backward critical ends. Its grazing entrance part is zero. The full-source measures I^ε *and* $\mathsf{G}^{\varepsilon,+}$*, obtained by* $c_\sigma = b_\sigma$*, give the common domination*

$$D_t R_c^\varepsilon \le \mathsf{I}^\varepsilon + \mathsf{G}^{\varepsilon,+} \tag{5.4}$$

for every Borel prefix, without regularity of its transverse graph.

Proof. On one sheet, integration by parts on $[a_\sigma, c_\sigma]$ gives, for $\phi \in C_c^1(\mathbb{R})$,

$$-\int_{a_\sigma}^{c_\sigma} f_\sigma^\varepsilon \phi' = f_\sigma^\varepsilon(a_\sigma+)\phi(a_\sigma) - f_\sigma^\varepsilon(c_\sigma-)\phi(c_\sigma) + \int_{a_\sigma}^{c_\sigma} (f_\sigma^\varepsilon)'\phi.$$

The endpoint values are justified by the absolute continuity and integrable derivative proved above. The fixed-count finite variation permits integration over σ. This proves the signed identity, with the common original coarea kernels. Splitting the derivative into its two signs before summation gives precisely (5.3), not an identification of either summand with the Jordan positive part of the projected derivative.

A prefix cut is the upper endpoint of an initial interval. Its coefficient is therefore negative, including cuts coinciding with physical forward ends. The lower endpoint a_σ is an actual maximal-segment end; chart assignments do not produce endpoints. The trace at a grazing entrance vanishes by Theorem 4.2. A critical directional trace is retained. Finally the selected positive measures are submeasures of the full positive measures. Discarding the two negative measures proves (5.4). □

5.1. **A comparison using only ancestors of the target fiber.** The common measure in (5.4) is convenient, but can charge segments which have already ended. The following sharper comparison keeps only the ancestors of the selected target. For a roof t, write $\mathcal{A}_c(t) = \{\sigma : a_\sigma < t < c_\sigma\}$ and $k_{r,t}(u) = (u - t + r)/r$ on $[t - r, t]$. Define

$$B_{c,r}^\varepsilon(t) = \int_{\mathcal{A}_c(t)} \mathbf{1}_{\{a_\sigma > t-r\}} k_{r,t}(a_\sigma) f_\sigma^\varepsilon(a_\sigma+)\, d\sigma,$$

$$G_{c,r}^{\varepsilon,+}(t) = \int_{\mathcal{A}_c(t)} \int_{\max(a_\sigma, t-r)}^{t} k_{r,t}(u)[(f_\sigma^\varepsilon)'(u)]_+ \, du\, d\sigma. \tag{5.5}$$

These are nonnegative Borel functions. In particular an entrance is counted only when its same selected segment reaches the target roof; there is no later re-entry or completed-prefix contribution.

Theorem 5.2 (Prescribed-scale pointwise comparison)**.** *For every fixed $r > 0$ and almost every roof,*

$$R_c^\varepsilon(t) \le \frac{1}{r}\int_{t-r}^{t} R_c^\varepsilon(s)\, ds + B_{c,r}^\varepsilon(t) + G_{c,r}^{\varepsilon,+}(t). \tag{5.6}$$

No integral in the starting roof is taken on the left. The radius is prescribed in advance and is independent of the collision count. The inequality holds for the full original source and the exact exceptional union at every pair of its cutoffs.

Proof. Fix a sheet in $\mathcal{A}_c(t)$. Integration of its zero extension from the left endpoint, followed by averaging the earlier roof, gives

$$f_\sigma^\varepsilon(t) = \frac{1}{r}\int_{t-r}^{t} \mathbf{1}_{\{s>a_\sigma\}} f_\sigma^\varepsilon(s)\, ds + \mathbf{1}_{\{a_\sigma > t-r\}} k_{r,t}(a_\sigma) f_\sigma^\varepsilon(a_\sigma+) + \int_{\max(a_\sigma, t-r)}^{t} k_{r,t}(u)(f_\sigma^\varepsilon)'(u)\, du.$$

The ramp is a Stieltjes averaging weight, not a physical Jacobian. All roofs between the entrance and t belong to the same selected prefix. Sum this identity over the target's ancestors, replace the last derivative by its positive part, and bound the earlier-fiber submeasure by the full earlier R_c^ε. The lost earlier prefixes are subtracted in the exact identity, not added as births. Fubini and the common kernels justify the assertion at almost every roof. No multiplicity bound enters the comparison. □

For later use let $B^{\varepsilon,\sharp}_{c,r}$ and $G^{\varepsilon,+,\sharp}_{c,r}$ denote the same expressions with the ramp replaced by one. The sheet identity between any earlier roof and t also proves

$$R^{\varepsilon}_{c}(s) \geq R^{\varepsilon}_{c}(t) - B^{\varepsilon,\sharp}_{c,r}(t) - G^{\varepsilon,+,\sharp}_{c,r}(t) \quad \text{for almost every } s \in (t-r,t). \tag{5.7}$$

Thus a value at least λ persists above $\lambda/2$ throughout the prescribed backward interval unless its actual ancestral entrance and positive-growth budget is at least $\lambda/2$. This is an exact physical dichotomy. An estimate proving the latter alternative uniformly small is still needed for unrestricted spike exclusion.

6. The estimated gain and the remaining upper-source measures

The positive-growth budget can be estimated on its entire noncritical part, uniformly in every previously fixed positive cutoff. This leaves actual ancestral entrances and critical positive growth, rather than an integrated oscillation norm, in the pointwise inequality.

For a fixed gradient threshold $\rho > 0$, let $G^{\varepsilon,<\rho}_{c,r}$ and $G^{\varepsilon,\geq\rho}_{c,r}$ be the terms in (5.5) restricted at the intermediate roof to $z < \rho$ and $z \geq \rho$, respectively. Define their sharp versions by removing the ramp. The cutoff is on the physical intermediate state, not on an initially transported surrogate source.

Theorem 6.1 (Complete noncritical positive gain)**.** *At every finite count and almost every roof,*

$$G^{\varepsilon,\geq\rho,\sharp}_{c,r}(t) \leq C(1+\varepsilon^{-1}) \int_{t-r}^{t} v^{\rho}(u)\, du, \tag{6.1}$$

where v^{ρ} is the actual truncated first-inverse-gradient source in Input 3.2. Consequently

$$\limsup_{m} \sup_{R,n,k,c} m^2 \|G^{\varepsilon,\geq\rho,\sharp}_{c,r}\|_{\infty} \leq C(1+\varepsilon^{-1})r. \tag{6.2}$$

The supremum over c ranges over all Borel prefixes at that count. The leading constant is independent of ρ, r and the two original exceptional-source cutoffs. Each $\rho, r > 0$ is fixed before the collision limit. The theorem does not assert its conclusion at $\rho = 0$.

Proof. Before applying the absolute bound of Theorem 4.2, keep the physical density in the product rule:

$$[(f^{\varepsilon}_{\sigma})'(u)]_{+} \leq C(1+\varepsilon^{-1}) A_{\sigma}(u) I_m(\Psi_{\sigma}(u))/z(\Psi_{\sigma}(u)).$$

For each intermediate roof u, ancestors of a fixed selected target form a subfamily of the original exact fiber, counted once by the Borel segment map. Integrating this density over that subfamily and then over $u \in (t-r,t)$ is bounded by the right side of (6.1). No trace is equated to unweighted flux, and no factor counting physical words is inserted. Apply the positive-cutoff estimate (3.4) with interval length r to prove (6.2). Its finite-count remainder depends on ρ; the legal ordered estimate has one leading constant for all previously fixed positive ρ. □

Combining Theorems 5.2 and 6.1 gives the following estimate on the original, unrenormalized source:

$$R^{\varepsilon}_{c}(t) \leq \frac{1}{r} \int_{t-r}^{t} R^{\varepsilon}_{c}(s)\, ds + B^{\varepsilon}_{c,r}(t) + G^{\varepsilon,<\rho}_{c,r}(t) + C(1+\varepsilon^{-1}) \int_{t-r}^{t} v^{\rho}(u)\, du. \tag{6.3}$$

Every term is formed at finite count. In particular the left side includes selected points with arbitrarily small initial gradient and arbitrarily short total residence.

6.1. **A source-specific upper-spike certificate.** For the sharp budgets put

$$H^{\varepsilon}_{c,r,\rho}(t) = B^{\varepsilon,\sharp}_{c,r}(t) + G^{\varepsilon,<\rho,\sharp}_{c,r}(t).$$

Fix a roof t and a level $\lambda > 0$. If $m^2 R^{\varepsilon}_c(t) \geq \lambda$ and

$$m^2 H^{\varepsilon}_{c,r,\rho}(t) + Cm^2(1+\varepsilon^{-1}) \int_{t-r}^{t} v^{\rho} \leq \lambda/2, \tag{6.4}$$

then (5.7) proves

$$m^2 R^{\varepsilon}_c(s) \geq \lambda/2 \quad \text{for almost every } s \in (t-r, t). \tag{6.5}$$

This is persistence on an entire prescribed interval, not only a weak bound for the measure of some superlevel set. It is quantitative and holds for the original selector, but condition (6.4) is not asserted uniformly at every roof.

For example, if a fixed interval containing $(t-r, t)$ has $\int m^2 R^{\varepsilon}_c < \lambda r/2$, then an upper spike forces the left side of (6.4) to exceed $\lambda/2$. Alternatively its critical measurable-set norm must be at least $(\lambda/2) r^{31/32}$ unless the same physical budget is large. The already proved source-mass and critical-norm decay therefore force any surviving high spike into a concentration of actual ancestral entrance traces or critical positive optical growth. The noncritical gain has already been paid by (6.2).

6.2. **An upper error estimate in the original topology.** For the full bad source take $c_\sigma = b_\sigma$ and omit c. On the original central target define

$$\begin{aligned} \mathfrak{I}_{\varepsilon,M}(r) &= \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}(M)} B^{\varepsilon}_r(t), \\ \mathfrak{C}_{\varepsilon,M}(r) &= \lim_{\rho \downarrow 0} \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}(M)} G^{\varepsilon,<\rho}_r(t). \end{aligned} \tag{6.6}$$

The latter limit is monotone and extended nonnegative. It is taken *after* the normalized essential supremum and collision limit. These positive quantities are not signed cancellations of exits.

Theorem 6.2 (The scalar upper half with its positive mechanisms separated)**.** *Under the retained analytic inputs, for all large fixed B and fixed $r > 0$, with $\varepsilon(B) = A_0 B^{-1/12}$,*

$$\mathcal{E}_M \leq C_M B^{-1/192} + \mathfrak{I}_{\varepsilon(B),M}(r) + \mathfrak{C}_{\varepsilon(B),M}(r) + C(1+\varepsilon(B)^{-1}) r. \tag{6.7}$$

The first and last terms are estimated. The two positive physical forcing terms are retained, not asserted to vanish. In particular, vanishing of their sum along a legal choice $r(B)$ with $(1+\varepsilon(B)^{-1}) r(B) \to 0$ is sufficient for the original pointwise assertion. Necessity of this additional forcing condition is not claimed.

Proof. Use (6.3) with the original full bad source. Its backward average is at most $2A_r b^{\varepsilon}$. Fix B, r, ρ and every auxiliary band before the collision limit. The fixed-width estimate gives $C\varepsilon^{1/16}$ for the normalized average, while Theorem 6.1 bounds the noncritical gain by $C(1+\varepsilon^{-1})r$. Only then let $\rho \downarrow 0$ in the remaining critical term. No integral or supremum is interchanged with this limit. Finally insert the positive scalar remainder and use $\varepsilon(B)^{1/16} = O(B^{-1/192})$. For example $r(B) = c\varepsilon(B)^2$ makes the explicit last term vanish, but does not estimate either forcing term. Every such width is fixed before its collision limit, with B outermost. □

The entrance measure may contain critical directional atoms; the critical growth may concentrate after the collision normalization. The uniform per-sheet modulus does not bound the complete transverse sum. These are precisely the reasons that (6.7) is not yet a charge-zero theorem. It is a pointwise source comparison, with signed physical losses and artificial prefix losses accounted for before the upper estimate, rather than another bounded normalization or finite-roof-norm conclusion.

7. The original endpoint and the retained consequences

The comparison just proved acts at almost every original roof, not only after integrating starting roofs. It also retains all true entrances and losses. An interior cutoff contributes a negative trace, so the original exceptional union does not acquire an uncontrolled positive cutoff-boundary term. These conclusions do not use the positive-reference denominator or suppress upper amplitudes by taking a reciprocal.

What is not proved is that the two positive forcing terms in (6.7) vanish. A uniform one-sheet modulus cannot be summed with an unestimated angular multiplicity. A critical point's zero area does not remove its possible directional entrance trace. The remaining task is therefore a complete-source estimate of these specific positive mechanisms or another source-specific upper bound. No new unrestricted charge-zero or pointwise likelihood assertion is made here.

All earlier weak, strong subcritical, logarithmic, diffusive-window, one-sided lower, reciprocal and conditioning theorems retain exactly their previous hypotheses. Both revision-88 variants remain distinct. The technical archive includes the full preceding article and every inherited proof; no alternate model, averaged target or pooled label has replaced (2.5).

Appendix A. Proof dependencies

The optical recurrence, normalization and logarithmic sensitivities are those of the retained positive optical polynomial theorem (module 182). The all-contact and original-guard derivatives come from the retained endpoint field and margin theorems (174, 177), and the full angular factor uses the uniform cubic action and angular-density results (180–181, 194). Lemma 4.1 is the new absolute use of that recurrence; it does not improve the logarithmic bound.

The measure parametrization and its incident-component convention are Theorem `thm:v85-segment-map` in module 196. The signed one-dimensional calculation is consistent with the distributional BV convention of [13]. Its full fixed-count integrability is proved in Theorem 4.2, using only qualitative finite geometry there, never a collision-uniform word count. The historical critical angular residue in module 176 is not represented as a new observation.

The positive-cutoff source bound is Proposition `prop:v85-cutoff`, especially equation `eq:v85-cutoff-uniform`, in module 196. Its finite-count remainder explains the retained order of the critical limit. The complete fixed-width bad-source and positive scalar remainder are the retained physical source theorem and module 108. The original exceptional union, mass bound and Borel-prefix representation are those of 204, 209, 212 and 215. Theorem 5.1 adds the sign and trace information rather than changing that selector. The hereditary spectral interfaces remain mathematical inputs, not certificates supplied by source hashes or finite diagnostics.

Part 12. One-sided lower inversion and positive-reference conditioning

Retained revision-92 abstract. We retain the pointwise local limit problem for the complete displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. The retained positive-remainder lower law holds pointwise and uniformly in radius and exact labels. Consequently the actual roof density has a uniform lower bound wherever the finite arithmetic reference has a prescribed positive floor; this conclusion does not require an upper-height estimate. Combining it with the unweakened critical weak limit yields strong convergence of the reciprocal density in every finite local norm. Under the retained positive path-remainder theorem, the same argument gives finite-norm convergence of the exact same-roof collision kernels and quantitative posterior bounds for the original exceptional union. Conditioning on arbitrary moving measurable roof sets with fixed positive reference mass is stable in the joint roof–path bounded-Lipschitz metric. The roof marginals also converge in both directions of relative entropy.

All source normalizations, exact labels and arithmetic classes are retained. Uniform exclusion of upper roof spikes, and hence the full pointwise theorem, remains a separate unproved requirement.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. The arithmetic reference and the target.

For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R, a_j), \quad H_j = -\nabla^2 \log\lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \geq aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} Hv/2}\, \mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\leq n\leq m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The pointwise conclusion and the conditioning conclusions.** The historical lower law, restated in Theorem 4.1, gives

$$\sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} [\mathcal{L}_{m,R} - m^2 p_{n,R}]_+ \longrightarrow 0.$$

This is a pointwise statement, deduced from the sign of the physical remainder rather than from a finite-norm embedding. It proves a uniform physical denominator floor on each quantified positive-reference class. It does not control the opposite sign of the raw error.

The ensuing normalization results retain that distinction. Theorem 4.3 proves every finite-norm limit for the bounded reciprocal error. Theorem 5.3 concerns the bounded-Lipschitz distance between the original same-roof collision path law and its retained bridge reference, not path-space total variation. Theorem 6.3 concerns relative entropy of the roof marginals, not entropy of path laws. Their full central-window versions use exactly $dt/\sqrt{m}$.

These results use neither a new source nor an unproved pointwise upper bound. In particular they do not replace (2.5) by a window theorem. They give additional consequences of the original positive source and its now-established critical weak convergence.

3. Positive-source inputs and the topology of conditioning

The following statements are used in precisely the forms proved in the retained source. They are not inferred from source preservation or numerical diagnostics. Write $\mathcal{C}(M) = \mathcal{C}_{m,R,n,k}(M)$, $P = m^2 p$, $G = \mathcal{L}_{m,R}$ and $Q = 32/31$. For $q > 1$ set

$$\mathcal{N}_q(u; J) = \sup_{E\subset J,\, 0<|E|<\infty} |E|^{-1+1/q} \int_E |u|.$$

This is the measurable-set norm of weak L^q, not strong L^q. The finite exponent Q is not the pointwise endpoint.

Input statement 3.1 (Positive scalar remainder)**.** For each sufficiently large fixed B,

$$P - G = b_B + e_B, \qquad b_B = m^2 b^{\varepsilon(B)} \geq 0, \qquad \varepsilon(B) = A_0 B^{-1/12}, \tag{3.1}$$

where

$$\limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \leq C_M B^{-1/192}.$$

The original finite-peak reference has $|G| \le D_M$ on the central class. The constants are uniform in radius and exact labels. All auxiliary bands are fixed before the collision limit and enlarged only afterwards, with B outermost.

This is Supplement Theorem 181.4, in the normalization restated in Supplement equation (3.5).

Input statement 3.2 (The unweakened critical limit and original union)**.** For each fixed $M, L > 0$,

$$\sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q((P-G)\mathbf{1}_{\mathcal{C}(M)}; J) \longrightarrow 0.$$

The same critical convergence holds after $t = m\bar{\tau}_R + \sqrt{m}\,u$ on $u \in [-M, M]$, with the central indicator retained. Thus the full-window measure is $dt/\sqrt{m}$. For $z = |\nabla_q F|$ and total maximal regular residence ℓ, define

$$U_{a,\delta}(t) = \int \mathbf{1}_{\{z<a \text{ or } \ell<\delta\}}\, d\mu_t.$$

At the cofinal cutoffs $a = h^{1/8}$, $\delta = h^{1/4}$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q(m^2 U_{a,\delta}\mathbf{1}_{\mathcal{C}(M)}; J) \le C_{M,L} h^{31/4096}. \tag{3.2}$$

Every original weighted selector with smaller cutoffs is dominated by this same nonnegative upper source, since $0 \le W^\varepsilon \le 1$. The definition and normalization of the weighted target are unchanged.

These are Supplement Theorem 4.4 and Theorem 5.2. Their proofs use the complete first-exit surface, including every positive residence scale, before taking the critical norm. The inherited finite-norm conclusions do not assume the original essential-height charge is zero.

Input statement 3.3 (The positive path remainder on common roofs)**.** Let $\mathcal{X} = C([0,1], \mathbb{R}^4)$. The actual collision path is tied to its own terminal record, with exactly the convention of Supplement Section 190. Its common conditional kernel is Q_t, and its centered Gaussian bridge reference is W_R. With $\mathbf{P} = P\mathsf{Q}_t$, the original guard restriction has a positive path kernel $0 \le \mathbf{b}_B \le \mathbf{P}$ with mass b_B. The same finite path measures satisfy

$$\mathbf{P} - G\mathsf{W}_R = \mathbf{b}_B + \mathbf{E}_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} \|\mathbf{E}_B\|_{\mathrm{BL}^*} \le C_M B^{-1/192}. \tag{3.3}$$

The bounded-Lipschitz class has supremum norm and Lipschitz constant at most one and includes the constant one. Thus the norm of a positive finite measure is its mass. The scalar remainder may be taken as $e_B = \mathbf{E}_B(1)$ when both inputs are used.

This is Supplement Theorem 190.3, with its cylinder inversion, variation tightness and endpoint-budget hypotheses retained. Supplement equation (76.3) records the elementary comparison subsequently used here. The path input is additional to the scalar input; it is not deduced from scalar weak convergence.

For each fixed m, R, n, k, the original Borel coarea construction fixes common roof kernels outside a null set. A countable path norming class makes the norms in (3.3) measurable on the same version. Parameter uniformity always means $\sup_R \operatorname{ess\,sup}_t$; no single null set for all radii is asserted. Conditional-kernel results below are restricted to $G \ge d > 0$, and moving-set results additionally require a fixed positive reference-mass floor. No statement replaces these quantified restrictions by mere positivity or assigns a normalized conditional law to an arithmetic zero class.

4. One-sided pointwise inversion and the physical denominator

Write $P = m^2 p_{n,R}(k,m,t)$, $G = \mathcal{L}_{m,R}(k_1,k_2,t,n)$ and $f = P - G$. The positive remainder has a consequence in essential supremum which does not require an upper bound for P. All suprema in this section retain the original radius and exact integer labels. The lower law itself is already Supplement Theorem 181.4, equation (181.8). We restate its short proof to make the denominator and normalization arguments self-contained; it is not a newly discovered lower law.

Theorem 4.1 (The pointwise lower half of raw inversion)**.** *Under Input 3.1, for each fixed $M > 0$,*

$$d_m(M) := \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} [G - P]_+ \longrightarrow 0. \tag{4.1}$$

For any fixed $d > 0$ put $D_m(d) = \mathcal{C}(M) \cap \{G \geq d\}$. For all sufficiently large m, uniformly over the parameters,

$$P \geq G - d_m(M) \geq d/2 \quad \text{a.e. on } D_m(d), \qquad \operatorname*{ess\,sup}_{D_m(d)} [G/P - 1]_+ \leq 2d_m(M)/d. \tag{4.2}$$

In particular the original conditional roof kernels have a positive physical denominator on these sets. No ratio is assigned on $G = 0$.

Proof. For each previously fixed reconstruction band B, the exact identity $f = b_B + e_B$, with $b_B = m^2 b^{\varepsilon(B)} \geq 0$, gives $[G - P]_+ \leq |e_B|$. Take the indicated supremum and collision limsup. The result is at most $C_M B^{-1/192}$. Taking $B \to \infty$ proves (4.1). This argument uses positivity, not weak-norm convergence or a pointwise embedding. For large m, $d_m(M) < d/2$; then $P \geq G - d_m(M)$ and $[G/P - 1]_+ = [G - P]_+/P \leq 2d_m(M)/d$. The coarea representatives and null sets are those of the original source. The order remains $\sup_R \operatorname{ess\,sup}_t$. □

The floor d in this theorem is indispensable to its uniform ratio statement. It is not replaced by the unquantified condition $G > 0$. The absolute lower bound (4.1), by contrast, retains all arithmetic classes. It says nothing about the positive part of $P - G$.

Lemma 4.2 (A bounded quantity with a critical weak bound)**.** *Let $q > 1$, $|J| = L < \infty$, $0 \leq u \leq H$ and $\mathcal{N}_q(u;J) \leq A$. For $1 \leq p < q$,*

$$\|u\|_{L^p(J)} \leq \left(\frac{q}{q-p}\right)^{1/p} A L^{1/p - 1/q}. \tag{4.3}$$

For $p > q$,

$$\|u\|_{L^p(J)} \leq \left(\frac{p}{p-q}\right)^{1/p} A^{q/p} H^{1-q/p}. \tag{4.4}$$

When $0 < A \leq HL^{1/q}$,

$$\|u\|_{L^q(J)}^q \leq A^q \left(1 + q \log \frac{HL^{1/q}}{A}\right). \tag{4.5}$$

Zero cases have their continuous interpretation; when $A > HL^{1/q}$, one may replace A by $HL^{1/q}$. In particular, bounded functions converging to zero in $\mathcal{N}_q$ converge strongly in every finite L^p.

Proof. For $\lambda > 0$, testing on $\{u > \lambda\}$ gives $|\{u > \lambda\}| \leq \min\{L, (A/\lambda)^q\}$. Integrate $p\lambda^{p-1}$ times this bound over $0 < \lambda < H$, splitting at $\lambda_0 = AL^{-1/q}$. For $p < q$ extending the upper integral to infinity gives (4.3). For $p > q$, the exact expression is at most $pA^q H^{p-q}/(p-q)$. At $p = q$ it is $A^q + qA^q \log(H/\lambda_0)$. These are the claimed bounds. The natural bound $\mathcal{N}_q(u;J) \leq HL^{1/q}$ justifies the replacement. All statements are for individual finite-count functions. □

Theorem 4.3 (The reciprocal density, without an upper-height premise)**.** *Assume also Input 3.2. On $D_m(d)$ define $r_m = |1 - G/P|$, and set it to zero outside this set. For all sufficiently large m,*

$$0 \le r_m \le 1, \qquad \mathcal{N}_Q(r_m; J) \le \frac{2}{d}\mathcal{N}_Q(f\mathbf{1}_{\mathcal{C}(M)}; J), \qquad Q = 32/31. \tag{4.6}$$

Consequently $r_m \to 0$ in every finite L^p on each fixed roof window, uniformly in its location and the exact parameters. The same conclusion holds on the complete central window in measure $dt/\sqrt{m}$. No strong critical-norm conclusion for the unbounded raw error *f is used or inferred.*

Proof. Equation (4.2) gives $r_m \le 2|f|/d$. If $P \ge G$ then $r_m = 1 - G/P \le 1$. If $P < G$ then $r_m \le 2d_m(M)/d \le 1$ for large m. The weak bound follows by positive domination. Apply Lemma 4.2 and the inherited unweakened critical convergence. For the diffusive window, use the same pointwise inequalities after the change $t = m\bar{\tau}_R + \sqrt{m}\,u$. The inherited critical norm there already includes its Jacobian. The domain in u is contained in $[-M, M]$, so the bounded-function lemma applies on that fixed interval. □

5. Positive physical disintegration and posterior stability

The lower denominator bound permits a limited, but unconditional, normalization theorem at positive-reference roofs. Its topology must be distinguished from uniform pointwise convergence of the conditional laws. The latter still requires exclusion of the upper spikes.

Let $\mathcal{X}$ and W_R have exactly the centered collision path convention of Input 3.3. Write

$$\mathbf{P} = P\mathsf{Q}_t, \qquad \mathfrak{d}_m(t) = \|\mathbf{P} - G\mathsf{W}_R\|_{\mathrm{BL}^*}.$$

The constant test belongs to the bounded-Lipschitz class. A countable norming class and the inherited common kernels fix measurability of all displayed norms before any supremum is taken.

Theorem 5.1 (Pointwise identification of the numerator excess)**.** *For every fixed M,*

$$0 \le \mathfrak{d}_m - |P - G|, \qquad \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} (\mathfrak{d}_m - |P - G|) \longrightarrow 0. \tag{5.1}$$

In particular $\mathfrak{d}_m$ tends to zero in the unweakened critical weak norm, locally and in the complete diffusive window. This theorem identifies the two discrepancies up to a uniform additive error; it does not assert that either discrepancy is uniformly small.

Proof. At every previously fixed B, the positive physical path remainder satisfies

$$\mathbf{P} - G\mathsf{W}_R = \mathbf{b}_B + \mathbf{E}_B, \qquad \|\mathbf{b}_B\|_{\mathrm{BL}^*} = \mathbf{b}_B(1) = b_B.$$

If $\eta_B(t) = \|\mathbf{E}_B(t)\|_{\mathrm{BL}^*}$, the constant test and positivity give

$$|P - G| \le \mathfrak{d}_m \le b_B + \eta_B \le |P - G| + 2\eta_B.$$

The ordered essential height of η_B is at most $C_M B^{-1/192}$. Take the collision limsup first, and then $B \to \infty$. This proves (5.1) without subtracting infinite limsup heights. The discrepancies are finite almost everywhere at each finite count. The critical-norm conclusion follows from the uniform additive error and Input 3.2, using the fixed measure of the local or rescaled window. □

Proposition 5.2 (The exact physical mixture)**.** *On $D_m(d)$, fix B sufficiently large and then m sufficiently large that $\eta_B \le d/2$. Put*

$$A_B = P - b_B, \qquad \beta_B = b_B/P.$$

The measure $\mathbf{A}_B = \mathbf{P} - \mathbf{b}_B$ is nonnegative, $A_B = G + \mathbf{E}_B(1) \ge d/2$, and

$$\mathsf{Q}_t = (1 - \beta_B)\mathsf{Q}_{B,t}^{\mathrm{good}} + \beta_B \mathsf{Q}_{B,t}^{\mathrm{bad}}, \qquad \mathsf{Q}_{B,t}^{\mathrm{good}} = \mathbf{A}_B/A_B. \tag{5.2}$$

The bad kernel can be assigned arbitrarily where $b_B = 0$. Moreover,

$$d_{\mathrm{BL}}(\mathsf{Q}_{B,t}^{\mathrm{good}}, \mathsf{W}_R) \le 4\eta_B/d, \tag{5.3}$$

$$d_{\mathrm{BL}}(\mathsf{Q}_t, \mathsf{W}_R) \le 2\beta_B + 4\eta_B/d, \tag{5.4}$$

$$|\beta_B - (1 - G/P)| \le 2\eta_B/d. \tag{5.5}$$

The weights in (5.2) *are the posterior masses of the original guard and its complement, not a new source normalization.*

Proof. The positive path remainder is the pushforward of the original guard restriction. Since $0 \le W^{\varepsilon(B)} \le 1$, its complement is a positive submeasure of $\mathbf{P}$. Subtracting that remainder gives $\mathbf{A}_B = G\mathsf{W}_R + \mathbf{E}_B$ and the mass identity. After normalization,

$$\|\mathsf{Q}_{B,t}^{\mathrm{good}} - \mathsf{W}_R\|_{\mathrm{BL}^*} \le \frac{\eta_B + |\mathbf{E}_B(1)|}{A_B} \le 4\eta_B/d.$$

Disintegration of the two positive restrictions proves the exact mixture, and the distance between two probability measures in this bounded-Lipschitz convention is at most two. Finally $\beta_B = 1 - G/P - \mathbf{E}_B(1)/P$ and $P \ge d/2$. All identities use the original common coarea version. □

Theorem 5.3 (Finite-norm stability of same-roof collision kernels)**.** *Under Inputs 3.1–3.3, for every fixed $d, M, L > 0$ and every finite $p \ge 1$,*

$$\sup_{R,n,k} \sup_{|J|=L} \left\| \mathbf{1}_{D_m(d)} d_{\mathrm{BL}}(\mathsf{Q}_t, \mathsf{W}_R) \right\|_{L^p(J)} \longrightarrow 0. \tag{5.6}$$

The same assertion holds on the complete central window with measure $dt/\sqrt{m}$. At each roof the distance already takes the supremum over all bounded-Lipschitz path tests; the exceptional roof is not chosen separately for each test. This is not path-space total variation, not essential-supremum convergence in the roof, and not a new collision-to-return clock theorem.

Proof. Let ρ_m be the supremum of the nonnegative gap in (5.1); then $\rho_m \to 0$. For $P > 0$,

$$P(\mathsf{Q}_t - \mathsf{W}_R) = (\mathbf{P} - G\mathsf{W}_R) + (G - P)\mathsf{W}_R.$$

On $D_m(d)$ this gives

$$d_{\mathrm{BL}}(\mathsf{Q}_t, \mathsf{W}_R) \le \min\{2,\, 2|1 - G/P| + 2\rho_m/d\}. \tag{5.7}$$

Apply Theorem 4.3; the additive error is uniform. The diffusive assertion is the same inequality in rescaled coordinates. The countable norming class fixes the discrepancy as one measurable function of the roof and therefore includes all path tests before integration. No independent numerator convergence premise is inserted at this step: it was deduced from the stated positive path remainder. □

5.1. **Posterior mass of the original exceptional union.** For arbitrary $\varepsilon > 0$ and positive cutoffs a, δ, let

$$H_{a,\delta}^{\varepsilon}(t) = m^2 \int W^{\varepsilon}(x) \mathbf{1}_{\{z(x) < a \text{ or } \ell(x) < \delta\}}\, d\mu_t(x).$$

On $D_m(d)$ its posterior mass is $\pi_{a,\delta}^{\varepsilon} = H_{a,\delta}^{\varepsilon}/P$. Extend this posterior by zero outside $D_m(d)$. It belongs to $[0, 1]$. Its two alternatives form one original union; the intersection is not counted twice.

Corollary 5.4 (All finite posterior moments of the same source)**.** *Put $a = h^{1/8}$ and $\delta = h^{1/4}$. For every finite $p \ge 1$,*

$$\limsup_{m} \sup_{R,n,k} \sup_{|J|=L} \sup_{\varepsilon > 0} \|\pi_{a,\delta}^{\varepsilon} \mathbf{1}_{D_m(d)}\|_{L^p(J)} \le C_{M,L,d,p} \begin{cases} h^{31/4096}, & p < Q, \\ h^{31/4096}(1 + |\log h|)^{1/Q}, & p = Q, \\ h^{1/(128p)}, & p > Q. \end{cases} \tag{5.8}$$

The same bound dominates every smaller pair of cutoffs, without assuming monotonicity of W^{ε} in ε. No conclusion at $p = \infty$ is asserted.

Proof. The original weighted union is bounded by the unweighted union U in Input 3.2, for every ε. Thus its posterior is at most $\min\{1, 2m^2U/d\}$ on $D_m(d)$. The critical norm of this common upper function has ordered bound $C_{M,L,d}h^{31/4096}$. Apply Lemma 4.2 to the upper function at every finite count, and then take the limsup. For $p > Q$ use $Q(31/4096) = 1/128$. At $p = Q$ the logarithm is kept. Positive domination also covers all smaller cutoffs simultaneously. □

6. Moving roof selections and relative entropy

We now distinguish an exact same-roof kernel from a law conditioned on a measurable set of roofs. The latter has a useful stability theorem even though uniform pointwise inversion has not been established. The scalar labels remain fixed throughout; no neighboring collision, displacement or occupation classes are averaged together.

There are two window conventions. Locally, K_m is any interval of fixed length L and $d\lambda_m = dt$. On the full central window, $K_m = [m\bar{\tau}_R - M\sqrt{m}, m\bar{\tau}_R + M\sqrt{m}]$ and $d\lambda_m = dt/\sqrt{m}$. In either case $\lambda_m(K_m) \le V$ for a fixed V. All sources below are restricted to the central class.

Fix $d, \zeta > 0$. Let $E_m \subset K_m \cap D_m(d)$ be any measurable set, possibly depending on every finite-count parameter, with

$$H_E := \int_{E_m} G\, d\lambda_m \ge \zeta. \tag{6.1}$$

Put $F_E = \int_{E_m} P\, d\lambda_m$. The two roof laws and two joint roof–path laws are

$$d\mathbb{P}_E^{\mathrm{roof}} = \mathbf{1}_{E_m} P\, d\lambda_m / F_E, \qquad d\mathbb{G}_E^{\mathrm{roof}} = \mathbf{1}_{E_m} G\, d\lambda_m / H_E,$$
$$d\mathbb{P}_E(t,x) = d\mathbb{P}_E^{\mathrm{roof}}(t)\, d\mathsf{Q}_t(x), \qquad d\mathbb{G}_E(t,x) = d\mathbb{G}_E^{\mathrm{roof}}(t)\, d\mathsf{W}_R(x).$$

The mass floor in (6.1) allows irregular and moving roof sets, but does not allow their reference mass to collapse to zero. No reference law is constructed on an arithmetic zero class.

Theorem 6.1 (Uniform stability under measurable roof conditioning)**.** *Under the three stated inputs, uniformly over all sets satisfying* (6.1)*, one has $F_E/H_E \to 1$ and*

$$\|\mathbb{P}_E^{\mathrm{roof}} - \mathbb{G}_E^{\mathrm{roof}}\|_{\mathrm{TV}} \longrightarrow 0. \tag{6.2}$$

For the joint laws, the discrepancy tested by every Borel $\Phi(t,x)$ whose path sections belong to the unit bounded-Lipschitz class also tends to zero, uniformly over that whole test class and over E_m. No regularity of the test in t is imposed. This joint conclusion is not total variation on path space.

Proof. Set

$$a_m = \int_{K_m \cap \mathcal{C}(M)} |P - G|\, d\lambda_m, \qquad v_m = \int_{K_m \cap \mathcal{C}(M)} \mathfrak{d}_m\, d\lambda_m.$$

Inputs 3.2 and Theorem 5.1 give $a_m, v_m \to 0$ uniformly in either window convention. In particular $|F_E - H_E| \le a_m$, so $F_E \ge \zeta/2$ eventually. Subtract the joint finite measures before normalizing. Their tested error is at most

$$\frac{v_m + |F_E - H_E|}{F_E} \le \frac{2}{\zeta}(v_m + a_m). \tag{6.3}$$

The same computation with scalar densities gives $\|\mathbb{P}_E^{\mathrm{roof}} - \mathbb{G}_E^{\mathrm{roof}}\|_{\mathrm{TV}} \le 2a_m/F_E$. These bounds do not depend on the geometry of E_m. Measurability follows from the original common kernels; alternatively a countable path norming class bounds every Borel path-section test by $\mathfrak{d}_m$ before the roof integral. The proof neither chooses a separate exceptional roof for each test nor normalizes any zero reference. □

Lemma 6.2 (Entropy bounds without an upper likelihood bound)**.** *For every $1 < p < 2$ there is $C_p < \infty$ such that*

$$x \log x - x + 1 \le C_p |x-1|^p \quad (x \ge 0), \qquad x - 1 - \log x \le C_p |x-1|^p \quad (x \ge 1/3). \tag{6.4}$$

The first expression is interpreted continuously at zero. The lower restriction in the second inequality is essential; no upper bound on x occurs in either inequality.

Proof. On $|x-1| \le 1/2$, Taylor's integral formula and a bound for the second derivatives give $C|x-1|^2 \le C|x-1|^p$. On the remaining compact intervals the ratios in (6.4) are continuous and bounded. For $x \ge 2$, the elementary bound $\log x \le x^{p-1}/(p-1)$ gives the first estimate after enlarging C_p. The second left side is at most x there, which is bounded by $2^p|x-1|^p$. This proves both assertions. □

Theorem 6.3 (Two-sided roof relative-entropy convergence)**.** *For the roof laws of Theorem 6.1, and any fixed $1 < p < Q = 32/31$, for all sufficiently large m,*

$$\begin{aligned} &\mathrm{KL}(\mathbb{P}_E^{\mathrm{roof}} \| \mathbb{G}_E^{\mathrm{roof}}) + \mathrm{KL}(\mathbb{G}_E^{\mathrm{roof}} \| \mathbb{P}_E^{\mathrm{roof}}) \\ &\qquad \le C_{M,V,d,\zeta,p} \int_{K_m \cap \mathcal{C}(M)} |P - G|^p \, d\lambda_m \longrightarrow 0. \end{aligned} \tag{6.5}$$

The bound is uniform over all permitted moving sets and exact labels. The statement concerns the one-dimensional roof marginals only. It asserts neither relative entropy between path laws nor uniform convergence of the forward likelihood.

Proof. Both roof densities are positive almost everywhere on E_m, by Theorem 4.1. Their forward likelihood is

$$R_E = \frac{P}{G} \frac{H_E}{F_E}.$$

Uniformly, $F_E/H_E \to 1$ and $P/G \ge 1 - d_m(M)/d$. Thus $R_E \ge 1/3$ for large m, while no upper likelihood bound is needed. Also

$$|R_E - 1| \le \frac{2}{d}|P - G| + \frac{2}{\zeta} a_m. \tag{6.6}$$

Since $G \le D_M$ on the central class and $H_E \ge \zeta$, Hölder on a set of measure at most V gives

$$\int |R_E - 1|^p \, d\mathbb{G}_E^{\mathrm{roof}} \le C_{M,V,d,\zeta,p} \int_{K_m \cap \mathcal{C}(M)} |P - G|^p \, d\lambda_m.$$

The right side tends to zero because unweakened weak L^Q convergence implies strong L^p convergence on the fixed local or rescaled window when $p < Q$.

The identity $\int R_E \, d\mathbb{G}_E^{\mathrm{roof}} = 1$ rewrites the two relative entropies as the integrals of $R_E \log R_E - R_E + 1$ and $R_E - 1 - \log R_E$, respectively. Apply Lemma 6.2. The same inequalities prove finiteness of these entropies for large m. No boundedness of R_E from above and no strong raw L^Q convergence is used. Since actual collision paths and the reference paths need not be mutually absolutely continuous, none of this argument is applied to path-space Radon–Nikodym derivatives. □

Remark 6.4 (The unremoved upper spike)*.* The pointwise lower bound, the bounded reciprocal, conditional-kernel finite norms, and roof entropy convergence are compatible with an upper spike of fixed height on a shrinking roof set. For example, $G = 1$ and $P_j = 1 + \mathbf{1}_{(0,\delta_j)}$, with $\delta_j \downarrow 0$, have all these properties while $\|P_j - G\|_\infty = 1$. This example is a topology check, not a physical counterexample. The remaining source-specific task is still to prove that the original upper exceptional-source essential-height charge is zero. None of the new conditional conclusions supplies that missing persistence theorem.

7. The remaining pointwise requirement

The retained estimate controls the absolute negative part in the original uniform essential-supremum norm. Thus the unresolved part of (2.5) is entirely an upper-excess problem. Its original weighted-source charge and its order of limits are unchanged. Neither Theorem 4.1 nor the conditional finite-norm and entropy results proves that this charge is zero. The upper-pulse example in Remark 6.4 makes the remaining distinction explicit.

Every reconstruction band is fixed before the collision limit and is sent to infinity last. Reference floors and reference-mass floors are also fixed, not chosen as shrinking functions of collision count. The conditional assertions are uniform on those stated classes, not on all sets of arbitrarily small positive mass. The actual-return clock and any uniform forward-likelihood assertion retain their separate hypotheses. No arithmetic zero class is normalized.

Appendix A. The active dependency map

The mathematical baseline is the complete revision-91 paper tree:

`5637513690e13d4abf1a46eedb568327648a0b1f`

The three inputs in Section 3 separate the scalar, critical-distribution and path assumptions. The restated pointwise lower bound uses only the first; no path theorem is needed to establish the scalar denominator floor.

Assertion used here	Exact retained source and role
Positive scalar error	Supplement Theorem 181.4; original nonnegative guard remainder and ordered uniform error.
Unweakened critical limit	Supplement Theorem 4.4 and Corollary 4.5; no logarithmic loss, with $dt/\sqrt{m}$ on the complete window.
Original union estimate	Supplement Theorem 5.2; the same critical-or-short-residence source, with one positive upper selector.
Positive path remainder	Supplement Theorem 190.3; actual path submeasure, cylinder inversion, endpoint budget and tightness.
Common path comparison	Supplement equation (76.3) and Theorem 72.1; common kernels, bounded-Lipschitz normalization and the original positive arithmetic restrictions.

Theorem 4.1 precedes every new ratio. The bounded-function lemma then permits finite-norm conditioning; it does not improve the regularity of the unbounded raw density. The roof entropy argument uses $1 < p < 32/31$ and a fixed reference floor. It does not assert absolute continuity of path probabilities or provide a return-clock theorem. Historical source snapshots and execution receipts are outside the mathematical proof. Their preservation is not an independent audit of the inherited continuum inputs.

Part 13. The unweakened critical limit and all-radius source control

Retained revision-91 abstract. We retain the pointwise local limit problem for the exact displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. We prove convergence of the original raw error in the unweakened critical weak norm of exponent 32/31, removing the previous logarithmic loss. The decisive step glues the complete surface estimate on roof windows matched to the exit width before taking the collision limit; the resulting surface norm decays as $r^{1/32}$ on every fixed window. A positive domination argument then gives the original critical-or-short-residence source a polynomial critical-norm bound, uniformly over its smaller cutoffs and its original weights. A maximal version controls every local averaging radius outside one quantified exceptional roof set. The full central-window limit is stated in

diffusive roof measure. All exact labels, arithmetic classes and prior proofs are retained. These finite-critical conclusions do not assert that the remaining essential-height charge is zero.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. The arithmetic reference and the target.

For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar{\tau}_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R, a_j), \quad H_j = -\nabla^2 \log\lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \geq aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} Hv/2}\, \mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\leq n\leq m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The unweakened critical limit.** The new critical surface decay is proved in Theorem 4.2. Its input is a scale-matched bound already valid for the complete one-sided first-exit surface. A finite partition into windows of that same length supplies the missing power on a fixed roof interval. Theorem 4.4 then proves

$$\sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{32/31}((m^2 p - \mathcal{L})\mathbf{1}_{\mathcal{C}(M)}; J) \longrightarrow 0$$

for every fixed $M, L > 0$, without a logarithmic denominator. This is a theorem at a finite distribution exponent, not a redefinition of (2.5). The same result holds on the complete central window with the explicit measure $dt/\sqrt{m}$.

For the original weighted exceptional source, the cofinal cutoffs $z < h^{1/8}$ or $\ell < h^{1/4}$ give a critical norm of order $h^{31/4096}$ in the ordered collision limit. The bound concerns the union itself, not just common survivors. The all-radius consequence estimates a single exceptional roof set, while leaving the absence of essential-height concentration as a separate pointwise obligation.

3. The retained inputs and the two endpoint topologies

All densities below are those of the original exact event. For $q > 1$, an interval J, and a measurable function f, write

$$\mathcal{N}_q(f; J) = \sup_{E\subset J,\, 0<|E|<\infty} |E|^{-1+1/q} \int_E |f(t)|\, dt. \tag{3.1}$$

This is the measurable-set norm for weak L^q, not the strong L^q norm. Set $Q = 32/31$, $\gamma = 1/32$ and $\alpha = 129/128$. The word "critical" below refers to this finite exponent, never to the essential-supremum target (2.5).

Input statement 3.1 (Original source comparison)**.** The physical stopped flow has $XF = 1$, with $F = L$, and preserves the endpoint-momentum angle. Its maximal regular segments have total residence $\ell = \tau^- + \tau^+$, gradient magnitude $z = |\nabla_q F|$, and physical angular density $\omega/(4\pi c_*\chi)$. Nongrazing outgoing faces have the same one-sided density, their incident components are retained, and their complete first-exit assignment is

$$\mathcal{F}_h(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<b-t\leq h,\ b-t<\tau^-(y)\}}\, d\zeta_+(y), \qquad \overline{\mathcal{F}}_r = \frac{1}{r}\int_0^r \mathcal{F}_h\, dh.$$

Here b denotes the outgoing roof, and $d\zeta_+$ is the physical outgoing flux. Put $I_m = 1+\sum_{i=1}^{m-1} q_0^{d_i}/c_i$ and

$$v^{\eta,a}(t)\,dt = F_{\#}(I_m z^{-\eta}\mathbf{1}_{\{0<z<a\}}\mathbf{1}_{\Pi}\nu_R^*), \quad v^{\eta}(t)\,dt = F_{\#}(I_m z^{-\eta}\mathbf{1}_{\Pi}\nu_R^*).$$

For $\eta = 1/4$, let $v = v^{1/4}$ and $\mathsf{P}_r v(t) = \int_0^r s^{-3/8} v(t+s)\,ds$. The complete source satisfies

$$\begin{aligned} b^{\varepsilon}(t) &\le 2A_r b^{\varepsilon}(t) + C\overline{\mathcal{F}}_r(t) + C\mathsf{P}_r v(t) \\ &\quad + C_{\Theta}(1+\varepsilon^{-1})\sqrt{r}\,A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\,du. \end{aligned} \tag{3.2}$$

This holds for each fixed $r > 0$ at almost every roof, without a positive incidence, gradient or residence cutoff.

This is the retained physical source-shift inequality combined with the two-sided optical payment of the actual first-exit surface. True exits and initial weights are included before the comparison; the surface is not a volume collar.

Input statement 3.2 (Local estimates and positive remainder)**.** For every previously fixed positive interval length and every fixed $\varepsilon > 0$,

$$\begin{gathered} \limsup_m \sup_{R,n,k} m^2 \|A_r b^{\varepsilon}\|_{\infty} \le C\varepsilon^{1/16}, \\ \limsup_m \sup_{R,n,k} m^2 \|A_r p\|_{\infty} \le C, \\ \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v \le CL, \\ \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{1/4,a} \le Ca^{\alpha-1/4}L, \\ \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J\int \mathbf{1}_{\{\ell<\delta\}}\,d\mu_t\,dt \le C\delta^{\gamma}L. \end{gathered} \tag{3.3}$$

The leading constants are independent of the previously fixed widths. The retained critical surface theorem, evaluated at window length r, gives

$$\limsup_m \sup_{R,n,k} \sup_{|I|=r} \mathcal{N}_Q(m^2\overline{\mathcal{F}}_r; I) \le Cr. \tag{3.4}$$

Its underlying estimate is $C\{L^{31/32}r^{1/32} + (L+4r)(1+r^{19/32})\}$ on windows of length L. It sums the complete capped residence tail at finite collision count. No logarithmic norm occurs in this bound.

For sufficiently large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.5}$$

The finite-peak reference satisfies $|\mathcal{L}| \le D_M$ on $\mathcal{C}(M)$, uniformly in all labels, radii and collision counts. Critical points have zero source measure.

The source comparison and moment inputs are restated in the retained article's geometric-input section. The matched estimate is precisely the conclusion of its critical-surface corollary, not a new surface spectral hypothesis. The new proofs below add only finite interval gluing, positive source comparison and real-variable maximal estimates. The full four-record spectral chain remains an inherited input.

For each m, R, n, k, the Borel maximal-segment construction fixes common roof kernels outside a null set and common roof–time kernels outside a joint null set. Every positive restriction

uses those kernels. The parameter convention is $\sup_R \operatorname{ess\,sup}_t$, not one common null set for all radii. In each inherited local theorem the auxiliary band is fixed before $m \to \infty$ and enlarged afterwards. In the new scale-gluing argument r and the number of its subintervals are also fixed before that limit; B in (3.5) remains outermost.

4. Scale gluing at the critical distribution exponent

The critical surface estimate has more information than a bound on one fixed window. Its constant is linear in the length of a window matched to the exit width. A finite partition retains this gain.

Lemma 4.1 (Gluing with the local scale retained)**.** *Let $q > 1$, $\beta = 1 - 1/q$, and partition an interval J into N measurable pieces, each contained in an interval on which $\mathcal{N}_q(f;\cdot) \le A$. Then*

$$\mathcal{N}_q(f;J) \le N^{1/q} A. \tag{4.1}$$

Consequently, for source families $f_{m,r}$, if

$$\limsup_m \sup_{R,n,k} \sup_{|I|=r} \mathcal{N}_q(f_{m,r};I) \le C r^{\zeta},$$

then, for every fixed $L, r > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(f_{m,r};J) \le C(L+r)^{1/q} r^{\zeta - 1/q}. \tag{4.2}$$

The interval partition is chosen before the collision limit.

Proof. For $E \subset J$ put $v_i = |E \cap J_i|$. At each finite count,

$$\int_E |f| \le A \sum_{i=1}^{N} v_i^{\beta} \le A N^{1-\beta} |E|^{\beta}.$$

This is Hölder's inequality on a finite counting measure. Divide by $|E|^{\beta}$ and take the measurable-set supremum. For the second assertion partition into $N = \lceil L/r \rceil$ pieces of length at most r and enlarge the last piece for its upper estimate. Now $N \le (L+r)/r$. The first inequality is valid before any parameter supremum, so it also holds for moving interval locations. For fixed L, r, N is independent of m. No infinite sum of collision limsups occurs. □

Theorem 4.2 (Complete critical surface decay)**.** *For every fixed $L > 0$ and sufficiently small fixed $r > 0$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q(m^2 \overline{\mathcal{F}}_r;J) \le C(L+r)^{31/32} r^{1/32}. \tag{4.3}$$

Every positive residence length, every incoming endpoint type and all one-sided incident physical components are included.

Proof. The retained critical surface inequality, evaluated with $|I| = r$, gives (3.4). Apply Lemma 4.1 with $q = Q$, $\zeta = 1$ and $f_{m,r} = m^2 \overline{\mathcal{F}}_r$. The local upper bound already sums all residence scales at finite collision count. The new partition concerns roof intervals only; it neither partitions the physical source into a selected word nor introduces an additional residence cutoff. □

Lemma 4.3 (The fractional potential at the same scale)**.** *For $v = v^{1/4}$ and $\mathsf{P}_r v(t) = \int_0^r s^{-3/8} v(t+s)\, ds$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \|m^2 \mathsf{P}_r v\|_{L^Q(J)} \le C(L+r)^{31/32} r^{5/8}. \tag{4.4}$$

Proof. On an interval I of length r, Minkowski's inequality gives

$$\|m^2\mathsf{P}_r v\|_{L^Q(I)} \le (1-3Q/8)^{-1/Q} r^{1/Q-3/8} m^2 \int_{I+(0,r)} v.$$

Here $3Q/8 = 12/31 < 1$. The fixed-width source estimate bounds the ordered mass on the right by Cr, uniformly in its location. Thus the local L^Q norm is at most $Cr^{1/Q+5/8}$ in the collision limsup. Partition J into at most $(L+r)/r$ such intervals, sum the Qth powers at finite count and take the limit. This proves (4.4). The width r stays fixed through the collision and auxiliary-band limits. □

Theorem 4.4 (The critical raw limit without logarithmic loss)**.** *Under the retained inputs, for every fixed $M, L > 0$,*

$$\sup_{R,n,k}\sup_{|J|=L} \mathcal{N}_Q((m^2 p - \mathcal{L})\mathbf{1}_{\mathcal{C}(M)}; J) \longrightarrow 0. \tag{4.5}$$

More precisely, for every fixed $\varepsilon > 0$,

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} \mathcal{N}_Q(m^2 b^\varepsilon; J) \le C L^{31/32}\varepsilon^{1/16}. \tag{4.6}$$

Neither conclusion asserts strong L^Q or essential-supremum convergence.

Proof. Apply $\mathcal{N}_Q$ to the positive source comparison (3.2). Its triangle inequality follows directly from its definition. For an essentially bounded function, $\mathcal{N}_Q(f;J) \le L^{1/Q}\|f\|_\infty$; also $\mathcal{N}_Q(f;J) \le \|f\|_{L^Q(J)}$. The fixed-width estimates, Theorem 4.2 and Lemma 4.3 therefore give

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} \mathcal{N}_Q(m^2 b^\varepsilon; J) \le C_\Theta (L+r)^{31/32}\{\varepsilon^{1/16} + r^{1/32} + r^{5/8} + (1+\varepsilon^{-1})\sqrt{r}\}. \tag{4.7}$$

All norms are applied to the complete source before limits. Let $r \downarrow 0$ only after the collision limit and the permitted auxiliary-band enlargement, with ε fixed. This proves (4.6).

Use the original positive scalar remainder with $\varepsilon(B) = A_0 B^{-1/12}$. Restricting to $\mathcal{C}(M)$ can only decrease a positive norm. The remainder has critical norm at most $C_M L^{31/32} B^{-1/192}$ in the ordered limit, as does the bad-source term. Send $B \to \infty$ last to prove (4.5). Equivalently choose $r(B) = c\varepsilon(B)^4$ in (4.7), fixed before $m \to \infty$; all its explicit errors are $O(\varepsilon(B)^{1/16})$. No frequency band depends on the collision count. □

Corollary 4.5 (The full central window)**.** *For $H = [-M, M]$ put*

$$\widetilde{f}_{m,R,n,k}(u) = (m^2 p - \mathcal{L})(m\bar{\tau}_R + \sqrt{m}\,u)\mathbf{1}_{\mathcal{C}(M)}(m\bar{\tau}_R + \sqrt{m}\,u).$$

Then $\sup_{R,n,k} \mathcal{N}_Q(\widetilde{f}_{m,R,n,k}; H) \to 0$. In roof coordinates the measure is $dt/\sqrt{m}$, not dt.

Proof. Partition the central roof window into at most $2M\sqrt{m}+2$ intervals of length at most one. If $d_m \to 0$ is the uniform local norm from (4.5), the glued norm is at most $d_m(2M\sqrt{m}+2)^{1/Q}$. Changing to u multiplies the norm by $m^{-1/(2Q)} = m^{-31/64}$. The product tends to zero. The growing number of intervals is paid explicitly, not treated as fixed. □

5. A power bound on the original exceptional union

The unweakened convergence can now be applied directly to positive subsources of the original density. This improves the logarithmic estimate of the exceptional union without modifying that union.

Lemma 5.1 (Domination by a bounded reference and a small error)**.** *Let $q > 1$, $\beta = 1 - 1/q$, and let $g, e \geq 0$ on an interval J. If $g \leq D + e$ almost everywhere, $D \geq 0$, and $\int_J g \leq \mu$, then*

$$\mathcal{N}_q(g; J) \leq D^\beta \mu^{1/q} + \mathcal{N}_q(e; J). \tag{5.1}$$

At $D = 0$ the first term is read as zero.

Proof. For $E \subset J$, $x = |E| > 0$, one has

$$\int_E g \leq \min\{\mu, Dx + \int_E e\} \leq \min(\mu, Dx) + \int_E e.$$

The supremum of $x^{-\beta} \min(\mu, Dx)$ over $x > 0$ is $D^\beta \mu^{1-\beta}$, attained at $x = \mu/D$ when $D\mu > 0$. Restricting to $x \leq |J|$ cannot increase it. The zero cases follow directly. This is a finite-function inequality, not an interchange of parameter suprema and limits. □

For positive a, δ retain the actual unweighted source

$$U_{a,\delta}(t) = \int \mathbf{1}_{\{z<a \text{ or } \ell<\delta\}}\, d\mu_t, \qquad z = |g|, \quad \ell = \tau^- + \tau^+. \tag{5.2}$$

The original weighted source $R^\varepsilon_{a,\delta}$ inserts W^ε in precisely this integral. Thus $0 \leq R^\varepsilon_{a,\delta} \leq U_{a,\delta} \leq p$. The unweighted union is used only for an upper comparison; it is not a replacement of the weighted target.

Theorem 5.2 (Unweakened critical smallness of the union)**.** *Put $\alpha = 129/128$ and $\gamma = 1/32$. For every fixed $M, L > 0$ and sufficiently small fixed $a, \delta > 0$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q(m^2 U_{a,\delta} \mathbf{1}_{\mathcal{C}(M)}; J) \leq C_{M,L}(a^\alpha + \delta^\gamma)^{1/Q}. \tag{5.3}$$

The same bound holds uniformly for every $\varepsilon > 0$ and every weighted source $R^\varepsilon_{a',\delta'}$ with $a' \leq a$, $\delta' \leq \delta$. In particular the original cofinal choice $a = h^{1/8}$, $\delta = h^{1/4}$ satisfies

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q(m^2 R^\varepsilon_{h^{1/8},h^{1/4}} \mathbf{1}_{\mathcal{C}(M)}; J) \leq C_{M,L} h^{31/4096}. \tag{5.4}$$

All cutoff choices precede the collision limit.

Proof. The fractional source input with $\eta = 1/4$ and $I_m \geq 1$ gives

$$\int_J \int \mathbf{1}_{\{0<z<a\}}\, d\mu_t\, dt \leq a^{1/4} \int_J v^{1/4,a}.$$

The critical points themselves have zero source measure. The leading ordered bound for this term is CLa^α. The complete short-residence input bounds the second part of the union by $CL\delta^\gamma$. Positivity therefore gives

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J U_{a,\delta} \leq C_L(a^\alpha + \delta^\gamma). \tag{5.5}$$

The definition uses the union once; the sum is solely an upper bound. Neither estimate pools exact occupation coefficients.

At finite count set $g_m = m^2 U_{a,\delta} \mathbf{1}_{\mathcal{C}(M)}$ and $e_m = |m^2 p - \mathcal{L}| \mathbf{1}_{\mathcal{C}(M)}$. The retained reference bound implies $g_m \leq C_M + e_m$. Apply Lemma 5.1, then Theorem 4.4 and (5.5). Since $1/Q = 31/32$, this proves (5.3). Monotonicity of the unweighted union bounds all smaller cutoffs and all original weights at once, without assuming monotonicity of Θ or of W^ε in ε. Finally $a^\alpha = h^{129/1024}$ and $\delta^\gamma = h^{1/128}$; the latter is the larger term for $h < 1$. Raising it to $1/Q$ gives (5.4). □

The gain is polynomial, rather than the earlier inverse power of a logarithm, and acts on the original exceptional source itself. The exponent 31/4096 is a consequence of the available mass exponent and the critical norm; no optimality is claimed. A fixed-height pulse on a shrinking roof interval still satisfies this topology. Thus (5.4) is not an essential-height estimate.

6. Simultaneous control of all local roof averages

The exceptional-source estimate admits a maximal form. Its conclusion is simultaneous over all roof averaging radii, but the exceptional roof set need not be empty. We keep this distinction explicit.

Lemma 6.1 (A local maximal inequality in the critical norm)**.** *Let $q > 1$ and extend $g \geq 0$ by zero outside a bounded interval K. Define*

$$\mathcal{M}g(t) = \sup_{I \ni t} \frac{1}{|I|} \int_I g(u)\, du,$$

where the supremum is over bounded intervals. If $\mathcal{N}_q(g; K) \leq A$, then

$$|\{\mathcal{M}g > \lambda\}| \leq 3(A/\lambda)^q, \qquad \lambda > 0. \tag{6.1}$$

For every bounded interval J, $\mathcal{N}_q(\mathcal{M}g; J) \leq 3^{1/q} q(q-1)^{-1} A$.

Proof. For a compact subset of $\{\mathcal{M}g > \lambda\}$ choose a finite interval cover with averages exceeding λ. Selecting intervals in decreasing order of length gives a disjoint subcollection $I_1, \ldots, I_N$ whose triple dilates cover that compact set. For $V = \bigcup I_i$, the norm definition and the zero extension give

$$\lambda |V| < \int_V g \leq A |V|^{1-1/q}.$$

Hence $|V| < (A/\lambda)^q$ and the compact set has measure at most $3(A/\lambda)^q$. Inner regularity proves (6.1). For $E \subset J$, integrate the bound $|E \cap \{\mathcal{M}g > \lambda\}| \leq \min(|E|, 3(A/\lambda)^q)$ in λ. Splitting at $3^{1/q} A |E|^{-1/q}$ gives the asserted norm bound. Rational interval endpoints suffice by local integrability, so all functions and exceptional sets can be chosen measurable. □

Theorem 6.2 (All-radius estimate on the original source)**.** *Fix $M, L > 0$. For each interval J of length L, let $K = J + (-1, 1)$ and put*

$$G_{m,a,\delta} = m^2 U_{a,\delta} \mathbf{1}_{\mathcal{C}(M) \cap K}.$$

For every fixed $\lambda > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J| = L} |\{t \in J : \mathcal{M}G_{m,a,\delta}(t) > \lambda\}| \leq C_{M,L} \lambda^{-Q} (a^\alpha + \delta^\gamma). \tag{6.2}$$

Outside this set, every interval average contained in K of every $m^2 R^\varepsilon_{a',\delta'} \mathbf{1}_{\mathcal{C}(M)}$, with $a' \leq a$, $\delta' \leq \delta$, is at most λ when the interval contains the starting roof. The same bound holds for the source values at their Lebesgue points. In the cofinal choice the right side of (6.2) *is $C_{M,L} \lambda^{-Q} h^{1/128}$.*

Proof. Apply Lemma 6.1 to the actual finite-count source $G_{m,a,\delta}$ and then take the specified suprema and collision limsup. Theorem 5.2 bounds its norm by $C_{M,L}(a^\alpha + \delta^\gamma)^{1/Q}$ in that limit. Raising to the Qth power proves (6.2). All smaller weighted selectors are dominated by this one unweighted union using the common coarea kernels. Their interval averages are therefore bounded simultaneously. The assertion on point values uses Lebesgue differentiation for each source; no common null set for all radii R is asserted. The cofinal exponents were computed in the preceding theorem. □

6.1. **The pointwise obligation that remains.** The organizing assertion is still (2.5). Its essential-height criterion is the vanishing, in the outer reconstruction limit, of the original weighted critical-or-short- residence source charge. Theorems 4.4 and 5.2 remove the logarithmic loss at the finite critical exponent. Theorem 6.2 estimates one exceptional roof set for all local averaging scales. None asserts that this set is empty or that its essential height is small.

For example, $\mathbf{1}_{(0,\delta)}$ has essential height one and critical norm $\delta^{1/Q}$. Its maximal superlevel sets shrink, and its $W^{s,1}$ norm tends to zero for $s < 1$, without any decline of its essential height. This is a logical test of the topology, not a proposed billiard counterexample. Thus the present critical convergence is not identified with a collision-uniform spike-width lower bound. The latter, or another genuinely pointwise physical estimate, is still needed for the original scalar theorem.

All actual labels and the original source normalization remain in these statements. Point-conditioned path ratios and likelihoods remain downstream of scalar pointwise closure and their retained positive-arithmetic hypotheses; arithmetic zero classes are never divided by. The historical integrated-oscillation proofs are retained in full, not replaced by the maximal statement.

7. Proof dependencies and preserved conclusions

The three new steps are scale-matched gluing, bounded-reference domination, and an interval covering argument. The first uses the complete physical surface theorem, including its capped residence sum. The second uses the resulting raw critical convergence and the actual critical/residence mass bounds. The third is applied to that same original union before the collision limit. None uses the pointwise theorem as an input.

The complete previous article and all inherited physical, spectral, arithmetic and source-oscillation proofs are retained in the technical supplement. The earlier logarithmic critical theorem remains valid but is strengthened by Theorem 4.4; its historical status is not edited. The original pointwise theorem still requires a uniform source-specific exclusion of collapsing high roofs. Independent verification of the inherited billiard and spectral inputs remains distinct from source preservation and finite checks.

Part 14. All-scale original-source oscillation

Retained revision-90 abstract. We retain the pointwise local limit problem for the complete displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. We prove finite-count oscillation estimates on the original collision source, including both true physical ends and backward critical entrances. A capped end-layer estimate and radial motion yield a uniform L^1 translation modulus for the full weighted roof density. A one-endpoint-per-segment argument then gives a modulus of order $h^{1/33}$ for every measurable initial segment restriction, uniformly in the endpoint location. This includes the original union of initial criticality and short total residence, with both cutoffs arbitrarily small. Its complete microscopic fractional-variation tail is bounded before the collision limit, and its fractional local norm has a quantitative vanishing bound. All exact labels, physical weights, arithmetic classes and previous proofs are retained. The new estimates control integrated roof oscillation; the unchanged essential-height charge still requires a uniform starting-roof estimate.

8. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{8.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{8.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{8.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

8.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{8.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{8.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

8.2. **An all-scale estimate on the original exceptional source.** Let τ^-,τ^+ be the stopped roof-flow lifetimes, $\ell=\tau^-+\tau^+$ and $z=|\nabla_q F|$. The original exceptional source is

$$R^{\varepsilon}_{a,\delta}(t)=\int W^{\varepsilon}(x)\mathbf{1}_{\{z(x)<a \text{ or } \ell(x)<\delta\}}\,d\mu_t(x).$$

Its union is counted once. Neither endpoint section membership nor a true boundary is removed. Our principal new assertion is the following.

Theorem 8.1 (Microscopic oscillation on the complete original union)**.** *Under the precise physical and spectral inputs stated below, for every fixed L and $0<\varepsilon\le 1$ there is $K_L<\infty$ such that*

$$\sup_{m\ge 2,R,n,k}\ \sup_{a,\delta>0}\ \sup_{|J|\le L} m^2\int_J |R^{\varepsilon}_{a,\delta}(t+h)-R^{\varepsilon}_{a,\delta}(t)|\,dt \le K_L\big(h^{1/33}+\varepsilon^{-1}\sqrt{h}\big)$$

for every sufficiently small $h>0$. The constant is obtained at one fixed auxiliary band. The supremum over finite collision counts and both cutoffs precedes $h\downarrow 0$. For $0<s<1/33$, the full integral of these differences against $h^{-1-s}dh$ down to zero has a uniform vanishing microscopic tail. For the cofinal choice $a=r^{1/8}$, $\delta=r^{1/4}$,

$$\limsup_{m}\ \sup_{R,n,k}\ \sup_{|J|=L} \|m^2 R^{\varepsilon}_{a,\delta}\|_{s,1;J} \le C_{L,s}(1+\varepsilon^{-1})^{33s} r^{(1-33s)/128},$$

where the fractional norm is defined in (12.1)*.*

The proof supplies a genuine pre-limit envelope, rather than a sum of fixed-scale limiting estimates. The complete density b^{ε} has the stronger translation exponent $1/32$. The loss to $1/33$ for the original union comes from the single new interior cutoff on a physical segment. Its location needs no regularity in angle.

These are oscillation statements in the roof variable, not another claim that a local mass estimate is a pointwise bound. In particular the integral over starting roofs is visible in the theorem. The exact pointwise objective remains (8.5); no unconditional point-conditioned law is deduced without its essential-height closure.

9. The physical and analytic inputs used in the proof

All new theorems use the following source-pinned assertions. They are stated in their original physical normalization, with finite-count remainders retained. The underlying anisotropic billiard literature [12] is background; the four-record and new source assertions are the repository's own explicitly inherited inputs.

Input statement 9.1 (Same-label physical segments)**.** Fix m,R,n,k. The regular exact event Ω is partitioned by a Borel bijection, with Borel inverse, into maximal roof intervals $a_\sigma<u<b_\sigma$ and disjoint angular charts. In endpoint angles $q=(\alpha_0,\alpha_m)$, put

$$F=L,\quad g=\nabla_q F=R(-p_0,p_m),\quad z=|g|,\quad K=D_q^2F,\quad \vartheta=\arg g,$$
$$\lambda_\circ=9\sqrt{19}/200,\quad \Lambda_\circ=47/6,\quad \omega=|\partial_{p_0}\alpha_m|^{-1},\quad \chi=|\det D_q(F,\vartheta)|.$$

The bounds $\lambda_\circ I \le K \le \Lambda_\circ I$ and $\lambda_\circ \le \chi \le \Lambda_\circ$ hold on all regular pieces. The field $X_q = K^{-1}g/(g^{\mathsf T}K^{-1}g)$ satisfies $XF = 1$, $X\vartheta = 0$, and

$$2\lambda_\circ h \le z(\Phi_h x)^2 - z(x)^2 \le 2\Lambda_\circ h. \tag{9.1}$$

A same-segment motion of roof length h has endpoint-curve length at most $C\sqrt{h}$. Contact and flight-margin first derivatives in endpoint angles are bounded by $Cq_0^{d_j}$ and $Cq_0^{d'_j}$. Thus every true forward end within h is in a layer

$$c_j \le C_E q_0^{d_j}\sqrt{h}, \quad \Delta_j \le C_E q_0^{d'_j}\sqrt{h}, \quad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{h}. \tag{9.2}$$

Forward critical termination cannot occur. The physical angular density and its derivative are

$$A_\sigma(u) = \frac{\omega(\Psi_\sigma(u))}{4\pi c_* \chi(\Psi_\sigma(u))}, \qquad |A'_\sigma(u)| \le C A_\sigma(u) I_m(u)/z(u), \qquad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i.$$

The Borel disintegration identifies $A_\sigma(u)du\,d\sigma$ with $d\bar\nu = \nu_R^*|_\Omega$, and its roof fibers with $d\mu_u$. It counts each interior physical point once, retaining its incident component. Artificial angular-chart seams are not ends.

These are the retained endpoint-field, all-contact, cubic-action and Borel-segment theorems. In particular the Borel map is Supplement Theorem 7.1. Its countable kernel convention is used throughout: for fixed m, R, n, k one fixes joint measurable fiber and time kernels, not a separate density version for each restriction. The parameter order remains $\sup_R \operatorname{ess\,sup}_t$; no common null set for every radius is asserted.

Input statement 9.2 (Finite-count positive local bounds)**.** At each fixed auxiliary band B_* there are constants C_{B_*} and $\rho_{B_*} < 1$ such that, uniformly in the interval J, finite collision count and exact labels, the following hold. A marked section strip $S_{R,s}$ of width Cs obeys

$$m^2\nu\{K_m^{\mathrm c} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le Cs^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2\rho_{B_*}^{m/2}. \tag{9.3}$$

The original bad source satisfies

$$m^2 \int_J b^\delta \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3\rho_{B_*}^{m/2}. \tag{9.4}$$

The extension to a fixed remaining compact range of δ is included. At parameter one, $b^1 = p$. The endpoint-only estimate has leading factor $\zeta^{129/128}$ and remainder $C_{B_*}(1 + B_*|J|)m^2\zeta\rho_{B_*}^{m/2}$. The actual fractional source

$$v(t)dt = F_\#(I_m z^{-1/4}\bar\nu)$$

is finite at every count; after enlarging constants its complete finite-count local bound has the form (10.3). No untruncated first inverse-gradient bound is part of this assertion.

The strip and bad-source results are Supplement Theorems 175.3 and 179.1. The endpoint-only bound is Supplement equation (4.2). For the complete finite-count fractional bound one may use Supplement equation (4.6) with $p = 33/32$ and $\eta = 1/4$, since $I_m \le I_m^p$ and $1/4 < 1 - p/2$. Its exponential remainder is enlarged to the common one in (10.3), not discarded. The actual negative-residence source and physical clearance cap are retained in module 205, compiled in the full preceding article.

Input statement 9.3 (Original source and scalar endpoint)**.** The bad-source density b^ε is nonnegative and compactly supported at finite count. For $\varepsilon(B) = A_0 B^{-1/12}$ and large fixed reconstruction band B,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{9.5}$$

For the original exceptional union in (11.4), let $\mathcal{H}_M(f) = \limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{\mathcal{C}(M)} f$. Its cofinal charge satisfies

$$\mathfrak{R}_{\varepsilon,M} = \inf_{a,\delta>0} \mathcal{H}_M(R^{\varepsilon}_{a,\delta}), \qquad \mathcal{E}_M = \lim_{B\to\infty} \mathfrak{R}_{\varepsilon(B),M}.$$

Neither charge is asserted zero by this input.

The scalar remainder is Supplement Theorem 181.4, and the exact original-source equivalence is Supplement Theorem 34.2. The critical weak boundedness, logarithmically weakened critical convergence and full central-window theorem of the preceding article are preserved. The last uses the diffusive measure $dt/\sqrt{m}$, not unnormalized dt.

10. Two-sided physical ends and oscillation of the original source

Fix an exact event, and use its Borel maximal-segment coordinates (σ, u), with $a_\sigma < u < b_\sigma$. The measure $d\sigma$ includes the disjoint angular charts and their incident components. Write $A_\sigma(u)$ for the physical density, $z = |g|$, and $\ell_\sigma = b_\sigma - a_\sigma$. In this section

$$\gamma = \frac{1}{32}, \qquad \eta = \frac{1}{4}, \qquad \kappa = \frac{1-\eta}{2} = \frac{3}{8}, \qquad v = v^{1/4}.$$

Every estimate is on the original collision source. In particular a backward end is an end of the roof flow in the same exact event, not a reversal or relabeling of the billiard record.

Lemma 10.1 (Both physical end layers)**.** *Let*

$$e^{\pm}_s(t) = \int \mathbf{1}_{\{\tau^{\pm}\le s\}}\, d\mu_t.$$

Backward critical termination is included in e^-_s. There are $s_0, C > 0$ such that

$$\bar{\nu}\{\tau^{\pm} \le s\} \le C\sqrt{s} \qquad (0 < s < s_0). \tag{10.1}$$

For every fixed auxiliary band B_*, after increasing its constants and choosing one $0 < \theta_{B_*} < 1$, both signs satisfy

$$\begin{aligned} m^2 \int_J e^{\pm}_s(t)\, dt \le{}& Cs^{\gamma}(|J| + B_*^{-1}) \\ &+ C_{B_*}(1 + B_*|J|)m^3 \min\{\theta^m_{B_*}, \sqrt{s}\}. \end{aligned} \tag{10.2}$$

The constants are uniform in all finite $m \ge 2$, radii and exact labels.

Proof. The forward inclusion is (9.2). For a backward segment of length at most s, apply the same contact-motion estimate to the segment with its earlier endpoint as the starting point. Its endpoint-curve length is at most $C\sqrt{s}$, and each contact and flight margin changes by at most its prescribed depth weight times that length. If the limiting backward end is regular and noncritical, strictly positive incidences, clearances and section-edge distances would continue both the word and the field, contradicting maximality. Thus the same three physical layers contain this case. If the limit is critical, radial growth gives instead

$$z(x)^2 \le 2\Lambda_\circ s.$$

These alternatives also cover simultaneous failures. No chart seam is used as an end.

The physical estimates used in the complete residence cap apply to each of the three layers: incidence has mass at most w^2, a section strip has mass at most Cw, and the image-collision clearance envelope has mass at most Cw. At widths $Cq_0^d\sqrt{s}$ their complete depth sum is $O(\sqrt{s})$. The additional critical set lies in $\{|p_0| \le C\sqrt{s}\}$, of invariant mass $O(\sqrt{s})$. Dropping the section tests on these positive upper comparisons costs exactly c_*^{-1}. This proves (10.1) for the original source, without changing its labels.

For the localized estimate, the incidence and clearance layers are contained in the original bad source at an auxiliary parameter $C\sqrt{s}$. The section layers use (9.3), summed at their depth widths. This is the forward finite-count exit proof and applies to the same layers on the backward side. For the critical alternative use the retained endpoint-only bound

$$m^2\bar{\nu}\{F \in J, z < \zeta\} \le C\zeta^{129/128}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2\zeta\rho_{B_*}^{m/2}.$$

At $\zeta = C\sqrt{s}$ its leading term is smaller than Cs^{γ}. This proves the uncapped version of (10.2). Apply $x \le P + D$, $x \le V \Rightarrow x \le P + \min(D, V)$ to that positive mass and (10.1). Since $m^2 \le m^3$, adjusting constants gives the displayed common cap. It is a cap on the positive estimate, not a claim that an isolated signed spectral remainder has the physical bound. □

For use below the complete sources p and v obey the retained finite-count estimate

$$m^2 \int_J (p + v) \le U_{m,B_*}(|J|), \quad U_{m,B_*}(L) = C(L + B_*^{-1}) + C_{B_*}(1 + B_*L)m^3\theta_{B_*}^m. \tag{10.3}$$

Its constants may be enlarged from those in (10.2). The stated fractional source, not an untruncated first inverse gradient, is used here.

Lemma 10.2 (A fixed-band envelope at every microscopic scale)**.** *For $0 < \theta < 1$ and $0 < s < 1$,*

$$\sup_{m \ge 2} m^3 \min\{\theta^m, \sqrt{s}\} \le C_\theta\sqrt{s}(1 + |\log s|)^3. \tag{10.4}$$

Consequently, at every one fixed admissible band and fixed L,

$$\sup_{m,R,n,k} \sup_{|J| \le L} m^2 \int_J e_s^{\pm} \le K_{B_*,L}s^{\gamma}, \qquad \sup_m U_{m,B_*}(L) < \infty. \tag{10.5}$$

These suprema precede $s \downarrow 0$.

Proof. Put $a = -\log\theta$ and $M_s = \max\{2, a^{-1}\log(s^{-1/2})\}$. For $m \le M_s$, use $m^3\sqrt{s}$. For $m > M_s$, write $m = M_s + y$ and use $(M_s + y)^3e^{-ay} \le C_a(1 + M_s)^3$; when $M_s = 2$ the same estimate follows after increasing C_a. This proves (10.4). Since $\gamma < 1/2$, $s^{1/2-\gamma}(1 + |\log s|)^3$ is bounded on $(0, 1)$. Substitution in (10.2) proves the first assertion of (10.5). Exponential decay proves the second. No Fourier band is allowed to grow with the collision count. □

Extend

$$f_\sigma^{\varepsilon}(u) = A_\sigma(u)W^{\varepsilon}(\Psi_\sigma(u))$$

by zero outside (a_σ, b_σ). Define the lifted oscillation

$$\Xi_h^{\varepsilon}(J) = \int_J \int |f_\sigma^{\varepsilon}(t + h) - f_\sigma^{\varepsilon}(t)|\, d\sigma\, dt. \tag{10.6}$$

This keeps the absolute value before the complete physical sum. It therefore dominates $\int_J |b^{\varepsilon}(t + h) - b^{\varepsilon}(t)|dt$.

Theorem 10.3 (All-scale oscillation of the original weighted source)**.** *For each fixed L and one fixed admissible band there is $K_{B_*,L}$ such that, for every $m \ge 2$, every exact event, every $0 < h < h_0$ and $0 < \varepsilon \le 1$,*

$$m^2\Xi_h^{\varepsilon}(J) \le K_{B_*,L}\big(h^{1/32} + \varepsilon^{-1}\sqrt{h}\big) \qquad (|J| \le L). \tag{10.7}$$

The estimate includes every true forward end, backward end and critical entrance. It is not restricted to common survivors. For p, the unweighted version has no guard-derivative term.

Proof. The all-contact first derivatives and the product rule for factors in $[0,1]$ give

$$|D_q W^\varepsilon| \le C\varepsilon^{-1} \sum_{d\ge 0} q_0^{3d/4} \le C_\Theta \varepsilon^{-1}.$$

This uses the original smooth guard and requires no monotonicity of Θ. Since $|X_q| \le C/z$, its derivative on each regular segment satisfies $|(W^\varepsilon)'| \le C_\Theta \varepsilon^{-1}/z$ almost everywhere. Together with the full angular-density derivative,

$$|(f_\sigma^\varepsilon)'| \le C A_\sigma I_m/z + C_\Theta \varepsilon^{-1} A_\sigma/z. \tag{10.8}$$

Local absolute continuity suffices; all subsequent integrals are first taken on compact subsegments.

For a pair $t, t+h$ on the same segment, integrate (10.8). In the subsequent integral over $t \in J$, an intermediate roof u is counted for a set of starting roofs of length at most $\min\{h, u - a_\sigma\}$. Radial growth from the lower endpoint gives $z(u)^2 \ge 2\lambda_\circ(u - a_\sigma)$. Thus

$$\frac{\min(h, u-a_\sigma)}{z(u)} \le C h^{5/8} z(u)^{-1/4},$$
$$\frac{\min(h, u-a_\sigma)}{z(u)} \le C\sqrt{h}.$$

Use the first inequality for optical variation and the second for the guard. By the Borel bijection their complete sums are bounded by

$$C h^{5/8} \int_{J+(0,h)} v(u)\, du + C_\Theta \varepsilon^{-1} \sqrt{h} \int_{J+(0,h)} p(u)\, du.$$

There is no physical-word multiplicity. Tonelli and positive exhaustion justify the sum even when the unweighted derivative $A_\sigma I_m/z$ is not globally integrable.

If only t is on a segment, its contribution is at most $e_h^+(t)$. If only $t+h$ is on it, its contribution is at most $e_h^-(t+h)$. These are the actual original end layers. Therefore the preceding drift bound plus $\int_J e_h^+ + \int_{J+h} e_h^-$ bounds $\Xi_h^\varepsilon(J)$. Apply (10.3) and (10.5). Since $h^{5/8} \le h^{1/32}$ for $h < 1$, the result follows. Negative translations follow by translating J and interchanging its two arguments. The common segment kernels fix the same representatives throughout; this is an integral identity for all shifts, not a supremum over exceptional fiber versions. □

11. Uniform oscillation at critical and residence cutoffs

The original exceptional source is not a common-survivor restriction. On a maximal segment its residence selector is constant, while its initial-gradient selector is an initial interval because z is strictly increasing with roof. We prove an estimate for every Borel initial interval of every physical segment, with no dependence on the position of its new endpoint.

Let $c_\sigma \in [a_\sigma, b_\sigma]$ be Borel; the choices $c_\sigma = a_\sigma$ and $c_\sigma = b_\sigma$ mean, respectively, no segment and the whole segment. Define

$$f_c^\varepsilon(t) = \int f_\sigma^\varepsilon(t) \mathbf{1}_{\{t<c_\sigma\}}\, d\sigma. \tag{11.1}$$

No regularity of c_σ in the angular coordinate is assumed. There is at most one new endpoint on each original segment.

Lemma 11.1 (Crossing one interior endpoint)**.** *For $0 < h \le \rho/4$ and $0 < \rho < \rho_0$, put*

$$C_{c,h}^\varepsilon(J) = \int_J \int f_\sigma^\varepsilon(t) \mathbf{1}_{\{t<c_\sigma\le t+h\}}\, d\sigma\, dt.$$

Then

$$C^{\varepsilon}_{c,h}(J) \le \int_J e^-_\rho(t)\,dt + C\frac{h}{\rho}\int_{J+(-\rho,h)} p(u)\,du + Ch\rho^{-3/8}\int_{J+(-\rho,h)} v(u)\,du. \tag{11.2}$$

The constants do not depend on c, ε, the collision count or the number of physical segments.

Proof. If $c_\sigma - a_\sigma < \rho$, every contributing starting roof t has $0 < t - a_\sigma < \rho$. Its original weighted mass is bounded by the first term. This includes endpoints arbitrarily close to a critical entrance; no trace there is assumed.

For $c_\sigma - a_\sigma \ge \rho$, choose an anchor $u_0 \in (c_\sigma - \rho/2, c_\sigma - \rho/4)$. Every contributing $t \in (c_\sigma - h, c_\sigma)$ lies after the anchor. On the entire interval between them, $u - a_\sigma \ge \rho/2$, so that $z(u)^2 \ge \lambda_\circ \rho$. The exact angular-density derivative therefore yields

$$A_\sigma(t) \le A_\sigma(u_0) + C\rho^{-3/8}\int_{c_\sigma-\rho/2}^{c_\sigma} A_\sigma(u) I_m(u) z(u)^{-1/4}\,du.$$

Average over the anchor interval of length $\rho/4$. Use $W^\varepsilon \le 1$ and integrate over a subset of the crossing interval of length at most h. The resulting two bounds are $4h/\rho$ times the segment's unweighted anchor mass and $Ch\rho^{-3/8}$ times its fractional source mass.

For every contributing segment $c_\sigma \in J + (0,h)$, so its anchor and comparison intervals lie in $J + (-\rho, h)$. Summing one such interval per original segment counts each interior physical point at most once under the Borel bijection. Thus the sums are submeasures of the original p and v sources on that roof interval. This proves (11.2). Compact exhaustion and Tonelli remove any endpoint truncations. The proof has used values inside the segment, not a postulated trace on the potentially irregular graph c_σ. □

Theorem 11.2 (A cutoff-uniform microscopic oscillation envelope)**.** *For every fixed L, every one fixed admissible band and all sufficiently small $h > 0$,*

$$\sup_{m,R,n,k,c}\ \sup_{|J|\le L} m^2 \int_J |f^\varepsilon_c(t+h) - f^\varepsilon_c(t)|\,dt \le K_{B_*,L}\big(h^{1/33} + \varepsilon^{-1}\sqrt{h}\big). \tag{11.3}$$

In particular, for every $a, \delta > 0$, the actual source

$$R^\varepsilon_{a,\delta}(t) = \int W^\varepsilon(x)\mathbf{1}_{\{z(x)<a \text{ or } \ell(x)<\delta\}}\,d\mu_t(x) \tag{11.4}$$

satisfies the same estimate, with constants independent of a and δ. The union is not counted twice. The supremum over all finite collision counts precedes the microscopic limit $h \downarrow 0$.

Proof. For the zero-extended segment density and $h > 0$, the elementary product difference gives

$$|f^\varepsilon_\sigma(t+h)\mathbf{1}_{\{t+h<c_\sigma\}} - f^\varepsilon_\sigma(t)\mathbf{1}_{\{t<c_\sigma\}}| \le |f^\varepsilon_\sigma(t+h) - f^\varepsilon_\sigma(t)| + f^\varepsilon_\sigma(t)\mathbf{1}_{\{t<c_\sigma\le t+h\}}.$$

Integrate before summing the complete source. The right side is $\Xi^\varepsilon_h(J) + C^\varepsilon_{c,h}(J)$. Use Theorem 10.3, Lemma 11.1 and the fixed-band finite-count envelopes (10.5)–(10.3). They bound it after normalization by

$$K_{B_*,L}\left(h^{1/32} + \varepsilon^{-1}\sqrt{h} + \rho^{1/32} + h/\rho + h\rho^{-3/8}\right).$$

Choose $\rho = 4h^{32/33}$. For sufficiently small fixed h this lies below ρ_0 and obeys $h \le \rho/4$. The three new powers are respectively $h^{1/33}$, $h^{1/33}$ and $h^{7/11}$, up to fixed factors. This proves (11.3), uniformly in every Borel choice of endpoints. Negative shifts follow as before.

For (11.4), take the full segment when $\ell_\sigma < \delta$. On every other segment take its initial interval on which $z < a$. Radial growth makes this interval empty, full or a prefix with a unique threshold

roof. Its endpoint is Borel: one may determine it from the values on the countable rational roofs of the existing segment parametrization. Thus it is a permitted c_σ. This represents the original union exactly, not an upper comparison to a transported or pooled source. No derivative in angle of this new graph, and no positive lower gradient or residence cutoff, was used in the uniform estimate. □

The estimate is a roof-translation modulus integrated over the starting roof. It is stronger than knowing only the mass of the selected union, and its constants remain uniform as both selectors tend to zero before any collision limit. It is not an essential-supremum modulus: the integral over J in (11.3) must not be replaced by a supremum over its starting roofs.

12. Microscopic variation of the original exceptional union

The new envelope is summable over all microscopic translation scales at finite collision count. This supplies a genuine regularity statement on the exceptional source itself. Its topology, and the remaining pointwise requirement, are specified separately.

For an interval J and $0 < s < 1$ set

$$[f]_{s,1;J} = \int_J \int_J \frac{|f(t) - f(u)|}{|t-u|^{1+s}}\, du\, dt, \qquad \|f\|_{s,1;J} = \|f\|_{L^1(J)} + [f]_{s,1;J}. \tag{12.1}$$

All integrals are actual nonnegative integrals; no sequence of fixed-scale collision limits defines this seminorm. Put $\chi_0 = 1/33$.

Theorem 12.1 (Complete microscopic oscillation tail)**.** *For $0 < s < \chi_0$, fixed L and one fixed auxiliary band,*

$$\sup_{m,R,n,k,a,\delta} \sup_{|J|\le L} m^2 \int_0^r h^{-1-s} \int_J |R^\varepsilon_{a,\delta}(t+h) - R^\varepsilon_{a,\delta}(t)|\, dt\, dh \le K_{B_*,L} \left(\frac{r^{\chi_0 - s}}{\chi_0 - s} + \varepsilon^{-1} \frac{r^{1/2-s}}{1/2-s} \right) \tag{12.2}$$

for all sufficiently small $r > 0$. In particular this complete microscopic oscillation tail vanishes uniformly before the collision limsup, with no positive lower translation scale. The corresponding statement for b^ε holds for $s < 1/32$ with $1/32$ in place of χ_0.

Proof. Apply (11.3) at finite count to a positive lower truncation of the integral in h. Integrate its two explicit powers. Monotone convergence removes the lower truncation because $s < \chi_0 < 1/2$. The same bound holds for every finite count and every selector, so the supremum may be taken afterwards. Use (10.7) for the last assertion. This order uses a pre-limit envelope and not an interchange of a collision limsup with an uncontrolled infinite series. □

Lemma 12.2 (Mass and an actual translation envelope)**.** *Suppose $f \in L^1(J)$ and for $0 < h < h_0$,*

$$\int_{J \cap (J-h)} |f(t+h) - f(t)|\, dt \le A h^\chi, \qquad 0 < s < \chi < 1.$$

If $\mu = \|f\|_{L^1(J)}$, then

$$\|f\|_{s,1;J} \le C_{s,\chi,h_0,|J|} \left(\mu + A^{s/\chi} \mu^{1-s/\chi} \right). \tag{12.3}$$

The convention for $\mu = 0$ or $A = 0$ is obtained by continuity.

Proof. The integral of a difference is also at most 2μ. Changing variables $u = t + h$ in (12.1) gives twice the integral of these differences against $h^{-1-s}dh$. Below h_0, use $\min\{2\mu, Ah^\chi\}$ and split at $(2\mu/A)^{1/\chi}$ when it lies below h_0. The two power integrals are bounded by a constant times $A^{s/\chi}\mu^{1-s/\chi}$. If the split lies above h_0, the bound is absorbed by a fixed multiple of μ. Above

h_0 use 2μ. This proves the formula, without an interpolation theorem for a different norm or a change of source representative. □

Retain exactly the source parameterization used in the revision-89 exceptional-source theorem:

$$R_r^{\varepsilon} = R^{\varepsilon}_{r^{1/8},r^{1/4}}. \tag{12.4}$$

Its complete interval mass satisfies the inherited ordered estimate

$$\limsup_m \sup_{R,n,k} \sup_{|J|\le L} m^2 \int_J R_r^{\varepsilon} \le C_L r^{1/128}. \tag{12.5}$$

It follows directly from the critical endpoint source bound and the capped residence estimate: their leading powers are $r^{129/1024}$ and $r^{1/128}$, respectively. Both sources retain the original normalization and exact label; their sum is only a positive upper bound for the union. All auxiliary bands are fixed before the collision limit and enlarged afterwards.

Theorem 12.3 (Fractional regularity and decay on the original union)**.** *For each fixed $\varepsilon > 0$, $L > 0$ and $0 < s < 1/33$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \|m^2 R_r^{\varepsilon}\|_{s,1;J} \le C_{L,s}(1+\varepsilon^{-1})^{33s} r^{(1-33s)/128} \tag{12.6}$$

for all sufficiently small $r > 0$. Moreover the norms $\|m^2 R^{\varepsilon}_{a,\delta}\|_{s,1;J}$ are bounded at every finite count by one constant independent of a, δ, m, R, n, k. These assertions act on the whole original weighted exceptional union, including its true ends and arbitrary collapsing cutoffs.

Proof. Choose one admissible band once. Equation (11.3) gives the finite-count translation hypothesis of Lemma 12.2 with $\chi = 1/33$ and $A = K_{B_*,L}(1+\varepsilon^{-1})$, uniformly for all selectors. The complete unweighted volume bound controls the finite-count L^1 mass since $0 \le R^{\varepsilon}_{a,\delta} \le p$. This proves the uniform norm assertion.

Apply the lemma at each finite count to the actual function $m^2 R_r^{\varepsilon}$, and only then take the parameter suprema and collision limsup. Use (12.5) for its mass; that mass estimate may enlarge its own fixed auxiliary band after the collision limit. The translation constant remains the constant of the one previously chosen band and has no dependence on r. Since $1 - s/\chi = 1 - 33s > 0$, the resulting bound is $C(r^{1/128} + r^{(1-33s)/128})$, as claimed. There is no infinite scale sum of limiting estimates in this proof: its entire small-scale part was already controlled by (12.2) at finite count. □

12.1. **High-roof sets and the unchanged essential-height assertion.** For a measurable $E \subset J$, define its regional variational cost

$$\mathrm{Cap}_{s,1}(E;J) = \inf\{\|u\|_{s,1;J} : u \ge 0,\ u \ge 1 \text{ a.e. on } E\}.$$

This definition is all that is needed here; no claim about values on Lebesgue-null sets is intended. Truncation is one-Lipschitz, so taking $u = \min\{1, m^2 R_r^{\varepsilon}/\lambda\}$ gives, for $\lambda > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathrm{Cap}_{s,1}(\{m^2 R_r^{\varepsilon} > \lambda\} \cap J; J) \le C_{\varepsilon,L,s} \lambda^{-1} r^{(1-33s)/128}. \tag{12.7}$$

Thus the new estimate measures the spatial oscillation cost of the large-value set, not just its Lebesgue measure.

It still does not bound the essential height of that set. For example, on a fixed interval the pulses $\mathbf{1}_{(0,d)}$ have height one, L^1 mass d, and fractional seminorm $O(d^{1-s})$ for $s < 1$. They are not a counterexample for the physical source; they explain why the new regularity topology cannot, by itself, be substituted for an essential-supremum theorem.

The original scalar target therefore continues to require

$$\lim_{B\to\infty} \mathfrak{R}_{\varepsilon(B),M} = 0.$$

Theorems 11.2 and 12.1 provide finite-count all-scale oscillation estimates on exactly that source. Their starting-roof integral has not been upgraded to a uniform starting-roof bound. For instance, choosing $r(B) = c\varepsilon(B)^N$ with $N > 4224s/(1-33s)$ makes the explicit bound (12.6) vanish along the legal outer limit. This is a cofinal fractional-variation conclusion, not charge-zero in essential height. The prior critical weak, logarithmic, and diffusive-window results remain proved with their original topologies and normalizations. No pointwise denominator, same-roof ratio or likelihood is promoted before scalar essential-height closure and the retained positive arithmetic hypotheses; zero arithmetic classes are not normalized.

13. Dependency and scalar-first order

The new argument has three steps. The two-sided end-layer theorem pays all true boundary changes, including critical backward ends. The weighted segment calculation controls surviving variation without an initial inverse-gradient cutoff. The prefix theorem pays the extra cut introduced by the original exceptional union without a derivative of its angular graph. Every estimate is summed over actual physical interiors using the same Borel disintegration.

The preceding critical-distribution and diffusive-window theorems remain in the complete preceding article in the supplement. They are not replaced, and the two revision-88 variants remain separately preserved. The new physical oscillation envelope does not establish zero essential-height charge. A further uniform starting-roof or comparable pointwise persistence bound is still required for that conclusion. Only after scalar closure may the retained positive-arithmetic normalization, same-roof bridge, clock transfer and likelihood arguments be used for unconditional point-conditioned laws.

Part 15. Critical distribution and the complete diffusive window

Retained revision-89 abstract. We study the pointwise local limit problem for the complete displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. Retaining the measure of a tested roof set through the complete residence sum yields a critical weak estimate at exponent 32/31 for the untruncated first-exit surface. Its finite-count remainder remains summable down to residence zero. The original raw density consequently has a uniform critical weak bound. Together with the established local mass convergence, this proves raw-error convergence at the critical exponent with every positive logarithmic loss. The same logarithmic estimate acts directly on the original critical-or-short-residence source. We also prove convergence over the full central window in diffusive roof measure, with the scale factor explicit. All exact labels, arithmetic classes, physical weights and earlier proofs are retained. The organizing pointwise assertion is unchanged; its residual essential-height charge is not inferred to vanish from these distribution bounds.

14. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K_{m,R}^{\mathrm{c}}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K_{m,R}^{\mathrm{c}} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{14.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{14.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{14.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

14.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{14.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{14.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

14.2. **Complete sources without a positive residence cutoff.** The pointwise error in (14.5) remains the organizing objective. The new assertions estimate the same raw density in finite local norms and the actual outgoing surface with all positive residence lengths retained. These statements are not conditional on a microscopic charge being zero.

For $1<q<\infty$ let $\mathcal{N}_q(f;J)$ be the measurable-set norm in (17.1), equivalent to weak L^q on J. For the original first-exit surface write $\overline{\mathcal{F}}_r$ for its average over exit widths between zero and r, as defined below.

Theorem 14.1 (Untruncated surface and raw-density estimates)**.** *Under the explicit inherited inputs of the next section, for every $1<q<32/31$ and fixed $L>0$,*

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} \mathcal{N}_q(m^2\overline{\mathcal{F}}_r;J)\le C_{q,L}r^{1/q-31/32}.$$

There is no lower gradient, residence or incidence cutoff in this estimate. For every fixed $M,L>0$, the raw scalar error satisfies

$$\sup_{R,n,k}\sup_{|J|=L}\mathcal{N}_q\Big((m^2p_{n,R}(k,m,\cdot)-\mathcal{L}_{m,R}(k_1,k_2,\cdot,n))\mathbf{1}_{\mathcal{C}_{m,R,n,k}(M)};J\Big)\longrightarrow 0.$$

The corresponding strong L^p convergence holds for every $1\le p<32/31$. All arithmetic classes remain unnormalized.

The proofs are Theorems 17.3 and 18.3. The new source theorem first establishes a negative moment of total regular residence by a capped layer cake at finite collision count. Its geometric trace application uses the whole same-label segment; the source term is not replaced by a boundary collar. A finite q is essential to the conclusion. The unchanged essential-supremum problem remains separate.

14.3. **Critical distribution and the original exceptional source.** Set $Q=32/31$ and $\gamma=1/32$. Theorem 19.2 proves a finite-count measurable-set estimate for the complete surface at the previously excluded exponent Q. Its proof keeps the factor $\min\{|E|,\ell\}$ until after the residence sum; it does not substitute an endpoint negative moment into a subcritical theorem. Proposition 20.1 then gives a critical weak bound for the raw density itself.

The logarithmic norm in (20.2) yields two further statements. The raw error tends to zero in that norm for every positive logarithmic exponent (Theorem 20.3). The original weighted union of initial criticality and short total residence has an explicit logarithmic modulus in the same critical norm (Theorem 20.4). The latter estimate acts on the exceptional source, not only its complementary surviving fibers. Theorem 21.2 gives the corresponding full central-window conclusions with measure $dt/\sqrt{m}$.

These statements preserve (14.5) as the pointwise objective. Neither a logarithmically weakened critical norm nor the normalized central-window theorem is identified with essential height. The exponent 32/31 comes from the available residence exponent, not a claim of physical optimality.

15. Explicit analytic and geometric inputs

We collect the indispensable inherited assertions in the exact form used by the new proofs. Their source theorems and operator dependencies are specified here and in Supplement Appendix A. The new potential and first-exit estimates do not assume a surface spectral insertion theorem or an unweighted pointwise local bound.

Input statement 15.1 (Physical flow and source comparison)**.** On the regular part Ω of each exact event, endpoint angles $q = (\alpha_0, \alpha_m)$ give $F = L$, $g = \nabla_q F = R(-p_0, p_m)$, $K = D_q^2 F$ and $\vartheta = \arg g$. Put

$$\lambda_\circ = 9\sqrt{19}/200, \qquad \Lambda_\circ = 47/6, \qquad \omega = |\partial_{p_0}\alpha_m|^{-1}, \qquad \chi = |\det D_q(F, \vartheta)|.$$

The bounds $\lambda_\circ I \le K \le \Lambda_\circ I$ and $\lambda_\circ \le \chi \le \Lambda_\circ$ hold uniformly. The explicit field

$$X_q = \frac{K^{-1} g}{g^{\mathsf{T}} K^{-1} g}$$

is stopped at true physical and section-decision boundaries, not at chart seams. Its flow satisfies $XF = 1$, $X\vartheta = 0$ and

$$2\lambda_\circ s \le |g(\Phi_s x)|^2 - |g(x)|^2 \le 2\Lambda_\circ s. \tag{15.1}$$

Its endpoint-curve length is at most $C\sqrt{s}$. For $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$ one has $|X\omega| \le C\omega I_m/|g|$ on regular survivors. At a genuine grazing exit some internal incidence tends to zero and ω tends to zero. Forward interior critical termination cannot occur.

Every starting point with forward lifetime at most r belongs to a layer of one of the following forms, at an actual collision or flight:

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \quad \Delta_j \le C_E q_0^{d'_j}\sqrt{r}, \quad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}. \tag{15.2}$$

The physical coarea covariance, proved in the appendix in the convention used here, has Jacobian $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$. Writing E_h^ε for all weighted starting exit mass and

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon(x)[1 - \omega(\Phi_h x)/\omega(x)]_+ \, d\mu_t(x),$$

the complete source satisfies, for each fixed r and almost every t,

$$b^\varepsilon(t) \le 2A_r b^\varepsilon(t) + \bar{E}_r^\varepsilon(t) + \bar{O}_r^\varepsilon(t) + C_\Theta(1 + \varepsilon^{-1})\sqrt{r}\, A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\, ds. \tag{15.3}$$

These are the endpoint-field, all-contact motion, uniform cubic-action and separated-volume results in retained modules 174, 177, 180–182. The guard term in (15.3) is the complete product guard, not a protective subword source. The derivative and continuation assertions are confined to regular physical trajectories.

Input statement 15.2 (Complete local source estimates)**.** For every fixed auxiliary band B_*, constants $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ give the following bounds uniformly in radius, exact labels, interval J and mark $0 \le j \le m$. Let $S_{R,s}$ be the union of strips of width Cs about the finitely many coordinate lines supporting ∂Y_R^*. For s small,

$$m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le C s^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \rho_{B_*}^{m/2}, \tag{15.4}$$

$$m^2 \int_J b^\delta(t)\, dt \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \rho_{B_*}^{m/2}. \tag{15.5}$$

The local extension to a fixed remaining compact range of δ is included. At auxiliary parameter one $W^1 = 1$, so $b^1 = p$.

Put $z = |g|$, $\alpha = 129/128$ and, for $0 < \eta < 1/2$, define the actual nonnegative source measures

$$v^{\eta,b}(t)\,dt = L_{\#}(I_m z^{-\eta}\mathbf{1}_{\{0<z<b\}}\mathbf{1}_{\Pi}\nu_R^*),$$
$$v^{\eta}(t)\,dt = L_{\#}(I_m z^{-\eta}\mathbf{1}_{\Pi}\nu_R^*). \tag{15.6}$$

They are finite and absolutely continuous at every fixed count, and for $0 < b < b_1$,

$$m^2\int_J v^{\eta,b}(t)\,dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|)m^2 b^{1/2-\eta}\rho_{B_*}^{m/4}. \tag{15.7}$$

Adding the bounded-gradient part gives, for every fixed $L > 0$,

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} m^2\int_J v^{\eta} \le C_\eta L, \qquad \limsup_m \sup_{R,n,k}\sup_{|J|=L} m^2\int_J v^{\eta,b} \le C_\eta b^{\alpha-\eta}L. \tag{15.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards; leading constants are independent of that band and of the previously fixed interval length.

Equation (15.4) is Supplement Theorem 175.3. The physical source bound (15.5) is Supplement Theorem 179.1, in particular its finite-window estimate. The fractional bounds are the positive layer-cake conclusions of retained module 187, stated in full above. The joint spectral and common-space assumptions used there are recorded in the appendix. No estimate for an untruncated first inverse gradient or for a singular roof convolution is part of this input.

Input statement 15.3 (The scalar positive remainder)**.** For large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{15.9}$$

The source b^ε is nonnegative and has a compactly supported BV density at fixed count. No collision-uniform variation bound is assumed. Supplement Theorem 181.4 gives this remainder with the same complete finite-peak reference.

15.1. **Maximal segments and the two existing charges.** The retained Borel parametrization, Supplement Theorem 7.1, is a bijection

$$\Psi : \coprod_j \{(\theta,u) : \theta \in E_j,\ a_j(\theta) < u < b_j(\theta)\} \longrightarrow \Omega,$$

with Borel inverse. Its exact density is $A_j(u,\theta) = \omega(\Psi_j(\theta,u))/(4\pi c_* \chi(\Psi_j(\theta,u)))$. Every segment remains in one regular exact event until a true boundary; Borel chart seams are not exits. Nongrazing outgoing faces retain their incident physical component and carry $d\zeta_+ = A_j(b_j-,\theta)\,d\theta$. The full angular variation is $|\partial_u A_j| \le C A_j I_m/z$. These are measure identities with the null-set convention already specified, not bounds with an unknown number of words.

The true first-exit assignment is

$$\mathcal{F}_h(t) = \int_{\Gamma_{\rm ng}^+} \mathbf{1}_{\{0<b-t\le h,\ b-t<\tau^-(y)\}}\,d\zeta_+(y), \qquad \overline{\mathcal{F}}_r = r^{-1}\int_0^r \mathcal{F}_h\,dh.$$

No lower gradient cutoff is imposed. Put $l = b - a$ and define

$$\mathcal{Z}_\delta(t) = \int_{\Gamma_{\rm ng}^+} \mathbf{1}_{\{0<l\le\delta\}}\mathbf{1}_{\{a<t<b\}}\,d\zeta_+, \qquad q_\delta(t)\,dt = F_{\#}(\mathbf{1}_{\Omega_+^{\rm ng}}\mathbf{1}_{\{l\le\delta\}}\bar\nu).$$

The second object is the original collision volume restricted to the same actual short segments, not a volume collar for a boundary. For source families set

$$\mathcal{H}_M(f)=\limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} f(t),\quad \mathfrak{B}_M=\lim_{\delta\downarrow 0}\mathcal{H}_M(\mathcal{Z}_\delta),\quad \mathfrak{Q}_M=\lim_{\delta\downarrow 0}\mathcal{H}_M(q_\delta).$$

These limits are monotone extended nonnegative numbers.

For $0<\eta<1/2$, with $\kappa=(1-\eta)/2$, retain

$$\mathsf{P}_{\eta,h}v^\eta(t)=\int_0^h s^{-\kappa}v^\eta(t+s)\,ds,\qquad \mathfrak{A}_{\eta,M}=\lim_{h\downarrow 0}\mathcal{H}_M(\mathsf{P}_{\eta,h}v^\eta).$$

Supplement Theorems 8.2 and 9.1 give

$$\mathfrak{B}_M\le \mathcal{H}_M(\overline{\mathcal{F}}_r)\le \mathfrak{B}_M+C_\eta r^{1/32}. \tag{15.10}$$

Their band sum is finite before the collision limsup at each positive residence cutoff. Its leading constant is cutoff independent; the complete zero-residence tail is still inside $\mathcal{H}_M$ before its cutoff tends to zero. The active offset correction to historical module 189 remains Supplement equation (6.1); the historical inclusion is explicitly marked superseded. None of these inputs asserts charge zero.

15.2. **The precise finite exit and surface inputs.** For total residence $\ell=\tau^-+\tau^+$, let $e_s(t)=\int \mathbf{1}_{\{\tau^+\le s\}}\,d\mu_t$. The finite-count statement of Supplement Theorem 5.1 is

$$m^2\int_J e_s\le Cs^{1/32}(|J|+B_*^{-1})+C_{B_*}(1+B_*|J|)m^3\rho_{B_*}^{m/2}. \tag{15.11}$$

It holds for all sufficiently small s, every interval J and all finite counts, with one fixed auxiliary band. The proof uses the actual layers in (15.2), the marked section estimate and the complete original guard, not a count of words. The one-flight envelope used in its geometric input is described in Supplement Lemma 178.1; its explicit coordinates also give the physical area cap proved below.

The actual untruncated surface and original weighted defects satisfy

$$E_h^\varepsilon+O_h^\varepsilon\le C\mathcal{F}_h+C_\eta\mathsf{P}_{\eta,h}v^\eta. \tag{15.12}$$

This is Supplement Theorem 5.2, with the same joint physical coarea kernels and initial weights. For the part $\mathcal{L}_r$ of $\overline{\mathcal{F}}_r$ having backward residence $\ell>2r$, Supplement Theorem 7.2 proves

$$\limsup_m \sup_{R,n,k} m^2\|\mathcal{L}_r\|_\infty\le Cr^{1/32}. \tag{15.13}$$

Its persistence proof precedes the mass-to-height deduction. We do not extend (15.13) to zero residence. The short part is estimated below in a different, explicitly finite norm.

All fiber identities have the following convention. For fixed m,R,n,k, the Borel maximal-segment parametrization fixes common kernels outside a null set in the roof, and common product kernels outside a joint null set for roof and time. Countably many widths share versions; the uniform order is $\sup_R \operatorname{ess\,sup}_t$, not a claim to one null set for all radii. All positive restrictions and infinite moment integrals below use these kernels.

16. A physical cap and negative residence moments

Throughout this section $\ell=\tau^-+\tau^+$ is the total residence of the original maximal regular exact-event segment. It is not the collision return time. Every point of Ω has $\ell>0$. Write $\gamma_0=1/32$ and $\bar\nu=\nu_R^*|_\Omega$. The following estimate integrates down to residence zero at finite collision count; its proof does not sum fixed-scale limiting bounds.

Lemma 16.1 (A count-independent physical cap)**.** *There are $s_0, C > 0$ such that, uniformly in the collision count, radius and exact labels,*

$$\bar\nu\{\ell \le s\} \le \bar\nu\{\tau^+ \le s\} \le Cs^{1/2}, \qquad 0 < s < s_0. \tag{16.1}$$

No interval in the roof or incidence corridor is imposed.

Proof. Use the actual starting layers in (15.2). Under invariant collision measure, $\nu\{c < w\} = 1 - \sqrt{1-w^2} \le w^2$. A w-neighborhood of the finitely many rectangular section edges has measure at most Cw, uniformly in R. For a flight-clearance layer use the image-collision envelope of Supplement Lemma 178.1, at collision $j+1$. Here is the area estimate, in addition to its retained stable-curve estimate. At fixed image angle α, each candidate disk in the finite set $\mathcal{D}$ has backward coordinates L_d, Z_d. On $L_d > 0$, $R \le |Z_d| \le R + w$, one has $L_d \ge \ell_* > 0$ and $\partial_\varphi Z_d = L_d$. The sinusoidal level equations have a bounded number of components. Their total φ length is at most Cw. Integrating $d\nu = (4\pi)^{-1}\cos\varphi\, d\alpha\, d\varphi$ proves that this envelope has measure at most Cw. The finite disk set and ℓ_* are uniform for the prescribed family.

Invariance permits these estimates at every actual marked collision. There are at most two contact marks, and two flight marks, at each depth. Thus the union bound at widths $C_E q_0^d \sqrt{s}$ is bounded by a constant times $\sum_{d\ge0}(q_0^{2d}s + q_0^d\sqrt{s}) \le C\sqrt{s}$. Dropping the initial and terminal section tests only on this positive upper comparison costs c_*^{-1} once. No exact labels are changed in the source being estimated. Finally $\ell \le s$ implies $\tau^+ \le s$. □

The retained finite-count exit theorem, restated in (15.11), and the physical cap give a stronger residence estimate than either bound separately. Namely,

$$\begin{aligned} m^2\bar\nu\{F \in J, \ell \le s\} &\le Cs^{\gamma_0}(|J| + B_*^{-1}) \\ &\quad + C_{B_*}(1 + B_*|J|)m^3 \min\{\rho_{B_*}^{m/2}, s^{1/2}\}. \end{aligned} \tag{16.2}$$

This holds for every interval and $0 < s < s_0$, with B_* fixed. Indeed $x \le A + D$ and $x \le V$ imply $x \le A + \min(D, V)$. Enlarging the common constant and using $m^2 \le m^3$ gives exactly the displayed remainder. This step is performed on the positive probability, not on an unsigned spectral remainder claimed to have the physical cap by itself.

Theorem 16.2 (Actual untruncated negative residence moments)**.** *For $0 < \upsilon < \gamma_0$ define the actual density*

$$q_{\upsilon,s}(t)\, dt = F_\#(\ell^{-\upsilon}\mathbf{1}_{\{0<\ell\le s\}}\bar\nu).$$

It is finite at every count. Uniformly in all the physical parameters, intervals, $m \ge 2$ and $0 < s < s_0$, at each fixed auxiliary band,

$$\begin{aligned} m^2\int_J q_{\upsilon,s}(t)\, dt &\le C_\upsilon s^{\gamma_0-\upsilon}(|J| + B_*^{-1}) \\ &\quad + C_{B_*,\upsilon}(1 + B_*|J|)m^3 \min\{s^{1/2-\upsilon}, \rho_{B_*}^{m(1-2\upsilon)/2}\}. \end{aligned} \tag{16.3}$$

Consequently, for every previously fixed $L, s > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J q_{\upsilon,s} \le C_\upsilon L s^{\gamma_0-\upsilon}. \tag{16.4}$$

The source with weight $\min(\ell, 1)^{-\upsilon}$ is also finite and has the corresponding fixed-width bound with a constant right side. There is no positive lower residence cutoff in these statements.

Proof. At the source level,

$$\ell^{-\upsilon}\mathbf{1}_{\{\ell\le s\}} = s^{-\upsilon}\mathbf{1}_{\{\ell\le s\}} + \upsilon\int_0^s v^{-\upsilon-1}\mathbf{1}_{\{\ell\le v\}}\, dv.$$

Use positive lower truncations first and then monotone convergence. The leading term of (16.2) integrates to $\gamma_0 s^{\gamma_0-\upsilon}/(\gamma_0-\upsilon)$. For $D>0$ the capped remainder obeys the exact identity

$$s^{-\upsilon}\min(D,s^{1/2})+\upsilon\int_0^s v^{-\upsilon-1}\min(D,v^{1/2})\,dv = \frac{1}{1-2\upsilon}\min\{s^{1/2-\upsilon},D^{1-2\upsilon}\}. \tag{16.5}$$

Split at D^2 when that point is less than s; the boundary terms cancel. Substitution of $D=\rho_{B_*}^{m/2}$ proves (16.3). An uncapped error independent of v would not be integrable here.

The physical cap alone also proves finite total mass for every $\upsilon<1/2$. Coarea on compact regular charts and positive exhaustion give absolute continuity for the exponents used above; the excluded critical set has zero source area. For the last assertion split at $s_0/2$: on its complement the weight is bounded, and the unweighted complete source estimate applies. Finally fix L,s,B_*, take the collision limsup, and only then enlarge B_*. The complete layer cake has already been evaluated at finite count. This proves (16.4). □

17. An uncut surface estimate from complete segment traces

The negative residence moment acts on the original volume source. To apply it to the outgoing surface one must still pay the change of the physical angular density. Integrating over the entire segment provides precisely the factor that makes this payment finite without an initial-gradient cutoff.

For $1<q<\infty$ define, for a measurable roof interval J,

$$\mathcal{N}_q(f;J)=\sup_{E\subset J,\ 0<|E|<\infty}|E|^{-1+1/q}\int_E|f(t)|\,dt. \tag{17.1}$$

This is a norm equivalent to the usual weak L^q norm. We shall use its subadditivity directly, not a change of density representative. Write $J^{[s]}=J+(-s,s)$.

Lemma 17.1 (A complete interval-pulse trace)**.** *Let a measure space Σ carry measurable finite intervals (a_σ,b_σ) of lengths $0<\ell_\sigma\le s$ and nonnegative weights T_σ. Put $f(t)=\int_\Sigma T_\sigma\mathbf{1}_{(a_\sigma,b_\sigma)}(t)\,d\sigma$. If $\upsilon=1-1/q$, then*

$$\mathcal{N}_q(f;J)\le\int_{\{(a_\sigma,b_\sigma)\cap J\neq\varnothing\}}T_\sigma\ell_\sigma^{1-\upsilon}\,d\sigma. \tag{17.2}$$

The assertion holds also for extended nonnegative integrals.

Proof. For $|E|=v$, Tonelli bounds the contribution of a segment by $T_\sigma\min(v,\ell_\sigma)$. In both cases $v\le\ell_\sigma$ and $v>\ell_\sigma$, $v^{-\upsilon}\min(v,\ell_\sigma)\le\ell_\sigma^{1-\upsilon}$. Integrate this inequality before taking the supremum over E. No disjointness of the roof intervals is assumed or needed. □

Theorem 17.2 (Weighted first-exit trace on every short segment)**.** *Use the complete Borel segment bijection in the original exact event and all its nongrazing outgoing endpoints. Let $T_j(\vartheta)=A_j(b_j-,\vartheta)$, $\ell_j=b_j-a_j$, and*

$$Z_s(t)=\sum_j\int_{E_j}T_j(\vartheta)\mathbf{1}_{\{0<\ell_j\le s,\ a_j<t<b_j\}}\,d\vartheta.$$

For $0<\eta<1/2$, $\kappa=(1-\eta)/2$ and $0<\upsilon<\min(\gamma_0,1-\kappa)$, set $q=(1-\upsilon)^{-1}$. At each finite count,

$$\mathcal{N}_q(Z_s;J)\le\int_{J^{[s]}}q_{\upsilon,s}(u)\,du+C_\eta s^{1-\kappa-\upsilon}\int_{J^{[s]}}v^\eta(u)\,du. \tag{17.3}$$

Thus for each fixed $L, s > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(m^2 Z_s; J) \le C_\upsilon (L+2s) s^{\gamma_0-\upsilon} + C_\eta (L+2s) s^{1-\kappa-\upsilon}. \tag{17.4}$$

All positive residence lengths, all initial gradients and all incoming endpoint types are included; no optical-sensitivity cutoff is used.

Proof. Fix one selected segment and suppress its angular index. Its exact physical density satisfies $|A'| \le CAI_m/z$. For $a < t < b$,

$$A(b-) \le A(t) + C\int_t^b A(u) I_m(u)/z(u)\, du.$$

Integrate over the entire interval (a, b), multiply by $\ell^{-\upsilon}$, and use Tonelli. This gives

$$A(b-)\ell^{1-\upsilon} \le \int_a^b A(t)\ell^{-\upsilon}\, dt + C\ell^{-\upsilon}\int_a^b (u-a)A(u)I_m(u)/z(u)\, du. \tag{17.5}$$

Forward radial growth, first from an interior point and then in the limit toward a, implies $z(u)^2 \ge 2\lambda_\circ(u-a)$. Hence

$$(u-a)z(u)^{-1} \le C_\eta (u-a)^{1-\kappa} z(u)^{-\eta} \le C_\eta \ell^{1-\kappa} z(u)^{-\eta}.$$

The second term of (17.5) is therefore at most $C_\eta s^{1-\kappa-\upsilon}\int_a^b AI_m z^{-\eta}$. This calculation pays the variation of both ω and χ in $A = \omega/(4\pi c_* \chi)$. It does not replace an inverse Jacobian in a first-exit formula by one.

Every selected segment that intersects J lies in $J^{[s]}$. Sum (17.5) over precisely these physical segments. The Borel bijection and its incident-component convention count each interior point once. The first sum is bounded by the actual source $q_{\upsilon,s}$; the second by the actual fractional source v^η. Nongrazing outgoing traces are used on the left; dropping that endpoint selector only on the two positive upper sources introduces neither a new normalization nor another label. Lemma 17.1 proves (17.3).

Initially all statements may be read as nonnegative extended inequalities on compact subsegments. The finite actual moments, Tonelli and positive exhaustion make the right side finite and justify the complete sum and endpoint limit. Apply (16.4) and (15.8), fixing both interval lengths and all bands before the collision limsup, to obtain (17.4). □

Theorem 17.3 (The complete untruncated first-exit surface)**.** *For every $1 < q < 32/31$, put $\upsilon = 1 - 1/q$ and $\nu_q = \gamma_0 - \upsilon = 1/q - 31/32 > 0$. For every fixed $L > 0$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(m^2 \overline{\mathcal{F}}_r; J) \le C_{q,L} r^{\nu_q} \tag{17.6}$$

for all sufficiently small previously fixed $r > 0$. This is the actual one-sided outgoing first-exit surface with no lower residence or gradient cutoff. For example $q = 64/63$ gives $\nu_q = 1/64$.

Proof. The complete residence kernel splits into its long part $\mathcal{L}_r$ with $\ell > 2r$ and a short part bounded by Z_{2r}. The retained long-residence height theorem (15.13) gives $\mathcal{N}_q(m^2\mathcal{L}_r; J) \le |J|^{1/q} m^2 \|\mathcal{L}_r\|_\infty$ and an ordered upper bound $CL^{1/q} r^{\gamma_0}$. Use Theorem 17.2 with $s = 2r$ for the complete short part. Take, for instance, $\eta = 1/4$. Then $1 - \kappa - \upsilon > \gamma_0 - \upsilon > 0$. Subadditivity of (17.1) proves the result. The short contribution was summed at finite count through an actual negative moment, not reconstructed from infinitely many band limsups. □

The weak exponent is finite. Equation (17.6) controls the distribution of heights and does not assert that the surface essential supremum tends to zero. The positive cutoff has been removed from this new norm theorem, not from the earlier essential-height theorem.

18. The original raw density in a complete local norm

The surface theorem applies to every first-exit component, including the zero-residence tail. Combined with the retained fractional source, it gives a norm estimate for the original bad density itself. The roof density is not averaged in the conclusion.

Lemma 18.1 (A truncated potential in a local norm)**.** *Let $0 < \kappa < 1$, $q > 1$, and $1/q > \kappa$. For a nonnegative locally integrable v, put $P_r v(t) = \int_0^r s^{-\kappa} v(t+s)\,ds$. Then*

$$\mathcal{N}_q(P_r v; J) \le \|P_r v\|_{L^q(J)} \le (1-\kappa q)^{-1/q} r^{1/q-\kappa} \int_{J+(0,r)} v(u)\,du. \tag{18.1}$$

Also, if $1 \le p < q$,

$$\|f\|_{L^p(J)} \le \left(\frac{q}{q-p}\right)^{1/p} |J|^{1/p-1/q} \mathcal{N}_q(f; J). \tag{18.2}$$

Proof. For fixed u, the kernel $\mathbf{1}_{\{0<u-t<r\}}(u-t)^{-\kappa}$ has $L^q(dt)$ norm at most $(1-\kappa q)^{-1/q} r^{1/q-\kappa}$. Minkowski's integral inequality over $u \in J + (0,r)$ gives the second inequality in (18.1); Hölder on each measurable $E \subset J$ gives the first. Apply this to truncated kernels and pass monotonically if needed. To prove (18.2), the definition of $\mathcal{N}_q$ implies $|\{|f| > \lambda\} \cap J| \le \min\{|J|, (\mathcal{N}_q/\lambda)^q\}$. Integrate the distribution function against $p\lambda^{p-1}$ and split at $\lambda = \mathcal{N}_q |J|^{-1/q}$. □

Theorem 18.2 (Complete original bad-source control)**.** *For every $1 < q < 32/31$ and fixed $L > 0$, under the explicitly retained inputs,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(m^2 b^\varepsilon; J) \le C_q L^{1/q} \varepsilon^{1/16} \tag{18.3}$$

for every fixed sufficiently small $\varepsilon > 0$. All initial momenta, positive total residences, collision incidences, terminal tests and exact labels are retained. No original source restriction is removed in forming b^ε.

Proof. The source-shift inequality (15.3) and the complete two-sided optical payment (15.12) give, for each fixed $r > 0$,

$$b^\varepsilon \le 2A_r b^\varepsilon + C\overline{\mathcal{F}}_r + C_\eta \mathsf{P}_{\eta,r} v^\eta + C_\Theta (1+\varepsilon^{-1})\sqrt{r}\, A_r p.$$

Every term is a complete source term, not a sum with an unknown physical-word multiplicity. Apply (17.1), then take the collision limsup with ε, r and all auxiliary bands fixed. The fixed-width source bounds in Input 15.2 give

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad \limsup_m \sup_{R,n,k} m^2 \|A_r p\|_\infty \le C.$$

Theorem 17.3 controls the surface by $C_{q,L} r^{\nu_q}$. Choose $\eta = 1/4$, so $\kappa = 3/8 < 1/q$; Lemma 18.1 and the actual fixed-width fractional input control the potential by $C_{q,\eta}(L+r) r^{1/q-\kappa}$. Thus the ordered bad-source norm is at most

$$C_q L^{1/q} \varepsilon^{1/16} + C_{q,L} r^{\nu_q} + C_{q,\eta}(L+r) r^{1/q-\kappa} + C_\Theta L^{1/q}(1+\varepsilon^{-1})\sqrt{r}.$$

Enlarge auxiliary bands only after the collision limit. Now let $r \downarrow 0$ with ε fixed. All three powers are positive. This proves (18.3). Averaging is an intermediate proof operation; the left side is the raw density. □

Theorem 18.3 (Exact-label raw local limit in finite local norms)**.** *Let*

$$f_{m,R,n,k}(t) = m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1, k_2, t, n).$$

For every fixed $M, L > 0$ and every $1 < q < 32/31$,

$$\sup_{R,n,k}\sup_{|J|=L}\mathcal{N}_q(f_{m,R,n,k}\mathbf{1}_{\mathcal{C}_{m,R,n,k}(M)};J)\longrightarrow 0. \tag{18.4}$$

For every $1 \le p < 32/31$ the corresponding $L^p(J\cap\mathcal{C}(M))$ norm also tends to zero, uniformly in the same parameters and interval locations. The reference includes every retained arithmetic peak and all zero classes; no reference density is divided by.

Proof. For each fixed large B use the original positive remainder (15.9). Theorem 18.2 and monotonicity under restriction to $\mathcal{C}(M)$ imply

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L}\mathcal{N}_q(f_{m,R,n,k}\mathbf{1}_{\mathcal{C}(M)};J)\le C_{q,M,L}B^{-1/192}.$$

Indeed $\varepsilon(B)^{1/16}=A_0^{1/16}B^{-1/192}$, and the uniform essential bound on e_B contributes at most $L^{1/q}C_MB^{-1/192}$. The left side is independent of B. Let $B\to\infty$ last. Nonnegativity proves the asserted convergence. For any stated p choose q strictly between $\max(1,p)$ and $32/31$ and use (18.2). □

The governing objective remains (14.5). The theorem above is a proved intermediate conclusion for its unchanged raw error, not a replacement target. Its uniformity is on every fixed length L, not automatically on an interval growing with m. Neither a finite weak norm nor these finite L^p norms control the normalized essential supremum. The exact original-source criterion of Supplement Theorem 34.2 is therefore retained: vanishing of its critical/zero-residence charge still has to be proved for the pointwise endpoint. Same-roof ratios and unconditional likelihood statements are not promoted on the strength of the local-norm theorem alone.

19. The critical surface exponent

Put $\gamma = 1/32$ and $Q = (1-\gamma)^{-1} = 32/31$. We fix $\eta = 1/4$, $\kappa = 3/8$ and $d = 1-\kappa = 5/8$ in this section. The endpoint estimate does not integrate a negative residence moment of order γ. Instead it retains the size of the tested roof set until after summing the complete residence distribution.

Lemma 19.1 (A capped dyadic pulse sum)**.** *Let $s_j = 2^{-j}s$, $0<\theta<1$ and $v, s > 0$. Then*

$$\sum_{j\ge0}\min\{1,2v/s_j\}s_j^\theta \le C_\theta\min\{v^\theta, s^\theta\}. \tag{19.1}$$

For $0<\gamma<a<1$ and $0<D\le1$, one also has

$$\sum_{j\ge0}\min\{1,2v/s_j\}\min\{D,s_j^a\}\le C_\gamma D^{1-\gamma/a}v^\gamma. \tag{19.2}$$

All sums are formed before taking a supremum over v.

Proof. For $v<s$, split at the first index for which $s_j\le v$. The later terms sum a geometric series of order v^θ. The earlier terms are bounded by $2vs_j^{\theta-1}$; their geometric sum has the same order. The full sum is also bounded by $\sum_j s_j^\theta = s^\theta/(1-2^{-\theta})$. This proves (19.1). Finally $\min(D,x^a)\le D^{1-\gamma/a}x^\gamma$ for every $x>0$. Apply the first assertion at exponent γ. □

Write Z_s for the complete outgoing pulse source defined in Theorem 17.2. Its traces are the actual one-sided nongrazing outgoing traces, including every incoming type. Let $J^{[s]} = J+(-s,s)$, $|J|=L$, and let $V_m(J^{[s]}) = m^2\int_{J^{[s]}} v^{1/4}(u)\,du$.

Theorem 19.2 (Finite-count critical distribution bound)**.** *For every fixed auxiliary band B_*, all $m \geq 2$ and $0 < s < s_0$, the full short surface satisfies*

$$\mathcal{N}_Q(m^2 Z_s; J) \leq C(L + 2s + B_*^{-1}) + Cs^{19/32} V_m(J^{[s]}) + C_{B_*}(1 + B_*(L + 2s))m^3 \rho_{B_*}^{15m/32}. \tag{19.3}$$

The bound holds uniformly over exact labels, radius and interval location. It includes the supremum over every measurable subset of J, even one depending on m, before either limit. Consequently

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q(m^2 Z_s; J) \leq C(L + 2s)(1 + s^{19/32}). \tag{19.4}$$

There is no positive lower residence, incidence or gradient cutoff.

Proof. For one maximal segment σ, put $T_\sigma = A_\sigma(b_\sigma -)$, $U_\sigma = \int_{a_\sigma}^{b_\sigma} A_\sigma$ and $V_\sigma = \int_{a_\sigma}^{b_\sigma} A_\sigma I_m z^{-1/4}$. The whole-segment calculation in (17.5), without a residence weight, gives

$$T_\sigma \ell_\sigma \leq U_\sigma + C\ell_\sigma^d V_\sigma. \tag{19.5}$$

It differentiates the complete density $\omega/(4\pi c_* \chi)$. The factor $u - a_\sigma$ obtained by integration over starting roofs is what permits the radial lower bound to be used at a critical incoming end. No inverse physical Jacobian is discarded.

Fix a measurable $E \subset J$ of measure $v > 0$. A segment in the band $s_j/2 < \ell_\sigma \leq s_j$ contributes at most $T_\sigma \min(v, \ell_\sigma)$ to $\int_E Z_s$. Its contribution is therefore at most

$$\min\{1, 2v/s_j\}\{U_\sigma + Cs_j^d V_\sigma\}.$$

Every such segment intersecting J lies in $J^{[s]}$. The Borel segment bijection counts its interior once. Thus the sum of U_σ in that band is at most the original volume mass $\bar{\nu}\{F \in J^{[s]}, \ell \leq s_j\}$, and the sum of V_σ is at most $\int_{J^{[s]}} v^{1/4}$. These are positive upper comparisons on the same exact event; no occupation coefficients are combined.

Insert (16.2) in the first sum. Its leading part is bounded by $C(L + 2s + B_*^{-1})v^\gamma$, by (19.1). In its remainder apply (19.2) with $a = 1/2$ and $D = \rho_{B_*}^{m/2}$. Since $1 - 2\gamma = 15/16$, its bound is

$$C_{B_*}(1 + B_*(L + 2s))m^3 \rho_{B_*}^{15m/32} v^\gamma.$$

This is the complete finite-count tail, not a sum of separate collision limsups. For the variation part use (19.1) at exponent d and $\min(v^d, s^d) \leq v^\gamma s^{d-\gamma}$. Here $d - \gamma = 19/32$. Dividing by v^γ and then taking the measurable-set supremum proves (19.3). Tonelli and monotone exhaustion justify the full sum, since all terms used in the upper comparison are nonnegative.

Fix L, s, B_* before the collision limsup. Apply (15.8) to the actual fractional source, and enlarge B_* only afterwards. The capped remainder vanishes, proving (19.4). □

Corollary 19.3 (The full surface at the critical exponent)**.** *For fixed $L > 0$ and sufficiently small fixed $r > 0$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q(m^2 \overline{\mathcal{F}}_r; J) \leq C\{L^{31/32} r^{1/32} + (L + 4r)(1 + r^{19/32})\}. \tag{19.6}$$

In particular the right side is $O(r)$ when $L = r$.

Proof. The surface with $\ell > 2r$ is the long-residence source in (15.13); its measurable-set norm is bounded by $L^{1/Q}$ times its essential height. The complementary surface is dominated by Z_{2r}. Apply Theorem 19.2 and subadditivity. The two classes exhaust the original first-exit assignment, retaining incident components and backward lifetime. □

The exponent 32/31 is inherited from the presently available residence exponent 1/32, not asserted to be physically sharp. The new assertion is endpoint boundedness. For a fixed L, its right side need not tend to zero with r; no endpoint convergence or essential-height bound is inferred from it.

20. A logarithmic endpoint on the original source

Keep $\gamma = 1/32$ and $Q = 32/31$. Endpoint boundedness can be combined with small mass without exchanging a microscopic supremum and a collision limit. The resulting estimate acts on the original critical-or-short-residence union, not only on its complement.

Proposition 20.1 (Uniform critical weak bound for the raw density)**.** *For every fixed $L > 0$,*

$$\limsup_{m} \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_Q(m^2 p_{n,R}(k,m,\cdot); J) \le K_L. \tag{20.1}$$

For the raw error $f_{m,R,n,k} = m^2 p - \mathcal{L}$, the same assertion holds with $f_{m,R,n,k}\mathbf{1}_{\mathcal{C}(M)}$ and a constant $K_{M,L}$.

Proof. Use the source comparison at auxiliary guard parameter one. Its first factor is zero since $c_0 \le 1$, so $W^1 = 1$ and $b^1 = p$. Equations (15.3) and (15.12) give

$$p \le (2 + C\sqrt{r}) A_r p + C\overline{\mathcal{F}}_r + C\mathsf{P}_{1/4,r} v^{1/4}.$$

Here $r > 0$ is fixed and small; every source on the right is complete. The fixed-width input bounds the first term in essential height. Corollary 19.3 bounds the second in $\mathcal{N}_Q$. Minkowski gives, since $\kappa Q < 1$,

$$\|\mathsf{P}_{1/4,r} v\|_{L^Q(J)} \le (1 - \kappa Q)^{-1/Q} r^{1/Q - \kappa} \int_{J+(0,r)} v.$$

Hölder implies $\mathcal{N}_Q \le \|\cdot\|_{L^Q}$. The actual fractional fixed-width input therefore bounds the last term. Take the collision limsup at fixed widths and auxiliary bands, and enlarge the bands afterwards. This proves the first assertion. For the error, use the positive scalar remainder at one fixed sufficiently large reconstruction band. Its positive term is bounded by $m^2 p$, and its other term has bounded central essential height. No pointwise estimate for p or division by the arithmetic reference is used. □

For $|J| = L$ and $\sigma > 0$ define the logarithmically weakened critical measurable-set norm

$$\mathcal{N}_{Q,\sigma}(f; J) = \sup_{E \subset J,\ 0 < |E| \le L} \frac{|E|^{-\gamma} \int_E |f|}{(1 + \log(L/|E|))^{\sigma}}. \tag{20.2}$$

The logarithmic denominator is part of the conclusion, not a change to the density, event, or physical guard.

Lemma 20.2 (Endpoint interpolation with the mass)**.** *Suppose $\mathcal{N}_Q(f; J) \le A$, $A > 0$, and put $t = \|f\|_{L^1(J)}/(AL^{\gamma})$. Then $0 \le t \le 1$. For $0 < t < 1$,*

$$\mathcal{N}_{Q,\sigma}(f; J) \le A \left\{ t^{1/2} + \left(1 + \frac{\log(1/t)}{2\gamma}\right)^{-\sigma} \right\}. \tag{20.3}$$

If $t = 0$, then $f = 0$ almost everywhere and the left side is zero.

Proof. Split the tested sets at measure $\delta = Lt^{1/(2\gamma)}$. For $|E| \ge \delta$, their numerator is at most $\delta^{-\gamma}\|f\|_1 = At^{1/2}$. For smaller sets use the bound A for the numerator and $1 + \log(L/|E|) \ge 1 + \log(1/t)/(2\gamma)$ in the denominator. The maximum of the two bounds is at most their sum. This is an inequality for each actual function before taking any limit. □

Theorem 20.3 (The raw error at the logarithmic critical endpoint)**.** *For every fixed $M, L, \sigma > 0$,*

$$\sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q,\sigma}(f_{m,R,n,k}\mathbf{1}_{\mathcal{C}(M)}; J) \longrightarrow 0. \tag{20.4}$$

In particular the critical exponent is attained with any positive logarithmic loss. Convergence in $\mathcal{N}_Q$ without that loss is not asserted.

Proof. The subcritical raw theorem includes uniform local L^1 convergence. Proposition 20.1 gives an eventually uniform bound $A = 2K_{M,L} + 1$ for the critical norm. Apply Lemma 20.2 to each actual error, then take the parameter and interval suprema. Their mass ratios tend uniformly to zero. Equivalently the inherited remainder gives an ordered L^1 bound $C_{M,L}B^{-1/192}$ for each fixed outer band; inserting it in the lemma and sending B to infinity last gives the same conclusion. No $B = B(m)$ is chosen. □

20.1. **The exceptional union itself.** Set $a_h = h^{1/8}$, $s_h = h^{1/4}$ and define the original weighted restriction

$$R_h^\varepsilon(t) = \int W^\varepsilon(x)\mathbf{1}_{\{z(x)<a_h \text{ or } \ell(x)<s_h\}}\, d\mu_t(x). \tag{20.5}$$

The two conditions form one union in this definition. All endpoint types and all original exact labels remain in the source.

Theorem 20.4 (A critical logarithmic estimate on the exceptional source)**.** *For every fixed $L, \sigma > 0$, uniformly in $0 < \varepsilon \le 1$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J R_h^\varepsilon \le C_L h^{1/128}, \tag{20.6}$$

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q,\sigma}(m^2 R_h^\varepsilon; J) \le C_{L,\sigma}(1 + |\log h|)^{-\sigma}. \tag{20.7}$$

All suprema over tested measurable sets are part of the norm before the collision limsup. The second estimate is not an essential-height bound and does not prove the residual charge zero.

Proof. On $0 < z < a$, one has $1 \le a^{1/4} I_m z^{-1/4}$. The critical fractional input therefore gives ordered interval mass at most $C_L a^\alpha$, with $\alpha = 129/128$. The complete short-residence bound gives at most $C_L s^\gamma$. Since $W^\varepsilon \le 1$, a positive union bound at $a = a_h$, $s = s_h$ proves

$$C_L(h^{129/1024} + h^{1/128}) \le C'_L h^{1/128}.$$

This upper comparison does not double count the source definition. Also $R_h^\varepsilon \le p$, so its critical weak norm is bounded by Proposition 20.1, independently of h and ε. Fix h before the collision limsup and apply Lemma 20.2. The mass ratio is at most $C_L h^{1/128}$ in that limit. Its power term is absorbed by the logarithmic right side in (20.7). The estimate applies to the exceptional union itself, including critical initial points in every positive neighborhood and arbitrarily short regular segments; it uses no common-survivor restriction. □

The cofinal limit of the essential heights of (20.5) is the same original charge as in the retained source-localization theorem: both positive thresholds decrease to zero, and the restriction is monotone in each threshold. Thus the governing pointwise assertion still requires that charge to vanish in the outer reconstruction limit. A logarithmic measurable-set estimate is an additional quantified restriction on concentration, not a proof that its essential height vanishes.

21. The complete central window in diffusive coordinates

Uniformity over the location of a fixed-length interval does not bound an unnormalized integral over a growing interval. It does, however, give a global theorem in the natural diffusive roof measure. We state that normalization explicitly.

For fixed $M > 0$ let $H = [-M, M]$ and put

$$\widetilde{f}_{m,R,n,k}(u) = f_{m,R,n,k}(m\bar{\tau}_R + \sqrt{m}\,u)\mathbf{1}_{\mathcal{C}_{m,R,n,k}(M)}(m\bar{\tau}_R + \sqrt{m}\,u). \tag{21.1}$$

There is no integration over neighboring displacement or occupation labels. The amplitude remains the original $m^2p - \mathcal{L}$.

Lemma 21.1 (Gluing local norms with their exact scale factor)**.** *Let a roof interval K be partitioned into N pieces, each contained in an interval of length one. If the local L^p norms of f are at most d_m, then*

$$\int_K |f|^p \le N d_m^p.$$

If its local critical measurable-set norms are at most A_m, then

$$\mathcal{N}_Q(f; K) \le A_m N^{1-\gamma}. \tag{21.2}$$

Under $t = t_0 + \sqrt{m}\,u$, the critical norm is multiplied by $m^{-(1-\gamma)/2}$ and the pth-power integral by $m^{-1/2}$.

Proof. The first assertion is finite additivity. For $E \subset K$ write v_j for the measures of its intersections with the partition. Then $\int_E |f| \le A_m \sum_j v_j^\gamma$. Hölder on the finite counting measure gives $\sum_j v_j^\gamma \le N^{1-\gamma}(\sum_j v_j)^\gamma$. Divide by $|E|^\gamma$ and take the supremum. The scale factors follow by change of variable, including the measure in the measurable-set normalization. □

Theorem 21.2 (A full diffusive-window raw limit)**.** *For every fixed $M > 0$, every $1 \le p < Q$ and every $\sigma > 0$,*

$$\sup_{R,n,k} \|\widetilde{f}_{m,R,n,k}\|_{L^p(H)} \longrightarrow 0, \tag{21.3}$$

$$\sup_{R,n,k} \mathcal{N}_{Q,\sigma}(\widetilde{f}_{m,R,n,k}; H) \longrightarrow 0, \tag{21.4}$$

$$\limsup_m \sup_{R,n,k} \mathcal{N}_Q(\widetilde{f}_{m,R,n,k}; H) \le C_M. \tag{21.5}$$

In original roof coordinates the first assertion is exactly

$$\sup_{R,n,k} \left\{ \frac{1}{\sqrt{m}} \int_{\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)|^p \, dt \right\}^{1/p} \longrightarrow 0. \tag{21.6}$$

No unnormalized growing-window norm or pointwise theorem is asserted.

Proof. The central roof interval is contained in $[m\bar{\tau}_R - M\sqrt{m}, m\bar{\tau}_R + M\sqrt{m}]$. Partition this interval into at most $2M\sqrt{m} + 2$ pieces as in Lemma 21.1. The inherited uniform local L^p error d_m tends to zero. After scaling, its pth-power integral is at most $(2M + 2/\sqrt{m})d_m^p$. This proves (21.3) and its exact coordinate form.

The local critical norm is eventually bounded by a common constant by Proposition 20.1. Its scaled glued bound is at most $A_m(2M + 2/\sqrt{m})^{1-\gamma}$, proving (21.5). Now use the global L^1 convergence already proved and Lemma 20.2 on the fixed interval H. This proves (21.4). The number of intervals is handled in the displayed scale factor, not treated as fixed before the collision limit. □

The residue concentration requirement for (14.5) is unchanged. The new estimates keep every measurable roof subset in the critical distribution norm before the collision limit, but their logarithmic loss still allows sufficiently narrow high peaks. Likewise $dt/\sqrt{m}$ in (21.6) cannot be replaced by dt without an additional collision-rate estimate. Neither a same-roof path ratio nor a forward likelihood follows merely from these global finite-norm conclusions. Their scalar pointwise denominator and positive-arithmetic restrictions remain the retained downstream hypotheses.

22. The unchanged pointwise endpoint

The new negative residence theorem is a finite-count assertion for an actual singular insertion. Its full layer cake includes arbitrarily short physical segments before the collision limit. The outgoing trace theorem similarly sums all incident components before applying a norm. Thus the complete surface weak-norm decay and raw local-norm limit proved here require no positive residence cutoff.

These results do not identify a finite local norm with essential height. The endpoint in (14.5) continues to require

$$\lim_{B\to\infty} \mathfrak{R}_{\varepsilon(B),M} = 0,$$

where $\mathfrak{R}$ is the original weighted-source concentration charge of Supplement Theorem 34.2. The critical distribution and logarithmic estimates constrain this same source but do not exclude narrow peaks in its essential supremum. The older sufficient pairs $(\mathfrak{A},\mathfrak{B})$ and $(\mathfrak{A},\mathfrak{Q})$ remain valid with their exact hypotheses in the complete archive. No charge-zero assertion is inferred here.

In particular, a strong local L^p theorem for $p < 32/31$ is not a uniform denominator theorem on positive arithmetic classes. The scalar-first path remainder, collision-to-return clock transfer and forward likelihood statements are preserved, not promoted to new unconditional same-roof ratios. Zero arithmetic classes are never normalized. All previous mathematical files and the complete preceding article remain in the supplement and provenance.

Part 16. Untruncated residence moments and the raw local norm

Retained revision-88 abstract. We study the pointwise local limit problem for the complete displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. A count-independent physical cap for very short regular residence can be combined with the finite-count spectral bound before taking either limit. This yields actual negative residence moments at every order below 1/32. Integrating the exact angular-density variation over a whole maximal segment converts those volume moments into a weighted outgoing trace without an initial-gradient cutoff. An interval-pulse inequality then proves decay in weak L^q for the complete, untruncated first-exit surface, for every $1 < q < 32/31$. The same range gives uniform local-norm convergence of the original raw exact-label density to its full arithmetic reference, with strong L^p convergence for $1 \le p < 32/31$. The original pointwise objective is unchanged: these complete finite-norm theorems do not yet establish vanishing of its critical and zero-residence essential-height charge.

23. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\ \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6,-1/2), \quad (-\theta_R,\sin\theta_R), \quad (\pi/3-\theta_R,\sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1-6q^2+q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{23.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{23.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{23.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

23.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0,0,\bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R,a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{23.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{23.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

23.2. **Complete sources without a positive residence cutoff.** The pointwise error in (23.5) remains the organizing objective. The new assertions estimate the same raw density in finite local norms and the actual outgoing surface with all positive residence lengths retained. These statements are not conditional on a microscopic charge being zero.

For $1<q<\infty$ let $\mathcal{N}_q(f;J)$ be the measurable-set norm in (26.1), equivalent to weak L^q on J. For the original first-exit surface write $\overline{\mathcal{F}}_r$ for its average over exit widths between zero and r, as defined below.

Theorem 23.1 (Untruncated surface and raw-density estimates)**.** *Under the explicit inherited inputs of the next section, for every $1<q<32/31$ and fixed $L>0$,*

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} \mathcal{N}_q(m^2\overline{\mathcal{F}}_r;J)\le C_{q,L}r^{1/q-31/32}.$$

There is no lower gradient, residence or incidence cutoff in this estimate. For every fixed $M,L>0$, the raw scalar error satisfies

$$\sup_{R,n,k}\sup_{|J|=L}\mathcal{N}_q\Big((m^2p_{n,R}(k,m,\cdot)-\mathcal{L}_{m,R}(k_1,k_2,\cdot,n))\mathbf{1}_{\mathcal{C}_{m,R,n,k}(M)};J\Big)\longrightarrow 0.$$

The corresponding strong L^p convergence holds for every $1\le p<32/31$. All arithmetic classes remain unnormalized.

The proofs are Theorems 26.3 and 27.3. The new source theorem first establishes a negative moment of total regular residence by a capped layer cake at finite collision count. Its geometric trace application uses the whole same-label segment; the source term is not replaced by a boundary collar. A finite q is essential to the conclusion. The unchanged essential-supremum problem remains separate.

24. Explicit analytic and geometric inputs

We collect the indispensable inherited assertions in the exact form used by the new proofs. Their source theorems and operator dependencies are specified here and in Supplement Appendix A. The new potential and first-exit estimates do not assume a surface spectral insertion theorem or an unweighted pointwise local bound.

Input statement 24.1 (Physical flow and source comparison)**.** On the regular part Ω of each exact event, endpoint angles $q=(\alpha_0,\alpha_m)$ give $F=L$, $g=\nabla_q F=R(-p_0,p_m)$, $K=D_q^2F$ and $\vartheta=\arg g$. Put

$$\lambda_\circ=9\sqrt{19}/200,\qquad \Lambda_\circ=47/6,\qquad \omega=|\partial_{p_0}\alpha_m|^{-1},\qquad \chi=|\det D_q(F,\vartheta)|.$$

The bounds $\lambda_\circ I\le K\le\Lambda_\circ I$ and $\lambda_\circ\le\chi\le\Lambda_\circ$ hold uniformly. The explicit field

$$X_q=\frac{K^{-1}g}{g^{\mathsf{T}}K^{-1}g}$$

is stopped at true physical and section-decision boundaries, not at chart seams. Its flow satisfies $XF=1$, $X\vartheta=0$ and

$$2\lambda_\circ s\le |g(\Phi_s x)|^2-|g(x)|^2\le 2\Lambda_\circ s. \tag{24.1}$$

Its endpoint-curve length is at most $C\sqrt{s}$. For $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$ one has $|X\omega| \le C\omega I_m/|g|$ on regular survivors. At a genuine grazing exit some internal incidence tends to zero and ω tends to zero. Forward interior critical termination cannot occur.

Every starting point with forward lifetime at most r belongs to a layer of one of the following forms, at an actual collision or flight:

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \quad \Delta_j \le C_E q_0^{d_j'}\sqrt{r}, \quad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}. \tag{24.2}$$

The physical coarea covariance, proved in the appendix in the convention used here, has Jacobian $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$. Writing E_h^ε for all weighted starting exit mass and

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon(x)[1 - \omega(\Phi_h x)/\omega(x)]_+ \, d\mu_t(x),$$

the complete source satisfies, for each fixed r and almost every t,

$$b^\varepsilon(t) \le 2A_r b^\varepsilon(t) + \bar{E}_r^\varepsilon(t) + \bar{O}_r^\varepsilon(t) + C_\Theta(1+\varepsilon^{-1})\sqrt{r}\, A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\, ds. \tag{24.3}$$

These are the endpoint-field, all-contact motion, uniform cubic-action and separated-volume results in retained modules 174, 177, 180–182. The guard term in (24.3) is the complete product guard, not a protective subword source. The derivative and continuation assertions are confined to regular physical trajectories.

Input statement 24.2 (Complete local source estimates)**.** For every fixed auxiliary band B_*, constants $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ give the following bounds uniformly in radius, exact labels, interval J and mark $0 \le j \le m$. Let $S_{R,s}$ be the union of strips of width Cs about the finitely many coordinate lines supporting ∂Y_R^*. For s small,

$$m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le Cs^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \rho_{B_*}^{m/2}, \tag{24.4}$$

$$m^2 \int_J b^\delta(t)\, dt \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \rho_{B_*}^{m/2}. \tag{24.5}$$

The local extension to a fixed remaining compact range of δ is included. At auxiliary parameter one $W^1 = 1$, so $b^1 = p$.

Put $z = |g|$, $\alpha = 129/128$ and, for $0 < \eta < 1/2$, define the actual nonnegative source measures

$$\begin{aligned} v^{\eta,b}(t)\, dt &= L_\#(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_\Pi \nu_R^*), \\ v^\eta(t)\, dt &= L_\#(I_m z^{-\eta} \mathbf{1}_\Pi \nu_R^*). \end{aligned} \tag{24.6}$$

They are finite and absolutely continuous at every fixed count, and for $0 < b < b_1$,

$$m^2 \int_J v^{\eta,b}(t)\, dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|)m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \tag{24.7}$$

Adding the bounded-gradient part gives, for every fixed $L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\eta \le C_\eta L, \qquad \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta,b} \le C_\eta b^{\alpha-\eta} L. \tag{24.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards; leading constants are independent of that band and of the previously fixed interval length.

Equation (24.4) is Supplement Theorem 175.3. The physical source bound (24.5) is Supplement Theorem 179.1, in particular its finite-window estimate. The fractional bounds are the positive layer-cake conclusions of retained module 187, stated in full above. The joint spectral and common-space assumptions used there are recorded in the appendix. No estimate for an untruncated first inverse gradient or for a singular roof convolution is part of this input.

Input statement 24.3 (The scalar positive remainder)**.** For large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{24.9}$$

The source b^ε is nonnegative and has a compactly supported BV density at fixed count. No collision-uniform variation bound is assumed. Supplement Theorem 181.4 gives this remainder with the same complete finite-peak reference.

24.1. **Maximal segments and the two existing charges.** The retained Borel parametrization, Supplement Theorem 7.1, is a bijection

$$\Psi : \coprod_j \{(\theta, u) : \theta \in E_j,\ a_j(\theta) < u < b_j(\theta)\} \longrightarrow \Omega,$$

with Borel inverse. Its exact density is $A_j(u,\theta) = \omega(\Psi_j(\theta,u))/(4\pi c_* \chi(\Psi_j(\theta,u)))$. Every segment remains in one regular exact event until a true boundary; Borel chart seams are not exits. Nongrazing outgoing faces retain their incident physical component and carry $d\zeta_+ = A_j(b_j-,\theta)\, d\theta$. The full angular variation is $|\partial_u A_j| \le C A_j I_m / z$. These are measure identities with the null-set convention already specified, not bounds with an unknown number of words.

The true first-exit assignment is

$$\mathcal{F}_h(t) = \int_{\Gamma_{\rm ng}^+} \mathbf{1}_{\{0<b-t\le h,\ b-t<\tau^-(y)\}}\, d\zeta_+(y), \qquad \overline{\mathcal{F}}_r = r^{-1}\int_0^r \mathcal{F}_h\, dh.$$

No lower gradient cutoff is imposed. Put $l = b - a$ and define

$$\mathcal{Z}_\delta(t) = \int_{\Gamma_{\rm ng}^+} \mathbf{1}_{\{0<l\le\delta\}} \mathbf{1}_{\{a<t<b\}}\, d\zeta_+, \qquad q_\delta(t)\, dt = F_\#(\mathbf{1}_{\Omega_+^{\rm ng}} \mathbf{1}_{\{l\le\delta\}} \bar\nu).$$

The second object is the original collision volume restricted to the same actual short segments, not a volume collar for a boundary. For source families set

$$\mathcal{H}_M(f) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} f(t), \quad \mathfrak{B}_M = \lim_{\delta\downarrow 0} \mathcal{H}_M(\mathcal{Z}_\delta), \quad \mathfrak{Q}_M = \lim_{\delta\downarrow 0} \mathcal{H}_M(q_\delta).$$

These limits are monotone extended nonnegative numbers.

For $0 < \eta < 1/2$, with $\kappa = (1-\eta)/2$, retain

$$\mathsf{P}_{\eta,h} v^\eta(t) = \int_0^h s^{-\kappa} v^\eta(t+s)\, ds, \qquad \mathfrak{A}_{\eta,M} = \lim_{h\downarrow 0} \mathcal{H}_M(\mathsf{P}_{\eta,h} v^\eta).$$

Supplement Theorems 8.2 and 9.1 give

$$\mathfrak{B}_M \le \mathcal{H}_M(\overline{\mathcal{F}}_r) \le \mathfrak{B}_M + C_\eta r^{1/32}. \tag{24.10}$$

Their band sum is finite before the collision limsup at each positive residence cutoff. Its leading constant is cutoff independent; the complete zero-residence tail is still inside $\mathcal{H}_M$ before its cutoff tends to zero. The active offset correction to historical module 189 remains Supplement equation (6.1); the historical inclusion is explicitly marked superseded. None of these inputs asserts charge zero.

24.2. **The precise finite exit and surface inputs.** For total residence $\ell = \tau^- + \tau^+$, let $e_s(t) = \int \mathbf{1}_{\{\tau^+ \le s\}}\, d\mu_t$. The finite-count statement of Supplement Theorem 5.1 is

$$m^2 \int_J e_s \le C s^{1/32}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \rho_{B_*}^{m/2}. \tag{24.11}$$

It holds for all sufficiently small s, every interval J and all finite counts, with one fixed auxiliary band. The proof uses the actual layers in (24.2), the marked section estimate and the complete original guard, not a count of words. The one-flight envelope used in its geometric input is described in Supplement Lemma 178.1; its explicit coordinates also give the physical area cap proved below.

The actual untruncated surface and original weighted defects satisfy

$$E_h^\varepsilon + O_h^\varepsilon \le C\mathcal{F}_h + C_\eta \mathsf{P}_{\eta,h} v^\eta. \tag{24.12}$$

This is Supplement Theorem 5.2, with the same joint physical coarea kernels and initial weights. For the part $\mathcal{L}_r$ of $\overline{\mathcal{F}}_r$ having backward residence $\ell > 2r$, Supplement Theorem 7.2 proves

$$\limsup_m \sup_{R,n,k} m^2 \|\mathcal{L}_r\|_\infty \le C r^{1/32}. \tag{24.13}$$

Its persistence proof precedes the mass-to-height deduction. We do not extend (24.13) to zero residence. The short part is estimated below in a different, explicitly finite norm.

All fiber identities have the following convention. For fixed m, R, n, k, the Borel maximal-segment parametrization fixes common kernels outside a null set in the roof, and common product kernels outside a joint null set for roof and time. Countably many widths share versions; the uniform order is $\sup_R \operatorname{ess\,sup}_t$, not a claim to one null set for all radii. All positive restrictions and infinite moment integrals below use these kernels.

25. A physical cap and negative residence moments

Throughout this section $\ell = \tau^- + \tau^+$ is the total residence of the original maximal regular exact-event segment. It is not the collision return time. Every point of Ω has $\ell > 0$. Write $\gamma_0 = 1/32$ and $\bar\nu = \nu_R^*|_\Omega$. The following estimate integrates down to residence zero at finite collision count; its proof does not sum fixed-scale limiting bounds.

Lemma 25.1 (A count-independent physical cap)**.** *There are $s_0, C > 0$ such that, uniformly in the collision count, radius and exact labels,*

$$\bar\nu\{\ell \le s\} \le \bar\nu\{\tau^+ \le s\} \le C s^{1/2}, \qquad 0 < s < s_0. \tag{25.1}$$

No interval in the roof or incidence corridor is imposed.

Proof. Use the actual starting layers in (24.2). Under invariant collision measure, $\nu\{c < w\} = 1 - \sqrt{1 - w^2} \le w^2$. A w-neighborhood of the finitely many rectangular section edges has measure at most Cw, uniformly in R. For a flight-clearance layer use the image-collision envelope of Supplement Lemma 178.1, at collision $j + 1$. Here is the area estimate, in addition to its retained stable-curve estimate. At fixed image angle α, each candidate disk in the finite set $\mathcal{D}$ has backward coordinates L_d, Z_d. On $L_d > 0$, $R \le |Z_d| \le R + w$, one has $L_d \ge \ell_* > 0$ and $\partial_\varphi Z_d = L_d$. The sinusoidal level equations have a bounded number of components. Their total φ length is at most Cw. Integrating $d\nu = (4\pi)^{-1} \cos\varphi\, d\alpha\, d\varphi$ proves that this envelope has measure at most Cw. The finite disk set and ℓ_* are uniform for the prescribed family.

Invariance permits these estimates at every actual marked collision. There are at most two contact marks, and two flight marks, at each depth. Thus the union bound at widths $C_E q_0^d \sqrt{s}$ is bounded by a constant times $\sum_{d \ge 0}(q_0^{2d} s + q_0^d \sqrt{s}) \le C\sqrt{s}$. Dropping the initial and terminal

section tests only on this positive upper comparison costs c_*^{-1} once. No exact labels are changed in the source being estimated. Finally $\ell \le s$ implies $\tau^+ \le s$. □

The retained finite-count exit theorem, restated in (24.11), and the physical cap give a stronger residence estimate than either bound separately. Namely,

$$m^2\bar{\nu}\{F \in J, \ell \le s\} \le Cs^{\gamma_0}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \min\{\rho_{B_*}^{m/2}, s^{1/2}\}. \tag{25.2}$$

This holds for every interval and $0 < s < s_0$, with B_* fixed. Indeed $x \le A + D$ and $x \le V$ imply $x \le A + \min(D, V)$. Enlarging the common constant and using $m^2 \le m^3$ gives exactly the displayed remainder. This step is performed on the positive probability, not on an unsigned spectral remainder claimed to have the physical cap by itself.

Theorem 25.2 (Actual untruncated negative residence moments)**.** *For $0 < \upsilon < \gamma_0$ define the actual density*

$$q_{\upsilon,s}(t)\,dt = F_{\#}(\ell^{-\upsilon}\mathbf{1}_{\{0<\ell\le s\}}\bar{\nu}).$$

It is finite at every count. Uniformly in all the physical parameters, intervals, $m \ge 2$ and $0 < s < s_0$, at each fixed auxiliary band,

$$m^2\int_J q_{\upsilon,s}(t)\,dt \le C_\upsilon s^{\gamma_0-\upsilon}(|J| + B_*^{-1}) + C_{B_*,\upsilon}(1 + B_*|J|)m^3 \min\{s^{1/2-\upsilon}, \rho_{B_*}^{m(1-2\upsilon)/2}\}. \tag{25.3}$$

Consequently, for every previously fixed $L, s > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J q_{\upsilon,s} \le C_\upsilon L s^{\gamma_0-\upsilon}. \tag{25.4}$$

The source with weight $\min(\ell, 1)^{-\upsilon}$ is also finite and has the corresponding fixed-width bound with a constant right side. There is no positive lower residence cutoff in these statements.

Proof. At the source level,

$$\ell^{-\upsilon}\mathbf{1}_{\{\ell\le s\}} = s^{-\upsilon}\mathbf{1}_{\{\ell\le s\}} + \upsilon\int_0^s v^{-\upsilon-1}\mathbf{1}_{\{\ell\le v\}}\,dv.$$

Use positive lower truncations first and then monotone convergence. The leading term of (25.2) integrates to $\gamma_0 s^{\gamma_0-\upsilon}/(\gamma_0 - \upsilon)$. For $D > 0$ the capped remainder obeys the exact identity

$$s^{-\upsilon}\min(D, s^{1/2}) + \upsilon\int_0^s v^{-\upsilon-1}\min(D, v^{1/2})\,dv = \frac{1}{1-2\upsilon}\min\{s^{1/2-\upsilon}, D^{1-2\upsilon}\}. \tag{25.5}$$

Split at D^2 when that point is less than s; the boundary terms cancel. Substitution of $D = \rho_{B_*}^{m/2}$ proves (25.3). An uncapped error independent of v would not be integrable here.

The physical cap alone also proves finite total mass for every $\upsilon < 1/2$. Coarea on compact regular charts and positive exhaustion give absolute continuity for the exponents used above; the excluded critical set has zero source area. For the last assertion split at $s_0/2$: on its complement the weight is bounded, and the unweighted complete source estimate applies. Finally fix L, s, B_*, take the collision limsup, and only then enlarge B_*. The complete layer cake has already been evaluated at finite count. This proves (25.4). □

26. An uncut surface estimate from complete segment traces

The negative residence moment acts on the original volume source. To apply it to the outgoing surface one must still pay the change of the physical angular density. Integrating over the entire segment provides precisely the factor that makes this payment finite without an initial-gradient cutoff.

For $1 < q < \infty$ define, for a measurable roof interval J,

$$\mathcal{N}_q(f;J) = \sup_{E\subset J,\ 0<|E|<\infty} |E|^{-1+1/q} \int_E |f(t)|\,dt. \tag{26.1}$$

This is a norm equivalent to the usual weak L^q norm. We shall use its subadditivity directly, not a change of density representative. Write $J^{[s]} = J + (-s, s)$.

Lemma 26.1 (A complete interval-pulse trace)**.** *Let a measure space Σ carry measurable finite intervals (a_σ, b_σ) of lengths $0 < \ell_\sigma \le s$ and nonnegative weights T_σ. Put $f(t) = \int_\Sigma T_\sigma \mathbf{1}_{(a_\sigma,b_\sigma)}(t)\,d\sigma$. If $\upsilon = 1 - 1/q$, then*

$$\mathcal{N}_q(f;J) \le \int_{\{(a_\sigma,b_\sigma)\cap J\neq\varnothing\}} T_\sigma \ell_\sigma^{1-\upsilon}\,d\sigma. \tag{26.2}$$

The assertion holds also for extended nonnegative integrals.

Proof. For $|E| = v$, Tonelli bounds the contribution of a segment by $T_\sigma \min(v, \ell_\sigma)$. In both cases $v \le \ell_\sigma$ and $v > \ell_\sigma$, $v^{-\upsilon}\min(v,\ell_\sigma) \le \ell_\sigma^{1-\upsilon}$. Integrate this inequality before taking the supremum over E. No disjointness of the roof intervals is assumed or needed. □

Theorem 26.2 (Weighted first-exit trace on every short segment)**.** *Use the complete Borel segment bijection in the original exact event and all its nongrazing outgoing endpoints. Let $T_j(\vartheta) = A_j(b_j-,\vartheta)$, $\ell_j = b_j - a_j$, and*

$$Z_s(t) = \sum_j \int_{E_j} T_j(\vartheta)\mathbf{1}_{\{0<\ell_j\le s,\ a_j<t<b_j\}}\,d\vartheta.$$

For $0 < \eta < 1/2$, $\kappa = (1-\eta)/2$ and $0 < \upsilon < \min(\gamma_0, 1-\kappa)$, set $q = (1-\upsilon)^{-1}$. At each finite count,

$$\mathcal{N}_q(Z_s;J) \le \int_{J^{[s]}} q_{\upsilon,s}(u)\,du + C_\eta s^{1-\kappa-\upsilon}\int_{J^{[s]}} v^\eta(u)\,du. \tag{26.3}$$

Thus for each fixed $L, s > 0$,

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} \mathcal{N}_q(m^2 Z_s;J) \le C_\upsilon(L+2s)s^{\gamma_0-\upsilon} + C_\eta(L+2s)s^{1-\kappa-\upsilon}. \tag{26.4}$$

All positive residence lengths, all initial gradients and all incoming endpoint types are included; no optical-sensitivity cutoff is used.

Proof. Fix one selected segment and suppress its angular index. Its exact physical density satisfies $|A'| \le CAI_m/z$. For $a < t < b$,

$$A(b-) \le A(t) + C\int_t^b A(u)I_m(u)/z(u)\,du.$$

Integrate over the entire interval (a,b), multiply by $\ell^{-\upsilon}$, and use Tonelli. This gives

$$A(b-)\ell^{1-\upsilon} \le \int_a^b A(t)\ell^{-\upsilon}\,dt + C\ell^{-\upsilon}\int_a^b (u-a)A(u)I_m(u)/z(u)\,du. \tag{26.5}$$

Forward radial growth, first from an interior point and then in the limit toward a, implies $z(u)^2 \ge 2\lambda_\circ(u-a)$. Hence

$$(u-a)z(u)^{-1} \le C_\eta (u-a)^{1-\kappa} z(u)^{-\eta} \le C_\eta \ell^{1-\kappa} z(u)^{-\eta}.$$

The second term of (26.5) is therefore at most $C_\eta s^{1-\kappa-\upsilon} \int_a^b A I_m z^{-\eta}$. This calculation pays the variation of both ω and χ in $A = \omega/(4\pi c_* \chi)$. It does not replace an inverse Jacobian in a first-exit formula by one.

Every selected segment that intersects J lies in $J^{[s]}$. Sum (26.5) over precisely these physical segments. The Borel bijection and its incident-component convention count each interior point once. The first sum is bounded by the actual source $q_{\upsilon,s}$; the second by the actual fractional source v^η. Nongrazing outgoing traces are used on the left; dropping that endpoint selector only on the two positive upper sources introduces neither a new normalization nor another label. Lemma 26.1 proves (26.3).

Initially all statements may be read as nonnegative extended inequalities on compact subsegments. The finite actual moments, Tonelli and positive exhaustion make the right side finite and justify the complete sum and endpoint limit. Apply (25.4) and (24.8), fixing both interval lengths and all bands before the collision limsup, to obtain (26.4). □

Theorem 26.3 (The complete untruncated first-exit surface)**.** *For every $1 < q < 32/31$, put $\upsilon = 1 - 1/q$ and $\nu_q = \gamma_0 - \upsilon = 1/q - 31/32 > 0$. For every fixed $L > 0$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(m^2 \overline{\mathcal{F}}_r; J) \le C_{q,L} r^{\nu_q} \tag{26.6}$$

for all sufficiently small previously fixed $r > 0$. This is the actual one-sided outgoing first-exit surface with no lower residence or gradient cutoff. For example $q = 64/63$ gives $\nu_q = 1/64$.

Proof. The complete residence kernel splits into its long part $\mathcal{L}_r$ with $\ell > 2r$ and a short part bounded by Z_{2r}. The retained long-residence height theorem (24.13) gives $\mathcal{N}_q(m^2 \mathcal{L}_r; J) \le |J|^{1/q} m^2 \|\mathcal{L}_r\|_\infty$ and an ordered upper bound $C L^{1/q} r^{\gamma_0}$. Use Theorem 26.2 with $s = 2r$ for the complete short part. Take, for instance, $\eta = 1/4$. Then $1 - \kappa - \upsilon > \gamma_0 - \upsilon > 0$. Subadditivity of (26.1) proves the result. The short contribution was summed at finite count through an actual negative moment, not reconstructed from infinitely many band limsups. □

The weak exponent is finite. Equation (26.6) controls the distribution of heights and does not assert that the surface essential supremum tends to zero. The positive cutoff has been removed from this new norm theorem, not from the earlier essential-height theorem.

27. The original raw density in a complete local norm

The surface theorem applies to every first-exit component, including the zero-residence tail. Combined with the retained fractional source, it gives a norm estimate for the original bad density itself. The roof density is not averaged in the conclusion.

Lemma 27.1 (A truncated potential in a local norm)**.** *Let $0 < \kappa < 1$, $q > 1$, and $1/q > \kappa$. For a nonnegative locally integrable v, put $P_r v(t) = \int_0^r s^{-\kappa} v(t+s)\, ds$. Then*

$$\mathcal{N}_q(P_r v; J) \le \|P_r v\|_{L^q(J)} \le (1-\kappa q)^{-1/q} r^{1/q - \kappa} \int_{J+(0,r)} v(u)\, du. \tag{27.1}$$

Also, if $1 \le p < q$,

$$\|f\|_{L^p(J)} \le \left(\frac{q}{q-p}\right)^{1/p} |J|^{1/p - 1/q} \mathcal{N}_q(f; J). \tag{27.2}$$

Proof. For fixed u, the kernel $\mathbf{1}_{\{0<u-t<r\}}(u-t)^{-\kappa}$ has $L^q(dt)$ norm at most $(1-\kappa q)^{-1/q}r^{1/q-\kappa}$. Minkowski's integral inequality over $u \in J+(0,r)$ gives the second inequality in (27.1); Hölder on each measurable $E \subset J$ gives the first. Apply this to truncated kernels and pass monotonically if needed. To prove (27.2), the definition of $\mathcal{N}_q$ implies $|\{|f| > \lambda\} \cap J| \le \min\{|J|, (\mathcal{N}_q/\lambda)^q\}$. Integrate the distribution function against $p\lambda^{p-1}$ and split at $\lambda = \mathcal{N}_q|J|^{-1/q}$. □

Theorem 27.2 (Complete original bad-source control)**.** *For every $1 < q < 32/31$ and fixed $L > 0$, under the explicitly retained inputs,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(m^2 b^\varepsilon; J) \le C_q L^{1/q}\varepsilon^{1/16} \tag{27.3}$$

for every fixed sufficiently small $\varepsilon > 0$. All initial momenta, positive total residences, collision incidences, terminal tests and exact labels are retained. No original source restriction is removed in forming b^ε.

Proof. The source-shift inequality (24.3) and the complete two-sided optical payment (24.12) give, for each fixed $r > 0$,

$$b^\varepsilon \le 2A_r b^\varepsilon + C\overline{\mathcal{F}}_r + C_\eta \mathsf{P}_{\eta,r} v^\eta + C_\Theta(1+\varepsilon^{-1})\sqrt{r}\, A_r p.$$

Every term is a complete source term, not a sum with an unknown physical-word multiplicity. Apply (26.1), then take the collision limsup with ε, r and all auxiliary bands fixed. The fixed-width source bounds in Input 24.2 give

$$\limsup_m \sup_{R,n,k} m^2 \|A_r b^\varepsilon\|_\infty \le C\varepsilon^{1/16}, \qquad \limsup_m \sup_{R,n,k} m^2 \|A_r p\|_\infty \le C.$$

Theorem 26.3 controls the surface by $C_{q,L} r^{\nu_q}$. Choose $\eta = 1/4$, so $\kappa = 3/8 < 1/q$; Lemma 27.1 and the actual fixed-width fractional input control the potential by $C_{q,\eta}(L+r)r^{1/q-\kappa}$. Thus the ordered bad-source norm is at most

$$C_q L^{1/q}\varepsilon^{1/16} + C_{q,L} r^{\nu_q} + C_{q,\eta}(L+r)r^{1/q-\kappa} + C_\Theta L^{1/q}(1+\varepsilon^{-1})\sqrt{r}.$$

Enlarge auxiliary bands only after the collision limit. Now let $r \downarrow 0$ with ε fixed. All three powers are positive. This proves (27.3). Averaging is an intermediate proof operation; the left side is the raw density. □

Theorem 27.3 (Exact-label raw local limit in finite local norms)**.** *Let*

$$f_{m,R,n,k}(t) = m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n).$$

For every fixed $M, L > 0$ and every $1 < q < 32/31$,

$$\sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(f_{m,R,n,k}\mathbf{1}_{\mathcal{C}_{m,R,n,k}(M)}; J) \longrightarrow 0. \tag{27.4}$$

For every $1 \le p < 32/31$ the corresponding $L^p(J \cap \mathcal{C}(M))$ norm also tends to zero, uniformly in the same parameters and interval locations. The reference includes every retained arithmetic peak and all zero classes; no reference density is divided by.

Proof. For each fixed large B use the original positive remainder (24.9). Theorem 27.2 and monotonicity under restriction to $\mathcal{C}(M)$ imply

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_q(f_{m,R,n,k}\mathbf{1}_{\mathcal{C}(M)}; J) \le C_{q,M,L} B^{-1/192}.$$

Indeed $\varepsilon(B)^{1/16} = A_0^{1/16} B^{-1/192}$, and the uniform essential bound on e_B contributes at most $L^{1/q} C_M B^{-1/192}$. The left side is independent of B. Let $B \to \infty$ last. Nonnegativity proves the asserted convergence. For any stated p choose q strictly between $\max(1,p)$ and $32/31$ and use (27.2). □

The governing objective remains (23.5). The theorem above is a proved intermediate conclusion for its unchanged raw error, not a replacement target. Its uniformity is on every fixed length L, not automatically on an interval growing with m. Neither a finite weak norm nor these finite L^p norms control the normalized essential supremum. The exact original-source criterion of Supplement Theorem 34.2 is therefore retained: vanishing of its critical/zero-residence charge still has to be proved for the pointwise endpoint. Same-roof ratios and unconditional likelihood statements are not promoted on the strength of the local-norm theorem alone.

28. The unchanged pointwise endpoint

The new negative residence theorem is a finite-count assertion for an actual singular insertion. Its full layer cake includes arbitrarily short physical segments before the collision limit. The outgoing trace theorem similarly sums all incident components before applying a norm. Thus the complete surface weak-norm decay and raw local-norm limit proved here require no positive residence cutoff.

These results do not identify a finite local norm with essential height. The endpoint in (23.5) continues to require

$$\lim_{B\to\infty} \mathfrak{R}_{\varepsilon(B),M} = 0,$$

where $\mathfrak{R}$ is the original weighted-source concentration charge of Supplement Theorem 34.2. The local-norm theorem constrains such concentration but does not exclude narrow peaks in its essential supremum. The older sufficient pairs $(\mathfrak{A},\mathfrak{B})$ and $(\mathfrak{A},\mathfrak{Q})$ remain valid with their exact hypotheses in the complete archive. No charge-zero assertion is inferred here.

In particular, a strong local L^p theorem for $p < 32/31$ is not a uniform denominator theorem on positive arithmetic classes. The scalar-first path remainder, collision-to-return clock transfer and forward likelihood statements are preserved, not promoted to new unconditional same-roof ratios. Zero arithmetic classes are never normalized. All previous mathematical files and the complete preceding article remain in the supplement and provenance.

Part 17. Complete segment traces and the original concentration source

Retained revision-87 abstract. We study the unchanged exact-label pointwise local limit problem for actual returns in a triangular Lorentz gas. A complete optical moment strictly above one yields a trace estimate for the physical angular density on maximal regular segments. Summation uses an exact Borel parametrization and has multiplicity one. The trace estimate gives a finite-count optical envelope on all initial states whose gradient and total residence exceed explicit powers of the microscopic roof width. The envelope includes every optical sensitivity and every positive incidence; for one choice its normalized bound is $Kh^{31/1408}$, uniformly before the collision limit. Inserting this estimate into the original first-exit argument localizes the full bad-source height to initial criticality and zero total residence. The organizing pointwise error is exactly the outer limit of this original weighted-source concentration charge. Its vanishing remains to be established; no surface cutoff or arithmetic class is removed from the theorem.

29. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\,\mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6,-1/2), \quad (-\theta_R,\sin\theta_R), \quad (\pi/3-\theta_R,\sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1-6q^2+q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{29.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{29.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{29.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

29.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R,a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{29.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{29.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

29.2. **Physical traces and a uniform microscopic envelope.** The sensitivity is the original additive function $I_m = 1+\sum_{i=1}^{m-1} q_0^{d_i}/c_i$. Write $z=|g|$ for the endpoint-gradient norm and ℓ for the total residence of its maximal regular exact-event segment. We use the common Borel coarea kernels μ_t. At fixed m,R,n,k the identities hold outside a null set in t, or a joint null set for product kernels; the uniform norm is $\sup_R \operatorname{ess\,sup}_t$, not a claim to one null set for all radii.

Theorem 29.1 (Microscopic envelope on a shrinking initial family)**.** *Set $a(h)=h^{1/128}$ and $\delta(h)=h^{1/64}$. On the original reachable target fibers define*

$$V_h^{\mathrm{reg}}(t)=\int_0^h\int \mathbf{1}_{\{\tau^-(y)>s,\ z(\Phi_{-s}y)\ge a(h),\ \ell(y)\ge\delta(h)\}}\frac{I_m(y)}{z(y)}\,d\mu_{t+s}(y)\,ds.$$

Under the explicitly retained inputs, there are fixed $K<\infty$ and $h_*>0$ such that

$$\sup_{m\ge 2,R,n,k} m^2\|V_h^{\mathrm{reg}}\|_\infty \le Kh^{31/1408} \qquad (0<h<h_*). \tag{29.6}$$

This includes all positive incidences and all values of I_m on the specified initial family. Both thresholds depend on the roof width, not on the collision count, and the estimate holds uniformly before any collision limit.

The proof is Theorems 32.2–33.2. It uses the actual moment $I_m^{33/32}$, obtained from the retained mixed optical source. The elementary trace inequality is proved in full, including its integration over complete physical segments.

The complementary initial mass is not discarded. Define

$$R_h^{\varepsilon}(t)=\int W^{\varepsilon}\mathbf{1}_{\{z<a(h)\ \text{or}\ \ell<\delta(h)\}}\,d\mu_t.$$

Theorem 34.2 identifies the original pointwise error with the outer limit of the ordered microscopic height of this source. The result does not assert that this last charge is zero, nor that previous sufficient optical charges have vanished. The original target, arithmetic reference and scalar-first downstream order remain unchanged.

30. Explicit analytic and geometric inputs

We collect the indispensable inherited assertions in the exact form used by the new proofs. Their source theorems and operator dependencies are specified here and in Appendix A. The new potential and first-exit estimates do not assume a surface spectral insertion theorem or an unweighted pointwise local bound.

Input statement 30.1 (Physical flow and source comparison)**.** On the regular part Ω of each exact event, endpoint angles $q=(\alpha_0,\alpha_m)$ give $F=L$, $g=\nabla_q F=R(-p_0,p_m)$, $K=D_q^2F$ and $\vartheta=\arg g$. Put

$$\lambda_\circ=9\sqrt{19}/200,\qquad \Lambda_\circ=47/6,\qquad \omega=|\partial_{p_0}\alpha_m|^{-1},\qquad \chi=|\det D_q(F,\vartheta)|.$$

The bounds $\lambda_\circ I\le K\le\Lambda_\circ I$ and $\lambda_\circ\le\chi\le\Lambda_\circ$ hold uniformly. The explicit field

$$X_q=\frac{K^{-1}g}{g^{\mathsf{T}}K^{-1}g}$$

is stopped at true physical and section-decision boundaries, not at chart seams. Its flow satisfies $XF = 1$, $X\vartheta = 0$ and

$$2\lambda_\circ s \le |g(\Phi_s x)|^2 - |g(x)|^2 \le 2\Lambda_\circ s. \tag{30.1}$$

Its endpoint-curve length is at most $C\sqrt{s}$. For $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$ one has $|X\omega| \le C\omega I_m/|g|$ on regular survivors. At a genuine grazing exit some internal incidence tends to zero and ω tends to zero. Forward interior critical termination cannot occur.

Every starting point with forward lifetime at most r belongs to a layer of one of the following forms, at an actual collision or flight:

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \quad \Delta_j \le C_E q_0^{d'_j}\sqrt{r}, \quad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}. \tag{30.2}$$

The physical coarea covariance, proved in the appendix in the convention used here, has Jacobian $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$. Writing E_h^ε for all weighted starting exit mass and

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon(x)[1 - \omega(\Phi_h x)/\omega(x)]_+ \, d\mu_t(x),$$

the complete source satisfies, for each fixed r and almost every t,

$$b^\varepsilon(t) \le 2A_r b^\varepsilon(t) + \bar{E}_r^\varepsilon(t) + \bar{O}_r^\varepsilon(t) + C_\Theta(1+\varepsilon^{-1})\sqrt{r}\, A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\, ds. \tag{30.3}$$

These are the endpoint-field, all-contact motion, uniform cubic-action and separated-volume results in retained modules 174, 177, 180–182. The guard term in (30.3) is the complete product guard, not a protective subword source. The derivative and continuation assertions are confined to regular physical trajectories.

Input statement 30.2 (Complete local source estimates)**.** For every fixed auxiliary band B_*, constants $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ give the following bounds uniformly in radius, exact labels, interval J and mark $0 \le j \le m$. Let $S_{R,s}$ be the union of strips of width Cs about the finitely many coordinate lines supporting ∂Y_R^*. For s small,

$$m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le Cs^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \rho_{B_*}^{m/2}, \tag{30.4}$$

$$m^2 \int_J b^\delta(t)\, dt \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \rho_{B_*}^{m/2}. \tag{30.5}$$

The local extension to a fixed remaining compact range of δ is included. At auxiliary parameter one $W^1 = 1$, so $b^1 = p$.

Put $z = |g|$, $\alpha = 129/128$ and, for $0 < \eta < 1/2$, define the actual nonnegative source measures

$$\begin{aligned} v^{\eta,b}(t)\, dt &= L_\#(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_\Pi \nu_R^*), \\ v^\eta(t)\, dt &= L_\#(I_m z^{-\eta} \mathbf{1}_\Pi \nu_R^*). \end{aligned} \tag{30.6}$$

They are finite and absolutely continuous at every fixed count, and for $0 < b < b_1$,

$$m^2 \int_J v^{\eta,b}(t)\, dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|)m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \tag{30.7}$$

Adding the bounded-gradient part gives, for every fixed $L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\eta \le C_\eta L, \qquad \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta,b} \le C_\eta b^{\alpha-\eta} L. \tag{30.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards; leading constants are independent of that band and of the previously fixed interval length.

Equation (30.4) is Supplement Theorem 175.3. The physical source bound (30.5) is Supplement Theorem 179.1, in particular its finite-window estimate. The fractional bounds are the positive layer-cake conclusions of retained module 187, stated in full above. The joint spectral and common-space assumptions used there are recorded in the appendix. No estimate for an untruncated first inverse gradient or for a singular roof convolution is part of this input.

Input statement 30.3 (The scalar positive remainder)**.** For large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{30.9}$$

The source b^ε is nonnegative and has a compactly supported BV density at fixed count. No collision-uniform variation bound is assumed. Supplement Theorem 181.4 gives this remainder with the same complete finite-peak reference.

30.1. **Maximal segments and the two existing charges.** The retained Borel parametrization, Supplement Theorem 7.1, is a bijection

$$\Psi : \coprod_j \{(\theta, u) : \theta \in E_j,\ a_j(\theta) < u < b_j(\theta)\} \longrightarrow \Omega,$$

with Borel inverse. Its exact density is $A_j(u,\theta) = \omega(\Psi_j(\theta,u))/(4\pi c_* \chi(\Psi_j(\theta,u)))$. Every segment remains in one regular exact event until a true boundary; Borel chart seams are not exits. Nongrazing outgoing faces retain their incident physical component and carry $d\zeta_+ = A_j(b_j-,\theta)\, d\theta$. The full angular variation is $|\partial_u A_j| \le C A_j I_m / z$. These are measure identities with the null-set convention already specified, not bounds with an unknown number of words.

The true first-exit assignment is

$$\mathcal{F}_h(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<b-t\le h,\ b-t<\tau^-(y)\}}\, d\zeta_+(y), \qquad \overline{\mathcal{F}}_r = r^{-1}\int_0^r \mathcal{F}_h\, dh.$$

No lower gradient cutoff is imposed. Put $l = b - a$ and define

$$\mathcal{Z}_\delta(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<l\le\delta\}} \mathbf{1}_{\{a<t<b\}}\, d\zeta_+, \qquad q_\delta(t)\, dt = F_\#(\mathbf{1}_{\Omega_+^{\mathrm{ng}}} \mathbf{1}_{\{l\le\delta\}} \bar{\nu}).$$

The second object is the original collision volume restricted to the same actual short segments, not a volume collar for a boundary. For source families set

$$\mathcal{H}_M(f) = \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} f(t), \quad \mathfrak{B}_M = \lim_{\delta\downarrow 0} \mathcal{H}_M(\mathcal{Z}_\delta), \quad \mathfrak{Q}_M = \lim_{\delta\downarrow 0} \mathcal{H}_M(q_\delta).$$

These limits are monotone extended nonnegative numbers.

For $0 < \eta < 1/2$, with $\kappa = (1-\eta)/2$, retain

$$\mathsf{P}_{\eta,h} v^\eta(t) = \int_0^h s^{-\kappa} v^\eta(t+s)\, ds, \qquad \mathfrak{A}_{\eta,M} = \lim_{h\downarrow 0} \mathcal{H}_M(\mathsf{P}_{\eta,h} v^\eta).$$

Supplement Theorems 8.2 and 9.1 give

$$\mathfrak{B}_M \le \mathcal{H}_M(\overline{\mathcal{F}}_r) \le \mathfrak{B}_M + C_\eta r^{1/32}. \tag{30.10}$$

Their band sum is finite before the collision limsup at each positive residence cutoff. Its leading constant is cutoff independent; the complete zero-residence tail is still inside $\mathcal{H}_M$ before its cutoff tends to zero. The active offset correction to historical module 189 remains Supplement equation (6.1); the historical inclusion is explicitly marked superseded. None of these inputs asserts charge zero.

31. Complete mixed optical moments at finite collision count

The finite-count cap, rather than the limiting Gaussian term alone, allows an optical moment strictly greater than one. This extra moment will be used before summing infinitely many microscopic roof scales. Write $z = |g|$, $\alpha = 129/128$, $\beta = 17/16$, and $\bar\nu = \nu_R^*|_\Omega$. All constants below are uniform in m, R, n, k. Dependence on the fixed moment exponents is permitted.

31.1. **The capped input.** The joint marked estimate retained in Supplement Theorem 25.3 has, after the fixed conversion from endpoint momentum to z, the form

$$m^2 \int \mathbf{1}_{\{F\in J, z<\zeta, c_i<s\}}\, d\bar\nu \le C\zeta^\alpha s^\beta (|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \min\{\zeta\rho_{B_*}^{m/2}, s^2\}. \tag{31.1}$$

It holds for $0 < \zeta < z_*$, $0 < s < s_*$ and every internal mark. The source retains its exact displacement and occupation coefficient; dropping endpoint section tests is used only in its positive upper comparison. Its endpoint-only counterpart is

$$m^2 \int \mathbf{1}_{\{F\in J, z<\zeta\}}\, d\bar\nu \le C\zeta^\alpha (|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \zeta\rho_{B_*}^{m/2}. \tag{31.2}$$

The stronger endpoint exponent in that proof implies the displayed one. Without a gradient restriction the retained marked-incidence estimate has exponent $5/4$ and cap $\min\{\rho_{B_*}^{m/2}, s^2\}$; see Supplement Theorem 35.3. These are explicitly inherited spectral inputs. Their four-record extensions are not being attributed to an external billiard theorem.

Lemma 31.1 (A capped power integral)**.** *For $0 < p < 2$, $0 < D \le 1$ and $0 < s \le 1$,*

$$s^{-p}\min(D, s^2) + p\int_0^s v^{-p-1}\min(D, v^2)\, dv \le \frac{2}{2-p}\min\{s^{2-p}, D^{1-p/2}\}. \tag{31.3}$$

Proof. For $s \le \sqrt{D}$ the left side is $2s^{2-p}/(2-p)$. For $s > \sqrt{D}$, split the integral at $\sqrt{D}$. The terms Ds^{-p} cancel, leaving $2D^{1-p/2}/(2-p)$. Thus the inequality is in fact an equality. The same computation justifies monotone removal of the lower integration endpoint. □

Theorem 31.2 (Actual mixed moments)**.** *Fix*

$$1 < p < \beta, \qquad \sigma_p = 1 - p/2, \qquad 0 < \eta < \sigma_p.$$

The actual densities

$$U_p^\zeta(t)\, dt = F_\#(I_m^p \mathbf{1}_{\{z<\zeta\}}\bar\nu), \qquad V_{p,\eta}^\zeta(t)\, dt = F_\#(I_m^p z^{-\eta}\mathbf{1}_{\{0<z<\zeta\}}\bar\nu)$$

are finite at each count. On every fixed auxiliary band,

$$m^2\int_J U_p^\zeta \le C_p\zeta^\alpha(|J| + B_*^{-1}) + C_{B_*,p}(1 + B_*|J|)m^2\zeta^{\sigma_p}\rho_{B_*}^{m\sigma_p/2}, \tag{31.4}$$

$$m^2\int_J V_{p,\eta}^\zeta \le C_{p,\eta}\zeta^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,p,\eta}(1 + B_*|J|)m^2\zeta^{\sigma_p-\eta}\rho_{B_*}^{m\sigma_p/2}. \tag{31.5}$$

The unrestricted actual source $V_{p,\eta}(t)\, dt = F_\#(I_m^p z^{-\eta}\bar\nu)$ satisfies

$$m^2\int_J V_{p,\eta} \le C_{p,\eta}(|J| + B_*^{-1}) + C_{B_*,p,\eta}(1 + B_*|J|)m^2\rho_{B_*}^{m\sigma_p/2}. \tag{31.6}$$

The estimates hold at finite count, for all intervals and thresholds in their stated ranges. The leading constants do not depend on the fixed band. No strong-space multiplier $M_{I_m^p z^{-\eta}}$ is assumed.

Proof. Put $S_m = 1+\sum_{i=1}^{m-1} q_0^{d_i} \le 1+2q_0/(1-q_0)$. Weighted Jensen gives the pointwise inequality

$$I_m^p \le S_m^{p-1}\left(1+\sum_{i=1}^{m-1} q_0^{d_i} c_i^{-p}\right). \tag{31.7}$$

For each marked power below s_* use the positive layer cake

$$c_i^{-p}\mathbf{1}_{\{c_i<s_*\}} = s_*^{-p}\mathbf{1}_{\{c_i<s_*\}} + p\int_0^{s_*} v^{-p-1}\mathbf{1}_{\{c_i<v\}}\,dv.$$

Insert (31.1) before any limit. The leading integral converges since $p<\beta$. Lemma 31.1, with $D=\zeta\rho_{B_*}^{m/2}$, bounds its entire remainder by $C_p\zeta^{\sigma_p}\rho_{B_*}^{m\sigma_p/2}$. The bounded complementary powers and the constant one use (31.2). Their errors are smaller, after adjusting fixed constants. Sum the geometric weights in (31.7); there is no physical-word count. This proves (31.4).

Now apply the positive gradient layer cake with weight I_m^p:

$$z^{-\eta}\mathbf{1}_{\{z<\zeta\}} = \zeta^{-\eta}\mathbf{1}_{\{z<\zeta\}} + \eta\int_0^{\zeta} q^{-\eta-1}\mathbf{1}_{\{z<q\}}\,dq.$$

Both terms of (31.4) are integrable at zero: $\alpha-\eta>0$ and $\sigma_p-\eta>0$. Their coefficients are, respectively, multiplied by $\alpha/(\alpha-\eta)$ and $\sigma_p/(\sigma_p-\eta)$. This proves (31.5) by monotone convergence of the actual source, including its finite-count remainder.

Split the unrestricted source at $z_*/2$. Above that threshold, $z^{-\eta}$ is bounded. Apply the unconditioned capped incidence input and the same Jensen and power integrals; $p<\beta<5/4$. Below it use (31.5). Enlarge $\rho_{B_*}<1$ if necessary to combine finitely many remainders. This proves (31.6). At fixed count the roof has bounded support, so taking a containing interval proves finite mass. Absolute continuity follows on compact regular charts by coarea, then on their increasing union by positivity; the removed critical set has zero source area. All densities use the common kernels specified at the first use of the segment map in this article. □

The strict inequalities on p and η are needed for two different integrals. In particular no untruncated first inverse gradient is obtained by substituting $\eta=1$. The new gain is a genuine power of the complete optical insertion above one, with a summable finite-count remainder.

32. A complete segment trace estimate

The additional optical moment can also be used without selecting a large-sensitivity sector. We apply a one-dimensional trace argument to an exact physical angular density, and sum the resulting energies through the Borel segment bijection. The power strictly above one is essential.

Fix $1<p<17/16$ and put $e_p = 1-1/p$. Write

$$u_p(t)\,dt = F_{\#}(I_m^p\bar{\nu}), \qquad M_p(J) = \int_J u_p(v)\,dv.$$

The endpoint gradient has a uniform upper bound $z\le z_{\max}$. Consequently $I_m^p \le z_{\max}^{\eta} I_m^p z^{-\eta}$ on the regular source, for any fixed $0<\eta<1-p/2$. The actual mixed-moment Theorem 31.2 therefore implies

$$m^2 M_p(J) \le C_p(|J|+B_*^{-1}) + C_{B_*,p}(1+B_*|J|)m^2\rho_{B_*}^{m(1-p/2)/2}. \tag{32.1}$$

This is a finite-count bound for every interval, not just an assertion about one previously fixed window. Its spectral inputs are exactly those of the retained mixed-moment theorem.

Lemma 32.1 (A positive-density trace inequality)**.** *Let A be positive and locally absolutely continuous on an interval, and let $I\ge 1$ be measurable. Suppose that $|A'|\le DAI$. Let J_0 be*

an interior interval of length $d > 0$, and let K be a set of target points such that the intervals joining J_0 to those points lie in an interval J_1 of length at most $8d$. Then

$$\sup_K A \le C_p\{d^{-1} + D^p d^{p-1}\} \int_{J_1} AI^p. \tag{32.2}$$

The inequality also holds for one-sided limits of target points when the right side is finite.

Proof. Set $f = A^{1/p}$. On every compact interior interval, $|f'| \le (D/p) f I$. There is a point $v_0 \in J_0$ at which $f(v_0) \le (d^{-1}\int_{J_0} A)^{1/p}$, or a sequence of points with this bound up to an arbitrarily small error. The fundamental theorem of calculus and Hölder's inequality give, for $v \in K$,

$$f(v) \le \left(d^{-1}\int_{J_0} A\right)^{1/p} + \frac{D}{p}(8d)^{1-1/p}\left(\int_{J_1} AI^p\right)^{1/p}.$$

Raise to the power p, use $(x+y)^p \le 2^{p-1}(x^p + y^p)$ and $I \ge 1$, and exhaust the joining intervals by compact subintervals. This proves the assertion. No bound for A^{-1} is used. □

For $0 < a, \delta \le 1$, define the following complete initial family on the roof t:

$$\mathcal{E}_t(a,\delta) = \{(j,\theta) : a_j(\theta) < t < b_j(\theta),\ z_j(t,\theta) \ge a,\ b_j(\theta) - a_j(\theta) \ge \delta\}. \tag{32.3}$$

All endpoint types are included. Here and below a without a subscript is a gradient threshold, and a_j is an endpoint roof. Set

$$d = d(a,\delta) = \min\{\delta/16,\ a^2/(32\Lambda_\circ),\ 1/16\}, \qquad J_t = (t-4d, t+4d). \tag{32.4}$$

For the physical constants in this article, $d \le a$. The source selection is a restriction of the original exact event, not a new normalization or a replacement of the full source.

Theorem 32.2 (Summed physical trace with multiplicity one)**.** *For almost every t at each fixed count and exact parameter set,*

$$S_t(a,\delta) := \sum_j \int_{\mathcal{E}_t(a,\delta)\cap(\{j\}\times E_j)} \sup_{t\le v<\min(t+d,b_j)} A_j(v,\theta)\, d\theta \le C_p d^{-1} M_p(J_t). \tag{32.5}$$

The same bound holds for the original unweighted source density restricted to $z \ge a$ and total regular residence at least δ. In particular, for every fixed positive a, δ,

$$\limsup_m \sup_{R,n,k} m^2 \|p^{a,\delta}\|_\infty \le C_p, \tag{32.6}$$

where the leading constant is independent of those fixed cutoffs. Equation (32.6) *does not remove them.*

Proof. At least one side of a selected segment has length at least $\delta/2 \ge 8d$ from t. Use the anchor interval $(t-3d, t-2d)$ on the backward side when that side is long, and $(t+2d, t+3d)$ otherwise. This is a Borel choice. Every path from an anchor to a target in (32.5) remains in the same physical segment and in J_t. Radial growth gives there

$$z_j(v)^2 \ge a^2 - 8\Lambda_\circ d \ge 3a^2/4.$$

The established inequality $|\partial_v A_j| \le C_0 A_j I_m / z_j$ thus permits Lemma 32.1 with $D = 2C_0/a$. Since $d \le a$, its coefficient is at most $C_p d^{-1}$.

Integrate over the selected angular parameters. At every interior roof the Borel bijection counts each physical point once, including its exact labels. Therefore the sum of the energies on all joining intervals is bounded by $M_p(J_t)$, with no word-count factor. Suprema over variable target intervals are measurable: continuity on each regular segment permits a countable dense set of roof offsets. The source moment is finite, so truncation and positive convergence justify

this integration also for countably many segment charts. The common physical kernels fix the almost-everywhere convention.

The density $p^{a,\delta}(t)$ is bounded by S_t. Insert (32.1), fix a, δ, B_*, take the collision limsup, and only then enlarge B_*. The length of J_t is $8d$, which cancels d^{-1}. This proves (32.6). The finite-count estimate still contains the explicit $d^{-1}B_*^{-1}$ and spectral remainder; it does not imply the height bound at zero cutoffs. □

33. An optical envelope on shrinking physical segment families

We now use the trace estimate before either the microscopic width or the collision count is sent to its limit. The source includes every positive incidence and every value of the optical sensitivity.

For $0 < h \le d(a,\delta)$ define

$$\mathcal{V}_h^{a,\delta}(t) = \sum_j \int_{\mathcal{E}_t(a,\delta)\cap(\{j\}\times E_j)} \int_0^{\min(h,b_j-t)} A_j(t+s,\theta)\frac{I_m(t+s,\theta)}{z_j(t+s,\theta)}\,ds\,d\theta. \tag{33.1}$$

The upper endpoint is interpreted from within the regular segment. Equivalently this is the reachable-target drift with $\tau^-(y) > s$, initial gradient $z(\Phi_{-s}y) \ge a$ and total residence at least δ. It never includes a later re-entry.

Theorem 33.1 (A finite-count trace-to-drift estimate)**.** *For $e_p = 1 - 1/p$ and $0 < h \le d(a,\delta)$,*

$$\mathcal{V}_h^{a,\delta}(t) \le \frac{h^{e_p}}{a} S_t(a,\delta)^{e_p} M_p((t,t+h))^{1/p}, \tag{33.2}$$

$$\le \frac{C_p}{a}(h/d)^{e_p} M_p(J_t). \tag{33.3}$$

Consequently, for each fixed auxiliary band B_*,

$$m^2\mathcal{V}_h^{a,\delta}(t) \le \frac{C_p}{a}(h/d)^{e_p}\big[C_p(8d + B_*^{-1}) + C_{B_*,p}(1+8B_*d)m^2\rho_{B_*}^{m(1-p/2)/2}\big]. \tag{33.4}$$

These inequalities hold at finite count, uniformly in every parameter in the displayed ranges, in the common essential-roof versions.

Proof. On a selected segment, forward radial monotonicity gives $z(t+s) \ge a$. Apply Hölder in its time variable, bounding the second factor by its length to the power e_p times the supremum of $A_j^{e_p}$. Then apply Hölder to the angular parameter and the disjoint union of segment charts. This gives (33.2); the first factor is precisely the summed trace, not a supremum over words. The energy in the second factor is a submeasure of the original I_m^p source on $(t,t+h)$. Insert Theorem 32.2 and the inclusion $(t,t+h) \subset J_t$. Since $e_p + 1/p = 1$, the powers of the larger energy sum to one, proving (33.3). Equation (32.1) now proves (33.4). Nonnegative exhaustion of regular segments and Tonelli preserve the first-exit selection throughout. □

Theorem 33.2 (Uniform microscopic control with vanishing cutoffs)**.** *Take $p = 33/32$ and set*

$$a(h) = h^{1/128}, \qquad \delta(h) = h^{1/64}, \qquad c_d = \min\{1/16, (32\Lambda_\circ)^{-1}\}.$$

There is a fixed $h_* > 0$ such that, for every $0 < h < h_*$, $d(a(h),\delta(h)) = c_d h^{1/64} \ge h$. For each fixed auxiliary band there is a finite K_{B_*} satisfying

$$\sup_{m\ge 2,R,n,k} m^2\|\mathcal{V}_h^{a(h),\delta(h)}\|_\infty \le K_{B_*}h^{31/1408}. \tag{33.5}$$

No bound on I_m, and no internal-incidence corridor, is imposed. The estimate also bounds, up to a fixed factor, the complete surviving optical loss and genuine grazing first-exit starting mass

restricted to this initial family. Both restrictions shrink to zero with h before the collision limit; the estimate is uniform for all finite collision counts.

Proof. Choose, for example, $h_* < \min\{1, c_d^{64/63}\}$. At fixed B_*,

$$\sup_{m\ge 2} m^2 \rho_{B_*}^{m(1-p/2)/2} < \infty.$$

The bracket in (33.4) is therefore uniformly bounded for $d \le 1/16$. Its outside power of h is exactly

$$-\frac{1}{128} + \frac{1}{33}\left(1 - \frac{1}{64}\right) = \frac{31}{1408} > 0.$$

This proves (33.5), with the fixed factor $c_d^{-1/33}$ included in K_{B_*}. One may keep B_* fixed at a single admissible value for this entire theorem. There is no growing Fourier band, no infinite sum of collision limsups and no discarded cutoff-dependent remainder.

For the stated optical consequence, on a surviving segment the positive decrease of ω is bounded by its negative variation. At a genuine grazing first exit its terminal value is zero and the same negative variation pays its full positive starting weight. These initial source classes are disjoint. Divide by the initial $4\pi c_* \chi$, use $W^\varepsilon \le 1$ and the two uniform bounds on χ, and use $|X\omega| \le C\omega I_m / z$. The result is bounded by $C\mathcal{V}_h^{a(h),\delta(h)}$. True nongrazing exit traces remain separate and are not set equal to zero. □

The power $31/1408$ is an explicit admissible choice, not an optimized exponent. More generally $a(h) = h^\xi$ and $\delta(h) = h^\upsilon$ work whenever $b = \max(2\xi, \upsilon) < 1$ and $e_p(1-b) - \xi > 0$. The proved mechanism is physical persistence of an interior density combined with an actual moment. Unlike the earlier high-sensitivity envelope, it controls all sensitivities on its stated geometric family. It does not include source states outside that family.

34. Concentration of the original source at criticality and zero residence

The finite-count envelope can be inserted directly into the scalar source-shift argument. The exceptional contribution can then be kept as its original weighted starting mass, instead of enlarged to an optical potential. No mass is removed from the original event.

Let $\ell(x) = \tau^-(x) + \tau^+(x)$ denote the total roof residence of its maximal regular exact-event segment. The Borel parametrization gives $\ell = b_j - a_j$. For $a(h), \delta(h)$ as in Theorem 33.2, define

$$R_h^\varepsilon(t) = \int W^\varepsilon(x) \mathbf{1}_{\{z(x)<a(h) \text{ or } \ell(x)<\delta(h)\}}\, d\mu_t(x), \qquad \mathfrak{R}_{\varepsilon,M} = \lim_{h\downarrow 0} \mathcal{H}_M(R_h^\varepsilon). \tag{34.1}$$

This limit exists in $[0,\infty]$: the source sets decrease with h. The two alternatives are a union, not a double-counted sum. Every boundary type, including grazing forward ends, is retained in this original-volume restriction. In particular it is not identical to the earlier source q_δ, which selected nongrazing forward ends. Cofinality shows that the same charge is obtained by taking the infimum over all positive pairs of gradient and total-residence thresholds, always after applying $\mathcal{H}_M$.

The fixed-positive-residence surface theorem retained in Supplement Theorem 8.2 states

$$\mathcal{H}_M(\overline{\mathcal{F}}_h^{\ge\delta}) \le C h^{1/32} \qquad (h, \delta > 0 \text{ fixed}). \tag{34.2}$$

Its leading constant is independent of the positive δ. Its proof sums finitely many physical residence bands before the collision limsup. We apply it with $\delta = \delta(h)$ for each fixed h, not at $\delta = 0$ or at a collision-dependent cutoff.

Theorem 34.1 (The original bad-source height is localized)**.** *For every fixed sufficiently small $\varepsilon > 0$ and M,*

$$\mathfrak{R}_{\varepsilon,M} \le \mathcal{H}_M(b^\varepsilon) \le \mathfrak{R}_{\varepsilon,M} + C\varepsilon^{1/16}. \tag{34.3}$$

More precisely, for each sufficiently small fixed $h > 0$,

$$\mathcal{H}_M(b^\varepsilon) \le \mathcal{H}_M(R_h^\varepsilon) + C\varepsilon^{1/16} + Ch^{1/32} + Kh^{31/1408} + C_\Theta(1+\varepsilon^{-1})\sqrt{h}. \tag{34.4}$$

All sources use the original section normalization and exact labels.

Proof. Fix the outer width h and its two initial cutoffs. Split the starting fiber into the exceptional set in (34.1) and its complement. For every $0 < s < h$, the exceptional contribution to $E_s^\varepsilon + O_s^\varepsilon$ is at most R_h^ε: exit and survivor are disjoint cases, and the positive inverse-twist loss on a survivor is at most its original weight.

On the complementary initial family, negative optical variation pays surviving loss and grazing first exits as in Theorem 33.2. For a nongrazing first exit, $\omega(t) \le \omega(b-) + \int_t^b [-\omega']_+$. Its endpoint term, after dividing by the initial angular normalization, is bounded by a fixed multiple of the actual outgoing flux on that same component. The full segment has residence at least $\delta(h)$, so dropping only the initial-gradient selector on this positive upper term gives $\mathcal{F}_s^{\ge\delta(h)}$. The variation is bounded by $C\mathcal{V}_h^{a(h),\delta(h)}$. Thus, before any limit,

$$\bar{E}_h^\varepsilon + \bar{O}_h^\varepsilon \le R_h^\varepsilon + C\overline{\mathcal{F}}_h^{\ge\delta(h)} + C\mathcal{V}_h^{a(h),\delta(h)}.$$

The averaging uses the same initial cutoffs throughout $0 < s < h$. The inverse physical Jacobian is paid by the optical variation; it has not been deleted from the first-exit formula.

Insert this inequality into (30.3). The fixed-width bad-source estimate bounds $2\mathcal{H}_M(A_h b^\varepsilon)$ by $C\varepsilon^{1/16}$, and its unweighted version bounds the guard term by $C_\Theta(1+\varepsilon^{-1})\sqrt{h}$. Equation (34.2) and the finite-count envelope give (34.4). For the latter one may use one fixed auxiliary band, so K does not change with h. All other auxiliary bands are enlarged only after the collision limit.

Since $0 \le R_h^\varepsilon \le b^\varepsilon$, the first inequality in (34.3) is immediate. For the second, let $h \downarrow 0$ in (34.4) only after its ordered height has been taken, with ε fixed. This argument remains valid for extended nonnegative heights. □

Theorem 34.2 (An exact original-source endpoint criterion)**.** *With $\varepsilon(B) = A_0 B^{-1/12}$ and the unchanged scalar remainder,*

$$\mathfrak{R}_{\varepsilon(B),M} \le \mathcal{E}_M + C_M B^{-1/192}, \qquad \mathcal{E}_M \le \mathfrak{R}_{\varepsilon(B),M} + C_M B^{-1/192}. \tag{34.5}$$

In particular

$$\mathcal{E}_M = \lim_{B\to\infty} \mathfrak{R}_{\varepsilon(B),M}. \tag{34.6}$$

The original full pointwise theorem is equivalent to the right side being zero for every M. This zero-charge assertion is not proved by the preceding trace and envelope theorems.

Proof. The positive remainder (30.9) and the triangle inequality give both

$$\mathcal{H}_M(b^{\varepsilon(B)}) \le \mathcal{E}_M + C_M B^{-1/192}, \qquad \mathcal{E}_M \le \mathcal{H}_M(b^{\varepsilon(B)}) + C_M B^{-1/192}.$$

Combine them with (34.3) and $\varepsilon(B)^{1/16} = A_0^{1/16} B^{-1/192}$. This proves (34.5) and then (34.6), also when the value is infinite. No monotonicity of the original smooth function Θ in its argument, or of the source in B, is needed for the squeeze. For every fixed B, all physical widths used above are fixed before the collision limit. The width tends to zero afterwards; B is last. No arithmetic class is normalized in this argument. □

The prior sufficient pairs $(\mathfrak{C}_M, \mathfrak{B}_M)$ and $(\mathfrak{C}_M, \mathfrak{Q}_M)$ remain valid in the preserved article. The new criterion is an equivalence for a restriction of the original bad source, not an assertion that either earlier, larger optical charge has vanished. It isolates collapsing initial gradient and collapsing total residence while estimating their complementary moving family uniformly at finite count. Source-specific control of the remaining restriction is still required. Scalar closure must precede every same-roof path ratio and every unconditional likelihood consequence.

35. The original endpoint and its remaining physical concentration

The microscopic estimate (33.5) is uniform at finite collision count. It does not restrict optical sensitivity, and it keeps the complete source family defined by the two shrinking initial thresholds. Its finite-count spectral remainder is retained in (33.4); the positive exponent makes the width limit legitimate without changing the auxiliary band.

The full pointwise problem has not been replaced by that subfamily. The exceptional initial mass is retained with the original weight W^{ε}, and the complete positive-cutoff outgoing source is paid by its independently proved residence theorem. The exact remaining assertion is

$$\lim_{B\to\infty} \mathfrak{R}_{\varepsilon(B),M} = 0 \quad \text{for every } M.$$

It concerns original source concentration near initial criticality or zero total regular residence, including every true endpoint type. The trace theorem does not estimate this last charge. It is not marked zero in the present revision.

Once the scalar endpoint is proved, Supplement Theorem 76.1 gives the same-roof path numerator. Normalization remains restricted to positive arithmetic reference classes. The actual-return clock and forward likelihood retain their separate inherited hypotheses; zero classes are never divided by. The complete preceding article and every inherited proof are retained in the supplement, rather than deleted from the record.

Appendix A. Physical kernels, first-exit traces and analytic dependencies

This appendix supplies the measure-theoretic interfaces used by the new arguments in one convention. It does not deduce a surface estimate from a volume multiplier theorem. The geometric and spectral estimates whose proofs are retained in the technical supplement are listed separately.

A.1. **Normalization and common kernels.** On a regular word use endpoint angles $q = (\alpha_0, \alpha_m)$ and oriented internal arclengths. Endpoint first variation gives $F_{\alpha_0} = -Rp_0$ and $F_{\alpha_m} = Rp_m$. Consequently

$$\omega = |\partial_{\alpha_m} p_0| = |F_{\alpha_0\alpha_m}|/R, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\, dq.$$

In the optical Schur complement the internal Hessian is $D_c A D_c$, where A has diagonal $t_{i-1} + t_i + 2/(Rc_i)$ and off-diagonal $-t_i$. The two angle-to-arclength factors are R^2; division by R in the first variation gives $\omega = Rc_0c_m/y_m$. For one flight $y_1 = \ell_0$, and for two flights $y_2 = \ell_0+\ell_1+2\ell_0\ell_1/(Rc_1)$. Changes of tangent orientation do not change these absolute identities.

Away from $g = 0$, the coordinates (F, ϑ) have Jacobian $\chi = e^{\perp\mathsf{T}} K e^{\perp}$ in absolute value, where $e = g/|g|$. Thus the angular expression for the original fiber is

$$d\mu_t = \frac{\omega(t,\vartheta)}{4\pi c_* \chi(t,\vartheta)}\, d\vartheta. \tag{A.1}$$

The sign of an oriented endpoint chart is irrelevant; all source Jacobians in this appendix are positive. The original source already contains its initial and terminal tests and is normalized only by c_*.

Proposition A.1 (Common graph covariance)**.** *At fixed m, R, n, k, the regular exact event has jointly measurable coarea kernels for which, for almost every (t, h) and every nonnegative Borel graph weight f,*

$$\int_{S_h(t)} f(x, h, \Phi_h x) J_h(x)\, d\mu_t(x) = \int_{\Phi_h S_h(t)} f(\Phi_{-h} y, h, y)\, d\mu_{t+h}(y). \tag{A.2}$$

The same versions can be used for countably many source restrictions, positive truncations and their increasing limits. Here $S_h(t)$ is the set of source points whose forward lifetime is greater than h.

Proof. At fixed count the regular physical words are disjoint open pieces of the collision cylinder. Their true boundaries and the removed critical points have zero source area. Take a countable cover by precompact flow boxes with lifted angular coordinate. Assign each point to its first containing box to obtain a Borel disjointization. This assignment is used only for integration: its seams do not stop the physical flow.

On the open survivor graph the map $(x, h) \mapsto (\Phi_h x, h)$ is injective, smooth and has physical Jacobian J_h. Its inverse on its image is the backward flow. Since $F(\Phi_h x) = F(x) + h$, ordinary change of variables followed by disintegration over (F, h) gives (A.2) on a countable determining algebra of Borel sets. Remove the union of the resulting null sets. Uniqueness of the kernel measures and monotone approximation extend equality to all nonnegative Borel weights outside this one joint null set. Countable disjoint summation gives the complete physical formula.

In endpoint coordinates the field is translation in F at fixed ϑ, so its endpoint Jacobian is $\chi(x)/\chi(\Phi_h x)$. Multiplication by the physical density ratio gives $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$, exactly the normalization in (A.2). Nonnegative graph selectors accommodate first-exit restrictions depending on (x, h) without new incompatible versions. Monotone convergence at fixed count permits the actual fractional insertions. No null set independent of the uncountable radius parameter is required: the ordered norm is $\sup_R \operatorname{ess\,sup}_t$. □

A.2. **True outgoing faces and exhaustion.** On a nongrazing limiting word all incidence limits are positive, so the stationary reflection equations extend smoothly to the boundary. This continuation is not asserted to remain the physical first-hit trajectory beyond that boundary. On a transverse outgoing arc write $F = b(\vartheta)$. Contraction of the area form $\omega(4\pi c_* \chi)^{-1} dF\, d\vartheta$ with ∂_F gives the positive flux in (5.4). A first-exit point is always counted together with its incident one-sided physical component. Lower-dimensional and tangent strata have zero transverse flux. Section decisions, competing obstacles and simultaneous failures are not suppressed by this description. An exit with a vanishing internal incidence belongs to the grazing class even when another condition fails there simultaneously; the remaining class is nongrazing.

Proposition A.2 (Initial mass and actual first-exit flux)**.** *For a nongrazing outgoing face, let $s = F(y) - t$ and $x = \Phi_{-s} y$. On $0 < s \le h$, $s < \tau^-(y)$, the contribution to initial weighted exit mass is*

$$\frac{W^\varepsilon(x)}{J_s(x)}\, d\zeta(y). \tag{A.3}$$

The boundary source (5.5) *has this same first-exit selection with weight one on the boundary, not with the factor in* (A.3)*. Both are defined on common almost-everywhere starting-roof versions.*

Proof. In one angular flow box put $a(u, \vartheta) = \omega(u, \vartheta)/(4\pi c_* \chi(u, \vartheta))$. The initial fiber mass is $W(x) a(t, \vartheta) d\vartheta$, whereas the outgoing flux is $a(b(\vartheta), \vartheta) d\vartheta$. The ratio of the latter density to the former unweighted density is $J_s(x)$. This proves (A.3). The backward-lifetime condition says precisely that the segment between the starting roof and this boundary has remained in that one-sided event; a later re-entry does not satisfy that condition.

Use precompact regular flow boxes, with rational rectangles in (F, ϑ), and increasing finite unions whose interiors cover the regular event. On overlaps within a physical component their coordinates agree up to a lifted angular constant and can be refined into finitely many pieces. Roof-constant faces are transverse; angle-constant faces have zero flux. The outside source plus the first-exit source for each such domain is the integral of the indicator that the whole forward segment is not contained in the domain. These indicators decrease to the indicator of a true exit. Every compact surviving segment eventually lies in an exhaustion domain. The usual endpoint convention at lifetime exactly h does not change the joint almost-everywhere formula.

For actual nongrazing faces, exhaust the regular one-sided arcs and their backward segments by such compact pieces. Disjoint assignment by the incident component and first-exit graph removes chart multiplicity. Apply the local formula and nonnegative convergence at fixed count. These operations also justify the variation payments in Theorem 5.2. No surface limit is commuted with the collision-normalized supremum. □

A.3. **The indispensable analytic chain.** Input 30.2 records the complete estimates used by this article, rather than requiring a reconstruction from historical status files. Its common-space operator dependencies are the collision norm definitions (Supplement equations (117.1)– (117.3)), the full-record fixed-band Doeblin–Fortet and spectral estimates (Supplement Theorem 157.3), and the finite moving-peak local inversion. The proof uses bounded transverse multipliers, not a bounded strong operator of multiplication by c^{-1} or $|g|^{-1}$.

For specificity, the joint endpoint/incidence estimate has leading factor $b^{129/128}s^{17/16}$ and finite-count cap $\min\{b\rho^{m/2}, s^2\}$. The endpoint test is the particular vector $P_b\mathbf{1}$ of strong norm $O(b^{7/8})$; that estimate is not a small operator norm on arbitrary vectors. Each continued rank-one peak is normalized using a nonzero smooth test against a physical peripheral vector, never by division by $\Pi\mathbf{1}$. The homogeneous trace estimates pass through bounded finite Cesaro approximants in the common strong space. These conventions and the two chronological characteristic pairings are proved in the retained module 186, not assumed for arbitrary L^∞ representatives.

Positive incidence layer cake integrates the cap first and yields a critical-volume remainder $b^{1/2}\rho^{m/4}$. A second layer cake against $b^{-1-\eta}$ converges at finite count only for $\eta < 1/2$, giving the actual $v^{\eta,b}$ estimate stated in Input 30.2. The leading exponent is $129/128 - \eta$. The first inverse-gradient remainder instead depends on its lower cutoff and is not used to remove that cutoff here. For the section layer only the all-depth marked-strip theorem is needed; for the incidence and clearance layers the original physical-guard local bound is used. All of these sources keep the forward occupation coefficient. Neither signed interface cancellation nor deletion of an arithmetic peak enters the new estimates.

Thus the new potential and surface theorems depend on explicitly stated volume estimates, but the surface source is bounded only after its actual first-exit graph and optical variation have been treated. The retained spectral and billiard estimates remain available for specialist verification. This appendix and native source compilation are not independent human or formal proof certification.

A.4. **The new trace interface.** The new proof uses the exact segment density and the inequality $|\partial_u A| \le CAI_m/z$, including both the optical and angular factors. Applying the fundamental theorem of calculus to $A^{1/p}$ requires only local absolute continuity and the actual integral of AI_m^p; it is not an anisotropic multiplier assertion. The reference interval is chosen on the longer side of each actual segment, with a Borel tie rule. Fubini then sums its energy with multiplicity one. The resulting finite-count bound permits the two initial thresholds to shrink with roof width. It does not extend through the unestimated critical/zero-residence source, which is retained in the final charge.

Part 18. Finite-count mixed moments and reachable optical concentration

Retained revision-86 abstract. We study the unchanged exact-label pointwise local limit problem for actual returns in a triangular Lorentz gas. A physically capped joint endpoint–incidence estimate gives complete mixed optical moments strictly above the first power, with a finite-count remainder integrable at every incidence and gradient scale. We use this additional moment to prove a uniform essential-height envelope for the entire microscopic sector whose optical sensitivity exceeds a prescribed inverse power of the roof offset. For explicit exponents its finite-count bound is $Kh^{1/8}$ and its ordered height is $Ch^{9/8}$. The infinite dyadic tail is summed before the collision limit. We also retain the exact backward-lifetime selection in the optical drift, so that only target states actually reachable from the starting roof contribute. This gives a smaller sufficient optical charge and removes the estimated high-sensitivity sector from it without changing the physical source. The original pointwise theorem is reduced to reachable moderate-sensitivity concentration and the original zero-residence volume concentration; their vanishing remains to be proved.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K_{m,R}^{\mathrm{c}}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K_{m,R}^{\mathrm{c}} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^m B_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **A finite-count microscopic envelope.** The optical sensitivity is the original additive function $I_m=1+\sum_{i=1}^{m-1}q_0^{d_i}/c_i$. Put $z=|g|$ on the regular exact event, using the endpoint gradient specified below. At the first use of a fiber expression we fix the common Borel kernels μ_t of that event. For each fixed m,R,n,k, measure identities hold outside a null set in t, and product identities outside a joint null set in (t,s). Countable selectors and dyadic scales use these same kernels. We do not require one null set for all radii.

Theorem 2.1 (Complete microscopic high-sensitivity control)**.** *Under the explicitly retained spectral inputs below, define*

$$D_h(t)=\int_0^h s^{-3/8}\int I_m(y)z(y)^{-1/4}\mathbf{1}_{\{I_m(y)>s^{-16}\}}\,d\mu_{t+s}(y)\,ds.$$

For each fixed auxiliary band B_* there is a finite K_{B_*} such that, for every $0<h<h_0$,

$$\sup_{m\ge2,R,n,k}m^2\|D_h\|_\infty\le K_{B_*}h^{1/8}, \tag{2.6}$$

$$\limsup_m\sup_{R,n,k}m^2\|D_h\|_\infty\le Ch^{9/8}. \tag{2.7}$$

The first bound is uniform before the collision limit. The second follows after that limit and enlargement of the fixed auxiliary band. Both include all actual words, exact labels and all microscopic scales down to zero in the stated sector.

Theorems 4.2 and 5.1 prove this statement. The extra moment is $I_m^{33/32} z^{-1/4}$, not an assumed bounded multiplier on an anisotropic space. The exponent $1/8$ makes the complete dyadic tail summable together with its finite-count error.

The theorem estimates a positive part of the original optical source; it does not replace that source by a protected one. Its complement remains explicit. Theorem 6.2 also retains the backward-lifetime selector in every optical comparison, replacing the unrestricted fractional majorant only by a proved reachable-fiber bound. The original pointwise target (2.5), and the scalar-first order of its downstream consequences, are unchanged.

3. Explicit analytic and geometric inputs

We collect the indispensable inherited assertions in the exact form used by the new proofs. Their source theorems and operator dependencies are specified here and in Appendix A. The new potential and first-exit estimates do not assume a surface spectral insertion theorem or an unweighted pointwise local bound.

Input statement 3.1 (Physical flow and source comparison)**.** On the regular part Ω of each exact event, endpoint angles $q = (\alpha_0, \alpha_m)$ give $F = L$, $g = \nabla_q F = R(-p_0, p_m)$, $K = D_q^2 F$ and $\vartheta = \arg g$. Put

$$\lambda_\circ = 9\sqrt{19}/200, \qquad \Lambda_\circ = 47/6, \qquad \omega = |\partial_{p_0}\alpha_m|^{-1}, \qquad \chi = |\det D_q(F, \vartheta)|.$$

The bounds $\lambda_\circ I \le K \le \Lambda_\circ I$ and $\lambda_\circ \le \chi \le \Lambda_\circ$ hold uniformly. The explicit field

$$X_q = \frac{K^{-1} g}{g^{\mathsf{T}} K^{-1} g}$$

is stopped at true physical and section-decision boundaries, not at chart seams. Its flow satisfies $XF = 1$, $X\vartheta = 0$ and

$$2\lambda_\circ s \le |g(\Phi_s x)|^2 - |g(x)|^2 \le 2\Lambda_\circ s. \tag{3.1}$$

Its endpoint-curve length is at most $C\sqrt{s}$. For $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$ one has $|X\omega| \le C\omega I_m/|g|$ on regular survivors. At a genuine grazing exit some internal incidence tends to zero and ω tends to zero. Forward interior critical termination cannot occur.

Every starting point with forward lifetime at most r belongs to a layer of one of the following forms, at an actual collision or flight:

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \quad \Delta_j \le C_E q_0^{d'_j}\sqrt{r}, \quad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}. \tag{3.2}$$

The physical coarea covariance, proved in the appendix in the convention used here, has Jacobian $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$. Writing E_h^ε for all weighted starting exit mass and

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon(x)[1 - \omega(\Phi_h x)/\omega(x)]_+ \, d\mu_t(x),$$

the complete source satisfies, for each fixed r and almost every t,

$$b^\varepsilon(t) \le 2A_r b^\varepsilon(t) + \bar{E}_r^\varepsilon(t) + \bar{O}_r^\varepsilon(t) + C_\Theta(1 + \varepsilon^{-1})\sqrt{r}\, A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\, ds. \tag{3.3}$$

These are the endpoint-field, all-contact motion, uniform cubic-action and separated-volume results in retained modules 174, 177, 180–182. The guard term in (3.3) is the complete product guard, not a protective subword source. The derivative and continuation assertions are confined to regular physical trajectories.

Input statement 3.2 (Complete local source estimates)**.** For every fixed auxiliary band B_*, constants $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ give the following bounds uniformly in radius, exact labels, interval J and mark $0 \le j \le m$. Let $S_{R,s}$ be the union of strips of width Cs about the finitely many coordinate lines supporting ∂Y_R^*. For s small,

$$m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le C s^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|) m^2 \rho_{B_*}^{m/2}, \tag{3.4}$$

$$m^2 \int_J b^\delta(t)\, dt \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|) m^3 \rho_{B_*}^{m/2}. \tag{3.5}$$

The local extension to a fixed remaining compact range of δ is included. At auxiliary parameter one $W^1 = 1$, so $b^1 = p$.

Put $z = |g|$, $\alpha = 129/128$ and, for $0 < \eta < 1/2$, define the actual nonnegative source measures

$$\begin{aligned} v^{\eta,b}(t)\, dt &= L_{\#}(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_\Pi \nu_R^*), \\ v^{\eta}(t)\, dt &= L_{\#}(I_m z^{-\eta} \mathbf{1}_\Pi \nu_R^*). \end{aligned} \tag{3.6}$$

They are finite and absolutely continuous at every fixed count, and for $0 < b < b_1$,

$$m^2 \int_J v^{\eta,b}(t)\, dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|) m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \tag{3.7}$$

Adding the bounded-gradient part gives, for every fixed $L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\eta \le C_\eta L, \qquad \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta,b} \le C_\eta b^{\alpha-\eta} L. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards; leading constants are independent of that band and of the previously fixed interval length.

Equation (3.4) is Supplement Theorem 175.3. The physical source bound (3.5) is Supplement Theorem 179.1, in particular its finite-window estimate. The fractional bounds are the positive layer-cake conclusions of retained module 187, stated in full above. The joint spectral and common-space assumptions used there are recorded in the appendix. No estimate for an untruncated first inverse gradient or for a singular roof convolution is part of this input.

Input statement 3.3 (The scalar positive remainder)**.** For large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.9}$$

The source b^ε is nonnegative and has a compactly supported BV density at fixed count. No collision-uniform variation bound is assumed. Supplement Theorem 181.4 gives this remainder with the same complete finite-peak reference.

3.1. **Maximal segments and the two existing charges.** The retained Borel parametrization, Supplement Theorem 7.1, is a bijection

$$\Psi : \coprod_j \{(\theta, u) : \theta \in E_j,\ a_j(\theta) < u < b_j(\theta)\} \longrightarrow \Omega,$$

with Borel inverse. Its exact density is $A_j(u, \theta) = \omega(\Psi_j(\theta, u))/(4\pi c_* \chi(\Psi_j(\theta, u)))$. Every segment remains in one regular exact event until a true boundary; Borel chart seams are not exits. Nongrazing outgoing faces retain their incident physical component and carry $d\zeta_+ = A_j(b_j-, \theta)\, d\theta$. The full angular variation is $|\partial_u A_j| \le C A_j I_m / z$. These are measure identities with the null-set convention already specified, not bounds with an unknown number of words.

The true first-exit assignment is

$$\mathcal{F}_h(t) = \int_{\Gamma^+_{\mathrm{ng}}} \mathbf{1}_{\{0<b-t\le h,\ b-t<\tau^-(y)\}}\, d\zeta_+(y), \qquad \overline{\mathcal{F}}_r = r^{-1}\int_0^r \mathcal{F}_h\, dh.$$

No lower gradient cutoff is imposed. Put $l = b - a$ and define

$$\mathcal{Z}_\delta(t) = \int_{\Gamma^+_{\mathrm{ng}}} \mathbf{1}_{\{0<l\le\delta\}}\mathbf{1}_{\{a<t<b\}}\, d\zeta_+, \qquad q_\delta(t)\, dt = F_{\#}(\mathbf{1}_{\Omega^{\mathrm{ng}}_+}\mathbf{1}_{\{l\le\delta\}}\bar\nu).$$

The second object is the original collision volume restricted to the same actual short segments, not a volume collar for a boundary. For source families set

$$\mathcal{H}_M(f) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} f(t), \quad \mathfrak{B}_M = \lim_{\delta\downarrow 0}\mathcal{H}_M(\mathcal{Z}_\delta), \quad \mathfrak{Q}_M = \lim_{\delta\downarrow 0}\mathcal{H}_M(q_\delta).$$

These limits are monotone extended nonnegative numbers.

For $0 < \eta < 1/2$, with $\kappa = (1-\eta)/2$, retain

$$\mathsf{P}_{\eta,h}v^\eta(t) = \int_0^h s^{-\kappa}v^\eta(t+s)\, ds, \qquad \mathfrak{A}_{\eta,M} = \lim_{h\downarrow 0}\mathcal{H}_M(\mathsf{P}_{\eta,h}v^\eta).$$

Supplement Theorems 8.2 and 9.1 give

$$\mathfrak{B}_M \le \mathcal{H}_M(\overline{\mathcal{F}}_r) \le \mathfrak{B}_M + C_\eta r^{1/32}. \tag{3.10}$$

Their band sum is finite before the collision limsup at each positive residence cutoff. Its leading constant is cutoff independent; the complete zero-residence tail is still inside $\mathcal{H}_M$ before its cutoff tends to zero. The active offset correction to historical module 189 remains Supplement equation (6.1); the historical inclusion is explicitly marked superseded. None of these inputs asserts charge zero.

4. Complete mixed optical moments at finite collision count

The finite-count cap, rather than the limiting Gaussian term alone, allows an optical moment strictly greater than one. This extra moment will be used before summing infinitely many microscopic roof scales. Write $z = |g|$, $\alpha = 129/128$, $\beta = 17/16$, and $\bar\nu = \nu^*_R|_\Omega$. All constants below are uniform in m, R, n, k. Dependence on the fixed moment exponents is permitted.

4.1. **The capped input.** The joint marked estimate retained in Supplement Theorem 25.3 has, after the fixed conversion from endpoint momentum to z, the form

$$\begin{aligned} m^2\int \mathbf{1}_{\{F\in J, z<\zeta, c_i<s\}}\, d\bar\nu \le{}& C\zeta^\alpha s^\beta(|J| + B_*^{-1}) \\ &+ C_{B_*}(1+B_*|J|)m^2\min\{\zeta\rho_{B_*}^{m/2}, s^2\}. \end{aligned} \tag{4.1}$$

It holds for $0 < \zeta < z_*$, $0 < s < s_*$ and every internal mark. The source retains its exact displacement and occupation coefficient; dropping endpoint section tests is used only in its positive upper comparison. Its endpoint-only counterpart is

$$m^2\int \mathbf{1}_{\{F\in J, z<\zeta\}}\, d\bar\nu \le C\zeta^\alpha(|J| + B_*^{-1}) + C_{B_*}(1+B_*|J|)m^2\zeta\rho_{B_*}^{m/2}. \tag{4.2}$$

The stronger endpoint exponent in that proof implies the displayed one. Without a gradient restriction the retained marked-incidence estimate has exponent $5/4$ and cap $\min\{\rho_{B_*}^{m/2}, s^2\}$; see Supplement Theorem 35.3. These are explicitly inherited spectral inputs. Their four-record extensions are not being attributed to an external billiard theorem.

Lemma 4.1 (A capped power integral)**.** *For $0 < p < 2$, $0 < D \le 1$ and $0 < s \le 1$,*

$$s^{-p}\min(D, s^2) + p\int_0^s v^{-p-1}\min(D, v^2)\,dv \le \frac{2}{2-p}\min\{s^{2-p}, D^{1-p/2}\}. \tag{4.3}$$

Proof. For $s \le \sqrt{D}$ the left side is $2s^{2-p}/(2-p)$. For $s > \sqrt{D}$, split the integral at $\sqrt{D}$. The terms Ds^{-p} cancel, leaving $2D^{1-p/2}/(2-p)$. Thus the inequality is in fact an equality. The same computation justifies monotone removal of the lower integration endpoint. □

Theorem 4.2 (Actual mixed moments)**.** *Fix*

$$1 < p < \beta, \qquad \sigma_p = 1 - p/2, \qquad 0 < \eta < \sigma_p.$$

The actual densities

$$U_p^\zeta(t)\,dt = F_\#(I_m^p \mathbf{1}_{\{z<\zeta\}}\bar{\nu}), \qquad V_{p,\eta}^\zeta(t)\,dt = F_\#(I_m^p z^{-\eta}\mathbf{1}_{\{0<z<\zeta\}}\bar{\nu})$$

are finite at each count. On every fixed auxiliary band,

$$m^2\int_J U_p^\zeta \le C_p\zeta^\alpha(|J| + B_*^{-1}) + C_{B_*,p}(1 + B_*|J|)m^2\zeta^{\sigma_p}\rho_{B_*}^{m\sigma_p/2}, \tag{4.4}$$

$$m^2\int_J V_{p,\eta}^\zeta \le C_{p,\eta}\zeta^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,p,\eta}(1 + B_*|J|)m^2\zeta^{\sigma_p-\eta}\rho_{B_*}^{m\sigma_p/2}. \tag{4.5}$$

The unrestricted actual source $V_{p,\eta}(t)\,dt = F_\#(I_m^p z^{-\eta}\bar{\nu})$ satisfies

$$m^2\int_J V_{p,\eta} \le C_{p,\eta}(|J| + B_*^{-1}) + C_{B_*,p,\eta}(1 + B_*|J|)m^2\rho_{B_*}^{m\sigma_p/2}. \tag{4.6}$$

The estimates hold at finite count, for all intervals and thresholds in their stated ranges. The leading constants do not depend on the fixed band. No strong-space multiplier $M_{I_m^p z^{-\eta}}$ is assumed.

Proof. Put $S_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i} \le 1 + 2q_0/(1-q_0)$. Weighted Jensen gives the pointwise inequality

$$I_m^p \le S_m^{p-1}\left(1 + \sum_{i=1}^{m-1} q_0^{d_i} c_i^{-p}\right). \tag{4.7}$$

For each marked power below s_* use the positive layer cake

$$c_i^{-p}\mathbf{1}_{\{c_i<s_*\}} = s_*^{-p}\mathbf{1}_{\{c_i<s_*\}} + p\int_0^{s_*} v^{-p-1}\mathbf{1}_{\{c_i<v\}}\,dv.$$

Insert (4.1) before any limit. The leading integral converges since $p < \beta$. Lemma 4.1, with $D = \zeta\rho_{B_*}^{m/2}$, bounds its entire remainder by $C_p\zeta^{\sigma_p}\rho_{B_*}^{m\sigma_p/2}$. The bounded complementary powers and the constant one use (4.2). Their errors are smaller, after adjusting fixed constants. Sum the geometric weights in (4.7); there is no physical-word count. This proves (4.4).

Now apply the positive gradient layer cake with weight I_m^p:

$$z^{-\eta}\mathbf{1}_{\{z<\zeta\}} = \zeta^{-\eta}\mathbf{1}_{\{z<\zeta\}} + \eta\int_0^\zeta q^{-\eta-1}\mathbf{1}_{\{z<q\}}\,dq.$$

Both terms of (4.4) are integrable at zero: $\alpha - \eta > 0$ and $\sigma_p - \eta > 0$. Their coefficients are, respectively, multiplied by $\alpha/(\alpha-\eta)$ and $\sigma_p/(\sigma_p-\eta)$. This proves (4.5) by monotone convergence of the actual source, including its finite-count remainder.

Split the unrestricted source at $z_*/2$. Above that threshold, $z^{-\eta}$ is bounded. Apply the unconditioned capped incidence input and the same Jensen and power integrals; $p < \beta < 5/4$. Below it use (4.5). Enlarge $\rho_{B_*} < 1$ if necessary to combine finitely many remainders. This proves (4.6). At fixed count the roof has bounded support, so taking a containing interval proves finite mass. Absolute continuity follows on compact regular charts by coarea, then on their

increasing union by positivity; the removed critical set has zero source area. All densities use the common kernels specified at the first use of the segment map in this article. □

The strict inequalities on p and η are needed for two different integrals. In particular no untruncated first inverse gradient is obtained by substituting $\eta = 1$. The new gain is a genuine power of the complete optical insertion above one, with a summable finite-count remainder.

5. A finite-count envelope on collapsing optical scales

Put $\kappa = (1-\eta)/2$ and fix admissible p, η as in Theorem 4.2. Choose

$$\gamma > \frac{\kappa}{p-1}, \qquad d = \gamma(p-1) - \kappa > 0. \tag{5.1}$$

The following source is defined directly through the common physical fiber kernels, not through separately chosen versions of a density:

$$\mathcal{D}_{\eta,\gamma,h}(t) = \int_0^h s^{-\kappa} \int I_m(y) z(y)^{-\eta} \mathbf{1}_{\{I_m(y) > s^{-\gamma}\}} \, d\mu_{t+s}(y) \, ds. \tag{5.2}$$

It contains every actual word of the fixed exact event. The selector may shrink with the microscopic offset because the input to which it will be compared is a finite-count estimate uniform in all intervals. The Fourier band remains fixed.

Theorem 5.1 (A complete microscopic large-sensitivity envelope)**.** *For $0 < h < h_0 < 1$, every $m \geq 2$ and every fixed band B_*,*

$$\begin{aligned} m^2 \|\mathcal{D}_{\eta,\gamma,h}\|_\infty &\leq C_{p,\eta,\gamma}(h^{d+1} + B_*^{-1} h^d) \\ &\quad + C_{B_*,p,\eta,\gamma}(1 + B_* h) m^2 \rho_{B_*}^{m\sigma_p/2} h^d. \end{aligned} \tag{5.3}$$

The essential supremum is global in the roof variable and uniform in radius and exact labels. In particular, for each fixed band there is a finite $K_{B_*,p,\eta,\gamma}$ such that

$$\sup_{m \geq 2, R, n, k} m^2 \|\mathcal{D}_{\eta,\gamma,h}\|_\infty \leq K_{B_*,p,\eta,\gamma} h^d. \tag{5.4}$$

After the collision limit and then enlargement of the auxiliary band,

$$\limsup_m \sup_{R,n,k} m^2 \|\mathcal{D}_{\eta,\gamma,h}\|_\infty \leq C_{p,\eta,\gamma} h^{d+1}. \tag{5.5}$$

Thus the entire infinite microscopic tail of this actual selected source has zero ordered concentration charge.

Proof. Let $\ell_j = 2^{-j} h$. On $\ell_j/2 < s \leq \ell_j$ the indicator in (5.2) implies $I_m > \ell_j^{-\gamma}$, and consequently

$$I_m \mathbf{1}_{\{I_m > s^{-\gamma}\}} \leq \ell_j^{\gamma(p-1)} I_m^p.$$

Also $s^{-\kappa} \leq 2^\kappa \ell_j^{-\kappa}$. Tonelli therefore gives, at finite count,

$$\mathcal{D}_{\eta,\gamma,h}(t) \leq 2^\kappa \sum_{j=0}^{\infty} \ell_j^d \int_{\ell_j/2}^{\ell_j} V_{p,\eta}(t+s) \, ds. \tag{5.6}$$

For a finite partial sum apply (4.6) to each actual interval $(t + \ell_j/2, t + \ell_j]$. The leading terms are bounded by $C\ell_j^d(\ell_j + B_*^{-1})$, and the remainders by $C_{B_*} m^2 \rho_{B_*}^{m\sigma_p/2} \ell_j^d (1 + B_* \ell_j)$. Both series are geometric since $d > 0$. Their sums give (5.3), first for all partial sums and then for the entire nonnegative source by monotone convergence. The countable dyadic comparisons use one common almost-everywhere fiber version. No infinite sum of separate collision limsups occurs.

At a fixed B_*, the supremum of $m^2\rho_{B_*}^{m\sigma_p/2}$ over $m \geq 2$ is finite. This proves the finite-count uniform envelope (5.4). For (5.5), first take the collision limsup with h and B_* fixed, and only then let $B_* \to \infty$. The leading constant is band independent. The same bounds apply to the complete tail below $2^{-N}h$ by replacing h with $2^{-N}h$. In particular its finite-count bound decays as 2^{-Nd} before any collision limit. □

Corollary 5.2 (Explicit exponents)**.** *One admissible choice is*

$$p = 33/32, \qquad \eta = 1/4, \qquad \sigma_p = 31/64, \qquad \gamma = 16, \qquad \kappa = 3/8, \qquad d = 1/8.$$

The finite-count microscopic envelope is $K_{B_*}h^{1/8}$, and its ordered essential-height bound is $Ch^{9/8}$.

Proof. Here $p < 17/16$, $\eta < 31/64$ and $16(33/32 - 1) - 3/8 = 1/8$. Substitute these identities in the theorem. □

5.1. **What is removed from the fractional charge.** Define $\mathsf{P}_{\eta,h}^{\mathrm{mod}}$ by replacing the indicator in (5.2) with $\mathbf{1}_{\{I_m \leq s^{-\gamma}\}}$. At finite count,

$$\mathsf{P}_{\eta,h}v^\eta = \mathsf{P}_{\eta,h}^{\mathrm{mod}} + \mathcal{D}_{\eta,\gamma,h}.$$

Each side uses the original source; this is an exact positive partition. With $\mathcal{H}_M$ as in the preceding section,

$$\mathfrak{A}_{\eta,M} = \lim_{h\downarrow 0} \mathcal{H}_M(\mathsf{P}_{\eta,h}^{\mathrm{mod}}). \tag{5.7}$$

Indeed positivity gives one inequality, and subadditivity of $\mathcal{H}_M$ with (5.5) gives the other. The equality holds also for infinite charges. The theorem removes a complete high-sensitivity microscopic sector, not the whole optical charge. It gives no pointwise estimate for its complementary sector.

6. Reachable fibers and the remaining pointwise endpoint

The original stopped transport reaches only part of the fiber at a translated roof. Retaining that selection yields a smaller sufficient optical charge than the unrestricted fractional majorant. No mass of the original source is removed: all source trajectories are included until their first true exit.

Let $\tau^-(y)$ be the backward lifetime in the same regular exact event. Define the actual reachable-fiber drift

$$\mathcal{V}_h(t) = \int_0^h \int \mathbf{1}_{\{\tau^-(y)>s\}} \frac{I_m(y)}{z(y)}\, d\mu_{t+s}(y)\, ds. \tag{6.1}$$

The integrand is defined on $z > 0$; values on the source-null critical set are immaterial. The Borel segment bijection makes its selection exactly $a_j(\theta) < t < t + s < b_j(\theta)$. All external variables use the common product kernels. Thus the formulas below hold outside one null set in (t, h) for each fixed m, R, n, k, and consequently for almost every t at each stated fixed width. Countably many widths share a null set; no null set common to all radii is asserted.

Theorem 6.1 (A source-level reachable-fiber comparison)**.** *For the original weighted exit and inverse-twist defects,*

$$E_h^\varepsilon(t) + O_h^\varepsilon(t) \leq C\mathcal{F}_h(t) + C\mathcal{V}_h(t). \tag{6.2}$$

For the original short-segment volume density q_δ and outgoing occupancy trace $\mathcal{Z}_\delta$,

$$|\mathcal{Z}_\delta(t) - q_\delta(t)| \leq C\mathcal{V}_\delta(t). \tag{6.3}$$

Moreover

$$\mathcal{V}_h(t) \leq C_\eta \mathsf{P}_{\eta,h}v^\eta(t). \tag{6.4}$$

No equality between the two potentials is claimed.

Proof. On one maximal segment write $w(u) = \omega(\Psi_j(\theta, u))$. On a survivor, $[w(t) - w(t+h)]_+$ is paid by the negative variation up to $t+h$. At a grazing first exit $w(b-) = 0$, so the same variation pays the entire positive starting weight. At a nongrazing first exit, $w(t) \le w(b-) + \int_t^b [-w']_+$. Divide by the initial angular normalization $4\pi c_* \chi(t)$, use $W^\varepsilon \le 1$ and $\lambda_\circ \le \chi \le \Lambda_\circ$, and sum these disjoint source cases. The bound $|X\omega| \le C\omega I_m / z$ and the Borel map identify their complete variation with a submeasure of (6.1). The endpoint terms are exactly the nongrazing first-exit assignment $\mathcal{F}_h$, up to the common ratio of angular bounds. This proves (6.2). A target y occurs only when its segment contains the starting roof, which is precisely $\tau^-(y) > s$; later re-entries do not occur.

For (6.3), use the full angular density $A_j = \omega/(4\pi c_* \chi)$ on each selected short segment. Its established variation bound is $|\partial_u A_j| \le C A_j I_m / z$. Integrate from t to its actual forward endpoint, whose distance is at most δ. At every intermediate roof its backward lifetime exceeds $s = u - t$. The same Borel bijection gives multiplicity one. Tonelli and positive exhaustion of compact regular subsegments prove both comparisons; no inverse physical Jacobian has been replaced by one.

If $\tau^-(y) > s$, then $y = \Phi_s x$ for a point x on the same exact event. Radial growth gives $z(y)^2 \ge 2\lambda_\circ s$. Thus $z(y)^{-1} \le (2\lambda_\circ)^{-\kappa} s^{-\kappa} z(y)^{-\eta}$. Discarding the reachable-fiber selector only on the upper side proves (6.4). It also supplies fixed-count local finiteness almost everywhere from the retained fractional potential theorem. □

Split $\mathcal{V}_h = \mathcal{V}_h^{\mathrm{mod}} + \mathcal{V}_h^{\mathrm{hi}}$ using $I_m(y) \le s^{-\gamma}$ and its complement, with the same fixed exponents as in Section 5. Then

$$\mathcal{V}_h^{\mathrm{hi}} \le C_\eta \mathcal{D}_{\eta,\gamma,h}. \tag{6.5}$$

The complete optical variation carried by this large-sensitivity sector therefore obeys the finite-count uniform envelope already proved; the source cases may include grazing first exits. The estimate concerns their selected variation contribution, not an unsupported claim that every such trajectory remains in that sector for its entire lifetime.

Theorem 6.2 (The smaller optical charge and its endpoint budget)**.** *The monotone limits*

$$\mathfrak{C}_M = \lim_{h \downarrow 0} \mathcal{H}_M(\mathcal{V}_h), \qquad \mathfrak{C}_M^{\mathrm{mod}} = \lim_{h \downarrow 0} \mathcal{H}_M(\mathcal{V}_h^{\mathrm{mod}})$$

exist in $[0, \infty]$ *and satisfy*

$$\mathfrak{C}_M = \mathfrak{C}_M^{\mathrm{mod}} \le C_\eta \mathfrak{A}_{\eta,M}, \qquad \mathfrak{B}_M \le \mathfrak{Q}_M + C\mathfrak{C}_M, \quad \mathfrak{Q}_M \le \mathfrak{B}_M + C\mathfrak{C}_M. \tag{6.6}$$

For all sufficiently large fixed B *and sufficiently small fixed* r*,*

$$\mathcal{E}_M \le C_M B^{-1/192} + C\mathfrak{B}_M + C\mathfrak{C}_M^{\mathrm{mod}} + C_\eta r^{1/32} + C_\Theta (1 + \varepsilon(B)^{-1}) \sqrt{r}, \tag{6.7}$$

$$\mathcal{E}_M \le C_M B^{-1/192} + C\mathfrak{Q}_M + C\mathfrak{C}_M^{\mathrm{mod}} + C_\eta r^{1/32} + C_\Theta (1 + \varepsilon(B)^{-1}) \sqrt{r}. \tag{6.8}$$

Constants in the second line may be enlarged. The conditions $\mathfrak{C}_M^{\mathrm{mod}} = \mathfrak{Q}_M = 0$ *for every* M *imply the original full pointwise target. Neither condition is asserted as proved by the large-sensitivity estimate.*

Proof. Each potential is monotone in its upper integration limit. Positivity, (6.5) and (5.5) give equality of the two causal charges, including infinite values. The remaining comparisons follow from (6.4) and the two positive inequalities in (6.3), with the complete tail inside $\mathcal{H}_M$ before taking its width to zero.

For $0 < \delta < r$, radial growth and the fractional comparison give

$$\mathcal{V}_r - \mathcal{V}_\delta \le C_\eta \int_\delta^r s^{-\kappa} v^\eta(t+s)\, ds.$$

Cover $[\delta, r]$ by finitely many dyadic annuli before the collision limsup. The fixed-width input bounds their normalized sum by $C_\eta r^{1-\kappa}$. Applying $\mathcal{H}_M$ first and then letting $\delta \downarrow 0$ proves

$$\mathcal{H}_M(\mathcal{V}_r) \le \mathfrak{C}_M + C_\eta r^{1-\kappa}. \tag{6.9}$$

No infinite physical sum is moved through a limsup in this step.

Average (6.2) in $0 < h < r$ and use monotonicity of $\mathcal{V}_h$. Insert the result into the complete source-shift input and then the scalar positive remainder. The fixed-width bad source contributes $O(B^{-1/192})$. The unweighted fixed-width bound pays the unchanged guard and angular term. The established surface sandwich gives $\mathcal{H}_M(\overline{\mathcal{F}}_r) \le \mathfrak{B}_M + C_\eta r^{1/32}$. Equation (6.9) and $1 - \kappa > 1/32$ prove (6.7); (6.6) proves (6.8). Under the stated zero-charge conditions, choose $r(B) = c\varepsilon(B)^\Gamma$ with $\Gamma > 2$. For each B the width and all auxiliary bands are fixed before the collision limit; auxiliary bands are enlarged afterwards and B is taken last. The original target follows, including its arithmetic zero classes without normalization. □

The new finite-count theorem excludes the entire large-sensitivity microscopic sector by a uniform essential-height envelope, rather than by fixed-scale limiting bounds. What remains is the reachable moderate-sensitivity optical concentration and the original zero-residence volume concentration. The reachability constraint is physical, not a chosen collar. Proving these last two quantities zero still requires a source-specific estimate beyond those proved here.

7. The scalar objective and the remaining concentration

The finite-count envelope (5.4) controls every microscopic scale of its selected large-sensitivity sector at once. It is not a reinterpretation of a fixed-width limit. Its proof keeps the physical cap through both singular integrations and through the complete dyadic sum. The complementary moderate-sensitivity sector, however, has no analogous uniform envelope in this revision.

The two sufficient pairs in the original endpoint are now $(\mathfrak{C}_M^{\mathrm{mod}}, \mathfrak{B}_M)$ and $(\mathfrak{C}_M^{\mathrm{mod}}, \mathfrak{Q}_M)$, as displayed in (6.7)–(6.8). The reachable optical charge is at most the preceding unrestricted fractional charge, but no converse is assumed. Both remaining vanishing requirements are kept explicit. The organizing theorem is not marked proved by the estimated sector alone.

After scalar closure, Supplement Theorem 76.1 gives the same-roof path numerator. Ratios remain restricted to positive arithmetic reference classes; the return clock and forward likelihood retain their separate inherited hypotheses. No zero class is normalized. The full preceding article and every inherited proof are preserved in the supplement, while source receipts and finite diagnostics remain outside the journal narrative.

Appendix A. Physical kernels, first-exit traces and analytic dependencies

This appendix supplies the measure-theoretic interfaces used by the new arguments in one convention. It does not deduce a surface estimate from a volume multiplier theorem. The geometric and spectral estimates whose proofs are retained in the technical supplement are listed separately.

A.1. **Normalization and common kernels.** On a regular word use endpoint angles $q = (\alpha_0, \alpha_m)$ and oriented internal arclengths. Endpoint first variation gives $F_{\alpha_0} = -Rp_0$ and $F_{\alpha_m} = Rp_m$. Consequently

$$\omega = |\partial_{\alpha_m} p_0| = |F_{\alpha_0\alpha_m}|/R, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\, dq.$$

In the optical Schur complement the internal Hessian is $D_c A D_c$, where A has diagonal $t_{i-1} + t_i + 2/(Rc_i)$ and off-diagonal $-t_i$. The two angle-to-arclength factors are R^2; division by R in the first

variation gives $\omega = Rc_0c_m/y_m$. For one flight $y_1 = \ell_0$, and for two flights $y_2 = \ell_0+\ell_1+2\ell_0\ell_1/(Rc_1)$. Changes of tangent orientation do not change these absolute identities.

Away from $g = 0$, the coordinates (F, ϑ) have Jacobian $\chi = e^{\perp\mathsf{T}}Ke^{\perp}$ in absolute value, where $e = g/|g|$. Thus the angular expression for the original fiber is

$$d\mu_t = \frac{\omega(t,\vartheta)}{4\pi c_*\chi(t,\vartheta)}\,d\vartheta. \tag{A.1}$$

The sign of an oriented endpoint chart is irrelevant; all source Jacobians in this appendix are positive. The original source already contains its initial and terminal tests and is normalized only by c_*.

Proposition A.1 (Common graph covariance)**.** *At fixed m, R, n, k, the regular exact event has jointly measurable coarea kernels for which, for almost every (t, h) and every nonnegative Borel graph weight f,*

$$\int_{S_h(t)} f(x,h,\Phi_h x)J_h(x)\,d\mu_t(x) = \int_{\Phi_h S_h(t)} f(\Phi_{-h}y,h,y)\,d\mu_{t+h}(y). \tag{A.2}$$

The same versions can be used for countably many source restrictions, positive truncations and their increasing limits. Here $S_h(t)$ is the set of source points whose forward lifetime is greater than h.

Proof. At fixed count the regular physical words are disjoint open pieces of the collision cylinder. Their true boundaries and the removed critical points have zero source area. Take a countable cover by precompact flow boxes with lifted angular coordinate. Assign each point to its first containing box to obtain a Borel disjointization. This assignment is used only for integration: its seams do not stop the physical flow.

On the open survivor graph the map $(x, h) \mapsto (\Phi_h x, h)$ is injective, smooth and has physical Jacobian J_h. Its inverse on its image is the backward flow. Since $F(\Phi_h x) = F(x) + h$, ordinary change of variables followed by disintegration over (F, h) gives (A.2) on a countable determining algebra of Borel sets. Remove the union of the resulting null sets. Uniqueness of the kernel measures and monotone approximation extend equality to all nonnegative Borel weights outside this one joint null set. Countable disjoint summation gives the complete physical formula.

In endpoint coordinates the field is translation in F at fixed ϑ, so its endpoint Jacobian is $\chi(x)/\chi(\Phi_h x)$. Multiplication by the physical density ratio gives $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$, exactly the normalization in (A.2). Nonnegative graph selectors accommodate first-exit restrictions depending on (x, h) without new incompatible versions. Monotone convergence at fixed count permits the actual fractional insertions. No null set independent of the uncountable radius parameter is required: the ordered norm is $\sup_R \operatorname{ess\,sup}_t$. □

A.2. **True outgoing faces and exhaustion.** On a nongrazing limiting word all incidence limits are positive, so the stationary reflection equations extend smoothly to the boundary. This continuation is not asserted to remain the physical first-hit trajectory beyond that boundary. On a transverse outgoing arc write $F = b(\vartheta)$. Contraction of the area form $\omega(4\pi c_*\chi)^{-1}dF\,d\vartheta$ with ∂_F gives the positive flux in (5.4). A first-exit point is always counted together with its incident one-sided physical component. Lower-dimensional and tangent strata have zero transverse flux. Section decisions, competing obstacles and simultaneous failures are not suppressed by this description. An exit with a vanishing internal incidence belongs to the grazing class even when another condition fails there simultaneously; the remaining class is nongrazing.

Proposition A.2 (Initial mass and actual first-exit flux)**.** *For a nongrazing outgoing face, let $s = F(y) - t$ and $x = \Phi_{-s}y$. On $0 < s \le h$, $s < \tau^-(y)$, the contribution to initial weighted exit*

mass is

$$\frac{W^{\varepsilon}(x)}{J_s(x)}\,d\zeta(y). \tag{A.3}$$

The boundary source (5.5) *has this same first-exit selection with weight one on the boundary, not with the factor in* (A.3)*. Both are defined on common almost-everywhere starting-roof versions.*

Proof. In one angular flow box put $a(u,\vartheta) = \omega(u,\vartheta)/(4\pi c_*\chi(u,\vartheta))$. The initial fiber mass is $W(x)a(t,\vartheta)d\vartheta$, whereas the outgoing flux is $a(b(\vartheta),\vartheta)d\vartheta$. The ratio of the latter density to the former unweighted density is $J_s(x)$. This proves (A.3). The backward-lifetime condition says precisely that the segment between the starting roof and this boundary has remained in that one-sided event; a later re-entry does not satisfy that condition.

Use precompact regular flow boxes, with rational rectangles in (F,ϑ), and increasing finite unions whose interiors cover the regular event. On overlaps within a physical component their coordinates agree up to a lifted angular constant and can be refined into finitely many pieces. Roof-constant faces are transverse; angle-constant faces have zero flux. The outside source plus the first-exit source for each such domain is the integral of the indicator that the whole forward segment is not contained in the domain. These indicators decrease to the indicator of a true exit. Every compact surviving segment eventually lies in an exhaustion domain. The usual endpoint convention at lifetime exactly h does not change the joint almost-everywhere formula.

For actual nongrazing faces, exhaust the regular one-sided arcs and their backward segments by such compact pieces. Disjoint assignment by the incident component and first-exit graph removes chart multiplicity. Apply the local formula and nonnegative convergence at fixed count. These operations also justify the variation payments in Theorem 5.2. No surface limit is commuted with the collision-normalized supremum. □

A.3. **The indispensable analytic chain.** Input 3.2 records the complete estimates used by this article, rather than requiring a reconstruction from historical status files. Its common-space operator dependencies are the collision norm definitions (Supplement equations (117.1)– (117.3)), the full-record fixed-band Doeblin–Fortet and spectral estimates (Supplement Theorem 157.3), and the finite moving-peak local inversion. The proof uses bounded transverse multipliers, not a bounded strong operator of multiplication by c^{-1} or $|g|^{-1}$.

For specificity, the joint endpoint/incidence estimate has leading factor $b^{129/128}s^{17/16}$ and finite-count cap $\min\{b\rho^{m/2}, s^2\}$. The endpoint test is the particular vector $P_b\mathbf{1}$ of strong norm $O(b^{7/8})$; that estimate is not a small operator norm on arbitrary vectors. Each continued rank-one peak is normalized using a nonzero smooth test against a physical peripheral vector, never by division by $\Pi\mathbf{1}$. The homogeneous trace estimates pass through bounded finite Cesaro approximants in the common strong space. These conventions and the two chronological characteristic pairings are proved in the retained module 186, not assumed for arbitrary L^∞ representatives.

Positive incidence layer cake integrates the cap first and yields a critical-volume remainder $b^{1/2}\rho^{m/4}$. A second layer cake against $b^{-1-\eta}$ converges at finite count only for $\eta < 1/2$, giving the actual $v^{\eta,b}$ estimate stated in Input 3.2. The leading exponent is $129/128 - \eta$. The first inverse-gradient remainder instead depends on its lower cutoff and is not used to remove that cutoff here. For the section layer only the all-depth marked-strip theorem is needed; for the incidence and clearance layers the original physical-guard local bound is used. All of these sources keep the forward occupation coefficient. Neither signed interface cancellation nor deletion of an arithmetic peak enters the new estimates.

Thus the new potential and surface theorems depend on explicitly stated volume estimates, but the surface source is bounded only after its actual first-exit graph and optical variation have been treated. The retained spectral and billiard estimates remain available for specialist

verification. This appendix and native source compilation are not independent human or formal proof certification.

Part 19. Complete residence bands and zero-residence source

Retained revision-85 abstract. We retain the exact-label pointwise local limit problem for actual returns in a triangular Lorentz gas. A Borel transversal parametrizes complete maximal regular roof segments and identifies their angular disintegration with the original collision measure, without a multiplicity factor. Residence persistence then controls the essential height of every fixed positive residence band, including critical and nongrazing boundary pairs. Summing any finite collection of such bands yields a full surface-height bound of order $r^{1/32}$ at every positive residence cutoff, with a leading constant independent of that cutoff. The short critical part has the sharper bound $r^{129/256}(1+|\log r|)$ in the same ordered norm. The complete zero-residence tail remains inside the collision-normalized height before the cutoff is removed. We identify it with the limiting surface height and compare its actual boundary traces to the original short-segment volume source by a paid fractional optical potential. The original pointwise objective, exact labels and arithmetic classes are unchanged. Its final budget retains two unproved microscopic vanishing requirements: fractional optical concentration and zero-residence concentration.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0)+\mathbb{Z}(1/2,\sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z})\times(-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\,\mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3-\theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1-6q^2+q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on

$[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_\#(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **Complete surface estimates on all positive residence scales.** An actual nongrazing outgoing face carries its backward regular lifetime $l = b - a$ in the same exact event. Let $\overline{\mathcal{F}}_r$ denote its full first-exit assignment, averaged over roof increments in $(0, r)$. A superscript $\ge \delta$ retains only faces with $l \ge \delta$. Among its short faces, $l \le 2r$, let $\mathcal{K}_r$ retain $|g(b-)|^2 < 8\Lambda_\circ r$.

Theorem 2.1 (All positive residence scales)**.** *Under the explicit retained inputs below, for every sufficiently small fixed $r > 0$ and every fixed $\delta > 0$,*

$$\limsup_m \sup_{R,n,k} m^2 \|\overline{\mathcal{F}}_r^{\ge\delta}\|_\infty \le C_\eta r^{1/32}, \tag{2.6}$$

$$\limsup_m \sup_{R,n,k} m^2 \|\mathcal{K}_r^{\ge\delta}\|_\infty \le C_\eta r^{129/256}(1 + |\log r|). \tag{2.7}$$

The constants are independent of the positive residence cutoff δ. No incoming-end type, physical word or exact label is removed within these selected families. The same first bound holds for their short noncritical paired incoming assignment.

Theorem 8.2 proves these assertions from persistence on each complete residence band. The underlying height bound is not inferred from local mass alone. Its measurable summation is supplied by the explicit Borel bijection in Theorem 7.1. Proposition 7.2 also reproduces the finite-count computation of the independent first-inverse-gradient cutoff constant used for paired transfer.

The uniform leading constants do not authorize removal of δ before the collision limit. Theorem 9.1 identifies the precise remaining zero-residence charge and proves that it is the small-width limit of the full surface height. Its comparison with an actual collision-source restriction is paid by optical variation, not by replacing a surface with a volume collar. Together with the canonical fractional charge, it gives the two-term microscopic budget in Theorem 9.3. Zero-charge estimates remain required for the original pointwise theorem.

3. Explicit analytic and geometric inputs

We collect the indispensable inherited assertions in the exact form used by the new proofs. Their source theorems and operator dependencies are specified here and in Appendix A. The new potential and first-exit estimates do not assume a surface spectral insertion theorem or an unweighted pointwise local bound.

Input statement 3.1 (Physical flow and source comparison)**.** On the regular part Ω of each exact event, endpoint angles $q = (\alpha_0, \alpha_m)$ give $F = L$, $g = \nabla_q F = R(-p_0, p_m)$, $K = D_q^2 F$ and $\vartheta = \arg g$. Put

$$\lambda_\circ = 9\sqrt{19}/200, \qquad \Lambda_\circ = 47/6, \qquad \omega = |\partial_{p_0}\alpha_m|^{-1}, \qquad \chi = |\det D_q(F, \vartheta)|.$$

The bounds $\lambda_\circ I \le K \le \Lambda_\circ I$ and $\lambda_\circ \le \chi \le \Lambda_\circ$ hold uniformly. The explicit field

$$X_q = \frac{K^{-1} g}{g^{\mathsf{T}} K^{-1} g}$$

is stopped at true physical and section-decision boundaries, not at chart seams. Its flow satisfies $XF = 1$, $X\vartheta = 0$ and

$$2\lambda_\circ s \le |g(\Phi_s x)|^2 - |g(x)|^2 \le 2\Lambda_\circ s. \tag{3.1}$$

Its endpoint-curve length is at most $C\sqrt{s}$. For $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$ one has $|X\omega| \le C\omega I_m/|g|$ on regular survivors. At a genuine grazing exit some internal incidence tends to zero and ω tends to zero. Forward interior critical termination cannot occur.

Every starting point with forward lifetime at most r belongs to a layer of one of the following forms, at an actual collision or flight:

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \quad \Delta_j \le C_E q_0^{d'_j}\sqrt{r}, \quad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}. \tag{3.2}$$

The physical coarea covariance, proved in the appendix in the convention used here, has Jacobian $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$. Writing E_h^ε for all weighted starting exit mass and

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon(x)[1 - \omega(\Phi_h x)/\omega(x)]_+ \, d\mu_t(x),$$

the complete source satisfies, for each fixed r and almost every t,

$$b^\varepsilon(t) \le 2A_r b^\varepsilon(t) + \bar{E}_r^\varepsilon(t) + \bar{O}_r^\varepsilon(t) + C_\Theta(1 + \varepsilon^{-1})\sqrt{r}\, A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\, ds. \tag{3.3}$$

These are the endpoint-field, all-contact motion, uniform cubic-action and separated-volume results in retained modules 174, 177, 180–182. The guard term in (3.3) is the complete product guard, not a protective subword source. The derivative and continuation assertions are confined to regular physical trajectories.

Input statement 3.2 (Complete local source estimates)**.** For every fixed auxiliary band B_*, constants $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ give the following bounds uniformly in radius, exact labels, interval J and mark $0 \le j \le m$. Let $S_{R,s}$ be the union of strips of width Cs about the finitely many coordinate lines supporting ∂Y_R^*. For s small,

$$m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le C s^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|) m^2 \rho_{B_*}^{m/2}, \tag{3.4}$$

$$m^2 \int_J b^{\delta}(t)\, dt \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|) m^3 \rho_{B_*}^{m/2}. \tag{3.5}$$

The local extension to a fixed remaining compact range of δ is included. At auxiliary parameter one $W^1 = 1$, so $b^1 = p$.

Put $z = |g|$, $\alpha = 129/128$ and, for $0 < \eta < 1/2$, define the actual nonnegative source measures

$$\begin{aligned} v^{\eta,b}(t)\, dt &= L_{\#}(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_{\Pi} \nu_R^*), \\ v^{\eta}(t)\, dt &= L_{\#}(I_m z^{-\eta} \mathbf{1}_{\Pi} \nu_R^*). \end{aligned} \tag{3.6}$$

They are finite and absolutely continuous at every fixed count, and for $0 < b < b_1$,

$$m^2 \int_J v^{\eta,b}(t)\, dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|) m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \tag{3.7}$$

Adding the bounded-gradient part gives, for every fixed $L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta} \le C_\eta L, \qquad \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta,b} \le C_\eta b^{\alpha-\eta} L. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards; leading constants are independent of that band and of the previously fixed interval length.

Equation (3.4) is Supplement Theorem 175.3. The physical source bound (3.5) is Supplement Theorem 179.1, in particular its finite-window estimate. The fractional bounds are the positive layer-cake conclusions of retained module 187, stated in full above. The joint spectral and common-space assumptions used there are recorded in the appendix. No estimate for an untruncated first inverse gradient or for a singular roof convolution is part of this input.

Input statement 3.3 (The scalar positive remainder)**.** For large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.9}$$

The source b^{ε} is nonnegative and has a compactly supported BV density at fixed count. No collision-uniform variation bound is assumed. Supplement Theorem 181.4 gives this remainder with the same complete finite-peak reference.

4. Untruncated optical transport and fractional potentials

Fix one of the original exact events. The field, optical density and coarea kernels in Input 3.1 are used throughout. No new field is chosen from the unknown roof density. Put $z = |g|$. For $0 < \eta < 1/2$ set

$$\kappa_\eta = \frac{1-\eta}{2}, \qquad Q_\eta = \kappa_\eta^{-1}, \qquad \mathsf{P}_{\eta,r} v(t) = \int_0^r s^{-\kappa_\eta} v(t+s)\, ds. \tag{4.1}$$

In particular $2 < Q_\eta < 4$. The source v^η below is the *actual*, untruncated source with insertion $I_m z^{-\eta}$ from Input 3.2; it is not a limit taken after the collision limit.

Write $E_h^{\varepsilon,\mathrm{gr}}$ for the starting mass of genuine grazing first exits and O_h^{ε} for the positive inverse-twist loss on survivors. These are disjoint classes of initial states. If $b > 0$, a subscript $< b$ restricts the initial source to $0 < z < b$. A bar denotes the average $r^{-1}\int_0^r \cdot\, dh$.

Theorem 4.1 (Optical transport without an initial lower cutoff)**.** *For almost every (t, h),*

$$O_h^{\varepsilon}(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \le C_\eta \mathsf{P}_{\eta,h} v^\eta(t). \tag{4.2}$$

The right side is locally integrable in t at every fixed count. If $0 < h \le r$ and $b_r = (b^2 + 2\Lambda_\circ r)^{1/2} < b_1$, then

$$(O_h^{\varepsilon} + E_h^{\varepsilon,\mathrm{gr}})_{<b}(t) \le C_\eta \mathsf{P}_{\eta,r} v^{\eta,b_r}(t). \tag{4.3}$$

The same bounds hold for the h-averaged sources. All internal incidences, all actual words and both section tests are retained.

Proof. Along a regular survivor, the optical sensitivity and radial speed give

$$|X\omega| \le C\omega I_m / z, \qquad z(\Phi_s x)^2 \ge z(x)^2 + 2\lambda_\circ s.$$

Consequently

$$|X\omega|(\Phi_s x) \le C_\eta s^{-\kappa_\eta} \omega(\Phi_s x) I_m(\Phi_s x) z(\Phi_s x)^{-\eta}. \tag{4.4}$$

For a surviving state, $[\omega(x) - \omega(\Phi_h x)]_+$ is bounded by the negative variation up to h. At a grazing first exit the limiting optical density is zero; hence its entire positive starting optical weight is bounded by negative variation up to the lifetime. This uses the one-word optical identity only to identify the zero limit, not to sum wordwise constants.

In an angular flow box the initial fiber density is $\omega(x)/(4\pi c_* \chi(x))\, d\vartheta$. Multiplying by $W^\varepsilon(x) \le 1$ and using

$$\frac{\omega(\Phi_s x)}{\omega(x)} = J_s(x) \frac{\chi(\Phi_s x)}{\chi(x)} \le C J_s(x)$$

transfers the integrand in (4.4) to a subset of the complete target fiber. The common-kernel covariance in Appendix A justifies this simultaneously for the disjoint survivor and first-exit selections. Tonelli then proves (4.2). Nonnegative truncation at fixed collision count removes all compact-chart restrictions.

For $z(x) < b$ the radial upper bound gives $z(\Phi_s x) < b_r$ before $h \le r$, proving (4.3). The kernel and sources are nonnegative, so averaging in h preserves the bounds. Finally $\kappa_\eta < 1$ and v^η is locally integrable with compact support. Fubini proves local integrability of each displayed potential. Neither an untruncated first inverse-gradient moment nor a pointwise bound for v^η has been used. □

For $Q > 1$, a measurable interval J and $f \ge 0$, define

$$\mathcal{N}_Q(f; J) = \sup_{E \subset J,\ 0<|E|<\infty} |E|^{-1+1/Q} \int_E f(t)\, dt, \tag{4.5}$$

where the supremum is over Lebesgue-measurable sets. This is a specific weak-L^Q norm; the definition fixes its normalization.

Lemma 4.2 (A localized potential bound)**.** *For every nonnegative locally integrable v, every $r > 0$, and any interval J, write $J_r = J + (0, r)$. Then*

$$\mathcal{N}_{Q_\eta}(\mathsf{P}_{\eta,r} v; J) \le \frac{1}{1 - \kappa_\eta} \int_{J_r} v(y)\, dy. \tag{4.6}$$

The inequality also holds for extended nonnegative quantities.

Proof. For fixed y and a measurable set E of length a, the decreasing kernel has

$$\int_E (y-t)^{-\kappa_\eta} \mathbf{1}_{\{0<y-t<r\}}\, dt \le \int_0^{\min(a,r)} s^{-\kappa_\eta}\, ds \le \frac{a^{1-\kappa_\eta}}{1-\kappa_\eta}.$$

The first inequality follows directly by integrating the measures of the kernel's superlevel intervals. Integrate against $v(y)$ and use Tonelli. Only $y \in J_r$ contributes. Divide by $a^{1-\kappa_\eta}$ and take the supremum over E. □

Set $D_r^\varepsilon = \bar{O}_r^\varepsilon + \bar{E}_r^{\varepsilon,\mathrm{gr}}$. All norms in the next theorem are taken before the collision limit.

Theorem 4.3 (Complete optical source bounds at the collision scale)**.** *For each previously fixed $r, L > 0$ and $0 < \eta < 1/2$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q_\eta}(m^2 D_r^\varepsilon; J) \le C_\eta (L + r), \tag{4.7}$$

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q_\eta}(m^2 D_{r,<b}^\varepsilon; J) \le C_\eta b_r^{\alpha-\eta}(L + r) \quad (b_r < b_1). \tag{4.8}$$

Here $\alpha = 129/128$. For $1 \le p < Q_\eta$, the left sides may be replaced by the corresponding $L^p(J)$ norms, provided the right sides are multiplied by $(Q_\eta/(Q_\eta - p))^{1/p} L^{1/p-1/Q_\eta}$. No essential-supremum conclusion is included.

Proof. Combine Theorem 4.1 with Lemma 4.2, then apply the fixed-width bounds for the actual sources v^η and v^{η,b_r} in Input 3.2 on the interval of length $L + r$. Its length and b_r are fixed before the collision limit. The auxiliary band is enlarged only afterwards.

If $\mathcal{N}_Q(f; J) \le A$, applying its definition to $\{f > \lambda\} \cap J$ gives $|\{f > \lambda\} \cap J| \le \min\{L, (A/\lambda)^Q\}$. Integration of this bound against $p\lambda^{p-1} d\lambda$ proves $\|f\|_{L^p(J)} \le (Q/(Q-p))^{1/p} A L^{1/p-1/Q}$. Apply it at each finite count before taking the limits. □

For example, $\eta = 1/4$ gives $Q_\eta = 8/3$ and $\alpha - \eta = 97/128$. On intervals of length r, the complete optical norm is $O(r)$; the part with initial $z < \sqrt{r}$ is $O(r^{1+97/256})$. The estimates concern the untruncated original source. They do not permit $p = \infty$, $\eta = 1/2$, or a collision-dependent roof width. Their role in the original pointwise problem is made explicit in Section 6.

5. The complete first-exit surface source

We now estimate a quantity defined on the actual outgoing surface, not a bounded multiplier substituted for that surface. The estimate uses both a complete starting-source bound and a two-sided payment of optical variation. It is local in roof mass; the distinction from pointwise height will be kept explicit.

Let $\tau(x)$ be the forward lifetime in the original regular exact event and put

$$e_h(t) = \int \mathbf{1}_{\{\tau(x) \le h\}}\, d\mu_t(x). \tag{5.1}$$

This is the *unweighted* starting mass; it keeps both section tests and the exact label. There is no renormalization after removal of a boundary layer.

Theorem 5.1 (All true exits in a roof interval)**.** *There is $r_0 > 0$ such that, for every fixed auxiliary band B_*, all $0 < r < r_0$, $m \ge 2$, and intervals J,*

$$m^2 \int_J e_r(t)\, dt \le C r^{1/32}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|) m^3 \rho_{B_*}^{m/2}. \tag{5.2}$$

The constants are uniform in R, n, k. Consequently, for fixed r, L,

$$\limsup_{m} \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_r(t)\,dt \le CLr^{1/32}. \tag{5.3}$$

The same conclusion holds for the h-average of e_h.

Proof. The all-contact motion estimate places every finite exit of lifetime at most r in the union of the starting layers

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \qquad \Delta_j \le C_E q_0^{d'_j}\sqrt{r}, \qquad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}.$$

The union includes every contact and flight, and all initial, internal and terminal decisions. A positive-margin endpoint would continue the physical word and field; a regular critical point cannot be a forward exit. Coordinate seams are not stopping faces.

The first two layer unions are contained in the set on which the original guard with auxiliary parameter $C\sqrt{r}$ has value zero: $q_0^d \le q_0^{d/4}$. Their source is therefore bounded by the complete physical bad source in Input 3.2. For the section layers use the marked transverse-strip estimate there at the actual collision time j, with width $C_E q_0^{d_j}\sqrt{r}$. Each positive upper comparison drops endpoint tests only on its upper side and costs exactly c_*^{-1}; the occupation and displacement coefficient remain (n, k).

There are at most two contact marks of each depth. The leading section terms sum as $Cr^{1/32}\sum_{d\ge 0} q_0^{d/16}$, and the physical guard term has the same power of r. The at most $m+1$ section remainders, together with the physical remainder, are bounded by the second line of (5.2). Thus no uncontrolled word count or sum of collision-dependent bandwidths occurs. Take the collision limit with r, L, B_* fixed and then enlarge B_*. Finally $e_h \le e_r$ for $h \le r$. □

5.1. **The physical surface measure and its starting-roof assignment.** Use one-sided nongrazing outgoing faces of the original exact event. On a transverse angular chart a face is $F = b(\vartheta)$, and its physical flux is

$$d\zeta = \frac{\omega(b(\vartheta),\vartheta)}{4\pi c_*\chi(b(\vartheta),\vartheta)}\,d\vartheta. \tag{5.4}$$

A countable disjoint refinement counts each face with its incident physical component. Tangential strata have zero transverse flux. Let $\tau^-(y)$ be the backward lifetime in that same one-sided event. Define

$$\mathcal{F}_h(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<L(y)-t\le h,\ L(y)-t<\tau^-(y)\}}\,d\zeta(y). \tag{5.5}$$

There is no initial lower bound on $|g|$. The selection in (5.5) excludes later re-entries. Competing-hit, section and other true exact-event faces remain present. This is the increasing limit of the original $\mathcal{F}_h^a$ as $a \downarrow 0$ at fixed count, on the common versions of Appendix A. The measure is initially allowed to be extended nonnegative.

Theorem 5.2 (Two-sided optical payment for the actual flux)**.** *For every $0 < \eta < 1/2$, jointly for almost every (t, h),*

$$\mathcal{F}_h(t) \le Ce_h(t) + C_\eta \mathsf{P}_{\eta,h} v^\eta(t), \tag{5.6}$$

$$E_h^\varepsilon(t) + O_h^\varepsilon(t) \le C\mathcal{F}_h(t) + C_\eta \mathsf{P}_{\eta,h} v^\eta(t). \tag{5.7}$$

In particular the full outgoing assignment is locally integrable in the starting roof at fixed count. The first inequality is a bound for the surface source itself, not just for its pullback cost.

Proof. Along a nongrazing first-exit orbit write $w(s) = \omega(\Phi_s x)$ and let $\tau \le h$ be its lifetime. In the angular parametrization, the starting fiber density is $w(0)/(4\pi c_* \chi(x))$ and the exit flux density is $w(\tau-)/(4\pi c_* \chi(\Phi_\tau x))$. Both limiting factors are positive at a nongrazing exit. The two inequalities

$$w(\tau-) \le w(0) + \int_0^\tau [w'(s)]_+ \, ds, \qquad w(0) \le w(\tau-) + \int_0^\tau [-w'(s)]_+ \, ds$$

are used in their respective directions. The uniform upper and lower bounds for χ compare their fiber normalizations.

For (5.6), sum the first inequality on the actual first-exit graph. Its starting term is at most Ce_h. Pay its positive variation by (4.4) and common *target-direction* coarea covariance. This yields the potential term. A boundary point is counted with precisely the initial states selected in (5.5); no multiplicity from a later return is introduced.

For (5.7), use the second inequality, with $W^\varepsilon(x) \le 1$, for nongrazing exits. At grazing exits $w(\tau-) = 0$ and negative variation pays the full starting weight. On survivors it pays the positive optical loss. These classes are disjoint, so their common target insertion is dominated by the full v^η source, not a word-cardinality multiple of it. Truncate to compact regular face pieces and flow boxes, and pass to the full nonnegative sums at fixed count. Appendix A gives the joint versions and first-exit identification. Local integrability follows from $e_h \le p$ and Theorem 4.1.

The exact pullback identity still contains the inverse forward Jacobian. The two variation inequalities account for its cost; they do not replace that Jacobian by one. □

Theorem 5.3 (A complete local-mass estimate for outgoing first exits)**.** *For every previously fixed $0 < r < r_0$, $L > 0$ and $0 < \eta < 1/2$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J \overline{\mathcal{F}}_r(t) \, dt \le CLr^{1/32} + C_\eta (L + r) r^{(1+\eta)/2}. \tag{5.8}$$

In particular, on roof intervals of length r, the right side is $O_\eta(r^{33/32})$. Every initial momentum and every nongrazing one-sided first-exit component is included. This is an estimate of local surface-assignment mass, not an essential-height estimate.

Proof. Average (5.6) and use $e_h \le e_r$ and $\mathsf{P}_{\eta,h} v^\eta \le \mathsf{P}_{\eta,r} v^\eta$. Fubini gives

$$\int_J \mathsf{P}_{\eta,r} v^\eta(t) \, dt \le \frac{r^{1-\kappa_\eta}}{1-\kappa_\eta} \int_{J+(0,r)} v^\eta(y) \, dy.$$

Use (5.3) for the first term and the fixed-width bound for the actual v^η source for the second. All intervals have fixed lengths before the collision limit. Since $1 - \kappa_\eta = (1+\eta)/2 > 1/32$, setting $L = r < 1$ proves the last assertion. The surface estimate has been proved through its first-exit graph and positive optical variation; no theorem for a bounded volume multiplier is being relabeled as a trace theorem. □

6. The ordered microscopic tail and the original pointwise law

The preceding estimates give complete source and surface controls in specified local norms. The original pointwise target is stronger. We now state the remaining implication on that target, and separate the microscopic tail before any interchange of limits could occur.

For each exact label retain the actual source v^η. For $r > 0$, put $\ell_j = 2^{-j} r$ and define

$$T_{\eta,r,N}(t) = \sum_{j=N}^{\infty} \ell_j^{-\kappa_\eta} \int_{\ell_j/2}^{\ell_j} v^\eta(t+s) \, ds. \tag{6.1}$$

The integration variable s is an offset from the starting roof. Equivalently one integrates $v^\eta(u)\,du$ over $(t+\ell_j/2, t+\ell_j]$; there is no second translation of the roof. All terms and the entire tail are formed at finite collision count.

Lemma 6.1 (An ordered decomposition, including the full tail)**.** *Pointwise as extended nonnegative quantities,*

$$T_{\eta,r,0}(t) \le \mathsf{P}_{\eta,r} v^\eta(t) \le 2^{\kappa_\eta} T_{\eta,r,0}(t). \tag{6.2}$$

Define

$$\mathfrak{T}_{\eta,M}(r) = \lim_{N\to\infty} \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} T_{\eta,r,N}(t), \tag{6.3}$$

$$\mathfrak{P}_{\eta,M}(r) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \mathsf{P}_{\eta,r} v^\eta(t). \tag{6.4}$$

The first limit exists monotonically in $[0,\infty]$*, and*

$$\mathfrak{T}_{\eta,M}(r) \le \mathfrak{P}_{\eta,M}(r) \le C_\eta r^{(1+\eta)/2} + 2^{\kappa_\eta}\mathfrak{T}_{\eta,M}(r). \tag{6.5}$$

Proof. On $(\ell_j/2, \ell_j]$, the kernel lies between $\ell_j^{-\kappa_\eta}$ and $2^{\kappa_\eta}\ell_j^{-\kappa_\eta}$. Tonelli proves (6.2). For any fixed N, the first N interval lengths are fixed before the collision limit. Input 3.2 bounds the normalized height of their integrals by $C_\eta \ell_j$. Hence their complete finite sum is at most $C_\eta \sum_{j<N} \ell_j^{1-\kappa_\eta} \le C_\eta r^{1-\kappa_\eta}$. Keep the whole remaining nonnegative tail inside the supremum and collision limsup. Only then let $N\to\infty$. Monotonicity of these tails proves existence and (6.5). This argument does not use the false reverse of Fatou's inequality for an infinite sum. □

Let the full, untruncated surface height be

$$\mathfrak{F}_M^{\mathrm{all}}(r) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \overline{\mathcal{F}}_r(t). \tag{6.6}$$

Its source is precisely (5.5).

Theorem 6.2 (Pointwise error with no initial-cutoff interchange)**.** *For each fixed* $0<\eta<1/2$*, all sufficiently large fixed* B *and all previously fixed* $0<r<r_0$*, the original pointwise error satisfies*

$$\begin{aligned}\mathcal{E}_M \le{}& C_M B^{-1/192} + C\mathfrak{F}_M^{\mathrm{all}}(r) + C_\eta \mathfrak{T}_{\eta,M}(r)\\ &+ C_\eta r^{(1+\eta)/2} + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.\end{aligned} \tag{6.7}$$

Thus the original assertion (2.5) *follows if the full surface height and the microscopic tail in this display tend to zero as* $r\downarrow 0$*. This sufficient condition is not asserted as proved by the local-norm estimates above.*

Proof. Use the complete-source shift inequality of Input 3.1 and replace its sum of true-exit and optical losses, at the source level, by the average of (5.7). The positive scalar remainder and the fixed-width bad-source bound then give

$$\mathcal{E}_M \le C_M B^{-1/192} + C\mathfrak{F}_M^{\mathrm{all}}(r) + C_\eta \mathfrak{P}_{\eta,M}(r) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.$$

Apply (6.5). Every source restriction has already been removed at fixed count, using actual fractional integrability; no first inverse-gradient cutoff is passed through a collision limit. To obtain the conditional conclusion, choose $r(B) = c\varepsilon(B)^\gamma$ with fixed $c>0$ and $\gamma>2$. For each B this is a fixed positive width. First take the collision limit, next enlarge auxiliary spectral bands, and take $B\to\infty$ last. Every displayed elementary error then vanishes, and the two stated conditions pay the remaining terms. The arithmetic reference and all zero classes are unchanged. □

Proposition 6.3 (A quantitative microscopic envelope)**.** *Suppose, as an additional estimate on the actual source, that for all sufficiently large m, all parameters and almost every central t,*

$$m^2 \int_0^s v^\eta(t+u)\,du \le H(s) \quad (0 < s < r_0), \qquad \int_0^{r_0} H(s) s^{-1-\kappa_\eta}\,ds < \infty, \tag{6.8}$$

with H nonnegative and nondecreasing. The exceptional set may be chosen independently of s by first using rational s and monotone limits. Then $\mathfrak{T}_{\eta,M}(r) = 0$ for every $r < r_0/2$. In particular a uniform bound $H(s) \le As^{\kappa_\eta+\delta}$, $\delta > 0$, gives the finite-count tail bound

$$m^2 T_{\eta,r,N}(t) \le \frac{Ar^\delta 2^{-N\delta}}{1-2^{-\delta}}. \tag{6.9}$$

The envelope is a uniform finite-count hypothesis, not a list of separate fixed-scale collision limsups.

Proof. Each normalized interval integral is at most $H(\ell_j)$. Moreover, monotonicity gives

$$H(\ell_j)\ell_j^{-\kappa_\eta} \le C_\eta \int_{\ell_j}^{2\ell_j} H(s) s^{-1-\kappa_\eta}\,ds.$$

The right intervals have disjoint interiors. Their tails tend to zero by the assumed integrability, uniformly before the collision limit. For the power envelope sum the geometric series directly. □

The fractional source lowers the sufficient envelope exponent from $1/2$ to $\kappa_\eta = (1-\eta)/2$, which can be any fixed number strictly between $1/4$ and $1/2$. The endpoint $1/4$ is not included. The complete surface mass bound (5.8) and the weak-L^{Q_η} optical bound do not establish (6.8) or (6.6). A uniformly small integral, even with these additional local controls, must not be substituted for an essential roof-height estimate. The original pointwise law, and scalar-first normalization of its path and likelihood consequences, remain exactly the organizing objective.

7. Measurable maximal segments and cutoff-uniform source bounds

All surfaces in this article are counted with their incident physical component. The following parametrization makes this convention, and the absence of a word-count factor, explicit. Fix m, R, n, k throughout its construction. Let Ω be the regular exact event, with the critical points of F removed, and let $\tau^\pm$ be the two lifetimes of X. Write $\bar\nu = \nu_R^*|_\Omega$.

Theorem 7.1 (A Borel parametrization of maximal regular segments)**.** *There are countably many Borel sets E_j in lifted angular intervals, Borel functions $a_j < b_j$, and a Borel bijection*

$$\Psi : \coprod_j \{(\theta,u) : \theta \in E_j,\ a_j(\theta) < u < b_j(\theta)\} \longrightarrow \Omega \tag{7.1}$$

with Borel inverse, such that each $\theta \in E_j$ labels one entire maximal segment, $F(\Psi_j(\theta,u)) = u$, and $\vartheta(\Psi_j(\theta,u)) = \theta$ modulo 2π. Its physical angular density is

$$A_j(u,\theta) = \frac{\omega(\Psi_j(\theta,u))}{4\pi c_* \chi(\Psi_j(\theta,u))}.$$

For every nonnegative Borel function f,

$$\int_\Omega f\,d\bar\nu = \sum_j \int_{E_j} \int_{a_j(\theta)}^{b_j(\theta)} f(\Psi_j(\theta,u)) A_j(u,\theta)\,du\,d\theta. \tag{7.2}$$

The same identity holds with any Borel selector of the segment and its interior point. For almost every t, disintegration gives the corresponding identity on μ_t, with $u = t$. Both identities are identities of measures, not estimates with an unspecified multiplicity.

Restrict to segments with a nongrazing forward endpoint. The endpoint map, retaining the incident one-sided component, is Borel and injective outside the transverse null sets of the first-exit construction. Its outgoing measure in these coordinates is

$$d\zeta_+ = A_j(b_j-,\theta)\,d\theta. \tag{7.3}$$

Each such face and an intermediate roof determine at most one physical interior state. In particular, summing a variation integral over these segments gives exactly the selected physical volume integral in (7.2).

Proof. Enumerate the rational numbers as $q_1, q_2, \ldots$. The lifetime functions of a smooth flow on an open set are lower semicontinuous; this also follows by covering each compact surviving segment with finitely many ordinary flow boxes. They are therefore Borel. Since $XF = 1$ and F is bounded at fixed count, the roof interval of every maximal segment is a bounded, nonempty open interval.

On the smooth transverse slice $\Sigma_i = \Omega \cap \{F = q_i\}$, keep the points whose maximal interval contains none of $q_1, \ldots, q_{i-1}$. These are Borel conditions on the lifetimes. Every maximal segment has exactly one representative in this union: use the first rational in its roof interval and strict monotonicity of F. On each slice, ϑ is a local coordinate, since $\det D_q(F, \vartheta) = \chi > 0$. Cover the slices by countably many angular charts, and disjointize their representatives by the first containing chart. Denote the resulting Borel angular subsets by E_j, and the associated representative by $\xi_j(\theta)$, at roof $q_{i(j)}$. Put

$$a_j = q_{i(j)} - \tau^-(\xi_j), \qquad b_j = q_{i(j)} + \tau^+(\xi_j), \qquad \Psi_j(\theta, u) = \Phi_{u-q_{i(j)}}\xi_j(\theta).$$

The flow is continuous on its open lifetime domain. This proves Borel measurability. The inverse is explicit: for x, select the first rational in its maximal roof interval, flow to that roof, and select the first angular chart containing the representative. All these selections are Borel. If two parameter points have the same image, uniqueness of the flow gives the same maximal segment and representative; then $F = u$ and the angular chart give identical parameters. This proves (7.1) without a quotient measurability assumption. The Borel seams assign representatives only; they are not new exits.

In a local angular flow box, change of variables from endpoint angles has density $\omega/(4\pi c_*\chi)$. The same formula at another roof follows by physical coarea covariance; the transverse angle is unchanged. Apply this local formula on countably many Borel-disjoint restrictions of the parametrization and use Tonelli. It gives (7.2) for every Borel selector. Disintegration over u and a countable determining algebra give equality of fiber measures outside one null set in t, then equality for all nonnegative Borel functions. Additional external variables use the product graph and Fubini, as in Proposition A.1.

The endpoint limits exist by the all-contact motion estimate. They are Borel as limits along the interior roofs $b_j - \min\{1/N, (b_j - a_j)/2\}$. A nongrazing forward endpoint has positive limiting gradient and a one-sided smooth stationary-word extension. The flow-box contraction of the physical area form gives (7.3). Two coincident geometric boundary points on different incident components are different faces by definition. On one incident component, uniqueness of the backward regular flow makes its interior segment unique. Tangent and lower-dimensional strata have the transverse-null convention of Proposition A.2; removing their flow saturations removes no area in (7.2). This proves the last assertions. No continuation past a true physical boundary is used, and a backward critical endpoint remains an endpoint of its segment, not a chart seam. □

7.1. **The lower cutoff and its finite-count remainder.** We record the quantitative step behind the cutoff-independent constant used in boundary transfer. It concerns an actual positive

volume source; it is not a surface insertion theorem. Write $z = |g|$, $\alpha = 129/128$, and

$$u^b(t)\,dt = F_{\#}(I_m \mathbf{1}_{\{z<b\}}\bar{\nu}), \quad v^{c,b}(t)\,dt = F_{\#}(I_m z^{-1}\mathbf{1}_{\{c\le z<b\}}\bar{\nu}), \quad v^c(t)\,dt = F_{\#}(I_m z^{-1}\mathbf{1}_{\{z\ge c\}}\bar{\nu}).$$

The source-pinned critical-volume estimate, including its remainder, is

$$m^2 \int_J u^b \le Cb^{\alpha}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 b^{1/2}\rho_{B_*}^{m/4}. \tag{7.4}$$

It is Supplement equation (26.2). Its dependence on the joint spectral theorem is listed in the dependency map; it is restated here, not independently recertified.

Proposition 7.2 (Uniform leading constant at every positive cutoff)**.** *For $0 < c < b < b_1$, the complete exact source satisfies*

$$m^2 \int_J v^{c,b} \le C\left(1 + \frac{1}{\alpha - 1}\right) b^{\alpha-1}(|J| + B_*^{-1}) + 2C_{B_*}(1 + B_*|J|)m^2 c^{-1/2}\rho_{B_*}^{m/4}. \tag{7.5}$$

Consequently, for every fixed $c, L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^c \le C_0 L, \qquad C_0 \text{ independent of } c \text{ and } L. \tag{7.6}$$

No assertion at $c = 0$, or at a collision-dependent c, is made.

Proof. Use the positive majorant

$$z^{-1}\mathbf{1}_{\{c\le z<b\}} \le b^{-1}\mathbf{1}_{\{z<b\}} + \int_c^b q^{-2}\mathbf{1}_{\{z<q\}}\,dq.$$

Insert (7.4) before any limit. The leading coefficient is at most $b^{\alpha-1} + \int_c^b q^{\alpha-2}\,dq \le (1 + (\alpha - 1)^{-1})b^{\alpha-1}$. The error coefficient is $b^{-1/2} + \int_c^b q^{-3/2}\,dq = 2c^{-1/2} - b^{-1/2} \le 2c^{-1/2}$. This proves (7.5). Split v^c at the fixed threshold $b_1/2$. Above it, use the complete depth-weighted source estimate of Supplement Corollary 36.2. Below it, apply (7.5), first fixing c, L, B_*, then taking the collision limsup, and then enlarging B_*. If $c \ge b_1/2$, positivity gives the same conclusion. The surviving leading constant contains only fixed physical thresholds, not c. The displayed finite-count error explains why removal of c before the collision limit is a separate question. □

Corollary 7.3 (The audited paired-transfer estimate)**.** *On any Borel subfamily of short segments with $b - a \le 2r$ and $z(b-)^2 \ge 8\Lambda_\circ r$, let S_r^+ and S_r^- use the same exit-window kernel and the outgoing and incoming angular traces, respectively. Then*

$$|S_r^+(t) - S_r^-(t)| \le C\int_{t-2r}^{t+2r} v^{c_r}(u)\,du, \qquad c_r = 2\sqrt{\Lambda_\circ r},$$

and their ordered global essential-height difference is at most $4CC_0 r$, with C_0 independent of $c_r > 0$. A grazing incoming trace is zero, so its entire outgoing contribution obeys this bound.

Proof. Radial growth keeps the full segment above c_r. Differentiate the physical angular density, retaining both factors: $|\partial_u A| \le CAI_m/z$, as proved in Supplement Lemma 8.1. At a grazing incoming end $\omega \to 0$ while $\chi \ge \lambda_\circ$; at any nongrazing end the one-sided regular extension gives its trace. Integrate the absolute variation between these traces. When its kernel is nonzero the full selected segment lies in $(t - 2r, t + 2r)$. Theorem 7.1 identifies the complete sum with a selected submeasure of $v^{c_r}(u)\,du$, with multiplicity one. Positive approximation proves the assertion for all selected sheets. Apply (7.6) with $L = 4r$. This supplies the named map, exact measure, and cutoff uniformity in the same argument. □

8. Essential heights on every positive residence scale

The residence argument need not stop at the long segments. On a fixed residence band the entire segment lies near any roof that it contains. A wider exit kernel therefore persists across the whole segment. This controls all endpoint types on that band, including the critical and two-nongrazing-end types left in the preceding reduction.

For a nongrazing outgoing face write $b = F(y)$, $l = \tau^-(y)$ and $a = b - l$. In the parametrization of Theorem 7.1, define

$$k_r(t;y) = \left(1 - \frac{b-t}{r}\right)_+ \mathbf{1}_{\{a<t<b\}},$$
$$\mathcal{B}_s(t) = \int_{\Gamma^+_{\mathrm{ng}}} \mathbf{1}_{\{s<l\le 2s\}} \mathbf{1}_{\{a<t<b\}}\, d\zeta(y). \tag{8.1}$$

Thus $\overline{\mathcal{F}}_r = \int k_r\, d\zeta$. The band source uses the full occupation indicator of the same maximal segment, not a renormalized boundary measure. A superscript $[b_0]$ will restrict an outgoing face to $z(b-) < b_0$.

Theorem 8.1 (Complete residence-band heights)**.** *For each fixed sufficiently small $s > 0$, almost everywhere in t,*

$$\mathcal{B}_s(t) \le \frac{2}{s} \int_{t-2s}^{t+2s} \overline{\mathcal{F}}_{4s}(v)\, dv. \tag{8.2}$$

Consequently, with $\sigma = 1/32$,

$$\limsup_m \sup_{R,n,k} m^2 \|\mathcal{B}_s\|_\infty \le C_\eta s^\sigma. \tag{8.3}$$

All nongrazing outgoing faces in the band are included, irrespective of their incoming endpoint or outgoing gradient. More precisely, for every fixed small $b_0 > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|\mathcal{B}_s^{[b_0]}\|_\infty \le C b_0^\alpha + C_\eta s^{(1+\eta)/2} b_0^{\alpha-\eta}. \tag{8.4}$$

It also satisfies (8.3)*. In particular the complete band with $z(b-)^2 < 8\Lambda_\circ s$ has essential height $O_\eta(s^{129/256})$, not merely local mass of that order.*

Proof. If $a < t < b$ and $s < l \le 2s$, then $(a,b) \subset (t-2s, t+2s)$. For every $v \in (a,b)$, $k_{4s}(v;y) = 1 - (b-v)/(4s) \ge 1/2$. Thus the integral of this kernel over that window is at least $l/2 > s/2$. Multiply by $2/s$ and integrate over the same faces. Tonelli proves (8.2); removing the band restriction only on the upper side preserves the actual source. Theorem 5.3, at width $4s$ and interval length $4s$, now gives $Cs^{1/32} + C_\eta s^{(1+\eta)/2}$ after division by s. This proves (8.3). No pointwise density bound has been inferred from mass without the preceding persistence.

Retain the outgoing restriction $z(b-) < b_0$ on both sides instead. Radial monotonicity keeps the same restriction at every interior point of these segments. The positive-variation surface comparison of Theorem 5.2, restricted to these faces, bounds the surface by the small-gradient starting source plus $C_\eta \mathsf{P}_{\eta,4s} v^{\eta,b_0}$. Its mass on a window of length $4s$ is at most $Csb_0^\alpha + C_\eta s^{1+(1+\eta)/2} b_0^{\alpha-\eta}$ in the prescribed ordered norm. The starting-source bound follows from (7.4); the potential bound follows by Fubini from (3.8). Division by s proves (8.4). When $b_0 = \sqrt{8\Lambda_\circ s}$, the exponents are $\alpha/2 = 129/256$ and $(1+\alpha)/2 = 257/256$. All widths and this outgoing restriction were fixed before the collision limit. Incoming grazing and backward critical ends require no deletion in this argument. □

Let $\overline{\mathcal{F}}_r^{\ge\delta}$ retain only $l \ge \delta$. Let $\mathcal{K}_r^{\ge\delta}$ further require $l \le 2r$ and $z(b-)^2 < 8\Lambda_\circ r$. These are restrictions of the original surface, not different surface definitions.

Theorem 8.2 (Full surface height with an arbitrary positive residence cutoff)**.** *For every fixed $r > 0$ sufficiently small and every fixed $\delta > 0$,*

$$\limsup_{m} \sup_{R,n,k} m^2 \|\overline{\mathcal{F}}_r^{\geq\delta}\|_\infty \leq C_\eta r^{1/32}, \tag{8.5}$$

$$\limsup_{m} \sup_{R,n,k} m^2 \|\mathcal{K}_r^{\geq\delta}\|_\infty \leq C_\eta r^{129/256}(1 + |\log r|). \tag{8.6}$$

The leading constants are independent of δ. The paired incoming assignment on short noncritical segments, restricted to $l \geq \delta$, also has ordered height at most $C_\eta r^{1/32}$. The statements fix δ before the collision limit; they do not assert these bounds for $\delta = 0$.

Proof. For $l > 2r$ use the long-residence persistence, reproduced here with notation distinct from the arithmetic reference:

$$\mathcal{R}_r(t) := \int_{\{l>2r\}} k_r(t;y)\, d\zeta(y) \leq \frac{2}{r} \int_{t-r}^{t} \overline{\mathcal{F}}_{2r}(v)\, dv.$$

Indeed, if $b - t \in (0, r)$, then $k_{2r}(t-u;y) \geq k_r(t;y)/2$ for $0 < u < r$. Its ordered height is $C_\eta r^{1/32}$.

For $\delta \leq l \leq 2r$ choose a finite N with $2^{-N} r < \delta$. The bands $(2^{-j}r, 2^{1-j}r]$, $0 \leq j \leq N$, cover these segments, and $k_r \leq \mathbf{1}_{(a,b)}$. Take the collision limsup of this finite sum and apply (8.3). Its leading constants sum to at most $C_\eta r^\sigma/(1 - 2^{-\sigma})$, independently of N. This proves (8.5). No infinite physical band sum has been moved through a limsup.

For the critical part put $b_r = \sqrt{8\Lambda_\circ r}$ and $s_j = 2^{-j}r$. Apply the smaller of (8.3) and (8.4) to each of the finitely many bands. Using $\min(x, y+z) \leq \min(x,y) + z$, their sum is at most

$$C_\eta \sum_{j\geq 0} \min\{s_j^\sigma, b_r^\alpha\} + C_\eta b_r^{\alpha-\eta} \sum_{j\geq 0} s_j^{(1+\eta)/2}.$$

These are sums of numerical upper bounds, not interchanged physical limits. Splitting the first series where s_j^σ crosses b_r^α bounds it by $C b_r^\alpha (1 + |\log r|)$ for small r. The second is $O_\eta(r^{(1+\alpha)/2})$. This proves (8.6). Finally apply Corollary 7.3 to the selected short noncritical family. Its outgoing source is bounded by (8.5); its paired incoming source differs by at most Cr. This proves the last height bound. □

The theorem controls the complete surface at every previously fixed positive residence cutoff, with a uniform leading constant. In particular it covers both kinds of short surfaces in the preceding report. The remaining issue is the complete mass carried by residence lengths tending to zero before the collision limit, not another unestimated positive residence band. A separate ordered tail must still be retained, just as for the fractional volume source.

9. Zero-residence concentration and the original pointwise source

The complete band estimate identifies exactly what the positive residence cutoff leaves behind. Its residual is a single monotone charge on the actual surface, and it has a direct comparison with an original collision-source restriction. Neither construction changes the exact event or assumes a surface spectral theorem.

For nonnegative source families put

$$\mathcal{H}_M(f) = \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} f_{m,R,n,k}(t).$$

At finite count define the short-segment occupancy trace

$$\mathcal{Z}_\delta(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<l\leq\delta\}} \mathbf{1}_{\{a<t<b\}}\, d\zeta(y), \qquad \mathfrak{B}_M = \lim_{\delta\downarrow 0} \mathcal{H}_M(\mathcal{Z}_\delta). \tag{9.1}$$

The limit exists monotonically in $[0,\infty]$. The complete residence tail is inside the height functional. At fixed count, $\mathcal{Z}_\delta \le 2\overline{\mathcal{F}}_{2\delta}$, so it is locally finite almost everywhere. Such fixed-count finiteness alone does not determine $\mathfrak{B}_M$.

Theorem 9.1 (The exact limiting surface charge)**.** *For every sufficiently small fixed $r>0$,*

$$\mathfrak{B}_M \le \mathfrak{F}_M^{\mathrm{all}}(r) \le \mathfrak{B}_M + C_\eta r^{1/32}. \tag{9.2}$$

In particular,

$$\lim_{r\downarrow 0} \mathfrak{F}_M^{\mathrm{all}}(r) = \mathfrak{B}_M. \tag{9.3}$$

Thus full surface-height decay is equivalent to zero charge of vanishing-residence segments. Every positive residence band has already been estimated in Theorem 8.1.

Proof. For $\delta < r$, a selected short segment satisfies $b-t<l\le\delta$ whenever it contains t. Therefore

$$(1-\delta/r)\mathcal{Z}_\delta(t) \le \overline{\mathcal{F}}_r(t), \qquad \overline{\mathcal{F}}_r(t) \le \mathcal{Z}_\delta(t) + \overline{\mathcal{F}}_r^{\ge\delta}(t).$$

A possible equality face $l=\delta$ on the upper side only enlarges the comparison. Apply $\mathcal{H}_M$ and Theorem 8.2, first fixing r,δ. Only afterwards let $\delta\downarrow 0$. This gives (9.2), including infinite charges by monotone inequalities. Its squeeze gives (9.3). No statement about small mass on a shrinking interval is substituted for the essential supremum. □

9.1. **A comparison on the original collision source.** Let Ω_+^{ng} consist of the regular states whose maximal forward segment has a nongrazing outgoing face. On a maximal segment put $l(x)=\tau^-(x)+\tau^+(x)$; this is constant along the segment. The Borel parametrization above makes the source

$$q_\delta(t)\,dt = F_\#(\mathbf{1}_{\Omega_+^{\mathrm{ng}}}\mathbf{1}_{\{l\le\delta\}}\bar\nu), \qquad \mathfrak{Q}_M = \lim_{\delta\downarrow 0}\mathcal{H}_M(q_\delta) \tag{9.4}$$

unambiguous. This is the original volume measure restricted to actual short maximal segments, with its sole original normalization. It is not a boundary collar. One has $0\le q_\delta\le e_\delta\le p$, where e_δ is the true-exit starting density of Theorem 5.1. In particular its fixed-width mass is at most $CL\delta^{1/32}$ in ordered normalization; this is not a proof that $\mathfrak{Q}_M=0$.

For the actual fractional source retain $0<\eta<1/2$ and define

$$\mathfrak{A}_{\eta,M} = \lim_{h\downarrow 0}\mathcal{H}_M(\mathsf{P}_{\eta,h}v^\eta). \tag{9.5}$$

This is the same canonical microscopic charge as in the preceding revision, with the offset convention (6.1).

Theorem 9.2 (Surface trace versus actual short-segment source)**.** *For every fixed $\delta>0$, almost everywhere,*

$$|\mathcal{Z}_\delta(t) - q_\delta(t)| \le C_\eta \mathsf{P}_{\eta,\delta}v^\eta(t). \tag{9.6}$$

Consequently, as extended nonnegative bounds,

$$\mathfrak{B}_M \le \mathfrak{Q}_M + C_\eta\mathfrak{A}_{\eta,M}, \qquad \mathfrak{Q}_M \le \mathfrak{B}_M + C_\eta\mathfrak{A}_{\eta,M}. \tag{9.7}$$

If the fractional concentration charge is zero, the actual surface charge equals the actual short-segment collision-source charge.

Proof. On one selected segment, the two fiber expressions use, respectively, $A_j(b_j-,\theta)$ and $A_j(t,\theta)$ with the same indicator $a_j<t<b_j$, $l_j\le\delta$. Absolute variation gives

$$|A_j(b_j-,\theta)-A_j(t,\theta)| \le C\int_t^{b_j} A_j(u,\theta)I_m(u,\theta)/z(u,\theta)\,du.$$

At every intermediate target $u=t+s$, radial growth yields $z(u)\ge\sqrt{2\lambda_\circ s}$, and hence $z(u)^{-1}\le C_\eta s^{-(1-\eta)/2}z(u)^{-\eta}$. Theorem 7.1 identifies the selected fiber sum at u with a submeasure of the

same actual source $v^{\eta}(u)$. The length to its first exit is at most δ. Tonelli, first on compact regular subsegments and then by positive convergence, proves (9.6). No inverse Jacobian is deleted: both physical angular densities and their full variation are retained. Local finiteness of both terms was proved above, so the difference does not subtract two undefined infinite assignments. Apply $\mathcal{H}_M$ to the two resulting positive inequalities and only then let $\delta \downarrow 0$. This proves (9.7). □

9.2. **The ordered pointwise theorem.** For clarity, the charge identities used below have their full width domain. With $\kappa = (1-\eta)/2$ and $0 < r, 2^{-j}r < r_0$,

$$\begin{aligned} \mathfrak{T}_{\eta,M}(2^{-j}r) &= \mathfrak{T}_{\eta,M}(r) && (j \in \mathbb{Z}), \\ \mathfrak{T}_{\eta,M}(r) &\le \mathfrak{A}_{\eta,M} \le 2^{\kappa}\mathfrak{T}_{\eta,M}(r), \\ \mathcal{H}_M(\mathsf{P}_{\eta,r}v^{\eta}) &\le C_{\eta}r^{1-\kappa} + 2^{\kappa}\mathfrak{A}_{\eta,M}. \end{aligned} \tag{9.8}$$

Indeed the dyadic annuli partition $(0, 2^{-N}r]$ at finite count. Apply the height functional to their comparison with the truncated potential before $N \to \infty$. The identity $T_{\eta,r/2,N} = T_{\eta,r,N+1}$ proves the first assertion on the displayed domain. A finite head is bounded using (3.8); only its complete remaining tail is then passed to $N \to \infty$. These are also direct proofs of (9.8).

Theorem 9.3 (The original endpoint after all positive residence scales)**.** *For all sufficiently large fixed B, fixed $0 < \eta < 1/2$ and small previously fixed $r > 0$,*

$$\begin{aligned} \mathcal{E}_M \le C_M B^{-1/192} + C\mathfrak{B}_M + C_{\eta}\mathfrak{A}_{\eta,M} \\ + C_{\eta}r^{1/32} + C_{\Theta}(1+\varepsilon(B)^{-1})\sqrt{r}. \end{aligned} \tag{9.9}$$

One may replace $C\mathfrak{B}_M$ by $C\mathfrak{Q}_M$ after enlarging the coefficient of $\mathfrak{A}_{\eta,M}$. Zero fractional charge and zero short-segment source charge therefore suffice for the unchanged pointwise assertion (2.5)*. Their vanishing is not asserted by the cutoff-uniform bounds proved in this article.*

Proof. Use the source-level positive-remainder comparison in the proof of Theorem 6.2 before separating its tail. Insert (9.2) and (9.8). Since $1 - \kappa > 1/2 > 1/32$, the explicit potential term is absorbed by $C_{\eta}r^{1/32}$. Equation (9.7) gives the source form. Under the two stated zero-charge conditions choose $r(B) = c\varepsilon(B)^{\gamma}$, with $\gamma > 2$. For each B all widths, every positive cutoff used in the finite comparisons, and all auxiliary bands are fixed before the collision limit. Enlarge the auxiliary bands afterwards and take $B \to \infty$ last. The explicit errors vanish. No zero arithmetic class is normalized. □

The estimates above close every fixed positive residence scale of both short-surface mechanisms in the referee report, with constants uniform in a subsequently chosen positive residence cutoff. They identify the remaining full surface height with one zero-residence charge and pay its comparison with the original volume restriction. The two remaining nonnegative requirements in (9.9) are genuine microscopic concentration assertions. They are not proved by fixed-count integrability, local weak norms, or exchanging a cutoff with the collision limsup. The principal target has not been replaced by any of those weaker statements.

10. The original scalar objective and downstream normalization

The governing assertion remains (2.5). The full surface is now estimated in essential height at every previously fixed positive residence cutoff, with one leading constant for all such cutoffs. The critical short surface has the stronger height estimate (2.7) under that same ordering. These are complete positive-scale theorems, not a sum over selected words.

The complete zero-residence tail, and the complete fractional optical tail, remain in the original normalized essential supremum before the collision limit. Neither vanishing assertion follows by letting a cutoff depend on collision count. Their exact role is (9.9); the surface-source comparison

(9.6) does not suppress a boundary flux or change the guard. The manuscript does not mark the organizing theorem proved until these microscopic estimates are established.

The active dyadic formula is the offset expression (6.1); its negative-shift domain is explicit in (9.8). The historical module 189 remains qualified at its inclusion in the supplement, with original bytes preserved. The supplement also contains a compact input-to-theorem, equation and proof-page map (Supplement page 3).

After scalar closure, the positive path-remainder equality in Supplement Theorem 76.1 supplies the same-roof numerator. Ratios are formed only on positive arithmetic reference classes. The actual-return clock and forward likelihood retain their separately stated hypotheses. No surface estimate is used to reverse this scalar-first logical order.

Appendix A. Physical kernels, first-exit traces and analytic dependencies

This appendix supplies the measure-theoretic interfaces used by the new arguments in one convention. It does not deduce a surface estimate from a volume multiplier theorem. The geometric and spectral estimates whose proofs are retained in the technical supplement are listed separately.

A.1. **Normalization and common kernels.** On a regular word use endpoint angles $q = (\alpha_0, \alpha_m)$ and oriented internal arclengths. Endpoint first variation gives $F_{\alpha_0} = -Rp_0$ and $F_{\alpha_m} = Rp_m$. Consequently

$$\omega = |\partial_{\alpha_m} p_0| = |F_{\alpha_0\alpha_m}|/R, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\, dq.$$

In the optical Schur complement the internal Hessian is $D_c A D_c$, where A has diagonal $t_{i-1} + t_i + 2/(Rc_i)$ and off-diagonal $-t_i$. The two angle-to-arclength factors are R^2; division by R in the first variation gives $\omega = Rc_0c_m/y_m$. For one flight $y_1 = \ell_0$, and for two flights $y_2 = \ell_0+\ell_1+2\ell_0\ell_1/(Rc_1)$. Changes of tangent orientation do not change these absolute identities.

Away from $g = 0$, the coordinates (F, ϑ) have Jacobian $\chi = e^{\perp\mathsf{T}} K e^{\perp}$ in absolute value, where $e = g/|g|$. Thus the angular expression for the original fiber is

$$d\mu_t = \frac{\omega(t,\vartheta)}{4\pi c_* \chi(t,\vartheta)}\, d\vartheta. \tag{A.1}$$

The sign of an oriented endpoint chart is irrelevant; all source Jacobians in this appendix are positive. The original source already contains its initial and terminal tests and is normalized only by c_*.

Proposition A.1 (Common graph covariance)**.** *At fixed m, R, n, k, the regular exact event has jointly measurable coarea kernels for which, for almost every (t, h) and every nonnegative Borel graph weight f,*

$$\int_{S_h(t)} f(x, h, \Phi_h x) J_h(x)\, d\mu_t(x) = \int_{\Phi_h S_h(t)} f(\Phi_{-h} y, h, y)\, d\mu_{t+h}(y). \tag{A.2}$$

The same versions can be used for countably many source restrictions, positive truncations and their increasing limits. Here $S_h(t)$ is the set of source points whose forward lifetime is greater than h.

Proof. At fixed count the regular physical words are disjoint open pieces of the collision cylinder. Their true boundaries and the removed critical points have zero source area. Take a countable cover by precompact flow boxes with lifted angular coordinate. Assign each point to its first containing box to obtain a Borel disjointization. This assignment is used only for integration: its seams do not stop the physical flow.

On the open survivor graph the map $(x,h)\mapsto(\Phi_h x,h)$ is injective, smooth and has physical Jacobian J_h. Its inverse on its image is the backward flow. Since $F(\Phi_h x)=F(x)+h$, ordinary change of variables followed by disintegration over (F,h) gives (A.2) on a countable determining algebra of Borel sets. Remove the union of the resulting null sets. Uniqueness of the kernel measures and monotone approximation extend equality to all nonnegative Borel weights outside this one joint null set. Countable disjoint summation gives the complete physical formula.

In endpoint coordinates the field is translation in F at fixed ϑ, so its endpoint Jacobian is $\chi(x)/\chi(\Phi_h x)$. Multiplication by the physical density ratio gives $J_h=(\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$, exactly the normalization in (A.2). Nonnegative graph selectors accommodate first-exit restrictions depending on (x,h) without new incompatible versions. Monotone convergence at fixed count permits the actual fractional insertions. No null set independent of the uncountable radius parameter is required: the ordered norm is $\sup_R \operatorname{ess\,sup}_t$. □

A.2. **True outgoing faces and exhaustion.** On a nongrazing limiting word all incidence limits are positive, so the stationary reflection equations extend smoothly to the boundary. This continuation is not asserted to remain the physical first-hit trajectory beyond that boundary. On a transverse outgoing arc write $F=b(\vartheta)$. Contraction of the area form $\omega(4\pi c_*\chi)^{-1}dF\,d\vartheta$ with ∂_F gives the positive flux in (5.4). A first-exit point is always counted together with its incident one-sided physical component. Lower-dimensional and tangent strata have zero transverse flux. Section decisions, competing obstacles and simultaneous failures are not suppressed by this description. An exit with a vanishing internal incidence belongs to the grazing class even when another condition fails there simultaneously; the remaining class is nongrazing.

Proposition A.2 (Initial mass and actual first-exit flux)**.** *For a nongrazing outgoing face, let $s=F(y)-t$ and $x=\Phi_{-s}y$. On $0<s\le h$, $s<\tau^-(y)$, the contribution to initial weighted exit mass is*

$$\frac{W^\varepsilon(x)}{J_s(x)}\,d\zeta(y). \tag{A.3}$$

The boundary source (5.5) *has this same first-exit selection with weight one on the boundary, not with the factor in* (A.3)*. Both are defined on common almost-everywhere starting-roof versions.*

Proof. In one angular flow box put $a(u,\vartheta)=\omega(u,\vartheta)/(4\pi c_*\chi(u,\vartheta))$. The initial fiber mass is $W(x)a(t,\vartheta)d\vartheta$, whereas the outgoing flux is $a(b(\vartheta),\vartheta)d\vartheta$. The ratio of the latter density to the former unweighted density is $J_s(x)$. This proves (A.3). The backward-lifetime condition says precisely that the segment between the starting roof and this boundary has remained in that one-sided event; a later re-entry does not satisfy that condition.

Use precompact regular flow boxes, with rational rectangles in (F,ϑ), and increasing finite unions whose interiors cover the regular event. On overlaps within a physical component their coordinates agree up to a lifted angular constant and can be refined into finitely many pieces. Roof-constant faces are transverse; angle-constant faces have zero flux. The outside source plus the first-exit source for each such domain is the integral of the indicator that the whole forward segment is not contained in the domain. These indicators decrease to the indicator of a true exit. Every compact surviving segment eventually lies in an exhaustion domain. The usual endpoint convention at lifetime exactly h does not change the joint almost-everywhere formula.

For actual nongrazing faces, exhaust the regular one-sided arcs and their backward segments by such compact pieces. Disjoint assignment by the incident component and first-exit graph removes chart multiplicity. Apply the local formula and nonnegative convergence at fixed count. These operations also justify the variation payments in Theorem 5.2. No surface limit is commuted with the collision-normalized supremum. □

A.3. **The indispensable analytic chain.** Input 3.2 records the complete estimates used by this article, rather than requiring a reconstruction from historical status files. Its common-space operator dependencies are the collision norm definitions (Supplement equations (117.1)– (117.3)), the full-record fixed-band Doeblin–Fortet and spectral estimates (Supplement Theorem 157.3), and the finite moving-peak local inversion. The proof uses bounded transverse multipliers, not a bounded strong operator of multiplication by c^{-1} or $|g|^{-1}$.

For specificity, the joint endpoint/incidence estimate has leading factor $b^{129/128}s^{17/16}$ and finite-count cap $\min\{b\rho^{m/2}, s^2\}$. The endpoint test is the particular vector $P_b\mathbf{1}$ of strong norm $O(b^{7/8})$; that estimate is not a small operator norm on arbitrary vectors. Each continued rank-one peak is normalized using a nonzero smooth test against a physical peripheral vector, never by division by $\Pi\mathbf{1}$. The homogeneous trace estimates pass through bounded finite Cesaro approximants in the common strong space. These conventions and the two chronological characteristic pairings are proved in the retained module 186, not assumed for arbitrary L^∞ representatives.

Positive incidence layer cake integrates the cap first and yields a critical-volume remainder $b^{1/2}\rho^{m/4}$. A second layer cake against $b^{-1-\eta}$ converges at finite count only for $\eta < 1/2$, giving the actual $v^{\eta,b}$ estimate stated in Input 3.2. The leading exponent is $129/128 - \eta$. The first inverse-gradient remainder instead depends on its lower cutoff and is not used to remove that cutoff here. For the section layer only the all-depth marked-strip theorem is needed; for the incidence and clearance layers the original physical-guard local bound is used. All of these sources keep the forward occupation coefficient. Neither signed interface cancellation nor deletion of an arithmetic peak enters the new estimates.

Thus the new potential and surface theorems depend on explicitly stated volume estimates, but the surface source is bounded only after its actual first-exit graph and optical variation have been treated. The retained spectral and billiard estimates remain available for specialist verification. This appendix and native source compilation are not independent human or formal proof certification.

Part 20. Residence persistence and paired physical boundary transfer

Retained revision-84 abstract. We study the original exact-label pointwise local limit problem for actual returns in a triangular Lorentz gas. The backward lifetime of a one-sided outgoing face gives a residence kernel for its complete first-exit assignment. This kernel converts the established local surface-mass theorem into an essential-height bound of order $r^{1/32}$ for all regular segments of length greater than $2r$. For shorter segments away from a critical outgoing gradient, the physical angular density transfers between the two true ends with an error of order r in the same collision-normalized essential norm. Grazing incoming ends contribute no unpaid boundary trace. The remaining short critical surface has a sharper local-mass bound. We also identify a canonical microscopic concentration charge and prove dyadic outer-scale rigidity of the fractional tail. The exact labels, original section, guard, arithmetic reference and first-exit selection remain unchanged. The final budget isolates paired short nongrazing boundaries, short critical segments and a microscopic charge; their remaining essential-height estimates are stated separately from the new proved bounds.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\ \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6,-1/2), \quad (-\theta_R,\sin\theta_R), \quad (\pi/3-\theta_R,\sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1-6q^2+q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R,a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The complete surface-height estimates proved here.** For each one-sided nongrazing outgoing face, let l be its backward lifetime in the original regular exact event. Write $\overline{\mathcal{F}}_r$ for the complete first-exit surface assignment averaged over roof increments in $(0,r)$. The source is not replaced by a boundary collar. Its restriction to $l>2r$ is denoted by $\mathcal{L}_r$. Among the remaining segments, let $\mathcal{S}_r^{\mathrm{nc}}$ have outgoing $|g|^2\ge 8\Lambda_\circ r$, and let $\mathcal{J}_r$ carry their incoming physical traces, paired with the same outgoing faces and the same exit-window kernel.

Theorem 2.1 (Residence and two-ended optical transfer)**.** *Under the explicit inherited inputs below, for every sufficiently small previously fixed $r>0$,*

$$\limsup_m \sup_{R,n,k} m^2\|\mathcal{L}_r\|_\infty \le Cr^{1/32}, \tag{2.6}$$

$$\limsup_m \sup_{R,n,k} m^2\|\mathcal{S}_r^{\mathrm{nc}}-\mathcal{J}_r\|_\infty \le Cr. \tag{2.7}$$

The complete short noncritical part with grazing incoming end also has height at most Cr. All comparisons retain the original exact label and every incident physical branch; no word-count multiplier is used.

The first conclusion is Theorem 7.2. It uses a persistence property of the actual surface kernel, not an inference from small mass alone. The second is Theorem 8.2; it estimates the change of physical angular density between the two ends, including both angular Jacobian factors. The positive lower gradient used inside that proof is a function of the fixed roof width only.

The preceding untruncated fractional transport and local surface-mass results are retained with their proofs. The new argument uses them to control the complementary long-residence source in essential height. The still unresolved sources are short paired nongrazing boundaries, short critical segments, and a canonical microscopic concentration charge. Theorem 9.3 gives their exact role in the unchanged pointwise assertion (2.5).

3. Explicit analytic and geometric inputs

We collect the indispensable inherited assertions in the exact form used by the new proofs. Their source theorems and operator dependencies are specified here and in Appendix A. The new potential and first-exit estimates do not assume a surface spectral insertion theorem or an unweighted pointwise local bound.

Input statement 3.1 (Physical flow and source comparison)**.** On the regular part Ω of each exact event, endpoint angles $q=(\alpha_0,\alpha_m)$ give $F=L$, $g=\nabla_q F=R(-p_0,p_m)$, $K=D_q^2F$ and $\vartheta=\arg g$. Put

$$\lambda_\circ=9\sqrt{19}/200,\qquad \Lambda_\circ=47/6,\qquad \omega=|\partial_{p_0}\alpha_m|^{-1},\qquad \chi=|\det D_q(F,\vartheta)|.$$

The bounds $\lambda_\circ I\le K\le\Lambda_\circ I$ and $\lambda_\circ\le\chi\le\Lambda_\circ$ hold uniformly. The explicit field

$$X_q=\frac{K^{-1}g}{g^{\mathsf{T}}K^{-1}g}$$

is stopped at true physical and section-decision boundaries, not at chart seams. Its flow satisfies $XF=1$, $X\vartheta=0$ and

$$2\lambda_\circ s\le |g(\Phi_s x)|^2-|g(x)|^2\le 2\Lambda_\circ s. \tag{3.1}$$

Its endpoint-curve length is at most $C\sqrt{s}$. For $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$ one has $|X\omega| \le C\omega I_m/|g|$ on regular survivors. At a genuine grazing exit some internal incidence tends to zero and ω tends to zero. Forward interior critical termination cannot occur.

Every starting point with forward lifetime at most r belongs to a layer of one of the following forms, at an actual collision or flight:

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \quad \Delta_j \le C_E q_0^{d'_j}\sqrt{r}, \quad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}. \tag{3.2}$$

The physical coarea covariance, proved in the appendix in the convention used here, has Jacobian $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$. Writing E_h^ε for all weighted starting exit mass and

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon(x)[1 - \omega(\Phi_h x)/\omega(x)]_+ \, d\mu_t(x),$$

the complete source satisfies, for each fixed r and almost every t,

$$b^\varepsilon(t) \le 2A_r b^\varepsilon(t) + \bar{E}_r^\varepsilon(t) + \bar{O}_r^\varepsilon(t) + C_\Theta(1+\varepsilon^{-1})\sqrt{r}\, A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\, ds. \tag{3.3}$$

These are the endpoint-field, all-contact motion, uniform cubic-action and separated-volume results in retained modules 174, 177, 180–182. The guard term in (3.3) is the complete product guard, not a protective subword source. The derivative and continuation assertions are confined to regular physical trajectories.

Input statement 3.2 (Complete local source estimates)**.** For every fixed auxiliary band B_*, constants $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ give the following bounds uniformly in radius, exact labels, interval J and mark $0 \le j \le m$. Let $S_{R,s}$ be the union of strips of width Cs about the finitely many coordinate lines supporting ∂Y_R^*. For s small,

$$m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le Cs^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \rho_{B_*}^{m/2}, \tag{3.4}$$

$$m^2 \int_J b^\delta(t)\, dt \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \rho_{B_*}^{m/2}. \tag{3.5}$$

The local extension to a fixed remaining compact range of δ is included. At auxiliary parameter one $W^1 = 1$, so $b^1 = p$.

Put $z = |g|$, $\alpha = 129/128$ and, for $0 < \eta < 1/2$, define the actual nonnegative source measures

$$\begin{aligned} v^{\eta,b}(t)\, dt &= L_\#(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_\Pi \nu_R^*), \\ v^\eta(t)\, dt &= L_\#(I_m z^{-\eta} \mathbf{1}_\Pi \nu_R^*). \end{aligned} \tag{3.6}$$

They are finite and absolutely continuous at every fixed count, and for $0 < b < b_1$,

$$m^2 \int_J v^{\eta,b}(t)\, dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|)m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \tag{3.7}$$

Adding the bounded-gradient part gives, for every fixed $L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^\eta \le C_\eta L, \qquad \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta,b} \le C_\eta b^{\alpha-\eta} L. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards; leading constants are independent of that band and of the previously fixed interval length.

Equation (3.4) is Supplement Theorem 175.3. The physical source bound (3.5) is Supplement Theorem 179.1, in particular its finite-window estimate. The fractional bounds are the positive layer-cake conclusions of retained module 187, stated in full above. The joint spectral and common-space assumptions used there are recorded in the appendix. No estimate for an untruncated first inverse gradient or for a singular roof convolution is part of this input.

Input statement 3.3 (The scalar positive remainder)**.** For large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.9}$$

The source b^ε is nonnegative and has a compactly supported BV density at fixed count. No collision-uniform variation bound is assumed. Supplement Theorem 181.4 gives this remainder with the same complete finite-peak reference.

4. Untruncated optical transport and fractional potentials

Fix one of the original exact events. The field, optical density and coarea kernels in Input 3.1 are used throughout. No new field is chosen from the unknown roof density. Put $z = |g|$. For $0 < \eta < 1/2$ set

$$\kappa_\eta = \frac{1-\eta}{2}, \qquad Q_\eta = \kappa_\eta^{-1}, \qquad \mathsf{P}_{\eta,r} v(t) = \int_0^r s^{-\kappa_\eta} v(t+s)\, ds. \tag{4.1}$$

In particular $2 < Q_\eta < 4$. The source v^η below is the *actual*, untruncated source with insertion $I_m z^{-\eta}$ from Input 3.2; it is not a limit taken after the collision limit.

Write $E_h^{\varepsilon,\mathrm{gr}}$ for the starting mass of genuine grazing first exits and O_h^ε for the positive inverse-twist loss on survivors. These are disjoint classes of initial states. If $b > 0$, a subscript $< b$ restricts the initial source to $0 < z < b$. A bar denotes the average $r^{-1}\int_0^r \cdot\, dh$.

Theorem 4.1 (Optical transport without an initial lower cutoff)**.** *For almost every (t,h),*

$$O_h^\varepsilon(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \le C_\eta \mathsf{P}_{\eta,h} v^\eta(t). \tag{4.2}$$

The right side is locally integrable in t at every fixed count. If $0 < h \le r$ and $b_r = (b^2 + 2\Lambda_\circ r)^{1/2} < b_1$, then

$$(O_h^\varepsilon + E_h^{\varepsilon,\mathrm{gr}})_{<b}(t) \le C_\eta \mathsf{P}_{\eta,r} v^{\eta,b_r}(t). \tag{4.3}$$

The same bounds hold for the h-averaged sources. All internal incidences, all actual words and both section tests are retained.

Proof. Along a regular survivor, the optical sensitivity and radial speed give

$$|X\omega| \le C\omega I_m / z, \qquad z(\Phi_s x)^2 \ge z(x)^2 + 2\lambda_\circ s.$$

Consequently

$$|X\omega|(\Phi_s x) \le C_\eta s^{-\kappa_\eta} \omega(\Phi_s x) I_m(\Phi_s x) z(\Phi_s x)^{-\eta}. \tag{4.4}$$

For a surviving state, $[\omega(x) - \omega(\Phi_h x)]_+$ is bounded by the negative variation up to h. At a grazing first exit the limiting optical density is zero; hence its entire positive starting optical weight is bounded by negative variation up to the lifetime. This uses the one-word optical identity only to identify the zero limit, not to sum wordwise constants.

In an angular flow box the initial fiber density is $\omega(x)/(4\pi c_* \chi(x))\, d\vartheta$. Multiplying by $W^\varepsilon(x) \le 1$ and using

$$\frac{\omega(\Phi_s x)}{\omega(x)} = J_s(x) \frac{\chi(\Phi_s x)}{\chi(x)} \le C J_s(x)$$

transfers the integrand in (4.4) to a subset of the complete target fiber. The common-kernel covariance in Appendix A justifies this simultaneously for the disjoint survivor and first-exit

selections. Tonelli then proves (4.2). Nonnegative truncation at fixed collision count removes all compact-chart restrictions.

For $z(x) < b$ the radial upper bound gives $z(\Phi_s x) < b_r$ before $h \le r$, proving (4.3). The kernel and sources are nonnegative, so averaging in h preserves the bounds. Finally $\kappa_\eta < 1$ and v^η is locally integrable with compact support. Fubini proves local integrability of each displayed potential. Neither an untruncated first inverse-gradient moment nor a pointwise bound for v^η has been used. □

For $Q > 1$, a measurable interval J and $f \ge 0$, define

$$\mathcal{N}_Q(f; J) = \sup_{E \subset J,\ 0<|E|<\infty} |E|^{-1+1/Q} \int_E f(t)\, dt, \tag{4.5}$$

where the supremum is over Lebesgue-measurable sets. This is a specific weak-L^Q norm; the definition fixes its normalization.

Lemma 4.2 (A localized potential bound)**.** *For every nonnegative locally integrable v, every $r > 0$, and any interval J, write $J_r = J + (0, r)$. Then*

$$\mathcal{N}_{Q_\eta}(\mathsf{P}_{\eta,r} v; J) \le \frac{1}{1-\kappa_\eta} \int_{J_r} v(y)\, dy. \tag{4.6}$$

The inequality also holds for extended nonnegative quantities.

Proof. For fixed y and a measurable set E of length a, the decreasing kernel has

$$\int_E (y-t)^{-\kappa_\eta} \mathbf{1}_{\{0<y-t<r\}}\, dt \le \int_0^{\min(a,r)} s^{-\kappa_\eta}\, ds \le \frac{a^{1-\kappa_\eta}}{1-\kappa_\eta}.$$

The first inequality follows directly by integrating the measures of the kernel's superlevel intervals. Integrate against $v(y)$ and use Tonelli. Only $y \in J_r$ contributes. Divide by $a^{1-\kappa_\eta}$ and take the supremum over E. □

Set $D_r^\varepsilon = \bar{O}_r^\varepsilon + \bar{E}_r^{\varepsilon,\mathrm{gr}}$. All norms in the next theorem are taken before the collision limit.

Theorem 4.3 (Complete optical source bounds at the collision scale)**.** *For each previously fixed $r, L > 0$ and $0 < \eta < 1/2$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q_\eta}(m^2 D_r^\varepsilon; J) \le C_\eta (L + r), \tag{4.7}$$

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q_\eta}(m^2 D_{r,<b}^\varepsilon; J) \le C_\eta b_r^{\alpha-\eta} (L + r) \quad (b_r < b_1). \tag{4.8}$$

Here $\alpha = 129/128$. For $1 \le p < Q_\eta$, the left sides may be replaced by the corresponding $L^p(J)$ norms, provided the right sides are multiplied by $(Q_\eta/(Q_\eta - p))^{1/p} L^{1/p-1/Q_\eta}$. No essential-supremum conclusion is included.

Proof. Combine Theorem 4.1 with Lemma 4.2, then apply the fixed-width bounds for the actual sources v^η and v^{η,b_r} in Input 3.2 on the interval of length $L + r$. Its length and b_r are fixed before the collision limit. The auxiliary band is enlarged only afterwards.

If $\mathcal{N}_Q(f; J) \le A$, applying its definition to $\{f > \lambda\} \cap J$ gives $|\{f > \lambda\} \cap J| \le \min\{L, (A/\lambda)^Q\}$. Integration of this bound against $p\lambda^{p-1} d\lambda$ proves $\|f\|_{L^p(J)} \le (Q/(Q-p))^{1/p} A L^{1/p-1/Q}$. Apply it at each finite count before taking the limits. □

For example, $\eta = 1/4$ gives $Q_\eta = 8/3$ and $\alpha - \eta = 97/128$. On intervals of length r, the complete optical norm is $O(r)$; the part with initial $z < \sqrt{r}$ is $O(r^{1+97/256})$. The estimates concern the untruncated original source. They do not permit $p = \infty$, $\eta = 1/2$, or a collision-dependent roof width. Their role in the original pointwise problem is made explicit in Section 6.

5. The complete first-exit surface source

We now estimate a quantity defined on the actual outgoing surface, not a bounded multiplier substituted for that surface. The estimate uses both a complete starting-source bound and a two-sided payment of optical variation. It is local in roof mass; the distinction from pointwise height will be kept explicit.

Let $\tau(x)$ be the forward lifetime in the original regular exact event and put

$$e_h(t) = \int \mathbf{1}_{\{\tau(x)\le h\}}\, d\mu_t(x). \tag{5.1}$$

This is the *unweighted* starting mass; it keeps both section tests and the exact label. There is no renormalization after removal of a boundary layer.

Theorem 5.1 (All true exits in a roof interval)**.** *There is $r_0 > 0$ such that, for every fixed auxiliary band B_*, all $0 < r < r_0$, $m \ge 2$, and intervals J,*

$$m^2 \int_J e_r(t)\, dt \le Cr^{1/32}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \rho_{B_*}^{m/2}. \tag{5.2}$$

The constants are uniform in R, n, k. Consequently, for fixed r, L,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J e_r(t)\, dt \le CLr^{1/32}. \tag{5.3}$$

The same conclusion holds for the h-average of e_h.

Proof. The all-contact motion estimate places every finite exit of lifetime at most r in the union of the starting layers

$$c_j \le C_E q_0^{d_j}\sqrt{r}, \qquad \Delta_j \le C_E q_0^{d'_j}\sqrt{r}, \qquad \operatorname{dist}(T_R^j x, \partial Y_R^*) \le C_E q_0^{d_j}\sqrt{r}.$$

The union includes every contact and flight, and all initial, internal and terminal decisions. A positive-margin endpoint would continue the physical word and field; a regular critical point cannot be a forward exit. Coordinate seams are not stopping faces.

The first two layer unions are contained in the set on which the original guard with auxiliary parameter $C\sqrt{r}$ has value zero: $q_0^d \le q_0^{d/4}$. Their source is therefore bounded by the complete physical bad source in Input 3.2. For the section layers use the marked transverse-strip estimate there at the actual collision time j, with width $C_E q_0^{d_j}\sqrt{r}$. Each positive upper comparison drops endpoint tests only on its upper side and costs exactly c_*^{-1}; the occupation and displacement coefficient remain (n, k).

There are at most two contact marks of each depth. The leading section terms sum as $Cr^{1/32}\sum_{d\ge 0} q_0^{d/16}$, and the physical guard term has the same power of r. The at most $m+1$ section remainders, together with the physical remainder, are bounded by the second line of (5.2). Thus no uncontrolled word count or sum of collision-dependent bandwidths occurs. Take the collision limit with r, L, B_* fixed and then enlarge B_*. Finally $e_h \le e_r$ for $h \le r$. □

5.1. **The physical surface measure and its starting-roof assignment.** Use one-sided nongrazing outgoing faces of the original exact event. On a transverse angular chart a face is $F = b(\vartheta)$, and its physical flux is

$$d\zeta = \frac{\omega(b(\vartheta),\vartheta)}{4\pi c_*\chi(b(\vartheta),\vartheta)}\, d\vartheta. \tag{5.4}$$

A countable disjoint refinement counts each face with its incident physical component. Tangential strata have zero transverse flux. Let $\tau^-(y)$ be the backward lifetime in that same one-sided event. Define

$$\mathcal{F}_h(t) = \int_{\Gamma^+_{\mathrm{ng}}} \mathbf{1}_{\{0<L(y)-t\le h,\ L(y)-t<\tau^-(y)\}}\, d\zeta(y). \tag{5.5}$$

There is no initial lower bound on $|g|$. The selection in (5.5) excludes later re-entries. Competing-hit, section and other true exact-event faces remain present. This is the increasing limit of the original $\mathcal{F}^a_h$ as $a \downarrow 0$ at fixed count, on the common versions of Appendix A. The measure is initially allowed to be extended nonnegative.

Theorem 5.2 (Two-sided optical payment for the actual flux)**.** *For every $0 < \eta < 1/2$, jointly for almost every (t, h),*

$$\mathcal{F}_h(t) \le Ce_h(t) + C_\eta \mathsf{P}_{\eta,h} v^\eta(t), \tag{5.6}$$

$$E^\varepsilon_h(t) + O^\varepsilon_h(t) \le C\mathcal{F}_h(t) + C_\eta \mathsf{P}_{\eta,h} v^\eta(t). \tag{5.7}$$

In particular the full outgoing assignment is locally integrable in the starting roof at fixed count. The first inequality is a bound for the surface source itself, not just for its pullback cost.

Proof. Along a nongrazing first-exit orbit write $w(s) = \omega(\Phi_s x)$ and let $\tau \le h$ be its lifetime. In the angular parametrization, the starting fiber density is $w(0)/(4\pi c_*\chi(x))$ and the exit flux density is $w(\tau-)/(4\pi c_*\chi(\Phi_\tau x))$. Both limiting factors are positive at a nongrazing exit. The two inequalities

$$w(\tau-) \le w(0) + \int_0^\tau [w'(s)]_+\, ds, \qquad w(0) \le w(\tau-) + \int_0^\tau [-w'(s)]_+\, ds$$

are used in their respective directions. The uniform upper and lower bounds for χ compare their fiber normalizations.

For (5.6), sum the first inequality on the actual first-exit graph. Its starting term is at most Ce_h. Pay its positive variation by (4.4) and common *target-direction* coarea covariance. This yields the potential term. A boundary point is counted with precisely the initial states selected in (5.5); no multiplicity from a later return is introduced.

For (5.7), use the second inequality, with $W^\varepsilon(x) \le 1$, for nongrazing exits. At grazing exits $w(\tau-) = 0$ and negative variation pays the full starting weight. On survivors it pays the positive optical loss. These classes are disjoint, so their common target insertion is dominated by the full v^η source, not a word-cardinality multiple of it. Truncate to compact regular face pieces and flow boxes, and pass to the full nonnegative sums at fixed count. Appendix A gives the joint versions and first-exit identification. Local integrability follows from $e_h \le p$ and Theorem 4.1.

The exact pullback identity still contains the inverse forward Jacobian. The two variation inequalities account for its cost; they do not replace that Jacobian by one. □

Theorem 5.3 (A complete local-mass estimate for outgoing first exits)**.** *For every previously fixed $0 < r < r_0$, $L > 0$ and $0 < \eta < 1/2$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J \overline{\mathcal{F}}_r(t)\, dt \le CLr^{1/32} + C_\eta(L + r)r^{(1+\eta)/2}. \tag{5.8}$$

In particular, on roof intervals of length r, the right side is $O_\eta(r^{33/32})$. Every initial momentum and every nongrazing one-sided first-exit component is included. This is an estimate of local surface-assignment mass, not an essential-height estimate.

Proof. Average (5.6) and use $e_h \le e_r$ and $\mathsf{P}_{\eta,h} v^\eta \le \mathsf{P}_{\eta,r} v^\eta$. Fubini gives

$$\int_J \mathsf{P}_{\eta,r} v^\eta(t)\, dt \le \frac{r^{1-\kappa_\eta}}{1-\kappa_\eta} \int_{J+(0,r)} v^\eta(y)\, dy.$$

Use (5.3) for the first term and the fixed-width bound for the actual v^η source for the second. All intervals have fixed lengths before the collision limit. Since $1-\kappa_\eta=(1+\eta)/2>1/32$, setting $L=r<1$ proves the last assertion. The surface estimate has been proved through its first-exit graph and positive optical variation; no theorem for a bounded volume multiplier is being relabeled as a trace theorem. □

6. The ordered microscopic tail and the original pointwise law

The preceding estimates give complete source and surface controls in specified local norms. The original pointwise target is stronger. We now state the remaining implication on that target, and separate the microscopic tail before any interchange of limits could occur.

For each exact label retain the actual source v^η. For $r>0$, put $\ell_j=2^{-j}r$ and define

$$T_{\eta,r,N}(t)=\sum_{j=N}^{\infty}\ell_j^{-\kappa_\eta}\int_{\ell_j/2}^{\ell_j}v^\eta(t+s)\,ds. \tag{6.1}$$

The integration variable s is an offset from the starting roof. Equivalently one integrates $v^\eta(u)\,du$ over $(t+\ell_j/2,t+\ell_j]$; there is no second translation of the roof. All terms and the entire tail are formed at finite collision count.

Lemma 6.1 (An ordered decomposition, including the full tail)**.** *Pointwise as extended nonnegative quantities,*

$$T_{\eta,r,0}(t)\le \mathsf{P}_{\eta,r}v^\eta(t)\le 2^{\kappa_\eta}T_{\eta,r,0}(t). \tag{6.2}$$

Define

$$\mathfrak{T}_{\eta,M}(r)=\lim_{N\to\infty}\limsup_{m}\sup_{R,n,k}m^2\operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}T_{\eta,r,N}(t), \tag{6.3}$$

$$\mathfrak{P}_{\eta,M}(r)=\limsup_{m}\sup_{R,n,k}m^2\operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}\mathsf{P}_{\eta,r}v^\eta(t). \tag{6.4}$$

The first limit exists monotonically in $[0,\infty]$*, and*

$$\mathfrak{T}_{\eta,M}(r)\le\mathfrak{P}_{\eta,M}(r)\le C_\eta r^{(1+\eta)/2}+2^{\kappa_\eta}\mathfrak{T}_{\eta,M}(r). \tag{6.5}$$

Proof. On $(\ell_j/2,\ell_j]$, the kernel lies between $\ell_j^{-\kappa_\eta}$ and $2^{\kappa_\eta}\ell_j^{-\kappa_\eta}$. Tonelli proves (6.2). For any fixed N, the first N interval lengths are fixed before the collision limit. Input 3.2 bounds the normalized height of their integrals by $C_\eta\ell_j$. Hence their complete finite sum is at most $C_\eta\sum_{j<N}\ell_j^{1-\kappa_\eta}\le C_\eta r^{1-\kappa_\eta}$. Keep the whole remaining nonnegative tail inside the supremum and collision limsup. Only then let $N\to\infty$. Monotonicity of these tails proves existence and (6.5). This argument does not use the false reverse of Fatou's inequality for an infinite sum. □

Let the full, untruncated surface height be

$$\mathfrak{F}_M^{\mathrm{all}}(r)=\limsup_{m}\sup_{R,n,k}m^2\operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}\overline{\mathcal{F}}_r(t). \tag{6.6}$$

Its source is precisely (5.5).

Theorem 6.2 (Pointwise error with no initial-cutoff interchange)**.** *For each fixed* $0<\eta<1/2$*, all sufficiently large fixed* B *and all previously fixed* $0<r<r_0$*, the original pointwise error satisfies*

$$\begin{aligned}\mathcal{E}_M\le{}& C_MB^{-1/192}+C\mathfrak{F}_M^{\mathrm{all}}(r)+C_\eta\mathfrak{T}_{\eta,M}(r)\\&+C_\eta r^{(1+\eta)/2}+C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.\end{aligned} \tag{6.7}$$

Thus the original assertion (2.5) *follows if the full surface height and the microscopic tail in this display tend to zero as* $r \downarrow 0$. *This sufficient condition is not asserted as proved by the local-norm estimates above.*

Proof. Use the complete-source shift inequality of Input 3.1 and replace its sum of true-exit and optical losses, at the source level, by the average of (5.7). The positive scalar remainder and the fixed-width bad-source bound then give

$$\mathcal{E}_M \le C_M B^{-1/192} + C\mathfrak{F}_M^{\mathrm{all}}(r) + C_\eta \mathfrak{P}_{\eta,M}(r) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.$$

Apply (6.5). Every source restriction has already been removed at fixed count, using actual fractional integrability; no first inverse-gradient cutoff is passed through a collision limit. To obtain the conditional conclusion, choose $r(B) = c\varepsilon(B)^\gamma$ with fixed $c > 0$ and $\gamma > 2$. For each B this is a fixed positive width. First take the collision limit, next enlarge auxiliary spectral bands, and take $B \to \infty$ last. Every displayed elementary error then vanishes, and the two stated conditions pay the remaining terms. The arithmetic reference and all zero classes are unchanged. □

Proposition 6.3 (A quantitative microscopic envelope)**.** *Suppose, as an additional estimate on the actual source, that for all sufficiently large* m, *all parameters and almost every central* t,

$$m^2 \int_0^s v^\eta(t+u)\,du \le H(s) \quad (0 < s < r_0), \qquad \int_0^{r_0} H(s) s^{-1-\kappa_\eta}\,ds < \infty, \tag{6.8}$$

with H *nonnegative and nondecreasing. The exceptional set may be chosen independently of* s *by first using rational* s *and monotone limits. Then* $\mathfrak{T}_{\eta,M}(r) = 0$ *for every* $r < r_0/2$. *In particular a uniform bound* $H(s) \le As^{\kappa_\eta+\delta}$, $\delta > 0$, *gives the finite-count tail bound*

$$m^2 T_{\eta,r,N}(t) \le \frac{Ar^\delta 2^{-N\delta}}{1-2^{-\delta}}. \tag{6.9}$$

The envelope is a uniform finite-count hypothesis, not a list of separate fixed-scale collision limsups.

Proof. Each normalized interval integral is at most $H(\ell_j)$. Moreover, monotonicity gives

$$H(\ell_j)\ell_j^{-\kappa_\eta} \le C_\eta \int_{\ell_j}^{2\ell_j} H(s) s^{-1-\kappa_\eta}\,ds.$$

The right intervals have disjoint interiors. Their tails tend to zero by the assumed integrability, uniformly before the collision limit. For the power envelope sum the geometric series directly. □

The fractional source lowers the sufficient envelope exponent from $1/2$ to $\kappa_\eta = (1-\eta)/2$, which can be any fixed number strictly between $1/4$ and $1/2$. The endpoint $1/4$ is not included. The complete surface mass bound (5.8) and the weak-L^{Q_η} optical bound do not establish (6.8) or (6.6). A uniformly small integral, even with these additional local controls, must not be substituted for an essential roof-height estimate. The original pointwise law, and scalar-first normalization of its path and likelihood consequences, remain exactly the organizing objective.

7. Backward residence and the height of the exit surface

A first-exit face carries more data than its roof coordinate. Its backward lifetime records the length of the maximal regular segment in the same exact event. Retaining that length yields an essential-height estimate for the complete long-residence part of the actual surface source.

Fix an exact event and its one-sided nongrazing outgoing faces Γ_{ng}^+. For y on such a face put

$$b(y) = F(y), \qquad l(y) = \tau^-(y), \qquad a(y) = b(y) - l(y).$$

Thus the maximal backward regular segment has roofs $(a(y), b(y))$. The endpoint a may be a true incoming face or a regular interior critical point removed from the flow domain. It is not an auxiliary chart seam. At fixed collision count the roof is bounded, so l is finite on every nonempty segment. All assertions below disregard only the transverse null sets already excluded from the first-exit kernel. Define

$$k_r(t; y) = \left(1 - \frac{b(y) - t}{r}\right)_+ \mathbf{1}_{\{a(y)<t<b(y)\}}. \tag{7.1}$$

The original surface normalization is in $d\zeta(y)$; it is not inserted again into k_r.

Lemma 7.1 (The complete residence kernel)**.** *At every fixed count, as a jointly measurable nonnegative kernel,*

$$\overline{\mathcal{F}}_r(t) = \int_{\Gamma^+_{\mathrm{ng}}} k_r(t; y)\, d\zeta(y). \tag{7.2}$$

The same identity holds after any Borel restriction of the outgoing face and its incident regular segment. No subsequent re-entry is counted.

Proof. For a fixed face put $s = b(y) - t$. The first-exit indicator is $\mathbf{1}_{\{0<s\le h,\ s<l(y)\}}$. Its average over $0 < h < r$ is $(1 - s/r)_+\mathbf{1}_{\{0<s<l(y)\}}$, which is (7.1). Tonelli proves the assertion, including the extended nonnegative case. The lifetime is taken in the original exact event, so each face remains attached to its incident physical component. The common flow-box kernels in Appendix A make these restrictions measurable. □

Split (7.2) as $\mathcal{L}_r + \mathcal{S}_r$ by $l(y) > 2r$ and $l(y) \le 2r$, respectively. Both summands are nonnegative. The letter $\mathcal{L}_r$ here denotes a surface assignment, not the arithmetic reference $\mathcal{L}_{m,R}$.

Theorem 7.2 (Long-residence essential height)**.** *For almost every t and every fixed sufficiently small $r > 0$,*

$$\mathcal{L}_r(t) \le \frac{2}{r} \int_{t-r}^{t} \overline{\mathcal{F}}_{2r}(v)\, dv. \tag{7.3}$$

Consequently the original exact-label surface has the complete bound

$$\limsup_m \sup_{R,n,k} m^2 \|\mathcal{L}_r\|_\infty \le C_\eta r^{1/32}. \tag{7.4}$$

This is an essential-height estimate, uniform over all actual words with backward residence greater than $2r$. It is not an estimate of only their total mass.

Proof. Suppose $k_r(t; y) > 0$, put $s = b(y) - t \in (0, r)$ and assume $l(y) > 2r$. For $0 < u < r$, one has $s + u < 2r < l(y)$ and

$$k_{2r}(t - u; y) = 1 - \frac{s + u}{2r} \ge \frac{1}{2}\left(1 - \frac{s}{r}\right) = \frac{1}{2} k_r(t; y).$$

The comparison follows the same face and the same exact event. Integrate first in u, then in $d\zeta(y)$. Removing the long-residence restriction only from the upper comparison gives (7.3).

Apply Theorem 5.3 with surface width $2r$ and interval length r. Division by r bounds the ordered right side by $Cr^{1/32} + C_\eta r^{(1+\eta)/2} \le C_\eta r^{1/32}$. Every interval length is fixed before the collision limit. The supremum over its location is already present in that theorem, so (7.3) gives an essential supremum, not merely an almost-everywhere pointwise limit. □

Let $\mathfrak{S}_M(r)$ denote the same ordered central essential height as (6.6), with $\mathcal{S}_r$ in place of $\overline{\mathcal{F}}_r$. Positivity and Theorem 7.2 give

$$\mathfrak{S}_M(r) \le \mathfrak{F}^{\mathrm{all}}_M(r) \le \mathfrak{S}_M(r) + C_\eta r^{1/32}. \tag{7.5}$$

Thus the full surface height tends to zero if and only if the height of its short maximal regular segments tends to zero. This exact localization uses a proved height bound for the complementary source. It does not assume that short segments have small height.

8. Optical transfer between the two true ends of a short segment

The preceding short-residence source can be resolved further. On a short segment whose outgoing gradient is not of order $\sqrt{r}$, all of the optical transfer between its two ends is estimated in essential height. In particular, a grazing incoming end creates no unestimated outgoing surface term.

For this argument we use two additional, precisely stated inherited volume bounds. With $z = |g|$ and I_m as before, let

$$v^c(t)\,dt = F_{\#}(I_m z^{-1}\mathbf{1}_{\{z\geq c\}}\mathbf{1}_{\Pi}\nu_R^*), \qquad u^b(t)\,dt = F_{\#}(I_m\mathbf{1}_{\{z<b\}}\mathbf{1}_{\Pi}\nu_R^*). \tag{8.1}$$

For every previously fixed $c, b, L > 0$, the source-pinned estimates are

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} m^2\int_J v^c \leq CL, \qquad \limsup_m \sup_{R,n,k}\sup_{|J|=L} m^2\int_J u^b \leq Cb^{\alpha}L. \tag{8.2}$$

The leading constant in the first bound is independent of $c > 0$; this is Supplement equation (26.9). The second is Supplement Theorem 26.1, with $\alpha = 129/128$ and $b < b_1$. The first source is actually truncated, at a positive number fixed before the collision limit. No assertion at $c = 0$ is used.

On each local angular sheet of a maximal regular segment write its roof as $u \in (a, b)$ and define its physical angular density by

$$A(u,\vartheta) = \frac{\omega(u,\vartheta)}{4\pi c_*\chi(u,\vartheta)}. \tag{8.3}$$

Then $d\mu_u = A(u,\vartheta)\,d\vartheta$ on that sheet. At a nongrazing outgoing end $d\zeta_+ = A(b-,\vartheta)\,d\vartheta$. Angular sheets are disjointized only to sum their kernels; no sheet edge is a new stopping boundary.

Lemma 8.1 (Variation of the physical angular density)**.** *Along every regular segment,*

$$|\partial_u A(u,\vartheta)| \leq CA(u,\vartheta)I_m(u,\vartheta)/z(u,\vartheta). \tag{8.4}$$

The constant is independent of collision count and all positive internal incidences. If the incoming endpoint has positive limiting z and a vanishing internal incidence, then $A(a+,\vartheta) = 0$. If all limiting incidences and z are positive, the incoming trace exists on the corresponding one-sided physical face.

Proof. The optical sensitivity gives $|X\omega| \leq C\omega I_m/z$. Since the momentum angle is preserved, $\chi = e_\perp^{\mathsf{T}} K e_\perp$ has derivative bounded by $T_*|X_q| \leq C/z$, using the uniform cubic estimate and $|X_q| \leq C/z$. Hence $|X\log\chi| \leq C/z$. Differentiate (8.3) and use $I_m \geq 1$.

At fixed count, vanishing of one internal incidence forces $\omega \to 0$ by the retained selected-incidence optical identity. The lower bound for χ proves the zero trace. Otherwise the stationary word extends smoothly to the limiting point by the regular reflection equations. This is a one-sided algebraic extension, not a continuation of the physical trajectory past a first-hit or section-decision boundary. Its positive noncritical trace is the incoming physical flux in these angular coordinates. □

Partition $\mathcal{S}_r$ into $\mathcal{S}_r^{\mathrm{nc}}$ and $\mathcal{K}_r$ by, respectively,

$$z(b-)^2 \geq 8\Lambda_\circ r, \qquad z(b-)^2 < 8\Lambda_\circ r. \tag{8.5}$$

The equality case belongs to the first part. Let $c_r = 2\sqrt{\Lambda_\circ r}$. Since $b - a \le 2r$ and $\partial_u z^2 \le 2\Lambda_\circ$, every segment in the first part has $z(u) \ge c_r$ throughout. In particular it has no critical incoming end.

For each such segment keep the same exit-window kernel $k_r(t; y)$, but replace its outgoing angular density $A(b-, \vartheta)$ by the incoming angular density $A(a+, \vartheta)$. Sum this nonnegative quantity over all angular sheets to define $\mathcal{J}_r(t)$. An incoming grazing end has zero trace and contributes zero. All remaining terms are attached to a pair of actual nongrazing incoming and outgoing faces of the *same maximal exact-event segment*. The kernel still uses the outgoing roof b, not the incoming roof a. Thus $\mathcal{J}_r$ is a paired boundary assignment, not the unrestricted flux of all incoming faces and not a volume surrogate.

Theorem 8.2 (A complete essential-height bound for boundary transfer)**.** *For almost every t,*

$$|\mathcal{S}_r^{\mathrm{nc}}(t) - \mathcal{J}_r(t)| \le C \int_{t-2r}^{t+2r} v^{c_r}(u)\, du. \tag{8.6}$$

Both assignments are locally finite almost everywhere at fixed count. Consequently

$$\limsup_m \sup_{R,n,k} m^2 \|\mathcal{S}_r^{\mathrm{nc}} - \mathcal{J}_r\|_\infty \le Cr. \tag{8.7}$$

The complete part of $\mathcal{S}_r^{\mathrm{nc}}$ whose incoming end is grazing satisfies the same bound Cr by itself.

Proof. On one selected segment the variation inequality gives

$$|A(b-, \vartheta) - A(a+, \vartheta)| \le C \int_a^b A(u, \vartheta) I_m(u, \vartheta)/z(u, \vartheta)\, du.$$

First use compact subsegments and then take the one-sided limits. Multiply by $k_r(t; y) \le 1$. Whenever it is nonzero, $a < t < b$; since $b - a \le 2r$, the full segment lies in $(t - 2r, t + 2r)$. At each roof u, the map from selected outgoing faces to their points on the same segment is injective in the physical collision source. Indeed two such faces with the same interior state would be the same first exit, by uniqueness of the flow. Different visits to the same angle on different physical sheets remain different states and are already included in μ_u.

The complete sum of the variation integrals is therefore bounded by the source in (8.1), with $z \ge c_r$. Tonelli and the common physical kernels prove (8.6), without a word-count factor. The source v^{c_r} is finite at every fixed count. The outgoing assignment is locally integrable by Theorem 5.3 and its fixed-count proof. The variation bound thus also proves local integrability of the incoming paired assignment; taking a difference has not subtracted two undefined infinities. Apply (8.2) with interval length $4r$ to obtain (8.7). Here c_r depends only on the previously fixed outer width, not on collision count. Finally restrict the same positive variation argument to incoming grazing ends, where the incoming trace is zero. □

Proposition 8.3 (Quantitative mass of the short critical part)**.** *Put $b_r = \sqrt{8\Lambda_\circ r} < b_1$. For fixed $r, L > 0$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J \mathcal{K}_r(t)\, dt \le CLb_r^\alpha + C_\eta (L + r) r^{(1+\eta)/2} b_r^{\alpha-\eta}. \tag{8.8}$$

In particular the mass on intervals of length r is $O_\eta(r^{385/256})$. This statement does not bound the essential height of $\mathcal{K}_r$.

Proof. The radial monotonicity implies $z(u) < b_r$ on every selected segment. Use the positive-variation direction of the proof of Theorem 5.2 on just these first exits. Its starting term is bounded by the unweighted exact source with $z < b_r$, which is at most u^{b_r} since $I_m \ge 1$. Its

drift term is bounded by $C_\eta \mathsf{P}_{\eta,r} v^{\eta,b_r}$: all intermediate targets of the selected segment still have $z < b_r$. The full source on the upper side is only enlarged, not redefined. Integrate over J, use Fubini and the bounds (8.2) and (3.8). This proves the display. For $L = r$, its exponents are $1 + \alpha/2 = 385/256$ and $1 + (1+\alpha)/2 = 513/256$, respectively. □

Write $\mathfrak{J}_M(r)$ and $\mathfrak{K}_M(r)$ for the ordered central essential heights of $\mathcal{J}_r$ and $\mathcal{K}_r$. The preceding theorems imply

$$\mathfrak{F}^{\mathrm{all}}_M(r) \le \mathfrak{J}_M(r) + \mathfrak{K}_M(r) + C_\eta r^{1/32}. \tag{8.9}$$

Conversely $\mathfrak{K}_M(r) \le \mathfrak{F}^{\mathrm{all}}_M(r)$ and $\mathfrak{J}_M(r) \le \mathfrak{F}^{\mathrm{all}}_M(r) + Cr$. Thus full surface-height decay is equivalent to decay of the two specified short-segment heights. Neither is inferred from its local mass. The newly controlled terms are all long-residence outgoing faces, all noncritical short grazing entrances, and the complete optical transfer between the two noncritical ends of every short segment.

9. The microscopic concentration charge and the exact-label endpoint

The microscopic limit has an additional rigidity: shrinking the outer width cannot gradually reduce a nonzero concentration charge. It is useful to state this directly, rather than to regard the tail height as an arbitrary modulus of that outer width.

Fix $0 < \eta < 1/2$, write $\kappa = \kappa_\eta$, and use exactly the actual source v^η and offset convention of (6.1). For a family of nonnegative sources f_m with the same parameter and label indices, abbreviate

$$\mathcal{H}_M(f) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} f_m(t).$$

This functional is monotone and subadditive. It is not assumed to commute with infinite sums. Define the canonical concentration charge by

$$\mathfrak{A}_{\eta,M} = \lim_{\delta \downarrow 0} \mathcal{H}_M(\mathsf{P}_{\eta,\delta} v^\eta). \tag{9.1}$$

The limit exists monotonically in $[0,\infty]$. The complete finite-count potential is inside $\mathcal{H}_M$; no density is truncated after taking its collision limit.

Theorem 9.1 (Outer-scale rigidity of the microscopic tail)**.** *For every fixed sufficiently small $r > 0$,*

$$\mathfrak{T}_{\eta,M}(r) \le \mathfrak{A}_{\eta,M} \le 2^\kappa \mathfrak{T}_{\eta,M}(r), \tag{9.2}$$

$$\mathfrak{T}_{\eta,M}(2^{-j} r) = \mathfrak{T}_{\eta,M}(r) \quad (j \in \mathbb{Z}), \tag{9.3}$$

$$\mathcal{H}_M(\mathsf{P}_{\eta,r} v^\eta) \le C_\eta r^{1-\kappa} + 2^\kappa \mathfrak{A}_{\eta,M}. \tag{9.4}$$

In (9.3) *both widths are required to be in the stated domain. The following conditions are equivalent: $\mathfrak{A}_{\eta,M} = 0$; the dyadic tail height vanishes at one positive width; it vanishes at every positive width; and $\mathfrak{T}_{\eta,M}(r) \to 0$ as $r \downarrow 0$.*

Proof. Put $\delta_N = 2^{-N} r$. At each finite collision count, the annuli in $T_{\eta,r,N}$ partition $(0, \delta_N]$, and hence

$$T_{\eta,r,N} \le \mathsf{P}_{\eta,\delta_N} v^\eta \le 2^\kappa T_{\eta,r,N}.$$

Apply $\mathcal{H}_M$ before passing to $N \to \infty$. Since $\delta_N \downarrow 0$, monotonicity identifies the middle limit with (9.1) and proves (9.2). The exact finite-count identity $T_{\eta,r/2,N} = T_{\eta,r,N+1}$ proves dyadic invariance. Iteration also proves the assertion for negative integers within the domain.

For a fixed N, split the dyadic sum at N. The head consists of finitely many intervals fixed before the collision limit. The fixed-width source input bounds its ordered height by $C_\eta \sum_{j<N} (2^{-j} r)^{1-\kappa} \le C_\eta r^{1-\kappa}$. Keep the entire remaining tail inside $\mathcal{H}_M$, and only then send

$N \to \infty$. Use (9.2) to obtain (9.4). Finally, a zero tail at one width forces the charge to be zero, and a zero charge forces all tails to vanish. If the tails tend to zero with r, the lower bound $\mathfrak{T}_{\eta,M}(r) \geq 2^{-\kappa}\mathfrak{A}_{\eta,M}$ forces the same conclusion. No claim of equality at arbitrary nondyadically related widths is needed. □

Remark 9.2 (The remaining estimate is uniform integrability). At any one fixed count, local integrability of v^η implies that its potential is finite for almost every roof and its truncated potential tends to zero there. This fact alone says nothing about (9.1). The charge retains the essential supremum and collision limsup before removing the microscopic width. The finite-count envelope of Proposition 6.3 is one sufficient way to prove that charge zero; it is not an additional theorem about the Lorentz source. In particular choosing a smaller outer reconstruction width does not pay a nonzero charge.

Theorem 9.3 (The endpoint after residence and boundary transfer)**.** *For the unchanged scalar positive remainder, every fixed $0 < \eta < 1/2$, all sufficiently large fixed B and sufficiently small previously fixed $r > 0$,*

$$\begin{aligned}\mathcal{E}_M \leq C_M B^{-1/192} + C\mathfrak{J}_M(r) + C\mathfrak{K}_M(r) + C_\eta\mathfrak{A}_{\eta,M}\\ + C_\eta r^{1/32} + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.\end{aligned} \tag{9.5}$$

Thus the original pointwise assertion follows if the canonical microscopic charge is zero and the paired nongrazing and short critical surface heights tend to zero. These conditions have not been inferred from local mass or weak-L^q bounds.

Proof. The source-level comparison in the proof of Theorem 6.2, prior to separating its dyadic tail, gives

$$\mathcal{E}_M \leq C_M B^{-1/192} + C\mathfrak{F}_M^{\mathrm{all}}(r) + C_\eta\mathcal{H}_M(\mathsf{P}_{\eta,r}v^\eta) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.$$

Apply (8.9) and (9.4). Since $1-\kappa > 1/2 > 1/32$ and $r < 1$, the explicit potential contribution is bounded by $C_\eta r^{1/32}$. This proves (9.5).

For the stated sufficient conclusion choose $r(B) = c\varepsilon(B)^\gamma$ with fixed $c > 0$ and $\gamma > 2$. At each B the width, its positive cutoff $c_{r(B)}$, and all auxiliary bands are fixed before the collision limit. Enlarge the auxiliary bands afterwards and take $B \to \infty$ last. The three proved explicit errors vanish, and the three stated conditions pay the remaining nonnegative quantities. No arithmetic class has been removed or normalized in this argument. □

The controlled surface height in Theorem 7.2 and the controlled optical transfer in Theorem 8.2 are genuine estimates on the original physical source. The unresolved geometric quantities in (9.5) are now supported on maximal regular segments of length at most $2r$, either with two nongrazing true ends or with outgoing gradient of order $\sqrt{r}$. The original pointwise theorem, rather than a local-norm substitute, remains the governing assertion. Scalar closure must still precede all same-roof path and likelihood normalizations.

10. The original scalar objective and downstream normalization

The governing assertion remains (2.5). The long-residence surface and the short noncritical optical transfer now have proved essential-height bounds. Their complements are not removed: the paired nongrazing trace and short critical surface remain inside the same exact-label norm. The latter has the quantitative local-mass bound (8.8), which is not its height bound. The canonical microscopic charge in (9.1) requires a separate vanishing estimate; dyadic outer-scale rigidity shows precisely why reducing the reconstruction width cannot replace that estimate.

The only active dyadic convention in this article is the offset formula (6.1). The retained module 189 is marked as superseded at its point of inclusion in the supplement, and its original

bytes remain in the historical source. Its ambiguous integration notation is not an active theorem used here.

After scalar closure, the positive path-remainder equality in Supplement Theorem 76.1 supplies the same-roof path numerator. Ratios are formed only on positive arithmetic reference classes; the actual-return clock and forward likelihood retain their separate inherited hypotheses. The unchanged scalar-first order is not reversed by any surface comparison.

Appendix A. Physical kernels, first-exit traces and analytic dependencies

This appendix supplies the measure-theoretic interfaces used by the new arguments in one convention. It does not deduce a surface estimate from a volume multiplier theorem. The geometric and spectral estimates whose proofs are retained in the technical supplement are listed separately.

A.1. **Normalization and common kernels.** On a regular word use endpoint angles $q = (\alpha_0, \alpha_m)$ and oriented internal arclengths. Endpoint first variation gives $F_{\alpha_0} = -Rp_0$ and $F_{\alpha_m} = Rp_m$. Consequently

$$\omega = |\partial_{\alpha_m} p_0| = |F_{\alpha_0\alpha_m}|/R, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\, dq.$$

In the optical Schur complement the internal Hessian is $D_c A D_c$, where A has diagonal $t_{i-1} + t_i + 2/(Rc_i)$ and off-diagonal $-t_i$. The two angle-to-arclength factors are R^2; division by R in the first variation gives $\omega = Rc_0c_m/y_m$. For one flight $y_1 = \ell_0$, and for two flights $y_2 = \ell_0+\ell_1+2\ell_0\ell_1/(Rc_1)$. Changes of tangent orientation do not change these absolute identities.

Away from $g = 0$, the coordinates (F, ϑ) have Jacobian $\chi = e^{\perp\mathsf{T}} K e^{\perp}$ in absolute value, where $e = g/|g|$. Thus the angular expression for the original fiber is

$$d\mu_t = \frac{\omega(t,\vartheta)}{4\pi c_* \chi(t,\vartheta)}\, d\vartheta. \tag{A.1}$$

The sign of an oriented endpoint chart is irrelevant; all source Jacobians in this appendix are positive. The original source already contains its initial and terminal tests and is normalized only by c_*.

Proposition A.1 (Common graph covariance)**.** *At fixed m, R, n, k, the regular exact event has jointly measurable coarea kernels for which, for almost every (t, h) and every nonnegative Borel graph weight f,*

$$\int_{S_h(t)} f(x, h, \Phi_h x) J_h(x)\, d\mu_t(x) = \int_{\Phi_h S_h(t)} f(\Phi_{-h} y, h, y)\, d\mu_{t+h}(y). \tag{A.2}$$

The same versions can be used for countably many source restrictions, positive truncations and their increasing limits. Here $S_h(t)$ is the set of source points whose forward lifetime is greater than h.

Proof. At fixed count the regular physical words are disjoint open pieces of the collision cylinder. Their true boundaries and the removed critical points have zero source area. Take a countable cover by precompact flow boxes with lifted angular coordinate. Assign each point to its first containing box to obtain a Borel disjointization. This assignment is used only for integration: its seams do not stop the physical flow.

On the open survivor graph the map $(x, h) \mapsto (\Phi_h x, h)$ is injective, smooth and has physical Jacobian J_h. Its inverse on its image is the backward flow. Since $F(\Phi_h x) = F(x) + h$, ordinary change of variables followed by disintegration over (F, h) gives (A.2) on a countable determining algebra of Borel sets. Remove the union of the resulting null sets. Uniqueness of the kernel

measures and monotone approximation extend equality to all nonnegative Borel weights outside this one joint null set. Countable disjoint summation gives the complete physical formula.

In endpoint coordinates the field is translation in F at fixed ϑ, so its endpoint Jacobian is $\chi(x)/\chi(\Phi_h x)$. Multiplication by the physical density ratio gives $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$, exactly the normalization in (A.2). Nonnegative graph selectors accommodate first-exit restrictions depending on (x, h) without new incompatible versions. Monotone convergence at fixed count permits the actual fractional insertions. No null set independent of the uncountable radius parameter is required: the ordered norm is $\sup_R \operatorname{ess\,sup}_t$. □

A.2. **True outgoing faces and exhaustion.** On a nongrazing limiting word all incidence limits are positive, so the stationary reflection equations extend smoothly to the boundary. This continuation is not asserted to remain the physical first-hit trajectory beyond that boundary. On a transverse outgoing arc write $F = b(\vartheta)$. Contraction of the area form $\omega(4\pi c_*\chi)^{-1} dF\, d\vartheta$ with ∂_F gives the positive flux in (5.4). A first-exit point is always counted together with its incident one-sided physical component. Lower-dimensional and tangent strata have zero transverse flux. Section decisions, competing obstacles and simultaneous failures are not suppressed by this description. An exit with a vanishing internal incidence belongs to the grazing class even when another condition fails there simultaneously; the remaining class is nongrazing.

Proposition A.2 (Initial mass and actual first-exit flux)**.** *For a nongrazing outgoing face, let $s = F(y) - t$ and $x = \Phi_{-s} y$. On $0 < s \le h$, $s < \tau^-(y)$, the contribution to initial weighted exit mass is*

$$\frac{W^\varepsilon(x)}{J_s(x)}\, d\zeta(y). \tag{A.3}$$

The boundary source (5.5) *has this same first-exit selection with weight one on the boundary, not with the factor in* (A.3)*. Both are defined on common almost-everywhere starting-roof versions.*

Proof. In one angular flow box put $a(u, \vartheta) = \omega(u, \vartheta)/(4\pi c_*\chi(u, \vartheta))$. The initial fiber mass is $W(x)a(t, \vartheta)d\vartheta$, whereas the outgoing flux is $a(b(\vartheta), \vartheta)d\vartheta$. The ratio of the latter density to the former unweighted density is $J_s(x)$. This proves (A.3). The backward-lifetime condition says precisely that the segment between the starting roof and this boundary has remained in that one-sided event; a later re-entry does not satisfy that condition.

Use precompact regular flow boxes, with rational rectangles in (F, ϑ), and increasing finite unions whose interiors cover the regular event. On overlaps within a physical component their coordinates agree up to a lifted angular constant and can be refined into finitely many pieces. Roof-constant faces are transverse; angle-constant faces have zero flux. The outside source plus the first-exit source for each such domain is the integral of the indicator that the whole forward segment is not contained in the domain. These indicators decrease to the indicator of a true exit. Every compact surviving segment eventually lies in an exhaustion domain. The usual endpoint convention at lifetime exactly h does not change the joint almost-everywhere formula.

For actual nongrazing faces, exhaust the regular one-sided arcs and their backward segments by such compact pieces. Disjoint assignment by the incident component and first-exit graph removes chart multiplicity. Apply the local formula and nonnegative convergence at fixed count. These operations also justify the variation payments in Theorem 5.2. No surface limit is commuted with the collision-normalized supremum. □

A.3. **The indispensable analytic chain.** Input 3.2 records the complete estimates used by this article, rather than requiring a reconstruction from historical status files. Its common-space operator dependencies are the collision norm definitions (Supplement equations (117.1)– (117.3)), the full-record fixed-band Doeblin–Fortet and spectral estimates (Supplement Theorem 157.3),

and the finite moving-peak local inversion. The proof uses bounded transverse multipliers, not a bounded strong operator of multiplication by c^{-1} or $|g|^{-1}$.

For specificity, the joint endpoint/incidence estimate has leading factor $b^{129/128}s^{17/16}$ and finite-count cap $\min\{b\rho^{m/2}, s^2\}$. The endpoint test is the particular vector $P_b\mathbf{1}$ of strong norm $O(b^{7/8})$; that estimate is not a small operator norm on arbitrary vectors. Each continued rank-one peak is normalized using a nonzero smooth test against a physical peripheral vector, never by division by $\Pi\mathbf{1}$. The homogeneous trace estimates pass through bounded finite Cesaro approximants in the common strong space. These conventions and the two chronological characteristic pairings are proved in the retained module 186, not assumed for arbitrary L^∞ representatives.

Positive incidence layer cake integrates the cap first and yields a critical-volume remainder $b^{1/2}\rho^{m/4}$. A second layer cake against $b^{-1-\eta}$ converges at finite count only for $\eta < 1/2$, giving the actual $v^{\eta,b}$ estimate stated in Input 3.2. The leading exponent is $129/128 - \eta$. The first inverse-gradient remainder instead depends on its lower cutoff and is not used to remove that cutoff here. For the section layer only the all-depth marked-strip theorem is needed; for the incidence and clearance layers the original physical-guard local bound is used. All of these sources keep the forward occupation coefficient. Neither signed interface cancellation nor deletion of an arithmetic peak enters the new estimates.

Thus the new potential and surface theorems depend on explicitly stated volume estimates, but the surface source is bounded only after its actual first-exit graph and optical variation have been treated. The retained spectral and billiard estimates remain available for specialist verification. This appendix and native source compilation are not independent human or formal proof certification.

Part 21. Untruncated optical potentials and complete first-exit surface mass

Retained revision-83 abstract. We study the pointwise local limit problem for the full displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. Actual fractional inverse-gradient insertions yield complete optical transport estimates without an initial critical cutoff. The optical loss and genuine grazing starting mass obey local weak-L^q bounds for every fixed $2 < q < 4$, with an additional power on the critical initial source. We estimate the actual nongrazing first-exit surface assignment by paying both signs of optical variation. Together with all-depth physical margin localization, this gives a collision-normalized $O(r^{33/32})$ bound for its mass on each previously fixed roof interval of length r. The exact labels, original guard and one-sided first-exit selection are retained. A separate ordered-tail argument identifies the microscopic estimate still required for the original essential-height law; local norm and surface mass bounds are not substituted for that pointwise estimate.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U=\{|\alpha-\pi/6|<3/50,\ |p|<3/20\},\qquad Y_R^*=Y_R\cup U,\qquad c_*=\frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*)=c_*$. We always use $\nu_R^*=c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L=L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R}=\sum_{j=0}^{m-1}\mathbf{1}_{Y_R^*}\circ T_R^j$. The event is

$$\Pi_{n,k,m,R}=Y_R^*\cap\{K^{\mathrm{c}}_{m,R}=k,\ A_{m,R}=n\}\cap T_R^{-m}Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j=\sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0\le\Theta\le1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0=47/53$, $d_j=\min(j,m-j)$ and $d_j'=\min(j,m-1-j)$, set

$$H^\varepsilon=\prod_{j=0}^{m}\Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right)\prod_{j=0}^{m-1}\Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right),\qquad W^\varepsilon=1-H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t=L_{\#}(W^\varepsilon\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^m B_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The complete bounds proved here.** The two positive transport mechanisms will be treated at the level of the full original source. Let O_h^ε denote positive optical loss on survivors, $E_h^{\varepsilon,\mathrm{gr}}$ the starting mass of genuine grazing first exits, and $\mathcal{F}_h$ the nongrazing outgoing first-exit flux assigned to the starting roof. Their precise definitions are given below and in (5.5). A bar denotes h-averaging over $(0,r)$. For $q>1$ use the normalization $\mathcal{N}_q$ in (4.5).

Theorem 2.1 (Complete optical and first-exit estimates)**.** *Under the explicit inherited inputs below, for every fixed $2<q<4$ and every previously fixed $r,L>0$,*

$$\limsup_m \sup_{R,n,k}\sup_{|J|=L} \mathcal{N}_q(m^2(\bar O_r^\varepsilon+\bar E_r^{\varepsilon,\mathrm{gr}});J)\le C_q(L+r), \tag{2.6}$$

$$\limsup_m \sup_{R,n,k}\sup_{|J|=r} m^2\int_J \overline{\mathcal{F}}_r(t)\,dt \le C_q r^{33/32}\quad (0<r<r_0). \tag{2.7}$$

Both sources are untruncated in the initial endpoint gradient and in all positive internal incidences. They retain the original exact label, initial and terminal tests, and true one-sided first-exit selection.

The first assertion is proved in Theorem 4.3 with $\eta=1-2/q$, and includes sharper bounds on a critical initial neighborhood. The second is Theorem 5.3. Their proofs are complete relative to the stated inputs. They do not assert (2.5), whose remaining essential-height estimate is formulated in Theorem 6.2. In particular the window r is never chosen as a function of the collision count.

3. Explicit analytic and geometric inputs

We collect the indispensable inherited assertions in the exact form used by the new proofs. Their source theorems and operator dependencies are specified here and in Appendix A. The new potential and first-exit estimates do not assume a surface spectral insertion theorem or an unweighted pointwise local bound.

Input statement 3.1 (Physical flow and source comparison)**.** On the regular part Ω of each exact event, endpoint angles $q=(\alpha_0,\alpha_m)$ give $F=L$, $g=\nabla_q F=R(-p_0,p_m)$, $K=D_q^2F$ and $\vartheta=\arg g$. Put

$$\lambda_\circ=9\sqrt{19}/200,\qquad \Lambda_\circ=47/6,\qquad \omega=|\partial_{p_0}\alpha_m|^{-1},\qquad \chi=|\det D_q(F,\vartheta)|.$$

The bounds $\lambda_\circ I\le K\le\Lambda_\circ I$ and $\lambda_\circ\le\chi\le\Lambda_\circ$ hold uniformly. The explicit field

$$X_q=\frac{K^{-1}g}{g^{\mathsf T}K^{-1}g}$$

is stopped at true physical and section-decision boundaries, not at chart seams. Its flow satisfies $XF=1$, $X\vartheta=0$ and

$$2\lambda_\circ s\le |g(\Phi_s x)|^2-|g(x)|^2\le 2\Lambda_\circ s. \tag{3.1}$$

Its endpoint-curve length is at most $C\sqrt{s}$. For $I_m=1+\sum_{i=1}^{m-1}q_0^{d_i}/c_i$ one has $|X\omega|\le C\omega I_m/|g|$ on regular survivors. At a genuine grazing exit some internal incidence tends to zero and ω tends to zero. Forward interior critical termination cannot occur.

Every starting point with forward lifetime at most r belongs to a layer of one of the following forms, at an actual collision or flight:

$$c_j\le C_E q_0^{d_j}\sqrt{r},\quad \Delta_j\le C_E q_0^{d_j'}\sqrt{r},\quad \operatorname{dist}(T_R^j x,\partial Y_R^*)\le C_E q_0^{d_j}\sqrt{r}. \tag{3.2}$$

The physical coarea covariance, proved in the appendix in the convention used here, has Jacobian $J_h=(\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$. Writing E_h^ε for all weighted starting exit mass and

$$O_h^\varepsilon(t)=\int_{S_h(t)} W^\varepsilon(x)[1-\omega(\Phi_h x)/\omega(x)]_+\,d\mu_t(x),$$

the complete source satisfies, for each fixed r and almost every t,

$$b^{\varepsilon}(t) \le 2A_r b^{\varepsilon}(t) + \bar{E}_r^{\varepsilon}(t) + \bar{O}_r^{\varepsilon}(t) + C_\Theta(1+\varepsilon^{-1})\sqrt{r}\, A_r p(t), \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(s)\, ds. \tag{3.3}$$

These are the endpoint-field, all-contact motion, uniform cubic-action and separated-volume results in retained modules 174, 177, 180–182. The guard term in (3.3) is the complete product guard, not a protective subword source. The derivative and continuation assertions are confined to regular physical trajectories.

Input statement 3.2 (Complete local source estimates)**.** For every fixed auxiliary band B_*, constants $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ give the following bounds uniformly in radius, exact labels, interval J and mark $0 \le j \le m$. Let $S_{R,s}$ be the union of strips of width Cs about the finitely many coordinate lines supporting ∂Y_R^*. For s small,

$$m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = n, L_m \in J, T_R^j x \in S_{R,s}\} \le Cs^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \rho_{B_*}^{m/2}, \tag{3.4}$$

$$m^2 \int_J b^{\delta}(t)\, dt \le C\delta^{1/16}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3 \rho_{B_*}^{m/2}. \tag{3.5}$$

The local extension to a fixed remaining compact range of δ is included. At auxiliary parameter one $W^1 = 1$, so $b^1 = p$.

Put $z = |g|$, $\alpha = 129/128$ and, for $0 < \eta < 1/2$, define the actual nonnegative source measures

$$\begin{aligned} v^{\eta,b}(t)\, dt &= L_{\#}(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_\Pi \nu_R^*), \\ v^{\eta}(t)\, dt &= L_{\#}(I_m z^{-\eta} \mathbf{1}_\Pi \nu_R^*). \end{aligned} \tag{3.6}$$

They are finite and absolutely continuous at every fixed count, and for $0 < b < b_1$,

$$m^2 \int_J v^{\eta,b}(t)\, dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|)m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \tag{3.7}$$

Adding the bounded-gradient part gives, for every fixed $L > 0$,

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta} \le C_\eta L, \qquad \limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J v^{\eta,b} \le C_\eta b^{\alpha-\eta} L. \tag{3.8}$$

Every auxiliary band is fixed before the collision limit and enlarged only afterwards; leading constants are independent of that band and of the previously fixed interval length.

Equation (3.4) is Supplement Theorem 175.3. The physical source bound (3.5) is Supplement Theorem 179.1, in particular its finite-window estimate. The fractional bounds are the positive layer-cake conclusions of retained module 187, stated in full above. The joint spectral and common-space assumptions used there are recorded in the appendix. No estimate for an untruncated first inverse gradient or for a singular roof convolution is part of this input.

Input statement 3.3 (The scalar positive remainder)**.** For large fixed B, with $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{3.9}$$

The source b^{ε} is nonnegative and has a compactly supported BV density at fixed count. No collision-uniform variation bound is assumed. Supplement Theorem 181.4 gives this remainder with the same complete finite-peak reference.

4. Untruncated optical transport and fractional potentials

Fix one of the original exact events. The field, optical density and coarea kernels in Input 3.1 are used throughout. No new field is chosen from the unknown roof density. Put $z = |g|$. For $0 < \eta < 1/2$ set

$$\kappa_\eta = \frac{1-\eta}{2}, \qquad Q_\eta = \kappa_\eta^{-1}, \qquad \mathsf{P}_{\eta,r} v(t) = \int_0^r s^{-\kappa_\eta} v(t+s)\, ds. \tag{4.1}$$

In particular $2 < Q_\eta < 4$. The source v^η below is the *actual*, untruncated source with insertion $I_m z^{-\eta}$ from Input 3.2; it is not a limit taken after the collision limit.

Write $E_h^{\varepsilon,\mathrm{gr}}$ for the starting mass of genuine grazing first exits and O_h^ε for the positive inverse-twist loss on survivors. These are disjoint classes of initial states. If $b > 0$, a subscript $< b$ restricts the initial source to $0 < z < b$. A bar denotes the average $r^{-1}\int_0^r \cdot\, dh$.

Theorem 4.1 (Optical transport without an initial lower cutoff)**.** *For almost every (t, h),*

$$O_h^\varepsilon(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \le C_\eta \mathsf{P}_{\eta,h} v^\eta(t). \tag{4.2}$$

The right side is locally integrable in t at every fixed count. If $0 < h \le r$ and $b_r = (b^2 + 2\Lambda_\circ r)^{1/2} < b_1$, then

$$(O_h^\varepsilon + E_h^{\varepsilon,\mathrm{gr}})_{<b}(t) \le C_\eta \mathsf{P}_{\eta,r} v^{\eta,b_r}(t). \tag{4.3}$$

The same bounds hold for the h-averaged sources. All internal incidences, all actual words and both section tests are retained.

Proof. Along a regular survivor, the optical sensitivity and radial speed give

$$|X\omega| \le C\omega I_m / z, \qquad z(\Phi_s x)^2 \ge z(x)^2 + 2\lambda_\circ s.$$

Consequently

$$|X\omega|(\Phi_s x) \le C_\eta s^{-\kappa_\eta} \omega(\Phi_s x) I_m(\Phi_s x) z(\Phi_s x)^{-\eta}. \tag{4.4}$$

For a surviving state, $[\omega(x) - \omega(\Phi_h x)]_+$ is bounded by the negative variation up to h. At a grazing first exit the limiting optical density is zero; hence its entire positive starting optical weight is bounded by negative variation up to the lifetime. This uses the one-word optical identity only to identify the zero limit, not to sum wordwise constants.

In an angular flow box the initial fiber density is $\omega(x)/(4\pi c_* \chi(x))\, d\vartheta$. Multiplying by $W^\varepsilon(x) \le 1$ and using

$$\frac{\omega(\Phi_s x)}{\omega(x)} = J_s(x) \frac{\chi(\Phi_s x)}{\chi(x)} \le C J_s(x)$$

transfers the integrand in (4.4) to a subset of the complete target fiber. The common-kernel covariance in Appendix A justifies this simultaneously for the disjoint survivor and first-exit selections. Tonelli then proves (4.2). Nonnegative truncation at fixed collision count removes all compact-chart restrictions.

For $z(x) < b$ the radial upper bound gives $z(\Phi_s x) < b_r$ before $h \le r$, proving (4.3). The kernel and sources are nonnegative, so averaging in h preserves the bounds. Finally $\kappa_\eta < 1$ and v^η is locally integrable with compact support. Fubini proves local integrability of each displayed potential. Neither an untruncated first inverse-gradient moment nor a pointwise bound for v^η has been used. □

For $Q > 1$, a measurable interval J and $f \ge 0$, define

$$\mathcal{N}_Q(f; J) = \sup_{E \subset J,\ 0 < |E| < \infty} |E|^{-1+1/Q} \int_E f(t)\, dt, \tag{4.5}$$

where the supremum is over Lebesgue-measurable sets. This is a specific weak-L^Q norm; the definition fixes its normalization.

Lemma 4.2 (A localized potential bound)**.** *For every nonnegative locally integrable v, every $r > 0$, and any interval J, write $J_r = J + (0, r)$. Then*

$$\mathcal{N}_{Q_\eta}(\mathsf{P}_{\eta,r} v; J) \le \frac{1}{1-\kappa_\eta} \int_{J_r} v(y)\, dy. \tag{4.6}$$

The inequality also holds for extended nonnegative quantities.

Proof. For fixed y and a measurable set E of length a, the decreasing kernel has

$$\int_E (y-t)^{-\kappa_\eta} \mathbf{1}_{\{0<y-t<r\}}\, dt \le \int_0^{\min(a,r)} s^{-\kappa_\eta}\, ds \le \frac{a^{1-\kappa_\eta}}{1-\kappa_\eta}.$$

The first inequality follows directly by integrating the measures of the kernel's superlevel intervals. Integrate against $v(y)$ and use Tonelli. Only $y \in J_r$ contributes. Divide by $a^{1-\kappa_\eta}$ and take the supremum over E. □

Set $D_r^\varepsilon = \bar{O}_r^\varepsilon + \bar{E}_r^{\varepsilon,\mathrm{gr}}$. All norms in the next theorem are taken before the collision limit.

Theorem 4.3 (Complete optical source bounds at the collision scale)**.** *For each previously fixed $r, L > 0$ and $0 < \eta < 1/2$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q_\eta}(m^2 D_r^\varepsilon; J) \le C_\eta (L + r), \tag{4.7}$$

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} \mathcal{N}_{Q_\eta}(m^2 D_{r,<b}^\varepsilon; J) \le C_\eta b_r^{\alpha-\eta}(L + r) \quad (b_r < b_1). \tag{4.8}$$

Here $\alpha = 129/128$. For $1 \le p < Q_\eta$, the left sides may be replaced by the corresponding $L^p(J)$ norms, provided the right sides are multiplied by $(Q_\eta/(Q_\eta - p))^{1/p} L^{1/p-1/Q_\eta}$. No essential-supremum conclusion is included.

Proof. Combine Theorem 4.1 with Lemma 4.2, then apply the fixed-width bounds for the actual sources v^η and v^{η,b_r} in Input 3.2 on the interval of length $L + r$. Its length and b_r are fixed before the collision limit. The auxiliary band is enlarged only afterwards.

If $\mathcal{N}_Q(f; J) \le A$, applying its definition to $\{f > \lambda\} \cap J$ gives $|\{f > \lambda\} \cap J| \le \min\{L, (A/\lambda)^Q\}$. Integration of this bound against $p\lambda^{p-1} d\lambda$ proves $\|f\|_{L^p(J)} \le (Q/(Q-p))^{1/p} A L^{1/p-1/Q}$. Apply it at each finite count before taking the limits. □

For example, $\eta = 1/4$ gives $Q_\eta = 8/3$ and $\alpha - \eta = 97/128$. On intervals of length r, the complete optical norm is $O(r)$; the part with initial $z < \sqrt{r}$ is $O(r^{1+97/256})$. The estimates concern the untruncated original source. They do not permit $p = \infty$, $\eta = 1/2$, or a collision-dependent roof width. Their role in the original pointwise problem is made explicit in Section 6.

5. The complete first-exit surface source

We now estimate a quantity defined on the actual outgoing surface, not a bounded multiplier substituted for that surface. The estimate uses both a complete starting-source bound and a two-sided payment of optical variation. It is local in roof mass; the distinction from pointwise height will be kept explicit.

Let $\tau(x)$ be the forward lifetime in the original regular exact event and put

$$e_h(t) = \int \mathbf{1}_{\{\tau(x) \le h\}}\, d\mu_t(x). \tag{5.1}$$

This is the *unweighted* starting mass; it keeps both section tests and the exact label. There is no renormalization after removal of a boundary layer.

Theorem 5.1 (All true exits in a roof interval)**.** *There is $r_0 > 0$ such that, for every fixed auxiliary band B_*, all $0 < r < r_0$, $m \geq 2$, and intervals J,*

$$m^2 \int_J e_r(t)\,dt \leq Cr^{1/32}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^3\rho_{B_*}^{m/2}. \tag{5.2}$$

The constants are uniform in R, n, k. Consequently, for fixed r, L,

$$\limsup_m \sup_{R,n,k\,|J|=L} m^2 \int_J e_r(t)\,dt \leq CLr^{1/32}. \tag{5.3}$$

The same conclusion holds for the h-average of e_h.

Proof. The all-contact motion estimate places every finite exit of lifetime at most r in the union of the starting layers

$$c_j \leq C_E q_0^{d_j}\sqrt{r}, \qquad \Delta_j \leq C_E q_0^{d'_j}\sqrt{r}, \qquad \operatorname{dist}(T_R^j x, \partial Y_R^*) \leq C_E q_0^{d_j}\sqrt{r}.$$

The union includes every contact and flight, and all initial, internal and terminal decisions. A positive-margin endpoint would continue the physical word and field; a regular critical point cannot be a forward exit. Coordinate seams are not stopping faces.

The first two layer unions are contained in the set on which the original guard with auxiliary parameter $C\sqrt{r}$ has value zero: $q_0^d \leq q_0^{d/4}$. Their source is therefore bounded by the complete physical bad source in Input 3.2. For the section layers use the marked transverse-strip estimate there at the actual collision time j, with width $C_E q_0^{d_j}\sqrt{r}$. Each positive upper comparison drops endpoint tests only on its upper side and costs exactly c_*^{-1}; the occupation and displacement coefficient remain (n, k).

There are at most two contact marks of each depth. The leading section terms sum as $Cr^{1/32}\sum_{d\geq 0} q_0^{d/16}$, and the physical guard term has the same power of r. The at most $m + 1$ section remainders, together with the physical remainder, are bounded by the second line of (5.2). Thus no uncontrolled word count or sum of collision-dependent bandwidths occurs. Take the collision limit with r, L, B_* fixed and then enlarge B_*. Finally $e_h \leq e_r$ for $h \leq r$. □

5.1. **The physical surface measure and its starting-roof assignment.** Use one-sided nongrazing outgoing faces of the original exact event. On a transverse angular chart a face is $F = b(\vartheta)$, and its physical flux is

$$d\zeta = \frac{\omega(b(\vartheta), \vartheta)}{4\pi c_* \chi(b(\vartheta), \vartheta)}\,d\vartheta. \tag{5.4}$$

A countable disjoint refinement counts each face with its incident physical component. Tangential strata have zero transverse flux. Let $\tau^-(y)$ be the backward lifetime in that same one-sided event. Define

$$\mathcal{F}_h(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<L(y)-t\leq h,\ L(y)-t<\tau^-(y)\}}\,d\zeta(y). \tag{5.5}$$

There is no initial lower bound on $|g|$. The selection in (5.5) excludes later re-entries. Competing-hit, section and other true exact-event faces remain present. This is the increasing limit of the original $\mathcal{F}_h^a$ as $a \downarrow 0$ at fixed count, on the common versions of Appendix A. The measure is initially allowed to be extended nonnegative.

Theorem 5.2 (Two-sided optical payment for the actual flux)**.** *For every $0 < \eta < 1/2$, jointly for almost every (t, h),*

$$\mathcal{F}_h(t) \le Ce_h(t) + C_\eta \mathsf{P}_{\eta,h} v^\eta(t), \tag{5.6}$$

$$E_h^\varepsilon(t) + O_h^\varepsilon(t) \le C\mathcal{F}_h(t) + C_\eta \mathsf{P}_{\eta,h} v^\eta(t). \tag{5.7}$$

In particular the full outgoing assignment is locally integrable in the starting roof at fixed count. The first inequality is a bound for the surface source itself, not just for its pullback cost.

Proof. Along a nongrazing first-exit orbit write $w(s) = \omega(\Phi_s x)$ and let $\tau \le h$ be its lifetime. In the angular parametrization, the starting fiber density is $w(0)/(4\pi c_* \chi(x))$ and the exit flux density is $w(\tau-)/(4\pi c_* \chi(\Phi_\tau x))$. Both limiting factors are positive at a nongrazing exit. The two inequalities

$$w(\tau-) \le w(0) + \int_0^\tau [w'(s)]_+ \, ds, \qquad w(0) \le w(\tau-) + \int_0^\tau [-w'(s)]_+ \, ds$$

are used in their respective directions. The uniform upper and lower bounds for χ compare their fiber normalizations.

For (5.6), sum the first inequality on the actual first-exit graph. Its starting term is at most Ce_h. Pay its positive variation by (4.4) and common *target-direction* coarea covariance. This yields the potential term. A boundary point is counted with precisely the initial states selected in (5.5); no multiplicity from a later return is introduced.

For (5.7), use the second inequality, with $W^\varepsilon(x) \le 1$, for nongrazing exits. At grazing exits $w(\tau-) = 0$ and negative variation pays the full starting weight. On survivors it pays the positive optical loss. These classes are disjoint, so their common target insertion is dominated by the full v^η source, not a word-cardinality multiple of it. Truncate to compact regular face pieces and flow boxes, and pass to the full nonnegative sums at fixed count. Appendix A gives the joint versions and first-exit identification. Local integrability follows from $e_h \le p$ and Theorem 4.1.

The exact pullback identity still contains the inverse forward Jacobian. The two variation inequalities account for its cost; they do not replace that Jacobian by one. □

Theorem 5.3 (A complete local-mass estimate for outgoing first exits)**.** *For every previously fixed $0 < r < r_0$, $L > 0$ and $0 < \eta < 1/2$,*

$$\limsup_m \sup_{R,n,k} \sup_{|J|=L} m^2 \int_J \overline{\mathcal{F}}_r(t) \, dt \le CLr^{1/32} + C_\eta (L + r) r^{(1+\eta)/2}. \tag{5.8}$$

In particular, on roof intervals of length r, the right side is $O_\eta(r^{33/32})$. Every initial momentum and every nongrazing one-sided first-exit component is included. This is an estimate of local surface-assignment mass, not an essential-height estimate.

Proof. Average (5.6) and use $e_h \le e_r$ and $\mathsf{P}_{\eta,h} v^\eta \le \mathsf{P}_{\eta,r} v^\eta$. Fubini gives

$$\int_J \mathsf{P}_{\eta,r} v^\eta(t) \, dt \le \frac{r^{1-\kappa_\eta}}{1-\kappa_\eta} \int_{J+(0,r)} v^\eta(y) \, dy.$$

Use (5.3) for the first term and the fixed-width bound for the actual v^η source for the second. All intervals have fixed lengths before the collision limit. Since $1 - \kappa_\eta = (1+\eta)/2 > 1/32$, setting $L = r < 1$ proves the last assertion. The surface estimate has been proved through its first-exit graph and positive optical variation; no theorem for a bounded volume multiplier is being relabeled as a trace theorem. □

6. The ordered microscopic tail and the original pointwise law

The preceding estimates give complete source and surface controls in specified local norms. The original pointwise target is stronger. We now state the remaining implication on that target, and separate the microscopic tail before any interchange of limits could occur.

For each exact label retain the actual source v^η. For $r > 0$, put $\ell_j = 2^{-j}r$ and define

$$T_{\eta,r,N}(t) = \sum_{j=N}^{\infty} \ell_j^{-\kappa_\eta} \int_{\ell_j/2}^{\ell_j} v^\eta(t+s)\, ds. \tag{6.1}$$

The integration variable s is an offset from the starting roof. Equivalently one integrates $v^\eta(u)\, du$ over $(t + \ell_j/2, t + \ell_j]$; there is no second translation of the roof. All terms and the entire tail are formed at finite collision count.

Lemma 6.1 (An ordered decomposition, including the full tail)**.** *Pointwise as extended nonnegative quantities,*

$$T_{\eta,r,0}(t) \le \mathsf{P}_{\eta,r} v^\eta(t) \le 2^{\kappa_\eta} T_{\eta,r,0}(t). \tag{6.2}$$

Define

$$\mathfrak{T}_{\eta,M}(r) = \lim_{N\to\infty} \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} T_{\eta,r,N}(t), \tag{6.3}$$

$$\mathfrak{P}_{\eta,M}(r) = \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \mathsf{P}_{\eta,r} v^\eta(t). \tag{6.4}$$

The first limit exists monotonically in $[0,\infty]$*, and*

$$\mathfrak{T}_{\eta,M}(r) \le \mathfrak{P}_{\eta,M}(r) \le C_\eta r^{(1+\eta)/2} + 2^{\kappa_\eta} \mathfrak{T}_{\eta,M}(r). \tag{6.5}$$

Proof. On $(\ell_j/2, \ell_j]$, the kernel lies between $\ell_j^{-\kappa_\eta}$ and $2^{\kappa_\eta}\ell_j^{-\kappa_\eta}$. Tonelli proves (6.2). For any fixed N, the first N interval lengths are fixed before the collision limit. Input 3.2 bounds the normalized height of their integrals by $C_\eta \ell_j$. Hence their complete finite sum is at most $C_\eta \sum_{j<N} \ell_j^{1-\kappa_\eta} \le C_\eta r^{1-\kappa_\eta}$. Keep the whole remaining nonnegative tail inside the supremum and collision limsup. Only then let $N \to \infty$. Monotonicity of these tails proves existence and (6.5). This argument does not use the false reverse of Fatou's inequality for an infinite sum. □

Let the full, untruncated surface height be

$$\mathfrak{F}_M^{\mathrm{all}}(r) = \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \overline{\mathcal{F}}_r(t). \tag{6.6}$$

Its source is precisely (5.5).

Theorem 6.2 (Pointwise error with no initial-cutoff interchange)**.** *For each fixed* $0 < \eta < 1/2$*, all sufficiently large fixed* B *and all previously fixed* $0 < r < r_0$*, the original pointwise error satisfies*

$$\begin{aligned} \mathcal{E}_M \le{}& C_M B^{-1/192} + C\mathfrak{F}_M^{\mathrm{all}}(r) + C_\eta \mathfrak{T}_{\eta,M}(r) \\ &+ C_\eta r^{(1+\eta)/2} + C_\Theta(1 + \varepsilon(B)^{-1})\sqrt{r}. \end{aligned} \tag{6.7}$$

Thus the original assertion (2.5) *follows if the full surface height and the microscopic tail in this display tend to zero as* $r \downarrow 0$*. This sufficient condition is not asserted as proved by the local-norm estimates above.*

Proof. Use the complete-source shift inequality of Input 3.1 and replace its sum of true-exit and optical losses, at the source level, by the average of (5.7). The positive scalar remainder and the fixed-width bad-source bound then give

$$\mathcal{E}_M \le C_M B^{-1/192} + C\mathfrak{F}_M^{\mathrm{all}}(r) + C_\eta \mathfrak{P}_{\eta,M}(r) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.$$

Apply (6.5). Every source restriction has already been removed at fixed count, using actual fractional integrability; no first inverse-gradient cutoff is passed through a collision limit. To obtain the conditional conclusion, choose $r(B) = c\varepsilon(B)^\gamma$ with fixed $c > 0$ and $\gamma > 2$. For each B this is a fixed positive width. First take the collision limit, next enlarge auxiliary spectral bands, and take $B \to \infty$ last. Every displayed elementary error then vanishes, and the two stated conditions pay the remaining terms. The arithmetic reference and all zero classes are unchanged. □

Proposition 6.3 (A quantitative microscopic envelope)**.** *Suppose, as an additional estimate on the actual source, that for all sufficiently large m, all parameters and almost every central t,*

$$m^2 \int_0^s v^\eta(t+u)\,du \le H(s) \quad (0 < s < r_0), \qquad \int_0^{r_0} H(s) s^{-1-\kappa_\eta}\,ds < \infty, \tag{6.8}$$

with H nonnegative and nondecreasing. The exceptional set may be chosen independently of s by first using rational s and monotone limits. Then $\mathfrak{T}_{\eta,M}(r) = 0$ for every $r < r_0/2$. In particular a uniform bound $H(s) \le As^{\kappa_\eta+\delta}$, $\delta > 0$, gives the finite-count tail bound

$$m^2 T_{\eta,r,N}(t) \le \frac{Ar^\delta 2^{-N\delta}}{1-2^{-\delta}}. \tag{6.9}$$

The envelope is a uniform finite-count hypothesis, not a list of separate fixed-scale collision limsups.

Proof. Each normalized interval integral is at most $H(\ell_j)$. Moreover, monotonicity gives

$$H(\ell_j)\ell_j^{-\kappa_\eta} \le C_\eta \int_{\ell_j}^{2\ell_j} H(s) s^{-1-\kappa_\eta}\,ds.$$

The right intervals have disjoint interiors. Their tails tend to zero by the assumed integrability, uniformly before the collision limit. For the power envelope sum the geometric series directly. □

The fractional source lowers the sufficient envelope exponent from $1/2$ to $\kappa_\eta = (1-\eta)/2$, which can be any fixed number strictly between $1/4$ and $1/2$. The endpoint $1/4$ is not included. The complete surface mass bound (5.8) and the weak-L^{Q_η} optical bound do not establish (6.8) or (6.6). A uniformly small integral, even with these additional local controls, must not be substituted for an essential roof-height estimate. The original pointwise law, and scalar-first normalization of its path and likelihood consequences, remain exactly the organizing objective.

7. The scalar objective and downstream normalization

The organizing assertion remains (2.5). The complete optical norm and first-exit surface mass are now quantitatively bounded without an initial critical cutoff. They are not essential-height bounds, and the two height terms in (6.7) remain distinct proof obligations. In particular a first-exit surface is not removed by an internal signed occupation difference, and a fixed-width volume theorem is not used at a collision-dependent microscopic width.

After scalar closure, the positive path-remainder equality in Supplement Theorem 76.1 supplies the same-roof path numerator. Ratios are formed only on positive arithmetic reference classes; the actual-return clock and forward likelihood retain their stated inherited hypotheses. Nothing in the new estimates divides by an arithmetic zero class or reverses this scalar-first order.

Appendix A. Physical kernels, first-exit traces and analytic dependencies

This appendix supplies the measure-theoretic interfaces used by the new arguments in one convention. It does not deduce a surface estimate from a volume multiplier theorem. The geometric and spectral estimates whose proofs are retained in the technical supplement are listed separately.

A.1. **Normalization and common kernels.** On a regular word use endpoint angles $q = (\alpha_0, \alpha_m)$ and oriented internal arclengths. Endpoint first variation gives $F_{\alpha_0} = -Rp_0$ and $F_{\alpha_m} = Rp_m$. Consequently

$$\omega = |\partial_{\alpha_m} p_0| = |F_{\alpha_0\alpha_m}|/R, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\, dq.$$

In the optical Schur complement the internal Hessian is $D_c A D_c$, where A has diagonal $t_{i-1} + t_i + 2/(Rc_i)$ and off-diagonal $-t_i$. The two angle-to-arclength factors are R^2; division by R in the first variation gives $\omega = Rc_0c_m/y_m$. For one flight $y_1 = \ell_0$, and for two flights $y_2 = \ell_0+\ell_1+2\ell_0\ell_1/(Rc_1)$. Changes of tangent orientation do not change these absolute identities.

Away from $g = 0$, the coordinates (F, ϑ) have Jacobian $\chi = e^{\perp\mathsf{T}} K e^{\perp}$ in absolute value, where $e = g/|g|$. Thus the angular expression for the original fiber is

$$d\mu_t = \frac{\omega(t,\vartheta)}{4\pi c_* \chi(t,\vartheta)}\, d\vartheta. \tag{A.1}$$

The sign of an oriented endpoint chart is irrelevant; all source Jacobians in this appendix are positive. The original source already contains its initial and terminal tests and is normalized only by c_*.

Proposition A.1 (Common graph covariance)**.** *At fixed m, R, n, k, the regular exact event has jointly measurable coarea kernels for which, for almost every (t, h) and every nonnegative Borel graph weight f,*

$$\int_{S_h(t)} f(x, h, \Phi_h x) J_h(x)\, d\mu_t(x) = \int_{\Phi_h S_h(t)} f(\Phi_{-h} y, h, y)\, d\mu_{t+h}(y). \tag{A.2}$$

The same versions can be used for countably many source restrictions, positive truncations and their increasing limits. Here $S_h(t)$ is the set of source points whose forward lifetime is greater than h.

Proof. At fixed count the regular physical words are disjoint open pieces of the collision cylinder. Their true boundaries and the removed critical points have zero source area. Take a countable cover by precompact flow boxes with lifted angular coordinate. Assign each point to its first containing box to obtain a Borel disjointization. This assignment is used only for integration: its seams do not stop the physical flow.

On the open survivor graph the map $(x, h) \mapsto (\Phi_h x, h)$ is injective, smooth and has physical Jacobian J_h. Its inverse on its image is the backward flow. Since $F(\Phi_h x) = F(x) + h$, ordinary change of variables followed by disintegration over (F, h) gives (A.2) on a countable determining algebra of Borel sets. Remove the union of the resulting null sets. Uniqueness of the kernel measures and monotone approximation extend equality to all nonnegative Borel weights outside this one joint null set. Countable disjoint summation gives the complete physical formula.

In endpoint coordinates the field is translation in F at fixed ϑ, so its endpoint Jacobian is $\chi(x)/\chi(\Phi_h x)$. Multiplication by the physical density ratio gives $J_h = (\omega(\Phi_h x)/\omega(x))\chi(x)/\chi(\Phi_h x)$, exactly the normalization in (A.2). Nonnegative graph selectors accommodate first-exit restrictions depending on (x, h) without new incompatible versions. Monotone convergence at fixed count permits the actual fractional insertions. No null set independent of the uncountable radius parameter is required: the ordered norm is $\sup_R \operatorname{ess\,sup}_t$. □

A.2. **True outgoing faces and exhaustion.** On a nongrazing limiting word all incidence limits are positive, so the stationary reflection equations extend smoothly to the boundary. This continuation is not asserted to remain the physical first-hit trajectory beyond that boundary. On a transverse outgoing arc write $F = b(\vartheta)$. Contraction of the area form $\omega(4\pi c_*\chi)^{-1}dF\,d\vartheta$ with ∂_F gives the positive flux in (5.4). A first-exit point is always counted together with its incident one-sided physical component. Lower-dimensional and tangent strata have zero transverse flux. Section decisions, competing obstacles and simultaneous failures are not suppressed by this description. An exit with a vanishing internal incidence belongs to the grazing class even when another condition fails there simultaneously; the remaining class is nongrazing.

Proposition A.2 (Initial mass and actual first-exit flux)**.** *For a nongrazing outgoing face, let $s = F(y) - t$ and $x = \Phi_{-s}y$. On $0 < s \le h$, $s < \tau^-(y)$, the contribution to initial weighted exit mass is*

$$\frac{W^\varepsilon(x)}{J_s(x)}\,d\zeta(y). \tag{A.3}$$

The boundary source (5.5) *has this same first-exit selection with weight one on the boundary, not with the factor in* (A.3)*. Both are defined on common almost-everywhere starting-roof versions.*

Proof. In one angular flow box put $a(u,\vartheta) = \omega(u,\vartheta)/(4\pi c_*\chi(u,\vartheta))$. The initial fiber mass is $W(x)a(t,\vartheta)d\vartheta$, whereas the outgoing flux is $a(b(\vartheta),\vartheta)d\vartheta$. The ratio of the latter density to the former unweighted density is $J_s(x)$. This proves (A.3). The backward-lifetime condition says precisely that the segment between the starting roof and this boundary has remained in that one-sided event; a later re-entry does not satisfy that condition.

Use precompact regular flow boxes, with rational rectangles in (F,ϑ), and increasing finite unions whose interiors cover the regular event. On overlaps within a physical component their coordinates agree up to a lifted angular constant and can be refined into finitely many pieces. Roof-constant faces are transverse; angle-constant faces have zero flux. The outside source plus the first-exit source for each such domain is the integral of the indicator that the whole forward segment is not contained in the domain. These indicators decrease to the indicator of a true exit. Every compact surviving segment eventually lies in an exhaustion domain. The usual endpoint convention at lifetime exactly h does not change the joint almost-everywhere formula.

For actual nongrazing faces, exhaust the regular one-sided arcs and their backward segments by such compact pieces. Disjoint assignment by the incident component and first-exit graph removes chart multiplicity. Apply the local formula and nonnegative convergence at fixed count. These operations also justify the variation payments in Theorem 5.2. No surface limit is commuted with the collision-normalized supremum. □

A.3. **The indispensable analytic chain.** Input 3.2 records the complete estimates used by this article, rather than requiring a reconstruction from historical status files. Its common-space operator dependencies are the collision norm definitions (Supplement equations (117.1)– (117.3)), the full-record fixed-band Doeblin–Fortet and spectral estimates (Supplement Theorem 157.3), and the finite moving-peak local inversion. The proof uses bounded transverse multipliers, not a bounded strong operator of multiplication by c^{-1} or $|g|^{-1}$.

For specificity, the joint endpoint/incidence estimate has leading factor $b^{129/128}s^{17/16}$ and finite-count cap $\min\{b\rho^{m/2}, s^2\}$. The endpoint test is the particular vector $P_b\mathbf{1}$ of strong norm $O(b^{7/8})$; that estimate is not a small operator norm on arbitrary vectors. Each continued rank-one peak is normalized using a nonzero smooth test against a physical peripheral vector, never by division by $\Pi\mathbf{1}$. The homogeneous trace estimates pass through bounded finite Cesaro approximants in the common strong space. These conventions and the two chronological characteristic pairings are proved in the retained module 186, not assumed for arbitrary L^∞ representatives.

Positive incidence layer cake integrates the cap first and yields a critical-volume remainder $b^{1/2}\rho^{m/4}$. A second layer cake against $b^{-1-\eta}$ converges at finite count only for $\eta < 1/2$, giving the actual $v^{\eta,b}$ estimate stated in Input 3.2. The leading exponent is $129/128 - \eta$. The first inverse-gradient remainder instead depends on its lower cutoff and is not used to remove that cutoff here. For the section layer only the all-depth marked-strip theorem is needed; for the incidence and clearance layers the original physical-guard local bound is used. All of these sources keep the forward occupation coefficient. Neither signed interface cancellation nor deletion of an arithmetic peak enters the new estimates.

Thus the new potential and surface theorems depend on explicitly stated volume estimates, but the surface source is bounded only after its actual first-exit graph and optical variation have been treated. The retained spectral and billiard estimates remain available for specialist verification. This appendix and native source compilation are not independent human or formal proof certification.

Part 22. Retained critical-source criterion

Retained revision-82 abstract.
Source clarification for the retained dyadic formula. The following historical article is reproduced without editing its source. In its module 189, the intervals named $I_j(t)$ are absolute roof intervals, whereas the integrand $v(t + s)$ uses an offset. The active revision resolves this notation by integrating s from $\ell_j/2$ to ℓ_j, or equivalently $v(u)$ over $(t + \ell_j/2, t + \ell_j]$. Its full microscopic tail is formed before the collision limsup. See the active article's ordered-tail theorem; no historical smallness condition is promoted by this clarification.

We retain the pointwise local limit problem for the full displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. Joint localization of the two endpoint momenta and one internal incidence yields a superlinear gain in both widths. The proof distinguishes a small initial vector from a bounded multiplier, and treats both chronological cases at every continued arithmetic peak. A physical cap on the finite-count remainder permits all inverse-incidence scales to be integrated. The resulting complete optical source has a critical-width gain greater than one, actual untruncated fractional inverse-gradient moments, and first inverse-gradient bounds uniform in a fixed positive lower cutoff. Applied on common exact-label coarea fibers, these estimates replace the earlier optical and grazing-exit bound Cr/a by Cr, and pay the same uniform compression cost at nongrazing exits. The complete critical initial defect and outgoing nongrazing flux remain the two roof-height estimates needed for the original pointwise theorem; no exchange of critical-cutoff and collision limits is inferred.

2. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1, 0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1, 1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0, 0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1 + q^2) - 4q(1 - q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{2.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k, m, t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m - j)$ and $d_j' = \min(j, m - 1 - j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{2.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{2.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

2.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R, \vartheta)$ and $\mathsf{P}_j(R, \vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{2.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1 \le n \le m,\, k \in \mathbb{Z}^2} \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{2.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2.2. **The analytic inputs in the form used here.**

Input statement 2.1. The unchanged source (2.3) has a nonnegative, compactly supported BV density at every fixed count. For every fixed $\varepsilon, \delta > 0$, with $A_\delta f(t) = (2\delta)^{-1} \int_{t-\delta}^{t+\delta} f(s)\,\mathrm{d}s$,

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_\delta b^\varepsilon\|_\infty \le C\varepsilon^{1/16}. \tag{2.6}$$

The constant is independent of the previously fixed width. For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{2.7}$$

All auxiliary spectral bands in these statements are fixed before the collision limit and enlarged only afterwards.

These are the complete-source tame theorem, the fixed-width complete-source bound, and the positive scalar remainder theorem in the technical supplement. Their proofs and historical hypotheses are retained there. The fixed-count source uses threshold integration [15], definable monotonicity [9], and the distributional characterization of BV [13]; it does not assume definability of the original smooth guard. They are restated as inputs to distinguish them from the new geometric and transport proofs. No long-time bound on a fixed-count variation constant is part of this statement.

2.3. **Precise inherited inputs.** The fixed-width estimate is Supplement Theorem 179.1, in particular Supplement equation (179.4). It uses Supplement Theorem 178.3, on the original exact displacement and occupation coefficient. For an interval of length $2r$, division by $2r$ gives (2.6). That proof fixes the auxiliary band, takes the collision limit, and only then enlarges the auxiliary band. Its leading constant is independent of the previously fixed interval length. The extension from small margins to a fixed remaining compact range is stated there explicitly. At auxiliary parameter one, $\Theta(c_0) = 0$ since $c_0 \le 1$, so $W^1 = 1$ and $b^1 = p$. This supplies the unweighted fixed-width estimate without asserting a raw pointwise bound. The positive remainder is the inherited scalar input (2.7), stated as Supplement Theorem 181.4; no surface-current estimate is included in any of these volume-source inputs.

2.4. **The complete-source estimate proved here.** The physical endpoint variables are $q = (\alpha_0, \alpha_m)$, $g = \nabla_q F = R(-p_0, p_m)$ and $K = D_q^2 F$. The retained canonical field satisfies $XF = 1$, $X \arg g = 0$ and $|g(\Phi_s x)|^2 \ge |g(x)|^2 + 2\lambda_\circ s$ up to its true lifetime. Its physical Jacobian factors into an inverse-twist ratio and an angular Hessian ratio. The latter and the original guard already have complete-source estimates.

The remaining optical drift is bounded by $C\omega I_m/|g|$, where

$$I_m = 1 + \sum_{j=1}^{m-1} q_0^{d_j}/c_j.$$

The preceding singular-insertion argument controlled I_m but replaced $|g|^{-1}$ by a^{-1} on initial $|g| \ge a$. The new argument keeps both components of g in the spectral pairing. With $\alpha = 129/128$, it proves, for the complete original source and each fixed roof width,

$$u^b(t)\,dt = L_\#(I_m \mathbf{1}_{\{|g|<b\}} \mathbf{1}_\Pi \nu_R^*), \qquad \limsup_{m} \sup_{R,n,k} m^2 \|A_r u^b\|_\infty \le C b^\alpha.$$

Theorem 12.2 integrates this bound with its finite-count remainder. It gives an actual untruncated mixed insertion for powers $|g|^{-\eta}$ with $0 < \eta < 1/2$, and a bound for $|g|^{-1}\mathbf{1}_{\{|g|\ge a\}}$ whose leading constant is independent of the previously fixed $a > 0$. Theorem 13.1 then proves a complete Cr optical and grazing starting-mass estimate on that initial source. Nongrazing exits

retain their exact outgoing flux; only the paid pullback cost is improved. These statements keep the original target (2.5), rather than replacing it by a local law for a different record.

3. A canonical flow from the endpoint momenta

The roof field can be specified on the whole regular exact event, without choosing critical disks or endpoint caps. The particular section in (2.1) is useful here: it bounds both endpoint momenta away from grazing, even though no internal incidence or clearance is bounded away from zero.

For one regular physical word use locally lifted endpoint angles $q = (\alpha_0, \alpha_m)$, and write $F(q)$ for its reduced length. The endpoint map is locally invertible, since the optical twist $B_m = \partial_{p_0}\alpha_m$ is nonzero. Set

$$g = \nabla_q F = R(-p_0, p_m), \qquad K = D_q^2 F, \qquad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}.$$

The terminal tangential velocity has the same value before and after reflection, which fixes the sign in the second component of g. All differentiations in this section are on actual regular words.

Lemma 3.1 (Uniform endpoint-angle Hessians on the exact event)**.** *At every regular physical trajectory,*

$$R\,\mathrm{diag}(c_0, c_m) \le K \le (2R_+^2/\tau_- + R_+)I_2. \tag{3.1}$$

On the original exact event, in particular,

$$\lambda_\circ I_2 \le K \le \Lambda_\circ I_2. \tag{3.2}$$

These are pointwise quadratic-form inequalities, uniform in m, R, n, k. They assert neither convexity of a disconnected physical domain nor uniform control of derivatives of K.

Proof. At a regular trajectory let w_0 be its outgoing first-flight velocity and w_m the negative of its incoming last-flight velocity. The polygonal length, minimized over the prescribed intermediate closed disks, is supported at this trajectory by

$$F(q_*) - w_0 \cdot (Q_0(q_0) - Q_0(q_{0,*})) - w_m \cdot (Q_m(q_m) - Q_m(q_{m,*})).$$

To see this directly, the internal gradient at contact i is $-2c_i n_i$. Its scalar product with a displacement into that disk is nonnegative. The convex gradient inequality for the polygonal length therefore gives the displayed support after minimizing the internal positions. Equality holds at the physical trajectory. The internal Hessian is positive definite, so the reduced roof is smooth on its regular neighborhood. Since $Q_i'' = -Rn_i$ and $w_i \cdot n_i = c_i$, differentiating the touching support proves the lower inequality in (3.1).

For the upper inequality keep the internal contacts fixed while varying the endpoints. The transverse second variation of a flight is its squared transverse displacement divided by its length. The two endpoint speeds have norm R. Their contribution is at most $2R^2|dq|^2/\tau_-$; this bound includes the case $m = 1$, when both displacements occur on the same flight. The endpoint curvature terms contribute at most $R|dq|^2$. The reduced Hessian is the Schur complement of the internal Hessian and cannot exceed this trial quadratic form. Taking the indicated extreme radii gives the upper bound, with $2R_+^2/\tau_- + R_+ = 47/6$.

In the five rectangles of Y_R^* one has $|p| < 9/10$. For the only moving momentum centers this follows from

$$\frac{2(53/100)}{1 + (53/100)^2} + \frac{1}{200} < \frac{9}{10};$$

the remaining centers and the added rectangle satisfy the bound immediately. Both the initial and terminal states belong to this same section. Thus $c_0, c_m > \sqrt{19}/10$, which proves (3.2). □

Away from $g = 0$ define the circle-valued physical observable

$$\vartheta(x) = \arg(-p_0(x), p_m(x)) \in \mathbb{R}/(2\pi\mathbb{Z}). \tag{3.3}$$

It is not the normal angle of an auxiliary endpoint cap. The orientation of each disk is the fixed orientation already used for the collision coordinates. Put $e = g/|g|$ and

$$\chi = e^{\mathsf{T}} \operatorname{cof}(K) e, \qquad v = K^{-1} g, \qquad s = g^{\mathsf{T}} K^{-1} g. \tag{3.4}$$

Here $\operatorname{cof}(K)$ is the adjugate matrix; K is symmetric. In dimension two its eigenvalues are those of K in reverse order. In particular $\lambda_\circ \le \chi \le \Lambda_\circ$ on the exact event.

Theorem 3.2 (An explicit critical-safe forward roof field)**.** *There is a uniquely specified smooth vector field X on the regular exact event with its interior critical points removed such that*

$$XL = 1, \qquad X\vartheta = 0. \tag{3.5}$$

In endpoint-angle coordinates it is

$$X_q = \frac{K^{-1} g}{s}. \tag{3.6}$$

In any local lift of ϑ on the original collision cylinder, it is equivalently

$$X = \frac{\vartheta_p \partial_\alpha - \vartheta_\alpha \partial_p}{L_\alpha \vartheta_p - L_p \vartheta_\alpha}.$$

No source density, guard, partition, or critical radius enters its choice. Stop its flow at the first exit from the regular exact event. For a forward survivor and $r_h = |g(\Phi_h x)|$, $r_0 = |g(x)|$, one has

$$2\lambda_\circ h \le r_h^2 - r_0^2 \le 2\Lambda_\circ h, \tag{3.7}$$

$$\operatorname{length}(q([0,h])) \le \frac{r_h - r_0}{\lambda_\circ} \le \frac{\sqrt{2\Lambda_\circ h}}{\lambda_\circ}, \tag{3.8}$$

$$|p_i(\Phi_h x) - p_i(x)| \le \frac{\sqrt{2\Lambda_\circ h}}{R_-}, \qquad i = 0, m. \tag{3.9}$$

The same estimates hold up to the limiting time of a finite lifetime. A forward trajectory cannot terminate at a regular interior critical point of L. The estimates do not assert survival through a true singular, first-hit, or section-decision boundary.

Proof. Writing $g^\perp = (-g_2, g_1)$ gives $d\vartheta = (g^\perp)^{\mathsf{T}} K\, dq/|g|^2$ and hence

$$\det D_q(F, \vartheta) = \frac{g^{\mathsf{T}} \operatorname{cof}(K) g}{|g|^2} = \chi > 0. \tag{3.10}$$

The endpoint map has Jacobian B_m, so $|\det D_{(\alpha,p)}(L, \vartheta)| = |B_m|\chi > 0$. The two equations (3.5) therefore specify a unique field. Formula (3.6) satisfies them: $g \cdot X_q = 1$ and $KX_q = g/s$ is radial in g. Endpoint lifts differ by translations by multiples of 2π; local lifts of ϑ also differ by constants. Both formulas therefore agree on overlaps and define the field without gluing.

Along the flow $\dot g = g/s$. Its direction is constant and $\dot r = r/s > 0$. Since $r^2/\Lambda_\circ \le s \le r^2/\lambda_\circ$, one has $2\lambda_\circ \le d(r^2)/dh \le 2\Lambda_\circ$. This proves (3.7). Using r as the parameter,

$$\frac{dq}{dr} = K^{-1} e.$$

Integration gives (3.8), including length rather than merely endpoint displacement. The direction of g is constant, so $|g(\Phi_h x) - g(x)| = r_h - r_0$ and the first-variation formula gives (3.9). In particular r_h cannot tend to zero in forward time. On a compact regular subset away from $g = 0$ the ordinary continuation theorem applies; any remaining finite exit is a true boundary of the regular exact event. □

Proposition 3.3 (The cancellation in the physical drift)**.** *Let $T = D^3F$ and let a be a locally Lipschitz density relative to dq. Away from $g = 0$, in the almost-everywhere sense,*

$$\operatorname{div}_q(aX_q) = \frac{\nabla a \cdot v}{s} + a\left\{\frac{T[v,v,v]}{s^2} - \frac{v \cdot \nabla \log\det K}{s}\right\}. \tag{3.11}$$

On a regular neighborhood where $\|T\| \le T_0$, this gives

$$|\operatorname{div}_q(aX_q)| \le \frac{1}{|g|}\left\{\frac{\Lambda_\circ}{\lambda_\circ}|\nabla a| + \left(\frac{\Lambda_\circ^2}{\lambda_\circ^3} + \frac{2\Lambda_\circ}{\lambda_\circ^2}\right)|a|T_0\right\}. \tag{3.12}$$

In particular the constant-density quadratic part has zero divergence. The bounds on T and ∇a in this assertion are local, not collision-uniform conclusions.

Proof. Differentiation of $v = K^{-1}g$ yields $\operatorname{div} v = 2 - v \cdot \nabla \log\det K$. Also $v \cdot \nabla s = 2s - T[v,v,v]$. The two terms $2/s$ cancel upon differentiating av/s, proving (3.11). Use $|v| \le |g|/\lambda_\circ$, $s \ge |g|^2/\Lambda_\circ$, and $|\nabla \log\det K| \le 2T_0/\lambda_\circ$ for the bound. Near a regular interior critical point the uniform positive Hessian makes $|g|$ comparable to distance, so the remaining displayed singularity is locally integrable in two dimensions. This last observation concerns a local integral only. □

4. All-contact motion on the original regular source

The endpoint field controls more than the two endpoints. Its effect on every contact decays with distance from the nearer endpoint, without imposing a positive lower bound on the internal incidences. This observation permits a uniform estimate for the original smooth guard, rather than a replacement by protected words.

Keep $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$. All endpoint derivatives below use $q = (\alpha_0, \alpha_m)$; internal contact displacements use arclength. Constants depend only on the fixed physical family and, where indicated, the originally chosen function Θ.

Lemma 4.1 (An incidence-normalized contact inverse)**.** *For the optical matrix A of the retained recursion, put $D_c = \operatorname{diag}(c_1, \dots, c_{m-1})$ and $C_A = R_+/[2(1-q_0)]$. For $1 \le i, j < m$,*

$$0 \le (A^{-1})_{ij} \le C_A \min(c_i, c_j) q_0^{|i-j|}, \qquad |(D_c^{-1}A^{-1})_{ij}| \le C_A q_0^{|i-j|}. \tag{4.1}$$

Consequently, at every regular physical word, its contact positions Q_j satisfy

$$\|D_q Q_j\| \le C_Q q_0^{d_j}, \qquad 0 \le j \le m, \qquad C_Q = R_+ + \frac{2C_A R_+}{\tau_- q_0}. \tag{4.2}$$

For $m = 1$ the internal assertion is empty and the endpoint estimate still holds. These are derivatives at actual regular trajectories; no common neighborhood of physical continuation is asserted.

Proof. Write $A = D(I - P)$, with D its diagonal. The entries of P are nonnegative, vanish except at neighboring indices, and have row sums at most

$$\frac{2/\tau_-}{2/\tau_- + 2/R_+} = q_0.$$

Also $D_{jj}^{-1} \le Rc_j/2$. In the Neumann series for $(I-P)^{-1}D^{-1}$, the (i,j) entry vanishes until the power $|i-j|$. Summing the remaining geometric series gives the bound with c_j. Symmetry gives the bound with c_i, and division by $c_i > 0$ proves the second inequality. This is the same normalized inverse underlying the retained tapered-contact construction; here only its first derivative is needed, at arbitrary regular endpoints.

The internal action Hessian in oriented arclengths is $D_c A D_c$. After division of its stationary equations by the internal cosines, endpoint forcing has only the first and last entries, of absolute

sizes $t_0 c_0\, du_0$ and $t_{m-1} c_m\, du_m$. Tangent signs may change these signs but not their sizes. Thus the displacement of internal arclength s_j obeys

$$|ds_j| \le \frac{C_A}{\tau_-}(q_0^{j-1}|du_0| + q_0^{m-1-j}|du_m|).$$

Both exponents are at least $d_j - 1$, and $|du_i| = R|d\alpha_i|$. The displayed choice of C_Q bounds the resulting Euclidean operator norm and the two endpoint derivatives. The Euclidean speed of a circle parameterized by arclength is one, which proves (4.2). □

Theorem 4.2 (Uniform motion of every physical margin)**.** *There is $C_D < \infty$, independent of m, R, n, k and of every positive incidence and clearance, with the following property. Along any regular endpoint curve of length ℓ_q in the exact event and in one physical word,*

$$|c_j(x) - c_j(y)| + |p_j(x) - p_j(y)| + \mathrm{dist}_{\mathbb{S}^1}(\alpha_j(x), \alpha_j(y)) \le C_D q_0^{d_j} \ell_q, \tag{4.3}$$

$$|\Delta_j(x) - \Delta_j(y)| \le C_D q_0^{d'_j} \ell_q. \tag{4.4}$$

In particular, for a forward survivor of the field in Theorem 3.2, the right sides are bounded by, respectively, $C_D C_X q_0^{d_j}\sqrt{h}$ and $C_D C_X q_0^{d'_j}\sqrt{h}$, where $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$.

Proof. Differentiate a flight direction:

$$dv_j = \ell_j^{-1}(I - v_j v_j^{\mathsf{T}})(dQ_{j+1} - dQ_j).$$

The depths of its two endpoints are at least d'_j. Its derivative is consequently bounded by $2C_Q q_0^{d'_j}/\tau_-$. The normal and tangent derivatives at a contact are bounded by $C_Q q_0^{d_j}/R_-$. Express c_j and p_j as scalar products of the appropriate unit flight velocity with these vectors. At the terminal contact use the incoming velocity, with the sign giving positive incidence; its tangential component is unchanged by reflection. Since adjacent depths differ by at most one, these formulas prove the first estimate with a fixed C_D. In particular we do not differentiate $\sqrt{1 - p_j^2}$ by dividing by c_j.

The Hausdorff displacement of a segment is no greater than the larger displacement of its two endpoints. Distance from that segment to a fixed third disk is a one-Lipschitz function of this Hausdorff displacement. Taking the minimum over third disks preserves the same bound. The minimizing disk may change; differentiability of that minimum is not required. The excluded incident disks are fixed on the word. Integration along the curve proves (4.4). Finally use the endpoint-curve length estimate of Theorem 3.2. □

Corollary 4.3 (The unchanged guard is uniformly controlled)**.** *For every forward survivor in the original exact event,*

$$|W^\varepsilon(\Phi_h x) - W^\varepsilon(x)| \le \min\{1, K_\Theta \varepsilon^{-1}\sqrt{h}\}, \qquad K_\Theta = \frac{4\,\mathrm{Lip}(\Theta) C_D C_X}{1 - q_0^{3/4}}. \tag{4.5}$$

No incidence cutoff, clearance cutoff, section-distance factor or monotonicity assumption on Θ is added.

Proof. For factors in $[0,1]$, the difference of their products is bounded by the sum of their absolute differences. A contact factor in (2.2) changes by at most $\mathrm{Lip}(\Theta) C_D C_X \varepsilon^{-1} q_0^{3d_j/4}\sqrt{h}$; the analogous flight factor has depth d'_j. There are at most two contacts and two flights at any depth. Their geometric series gives (4.5). Both weights belong to $[0,1]$, proving the additional bound by one. The argument uses only the Lipschitz constant and range of the original smooth function. □

Proposition 4.4 (Where a finite forward exit can occur)**.** *Let s_j be distance from the collision state (α_j, p_j) to the boundary of Y_R^* in the product metric. There is a common C_E such that any source point whose forward lifetime in the regular exact event is at most h belongs to the union*

$$\mathcal{B}_h = \bigcup_{j=0}^{m} \{c_j \le C_E q_0^{d_j} \sqrt{h}\} \ \cup \ \bigcup_{j=0}^{m-1} \{\Delta_j \le C_E q_0^{d'_j} \sqrt{h}\} \\ \cup \ \bigcup_{j=0}^{m} \{s_j \le C_E q_0^{d_j} \sqrt{h}\}. \tag{4.6}$$

Consequently its exit mass is bounded above by $\int W^{\varepsilon} \mathbf{1}_{\mathcal{B}_h} \, d\mu_t$. This is a source inclusion, not a pointwise concentration estimate.

Proof. The finite endpoint-curve length gives limiting contact data up to a finite lifetime. A forward trajectory cannot reach an interior critical point, by Theorem 3.2. If all limiting incidences, clearances and distances to section edges were positive, the regular reflection equations, strict first-hit tests and locally constant section decisions would continue the same physical word and exact event. The endpoint Jacobian remains nonzero there, so the roof field also continues, a contradiction. Thus some listed margin tends to zero. Apply Theorem 4.2, including the one-Lipschitz property of distance to a closed edge set. Taking the union over the possible margins and retaining the original weight proves the mass bound. A section edge is charged whether or not another simultaneous change might preserve the occupation; this can enlarge the upper bound but cannot omit a true exit. □

5. A uniform cubic estimate for the reduced action

The second derivatives of the internal contact map need not be uniformly bounded near grazing. They are not needed to estimate the third derivative of the reduced roof. Stationarity cancels them. Throughout this section derivatives are evaluated at actual regular physical trajectories; no continuation through a singularity is made.

Lemma 5.1 (Cubic differentiation of a stationary action)**.** *Let $\mathcal{S}(q, z)$ be smooth and suppose that $\partial_z \mathcal{S}(q, z(q)) = 0$, with invertible internal Hessian. Set $F(q) = \mathcal{S}(q, z(q))$ and $U_u = (u, Dz(q)u)$. Then*

$$D^3 F(q)[u, v, w] = D^3 \mathcal{S}(q, z(q))[U_u, U_v, U_w]. \tag{5.1}$$

The equality is pointwise and requires no bound for the inverse internal Hessian or for $D^2 z$.

Proof. The differentiated stationary equation says $D^2 \mathcal{S}[U_u, (0, \zeta)] = 0$ for every internal vector ζ. Consequently $D^2 F[u, v] = D^2 \mathcal{S}[U_u, U_v]$. Differentiate this identity in direction w. The two terms in which U_u or U_v is differentiated are contractions of $D^2 \mathcal{S}$ with an internal vector and vanish by the same stationary equation. The remaining term is (5.1). This calculation is local, where the implicit function theorem gives a smooth $z(q)$. □

Use arclength at every contact, including $s_0 = R\alpha_0$ and $s_m = R\alpha_m$, and write

$$\mathcal{S}(s_0, \dots, s_m) = \sum_{j=0}^{m-1} |Q_{j+1}(s_{j+1}) - Q_j(s_j)|.$$

The endpoints are linear functions of $q = (\alpha_0, \alpha_m)$. At a reflection the internal arclength derivative of $\mathcal{S}$ is zero. Its internal Hessian is $D_c A D_c$, with A as in (7.1); this is positive definite at every regular word. In particular Lemma 5.1 applies even when some positive incidence is arbitrarily small.

Theorem 5.2 (Collision-uniform third derivatives)**.** *Let C_Q be the explicit constant in* (4.2)*, and put*

$$C_3 = \frac{6}{\tau_-^2} + \frac{3}{R_-\tau_-} + \frac{1}{R_-^2}, \qquad T_* = \frac{16 C_3 C_Q^3}{1 - q_0^3}. \tag{5.2}$$

For every regular physical word, its reduced roof in endpoint angles satisfies

$$|D^3 F(q)[u, v, w]| \le T_* |u|\,|v|\,|w|. \tag{5.3}$$

The constant is independent of m, R, n, k, all positive internal incidences and all positive third-disk clearances. In particular, on any regular endpoint curve γ,

$$\|K(\gamma(b)) - K(\gamma(a))\| \le T_* \,\mathrm{length}(\gamma). \tag{5.4}$$

This does not assert a common continuation neighborhood for the words.

Proof. For $N(z) = |z|$ at a nonzero vector of length at least τ_-,

$$\|DN\| \le 1, \qquad \|D^2 N\| \le \tau_-^{-1}, \qquad \|D^3 N\| \le 6\tau_-^{-2}.$$

A circular arclength parametrization has derivative norms $1, R^{-1}, R^{-2}$ through order three. The chain rule for one flight therefore bounds its third derivative on contact vectors U, V, W by

$$C_3(|U_j| + |U_{j+1}|)(|V_j| + |V_{j+1}|)(|W_j| + |W_{j+1}|).$$

These are derivatives of the action on the product of circles, not third derivatives of the contact map.

For a lifted endpoint direction u, Lemma 4.1 gives $|(U_u)_j| \le C_Q q_0^{d_j} |u|$. Indeed a circle has unit speed in arclength, so its contact-position derivative has precisely the absolute value of its arclength derivative. The two contact depths of flight j are at least d'_j. Thus this flight contributes at most $8 C_3 C_Q^3 q_0^{3d'_j} |u||v||w|$ to (5.1). There are at most two flights at every depth. Summation of the geometric series proves (5.3), including $m = 1$, when there are no internal variables. Integrating $DK = D^3 F$ along γ proves (5.4). □

Corollary 5.3 (Uniform angular Jacobian variation)**.** *Let Φ_h be the canonical forward flow of Theorem 3.2. On its surviving exact-label trajectories, with $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$,*

$$|\chi(\Phi_h x) - \chi(x)| \le T_* C_X \sqrt{h}, \qquad \left|\log \frac{\chi(\Phi_h x)}{\chi(x)}\right| \le \frac{T_* C_X}{\lambda_\circ} \sqrt{h}. \tag{5.5}$$

In endpoint coordinates the Jacobian of the flow relative to dq is

$$J_h^q(x) = \frac{\chi(x)}{\chi(\Phi_h x)}. \tag{5.6}$$

The physical source Jacobian is different and is computed below.

Proof. In dimension two $\chi = (e^\perp)^{\mathsf{T}} K e^\perp$, and e is constant along the canonical flow. Apply (5.4), the forward length bound, and $\chi \ge \lambda_\circ$. The map (F, ϑ) has Jacobian χ and conjugates the flow to $(t, \theta) \mapsto (t + h, \theta)$. Ordinary change of variables proves (5.6). No estimate on a derivative of the source density was used. □

6. The angular volume estimate on the complete source

Fix one original exact event, and keep its source measure $\bar\nu = (4\pi c_*)^{-1} d\alpha\, dp$. On its regular part set

$$\omega(x) = |B_m(x)|^{-1}, \qquad d\bar\nu = \frac{\omega(q)}{4\pi c_*}\, dq, \qquad d\mu_t = \frac{1}{4\pi c_* |\nabla L|}\, d\mathcal{H}^1|_{\{L=t\}}.$$

Thus $p(t) = \mu_t(\Pi)$ and $b^\varepsilon(t) = \int W^\varepsilon\, d\mu_t$. All fibers below are those of this same event. The gradient in the coarea formula is in the original collision coordinates.

Let $S_h(t)$ be the starting points on this fiber whose canonical forward flow survives until h. Its physical area Jacobian and its optical ratio are denoted by J_h and η_h, respectively. Define three nonnegative quantities before any collision limit:

$$E_h^\varepsilon(t) = \int_{\{L=t\}\setminus S_h(t)} W^\varepsilon\, d\mu_t, \tag{6.1}$$

$$V_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - J_h]_+\, d\mu_t, \tag{6.2}$$

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - \eta_h]_+\, d\mu_t. \tag{6.3}$$

The first is the unchanged initial mass of killed trajectories. The second is the full physical volume loss. The third isolates the optical inverse-twist loss. Neither E_h nor O_h omits a physical word. We write $\bar E_r = r^{-1}\int_0^r E_h\, dh$, and similarly for V and O. All assertions use joint coarea versions in (t, h).

Theorem 6.1 (Exact Jacobian factorization and a complete angular bound)**.** *On every survivor,*

$$\eta_h(x) = \frac{\omega(\Phi_h x)}{\omega(x)}, \qquad J_h(x) = \eta_h(x)\frac{\chi(x)}{\chi(\Phi_h x)}. \tag{6.4}$$

There is a uniform C_{ang} such that, for each fixed $r > 0$,

$$|\bar V_r^\varepsilon(t) - \bar O_r^\varepsilon(t)| \le C_{\mathrm{ang}}\sqrt{r}\, A_r p(t) \tag{6.5}$$

for almost every t. Consequently, from the stated fixed-width analytic input,

$$\limsup_m \sup_{R,n,k} m^2 \|\bar V_r^\varepsilon - \bar O_r^\varepsilon\|_\infty \le C\sqrt{r}. \tag{6.6}$$

The constants are independent of ε. This is a bound for the complete source, not a per-word estimate multiplied by a word count.

Proof. Multiplying the endpoint-coordinate Jacobian (5.6) by the ratio of the physical area densities gives (6.4). Put $k_h = \chi(x)/\chi(\Phi_h x)$. The positive-part function is one-Lipschitz, so

$$|[1 - \eta_h k_h]_+ - [1 - \eta_h]_+| \le \eta_h |1 - k_h|.$$

Corollary 5.3 bounds $|1 - k_h|$ by $C\sqrt{h}$. Moreover $\eta_h \le (\Lambda_\circ/\lambda_\circ) J_h$. Coarea covariance in the target direction therefore yields

$$|V_h^\varepsilon(t) - O_h^\varepsilon(t)| \le C\sqrt{h}\int_{S_h(t)} J_h\, d\mu_t \le C\sqrt{h}\, p(t+h).$$

The target sum contains all its actual words, and $W^\varepsilon \le 1$ was used only for an upper bound. Average over $0 < h < r$ and bound the forward average of the nonnegative density by $2A_r p$. This proves (6.5).

For clarity, the unweighted input used in the last step is

$$\limsup_m \sup_{R,n,k} m^2 \|A_r p\|_\infty \le C \quad \text{for every previously fixed } r > 0.$$

It follows from Input statement 2.1: for the auxiliary parameter 1 the initial factor is $\Theta(c_0) = 0$, whence $W^1 = 1$ and $b^1 = p$. The precise inherited source propositions are pinned in Section 2.3. No raw pointwise bound for p or collision-dependent averaging width is used. □

Define the separate ordered heights

$$\begin{aligned} \mathfrak{X}_{\varepsilon,M}(r) &= \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{E}^{\varepsilon}_{r}, \\ \mathfrak{V}_{\varepsilon,M}(r) &= \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{V}^{\varepsilon}_{r}, \\ \mathfrak{O}_{\varepsilon,M}(r) &= \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{O}^{\varepsilon}_{r}. \end{aligned} \tag{6.7}$$

Equation (6.6) gives both $\mathfrak{V} \le \mathfrak{O} + C\sqrt{r}$ and $\mathfrak{O} \le \mathfrak{V} + C\sqrt{r}$ as extended nonnegative inequalities. Thus the unestimated interior volume contribution has been reduced to inverse-twist loss, with no unestimated third derivative of the reduced roof left in this comparison.

Theorem 6.2 (The unchanged pointwise error with separated defects)**.** *Under Input statement 2.1, for sufficiently large fixed B and every fixed $r > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + C_\Theta(1 + \varepsilon(B)^{-1})\sqrt{r}. \tag{6.8}$$

Here $\varepsilon(B) = A_0 B^{-1/12}$. The order is

$$B,\ \varepsilon(B),\ r,\ \textit{auxiliary band fixed};\quad m \to \infty;\quad \textit{auxiliary band} \to \infty;\quad B \to \infty.$$

In particular, separate bounds

$$\mathfrak{X}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_X} r^{\beta_X}, \qquad \mathfrak{O}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_O} r^{\beta_O} \tag{6.9}$$

with finite nonnegative q_X, q_O and positive β_X, β_O imply the original pointwise assertion. One may choose $r = c\varepsilon(B)^\gamma$ with $\gamma > \max\{2, q_X/\beta_X, q_O/\beta_O\}$. The two estimates in (6.9) *are proof obligations, not conclusions of the present theorem.*

Proof. The stopped subcoupling has density $\min\{W(x), W(\Phi_h x)J_h(x)\}$ on survivors. Its two marginals are dominated by the original weighted fibers. The elementary bound $[w - w'J]_+ \le w[1 - J]_+ + J|w - w'|$, the complete guard modulus of Corollary 4.3, and target coarea covariance give

$$b^\varepsilon(t) \le b^\varepsilon(t + h) + E^\varepsilon_h(t) + V^\varepsilon_h(t) + K_\Theta \varepsilon^{-1}\sqrt{h}\, p(t + h).$$

Average at the previously fixed width and apply (6.5). The averaged bad source is bounded at the ordered height by $2C\varepsilon^{1/16}$; the unweighted target is bounded as in the preceding proof. Inserting the positive scalar remainder gives (6.8). All inequalities precede the suprema and the collision limit. The displayed choice of γ makes each of the three explicit powers positive, proving the implication. □

Corollary 6.3 (An exact separated criterion)**.** *Under the same analytic input, the original pointwise assertion is equivalent to, for every M and every fixed $r_0 > 0$,*

$$\lim_{B\to\infty} \inf_{0<r<r_0} \left\{ \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + (1 + \varepsilon(B)^{-1})\sqrt{r} \right\} = 0. \tag{6.10}$$

Proof. Sufficiency follows by approximating the infimum before the collision limit. For necessity, the positive remainder and the pointwise assertion give ordered height at most $C_M B^{-1/192}$ for $b^{\varepsilon(B)}$. Both E^ε_h and O^ε_h are bounded by this same positive density at the starting roof. Their separate ordered heights are therefore at most this bound, uniformly in r. Take the infimum and then $B \to \infty$. This argument does not claim that the infimum is attained or select a width depending on m. □

7. Optical sensitivity and source-weighted exit flux

The two terms in (6.8) have distinct geometric mechanisms. We first give a positive, collision-uniform sensitivity formula for the optical term, and then an exact boundary formula for the initial source mass which exits. Neither formula identifies a surface insertion with a bounded volume multiplier.

7.1. **A positive optical polynomial.** For a regular word put $t_j = \ell_j^{-1}$, $V_i = 2/(Rc_i)$ and

$$A_{ii} = t_{i-1} + t_i + V_i, \qquad A_{i,i+1} = A_{i+1,i} = -t_i. \tag{7.1}$$

The matrix A is the incidence-normalized optical matrix; the actual internal arclength Hessian is $D_c A D_c$. In the Neumann estimate it is $A = D_A(I - P)$, where D_A is its diagonal and the row sums of P are at most q_0. These two diagonal normalizations are different.

Define $y_0 = 0$, $f_0 = 1$, $y_1 = \ell_0$, and

$$f_i = f_{i-1} + V_i y_i, \qquad y_{i+1} = y_i + \ell_i f_i \quad (1 \le i < m). \tag{7.2}$$

The continuant and corner Schur-complement identities give

$$y_m = \left(\prod_{j=0}^{m-1} \ell_j \right) \det A, \qquad \omega = \frac{Rc_0 c_m}{y_m}. \tag{7.3}$$

For $m = 1$ the determinant is empty and $y_1 = \ell_0$. Indeed eliminating f_i gives $y_{i+1} = (1 + \ell_i/\ell_{i-1} + \ell_i V_i) y_i - (\ell_i/\ell_{i-1}) y_{i-1}$, the continuant recurrence with the same initial values. In the mixed endpoint Schur complement all internal incidence squares cancel against $\det(D_c A D_c)$, leaving $|F_{uv}| = c_0 c_m \prod t_j / \det A$ in endpoint arclengths. The change to endpoint angles gives $\omega = R|F_{uv}|$ as stated.

Proposition 7.1 (Optical logarithmic sensitivities)**.** *The polynomial y_m has nonnegative coefficients and is of degree at most one in each of the separate variables ℓ_j, V_i. Thus*

$$a_i = V_i \partial_{V_i} \log y_m \in [0,1], \qquad b_j = \ell_j \partial_{\ell_j} \log y_m \in [0,1]. \tag{7.4}$$

At fixed R its differential on any actual regular word is

$$d \log \omega = d \log c_0 + d \log c_m + \sum_{i=1}^{m-1} a_i \, d \log c_i - \sum_{j=0}^{m-1} b_j \, d \log \ell_j. \tag{7.5}$$

In endpoint-angle coordinates, for a fixed uniform C_O,

$$\|\nabla_q \log \omega\| \le C_O \left(1 + \sum_{i=1}^{m-1} \frac{q_0^{d_i}}{c_i} \right). \tag{7.6}$$

No third derivative or sum of absolute word multiplicities occurs in this bound.

Proof. Each new V_i enters the positive recursion once, and each new ℓ_i enters once after that step. Later operations are linear in all previous variables. Induction proves positivity and separate affineness. For a positive separately affine polynomial $Y = A + Bz$, $z\partial_z \log Y = Bz/(A + Bz) \in [0,1]$. Applying this identity to each variable gives (7.4); differentiating (7.3) gives (7.5).

The endpoint incidences are bounded below by $\sqrt{19}/10$. Theorem 4.2 bounds $\|D_q c_i\|$ by $C_D q_0^{d_i}$. Also $|d\ell_j| \le |dQ_j| + |dQ_{j+1}|$ and $\ell_j \ge \tau_-$, so $\|D_q \log \ell_j\| \le 2C_Q q_0^{d'_j}/\tau_-$. The sum of these flight bounds is uniform in m. Inserting the sensitivity weights proves (7.6). The inverse-incidence terms are retained; positivity does not make their complete source sum finite. □

7.2. **A quantified regular part of the optical loss.** For $0 < \delta \le 1$ let $S_h^\delta(t) \subset S_h(t)$ consist of survivors satisfying, for every internal contact and every $0 \le s \le h$,

$$c_i(\Phi_s x) \ge \delta q_0^{d_i/2}. \tag{7.7}$$

Write $O_h^{\varepsilon,\mathrm{reg}(\delta)}$ for (6.3) restricted to this set, and $O_h^{\varepsilon,\mathrm{thin}(\delta)}$ for its nonnegative complement. This is a decomposition of the original weighted source, not a new guard or a renormalized probability law.

Theorem 7.2 (Uniform regular optical distortion)**.** *There are uniform positive c_1, C_1, C_2 such that, whenever $0 < r \le c_1\delta^2$,*

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,\mathrm{reg}(\delta)}\|_\infty \le C_1 \delta^{-1}\sqrt{r}, \tag{7.8}$$

$$\bar{O}_r^{\varepsilon,\mathrm{thin}(\delta)}(t) \le b^{C_2\delta}(t) \quad \textit{for almost every } t. \tag{7.9}$$

The first estimate uses only the stated fixed-width analytic input. The second is a pointwise source domination and is not a bound for the normalized essential height of its right side.

Proof. On the corridor, the sum in (7.6) is at most $2\delta^{-1}/(1-\sqrt{q_0})$. The forward endpoint-curve length is at most $C_X\sqrt{h}$. Hence $|\log \eta_h| \le C\delta^{-1}\sqrt{h}$. Choose c_1 so that this upper bound is at most one for $h \le r$. Then $[1-\eta_h]_+ \le C\delta^{-1}\sqrt{h}\,\eta_h$. As in the proof of Theorem 6.1, $\eta_h \le (\Lambda_\circ/\lambda_\circ)J_h$ and target coarea gives

$$O_h^{\varepsilon,\mathrm{reg}(\delta)}(t) \le C\delta^{-1}\sqrt{h}\,p(t+h).$$

Averaging first proves (7.8).

If the corridor fails at time $s \le h \le r$, the margin motion bound implies at the starting point

$$c_i(x) \le \delta q_0^{d_i/2} + C_D C_X q_0^{d_i}\sqrt{r} \le C_2\delta q_0^{d_i/2}.$$

Since $q_0^{d_i/2} \le q_0^{d_i/4}$, the corresponding factor of the original guard at auxiliary parameter $C_2\delta$ is zero. Thus $W^{C_2\delta}(x) = 1$ on the entire thin starting set. Its optical loss is bounded by its original weighted source, and $W^\varepsilon \le 1$. This proves (7.9) before averaging, hence afterwards. There are no internal contacts when $m = 1$ and the thin set is empty. □

The theorem quantitatively controls the regular optical contribution on the full exact event. It does not allow replacement of $b^{C_2\delta}$ by its fixed-width average at a starting roof. That replacement would be exactly the missing pointwise step.

7.3. **Exit mass in the correct flux convention.** The initial mass of trajectories which exit need not equal the physical flux measured at their exit. Compression between the starting roof and the exit must be retained. We state the exact formula on regular subdomains and recover the full stopped event by exhaustion.

Let D be a relatively compact regular domain in the exact event, away from $g = 0$, with piecewise smooth boundary. We take the boundary transverse to X except for pieces and lower-dimensional strata whose images in the angular coordinate have zero length. Such domains are obtained from finite flow boxes; increasing exhaustions can be formed by refining the boxes. Let Γ_D^+ denote its outward boundary, N its outward unit normal in the collision coordinates, and $\tau_D^-(y)$ the first backward lifetime from y inside D. Use the physical flux measure

$$d\zeta_D(y) = \frac{[X(y)\cdot N(y)]_+}{4\pi c_*}\, d\mathcal{H}^1(y). \tag{7.10}$$

Proposition 7.3 (First-exit flux with the initial source weight)**.** *Let $E^{\varepsilon}_{D,h}(t)$ be the weighted initial mass on $L = t$ starting in D and exiting D within h. For almost every (t, h),*

$$E^{\varepsilon}_{D,h}(t) = \int_{\Gamma_D^+} \mathbf{1}_{\{0<L(y)-t\le h,\ L(y)-t<\tau_D^-(y)\}} \frac{W^{\varepsilon}(x_{t,y})}{J_{L(y)-t}(x_{t,y})}\, d\zeta_D(y), \quad x_{t,y} = \Phi_{t-L(y)}y. \tag{7.11}$$

For an increasing regular exhaustion D_ℓ of the full regular exact event, its original true-exit mass satisfies

$$E^{\varepsilon}_{h}(t) = \lim_{\ell\to\infty} \left\{ \int_{\{L=t\}\setminus D_\ell} W^{\varepsilon}\, d\mu_t + E^{\varepsilon}_{D_\ell,h}(t) \right\}. \tag{7.12}$$

The bracketed nonnegative quantities decrease to the left side. The limit is at fixed collision count; it is not interchanged with a normalized collision limsup.

Proof. In a flow box use $(u, \theta) = (L, \vartheta)$, so that $X = \partial_u$. The physical source has area density $a(u, \theta) = \omega/(4\pi c_*\chi)$ relative to $du\, d\theta$. On a transverse outward boundary $u = b(\theta)$, its flux is $a(b(\theta), \theta)|d\theta|$. At the starting roof its coarea mass is $a(t, \theta)|d\theta|$, and

$$J_{b(\theta)-t}(x_{t,y}) = a(b(\theta), \theta)/a(t, \theta).$$

Multiplication by the initial weight proves (7.11) on the box. The backward lifetime selects the first forward exit rather than a later re-entry. Uniqueness of the flow makes the source pieces disjoint; a measurable refinement of the boxes sums them without multiplicity. The excluded boundary strata have zero angular measure and hence zero source-fiber measure. Coarea supplies the joint almost-everywhere statement.

For each starting point, declare the D_ℓ-lifetime zero when it starts outside D_ℓ. These lifetimes increase to the lifetime in the full event: every compact surviving flow segment eventually lies in some D_ℓ. The indicators of lifetime at most h therefore decrease to the corresponding full-event indicator. Dominated convergence on each finite coarea fiber proves (7.12). The exceptional (t, h) set can be chosen jointly by disintegration. Chart boundaries in the exhaustion are not declared physical; their effects disappear only through this source-level limiting identity. □

The inverse Jacobian in (7.11) is essential. Along a limiting grazing boundary the physical density may tend to zero while this inverse factor diverges. Vanishing unweighted exit flux therefore does not make the original initial exit mass zero. The formula retains grazing, competing-hit and section exits. If several margins first vanish simultaneously, a fixed ordering partitions the initial source into disjoint exit types; their sum is E^{ε}_{h}. The internal occupation-current identity in the supplement is signed. It has not been used to subtract any of these positive exit masses.

8. Homogeneity and a superlinear incidence estimate

The inverse incidences in (7.6) can be summed on the complete source. The required improvement is not a bound for multiplication by c^{-1} on the strong space. It is a stronger estimate for the real-frequency physical powers, followed by a summable, width-dependent remainder.

8.1. **The precise spectral input.** Write $Q = \mathscr{T}_{R,\xi}$ for the uncentered four-coordinate operator, with $\xi = (u_1, u_2, b, v)$ real. The densities in the collision spaces are relative to ν; the initial vector is $\mathbf{1}$, and

$$Q^j\mathbf{1}(x) = \exp\big(i\xi \cdot S_j f(T_R^{-j}x)\big), \qquad |Q^j\mathbf{1}| = 1 \tag{8.1}$$

on the regular physical realization. Here $S_j f$ denotes the full record, not the roof alone. This equality would not justify a cosine factor in the stable-curve norm.

We use the following source-pinned statements. The norm definitions are Supplement equations (117.1)–(117.3); the full-occupation Doeblin–Fortet inequality and fixed-band spectrum are

in Supplement Theorem 157.3 and the moving-peak construction used in Supplement Lemma 175.2. The incidence multiplier $M_s = M_{\mathbf{1}_{\{c<s\}}}$ satisfies

$$\|M_s h\| \le C\|h\|, \qquad |M_s h|_w \le C s^{1/8}\|h\|. \tag{8.2}$$

This is Supplement Proposition 178.2. On a fixed finite collection of peak charts one has

$$\begin{aligned} &Q^j = \lambda^j \Pi + N^j, && \|N^j\| \le C\rho^j, && \|Q^j\| \le C, \\ &\|Q^j h\| \le C\sigma^j\|h\| + C(1+j)|h|_w, && |Q^j h|_w \le C(1+j)|h|_w, \end{aligned} \tag{8.3}$$

where $\rho, \sigma < 1$. The fixed-band complement has a uniform spectral gap. The four-frequency Gaussian integral on each peak is $O(m^{-2})$. These are inherited spectral theorems, not consequences of the coarea argument.

After shrinking the neighborhoods of unit-modulus points, we may take $|\lambda| \ge r_*$ on the retained charts with

$$\frac{\log(1/r_*)}{\log(1/\sigma)} < \frac{1}{32}, \qquad \frac{\log(1/r_*)}{\log(1/\rho)} < \frac{1}{24}. \tag{8.4}$$

The removed compact portions have a spectral gap and join the complement. This is a fixed refinement, independent of m, s and of the eventual outer band. No arithmetic branch is discarded.

Lemma 8.1 (Real powers on homogeneous grazing strips)**.** *For all $j \ge 0$ and sufficiently small $s > 0$, uniformly on the retained real peak charts,*

$$|M_s Q^j \mathbf{1}|_w \le C s^{3/2}. \tag{8.5}$$

The exponent comes from the length of a homogeneous stable curve, not from replacing the density convention by Lebesgue density.

Proof. The quadratic homogeneity strips have angular widths comparable to k^{-3} and incidences comparable to k^{-2}. In the standard coordinates they lie between $\pi/2 - (k+1)^{-2}$ and $\pi/2 - k^{-2}$, or their negative counterparts. A stable graph has slope bounded away from zero and infinity. Consequently a homogeneous stable curve in the kth strip has length at most Ck^{-3}. If that strip meets $\{c < s\}$, then $k \ge C^{-1}s^{-1/2}$; the same estimate includes a threshold crossing the strip. Thus the entire tested intersection has length at most $Cs^{3/2}$. For small s, the central strips do not meet it. Equation (8.1) and the supremum bound of a weak test now prove (8.5). This proof applies to the physical branchwise realization; the bounded multiplier and physical realization in (8.2) give the same element of the weak completion. □

Lemma 8.2 (Interpolation close to the weak endpoint)**.** *On the fixed charts chosen above,*

$$\|\Pi h\| + \sup_{a \ge 0} |\ell(Q^a h)| \le C\|h\|^{1/8}|h|_w^{7/8}. \tag{8.6}$$

Moreover

$$|M_s \Pi \mathbf{1}|_w \le C s^{23/16}. \tag{8.7}$$

Proof. Normalize $\|h\| = 1$ and put $t = |h|_w$. For $0 < t < 1/2$ choose L with $\sigma^L \le t < \sigma^{L-1}$. Commutation and (8.3) give

$$\|\Pi h\| \le C r_*^{-L}\{\sigma^L + (1+L)t\} \le C(1 + |\log t|)t^{31/32} \le C t^{7/8}.$$

For $a \le L$ the weak power bound controls the terminal pairing; for $a > L$ apply the strong inequality at L and power boundedness afterwards. Both costs are at most $C(1+L)t \le Ct^{7/8}$. The cases of t bounded below and $t = 0$ follow, respectively, by power boundedness and by letting $L \to \infty$. Homogeneity gives (8.6).

For (8.7), choose instead $L = \lceil (3/2)\log(1/s)/\log(1/\rho)\rceil$. The exact identity

$$M_s\Pi\mathbf{1} = \lambda^{-L}M_sQ^L\mathbf{1} - \lambda^{-L}M_sN^L\mathbf{1}$$

uses $N^L = Q^L - \lambda^L\Pi$. Its weak norm is at most $Cr_*^{-L}(s^{3/2} + s^{1/8}\rho^L)$. Here $\rho^L \le s^{3/2}$ and $r_*^{-L} \le Cs^{-1/16}$ by (8.4). This proves the assertion. The strong norm of $M_s\Pi\mathbf{1}$ remains bounded. This argument does not assign modulus one to the continued eigenvector. □

Theorem 8.3 (A capped superlinear marked-incidence bound)**.** *There is C such that, for every fixed auxiliary roof band $B_* \ge B_0$, there are $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ satisfying*

$$\begin{aligned} &m^2\nu\{K_m^{\mathrm{c}} = k, A_m = l, L_m \in J,\ c(T_R^j x) < s\} \\ &\quad \le Cs^{5/4}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2\min\{\rho_{B_*}^{m/2}, s^2\}. \end{aligned} \tag{8.8}$$

The estimate is uniform in R, $m \ge 2$, $0 \le j \le m$, exact labels, intervals J and $0 < s < s_0$. In particular the mark and threshold may vary with m. The leading constant is independent of the previously fixed outer band.

Proof. The exact characteristic pairing is $\ell(Q^{m-j}M_sQ^j\mathbf{1})$. If the left block is long, expand that block and apply (8.6) and (8.5) to $M_sQ^j\mathbf{1}$. The leading amplitude is at most $Cs^{21/16}$. If the right block is long, its leading term is controlled by (8.6) and (8.7); its amplitude is at most $Cs^{161/128}$. Both are bounded by $Cs^{5/4}$, since $161/128 > 5/4$. The complementary term is at most $C\rho^{m/2}$. On a fixed-band complement a long block gives $C_{B_*}\rho_{B_*}^{m/2}$.

There is also the independent physical bound at every real frequency,

$$|\ell(Q^{m-j}M_sQ^j\mathbf{1})| \le \nu\{c < s\} = 1 - \sqrt{1 - s^2} \le s^2.$$

If $|z| \le A + D$ and $|z| \le V$, then $|z| \le A + \min(D, V)$. Apply this with the Gaussian leading term as A. Its four-frequency integral is $Cs^{5/4}m^{-2}$; the remainder retains the cap $\min(C_{B_*}\rho_{B_*}^{m/2}, Cs^2)$. The positive interval majorant has Fourier supremum at most $|J| + 2\pi/B_*$ and roof support in $[-B_*, B_*]$. Torus coefficient extraction and integration over that fixed band therefore give (8.8). The cap is essential: an error independent of s could not be summed against an inverse incidence. □

9. Unbounded insertions and the optical source

The cap in (8.8) permits summation all the way to zero incidence. In particular the thin component is no longer estimated by the pointwise height of another bad source.

For $0 < a < 5/4$ and $0 < s < s_0$ put $h_{a,s}(x) = c(x)^{-a}\mathbf{1}_{\{c(x)<s\}}$. The value on the collision-null grazing set is immaterial.

Theorem 9.1 (A real-frequency unbounded insertion estimate)**.** *For every fixed auxiliary band B_*, uniformly in the mark and the original exact collision labels,*

$$\begin{aligned} &m^2\int h_{a,s}(T_R^j x)\mathbf{1}_{\{K_m^{\mathrm{c}}=k, A_m=l, L_m\in J\}}\,d\nu(x) \\ &\quad \le C_a s^{5/4-a}(|J| + B_*^{-1}) + C_{B_*,a}(1 + B_*|J|)m^2\min\{s^{2-a}, \rho_{B_*}^{m(2-a)/4}\}. \end{aligned} \tag{9.1}$$

No bounded strong-space multiplication operator $M_{c^{-a}}$ is asserted. The formula concerns the actual physical insertion; all thresholds are summed before either the collision or band limit.

Proof. The positive layer-cake identity is

$$c^{-a}\mathbf{1}_{\{c<s\}} = s^{-a}\mathbf{1}_{\{c<s\}} + a\int_0^s v^{-a-1}\mathbf{1}_{\{c<v\}}\,dv.$$

First truncate the integral at a positive lower endpoint. Apply Theorem 8.3 and then monotone convergence. The leading integral is finite since $a < 5/4$. For $D \in (0,1]$ its remainder satisfies

$$s^{-a}\min(D,s^2) + a\int_0^s v^{-a-1}\min(D,v^2)\,dv \le C_a \min\{s^{2-a}, D^{1-a/2}\}. \tag{9.2}$$

For $s \le \sqrt{D}$ integrate v^{1-a} directly; for larger s split at $\sqrt{D}$. This proves the inequality for all $a < 2$ and, with $D = \rho_{B_*}^{m/2}$, proves (9.1). Physical finiteness is consistent with $\nu\{c < v\} \le v^2$ and invariance at the marked time. No infinite sum of an uncapped spectral error is taken. □

Retain $d_i = \min(i, m-i)$ and define the additive optical insertion on the original m-collision orbit by

$$I_m(x) = 1 + \sum_{i=1}^{m-1} \frac{q_0^{d_i}}{c_i(x)}, \qquad u_{n,k,m,R}(t)\,dt = L_{\#}(I_m \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{9.3}$$

There is no product over all inverse incidences in this definition. The source keeps its initial and terminal tests. For each finite count it is finite and absolutely continuous: on compact regular charts this is ordinary coarea, and the removed critical points have zero source measure. Alternatively it is dominated by a constant multiple of the complete inverse-incidence source in Supplement Theorem 97.3.

Corollary 9.2 (Complete depth-weighted source concentration)**.** *For every previously fixed $r > 0$,*

$$\limsup_m \sup_{R,n,k} m^2 \|A_r u_{n,k,m,R}\|_\infty \le C. \tag{9.4}$$

If only the summands with $c_i < s$ are retained in I_m, the right side is at most $Cs^{1/4}$. The leading constants do not depend on r, s or the collision count.

Proof. Apply (9.1) with $a = 1$ and sum with weights $q_0^{d_i}$. Their sum is at most $2q_0/(1-q_0)$. The truncated terms $c_i^{-1}\mathbf{1}_{\{c_i \ge s_0\}}$ and the constant one use the inherited unweighted concentration estimate. For each positive marked integral, dropping the endpoint tests gives an upper bound with initial measure ν/c_* and the *same* exact coefficient (k,n); its inverse-incidence weight is retained. This upper bound does not alter the source definition. No neighboring occupation labels are pooled. Take $J = [t-r, t+r]$, divide by $2r$, take the collision limsup with B_* fixed, and only then let $B_* \to \infty$. The capped remainders decay exponentially after the complete sum. The thin version follows from the same formula at threshold s. □

9.1. **The physical normalization.** The convention of (7.3) can also be checked directly from the endpoint first variation. In endpoint *angles*, $F_{\alpha_0} = -Rp_0$, and hence

$$\omega = |\partial_{\alpha_m} p_0| = \frac{|K_{0m}|}{R}, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\,d\alpha_0\,d\alpha_m, \qquad \omega = \frac{Rc_0c_m}{y_m}. \tag{9.5}$$

In internal *arclengths* the Hessian is $D_c A D_c$. The corner Schur complement cancels its internal incidence squares; the two endpoint angle changes supply R^2, and the first-variation conversion divides by R. Thus only the one factor R in (9.5) remains. For one flight $y_1 = \ell_0$; for two flights $y_2 = \ell_0 + \ell_1 + 2\ell_0\ell_1/(Rc_1)$. These agree with the direct one- and two-flight Schur complements. Tangent orientation changes conjugate the internal matrix by diagonal signs and leave these absolute identities unchanged.

The cubic cancellation in Lemma 5.1 uses these same mixed coordinates. For an internal vector ζ, differentiated stationarity gives $D^2S[(u, Dz\,u),(0,\zeta)] = 0$. Every derivative of a lifted contact direction in D^2F is purely internal, so these terms vanish before any estimate is taken. At an endpoint the lift has arclength component Ru_i; the circle's second and third arclength

derivatives have norms R^{-1} and R^{-2}. This accounts for the factors in C_3 and T_* without treating the endpoint angles as arclengths.

9.2. **All incidences in the noncritical optical estimate.** Let $g = \nabla_q F$ and let $\tau(x)$ be the forward lifetime of the canonical field. For $a > 0$, a superscript a means restriction of an *initial* source to $|g(x)| \geq a$. Define $O_h^{\varepsilon,a}$ by this restriction in (6.3). No internal incidence corridor is imposed.

A finite true exit is called grazing here when at least one internal incidence tends to zero as the lifetime is approached. All contacts have limits by the all-contact motion estimate. Assign a simultaneous failure to this grazing class whenever an incidence vanishes; every other finite exit belongs to the disjoint nongrazing class. Let $E_h^{\varepsilon,\mathrm{gr},a}$ be its weighted starting mass with $\tau \leq h$ and $|g| \geq a$.

Theorem 9.3 (Complete optical loss and genuine grazing exits)**.** *For each $a, r > 0$,*

$$O_h^{\varepsilon,a}(t) + E_h^{\varepsilon,\mathrm{gr},a}(t) \leq \frac{C}{a} \int_0^h u(t+s)\, ds, \tag{9.6}$$

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,a} + \bar{E}_r^{\varepsilon,\mathrm{gr},a}\|_\infty \leq \frac{Cr}{a}. \tag{9.7}$$

The first assertion holds jointly for almost every (t, h). Constants are independent of ε and of all positive internal incidences and clearances. The estimate includes every actual word of the original exact event with the stated initial noncritical restriction.

Proof. The field and (7.6) give, along a survivor,

$$|X\omega| \leq C \frac{\omega I_m}{|g|}. \tag{9.8}$$

The radial speed implies $|g(\Phi_s x)| \geq |g(x)| \geq a$. For a surviving orbit the positive decrease of ω up to h is bounded by the integral of $[-X\omega]_+$ over that segment. At a grazing exit, (7.3) and the selected-incidence determinant bound imply $\omega(\Phi_s x) \to 0$. This last assertion needs only a finite-count bound $\omega \leq C_m c_i$; it does not sum such constants over words. Thus the entire starting optical weight at such an exit is bounded by the same negative-variation integral up to τ. The two source cases are disjoint. Other exits contribute zero to the left.

In a local angular chart the initial fiber density is $\omega(x)/(4\pi c_* \chi(x))\, d\vartheta$. At time s, the ratio $\omega(\Phi_s x)/\omega(x)$ is at most $(\Lambda_\circ/\lambda_\circ) J_s(x)$. With $W^\varepsilon(x) \leq 1$, coarea covariance in the *target* direction bounds the integral of the drift in (9.8) by $Ca^{-1}u(t+s)$. The target image is a subset of the same exact fiber. The common-kernel construction below justifies this comparison simultaneously on all physical words and at joint almost-everywhere levels. Tonelli proves (9.6). Averaging over $0 < h < r$ gives an upper bound $2Cra^{-1}A_r u(t)$. Apply Corollary 9.2. In particular the initial mass of a grazing exit has been paid by optical drift; it has not been declared zero because its boundary optical density vanishes. □

Without the initial lower bound, the same proof gives the extended nonnegative inequality

$$O_h^{\varepsilon}(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \leq C \int_0^h s^{-1/2} u(t+s)\, ds. \tag{9.9}$$

It uses $|g(\Phi_s x)|^2 \geq 2\lambda_\circ s$. A bound for every fixed average of u is not a bound for this singular convolution at its initial roof, uniformly in m. Equation (9.9) is therefore not used to remove the critical restriction in (9.7).

10. Common fibers and the remaining boundary source

The singular insertion estimate acts on the original volume source. Its use for optical drift requires common versions of that source on all words. For a nongrazing exit it also identifies which part of the inverse-Jacobian pullback has been paid by the drift estimate. Neither step changes the exact event.

Lemma 10.1 (A common physical coarea version)**.** *At a fixed collision count, radius and exact label, the original regular source admits jointly measurable coarea kernels μ_t and a survivor graph for which*

$$\int_{S_h(t)} f(\Phi_h x) J_h(x)\, d\mu_t(x) = \int_{\Phi_h S_h(t)} f(y)\, d\mu_{t+h}(y) \tag{10.1}$$

holds as an identity of kernels for almost every (t,h) and every nonnegative Borel f. There is also a precompact regular exhaustion $D_j \uparrow \Omega$ for which the source-level first-exit formula of Proposition 7.3 holds monotonically. Auxiliary chart seams are not exits of Ω.

Proof. The regular physical words are disjoint open pieces of the collision source. Their boundaries and critical points have zero source area. Off these sets, (F,ϑ) is a local diffeomorphism with positive Jacobian χ in endpoint coordinates. Choose countably many lifted angular intervals and flow boxes whose closures lie in these regular pieces and whose (F,ϑ) images are rectangles with rational endpoints. They can be chosen to cover Ω. Disjointize the boxes by assigning a point to its first containing box. This is a Borel assignment for summation only; the flow does not stop at the resulting Borel seams.

On the open survivor set in $\Omega \times (0,\infty)$ the map $(x,h) \mapsto (\Phi_h x, h)$ is one-to-one and smooth, with Jacobian J_h relative to the physical source times dh. Its inverse is the backward flow on its image. Apply change of variables in each box and disintegrate with respect to (F,h). A countable determining algebra for the Borel sets of the collision cylinder identifies both kernel measures outside one null set in (t,h). Equality for all nonnegative Borel functions then follows by monotone approximation. Countable disjoint summation gives (10.1). The same kernels accommodate a source restriction depending on (x,h) by using the corresponding graph measure. Thus survivor restrictions, truncated optical insertions and their increasing limits do not require incompatible density representatives.

For the exhaustion take the union of the first j open precompact rational rectangles, arranging that their interiors cover Ω. Finite overlaps can be refined in the common (F,ϑ) coordinates on each physical branch. The boundary faces are roof-constant or angle-constant, with finitely many corners after refinement. The first kind is transverse to X; the second has zero flux because $X\vartheta = 0$. Corners have zero transverse measure. The ordinary flow-box exit formula applies, with outward orientation and backward lifetime selecting the first exit. Its value plus the initial mass outside D_j is the integral of the indicator that the forward segment of length h is not contained in D_j. These indicators decrease. A surviving segment is compactly contained in Ω and is therefore contained in some D_j. A segment ending at a true exit is never contained in any such precompact set. This proves the fixed-count monotone limit, with the usual harmless endpoint convention at lifetime h. No exhaustion limit is moved through the collision-normalized supremum. □

10.1. **Nongrazing first-exit flux.** For this paragraph initial points satisfy $|g| \geq a > 0$. Along a nongrazing finite exit all incidences have positive limits. The selected stationary word consequently has a smooth local extension to the limiting boundary point; this is an algebraic extension, not an assertion that the same trajectory remains the physical first-hit orbit beyond that point. The limiting ω and χ are positive. Section and competing-hit faces remain true boundaries.

Use the one-sided outgoing nongrazing faces of the exact event, with their physical flux measure $d\zeta$ from (7.10). Boundary points are counted with their incident physical component. In a regular angular flow box each transverse outgoing face is a graph $F = b(\vartheta)$ and its measure is $\omega(b,\vartheta)/(4\pi c_*\chi(b,\vartheta))\,d\vartheta$. The finite-word tame stratification gives these graphs after partition into regular arcs; angular critical and tangent strata have zero transverse measure. Equivalently one can use a countable disjoint refinement of the one-sided graphs. First-exit selection, not graph multiplicity, defines the measure below.

Define the unweighted outgoing flux assigned to an initial roof by

$$\mathcal{F}_h^a(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<s=L(y)-t\le h,\ s<\tau^-(y),\ |g(\Phi_{-s}y)|\ge a\}}\, d\zeta(y). \tag{10.2}$$

Here τ^- is the backward lifetime in the same one-sided regular exact event. The integral is an extended nonnegative quantity. It contains initial, terminal and internal section decisions and competing-hit exits whenever they are nongrazing. The guard is not inserted at the boundary and no labels are pooled.

Theorem 10.2 (Paying the pullback by singular volume drift)**.** *For almost every (t,h),*

$$E_h^{\varepsilon,a}(t) + O_h^{\varepsilon,a}(t) \le C\mathcal{F}_h^a(t) + \frac{C}{a}\int_0^h u(t+s)\,ds. \tag{10.3}$$

The boundary term has no inverse Jacobian because its missing pullback cost is included in the proved volume-drift term. It is not obtained by deleting the inverse Jacobian from the first-exit identity.

Proof. In an angular chart write the optical weight along a starting orbit as $w(s) = \omega(\Phi_s x)$. For a nongrazing exit at $\tau \le h$,

$$w(0) \le w(\tau-) + \int_0^\tau [-w'(s)]_+\,ds.$$

Divide by $4\pi c_*\chi(x)$ and use $W^\varepsilon(x) \le 1$. The first term is at most $(\Lambda_\circ/\lambda_\circ)$ times the outgoing physical flux on the corresponding first-exit graph. The restriction in (10.2) is exactly its initial-source selection. At a grazing exit $w(\tau-) = 0$, but its full initial mass is paid by the negative-variation integral, as in Theorem 9.3. On a surviving trajectory the optical loss is bounded by that integral up to h. These three classes of starting states are disjoint. Sum their negative variations and apply (9.8), the monotonicity $|g(\Phi_s x)| \ge a$, and Lemma 10.1. The result is bounded by the second term of (10.3). Truncate to regular subsets and then use Tonelli and monotone convergence for the nonnegative drift and one-sided face integrals. This proves the assertion also when the right side is infinite. □

10.2. **The original scalar objective.** Let $C_h^{\varepsilon,a}$ be the sum of the original true-exit and optical defects restricted to initial $|g| < a$. Critical points $g = 0$ themselves have zero source measure and are not assigned an artificial boundary mass. Write

$$\mathfrak{C}_{\varepsilon,M}(a,r) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \frac{1}{r}\int_0^r C_h^{\varepsilon,a}(t)\,dh,$$

$$\mathfrak{F}_M(a,r) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \frac{1}{r}\int_0^r \mathcal{F}_h^a(t)\,dh.$$

The distinction between a critical neighborhood and a grazing incidence is important: the latter has already been summed in (9.4), including arbitrarily small incidences.

Corollary 10.3 (The improved complete-source endpoint budget)**.** *With the inherited scalar input and $\varepsilon(B) = A_0 B^{-1/12}$, for all previously fixed $a, r > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{C}_{\varepsilon(B),M}(a,r) + C\mathfrak{F}_M(a,r) + Cr/a + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}. \tag{10.4}$$

All exact arithmetic classes and the original source are retained.

Proof. Apply the source-shift inequality in the proof of Theorem 6.2, before taking any supremum, and split its positive sum $E_h + O_h$ by the initial condition $|g| < a$ or $|g| \ge a$. Use Theorem 10.2 in the second part, average, and then use (9.4). Taking the collision limsup only after B, a, r and the auxiliary spectral band are fixed proves (10.4). The separate suprema of E and O need not be replaced by a smaller joint supremum: the argument uses their sum at the source level. □

This budget has a newly estimated all-incidence drift term, not a new pointwise bad-source premise for the thin corridor. To finish the original pointwise theorem one must still bound the critical initial source and the nongrazing outgoing flux in the stated order. For example, outer choices $a(B), r(B) > 0$ suffice if their two heights tend to zero, $r(B)/a(B) \to 0$, and $(1+\varepsilon(B)^{-1})\sqrt{r(B)} \to 0$. Those choices must be made before the collision limit. Neither the Abel representation (9.9) nor the surface representation (10.2) establishes these two decay statements. No surface-current spectral theorem is being inferred from the singular *volume* insertion proved here.

11. Joint endpoint and incidence localization

The critical variable $g = R(-p_0, p_m)$ involves both endpoints. A one-endpoint estimate loses this geometry. We retain both small momenta in the characteristic pairing and obtain a gain greater than one in their common width, at the same time as a superlinear incidence gain. The two chronological cases require different estimates; neither reverses the occupation record.

For $0 < b < b_0$ let $P_b = M_{\mathbf{1}_{\{|p|<b\}}}$ and retain $M_s = M_{\mathbf{1}_{\{c<s\}}}$. Take b_0, s_0 small and fixed. All norms below are the density-relative-to-ν norms of Section 8, with stable length exponent $1/8$ and unstable exponent $\gamma_0 \le 1/100$. In particular the stable tests carry no additional incidence density; compare [12, Section 3.1, equations (10)–(12)].

Lemma 11.1 (A small vector, not a small multiplier)**.** *Uniformly in b,*

$$\|P_b\mathbf{1}\| \le Cb^{7/8}, \qquad |P_b\mathbf{1}|_w \le Cb, \tag{11.1}$$

$$\|P_b h\| \le C\|h\|, \qquad |P_b h|_w \le Cb^{1/8}\|h\|, \tag{11.2}$$

$$|P_b h|_w \le C|h|_w, \qquad |\ell(P_b h)| \le Cb^{1/8}\|h\|. \tag{11.3}$$

The small strong norm in (11.1) *is asserted only for the particular vector $P_b\mathbf{1}$.*

Proof. Near $p = 0$, $p = \sin\varphi$ has uniformly bounded derivative and inverse derivative. A stable graph crosses $|p| < b$ in at most one interval of length Cb. Its strong stable integral is bounded by $C\min\{|W|, b\}|W|^{-1/8} \le Cb^{7/8}$. For two matched graphs at distance d, the horizontal cuts move by at most Cd. The tests agree in graph coordinates on the common interval. The difference from unmatched intervals and arclength Jacobians is at most $C\min(b,d)$, using the separate bound Cb when $d \ge b$. Hence the unstable seminorm is at most $C\sup_d d^{-\gamma_0}\min(b,d) = Cb^{1-\gamma_0}$. This proves the strong vector estimate; the weak estimate is the unweighted intersection-length bound.

The two transverse horizontal cuts satisfy the uniform piecewise multiplier criterion [12, Lemma 3.3(b)]. Restricting a weak test to its retained interval and using the strong stable length

normalization gives the second estimate in (11.2). The same restriction is an admissible weak test on a shorter stable curve, so multiplication is also weakly bounded. Mass evaluation is weakly continuous, giving the last estimate. Smooth averages of the level indicators give the stated elements and operations in the completions. For the vector convergence, the same calculation on bands of width tending to zero gives convergence in the strong norm. No derivative of order b^{-1} is assigned to the limiting multiplier. □

11.1. **Peak vectors with a physical reference.** Refine the fixed peak charts, without changing the arithmetic reference, so that

$$\frac{\log(1/r_*)}{\log(1/\sigma)} < \frac{1}{256}, \qquad \frac{\log(1/r_*)}{\log(1/\rho)} < \frac{1}{256}. \tag{11.4}$$

Here σ, ρ are the constants in (8.3). The removed compact annuli join the fixed-band spectral complement. All unit-modulus points and their continued peaks remain represented.

On a chart write $\Pi h = v_\xi \ell_\xi(h)$, with uniformly bounded vectors and functionals. There is no assumption that $\Pi\mathbf{1}$ is nonzero. Indeed a nontrivial arithmetic peak can annihilate that particular initial vector.

Lemma 11.2 (Localized peak vector and endpoint tests)**.** *Put $\theta = 31/32$. The chart normalization can be chosen so that*

$$\|\Pi h\| + \sup_{d\ge 0} |\ell(P_b Q^d h)| \le C\|h\|^{1-\theta}|h|_w^{\theta}, \tag{11.5}$$

$$|P_b v_\xi|_w \le C b^{63/64}, \qquad |M_s v_\xi|_w \le C s^{47/32}, \qquad \|M_s v_\xi\| \le C. \tag{11.6}$$

The constants are independent of the two widths and the length d.

Proof. The interpolation proof of Lemma 8.2 applies with the stricter refinement (11.4) and exponent θ. In the short-power case use $|\ell(P_b Q^d h)| \le C|Q^d h|_w \le C(1+d)|h|_w$; in the long-power case apply the strong inequality at the balancing time and use power boundedness afterwards. The factor $1 + |\log t|$ is absorbed between the exponents $255/256$ and $31/32$.

For completeness we specify the physical vector used to localize v_ξ. At a unit-modulus point the inherited physical peripheral theorem, Supplement Theorem 157.4, gives a strong-space eigenvector u_* with $|u_*| = 1$ almost everywhere. Choose a weakly continuous smooth test whose pairing with u_* is nonzero. Weak continuity of the continued projection makes $\Pi_\xi u_*$ nonzero on a smaller chart, with that pairing bounded away from zero. Normalize by this pairing. A finite cover supplies common constants. The invariant coordinate cylinder is the same for the circular family, so u_* remains a physical bounded function when used at neighboring parameters. For all real frequencies, $|Q^L u_*| = 1$ in the physical realization. The weak estimates are justified by bounded strong approximants, not by assigning arbitrary stable-curve traces to an L^∞ representative. The physical Cesaro construction in the cited peripheral theorem realizes u_* as a strong limit of uniformly bounded finite Cesaro sums of powers at the reference parameter applied to one fixed smooth bounded vector, followed by a fixed normalization. At neighboring real parameters, applying a fixed power Q^L to these finite physical sums preserves their uniform modulus bound. Endpoint and homogeneous-incidence intersections therefore satisfy the same length bounds on every approximant. Boundedness of the fixed power and multipliers on the common strong space permits passage to the weak limit. Only after that passage is the integer L chosen below. Thus the length estimates do not require a trace assertion for arbitrary bounded measurable densities.

Use

$$\Pi_\xi u_* = \lambda^{-L} Q^L u_* - \lambda^{-L} N^L u_*.$$

For P_b choose $L = \lceil \log(1/b)/\log(1/\rho) \rceil$. The weak norm is at most $C r_*^{-L}(b + b^{1/8}\rho^L)$, which is bounded by $C b^{63/64}$ under (11.4). For M_s choose instead $L = \lceil (3/2)\log(1/s)/\log(1/\rho) \rceil$. The

homogeneous length estimate, valid for every physical vector of modulus at most one, gives $Cr_*^{-L}(s^{3/2}+s^{1/8}\rho^L)\le Cs^{47/32}$. The bounded strong multiplier gives the last assertion. This argument neither assigns physical modulus one to a continued nonperipheral eigenvector nor divides by $\Pi\mathbf{1}$. □

Set

$$\alpha=\frac{129}{128},\qquad \beta=\frac{17}{16},\qquad \delta=\alpha-1=\frac{1}{128}. \tag{11.7}$$

Let $D_b=\{|p_0|<b,\ |p_m|<b\}$. In the next theorem the endpoint marks are retained even when the initial and terminal section tests are dropped in a positive upper comparison.

Theorem 11.3 (Two-endpoint, one-incidence local bound)**.** *For every fixed auxiliary band B_* there are $C_{B_*}<\infty$ and $\rho_{B_*}<1$ such that, uniformly in $R,m\ge 2$, $0\le j\le m$, exact labels, intervals J, and $0<b<b_0$, $0<s<s_0$,*

$$\begin{aligned} &m^2\nu\{K_m^{\mathrm{c}}=k,A_m=l,L_m\in J,\ D_b,\ c_j<s\}\\ &\quad\le Cb^{\alpha}s^{\beta}(|J|+B_*^{-1})+C_{B_*}(1+B_*|J|)m^2\min\{b\rho_{B_*}^{m/2},s^2\}. \end{aligned} \tag{11.8}$$

The leading constant is independent of the previously fixed band. The estimate is on the original occupation coefficient, not on a reversed or neighboring occupation class.

Proof. The characteristic pairing is exactly

$$\ell(P_bQ^{m-j}M_sQ^jP_b\mathbf{1}).$$

Suppose first $m-j\ge m/2$. Put $h=M_sQ^jP_b\mathbf{1}$. Power boundedness and (11.1) give $\|h\|\le Cb^{7/8}$. Its physical realization has modulus at most one, so $|h|_w\le Cs^{3/2}$. The long-block leading term has amplitude at most

$$Cb^{63/64+(7/8)(1/32)}s^{(3/2)(31/32)}=Cb^{259/256}s^{93/64}.$$

Its remainder is bounded by $Cb\rho^{m/2}$, using the terminal factor $b^{1/8}$ as well as the initial strong factor $b^{7/8}$.

If $j\ge m/2$, expand $Q^jP_b\mathbf{1}$ instead. Its projected coefficient is at most $Cb^{255/256}$ by (11.5). Set $h=M_sv_\xi$ and $d=m-j$. There are two bounds for the remaining terminal pairing:

$$|\ell(P_bQ^dh)|\le Cb^{1/8},\qquad |\ell(P_bQ^dh)|\le Cs^{1457/1024}.$$

The first uses strong power boundedness and (11.3); the second uses (11.5) and (11.6). Their geometric interpolation with powers $1/4$ and $3/4$ gives $Cb^{1/32}s^{4371/4096}$. Thus the leading amplitude in this case is at most $Cb^{263/256}s^{4371/4096}$. The remainder is again $Cb\rho^{m/2}$. Both leading bounds imply $Cb^{\alpha}s^{\beta}$, since $259/256>129/128$ and $4371/4096>17/16$.

The long spectral factor in either case has a four-frequency Gaussian integral of order m^{-2}. On the fixed-band complement the long block has an exponential gap and the same endpoint norm factors. Independently, the physical characteristic pairing is at most $\nu(c_j<s)\le s^2$ at every real frequency. Applying $|z|\le A+D$, $|z|\le V\Rightarrow|z|\le A+\min(D,V)$ retains the cap in (11.8). The positive interval majorant and exact torus coefficient extraction now give that estimate. No number of physical words occurs in the argument. □

12. Mixed critical volume insertions

The joint estimate has a critical-width gain greater than one. It supplies a uniform estimate for the first inverse gradient with a fixed positive truncation, and actual untruncated fractional inverse-gradient insertions. The distinction is visible in the finite-count remainder and will be retained in the roof-height argument.

Keep the original additive insertion I_m of (9.3). Write $z = |g|$ on the regular exact event and, for a small fixed $b > 0$, define

$$u^b(t)\,dt = L_{\#}(I_m \mathbf{1}_{\{z<b\}} \mathbf{1}_{\Pi} \nu_R^*). \tag{12.1}$$

The threshold b now measures $|g|$, not the momentum coordinate. Since $\{z < b\} \subset D_{b/R_-}$, the difference changes only uniform constants. There is a fixed $b_1 > 0$ for which the following statements hold for $b < b_1$.

Theorem 12.1 (Critical localization of the complete optical source)**.** *For every fixed band B_*,*

$$\begin{aligned} m^2 \int_J u^b(t)\,dt &\le C b^{\alpha}(|J| + B_*^{-1}) \\ &\quad + C_{B_*}(1 + B_*|J|) m^2 b^{1/2} \rho_{B_*}^{m/4}. \end{aligned} \tag{12.2}$$

Consequently, for every previously fixed roof width $r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r u^b\|_\infty \le C b^{\alpha}. \tag{12.3}$$

Every inverse-incidence scale and every actual word is included before the collision limit.

Proof. Insert the positive layer cake for $c_j^{-1} \mathbf{1}_{\{c_j < s_0\}}$ into (11.8). Its leading integral is finite because $\beta > 1$. In the capped remainder put $D = b\rho_{B_*}^{m/2}$. Equation (9.2), with exponent one, bounds the entire remainder by $C\sqrt{D}$; the cap is integrated before the collision limit. This gives the stated $b^{1/2}$ error.

For the bounded complementary part of c_j^{-1} and the constant one, the endpoint-only pairing is $\ell(P_b Q^m P_b \mathbf{1})$. Its leading amplitude is at most $C b^{7/8+63/64}$, and its remainder is at most $C b \rho^m$. These are bounded by the respective terms in (12.2). Sum the internal marks with weights $q_0^{d_j}$, whose sum is uniformly bounded. Dropping only the section tests in each positive comparison retains D_{b/R_-}, the mark and the same exact (k, n, m) coefficient, with the sole normalization factor c_*^{-1}. It does not alter the defined source. Finally divide by $2r$, take the collision limit with B_* fixed, and enlarge that band afterwards. This proves (12.3). □

Theorem 12.2 (Actual fractional and truncated first inverse gradients)**.** *For $0 < \eta < 1/2$, the actual measure*

$$v^{\eta,b}(t)\,dt = L_{\#}(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_{\Pi} \nu_R^*) \tag{12.4}$$

is finite at each count and satisfies

$$\begin{aligned} m^2 \int_J v^{\eta,b}(t)\,dt &\le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) \\ &\quad + C_{B_*,\eta}(1 + B_*|J|) m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \end{aligned} \tag{12.5}$$

In particular its fixed-width collision-normalized height is at most $C_\eta b^{\alpha-\eta}$.

For $0 < a < b < b_1$, put

$$v^{a,b}(t)\,dt = L_{\#}(I_m z^{-1} \mathbf{1}_{\{a\le z<b\}} \mathbf{1}_{\Pi} \nu_R^*). \tag{12.6}$$

This source obeys

$$\begin{aligned} m^2 \int_J v^{a,b}(t)\,dt &\le C b^{\delta}(|J| + B_*^{-1}) \\ &\quad + C_{B_*}(1 + B_*|J|) m^2 a^{-1/2} \rho_{B_*}^{m/4}. \end{aligned} \tag{12.7}$$

Thus, for each fixed $a, b, r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r v^{a,b}\|_\infty \le C b^{\delta}, \qquad \delta = 1/128, \tag{12.8}$$

with a leading constant independent of the lower truncation a.

Proof. For the fractional assertion use

$$z^{-\eta}\mathbf{1}_{\{z<b\}} = b^{-\eta}\mathbf{1}_{\{z<b\}} + \eta\int_0^b q^{-\eta-1}\mathbf{1}_{\{z<q\}}\,dq$$

on $z>0$. Apply (12.2) first with a positive lower integration limit, and then use monotone convergence. Both integrals are finite: their exponents are $\alpha-\eta>0$ and $1/2-\eta>0$. This proves physical finiteness as well as (12.5), not just convergence of a formal leading term. The removed set $g=0$ has zero source area; no value assigned there affects the measure.

For the first inverse power a convenient positive majorant is

$$z^{-1}\mathbf{1}_{\{a\le z<b\}} \le b^{-1}\mathbf{1}_{\{z<b\}} + \int_a^b q^{-2}\mathbf{1}_{\{z<q\}}\,dq.$$

The leading term in (12.2) gives $C\{b^{\alpha-1}+\int_a^b q^{\alpha-2}\,dq\}\le Cb^{\delta}$. The error gives $C\{b^{-1/2}+\int_a^b q^{-3/2}\,dq\}\le Ca^{-1/2}$. This proves (12.7). The ordered fixed-width limits prove the remaining assertions. □

Let v^a be the same exact source with insertion $I_m z^{-1}\mathbf{1}_{\{z\ge a\}}$. Split it at $b_1/2$, and use (9.4) on the part above that threshold. For every fixed $a,r>0$ this gives

$$\limsup_m \sup_{R,n,k} m^2\|A_r v^a\|_\infty \le C, \tag{12.9}$$

where C is independent of a. This statement does not assert the same conclusion for the untruncated first inverse gradient. The factor $a^{-1/2}$ in (12.7) prevents taking $a\downarrow 0$ before the collision limit by this proof. By contrast, (12.5) is an actual untruncated insertion throughout the stated range $\eta<1/2$.

13. Optical transport uniform in the initial critical cutoff

We now use the joint volume estimate on the same survivor graph as before. The factor a^{-1} in the earlier all-incidence estimate can be removed without changing the initial event or discarding internal grazing scales. This improvement also applies to the compression cost of the nongrazing boundary pullback.

The definitions of $E_h^{\varepsilon,a}$, $E_h^{\varepsilon,\mathrm{gr},a}$, $O_h^{\varepsilon,a}$ and $\mathcal{F}_h^a$ are unchanged. In particular a is an initial lower cutoff for $|g|$, and the flux keeps the first-exit selection and each incident physical component.

Theorem 13.1 (A cutoff-independent optical height bound)**.** *For each fixed $a,r>0$,*

$$\limsup_m \sup_{R,n,k} m^2\|\bar O_r^{\varepsilon,a} + \bar E_r^{\varepsilon,\mathrm{gr},a}\|_\infty \le Cr, \tag{13.1}$$

with a constant independent of a and ε. For initial sources in $a\le|g|<b$, if $b'=(b^2+2\Lambda_\circ r)^{1/2}<b_1$, the stronger estimate is

$$\limsup_m \sup_{R,n,k} m^2\|\bar O_r^{\varepsilon,[a,b)} + \bar E_r^{\varepsilon,\mathrm{gr},[a,b)}\|_\infty \le Cr(b')^{\delta}. \tag{13.2}$$

All parameters in these statements are fixed before the collision limit. The constants are uniform over those choices.

Proof. Repeat the negative-optical-variation argument of Theorem 9.3, retaining $|g|^{-1}$ in the target insertion rather than replacing it by a^{-1}. The radial monotonicity gives $|g(\Phi_s x)|\ge a$. The common coarea kernels of Lemma 10.1 therefore yield

$$O_h^{\varepsilon,a}(t) + E_h^{\varepsilon,\mathrm{gr},a}(t) \le C\int_0^h v^a(t+s)\,ds \tag{13.3}$$

for almost every (t,h). The entire positive starting mass of a grazing exit is paid by negative optical variation as before; a vanishing boundary density is not used to erase that mass. Averaging gives $2CrA_r v^a(t)$, and (12.9) proves (13.1).

For the annulus, the radial speed additionally gives $a \le |g(\Phi_s x)| < b'$ for $s < r$ before the first exit. Replace v^a by $v^{a,b'}$ in the comparison, and apply (12.8). This proves (13.2). The comparison is on the full source and its injective survivor graph, not on an estimate for each word followed by a cardinality bound. □

Theorem 13.2 (Nongrazing pullback with a uniform paid drift)**.** *At almost every (t,h),*

$$E_h^{\varepsilon,a}(t) + O_h^{\varepsilon,a}(t) \le C\mathcal{F}_h^a(t) + C\int_0^h v^a(t+s)\,ds. \tag{13.4}$$

Consequently the original scalar endpoint budget improves to

$$\begin{aligned}\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{C}_{\varepsilon(B),M}(a,r) + C\mathfrak{F}_M(a,r)\\ + Cr + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.\end{aligned} \tag{13.5}$$

The complete critical initial defect and nongrazing flux are exactly those in (10.4)*; only their paid drift has changed. The leading constant no longer requires $r/a \to 0$.*

Proof. For a nongrazing first exit use $\omega(0) \le \omega(\tau-) + \int_0^\tau [-X\omega]_+\,ds$, dividing by the initial angular Jacobian. The terminal term is at most the fixed angular-ratio constant times the assigned outgoing flux. The integral is compared with v^a using the physical Jacobian and the same target coarea kernels. Grazing exits and surviving optical losses are disjoint source cases and use (13.3). Thus the inverse-Jacobian compression cost is paid, not set equal to one, and (13.4) follows.

Apply the original source-shift inequality before taking any supremum, split its positive defect by initial $|g| < a$ or $|g| \ge a$, and average in h. The second part uses the new comparison and (12.9); the first part is precisely $\mathfrak{C}$. The already proved guard and angular errors are unchanged. The collision limit is taken after fixing $B, \varepsilon(B), a, r$ and every auxiliary band; the auxiliary band is enlarged afterwards. This gives (13.5). □

13.1. **The untruncated critical comparison.** There is also a genuine improvement without a lower critical cutoff, but its kernel is still singular. Let v^η denote the complete source with insertion $I_m|g|^{-\eta}$, obtained by adding the bounded-away-from-zero part to (12.4). For $0 < \eta < 1/2$, its fixed-width normalized height is bounded. Since $|g(\Phi_s x)|^2 \ge 2\lambda_\circ s$, the same positive variation proof gives the extended nonnegative inequality

$$O_h^\varepsilon(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \le C_\eta \int_0^h s^{-(1-\eta)/2} v^\eta(t+s)\,ds. \tag{13.6}$$

This uses an actual untruncated mixed insertion, not a formal integration of an uncapped spectral remainder.

Neither (13.6) nor the uniformity in a of (13.1) justifies removing the initial cutoff inside the collision-normalized limsup. At fixed count, monotone convergence recovers the full source as $a \downarrow 0$. That fixed-count assertion is not an exchange of the two limits. In particular the critical tail can concentrate at roofs depending on the collision count even though all its fixed-width averages are controlled. The outgoing nongrazing flux also remains a surface source, not a consequence of the volume insertion theorem.

The organizing assertion remains (2.5). To finish it through (13.5), it remains to control the two displayed complete heights with positive outer choices $a(B), r(B)$, while $(1+\varepsilon(B)^{-1})\sqrt{r(B)} \to 0$. No further condition $r(B)/a(B) \to 0$ is imposed by the now estimated optical drift. Scalar closure

continues to precede every path normalization and every likelihood consequence on positive arithmetic reference classes.

14. The remaining scalar estimates and their consequences

Equation (13.5) is the revised error budget for the original pointwise assertion. It retains the complete critical initial defect and the nongrazing outgoing flux. The uniformity of the new estimate removes the former restriction $r/a \to 0$ from the paid drift; it does not remove those two sources. In particular the inverse-Jacobian cost has been estimated rather than erased from the first-exit formula.

The quantitative distinction between a truncated and an untruncated first inverse gradient is explicit in (12.7). The actual fractional insertion (12.5) integrates both singular parameters before the collision limit, whereas the first inverse power still requires its positive cutoff to be fixed. The full pointwise theorem is not asserted until the two remaining roof-height bounds are proved in the stated order. All arithmetic peaks and zero classes remain in the same reference.

After scalar closure, Supplement Theorem 76.1 and the positive path remainder give the same-roof path numerator. Normalization is only on positive arithmetic reference classes. Collision-to-return clock transfer and forward likelihood retain their separate inherited hypotheses. The complete preceding article and every inherited proof are retained in the supplement.

Superseded dyadic notation. The next module is retained as a historical source, not as an active reference for its dyadic integration formula. The active convention is

$$\ell_j = 2^{-j} r, \qquad \int_{\ell_j/2}^{\ell_j} v(t+s)\,ds = \int_{t+\ell_j/2}^{t+\ell_j} v(u)\,du.$$

The variable in the first integral is an offset. The complete microscopic tail is formed before the collision limsup. This corrected convention, used in the active article, supersedes the ambiguous historical display; no decay conclusion is inferred from this editorial correction.

15. Dyadic critical-source localization at the original roof

This section isolates a quantitative endpoint criterion without changing the physical source, the exact displacement and occupation coefficient, or the terminal membership test. All statements below are at fixed collision count until the order of limits is specified. The purpose of the decomposition is to avoid exchanging fixed-width concentration with an unbounded Abel kernel.

Let $u = u_{n,k,m,R}$ be the nonnegative complete inverse-incidence density from (9.3), and let $v^\eta\,dt = L_{\#}(I_m|g|^{-\eta}\mathbf{1}_\Pi \nu_R^*)$ for $0 < \eta < 1/2$. The inherited common coarea kernels give measurable nonnegative versions. Put $I_j(t) = (t + 2^{-j-1}r, t + 2^{-j}r]$ for $j \ge 0$.

Lemma 15.1 (Exact dyadic majorant for the critical Abel source)**.** *For any nonnegative locally integrable v, any $0 < r < \infty$, and almost every t,*

$$\int_0^r s^{-1/2} v(t+s)\,ds \le \sqrt{2} \sum_{j=0}^{\infty} (2^{-j-1}r)^{-1/2} \int_{I_j(t)} v(t+s)\,ds. \tag{15.1}$$

*The inequality holds in $[0,\infty]$, with no regularity assumption on v. In particular, if for each fixed outer parameter ε one proves a collision-uniform *local* bound*

$$\limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}(M)} \int_0^\ell v(t+s)\,ds \le H_{\varepsilon,M}(\ell) \quad (0 < \ell < r_0), \tag{15.2}$$

and the limsup is uniform over the finite partial sums in (15.1) *including their tails, then the ordered Abel height is bounded by*

$$C\sum_{j\ge 0}(2^{-j}r)^{-1/2}H_{\varepsilon,M}(2^{-j}r). \tag{15.3}$$

For $H_{\varepsilon,M}(\ell)\le A_M\varepsilon^{-q}\ell^{1/2+\delta}$, $\delta>0$, this is $O_M(\varepsilon^{-q}r^{\delta})$. The tail-uniformity clause cannot be omitted: pointwise limsups of every fixed scale do not control the limsup of an infinite sum of nonnegative terms.

Proof. Partition $(0,r]$ into the half-open intervals $(2^{-j-1}r,2^{-j}r]$. On each interval $s^{-1/2}\le (2^{-j-1}r)^{-1/2}$; Tonelli proves (15.1). Each interval integral is bounded by the cumulative positive integral up to $2^{-j}r$. For any finite partial sum the limsup is bounded by the sum of the finitely many corresponding limsups. A uniform bound on the remaining tails permits passage from these partial sums to the infinite sum, with no exchange of collision limit and an unbounded kernel. The geometric series with ratio $2^{-\delta}$ proves the last estimate. □

Proposition 15.2 (The exact-label surface obstruction remains independent)**.** *The volume criterion* (15.2)*, even if established with its tail uniformity, does not by itself bound the nongrazing first-exit surface height $\mathfrak{F}_M(a,r)$ in* (10.4)*. Its first-exit selection, one-sided physical components, backward lifetime and exact section decisions remain part of a separate positive measure. Any surface local theorem used in the endpoint budget must be established for this actual measure, not inferred from bounded volume multipliers or from cancellation of internal section currents.*

Proof. The coarea source entering (15.2) is a two-dimensional volume pushforward. The outgoing measure in (10.2) is a one-dimensional transverse flux, supported on true boundary faces, with first-exit selection. An estimate for the former supplies no absolute-continuity or uniform trace estimate for the latter. The assertion is a precise separation of hypotheses, not a claim of new decay for the surface measure. □

Part 23. Joint endpoint localization and critical optical insertions

Retained revision-81 abstract. We retain the pointwise local limit problem for the full displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. Joint localization of the two endpoint momenta and one internal incidence yields a superlinear gain in both widths. The proof distinguishes a small initial vector from a bounded multiplier, and treats both chronological cases at every continued arithmetic peak. A physical cap on the finite-count remainder permits all inverse-incidence scales to be integrated. The resulting complete optical source has a critical-width gain greater than one, actual untruncated fractional inverse-gradient moments, and first inverse-gradient bounds uniform in a fixed positive lower cutoff. Applied on common exact-label coarea fibers, these estimates replace the earlier optical and grazing-exit bound Cr/a by Cr, and pay the same uniform compression cost at nongrazing exits. The complete critical initial defect and outgoing nongrazing flux remain the two roof-height estimates needed for the original pointwise theorem; no exchange of critical-cutoff and collision limits is inferred.

16. The unchanged physical problem

Put $R_-=9/20$ and $R_+=47/100$, and let $R\in I=[R_-,R_+]$. Circular scatterers have centers in $\Lambda=\mathbb{Z}(1,0)+\mathbb{Z}(1/2,\sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha,p)\in(\mathbb{R}/2\pi\mathbb{Z})\times(-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu=\mathrm{d}\alpha\,\mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_-\le\tau_R\le\tau_+$, where $\tau_-=3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6,-1/2), \quad (-\theta_R,\sin\theta_R), \quad (\pi/3-\theta_R,\sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1-6q^2+q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{16.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{16.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{16.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

16.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0,0,\bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R,a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{16.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow 0. \tag{16.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

16.2. **The analytic inputs in the form used here.**

Input statement 16.1. The unchanged source (16.3) has a nonnegative, compactly supported BV density at every fixed count. For every fixed $\varepsilon,\delta>0$, with $A_\delta f(t)=(2\delta)^{-1}\int_{t-\delta}^{t+\delta} f(s)\,\mathrm{d}s$,

$$\limsup_m \sup_{R,n,k} m^2\|A_\delta b^\varepsilon\|_\infty \le C\varepsilon^{1/16}. \tag{16.6}$$

The constant is independent of the previously fixed width. For large fixed B and $\varepsilon(B)=A_0B^{-1/12}$,

$$m^2p-\mathcal{L}=m^2b^{\varepsilon(B)}+e_B,\qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)}|e_B|\le C_MB^{-1/192}. \tag{16.7}$$

All auxiliary spectral bands in these statements are fixed before the collision limit and enlarged only afterwards.

These are the complete-source tame theorem, the fixed-width complete-source bound, and the positive scalar remainder theorem in the technical supplement. Their proofs and historical hypotheses are retained there. The fixed-count source uses threshold integration [15], definable monotonicity [9], and the distributional characterization of BV [13]; it does not assume definability of the original smooth guard. They are restated as inputs to distinguish them from the new geometric and transport proofs. No long-time bound on a fixed-count variation constant is part of this statement.

16.3. **Precise inherited inputs.** The fixed-width estimate is Supplement Theorem 179.1, in particular Supplement equation (179.4). It uses Supplement Theorem 178.3, on the original exact displacement and occupation coefficient. For an interval of length $2r$, division by $2r$ gives (16.6). That proof fixes the auxiliary band, takes the collision limit, and only then enlarges the auxiliary band. Its leading constant is independent of the previously fixed interval length. The extension from small margins to a fixed remaining compact range is stated there explicitly. At auxiliary parameter one, $\Theta(c_0)=0$ since $c_0\le 1$, so $W^1=1$ and $b^1=p$. This supplies the unweighted fixed-width estimate without asserting a raw pointwise bound. The positive remainder is the inherited scalar input (16.7), stated as Supplement Theorem 181.4; no surface-current estimate is included in any of these volume-source inputs.

16.4. **The complete-source estimate proved here.** The physical endpoint variables are $q=(\alpha_0,\alpha_m)$, $g=\nabla_q F=R(-p_0,p_m)$ and $K=D_q^2F$. The retained canonical field satisfies $XF=1$, $X\arg g=0$ and $|g(\Phi_s x)|^2\ge|g(x)|^2+2\lambda_\circ s$ up to its true lifetime. Its physical Jacobian factors into an inverse-twist ratio and an angular Hessian ratio. The latter and the original guard already have complete-source estimates.

The remaining optical drift is bounded by $C\omega I_m/|g|$, where

$$I_m=1+\sum_{j=1}^{m-1} q_0^{d_j}/c_j.$$

The preceding singular-insertion argument controlled I_m but replaced $|g|^{-1}$ by a^{-1} on initial $|g|\ge a$. The new argument keeps both components of g in the spectral pairing. With $\alpha=129/128$,

it proves, for the complete original source and each fixed roof width,

$$u^b(t)\,dt = L_{\#}(I_m \mathbf{1}_{\{|g|<b\}} \mathbf{1}_{\Pi} \nu_R^*), \qquad \limsup_m \sup_{R,n,k} m^2 \|A_r u^b\|_\infty \le C b^\alpha.$$

Theorem 26.2 integrates this bound with its finite-count remainder. It gives an actual untruncated mixed insertion for powers $|g|^{-\eta}$ with $0 < \eta < 1/2$, and a bound for $|g|^{-1}\mathbf{1}_{\{|g|\ge a\}}$ whose leading constant is independent of the previously fixed $a > 0$. Theorem 27.1 then proves a complete Cr optical and grazing starting-mass estimate on that initial source. Nongrazing exits retain their exact outgoing flux; only the paid pullback cost is improved. These statements keep the original target (16.5), rather than replacing it by a local law for a different record.

17. A canonical flow from the endpoint momenta

The roof field can be specified on the whole regular exact event, without choosing critical disks or endpoint caps. The particular section in (16.1) is useful here: it bounds both endpoint momenta away from grazing, even though no internal incidence or clearance is bounded away from zero.

For one regular physical word use locally lifted endpoint angles $q = (\alpha_0, \alpha_m)$, and write $F(q)$ for its reduced length. The endpoint map is locally invertible, since the optical twist $B_m = \partial_{p_0}\alpha_m$ is nonzero. Set

$$g = \nabla_q F = R(-p_0, p_m), \qquad K = D_q^2 F, \qquad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}.$$

The terminal tangential velocity has the same value before and after reflection, which fixes the sign in the second component of g. All differentiations in this section are on actual regular words.

Lemma 17.1 (Uniform endpoint-angle Hessians on the exact event)**.** *At every regular physical trajectory,*

$$R\,\mathrm{diag}(c_0, c_m) \le K \le (2R_+^2/\tau_- + R_+) I_2. \tag{17.1}$$

On the original exact event, in particular,

$$\lambda_\circ I_2 \le K \le \Lambda_\circ I_2. \tag{17.2}$$

These are pointwise quadratic-form inequalities, uniform in m, R, n, k. They assert neither convexity of a disconnected physical domain nor uniform control of derivatives of K.

Proof. At a regular trajectory let w_0 be its outgoing first-flight velocity and w_m the negative of its incoming last-flight velocity. The polygonal length, minimized over the prescribed intermediate closed disks, is supported at this trajectory by

$$F(q_*) - w_0 \cdot (Q_0(q_0) - Q_0(q_{0,*})) - w_m \cdot (Q_m(q_m) - Q_m(q_{m,*})).$$

To see this directly, the internal gradient at contact i is $-2c_i n_i$. Its scalar product with a displacement into that disk is nonnegative. The convex gradient inequality for the polygonal length therefore gives the displayed support after minimizing the internal positions. Equality holds at the physical trajectory. The internal Hessian is positive definite, so the reduced roof is smooth on its regular neighborhood. Since $Q_i'' = -Rn_i$ and $w_i \cdot n_i = c_i$, differentiating the touching support proves the lower inequality in (17.1).

For the upper inequality keep the internal contacts fixed while varying the endpoints. The transverse second variation of a flight is its squared transverse displacement divided by its length. The two endpoint speeds have norm R. Their contribution is at most $2R^2|dq|^2/\tau_-$; this bound includes the case $m = 1$, when both displacements occur on the same flight. The endpoint curvature terms contribute at most $R|dq|^2$. The reduced Hessian is the Schur complement of the

internal Hessian and cannot exceed this trial quadratic form. Taking the indicated extreme radii gives the upper bound, with $2R_+^2/\tau_- + R_+ = 47/6$.

In the five rectangles of Y_R^* one has $|p| < 9/10$. For the only moving momentum centers this follows from

$$\frac{2(53/100)}{1+(53/100)^2} + \frac{1}{200} < \frac{9}{10};$$

the remaining centers and the added rectangle satisfy the bound immediately. Both the initial and terminal states belong to this same section. Thus $c_0, c_m > \sqrt{19}/10$, which proves (17.2). □

Away from $g = 0$ define the circle-valued physical observable

$$\vartheta(x) = \arg(-p_0(x), p_m(x)) \in \mathbb{R}/(2\pi\mathbb{Z}). \tag{17.3}$$

It is not the normal angle of an auxiliary endpoint cap. The orientation of each disk is the fixed orientation already used for the collision coordinates. Put $e = g/|g|$ and

$$\chi = e^{\mathsf{T}} \operatorname{cof}(K) e, \qquad v = K^{-1} g, \qquad s = g^{\mathsf{T}} K^{-1} g. \tag{17.4}$$

Here $\operatorname{cof}(K)$ is the adjugate matrix; K is symmetric. In dimension two its eigenvalues are those of K in reverse order. In particular $\lambda_\circ \le \chi \le \Lambda_\circ$ on the exact event.

Theorem 17.2 (An explicit critical-safe forward roof field)**.** *There is a uniquely specified smooth vector field X on the regular exact event with its interior critical points removed such that*

$$XL = 1, \qquad X\vartheta = 0. \tag{17.5}$$

In endpoint-angle coordinates it is

$$X_q = \frac{K^{-1} g}{s}. \tag{17.6}$$

In any local lift of ϑ on the original collision cylinder, it is equivalently

$$X = \frac{\vartheta_p \partial_\alpha - \vartheta_\alpha \partial_p}{L_\alpha \vartheta_p - L_p \vartheta_\alpha}.$$

No source density, guard, partition, or critical radius enters its choice. Stop its flow at the first exit from the regular exact event. For a forward survivor and $r_h = |g(\Phi_h x)|$, $r_0 = |g(x)|$, one has

$$2\lambda_\circ h \le r_h^2 - r_0^2 \le 2\Lambda_\circ h, \tag{17.7}$$

$$\operatorname{length}(q([0,h])) \le \frac{r_h - r_0}{\lambda_\circ} \le \frac{\sqrt{2\Lambda_\circ h}}{\lambda_\circ}, \tag{17.8}$$

$$|p_i(\Phi_h x) - p_i(x)| \le \frac{\sqrt{2\Lambda_\circ h}}{R_-}, \qquad i = 0, m. \tag{17.9}$$

The same estimates hold up to the limiting time of a finite lifetime. A forward trajectory cannot terminate at a regular interior critical point of L. The estimates do not assert survival through a true singular, first-hit, or section-decision boundary.

Proof. Writing $g^\perp = (-g_2, g_1)$ gives $d\vartheta = (g^\perp)^{\mathsf{T}} K\, dq/|g|^2$ and hence

$$\det D_q(F, \vartheta) = \frac{g^{\mathsf{T}} \operatorname{cof}(K) g}{|g|^2} = \chi > 0. \tag{17.10}$$

The endpoint map has Jacobian B_m, so $|\det D_{(\alpha,p)}(L, \vartheta)| = |B_m|\chi > 0$. The two equations (17.5) therefore specify a unique field. Formula (17.6) satisfies them: $g \cdot X_q = 1$ and $KX_q = g/s$ is radial in g. Endpoint lifts differ by translations by multiples of 2π; local lifts of ϑ also differ by constants. Both formulas therefore agree on overlaps and define the field without gluing.

Along the flow $\dot g = g/s$. Its direction is constant and $\dot r = r/s > 0$. Since $r^2/\Lambda_\circ \le s \le r^2/\lambda_\circ$, one has $2\lambda_\circ \le d(r^2)/dh \le 2\Lambda_\circ$. This proves (17.7). Using r as the parameter,

$$\frac{dq}{dr} = K^{-1}e.$$

Integration gives (17.8), including length rather than merely endpoint displacement. The direction of g is constant, so $|g(\Phi_h x) - g(x)| = r_h - r_0$ and the first-variation formula gives (17.9). In particular r_h cannot tend to zero in forward time. On a compact regular subset away from $g = 0$ the ordinary continuation theorem applies; any remaining finite exit is a true boundary of the regular exact event. □

Proposition 17.3 (The cancellation in the physical drift)**.** *Let $T = D^3F$ and let a be a locally Lipschitz density relative to dq. Away from $g = 0$, in the almost-everywhere sense,*

$$\operatorname{div}_q(aX_q) = \frac{\nabla a \cdot v}{s} + a\left\{\frac{T[v,v,v]}{s^2} - \frac{v \cdot \nabla \log\det K}{s}\right\}. \tag{17.11}$$

On a regular neighborhood where $\|T\| \le T_0$, this gives

$$|\operatorname{div}_q(aX_q)| \le \frac{1}{|g|}\left\{\frac{\Lambda_\circ}{\lambda_\circ}|\nabla a| + \left(\frac{\Lambda_\circ^2}{\lambda_\circ^3} + \frac{2\Lambda_\circ}{\lambda_\circ^2}\right)|a|T_0\right\}. \tag{17.12}$$

In particular the constant-density quadratic part has zero divergence. The bounds on T and ∇a in this assertion are local, not collision-uniform conclusions.

Proof. Differentiation of $v = K^{-1}g$ yields $\operatorname{div} v = 2 - v \cdot \nabla \log\det K$. Also $v \cdot \nabla s = 2s - T[v,v,v]$. The two terms $2/s$ cancel upon differentiating av/s, proving (17.11). Use $|v| \le |g|/\lambda_\circ$, $s \ge |g|^2/\Lambda_\circ$, and $|\nabla \log\det K| \le 2T_0/\lambda_\circ$ for the bound. Near a regular interior critical point the uniform positive Hessian makes $|g|$ comparable to distance, so the remaining displayed singularity is locally integrable in two dimensions. This last observation concerns a local integral only. □

18. All-contact motion on the original regular source

The endpoint field controls more than the two endpoints. Its effect on every contact decays with distance from the nearer endpoint, without imposing a positive lower bound on the internal incidences. This observation permits a uniform estimate for the original smooth guard, rather than a replacement by protected words.

Keep $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$. All endpoint derivatives below use $q = (\alpha_0, \alpha_m)$; internal contact displacements use arclength. Constants depend only on the fixed physical family and, where indicated, the originally chosen function Θ.

Lemma 18.1 (An incidence-normalized contact inverse)**.** *For the optical matrix A of the retained recursion, put $D_c = \operatorname{diag}(c_1, \dots, c_{m-1})$ and $C_A = R_+/[2(1-q_0)]$. For $1 \le i, j < m$,*

$$0 \le (A^{-1})_{ij} \le C_A \min(c_i, c_j) q_0^{|i-j|}, \qquad |(D_c^{-1}A^{-1})_{ij}| \le C_A q_0^{|i-j|}. \tag{18.1}$$

Consequently, at every regular physical word, its contact positions Q_j satisfy

$$\|D_q Q_j\| \le C_Q q_0^{d_j}, \qquad 0 \le j \le m, \qquad C_Q = R_+ + \frac{2C_A R_+}{\tau_- q_0}. \tag{18.2}$$

For $m = 1$ the internal assertion is empty and the endpoint estimate still holds. These are derivatives at actual regular trajectories; no common neighborhood of physical continuation is asserted.

Proof. Write $A = D(I-P)$, with D its diagonal. The entries of P are nonnegative, vanish except at neighboring indices, and have row sums at most

$$\frac{2/\tau_-}{2/\tau_- + 2/R_+} = q_0.$$

Also $D_{jj}^{-1} \le Rc_j/2$. In the Neumann series for $(I-P)^{-1}D^{-1}$, the (i,j) entry vanishes until the power $|i-j|$. Summing the remaining geometric series gives the bound with c_j. Symmetry gives the bound with c_i, and division by $c_i > 0$ proves the second inequality. This is the same normalized inverse underlying the retained tapered-contact construction; here only its first derivative is needed, at arbitrary regular endpoints.

The internal action Hessian in oriented arclengths is D_cAD_c. After division of its stationary equations by the internal cosines, endpoint forcing has only the first and last entries, of absolute sizes $t_0c_0\,du_0$ and $t_{m-1}c_m\,du_m$. Tangent signs may change these signs but not their sizes. Thus the displacement of internal arclength s_j obeys

$$|ds_j| \le \frac{C_A}{\tau_-}(q_0^{j-1}|du_0| + q_0^{m-1-j}|du_m|).$$

Both exponents are at least $d_j - 1$, and $|du_i| = R|d\alpha_i|$. The displayed choice of C_Q bounds the resulting Euclidean operator norm and the two endpoint derivatives. The Euclidean speed of a circle parameterized by arclength is one, which proves (18.2). □

Theorem 18.2 (Uniform motion of every physical margin)**.** *There is $C_D < \infty$, independent of m, R, n, k and of every positive incidence and clearance, with the following property. Along any regular endpoint curve of length ℓ_q in the exact event and in one physical word,*

$$|c_j(x) - c_j(y)| + |p_j(x) - p_j(y)| + \operatorname{dist}_{\mathbb{S}^1}(\alpha_j(x), \alpha_j(y)) \le C_D q_0^{d_j}\ell_q, \tag{18.3}$$

$$|\Delta_j(x) - \Delta_j(y)| \le C_D q_0^{d'_j}\ell_q. \tag{18.4}$$

In particular, for a forward survivor of the field in Theorem 17.2, the right sides are bounded by, respectively, $C_DC_Xq_0^{d_j}\sqrt{h}$ and $C_DC_Xq_0^{d'_j}\sqrt{h}$, where $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$.

Proof. Differentiate a flight direction:

$$dv_j = \ell_j^{-1}(I - v_jv_j^{\mathsf{T}})(dQ_{j+1} - dQ_j).$$

The depths of its two endpoints are at least d'_j. Its derivative is consequently bounded by $2C_Qq_0^{d'_j}/\tau_-$. The normal and tangent derivatives at a contact are bounded by $C_Qq_0^{d_j}/R_-$. Express c_j and p_j as scalar products of the appropriate unit flight velocity with these vectors. At the terminal contact use the incoming velocity, with the sign giving positive incidence; its tangential component is unchanged by reflection. Since adjacent depths differ by at most one, these formulas prove the first estimate with a fixed C_D. In particular we do not differentiate $\sqrt{1-p_j^2}$ by dividing by c_j.

The Hausdorff displacement of a segment is no greater than the larger displacement of its two endpoints. Distance from that segment to a fixed third disk is a one-Lipschitz function of this Hausdorff displacement. Taking the minimum over third disks preserves the same bound. The minimizing disk may change; differentiability of that minimum is not required. The excluded incident disks are fixed on the word. Integration along the curve proves (18.4). Finally use the endpoint-curve length estimate of Theorem 17.2. □

Corollary 18.3 (The unchanged guard is uniformly controlled)**.** *For every forward survivor in the original exact event,*

$$|W^\varepsilon(\Phi_h x) - W^\varepsilon(x)| \le \min\{1, K_\Theta \varepsilon^{-1}\sqrt{h}\}, \qquad K_\Theta = \frac{4\,\mathrm{Lip}(\Theta) C_D C_X}{1 - q_0^{3/4}}. \tag{18.5}$$

No incidence cutoff, clearance cutoff, section-distance factor or monotonicity assumption on Θ *is added.*

Proof. For factors in $[0,1]$, the difference of their products is bounded by the sum of their absolute differences. A contact factor in (16.2) changes by at most $\mathrm{Lip}(\Theta) C_D C_X \varepsilon^{-1} q_0^{3d_j/4}\sqrt{h}$; the analogous flight factor has depth d'_j. There are at most two contacts and two flights at any depth. Their geometric series gives (18.5). Both weights belong to $[0,1]$, proving the additional bound by one. The argument uses only the Lipschitz constant and range of the original smooth function. □

Proposition 18.4 (Where a finite forward exit can occur)**.** *Let* s_j *be distance from the collision state* (α_j, p_j) *to the boundary of* Y_R^* *in the product metric. There is a common* C_E *such that any source point whose forward lifetime in the regular exact event is at most* h *belongs to the union*

$$\mathcal{B}_h = \bigcup_{j=0}^{m} \{c_j \le C_E q_0^{d_j}\sqrt{h}\} \;\cup\; \bigcup_{j=0}^{m-1} \{\Delta_j \le C_E q_0^{d'_j}\sqrt{h}\} \;\cup\; \bigcup_{j=0}^{m} \{s_j \le C_E q_0^{d_j}\sqrt{h}\}. \tag{18.6}$$

Consequently its exit mass is bounded above by $\int W^\varepsilon \mathbf{1}_{\mathcal{B}_h}\, d\mu_t$*. This is a source inclusion, not a pointwise concentration estimate.*

Proof. The finite endpoint-curve length gives limiting contact data up to a finite lifetime. A forward trajectory cannot reach an interior critical point, by Theorem 17.2. If all limiting incidences, clearances and distances to section edges were positive, the regular reflection equations, strict first-hit tests and locally constant section decisions would continue the same physical word and exact event. The endpoint Jacobian remains nonzero there, so the roof field also continues, a contradiction. Thus some listed margin tends to zero. Apply Theorem 18.2, including the one-Lipschitz property of distance to a closed edge set. Taking the union over the possible margins and retaining the original weight proves the mass bound. A section edge is charged whether or not another simultaneous change might preserve the occupation; this can enlarge the upper bound but cannot omit a true exit. □

19. A uniform cubic estimate for the reduced action

The second derivatives of the internal contact map need not be uniformly bounded near grazing. They are not needed to estimate the third derivative of the reduced roof. Stationarity cancels them. Throughout this section derivatives are evaluated at actual regular physical trajectories; no continuation through a singularity is made.

Lemma 19.1 (Cubic differentiation of a stationary action)**.** *Let* $\mathcal{S}(q,z)$ *be smooth and suppose that* $\partial_z \mathcal{S}(q, z(q)) = 0$*, with invertible internal Hessian. Set* $F(q) = \mathcal{S}(q, z(q))$ *and* $U_u = (u, Dz(q)u)$*. Then*

$$D^3F(q)[u,v,w] = D^3\mathcal{S}(q, z(q))[U_u, U_v, U_w]. \tag{19.1}$$

The equality is pointwise and requires no bound for the inverse internal Hessian or for $D^2 z$*.*

Proof. The differentiated stationary equation says $D^2\mathcal{S}[U_u,(0,\zeta)]=0$ for every internal vector ζ. Consequently $D^2F[u,v]=D^2\mathcal{S}[U_u,U_v]$. Differentiate this identity in direction w. The two terms in which U_u or U_v is differentiated are contractions of $D^2\mathcal{S}$ with an internal vector and vanish by the same stationary equation. The remaining term is (19.1). This calculation is local, where the implicit function theorem gives a smooth $z(q)$. □

Use arclength at every contact, including $s_0=R\alpha_0$ and $s_m=R\alpha_m$, and write

$$\mathcal{S}(s_0,\dots,s_m)=\sum_{j=0}^{m-1}|Q_{j+1}(s_{j+1})-Q_j(s_j)|.$$

The endpoints are linear functions of $q=(\alpha_0,\alpha_m)$. At a reflection the internal arclength derivative of $\mathcal{S}$ is zero. Its internal Hessian is D_cAD_c, with A as in (21.1); this is positive definite at every regular word. In particular Lemma 19.1 applies even when some positive incidence is arbitrarily small.

Theorem 19.2 (Collision-uniform third derivatives)**.** *Let C_Q be the explicit constant in* (18.2)*, and put*

$$C_3=\frac{6}{\tau_-^2}+\frac{3}{R_-\tau_-}+\frac{1}{R_-^2},\qquad T_*=\frac{16C_3C_Q^3}{1-q_0^3}. \tag{19.2}$$

For every regular physical word, its reduced roof in endpoint angles satisfies

$$|D^3F(q)[u,v,w]|\le T_*|u|\,|v|\,|w|. \tag{19.3}$$

The constant is independent of m,R,n,k, all positive internal incidences and all positive third-disk clearances. In particular, on any regular endpoint curve γ,

$$\|K(\gamma(b))-K(\gamma(a))\|\le T_*\,\mathrm{length}(\gamma). \tag{19.4}$$

This does not assert a common continuation neighborhood for the words.

Proof. For $N(z)=|z|$ at a nonzero vector of length at least τ_-,

$$\|DN\|\le 1,\qquad \|D^2N\|\le\tau_-^{-1},\qquad \|D^3N\|\le 6\tau_-^{-2}.$$

A circular arclength parametrization has derivative norms $1,R^{-1},R^{-2}$ through order three. The chain rule for one flight therefore bounds its third derivative on contact vectors U,V,W by

$$C_3(|U_j|+|U_{j+1}|)(|V_j|+|V_{j+1}|)(|W_j|+|W_{j+1}|).$$

These are derivatives of the action on the product of circles, not third derivatives of the contact map.

For a lifted endpoint direction u, Lemma 18.1 gives $|(U_u)_j|\le C_Qq_0^{d_j}|u|$. Indeed a circle has unit speed in arclength, so its contact-position derivative has precisely the absolute value of its arclength derivative. The two contact depths of flight j are at least d'_j. Thus this flight contributes at most $8C_3C_Q^3q_0^{3d'_j}|u||v||w|$ to (19.1). There are at most two flights at every depth. Summation of the geometric series proves (19.3), including $m=1$, when there are no internal variables. Integrating $DK=D^3F$ along γ proves (19.4). □

Corollary 19.3 (Uniform angular Jacobian variation)**.** *Let Φ_h be the canonical forward flow of Theorem 17.2. On its surviving exact-label trajectories, with $C_X=\sqrt{2\Lambda_\circ}/\lambda_\circ$,*

$$|\chi(\Phi_hx)-\chi(x)|\le T_*C_X\sqrt{h},\qquad \left|\log\frac{\chi(\Phi_hx)}{\chi(x)}\right|\le\frac{T_*C_X}{\lambda_\circ}\sqrt{h}. \tag{19.5}$$

In endpoint coordinates the Jacobian of the flow relative to dq is

$$J_h^q(x) = \frac{\chi(x)}{\chi(\Phi_h x)}. \tag{19.6}$$

The physical source Jacobian is different and is computed below.

Proof. In dimension two $\chi = (e^\perp)^{\mathsf{T}} K e^\perp$, and e is constant along the canonical flow. Apply (19.4), the forward length bound, and $\chi \geq \lambda_\circ$. The map (F, ϑ) has Jacobian χ and conjugates the flow to $(t, \theta) \mapsto (t + h, \theta)$. Ordinary change of variables proves (19.6). No estimate on a derivative of the source density was used. □

20. The angular volume estimate on the complete source

Fix one original exact event, and keep its source measure $\bar\nu = (4\pi c_*)^{-1} d\alpha\, dp$. On its regular part set

$$\omega(x) = |B_m(x)|^{-1}, \qquad d\bar\nu = \frac{\omega(q)}{4\pi c_*}\, dq, \qquad d\mu_t = \frac{1}{4\pi c_* |\nabla L|}\, d\mathcal{H}^1|_{\{L=t\}}.$$

Thus $p(t) = \mu_t(\Pi)$ and $b^\varepsilon(t) = \int W^\varepsilon\, d\mu_t$. All fibers below are those of this same event. The gradient in the coarea formula is in the original collision coordinates.

Let $S_h(t)$ be the starting points on this fiber whose canonical forward flow survives until h. Its physical area Jacobian and its optical ratio are denoted by J_h and η_h, respectively. Define three nonnegative quantities before any collision limit:

$$E_h^\varepsilon(t) = \int_{\{L=t\}\setminus S_h(t)} W^\varepsilon\, d\mu_t, \tag{20.1}$$

$$V_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - J_h]_+\, d\mu_t, \tag{20.2}$$

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - \eta_h]_+\, d\mu_t. \tag{20.3}$$

The first is the unchanged initial mass of killed trajectories. The second is the full physical volume loss. The third isolates the optical inverse-twist loss. Neither E_h nor O_h omits a physical word. We write $\bar E_r = r^{-1} \int_0^r E_h\, dh$, and similarly for V and O. All assertions use joint coarea versions in (t, h).

Theorem 20.1 (Exact Jacobian factorization and a complete angular bound)**.** *On every survivor,*

$$\eta_h(x) = \frac{\omega(\Phi_h x)}{\omega(x)}, \qquad J_h(x) = \eta_h(x) \frac{\chi(x)}{\chi(\Phi_h x)}. \tag{20.4}$$

There is a uniform C_{ang} such that, for each fixed $r > 0$,

$$|\bar V_r^\varepsilon(t) - \bar O_r^\varepsilon(t)| \leq C_{\mathrm{ang}} \sqrt{r}\, A_r p(t) \tag{20.5}$$

for almost every t. Consequently, from the stated fixed-width analytic input,

$$\limsup_m \sup_{R,n,k} m^2 \|\bar V_r^\varepsilon - \bar O_r^\varepsilon\|_\infty \leq C\sqrt{r}. \tag{20.6}$$

The constants are independent of ε. This is a bound for the complete source, not a per-word estimate multiplied by a word count.

Proof. Multiplying the endpoint-coordinate Jacobian (19.6) by the ratio of the physical area densities gives (20.4). Put $k_h = \chi(x)/\chi(\Phi_h x)$. The positive-part function is one-Lipschitz, so

$$|[1 - \eta_h k_h]_+ - [1 - \eta_h]_+| \leq \eta_h |1 - k_h|.$$

Corollary 19.3 bounds $|1-k_h|$ by $C\sqrt{h}$. Moreover $\eta_h \le (\Lambda_\circ/\lambda_\circ)J_h$. Coarea covariance in the target direction therefore yields

$$|V_h^\varepsilon(t) - O_h^\varepsilon(t)| \le C\sqrt{h}\int_{S_h(t)} J_h \, d\mu_t \le C\sqrt{h}\, p(t+h).$$

The target sum contains all its actual words, and $W^\varepsilon \le 1$ was used only for an upper bound. Average over $0 < h < r$ and bound the forward average of the nonnegative density by $2A_r p$. This proves (20.5).

For clarity, the unweighted input used in the last step is

$$\limsup_m \sup_{R,n,k} m^2 \|A_r p\|_\infty \le C \quad \text{for every previously fixed } r > 0.$$

It follows from Input statement 16.1: for the auxiliary parameter 1 the initial factor is $\Theta(c_0) = 0$, whence $W^1 = 1$ and $b^1 = p$. The precise inherited source propositions are pinned in Section 16.3. No raw pointwise bound for p or collision-dependent averaging width is used. □

Define the separate ordered heights

$$\begin{aligned}
\mathfrak{X}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{E}_r^\varepsilon,\\
\mathfrak{V}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{V}_r^\varepsilon,\\
\mathfrak{O}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{O}_r^\varepsilon.
\end{aligned} \tag{20.7}$$

Equation (20.6) gives both $\mathfrak{V} \le \mathfrak{O} + C\sqrt{r}$ and $\mathfrak{O} \le \mathfrak{V} + C\sqrt{r}$ as extended nonnegative inequalities. Thus the unestimated interior volume contribution has been reduced to inverse-twist loss, with no unestimated third derivative of the reduced roof left in this comparison.

Theorem 20.2 (The unchanged pointwise error with separated defects)**.** *Under Input statement 16.1, for sufficiently large fixed B and every fixed $r > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}. \tag{20.8}$$

Here $\varepsilon(B) = A_0 B^{-1/12}$. The order is

$$B,\ \varepsilon(B),\ r,\ \textit{auxiliary band fixed};\quad m \to \infty;\quad \textit{auxiliary band} \to \infty;\quad B \to \infty.$$

In particular, separate bounds

$$\mathfrak{X}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_X} r^{\beta_X}, \qquad \mathfrak{O}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_O} r^{\beta_O} \tag{20.9}$$

with finite nonnegative q_X, q_O and positive β_X, β_O imply the original pointwise assertion. One may choose $r = c\varepsilon(B)^\gamma$ with $\gamma > \max\{2, q_X/\beta_X, q_O/\beta_O\}$. The two estimates in (20.9) *are proof obligations, not conclusions of the present theorem.*

Proof. The stopped subcoupling has density $\min\{W(x), W(\Phi_h x)J_h(x)\}$ on survivors. Its two marginals are dominated by the original weighted fibers. The elementary bound $[w - w'J]_+ \le w[1-J]_+ + J|w-w'|$, the complete guard modulus of Corollary 18.3, and target coarea covariance give

$$b^\varepsilon(t) \le b^\varepsilon(t+h) + E_h^\varepsilon(t) + V_h^\varepsilon(t) + K_\Theta \varepsilon^{-1}\sqrt{h}\, p(t+h).$$

Average at the previously fixed width and apply (20.5). The averaged bad source is bounded at the ordered height by $2C\varepsilon^{1/16}$; the unweighted target is bounded as in the preceding proof. Inserting the positive scalar remainder gives (20.8). All inequalities precede the suprema and the collision limit. The displayed choice of γ makes each of the three explicit powers positive, proving the implication. □

Corollary 20.3 (An exact separated criterion)**.** *Under the same analytic input, the original pointwise assertion is equivalent to, for every M and every fixed $r_0 > 0$,*

$$\lim_{B\to\infty} \inf_{0<r<r_0} \left\{ \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + (1+\varepsilon(B)^{-1})\sqrt{r} \right\} = 0. \tag{20.10}$$

Proof. Sufficiency follows by approximating the infimum before the collision limit. For necessity, the positive remainder and the pointwise assertion give ordered height at most $C_M B^{-1/192}$ for $b^{\varepsilon(B)}$. Both E_h^{ε} and O_h^{ε} are bounded by this same positive density at the starting roof. Their separate ordered heights are therefore at most this bound, uniformly in r. Take the infimum and then $B \to \infty$. This argument does not claim that the infimum is attained or select a width depending on m. □

21. Optical sensitivity and source-weighted exit flux

The two terms in (20.8) have distinct geometric mechanisms. We first give a positive, collision-uniform sensitivity formula for the optical term, and then an exact boundary formula for the initial source mass which exits. Neither formula identifies a surface insertion with a bounded volume multiplier.

21.1. **A positive optical polynomial.** For a regular word put $t_j = \ell_j^{-1}$, $V_i = 2/(Rc_i)$ and

$$A_{ii} = t_{i-1} + t_i + V_i, \qquad A_{i,i+1} = A_{i+1,i} = -t_i. \tag{21.1}$$

The matrix A is the incidence-normalized optical matrix; the actual internal arclength Hessian is $D_c A D_c$. In the Neumann estimate it is $A = D_A(I - P)$, where D_A is its diagonal and the row sums of P are at most q_0. These two diagonal normalizations are different.

Define $y_0 = 0$, $f_0 = 1$, $y_1 = \ell_0$, and

$$f_i = f_{i-1} + V_i y_i, \qquad y_{i+1} = y_i + \ell_i f_i \quad (1 \le i < m). \tag{21.2}$$

The continuant and corner Schur-complement identities give

$$y_m = \left(\prod_{j=0}^{m-1} \ell_j \right) \det A, \qquad \omega = \frac{Rc_0 c_m}{y_m}. \tag{21.3}$$

For $m = 1$ the determinant is empty and $y_1 = \ell_0$. Indeed eliminating f_i gives $y_{i+1} = (1 + \ell_i/\ell_{i-1} + \ell_i V_i) y_i - (\ell_i/\ell_{i-1}) y_{i-1}$, the continuant recurrence with the same initial values. In the mixed endpoint Schur complement all internal incidence squares cancel against $\det(D_c A D_c)$, leaving $|F_{uv}| = c_0 c_m \prod t_j / \det A$ in endpoint arclengths. The change to endpoint angles gives $\omega = R|F_{uv}|$ as stated.

Proposition 21.1 (Optical logarithmic sensitivities)**.** *The polynomial y_m has nonnegative coefficients and is of degree at most one in each of the separate variables ℓ_j, V_i. Thus*

$$a_i = V_i \partial_{V_i} \log y_m \in [0,1], \qquad b_j = \ell_j \partial_{\ell_j} \log y_m \in [0,1]. \tag{21.4}$$

At fixed R its differential on any actual regular word is

$$d\log\omega = d\log c_0 + d\log c_m + \sum_{i=1}^{m-1} a_i \, d\log c_i - \sum_{j=0}^{m-1} b_j \, d\log \ell_j. \tag{21.5}$$

In endpoint-angle coordinates, for a fixed uniform C_O,

$$\|\nabla_q \log\omega\| \le C_O \left(1 + \sum_{i=1}^{m-1} \frac{q_0^{d_i}}{c_i} \right). \tag{21.6}$$

No third derivative or sum of absolute word multiplicities occurs in this bound.

Proof. Each new V_i enters the positive recursion once, and each new ℓ_i enters once after that step. Later operations are linear in all previous variables. Induction proves positivity and separate affineness. For a positive separately affine polynomial $Y = A + Bz$, $z\partial_z \log Y = Bz/(A + Bz) \in [0, 1]$. Applying this identity to each variable gives (21.4); differentiating (21.3) gives (21.5).

The endpoint incidences are bounded below by $\sqrt{19}/10$. Theorem 18.2 bounds $\|D_q c_i\|$ by $C_D q_0^{d_i}$. Also $|d\ell_j| \le |dQ_j| + |dQ_{j+1}|$ and $\ell_j \ge \tau_-$, so $\|D_q \log \ell_j\| \le 2C_Q q_0^{d'_j}/\tau_-$. The sum of these flight bounds is uniform in m. Inserting the sensitivity weights proves (21.6). The inverse-incidence terms are retained; positivity does not make their complete source sum finite. □

21.2. **A quantified regular part of the optical loss.** For $0 < \delta \le 1$ let $S_h^\delta(t) \subset S_h(t)$ consist of survivors satisfying, for every internal contact and every $0 \le s \le h$,

$$c_i(\Phi_s x) \ge \delta q_0^{d_i/2}. \tag{21.7}$$

Write $O_h^{\varepsilon,\mathrm{reg}(\delta)}$ for (20.3) restricted to this set, and $O_h^{\varepsilon,\mathrm{thin}(\delta)}$ for its nonnegative complement. This is a decomposition of the original weighted source, not a new guard or a renormalized probability law.

Theorem 21.2 (Uniform regular optical distortion)**.** *There are uniform positive c_1, C_1, C_2 such that, whenever $0 < r \le c_1\delta^2$,*

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,\mathrm{reg}(\delta)}\|_\infty \le C_1 \delta^{-1}\sqrt{r}, \tag{21.8}$$

$$\bar{O}_r^{\varepsilon,\mathrm{thin}(\delta)}(t) \le b^{C_2\delta}(t) \quad \textit{for almost every } t. \tag{21.9}$$

The first estimate uses only the stated fixed-width analytic input. The second is a pointwise source domination and is not a bound for the normalized essential height of its right side.

Proof. On the corridor, the sum in (21.6) is at most $2\delta^{-1}/(1 - \sqrt{q_0})$. The forward endpoint-curve length is at most $C_X\sqrt{h}$. Hence $|\log \eta_h| \le C\delta^{-1}\sqrt{h}$. Choose c_1 so that this upper bound is at most one for $h \le r$. Then $[1 - \eta_h]_+ \le C\delta^{-1}\sqrt{h}\,\eta_h$. As in the proof of Theorem 20.1, $\eta_h \le (\Lambda_\circ/\lambda_\circ)J_h$ and target coarea gives

$$O_h^{\varepsilon,\mathrm{reg}(\delta)}(t) \le C\delta^{-1}\sqrt{h}\,p(t + h).$$

Averaging first proves (21.8).

If the corridor fails at time $s \le h \le r$, the margin motion bound implies at the starting point

$$c_i(x) \le \delta q_0^{d_i/2} + C_D C_X q_0^{d_i}\sqrt{r} \le C_2\delta q_0^{d_i/2}.$$

Since $q_0^{d_i/2} \le q_0^{d_i/4}$, the corresponding factor of the original guard at auxiliary parameter $C_2\delta$ is zero. Thus $W^{C_2\delta}(x) = 1$ on the entire thin starting set. Its optical loss is bounded by its original weighted source, and $W^\varepsilon \le 1$. This proves (21.9) before averaging, hence afterwards. There are no internal contacts when $m = 1$ and the thin set is empty. □

The theorem quantitatively controls the regular optical contribution on the full exact event. It does not allow replacement of $b^{C_2\delta}$ by its fixed-width average at a starting roof. That replacement would be exactly the missing pointwise step.

21.3. **Exit mass in the correct flux convention.** The initial mass of trajectories which exit need not equal the physical flux measured at their exit. Compression between the starting roof and the exit must be retained. We state the exact formula on regular subdomains and recover the full stopped event by exhaustion.

Let D be a relatively compact regular domain in the exact event, away from $g = 0$, with piecewise smooth boundary. We take the boundary transverse to X except for pieces and

lower-dimensional strata whose images in the angular coordinate have zero length. Such domains are obtained from finite flow boxes; increasing exhaustions can be formed by refining the boxes. Let Γ_D^+ denote its outward boundary, N its outward unit normal in the collision coordinates, and $\tau_D^-(y)$ the first backward lifetime from y inside D. Use the physical flux measure

$$d\zeta_D(y) = \frac{[X(y)\cdot N(y)]_+}{4\pi c_*}\, d\mathcal{H}^1(y). \tag{21.10}$$

Proposition 21.3 (First-exit flux with the initial source weight)**.** *Let $E^{\varepsilon}_{D,h}(t)$ be the weighted initial mass on $L = t$ starting in D and exiting D within h. For almost every (t, h),*

$$E^{\varepsilon}_{D,h}(t) = \int_{\Gamma_D^+} \mathbf{1}_{\{0<L(y)-t\le h,\ L(y)-t<\tau_D^-(y)\}} \frac{W^{\varepsilon}(x_{t,y})}{J_{L(y)-t}(x_{t,y})}\, d\zeta_D(y), \quad x_{t,y} = \Phi_{t-L(y)}y. \tag{21.11}$$

For an increasing regular exhaustion D_ℓ of the full regular exact event, its original true-exit mass satisfies

$$E^{\varepsilon}_h(t) = \lim_{\ell\to\infty} \left\{ \int_{\{L=t\}\setminus D_\ell} W^{\varepsilon}\, d\mu_t + E^{\varepsilon}_{D_\ell,h}(t) \right\}. \tag{21.12}$$

The bracketed nonnegative quantities decrease to the left side. The limit is at fixed collision count; it is not interchanged with a normalized collision limsup.

Proof. In a flow box use $(u, \theta) = (L, \vartheta)$, so that $X = \partial_u$. The physical source has area density $a(u, \theta) = \omega/(4\pi c_*\chi)$ relative to $du\, d\theta$. On a transverse outward boundary $u = b(\theta)$, its flux is $a(b(\theta), \theta)|d\theta|$. At the starting roof its coarea mass is $a(t, \theta)|d\theta|$, and

$$J_{b(\theta)-t}(x_{t,y}) = a(b(\theta), \theta)/a(t, \theta).$$

Multiplication by the initial weight proves (21.11) on the box. The backward lifetime selects the first forward exit rather than a later re-entry. Uniqueness of the flow makes the source pieces disjoint; a measurable refinement of the boxes sums them without multiplicity. The excluded boundary strata have zero angular measure and hence zero source-fiber measure. Coarea supplies the joint almost-everywhere statement.

For each starting point, declare the D_ℓ-lifetime zero when it starts outside D_ℓ. These lifetimes increase to the lifetime in the full event: every compact surviving flow segment eventually lies in some D_ℓ. The indicators of lifetime at most h therefore decrease to the corresponding full-event indicator. Dominated convergence on each finite coarea fiber proves (21.12). The exceptional (t, h) set can be chosen jointly by disintegration. Chart boundaries in the exhaustion are not declared physical; their effects disappear only through this source-level limiting identity. □

The inverse Jacobian in (21.11) is essential. Along a limiting grazing boundary the physical density may tend to zero while this inverse factor diverges. Vanishing unweighted exit flux therefore does not make the original initial exit mass zero. The formula retains grazing, competing-hit and section exits. If several margins first vanish simultaneously, a fixed ordering partitions the initial source into disjoint exit types; their sum is E^{ε}_h. The internal occupation-current identity in the supplement is signed. It has not been used to subtract any of these positive exit masses.

22. Homogeneity and a superlinear incidence estimate

The inverse incidences in (21.6) can be summed on the complete source. The required improvement is not a bound for multiplication by c^{-1} on the strong space. It is a stronger estimate for the real-frequency physical powers, followed by a summable, width-dependent remainder.

22.1. **The precise spectral input.** Write $Q = \mathscr{T}_{R,\xi}$ for the uncentered four-coordinate operator, with $\xi = (u_1, u_2, b, v)$ real. The densities in the collision spaces are relative to ν; the initial vector is $\mathbf{1}$, and

$$Q^j\mathbf{1}(x) = \exp(i\xi \cdot S_j f(T_R^{-j}x)), \qquad |Q^j\mathbf{1}| = 1 \tag{22.1}$$

on the regular physical realization. Here $S_j f$ denotes the full record, not the roof alone. This equality would not justify a cosine factor in the stable-curve norm.

We use the following source-pinned statements. The norm definitions are Supplement equations (117.1)–(117.3); the full-occupation Doeblin–Fortet inequality and fixed-band spectrum are in Supplement Theorem 157.3 and the moving-peak construction used in Supplement Lemma 175.2. The incidence multiplier $M_s = M_{\mathbf{1}_{\{c<s\}}}$ satisfies

$$\|M_s h\| \le C\|h\|, \qquad |M_s h|_w \le Cs^{1/8}\|h\|. \tag{22.2}$$

This is Supplement Proposition 178.2. On a fixed finite collection of peak charts one has

$$\begin{aligned} &Q^j = \lambda^j \Pi + N^j, && \|N^j\| \le C\rho^j, && \|Q^j\| \le C, \\ &\|Q^j h\| \le C\sigma^j\|h\| + C(1+j)|h|_w, && |Q^j h|_w \le C(1+j)|h|_w, \end{aligned} \tag{22.3}$$

where $\rho, \sigma < 1$. The fixed-band complement has a uniform spectral gap. The four-frequency Gaussian integral on each peak is $O(m^{-2})$. These are inherited spectral theorems, not consequences of the coarea argument.

After shrinking the neighborhoods of unit-modulus points, we may take $|\lambda| \ge r_*$ on the retained charts with

$$\frac{\log(1/r_*)}{\log(1/\sigma)} < \frac{1}{32}, \qquad \frac{\log(1/r_*)}{\log(1/\rho)} < \frac{1}{24}. \tag{22.4}$$

The removed compact portions have a spectral gap and join the complement. This is a fixed refinement, independent of m, s and of the eventual outer band. No arithmetic branch is discarded.

Lemma 22.1 (Real powers on homogeneous grazing strips)**.** *For all $j \ge 0$ and sufficiently small $s > 0$, uniformly on the retained real peak charts,*

$$|M_s Q^j\mathbf{1}|_w \le Cs^{3/2}. \tag{22.5}$$

The exponent comes from the length of a homogeneous stable curve, not from replacing the density convention by Lebesgue density.

Proof. The quadratic homogeneity strips have angular widths comparable to k^{-3} and incidences comparable to k^{-2}. In the standard coordinates they lie between $\pi/2 - (k+1)^{-2}$ and $\pi/2 - k^{-2}$, or their negative counterparts. A stable graph has slope bounded away from zero and infinity. Consequently a homogeneous stable curve in the kth strip has length at most Ck^{-3}. If that strip meets $\{c < s\}$, then $k \ge C^{-1}s^{-1/2}$; the same estimate includes a threshold crossing the strip. Thus the entire tested intersection has length at most $Cs^{3/2}$. For small s, the central strips do not meet it. Equation (22.1) and the supremum bound of a weak test now prove (22.5). This proof applies to the physical branchwise realization; the bounded multiplier and physical realization in (22.2) give the same element of the weak completion. □

Lemma 22.2 (Interpolation close to the weak endpoint)**.** *On the fixed charts chosen above,*

$$\|\Pi h\| + \sup_{a \ge 0} |\ell(Q^a h)| \le C\|h\|^{1/8}|h|_w^{7/8}. \tag{22.6}$$

Moreover

$$|M_s \Pi\mathbf{1}|_w \le Cs^{23/16}. \tag{22.7}$$

Proof. Normalize $\|h\| = 1$ and put $t = |h|_w$. For $0 < t < 1/2$ choose L with $\sigma^L \le t < \sigma^{L-1}$. Commutation and (22.3) give

$$\|\Pi h\| \le C r_*^{-L}\{\sigma^L + (1+L)t\} \le C(1 + |\log t|)t^{31/32} \le Ct^{7/8}.$$

For $a \le L$ the weak power bound controls the terminal pairing; for $a > L$ apply the strong inequality at L and power boundedness afterwards. Both costs are at most $C(1+L)t \le Ct^{7/8}$. The cases of t bounded below and $t = 0$ follow, respectively, by power boundedness and by letting $L \to \infty$. Homogeneity gives (22.6).

For (22.7), choose instead $L = \lceil (3/2)\log(1/s)/\log(1/\rho)\rceil$. The exact identity

$$M_s \Pi \mathbf{1} = \lambda^{-L} M_s Q^L \mathbf{1} - \lambda^{-L} M_s N^L \mathbf{1}$$

uses $N^L = Q^L - \lambda^L \Pi$. Its weak norm is at most $C r_*^{-L}(s^{3/2} + s^{1/8}\rho^L)$. Here $\rho^L \le s^{3/2}$ and $r_*^{-L} \le C s^{-1/16}$ by (22.4). This proves the assertion. The strong norm of $M_s\Pi\mathbf{1}$ remains bounded. This argument does not assign modulus one to the continued eigenvector. □

Theorem 22.3 (A capped superlinear marked-incidence bound)**.** *There is C such that, for every fixed auxiliary roof band $B_* \ge B_0$, there are $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ satisfying*

$$\begin{aligned} m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = l, L_m \in J,\ c(T_R^j x) < s\} \\ \le C s^{5/4}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 \min\{\rho_{B_*}^{m/2}, s^2\}. \end{aligned} \tag{22.8}$$

The estimate is uniform in R, $m \ge 2$, $0 \le j \le m$, exact labels, intervals J and $0 < s < s_0$. In particular the mark and threshold may vary with m. The leading constant is independent of the previously fixed outer band.

Proof. The exact characteristic pairing is $\ell(Q^{m-j} M_s Q^j \mathbf{1})$. If the left block is long, expand that block and apply (22.6) and (22.5) to $M_s Q^j \mathbf{1}$. The leading amplitude is at most $Cs^{21/16}$. If the right block is long, its leading term is controlled by (22.6) and (22.7); its amplitude is at most $Cs^{161/128}$. Both are bounded by $Cs^{5/4}$, since $161/128 > 5/4$. The complementary term is at most $C\rho^{m/2}$. On a fixed-band complement a long block gives $C_{B_*}\rho_{B_*}^{m/2}$.

There is also the independent physical bound at every real frequency,

$$|\ell(Q^{m-j} M_s Q^j \mathbf{1})| \le \nu\{c < s\} = 1 - \sqrt{1 - s^2} \le s^2.$$

If $|z| \le A + D$ and $|z| \le V$, then $|z| \le A + \min(D, V)$. Apply this with the Gaussian leading term as A. Its four-frequency integral is $Cs^{5/4}m^{-2}$; the remainder retains the cap $\min(C_{B_*}\rho_{B_*}^{m/2}, Cs^2)$. The positive interval majorant has Fourier supremum at most $|J| + 2\pi/B_*$ and roof support in $[-B_*, B_*]$. Torus coefficient extraction and integration over that fixed band therefore give (22.8). The cap is essential: an error independent of s could not be summed against an inverse incidence. □

23. Unbounded insertions and the optical source

The cap in (22.8) permits summation all the way to zero incidence. In particular the thin component is no longer estimated by the pointwise height of another bad source.

For $0 < a < 5/4$ and $0 < s < s_0$ put $h_{a,s}(x) = c(x)^{-a}\mathbf{1}_{\{c(x)<s\}}$. The value on the collision-null grazing set is immaterial.

Theorem 23.1 (A real-frequency unbounded insertion estimate)**.** *For every fixed auxiliary band B_*, uniformly in the mark and the original exact collision labels,*

$$m^2 \int h_{a,s}(T_R^j x)\mathbf{1}_{\{K_m^c=k,A_m=l,L_m\in J\}}\,d\nu(x) \\ \le C_a s^{5/4-a}(|J|+B_*^{-1}) + C_{B_*,a}(1+B_*|J|)m^2\min\{s^{2-a},\rho_{B_*}^{m(2-a)/4}\}. \tag{23.1}$$

No bounded strong-space multiplication operator $M_{c^{-a}}$ is asserted. The formula concerns the actual physical insertion; all thresholds are summed before either the collision or band limit.

Proof. The positive layer-cake identity is

$$c^{-a}\mathbf{1}_{\{c<s\}} = s^{-a}\mathbf{1}_{\{c<s\}} + a\int_0^s v^{-a-1}\mathbf{1}_{\{c<v\}}\,dv.$$

First truncate the integral at a positive lower endpoint. Apply Theorem 22.3 and then monotone convergence. The leading integral is finite since $a<5/4$. For $D\in(0,1]$ its remainder satisfies

$$s^{-a}\min(D,s^2) + a\int_0^s v^{-a-1}\min(D,v^2)\,dv \le C_a\min\{s^{2-a},D^{1-a/2}\}. \tag{23.2}$$

For $s\le\sqrt{D}$ integrate v^{1-a} directly; for larger s split at $\sqrt{D}$. This proves the inequality for all $a<2$ and, with $D=\rho_{B_*}^{m/2}$, proves (23.1). Physical finiteness is consistent with $\nu\{c<v\}\le v^2$ and invariance at the marked time. No infinite sum of an uncapped spectral error is taken. □

Retain $d_i=\min(i,m-i)$ and define the additive optical insertion on the original m-collision orbit by

$$I_m(x) = 1+\sum_{i=1}^{m-1}\frac{q_0^{d_i}}{c_i(x)}, \qquad u_{n,k,m,R}(t)\,dt = L_{\#}(I_m\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*). \tag{23.3}$$

There is no product over all inverse incidences in this definition. The source keeps its initial and terminal tests. For each finite count it is finite and absolutely continuous: on compact regular charts this is ordinary coarea, and the removed critical points have zero source measure. Alternatively it is dominated by a constant multiple of the complete inverse-incidence source in Supplement Theorem 97.3.

Corollary 23.2 (Complete depth-weighted source concentration)**.** *For every previously fixed $r>0$,*

$$\limsup_m \sup_{R,n,k} m^2\|A_r u_{n,k,m,R}\|_\infty \le C. \tag{23.4}$$

If only the summands with $c_i<s$ are retained in I_m, the right side is at most $Cs^{1/4}$. The leading constants do not depend on r, s or the collision count.

Proof. Apply (23.1) with $a=1$ and sum with weights $q_0^{d_i}$. Their sum is at most $2q_0/(1-q_0)$. The truncated terms $c_i^{-1}\mathbf{1}_{\{c_i\ge s_0\}}$ and the constant one use the inherited unweighted concentration estimate. For each positive marked integral, dropping the endpoint tests gives an upper bound with initial measure ν/c_* and the *same* exact coefficient (k,n); its inverse-incidence weight is retained. This upper bound does not alter the source definition. No neighboring occupation labels are pooled. Take $J=[t-r,t+r]$, divide by $2r$, take the collision limsup with B_* fixed, and only then let $B_*\to\infty$. The capped remainders decay exponentially after the complete sum. The thin version follows from the same formula at threshold s. □

23.1. **The physical normalization.** The convention of (21.3) can also be checked directly from the endpoint first variation. In endpoint *angles*, $F_{\alpha_0} = -Rp_0$, and hence

$$\omega = |\partial_{\alpha_m} p_0| = \frac{|K_{0m}|}{R}, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\, d\alpha_0\, d\alpha_m, \qquad \omega = \frac{Rc_0 c_m}{y_m}. \tag{23.5}$$

In internal *arclengths* the Hessian is $D_c A D_c$. The corner Schur complement cancels its internal incidence squares; the two endpoint angle changes supply R^2, and the first-variation conversion divides by R. Thus only the one factor R in (23.5) remains. For one flight $y_1 = \ell_0$; for two flights $y_2 = \ell_0 + \ell_1 + 2\ell_0\ell_1/(Rc_1)$. These agree with the direct one- and two-flight Schur complements. Tangent orientation changes conjugate the internal matrix by diagonal signs and leave these absolute identities unchanged.

The cubic cancellation in Lemma 19.1 uses these same mixed coordinates. For an internal vector ζ, differentiated stationarity gives $D^2 S[(u, Dz\, u), (0, \zeta)] = 0$. Every derivative of a lifted contact direction in $D^2 F$ is purely internal, so these terms vanish before any estimate is taken. At an endpoint the lift has arclength component Ru_i; the circle's second and third arclength derivatives have norms R^{-1} and R^{-2}. This accounts for the factors in C_3 and T_* without treating the endpoint angles as arclengths.

23.2. **All incidences in the noncritical optical estimate.** Let $g = \nabla_q F$ and let $\tau(x)$ be the forward lifetime of the canonical field. For $a > 0$, a superscript a means restriction of an *initial* source to $|g(x)| \geq a$. Define $O_h^{\varepsilon,a}$ by this restriction in (20.3). No internal incidence corridor is imposed.

A finite true exit is called grazing here when at least one internal incidence tends to zero as the lifetime is approached. All contacts have limits by the all-contact motion estimate. Assign a simultaneous failure to this grazing class whenever an incidence vanishes; every other finite exit belongs to the disjoint nongrazing class. Let $E_h^{\varepsilon,\mathrm{gr},a}$ be its weighted starting mass with $\tau \leq h$ and $|g| \geq a$.

Theorem 23.3 (Complete optical loss and genuine grazing exits)**.** *For each $a, r > 0$,*

$$O_h^{\varepsilon,a}(t) + E_h^{\varepsilon,\mathrm{gr},a}(t) \leq \frac{C}{a} \int_0^h u(t+s)\, ds, \tag{23.6}$$

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,a} + \bar{E}_r^{\varepsilon,\mathrm{gr},a}\|_\infty \leq \frac{Cr}{a}. \tag{23.7}$$

The first assertion holds jointly for almost every (t, h). Constants are independent of ε and of all positive internal incidences and clearances. The estimate includes every actual word of the original exact event with the stated initial noncritical restriction.

Proof. The field and (21.6) give, along a survivor,

$$|X\omega| \leq C \frac{\omega I_m}{|g|}. \tag{23.8}$$

The radial speed implies $|g(\Phi_s x)| \geq |g(x)| \geq a$. For a surviving orbit the positive decrease of ω up to h is bounded by the integral of $[-X\omega]_+$ over that segment. At a grazing exit, (21.3) and the selected-incidence determinant bound imply $\omega(\Phi_s x) \to 0$. This last assertion needs only a finite-count bound $\omega \leq C_m c_i$; it does not sum such constants over words. Thus the entire starting optical weight at such an exit is bounded by the same negative-variation integral up to τ. The two source cases are disjoint. Other exits contribute zero to the left.

In a local angular chart the initial fiber density is $\omega(x)/(4\pi c_* \chi(x))\, d\vartheta$. At time s, the ratio $\omega(\Phi_s x)/\omega(x)$ is at most $(\Lambda_\circ/\lambda_\circ) J_s(x)$. With $W^\varepsilon(x) \leq 1$, coarea covariance in the *target* direction bounds the integral of the drift in (23.8) by $Ca^{-1}u(t+s)$. The target image is a subset of the

same exact fiber. The common-kernel construction below justifies this comparison simultaneously on all physical words and at joint almost-everywhere levels. Tonelli proves (23.6). Averaging over $0 < h < r$ gives an upper bound $2Cra^{-1}A_r u(t)$. Apply Corollary 23.2. In particular the initial mass of a grazing exit has been paid by optical drift; it has not been declared zero because its boundary optical density vanishes. □

Without the initial lower bound, the same proof gives the extended nonnegative inequality

$$O_h^{\varepsilon}(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \le C \int_0^h s^{-1/2} u(t+s)\, ds. \tag{23.9}$$

It uses $|g(\Phi_s x)|^2 \ge 2\lambda_\circ s$. A bound for every fixed average of u is not a bound for this singular convolution at its initial roof, uniformly in m. Equation (23.9) is therefore not used to remove the critical restriction in (23.7).

24. Common fibers and the remaining boundary source

The singular insertion estimate acts on the original volume source. Its use for optical drift requires common versions of that source on all words. For a nongrazing exit it also identifies which part of the inverse-Jacobian pullback has been paid by the drift estimate. Neither step changes the exact event.

Lemma 24.1 (A common physical coarea version)**.** *At a fixed collision count, radius and exact label, the original regular source admits jointly measurable coarea kernels μ_t and a survivor graph for which*

$$\int_{S_h(t)} f(\Phi_h x) J_h(x)\, d\mu_t(x) = \int_{\Phi_h S_h(t)} f(y)\, d\mu_{t+h}(y) \tag{24.1}$$

holds as an identity of kernels for almost every (t,h) and every nonnegative Borel f. There is also a precompact regular exhaustion $D_j \uparrow \Omega$ for which the source-level first-exit formula of Proposition 21.3 holds monotonically. Auxiliary chart seams are not exits of Ω.

Proof. The regular physical words are disjoint open pieces of the collision source. Their boundaries and critical points have zero source area. Off these sets, (F,ϑ) is a local diffeomorphism with positive Jacobian χ in endpoint coordinates. Choose countably many lifted angular intervals and flow boxes whose closures lie in these regular pieces and whose (F,ϑ) images are rectangles with rational endpoints. They can be chosen to cover Ω. Disjointize the boxes by assigning a point to its first containing box. This is a Borel assignment for summation only; the flow does not stop at the resulting Borel seams.

On the open survivor set in $\Omega \times (0,\infty)$ the map $(x,h) \mapsto (\Phi_h x, h)$ is one-to-one and smooth, with Jacobian J_h relative to the physical source times dh. Its inverse is the backward flow on its image. Apply change of variables in each box and disintegrate with respect to (F,h). A countable determining algebra for the Borel sets of the collision cylinder identifies both kernel measures outside one null set in (t,h). Equality for all nonnegative Borel functions then follows by monotone approximation. Countable disjoint summation gives (24.1). The same kernels accommodate a source restriction depending on (x,h) by using the corresponding graph measure. Thus survivor restrictions, truncated optical insertions and their increasing limits do not require incompatible density representatives.

For the exhaustion take the union of the first j open precompact rational rectangles, arranging that their interiors cover Ω. Finite overlaps can be refined in the common (F,ϑ) coordinates on each physical branch. The boundary faces are roof-constant or angle-constant, with finitely many corners after refinement. The first kind is transverse to X; the second has zero flux because $X\vartheta = 0$. Corners have zero transverse measure. The ordinary flow-box exit formula applies,

with outward orientation and backward lifetime selecting the first exit. Its value plus the initial mass outside D_j is the integral of the indicator that the forward segment of length h is not contained in D_j. These indicators decrease. A surviving segment is compactly contained in Ω and is therefore contained in some D_j. A segment ending at a true exit is never contained in any such precompact set. This proves the fixed-count monotone limit, with the usual harmless endpoint convention at lifetime h. No exhaustion limit is moved through the collision-normalized supremum. □

24.1. **Nongrazing first-exit flux.** For this paragraph initial points satisfy $|g| \geq a > 0$. Along a nongrazing finite exit all incidences have positive limits. The selected stationary word consequently has a smooth local extension to the limiting boundary point; this is an algebraic extension, not an assertion that the same trajectory remains the physical first-hit orbit beyond that point. The limiting ω and χ are positive. Section and competing-hit faces remain true boundaries.

Use the one-sided outgoing nongrazing faces of the exact event, with their physical flux measure $d\zeta$ from (21.10). Boundary points are counted with their incident physical component. In a regular angular flow box each transverse outgoing face is a graph $F = b(\vartheta)$ and its measure is $\omega(b,\vartheta)/(4\pi c_* \chi(b,\vartheta))\, d\vartheta$. The finite-word tame stratification gives these graphs after partition into regular arcs; angular critical and tangent strata have zero transverse measure. Equivalently one can use a countable disjoint refinement of the one-sided graphs. First-exit selection, not graph multiplicity, defines the measure below.

Define the unweighted outgoing flux assigned to an initial roof by

$$\mathcal{F}_h^a(t) = \int_{\Gamma_{\mathrm{ng}}^+} \mathbf{1}_{\{0<s=L(y)-t\leq h,\ s<\tau^-(y),\ |g(\Phi_{-s}y)|\geq a\}}\, d\zeta(y). \tag{24.2}$$

Here τ^- is the backward lifetime in the same one-sided regular exact event. The integral is an extended nonnegative quantity. It contains initial, terminal and internal section decisions and competing-hit exits whenever they are nongrazing. The guard is not inserted at the boundary and no labels are pooled.

Theorem 24.2 (Paying the pullback by singular volume drift)**.** *For almost every (t,h),*

$$E_h^{\varepsilon,a}(t) + O_h^{\varepsilon,a}(t) \leq C\mathcal{F}_h^a(t) + \frac{C}{a}\int_0^h u(t+s)\, ds. \tag{24.3}$$

The boundary term has no inverse Jacobian because its missing pullback cost is included in the proved volume-drift term. It is not obtained by deleting the inverse Jacobian from the first-exit identity.

Proof. In an angular chart write the optical weight along a starting orbit as $w(s) = \omega(\Phi_s x)$. For a nongrazing exit at $\tau \leq h$,

$$w(0) \leq w(\tau-) + \int_0^\tau [-w'(s)]_+\, ds.$$

Divide by $4\pi c_* \chi(x)$ and use $W^\varepsilon(x) \leq 1$. The first term is at most $(\Lambda_\circ/\lambda_\circ)$ times the outgoing physical flux on the corresponding first-exit graph. The restriction in (24.2) is exactly its initial-source selection. At a grazing exit $w(\tau-) = 0$, but its full initial mass is paid by the negative-variation integral, as in Theorem 23.3. On a surviving trajectory the optical loss is bounded by that integral up to h. These three classes of starting states are disjoint. Sum their negative variations and apply (23.8), the monotonicity $|g(\Phi_s x)| \geq a$, and Lemma 24.1. The result is bounded by the second term of (24.3). Truncate to regular subsets and then use Tonelli and monotone convergence for the nonnegative drift and one-sided face integrals. This proves the assertion also when the right side is infinite. □

24.2. **The original scalar objective.** Let $C_h^{\varepsilon,a}$ be the sum of the original true-exit and optical defects restricted to initial $|g| < a$. Critical points $g = 0$ themselves have zero source measure and are not assigned an artificial boundary mass. Write

$$\mathfrak{C}_{\varepsilon,M}(a,r) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \frac{1}{r}\int_0^r C_h^{\varepsilon,a}(t)\,dh,$$

$$\mathfrak{F}_M(a,r) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \frac{1}{r}\int_0^r \mathcal{F}_h^a(t)\,dh.$$

The distinction between a critical neighborhood and a grazing incidence is important: the latter has already been summed in (23.4), including arbitrarily small incidences.

Corollary 24.3 (The improved complete-source endpoint budget)**.** *With the inherited scalar input and $\varepsilon(B) = A_0 B^{-1/12}$, for all previously fixed $a, r > 0$,*

$$\begin{aligned}\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{C}_{\varepsilon(B),M}(a,r) + C\mathfrak{F}_M(a,r)\\ + Cr/a + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.\end{aligned} \tag{24.4}$$

All exact arithmetic classes and the original source are retained.

Proof. Apply the source-shift inequality in the proof of Theorem 20.2, before taking any supremum, and split its positive sum $E_h + O_h$ by the initial condition $|g| < a$ or $|g| \ge a$. Use Theorem 24.2 in the second part, average, and then use (23.4). Taking the collision limsup only after B, a, r and the auxiliary spectral band are fixed proves (24.4). The separate suprema of E and O need not be replaced by a smaller joint supremum: the argument uses their sum at the source level. □

This budget has a newly estimated all-incidence drift term, not a new pointwise bad-source premise for the thin corridor. To finish the original pointwise theorem one must still bound the critical initial source and the nongrazing outgoing flux in the stated order. For example, outer choices $a(B), r(B) > 0$ suffice if their two heights tend to zero, $r(B)/a(B) \to 0$, and $(1+\varepsilon(B)^{-1})\sqrt{r(B)} \to 0$. Those choices must be made before the collision limit. Neither the Abel representation (23.9) nor the surface representation (24.2) establishes these two decay statements. No surface-current spectral theorem is being inferred from the singular *volume* insertion proved here.

25. Joint endpoint and incidence localization

The critical variable $g = R(-p_0, p_m)$ involves both endpoints. A one-endpoint estimate loses this geometry. We retain both small momenta in the characteristic pairing and obtain a gain greater than one in their common width, at the same time as a superlinear incidence gain. The two chronological cases require different estimates; neither reverses the occupation record.

For $0 < b < b_0$ let $P_b = M_{\mathbf{1}_{\{|p|<b\}}}$ and retain $M_s = M_{\mathbf{1}_{\{c<s\}}}$. Take b_0, s_0 small and fixed. All norms below are the density-relative-to-ν norms of Section 22, with stable length exponent $1/8$ and unstable exponent $\gamma_0 \le 1/100$. In particular the stable tests carry no additional incidence density; compare [12, Section 3.1, equations (10)–(12)].

Lemma 25.1 (A small vector, not a small multiplier)**.** *Uniformly in b,*

$$\|P_b\mathbf{1}\| \le Cb^{7/8}, \qquad |P_b\mathbf{1}|_w \le Cb, \tag{25.1}$$

$$\|P_b h\| \le C\|h\|, \qquad |P_b h|_w \le Cb^{1/8}\|h\|, \tag{25.2}$$

$$|P_b h|_w \le C|h|_w, \qquad |\ell(P_b h)| \le Cb^{1/8}\|h\|. \tag{25.3}$$

The small strong norm in (25.1) *is asserted only for the particular vector $P_b\mathbf{1}$.*

Proof. Near $p = 0$, $p = \sin\varphi$ has uniformly bounded derivative and inverse derivative. A stable graph crosses $|p| < b$ in at most one interval of length Cb. Its strong stable integral is bounded by $C\min\{|W|, b\}|W|^{-1/8} \le Cb^{7/8}$. For two matched graphs at distance d, the horizontal cuts move by at most Cd. The tests agree in graph coordinates on the common interval. The difference from unmatched intervals and arclength Jacobians is at most $C\min(b, d)$, using the separate bound Cb when $d \ge b$. Hence the unstable seminorm is at most $C\sup_d d^{-\gamma_0}\min(b, d) = Cb^{1-\gamma_0}$. This proves the strong vector estimate; the weak estimate is the unweighted intersection-length bound.

The two transverse horizontal cuts satisfy the uniform piecewise multiplier criterion [12, Lemma 3.3(b)]. Restricting a weak test to its retained interval and using the strong stable length normalization gives the second estimate in (25.2). The same restriction is an admissible weak test on a shorter stable curve, so multiplication is also weakly bounded. Mass evaluation is weakly continuous, giving the last estimate. Smooth averages of the level indicators give the stated elements and operations in the completions. For the vector convergence, the same calculation on bands of width tending to zero gives convergence in the strong norm. No derivative of order b^{-1} is assigned to the limiting multiplier. □

25.1. **Peak vectors with a physical reference.** Refine the fixed peak charts, without changing the arithmetic reference, so that

$$\frac{\log(1/r_*)}{\log(1/\sigma)} < \frac{1}{256}, \qquad \frac{\log(1/r_*)}{\log(1/\rho)} < \frac{1}{256}. \tag{25.4}$$

Here σ, ρ are the constants in (22.3). The removed compact annuli join the fixed-band spectral complement. All unit-modulus points and their continued peaks remain represented.

On a chart write $\Pi h = v_\xi \ell_\xi(h)$, with uniformly bounded vectors and functionals. There is no assumption that $\Pi\mathbf{1}$ is nonzero. Indeed a nontrivial arithmetic peak can annihilate that particular initial vector.

Lemma 25.2 (Localized peak vector and endpoint tests)**.** *Put $\theta = 31/32$. The chart normalization can be chosen so that*

$$\|\Pi h\| + \sup_{d\ge 0} |\ell(P_b Q^d h)| \le C\|h\|^{1-\theta}|h|_w^{\theta}, \tag{25.5}$$

$$|P_b v_\xi|_w \le Cb^{63/64}, \qquad |M_s v_\xi|_w \le Cs^{47/32}, \qquad \|M_s v_\xi\| \le C. \tag{25.6}$$

The constants are independent of the two widths and the length d.

Proof. The interpolation proof of Lemma 22.2 applies with the stricter refinement (25.4) and exponent θ. In the short-power case use $|\ell(P_b Q^d h)| \le C|Q^d h|_w \le C(1+d)|h|_w$; in the long-power case apply the strong inequality at the balancing time and use power boundedness afterwards. The factor $1 + |\log t|$ is absorbed between the exponents $255/256$ and $31/32$.

For completeness we specify the physical vector used to localize v_ξ. At a unit-modulus point the inherited physical peripheral theorem, Supplement Theorem 157.4, gives a strong-space eigenvector u_* with $|u_*| = 1$ almost everywhere. Choose a weakly continuous smooth test whose pairing with u_* is nonzero. Weak continuity of the continued projection makes $\Pi_\xi u_*$ nonzero on a smaller chart, with that pairing bounded away from zero. Normalize by this pairing. A finite cover supplies common constants. The invariant coordinate cylinder is the same for the circular family, so u_* remains a physical bounded function when used at neighboring parameters. For all real frequencies, $|Q^L u_*| = 1$ in the physical realization. The weak estimates are justified by bounded strong approximants, not by assigning arbitrary stable-curve traces to an L^∞ representative. The physical Cesaro construction in the cited peripheral theorem realizes u_* as a strong limit of uniformly bounded finite Cesaro sums of powers at the reference parameter applied to one

fixed smooth bounded vector, followed by a fixed normalization. At neighboring real parameters, applying a fixed power Q^L to these finite physical sums preserves their uniform modulus bound. Endpoint and homogeneous-incidence intersections therefore satisfy the same length bounds on every approximant. Boundedness of the fixed power and multipliers on the common strong space permits passage to the weak limit. Only after that passage is the integer L chosen below. Thus the length estimates do not require a trace assertion for arbitrary bounded measurable densities.

Use

$$\Pi_\xi u_* = \lambda^{-L} Q^L u_* - \lambda^{-L} N^L u_*.$$

For P_b choose $L = \lceil \log(1/b)/\log(1/\rho) \rceil$. The weak norm is at most $C r_*^{-L}(b + b^{1/8}\rho^L)$, which is bounded by $Cb^{63/64}$ under (25.4). For M_s choose instead $L = \lceil (3/2)\log(1/s)/\log(1/\rho) \rceil$. The homogeneous length estimate, valid for every physical vector of modulus at most one, gives $C r_*^{-L}(s^{3/2} + s^{1/8}\rho^L) \le C s^{47/32}$. The bounded strong multiplier gives the last assertion. This argument neither assigns physical modulus one to a continued nonperipheral eigenvector nor divides by $\Pi\mathbf{1}$. □

Set

$$\alpha = \frac{129}{128}, \qquad \beta = \frac{17}{16}, \qquad \delta = \alpha - 1 = \frac{1}{128}. \tag{25.7}$$

Let $D_b = \{|p_0| < b,\ |p_m| < b\}$. In the next theorem the endpoint marks are retained even when the initial and terminal section tests are dropped in a positive upper comparison.

Theorem 25.3 (Two-endpoint, one-incidence local bound)**.** *For every fixed auxiliary band B_* there are $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ such that, uniformly in $R, m \ge 2$, $0 \le j \le m$, exact labels, intervals J, and $0 < b < b_0$, $0 < s < s_0$,*

$$\begin{aligned} &m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = l, L_m \in J,\ D_b,\ c_j < s\} \\ &\quad \le C b^\alpha s^\beta (|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|) m^2 \min\{b\rho_{B_*}^{m/2}, s^2\}. \end{aligned} \tag{25.8}$$

The leading constant is independent of the previously fixed band. The estimate is on the original occupation coefficient, not on a reversed or neighboring occupation class.

Proof. The characteristic pairing is exactly

$$\ell(P_b Q^{m-j} M_s Q^j P_b \mathbf{1}).$$

Suppose first $m - j \ge m/2$. Put $h = M_s Q^j P_b \mathbf{1}$. Power boundedness and (25.1) give $\|h\| \le C b^{7/8}$. Its physical realization has modulus at most one, so $|h|_w \le C s^{3/2}$. The long-block leading term has amplitude at most

$$C b^{63/64 + (7/8)(1/32)} s^{(3/2)(31/32)} = C b^{259/256} s^{93/64}.$$

Its remainder is bounded by $Cb\rho^{m/2}$, using the terminal factor $b^{1/8}$ as well as the initial strong factor $b^{7/8}$.

If $j \ge m/2$, expand $Q^j P_b \mathbf{1}$ instead. Its projected coefficient is at most $C b^{255/256}$ by (25.5). Set $h = M_s v_\xi$ and $d = m - j$. There are two bounds for the remaining terminal pairing:

$$|\ell(P_b Q^d h)| \le C b^{1/8}, \qquad |\ell(P_b Q^d h)| \le C s^{1457/1024}.$$

The first uses strong power boundedness and (25.3); the second uses (25.5) and (25.6). Their geometric interpolation with powers $1/4$ and $3/4$ gives $C b^{1/32} s^{4371/4096}$. Thus the leading amplitude in this case is at most $C b^{263/256} s^{4371/4096}$. The remainder is again $Cb\rho^{m/2}$. Both leading bounds imply $Cb^\alpha s^\beta$, since $259/256 > 129/128$ and $4371/4096 > 17/16$.

The long spectral factor in either case has a four-frequency Gaussian integral of order m^{-2}. On the fixed-band complement the long block has an exponential gap and the same endpoint norm factors. Independently, the physical characteristic pairing is at most $\nu(c_j < s) \le s^2$ at

every real frequency. Applying $|z| \le A + D$, $|z| \le V \Rightarrow |z| \le A + \min(D, V)$ retains the cap in (25.8). The positive interval majorant and exact torus coefficient extraction now give that estimate. No number of physical words occurs in the argument. □

26. Mixed critical volume insertions

The joint estimate has a critical-width gain greater than one. It supplies a uniform estimate for the first inverse gradient with a fixed positive truncation, and actual untruncated fractional inverse-gradient insertions. The distinction is visible in the finite-count remainder and will be retained in the roof-height argument.

Keep the original additive insertion I_m of (23.3). Write $z = |g|$ on the regular exact event and, for a small fixed $b > 0$, define

$$u^b(t)\,dt = L_{\#}(I_m \mathbf{1}_{\{z<b\}} \mathbf{1}_{\Pi} \nu_R^*). \tag{26.1}$$

The threshold b now measures $|g|$, not the momentum coordinate. Since $\{z < b\} \subset D_{b/R_-}$, the difference changes only uniform constants. There is a fixed $b_1 > 0$ for which the following statements hold for $b < b_1$.

Theorem 26.1 (Critical localization of the complete optical source)**.** *For every fixed band B_*,*

$$m^2 \int_J u^b(t)\,dt \le C b^{\alpha}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|) m^2 b^{1/2} \rho_{B_*}^{m/4}. \tag{26.2}$$

Consequently, for every previously fixed roof width $r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r u^b\|_\infty \le C b^{\alpha}. \tag{26.3}$$

Every inverse-incidence scale and every actual word is included before the collision limit.

Proof. Insert the positive layer cake for $c_j^{-1} \mathbf{1}_{\{c_j < s_0\}}$ into (25.8). Its leading integral is finite because $\beta > 1$. In the capped remainder put $D = b\rho_{B_*}^{m/2}$. Equation (23.2), with exponent one, bounds the entire remainder by $C\sqrt{D}$; the cap is integrated before the collision limit. This gives the stated $b^{1/2}$ error.

For the bounded complementary part of c_j^{-1} and the constant one, the endpoint-only pairing is $\ell(P_b Q^m P_b \mathbf{1})$. Its leading amplitude is at most $C b^{7/8+63/64}$, and its remainder is at most $C b \rho^m$. These are bounded by the respective terms in (26.2). Sum the internal marks with weights $q_0^{d_j}$, whose sum is uniformly bounded. Dropping only the section tests in each positive comparison retains D_{b/R_-}, the mark and the same exact (k, n, m) coefficient, with the sole normalization factor c_*^{-1}. It does not alter the defined source. Finally divide by $2r$, take the collision limit with B_* fixed, and enlarge that band afterwards. This proves (26.3). □

Theorem 26.2 (Actual fractional and truncated first inverse gradients)**.** *For $0 < \eta < 1/2$, the actual measure*

$$v^{\eta,b}(t)\,dt = L_{\#}(I_m z^{-\eta} \mathbf{1}_{\{0<z<b\}} \mathbf{1}_{\Pi} \nu_R^*) \tag{26.4}$$

is finite at each count and satisfies

$$m^2 \int_J v^{\eta,b}(t)\,dt \le C_\eta b^{\alpha-\eta}(|J| + B_*^{-1}) + C_{B_*,\eta}(1 + B_*|J|) m^2 b^{1/2-\eta} \rho_{B_*}^{m/4}. \tag{26.5}$$

In particular its fixed-width collision-normalized height is at most $C_\eta b^{\alpha-\eta}$.

For $0 < a < b < b_1$, put

$$v^{a,b}(t)\,dt = L_{\#}(I_m z^{-1}\mathbf{1}_{\{a\le z<b\}}\mathbf{1}_{\Pi}\nu_R^*). \tag{26.6}$$

This source obeys

$$m^2\int_J v^{a,b}(t)\,dt \le Cb^{\delta}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2 a^{-1/2}\rho_{B_*}^{m/4}. \tag{26.7}$$

Thus, for each fixed $a, b, r > 0$,

$$\limsup_m \sup_{R,n,k} m^2\|A_r v^{a,b}\|_\infty \le Cb^{\delta}, \qquad \delta = 1/128, \tag{26.8}$$

with a leading constant independent of the lower truncation a.

Proof. For the fractional assertion use

$$z^{-\eta}\mathbf{1}_{\{z<b\}} = b^{-\eta}\mathbf{1}_{\{z<b\}} + \eta\int_0^b q^{-\eta-1}\mathbf{1}_{\{z<q\}}\,dq$$

on $z > 0$. Apply (26.2) first with a positive lower integration limit, and then use monotone convergence. Both integrals are finite: their exponents are $\alpha - \eta > 0$ and $1/2 - \eta > 0$. This proves physical finiteness as well as (26.5), not just convergence of a formal leading term. The removed set $g = 0$ has zero source area; no value assigned there affects the measure.

For the first inverse power a convenient positive majorant is

$$z^{-1}\mathbf{1}_{\{a\le z<b\}} \le b^{-1}\mathbf{1}_{\{z<b\}} + \int_a^b q^{-2}\mathbf{1}_{\{z<q\}}\,dq.$$

The leading term in (26.2) gives $C\{b^{\alpha-1} + \int_a^b q^{\alpha-2}\,dq\} \le Cb^{\delta}$. The error gives $C\{b^{-1/2} + \int_a^b q^{-3/2}\,dq\} \le Ca^{-1/2}$. This proves (26.7). The ordered fixed-width limits prove the remaining assertions. □

Let v^a be the same exact source with insertion $I_m z^{-1}\mathbf{1}_{\{z\ge a\}}$. Split it at $b_1/2$, and use (23.4) on the part above that threshold. For every fixed $a, r > 0$ this gives

$$\limsup_m \sup_{R,n,k} m^2\|A_r v^a\|_\infty \le C, \tag{26.9}$$

where C is independent of a. This statement does not assert the same conclusion for the untruncated first inverse gradient. The factor $a^{-1/2}$ in (26.7) prevents taking $a \downarrow 0$ before the collision limit by this proof. By contrast, (26.5) is an actual untruncated insertion throughout the stated range $\eta < 1/2$.

27. Optical transport uniform in the initial critical cutoff

We now use the joint volume estimate on the same survivor graph as before. The factor a^{-1} in the earlier all-incidence estimate can be removed without changing the initial event or discarding internal grazing scales. This improvement also applies to the compression cost of the nongrazing boundary pullback.

The definitions of $E_h^{\varepsilon,a}$, $E_h^{\varepsilon,\mathrm{gr},a}$, $O_h^{\varepsilon,a}$ and $\mathcal{F}_h^a$ are unchanged. In particular a is an initial lower cutoff for $|g|$, and the flux keeps the first-exit selection and each incident physical component.

Theorem 27.1 (A cutoff-independent optical height bound)**.** *For each fixed $a, r > 0$,*

$$\limsup_m \sup_{R,n,k} m^2\|\bar{O}_r^{\varepsilon,a} + \bar{E}_r^{\varepsilon,\mathrm{gr},a}\|_\infty \le Cr, \tag{27.1}$$

with a constant independent of a and ε. For initial sources in $a \le |g| < b$, if $b' = (b^2 + 2\Lambda_\circ r)^{1/2} < b_1$, the stronger estimate is

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,[a,b)} + \bar{E}_r^{\varepsilon,\mathrm{gr},[a,b)}\|_\infty \le Cr(b')^\delta. \tag{27.2}$$

All parameters in these statements are fixed before the collision limit. The constants are uniform over those choices.

Proof. Repeat the negative-optical-variation argument of Theorem 23.3, retaining $|g|^{-1}$ in the target insertion rather than replacing it by a^{-1}. The radial monotonicity gives $|g(\Phi_s x)| \ge a$. The common coarea kernels of Lemma 24.1 therefore yield

$$O_h^{\varepsilon,a}(t) + E_h^{\varepsilon,\mathrm{gr},a}(t) \le C \int_0^h v^a(t+s)\, ds \tag{27.3}$$

for almost every (t, h). The entire positive starting mass of a grazing exit is paid by negative optical variation as before; a vanishing boundary density is not used to erase that mass. Averaging gives $2CrA_r v^a(t)$, and (26.9) proves (27.1).

For the annulus, the radial speed additionally gives $a \le |g(\Phi_s x)| < b'$ for $s < r$ before the first exit. Replace v^a by $v^{a,b'}$ in the comparison, and apply (26.8). This proves (27.2). The comparison is on the full source and its injective survivor graph, not on an estimate for each word followed by a cardinality bound. □

Theorem 27.2 (Nongrazing pullback with a uniform paid drift)**.** *At almost every (t, h),*

$$E_h^{\varepsilon,a}(t) + O_h^{\varepsilon,a}(t) \le C\mathcal{F}_h^a(t) + C \int_0^h v^a(t+s)\, ds. \tag{27.4}$$

Consequently the original scalar endpoint budget improves to

$$\begin{aligned} \mathcal{E}_M \le{}& C_M B^{-1/192} + \mathfrak{C}_{\varepsilon(B),M}(a,r) + C\mathfrak{F}_M(a,r) \\ &+ Cr + C_\Theta(1 + \varepsilon(B)^{-1})\sqrt{r}. \end{aligned} \tag{27.5}$$

The complete critical initial defect and nongrazing flux are exactly those in (24.4)*; only their paid drift has changed. The leading constant no longer requires $r/a \to 0$.*

Proof. For a nongrazing first exit use $\omega(0) \le \omega(\tau-) + \int_0^\tau [-X\omega]_+\, ds$, dividing by the initial angular Jacobian. The terminal term is at most the fixed angular-ratio constant times the assigned outgoing flux. The integral is compared with v^a using the physical Jacobian and the same target coarea kernels. Grazing exits and surviving optical losses are disjoint source cases and use (27.3). Thus the inverse-Jacobian compression cost is paid, not set equal to one, and (27.4) follows.

Apply the original source-shift inequality before taking any supremum, split its positive defect by initial $|g| < a$ or $|g| \ge a$, and average in h. The second part uses the new comparison and (26.9); the first part is precisely $\mathfrak{C}$. The already proved guard and angular errors are unchanged. The collision limit is taken after fixing $B, \varepsilon(B), a, r$ and every auxiliary band; the auxiliary band is enlarged afterwards. This gives (27.5). □

27.1. **The untruncated critical comparison.** There is also a genuine improvement without a lower critical cutoff, but its kernel is still singular. Let v^η denote the complete source with insertion $I_m|g|^{-\eta}$, obtained by adding the bounded-away-from-zero part to (26.4). For $0 < \eta < 1/2$, its fixed-width normalized height is bounded. Since $|g(\Phi_s x)|^2 \ge 2\lambda_\circ s$, the same positive variation proof gives the extended nonnegative inequality

$$O_h^\varepsilon(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \le C_\eta \int_0^h s^{-(1-\eta)/2} v^\eta(t+s)\, ds. \tag{27.6}$$

This uses an actual untruncated mixed insertion, not a formal integration of an uncapped spectral remainder.

Neither (27.6) nor the uniformity in a of (27.1) justifies removing the initial cutoff inside the collision-normalized limsup. At fixed count, monotone convergence recovers the full source as $a \downarrow 0$. That fixed-count assertion is not an exchange of the two limits. In particular the critical tail can concentrate at roofs depending on the collision count even though all its fixed-width averages are controlled. The outgoing nongrazing flux also remains a surface source, not a consequence of the volume insertion theorem.

The organizing assertion remains (16.5). To finish it through (27.5), it remains to control the two displayed complete heights with positive outer choices $a(B), r(B)$, while $(1+\varepsilon(B)^{-1})\sqrt{r(B)} \to 0$. No further condition $r(B)/a(B) \to 0$ is imposed by the now estimated optical drift. Scalar closure continues to precede every path normalization and every likelihood consequence on positive arithmetic reference classes.

28. The remaining scalar estimates and their consequences

Equation (27.5) is the revised error budget for the original pointwise assertion. It retains the complete critical initial defect and the nongrazing outgoing flux. The uniformity of the new estimate removes the former restriction $r/a \to 0$ from the paid drift; it does not remove those two sources. In particular the inverse-Jacobian cost has been estimated rather than erased from the first-exit formula.

The quantitative distinction between a truncated and an untruncated first inverse gradient is explicit in (26.7). The actual fractional insertion (26.5) integrates both singular parameters before the collision limit, whereas the first inverse power still requires its positive cutoff to be fixed. The full pointwise theorem is not asserted until the two remaining roof-height bounds are proved in the stated order. All arithmetic peaks and zero classes remain in the same reference.

After scalar closure, Supplement Theorem 76.1 and the positive path remainder give the same-roof path numerator. Normalization is only on positive arithmetic reference classes. Collision-to-return clock transfer and forward likelihood retain their separate inherited hypotheses. The complete preceding article and every inherited proof are retained in the supplement.

Part 24. Singular insertions and common exact-label fibers

Retained revision-80 abstract. We study the pointwise local limit problem for the complete displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. Homogeneous grazing strips give a superlinear weak estimate for real-frequency physical powers. Combining it with a fixed refinement of the spectral peaks and a physical cap on the remainder yields a marked-incidence bound which can be integrated against inverse incidence. This proves a complete fixed-width local estimate for unbounded insertions, without assuming that they are bounded strong-space multipliers. Applied to the canonical optical flow, it controls inverse-twist loss and the initial mass of genuine grazing exits at every internal incidence scale, on sources with noncritical initial endpoint momentum. The same drift estimate pays the compression cost in the nongrazing first-exit pullback. Common coarea kernels and a flow-box exhaustion keep all exact labels and true boundaries visible. The original scalar endpoint is reduced to the critical initial source and the remaining nongrazing boundary flux; their collision-uniform height estimates are not inferred from fixed-width volume concentration.

29. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$

for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\,\mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K_{m,R}^{\mathrm{c}}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K_{m,R}^{\mathrm{c}} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{29.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{29.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{29.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

29.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{29.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that

analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{29.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

29.2. **The analytic inputs in the form used here.**

Input statement 29.1. The unchanged source (29.3) has a nonnegative, compactly supported BV density at every fixed count. For every fixed $\varepsilon,\delta>0$, with $A_\delta f(t)=(2\delta)^{-1}\int_{t-\delta}^{t+\delta} f(s)\,\mathrm{d}s$,

$$\limsup_m \sup_{R,n,k} m^2\|A_\delta b^\varepsilon\|_\infty \le C\varepsilon^{1/16}. \tag{29.6}$$

The constant is independent of the previously fixed width. For large fixed B and $\varepsilon(B)=A_0B^{-1/12}$,

$$m^2p-\mathcal{L}=m^2b^{\varepsilon(B)}+e_B,\qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B|\le C_M B^{-1/192}. \tag{29.7}$$

All auxiliary spectral bands in these statements are fixed before the collision limit and enlarged only afterwards.

These are the complete-source tame theorem, the fixed-width complete-source bound, and the positive scalar remainder theorem in the technical supplement. Their proofs and historical hypotheses are retained there. The fixed-count source uses threshold integration [15], definable monotonicity [9], and the distributional characterization of BV [13]; it does not assume definability of the original smooth guard. They are restated as inputs to distinguish them from the new geometric and transport proofs. No long-time bound on a fixed-count variation constant is part of this statement.

29.3. **Precise inherited inputs.** The fixed-width estimate is Supplement Theorem 179.1, in particular Supplement equation (179.4). It uses Supplement Theorem 178.3, on the original exact displacement and occupation coefficient. For an interval of length $2r$, division by $2r$ gives (29.6). That proof fixes the auxiliary band, takes the collision limit, and only then enlarges the auxiliary band. Its leading constant is independent of the previously fixed interval length. The extension from small margins to a fixed remaining compact range is stated there explicitly. At auxiliary parameter one, $\Theta(c_0)=0$ since $c_0\le 1$, so $W^1=1$ and $b^1=p$. This supplies the unweighted fixed-width estimate without asserting a raw pointwise bound. The positive remainder is the inherited scalar input (29.7), stated as Supplement Theorem 181.4; no surface-current estimate is included in any of these volume-source inputs.

29.4. **Singular insertions and physical optical transport.** The first-variation coordinates are $q=(\alpha_0,\alpha_m)$, $g=\nabla_q F=R(-p_0,p_m)$, $K=D_q^2F$ and $\vartheta=\arg(-p_0,p_m)$. The canonical field satisfies $XF=1$, $X\vartheta=0$, and is stopped at the true boundary of the regular exact event. Put $\omega=|\partial_{p_0}\alpha_m|^{-1}$ and $\chi=\det D_q(F,\vartheta)$. The retained geometric proofs below give uniform positive bounds for K and χ, a uniform cubic bound for F, and the physical Jacobian factorization

$$J_h=\frac{\omega(\Phi_h x)}{\omega(x)}\frac{\chi(x)}{\chi(\Phi_h x)}.$$

The angular ratio and the original guard have already been controlled on the complete source. The remaining optical sensitivity contains $\sum_i q_0^{d_i}/c_i$, which is singular at small internal incidences.

The new estimate for this sensitivity is a local concentration theorem for the actual unbounded volume insertion

$$I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i} c_i^{-1}, \qquad u(t)\,dt = L_{\#}(I_m \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*).$$

Corollary 36.2 proves, for each fixed $r > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r u\|_\infty \le C.$$

The proof uses a capped superlinear marked-incidence estimate, not a pointwise bound for another bad source and not a bounded multiplier assertion for c^{-1}. All grazing thresholds are summed before the collision limit, while the auxiliary Fourier band remains fixed.

Let E_h^ε be the original weighted true-exit mass, and let O_h^ε be the positive inverse-twist loss on survivors, as defined in Section 33. For a fixed $a > 0$, restrict their initial source to $|g| \ge a$; use gr for a true exit with some internal incidence tending to zero. Theorem 36.3 gives the full estimate

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,a} + \bar{E}_r^{\varepsilon,\mathrm{gr},a}\|_\infty \le Cr/a.$$

This includes every positive internal incidence, without an incidence corridor. The zero optical density at a grazing exit does not erase its positive starting mass: that mass is paid by the negative optical variation. The corresponding nongrazing boundary inequality is Theorem 37.2; it retains the outgoing physical flux and explicitly pays the missing pullback by volume drift.

Corollary 37.3 inserts these estimates into the original pointwise objective. The initial critical neighborhood and the nongrazing boundary flux still require their own height estimates. An integrable singular insertion at every fixed width is not a uniform pointwise bound for a critical Abel convolution, nor is it a theorem about a surface-current insertion.

29.5. **Contact estimates and the literal guard.** The constant in each entry below is uniform in the collision count. Here ℓ_q is the length of a regular endpoint curve and the displayed powers refer to the contact and flight depths defined above.

Quantity	Change bounded by	Original guard factor
Incidence c_j	$C q_0^{d_j} \ell_q$	Yes
Third-disk clearance Δ_j	$C q_0^{d'_j} \ell_q$	Yes
Section-edge distance	$C q_0^{d_j} \ell_q$	No
Flight length ℓ_j	$C q_0^{d'_j} \ell_q$	No

Section decisions remain exact event tests. They are not new smooth guard factors. The canonical field crosses coordinate cuts by change of chart; it stops at the prescribed physical and section-decision boundaries. Exhaustion boundaries enter only the limiting flux identity in Proposition 34.3.

30. A canonical flow from the endpoint momenta

The roof field can be specified on the whole regular exact event, without choosing critical disks or endpoint caps. The particular section in (29.1) is useful here: it bounds both endpoint momenta away from grazing, even though no internal incidence or clearance is bounded away from zero.

For one regular physical word use locally lifted endpoint angles $q = (\alpha_0, \alpha_m)$, and write $F(q)$ for its reduced length. The endpoint map is locally invertible, since the optical twist $B_m = \partial_{p_0} \alpha_m$

is nonzero. Set

$$g = \nabla_q F = R(-p_0, p_m), \qquad K = D_q^2 F, \qquad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}.$$

The terminal tangential velocity has the same value before and after reflection, which fixes the sign in the second component of g. All differentiations in this section are on actual regular words.

Lemma 30.1 (Uniform endpoint-angle Hessians on the exact event)**.** *At every regular physical trajectory,*

$$R\,\mathrm{diag}(c_0, c_m) \le K \le (2R_+^2/\tau_- + R_+)I_2. \tag{30.1}$$

On the original exact event, in particular,

$$\lambda_\circ I_2 \le K \le \Lambda_\circ I_2. \tag{30.2}$$

These are pointwise quadratic-form inequalities, uniform in m, R, n, k. They assert neither convexity of a disconnected physical domain nor uniform control of derivatives of K.

Proof. At a regular trajectory let w_0 be its outgoing first-flight velocity and w_m the negative of its incoming last-flight velocity. The polygonal length, minimized over the prescribed intermediate closed disks, is supported at this trajectory by

$$F(q_*) - w_0 \cdot (Q_0(q_0) - Q_0(q_{0,*})) - w_m \cdot (Q_m(q_m) - Q_m(q_{m,*})).$$

To see this directly, the internal gradient at contact i is $-2c_i n_i$. Its scalar product with a displacement into that disk is nonnegative. The convex gradient inequality for the polygonal length therefore gives the displayed support after minimizing the internal positions. Equality holds at the physical trajectory. The internal Hessian is positive definite, so the reduced roof is smooth on its regular neighborhood. Since $Q_i'' = -Rn_i$ and $w_i \cdot n_i = c_i$, differentiating the touching support proves the lower inequality in (30.1).

For the upper inequality keep the internal contacts fixed while varying the endpoints. The transverse second variation of a flight is its squared transverse displacement divided by its length. The two endpoint speeds have norm R. Their contribution is at most $2R^2|dq|^2/\tau_-$; this bound includes the case $m = 1$, when both displacements occur on the same flight. The endpoint curvature terms contribute at most $R|dq|^2$. The reduced Hessian is the Schur complement of the internal Hessian and cannot exceed this trial quadratic form. Taking the indicated extreme radii gives the upper bound, with $2R_+^2/\tau_- + R_+ = 47/6$.

In the five rectangles of Y_R^* one has $|p| < 9/10$. For the only moving momentum centers this follows from

$$\frac{2(53/100)}{1 + (53/100)^2} + \frac{1}{200} < \frac{9}{10};$$

the remaining centers and the added rectangle satisfy the bound immediately. Both the initial and terminal states belong to this same section. Thus $c_0, c_m > \sqrt{19}/10$, which proves (30.2). □

Away from $g = 0$ define the circle-valued physical observable

$$\vartheta(x) = \arg(-p_0(x), p_m(x)) \in \mathbb{R}/(2\pi\mathbb{Z}). \tag{30.3}$$

It is not the normal angle of an auxiliary endpoint cap. The orientation of each disk is the fixed orientation already used for the collision coordinates. Put $e = g/|g|$ and

$$\chi = e^{\mathsf{T}} \operatorname{cof}(K) e, \qquad v = K^{-1} g, \qquad s = g^{\mathsf{T}} K^{-1} g. \tag{30.4}$$

Here $\operatorname{cof}(K)$ is the adjugate matrix; K is symmetric. In dimension two its eigenvalues are those of K in reverse order. In particular $\lambda_\circ \le \chi \le \Lambda_\circ$ on the exact event.

Theorem 30.2 (An explicit critical-safe forward roof field)**.** *There is a uniquely specified smooth vector field X on the regular exact event with its interior critical points removed such that*

$$XL = 1, \qquad X\vartheta = 0. \tag{30.5}$$

In endpoint-angle coordinates it is

$$X_q = \frac{K^{-1}g}{s}. \tag{30.6}$$

In any local lift of ϑ on the original collision cylinder, it is equivalently

$$X = \frac{\vartheta_p \partial_\alpha - \vartheta_\alpha \partial_p}{L_\alpha \vartheta_p - L_p \vartheta_\alpha}.$$

No source density, guard, partition, or critical radius enters its choice. Stop its flow at the first exit from the regular exact event. For a forward survivor and $r_h = |g(\Phi_h x)|$, $r_0 = |g(x)|$, one has

$$2\lambda_\circ h \le r_h^2 - r_0^2 \le 2\Lambda_\circ h, \tag{30.7}$$

$$\operatorname{length}(q([0,h])) \le \frac{r_h - r_0}{\lambda_\circ} \le \frac{\sqrt{2\Lambda_\circ h}}{\lambda_\circ}, \tag{30.8}$$

$$|p_i(\Phi_h x) - p_i(x)| \le \frac{\sqrt{2\Lambda_\circ h}}{R_-}, \qquad i = 0, m. \tag{30.9}$$

The same estimates hold up to the limiting time of a finite lifetime. A forward trajectory cannot terminate at a regular interior critical point of L. The estimates do not assert survival through a true singular, first-hit, or section-decision boundary.

Proof. Writing $g^\perp = (-g_2, g_1)$ gives $d\vartheta = (g^\perp)^{\mathsf{T}} K\, dq/|g|^2$ and hence

$$\det D_q(F, \vartheta) = \frac{g^{\mathsf{T}} \operatorname{cof}(K) g}{|g|^2} = \chi > 0. \tag{30.10}$$

The endpoint map has Jacobian B_m, so $|\det D_{(\alpha,p)}(L,\vartheta)| = |B_m|\chi > 0$. The two equations (30.5) therefore specify a unique field. Formula (30.6) satisfies them: $g \cdot X_q = 1$ and $KX_q = g/s$ is radial in g. Endpoint lifts differ by translations by multiples of 2π; local lifts of ϑ also differ by constants. Both formulas therefore agree on overlaps and define the field without gluing.

Along the flow $\dot g = g/s$. Its direction is constant and $\dot r = r/s > 0$. Since $r^2/\Lambda_\circ \le s \le r^2/\lambda_\circ$, one has $2\lambda_\circ \le d(r^2)/dh \le 2\Lambda_\circ$. This proves (30.7). Using r as the parameter,

$$\frac{dq}{dr} = K^{-1} e.$$

Integration gives (30.8), including length rather than merely endpoint displacement. The direction of g is constant, so $|g(\Phi_h x) - g(x)| = r_h - r_0$ and the first-variation formula gives (30.9). In particular r_h cannot tend to zero in forward time. On a compact regular subset away from $g = 0$ the ordinary continuation theorem applies; any remaining finite exit is a true boundary of the regular exact event. □

Proposition 30.3 (The cancellation in the physical drift)**.** *Let $T = D^3F$ and let a be a locally Lipschitz density relative to dq. Away from $g = 0$, in the almost-everywhere sense,*

$$\operatorname{div}_q(aX_q) = \frac{\nabla a \cdot v}{s} + a\left\{\frac{T[v,v,v]}{s^2} - \frac{v \cdot \nabla \log\det K}{s}\right\}. \tag{30.11}$$

On a regular neighborhood where $\|T\| \le T_0$, this gives

$$|\operatorname{div}_q(aX_q)| \le \frac{1}{|g|}\left\{\frac{\Lambda_\circ}{\lambda_\circ}|\nabla a| + \left(\frac{\Lambda_\circ^2}{\lambda_\circ^3} + \frac{2\Lambda_\circ}{\lambda_\circ^2}\right)|a| T_0\right\}. \tag{30.12}$$

In particular the constant-density quadratic part has zero divergence. The bounds on T and ∇a in this assertion are local, not collision-uniform conclusions.

Proof. Differentiation of $v = K^{-1}g$ yields $\operatorname{div} v = 2 - v \cdot \nabla \log\det K$. Also $v \cdot \nabla s = 2s - T[v,v,v]$. The two terms $2/s$ cancel upon differentiating av/s, proving (30.11). Use $|v| \le |g|/\lambda_\circ$, $s \ge |g|^2/\Lambda_\circ$, and $|\nabla \log\det K| \le 2T_0/\lambda_\circ$ for the bound. Near a regular interior critical point the uniform positive Hessian makes $|g|$ comparable to distance, so the remaining displayed singularity is locally integrable in two dimensions. This last observation concerns a local integral only. □

31. All-contact motion on the original regular source

The endpoint field controls more than the two endpoints. Its effect on every contact decays with distance from the nearer endpoint, without imposing a positive lower bound on the internal incidences. This observation permits a uniform estimate for the original smooth guard, rather than a replacement by protected words.

Keep $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$. All endpoint derivatives below use $q = (\alpha_0, \alpha_m)$; internal contact displacements use arclength. Constants depend only on the fixed physical family and, where indicated, the originally chosen function Θ.

Lemma 31.1 (An incidence-normalized contact inverse)**.** *For the optical matrix A of the retained recursion, put $D_c = \operatorname{diag}(c_1, \dots, c_{m-1})$ and $C_A = R_+/[2(1-q_0)]$. For $1 \le i, j < m$,*

$$0 \le (A^{-1})_{ij} \le C_A \min(c_i, c_j) q_0^{|i-j|}, \qquad |(D_c^{-1}A^{-1})_{ij}| \le C_A q_0^{|i-j|}. \tag{31.1}$$

Consequently, at every regular physical word, its contact positions Q_j satisfy

$$\|D_q Q_j\| \le C_Q q_0^{d_j}, \qquad 0 \le j \le m, \qquad C_Q = R_+ + \frac{2C_A R_+}{\tau_- q_0}. \tag{31.2}$$

For $m = 1$ the internal assertion is empty and the endpoint estimate still holds. These are derivatives at actual regular trajectories; no common neighborhood of physical continuation is asserted.

Proof. Write $A = D(I - P)$, with D its diagonal. The entries of P are nonnegative, vanish except at neighboring indices, and have row sums at most

$$\frac{2/\tau_-}{2/\tau_- + 2/R_+} = q_0.$$

Also $D_{jj}^{-1} \le Rc_j/2$. In the Neumann series for $(I-P)^{-1}D^{-1}$, the (i,j) entry vanishes until the power $|i-j|$. Summing the remaining geometric series gives the bound with c_j. Symmetry gives the bound with c_i, and division by $c_i > 0$ proves the second inequality. This is the same normalized inverse underlying the retained tapered-contact construction; here only its first derivative is needed, at arbitrary regular endpoints.

The internal action Hessian in oriented arclengths is $D_c A D_c$. After division of its stationary equations by the internal cosines, endpoint forcing has only the first and last entries, of absolute sizes $t_0 c_0\, du_0$ and $t_{m-1} c_m\, du_m$. Tangent signs may change these signs but not their sizes. Thus the displacement of internal arclength s_j obeys

$$|ds_j| \le \frac{C_A}{\tau_-}(q_0^{j-1}|du_0| + q_0^{m-1-j}|du_m|).$$

Both exponents are at least $d_j - 1$, and $|du_i| = R|d\alpha_i|$. The displayed choice of C_Q bounds the resulting Euclidean operator norm and the two endpoint derivatives. The Euclidean speed of a circle parameterized by arclength is one, which proves (31.2). □

Theorem 31.2 (Uniform motion of every physical margin)**.** *There is $C_D < \infty$, independent of m, R, n, k and of every positive incidence and clearance, with the following property. Along any regular endpoint curve of length ℓ_q in the exact event and in one physical word,*

$$|c_j(x) - c_j(y)| + |p_j(x) - p_j(y)| + \operatorname{dist}_{\mathbb{S}^1}(\alpha_j(x), \alpha_j(y)) \le C_D q_0^{d_j} \ell_q, \tag{31.3}$$

$$|\Delta_j(x) - \Delta_j(y)| \le C_D q_0^{d'_j} \ell_q. \tag{31.4}$$

In particular, for a forward survivor of the field in Theorem 30.2, the right sides are bounded by, respectively, $C_D C_X q_0^{d_j}\sqrt{h}$ and $C_D C_X q_0^{d'_j}\sqrt{h}$, where $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$.

Proof. Differentiate a flight direction:

$$dv_j = \ell_j^{-1}(I - v_j v_j^{\mathsf{T}})(dQ_{j+1} - dQ_j).$$

The depths of its two endpoints are at least d'_j. Its derivative is consequently bounded by $2C_Q q_0^{d'_j}/\tau_-$. The normal and tangent derivatives at a contact are bounded by $C_Q q_0^{d_j}/R_-$. Express c_j and p_j as scalar products of the appropriate unit flight velocity with these vectors. At the terminal contact use the incoming velocity, with the sign giving positive incidence; its tangential component is unchanged by reflection. Since adjacent depths differ by at most one, these formulas prove the first estimate with a fixed C_D. In particular we do not differentiate $\sqrt{1 - p_j^2}$ by dividing by c_j.

The Hausdorff displacement of a segment is no greater than the larger displacement of its two endpoints. Distance from that segment to a fixed third disk is a one-Lipschitz function of this Hausdorff displacement. Taking the minimum over third disks preserves the same bound. The minimizing disk may change; differentiability of that minimum is not required. The excluded incident disks are fixed on the word. Integration along the curve proves (31.4). Finally use the endpoint-curve length estimate of Theorem 30.2. □

Corollary 31.3 (The unchanged guard is uniformly controlled)**.** *For every forward survivor in the original exact event,*

$$|W^\varepsilon(\Phi_h x) - W^\varepsilon(x)| \le \min\{1, K_\Theta \varepsilon^{-1}\sqrt{h}\}, \qquad K_\Theta = \frac{4\operatorname{Lip}(\Theta) C_D C_X}{1 - q_0^{3/4}}. \tag{31.5}$$

No incidence cutoff, clearance cutoff, section-distance factor or monotonicity assumption on Θ is added.

Proof. For factors in $[0,1]$, the difference of their products is bounded by the sum of their absolute differences. A contact factor in (29.2) changes by at most $\operatorname{Lip}(\Theta) C_D C_X \varepsilon^{-1} q_0^{3d_j/4}\sqrt{h}$; the analogous flight factor has depth d'_j. There are at most two contacts and two flights at any depth. Their geometric series gives (31.5). Both weights belong to $[0,1]$, proving the additional bound by one. The argument uses only the Lipschitz constant and range of the original smooth function. □

Proposition 31.4 (Where a finite forward exit can occur)**.** *Let s_j be distance from the collision state (α_j, p_j) to the boundary of Y_R^* in the product metric. There is a common C_E such that any source point whose forward lifetime in the regular exact event is at most h belongs to the union*

$$\mathcal{B}_h = \bigcup_{j=0}^{m} \{c_j \le C_E q_0^{d_j}\sqrt{h}\} \cup \bigcup_{j=0}^{m-1} \{\Delta_j \le C_E q_0^{d'_j}\sqrt{h}\} \cup \bigcup_{j=0}^{m} \{s_j \le C_E q_0^{d_j}\sqrt{h}\}. \tag{31.6}$$

Consequently its exit mass is bounded above by $\int W^{\varepsilon}\mathbf{1}_{\mathcal{B}_h}\,d\mu_t$. This is a source inclusion, not a pointwise concentration estimate.

Proof. The finite endpoint-curve length gives limiting contact data up to a finite lifetime. A forward trajectory cannot reach an interior critical point, by Theorem 30.2. If all limiting incidences, clearances and distances to section edges were positive, the regular reflection equations, strict first-hit tests and locally constant section decisions would continue the same physical word and exact event. The endpoint Jacobian remains nonzero there, so the roof field also continues, a contradiction. Thus some listed margin tends to zero. Apply Theorem 31.2, including the one-Lipschitz property of distance to a closed edge set. Taking the union over the possible margins and retaining the original weight proves the mass bound. A section edge is charged whether or not another simultaneous change might preserve the occupation; this can enlarge the upper bound but cannot omit a true exit. □

32. A uniform cubic estimate for the reduced action

The second derivatives of the internal contact map need not be uniformly bounded near grazing. They are not needed to estimate the third derivative of the reduced roof. Stationarity cancels them. Throughout this section derivatives are evaluated at actual regular physical trajectories; no continuation through a singularity is made.

Lemma 32.1 (Cubic differentiation of a stationary action)**.** *Let $\mathcal{S}(q,z)$ be smooth and suppose that $\partial_z\mathcal{S}(q,z(q))=0$, with invertible internal Hessian. Set $F(q)=\mathcal{S}(q,z(q))$ and $U_u=(u,Dz(q)u)$. Then*

$$D^3F(q)[u,v,w]=D^3\mathcal{S}(q,z(q))[U_u,U_v,U_w]. \tag{32.1}$$

The equality is pointwise and requires no bound for the inverse internal Hessian or for D^2z.

Proof. The differentiated stationary equation says $D^2\mathcal{S}[U_u,(0,\zeta)]=0$ for every internal vector ζ. Consequently $D^2F[u,v]=D^2\mathcal{S}[U_u,U_v]$. Differentiate this identity in direction w. The two terms in which U_u or U_v is differentiated are contractions of $D^2\mathcal{S}$ with an internal vector and vanish by the same stationary equation. The remaining term is (32.1). This calculation is local, where the implicit function theorem gives a smooth $z(q)$. □

Use arclength at every contact, including $s_0=R\alpha_0$ and $s_m=R\alpha_m$, and write

$$\mathcal{S}(s_0,\dots,s_m)=\sum_{j=0}^{m-1}|Q_{j+1}(s_{j+1})-Q_j(s_j)|.$$

The endpoints are linear functions of $q=(\alpha_0,\alpha_m)$. At a reflection the internal arclength derivative of $\mathcal{S}$ is zero. Its internal Hessian is D_cAD_c, with A as in (34.1); this is positive definite at every regular word. In particular Lemma 32.1 applies even when some positive incidence is arbitrarily small.

Theorem 32.2 (Collision-uniform third derivatives)**.** *Let C_Q be the explicit constant in* (31.2)*, and put*

$$C_3=\frac{6}{\tau_-^2}+\frac{3}{R_-\tau_-}+\frac{1}{R_-^2},\qquad T_*=\frac{16C_3C_Q^3}{1-q_0^3}. \tag{32.2}$$

For every regular physical word, its reduced roof in endpoint angles satisfies

$$|D^3F(q)[u,v,w]|\le T_*|u|\,|v|\,|w|. \tag{32.3}$$

The constant is independent of m,R,n,k, all positive internal incidences and all positive third-disk clearances. In particular, on any regular endpoint curve γ,

$$\|K(\gamma(b))-K(\gamma(a))\|\le T_*\,\mathrm{length}(\gamma). \tag{32.4}$$

This does not assert a common continuation neighborhood for the words.

Proof. For $N(z) = |z|$ at a nonzero vector of length at least τ_-,

$$\|DN\| \le 1, \qquad \|D^2N\| \le \tau_-^{-1}, \qquad \|D^3N\| \le 6\tau_-^{-2}.$$

A circular arclength parametrization has derivative norms $1, R^{-1}, R^{-2}$ through order three. The chain rule for one flight therefore bounds its third derivative on contact vectors U, V, W by

$$C_3(|U_j| + |U_{j+1}|)(|V_j| + |V_{j+1}|)(|W_j| + |W_{j+1}|).$$

These are derivatives of the action on the product of circles, not third derivatives of the contact map.

For a lifted endpoint direction u, Lemma 31.1 gives $|(U_u)_j| \le C_Q q_0^{d_j}|u|$. Indeed a circle has unit speed in arclength, so its contact-position derivative has precisely the absolute value of its arclength derivative. The two contact depths of flight j are at least d'_j. Thus this flight contributes at most $8C_3C_Q^3 q_0^{3d'_j}|u||v||w|$ to (32.1). There are at most two flights at every depth. Summation of the geometric series proves (32.3), including $m = 1$, when there are no internal variables. Integrating $DK = D^3F$ along γ proves (32.4). □

Corollary 32.3 (Uniform angular Jacobian variation)**.** *Let Φ_h be the canonical forward flow of Theorem 30.2. On its surviving exact-label trajectories, with $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$,*

$$|\chi(\Phi_h x) - \chi(x)| \le T_* C_X \sqrt{h}, \qquad \left|\log \frac{\chi(\Phi_h x)}{\chi(x)}\right| \le \frac{T_* C_X}{\lambda_\circ}\sqrt{h}. \tag{32.5}$$

In endpoint coordinates the Jacobian of the flow relative to dq is

$$J_h^q(x) = \frac{\chi(x)}{\chi(\Phi_h x)}. \tag{32.6}$$

The physical source Jacobian is different and is computed below.

Proof. In dimension two $\chi = (e^\perp)^{\mathsf{T}} K e^\perp$, and e is constant along the canonical flow. Apply (32.4), the forward length bound, and $\chi \ge \lambda_\circ$. The map (F, ϑ) has Jacobian χ and conjugates the flow to $(t, \theta) \mapsto (t + h, \theta)$. Ordinary change of variables proves (32.6). No estimate on a derivative of the source density was used. □

33. The angular volume estimate on the complete source

Fix one original exact event, and keep its source measure $\bar\nu = (4\pi c_*)^{-1} d\alpha\, dp$. On its regular part set

$$\omega(x) = |B_m(x)|^{-1}, \qquad d\bar\nu = \frac{\omega(q)}{4\pi c_*}\, dq, \qquad d\mu_t = \frac{1}{4\pi c_* |\nabla L|}\, d\mathcal{H}^1|_{\{L=t\}}.$$

Thus $p(t) = \mu_t(\Pi)$ and $b^\varepsilon(t) = \int W^\varepsilon\, d\mu_t$. All fibers below are those of this same event. The gradient in the coarea formula is in the original collision coordinates.

Let $S_h(t)$ be the starting points on this fiber whose canonical forward flow survives until h. Its physical area Jacobian and its optical ratio are denoted by J_h and η_h, respectively. Define three nonnegative quantities before any collision limit:

$$E_h^\varepsilon(t) = \int_{\{L=t\}\setminus S_h(t)} W^\varepsilon\, d\mu_t, \tag{33.1}$$

$$V_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - J_h]_+\, d\mu_t, \tag{33.2}$$

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - \eta_h]_+\, d\mu_t. \tag{33.3}$$

The first is the unchanged initial mass of killed trajectories. The second is the full physical volume loss. The third isolates the optical inverse-twist loss. Neither E_h nor O_h omits a physical word. We write $\bar{E}_r = r^{-1}\int_0^r E_h\,dh$, and similarly for V and O. All assertions use joint coarea versions in (t,h).

Theorem 33.1 (Exact Jacobian factorization and a complete angular bound)**.** *On every survivor,*

$$\eta_h(x) = \frac{\omega(\Phi_h x)}{\omega(x)}, \qquad J_h(x) = \eta_h(x)\frac{\chi(x)}{\chi(\Phi_h x)}. \tag{33.4}$$

There is a uniform C_{ang} such that, for each fixed $r>0$,

$$|\bar{V}_r^{\varepsilon}(t) - \bar{O}_r^{\varepsilon}(t)| \le C_{\mathrm{ang}}\sqrt{r}\,A_r p(t) \tag{33.5}$$

for almost every t. Consequently, from the stated fixed-width analytic input,

$$\limsup_m \sup_{R,n,k} m^2\|\bar{V}_r^{\varepsilon} - \bar{O}_r^{\varepsilon}\|_\infty \le C\sqrt{r}. \tag{33.6}$$

The constants are independent of ε. This is a bound for the complete source, not a per-word estimate multiplied by a word count.

Proof. Multiplying the endpoint-coordinate Jacobian (32.6) by the ratio of the physical area densities gives (33.4). Put $k_h = \chi(x)/\chi(\Phi_h x)$. The positive-part function is one-Lipschitz, so

$$|[1-\eta_h k_h]_+ - [1-\eta_h]_+| \le \eta_h|1-k_h|.$$

Corollary 32.3 bounds $|1-k_h|$ by $C\sqrt{h}$. Moreover $\eta_h \le (\Lambda_\circ/\lambda_\circ)J_h$. Coarea covariance in the target direction therefore yields

$$|V_h^{\varepsilon}(t) - O_h^{\varepsilon}(t)| \le C\sqrt{h}\int_{S_h(t)} J_h\,d\mu_t \le C\sqrt{h}\,p(t+h).$$

The target sum contains all its actual words, and $W^{\varepsilon}\le 1$ was used only for an upper bound. Average over $0<h<r$ and bound the forward average of the nonnegative density by $2A_r p$. This proves (33.5).

For clarity, the unweighted input used in the last step is

$$\limsup_m \sup_{R,n,k} m^2\|A_r p\|_\infty \le C \quad \text{for every previously fixed } r>0.$$

It follows from Input statement 29.1: for the auxiliary parameter 1 the initial factor is $\Theta(c_0)=0$, whence $W^1=1$ and $b^1=p$. The precise inherited source propositions are pinned in Section 29.3. No raw pointwise bound for p or collision-dependent averaging width is used. □

Define the separate ordered heights

$$\begin{aligned}
\mathfrak{X}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{E}_r^{\varepsilon},\\
\mathfrak{V}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{V}_r^{\varepsilon},\\
\mathfrak{O}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{O}_r^{\varepsilon}.
\end{aligned} \tag{33.7}$$

Equation (33.6) gives both $\mathfrak{V}\le\mathfrak{O}+C\sqrt{r}$ and $\mathfrak{O}\le\mathfrak{V}+C\sqrt{r}$ as extended nonnegative inequalities. Thus the unestimated interior volume contribution has been reduced to inverse-twist loss, with no unestimated third derivative of the reduced roof left in this comparison.

Theorem 33.2 (The unchanged pointwise error with separated defects)**.** *Under Input statement 29.1, for sufficiently large fixed B and every fixed $r > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}. \tag{33.8}$$

Here $\varepsilon(B) = A_0 B^{-1/12}$. The order is

$$B,\ \varepsilon(B),\ r,\ \textit{auxiliary band fixed};\quad m\to\infty;\quad \textit{auxiliary band}\to\infty;\quad B\to\infty.$$

In particular, separate bounds

$$\mathfrak{X}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_X} r^{\beta_X}, \qquad \mathfrak{O}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_O} r^{\beta_O} \tag{33.9}$$

with finite nonnegative q_X, q_O and positive β_X, β_O imply the original pointwise assertion. One may choose $r = c\varepsilon(B)^\gamma$ with $\gamma > \max\{2, q_X/\beta_X, q_O/\beta_O\}$. The two estimates in (33.9) *are proof obligations, not conclusions of the present theorem.*

Proof. The stopped subcoupling has density $\min\{W(x), W(\Phi_h x)J_h(x)\}$ on survivors. Its two marginals are dominated by the original weighted fibers. The elementary bound $[w - w'J]_+ \le w[1-J]_+ + J|w-w'|$, the complete guard modulus of Corollary 31.3, and target coarea covariance give

$$b^\varepsilon(t) \le b^\varepsilon(t+h) + E_h^\varepsilon(t) + V_h^\varepsilon(t) + K_\Theta \varepsilon^{-1}\sqrt{h}\, p(t+h).$$

Average at the previously fixed width and apply (33.5). The averaged bad source is bounded at the ordered height by $2C\varepsilon^{1/16}$; the unweighted target is bounded as in the preceding proof. Inserting the positive scalar remainder gives (33.8). All inequalities precede the suprema and the collision limit. The displayed choice of γ makes each of the three explicit powers positive, proving the implication. □

Corollary 33.3 (An exact separated criterion)**.** *Under the same analytic input, the original pointwise assertion is equivalent to, for every M and every fixed $r_0 > 0$,*

$$\lim_{B\to\infty} \inf_{0<r<r_0} \left\{ \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + (1+\varepsilon(B)^{-1})\sqrt{r} \right\} = 0. \tag{33.10}$$

Proof. Sufficiency follows by approximating the infimum before the collision limit. For necessity, the positive remainder and the pointwise assertion give ordered height at most $C_M B^{-1/192}$ for $b^{\varepsilon(B)}$. Both E_h^ε and O_h^ε are bounded by this same positive density at the starting roof. Their separate ordered heights are therefore at most this bound, uniformly in r. Take the infimum and then $B \to \infty$. This argument does not claim that the infimum is attained or select a width depending on m. □

34. Optical sensitivity and source-weighted exit flux

The two terms in (33.8) have distinct geometric mechanisms. We first give a positive, collision-uniform sensitivity formula for the optical term, and then an exact boundary formula for the initial source mass which exits. Neither formula identifies a surface insertion with a bounded volume multiplier.

34.1. **A positive optical polynomial.** For a regular word put $t_j = \ell_j^{-1}$, $V_i = 2/(Rc_i)$ and

$$A_{ii} = t_{i-1} + t_i + V_i, \qquad A_{i,i+1} = A_{i+1,i} = -t_i. \tag{34.1}$$

The matrix A is the incidence-normalized optical matrix; the actual internal arclength Hessian is $D_c A D_c$. In the Neumann estimate it is $A = D_A(I - P)$, where D_A is its diagonal and the row sums of P are at most q_0. These two diagonal normalizations are different.

Define $y_0 = 0$, $f_0 = 1$, $y_1 = \ell_0$, and

$$f_i = f_{i-1} + V_i y_i, \qquad y_{i+1} = y_i + \ell_i f_i \quad (1 \le i < m). \tag{34.2}$$

The continuant and corner Schur-complement identities give

$$y_m = \left(\prod_{j=0}^{m-1} \ell_j\right) \det A, \qquad \omega = \frac{Rc_0c_m}{y_m}. \tag{34.3}$$

For $m = 1$ the determinant is empty and $y_1 = \ell_0$. Indeed eliminating f_i gives $y_{i+1} = (1+\ell_i/\ell_{i-1}+\ell_iV_i)y_i - (\ell_i/\ell_{i-1})y_{i-1}$, the continuant recurrence with the same initial values. In the mixed endpoint Schur complement all internal incidence squares cancel against $\det(D_cAD_c)$, leaving $|F_{uv}| = c_0c_m\prod t_j/\det A$ in endpoint arclengths. The change to endpoint angles gives $\omega = R|F_{uv}|$ as stated.

Proposition 34.1 (Optical logarithmic sensitivities)**.** *The polynomial y_m has nonnegative coefficients and is of degree at most one in each of the separate variables ℓ_j, V_i. Thus*

$$a_i = V_i\partial_{V_i}\log y_m \in [0,1], \qquad b_j = \ell_j\partial_{\ell_j}\log y_m \in [0,1]. \tag{34.4}$$

At fixed R its differential on any actual regular word is

$$d\log\omega = d\log c_0 + d\log c_m + \sum_{i=1}^{m-1} a_i\, d\log c_i - \sum_{j=0}^{m-1} b_j\, d\log\ell_j. \tag{34.5}$$

In endpoint-angle coordinates, for a fixed uniform C_O,

$$\|\nabla_q \log\omega\| \le C_O\left(1 + \sum_{i=1}^{m-1}\frac{q_0^{d_i}}{c_i}\right). \tag{34.6}$$

No third derivative or sum of absolute word multiplicities occurs in this bound.

Proof. Each new V_i enters the positive recursion once, and each new ℓ_i enters once after that step. Later operations are linear in all previous variables. Induction proves positivity and separate affineness. For a positive separately affine polynomial $Y = A + Bz$, $z\partial_z\log Y = Bz/(A+Bz) \in [0,1]$. Applying this identity to each variable gives (34.4); differentiating (34.3) gives (34.5).

The endpoint incidences are bounded below by $\sqrt{19}/10$. Theorem 31.2 bounds $\|D_qc_i\|$ by $C_Dq_0^{d_i}$. Also $|d\ell_j| \le |dQ_j| + |dQ_{j+1}|$ and $\ell_j \ge \tau_-$, so $\|D_q\log\ell_j\| \le 2C_Qq_0^{d'_j}/\tau_-$. The sum of these flight bounds is uniform in m. Inserting the sensitivity weights proves (34.6). The inverse-incidence terms are retained; positivity does not make their complete source sum finite. □

34.2. **A quantified regular part of the optical loss.** For $0 < \delta \le 1$ let $S_h^\delta(t) \subset S_h(t)$ consist of survivors satisfying, for every internal contact and every $0 \le s \le h$,

$$c_i(\Phi_s x) \ge \delta q_0^{d_i/2}. \tag{34.7}$$

Write $O_h^{\varepsilon,\mathrm{reg}(\delta)}$ for (33.3) restricted to this set, and $O_h^{\varepsilon,\mathrm{thin}(\delta)}$ for its nonnegative complement. This is a decomposition of the original weighted source, not a new guard or a renormalized probability law.

Theorem 34.2 (Uniform regular optical distortion)**.** *There are uniform positive c_1, C_1, C_2 such that, whenever $0 < r \le c_1\delta^2$,*

$$\limsup_m \sup_{R,n,k} m^2\|\bar O_r^{\varepsilon,\mathrm{reg}(\delta)}\|_\infty \le C_1\delta^{-1}\sqrt{r}, \tag{34.8}$$

$$\bar O_r^{\varepsilon,\mathrm{thin}(\delta)}(t) \le b^{C_2\delta}(t) \quad \textit{for almost every } t. \tag{34.9}$$

The first estimate uses only the stated fixed-width analytic input. The second is a pointwise source domination and is not a bound for the normalized essential height of its right side.

Proof. On the corridor, the sum in (34.6) is at most $2\delta^{-1}/(1-\sqrt{q_0})$. The forward endpoint-curve length is at most $C_X\sqrt{h}$. Hence $|\log \eta_h| \le C\delta^{-1}\sqrt{h}$. Choose c_1 so that this upper bound is at most one for $h \le r$. Then $[1-\eta_h]_+ \le C\delta^{-1}\sqrt{h}\,\eta_h$. As in the proof of Theorem 33.1, $\eta_h \le (\Lambda_\circ/\lambda_\circ)J_h$ and target coarea gives

$$O_h^{\varepsilon,\mathrm{reg}(\delta)}(t) \le C\delta^{-1}\sqrt{h}\,p(t+h).$$

Averaging first proves (34.8).

If the corridor fails at time $s \le h \le r$, the margin motion bound implies at the starting point

$$c_i(x) \le \delta q_0^{d_i/2} + C_D C_X q_0^{d_i}\sqrt{r} \le C_2\delta q_0^{d_i/2}.$$

Since $q_0^{d_i/2} \le q_0^{d_i/4}$, the corresponding factor of the original guard at auxiliary parameter $C_2\delta$ is zero. Thus $W^{C_2\delta}(x) = 1$ on the entire thin starting set. Its optical loss is bounded by its original weighted source, and $W^\varepsilon \le 1$. This proves (34.9) before averaging, hence afterwards. There are no internal contacts when $m = 1$ and the thin set is empty. □

The theorem quantitatively controls the regular optical contribution on the full exact event. It does not allow replacement of $b^{C_2\delta}$ by its fixed-width average at a starting roof. That replacement would be exactly the missing pointwise step.

34.3. **Exit mass in the correct flux convention.** The initial mass of trajectories which exit need not equal the physical flux measured at their exit. Compression between the starting roof and the exit must be retained. We state the exact formula on regular subdomains and recover the full stopped event by exhaustion.

Let D be a relatively compact regular domain in the exact event, away from $g = 0$, with piecewise smooth boundary. We take the boundary transverse to X except for pieces and lower-dimensional strata whose images in the angular coordinate have zero length. Such domains are obtained from finite flow boxes; increasing exhaustions can be formed by refining the boxes. Let Γ_D^+ denote its outward boundary, N its outward unit normal in the collision coordinates, and $\tau_D^-(y)$ the first backward lifetime from y inside D. Use the physical flux measure

$$d\zeta_D(y) = \frac{[X(y)\cdot N(y)]_+}{4\pi c_*}\, d\mathcal{H}^1(y). \tag{34.10}$$

Proposition 34.3 (First-exit flux with the initial source weight)**.** *Let $E_{D,h}^\varepsilon(t)$ be the weighted initial mass on $L = t$ starting in D and exiting D within h. For almost every (t,h),*

$$E_{D,h}^\varepsilon(t) = \int_{\Gamma_D^+} \mathbf{1}_{\{0<L(y)-t\le h,\ L(y)-t<\tau_D^-(y)\}} \frac{W^\varepsilon(x_{t,y})}{J_{L(y)-t}(x_{t,y})}\, d\zeta_D(y), \quad x_{t,y} = \Phi_{t-L(y)}y. \tag{34.11}$$

For an increasing regular exhaustion D_ℓ of the full regular exact event, its original true-exit mass satisfies

$$E_h^\varepsilon(t) = \lim_{\ell\to\infty}\left\{\int_{\{L=t\}\setminus D_\ell} W^\varepsilon\, d\mu_t + E_{D_\ell,h}^\varepsilon(t)\right\}. \tag{34.12}$$

The bracketed nonnegative quantities decrease to the left side. The limit is at fixed collision count; it is not interchanged with a normalized collision limsup.

Proof. In a flow box use $(u,\theta) = (L,\vartheta)$, so that $X = \partial_u$. The physical source has area density $a(u,\theta) = \omega/(4\pi c_*\chi)$ relative to $du\,d\theta$. On a transverse outward boundary $u = b(\theta)$, its flux is $a(b(\theta),\theta)|d\theta|$. At the starting roof its coarea mass is $a(t,\theta)|d\theta|$, and

$$J_{b(\theta)-t}(x_{t,y}) = a(b(\theta),\theta)/a(t,\theta).$$

Multiplication by the initial weight proves (34.11) on the box. The backward lifetime selects the first forward exit rather than a later re-entry. Uniqueness of the flow makes the source pieces disjoint; a measurable refinement of the boxes sums them without multiplicity. The excluded boundary strata have zero angular measure and hence zero source-fiber measure. Coarea supplies the joint almost-everywhere statement.

For each starting point, declare the D_ℓ-lifetime zero when it starts outside D_ℓ. These lifetimes increase to the lifetime in the full event: every compact surviving flow segment eventually lies in some D_ℓ. The indicators of lifetime at most h therefore decrease to the corresponding full-event indicator. Dominated convergence on each finite coarea fiber proves (34.12). The exceptional (t,h) set can be chosen jointly by disintegration. Chart boundaries in the exhaustion are not declared physical; their effects disappear only through this source-level limiting identity. □

The inverse Jacobian in (34.11) is essential. Along a limiting grazing boundary the physical density may tend to zero while this inverse factor diverges. Vanishing unweighted exit flux therefore does not make the original initial exit mass zero. The formula retains grazing, competing-hit and section exits. If several margins first vanish simultaneously, a fixed ordering partitions the initial source into disjoint exit types; their sum is E_h^ε. The internal occupation-current identity in the supplement is signed. It has not been used to subtract any of these positive exit masses.

35. Homogeneity and a superlinear incidence estimate

The inverse incidences in (34.6) can be summed on the complete source. The required improvement is not a bound for multiplication by c^{-1} on the strong space. It is a stronger estimate for the real-frequency physical powers, followed by a summable, width-dependent remainder.

35.1. **The precise spectral input.** Write $Q = \mathscr{T}_{R,\xi}$ for the uncentered four-coordinate operator, with $\xi = (u_1, u_2, b, v)$ real. The densities in the collision spaces are relative to ν; the initial vector is $\mathbf{1}$, and

$$Q^j\mathbf{1}(x) = \exp\big(i\xi \cdot S_j f(T_R^{-j}x)\big), \qquad |Q^j\mathbf{1}| = 1 \tag{35.1}$$

on the regular physical realization. Here $S_j f$ denotes the full record, not the roof alone. This equality would not justify a cosine factor in the stable-curve norm.

We use the following source-pinned statements. The norm definitions are Supplement equations (117.1)–(117.3); the full-occupation Doeblin–Fortet inequality and fixed-band spectrum are in Supplement Theorem 157.3 and the moving-peak construction used in Supplement Lemma 175.2. The incidence multiplier $M_s = M_{\mathbf{1}_{\{c<s\}}}$ satisfies

$$\|M_s h\| \le C\|h\|, \qquad |M_s h|_w \le Cs^{1/8}\|h\|. \tag{35.2}$$

This is Supplement Proposition 178.2. On a fixed finite collection of peak charts one has

$$\begin{aligned} &Q^j = \lambda^j \Pi + N^j, && \|N^j\| \le C\rho^j, && \|Q^j\| \le C, \\ &\|Q^j h\| \le C\sigma^j\|h\| + C(1+j)|h|_w, && |Q^j h|_w \le C(1+j)|h|_w, \end{aligned} \tag{35.3}$$

where $\rho, \sigma < 1$. The fixed-band complement has a uniform spectral gap. The four-frequency Gaussian integral on each peak is $O(m^{-2})$. These are inherited spectral theorems, not consequences of the coarea argument.

After shrinking the neighborhoods of unit-modulus points, we may take $|\lambda| \ge r_*$ on the retained charts with

$$\frac{\log(1/r_*)}{\log(1/\sigma)} < \frac{1}{32}, \qquad \frac{\log(1/r_*)}{\log(1/\rho)} < \frac{1}{24}. \tag{35.4}$$

The removed compact portions have a spectral gap and join the complement. This is a fixed refinement, independent of m, s and of the eventual outer band. No arithmetic branch is discarded.

Lemma 35.1 (Real powers on homogeneous grazing strips)**.** *For all $j \geq 0$ and sufficiently small $s > 0$, uniformly on the retained real peak charts,*

$$|M_s Q^j \mathbf{1}|_w \leq C s^{3/2}. \tag{35.5}$$

The exponent comes from the length of a homogeneous stable curve, not from replacing the density convention by Lebesgue density.

Proof. The quadratic homogeneity strips have angular widths comparable to k^{-3} and incidences comparable to k^{-2}. In the standard coordinates they lie between $\pi/2 - (k+1)^{-2}$ and $\pi/2 - k^{-2}$, or their negative counterparts. A stable graph has slope bounded away from zero and infinity. Consequently a homogeneous stable curve in the kth strip has length at most Ck^{-3}. If that strip meets $\{c < s\}$, then $k \geq C^{-1} s^{-1/2}$; the same estimate includes a threshold crossing the strip. Thus the entire tested intersection has length at most $Cs^{3/2}$. For small s, the central strips do not meet it. Equation (35.1) and the supremum bound of a weak test now prove (35.5). This proof applies to the physical branchwise realization; the bounded multiplier and physical realization in (35.2) give the same element of the weak completion. □

Lemma 35.2 (Interpolation close to the weak endpoint)**.** *On the fixed charts chosen above,*

$$\|\Pi h\| + \sup_{a \geq 0} |\ell(Q^a h)| \leq C \|h\|^{1/8} |h|_w^{7/8}. \tag{35.6}$$

Moreover

$$|M_s \Pi \mathbf{1}|_w \leq C s^{23/16}. \tag{35.7}$$

Proof. Normalize $\|h\| = 1$ and put $t = |h|_w$. For $0 < t < 1/2$ choose L with $\sigma^L \leq t < \sigma^{L-1}$. Commutation and (35.3) give

$$\|\Pi h\| \leq C r_*^{-L} \{\sigma^L + (1+L)t\} \leq C(1 + |\log t|) t^{31/32} \leq C t^{7/8}.$$

For $a \leq L$ the weak power bound controls the terminal pairing; for $a > L$ apply the strong inequality at L and power boundedness afterwards. Both costs are at most $C(1+L)t \leq Ct^{7/8}$. The cases of t bounded below and $t = 0$ follow, respectively, by power boundedness and by letting $L \to \infty$. Homogeneity gives (35.6).

For (35.7), choose instead $L = \lceil (3/2) \log(1/s) / \log(1/\rho) \rceil$. The exact identity

$$M_s \Pi \mathbf{1} = \lambda^{-L} M_s Q^L \mathbf{1} - \lambda^{-L} M_s N^L \mathbf{1}$$

uses $N^L = Q^L - \lambda^L \Pi$. Its weak norm is at most $C r_*^{-L} (s^{3/2} + s^{1/8} \rho^L)$. Here $\rho^L \leq s^{3/2}$ and $r_*^{-L} \leq C s^{-1/16}$ by (35.4). This proves the assertion. The strong norm of $M_s \Pi \mathbf{1}$ remains bounded. This argument does not assign modulus one to the continued eigenvector. □

Theorem 35.3 (A capped superlinear marked-incidence bound)**.** *There is C such that, for every fixed auxiliary roof band $B_* \geq B_0$, there are $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ satisfying*

$$\begin{aligned} m^2 \nu\{K_m^{\mathrm{c}} = k, A_m = l, L_m \in J,\ c(T_R^j x) < s\} \\ \leq C s^{5/4} (|J| + B_*^{-1}) + C_{B_*} (1 + B_* |J|) m^2 \min\{\rho_{B_*}^{m/2}, s^2\}. \end{aligned} \tag{35.8}$$

The estimate is uniform in R, $m \geq 2$, $0 \leq j \leq m$, exact labels, intervals J and $0 < s < s_0$. In particular the mark and threshold may vary with m. The leading constant is independent of the previously fixed outer band.

Proof. The exact characteristic pairing is $\ell(Q^{m-j}M_sQ^j\mathbf{1})$. If the left block is long, expand that block and apply (35.6) and (35.5) to $M_sQ^j\mathbf{1}$. The leading amplitude is at most $Cs^{21/16}$. If the right block is long, its leading term is controlled by (35.6) and (35.7); its amplitude is at most $Cs^{161/128}$. Both are bounded by $Cs^{5/4}$, since $161/128 > 5/4$. The complementary term is at most $C\rho^{m/2}$. On a fixed-band complement a long block gives $C_{B_*}\rho_{B_*}^{m/2}$.

There is also the independent physical bound at every real frequency,

$$|\ell(Q^{m-j}M_sQ^j\mathbf{1})| \le \nu\{c < s\} = 1 - \sqrt{1-s^2} \le s^2.$$

If $|z| \le A + D$ and $|z| \le V$, then $|z| \le A + \min(D, V)$. Apply this with the Gaussian leading term as A. Its four-frequency integral is $Cs^{5/4}m^{-2}$; the remainder retains the cap $\min(C_{B_*}\rho_{B_*}^{m/2}, Cs^2)$. The positive interval majorant has Fourier supremum at most $|J| + 2\pi/B_*$ and roof support in $[-B_*, B_*]$. Torus coefficient extraction and integration over that fixed band therefore give (35.8). The cap is essential: an error independent of s could not be summed against an inverse incidence. □

36. Unbounded insertions and the optical source

The cap in (35.8) permits summation all the way to zero incidence. In particular the thin component is no longer estimated by the pointwise height of another bad source.

For $0 < a < 5/4$ and $0 < s < s_0$ put $h_{a,s}(x) = c(x)^{-a}\mathbf{1}_{\{c(x)<s\}}$. The value on the collision-null grazing set is immaterial.

Theorem 36.1 (A real-frequency unbounded insertion estimate)**.** *For every fixed auxiliary band B_*, uniformly in the mark and the original exact collision labels,*

$$\begin{aligned} m^2 \int h_{a,s}(T_R^j x)\mathbf{1}_{\{K_m^c=k, A_m=l, L_m\in J\}}\, d\nu(x) \\ \le C_a s^{5/4-a}(|J| + B_*^{-1}) + C_{B_*,a}(1 + B_*|J|)m^2 \min\{s^{2-a}, \rho_{B_*}^{m(2-a)/4}\}. \end{aligned} \tag{36.1}$$

No bounded strong-space multiplication operator $M_{c^{-a}}$ is asserted. The formula concerns the actual physical insertion; all thresholds are summed before either the collision or band limit.

Proof. The positive layer-cake identity is

$$c^{-a}\mathbf{1}_{\{c<s\}} = s^{-a}\mathbf{1}_{\{c<s\}} + a\int_0^s v^{-a-1}\mathbf{1}_{\{c<v\}}\, dv.$$

First truncate the integral at a positive lower endpoint. Apply Theorem 35.3 and then monotone convergence. The leading integral is finite since $a < 5/4$. For $D \in (0, 1]$ its remainder satisfies

$$s^{-a}\min(D, s^2) + a\int_0^s v^{-a-1}\min(D, v^2)\, dv \le C_a \min\{s^{2-a}, D^{1-a/2}\}. \tag{36.2}$$

For $s \le \sqrt{D}$ integrate v^{1-a} directly; for larger s split at $\sqrt{D}$. This proves the inequality for all $a < 2$ and, with $D = \rho_{B_*}^{m/2}$, proves (36.1). Physical finiteness is consistent with $\nu\{c < v\} \le v^2$ and invariance at the marked time. No infinite sum of an uncapped spectral error is taken. □

Retain $d_i = \min(i, m-i)$ and define the additive optical insertion on the original m-collision orbit by

$$I_m(x) = 1 + \sum_{i=1}^{m-1} \frac{q_0^{d_i}}{c_i(x)}, \qquad u_{n,k,m,R}(t)\, dt = L_\#(I_m\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*). \tag{36.3}$$

There is no product over all inverse incidences in this definition. The source keeps its initial and terminal tests. For each finite count it is finite and absolutely continuous: on compact regular charts this is ordinary coarea, and the removed critical points have zero source measure.

Alternatively it is dominated by a constant multiple of the complete inverse-incidence source in Supplement Theorem 97.3.

Corollary 36.2 (Complete depth-weighted source concentration)**.** *For every previously fixed* $r > 0$*,*

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_r u_{n,k,m,R}\|_\infty \le C. \tag{36.4}$$

If only the summands with $c_i < s$ *are retained in* I_m*, the right side is at most* $Cs^{1/4}$*. The leading constants do not depend on* r*,* s *or the collision count.*

Proof. Apply (36.1) with $a = 1$ and sum with weights $q_0^{d_i}$. Their sum is at most $2q_0/(1-q_0)$. The truncated terms $c_i^{-1}\mathbf{1}_{\{c_i \ge s_0\}}$ and the constant one use the inherited unweighted concentration estimate. For each positive marked integral, dropping the endpoint tests gives an upper bound with initial measure ν/c_* and the *same* exact coefficient (k, n); its inverse-incidence weight is retained. This upper bound does not alter the source definition. No neighboring occupation labels are pooled. Take $J = [t - r, t + r]$, divide by $2r$, take the collision limsup with B_* fixed, and only then let $B_* \to \infty$. The capped remainders decay exponentially after the complete sum. The thin version follows from the same formula at threshold s. □

36.1. **The physical normalization.** The convention of (34.3) can also be checked directly from the endpoint first variation. In endpoint *angles*, $F_{\alpha_0} = -Rp_0$, and hence

$$\omega = |\partial_{\alpha_m} p_0| = \frac{|K_{0m}|}{R}, \qquad d\bar{\nu} = \frac{\omega}{4\pi c_*}\, d\alpha_0\, d\alpha_m, \qquad \omega = \frac{Rc_0 c_m}{y_m}. \tag{36.5}$$

In internal *arclengths* the Hessian is $D_c A D_c$. The corner Schur complement cancels its internal incidence squares; the two endpoint angle changes supply R^2, and the first-variation conversion divides by R. Thus only the one factor R in (36.5) remains. For one flight $y_1 = \ell_0$; for two flights $y_2 = \ell_0 + \ell_1 + 2\ell_0\ell_1/(Rc_1)$. These agree with the direct one- and two-flight Schur complements. Tangent orientation changes conjugate the internal matrix by diagonal signs and leave these absolute identities unchanged.

The cubic cancellation in Lemma 32.1 uses these same mixed coordinates. For an internal vector ζ, differentiated stationarity gives $D^2 S[(u, Dz\, u), (0, \zeta)] = 0$. Every derivative of a lifted contact direction in $D^2 F$ is purely internal, so these terms vanish before any estimate is taken. At an endpoint the lift has arclength component Ru_i; the circle's second and third arclength derivatives have norms R^{-1} and R^{-2}. This accounts for the factors in C_3 and T_* without treating the endpoint angles as arclengths.

36.2. **All incidences in the noncritical optical estimate.** Let $g = \nabla_q F$ and let $\tau(x)$ be the forward lifetime of the canonical field. For $a > 0$, a superscript a means restriction of an *initial* source to $|g(x)| \ge a$. Define $O_h^{\varepsilon,a}$ by this restriction in (33.3). No internal incidence corridor is imposed.

A finite true exit is called grazing here when at least one internal incidence tends to zero as the lifetime is approached. All contacts have limits by the all-contact motion estimate. Assign a simultaneous failure to this grazing class whenever an incidence vanishes; every other finite exit belongs to the disjoint nongrazing class. Let $E_h^{\varepsilon,\mathrm{gr},a}$ be its weighted starting mass with $\tau \le h$ and $|g| \ge a$.

Theorem 36.3 (Complete optical loss and genuine grazing exits)**.** *For each $a, r > 0$,*

$$O_h^{\varepsilon,a}(t) + E_h^{\varepsilon,\mathrm{gr},a}(t) \le \frac{C}{a}\int_0^h u(t+s)\,ds, \tag{36.6}$$

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,a} + \bar{E}_r^{\varepsilon,\mathrm{gr},a}\|_\infty \le \frac{Cr}{a}. \tag{36.7}$$

The first assertion holds jointly for almost every (t,h). Constants are independent of ε and of all positive internal incidences and clearances. The estimate includes every actual word of the original exact event with the stated initial noncritical restriction.

Proof. The field and (34.6) give, along a survivor,

$$|X\omega| \le C\frac{\omega I_m}{|g|}. \tag{36.8}$$

The radial speed implies $|g(\Phi_s x)| \ge |g(x)| \ge a$. For a surviving orbit the positive decrease of ω up to h is bounded by the integral of $[-X\omega]_+$ over that segment. At a grazing exit, (34.3) and the selected-incidence determinant bound imply $\omega(\Phi_s x) \to 0$. This last assertion needs only a finite-count bound $\omega \le C_m c_i$; it does not sum such constants over words. Thus the entire starting optical weight at such an exit is bounded by the same negative-variation integral up to τ. The two source cases are disjoint. Other exits contribute zero to the left.

In a local angular chart the initial fiber density is $\omega(x)/(4\pi c_*\chi(x))\,d\vartheta$. At time s, the ratio $\omega(\Phi_s x)/\omega(x)$ is at most $(\Lambda_\circ/\lambda_\circ)J_s(x)$. With $W^\varepsilon(x) \le 1$, coarea covariance in the *target* direction bounds the integral of the drift in (36.8) by $Ca^{-1}u(t+s)$. The target image is a subset of the same exact fiber. The common-kernel construction below justifies this comparison simultaneously on all physical words and at joint almost-everywhere levels. Tonelli proves (36.6). Averaging over $0 < h < r$ gives an upper bound $2Cra^{-1}A_r u(t)$. Apply Corollary 36.2. In particular the initial mass of a grazing exit has been paid by optical drift; it has not been declared zero because its boundary optical density vanishes. □

Without the initial lower bound, the same proof gives the extended nonnegative inequality

$$O_h^{\varepsilon}(t) + E_h^{\varepsilon,\mathrm{gr}}(t) \le C\int_0^h s^{-1/2}u(t+s)\,ds. \tag{36.9}$$

It uses $|g(\Phi_s x)|^2 \ge 2\lambda_\circ s$. A bound for every fixed average of u is not a bound for this singular convolution at its initial roof, uniformly in m. Equation (36.9) is therefore not used to remove the critical restriction in (36.7).

37. Common fibers and the remaining boundary source

The singular insertion estimate acts on the original volume source. Its use for optical drift requires common versions of that source on all words. For a nongrazing exit it also identifies which part of the inverse-Jacobian pullback has been paid by the drift estimate. Neither step changes the exact event.

Lemma 37.1 (A common physical coarea version)**.** *At a fixed collision count, radius and exact label, the original regular source admits jointly measurable coarea kernels μ_t and a survivor graph for which*

$$\int_{S_h(t)} f(\Phi_h x)J_h(x)\,d\mu_t(x) = \int_{\Phi_h S_h(t)} f(y)\,d\mu_{t+h}(y) \tag{37.1}$$

holds as an identity of kernels for almost every (t,h) and every nonnegative Borel f. There is also a precompact regular exhaustion $D_j \uparrow \Omega$ for which the source-level first-exit formula of Proposition 34.3 holds monotonically. Auxiliary chart seams are not exits of Ω.

Proof. The regular physical words are disjoint open pieces of the collision source. Their boundaries and critical points have zero source area. Off these sets, (F,ϑ) is a local diffeomorphism with positive Jacobian χ in endpoint coordinates. Choose countably many lifted angular intervals and flow boxes whose closures lie in these regular pieces and whose (F,ϑ) images are rectangles with rational endpoints. They can be chosen to cover Ω. Disjointize the boxes by assigning a point to its first containing box. This is a Borel assignment for summation only; the flow does not stop at the resulting Borel seams.

On the open survivor set in $\Omega\times(0,\infty)$ the map $(x,h)\mapsto(\Phi_h x,h)$ is one-to-one and smooth, with Jacobian J_h relative to the physical source times dh. Its inverse is the backward flow on its image. Apply change of variables in each box and disintegrate with respect to (F,h). A countable determining algebra for the Borel sets of the collision cylinder identifies both kernel measures outside one null set in (t,h). Equality for all nonnegative Borel functions then follows by monotone approximation. Countable disjoint summation gives (37.1). The same kernels accommodate a source restriction depending on (x,h) by using the corresponding graph measure. Thus survivor restrictions, truncated optical insertions and their increasing limits do not require incompatible density representatives.

For the exhaustion take the union of the first j open precompact rational rectangles, arranging that their interiors cover Ω. Finite overlaps can be refined in the common (F,ϑ) coordinates on each physical branch. The boundary faces are roof-constant or angle-constant, with finitely many corners after refinement. The first kind is transverse to X; the second has zero flux because $X\vartheta=0$. Corners have zero transverse measure. The ordinary flow-box exit formula applies, with outward orientation and backward lifetime selecting the first exit. Its value plus the initial mass outside D_j is the integral of the indicator that the forward segment of length h is not contained in D_j. These indicators decrease. A surviving segment is compactly contained in Ω and is therefore contained in some D_j. A segment ending at a true exit is never contained in any such precompact set. This proves the fixed-count monotone limit, with the usual harmless endpoint convention at lifetime h. No exhaustion limit is moved through the collision-normalized supremum. □

37.1. **Nongrazing first-exit flux.** For this paragraph initial points satisfy $|g|\ge a>0$. Along a nongrazing finite exit all incidences have positive limits. The selected stationary word consequently has a smooth local extension to the limiting boundary point; this is an algebraic extension, not an assertion that the same trajectory remains the physical first-hit orbit beyond that point. The limiting ω and χ are positive. Section and competing-hit faces remain true boundaries.

Use the one-sided outgoing nongrazing faces of the exact event, with their physical flux measure $d\zeta$ from (34.10). Boundary points are counted with their incident physical component. In a regular angular flow box each transverse outgoing face is a graph $F=b(\vartheta)$ and its measure is $\omega(b,\vartheta)/(4\pi c_*\chi(b,\vartheta))\,d\vartheta$. The finite-word tame stratification gives these graphs after partition into regular arcs; angular critical and tangent strata have zero transverse measure. Equivalently one can use a countable disjoint refinement of the one-sided graphs. First-exit selection, not graph multiplicity, defines the measure below.

Define the unweighted outgoing flux assigned to an initial roof by

$$\mathcal{F}_h^a(t)=\int_{\Gamma_{\mathrm{ng}}^+}\mathbf{1}_{\{0<s=L(y)-t\le h,\ s<\tau^-(y),\ |g(\Phi_{-s}y)|\ge a\}}\,d\zeta(y). \tag{37.2}$$

Here τ^- is the backward lifetime in the same one-sided regular exact event. The integral is an extended nonnegative quantity. It contains initial, terminal and internal section decisions and competing-hit exits whenever they are nongrazing. The guard is not inserted at the boundary and no labels are pooled.

Theorem 37.2 (Paying the pullback by singular volume drift)**.** *For almost every* (t,h)*,*

$$E_h^{\varepsilon,a}(t)+O_h^{\varepsilon,a}(t)\le C\mathcal{F}_h^a(t)+\frac{C}{a}\int_0^h u(t+s)\,ds. \tag{37.3}$$

The boundary term has no inverse Jacobian because its missing pullback cost is included in the proved volume-drift term. It is not obtained by deleting the inverse Jacobian from the first-exit identity.

Proof. In an angular chart write the optical weight along a starting orbit as $w(s)=\omega(\Phi_s x)$. For a nongrazing exit at $\tau\le h$,

$$w(0)\le w(\tau-)+\int_0^\tau [-w'(s)]_+\,ds.$$

Divide by $4\pi c_*\chi(x)$ and use $W^\varepsilon(x)\le 1$. The first term is at most $(\Lambda_\circ/\lambda_\circ)$ times the outgoing physical flux on the corresponding first-exit graph. The restriction in (37.2) is exactly its initial-source selection. At a grazing exit $w(\tau-)=0$, but its full initial mass is paid by the negative-variation integral, as in Theorem 36.3. On a surviving trajectory the optical loss is bounded by that integral up to h. These three classes of starting states are disjoint. Sum their negative variations and apply (36.8), the monotonicity $|g(\Phi_s x)|\ge a$, and Lemma 37.1. The result is bounded by the second term of (37.3). Truncate to regular subsets and then use Tonelli and monotone convergence for the nonnegative drift and one-sided face integrals. This proves the assertion also when the right side is infinite. □

37.2. **The original scalar objective.** Let $C_h^{\varepsilon,a}$ be the sum of the original true-exit and optical defects restricted to initial $|g|<a$. Critical points $g=0$ themselves have zero source measure and are not assigned an artificial boundary mass. Write

$$\mathfrak{C}_{\varepsilon,M}(a,r)=\limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \frac{1}{r}\int_0^r C_h^{\varepsilon,a}(t)\,dh,$$

$$\mathfrak{F}_M(a,r)=\limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \frac{1}{r}\int_0^r \mathcal{F}_h^a(t)\,dh.$$

The distinction between a critical neighborhood and a grazing incidence is important: the latter has already been summed in (36.4), including arbitrarily small incidences.

Corollary 37.3 (The improved complete-source endpoint budget)**.** *With the inherited scalar input and* $\varepsilon(B)=A_0B^{-1/12}$*, for all previously fixed* $a,r>0$*,*

$$\begin{aligned}\mathcal{E}_M \le{}& C_M B^{-1/192}+\mathfrak{C}_{\varepsilon(B),M}(a,r)+C\mathfrak{F}_M(a,r)\\ &+Cr/a+C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.\end{aligned} \tag{37.4}$$

All exact arithmetic classes and the original source are retained.

Proof. Apply the source-shift inequality in the proof of Theorem 33.2, before taking any supremum, and split its positive sum E_h+O_h by the initial condition $|g|<a$ or $|g|\ge a$. Use Theorem 37.2 in the second part, average, and then use (36.4). Taking the collision limsup only after B,a,r and the auxiliary spectral band are fixed proves (37.4). The separate suprema of E and O need not be replaced by a smaller joint supremum: the argument uses their sum at the source level. □

This budget has a newly estimated all-incidence drift term, not a new pointwise bad-source premise for the thin corridor. To finish the original pointwise theorem one must still bound the critical initial source and the nongrazing outgoing flux in the stated order. For example, outer choices $a(B),r(B)>0$ suffice if their two heights tend to zero, $r(B)/a(B)\to 0$, and $(1+\varepsilon(B)^{-1})\sqrt{r(B)}\to 0$. Those choices must be made before the collision limit. Neither the Abel

representation (36.9) nor the surface representation (37.2) establishes these two decay statements. No surface-current spectral theorem is being inferred from the singular *volume* insertion proved here.

38. The remaining scalar estimates and their consequences

The organizing assertion remains (29.5), with the complete arithmetic transition, original normalization, exact labels and original smooth guard. The new singular insertion theorem closes the incidence-scale summation needed for optical drift away from the initial critical set. The boundary inequality estimates the same pullback cost for nongrazing exits, but does not bound their outgoing flux height. The two residual sources are displayed in (37.4); neither is suppressed by changing the event or by a signed occupation-current cancellation.

All widths, including the initial critical threshold, are fixed before the collision limit. After that limit the auxiliary spectral band is enlarged, and the outer reconstruction parameter is taken last. The full scalar theorem is not asserted until the two remaining height estimates have been proved in this order. The new capped remainder, common-kernel argument and singular-volume estimate do not license exchanging these limits.

After scalar closure, Supplement Theorem 76.1 and the positive path remainder give the same-roof path numerator. Normalization is made only on positive arithmetic reference classes. Collision-to-return clock transfer and forward likelihood retain their separate inherited hypotheses; arithmetic zero classes are never divided by. The complete preceding article and every inherited proof remain in the supplement.

Part 25. Uniform cubic geometry and separated optical transport

Retained revision-79 abstract. We study the pointwise local limit problem for the full displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. We prove a collision-uniform third-derivative bound for the reduced roof at every regular physical word, without a lower internal-incidence cutoff. A stationary-action identity eliminates all second derivatives of the internal contact map. The physical Jacobian of the canonical endpoint-momentum flow then splits into an angular factor and an inverse-twist ratio. The complete angular-volume error is bounded by a constant times the square root of the previously fixed roof width at the required collision normalization. Positive optical sensitivities give the same type of estimate on incidence corridors, with the remaining thin part charged to the original source. We also express first-exit mass through its initial-weight pullback to outgoing boundaries. The resulting estimate separates true exits from optical loss while retaining every exact label and arithmetic class. The uniform roof-height bounds for these two remaining terms are stated separately; they are not inferred from the controlled angular factor.

39. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U=\{|\alpha-\pi/6|<3/50,\ |p|<3/20\},\qquad Y_R^*=Y_R\cup U,\qquad c_*=\frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*)=c_*$. We always use $\nu_R^*=c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L=L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R}=\sum_{j=0}^{m-1}\mathbf{1}_{Y_R^*}\circ T_R^j$. The event is

$$\Pi_{n,k,m,R}=Y_R^*\cap\{K^{\mathrm{c}}_{m,R}=k,\ A_{m,R}=n\}\cap T_R^{-m}Y_R^*. \tag{39.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j=\sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0\le\Theta\le1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0=47/53$, $d_j=\min(j,m-j)$ and $d_j'=\min(j,m-1-j)$, set

$$H^\varepsilon=\prod_{j=0}^{m}\Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right)\prod_{j=0}^{m-1}\Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right),\qquad W^\varepsilon=1-H^\varepsilon, \tag{39.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t=L_{\#}(W^\varepsilon\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*). \tag{39.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

39.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^m B_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right]. \tag{39.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow0. \tag{39.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

39.2. **The analytic inputs in the form used here.**

Input statement 39.1. The unchanged source (39.3) has a nonnegative, compactly supported BV density at every fixed count. For every fixed $\varepsilon, \delta > 0$, with $A_\delta f(t) = (2\delta)^{-1} \int_{t-\delta}^{t+\delta} f(s)\,\mathrm{d}s$,

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_\delta b^\varepsilon\|_\infty \le C\varepsilon^{1/16}. \tag{39.6}$$

The constant is independent of the previously fixed width. For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{39.7}$$

All auxiliary spectral bands in these statements are fixed before the collision limit and enlarged only afterwards.

These are the complete-source tame theorem, the fixed-width complete-source bound, and the positive scalar remainder theorem in the technical supplement. Their proofs and historical hypotheses are retained there. The fixed-count source uses threshold integration [15], definable monotonicity [9], and the distributional characterization of BV [13]; it does not assume definability of the original smooth guard. They are restated as inputs to distinguish them from the new geometric and transport proofs. No long-time bound on a fixed-count variation constant is part of this statement.

39.3. **Precise inherited inputs.** The fixed-width estimate is Supplement Theorem 179.1, in particular Supplement equation (179.4). It uses Supplement Theorem 178.3, on the original exact displacement and occupation coefficient. For an interval of length $2r$, division by $2r$ gives (39.6). That proof fixes the auxiliary band, takes the collision limit, and only then enlarges the auxiliary band. Its leading constant is independent of the previously fixed interval length. The extension from small margins to a fixed remaining compact range is stated there explicitly. At auxiliary parameter one, $\Theta(c_0) = 0$ since $c_0 \le 1$, so $W^1 = 1$ and $b^1 = p$. This supplies the unweighted fixed-width estimate without asserting a raw pointwise bound. The positive remainder is the inherited scalar input (39.7), stated as Supplement Theorem 181.4; no surface-current estimate is included in any of these volume-source inputs.

39.4. **The principal geometric estimates.** Let $q = (\alpha_0, \alpha_m)$ be local endpoint angles, F the reduced roof, $K = D_q^2 F$, and $\vartheta = \arg(-p_0, p_m)$. The canonical field satisfies $XL = 1$ and $X\vartheta = 0$ and is stopped at the boundary of the regular exact event. Set $\omega = |\partial_{p_0}\alpha_m|^{-1}$ and $\chi = \det D_q(F, \vartheta)$. These quantities and their positive uniform bounds are derived in Section 40. Theorem 42.2 proves

$$\|D_q^3 F\| \le T_*$$

with one constant for all regular physical words. It is a consequence of the local stationary equations and the all-contact first derivatives, not of a uniform implicit-function radius near grazing.

To state the endpoint consequence precisely, let $d\mu_t$ be the unweighted coarea fiber of the original normalized source, $S_h(t)$ the survivors, and J_h the area Jacobian relative to that source. The two remaining quantities at a fixed count are

$$E_h^\varepsilon(t) = \int_{\{L=t\}\setminus S_h(t)} W^\varepsilon \, d\mu_t, \qquad O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon \left[1 - \frac{\omega(\Phi_h x)}{\omega(x)}\right]_+ d\mu_t.$$

The physical Jacobian is $J_h = (\omega \circ \Phi_h/\omega)(\chi/\chi \circ \Phi_h)$. Theorem 43.1 controls, on the complete source, the difference between the physical volume loss and O_h. Averaging at a width fixed before

the collision limit gives an $O(\sqrt{r})$ bound after the m^2 normalization. If $\mathfrak{X}$ and $\mathfrak{O}$ denote the separate ordered heights of the averages of E_h and O_h, respectively, Theorem 43.2 gives

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}.$$

The order is: fix B, $\varepsilon(B)$, r and the auxiliary band; then take the collision limit; enlarge the auxiliary band afterwards; and take $B \to \infty$ last. No positive exit mass has been removed by an internal occupation-current cancellation. Such a cancellation is a signed statement, with nonzero occupation-frequency phases still present.

39.5. **Contact estimates and the literal guard.** The constant in each entry below is uniform in the collision count. Here ℓ_q is the length of a regular endpoint curve and the displayed powers refer to the contact and flight depths defined above.

Quantity	Change bounded by	Original guard factor
Incidence c_j	$Cq_0^{d_j}\ell_q$	Yes
Third-disk clearance Δ_j	$Cq_0^{d'_j}\ell_q$	Yes
Section-edge distance	$Cq_0^{d_j}\ell_q$	No
Flight length ℓ_j	$Cq_0^{d'_j}\ell_q$	No

Section decisions remain exact event tests. They are not new smooth guard factors. The canonical field crosses coordinate cuts by change of chart; it stops at the prescribed physical and section-decision boundaries. Exhaustion boundaries enter only the limiting flux identity in Proposition 44.3.

40. A canonical flow from the endpoint momenta

The roof field can be specified on the whole regular exact event, without choosing critical disks or endpoint caps. The particular section in (39.1) is useful here: it bounds both endpoint momenta away from grazing, even though no internal incidence or clearance is bounded away from zero.

For one regular physical word use locally lifted endpoint angles $q = (\alpha_0, \alpha_m)$, and write $F(q)$ for its reduced length. The endpoint map is locally invertible, since the optical twist $B_m = \partial_{p_0}\alpha_m$ is nonzero. Set

$$g = \nabla_q F = R(-p_0, p_m), \qquad K = D_q^2 F, \qquad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}.$$

The terminal tangential velocity has the same value before and after reflection, which fixes the sign in the second component of g. All differentiations in this section are on actual regular words.

Lemma 40.1 (Uniform endpoint-angle Hessians on the exact event)**.** *At every regular physical trajectory,*

$$R\,\mathrm{diag}(c_0, c_m) \le K \le (2R_+^2/\tau_- + R_+)I_2. \tag{40.1}$$

On the original exact event, in particular,

$$\lambda_\circ I_2 \le K \le \Lambda_\circ I_2. \tag{40.2}$$

These are pointwise quadratic-form inequalities, uniform in m, R, n, k. They assert neither convexity of a disconnected physical domain nor uniform control of derivatives of K.

Proof. At a regular trajectory let w_0 be its outgoing first-flight velocity and w_m the negative of its incoming last-flight velocity. The polygonal length, minimized over the prescribed intermediate closed disks, is supported at this trajectory by

$$F(q_*) - w_0 \cdot (Q_0(q_0) - Q_0(q_{0,*})) - w_m \cdot (Q_m(q_m) - Q_m(q_{m,*})).$$

To see this directly, the internal gradient at contact i is $-2c_in_i$. Its scalar product with a displacement into that disk is nonnegative. The convex gradient inequality for the polygonal length therefore gives the displayed support after minimizing the internal positions. Equality holds at the physical trajectory. The internal Hessian is positive definite, so the reduced roof is smooth on its regular neighborhood. Since $Q_i'' = -Rn_i$ and $w_i \cdot n_i = c_i$, differentiating the touching support proves the lower inequality in (40.1).

For the upper inequality keep the internal contacts fixed while varying the endpoints. The transverse second variation of a flight is its squared transverse displacement divided by its length. The two endpoint speeds have norm R. Their contribution is at most $2R^2|dq|^2/\tau_-$; this bound includes the case $m = 1$, when both displacements occur on the same flight. The endpoint curvature terms contribute at most $R|dq|^2$. The reduced Hessian is the Schur complement of the internal Hessian and cannot exceed this trial quadratic form. Taking the indicated extreme radii gives the upper bound, with $2R_+^2/\tau_- + R_+ = 47/6$.

In the five rectangles of Y_R^* one has $|p| < 9/10$. For the only moving momentum centers this follows from

$$\frac{2(53/100)}{1 + (53/100)^2} + \frac{1}{200} < \frac{9}{10};$$

the remaining centers and the added rectangle satisfy the bound immediately. Both the initial and terminal states belong to this same section. Thus $c_0, c_m > \sqrt{19}/10$, which proves (40.2). □

Away from $g = 0$ define the circle-valued physical observable

$$\vartheta(x) = \arg(-p_0(x), p_m(x)) \in \mathbb{R}/(2\pi\mathbb{Z}). \tag{40.3}$$

It is not the normal angle of an auxiliary endpoint cap. The orientation of each disk is the fixed orientation already used for the collision coordinates. Put $e = g/|g|$ and

$$\chi = e^{\mathsf{T}} \operatorname{cof}(K) e, \qquad v = K^{-1} g, \qquad s = g^{\mathsf{T}} K^{-1} g. \tag{40.4}$$

Here $\operatorname{cof}(K)$ is the adjugate matrix; K is symmetric. In dimension two its eigenvalues are those of K in reverse order. In particular $\lambda_\circ \le \chi \le \Lambda_\circ$ on the exact event.

Theorem 40.2 (An explicit critical-safe forward roof field)**.** *There is a uniquely specified smooth vector field X on the regular exact event with its interior critical points removed such that*

$$XL = 1, \qquad X\vartheta = 0. \tag{40.5}$$

In endpoint-angle coordinates it is

$$X_q = \frac{K^{-1} g}{s}. \tag{40.6}$$

In any local lift of ϑ on the original collision cylinder, it is equivalently

$$X = \frac{\vartheta_p \partial_\alpha - \vartheta_\alpha \partial_p}{L_\alpha \vartheta_p - L_p \vartheta_\alpha}.$$

No source density, guard, partition, or critical radius enters its choice. Stop its flow at the first exit from the regular exact event. For a forward survivor and $r_h = |g(\Phi_h x)|$, $r_0 = |g(x)|$, one has

$$2\lambda_\circ h \le r_h^2 - r_0^2 \le 2\Lambda_\circ h, \tag{40.7}$$

$$\operatorname{length}(q([0,h])) \le \frac{r_h - r_0}{\lambda_\circ} \le \frac{\sqrt{2\Lambda_\circ h}}{\lambda_\circ}, \tag{40.8}$$

$$|p_i(\Phi_h x) - p_i(x)| \le \frac{\sqrt{2\Lambda_\circ h}}{R_-}, \qquad i = 0, m. \tag{40.9}$$

The same estimates hold up to the limiting time of a finite lifetime. A forward trajectory cannot terminate at a regular interior critical point of L. The estimates do not assert survival through a true singular, first-hit, or section-decision boundary.

Proof. Writing $g^{\perp} = (-g_2, g_1)$ gives $d\vartheta = (g^{\perp})^{\mathsf{T}} K\, dq/|g|^2$ and hence

$$\det D_q(F, \vartheta) = \frac{g^{\mathsf{T}} \operatorname{cof}(K) g}{|g|^2} = \chi > 0. \tag{40.10}$$

The endpoint map has Jacobian B_m, so $|\det D_{(\alpha,p)}(L, \vartheta)| = |B_m|\chi > 0$. The two equations (40.5) therefore specify a unique field. Formula (40.6) satisfies them: $g \cdot X_q = 1$ and $KX_q = g/s$ is radial in g. Endpoint lifts differ by translations by multiples of 2π; local lifts of ϑ also differ by constants. Both formulas therefore agree on overlaps and define the field without gluing.

Along the flow $\dot{g} = g/s$. Its direction is constant and $\dot{r} = r/s > 0$. Since $r^2/\Lambda_\circ \le s \le r^2/\lambda_\circ$, one has $2\lambda_\circ \le d(r^2)/dh \le 2\Lambda_\circ$. This proves (40.7). Using r as the parameter,

$$\frac{dq}{dr} = K^{-1} e.$$

Integration gives (40.8), including length rather than merely endpoint displacement. The direction of g is constant, so $|g(\Phi_h x) - g(x)| = r_h - r_0$ and the first-variation formula gives (40.9). In particular r_h cannot tend to zero in forward time. On a compact regular subset away from $g = 0$ the ordinary continuation theorem applies; any remaining finite exit is a true boundary of the regular exact event. □

Proposition 40.3 (The cancellation in the physical drift)**.** *Let $T = D^3F$ and let a be a locally Lipschitz density relative to dq. Away from $g = 0$, in the almost-everywhere sense,*

$$\operatorname{div}_q(aX_q) = \frac{\nabla a \cdot v}{s} + a\left\{\frac{T[v, v, v]}{s^2} - \frac{v \cdot \nabla \log\det K}{s}\right\}. \tag{40.11}$$

On a regular neighborhood where $\|T\| \le T_0$, this gives

$$|\operatorname{div}_q(aX_q)| \le \frac{1}{|g|}\left\{\frac{\Lambda_\circ}{\lambda_\circ}|\nabla a| + \left(\frac{\Lambda_\circ^2}{\lambda_\circ^3} + \frac{2\Lambda_\circ}{\lambda_\circ^2}\right)|a| T_0\right\}. \tag{40.12}$$

In particular the constant-density quadratic part has zero divergence. The bounds on T and ∇a in this assertion are local, not collision-uniform conclusions.

Proof. Differentiation of $v = K^{-1}g$ yields $\operatorname{div} v = 2 - v \cdot \nabla \log\det K$. Also $v \cdot \nabla s = 2s - T[v, v, v]$. The two terms $2/s$ cancel upon differentiating av/s, proving (40.11). Use $|v| \le |g|/\lambda_\circ$, $s \ge |g|^2/\Lambda_\circ$, and $|\nabla \log\det K| \le 2T_0/\lambda_\circ$ for the bound. Near a regular interior critical point the uniform positive Hessian makes $|g|$ comparable to distance, so the remaining displayed singularity is locally integrable in two dimensions. This last observation concerns a local integral only. □

41. All-contact motion on the original regular source

The endpoint field controls more than the two endpoints. Its effect on every contact decays with distance from the nearer endpoint, without imposing a positive lower bound on the internal incidences. This observation permits a uniform estimate for the original smooth guard, rather than a replacement by protected words.

Keep $q_0 = 47/53$, $d_j = \min(j, m - j)$ and $d'_j = \min(j, m - 1 - j)$. All endpoint derivatives below use $q = (\alpha_0, \alpha_m)$; internal contact displacements use arclength. Constants depend only on the fixed physical family and, where indicated, the originally chosen function Θ.

Lemma 41.1 (An incidence-normalized contact inverse)**.** *For the optical matrix A of the retained recursion, put $D_c = \operatorname{diag}(c_1, \dots, c_{m-1})$ and $C_A = R_+/[2(1-q_0)]$. For $1 \le i, j < m$,*

$$0 \le (A^{-1})_{ij} \le C_A \min(c_i, c_j) q_0^{|i-j|}, \qquad |(D_c^{-1} A^{-1})_{ij}| \le C_A q_0^{|i-j|}. \tag{41.1}$$

Consequently, at every regular physical word, its contact positions Q_j satisfy

$$\|D_q Q_j\| \le C_Q q_0^{d_j}, \qquad 0 \le j \le m, \qquad C_Q = R_+ + \frac{2C_A R_+}{\tau_- q_0}. \tag{41.2}$$

For $m = 1$ the internal assertion is empty and the endpoint estimate still holds. These are derivatives at actual regular trajectories; no common neighborhood of physical continuation is asserted.

Proof. Write $A = D(I - P)$, with D its diagonal. The entries of P are nonnegative, vanish except at neighboring indices, and have row sums at most

$$\frac{2/\tau_-}{2/\tau_- + 2/R_+} = q_0.$$

Also $D_{jj}^{-1} \le Rc_j/2$. In the Neumann series for $(I-P)^{-1}D^{-1}$, the (i,j) entry vanishes until the power $|i-j|$. Summing the remaining geometric series gives the bound with c_j. Symmetry gives the bound with c_i, and division by $c_i > 0$ proves the second inequality. This is the same normalized inverse underlying the retained tapered-contact construction; here only its first derivative is needed, at arbitrary regular endpoints.

The internal action Hessian in oriented arclengths is $D_c A D_c$. After division of its stationary equations by the internal cosines, endpoint forcing has only the first and last entries, of absolute sizes $t_0 c_0\, du_0$ and $t_{m-1} c_m\, du_m$. Tangent signs may change these signs but not their sizes. Thus the displacement of internal arclength s_j obeys

$$|ds_j| \le \frac{C_A}{\tau_-}(q_0^{j-1}|du_0| + q_0^{m-1-j}|du_m|).$$

Both exponents are at least $d_j - 1$, and $|du_i| = R|d\alpha_i|$. The displayed choice of C_Q bounds the resulting Euclidean operator norm and the two endpoint derivatives. The Euclidean speed of a circle parameterized by arclength is one, which proves (41.2). □

Theorem 41.2 (Uniform motion of every physical margin)**.** *There is $C_D < \infty$, independent of m, R, n, k and of every positive incidence and clearance, with the following property. Along any regular endpoint curve of length ℓ_q in the exact event and in one physical word,*

$$|c_j(x) - c_j(y)| + |p_j(x) - p_j(y)| + \operatorname{dist}_{\mathbb{S}^1}(\alpha_j(x), \alpha_j(y)) \le C_D q_0^{d_j} \ell_q, \tag{41.3}$$

$$|\Delta_j(x) - \Delta_j(y)| \le C_D q_0^{d'_j} \ell_q. \tag{41.4}$$

In particular, for a forward survivor of the field in Theorem 40.2, the right sides are bounded by, respectively, $C_D C_X q_0^{d_j}\sqrt{h}$ and $C_D C_X q_0^{d'_j}\sqrt{h}$, where $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$.

Proof. Differentiate a flight direction:

$$dv_j = \ell_j^{-1}(I - v_j v_j^{\mathsf{T}})(dQ_{j+1} - dQ_j).$$

The depths of its two endpoints are at least d'_j. Its derivative is consequently bounded by $2C_Q q_0^{d'_j}/\tau_-$. The normal and tangent derivatives at a contact are bounded by $C_Q q_0^{d_j}/R_-$. Express c_j and p_j as scalar products of the appropriate unit flight velocity with these vectors. At the terminal contact use the incoming velocity, with the sign giving positive incidence; its tangential component is unchanged by reflection. Since adjacent depths differ by at most one,

these formulas prove the first estimate with a fixed C_D. In particular we do not differentiate $\sqrt{1-p_j^2}$ by dividing by c_j.

The Hausdorff displacement of a segment is no greater than the larger displacement of its two endpoints. Distance from that segment to a fixed third disk is a one-Lipschitz function of this Hausdorff displacement. Taking the minimum over third disks preserves the same bound. The minimizing disk may change; differentiability of that minimum is not required. The excluded incident disks are fixed on the word. Integration along the curve proves (41.4). Finally use the endpoint-curve length estimate of Theorem 40.2. □

Corollary 41.3 (The unchanged guard is uniformly controlled)**.** *For every forward survivor in the original exact event,*

$$|W^\varepsilon(\Phi_h x) - W^\varepsilon(x)| \le \min\{1, K_\Theta \varepsilon^{-1}\sqrt{h}\}, \qquad K_\Theta = \frac{4\,\mathrm{Lip}(\Theta) C_D C_X}{1 - q_0^{3/4}}. \tag{41.5}$$

No incidence cutoff, clearance cutoff, section-distance factor or monotonicity assumption on Θ is added.

Proof. For factors in $[0,1]$, the difference of their products is bounded by the sum of their absolute differences. A contact factor in (39.2) changes by at most $\mathrm{Lip}(\Theta) C_D C_X \varepsilon^{-1} q_0^{3d_j/4}\sqrt{h}$; the analogous flight factor has depth d'_j. There are at most two contacts and two flights at any depth. Their geometric series gives (41.5). Both weights belong to $[0,1]$, proving the additional bound by one. The argument uses only the Lipschitz constant and range of the original smooth function. □

Proposition 41.4 (Where a finite forward exit can occur)**.** *Let s_j be distance from the collision state (α_j, p_j) to the boundary of Y_R^* in the product metric. There is a common C_E such that any source point whose forward lifetime in the regular exact event is at most h belongs to the union*

$$\mathcal{B}_h = \bigcup_{j=0}^{m} \{c_j \le C_E q_0^{d_j}\sqrt{h}\} \;\cup\; \bigcup_{j=0}^{m-1} \{\Delta_j \le C_E q_0^{d'_j}\sqrt{h}\} \;\cup\; \bigcup_{j=0}^{m} \{s_j \le C_E q_0^{d_j}\sqrt{h}\}. \tag{41.6}$$

Consequently its exit mass is bounded above by $\int W^\varepsilon \mathbf{1}_{\mathcal{B}_h}\, d\mu_t$. This is a source inclusion, not a pointwise concentration estimate.

Proof. The finite endpoint-curve length gives limiting contact data up to a finite lifetime. A forward trajectory cannot reach an interior critical point, by Theorem 40.2. If all limiting incidences, clearances and distances to section edges were positive, the regular reflection equations, strict first-hit tests and locally constant section decisions would continue the same physical word and exact event. The endpoint Jacobian remains nonzero there, so the roof field also continues, a contradiction. Thus some listed margin tends to zero. Apply Theorem 41.2, including the one-Lipschitz property of distance to a closed edge set. Taking the union over the possible margins and retaining the original weight proves the mass bound. A section edge is charged whether or not another simultaneous change might preserve the occupation; this can enlarge the upper bound but cannot omit a true exit. □

42. A uniform cubic estimate for the reduced action

The second derivatives of the internal contact map need not be uniformly bounded near grazing. They are not needed to estimate the third derivative of the reduced roof. Stationarity cancels

them. Throughout this section derivatives are evaluated at actual regular physical trajectories; no continuation through a singularity is made.

Lemma 42.1 (Cubic differentiation of a stationary action)**.** *Let $\mathcal{S}(q,z)$ be smooth and suppose that $\partial_z\mathcal{S}(q,z(q))=0$, with invertible internal Hessian. Set $F(q)=\mathcal{S}(q,z(q))$ and $U_u=(u,Dz(q)u)$. Then*

$$D^3F(q)[u,v,w]=D^3\mathcal{S}(q,z(q))[U_u,U_v,U_w]. \tag{42.1}$$

The equality is pointwise and requires no bound for the inverse internal Hessian or for D^2z.

Proof. The differentiated stationary equation says $D^2\mathcal{S}[U_u,(0,\zeta)]=0$ for every internal vector ζ. Consequently $D^2F[u,v]=D^2\mathcal{S}[U_u,U_v]$. Differentiate this identity in direction w. The two terms in which U_u or U_v is differentiated are contractions of $D^2\mathcal{S}$ with an internal vector and vanish by the same stationary equation. The remaining term is (42.1). This calculation is local, where the implicit function theorem gives a smooth $z(q)$. □

Use arclength at every contact, including $s_0=R\alpha_0$ and $s_m=R\alpha_m$, and write

$$\mathcal{S}(s_0,\dots,s_m)=\sum_{j=0}^{m-1}|Q_{j+1}(s_{j+1})-Q_j(s_j)|.$$

The endpoints are linear functions of $q=(\alpha_0,\alpha_m)$. At a reflection the internal arclength derivative of $\mathcal{S}$ is zero. Its internal Hessian is D_cAD_c, with A as in (44.1); this is positive definite at every regular word. In particular Lemma 42.1 applies even when some positive incidence is arbitrarily small.

Theorem 42.2 (Collision-uniform third derivatives)**.** *Let C_Q be the explicit constant in* (41.2)*, and put*

$$C_3=\frac{6}{\tau_-^2}+\frac{3}{R_-\tau_-}+\frac{1}{R_-^2},\qquad T_*=\frac{16C_3C_Q^3}{1-q_0^3}. \tag{42.2}$$

For every regular physical word, its reduced roof in endpoint angles satisfies

$$|D^3F(q)[u,v,w]|\le T_*|u|\,|v|\,|w|. \tag{42.3}$$

The constant is independent of m,R,n,k, all positive internal incidences and all positive third-disk clearances. In particular, on any regular endpoint curve γ,

$$\|K(\gamma(b))-K(\gamma(a))\|\le T_*\,\mathrm{length}(\gamma). \tag{42.4}$$

This does not assert a common continuation neighborhood for the words.

Proof. For $N(z)=|z|$ at a nonzero vector of length at least τ_-,

$$\|DN\|\le 1,\qquad \|D^2N\|\le\tau_-^{-1},\qquad \|D^3N\|\le 6\tau_-^{-2}.$$

A circular arclength parametrization has derivative norms $1,R^{-1},R^{-2}$ through order three. The chain rule for one flight therefore bounds its third derivative on contact vectors U,V,W by

$$C_3(|U_j|+|U_{j+1}|)(|V_j|+|V_{j+1}|)(|W_j|+|W_{j+1}|).$$

These are derivatives of the action on the product of circles, not third derivatives of the contact map.

For a lifted endpoint direction u, Lemma 41.1 gives $|(U_u)_j|\le C_Qq_0^{d_j}|u|$. Indeed a circle has unit speed in arclength, so its contact-position derivative has precisely the absolute value of its arclength derivative. The two contact depths of flight j are at least d'_j. Thus this flight contributes at most $8C_3C_Q^3q_0^{3d'_j}|u||v||w|$ to (42.1). There are at most two flights at every depth. Summation of the geometric series proves (42.3), including $m=1$, when there are no internal variables. Integrating $DK=D^3F$ along γ proves (42.4). □

Corollary 42.3 (Uniform angular Jacobian variation)**.** *Let Φ_h be the canonical forward flow of Theorem 40.2. On its surviving exact-label trajectories, with $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$,*

$$|\chi(\Phi_h x) - \chi(x)| \le T_* C_X \sqrt{h}, \qquad \left|\log \frac{\chi(\Phi_h x)}{\chi(x)}\right| \le \frac{T_* C_X}{\lambda_\circ}\sqrt{h}. \tag{42.5}$$

In endpoint coordinates the Jacobian of the flow relative to dq is

$$J_h^q(x) = \frac{\chi(x)}{\chi(\Phi_h x)}. \tag{42.6}$$

The physical source Jacobian is different and is computed below.

Proof. In dimension two $\chi = (e^\perp)^{\mathsf{T}} K e^\perp$, and e is constant along the canonical flow. Apply (42.4), the forward length bound, and $\chi \ge \lambda_\circ$. The map (F, ϑ) has Jacobian χ and conjugates the flow to $(t, \theta) \mapsto (t + h, \theta)$. Ordinary change of variables proves (42.6). No estimate on a derivative of the source density was used. □

43. The angular volume estimate on the complete source

Fix one original exact event, and keep its source measure $\bar\nu = (4\pi c_*)^{-1} d\alpha\, dp$. On its regular part set

$$\omega(x) = |B_m(x)|^{-1}, \qquad d\bar\nu = \frac{\omega(q)}{4\pi c_*}\, dq, \qquad d\mu_t = \frac{1}{4\pi c_* |\nabla L|}\, d\mathcal{H}^1|_{\{L=t\}}.$$

Thus $p(t) = \mu_t(\Pi)$ and $b^\varepsilon(t) = \int W^\varepsilon\, d\mu_t$. All fibers below are those of this same event. The gradient in the coarea formula is in the original collision coordinates.

Let $S_h(t)$ be the starting points on this fiber whose canonical forward flow survives until h. Its physical area Jacobian and its optical ratio are denoted by J_h and η_h, respectively. Define three nonnegative quantities before any collision limit:

$$E_h^\varepsilon(t) = \int_{\{L=t\}\setminus S_h(t)} W^\varepsilon\, d\mu_t, \tag{43.1}$$

$$V_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - J_h]_+\, d\mu_t, \tag{43.2}$$

$$O_h^\varepsilon(t) = \int_{S_h(t)} W^\varepsilon [1 - \eta_h]_+\, d\mu_t. \tag{43.3}$$

The first is the unchanged initial mass of killed trajectories. The second is the full physical volume loss. The third isolates the optical inverse-twist loss. Neither E_h nor O_h omits a physical word. We write $\bar E_r = r^{-1}\int_0^r E_h\, dh$, and similarly for V and O. All assertions use joint coarea versions in (t, h).

Theorem 43.1 (Exact Jacobian factorization and a complete angular bound)**.** *On every survivor,*

$$\eta_h(x) = \frac{\omega(\Phi_h x)}{\omega(x)}, \qquad J_h(x) = \eta_h(x)\frac{\chi(x)}{\chi(\Phi_h x)}. \tag{43.4}$$

There is a uniform C_{ang} such that, for each fixed $r > 0$,

$$|\bar V_r^\varepsilon(t) - \bar O_r^\varepsilon(t)| \le C_{\mathrm{ang}} \sqrt{r}\, A_r p(t) \tag{43.5}$$

for almost every t. Consequently, from the stated fixed-width analytic input,

$$\limsup_m \sup_{R,n,k} m^2 \|\bar V_r^\varepsilon - \bar O_r^\varepsilon\|_\infty \le C\sqrt{r}. \tag{43.6}$$

The constants are independent of ε. This is a bound for the complete source, not a per-word estimate multiplied by a word count.

Proof. Multiplying the endpoint-coordinate Jacobian (42.6) by the ratio of the physical area densities gives (43.4). Put $k_h = \chi(x)/\chi(\Phi_h x)$. The positive-part function is one-Lipschitz, so

$$|[1-\eta_h k_h]_+ - [1-\eta_h]_+| \le \eta_h |1-k_h|.$$

Corollary 42.3 bounds $|1-k_h|$ by $C\sqrt{h}$. Moreover $\eta_h \le (\Lambda_\circ/\lambda_\circ)J_h$. Coarea covariance in the target direction therefore yields

$$|V_h^\varepsilon(t) - O_h^\varepsilon(t)| \le C\sqrt{h}\int_{S_h(t)} J_h \, d\mu_t \le C\sqrt{h}\, p(t+h).$$

The target sum contains all its actual words, and $W^\varepsilon \le 1$ was used only for an upper bound. Average over $0 < h < r$ and bound the forward average of the nonnegative density by $2A_r p$. This proves (43.5).

For clarity, the unweighted input used in the last step is

$$\limsup_m \sup_{R,n,k} m^2 \|A_r p\|_\infty \le C \quad \text{for every previously fixed } r > 0.$$

It follows from Input statement 39.1: for the auxiliary parameter 1 the initial factor is $\Theta(c_0) = 0$, whence $W^1 = 1$ and $b^1 = p$. The precise inherited source propositions are pinned in Section 39.3. No raw pointwise bound for p or collision-dependent averaging width is used. □

Define the separate ordered heights

$$\begin{aligned}
\mathfrak{X}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{E}_r^\varepsilon,\\
\mathfrak{V}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{V}_r^\varepsilon,\\
\mathfrak{O}_{\varepsilon,M}(r) &= \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} \bar{O}_r^\varepsilon.
\end{aligned} \tag{43.7}$$

Equation (43.6) gives both $\mathfrak{V} \le \mathfrak{O} + C\sqrt{r}$ and $\mathfrak{O} \le \mathfrak{V} + C\sqrt{r}$ as extended nonnegative inequalities. Thus the unestimated interior volume contribution has been reduced to inverse-twist loss, with no unestimated third derivative of the reduced roof left in this comparison.

Theorem 43.2 (The unchanged pointwise error with separated defects)**.** *Under Input statement 39.1, for sufficiently large fixed B and every fixed $r > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + C_\Theta(1+\varepsilon(B)^{-1})\sqrt{r}. \tag{43.8}$$

Here $\varepsilon(B) = A_0 B^{-1/12}$. The order is

$$B,\ \varepsilon(B),\ r,\ \textit{auxiliary band fixed};\quad m \to \infty;\quad \textit{auxiliary band} \to \infty;\quad B \to \infty.$$

In particular, separate bounds

$$\mathfrak{X}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_X} r^{\beta_X}, \qquad \mathfrak{O}_{\varepsilon,M}(r) \le C_M \varepsilon^{-q_O} r^{\beta_O} \tag{43.9}$$

with finite nonnegative q_X, q_O and positive β_X, β_O imply the original pointwise assertion. One may choose $r = c\varepsilon(B)^\gamma$ with $\gamma > \max\{2, q_X/\beta_X, q_O/\beta_O\}$. The two estimates in (43.9) *are proof obligations, not conclusions of the present theorem.*

Proof. The stopped subcoupling has density $\min\{W(x), W(\Phi_h x)J_h(x)\}$ on survivors. Its two marginals are dominated by the original weighted fibers. The elementary bound $[w - w'J]_+ \le w[1-J]_+ + J|w-w'|$, the complete guard modulus of Corollary 41.3, and target coarea covariance give

$$b^\varepsilon(t) \le b^\varepsilon(t+h) + E_h^\varepsilon(t) + V_h^\varepsilon(t) + K_\Theta \varepsilon^{-1}\sqrt{h}\, p(t+h).$$

Average at the previously fixed width and apply (43.5). The averaged bad source is bounded at the ordered height by $2C\varepsilon^{1/16}$; the unweighted target is bounded as in the preceding proof.

Inserting the positive scalar remainder gives (43.8). All inequalities precede the suprema and the collision limit. The displayed choice of γ makes each of the three explicit powers positive, proving the implication. □

Corollary 43.3 (An exact separated criterion)**.** *Under the same analytic input, the original pointwise assertion is equivalent to, for every M and every fixed $r_0 > 0$,*

$$\lim_{B\to\infty} \inf_{0<r<r_0} \left\{ \mathfrak{X}_{\varepsilon(B),M}(r) + \mathfrak{O}_{\varepsilon(B),M}(r) + (1+\varepsilon(B)^{-1})\sqrt{r} \right\} = 0. \tag{43.10}$$

Proof. Sufficiency follows by approximating the infimum before the collision limit. For necessity, the positive remainder and the pointwise assertion give ordered height at most $C_M B^{-1/192}$ for $b^{\varepsilon(B)}$. Both E_h^ε and O_h^ε are bounded by this same positive density at the starting roof. Their separate ordered heights are therefore at most this bound, uniformly in r. Take the infimum and then $B \to \infty$. This argument does not claim that the infimum is attained or select a width depending on m. □

44. Optical sensitivity and source-weighted exit flux

The two terms in (43.8) have distinct geometric mechanisms. We first give a positive, collision-uniform sensitivity formula for the optical term, and then an exact boundary formula for the initial source mass which exits. Neither formula identifies a surface insertion with a bounded volume multiplier.

44.1. **A positive optical polynomial.** For a regular word put $t_j = \ell_j^{-1}$, $V_i = 2/(Rc_i)$ and

$$A_{ii} = t_{i-1} + t_i + V_i, \qquad A_{i,i+1} = A_{i+1,i} = -t_i. \tag{44.1}$$

The matrix A is the incidence-normalized optical matrix; the actual internal arclength Hessian is $D_c A D_c$. In the Neumann estimate it is $A = D_A(I - P)$, where D_A is its diagonal and the row sums of P are at most q_0. These two diagonal normalizations are different.

Define $y_0 = 0$, $f_0 = 1$, $y_1 = \ell_0$, and

$$f_i = f_{i-1} + V_i y_i, \qquad y_{i+1} = y_i + \ell_i f_i \quad (1 \le i < m). \tag{44.2}$$

The continuant and corner Schur-complement identities give

$$y_m = \left(\prod_{j=0}^{m-1} \ell_j \right) \det A, \qquad \omega = \frac{Rc_0c_m}{y_m}. \tag{44.3}$$

For $m = 1$ the determinant is empty and $y_1 = \ell_0$. Indeed eliminating f_i gives $y_{i+1} = (1+\ell_i/\ell_{i-1}+\ell_i V_i)y_i - (\ell_i/\ell_{i-1})y_{i-1}$, the continuant recurrence with the same initial values. In the mixed endpoint Schur complement all internal incidence squares cancel against $\det(D_c A D_c)$, leaving $|F_{uv}| = c_0 c_m \prod t_j / \det A$ in endpoint arclengths. The change to endpoint angles gives $\omega = R|F_{uv}|$ as stated.

Proposition 44.1 (Optical logarithmic sensitivities)**.** *The polynomial y_m has nonnegative coefficients and is of degree at most one in each of the separate variables ℓ_j, V_i. Thus*

$$a_i = V_i \partial_{V_i} \log y_m \in [0,1], \qquad b_j = \ell_j \partial_{\ell_j} \log y_m \in [0,1]. \tag{44.4}$$

At fixed R its differential on any actual regular word is

$$d\log\omega = d\log c_0 + d\log c_m + \sum_{i=1}^{m-1} a_i \, d\log c_i - \sum_{j=0}^{m-1} b_j \, d\log \ell_j. \tag{44.5}$$

In endpoint-angle coordinates, for a fixed uniform C_O,

$$\|\nabla_q \log \omega\| \le C_O \left(1 + \sum_{i=1}^{m-1} \frac{q_0^{d_i}}{c_i}\right). \tag{44.6}$$

No third derivative or sum of absolute word multiplicities occurs in this bound.

Proof. Each new V_i enters the positive recursion once, and each new ℓ_i enters once after that step. Later operations are linear in all previous variables. Induction proves positivity and separate affineness. For a positive separately affine polynomial $Y = A + Bz$, $z\partial_z \log Y = Bz/(A + Bz) \in [0, 1]$. Applying this identity to each variable gives (44.4); differentiating (44.3) gives (44.5).

The endpoint incidences are bounded below by $\sqrt{19}/10$. Theorem 41.2 bounds $\|D_q c_i\|$ by $C_D q_0^{d_i}$. Also $|d\ell_j| \le |dQ_j| + |dQ_{j+1}|$ and $\ell_j \ge \tau_-$, so $\|D_q \log \ell_j\| \le 2C_Q q_0^{d'_j}/\tau_-$. The sum of these flight bounds is uniform in m. Inserting the sensitivity weights proves (44.6). The inverse-incidence terms are retained; positivity does not make their complete source sum finite. □

44.2. **A quantified regular part of the optical loss.** For $0 < \delta \le 1$ let $S_h^\delta(t) \subset S_h(t)$ consist of survivors satisfying, for every internal contact and every $0 \le s \le h$,

$$c_i(\Phi_s x) \ge \delta q_0^{d_i/2}. \tag{44.7}$$

Write $O_h^{\varepsilon,\mathrm{reg}(\delta)}$ for (43.3) restricted to this set, and $O_h^{\varepsilon,\mathrm{thin}(\delta)}$ for its nonnegative complement. This is a decomposition of the original weighted source, not a new guard or a renormalized probability law.

Theorem 44.2 (Uniform regular optical distortion)**.** *There are uniform positive c_1, C_1, C_2 such that, whenever $0 < r \le c_1\delta^2$,*

$$\limsup_m \sup_{R,n,k} m^2 \|\bar{O}_r^{\varepsilon,\mathrm{reg}(\delta)}\|_\infty \le C_1 \delta^{-1}\sqrt{r}, \tag{44.8}$$

$$\bar{O}_r^{\varepsilon,\mathrm{thin}(\delta)}(t) \le b^{C_2\delta}(t) \quad \textit{for almost every } t. \tag{44.9}$$

The first estimate uses only the stated fixed-width analytic input. The second is a pointwise source domination and is not a bound for the normalized essential height of its right side.

Proof. On the corridor, the sum in (44.6) is at most $2\delta^{-1}/(1 - \sqrt{q_0})$. The forward endpoint-curve length is at most $C_X\sqrt{h}$. Hence $|\log \eta_h| \le C\delta^{-1}\sqrt{h}$. Choose c_1 so that this upper bound is at most one for $h \le r$. Then $[1 - \eta_h]_+ \le C\delta^{-1}\sqrt{h}\,\eta_h$. As in the proof of Theorem 43.1, $\eta_h \le (\Lambda_\circ/\lambda_\circ)J_h$ and target coarea gives

$$O_h^{\varepsilon,\mathrm{reg}(\delta)}(t) \le C\delta^{-1}\sqrt{h}\, p(t + h).$$

Averaging first proves (44.8).

If the corridor fails at time $s \le h \le r$, the margin motion bound implies at the starting point

$$c_i(x) \le \delta q_0^{d_i/2} + C_D C_X q_0^{d_i}\sqrt{r} \le C_2\delta q_0^{d_i/2}.$$

Since $q_0^{d_i/2} \le q_0^{d_i/4}$, the corresponding factor of the original guard at auxiliary parameter $C_2\delta$ is zero. Thus $W^{C_2\delta}(x) = 1$ on the entire thin starting set. Its optical loss is bounded by its original weighted source, and $W^\varepsilon \le 1$. This proves (44.9) before averaging, hence afterwards. There are no internal contacts when $m = 1$ and the thin set is empty. □

The theorem quantitatively controls the regular optical contribution on the full exact event. It does not allow replacement of $b^{C_2\delta}$ by its fixed-width average at a starting roof. That replacement would be exactly the missing pointwise step.

44.3. **Exit mass in the correct flux convention.** The initial mass of trajectories which exit need not equal the physical flux measured at their exit. Compression between the starting roof and the exit must be retained. We state the exact formula on regular subdomains and recover the full stopped event by exhaustion.

Let D be a relatively compact regular domain in the exact event, away from $g = 0$, with piecewise smooth boundary. We take the boundary transverse to X except for pieces and lower-dimensional strata whose images in the angular coordinate have zero length. Such domains are obtained from finite flow boxes; increasing exhaustions can be formed by refining the boxes. Let Γ_D^+ denote its outward boundary, N its outward unit normal in the collision coordinates, and $\tau_D^-(y)$ the first backward lifetime from y inside D. Use the physical flux measure

$$d\zeta_D(y) = \frac{[X(y) \cdot N(y)]_+}{4\pi c_*}\, d\mathcal{H}^1(y). \tag{44.10}$$

Proposition 44.3 (First-exit flux with the initial source weight)**.** *Let $E^{\varepsilon}_{D,h}(t)$ be the weighted initial mass on $L = t$ starting in D and exiting D within h. For almost every (t, h),*

$$E^{\varepsilon}_{D,h}(t) = \int_{\Gamma_D^+} \mathbf{1}_{\{0<L(y)-t\le h,\ L(y)-t<\tau_D^-(y)\}} \frac{W^{\varepsilon}(x_{t,y})}{J_{L(y)-t}(x_{t,y})}\, d\zeta_D(y), \quad x_{t,y} = \Phi_{t-L(y)}y. \tag{44.11}$$

For an increasing regular exhaustion D_ℓ of the full regular exact event, its original true-exit mass satisfies

$$E^{\varepsilon}_h(t) = \lim_{\ell\to\infty} \left\{ \int_{\{L=t\}\setminus D_\ell} W^{\varepsilon}\, d\mu_t + E^{\varepsilon}_{D_\ell,h}(t) \right\}. \tag{44.12}$$

The bracketed nonnegative quantities decrease to the left side. The limit is at fixed collision count; it is not interchanged with a normalized collision limsup.

Proof. In a flow box use $(u, \theta) = (L, \vartheta)$, so that $X = \partial_u$. The physical source has area density $a(u, \theta) = \omega/(4\pi c_*\chi)$ relative to $du\, d\theta$. On a transverse outward boundary $u = b(\theta)$, its flux is $a(b(\theta), \theta)|d\theta|$. At the starting roof its coarea mass is $a(t, \theta)|d\theta|$, and

$$J_{b(\theta)-t}(x_{t,y}) = a(b(\theta), \theta)/a(t, \theta).$$

Multiplication by the initial weight proves (44.11) on the box. The backward lifetime selects the first forward exit rather than a later re-entry. Uniqueness of the flow makes the source pieces disjoint; a measurable refinement of the boxes sums them without multiplicity. The excluded boundary strata have zero angular measure and hence zero source-fiber measure. Coarea supplies the joint almost-everywhere statement.

For each starting point, declare the D_ℓ-lifetime zero when it starts outside D_ℓ. These lifetimes increase to the lifetime in the full event: every compact surviving flow segment eventually lies in some D_ℓ. The indicators of lifetime at most h therefore decrease to the corresponding full-event indicator. Dominated convergence on each finite coarea fiber proves (44.12). The exceptional (t, h) set can be chosen jointly by disintegration. Chart boundaries in the exhaustion are not declared physical; their effects disappear only through this source-level limiting identity. □

The inverse Jacobian in (44.11) is essential. Along a limiting grazing boundary the physical density may tend to zero while this inverse factor diverges. Vanishing unweighted exit flux therefore does not make the original initial exit mass zero. The formula retains grazing, competing-hit and section exits. If several margins first vanish simultaneously, a fixed ordering partitions the initial source into disjoint exit types; their sum is E^{ε}_h. The internal occupation-current identity in the supplement is signed. It has not been used to subtract any of these positive exit masses.

45. Completion of the original pointwise theorem

The original assertion is (39.5), with its full arithmetic transition and all exact labels. The new uniform cubic estimate and the complete angular-volume estimate require no new statistical hypothesis beyond the stated fixed-width input. The incidence-corridor theorem also estimates a complete nonnegative part of the optical loss; its thin complement remains charged to the original source rather than omitted.

The two quantitative bounds in (43.9), or the separated qualitative criterion (43.10), are still required to conclude the organizing theorem. The first-exit formula shows why ordinary surface flux without the inverse Jacobian is not such a bound. The optical sensitivity formula shows which inverse-incidence factors survive after the uniform cubic estimate. No theorem for bounded volume multipliers is promoted to a theorem for these surface or singular insertions. Thus the pointwise law and its unconditional consequences are not asserted on the basis of those formulas.

After scalar closure, Supplement Theorem 76.1 and the retained positive path remainder supply the same-roof path numerator; normalization is made only on positive arithmetic reference classes. Collision-to-return clock transfer and forward likelihood retain their stated inherited hypotheses. Zero classes are never divided by. The complete earlier arguments, including the angular recoupling and occupation-current proofs, are retained in the technical supplement. They do not replace the two remaining estimates.

Part 26. Complete guard distortion and exact occupation currents

Retained revision-78 abstract. We study the pointwise local limit problem for the complete displacement, collision-count, occupation and roof record of actual returns in a triangular Lorentz gas. The physical endpoint-momentum field gives collision-uniform control of every contact, with an exponential loss of influence toward the middle of the word. This holds on all regular surviving trajectories, without a lower incidence cutoff. It yields a uniform square-root modulus for the original full guard. By transporting its error to the target fiber and averaging before the collision limit, we estimate the complete guard-distortion contribution at the required normalization using only fixed-width concentration. The remaining geometric contribution retains true exits and physical coarea-volume loss. We also prove an exact discrete-divergence identity for internal section-decision currents, with a peak-aware Fourier estimate that keeps nontrivial arithmetic phases. These results advance the original pointwise theorem without replacing its exact labels or its source; the complete true-exit and geometric-distortion estimate is the quantitative step still to be established.

46. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U=\{|\alpha-\pi/6|<3/50,\ |p|<3/20\},\qquad Y_R^*=Y_R\cup U,\qquad c_*=\frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*)=c_*$. We always use $\nu_R^*=c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L=L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R}=\sum_{j=0}^{m-1}\mathbf{1}_{Y_R^*}\circ T_R^j$. The event is

$$\Pi_{n,k,m,R}=Y_R^*\cap\{K^{\mathrm{c}}_{m,R}=k,\ A_{m,R}=n\}\cap T_R^{-m}Y_R^*. \tag{46.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j=\sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0\le\Theta\le1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0=47/53$, $d_j=\min(j,m-j)$ and $d_j'=\min(j,m-1-j)$, set

$$H^\varepsilon=\prod_{j=0}^{m}\Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right)\prod_{j=0}^{m-1}\Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right),\qquad W^\varepsilon=1-H^\varepsilon, \tag{46.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t=L_{\#}(W^\varepsilon\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*). \tag{46.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

46.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^m B_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right]. \tag{46.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow0. \tag{46.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

46.2. **The analytic inputs in the form used here.**

Input statement 46.1. The unchanged source (46.3) has a nonnegative, compactly supported BV density at every fixed count. For every fixed $\varepsilon, \delta > 0$, with $A_\delta f(t) = (2\delta)^{-1} \int_{t-\delta}^{t+\delta} f(s)\,\mathrm{d}s$,

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_\delta b^\varepsilon\|_\infty \le C\varepsilon^{1/16}. \tag{46.6}$$

The constant is independent of the previously fixed width. For large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{46.7}$$

All auxiliary spectral bands in these statements are fixed before the collision limit and enlarged only afterwards.

These are the complete-source tame theorem, the fixed-width complete-source bound, and the positive scalar remainder theorem in the technical supplement. Their proofs and historical hypotheses are retained there. The fixed-count source uses threshold integration [15], definable monotonicity [9], and the distributional characterization of BV [13]; it does not assume definability of the original smooth guard. They are restated as inputs to distinguish them from the new geometric and transport proofs. No long-time bound on a fixed-count variation constant is part of this statement.

46.3. **The physical construction and the new estimate.** The endpoint-momentum field and complete angular recoupling are retained below with their proofs. The new argument then estimates one full dynamical contribution, rather than replacing it by a per-word bound. All-contact optical response gives a uniform modulus for the unchanged guard, even with arbitrarily small internal incidences. Its transported error is measured against a translated unweighted target; averaging that target permits the fixed-width analytic input to be used without a pointwise density assumption.

For the remaining physical geometry, internal section interfaces admit an exact signed difference in the occupation index. That identity is stated before taking absolute values, and its Fourier consequence is formulated at every peak rather than only at the origin. True singular and outer section boundaries remain explicit.

47. A canonical flow from the endpoint momenta

The roof field can be specified on the whole regular exact event, without choosing critical disks or endpoint caps. The particular section in (46.1) is useful here: it bounds both endpoint momenta away from grazing, even though no internal incidence or clearance is bounded away from zero.

For one regular physical word use locally lifted endpoint angles $q = (\alpha_0, \alpha_m)$, and write $F(q)$ for its reduced length. The endpoint map is locally invertible, since the optical twist $B_m = \partial_{p_0}\alpha_m$ is nonzero. Set

$$g = \nabla_q F = R(-p_0, p_m), \qquad K = D_q^2 F, \qquad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}.$$

The terminal tangential velocity has the same value before and after reflection, which fixes the sign in the second component of g. All differentiations in this section are on actual regular words.

Lemma 47.1 (Uniform endpoint-angle Hessians on the exact event)**.** *At every regular physical trajectory,*

$$R\,\mathrm{diag}(c_0, c_m) \le K \le (2R_+^2/\tau_- + R_+) I_2. \tag{47.1}$$

On the original exact event, in particular,

$$\lambda_\circ I_2 \le K \le \Lambda_\circ I_2. \tag{47.2}$$

These are pointwise quadratic-form inequalities, uniform in m, R, n, k. They assert neither convexity of a disconnected physical domain nor uniform control of derivatives of K.

Proof. At a regular trajectory let w_0 be its outgoing first-flight velocity and w_m the negative of its incoming last-flight velocity. The polygonal length, minimized over the prescribed intermediate closed disks, is supported at this trajectory by

$$F(q_*) - w_0 \cdot (Q_0(q_0) - Q_0(q_{0,*})) - w_m \cdot (Q_m(q_m) - Q_m(q_{m,*})).$$

To see this directly, the internal gradient at contact i is $-2c_i n_i$. Its scalar product with a displacement into that disk is nonnegative. The convex gradient inequality for the polygonal length therefore gives the displayed support after minimizing the internal positions. Equality holds at the physical trajectory. The internal Hessian is positive definite, so the reduced roof is smooth on its regular neighborhood. Since $Q_i'' = -Rn_i$ and $w_i \cdot n_i = c_i$, differentiating the touching support proves the lower inequality in (47.1).

For the upper inequality keep the internal contacts fixed while varying the endpoints. The transverse second variation of a flight is its squared transverse displacement divided by its length. The two endpoint speeds have norm R. Their contribution is at most $2R^2|dq|^2/\tau_-$; this bound includes the case $m = 1$, when both displacements occur on the same flight. The endpoint curvature terms contribute at most $R|dq|^2$. The reduced Hessian is the Schur complement of the internal Hessian and cannot exceed this trial quadratic form. Taking the indicated extreme radii gives the upper bound, with $2R_+^2/\tau_- + R_+ = 47/6$.

In the five rectangles of Y_R^* one has $|p| < 9/10$. For the only moving momentum centers this follows from

$$\frac{2(53/100)}{1 + (53/100)^2} + \frac{1}{200} < \frac{9}{10};$$

the remaining centers and the added rectangle satisfy the bound immediately. Both the initial and terminal states belong to this same section. Thus $c_0, c_m > \sqrt{19}/10$, which proves (47.2). □

Away from $g = 0$ define the circle-valued physical observable

$$\vartheta(x) = \arg(-p_0(x), p_m(x)) \in \mathbb{R}/(2\pi\mathbb{Z}). \tag{47.3}$$

It is not the normal angle of an auxiliary endpoint cap. The orientation of each disk is the fixed orientation already used for the collision coordinates. Put $e = g/|g|$ and

$$\chi = e^{\mathsf{T}} \operatorname{cof}(K) e, \qquad v = K^{-1} g, \qquad s = g^{\mathsf{T}} K^{-1} g. \tag{47.4}$$

Here $\operatorname{cof}(K)$ is the adjugate matrix; K is symmetric. In dimension two its eigenvalues are those of K in reverse order. In particular $\lambda_\circ \le \chi \le \Lambda_\circ$ on the exact event.

Theorem 47.2 (An explicit critical-safe forward roof field)**.** *There is a uniquely specified smooth vector field X on the regular exact event with its interior critical points removed such that*

$$XL = 1, \qquad X\vartheta = 0. \tag{47.5}$$

In endpoint-angle coordinates it is

$$X_q = \frac{K^{-1} g}{s}. \tag{47.6}$$

In any local lift of ϑ on the original collision cylinder, it is equivalently

$$X = \frac{\vartheta_p \partial_\alpha - \vartheta_\alpha \partial_p}{L_\alpha \vartheta_p - L_p \vartheta_\alpha}.$$

No source density, guard, partition, or critical radius enters its choice. Stop its flow at the first exit from the regular exact event. For a forward survivor and $r_h = |g(\Phi_h x)|$, $r_0 = |g(x)|$, one has

$$2\lambda_\circ h \le r_h^2 - r_0^2 \le 2\Lambda_\circ h, \tag{47.7}$$

$$\operatorname{length}(q([0,h])) \le \frac{r_h - r_0}{\lambda_\circ} \le \frac{\sqrt{2\Lambda_\circ h}}{\lambda_\circ}, \tag{47.8}$$

$$|p_i(\Phi_h x) - p_i(x)| \le \frac{\sqrt{2\Lambda_\circ h}}{R_-}, \qquad i = 0, m. \tag{47.9}$$

The same estimates hold up to the limiting time of a finite lifetime. A forward trajectory cannot terminate at a regular interior critical point of L. The estimates do not assert survival through a true singular, first-hit, or section-decision boundary.

Proof. Writing $g^\perp = (-g_2, g_1)$ gives $d\vartheta = (g^\perp)^{\mathsf{T}} K\, dq/|g|^2$ and hence

$$\det D_q(F, \vartheta) = \frac{g^{\mathsf{T}} \operatorname{cof}(K) g}{|g|^2} = \chi > 0. \tag{47.10}$$

The endpoint map has Jacobian B_m, so $|\det D_{(\alpha,p)}(L, \vartheta)| = |B_m|\chi > 0$. The two equations (47.5) therefore specify a unique field. Formula (47.6) satisfies them: $g \cdot X_q = 1$ and $KX_q = g/s$ is radial in g. Endpoint lifts differ by translations by multiples of 2π; local lifts of ϑ also differ by constants. Both formulas therefore agree on overlaps and define the field without gluing.

Along the flow $\dot g = g/s$. Its direction is constant and $\dot r = r/s > 0$. Since $r^2/\Lambda_\circ \le s \le r^2/\lambda_\circ$, one has $2\lambda_\circ \le d(r^2)/dh \le 2\Lambda_\circ$. This proves (47.7). Using r as the parameter,

$$\frac{dq}{dr} = K^{-1} e.$$

Integration gives (47.8), including length rather than merely endpoint displacement. The direction of g is constant, so $|g(\Phi_h x) - g(x)| = r_h - r_0$ and the first-variation formula gives (47.9). In particular r_h cannot tend to zero in forward time. On a compact regular subset away from $g = 0$ the ordinary continuation theorem applies; any remaining finite exit is a true boundary of the regular exact event. □

Proposition 47.3 (The cancellation in the physical drift)**.** *Let $T = D^3F$ and let a be a locally Lipschitz density relative to dq. Away from $g = 0$, in the almost-everywhere sense,*

$$\operatorname{div}_q(aX_q) = \frac{\nabla a \cdot v}{s} + a\left\{\frac{T[v,v,v]}{s^2} - \frac{v \cdot \nabla \log\det K}{s}\right\}. \tag{47.11}$$

On a regular neighborhood where $\|T\| \le T_0$, this gives

$$|\operatorname{div}_q(aX_q)| \le \frac{1}{|g|}\left\{\frac{\Lambda_\circ}{\lambda_\circ}|\nabla a| + \left(\frac{\Lambda_\circ^2}{\lambda_\circ^3} + \frac{2\Lambda_\circ}{\lambda_\circ^2}\right)|a| T_0\right\}. \tag{47.12}$$

In particular the constant-density quadratic part has zero divergence. The bounds on T and ∇a in this assertion are local, not collision-uniform conclusions.

Proof. Differentiation of $v = K^{-1}g$ yields $\operatorname{div} v = 2 - v \cdot \nabla \log\det K$. Also $v \cdot \nabla s = 2s - T[v,v,v]$. The two terms $2/s$ cancel upon differentiating av/s, proving (47.11). Use $|v| \le |g|/\lambda_\circ$, $s \ge |g|^2/\Lambda_\circ$, and $|\nabla \log\det K| \le 2T_0/\lambda_\circ$ for the bound. Near a regular interior critical point the uniform positive Hessian makes $|g|$ comparable to distance, so the remaining displayed singularity is locally integrable in two dimensions. This last observation concerns a local integral only. □

48. Recoupling the complete physical source at fixed momentum angle

Use the unchanged $W = W^{\varepsilon}$ and one fixed exact label (m, R, n, k). Let Ω be its regular event, with the interior critical points removed. The removed sets have zero original area. Write μ_t for the coarea fiber of $d\alpha\, dp/(4\pi c_*)$ on this event. Thus $b^{\varepsilon}(t) = \int W\, d\mu_t$.

Theorem 48.1 (A complete, cap-free two-observable disintegration)**.** *The physical measure $(L, \vartheta)_{\#}(W\mathbf{1}_{\Omega}\nu_R^*)$ has a nonnegative density $A^{\varepsilon}(t, \theta)$ relative to $dt\, d\theta$ on $\mathbb{R} \times \mathbb{S}^1$. For almost every (t, θ),*

$$A^{\varepsilon}(t, \theta) = \frac{1}{4\pi c_*} \sum_{x \in \Omega:\ L(x)=t,\ \vartheta(x)=\theta} \frac{W(x)}{|B_m(x)|\chi(x)}. \tag{48.1}$$

The sum is over actual initial states, with every physical word and the exact occupation, displacement, initial and terminal tests retained. At fixed m its fibers are finite; no estimate for their cardinality as m grows is used. One has

$$b^{\varepsilon}(t) = \int_{\mathbb{S}^1} A^{\varepsilon}(t, \theta)\, d\theta, \tag{48.2}$$

$$\int_{\mathbb{R}} \int_{\mathbb{S}^1} A^{\varepsilon}(t, \theta)\, d\theta\, dt = \int_{\Pi_{n,k,m,R}} W\, d\nu_R^* \le 1. \tag{48.3}$$

These equalities do not normalize the retained source a second time.

Proof. By (47.10) the absolute Jacobian of the physical map $(\alpha, p) \mapsto (L, \vartheta)$ is $|B_m|\chi$. The area formula, on a finite circle atlas and a disjoint refinement of regular inverse charts, proves (48.1). Overlaps are removed at the level of physical initial states. The formula is consequently unchanged by relabeling a word chart or a cap. Tonelli and ordinary disintegration prove the two identities.

For finiteness of the fibers at this fixed count, the explicit finite physical graph retained in the supplement describes contacts, first-hit tests, section decisions, and endpoint momenta by finitely many semialgebraic conditions in rational angular charts. Add $g = r(\cos\theta, \sin\theta)$, $r > 0$, and $L = t$. The nonzero Jacobian makes each such fiber discrete. A definable discrete set is finite, uniformly in parameters for this fixed formula. This assertion is a fixed-count fact; it is not a pressure estimate for an increasing family of words. □

The inverse twist in this formula retains the optical bound

$$\frac{1}{|B_m|} \le \frac{Rc_0c_m}{\tau_-} \frac{\sinh\eta}{\sinh(m\eta)} \prod_{i \in S} \frac{53c_i}{6 + 47c_i}, \qquad \eta = \operatorname{arcosh}(53/47), \tag{48.4}$$

for any selected internal set S. For clarity, its denominator is obtained from $y_0 = 0$, $f_0 = 1$, $y_1 = \ell_0$ and $f_i = f_{i-1} + 2y_i/(Rc_i)$, $y_{i+1} = y_i + \ell_i f_i$. The corner Schur complement gives $|B_m|^{-1} = Rc_0c_m/y_m$. Lowering lengths to τ_- and potentials to $2/R_+$ gives the hyperbolic-sine solution. Restoring selected potentials one at a time gives the factors in (48.4) by their positive scalar Schur complements. The complete proof is retained in the supplement. We use its per-word estimate, not an unproved exponential word-count bound. Identity (48.3) is an actual complete weighted summability identity, but only in integrated form.

Disintegrate the weighted level source also over θ:

$$W\mu_t = \int_{\mathbb{S}^1} A^{\varepsilon}(t, \theta)\, \rho_{t,\theta}\, d\theta, \tag{48.5}$$

where $\rho_{t,\theta}$ is a probability on physical initial states with this roof, angle, and exact label whenever its coefficient is positive. In the finite regular fiber it assigns weights proportional to the summands of (48.1). Its value where the coefficient is zero is immaterial.

Theorem 48.2 (Maximal recoupling with the physical angle retained)**.** *For almost every (t,h) with $h>0$, the measure*

$$\Gamma_{t,h}=\int_{\mathbb{S}^1}\min\{A^{\varepsilon}(t,\theta),A^{\varepsilon}(t+h,\theta)\}(\rho_{t,\theta}\otimes\rho_{t+h,\theta})\,d\theta \tag{48.6}$$

is a subcoupling of the complete physical weighted fibers at t and $t+h$. Among all subcouplings which preserve ϑ, its matched mass is maximal. Its unmatched initial mass is exactly

$$\Delta_h^{\varepsilon}(t)=\int_{\mathbb{S}^1}[A^{\varepsilon}(t,\theta)-A^{\varepsilon}(t+h,\theta)]_+\,d\theta. \tag{48.7}$$

It follows that

$$[b^{\varepsilon}(t)-b^{\varepsilon}(t+h)]_+\le\Delta_h^{\varepsilon}(t)\le R^+_{X,h}(t), \tag{48.8}$$

where $R^+_{X,h}$ is the unmatched mass of the stopped physical-flow subcoupling for the explicit field in Theorem 47.2.

Proof. Each marginal of (48.6) is dominated by the corresponding measure in (48.5). For any other angle-preserving subcoupling, push its matched measure to the common angle. Its density is bounded above by both $A^{\varepsilon}(t,\theta)$ and $A^{\varepsilon}(t+h,\theta)$, and therefore by their minimum. Formula (48.6) attains this upper bound. Subtracting its mass from the source mass gives (48.7), and target domination proves the first inequality in (48.8).

The stopped-flow graph for X preserves ϑ by (47.5). On its surviving part its source weight is the minimum of the source density and the transported target density, with the physical coarea Jacobian. The usual flow change of variables gives domination of both physical marginals. It is therefore one admissible competitor in the preceding maximality argument. Its unmatched mass is $R^+_{X,h}$, which proves the last inequality. Joint disintegration supplies almost-everywhere versions in (t,h); no pointwise selection on an exceptional roof is required. □

The optimization in this theorem is not over unrestricted scalar couplings. It retains the explicitly specified physical angle. It may pair two different regular words, but only when both are actual states of the same original exact event at their respective roofs. It does not splice a nonphysical orbit or pool different labels. The full sum in (48.1) is formed before its positive difference is taken. Thus a true word exit may be balanced by actual incoming mass at the same angle and label; any remaining loss is still charged. Such balancing was unavailable to a stopped single-orbit graph. For example, two nonnegative angular profiles supported on successive roof intervals can have a continuous sum although the first profile has an exit. This is an algebraic illustration of the distinction, not a claim about a particular billiard trajectory. No comparison with every earlier scalar short-excursion modulus is asserted.

Proposition 48.3 (The angular residue at an interior critical roof)**.** *Let x_* be a regular interior critical point with a neighborhood in the exact event, $F_*=F(x_*)$, and $K_*=D^2F(x_*)$. Write $a_*=W(x_*)/(4\pi c_*|B_m(x_*)|)$. Its local angular source has right-minus-left jump*

$$j_*(\theta)=\frac{a_*}{e_\theta^{\mathsf T}\operatorname{cof}(K_*)e_\theta}\ge 0,\qquad \int_{\mathbb{S}^1}j_*(\theta)\,d\theta=\frac{W(x_*)}{2c_*|B_m(x_*)|\sqrt{\det K_*}}. \tag{48.9}$$

Every direction is an entrance, not a forward exit. This is a local charge statement; no sign is assigned to other physical boundaries at the same roof.

Proof. Translate q_* to zero. The inverse function theorem for $g=\nabla F$ gives a smooth inverse $q(re_\theta)$ for small r. Along it $dF/dr=re_\theta^{\mathsf T}K^{-1}e_\theta>0$. Thus, for each angle, the local branch starts at F_* from above. The density in (48.1) tends to $a_*/(e_\theta^{\mathsf T}\operatorname{cof}(K_*)e_\theta)$. The convergence is uniform in angle on this fixed neighborhood. Diagonalizing the positive matrix K_* and integrating $(a\cos^2\theta+b\sin^2\theta)^{-1}$ gives $2\pi/\sqrt{ab}$, which proves the formula. The neighborhood is used only

to calculate the residue. The globally defined field and its uniform motion bounds use no such radius. □

49. The signed angular current and the pointwise endpoint

The angular source admits a finite signed roof derivative at each fixed collision count. The differentiation is performed after the complete exact-label sum, not separately with absolute values on individual words.

Theorem 49.1 (Finite variation before projection to the scalar roof)**.** *For each fixed $m \geq 2$ there is $C_m < \infty$, independent of R, ε, n, k, such that for almost every θ,*

$$A^\varepsilon(\,\cdot\,,\theta) \in BV(\mathbb{R}), \qquad \|D_t A^\varepsilon(\,\cdot\,,\theta)\|_{\mathrm{TV}} \leq C_m. \tag{49.1}$$

Consequently $\mathscr{J}^\varepsilon = D_t A^\varepsilon$ is a finite signed measure on $\mathbb{R}\times\mathbb{S}^1$. It and its Jordan parts disintegrate over $d\theta$ as the corresponding one-dimensional roof derivatives and Jordan parts. Its projection to the roof is Db^ε. No estimate for $m^2 C_m$ is asserted.

Proof. We give the finite-count details because the original smooth guard need not be definable. On the finite physical graph, enumerate the $N = 2m+1$ normalized incidence and clearance margins as a_i. For $y \in [1,2]^N$ let $W_y = 1 - \prod_{i=1}^N \mathbf{1}_{\{a_i > y_i\}}$ and let A_y be its angular density. By the nonzero Jacobian and the explicit finite physical graph, the inverse fibers in (48.1) are definable discrete sets. Uniform finiteness gives a bound N_m for their size, uniform in (R, n, k, t, e_θ) at this fixed m. The maps B_m, K, and χ on a regular sheet are definable: their derivatives can be taken on the rational angular charts of the physical graph, followed by the rational change from chart derivatives to angular derivatives. Finite definable sums and threshold tests show that A_y has a definable representative.

The empty-set optical bound and $\chi \geq \lambda_\circ$ give

$$0 \leq A_y(t,\theta) \leq H_m := \frac{N_m R_+}{4\pi c_* \tau_- \lambda_\circ} \frac{\sinh\eta}{\sinh(m\eta)}.$$

Take the value zero on exceptional graph cells. On a fixed finite atlas of the angle circle, the uniform monotonicity theorem for definable families [9] supplies $J_m < \infty$ monotonicity intervals and point cells in the roof variable, with all remaining variables treated as parameters. Extension by zero outside $[m\tau_-, m\tau_+]$ therefore gives $\|D_t A_y(\,\cdot\,,\theta)\|_{\mathrm{TV}} \leq 4(J_m+1)H_m$. Both N_m and J_m are constants of a fixed finite formula, not controlled word-pressure constants.

Put $\sigma = d\Theta$ and $V_\Theta = \|\sigma\|_{\mathrm{TV}}$. The exact identity

$$W^\varepsilon = \int_{[1,2]^N} W_y \, d\sigma^{\otimes N}(y)$$

uses $\sigma(\mathbb{R}) = 1$ and is valid even when Θ is not monotone. Coarea and finite signed Fubini give $A^\varepsilon = \int A_y \, d\sigma^{\otimes N}$ almost everywhere. The distributional characterization of BV [13] proves (49.1) with $C_m = 4(J_m+1)H_m V_\Theta^N$.

Integrating the derivative kernels over the circle defines a finite signed measure of variation at most $2\pi C_m$. Measurability follows first for smooth test functions from Fubini, and then for the variation and the Jordan kernels by the countable test-function characterization of variation. This identifies the measure with $D_t A^\varepsilon$ and its Jordan disintegrations. Testing with a function independent of θ and using (48.2) proves the projection identity. Jordan decomposition need not commute with this final projection. □

49.1. **The physical meaning of this current.** The finite-current argument just used is an application of the inherited threshold method to the two-observable density. It is not a new proof of collision-uniform weighted summability. Its geometric computation can nevertheless be made entirely on actual regular states. On a compact regular endpoint domain D write $a = W/(4\pi c_*|B_m|)$ and $Y = X_q$. For a smooth test $\psi(t,\theta)$, since $YF = 1$ and $Y\vartheta = 0$, the divergence theorem gives

$$D_t(F,\vartheta)_{\#}(a\mathbf{1}_D\,dq) = (F,\vartheta)_{\#}\left(\operatorname{div}(aY)\,dq|_D - (aY\cdot n)\,d\mathcal{H}^1|_{\partial D}\right). \tag{49.2}$$

On the left, the derivative is that of the corresponding density. For a locally Lipschitz weight the same identity follows by smooth approximation on the compact domain. The bulk term is explicitly (47.11). At an artificial common boundary with the same physical source, the two oriented traces cancel. A true source jump leaves its jump flux. A limiting interior critical boundary has the positive residue (48.9).

Regular exhaustions converge to $\mathscr{J}^{\varepsilon}$ in distributions: the omitted source mass tends to zero, and its action is bounded by that mass times $\|\partial_t\psi\|_\infty$. No convergence of the separately absolute-valued bulk and boundary terms is asserted. The full finite measure is secured by Theorem 49.1, not by an unproved uniform bound on an exhaustion. True singularities, first-hit switches, and section decisions thus remain in the signed physical current. Only actual cancellation at the same angle and exact label is credited before taking its negative part.

Set

$$\Delta_r^{\varepsilon}(t) = \operatorname*{ess\,sup}_{0<h<r} \Delta_h^{\varepsilon}(t), \tag{49.3}$$

$$V_r^{\varepsilon}(t) = (\mathscr{J}^{\varepsilon})^{-}([t,t+r]\times\mathbb{S}^1). \tag{49.4}$$

The closed interval is harmless for the essential bounds and retains all possible boundary atoms. All translated roofs remain on the full roof line, not on a truncated central target.

Theorem 49.2 (Complete angular persistence)**.** *For every fixed $r>0$ and almost every t,*

$$[b^{\varepsilon}(t) - 2A_r b^{\varepsilon}(t)]_+ \le \Delta_r^{\varepsilon}(t) \le V_r^{\varepsilon}(t). \tag{49.5}$$

The defect on the left comparison is no larger than the maximal forward stopped-flow defect for the explicit field of Theorem 47.2. Positive interior critical entrances are not charged as forward losses.

Proof. For a BV function f, $[f(t)-f(t+h)]_+ \le (Df)^-([t,t+h])$ for almost every (t,h). Apply this to $A^{\varepsilon}(\,\cdot\,,\theta)$ and integrate in θ. The Jordan disintegration proves $\Delta_h^{\varepsilon}(t) \le V_r^{\varepsilon}(t)$ when $0<h<r$, and therefore the second inequality. The first follows by averaging $b^{\varepsilon}(t) \le b^{\varepsilon}(t+h) + \Delta_r^{\varepsilon}(t)$ over $0<h<r$. Nonnegativity bounds this forward average by $2A_r b^{\varepsilon}(t)$. The stopped-flow comparison follows from (48.8). The critical assertion follows from its positive angular residue; unrelated true boundary terms are still present in the signed sum. □

Define the ordered complete angular defect

$$\mathcal{A}_{\varepsilon,M}(r) = \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \Delta_r^{\varepsilon}(t). \tag{49.6}$$

There is no supremum over selected physical words in this definition; their complete weighted sum has already been taken.

Theorem 49.3 (Ordered implication for the unchanged pointwise law)**.** *Assume Input statement 46.1. For each sufficiently large fixed B and fixed $r>0$,*

$$\mathcal{E}_M \le C'_M B^{-1/192} + \mathcal{A}_{\varepsilon(B),M}(r). \tag{49.7}$$

In particular the original target (46.5) *follows if, for some* $A_M < \infty$, $q \geq 0$, $\beta > 0$ *and fixed* $\varepsilon_0, r_0 > 0$, *the complete source satisfies*

$$\mathcal{A}_{\varepsilon,M}(r) \leq A_M \varepsilon^{-q} r^{\beta} \quad (0 < \varepsilon < \varepsilon_0,\ 0 < r < r_0). \tag{49.8}$$

The same bound on the ordered norm of V_r^{ε} *is sufficient. Estimate* (49.8) *is an additional dynamical estimate; it is not proved by the fixed-count variation theorem or the integrated identity* (48.3).

Proof. Use the positive remainder in (46.7) and (49.5) to bound the pointwise error by $2m^2 A_r b^{\varepsilon(B)} + m^2 \Delta_r^{\varepsilon(B)} + |e_B|$. Fix B, $\varepsilon(B)$, r, and each auxiliary spectral band before the collision limsup. The fixed-width complete-source bound then gives

$$\mathcal{E}_M \leq 2C\varepsilon(B)^{1/16} + \mathcal{A}_{\varepsilon(B),M}(r) + C_M B^{-1/192}.$$

Since $\varepsilon(B) = A_0 B^{-1/12}$, this proves (49.7). Under (49.8), take $r(B) = c\varepsilon(B)^{(q+a)/\beta}$ with $a > 0$ and a fixed $c > 0$ small enough that $r(B) < r_0$. This is fixed before the collision limit. Enlarge the auxiliary spectral band only afterwards and let $B \to \infty$ last. The two errors tend to zero. □

49.2. **The same scalar source in path normalization.** The retained path input uses a common finite positive path remainder $\mathsf{B}_B(t)$ of mass $m^2 b^{\varepsilon(B)}(t)$, and

$$\mathsf{P}_m(t) - \mathsf{L}_m(t) = \mathsf{B}_B(t) + \mathsf{e}_B(t), \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} \|\mathsf{e}_B(t)\|_{\mathrm{BL}^*} \leq C_M B^{-1/192}.$$

The bounded-Lipschitz class contains the constant one, so the norm of the positive remainder is its mass. Under (49.8), the preceding proof makes this mass vanish in the same ordered limits. Path-numerator convergence then follows. Ratios may be formed only on central arithmetic classes with scalar reference bounded below by a fixed positive constant. No arithmetic zero class is divided by. Collision-to-return clock transfer and forward likelihood retain their own inherited hypotheses. None of these conclusions is used to establish the scalar estimate.

50. All-contact motion on the original regular source

The endpoint field controls more than the two endpoints. Its effect on every contact decays with distance from the nearer endpoint, without imposing a positive lower bound on the internal incidences. This observation permits a uniform estimate for the original smooth guard, rather than a replacement by protected words.

Keep $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$. All endpoint derivatives below use $q = (\alpha_0, \alpha_m)$; internal contact displacements use arclength. Constants depend only on the fixed physical family and, where indicated, the originally chosen function Θ.

Lemma 50.1 (An incidence-normalized contact inverse)**.** *For the optical matrix* A *of the retained recursion, put* $D_c = \operatorname{diag}(c_1, \ldots, c_{m-1})$ *and* $C_A = R_+/[2(1-q_0)]$. *For* $1 \leq i, j < m$,

$$0 \leq (A^{-1})_{ij} \leq C_A \min(c_i, c_j) q_0^{|i-j|}, \qquad |(D_c^{-1} A^{-1})_{ij}| \leq C_A q_0^{|i-j|}. \tag{50.1}$$

Consequently, at every regular physical word, its contact positions Q_j *satisfy*

$$\|D_q Q_j\| \leq C_Q q_0^{d_j}, \qquad 0 \leq j \leq m, \qquad C_Q = R_+ + \frac{2C_A R_+}{\tau_- q_0}. \tag{50.2}$$

For $m = 1$ *the internal assertion is empty and the endpoint estimate still holds. These are derivatives at actual regular trajectories; no common neighborhood of physical continuation is asserted.*

Proof. Write $A = D(I - P)$, with D its diagonal. The entries of P are nonnegative, vanish except at neighboring indices, and have row sums at most

$$\frac{2/\tau_-}{2/\tau_- + 2/R_+} = q_0.$$

Also $D_{jj}^{-1} \le Rc_j/2$. In the Neumann series for $(I - P)^{-1}D^{-1}$, the (i, j) entry vanishes until the power $|i - j|$. Summing the remaining geometric series gives the bound with c_j. Symmetry gives the bound with c_i, and division by $c_i > 0$ proves the second inequality. This is the same normalized inverse underlying the retained tapered-contact construction; here only its first derivative is needed, at arbitrary regular endpoints.

The internal action Hessian in oriented arclengths is D_cAD_c. After division of its stationary equations by the internal cosines, endpoint forcing has only the first and last entries, of absolute sizes $t_0c_0\,du_0$ and $t_{m-1}c_m\,du_m$. Tangent signs may change these signs but not their sizes. Thus the displacement of internal arclength s_j obeys

$$|ds_j| \le \frac{C_A}{\tau_-}(q_0^{j-1}|du_0| + q_0^{m-1-j}|du_m|).$$

Both exponents are at least $d_j - 1$, and $|du_i| = R|d\alpha_i|$. The displayed choice of C_Q bounds the resulting Euclidean operator norm and the two endpoint derivatives. The Euclidean speed of a circle parameterized by arclength is one, which proves (50.2). □

Theorem 50.2 (Uniform motion of every physical margin)**.** *There is $C_D < \infty$, independent of m, R, n, k and of every positive incidence and clearance, with the following property. Along any regular endpoint curve of length ℓ_q in the exact event and in one physical word,*

$$|c_j(x) - c_j(y)| + |p_j(x) - p_j(y)| + \mathrm{dist}_{\mathbb{S}^1}(\alpha_j(x), \alpha_j(y)) \le C_D q_0^{d_j}\ell_q, \tag{50.3}$$

$$|\Delta_j(x) - \Delta_j(y)| \le C_D q_0^{d'_j}\ell_q. \tag{50.4}$$

In particular, for a forward survivor of the field in Theorem 47.2, the right sides are bounded by, respectively, $C_DC_Xq_0^{d_j}\sqrt{h}$ and $C_DC_Xq_0^{d'_j}\sqrt{h}$, where $C_X = \sqrt{2\Lambda_\circ}/\lambda_\circ$.

Proof. Differentiate a flight direction:

$$dv_j = \ell_j^{-1}(I - v_jv_j^{\mathsf{T}})(dQ_{j+1} - dQ_j).$$

The depths of its two endpoints are at least d'_j. Its derivative is consequently bounded by $2C_Qq_0^{d'_j}/\tau_-$. The normal and tangent derivatives at a contact are bounded by $C_Qq_0^{d_j}/R_-$. Express c_j and p_j as scalar products of the appropriate unit flight velocity with these vectors. At the terminal contact use the incoming velocity, with the sign giving positive incidence; its tangential component is unchanged by reflection. Since adjacent depths differ by at most one, these formulas prove the first estimate with a fixed C_D. In particular we do not differentiate $\sqrt{1 - p_j^2}$ by dividing by c_j.

The Hausdorff displacement of a segment is no greater than the larger displacement of its two endpoints. Distance from that segment to a fixed third disk is a one-Lipschitz function of this Hausdorff displacement. Taking the minimum over third disks preserves the same bound. The minimizing disk may change; differentiability of that minimum is not required. The excluded incident disks are fixed on the word. Integration along the curve proves (50.4). Finally use the endpoint-curve length estimate of Theorem 47.2. □

Corollary 50.3 (The unchanged guard is uniformly controlled)**.** *For every forward survivor in the original exact event,*

$$|W^\varepsilon(\Phi_h x) - W^\varepsilon(x)| \le \min\{1, K_\Theta \varepsilon^{-1}\sqrt{h}\}, \qquad K_\Theta = \frac{4\,\mathrm{Lip}(\Theta) C_D C_X}{1 - q_0^{3/4}}. \tag{50.5}$$

No incidence cutoff, clearance cutoff, section-distance factor or monotonicity assumption on Θ *is added.*

Proof. For factors in $[0,1]$, the difference of their products is bounded by the sum of their absolute differences. A contact factor in (46.2) changes by at most $\mathrm{Lip}(\Theta) C_D C_X \varepsilon^{-1} q_0^{3d_j/4}\sqrt{h}$; the analogous flight factor has depth d'_j. There are at most two contacts and two flights at any depth. Their geometric series gives (50.5). Both weights belong to $[0,1]$, proving the additional bound by one. The argument uses only the Lipschitz constant and range of the original smooth function. □

Proposition 50.4 (Where a finite forward exit can occur)**.** *Let* s_j *be distance from the collision state* (α_j, p_j) *to the boundary of* Y_R^* *in the product metric. There is a common* C_E *such that any source point whose forward lifetime in the regular exact event is at most* h *belongs to the union*

$$\mathcal{B}_h = \bigcup_{j=0}^{m} \{c_j \le C_E q_0^{d_j}\sqrt{h}\} \;\cup\; \bigcup_{j=0}^{m-1} \{\Delta_j \le C_E q_0^{d'_j}\sqrt{h}\} \;\cup\; \bigcup_{j=0}^{m} \{s_j \le C_E q_0^{d_j}\sqrt{h}\}. \tag{50.6}$$

Consequently its exit mass is bounded above by $\int W^\varepsilon \mathbf{1}_{\mathcal{B}_h}\, d\mu_t$*. This is a source inclusion, not a pointwise concentration estimate.*

Proof. The finite endpoint-curve length gives limiting contact data up to a finite lifetime. A forward trajectory cannot reach an interior critical point, by Theorem 47.2. If all limiting incidences, clearances and distances to section edges were positive, the regular reflection equations, strict first-hit tests and locally constant section decisions would continue the same physical word and exact event. The endpoint Jacobian remains nonzero there, so the roof field also continues, a contradiction. Thus some listed margin tends to zero. Apply Theorem 50.2, including the one-Lipschitz property of distance to a closed edge set. Taking the union over the possible margins and retaining the original weight proves the mass bound. A section edge is charged whether or not another simultaneous change might preserve the occupation; this can enlarge the upper bound but cannot omit a true exit. □

51. A complete-source estimate for guard distortion

The preceding pointwise control of the guard can be used at the collision scale without assuming a pointwise bound for the unweighted roof density. The essential step is to average the translated target before applying the fixed-width concentration estimate.

Fix one exact event and write $p(t) = p_{n,R}(k,m,t)$ for its unweighted roof density. Let $S_h(t)$ be the surviving part of its fiber under the explicit forward field, J_h the area Jacobian relative to the original collision source, and $E_h^\varepsilon(t)$ the original weighted mass killed before time h. Define

$$G_h^\varepsilon(t) = E_h^\varepsilon(t) + \int_{S_h(t)} W^\varepsilon(x)[1 - J_h(x)]_+\, d\mu_t(x), \qquad \overline{G}_r^\varepsilon(t) = \frac{1}{r}\int_0^r G_h^\varepsilon(t)\, dh. \tag{51.1}$$

The first term contains every true exit. The second contains the loss of physical coarea volume, not the change of the guard. In particular $0 \le G_h^\varepsilon(t) \le b^\varepsilon(t)$. These quantities are defined by the full physical source, not by summing estimates of word cardinalities.

Theorem 51.1 (Averaged separation of geometry and guard)**.** *For each fixed $r > 0$ and almost every t,*

$$b^{\varepsilon}(t) \le 2A_r b^{\varepsilon}(t) + \overline{G}_r^{\varepsilon}(t) + 2K_{\Theta}\varepsilon^{-1}\sqrt{r}\, A_r p(t). \tag{51.2}$$

The guard contribution on the right has the unconditional ordered bound, relative to Input statement 46.1,

$$\limsup_{m} \sup_{R,n,k} m^2 \left\| 2K_{\Theta}\varepsilon^{-1}\sqrt{r}\, A_r p \right\|_{\infty} \le C_{\Theta}\varepsilon^{-1}\sqrt{r}. \tag{51.3}$$

All incidences, clearances, terminal tests and exact labels remain present in this statement. No estimate of $m^2\|p\|_{\infty}$ is used.

Proof. The stopped physical graph couples source weight $\min\{W(x), W(\Phi_h x)J_h(x)\}$ to the target fiber. For any $w, w' \in [0,1]$ and $J \ge 0$,

$$[w - w'J]_+ \le w[1-J]_+ + J|w - w'|.$$

Apply this identity on survivors and use Corollary 50.3. Coarea covariance gives

$$\int_{S_h(t)} J_h(x)\, d\mu_t(x) = \mu_{t+h}(\Phi_h S_h(t)) \le p(t+h).$$

Here μ is the *unweighted* coarea measure on the exact event; hence its full mass is precisely p. The Jacobian is in the target direction in this formula. Target domination of the subcoupling now yields

$$b^{\varepsilon}(t) \le b^{\varepsilon}(t+h) + G_h^{\varepsilon}(t) + K_{\Theta}\varepsilon^{-1}\sqrt{h}\, p(t+h)$$

for almost every (t,h). Average over $0 < h < r$, bound $\sqrt{h} \le \sqrt{r}$, and bound each forward average by twice the centered average of its nonnegative density. This proves (51.2).

The unweighted fixed-width estimate is already contained in the analytic input as stated. Indeed with auxiliary guard parameter one, the first incidence factor is $\Theta(c_0) = 0$, so that $H^1 = 0$, $W^1 = 1$, and $b^1 = p$. Thus

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_r p\|_{\infty} \le C$$

for each fixed r, with the same constant independent of that previously fixed width. This application does not change the guard W^{ε} in the source being proved; it supplies only an upper bound for its translated target. Substitution proves (51.3). Equivalently one may use the retained complete-source fixed-width concentration theorem with weight one. The averaging must precede the collision limit; replacing $A_r p$ by p would be an unproved step. □

Set

$$\mathcal{G}_{\varepsilon,M}(r) = \limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} \overline{G}_r^{\varepsilon}(t). \tag{51.4}$$

Theorem 51.2 (The original endpoint with the guard term estimated)**.** *Under Input statement 46.1, for all sufficiently large fixed B and fixed $r > 0$,*

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathcal{G}_{\varepsilon(B),M}(r) + C_{\Theta}\varepsilon(B)^{-1}\sqrt{r}. \tag{51.5}$$

In particular the original pointwise assertion follows from $\mathcal{G}_{\varepsilon,M}(r) \le A_M \varepsilon^{-q} r^{\beta}$ for finite $q \ge 0$ and $\beta > 0$. More precisely it is equivalent, under the same analytic input, to

$$\lim_{B\to\infty} \inf_{0<r<r_0} \left\{ \mathcal{G}_{\varepsilon(B),M}(r) + \varepsilon(B)^{-1}\sqrt{r} \right\} = 0 \quad \textit{for every fixed } r_0 > 0. \tag{51.6}$$

Neither the geometric bound nor (51.6) *is asserted as proved here. Unlike the guard contribution, these terms still contain true exits and physical volume distortion.*

Proof. Combine (51.2) with the positive remainder, and take the collision limsup only after fixing B, $\varepsilon(B)$, r and the auxiliary spectral bands. The fixed-width bad-source term is at most $2C\varepsilon(B)^{1/16} = O(B^{-1/192})$. Equation (51.3) proves (51.5). Under the power bound choose $r = c\varepsilon(B)^{\gamma}$, with $\gamma > \max\{2, q/\beta\}$ and fixed sufficiently small $c > 0$. Both terms then vanish. The auxiliary spectral band is enlarged after the collision limit and B last.

Condition (51.6) is sufficient by (51.5); approximate the infimum with a positive width fixed for each B before taking the collision limit. Conversely, if the original endpoint holds, the positive remainder implies

$$\limsup_{m} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} b^{\varepsilon(B)} \le C_M B^{-1/192}.$$

Since $\overline{G}_r^{\varepsilon} \le b^{\varepsilon}$ for every r, the same bound holds for $\mathcal{G}$ independently of r. Taking the infimum over r and then $B \to \infty$ proves necessity. No collision-dependent Fourier band or density representative is selected by this argument. □

The original angular recoupling remains available and can only improve its own stopped-flow comparison. The estimate just proved isolates and controls one complete contribution without claiming that the angular current, the true boundary flux or the full optical volume distortion has thereby been bounded. Scalar closure must still precede every same-roof path normalization and every likelihood conclusion, as in the retained path remainder theorem.

52. Exact occupation differences at regular decision boundaries

The original guard contains no section-distance factor. Internal section decisions therefore cannot be discarded by the guard estimate. Their signed current has an exact algebraic structure which should be retained before absolute values are taken. This structure does not require pooling the exact occupation classes.

Work first on a compact regular endpoint domain D, away from $g = 0$, within one physical word and one fixed displacement. Choose D with a finite piecewise smooth stratification compatible with the internal section interfaces; compact definable subdomains of the physical graph provide such choices. The initial and terminal section tests hold throughout D. Its cells have constant internal section decisions, up to lower-dimensional strata. Write

$$a(q) = \frac{W^{\varepsilon}(q)}{4\pi c_* |B_m(q)|}, \qquad Y = X_q, \qquad A(q) = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*}(T_R^j q).$$

The notation $T_R^j q$ means the physical trajectory represented by these endpoints. The functions a, Y, F, ϑ have the same traces on both sides of an internal section interface; only the occupation changes. The locally Lipschitz guard is sufficient for the divergence identity by approximation.

Theorem 52.1 (An exact discrete divergence for internal decisions)**.** *For a variable z, form the polynomial-valued density $(F, \vartheta)_{\#}(az^A \mathbf{1}_D\, dq)$. The contribution of its* internal section interfaces *to its roof derivative is*

$$\mathscr{S}_D(z) = (z-1)\mathscr{F}_D(z). \tag{52.1}$$

Both sides are finite polynomial-valued signed measures on $\mathbb{R} \times \mathbb{S}^1$. If $\mathscr{F}_D(z) = \sum_n z^n \mathscr{F}_{D,n}$, then the contribution to the original exact occupation n is precisely

$$\mathscr{S}_{D,n} = \mathscr{F}_{D,n-1} - \mathscr{F}_{D,n}. \tag{52.2}$$

The displacement, collision count, initial and terminal tests and original source normalization are unchanged in every coefficient.

Proof. Orient an interface Σ from the cell with occupation a_- to the cell with occupation a_+, and let N be that normal. The negative outward flux from the first cell and that from the second sum to

$$(F,\vartheta)_{\#}\left(a(Y\cdot N)(z^{a_+}-z^{a_-})\,d\mathcal{H}^1|_\Sigma\right).$$

There is no absolute value on this signed expression. For nonnegative integers u, v, define

$$P_{u,v}(z)=\begin{cases}\sum_{j=u}^{v-1} z^j, & v>u,\\ 0, & v=u,\\ -\sum_{j=v}^{u-1} z^j, & v<u.\end{cases}\qquad z^v-z^u=(z-1)P_{u,v}(z).$$

Sum the pushforwards with P_{a_-,a_+} in place of the difference to define $\mathscr{F}_D$. Changing the chosen normal exchanges a_-, a_+ and changes both signs, so this definition is independent of orientation. A coincident interface at which several visits change is covered by the same identity; a net zero change contributes zero. Intersections of distinct smooth interfaces of lower dimension carry no surface flux. Compact regularity makes all these signed measures finite. Taking the coefficient of z^n proves (52.2). □

Remark 52.2 (Outer tests and true singularities are separate). An initial or terminal section boundary need not have an admissible source on both sides. Its flux is therefore not covered by (52.1). Nor does the theorem identify a first-hit switch or a grazing word transition with a regular section interface. All such contributions remain in the full physical current and in the exit term of (51.1). On an exhaustion the polynomial identity holds at every compact stage. Passage of the complete current to distributions follows from source-mass convergence, as in the retained angular current theorem. Convergence or a uniform bound for the isolated primitive $\mathscr{F}_D$ does not follow from that fact and is not assumed.

52.1. **The arithmetic phase of a discrete difference.** A factor $z-1$ is not small near a general spectral peak. The following estimate records exactly what such a factor can and cannot supply. Let $\xi=(u_1,u_2,b,v)$ be the four Fourier coordinates. In one fixed peak chart centered at a_j with occupation coordinate v_j, suppose a scalar amplitude satisfies

$$|H_{m,j}(a_j+\delta)|\le Ce^{-cm|\delta|^2}$$

throughout that chart, with constants uniform in the permitted parameters. Let $F_{m,j}(y)$ be its Fourier integral with factor $(2\pi)^{-4}$, and let $S_{m,j}(y)$ be the same integral with multiplier $e^{iv}-1$. Here $y=(k_1,k_2,t,n)$; no division by an arithmetic reference is involved.

Proposition 52.3 (Peak-aware coefficient difference)**.** *Under the displayed amplitude bound,*

$$S_{m,j}(y)=(e^{iv_j}-1)F_{m,j}(y)+O(m^{-5/2}),\tag{52.3}$$

uniformly in all targets. Equivalently the covariant difference $F_{m,j}(k,t,n-1)-e^{iv_j}F_{m,j}(k,t,n)$ is $O(m^{-5/2})$. At a peak with $v_j=0$ modulo 2π, the amplitude hypothesis alone gives the same bound for the ordinary difference. At any other peak that conclusion requires a separate $O(m^{-5/2})$ bound for the retained leading term; it need not vanish identically.

Proof. On the chart,

$$|(e^{iv}-1)-(e^{iv_j}-1)|\le|v-v_j|\le|\delta|.$$

The absolute value of the remaining Fourier integral is bounded by $C\int_{\mathbb{R}^4}|\delta|e^{-cm|\delta|^2}\,d\delta$. The change of variable $\delta=m^{-1/2}w$ gives $O(m^{-5/2})$. Multiplication by e^{iv} shifts the extracted occupation from n to $n-1$, proving the second formulation. A moving center does not change the estimate when the displayed constants are uniform. □

The hypothesis of this proposition is a scalar bound for the *surface-current primitive*, not just for the original volume source. No such uniform bound for all physical interfaces is imported from a theorem about bounded volume multipliers. Furthermore a nonzero occupation peak retains the first term in (52.3), including at transition radii. This prevents a spurious cancellation of nontrivial arithmetic classes. The exact interface identity and its peak-aware estimate provide a specific signed summation mechanism for the still required geometric bound; they do not certify that bound or remove any Fourier peak.

53. The complete geometric estimate and the original theorem

The pointwise assertion remains (46.5), with the full arithmetic transition, the same normalization and all zero classes. Theorem 51.1 proves the complete guard-distortion bound (51.3); the constant is uniform in collision count and no selected physical word has replaced the source. Proposition 50.4 locates every true forward exit, but does not bound concentration on the starting roof. Theorem 52.1 computes the signed internal section contribution without suppressing its exact-label coefficients.

To conclude the organizing theorem, one must still estimate the complete geometric term in (51.4), or the complete angular defect in (49.6). The new guard estimate does not control the optical Jacobian loss. An integrated boundary-layer estimate is not a pointwise source estimate, and a spectral estimate for volume multipliers is not automatically an estimate for surface-current insertions. In particular (52.3) retains nonzero arithmetic leading terms. The proof ledger records these distinctions explicitly.

The organizing pointwise theorem is therefore not marked proved in this revision. Its path and likelihood consequences retain the same scalar-first logical order. The complete inherited mathematical source and the preceding journal article are retained in the technical supplement and provenance, rather than deleted or recast as a different problem.

Part 27. Endpoint momenta and complete angular recoupling

Retained revision-77 abstract. We retain the pointwise local limit problem for the full displacement, collision count, occupation, and roof record of actual returns in a triangular Lorentz gas. The exact initial and terminal section tests give collision-uniform positive bounds for the Hessian in physical endpoint angles. They determine an explicit roof flow preserving the angle of the endpoint momenta. Its forward motion is uniformly square-root controlled and never reaches an interior critical point; no Morse collars or cap-dependent patching are needed. We disintegrate the complete original source jointly over roof and momentum angle, and construct a maximal angle-preserving subcoupling after summing all actual words with the same exact label. This recoupling can recover mass lost by an individual stopped orbit. A finite signed angular current retains true boundary flux and the positive critical residues. These results sharpen the complete-source persistence estimate for the original pointwise law. The collision-normalized concentration bound for the complete angular defect remains the quantitative estimate required to close that law.

54. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1, 1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6,-1/2), \quad (-\theta_R,\sin\theta_R), \quad (\pi/3-\theta_R,\sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1-6q^2+q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{54.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty,1]$ and one on $[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{54.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{54.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

54.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R,a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R,a_j), \quad H_j = -\nabla^2 \log\lambda_j(R,a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{54.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow 0. \tag{54.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

54.2. **The analytic inputs in the form used here.**

Input statement 54.1. The unchanged source (54.3) has a nonnegative, compactly supported BV density at every fixed count. For every fixed $\varepsilon,\delta>0$, with $A_\delta f(t)=(2\delta)^{-1}\int_{t-\delta}^{t+\delta} f(s)\,\mathrm{d}s$,

$$\limsup_m \sup_{R,n,k} m^2\|A_\delta b^\varepsilon\|_\infty \le C\varepsilon^{1/16}. \tag{54.6}$$

The constant is independent of the previously fixed width. For large fixed B and $\varepsilon(B)=A_0B^{-1/12}$,

$$m^2p-\mathcal{L}=m^2b^{\varepsilon(B)}+e_B,\qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)}|e_B|\le C_MB^{-1/192}. \tag{54.7}$$

All auxiliary spectral bands in these statements are fixed before the collision limit and enlarged only afterwards.

These are the complete-source tame theorem, the fixed-width complete-source bound, and the positive scalar remainder theorem in the technical supplement. Their proofs and historical hypotheses are retained there. The fixed-count source uses threshold integration [15], definable monotonicity [9], and the distributional characterization of BV [13]; it does not assume definability of the original smooth guard. They are restated as inputs to distinguish them from the new geometric and transport proofs. No long-time bound on a fixed-count variation constant is part of this statement.

54.3. **The new physical construction.** The retained convex-support and optical arguments provide genuine information on every regular physical word. They do not on their own control the complete weighted sum on an exact roof. The present construction uses a different observable from the auxiliary cap normal: the angle of the two physical endpoint momenta. The fixed section makes its coarea Jacobian uniformly positive after the optical twist is factored out. The resulting field is explicit and requires no collision-dependent critical neighborhoods.

The new subcoupling is constructed after summing all actual words of one exact event. It may match mass between different true words, while preserving their exact displacement, occupation, terminal test, and endpoint-momentum angle. This is more permissive than continuation of a single physical orbit and more restrictive than unconstrained scalar matching. Its finite-current estimate is an unconditional fixed-count statement. Its collision-normalized concentration remains a distinct dynamical question.

55. A canonical flow from the endpoint momenta

The roof field can be specified on the whole regular exact event, without choosing critical disks or endpoint caps. The particular section in (54.1) is useful here: it bounds both endpoint momenta away from grazing, even though no internal incidence or clearance is bounded away from zero.

For one regular physical word use locally lifted endpoint angles $q=(\alpha_0,\alpha_m)$, and write $F(q)$ for its reduced length. The endpoint map is locally invertible, since the optical twist $B_m=\partial_{p_0}\alpha_m$ is nonzero. Set

$$g=\nabla_q F=R(-p_0,p_m),\qquad K=D_q^2F,\qquad \lambda_\circ=\frac{9\sqrt{19}}{200},\quad \Lambda_\circ=\frac{47}{6}.$$

The terminal tangential velocity has the same value before and after reflection, which fixes the sign in the second component of g. All differentiations in this section are on actual regular words.

Lemma 55.1 (Uniform endpoint-angle Hessians on the exact event)**.** *At every regular physical trajectory,*

$$R\,\mathrm{diag}(c_0, c_m) \le K \le (2R_+^2/\tau_- + R_+)I_2. \tag{55.1}$$

On the original exact event, in particular,

$$\lambda_\circ I_2 \le K \le \Lambda_\circ I_2. \tag{55.2}$$

These are pointwise quadratic-form inequalities, uniform in m, R, n, k. They assert neither convexity of a disconnected physical domain nor uniform control of derivatives of K.

Proof. At a regular trajectory let w_0 be its outgoing first-flight velocity and w_m the negative of its incoming last-flight velocity. The polygonal length, minimized over the prescribed intermediate closed disks, is supported at this trajectory by

$$F(q_*) - w_0 \cdot (Q_0(q_0) - Q_0(q_{0,*})) - w_m \cdot (Q_m(q_m) - Q_m(q_{m,*})).$$

To see this directly, the internal gradient at contact i is $-2c_i n_i$. Its scalar product with a displacement into that disk is nonnegative. The convex gradient inequality for the polygonal length therefore gives the displayed support after minimizing the internal positions. Equality holds at the physical trajectory. The internal Hessian is positive definite, so the reduced roof is smooth on its regular neighborhood. Since $Q_i'' = -Rn_i$ and $w_i \cdot n_i = c_i$, differentiating the touching support proves the lower inequality in (55.1).

For the upper inequality keep the internal contacts fixed while varying the endpoints. The transverse second variation of a flight is its squared transverse displacement divided by its length. The two endpoint speeds have norm R. Their contribution is at most $2R^2|dq|^2/\tau_-$; this bound includes the case $m = 1$, when both displacements occur on the same flight. The endpoint curvature terms contribute at most $R|dq|^2$. The reduced Hessian is the Schur complement of the internal Hessian and cannot exceed this trial quadratic form. Taking the indicated extreme radii gives the upper bound, with $2R_+^2/\tau_- + R_+ = 47/6$.

In the five rectangles of Y_R^* one has $|p| < 9/10$. For the only moving momentum centers this follows from

$$\frac{2(53/100)}{1 + (53/100)^2} + \frac{1}{200} < \frac{9}{10};$$

the remaining centers and the added rectangle satisfy the bound immediately. Both the initial and terminal states belong to this same section. Thus $c_0, c_m > \sqrt{19}/10$, which proves (55.2). □

Away from $g = 0$ define the circle-valued physical observable

$$\vartheta(x) = \arg(-p_0(x), p_m(x)) \in \mathbb{R}/(2\pi\mathbb{Z}). \tag{55.3}$$

It is not the normal angle of an auxiliary endpoint cap. The orientation of each disk is the fixed orientation already used for the collision coordinates. Put $e = g/|g|$ and

$$\chi = e^{\mathsf{T}} \operatorname{cof}(K) e, \qquad v = K^{-1} g, \qquad s = g^{\mathsf{T}} K^{-1} g. \tag{55.4}$$

Here $\operatorname{cof}(K)$ is the adjugate matrix; K is symmetric. In dimension two its eigenvalues are those of K in reverse order. In particular $\lambda_\circ \le \chi \le \Lambda_\circ$ on the exact event.

Theorem 55.2 (An explicit critical-safe forward roof field)**.** *There is a uniquely specified smooth vector field X on the regular exact event with its interior critical points removed such that*

$$XL = 1, \qquad X\vartheta = 0. \tag{55.5}$$

In endpoint-angle coordinates it is

$$X_q = \frac{K^{-1}g}{s}. \tag{55.6}$$

In any local lift of ϑ on the original collision cylinder, it is equivalently

$$X = \frac{\vartheta_p \partial_\alpha - \vartheta_\alpha \partial_p}{L_\alpha \vartheta_p - L_p \vartheta_\alpha}.$$

No source density, guard, partition, or critical radius enters its choice. Stop its flow at the first exit from the regular exact event. For a forward survivor and $r_h = |g(\Phi_h x)|$, $r_0 = |g(x)|$, one has

$$2\lambda_\circ h \le r_h^2 - r_0^2 \le 2\Lambda_\circ h, \tag{55.7}$$

$$\operatorname{length}(q([0,h])) \le \frac{r_h - r_0}{\lambda_\circ} \le \frac{\sqrt{2\Lambda_\circ h}}{\lambda_\circ}, \tag{55.8}$$

$$|p_i(\Phi_h x) - p_i(x)| \le \frac{\sqrt{2\Lambda_\circ h}}{R_-}, \qquad i = 0, m. \tag{55.9}$$

The same estimates hold up to the limiting time of a finite lifetime. A forward trajectory cannot terminate at a regular interior critical point of L. The estimates do not assert survival through a true singular, first-hit, or section-decision boundary.

Proof. Writing $g^\perp = (-g_2, g_1)$ gives $d\vartheta = (g^\perp)^{\mathsf{T}} K\, dq/|g|^2$ and hence

$$\det D_q(F, \vartheta) = \frac{g^{\mathsf{T}} \operatorname{cof}(K) g}{|g|^2} = \chi > 0. \tag{55.10}$$

The endpoint map has Jacobian B_m, so $|\det D_{(\alpha,p)}(L, \vartheta)| = |B_m|\chi > 0$. The two equations (55.5) therefore specify a unique field. Formula (55.6) satisfies them: $g \cdot X_q = 1$ and $KX_q = g/s$ is radial in g. Endpoint lifts differ by translations by multiples of 2π; local lifts of ϑ also differ by constants. Both formulas therefore agree on overlaps and define the field without gluing.

Along the flow $\dot g = g/s$. Its direction is constant and $\dot r = r/s > 0$. Since $r^2/\Lambda_\circ \le s \le r^2/\lambda_\circ$, one has $2\lambda_\circ \le d(r^2)/dh \le 2\Lambda_\circ$. This proves (55.7). Using r as the parameter,

$$\frac{dq}{dr} = K^{-1} e.$$

Integration gives (55.8), including length rather than merely endpoint displacement. The direction of g is constant, so $|g(\Phi_h x) - g(x)| = r_h - r_0$ and the first-variation formula gives (55.9). In particular r_h cannot tend to zero in forward time. On a compact regular subset away from $g = 0$ the ordinary continuation theorem applies; any remaining finite exit is a true boundary of the regular exact event. □

Proposition 55.3 (The cancellation in the physical drift)**.** *Let $T = D^3F$ and let a be a locally Lipschitz density relative to dq. Away from $g = 0$, in the almost-everywhere sense,*

$$\operatorname{div}_q(aX_q) = \frac{\nabla a \cdot v}{s} + a\left\{\frac{T[v,v,v]}{s^2} - \frac{v \cdot \nabla \log\det K}{s}\right\}. \tag{55.11}$$

On a regular neighborhood where $\|T\| \le T_0$, this gives

$$|\operatorname{div}_q(aX_q)| \le \frac{1}{|g|}\left\{\frac{\Lambda_\circ}{\lambda_\circ}|\nabla a| + \left(\frac{\Lambda_\circ^2}{\lambda_\circ^3} + \frac{2\Lambda_\circ}{\lambda_\circ^2}\right)|a| T_0\right\}. \tag{55.12}$$

In particular the constant-density quadratic part has zero divergence. The bounds on T and ∇a in this assertion are local, not collision-uniform conclusions.

Proof. Differentiation of $v = K^{-1}g$ yields $\operatorname{div} v = 2 - v \cdot \nabla \log\det K$. Also $v \cdot \nabla s = 2s - T[v,v,v]$. The two terms $2/s$ cancel upon differentiating av/s, proving (55.11). Use $|v| \le |g|/\lambda_\circ$, $s \ge |g|^2/\Lambda_\circ$, and $|\nabla \log\det K| \le 2T_0/\lambda_\circ$ for the bound. Near a regular interior critical point the uniform positive Hessian makes $|g|$ comparable to distance, so the remaining displayed singularity is locally integrable in two dimensions. This last observation concerns a local integral only. □

56. Recoupling the complete physical source at fixed momentum angle

Use the unchanged $W = W^\varepsilon$ and one fixed exact label (m, R, n, k). Let Ω be its regular event, with the interior critical points removed. The removed sets have zero original area. Write μ_t for the coarea fiber of $d\alpha\, dp/(4\pi c_*)$ on this event. Thus $b^\varepsilon(t) = \int W\, d\mu_t$.

Theorem 56.1 (A complete, cap-free two-observable disintegration)**.** *The physical measure $(L,\vartheta)_\#(W\mathbf{1}_\Omega \nu_R^*)$ has a nonnegative density $A^\varepsilon(t,\theta)$ relative to $dt\, d\theta$ on $\mathbb{R}\times\mathbb{S}^1$. For almost every (t,θ),*

$$A^\varepsilon(t,\theta) = \frac{1}{4\pi c_*} \sum_{x\in\Omega:\ L(x)=t,\ \vartheta(x)=\theta} \frac{W(x)}{|B_m(x)|\chi(x)}. \tag{56.1}$$

The sum is over actual initial states, with every physical word and the exact occupation, displacement, initial and terminal tests retained. At fixed m its fibers are finite; no estimate for their cardinality as m grows is used. One has

$$b^\varepsilon(t) = \int_{\mathbb{S}^1} A^\varepsilon(t,\theta)\, d\theta, \tag{56.2}$$

$$\int_{\mathbb{R}}\int_{\mathbb{S}^1} A^\varepsilon(t,\theta)\, d\theta\, dt = \int_{\Pi_{n,k,m,R}} W\, d\nu_R^* \le 1. \tag{56.3}$$

These equalities do not normalize the retained source a second time.

Proof. By (55.10) the absolute Jacobian of the physical map $(\alpha,p)\mapsto(L,\vartheta)$ is $|B_m|\chi$. The area formula, on a finite circle atlas and a disjoint refinement of regular inverse charts, proves (56.1). Overlaps are removed at the level of physical initial states. The formula is consequently unchanged by relabeling a word chart or a cap. Tonelli and ordinary disintegration prove the two identities.

For finiteness of the fibers at this fixed count, the explicit finite physical graph retained in the supplement describes contacts, first-hit tests, section decisions, and endpoint momenta by finitely many semialgebraic conditions in rational angular charts. Add $g = r(\cos\theta, \sin\theta)$, $r > 0$, and $L = t$. The nonzero Jacobian makes each such fiber discrete. A definable discrete set is finite, uniformly in parameters for this fixed formula. This assertion is a fixed-count fact; it is not a pressure estimate for an increasing family of words. □

The inverse twist in this formula retains the optical bound

$$\frac{1}{|B_m|} \le \frac{Rc_0c_m}{\tau_-}\frac{\sinh\eta}{\sinh(m\eta)}\prod_{i\in S}\frac{53c_i}{6+47c_i}, \qquad \eta = \operatorname{arcosh}(53/47), \tag{56.4}$$

for any selected internal set S. For clarity, its denominator is obtained from $y_0 = 0$, $f_0 = 1$, $y_1 = \ell_0$ and $f_i = f_{i-1} + 2y_i/(Rc_i)$, $y_{i+1} = y_i + \ell_i f_i$. The corner Schur complement gives $|B_m|^{-1} = Rc_0c_m/y_m$. Lowering lengths to τ_- and potentials to $2/R_+$ gives the hyperbolic-sine solution. Restoring selected potentials one at a time gives the factors in (56.4) by their positive scalar Schur complements. The complete proof is retained in the supplement. We use its per-word estimate, not an unproved exponential word-count bound. Identity (56.3) is an actual complete weighted summability identity, but only in integrated form.

Disintegrate the weighted level source also over θ:

$$W\mu_t = \int_{\mathbb{S}^1} A^\varepsilon(t,\theta)\,\rho_{t,\theta}\,d\theta, \tag{56.5}$$

where $\rho_{t,\theta}$ is a probability on physical initial states with this roof, angle, and exact label whenever its coefficient is positive. In the finite regular fiber it assigns weights proportional to the summands of (56.1). Its value where the coefficient is zero is immaterial.

Theorem 56.2 (Maximal recoupling with the physical angle retained)**.** *For almost every (t,h) with $h>0$, the measure*

$$\Gamma_{t,h} = \int_{\mathbb{S}^1} \min\{A^\varepsilon(t,\theta), A^\varepsilon(t+h,\theta)\}(\rho_{t,\theta}\otimes\rho_{t+h,\theta})\,d\theta \tag{56.6}$$

is a subcoupling of the complete physical weighted fibers at t and $t+h$. Among all subcouplings which preserve ϑ, its matched mass is maximal. Its unmatched initial mass is exactly

$$\Delta_h^\varepsilon(t) = \int_{\mathbb{S}^1} [A^\varepsilon(t,\theta) - A^\varepsilon(t+h,\theta)]_+\,d\theta. \tag{56.7}$$

It follows that

$$[b^\varepsilon(t) - b^\varepsilon(t+h)]_+ \le \Delta_h^\varepsilon(t) \le R^+_{X,h}(t), \tag{56.8}$$

where $R^+_{X,h}$ is the unmatched mass of the stopped physical-flow subcoupling for the explicit field in Theorem 55.2.

Proof. Each marginal of (56.6) is dominated by the corresponding measure in (56.5). For any other angle-preserving subcoupling, push its matched measure to the common angle. Its density is bounded above by both $A^\varepsilon(t,\theta)$ and $A^\varepsilon(t+h,\theta)$, and therefore by their minimum. Formula (56.6) attains this upper bound. Subtracting its mass from the source mass gives (56.7), and target domination proves the first inequality in (56.8).

The stopped-flow graph for X preserves ϑ by (55.5). On its surviving part its source weight is the minimum of the source density and the transported target density, with the physical coarea Jacobian. The usual flow change of variables gives domination of both physical marginals. It is therefore one admissible competitor in the preceding maximality argument. Its unmatched mass is $R^+_{X,h}$, which proves the last inequality. Joint disintegration supplies almost-everywhere versions in (t,h); no pointwise selection on an exceptional roof is required. □

The optimization in this theorem is not over unrestricted scalar couplings. It retains the explicitly specified physical angle. It may pair two different regular words, but only when both are actual states of the same original exact event at their respective roofs. It does not splice a nonphysical orbit or pool different labels. The full sum in (56.1) is formed before its positive difference is taken. Thus a true word exit may be balanced by actual incoming mass at the same angle and label; any remaining loss is still charged. Such balancing was unavailable to a stopped single-orbit graph. For example, two nonnegative angular profiles supported on successive roof intervals can have a continuous sum although the first profile has an exit. This is an algebraic illustration of the distinction, not a claim about a particular billiard trajectory. No comparison with every earlier scalar short-excursion modulus is asserted.

Proposition 56.3 (The angular residue at an interior critical roof)**.** *Let x_* be a regular interior critical point with a neighborhood in the exact event, $F_* = F(x_*)$, and $K_* = D^2F(x_*)$. Write $a_* = W(x_*)/(4\pi c_*|B_m(x_*)|)$. Its local angular source has right-minus-left jump*

$$j_*(\theta) = \frac{a_*}{e_\theta^{\mathsf{T}}\operatorname{cof}(K_*)e_\theta} \ge 0, \qquad \int_{\mathbb{S}^1} j_*(\theta)\,d\theta = \frac{W(x_*)}{2c_*|B_m(x_*)|\sqrt{\det K_*}}. \tag{56.9}$$

Every direction is an entrance, not a forward exit. This is a local charge statement; no sign is assigned to other physical boundaries at the same roof.

Proof. Translate q_* to zero. The inverse function theorem for $g = \nabla F$ gives a smooth inverse $q(re_\theta)$ for small r. Along it $dF/dr = re_\theta^{\mathsf{T}} K^{-1} e_\theta > 0$. Thus, for each angle, the local branch starts at F_* from above. The density in (56.1) tends to $a_*/(e_\theta^{\mathsf{T}} \operatorname{cof}(K_*) e_\theta)$. The convergence is uniform in angle on this fixed neighborhood. Diagonalizing the positive matrix K_* and integrating $(a\cos^2\theta + b\sin^2\theta)^{-1}$ gives $2\pi/\sqrt{ab}$, which proves the formula. The neighborhood is used only to calculate the residue. The globally defined field and its uniform motion bounds use no such radius. □

57. The signed angular current and the pointwise endpoint

The angular source admits a finite signed roof derivative at each fixed collision count. The differentiation is performed after the complete exact-label sum, not separately with absolute values on individual words.

Theorem 57.1 (Finite variation before projection to the scalar roof)**.** *For each fixed $m \geq 2$ there is $C_m < \infty$, independent of R, ε, n, k, such that for almost every θ,*

$$A^\varepsilon(\,\cdot\,,\theta) \in BV(\mathbb{R}), \qquad \|D_t A^\varepsilon(\,\cdot\,,\theta)\|_{\mathrm{TV}} \leq C_m. \tag{57.1}$$

Consequently $\mathscr{J}^\varepsilon = D_t A^\varepsilon$ is a finite signed measure on $\mathbb{R}\times\mathbb{S}^1$. It and its Jordan parts disintegrate over $d\theta$ as the corresponding one-dimensional roof derivatives and Jordan parts. Its projection to the roof is Db^ε. No estimate for $m^2 C_m$ is asserted.

Proof. We give the finite-count details because the original smooth guard need not be definable. On the finite physical graph, enumerate the $N = 2m+1$ normalized incidence and clearance margins as a_i. For $y \in [1,2]^N$ let $W_y = 1 - \prod_{i=1}^N \mathbf{1}_{\{a_i > y_i\}}$ and let A_y be its angular density. By the nonzero Jacobian and the explicit finite physical graph, the inverse fibers in (56.1) are definable discrete sets. Uniform finiteness gives a bound N_m for their size, uniform in (R, n, k, t, e_θ) at this fixed m. The maps B_m, K, and χ on a regular sheet are definable: their derivatives can be taken on the rational angular charts of the physical graph, followed by the rational change from chart derivatives to angular derivatives. Finite definable sums and threshold tests show that A_y has a definable representative.

The empty-set optical bound and $\chi \geq \lambda_\circ$ give

$$0 \leq A_y(t,\theta) \leq H_m := \frac{N_m R_+}{4\pi c_* \tau_- \lambda_\circ} \frac{\sinh\eta}{\sinh(m\eta)}.$$

Take the value zero on exceptional graph cells. On a fixed finite atlas of the angle circle, the uniform monotonicity theorem for definable families [9] supplies $J_m < \infty$ monotonicity intervals and point cells in the roof variable, with all remaining variables treated as parameters. Extension by zero outside $[m\tau_-, m\tau_+]$ therefore gives $\|D_t A_y(\,\cdot\,,\theta)\|_{\mathrm{TV}} \leq 4(J_m+1)H_m$. Both N_m and J_m are constants of a fixed finite formula, not controlled word-pressure constants.

Put $\sigma = d\Theta$ and $V_\Theta = \|\sigma\|_{\mathrm{TV}}$. The exact identity

$$W^\varepsilon = \int_{[1,2]^N} W_y \, d\sigma^{\otimes N}(y)$$

uses $\sigma(\mathbb{R}) = 1$ and is valid even when Θ is not monotone. Coarea and finite signed Fubini give $A^\varepsilon = \int A_y \, d\sigma^{\otimes N}$ almost everywhere. The distributional characterization of BV [13] proves (57.1) with $C_m = 4(J_m+1)H_m V_\Theta^N$.

Integrating the derivative kernels over the circle defines a finite signed measure of variation at most $2\pi C_m$. Measurability follows first for smooth test functions from Fubini, and then for

the variation and the Jordan kernels by the countable test-function characterization of variation. This identifies the measure with $D_t A^\varepsilon$ and its Jordan disintegrations. Testing with a function independent of θ and using (56.2) proves the projection identity. Jordan decomposition need not commute with this final projection. □

57.1. **The physical meaning of this current.** The finite-current argument just used is an application of the inherited threshold method to the two-observable density. It is not a new proof of collision-uniform weighted summability. Its geometric computation can nevertheless be made entirely on actual regular states. On a compact regular endpoint domain D write $a = W/(4\pi c_*|B_m|)$ and $Y = X_q$. For a smooth test $\psi(t,\theta)$, since $YF = 1$ and $Y\vartheta = 0$, the divergence theorem gives

$$D_t(F,\vartheta)_\#(a\mathbf{1}_D\,dq) = (F,\vartheta)_\#\left(\operatorname{div}(aY)\,dq|_D - (aY\cdot n)\,d\mathcal{H}^1|_{\partial D}\right). \tag{57.2}$$

On the left, the derivative is that of the corresponding density. For a locally Lipschitz weight the same identity follows by smooth approximation on the compact domain. The bulk term is explicitly (55.11). At an artificial common boundary with the same physical source, the two oriented traces cancel. A true source jump leaves its jump flux. A limiting interior critical boundary has the positive residue (56.9).

Regular exhaustions converge to $\mathscr{J}^\varepsilon$ in distributions: the omitted source mass tends to zero, and its action is bounded by that mass times $\|\partial_t\psi\|_\infty$. No convergence of the separately absolute-valued bulk and boundary terms is asserted. The full finite measure is secured by Theorem 57.1, not by an unproved uniform bound on an exhaustion. True singularities, first-hit switches, and section decisions thus remain in the signed physical current. Only actual cancellation at the same angle and exact label is credited before taking its negative part.

Set

$$\Delta_r^\varepsilon(t) = \operatorname*{ess\,sup}_{0<h<r} \Delta_h^\varepsilon(t), \tag{57.3}$$

$$V_r^\varepsilon(t) = (\mathscr{J}^\varepsilon)^-([t,t+r]\times\mathbb{S}^1). \tag{57.4}$$

The closed interval is harmless for the essential bounds and retains all possible boundary atoms. All translated roofs remain on the full roof line, not on a truncated central target.

Theorem 57.2 (Complete angular persistence)**.** *For every fixed $r>0$ and almost every t,*

$$[b^\varepsilon(t) - 2A_r b^\varepsilon(t)]_+ \le \Delta_r^\varepsilon(t) \le V_r^\varepsilon(t). \tag{57.5}$$

The defect on the left comparison is no larger than the maximal forward stopped-flow defect for the explicit field of Theorem 55.2. Positive interior critical entrances are not charged as forward losses.

Proof. For a BV function f, $[f(t)-f(t+h)]_+ \le (Df)^-([t,t+h])$ for almost every (t,h). Apply this to $A^\varepsilon(\,\cdot\,,\theta)$ and integrate in θ. The Jordan disintegration proves $\Delta_h^\varepsilon(t) \le V_r^\varepsilon(t)$ when $0<h<r$, and therefore the second inequality. The first follows by averaging $b^\varepsilon(t) \le b^\varepsilon(t+h) + \Delta_r^\varepsilon(t)$ over $0<h<r$. Nonnegativity bounds this forward average by $2A_r b^\varepsilon(t)$. The stopped-flow comparison follows from (56.8). The critical assertion follows from its positive angular residue; unrelated true boundary terms are still present in the signed sum. □

Define the ordered complete angular defect

$$\mathcal{A}_{\varepsilon,M}(r) = \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \Delta_r^\varepsilon(t). \tag{57.6}$$

There is no supremum over selected physical words in this definition; their complete weighted sum has already been taken.

Theorem 57.3 (Ordered implication for the unchanged pointwise law)**.** *Assume Input statement 54.1. For each sufficiently large fixed B and fixed $r > 0$,*

$$\mathcal{E}_M \le C'_M B^{-1/192} + \mathcal{A}_{\varepsilon(B),M}(r). \tag{57.7}$$

In particular the original target (54.5) *follows if, for some $A_M < \infty$, $q \ge 0$, $\beta > 0$ and fixed $\varepsilon_0, r_0 > 0$, the complete source satisfies*

$$\mathcal{A}_{\varepsilon,M}(r) \le A_M \varepsilon^{-q} r^{\beta} \quad (0 < \varepsilon < \varepsilon_0,\ 0 < r < r_0). \tag{57.8}$$

The same bound on the ordered norm of V_r^{ε} is sufficient. Estimate (57.8) *is an additional dynamical estimate; it is not proved by the fixed-count variation theorem or the integrated identity* (56.3)*.*

Proof. Use the positive remainder in (54.7) and (57.5) to bound the pointwise error by $2m^2 A_r b^{\varepsilon(B)} + m^2 \Delta_r^{\varepsilon(B)} + |e_B|$. Fix B, $\varepsilon(B)$, r, and each auxiliary spectral band before the collision limsup. The fixed-width complete-source bound then gives

$$\mathcal{E}_M \le 2C\varepsilon(B)^{1/16} + \mathcal{A}_{\varepsilon(B),M}(r) + C_M B^{-1/192}.$$

Since $\varepsilon(B) = A_0 B^{-1/12}$, this proves (57.7). Under (57.8), take $r(B) = c\varepsilon(B)^{(q+a)/\beta}$ with $a > 0$ and a fixed $c > 0$ small enough that $r(B) < r_0$. This is fixed before the collision limit. Enlarge the auxiliary spectral band only afterwards and let $B \to \infty$ last. The two errors tend to zero. □

57.2. **The same scalar source in path normalization.** The retained path input uses a common finite positive path remainder $\mathsf{B}_B(t)$ of mass $m^2 b^{\varepsilon(B)}(t)$, and

$$\mathsf{P}_m(t) - \mathsf{L}_m(t) = \mathsf{B}_B(t) + \mathsf{e}_B(t), \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} \|\mathsf{e}_B(t)\|_{\mathrm{BL}^*} \le C_M B^{-1/192}.$$

The bounded-Lipschitz class contains the constant one, so the norm of the positive remainder is its mass. Under (57.8), the preceding proof makes this mass vanish in the same ordered limits. Path-numerator convergence then follows. Ratios may be formed only on central arithmetic classes with scalar reference bounded below by a fixed positive constant. No arithmetic zero class is divided by. Collision-to-return clock transfer and forward likelihood retain their own inherited hypotheses. None of these conclusions is used to establish the scalar estimate.

58. The remaining complete-source estimate

The principal assertion remains (54.5), with its full arithmetic transition and all zero classes. The new uniform bounds concern the actual endpoint Hessian and the forward motion of an explicit physical field. They remove arbitrary field choice and Morse-radius patching from that construction. The exact angular sum and its maximal recoupling also avoid charging a separate loss when another actual word supplies matching mass at the same angle and label. True boundaries are neither declared null in flux nor removed from the original guard.

What is still needed is (57.8), or a direct estimate of comparable strength for the original complete source. Neither $\int A^{\varepsilon} \le 1$ nor the finite constants N_m, J_m, C_m control concentration at one roof after multiplication by m^2. The explicit drift formula also involves third derivatives and source derivatives for which no collision-uniform summed bound has been obtained here. Thus the organizing pointwise theorem and its unconditional path and likelihood consequences are not claimed as proved. All earlier arguments, including their exact hypotheses, are retained in the technical supplement; the target is not replaced by a smoothed law or a different record.

Part 28. Stopped physical transport and optical persistence

Retained revision-76 abstract. We retain the pointwise local limit problem for the full displacement, collision count, occupation, and roof record of actual returns in a triangular Lorentz gas. We replace matching of coordinate-dependent Gauss traces by a stopped transport of the original physical level measures. The resulting identity charges only genuine exits and relative source distortion; changes of endpoint caps make no contribution. A positive transfer-matrix comparison bounds the inverse twist of every regular word by an explicit hyperbolic-sine factor, while retaining simultaneous incidence gains. At an interior critical roof, a Morse-adapted transport has zero planar Jacobian distortion and a square-root source defect, without charging the positive jump. These results give a complete-source persistence inequality and a self-contained implication to the original arithmetic pointwise law. The collision-normalized sum of the true exit and distortion defects is the additional estimate required for that implication; it is not inferred from the single-word comparison or from fixed-count collars.

59. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{59.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{59.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{59.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

59.1. **The arithmetic reference and the target.** For $y=(k_1,k_2,t,n)$ put

$$\mu_R=(0,0,\bar\tau_R,c_*),\quad Z=(y-m\mu_R)/\sqrt{m},\quad \mathcal{C}_{m,R,n,k}(M)=\{t:|Z|\le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1,\kappa_2,\tau,\mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j=\lambda_j(R,a_j),\quad \mu_j=i^{-1}\nabla\log\lambda_j(R,a_j),\quad H_j=-\nabla^2\log\lambda_j(R,a_j),$$

with $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. If $\eta_R=\mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j=\chi_j(R,a_j)\ell(M_{\eta_R}\mathsf{P}_j(R,a_j)(\eta_R\nu))$. Define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf{T}}Hv/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\left[c_*^{-1}\sum_j e^{-ia_j\cdot y}z_j^mB_jG_{H_j}\left((y-m\mu_j)/\sqrt{m}\right)\right].\tag{59.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\,1\le n\le m,\,k\in\mathbb{Z}^2}\ \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)}|m^2p_{n,R}(k,m,t)-\mathcal{L}_{m,R}(k_1,k_2,t,n)|\longrightarrow 0.\tag{59.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R\operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

59.2. **The analytic inputs in the form used here.**

Input statement 59.1. The unchanged source (59.3) has a nonnegative, compactly supported BV density at every fixed count. For every fixed $\varepsilon,\delta>0$, with $A_\delta f(t)=(2\delta)^{-1}\int_{t-\delta}^{t+\delta}f(s)\,\mathrm{d}s$,

$$\limsup_m\sup_{R,n,k}m^2\|A_\delta b^\varepsilon\|_\infty\le C\varepsilon^{1/16}.\tag{59.6}$$

The constant is independent of the previously fixed width. For large fixed B and $\varepsilon(B)=A_0B^{-1/12}$,

$$m^2p-\mathcal{L}=m^2b^{\varepsilon(B)}+e_B,\qquad \limsup_m\sup_{R,n,k}\operatorname*{ess\,sup}_{\mathcal{C}(M)}|e_B|\le C_MB^{-1/192}.\tag{59.7}$$

All auxiliary spectral bands in these statements are fixed before the collision limit and enlarged only afterwards.

These are the complete-source tame theorem, the fixed-width complete-source bound, and the positive scalar remainder theorem in the technical supplement. Their proofs and historical hypotheses are retained there. The fixed-count source uses threshold integration [15], definable monotonicity [9], and the distributional characterization of BV [13]; it does not assume definability of the original smooth guard. They are restated as inputs to distinguish them from the new geometric and transport proofs. No long-time bound on a fixed-count variation constant is part of this statement.

For context, the preceding geometric theorem supplies at most 256 endpoint-cap rectangles for each physical word, with one strongly convex supporting action per rectangle. On every regular physical piece its Hessian lies between $(50/47)I_2$ and $(1360/9)I_2$. All its level arcs together have coarea integral at most $2\pi/(50/47)$. Its positive Gauss formula is exact, but comparing the

numerical normal angle at different roofs depends on the chosen caps. The construction below instead takes place on the original physical collision manifold. Its dependence on the chosen smooth roof vector field is explicit; it has no dependence on a cap partition used merely to calculate its measures.

60. Stopped transport of the physical level measure

Fix m, R, n, k. Remove the finite-time singularities, the preimages of section boundaries, and the critical points of L from the regular exact event. Its open part is denoted by Ω. These removals have zero collision-source measure. At almost every roof they have zero source in the coarea disintegration as well. For critical points this follows, for example, from the finite endpoint-cap cover and strict convexity: each word has finitely many such points. The boundaries are not suppressed in a transport; a trajectory reaching them is stopped.

Use the original cylinder area $\mathrm{d}\bar{\nu}_R = \mathrm{d}\alpha\ \mathrm{d}p/(4\pi c_*)$, restricted to Ω. It agrees with ν_R^* there. Give the cylinder its flat metric. On $\{L = t\} \cap \Omega$ write

$$\mathrm{d}\mu_t(x) = \frac{1}{4\pi c_* |\nabla L(x)|}\, \mathrm{d}\mathcal{H}^1(x), \qquad b_W(t) = \int W\, \mathrm{d}\mu_t. \tag{60.1}$$

We allow $0 \le W \le 1$ locally Lipschitz on Ω; the complete weight W^ε has this property. The least clearance is a minimum of finitely many smooth distance branches locally, and therefore is locally Lipschitz even at a switch of minimizing disk. The unweighted fibers are finite for almost every t by coarea and finiteness of the restricted cylinder area.

Choose a smooth vector field X on Ω such that $XL = 1$. The normalized gradient is one choice. Other choices, including the Morse-adapted choice below, are allowed. For a fixed geometric field, coordinate changes do not change the construction. Let Φ_h be its maximal flow, killed at its lifetime in Ω. No continuation through an excluded physical boundary is made. For $\sigma \in \{-1, 1\}$ and $h > 0$ put

$$S_h^\sigma(t) = \{x \in \Omega : L(x) = t,\ \Phi_{\sigma s} x \in \Omega \text{ and exists for every } 0 \le s \le h\}.$$

On this set $L(\Phi_{\sigma s} x) = t + \sigma s$. Define

$$J_h^\sigma(x) = \exp\left(\int_0^h \sigma \operatorname{div}_{\bar{\nu}_R} X(\Phi_{\sigma s} x)\, \mathrm{d}s \right). \tag{60.2}$$

This is an area Jacobian relative to the original source, not a Jacobian for a cap label.

Lemma 60.1 (Physical coarea covariance)**.** *For every nonnegative Borel a and almost every pair (t, h),*

$$\int_{\Phi_{\sigma h} S_h^\sigma(t)} a(y)\, \mathrm{d}\mu_{t+\sigma h}(y) = \int_{S_h^\sigma(t)} a(\Phi_{\sigma h} x) J_h^\sigma(x)\, \mathrm{d}\mu_t(x). \tag{60.3}$$

The formula includes the entire exact event and its original factor c_*^{-1}.

Proof. Work first on a relatively compact regular flow box. The ordinary change of variables for the smooth flow gives $\Phi_{\sigma h}^* \bar{\nu}_R = J_h^\sigma \bar{\nu}_R$ by the Liouville formula. Disintegrate both sides using L; its translation has derivative one. This gives (60.3), including the coarea speed factor. Equivalently, contracting the oriented area form with X gives the level form, and the flow preserves X. Uniqueness of solutions makes the flow injective wherever it exists. A countable exhaustion by flow boxes, a disjoint measurable refinement, and monotone convergence give the formula on the surviving domain. Fubini supplies common almost-everywhere versions in (t, h). □

The actual source mass lost under the resulting subcoupling is

$$E_h^\sigma(t) = \int_{\{L=t\}\cap\Omega\setminus S_h^\sigma(t)} W \,\mathrm{d}\mu_t,$$
$$D_h^\sigma(t) = \int_{S_h^\sigma(t)} [W(x) - W(\Phi_{\sigma h}x)J_h^\sigma(x)]_+ \,\mathrm{d}\mu_t(x),$$
$$R_h^\sigma(t) = E_h^\sigma(t) + D_h^\sigma(t). \tag{60.4}$$

Here D denotes a transport defect, not differentiation or a word-alphabet size. All physical words with this one exact label are included in each integral.

Theorem 60.2 (A label-preserving subcoupling without cap losses)**.** *The measure on the graph $x \mapsto \Phi_{\sigma h}x$ with source marginal*

$$\mathbf{1}_{S_h^\sigma(t)} \min\{W(x), W(\Phi_{\sigma h}x)J_h^\sigma(x)\} \,\mathrm{d}\mu_t(x)$$

is dominated in its two marginals by the physical weighted level measures at t and $t+\sigma h$. Its unmatched initial mass is exactly $R_h^\sigma(t)$. Consequently

$$[b_W(t) - b_W(t+\sigma h)]_+ \le R_h^\sigma(t) \qquad \text{for almost every } (t,h). \tag{60.5}$$

Refining, changing, or removing an auxiliary endpoint-cap partition changes neither this coupling nor its defect. The coupling never changes an exact displacement, occupation, terminal test, return index, or collision count.

Proof. The first marginal domination is immediate. For the second, divide by the positive Jacobian on the surviving graph and use Lemma 60.1; the transported density is at most W at its image. The image is a subset of the target exact event, not necessarily the whole target. Subtract the coupled mass from $b_W(t)$. On the nonsurvivors the result is E_h^σ; on the survivors it is D_h^σ. Target marginal domination gives (60.5). This proves the assertion without dividing by W, which may vanish.

For an auxiliary measurable partition, disintegrate the very same source measure on its pieces and then sum. A change of chart changes both the displayed density and its Jacobian by inverse coordinate factors; the measure and the graph are unchanged. In particular the image of a point is assigned its target chart, not killed when it ceases to have its starting chart. Thus no zero extension at a first-admissible cap boundary enters (60.4). In contrast, reaching a true boundary of Ω does enter E_h^σ. All retained labels are locally constant in Ω and persist throughout the stopped orbit. Different words cannot produce multiplicity: the underlying physical initial states, and their flow images by uniqueness, are disjoint. □

60.1. **An exact source for the distortion loss.** Weak derivatives are sufficient for W. Define the nonnegative locally integrable function

$$g_\sigma = [-\sigma(XW + W \operatorname{div}_{\bar\nu_R} X)]_+, \qquad G_\sigma(u) = \int g_\sigma \,\mathrm{d}\mu_u. \tag{60.6}$$

The latter may have infinite integral on a large roof interval; no global finiteness is being assumed.

Proposition 60.3 (Survivor distortion as a physical drift integral)**.** *For almost every (t,h),*

$$D_h^\sigma(t) \le \int_0^h G_\sigma(t+\sigma s) \,\mathrm{d}s. \tag{60.7}$$

This is an extended nonnegative inequality. On compact regular flow boxes both sides are finite.

Proof. Along an almost-everywhere flow line the locally Lipschitz function W obeys the chain rule. Differentiating its transported density gives

$$\frac{\mathrm{d}}{\mathrm{d}s}\big(W(\Phi_{\sigma s}x)J_s^\sigma(x)\big) = \sigma(XW + W\,\mathrm{div}_{\bar\nu_R}X)(\Phi_{\sigma s}x)J_s^\sigma(x).$$

The positive part of the net decrease is bounded by the integral of the negative derivative. Integrate over $S_h^\sigma(t)$, use $S_h^\sigma(t) \subset S_s^\sigma(t)$ for $s \le h$, and apply (60.3). The image at time s is a subset of the full fiber at $t + \sigma s$, so positivity gives the bound by G_σ. Tonelli and compact exhaustions prove the extended statement. No cardinality estimate for the word set is used. □

61. Optical weights and transport at a critical roof

The next estimate is on the actual regular word, without a lower incidence cutoff. It improves an exponential cardinality majorant only at the level of its optical weights; it does not estimate the complete word sum.

For a regular word of m flights let u, v be oriented endpoint arclengths, F its reduced length, $c_i > 0$ its incidences, and $\ell_i \ge \tau_-$ its flight lengths. Put $t_i = \ell_i^{-1}$ and $V_i = 2/(Rc_i)$ for $1 \le i < m$. The internal optical matrix is

$$A_{ii} = t_{i-1} + t_i + V_i, \qquad A_{i,i+1} = A_{i+1,i} = -t_i. \tag{61.1}$$

Signs of internal tangents can be chosen successively to give this form. The full internal action Hessian is D_cAD_c, with $D_c = \mathrm{diag}(c_1, \dots, c_{m-1})$.

Lemma 61.1 (Transfer denominator and monotonicity)**.** *Define $y_0 = 0$, $f_0 = 1$, $y_1 = \ell_0$, and, for $1 \le i < m$,*

$$f_i = f_{i-1} + V_i y_i, \qquad y_{i+1} = y_i + \ell_i f_i. \tag{61.2}$$

Then

$$y_m = \left(\prod_{i=0}^{m-1} \ell_i\right) \det A, \qquad |F_{uv}| = \frac{c_0 c_m}{y_m}. \tag{61.3}$$

The empty determinant is one. The positive polynomial y_m is nondecreasing in every flight length and every internal potential.

Proof. Eliminating f_i gives

$$y_{i+1} = (1 + \ell_i/\ell_{i-1} + \ell_i V_i)y_i - (\ell_i/\ell_{i-1})y_{i-1}.$$

This is exactly the continuant recursion for $(\prod_{j=0}^{i} \ell_j) \det A_{[1,i]}$, with the same initial values. It proves the first identity, including $m = 1$. The mixed endpoint derivative is the corner Schur complement of the contact Hessian. Its numerator is the product of neighboring mixed entries, and the internal factors c_i^2 cancel from its determinant. Thus $|F_{uv}| = c_0 c_m \prod t_i / \det A$, proving the second identity. The positive recursion (61.2), rather than its subtractive second-order form, proves the monotonicity and the positive-polynomial assertion by induction. □

Put

$$\eta = \operatorname{arcosh}(53/47), \qquad \rho = e^{-\eta} = \frac{53 - 10\sqrt{6}}{47} < 1, \qquad Q(c) = \frac{53c}{6 + 47c}. \tag{61.4}$$

The guard parameter $q_0 = 47/53$ has not been changed to ρ.

Theorem 61.2 (Explicit inverse twist with simultaneous incidence gains)**.** *Let $B_m = \partial_{p_0}\alpha_m$. For every subset $S \subset \{1, \dots, m-1\}$,*

$$\frac{1}{|B_m|} = R|F_{uv}| \le \frac{Rc_0c_m}{\tau_-} \frac{\sinh\eta}{\sinh(m\eta)} \prod_{i \in S} Q(c_i). \tag{61.5}$$

All constants are independent of the physical word, its length, positive incidences, and positive clearances. In particular each bounded-source Gauss summand of the preceding geometric theorem is bounded by a fixed constant times

$$W c_0 c_m \frac{\sinh \eta}{\sinh(m\eta)} \prod_{i \in S} Q(c_i),$$

with its exact event indicator still present.

Proof. First take S empty. Monotonicity in Lemma 61.1 allows all ℓ_i to be replaced by τ_- and all V_i by $2/R_+$. The resulting recurrence is $y_{i+1} = 2(1 + \tau_-/R_+)y_i - y_{i-1}$, with $y_0 = 0$ and $y_1 = \tau_-$. Since $1 + \tau_-/R_+ = 53/47$, its solution is

$$y_m^{\min} = \tau_- \frac{\sinh(m\eta)}{\sinh \eta}.$$

The inverse-coordinate Jacobian and $u = R\alpha_0$, $v = R\alpha_m$ give $|B_m|^{-1} = R|F_{uv}|$. This proves the empty-set estimate.

For general S, let $A^{(S)}$ replace c_i by one in the selected diagonal potentials only. Restore the selected incidences one at a time. Before restoring c_i, its scalar Schur complement s_i is positive and at most $2/\tau_- + 2/R$. The determinant ratio for that step is

$$\frac{s_i}{s_i + (2/R)(c_i^{-1} - 1)} \le \frac{(1 + R/\tau_-)c_i}{1 + (R/\tau_-)c_i} \le Q(c_i).$$

Each intermediate matrix is positive definite, so multiplication gives $\det A^{(S)} / \det A \le \prod_{i \in S} Q(c_i)$. The empty-set transfer bound also applies to $A^{(S)}$, since all its potentials remain at least $2/R_+$. This proves (61.5). Finally the preceding Gauss formula bounds its summand by $C_G W \mathbf{1}_\Pi \mathbf{1}_a / |B_m|$, where $C_G = 2R_+/(\pi c_* R_-^2)$. Substitution gives the last assertion. No comparison matrix has been assigned a physical orbit or source. □

61.1. **Interior critical points and the direction of the jump.** A regular interior critical point of the roof has $p_0 = p_m = 0$ by the first variation. Let $K = D^2_{(\alpha_0, \alpha_m)} F$ there. At that point

$$K \ge R I_2. \tag{61.6}$$

Indeed the affine support supplied by its own physical trajectory restricts to the two endpoint circles as $F_* + R(1 - \cos(\alpha_0 - \alpha_{0,*})) + R(1 - \cos(\alpha_m - \alpha_{m,*}))$. It touches the physical roof from below. Differentiating twice at the contact proves (61.6). Internal strict positivity gives smoothness on a regular physical neighborhood.

Proposition 61.3 (An intrinsic positive critical charge)**.** *At a regular interior critical point x_* whose neighborhood stays in the exact event, the local density for a continuous locally Lipschitz nonnegative weight W has right-minus-left jump*

$$J_*^W = \frac{W(x_*)}{2c_* |B_m(x_*)| \sqrt{\det K}} \ge 0. \tag{61.7}$$

For every selected internal set S it satisfies

$$J_*^W \le \frac{W(x_*)}{2c_* \tau_-} \frac{\sinh \eta}{\sinh(m\eta)} \prod_{i \in S} Q(c_i(x_*)). \tag{61.8}$$

These are local charges. Other physical or decision-boundary contributions at the same roof are not asserted to have this sign.

Proof. In endpoint angles the source density is $a_\alpha = W/(4\pi c_*|B_m|)$. Smooth Morse coordinates $z \in \mathbb{R}^2$ put $F = F_* + |z|^2/2$; their inverse Jacobian at zero is $(\det K)^{-1/2}$. Coarea in polar coordinates shows that the right limiting density is $2\pi a_\alpha(x_*)/\sqrt{\det K}$, while the local source below F_* is zero. A cutoff equal to one near x_* does not change this jump. This proves (61.7). Since $\sqrt{\det K} \geq R$, $c_0 = c_m = 1$, and (61.5) applies, the stated bound follows. The assertion concerns the local nonnegative source, not the Jordan decomposition of unrelated terms at the same roof. □

Proposition 61.4 (Morse transport does not charge a positive jump)**.** *Use the preceding Morse coordinates on a disk $|z| < r_*$, and write the full weighted physical area there as $a(z)\,\mathrm{d}z$, where a is nonnegative and Lipschitz with constant L_a. The forward roof flow*

$$z \longmapsto z_h = \sqrt{1 + 2h/|z|^2}\, z \tag{61.9}$$

has Euclidean Jacobian one and increases F by exactly h. If $0 < t - F_* < r_*^2/8$ and $0 < h < r_*^2/8$, it stays in this physical disk and its unmatched initial mass satisfies

$$R_h^+(t) \leq 2\pi L_a \sqrt{2h}. \tag{61.10}$$

Here the level source is the local disk source, so no exterior trajectory is included. In particular the coefficient J_*^W itself is not an exit or distortion charge in this direction.

Proof. Put $r = |z|$ and $r_h = \sqrt{r^2 + 2h}$. The radial and tangential derivative factors are r/r_h and r_h/r, respectively, and their product is one. The full circle at the starting roof and its image lie in the disk under the stated restrictions. Polar coarea for $|z|^2/2$ is $\mathrm{d}\theta$. The unmatched mass is consequently

$$\int_0^{2\pi} [a(re_\theta) - a(r_h e_\theta)]_+ \,\mathrm{d}\theta \leq 2\pi L_a (r_h - r) \leq 2\pi L_a \sqrt{2h}.$$

The constant part of a makes zero contribution, including its positive jump at F_*. Push this vector field back to the original collision manifold. It satisfies $XL = 1$ away from the critical point, so Theorem 60.2 applies. A smooth partition of unity glues this field to the normalized gradient outside a smaller disk, since a convex combination of fields satisfying $XL = 1$ still satisfies that identity. There are finitely many interior critical points at fixed count, and disjoint neighborhoods can be used. Neither their radii nor the sum of their Lipschitz constants is asserted uniform in the collision count. □

62. Complete-source persistence and the original endpoint

Apply the physical construction to $W = W^\varepsilon$ without altering the guard. For a fixed choice of the physical roof field at each exact event, define

$$Q_r^\sigma(t) = \operatorname*{ess\,sup}_{0<h<r} R_h^\sigma(t), \qquad Q_r(t) = \min\{Q_r^-(t), Q_r^+(t)\}, \tag{62.1}$$

$$\mathcal{Q}_{\varepsilon,M}(r) = \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} Q_r(t). \tag{62.2}$$

The fibers at translated roofs are on the full roof line. They are not cut off at the central target. All word and exact-source contributions are included before any supremum or collision limit. The field is part of the specified transport, but an auxiliary cap partition is not.

Theorem 62.1 (Persistence of the complete positive source)**.** *For almost every t and every fixed $r > 0$,*

$$[b^\varepsilon(t) - 2A_r b^\varepsilon(t)]_+ \leq Q_r(t). \tag{62.3}$$

This holds without a count-uniform exit, drift, or variation bound.

Proof. For either sign, Theorem 60.2 gives $b^{\varepsilon}(t) \le b^{\varepsilon}(t+\sigma h) + Q_r^{\sigma}(t)$ for almost every $h \in (0,r)$. Integrate in h. The one-sided average of a nonnegative density is at most twice its centered average. Therefore $b^{\varepsilon}(t) \le 2A_r b^{\varepsilon}(t) + Q_r^{\sigma}(t)$ for each sign, and taking their minimum proves the assertion. The common almost-everywhere versions follow from the joint coarea formulation; no supremum over a null exceptional set is used. □

The defect can be bounded directly by physical sources. Put

$$\mathcal{X}_r^{\sigma}(t) = \int_{\{L=t\}\cap\Omega} W^{\varepsilon}(x)\mathbf{1}_{\{\text{lifetime in direction } \sigma<r\}}\,\mathrm{d}\mu_t(x).$$

An exit exactly at r is immaterial to an essential supremum over $h < r$. Monotonicity of the stopped domains and Proposition 60.3 give

$$Q_r(t) \le \min_{\sigma\in\{-1,1\}} \left\{ \mathcal{X}_r^{\sigma}(t) + \int_0^r G_{\sigma}(t+\sigma s)\,\mathrm{d}s \right\}. \tag{62.4}$$

The expression is permitted to be infinite. It is not declared small on the basis of a coordinate cover. In a Morse neighborhood one may use the sharper direct defect (61.10) instead of this drift majorant.

Theorem 62.2 (A quantitative closure implication on the original record)**.** *Assume Input statement 59.1. Suppose, in addition, that the complete stopped physical transports satisfy, for some $A_M < \infty$, $q \ge 0$, $\beta > 0$ and fixed $\varepsilon_0, r_0 > 0$,*

$$\mathcal{Q}_{\varepsilon,M}(r) \le A_M \varepsilon^{-q} r^{\beta} \qquad (0 < \varepsilon < \varepsilon_0,\ 0 < r < r_0). \tag{62.5}$$

Then the original exact-label pointwise law (59.5) *holds. More precisely, for every sufficiently large fixed B and fixed $r < r_0$,*

$$\mathcal{E}_M \le C'_M B^{-1/192} + A_M \varepsilon(B)^{-q} r^{\beta}. \tag{62.6}$$

The bound (62.5) *is an additional dynamical premise, not a conclusion of the optical or local Morse theorems.*

Proof. Combine (59.7) with Theorem 62.1. Positivity gives

$$|m^2 p - \mathcal{L}| \le 2m^2 A_r b^{\varepsilon(B)} + m^2 Q_r + |e_B|.$$

Fix B, then $\varepsilon(B)$, then r, and the auxiliary band in (59.6), before taking the collision limsup. Enlarge that auxiliary band only afterwards. The result is

$$\mathcal{E}_M \le 2C\varepsilon(B)^{1/16} + \mathcal{Q}_{\varepsilon(B),M}(r) + C_M B^{-1/192}.$$

Since $\varepsilon(B)^{1/16} = A_0^{1/16} B^{-1/192}$, (62.5) proves (62.6). Choose $r(B) = \varepsilon(B)^{(q+a)/\beta}$ for any $a > 0$, reducing it by a fixed constant if necessary. It is fixed before the collision limit. Finally let $B \to \infty$. Both terms vanish. No band, roof width, critical collar, or mesh is chosen as a function of m inside a spectral estimate. □

62.1. **Downstream statements require scalar closure.** For precision, the retained path input has the following form. At each exact label and roof there is a finite positive path remainder $\mathsf{B}_B(t)$ with mass $\mathsf{B}_B(t)(1) = m^2 b^{\varepsilon(B)}(t)$. If $\mathsf{P}_m(t)$ is the normalized path-numerator density and $\mathsf{L}_m(t)$ its complete arithmetic reference measure, then

$$\mathsf{P}_m - \mathsf{L}_m = \mathsf{B}_B + \mathsf{e}_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} \|\mathsf{e}_B\|_{\mathrm{BL}^*} \le C_M B^{-1/192}.$$

The bounded-Lipschitz test class includes the constant one. These are measures for the same scalar disintegration, not separately selected density versions for different tests.

Corollary 62.3 (The same-roof normalization implication)**.** *If* (62.5) *and the preceding path input hold, the path-numerator error vanishes in the same ordered limits. Conditional path ratios may then be formed on central arithmetic classes whose scalar reference is bounded below by a fixed* $d > 0$*. Arithmetic zero classes are never divided by. Any actual-return clock-transfer or forward-likelihood conclusion additionally retains its original hypotheses; none becomes unconditional from a single-word optical estimate.*

Proof. The dual bounded-Lipschitz norm of a positive measure is its mass. The proof of Theorem 62.2 bounds that mass after the same limits and therefore bounds the path error. On a class with reference at least d, scalar convergence makes the actual normalized density at least $d/2$ for large m. Subtract the two ratios, use this lower bound and the convergent numerator and denominator errors. This proves the conditional implication on that class only. Clock transfer and likelihood are not proved by this algebra and require their stated retained inputs. □

63. The mathematical boundary of the present argument

The optical comparison controls every regular word and every selected set of incidence defects. The stopped transport is a complete-source identity, not a protected-source replacement. Its losses distinguish physical exits from the changes of coordinates that contributed to the earlier sufficient trace modulus. The local Morse calculation also avoids assigning the height of a positive critical jump to a forward transport defect.

Neither statement bounds the complete normalized exit and distortion budget in Theorem 62.2. There may be many words, small true physical domains, and small critical neighborhoods. Their weighted concentration on an exact-label roof interval must still be estimated. The fixed-count Morse radii are not used as widths in a collision limit. Likewise, the inverse-twist comparison is not summed by replacing the weighted physical word set with its cardinality. Proving this remaining dynamical estimate, on the original source and in the stated order of limits, continues to be the route to (59.5); the target has not been replaced by a different record or by a smoothed law.

Part 29. Convex support and the complete physical Gauss trace

Retained revision-75 abstract. We study exact displacement, collision count and flight time for actual returns in a triangular periodic Lorentz gas. For every regular physical word we construct, on a fixed finite cover of endpoint projection coordinates, a strongly convex supporting action which agrees with the original roof on its entire physical endpoint image. Both local curvature bounds are independent of the collision count and of every positive incidence and clearance. The convex comparison controls all physical level pieces together: their coarea integral has a count-independent bound and their Gauss map has at most one inverse per normal direction. This removes the wordwise level-complexity loss from the complete inverse-incidence height estimate. We obtain an exact positive Gauss coarea formula retaining every original return label and physical source. The remaining collision-uniform short-excursion estimate is expressed on this explicit inverse-twist trace. The geometric estimates are proved without assuming that weighted dynamical estimate.

64. The original record and the geometric estimate

Let $R \in I = [9/20, 47/100]$, and let $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ be the actual return record under the original section probability $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, where $c_* = 91/(10000\pi)$. The circular scatterers have centers in the triangular lattice $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. In collision coordinates (α, p),

$d\nu = d\alpha\, dp/(4\pi)$ and p is the tangential component of the outgoing unit velocity. The section, including its moving centers, is specified without alteration in the supplement's Section 69.

At m flights let $L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ the accumulated displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$ the occupation. The exact event used here is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*.$$

Thus the initial visit is counted, the terminal visit is tested separately, and the actual nth return has collision count m on this event. The density of its roof under ν_R^* is $p_{n,R}(k,m,t)$. Set

$$Z_{m,R}(k,t,n) = \frac{(k_1, k_2, t - m\bar{\tau}_R, n - mc_*)}{\sqrt{m}}, \qquad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z_{m,R}(k,t,n)| \le M\}.$$

The governing pointwise assertion remains, for every finite M,

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{64.1}$$

The reference $\mathcal{L}_{m,R}$ is the complete arithmetic transition reference in the supplement, not a Gaussian obtained by pooling labels. Every arithmetic zero class is retained. Denote the collision limsup of the left side by $\mathcal{E}_M$.

64.1. **The complete positive source and the imported statements.** Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the actual third-disk clearance of flight j. With $q_0 = 47/53$, set $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$. The original guard is

$$H^{\varepsilon}_{m,R} = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right),$$

where the unchanged smooth cutoff satisfies $0 \le \Theta \le 1$, $\Theta = 0$ on $(-\infty, 1]$ and $\Theta = 1$ on $[2, \infty)$. No definability of this smooth cutoff is assumed. Define

$$b^{\varepsilon}(t)\, dt = (L_{m,R})_{\#}\big((1 - H^{\varepsilon}_{m,R}) \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*\big). \tag{64.2}$$

Every incidence and clearance factor remains in this definition. In particular the first-clearance source outside selected critical disks and the first-incidence source are both included.

Input statement 64.1 (Analytic interfaces on the unchanged source)**.** For each fixed m, R, ε, n, k, the density in (64.2) is nonnegative, compactly supported and BV. For each fixed $\varepsilon, \delta > 0$, with $A_\delta f(t) = (2\delta)^{-1} \int_{t-\delta}^{t+\delta} f(s)\, ds$,

$$\limsup_{m} \sup_{R,n,k} m^2 \|A_\delta b^{\varepsilon}\|_\infty \le C\varepsilon^{1/16}. \tag{64.3}$$

For sufficiently large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$,

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_{m} \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{64.4}$$

The constant in (64.3) is independent of the previously fixed width. Its auxiliary spectral band is fixed before the collision limit and enlarged only afterwards. For nonnegative compactly supported BV densities,

$$[f - 2A_\delta f]_+ \le S_\delta f, \qquad S_\delta f(t) = \frac{1}{\delta} \int_0^\delta T_f(t,r)\, dr,$$

where $T_f(t,r)$ counts the height levels whose essential superlevel component through t has length less than r.

The exact sources are the supplement's Theorem 69.2, (79.8), (181.7), and Theorem 70.2 with its length-tail identity. The positivity comparison yields

$$\mathcal{E}_M \le \mathcal{S}_M(B,\delta) + C_M B^{-1/192}, \qquad \mathcal{S}_M(B,\delta) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\mathcal{C}(M)} S_\delta b^{\varepsilon(B)}.$$

The full two-sided comparison is Theorem 71.1. The new geometric proofs below do not use any collision-uniform variation or tail bound.

For downstream path conclusions we additionally retain Theorem 190.3: the path-numerator error has a positive remainder with exactly the scalar mass $m^2 b^{\varepsilon(B)}$ and a bounded-Lipschitz dual error with the same vanishing outer-band bound. The ordered path and scalar errors then agree by Theorem 76.1. Conditional ratios are formed only on central reference classes bounded below by a positive constant; arithmetic zero classes are not normalized. Actual-return path statements further retain the original clock-transfer hypotheses. These analytic and path interfaces are source-pinned inputs, not independent spectral or clock proofs supplied by the present argument.

64.2. **The new physical mechanism.** A positive Hessian on the physical endpoint image is not enough: that image may have many components separated by first-hit and exact-label boundaries. We instead construct one convex supporting action on an entire endpoint rectangle. A fixed 16-direction cover at each endpoint accommodates all positive incidences, even near grazing. On the physical pieces its Hessian obeys

$$\frac{50}{47} I_2 \le D^2 F_w \le \frac{1360}{9} I_2.$$

Theorem 65.3 proves this statement. Lemma 66.1 then bounds all physical level pieces of a word together by $2\pi/\lambda$, with at most one inverse for each outward normal. Theorem 66.2 uses that inverse to express the complete exact-label density as a positive Gauss coarea integral. This removes a collision-dependent wordwise level-complexity loss, rather than concealing it in a constant.

The remaining sum of inverse-twist weights over physical words is not thereby controlled at the m^{-2} scale. Section 67 states precisely the paired modulus of that trace sufficient to complete (64.1), and distinguishes it from the uniform per-word geometry actually proved here. The short article and the complete supplement retain the same original pointwise problem.

65. Convex support on endpoint caps

Put $R_- = 9/20$, $R_+ = 47/100$ and $\tau_- = 3/50$. A word $w = (C_0, \dots, C_m)$ records the actual successive disk centers, with $C_0 = 0$. A regular word has positive flights, positive incidences, and the original first-hit exclusions. Its physical endpoint image need not be connected or convex.

For a regular trajectory write $Q_i = C_i + Rn_i$, let v_i be its outgoing unit velocity, and set

$$w_0 = v_0, \qquad w_m = -v_{m-1}, \qquad n_0 \cdot w_0 = c_0 > 0, \quad n_m \cdot w_m = c_m > 0.$$

Here v_{m-1} is the velocity on the last flight, before its terminal reflection. Thus the endpoint gradients of the polygonal length are $-w_0$ and $-w_m$.

Lemma 65.1 (A fixed cover of regular endpoint data)**.** *Let $e_j = (\cos(2\pi j/16), \sin(2\pi j/16))$, $0 \le j < 16$. For every pair of unit vectors n, w with $n \cdot w > 0$, some j satisfies $n \cdot e_j > 1/2$ and $w \cdot e_j > 1/2$. On that cap the coordinate*

$$Q_j(s) = C + s e_j^{\perp} + \sqrt{R^2 - s^2}\, e_j, \qquad |s| < \sqrt{3}R/2, \tag{65.1}$$

is single-valued, with $|Q_j'| \le 2$ and $|Q_j''| \le 8/R_-$. Choosing the first admissible j at each endpoint gives a Borel partition with at most $16^2 = 256$ parts, independent of m and R.

Proof. The shorter angular distance between n and w is less than $\pi/2$. Its midpoint is within $\pi/4$ of both. A closest grid direction is within $\pi/16$ of the midpoint; hence both distances are less than $5\pi/16 < \pi/3$. The two scalar products exceed $1/2$. Formula (65.1) parameterizes the stated semicircle cap. Its derivatives are

$$Q_j' = e_j^{\perp} - \frac{s}{\sqrt{R^2 - s^2}} e_j, \qquad Q_j'' = -\frac{R^2}{(R^2 - s^2)^{3/2}} e_j.$$

These give the bounds and the measurable first-admissible partition. No lower bound on the physical incidence $n \cdot w$ is used. □

65.1. **Supporting actions, not a continuation of the orbit.** For the prescribed word define the auxiliary endpoint value

$$V_w(q_0, q_m) = \min_{q_i \in \overline{D(C_i,R)},\ 0<i<m} \sum_{i=0}^{m-1} |q_{i+1} - q_i|. \tag{65.2}$$

For $m = 1$ this is just $|q_m - q_0|$. The minimization is a comparison in Euclidean space. Its other minimizers are not declared to be billiard trajectories; no measure or label is assigned to them.

Lemma 65.2 (A global supporting inequality)**.** *At a regular physical trajectory x of word w, with roof $L(x)$,*

$$\begin{aligned} &V_w(q_0, q_m) \geq \ell_x(q_0, q_m), \\ &\ell_x(q_0, q_m) = L(x) - w_0(x) \cdot (q_0 - Q_0(x)) - w_m(x) \cdot (q_m - Q_m(x)). \end{aligned} \tag{65.3}$$

Equality holds at its own endpoints. A regular physical trajectory of the word is uniquely determined by its endpoints.

Proof. The polygonal length is jointly convex in all contact positions. At an internal reflected contact its gradient is $-2c_i n_i$. For q_i' in the corresponding closed disk, $-2c_i n_i \cdot (q_i' - Q_i) \geq 0$, strictly if $q_i' \neq Q_i$, because $c_i > 0$ and the disk is strictly convex. The convex gradient inequality, followed by minimization over the internal disks, gives (65.3). With the endpoints unchanged it also shows that the physical internal tuple is the unique minimizer. Two regular trajectories of the same word and endpoints must therefore coincide. The one-flight case follows from its straight segment. The minimum in (65.2) exists by compactness. It is convex in the Cartesian endpoints: interpolate both endpoint pairs and their minimizing internal tuples, and apply convexity of the polygonal length. The supporting inequality, rather than any physical interpretation of interpolated tuples, is what will be used. □

Fix a pair $a = (j_0, j_m)$ of admissible caps and let $\mathcal{U}_{w,a}$ be the full regular physical endpoint image satisfying both strict cap tests. It is considered before the first-admissible partition and before restriction to any return label or source weight. In the open rectangle $Q_R = (-\sqrt{3}R/2, \sqrt{3}R/2)^2$ define

$$\Phi_{w,a}(s_0, s_m) = \sup_{x \in \mathcal{U}_{w,a}} \ell_x(Q_{j_0}(s_0), Q_{j_m}(s_m)). \tag{65.4}$$

An empty $\mathcal{U}_{w,a}$ is omitted.

Theorem 65.3 (Uniform convex support for every physical word)**.** *The function* (65.4) *is finite and λ-strongly convex on Q_R, where*

$$\lambda = \frac{1}{2R_+} = \frac{50}{47}, \qquad \Lambda = \frac{8}{\tau_-} + \frac{8}{R_-} = \frac{1360}{9}. \tag{65.5}$$

It equals the actual reduced roof F_w on $\mathcal{U}_{w,a}$. On every regular open piece of that image,

$$\lambda I_2 \leq D^2 F_w \leq \Lambda I_2. \tag{65.6}$$

Both constants are independent of the word length, every internal incidence, and every positive physical clearance. No uniform size of a physical neighborhood and no bound on higher derivatives is asserted.

Proof. Each supporting function in (65.4) has zero mixed second derivative in these coordinates. At an endpoint its second derivative is

$$(w_i(x)\cdot e_{j_i})\frac{R^2}{(R^2-s_i^2)^{3/2}}\geq\frac{1}{2R_+}.$$

Subtracting $\lambda|s|^2/2$ therefore makes every support convex; their supremum is convex. For fixed m all support values are bounded, since $L(x)\leq m\tau_+$ and the endpoint displacements are at most $2R$ each. The envelope is finite. By Lemma 65.2 every support is at most $V_w(Q_{j_0}(s_0),Q_{j_m}(s_m))$. At a physical point its own support attains the roof, which equals that value. Thus the envelope agrees with the roof throughout the open physical cap image. Endpoint injectivity follows from the same lemma. Local smoothness of the reduced action follows from the regular reflection equations and the positive definite internal contact Hessian; this is also the endpoint construction in the inherited curvature lemma. Strong convexity of the envelope proves the lower inequality.

For the upper inequality one can keep all internal contacts fixed while the two endpoints vary. The resulting trial length touches the reduced action from above. Equivalently, the reduced Hessian is the Schur complement of the internal contact Hessian. For an endpoint variation $h=(h_0,h_m)$ the transverse second variation of a segment is its squared transverse displacement divided by its flight length. At most two endpoint displacements occur. Using $|Q_j'|\leq 2$ and $\ell_i\geq\tau_-$ bounds their total by $8|h|^2/\tau_-$, including $m=1$, when they are on the same segment. The endpoint curvature terms are at most $8|h|^2/R_-$. Minimizing over internal variations cannot increase this trial quadratic form. This proves the upper inequality. The argument is local at a regular trajectory; it makes no continuation across a grazing or first-hit boundary. □

The use of supporting actions is essential. Positivity of a Hessian on a disconnected physical image alone would not bound the total turning of all its level components. The single convex envelope on the entire endpoint rectangle supplies that missing global comparison.

66. A wordwise coarea estimate without a complexity factor

All measures below are on the original physical trajectories. The convex envelope is used only to bound a level integral. For a regular endpoint level let t be its unit tangent and put $H_T=\mathsf{t}^{\mathsf{T}}D^2F_w\mathsf{t}$. The normal angle θ refers to the two-dimensional endpoint coordinate plane of the fixed cap pair, not to a collision angle.

Lemma 66.1 (Total turning of all physical level pieces)**.** *For a fixed word and cap pair, and almost every roof value t,*

$$\int_{\mathcal{U}_{w,a}\cap\{F_w=t\}}\frac{d\mathcal{H}^1}{|\nabla F_w|}\leq\frac{2\pi}{\lambda}.\tag{66.1}$$

Every Borel subset of the physical endpoint image satisfies the same bound. On its regular level pieces the outward Gauss map is one-to-one, apart from null sets, and

$$\int_{\{F_w=t\}}\frac{g}{|\nabla F_w|}\,d\mathcal{H}^1=\int_{\Theta_{w,a}(t)}\frac{g(s_{w,a}(t,\theta))}{H_T(s_{w,a}(t,\theta))}\,d\theta\tag{66.2}$$

for every nonnegative Borel weight on any such subset.

Proof. Intersect the convex sublevel set $\{\Phi_{w,a}\leq t\}$ with Q_R. A physical point at which $\nabla F_w\neq 0$ lies on its boundary, with outward normal $\nabla F_w/|\nabla F_w|$. On every smooth physical arc the curvature is $\kappa=H_T/|\nabla F_w|>0$. Two distinct points of a convex boundary with the same

supporting normal would lie on one exposed segment. A positive-curvature point in an open physical arc cannot belong to that segment. Thus the Gauss map is injective on all these arcs together. Its image has angular measure at most 2π, even when physical exclusions split the level into many pieces. The one-dimensional change of variables $d\theta = \kappa\, d\mathcal{H}^1$ gives (66.2), and $H_T \geq \lambda$ gives (66.1). Apply the formulas on compact regular exhaustions and then use monotone convergence. There is at most one stationary endpoint point in a convex cap image, so its roof value causes no difficulty for an almost-everywhere statement. The rectangle boundary and nonsmooth portions of the auxiliary convex boundary are not added to the physical source. □

66.1. **The unchanged exact-label density.** Let $B_m = \partial_{p_0}\alpha_m$ on a regular word. It is nonzero. For a source weight $W \geq 0$ on the original exact event $\Pi = \Pi_{n,k,m,R}$, use the first-admissible endpoint partition from Lemma 65.1. At a selected physical point $s = s_{w,a}(t,\theta)$ define

$$\gamma^W_{w,a}(t,\theta) = \frac{W(x)\mathbf{1}_\Pi(x)\mathbf{1}_a(x)}{4\pi c_* R^2 (n_0 \cdot e_{j_0})(n_m \cdot e_{j_m})|B_m(x)|H_T(s)}. \tag{66.3}$$

Set this expression to zero when no selected physical point exists, and put $\Gamma^W = \sum_{w,a} \gamma^W_{w,a}$. Every word is a physical m-flight word. Its terminal center fixes its displacement; its actual occupation and terminal section test remain in $\mathbf{1}_\Pi$.

Theorem 66.2 (Positive Gauss coarea at every original label)**.** *The pushforward of $W\mathbf{1}_\Pi \nu^*_R$ has density*

$$p_W(t) = \int_0^{2\pi} \Gamma^W(t,\theta)\, d\theta. \tag{66.4}$$

The formula holds for nonnegative weights by monotone limits; for bounded signed or complex insertions it holds by linearity. Moreover,

$$\gamma^W_{w,a}(t,\theta) \leq C_G \frac{W(x)\mathbf{1}_\Pi(x)\mathbf{1}_a(x)}{|B_m(x)|}, \qquad C_G = \frac{2R_+}{\pi c_* R_-^2}. \tag{66.5}$$

The constant and the number of cap pairs are independent of m. There is no semialgebraic level-complexity or monotonicity factor in these two formulas.

Proof. Endpoint injectivity and the nonzero twist give $d\alpha_0\, dp_0 = |B_m|^{-1} d\alpha_0\, d\alpha_m$. Since $d\alpha_i = ds_i/[R(n_i \cdot e_{j_i})]$, the original section probability $d\alpha_0\, dp_0/(4\pi c_*)$ becomes the numerator of (66.3) without its factor H_T. Coarea followed by (66.2) proves the formula on each selected physical piece. The pieces are disjoint at the source. Sum them by Tonelli, keeping $\mathbf{1}_\Pi$ throughout. The two cap scalar products exceed $1/2$ and $H_T \geq \lambda$; these inequalities give (66.5).

For completeness the inverse $(t,\theta) \mapsto s_{w,a}$ can be chosen Borel. On each regular piece the roof and its Gauss map are smooth, and the pair is injective: two points with the same roof and normal were excluded in Lemma 66.1. The inverse of an injective Borel map between standard Borel spaces is Borel on its image. A finite disjoint refinement of physical and cap pieces, or a countable compact exhaustion, gives the stated version. There are finitely many words at fixed m; exceptional roof sets may depend on R. The order remains $\sup_R \operatorname{ess\,sup}_t$. □

66.2. **A sharper complete inverse-incidence estimate.** Let $\mathcal{I}_m = \prod_{i=0}^m c_i^{-1}$. Denote by D the cardinality of a fixed finite alphabet containing every next-center increment. This D is a cardinality, not a displacement coordinate. The inherited contact-determinant identity states, in endpoint arclengths (u,v),

$$\mathcal{I}_m |F_{uv}| \leq (R/2)^{m-1} \prod_{i=0}^{m-1} \ell_i^{-1}. \tag{66.6}$$

Its proof factors the internal contact matrix into its positive Dirichlet Laplacian and the diagonal $2/(Rc_i)$; it requires no incidence margin.

Corollary 66.3 (No level-complexity loss in the complete height)**.** *Under the full collision probability, the inverse-incidence roof density satisfies*

$$\|\rho^{\mathcal{I}}_{m,R}\|_\infty \le \frac{1024R_+}{R_-\tau_-}\,D^m\left(\frac{R_+}{2\tau_-}\right)^{m-1}. \tag{66.7}$$

A positive exact-return restriction has this upper bound divided by c_*; a bounded insertion costs its supremum norm. The geometric factor formerly needed to count or turn all level components of a word is absent. The remaining word sum is not asserted to have central-scale order m^{-2}.

Proof. In endpoint arclengths the collision density is $|F_{uv}|/(4\pi R)$. On the cap coordinates, $du\,dv = ds_0\,ds_m/[(n_0\cdot e_{j_0})(n_m\cdot e_{j_m})]$, so (66.6) bounds the weighted source density by $(R_+/2)^{m-1}\tau_-^{-m}/(\pi R_-)$. Use (66.1), sum at most 256 cap pairs and at most D^m words, and insert $\lambda^{-1}=2R_+$. Truncation and monotone convergence prove finiteness for the unbounded insertion as well. Exact-label and bounded-weight assertions follow by positive restriction at the source, not by replacing their labels. □

67. Short excursions of the complete physical trace

Apply Theorem 66.2 with the original weight $W=1-H^\varepsilon$ and write its complete trace as $\Gamma^\varepsilon_{m,R,n,k}$. Thus $b^\varepsilon(t)=\int\Gamma^\varepsilon(t,\theta)\,d\theta$. This includes incidence, every competing-hit clearance, and the first-clearance source outside the previously selected disks. The factor c_*^{-1} already occurs in (66.3).

For a nonnegative trace Γ and $r>0$ define

$$\omega_r^{\pm}\Gamma(t)=\operatorname*{ess\,sup}_{0<h<r}\int_0^{2\pi}[\Gamma(t,\theta)-\Gamma(t\pm h,\theta)]_+\,d\theta, \tag{67.1}$$

$$\mathfrak{D}_r\Gamma(t)=\min\{\omega_r^-\Gamma(t),\omega_r^+\Gamma(t)\}. \tag{67.2}$$

The entire word sum is taken before the positive part and the two essential suprema. No assertion that a short-excursion functional commutes with a positive sum is needed.

Lemma 67.1 (From a physical trace to its level-component tail)**.** *If $f(t)=\int\Gamma(t,\theta)\,d\theta$ is nonnegative, compactly supported and BV, then, for almost every t,*

$$T_f(t,r)\le\mathfrak{D}_r\Gamma(t),\qquad S_\delta f(t)\le\frac{1}{\delta}\int_0^\delta\mathfrak{D}_r\Gamma(t)\,dr. \tag{67.3}$$

Here $T_f(t,r)$ is the height measure of levels $0<u<f(t)$ whose essential superlevel component through t has length less than r.

Proof. At almost every such level the essential superlevel set is a finite union of intervals. If its component (a,b) through t has length less than r, its two endpoints lie in $(t-r,t+r)$. Immediately beyond each endpoint there is a set of positive measure on which $f\le u$; otherwise it is not an endpoint of that essential component. Consequently

$$u\ge\max\left\{\operatorname*{ess\,inf}_{0<h<r}f(t-h),\operatorname*{ess\,inf}_{0<h<r}f(t+h)\right\}.$$

The measure of all contributing heights is therefore at most the minimum of the two essential suprema of $[f(t)-f(t\pm h)]_+$. For each sign, the exact trace formula and positivity bound this last difference by the integral in (67.1). Fubini fixes the needed versions for almost every (t,h). This proves the first inequality. The second follows from the inherited identity $S_\delta f=\delta^{-1}\int_0^\delta T_f(t,r)\,dr$. All operations involve the complete scalar density, not components formed separately on individual words. □

Set, with empty central targets contributing zero,

$$\mathcal{D}_{\varepsilon,M}(r) = \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \mathfrak{D}_r \Gamma^{\varepsilon}_{m,R,n,k}(t). \tag{67.4}$$

The trace in (67.1) uses the full roof line; it is not cut off at the boundary of the central target.

Proposition 67.2 (A physical trace estimate sufficient for the endpoint)**.** *Suppose the complete Lorentz traces obey, with $A < \infty$, $q \geq 0$, $\beta > 0$ and fixed positive ε_0, r_0,*

$$\mathcal{D}_{\varepsilon,M}(r) \leq A\varepsilon^{-q} r^{\beta} \quad (0 < \varepsilon < \varepsilon_0,\ 0 < r < r_0). \tag{67.5}$$

Then the original exact-label pointwise theorem follows from the source-pinned analytic inputs in the introduction. In particular, for large fixed B, $\varepsilon(B) = A_0 B^{-1/12}$ and $\delta < r_0$,

$$\mathcal{E}_M \leq C_M B^{-1/192} + \frac{A}{\beta+1}\varepsilon(B)^{-q}\delta^{\beta}. \tag{67.6}$$

Equation (67.5) *is a further dynamical hypothesis, not a conclusion of the wordwise coarea theorem.*

Proof. Lemma 67.1 gives $\mathcal{T}_{\varepsilon,M}(r) \leq \mathcal{D}_{\varepsilon,M}(r)$. For clarity the ordered integral step can also be done directly. The function $\mathfrak{D}_r\Gamma$ is nondecreasing in r. For every fixed integer N,

$$S_\delta b \leq \frac{1}{N}\sum_{j=1}^{N} \mathfrak{D}_{j\delta/N}\Gamma.$$

Fix $B, \varepsilon(B), \delta, N$ and the auxiliary spectral band before the collision limsup. The finite sum and (67.5) bound the resulting limsup by $A\varepsilon(B)^{-q}\delta^{\beta}N^{-1}\sum_{j=1}^{N}(j/N)^{\beta}$. Only afterwards refine N, giving $1/(\beta+1)$. The inherited positive-remainder and fixed-width estimates give $\mathcal{E}_M \leq \mathcal{S}_M(B,\delta) + C_M B^{-1/192}$, proving (67.6). Choose $\delta(B) = \varepsilon(B)^{(q+a)/\beta}$ for any $a > 0$, before $m \to \infty$, and finally let $B \to \infty$. The scalar error vanishes. The inherited common positive path remainder then gives the path-numerator conclusion; normalization is performed only on positive arithmetic classes and the return-clock hypotheses are retained. No growing-band spectral estimate is used. □

67.1. **The remaining weighted dynamical estimate.** Theorems 65.3 and 66.2 prove a collision-independent geometric bound on each word, even at arbitrarily small positive incidence. They also replace the exponential wordwise level-multiplicity bound by one inverse per normal direction. They do not bound the sum of inverse twists with the exact occupation and roof constraints, nor its paired trace modulus in (67.4). In particular (66.7) is still exponential in m. It cannot be inserted as an m^{-2} bound in (67.5).

The quantitative issue is now on a fully specified physical trace: control the creation, disappearance and change of its exact-label inverse-twist weights as the roof varies, after summing all physical words. A cap change or a physical exit is represented by the zero-extended term in (66.3), not suppressed. The supporting envelope contributes no substitute trajectory or extra source. No collision-uniform short-excursion decay, full pointwise raw law, or unconditional same-roof normalization is claimed without that remaining estimate. The pointwise theorem (64.1), with all its original labels, remains the governing assertion.

Part 30. The retained complete source and short excursions

Retained revision-74 abstract. We study exact displacement, collision count and flight time for actual returns in a triangular periodic Lorentz gas. For the complete physical boundary source, we give an explicit tame integration family, including the moving section and every exact label. We then match incoming and outgoing current on the same superlevel component. This yields a

pointwise short-excursion bound sharper than the minimum of the two Jordan masses, a sharp integrated estimate, and an exact length-tail identity. The ordered short-excursion budget has the same outer limit as the original arithmetic pointwise error and, under the common positive path-remainder theorem, its bounded-Lipschitz path-numerator error. A collision-uniform length-tail estimate sufficient for the full raw pointwise law is isolated with its order of limits. That dynamical estimate is not inferred from fixed-count finite variation.

68. The physical record and the pointwise theorem

Let $R \in I = [9/20, 47/100]$. Under the original section probability $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, with $c_* = 91/(10000\pi)$, write $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ for the actual return record. The density at the exact displacement k and collision count m is $p_{n,R}(k, m, t)$. Occupation counts collisions $0, \dots, m-1$; terminal section membership is tested at collision m. Set

$$Z_{m,R}(k,t,n) = \frac{(k_1, k_2, t - m\bar{\tau}_R, n - mc_*)}{\sqrt{m}}, \qquad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z_{m,R}(k,t,n)| \le M\}.$$

The organizing pointwise assertion is, for every finite M,

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{68.1}$$

Here $\mathcal{L}_{m,R}$ is the full arithmetic transition reference of the technical supplement, including every zero class. Denote the collision limsup of the left side by $\mathcal{E}_M$. No discrete coordinate is pooled and no roof set is removed beyond the null sets inherent in density versions.

For a fixed guard width $\varepsilon > 0$, the complete positive source is

$$b^\varepsilon(t)\, dt = L_\#\big((1 - H^\varepsilon)\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*\big). \tag{68.2}$$

The guard contains all original incidence and clearance factors. In the earlier source decomposition it is $b = s_{71,1} + d_{71,1} + b^{\varepsilon,\mathrm{inc}} + e_{70}$, where e_{70} is first clearance outside the selected disks, with first incidence separate. Every term remains present. Define, before any averaging argument,

$$A_\delta f(t) = \frac{1}{2\delta}\int_{t-\delta}^{t+\delta} f(s)\, ds, \qquad \mathcal{H}(f) = \limsup_{m\to\infty} \sup_{R,n,k} m^2\|f\|_\infty.$$

Widths are fixed before this collision limit.

68.1. **The precise inherited inputs.** The technical supplement contains all 163 preceding proof modules, their appendices and bibliography. Its theorem numbering is distinct from the present article. The following statement identifies exactly which interfaces are used; the subsequent arguments do not re-prove the full spectral, arithmetic or return-clock chain.

Input statement 68.1 (Source-pinned analytic interfaces)**.** The complete fixed-count height theorem gives $\|\rho_{m,R}\|_\infty \le CA^m$, uniformly in R, and hence bounds every positive exact-label restriction by $L_m = CA^m/c_*$. For each fixed $\varepsilon, \delta > 0$, the complete-source fixed-width theorem gives

$$\mathcal{H}(A_\delta b^\varepsilon) \le C\varepsilon^{1/16}. \tag{68.3}$$

For sufficiently large fixed B and $\varepsilon(B) = A_0 B^{-1/12}$, the positive scalar remainder satisfies

$$m^2 p - \mathcal{L} = m^2 b^{\varepsilon(B)} + e_B, \qquad \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{\mathcal{C}(M)} |e_B| \le C_M B^{-1/192}. \tag{68.4}$$

The path consequence uses additionally the common positive path-remainder theorem with this same scalar mass, and actual-return path consequences additionally use the original clock-transfer hypotheses on positive arithmetic classes.

The exact sources are respectively the supplement's Theorem 167.3, (79.8), (181.7), and Theorem 190.3. The common-kernel version convention and the scalar/path comparison are recorded there in Lemma 73.1 and Theorem 76.1. In (68.3), the auxiliary band is fixed before the collision limit and enlarged afterwards. The resulting constant is independent of the previously fixed width. These are the precise interfaces used in the main error comparison.

68.2. **The new estimate.** For a nonnegative compactly supported BV density f, let $I_u(t)$ be the component of the essential superlevel set $\{f > u\}$ containing t. Define its short-excursion budget by

$$S_\delta f(t) = \int_0^{f(t)} (1 - |I_u(t)|/\delta)_+ \, du.$$

Section 70 specifies the measurable versions and proves

$$[f - 2A_\delta f]_+ \le S_\delta f \le \min\{(Df)^+([t-\delta, t]), (Df)^-([t, t+\delta])\}. \tag{68.5}$$

Unlike the last minimum, $S_\delta f$ matches the entrance and exit on the same height layer and the same component through the roof. It can vanish when both of those Jordan masses are positive. A narrow pulse remains charged with its exact excess. Moreover,

$$\int S_\delta f \le \frac{\delta}{8} \|Df\|_{\mathrm{TV}},$$

with sharp constant. This integrated inequality is not used as a pointwise density bound.

Theorem 69.2 supplies the finite current of the complete original source with an explicit moving-section graph. Theorem 71.1 then shows that its ordered short-excursion budget tends, as the outer band grows, to exactly $\mathcal{E}_M$. Theorem 71.2 proves a sufficient quantitative length-tail criterion, with every limit in the required order. The Lorentz-specific uniform decay in that criterion is still to be established; the original pointwise assertion (68.1) is therefore not claimed as a consequence of the present fixed-count and deterministic results alone. The present article, its technical supplement and the frozen provenance are separate parts of one source-preserving revision, not different mathematical topics.

69. The parameterized physical source

We record explicitly the finite-dimensional family to which integration and monotonicity are applied. The integer m is fixed throughout this section. A parameter uniformity assertion in this section is not an assertion uniform in m.

69.1. **The moving section.** Let $a = (1, 0)$, $b = (1/2, \sqrt{3}/2)$ and $\Lambda = \mathbb{Z}a + \mathbb{Z}b$. In collision coordinates (α, p) the invariant probability is $d\alpha\, dp/(4\pi)$. Put $h_0 = 1/200$. The four squares defining Y_R have half-side h_0 and centers

$$(0, 0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

The actual section is $Y_R^* = Y_R \cup U$, where

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad c_* := \nu(Y_R^*) = \frac{91}{10000\pi}.$$

Angles are taken modulo 2π. These five rectangles are disjoint. The probability used below is always $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$.

Lemma 69.1 (An algebraic coordinate for the moving centers)**.** *For $R \in I = [9/20, 47/100]$, the number $q_R = \tan(\theta_R/2)$ is the unique zero in $(49/100, 53/100)$ of*

$$P(R, q) = 8Rq(1 + q^2) - 4q(1 - q^2) + \sqrt{3}(1 - 6q^2 + q^4). \tag{69.1}$$

In particular the normals and the p coordinates of all five section centers are semialgebraic functions of R.

Proof. The defining equation for the physical winding orbit is $2R\sin\theta - \sin\theta\cos\theta + (\sqrt{3}/2)\cos(2\theta) = 0$. Substitute $\sin\theta = 2q/(1+q^2)$ and $\cos\theta = (1-q^2)/(1+q^2)$ and multiply by $2(1+q^2)^2$. This gives (69.1). On the displayed rectangle,

$$\partial_q P = 8R(1+3q^2) - 4 + 12q^2 + \sqrt{3}(-12q + 4q^3) < -2.$$

For example, the positive terms are bounded above at $R = 47/100$, $q = 53/100$, and the last term above by $-(1732/1000)(12(49/100) - 4(49/100)^3)$. Also $P(R, 49/100) > 0$ and $P(R, 53/100) < 0$ throughout I. Both endpoint signs follow by monotonicity in R and the rational bounds $1732/1000 < \sqrt{3} < 1733/1000$. Thus the zero exists and is unique. The physical root, whose angle lies in $(0.92, 0.967)$, has its half-angle in this interval and hence is this zero. The implicit derivative is nonzero, and the graph is semialgebraic. The displayed rational sine and cosine, followed by fixed rotations, give the asserted center coordinates. □

An angular interval of half-width $h < \pi$ and center normal n_* can be tested by $n \cdot n_* > \cos h$. Its companion p interval uses two linear inequalities. Consequently the five rectangle tests are semialgebraic in R, n, p. The fixed real constants $\cos h_0$ and $\cos(3/50)$ are coefficients; their presence does not require an algebraic value of either angle.

69.2. **A finite graph with the exact labels.** Fix a uniform upper flight bound τ_+ from the finite-horizon geometry; the lower bound is $\tau_- = 3/50$. Set $R_+ = 47/100$ and

$$\mathcal{A} = \{z \in \Lambda \setminus \{0\} : |z| \le \tau_+ + 2R_+ + 1\}, \qquad D = \max_{z = z_1 a + z_2 b \in \mathcal{A}} \max(|z_1|, |z_2|).$$

The set $\mathcal{A}$ is finite. It contains every next-center increment and every center needed for first-hit and minimal third-disk clearance tests. A center whose disk meets a segment is within $\tau_+ + 2R_+$ of its starting center. For the clearance, at least one of the six nearest neighbors of that starting center is not the terminal center. Its distance to the initial contact is at most $1 + R$. Thus the minimum third-center distance to the segment is at most $1 + R$. Any center attaining that minimum is within $\tau_+ + 2R_+ + 1$ of the starting center. Centers outside $\mathcal{A}$ cannot change the exact minimum. The starting disk is tested separately for first-hit conditions. Thus a finite set containing all attainable exact labels is

$$\mathscr{L}_m = \{(n, k) : 1 \le n \le m,\ k \in \mathbb{Z}^2, \max(|k_1|, |k_2|) \le mD\}. \tag{69.2}$$

Unattainable labels in this set give the zero source. No labels are pooled in the integrations which follow.

Use a finite disjoint refinement of rational angular charts

$$n(\alpha) = \mathrm{Rot}_{\alpha_*}\left(\frac{1-x^2}{1+x^2}, \frac{2x}{1+x^2}\right), \qquad |x| < 1.$$

Overlap boundaries are assigned by a fixed convention. For a center word $C_0 = 0, C_1, \dots, C_m$, keep as auxiliary variables the contact positions Q_j, normals n_j, velocities v_j and flight lengths τ_j. The graph includes

$$\begin{aligned} Q_j &= C_j + Rn_j, & |n_j|^2 &= |v_j|^2 = 1, \\ Q_{j+1} &= Q_j + \tau_j v_j, & \tau_- &\le \tau_j \le \tau_+, \\ v_j &= v_{j-1} - 2(v_{j-1}\cdot n_j)n_j, & v_{j-1}\cdot n_j &< 0 \quad (1 \le j \le m). \end{aligned}$$

At time zero, $v_0 = c_0 n_0 + p\, n_0^{\perp}$, with $c_0 > 0$ and $c_0^2 + p^2 = 1$. Exclusion of an earlier collision is imposed for every candidate disk. These are finite semialgebraic conditions: the minimum

distance to a segment is obtained at an endpoint or at the orthogonal projection when that projection lies inside the segment. Projection denominators are positive. The intended endpoint disks retain the first-hit sign conditions; no trajectory is continued through grazing. This is the same physical first-hit graph as in the complete-height theorem.

The incidence margins are positive square roots or dot products on this graph. The original clearance margins are the minima of the corresponding finitely many candidate-distance branches, with their branch conditions retained. Distances are encoded by a nonnegative variable and its squared-distance equation. Hence the original margins c_j, Δ_j, not substitute margins, are semialgebraic. The first-hit inequalities and regularity conditions remain part of the domain when a margin tends to zero.

Write $\eta_j = \mathbf{1}_{Y_R^*}(T_R^j x)$. The exact event requires

$$\eta_0 = \eta_m = 1, \qquad \sum_{j=0}^{m-1} \eta_j = n, \qquad C_m = k_1 a + k_2 b.$$

The terminal test is not an occupation summand. The physical roof is $L = \sum_{j=0}^{m-1} \tau_j$. The graph projects to the two original initial variables (x, p); over a regular initial state the physical successor is unique. Distinct physical center words therefore give disjoint projected pieces, up to the already excluded singular null set. Integration below is in these two initial variables, not volume integration on the auxiliary graph.

69.3. **The exact integration family.** Put $N = 2m + 1$, $q_0 = 47/53$, $e_j = q_0^{\min(j,m-j)/4}$ and $e'_j = q_0^{\min(j,m-1-j)/4}$. The normalized margins are $c_j/(\varepsilon e_j)$ and $\Delta_j/(\varepsilon e'_j)$. Enumerate them as a_i, and for $y \in [1,2]^N$ set

$$W_y = 1 - \prod_{i=1}^{N} \mathbf{1}_{\{a_i > y_i\}}.$$

Here the parameter domain is $R \in I$, $\varepsilon > 0$, $y \in [1,2]^N$, $t \in \mathbb{R}$, and $(n,k) \in \mathscr{L}_m$. The physical application subsequently restricts ε to its original small range. Finiteness of this fixed-formula family holds on the stated, larger positive domain.

Theorem 69.2 (An explicit tame disintegration family)**.** *For each fixed m and exact label, the sublevel mass*

$$M(R, \varepsilon, y, t) = \int \mathbf{1}_{\Pi_{n,k,m,R}} W_y \mathbf{1}_{\{L \le t\}} \, d\nu_R^* \tag{69.3}$$

is constructible. Its almost-everywhere roof derivative has a representative definable in $\mathbb{R}_{\mathrm{an,exp}}$, bounded between zero and $L_m = CA^m/c_*$. There is $J_m < \infty$, uniform over the stated continuous parameters and the finite set (69.2)*, which bounds the number of monotonicity intervals and point cells of that representative. The complete original smooth-guard density belongs to $BV(\mathbb{R})$ and satisfies*

$$\|Db^\varepsilon\|_{\mathrm{TV}} \le 4(J_m + 1) L_m \|d\Theta\|_{\mathrm{TV}}^{2m+1}. \tag{69.4}$$

Proof. On each disjoint initial chart the integrand supplied to the parameterized integration theorem is exactly

$$\frac{\mathbf{1}_{\Pi_{n,k,m,R}} W_y \mathbf{1}_{\{L \le t\}}}{2\pi c_* (1 + x^2)},$$

extended by zero off the projected regular physical graph. All tests in this formula have just been specified. Projection preserves semialgebraicity, and the deterministic graph makes the roof and the tests single-valued. The integrand is globally subanalytic and integrable in (x, p) for every parameter. The stability theorem for constructible functions under integration [15, Theorem 1.3]

proves the assertion about M after the finite disjoint sum over charts and words. It does not assert that M itself is semialgebraic.

Positive domination by the complete density gives a Lipschitz constant L_m for each t section. Define $g = \partial_t M$ where the finite derivative exists, and $g = 0$ elsewhere. Differentiability and its derivative are definable by quantified inequalities in $\mathbb{R}_{\mathrm{an,exp}}$. Thus g is definable, and absolute continuity identifies it with the positive threshold density almost everywhere. Positivity and the Lipschitz bound give $0 \le g \le L_m$. The uniform monotonicity theorem for definable families [9] gives finitely many cells with the roof coordinate last. Take the maximum over the finite labels. This defines J_m for this fixed formula, without asserting an estimate as its collision length grows.

Let $\sigma = d\Theta$. Although σ may be signed, every W_y is positive. The exact identity $1 - H^{\varepsilon} = \int W_y \, d\sigma^{\otimes N}(y)$ follows from $\Theta(a) = \int \mathbf{1}_{\{y<a\}} \, d\sigma(y)$ and $\sigma(\mathbb{R}) = 1$. A bounded monotone-piece density extended by zero has variation at most $4(J_m + 1)L_m$. Integrating its distributional derivative against the total variation of $\sigma^{\otimes N}$ proves (69.4) by the distributional characterization of BV [13]. This is the derivative of the full positive smooth-guard source. Its support is contained in $[m\tau_-, m\tau_+]$, and its total mass is zero. The support is compact at each fixed count; it is not one compact interval independent of m. □

The theorem supplies the finite current needed below, including incidence and clearance outside the selected disks. It supplies no collision-normalized concentration estimate. In particular neither J_m nor the variation bound in (69.4) is used as an m^{-2} estimate.

70. Level-matched short excursions

The two Jordan masses near a roof need not describe the same excursion. We pair entrances and exits on the same component of a superlevel set. This construction is made after summing the entire physical density, rather than separately on words or threshold sources.

Let $f \ge 0$ be compactly supported and in $BV(\mathbb{R})$. For almost every $u > 0$, the set $E_u = \{f > u\}$ agrees modulo null sets with a finite disjoint union of bounded open intervals. Use that essential interval representative. If t lies in its component $I_u(t)$, write $\ell_u(t) = |I_u(t)|$. For almost every (u, t) with $0 < u < f(t)$, this length is finite and positive. Define

$$S_\delta f(t) = \int_0^{f(t)} \left(1 - \frac{\ell_u(t)}{\delta}\right)_+ du, \qquad \delta > 0. \tag{70.1}$$

Exceptional levels and pairs are assigned zero. All identities and norms below are almost-everywhere identities and essential norms.

Lemma 70.1 (Measurability and oriented level decomposition)**.** *The function in* (70.1) *is measurable. As finite measures,*

$$(Df)^+ = \int_0^\infty \sum_{(a,b)\subset E_u} \delta_a \, du, \qquad (Df)^- = \int_0^\infty \sum_{(a,b)\subset E_u} \delta_b \, du. \tag{70.2}$$

In each sum, (a, b) runs over the essential interval components.

Proof. The one-dimensional BV coarea theorem gives finite perimeter of E_u for almost every u, and the finite-interval description follows since these sets are bounded. Endpoints can be chosen measurably: for an interior pair (u, t) take the infimum of rational $a < t$ for which $\int_a^t \mathbf{1}_{\{f\le u\}} = 0$, and the supremum of rational $b > t$ for which $\int_t^b \mathbf{1}_{\{f\le u\}} = 0$. These are the two component endpoints. The tests are measurable by parameterized integration, so their difference is measurable on the full-measure set of relevant pairs. Fubini justifies the conventions elsewhere.

The layer-cake identity gives $Df = \int D\mathbf{1}_{E_u} \, du$ as distributions. For a finite interval union, $D\mathbf{1}_{E_u} = \sum(\delta_a - \delta_b)$. The localized coarea identity $|Df| = \int |D\mathbf{1}_{E_u}| \, du$ identifies finite measures,

not only total masses. Adding and subtracting these two identities proves (70.2); see [13, Chapter 3]. □

Theorem 70.2 (A level-matched height bound)**.** *For almost every t,*

$$[f(t) - 2A_\delta f(t)]_+ \le S_\delta f(t) \le \min\{f(t), (Df)^+([t-\delta, t]), (Df)^-([t, t+\delta])\}, \tag{70.3}$$

where $A_\delta f(t) = (2\delta)^{-1} \int_{t-\delta}^{t+\delta} f(s)\,ds$. Furthermore $S_\delta(cf) = cS_\delta f$ for $c \ge 0$, and $S_\delta f$ is nondecreasing as a function of δ. The reserve two is optimal among bounds of the first form which charge only components shorter than δ.

Proof. If $t \in I = (a, b)$, then

$$|I \cap (t-\delta, t+\delta)| \ge \min\{\delta, b-a\}.$$

For a level not containing t, the contribution to $\mathbf{1}_{E_u}(t) - \delta^{-1}|E_u \cap (t-\delta, t+\delta)|$ is nonpositive. For a level containing t, the preceding inequality bounds that contribution by $(1 - \ell_u(t)/\delta)_+$. Integration in u and layer cake give $f - 2A_\delta f \le S_\delta f$. The right side is nonnegative, so this also bounds the positive part.

The bound $S_\delta f \le f$ is immediate. A component contributing positively has length less than δ; both its entrance and its exit lie in the corresponding one-sided windows about t. Each contributing weight is at most one. Equation (70.2) proves the other two bounds. The substitution $u = cv$ proves homogeneity for $c > 0$, and $c = 0$ is immediate. Monotonicity in δ follows from the integrand.

For optimality of the reserve, take $f = \mathbf{1}_{(0,L)}$ with $L > 2\delta$. Its short-excursion function vanishes. For $0 < t < \delta$, $A_\delta f(t) = (t+\delta)/(2\delta)$. Every reserve $K < 2$ leaves $f - KA_\delta f > 0$ on an interval of positive length next to zero. Thus it cannot replace two in this component-only estimate. □

Proposition 70.3 (Sharp integrated identity and length tails)**.** *One has*

$$\int_{\mathbb{R}} S_\delta f(t)\,dt = \int_0^\infty \sum_{I \subset E_u} |I|(1 - |I|/\delta)_+\,du \le \frac{\delta}{8}\|Df\|_{\mathrm{TV}}. \tag{70.4}$$

The constant $1/8$ is sharp. Define

$$T_f(t, r) = \int_0^{f(t)} \mathbf{1}_{\{\ell_u(t) < r\}}\,du, \qquad \Lambda_\beta f(t) = \int_0^{f(t)} \ell_u(t)^{-\beta}\,du.$$

For $\beta > 0$, allowing an infinite moment,

$$S_\delta f(t) = \frac{1}{\delta}\int_0^\delta T_f(t, r)\,dr \le \frac{\beta^\beta}{(\beta+1)^{\beta+1}}\delta^\beta \Lambda_\beta f(t). \tag{70.5}$$

Proof. Tonelli gives the identity in (70.4) by integrating first on each interval. The bound $x(1 - x/\delta)_+ \le \delta/4$ and $\int_0^\infty \#\{I \subset E_u\}\,du = \|Df\|_{\mathrm{TV}}/2$ prove its inequality. A pulse of width $\delta/2$ gives equality. Next $(1 - \ell/\delta)_+ = \delta^{-1}\int_0^\delta \mathbf{1}_{\{\ell < r\}}\,dr$, which proves the tail identity. Finally the maximum of $x^\beta(1-x)$ on $[0,1]$ is $\beta^\beta/(\beta+1)^{\beta+1}$. Apply this with $x = \ell/\delta$ when $\ell < \delta$, and integrate. □

These bounds are strictly more selective than unpaired Jordan masses. For $\delta = 1$ take

$$f = \mathbf{1}_{(-1/2,2)} + \mathbf{1}_{(1/2,3/4)}.$$

At $t = 0$ the incoming positive and outgoing negative Jordan masses are both one. The levels containing t all have component length $5/2$, however, so $S_1 f(0) = 0$. The same strict separation holds on a neighborhood of zero. The two nearby atoms lie on different height layers and do not form a short excursion through that roof. Conversely, for a pulse $f = h\mathbf{1}_{(a,b)}$ with $b - a < \delta$,

$$S_\delta f(t) = h(1 - (b-a)/\delta) = [f(t) - 2A_\delta f(t)]_+ \quad (a < t < b).$$

A genuine narrow peak is therefore retained, not canceled. Neither example is asserted to be a physical Lorentz source. The integrated bound (70.4) is an L^1 bound only; it is not converted into a pointwise collision-normalized estimate.

71. The exact-label short-excursion budget

We now apply the preceding theorem to the complete original scalar source $b = b^{\varepsilon}_{n,k,m,R}$. In particular the level components are formed after all physical contributions have been summed. The threshold superposition is used to prove BV membership only; no level component is matched separately inside that signed superposition.

For fixed $M, B, \delta > 0$ let

$$\mathcal{S}_M(B,\delta) = \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} S_\delta b^{\varepsilon(B)}(t), \tag{71.1}$$

$$\mathcal{T}_{\varepsilon,M}(r) = \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} T_{b^\varepsilon}(t,r). \tag{71.2}$$

Here $\mathcal{C}_{m,R,n,k}(M) = \{t : |Z_{m,R}(k,t,n)| \le M\}$; an empty set contributes zero. These are extended nonnegative quantities. The radius supremum is outside the roof essential supremum. No common exceptional roof set over uncountably many radii is required.

Theorem 71.1 (Comparison with the original pointwise error)**.** *For sufficiently large fixed B, $\varepsilon(B) = A_0 B^{-1/12}$ and every fixed $\delta > 0$,*

$$\mathcal{S}_M(B,\delta) \le \mathcal{E}_M + C_M B^{-1/192},$$
$$\mathcal{E}_M \le \mathcal{S}_M(B,\delta) + C_M B^{-1/192}. \tag{71.3}$$

The constants can be chosen independent of δ. Consequently, for every choice $\delta(B) > 0$ made before the collision limit,

$$\lim_{B\to\infty} \mathcal{S}_M(B,\delta(B)) = \mathcal{E}_M = \mathcal{E}_M^{\mathrm{path}}, \tag{71.4}$$

where equality with the path error uses the common positive path-remainder theorem in the dependency statement. The limit is understood in the extended nonnegative reals. It refers to the original exact labels and the full arithmetic reference, including its zero classes.

Proof. Write $P = m^2 p$, $G = \mathcal{L}_{m,R}$ and $P - G = m^2 b^{\varepsilon(B)} + e_B$, with the scalar error from the positive-remainder identity. Since $0 \le S_\delta b \le b$,

$$m^2 S_\delta b \le m^2 b \le |P - G| + |e_B|.$$

This proves the first inequality after the indicated limits. Conversely, Theorem 70.2 gives $b \le 2A_\delta b + S_\delta b$. The inherited fixed-width bound and $\varepsilon(B)^{1/16} = A_0^{1/16} B^{-1/192}$ yield

$$|P - G| \le m^2 S_\delta b + 2m^2 A_\delta b + |e_B|,$$

which proves the second inequality. The auxiliary spectral band used in the fixed-width bound is held fixed before $m \to \infty$ and is enlarged only afterwards. This is valid for each previously chosen $\delta(B)$ and does not evaluate a spectral estimate at a collision-dependent width.

For finite $\mathcal{E}_M$ the two bounds squeeze $\mathcal{S}_M$ to that value. If $\mathcal{E}_M = +\infty$, the second bound forces $\mathcal{S}_M(B,\delta) = +\infty$ for every such fixed B, δ, which proves the extended assertion. Equality with the path error is proved in Section 72. □

Theorem 71.2 (A length-tail estimate sufficient for closure)**.** *Suppose that for the original Lorentz source there exist $A < \infty$, $q \geq 0$, $\beta > 0$, and fixed $\varepsilon_0, r_0 > 0$ such that*

$$\mathcal{T}_{\varepsilon,M}(r) \leq A\varepsilon^{-q} r^{\beta} \quad (0 < \varepsilon < \varepsilon_0,\ 0 < r < r_0). \tag{71.5}$$

Then, for every $\delta < r_0$,

$$\mathcal{S}_M(B,\delta) \leq \frac{A}{\beta+1}\varepsilon(B)^{-q}\delta^{\beta}. \tag{71.6}$$

For any $a > 0$, the legal choice $\delta(B) = \varepsilon(B)^{(q+a)/\beta}$, for sufficiently large B, gives

$$\mathcal{E}_M \leq C_M B^{-1/192} + C_{M,a} B^{-a/12},$$

and hence the original scalar endpoint and the same-roof path numerator. The displayed power estimate is a hypothesis of this theorem, not a conclusion about the Lorentz family.

Proof. We justify the order of limits rather than invoking an unjustified interchange of a limsup and an integral. The function $T_f(t,r)$ is nondecreasing in r. For every positive integer N,

$$S_\delta f(t) \leq \frac{1}{N}\sum_{j=1}^{N} T_f(t, j\delta/N).$$

Fix N, δ, ε before m. Taking the essential suprema and then the collision limsup, the finite sum gives

$$\mathcal{S}_M(B,\delta) \leq \frac{1}{N}\sum_{j=1}^{N} \mathcal{T}_{\varepsilon(B),M}(j\delta/N) \leq A\varepsilon(B)^{-q}\delta^{\beta}\frac{1}{N}\sum_{j=1}^{N}(j/N)^{\beta}.$$

Now, and only now, let $N \to \infty$. Its Riemann sum tends to $1/(\beta+1)$, proving (71.6) without an integrability assumption on a collision-dependent envelope. Substitute the chosen width and apply (71.3). Finally enlarge B. □

There is also a moment version: an ordered bound $\limsup_m \sup_{R,n,k} m^2 \operatorname{ess\,sup}_{\mathcal{C}(M)} \Lambda_\beta b^\varepsilon \leq A\varepsilon^{-q}$ implies (71.6) with its constant replaced by $A\beta^\beta/(\beta+1)^{\beta+1}$, directly from (70.5). This too is a sufficient dynamical hypothesis, not a consequence of fixed-count finite variation.

Proposition 71.3 (The controlled source remains present)**.** *On the common physical roof kernel write $b = s + r$, where $s = s_{71,1}$ and $r = d_{71,1} + b^{\varepsilon,\mathrm{inc}} + e_{70}$. There are complementary physical weights retaining s in the controlled part, whose residual density is bounded by $S_\delta b$. All exact labels and every original physical contribution are retained.*

Proof. Put $q_\delta = A_\delta b$ and, on $\{r > 0\}$, set $\alpha = \min\{1, (2q_\delta - s)_+/r\}$; set $\alpha = 0$ elsewhere. The new densities are $s + \alpha r$ and $(1-\alpha)r$. Their sum is b, and

$$0 \leq (1-\alpha)r \leq [b - 2q_\delta]_+ \leq S_\delta b.$$

The coefficient is Borel. Multiplying the original residual weight by $\alpha(L)$ and $1-\alpha(L)$ realizes these densities on the same trajectory space and common kernel. No orbit, label or conditioning event changes. The term e_{70} is first clearance outside the selected disks; it does not include first incidence. The allocated weights are not asserted to have BV densities. □

The comparison (71.3) is a theorem for the full source. It eliminates the charge for mismatched height layers and identifies exactly the short excursions relevant to the pointwise error. Neither (71.5) nor its moment alternative has been derived here from collision dynamics. In particular the fixed-count constant in (69.4) and the integrated bound (70.4) do not prove either hypothesis. The original pointwise theorem remains the organizing target; no interval theorem or reduced label is substituted for it.

72. The common path numerator and positive arithmetic classes

The scalar source estimate need not be repeated for each bounded path mark. We give the precise downstream implication, keeping its analytic input visible.

Let $\mathcal{X} = C([0,1], \mathbb{R}^4)$, and use the bounded-Lipschitz class with supremum norm and Lipschitz constant at most one, including the constant function one. The notation $\|\cdot\|_{\mathrm{BL}^*}$ denotes its dual norm. The common roof kernel and a countable norming class fix all versions before taking essential suprema. Write

$$\mathbf{P}(t) = m^2 p(t) \mathsf{Q}_t, \qquad P(t) = \mathbf{P}(t)(1), \qquad G(t) = \mathcal{L}_{m,R}(k_1, k_2, t, n).$$

The reference path probability W_R is the one in the inherited path theorem, with its original covariance and arithmetic coefficient. For each fixed B the assumed, source-pinned theorem states

$$\mathbf{P} - G\mathsf{W}_R = \mathbf{b}_B + \mathbf{E}_B, \qquad \mathbf{b}_B \geq 0, \quad \mathbf{b}_B(1) = m^2 b^{\varepsilon(B)}.$$

The ordered error norm of $\mathbf{E}_B$ is at most $C_M B^{-1/192}$. This is the inherited positive path-remainder theorem, including its cylinder inversion and tightness inputs; it is not inferred from the scalar BV theorem.

Theorem 72.1 (Equality and normalization on the same roofs)**.** *Under this positive path-remainder input, $\mathcal{E}_M^{\mathrm{path}} = \mathcal{E}_M$, in the extended nonnegative reals. If the scalar endpoint vanishes, then on every original central target with $G \geq d > 0$,*

$$d_{\mathrm{BL}}(\mathsf{Q}_t, \mathsf{W}_R) \longrightarrow 0, \qquad P/G \longrightarrow 1, \qquad G/P \longrightarrow 1$$

uniformly in essential supremum, radius and exact labels within that target. On any positive-length roof window in the same target, the forward normalized arithmetic likelihood tends to one in essential supremum. No normalization is made on zero arithmetic classes. The path assertion is in bounded-Lipschitz dual norm, not total variation on path space.

Proof. Put $e_B = \|\mathbf{E}_B\|_{\mathrm{BL}^*}$. Positivity and the constant test give $\|\mathbf{b}_B\|_{\mathrm{BL}^*} = \mathbf{b}_B(1)$, and hence

$$|P - G| \leq \|\mathbf{P} - G\mathsf{W}_R\|_{\mathrm{BL}^*} \leq |P - G| + 2e_B.$$

Take the collision limsup at fixed B, then let $B \to \infty$. This proves the equality, including the infinite case by the left inequality.

Let $a = |P - G|$ and $D = \|\mathbf{P} - G\mathsf{W}_R\|_{\mathrm{BL}^*}$. When $G \geq d$ and $a \leq d/2$, one has $P \geq d/2$ and

$$d_{\mathrm{BL}}(\mathsf{Q}_t, \mathsf{W}_R) \leq \frac{D + a}{P} \leq \frac{2(D + a)}{d}, \qquad |P/G - 1| \leq a/d, \qquad |G/P - 1| \leq 2a/d.$$

These inequalities imply the stated limits. For a roof window J of positive length put $F_J = \int_J P$, $H_J = \int_J G$ and $r = \|P - G\|_{L^\infty(J)}/d$. Then $|F_J/H_J - 1| \leq r$. For $r < 1$ the normalized original roof laws satisfy

$$\left\| \frac{d\mathbb{P}_J}{d\mathbb{Q}_J} - 1 \right\|_\infty = \left\| \frac{P/G}{F_J/H_J} - 1 \right\|_\infty \leq \frac{2r}{1 - r}.$$

All ratios are taken only where the reference floor holds. No absolute continuity between path-space probabilities is asserted. □

The actual-return path additionally uses the inherited collision-to-return clock theorem on these same positive classes. The preceding argument supplies its collision numerator after the scalar endpoint; it introduces no new clock. Thus a proof of the full-source short-excursion decay in (71.4), for every central M, would propagate to the original downstream conclusions. The decay and an independent specialist audit of the inherited spectral and clock inputs are not supplied by the new deterministic inequalities.

Part 31. Finite physical currents and the pointwise error

73. Common roof versions and oriented physical interfaces

Throughout this part, $I = [9/20, 47/100]$, $m \geq 2$, $R \in I$, $1 \leq n \leq m$ and $k \in \mathbb{Z}^2$. The exact event is $\Pi = \Pi_{n,k,m,R}$, the roof is $L = L_{m,R}$, and the source is $\mu = \mathbf{1}_{\Pi}\nu_R^*$. Thus the factor c_*^{-1}, where $c_* = 91/(10000\pi)$, occurs once. The occupation count uses times $0, \dots, m-1$ and terminal membership uses time m. All parameters introduced in a collision limsup are fixed before that limsup unless an outer dependence is explicitly displayed.

Lemma 73.1 (One finite kernel for the original source)**.** *There is a Borel kernel $t \mapsto \mathsf{D}_t$ of finite positive measures on the physical source space such that*

$$\mu(A \cap L^{-1}(J)) = \int_J \mathsf{D}_t(A)\, dt, \qquad \mathsf{D}_t(1) = p(t), \qquad \mathsf{D}_t(L = t) = p(t) \tag{73.1}$$

for every Borel A, J, with the last two identities outside one Lebesgue-null set. It may be taken to be zero where $p = 0$. For a chosen Borel physical weight W, define $p_W(t) = \mathsf{D}_t(W)$. Every pointwise identity or order relation between the chosen physical weights is then inherited by their roof densities on this same kernel. In particular all the original positive-source decompositions and all bounded mark dominations hold simultaneously in this representation.

Proof. The collision space is standard Borel. Disintegrate μ over L with respect to its marginal $p(t)\, dt$, whose absolute continuity is Theorem 167.3. If Q_t is the resulting probability kernel, put $\mathsf{D}_t = p(t)Q_t$ and set it to zero on the remaining null set and on $\{p = 0\}$. A countable basis of roof intervals gives the fiber assertion in (73.1). Integration of an actual Borel weight against this kernel is a version of its pushforward density. Linearity and positivity prove the remaining assertions; in particular $|p_{Ww}| \leq Mp_W$ whenever $W \geq 0$ and $|w| \leq M$.

The assertion is about a common kernel and specified Borel weights, not a simultaneous choice of representatives for every equivalence class of measurable functions. A source identity holding only μ-almost everywhere is first imposed on a common full-measure set for the finite or countable collection of identities being used. No union over an uncountable set of arbitrary source-null sets is required. □

Apply this construction to the existing weights with densities

$$b = s + r, \qquad s = s_{71,1}, \qquad r = d_{71,1} + b^{\varepsilon,\mathrm{inc}} + e_{70}. \tag{73.2}$$

Here e_{70} is the original first-clearance source outside the selected disks. It does not include first incidence. Every term in (73.2) has its original exact labels. For a fixed $\delta > 0$, the function $q = A_\delta b$ is continuous and independent of the representative of b: it is a difference of two values of the absolutely continuous indefinite integral of b. Set $\alpha = \min\{1, (Kq - s)_+/r\}$ where $r > 0$, and set $\alpha = 0$ elsewhere. This is a Borel function after modification on the fixed exceptional set. If W_s, W_r are the original weights, the complementary replacements are exactly

$$W_s + \alpha(L)W_r, \qquad (1 - \alpha(L))W_r. \tag{73.3}$$

The fiber assertion in Lemma 73.1 proves that their densities are $s + \alpha r$ and $(1 - \alpha)r$. Where $r = 0$, positivity gives $\mathsf{D}_t(W_r) = 0$, so the convention for α has no effect. This verifies the

allocation on the original space, without transport, a new conditioning event, or a regularity claim for α.

If X is the original path observable with values in a Polish path space, use $X_{\#}\mathsf{D}_t$ and its weighted counterparts. A countable bounded-Lipschitz norming class fixes the common exceptional set for the path inequalities. For different radii the exceptional sets may be different. Every norm below has the order $\sup_R \operatorname{ess\,sup}_t$, not an essential supremum after an uncountable union of null sets.

Proposition 73.2 (The interface ledger)**.** *On regular physical patches, combine coarea fluxes before taking Jordan parts. An artificial interface cancels when the physical roof and source agree on its two sides. An actual source jump, physical itinerary boundary or exact-label boundary is retained. The compactly supported window $f\mathbf{1}_{(a,b)}$, with $f \in BV(a,b)$ and finite one-sided traces, contributes both atoms $f(a+)\delta_a - f(b-)\delta_b$ in addition to its interior derivative.*

Proof. In common coordinates let an interface normal n point from Ω_- to Ω_+. Write $F_\pm$ for the roofs and $V_\pm = \nabla F_\pm/|\nabla F_\pm|^2$. The boundary contributions of Proposition 78.2 are

$$-(F_-)_{\#}(w_- V_- \cdot n\, d\mathcal{H}^1) + (F_+)_{\#}(w_+ V_+ \cdot n\, d\mathcal{H}^1).$$

When both patches describe one smooth physical source, their traces coincide and these measures cancel. With a common roof but a true jump, the surviving flux is the pushforward of $(w_+V_+ - w_-V_-)\cdot n\, d\mathcal{H}^1$. When the roofs do not coincide they are not identified. A boundary with no patch in the same exact event retains its one-sided contribution. This covers physical-word and exact-label boundaries, not only chart cuts.

One-dimensional BV integration by parts gives

$$D(f\mathbf{1}_{(a,b)}) = (Df)|_{(a,b)} + f(a+)\delta_a - f(b-)\delta_b.$$

Adjacent pieces therefore cancel artificial traces and count a true jump once. The derivative has total mass zero when the function is compactly supported. At critical or singular sets the full current is the distributional limit specified in Proposition 78.2. Cancellation is asserted only for matching artificial interfaces. Neither that exhaustion nor the existence of a finite full current below asserts convergence of arbitrary patch currents in total variation. □

74. Finite variation of the complete physical current

We prove that the original full current is a finite measure at each collision count. The argument uses the complete finite-count height bound, not a bound on separately differentiated word charts. It also avoids an unproved definability assertion for a smooth cutoff.

74.1. **Thresholds on the unchanged finite collision graph.** The physical guard in (176.1) is

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon e_j}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon e'_j}\right), \quad e_j = q_0^{\min(j,m-j)/4}, \quad e'_j = q_0^{\min(j,m-1-j)/4}, \quad q_0 = 47/53.$$

Here Θ is the original fixed smooth cutoff, equal to zero on $(-\infty, 1]$ and one on $[2,\infty)$. Its values lie in $[0,1]$. Let $N = 2m+1$ and enumerate the displayed normalized margins as $a_1, \dots, a_N$. Put $\sigma = d\Theta$ and $V_\Theta = \|\sigma\|_{\mathrm{TV}} < \infty$. This is a finite signed measure of mass one, supported in $[1,2]$. The proof does not require monotonicity; for a nondecreasing cutoff $V_\Theta = 1$. For $y \in [1,2]^N$ define the positive threshold source

$$W_y(x) = 1 - \prod_{i=1}^{N} \mathbf{1}_{\{a_i(x) > y_i\}}, \qquad b_y(t)\, dt = L_{\#}(W_y \mu). \tag{74.1}$$

Lemma 74.1 (Exact threshold representation)**.** *The original complete source, with its original smooth guard, satisfies*

$$1 - H^{\varepsilon} = \int W_y \, d\sigma^{\otimes N}(y), \qquad b = \int b_y \, d\sigma^{\otimes N}(y) \quad \text{in } L^1(\mathbb{R}). \tag{74.2}$$

Each b_y is nonnegative and is dominated by the full exact-label density. Uniformly in R, ε, y, n, k at this count,

$$0 \le b_y \le L_m \quad a.e., \qquad L_m = CA^m/c_*. \tag{74.3}$$

Proof. The fundamental theorem of calculus gives $\Theta(a) = \int \mathbf{1}_{\{y<a\}} \, d\sigma(y)$ for every real a. Multiplication and the mass-one identity for $\sigma^{\otimes N}$ give the first equality. Finite signed Fubini applies because $0 \le W_y \le 1$ and μ is finite. It gives the equality of pushforward measures. The common kernel of Lemma 73.1, or joint Radon–Nikodym versions, gives the displayed L^1 identity. Finally Theorem 167.3 bounds every positive restriction of the complete source by (74.3). The thresholds are auxiliary integration parameters, not a random perturbation or a change to an actual orbit. □

Lemma 74.2 (A uniform finite number of monotonicity intervals)**.** *For each fixed m there is an integer $J_m < \infty$ such that every b_y has a representative which is monotone on at most J_m open intervals and has at most J_m intervening point cells. The same integer works for $R \in I$, all positive ε, all thresholds, and every attainable exact label at this count. No growth bound on J_m as $m \to \infty$ is asserted.*

Proof. Use the rational collision charts and the auxiliary physical graph of Lemma 167.2. At fixed m there are finitely many physical words, charts and exact labels. Circle, flight and reflection equations, positive incidence and flight length, and exclusion of an earlier hit are semialgebraic conditions on this graph. Clearance is the minimum over the finite physical candidate list, with the branch inequalities retained. Adding $a_i > y_i$ therefore gives a semialgebraic threshold condition. The original section is a finite union of rectangular chart pieces; its initial, occupation and terminal tests are finite Boolean conditions on this same graph. Displacement and occupation are fixed at (k, n), not summed in the pushforward under consideration.

The integration variables are the two initial coordinates, not the ambient variables of the auxiliary graph. In a rational angular chart $\alpha = \alpha_* + 2\arctan x$, the initial density is a constant multiple of $(1 + x^2)^{-1} dx \, dp$, with the section factor c_*^{-1} inserted once. The unique physical collision graph and projection show that its integrand, restricted by the exact event, the threshold source and $L \le t$, is globally subanalytic. The finite candidate alphabet follows from the uniform finite horizon. Positive denominators and first-hit comparisons are retained under projection; no continuation through a singularity is used.

It follows from the parameterized integration theorem [15, Theorem 1.3] that

$$M(R, \varepsilon, y, t) = \int W_y \mathbf{1}_{\{L \le t\}} \, d\mu$$

is constructible, after summing the finitely many disjoint initial chart pieces. Constructible functions are definable in $\mathbb{R}_{\mathrm{an,exp}}$. By (74.3), every t-section of M is L_m-Lipschitz. Define $g = \partial_t M$ where the finite derivative exists and $g = 0$ otherwise. The definition of a derivative by quantified inequalities shows that g is definable in that same o-minimal structure. We need no assertion that a constructible derivative is itself constructible. Absolute continuity gives $g = b_y$ almost everywhere, and $0 \le g \le L_m$ everywhere with this convention.

Uniform cell decomposition and the one-variable monotonicity theorem in an o-minimal structure [9] give a finite partition, with the roof coordinate last, into cells on which these sections are continuous and monotone or constant. The number of cells is uniform in the displayed parameters for the fixed formula at this m. Taking the maximum over the finitely many labels

and chart unions gives J_m. This is a finiteness statement for each fixed formula, not a bound uniform over formulas of increasing collision length. □

Theorem 74.3 (The complete scalar current is a finite measure)**.** *For every fixed $m \geq 2$, the original full density $b = b^{\varepsilon}_{n,k,m,R}$ belongs to $BV(\mathbb{R})$. Its canonical current of Theorem 78.1 is a finite signed measure, and for some finite J_m as above,*

$$\|Db\|_{\mathrm{TV}} \leq 4(J_m + 1)L_m V_{\Theta}^{2m+1} =: C_m. \tag{74.4}$$

The bound is uniform in R, ε and exact labels at this fixed count. The current is compactly supported and has total mass zero. It is the derivative of the complete original source, including first incidence and original clearance outside the selected disks.

Proof. Each monotone interval in Lemma 74.2 has variation at most L_m, and its traces lie in $[0, L_m]$. Joining the finitely many pieces and extending by zero outside the finite physical roof range gives $\|Db_y\|_{\mathrm{TV}} \leq 4(J_m + 1)L_m$. Isolated choices of representative do not affect the distributional derivative. For $\phi \in C_c^{\infty}(\mathbb{R})$, signed Fubini and (74.2) yield

$$|\langle Db, \phi\rangle| \leq \int |\langle Db_y, \phi\rangle|\, d|\sigma|^{\otimes N}(y) \leq 4(J_m + 1)L_m V_{\Theta}^{N} \|\phi\|_{\infty}.$$

The order-zero representation theorem identifies Db with a finite signed measure and proves (74.4); equivalently this is the distributional characterization of BV [13]. It is the same canonical derivative as before because both act as $-\int b\phi'$. A smooth function equal to one on the physical roof range shows that this derivative has total mass zero. □

The theorem establishes a property of the full physical sum. It does not assert that a source multiplied by an arbitrary bounded mark, nor an allocated subsource in (73.3), has a finite derivative. Nor does (74.4) control $m^2\|Db\|_{\mathrm{TV}}$ as m grows. The height L_m is exponential and the monotonicity number J_m has no growth estimate here. Finite variation supplies legitimate Jordan parts for the next argument; their collision-uniform concentration remains a separate quantitative question.

75. Paired directed flux at a common roof

A bound on the sum of directed Jordan masses charges an isolated entrance atom even when the neighboring average already pays for its height. A reserve of two permits a different estimate. It requires incoming and outgoing flux near the same roof, and hence does not demand that each genuine physical jump disappear.

Lemma 75.1 (A two-sided averaging inequality)**.** *Let $f \geq 0$ be compactly supported and in $BV(\mathbb{R})$, and let $\delta > 0$. Then for almost every t,*

$$[f(t) - 2A_\delta f(t)]_+ \leq \min\{(Df)^+([t-\delta, t]), (Df)^-([t, t+\delta])\}. \tag{75.1}$$

The same upper bound applies to $[f - KA_\delta f]_+$ for $K \geq 2$.

Proof. Use the usual BV representative and exclude its countable jump set. Put

$$L_\delta f(t) = \delta^{-1}\int_{t-\delta}^{t} f(u)\, du, \qquad R_\delta f(t) = \delta^{-1}\int_{t}^{t+\delta} f(u)\, du.$$

Integration of the increment formula for f gives

$$f(t) - L_\delta f(t) = \int_{(t-\delta, t)} \frac{u - t + \delta}{\delta}\, Df(du),$$

$$f(t) - R_\delta f(t) = -\int_{(t, t+\delta)} \frac{t + \delta - u}{\delta}\, Df(du).$$

The weights lie in $[0,1]$. Since $L_\delta f, R_\delta f \ge 0$ and $2A_\delta f = L_\delta f + R_\delta f$, the positive part of $f - 2A_\delta f$ is at most either of the two positive discrepancies on the left. Their respective bounds are the two Jordan masses in (75.1). Increasing K only decreases the positive part. The excluded set does not affect essential norms. □

For the complete current $\mathfrak{D}^\varepsilon = Db$ of Theorem 74.3, define

$$\mathcal{P}_\varepsilon(\delta) = \limsup_{m\to\infty} \sup_{R,n,k} \operatorname*{ess\,sup}_{t\in\mathbb{R}} m^2 \min\{(Db)^+([t-\delta,t]), (Db)^-([t,t+\delta])\}. \tag{75.2}$$

The interval masses are measurable functions of t. Their measures are the Jordan parts of the complete physically weighted current, not sums of variations of words, witnesses or threshold sources. The value is allowed to be infinite. Existence of these Jordan parts is now a conclusion for the original Lorentz source, rather than an additional finite-measure assumption.

Theorem 75.2 (A full-source paired-flux budget)**.** *For the original source and every fixed $\varepsilon, \delta > 0$,*

$$\mathcal{U}_{\varepsilon,\delta}(K) \le \mathcal{P}_\varepsilon(\delta) \le \tfrac{1}{2}\mathcal{V}_\varepsilon(\delta), \qquad K \ge 2. \tag{75.3}$$

For sufficiently large fixed B, put $\varepsilon(B) = A_0 B^{-1/12}$ and choose any $\delta(B) > 0$ before the collision limsup. Then

$$\mathcal{E}_M \le C_M B^{-1/192} + \mathcal{P}_{\varepsilon(B)}(\delta(B)). \tag{75.4}$$

The arithmetic main term, including every zero class, is unchanged. If one establishes, for some $q \ge 0$ and $\beta > 0$, the additional quantitative bound

$$\mathcal{P}_\varepsilon(\delta) \le A\varepsilon^{-q}\delta^\beta, \tag{75.5}$$

then for any $a > 0$ the choice $\delta(B) = \varepsilon(B)^{(q+a)/\beta}$ gives $\mathcal{E}_M \le C_M B^{-1/192} + C_a B^{-a/12}$.

Proof. Apply Lemma 75.1 to the full source, whose finite variation was proved in the preceding section. Taking the indicated essential suprema and collision limsup gives the first inequality of (75.3). The second follows from $\min(x,y) \le (x+y)/2$ and the definition of $\mathcal{V}_\varepsilon(\delta)$ in (79.11).

For fixed δ, the finite interval estimate (79.8), first with fixed auxiliary band B_* and then with $B_* \to \infty$ after the collision limit, gives $\mathcal{H}(A_\delta b) \le C\varepsilon^{1/16}$. The pointwise inequality $b \le 2A_\delta b + [b - 2A_\delta b]_+$ therefore yields $\mathcal{H}(b) \le 2C\varepsilon^{1/16} + \mathcal{P}_\varepsilon(\delta)$. Insert it in the positive-remainder error bound (181.7). This proves (75.4). Alternatively the source-preserving allocation of Lemma 79.1, with $K = 2$, retains $s_{71,1}$ and gives the same bound, since $B^{-1/4} \le B^{-1/192}$ for $B \ge 1$. The substitution in (75.5) gives $\mathcal{P}_{\varepsilon(B)}(\delta(B)) \le A\varepsilon(B)^a$. All choices depending on B precede $m \to \infty$; only afterwards is B enlarged. No estimate is evaluated at a collision-dependent spectral band. □

The pairing is consequential even at a fixed count. For $f = h\mathbf{1}_{(a,b)}$, the two genuine boundary atoms are $h\delta_a - h\delta_b$. If $b - a > 2\delta$, the two Jordan intervals in (75.1) cannot both contain their respective atoms, except at irrelevant endpoints. Thus the paired bound is zero although the directed sum has supremum at least h. If $b - a < \delta$, a roof interval can see both edges and the paired bound retains their height; at interior roofs $[f - 2A_\delta f]_+ = h(1 - (b-a)/\delta)$. The estimate cancels no physical atom. It uses the positive averaging capacity to distinguish an isolated edge from a narrow positive excursion.

The power bound (75.5) is a stated sufficient estimate, not a proved collision-uniform Lorentz bound. The constant C_m of (74.4) does not establish it. In particular finite variation and vanishing mass of a thin source are not substituted for a bound on its normalized essential height. The remaining problem is quantitative control of paired physical flux at the original exact labels.

76. The scalar and path-valued pointwise errors coincide

The full-source estimate should be used for all bounded-Lipschitz path tests on a common version, rather than proved anew for each numerator. The inherited positive path-remainder theorem gives an exact comparison of the two asymptotic errors. This comparison keeps the scalar endpoint as the target; it is not an assertion that the error has already vanished.

Let $\mathcal{C} = C([0,1],\mathbb{R}^4)$ and use the bounded-Lipschitz class $\mathcal{F}$ of Section 190, including the constant function one. At a fixed roof put

$$\mathbf{P}(t) = m^2 p(t)\mathsf{Q}_t, \qquad P(t) = \mathbf{P}(t)(1), \qquad G(t) = \mathcal{L}_{m,R}(k_1,k_2,t,n).$$

The reference path law is W_R, with the same covariance and finite arithmetic coefficient as in Theorem 190.3. For each fixed B let $\mathbf{b}_B$ be the kernel obtained from the original physical weight $1 - H^{\varepsilon(B)}$. Its mass is $m^2 b^{\varepsilon(B)}$. All kernels are those of Lemma 73.1.

Define the original central roof set and two ordered errors by

$$\begin{aligned} \mathcal{C}_{m,R,n,k}(M) &= \{t : |Z_{m,R}(k,t,n)| \le M\},\\ E_{\mathrm{s}}(M) &= \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |P(t) - G(t)|,\\ E_{\mathrm{p}}(M) &= \limsup_m \sup_{R,n,k} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \|\mathbf{P}(t) - G(t)\mathsf{W}_R\|_{\mathrm{BL}^*}. \end{aligned} \tag{76.1}$$

An empty target contributes zero. In particular $E_{\mathrm{s}}(M)$ is the original $\mathcal{E}_M$, not an interval-averaged error.

Theorem 76.1 (Equality of the ordered pointwise errors)**.** *On the hypotheses and conclusions of the inherited Theorem 190.3, for every $M < \infty$,*

$$E_{\mathrm{p}}(M) = E_{\mathrm{s}}(M). \tag{76.2}$$

Consequently vanishing of the original scalar pointwise error is equivalent to vanishing of the full path-numerator error in the bounded-Lipschitz dual norm at the same roofs and exact labels. This statement does not assert total-variation convergence on path space.

Proof. For fixed B write $\mathbf{E}_B = \mathbf{P} - G\mathsf{W}_R - \mathbf{b}_B$ and let $e_B = \|\mathbf{E}_B\|_{\mathrm{BL}^*}$. Positivity gives $\|\mathbf{b}_B\|_{\mathrm{BL}^*} = \mathbf{b}_B(1)$. The constant test one and the triangle inequality therefore give pointwise, on the common versions,

$$|P - G| \le \|\mathbf{P} - G\mathsf{W}_R\|_{\mathrm{BL}^*} \le |P - G| + 2e_B. \tag{76.3}$$

Indeed the middle term is at most $\mathbf{b}_B(1) + e_B$, whereas $|P - G| = |\mathbf{b}_B(1) + \mathbf{E}_B(1)| \ge \mathbf{b}_B(1) - e_B$. The inherited theorem bounds the collision limsup of the global essential supremum of e_B by $CB^{-1/192}$. Restricting to the original central set, taking the collision limit at fixed B, and then sending B to infinity proves (76.2). If the scalar error is infinite the lower inequality already gives equality in the extended sense. No interchange of a supremum over uncountably many path versions is made: one common kernel and a countable norming class were fixed first. □

The input to this theorem includes the inherited cylinder inversion, variation tightness and endpoint budget in modules 110 and 112. The equality neither reconstructs that spectral chain nor replaces its specialist audit. It shows that, once that chain is used as written, a second independent vanishing-height estimate for the path numerator is unnecessary.

Corollary 76.2 (Central excess and positive arithmetic classes)**.** *For fixed B, δ, K, define*

$$U_M(B,\delta,K) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} [b^{\varepsilon(B)} - KA_\delta b^{\varepsilon(B)}]_+(t).$$

For $K \geq 1$ there is C_M independent of B, δ, K such that

$$U_M(B,\delta,K) \leq E_{\mathrm{s}}(M) + C_M B^{-1/192},$$
$$E_{\mathrm{s}}(M) \leq U_M(B,\delta,K) + C_M(1+K)B^{-1/192}. \tag{76.4}$$

Thus if $K(B)B^{-1/192} \to 0$, the central excess has the same outer limit as the scalar and path errors, whenever these errors are finite. In particular their vanishing is equivalent. The paired flux estimate of Theorem 75.2 is a sufficient bound for this excess with $K = 2$.

On any of these central targets with $G \geq d > 0$, vanishing of $E_{\mathrm{s}}(M)$ gives, uniformly in essential supremum,

$$d_{\mathrm{BL}}(\mathsf{Q}_t, \mathsf{W}_R) \longrightarrow 0, \qquad P/G \longrightarrow 1, \qquad G/P \longrightarrow 1.$$

The unnormalized error comparison includes zero arithmetic classes; none of the displayed ratios or conditional laws is assigned there. On each original roof window contained in this central target with the same reference floor, the forward normalized arithmetic likelihood of Section 184 also tends to one in essential supremum.

Proof. The scalar test in the positive path-remainder theorem gives $P - G = m^2 b + \mathbf{E}_B(1)$. Hence the ordered central height of b is at most $E_{\mathrm{s}}(M) + C_M B^{-1/192}$, proving the first inequality. Conversely $b \leq KA_\delta b + [b - KA_\delta b]_+$, and the fixed-width estimate $\mathcal{H}(A_\delta b) \leq C\varepsilon(B)^{1/16}$ proves the second. All widths and reserves are fixed before the collision limit. This is localization to the target already in the original pointwise theorem, not deletion of any physical source or label.

For the conditional assertion let $D = \|\mathbf{P} - G\mathsf{W}_R\|_{\mathrm{BL}^*}$ and $a = |P - G|$. Where $G \geq d$ and $a \leq d/2$, $P \geq d/2$ and

$$d_{\mathrm{BL}}(\mathsf{Q}_t, \mathsf{W}_R) \leq \frac{D+a}{P} \leq \frac{2(D+a)}{d}, \qquad |P/G - 1| \leq a/d, \qquad |G/P - 1| \leq 2a/d.$$

Theorem 76.1 gives the required uniform limits. These are pointwise ratios of the original scalar densities; no absolute continuity of one path law with respect to another is asserted.

For the last assertion take a roof window J of positive length on which $G \geq d$. Write $F_J = \int_J P$, $H_J = \int_J G$, and put $r = \|P - G\|_{L^\infty(J)}/d$. Then $|F_J/H_J - 1| \leq r$. For $r < 1$ the original normalized roof laws satisfy

$$\left\| \frac{d\mathbb{P}}{d\mathbb{Q}} - 1 \right\|_\infty = \left\| \frac{P/G}{F_J/H_J} - 1 \right\|_\infty \leq \frac{2r}{1-r}.$$

Thus the forward essential-likelihood consequence requires no additional numerator estimate after the scalar endpoint. This is a consequence for the original window-normalized likelihood, not a replacement of the pointwise theorem by a window law. □

The actual-return path continues to require the inherited collision-to- return clock transfer on the same positive arithmetic classes. The new comparison supplies its same-roof collision numerator once the scalar endpoint is closed; it introduces neither a new clock nor an unproved conditional normalization. At present the quantitative paired-flux decay for the unrestricted Lorentz family remains to be established, so the unconditional scalar and return-path endpoint is not asserted by this comparison.

Part 32. The complete physical current and source recovery

77. The distributional current of the moving chart window

The radial balance in Section 81 is an identity of functions at a common roof. To differentiate that identity one must also differentiate the active window. We construct the resulting distribution,

including its entrance and exit terms. The construction does not assume finite variation at the outer edge of a selected disk.

Fix $m, R, \lambda = (n, k), \chi, \varepsilon$, and write $h = h_\chi$, $j = h/2$, $L_z = L_{z,\lambda}$ and $\overline{L}_z = j^{-1}\int_0^j L_z(v)\,dv$. All sums below run over the same finite selected family at this fixed exact record. Translation of a measure or distribution by a is denoted by τ_a. For $j < H < h$ put

$$\begin{aligned} X_H(t) &= \sum_z J_z \mathbf{1}_{(0,H)}(t - t_z) L_z(t - t_z), \\ Y_H(t) &= \sum_z J_z \overline{L}_z \mathbf{1}_{(0,H)}(t - t_z), \qquad C_H = X_H - Y_H. \end{aligned} \tag{77.1}$$

The cutoff is an auxiliary exhaustion, not a deletion from the physical source. The full functions X, Y, C use $H = h$ and agree with $\mathsf{X}, \mathsf{Y}, \mathsf{C}$ of the preceding section.

Theorem 77.1 (Entrance and exit currents)**.** *Let $H \in (j, h)$ be a continuity point of every L_z. The following identities hold as finite signed Radon measures on $\mathbb{R}$:*

$$DX_H = \sum_z J_z\big\{\tau_{t_z}(DL_z|_{(0,H)}) + L_z(0+)\delta_{t_z} - L_z(H-)\delta_{t_z+H}\big\}, \tag{77.2}$$

$$DY_H = \sum_z J_z \overline{L}_z(\delta_{t_z} - \delta_{t_z+H}), \tag{77.3}$$

$$\begin{aligned} DC_H = \sum_z J_z\big\{&\tau_{t_z}(DL_z|_{(0,H)}) + (L_z(0+) - \overline{L}_z)\delta_{t_z} \\ &- (L_z(H-) - \overline{L}_z)\delta_{t_z+H}\big\}. \end{aligned} \tag{77.4}$$

Each of the three measures has total mass zero. As $H \uparrow h$ through such values, the functions converge in $L^1(\mathbb{R})$ to X, Y, C, and their derivatives converge in distributions to DX, DY, DC. In particular DX and DC are well-defined compactly supported distributions of order at most one. They are not asserted to be finite measures without an additional outer-edge variation bound.

Proof. Theorem 83.1 gives $L_z \in BV(0, H')$ for every $H' < h$, including finite variation near zero. Thus the traces in the assertion exist. There are only countably many excluded values of H, even after the finite sum over z. For $\phi \in C_c^\infty(\mathbb{R})$, integration by parts for BV functions gives

$$-\int_0^H L_z(u)\phi'(t_z + u)\,du = \int_{(0,H)} \phi(t_z + u)\,dDL_z(u) + L_z(0+)\phi(t_z) - L_z(H-)\phi(t_z + H).$$

Multiply by J_z and sum. A constant profile gives the second formula, and subtraction gives the third. Testing against a smooth function equal to one on the common compact support proves zero total mass. Equivalently, the interior mass is $L_z(H-) - L_z(0+)$ and cancels the two traces.

Since $0 \le L_z, \overline{L}_z \le 1$,

$$\|X - X_H\|_1 + \|Y - Y_H\|_1 \le 2(h - H)\sum_z J_z.$$

The sum is finite at the fixed count. The same bound holds for $\|C - C_H\|_1$. Distributional differentiation is continuous under L^1 convergence: the error against ϕ is bounded by the corresponding L^1 error times $\|\phi'\|_\infty$. This also proves independence of the exhausting sequence. No collision-uniform bound on $\sum_z J_z$ is used or obtained. □

The exit terms in (77.4) are indispensable. For example, if $L_z = \mathbf{1}_{(a,h)}$ with $j < a < H$, then $\overline{L}_z = 0$ and this chart contributes $J_z(\delta_{t_z+a} - \delta_{t_z+H})$. Retaining only the interior birth would give a measure of nonzero total mass, which cannot be the derivative of this compactly supported function. This is a scalar check of the boundary formula, not a claim about realizable Lorentz words. If the outer trace does not exist, the complete right side of (77.4) has a distributional limit; the exit terms need not have a separate measure limit.

Proposition 77.2 (Gluing and Fourier representation)**.** *Artificial cuts of one profile cancel with their orientations before any variation is taken. At a genuine jump the surviving atom is its actual jump. With the Fourier convention $\widehat{f}(\xi) = \int e^{-i\xi t} f(t)\,dt$, one has*

$$\widehat{DX_H}(\xi) = \sum_z J_z e^{-i\xi t_z} \Big\{ \int_{(0,H)} e^{-i\xi u}\, dDL_z(u) + L_z(0+) - e^{-i\xi H} L_z(H-) \Big\} = i\xi \widehat{X}_H(\xi). \quad (77.5)$$

The analogous identities hold for Y_H, C_H and for their full window limits as distributions. For $\xi \neq 0$ the right side can be divided by $i\xi$; at zero the Fourier transform of the density, not that quotient as an undefined expression, is used.

Proof. Split $(0, H)$ at a. The left piece has exit trace $-L_z(a-)$; the right piece has entrance trace $L_z(a+)$. Their sum is the actual jump. The interior derivatives of the two open pieces exclude a, so this jump is counted once. At continuity it is zero. This proves the gluing assertion. Test (77.2) against $e^{-i\xi t}$ on the compact support, or apply the Fourier transform to the distributional derivative. Passage to the full window is justified by the L^1 convergence above. In particular $|\widehat{DX}(\xi)| \leq |\xi| \|X\|_1$ at each fixed count. This is a finite-count bound, not a local-limit estimate. □

One must not identify DC with the old scalar function C. Nor may the present identities be called an operator-resolvent bound. They supply the global derivative which such an argument would have to use, with every active-window term retained. All translations and physical coefficients occur before combining signed measures. No cancellation between different m, R, n, k is permitted.

78. A current on the complete original physical source

A representation on the selected disks alone leaves the first-incidence and outside sources untouched. We instead differentiate their positive sum on the original collision space. The result has no dependence on a choice of Morse disks or on an ordering of artificial chart cuts.

Fix $m, R, \lambda = (n, k)$ and $\varepsilon > 0$. Let $\Pi_{n,k,m,R}$ be the original exact-return event. Its occupation count uses collisions $0, \ldots, m-1$, and its terminal section test is made at collision m. On this event the original roof is $L_{m,R} = T_{n,R}$. Write

$$b = b^{\varepsilon}_{n,k,m,R} = \frac{d\mu^{\varepsilon}_{n,k,m,R}}{dt}, \qquad \mu^{\varepsilon}_{n,k,m,R} = (L_{m,R})_{\#}\big((1 - H^{\varepsilon}_{m,R})\mathbf{1}_{\Pi_{n,k,m,R}} \nu^*_R\big). \quad (78.1)$$

Here the first quotient denotes the Radon–Nikodym density of the displayed measure, not its distributional derivative. The density exists by the inherited coarea construction. It is nonnegative, integrable and supported in the finite physical roof range at this count. The probability ν^*_R already contains c_*^{-1}.

Theorem 78.1 (Complete physical current and partition independence)**.** *There is a canonical compactly supported distribution $\mathfrak{D}^{\varepsilon}_{n,k,m,R} = Db$ given by*

$$\langle \mathfrak{D}^{\varepsilon}_{n,k,m,R}, \phi \rangle = -\int_{\Pi_{n,k,m,R}} (1 - H^{\varepsilon}_{m,R}(x)) \phi'(L_{m,R}(x))\, d\nu^*_R(x). \quad (78.2)$$

It has order at most one. On every common source disintegration,

$$\begin{aligned} \mathfrak{D}^{\varepsilon} &= Db^{\varepsilon,\mathrm{inc}} + Db^{\varepsilon,\mathrm{clr}} \\ &= Ds_{71,1} + Dd_{71,1} + Db^{\varepsilon,\mathrm{inc}} + De_{70}. \end{aligned} \quad (78.3)$$

These are distributional identities on the whole roof line. In particular artificial disk boundaries and first-defect subdivisions cannot create an additional current in their sum. Physical itinerary boundaries and boundaries of the exact-label event are not erased.

Proof. The right side of (78.2) is bounded in absolute value by $\|\phi'\|_\infty$ times the mass of the displayed positive source. Pushforward gives $-\int b(t)\phi'(t)\,dt$, which is precisely Db. This proves existence, the support assertion, and the order bound. The two source partitions are (181.4) and (80.7), respectively, with the unchanged incidence source added. Differentiate their exact equalities. The result depends only on the original source in (78.1), not on any partition used to compute it. For bounded measurable insertions the same construction holds by linearity and domination in variation; no regularity of such an insertion is inferred. □

We record how the distribution is computed geometrically without continuing an orbit through a singularity. This also fixes the orientation of its boundary terms.

Proposition 78.2 (Regular coarea flux and exhaustion)**.** *In a regular coordinate patch of the original exact-label event, let $F = L_{m,R}$, let Ω be a bounded C^1 subdomain with compact closure and $|\nabla F| > 0$, and let w be its C^1 physical source density. Put $V = \nabla F/|\nabla F|^2$ and let b_Ω be the density of $F_\#(w\mathbf{1}_\Omega\,dx)$. Then*

$$Db_\Omega = F_\#\big(\operatorname{div}(wV)\,dx|_\Omega - (wV\cdot n)\,d\mathcal{H}^1|_{\partial\Omega}\big). \tag{78.4}$$

The same formula holds piecewise after common interfaces are combined with their orientations. Regular compact exhaustions of the original source converge to (78.2) *in distributions. This passage requires only convergence of source mass; it does not assert convergence in total variation of the currents.*

Proof. Since $V\cdot\nabla F = 1$, the divergence theorem applied to $\phi(F)wV$ gives

$$\int_\Omega \phi'(F)w\,dx = \int_{\partial\Omega}\phi(F)wV\cdot n\,d\mathcal{H}^1 - \int_\Omega \phi(F)\operatorname{div}(wV)\,dx.$$

Its negative is the action of Db_Ω, proving the formula. On a common interface carrying the same physical roof and density, the two outward normals have opposite signs. The interface terms therefore cancel. If the physical density has a true jump, their sum is instead the jump flux and must be retained. This assertion never identifies roofs or sources belonging to different exact labels.

For the exhaustion, first restrict the physical measure to regular coordinate patches and remove neighborhoods of the critical set. The singular set and the critical set have collision measure zero in the inherited finite-word coarea construction. On the remaining open cells, regular compact subdomains and smooth density cutoffs approximate the source in L^1. Piecewise smooth guards may be approximated there in the same way. All cutoffs are applied to actual physical states, not to their nonphysical continuations. If b_q is the resulting roof density, positive domination, or the contraction of variation under pushforward, gives $\|b_q - b\|_1 \to 0$. Consequently

$$|\langle Db_q - Db, \phi\rangle| \le \|b_q - b\|_1\|\phi'\|_\infty \longrightarrow 0.$$

This proves the asserted distributional limit and its independence of the exhaustion. Boundary flux near a critical point is retained in this limit, even when the corresponding source mass tends to zero. The entrance atom of a Morse step is the elementary example. Neither a uniform bound on $\|Db_q\|_{\mathrm{TV}}$ nor a limit exchange with $m\to\infty$ has been made. □

In particular, for smooth physical factors $a_1,\dots,a_N$ the first-defect identity and its derivative are

$$\sum_{i=1}^N (1-a_i)\prod_{r<i}a_r = 1 - \prod_{i=1}^N a_i, \qquad D\left(\sum_{i=1}^N (1-a_i)\prod_{r<i}a_r\right) = -D\left(\prod_{i=1}^N a_i\right).$$

Thus derivatives of an artificial witness subdivision are combined before a Jordan decomposition. This does not permit cancellation of positive incidence and clearance densities: it is cancellation in the derivative of their unchanged positive sum.

78.1. **A fixed-width primitive of the complete current.** For $\delta > 0$ set

$$\rho_\delta(t) = \frac{\mathbf{1}_{(-\delta,\delta)}(t)}{2\delta}, \qquad A_\delta b = \rho_\delta * b, \qquad \kappa_\delta(t) = \begin{cases} -(\delta + t)/(2\delta), & -\delta < t < 0, \\ (\delta - t)/(2\delta), & 0 < t < \delta, \\ 0, & |t| \geq \delta. \end{cases} \tag{78.5}$$

Its value at zero is irrelevant to the distribution. Notice its unit upward jump there.

Theorem 78.3 (Exact current primitive at the original roof)**.** *Convolution in distributions gives the L^1 function*

$$\kappa_\delta * \mathfrak{D}^\varepsilon = b - A_\delta b. \tag{78.6}$$

Thus the equality holds almost everywhere on the original roof line. Its Fourier multiplier is

$$\widehat{\kappa}_\delta(\xi) = \frac{1 - \sin(\delta\xi)/(\delta\xi)}{i\xi} \quad (\xi \neq 0), \qquad \widehat{\kappa}_\delta(0) = 0. \tag{78.7}$$

In particular $|\widehat{\kappa}_\delta(\xi)| \leq \min\{\delta^2|\xi|/6, 2/|\xi|\}$ for $\xi \neq 0$.

Proof. Direct distributional differentiation gives $D\kappa_\delta = \delta_0 - \rho_\delta\, dt$. The unit atom is essential. Both κ_δ and b have compact support at the fixed count, so differentiation commutes with their convolution:

$$\kappa_\delta * Db = (D\kappa_\delta) * b = b - \rho_\delta * b.$$

The right side is in L^1 and fixes its almost-everywhere version. The Fourier transform of the box average is $\sin(\delta\xi)/(\delta\xi)$. This proves the multiplier formula away from zero. Its limit at zero is zero. Finally $1 - \sin x/x = \int_0^1 (1 - \cos(rx))\, dr$ is nonnegative and at most $x^2/6$, and its absolute value is at most two. □

This identity is valid even if Db is not a finite measure. It can also be read by first convolving Db with a smooth approximate identity at each fixed count, and then taking that approximation parameter to zero. A positive part in the next section is taken only after (78.6) has produced an actual function. The factor $1/|\xi|$ alone supplies no integrable high-frequency majorant and no transfer-operator resolvent estimate.

79. Recovery of every physical defect and a quantitative height bound

We now apply a receiving capacity to the complete physical source, not only to the selected disks. The preceding recovered source is retained. The new allocation acts also on first incidence and on the outside source, using only a bounded weight on the same original exact-label event.

For a family of exact-label densities, the ordered height used throughout this part is

$$\mathcal{H}(f) = \limsup_{m\to\infty} \sup_{\substack{R\in I,\, 1\leq n\leq m \\ k\in\mathbb{Z}^2}} m^2 \|f_{n,k,m,R}\|_{L^\infty(\mathbb{R})}, \qquad I = [9/20, 47/100]. \tag{79.1}$$

Let $\mathcal{E}_M$ denote the collision limsup of the left side of (Q.1). Every density norm is an essential supremum. Suppressed width and reserve parameters are fixed before this collision limsup. The term e_{70} always refers to the original first-clearance source outside the selected disks; incidence is retained separately, not counted in that term.

79.1. **A fixed-width capacity on the full source.** Let $b = b^{\varepsilon}_{n,k,m,R}$ be (78.1), set $s = s_{71,1}$, and write

$$r = b - s = d_{71,1} + b^{\varepsilon,\mathrm{inc}} + e_{70} \geq 0. \tag{79.2}$$

Fix $\delta > 0$ and $K \geq 1$ before the collision limit, and put $q = A_\delta b$. At almost every roof define

$$\alpha(t) = \begin{cases} \min\{1, (Kq(t) - s(t))_+ / r(t)\}, & r(t) > 0, \\ 0, & r(t) = 0, \end{cases} \qquad s_{72,K} = s + \alpha r, \quad r_{72,K} = (1-\alpha) r. \tag{79.3}$$

The values on exceptional coarea levels are immaterial.

Lemma 79.1 (Recovery preserving an already controlled source)**.** *For nonnegative finite b, s, q with $s \leq b$, the preceding allocation satisfies*

$$s_{72,K} = \min\{b, \max(s, Kq)\}, \qquad s \leq s_{72,K} \leq s + Kq, \tag{79.4}$$

$$r_{72,K} = [b - \max(s, Kq)]_+, \qquad 0 \leq r_{72,K} \leq \min\{r, [b - Kq]_+\}. \tag{79.5}$$

Their sum is exactly b. The controlled source is nondecreasing in K and the residual is nonincreasing. Where $r > 0$, α is maximal among $a \in [0,1]$ for which $s + ar \leq \max(s, Kq)$. At $r = 0$ its value is immaterial.

Proof. If $Kq \leq s$, no residual is recovered. If $s < Kq < b$, the recovered amount is $Kq - s$. If $Kq \geq b$, the entire residual is recovered. These three cases prove both identities and all bounds. They also prove monotonicity and the stated scalar maximality. □

The allocation is realized on the physical source. Let W_s, W_r be its existing complementary weights, so that their pushforwards have densities s, r. Replace them by $W_s + \alpha(L_{m,R}) W_r$ and $(1 - \alpha(L_{m,R}))W_r$. These are nonnegative complementary weights with the same sum, the same section normalization and the same exact labels. In particular

$$r_{72,K} = (1-\alpha) d_{71,1} + (1-\alpha) b^{\varepsilon,\mathrm{inc}} + (1-\alpha) e_{70}. \tag{79.6}$$

Every original physical state is still accounted for. The three terms on the right need not vanish. The allocation borrows no capacity from a different m, R, n, k and makes no transport of physical trajectories. The interval average is an auxiliary upper capacity; the law being estimated and the recovered source are not roof-window laws.

Theorem 79.2 (Ordered recovery on the complete physical source)**.** *For fixed $0 < \varepsilon \leq \chi/4$, $0 < \chi \leq 1/10$, $\delta > 0$ and $K \geq 1$, the sources in* (79.3) *satisfy*

$$\mathcal{H}(s_{72,K}) \leq C\chi^6 + CK\varepsilon^{1/16}. \tag{79.7}$$

The constants do not depend on δ, the collision count, the exact label, the number of words, or the number of first-defect witnesses. Here $\varepsilon, \chi, \delta, K$ are fixed before the collision limsup.

Proof. For each fixed auxiliary band B_*, apply (179.2) with interval length 2δ and divide by 2δ. It gives the finite-count bound

$$\begin{aligned} \sup_{R,n,k} m^2 \|q\|_\infty \leq C\varepsilon^{1/16} \left(1 + \frac{1}{2\delta B_*}\right) \\ + C_{B_*} \frac{1 + 2\delta B_*}{2\delta} m^3 \rho_{B_*}^{m/2}. \end{aligned} \tag{79.8}$$

First let $m \to \infty$ with δ, B_*, ε fixed. Then let $B_* \to \infty$, still with δ fixed. This proves $\mathcal{H}(q) \leq C\varepsilon^{1/16}$. This is only a bound for the average, not for b. Theorem 80.2 with reserve one gives $\mathcal{H}(s) \leq C\chi^6$. Finally use $s_{72,K} \leq s + Kq$ from (79.4). The averaging coefficient is used only in a positive source comparison; no spectral norm of that coefficient is required. □

79.2. **The complete excess as a global current.** The quantity

$$\mathcal{U}_{\varepsilon,\delta}(K)=\limsup_{m\to\infty}\sup_{R,n,k} m^2\left\|\left[\kappa_\delta * \mathfrak{D}^{\varepsilon}_{n,k,m,R}-(K-1)A_\delta b\right]_+\right\|_\infty \tag{79.9}$$

is well-defined as an extended nonnegative number. Its integrand is $[b-KA_\delta b]_+$ by Theorem 78.3. Thus every signed physical flux, including all moving boundaries, is combined at the same roof before a positive part is taken. This definition is not a claim that the ordered excess is small.

Theorem 79.3 (A complete physical-current budget)**.** *For every sufficiently large fixed B, choose*

$$\varepsilon(B)=A_0B^{-1/12},\qquad \chi(B)=\sqrt{\varepsilon(B)},\qquad \delta(B)>0,\qquad K(B)\ge 1.$$

All four choices precede the collision limsup. Then the original arithmetic raw error satisfies

$$\mathcal{E}_M\le C_MB^{-1/192}+CB^{-1/4}+CK(B)B^{-1/192}+\mathcal{U}_{\varepsilon(B),\delta(B)}(K(B)). \tag{79.10}$$

Consequently the unrestricted two-sided arithmetic pointwise law follows if $K(B)B^{-1/192}\to 0$ and the last term tends to zero. This is a sufficient condition on the complete original source, not an assertion that the Lorentz family already satisfies it.

Proof. The exact positive-remainder representation (181.7) bounds the original error by $C_MB^{-1/192}+\mathcal{H}(b)$. Use $b=s_{72,K}+r_{72,K}$, Theorem 79.2, and the pointwise inequality $r_{72,K}\le[b-KA_\delta b]_+$. The stated substitutions give $\chi^6=A_0^3B^{-1/4}$ and $\varepsilon^{1/16}=A_0^{1/16}B^{-1/192}$. This proves the budget. Only after this fixed-band limit may $B\to\infty$. The original finite arithmetic transition kernel, including its zero classes, has not been changed. □

There is no assertion that (79.10) is numerically smaller than the three-term budget of Corollary 81.3. Its gain is different: it recovers portions of all three physical remainders, preserves the old controlled source, and gives one partition-independent current on which cancellations can be studied. Taking $K=1$ retains the $B^{-1/192}$ controlled rate. A growing reserve must now pay that rate, not the selected-disk rate $B^{-1/4}$.

79.3. **A quantitative local-flux sufficient estimate.** Suppose, additionally, that $\mathfrak{D}^{\varepsilon}_{n,k,m,R}$ is a finite signed measure for each indicated finite count. Define

$$\begin{aligned}\mathcal{V}_\varepsilon(\delta)=\limsup_{m}\sup_{R,n,k,t} m^2\big[&(\mathfrak{D}^{\varepsilon})^+([t-\delta,t])\\ &+(\mathfrak{D}^{\varepsilon})^-([t,t+\delta])\big].\end{aligned} \tag{79.11}$$

The Jordan parts here are those of the complete physically weighted current, not sums of wordwise variations. The finite-measure hypothesis is explicit and is not inferred from the distributional construction.

Proposition 79.4 (Local flux controls essential height)**.** *Under this finite-measure hypothesis,*

$$\mathcal{U}_{\varepsilon,\delta}(K)\le\mathcal{U}_{\varepsilon,\delta}(1)\le\tfrac12\mathcal{V}_\varepsilon(\delta)\qquad(K\ge1). \tag{79.12}$$

If, for some $A<\infty$, $q\ge0$ and $\beta>0$, one proves

$$\mathcal{V}_\varepsilon(\delta)\le A\varepsilon^{-q}\delta^{\beta} \tag{79.13}$$

for every sufficiently small fixed ε,δ, then for any $a>0$ the choice $K(B)=1$, $\delta(B)=\varepsilon(B)^{(q+a)/\beta}$ gives

$$\mathcal{E}_M\le C_MB^{-1/192}+CB^{-1/4}+C_aB^{-a/12}. \tag{79.14}$$

The limit order remains fixed B, then $m\to\infty$, then $B\to\infty$.

Proof. A finite measure Db makes b a BV function. At its continuity points, the convolution in (78.6) has positive kernel on currents below t and negative kernel on currents above t. Both kernel magnitudes are at most $1/2$. The positive part is therefore at most

$$\tfrac{1}{2}\big[(Db)^+([t-\delta,t])+(Db)^-([t,t+\delta])\big].$$

The excluded jump points are countable. The inequality thus bounds the essential height; common versions are fixed before taking the suprema. Subtracting the nonnegative $(K-1)A_\delta b$ can only decrease the positive part. This proves (79.12). Under (79.13), the proposed width gives $\mathcal{V}_{\varepsilon(B)}(\delta(B))\le A\varepsilon(B)^a$. Insert this bound in (79.10). □

Neither the finite-measure hypothesis for the full source nor (79.13) is proved for the unrestricted Lorentz family here. A bounded variation statement for each truncated word does not supply either missing ordered estimate. For example, the scalar densities $m^{-2}\mathbf{1}_{(0,e^{-m})}$ have vanishing normalized mass on each fixed window but normalized essential height one; their two boundary atoms retain order-one normalized local variation. This is a test against an invalid limit exchange, not a Lorentz counterexample.

Thus the complete first-incidence and outside-source requests remain part of the same full problem. Their recovered portions now have the unconditional height estimate (79.7); their remaining positive portions are displayed in (79.6). No source is discarded and no unrestricted height is declared proved. Same-roof bridges, forward essential likelihoods and pointwise roof-conditioned path laws still require the remaining scalar estimate and their inherited numerator hypotheses. Conditioning on a zero arithmetic reference class remains undefined. The full pointwise target is retained throughout.

Part 33. Physical capacity recovery and signed radial balance

80. Recovery of angular loss by a common physical collar

A chart may lose angular occupation while another chart of the same exact record gains it. The original comparison treats the losses separately. We retain its controlled source and use the remaining receiving capacity before declaring a loss. The operation is a scalar weighting of the original probability space. It neither joins orbits nor exchanges their integer labels.

80.1. **A positive recovery principle.**

Lemma 80.1 (Maximal recovery with a prescribed capacity)**.** *Let X,O,Y,S,D be nonnegative finite numbers, with*

$$O\le\min(X,Y),\qquad S\le EO,\qquad D\le E(X-O),\qquad E>0.$$

For $K\ge1$, define

$$\theta_K=\begin{cases}\min\{1,(KY-O)/(X-O)\}, & X>O,\\ 1, & X=O.\end{cases}\tag{80.1}$$

Then $0\le\theta_K\le1$ and

$$S_K:=S+\theta_KD\le E\min(X,KY),\tag{80.2}$$

$$D_K:=(1-\theta_K)D\le E(X-KY)_+,\qquad S_K+D_K=S+D.\tag{80.3}$$

Moreover $S_K\ge S$ and $0\le D_K\le D$. When $X>O$, θ_K is the largest $\theta\in[0,1]$ satisfying $O+\theta(X-O)\le KY$. For fixed data, D_K is nonincreasing in K.

Proof. Since $KY \geq O$, the coefficient is nonnegative. Directly,

$$O + \theta_K(X - O) = \min(X, KY), \qquad (1 - \theta_K)(X - O) = (X - KY)_+.$$

These identities include $X = O$, when $D = 0$ and $KY \geq X$. The bounds for S, D prove the two estimates. The remaining statements follow from the defining inequality and from monotonicity in K. □

The optimality here is for the displayed scalar capacity constraint. It is not an optimal-transport statement about the physical measure. In particular unequal guards or physical Jacobians have not been replaced by equal ones; E bounds them in the source comparison.

80.2. **Exactly labelled capacities at a common roof.** Fix m, R, $\lambda = (n, k)$, $0 < \chi \leq 1/10$ and $0 < \varepsilon \leq \chi/4$. Set

$$h = h_\chi, \qquad j = h/2, \qquad J = (0, j), \qquad E_\chi = \exp(K_\chi r_\chi).$$

Every angular profile below has the same exact label λ. For a selected chart put

$$\overline{L}_{z,\lambda} = j^{-1} \int_0^j L_{z,\lambda}(v)\, dv, \qquad o_{z,\lambda}(u) = j^{-1} \int_0^j \min\{L_{z,\lambda}(u), L_{z,\lambda}(v)\}\, dv.$$

Here $L_{z,\lambda}(u) = \ell_{z,\lambda}(\sqrt{2u})$ is the actual angular fraction, including every physical and section test. For a roof t write

$$\mathcal{Z}_\lambda(t) = \{z : 0 < u_z(t) := t - t_z < h\}.$$

The selected chart set depends on m, R; it is finite at each count. Define the three capacities

$$\begin{aligned} \mathsf{X}_\lambda(t) &= \sum_{z \in \mathcal{Z}_\lambda(t)} J_z L_{z,\lambda}(u_z(t)), \\ \mathsf{O}_\lambda(t) &= \sum_{z \in \mathcal{Z}_\lambda(t)} J_z o_{z,\lambda}(u_z(t)), \\ \mathsf{Y}_\lambda(t) &= \sum_{z \in \mathcal{Z}_\lambda(t)} J_z \overline{L}_{z,\lambda}. \end{aligned} \tag{80.4}$$

The dependence on m, R, χ, ε is suppressed, not averaged. The normalization c_*^{-1} is already included in each J_z. There is no summation over different return indices or displacements.

Use the unchanged s_{70}, d_{70} of (84.1). On common coarea versions,

$$0 \leq \mathsf{O}_\lambda \leq \min(\mathsf{X}_\lambda, \mathsf{Y}_\lambda), \quad s_{70} \leq E_\chi \mathsf{O}_\lambda, \quad d_{70} \leq E_\chi(\mathsf{X}_\lambda - \mathsf{O}_\lambda). \tag{80.5}$$

Indeed $b_z(t_z + u) \leq E_\chi J_z L_{z,\lambda}(u)$, since the original guard is at most one. Multiply by the old overlap and loss weights, and use (82.3). This also proves that zero angular capacity produces zero source; no quotient is taken there. These estimates hold for the actual nonconstant guards.

Insert $(\mathsf{X}_\lambda, \mathsf{O}_\lambda, \mathsf{Y}_\lambda)$ in (80.1), and write the resulting measurable coefficient as $\theta_{K,m,R,\lambda}(t)$. Define

$$s_{71,K} = s_{70} + \theta_{K,m,R,\lambda} d_{70}, \qquad d_{71,K} = (1 - \theta_{K,m,R,\lambda}) d_{70}. \tag{80.6}$$

This is a weighting on the original source: on an actual chart state x use $\theta_{K,m,R,\lambda}(L_{m,R}(x))$ in its original loss weight. Every multiplier lies in $[0, 1]$. The densities in (80.6) therefore are the pushforwards of these source weights, not merely an algebraic replacement of densities. The weights may depend measurably on the whole exactly labelled profile; no spectral norm of that insertion is used.

Theorem 80.2 (Ordered recovery on the complete selected disks)**.** *Fix $K \geq 1$, χ and ε before the collision limit. The complete original clearance source has the positive identity*

$$b^{\varepsilon,\mathrm{clr}} = s_{71,K} + d_{71,K} + e_{70}. \tag{80.7}$$

The outside source is exactly the one in (95.8)*. On every selected disk, including its outer annuli,*

$$s_{71,K} \geq s_{70}, \qquad 0 \leq d_{71,K} \leq d_{70}, \tag{80.8}$$

$$d_{71,K}(t) \leq E_\chi(\mathsf{X}_\lambda(t) - K\mathsf{Y}_\lambda(t))_+. \tag{80.9}$$

Moreover

$$\mathcal{H}(s_{71,K}) \leq \frac{3}{2}CKh_\chi \exp\{(1+1/\sqrt{2})K_\chi r_\chi\} \leq C'K\chi^6. \tag{80.10}$$

The constants are independent of the collision count, the exact label and the number of charts. Selected-chart and outside pieces are disjoint events; controlled and residual chart pieces are complementary weights, not disjoint events.

Proof. Apply Lemma 80.1 to (80.5). It gives the pointwise claims and $s_{71,K} \leq E_\chi K\mathsf{Y}_\lambda$. Adding the unchanged outside source proves the complete identity. For the ordered estimate, the lower unguarded physical profile on $0 < v < j$ gives

$$J_z\overline{L}_{z,\lambda} \leq e^{K_\chi\sqrt{2j}}\widehat{q}_{z,j;\lambda}.$$

At roof t all contributing centers belong to $[t-h,t]$. Thus the sum of these positive collar averages is bounded by Theorem 85.2, with $w = h$ and $s = j$. The interval length plus collar width is $h + j = 3h/2$; the receiving average has already been divided by j. This proves the first bound, since $\sqrt{2j} = r_\chi/\sqrt{2}$. The exponential is uniformly bounded because $K_\chi r_\chi = Cr_0\chi$ and $h_\chi = r_0^2\chi^6/2$.

The estimate is applied to one positive physical event after the sum over centers. In particular the dependence of θ on t or on all the charts introduces no chart-count or label-count factor. All widths remain fixed throughout that collision limit. □

Corollary 80.3 (Complete selected-source height with a receiving reserve)**.** *Suppose $\mathsf{X}_\lambda(t) \leq K\mathsf{Y}_\lambda(t)$ for almost every roof, uniformly in R, λ at all sufficiently large counts. Then the entire original selected-chart source has height at most $C'K\chi^6$. No individual angular profile is required to be monotone, or to have a bounded variation independent of the word. In particular the old loss d_{70} has this same height bound.*

Proof. The residual in (80.9) vanishes. The entire selected source equals $s_{71,K}$, and d_{70} is its positive subsource. Apply (80.10). □

This is a condition on the actual, physically weighted capacities; it is not asserted for every Lorentz record. It is strictly weaker than requiring each chart to have no loss to its receiving collar. For example, as a scalar diagnostic, take two equal weights and equal centers with profiles $L_1(u) = \mathbf{1}_{(j,h)}(u)$ and $L_2(u) = \mathbf{1}_{(0,j)}(u)$. At $u > j$ the first profile is a late birth and its old source is wholly loss, whereas $\mathsf{X} = \mathsf{Y}$ and the new residual is zero. More generally complementary profiles with equal weights have this property, regardless of their number of radial changes. These are capacity examples, not assertions that arbitrary profiles occur as Lorentz words. Different labels, radii or collision counts cannot be pooled. For an isolated late birth with $\mathsf{Y} = 0 < \mathsf{X}$, no finite K recovers its source. No exhaustion claim is hidden in the parameter K.

81. The signed two-sided current and the complete raw budget

For a receiving collar below part of the source interval, radial births contribute to the deficit as well as radial deaths. We now identify their exact signed combination. The positive part will be taken only after the physical weights, the common-roof translations and all charts of one exact record have been combined.

81.1. **A two-sided BV identity.** Let $0 < j < h$, let $L \in BV(0, S)$ for every $S < h$, with finite variation near zero, and suppose $0 \le L \le 1$. Write $\mu = DL$ and $\overline{L} = j^{-1} \int_0^j L(v)\, dv$. Define

$$F_j(s) = \min\{s/j, 1\} \quad (0 < s < h), \qquad k_j(u, s) = \begin{cases} F_j(s), & 0 < s < u, \\ -(1 - F_j(s)), & u < s < h. \end{cases} \tag{81.1}$$

The value at $s = u$ is immaterial at continuity points of L. For each $u < h$ the kernel is supported below $\max(u, j) < h$, so its integral uses a finite signed measure on a fixed compact radial range together with the finite-variation neighborhood of zero. No assertion of finite total variation up to h is needed.

Lemma 81.1 (Signed collar balance)**.** *For almost every $u \in (0, h)$,*

$$L(u) - \overline{L} = \int_{(0,h)} k_j(u, s)\, d\mu(s). \tag{81.2}$$

In particular

$$(L(u) - \overline{L})_+ \le \int_{(0,u)} F_j(s)\, d\mu^+(s) + \int_{(u,h)} (1 - F_j(s))\, d\mu^-(s). \tag{81.3}$$

Both formulas retain atoms of DL. They introduce no endpoint mass at 0 or h.

Proof. Choose a continuity point u, and then S with $\max(u, j) < S < h$. For almost every $v \in (0, j)$, the fundamental theorem for the precise BV representative expresses $L(u) - L(v)$ as $\mu((v, u))$ if $v < u$, and as $-\mu((u, v))$ if $v > u$. The omitted discontinuities are countable. Average these identities in v. A point $s < u$ is traversed by a fraction $|(0, j) \cap (0, s)|/j = F_j(s)$ of the receiving interval. A point $s > u$ is traversed in the opposite direction by a fraction $|(0, j) \cap (s, h)|/j = 1 - F_j(s)$. Fubini for the finite variation measure on $(0, S)$ proves (81.2). Its right side is independent of S. The Jordan decomposition and the signs of k_j give (81.3). □

For a jump of size a at s_0, its exact contribution is $aF_j(s_0)$ when $s_0 < u$, and $-a(1 - F_j(s_0))$ when $s_0 > u$. Thus a positive-radius birth below u has a positive cost. A death above u also has a positive cost. The reverse orientations give credits rather than additional loss. In the isolated late-birth example of Section 80, take the birth at $s_0 > j$. For $u > s_0$ the current is a positive unit atom, $\mu^- = 0$, but $L(u) - \overline{L} = 1$. A bound using only negative radial variation cannot control this whole-disk comparison. This does not alter the valid inner-to-outer formula (83.3), whose receiving geometry is different.

81.2. **The physically weighted net current.** Return to $h = h_\chi$, $j = h/2$ and the exact label λ. Theorem 83.1 applies to $\mu_{z,\lambda} = DL_{z,\lambda}$ on every required radial range and identifies it as

$$\mu_{z,\lambda} = -\frac{1}{2\pi} \pi_\# (n_u\, \mathcal{H}^1|_{\partial^* E_{z,\lambda}}). \tag{81.4}$$

Use the actual Boolean physical indicator and attribute a coincident primitive interface once, as in Section 83. Define the signed scalar function

$$\mathsf{C}_\lambda(t) = \sum_{z \in \mathcal{Z}_\lambda(t)} J_z \int_{(0,h)} k_j(t - t_z, s)\, d\mu_{z,\lambda}(s). \tag{81.5}$$

This is a common-roof evaluation of signed radial currents. It is not asserted to be the distributional derivative of the total source as a function of t; no derivative of the moving active-chart set is taken.

Theorem 81.2 (Exact net-current identity and recovered loss)**.** *On common coarea versions,*

$$\mathsf{C}_\lambda(t) = \mathsf{X}_\lambda(t) - \mathsf{Y}_\lambda(t). \tag{81.6}$$

Consequently, for every fixed $K \geq 1$,

$$d_{71,K}(t) \leq E_\chi[\mathsf{C}_\lambda(t) - (K-1)\mathsf{Y}_\lambda(t)]_+. \tag{81.7}$$

In particular this retains cancellation between births and deaths, and between different selected charts of the same exact label. A sufficient, generally larger, upper bound at $K = 1$ is

$$\begin{aligned} d_{71,1}(t) \leq E_\chi \sum_{z \in \mathcal{Z}_\lambda(t)} J_z \Big[& \int_{(0,u_z(t))} F_j(s)\, d\mu^+_{z,\lambda}(s) \\ & + \int_{(u_z(t),h)} (1 - F_j(s))\, d\mu^-_{z,\lambda}(s) \Big]. \end{aligned} \tag{81.8}$$

The exact formula (81.7)*, not the sum of absolute interface currents, is used in the endpoint budget.*

Proof. Apply Lemma 81.1 to each active chart at $u = t - t_z$, multiply by its actual J_z, and sum. There are finitely many charts at a fixed count; their translated exceptional sets have zero roof measure. The resulting identity is exactly (81.6). Apply (80.9) to obtain (81.7). Finally the positive part of a finite sum is no larger than the sum of its positive parts. Apply (81.3) to get (81.8). All signs and atoms in (81.4) are retained until these explicit inequalities are taken. □

The common-roof translation is essential. Profiles with different critical values are evaluated at different radial coordinates. Likewise the physical weights cannot be suppressed when taking cancellation. The ordered first-clearance witness and its next-collision mark remain inside s_{70}, d_{70}; the recovery does not redefine either witness.

81.3. **An ordered complete-source estimate.** For parameters fixed before the collision limit, put

$$\mathcal{N}_\chi(K) = \limsup_{m\to\infty} \sup_{R,\lambda} m^2 \big\| [\mathsf{C}_\lambda - (K-1)\mathsf{Y}_\lambda]_+ \big\|_\infty. \tag{81.9}$$

Any dependence of the selected source on ε is retained at its fixed value; the subscript does not average over that parameter. This extended nonnegative quantity uses essential height, not mass or total variation. The current identity defines its finite-count integrand but does not bound the ordered quantity. We have proved

$$\mathcal{H}(d_{71,K}) \leq E_\chi \mathcal{N}_\chi(K), \qquad \mathcal{H}(b^{\varepsilon,\chi,\mathrm{chart}}) \leq C' K \chi^6 + E_\chi \mathcal{N}_\chi(K). \tag{81.10}$$

Since d_{70} is a positive part of the complete selected source, the second bound also bounds its ordered height. In particular an ordered estimate for the pooled net current is a direct sufficient estimate for the original angular-loss request, not a change of the source whose height is sought.

Corollary 81.3 (The original pointwise budget with recovered loss)**.** *For sufficiently large fixed B, choose*

$$\varepsilon(B) = A_0 B^{-1/12}, \qquad \chi(B) = \sqrt{\varepsilon(B)}, \qquad K(B) \geq 1.$$

Every choice is made before the collision limsup. Then

$$\begin{aligned} \mathcal{E}_M \leq\ & C_M B^{-1/192} + C' K(B) B^{-1/4} + C'' \mathcal{N}_{\chi(B)}(K(B)) \\ & + \mathcal{H}(b^{\varepsilon(B),\mathrm{inc}}) + \mathcal{H}(e_{70}). \end{aligned} \tag{81.11}$$

Here e_{70} is formed at these same fixed parameters. Thus the unrestricted two-sided arithmetic raw-density theorem follows if

$$K(B)B^{-1/4} \longrightarrow 0, \qquad \mathcal{N}_{\chi(B)}(K(B)) \longrightarrow 0, \\ \mathcal{H}(b^{\varepsilon(B),\mathrm{inc}}) + \mathcal{H}(e_{70}) \longrightarrow 0. \tag{81.12}$$

The finite arithmetic transition kernel and its zero classes are unchanged.

Proof. Insert (80.7) and the unchanged incidence source into Theorem 181.4. Use (80.10) and (81.7). The parameter choices give $\chi^6 = A_0^3 B^{-1/4}$ and a uniform bound for E_χ. This proves (81.11). Only after obtaining the fixed-band limsup may B tend to infinity. Under (81.12) every term vanishes. □

For instance $K = 1$ retains the controlled $O(B^{-1/4})$ rate; $K(B) = B^\eta$, $0 < \eta < 1/4$, trades it for $O(B^{\eta-1/4})$ and tests only excess current above the larger receiving reserve. Neither choice depends on m. The positive part in (81.9) can be much smaller than the physically weighted sum of Jordan variations. It need not be small merely because each word has finite perimeter.

No unconditional estimate for the last three quantities in (81.11) is proved here. The outside source still contains failed taper, selected grazing, noncritical roof-rank pieces and uncovered physical states. The bound $CA^m\varepsilon$ in Corollary 97.4 remains a finite-count estimate. It cannot be substituted for its fixed-band ordered height. No roof set is removed and no cutoff is chosen exponentially in m.

The same-roof conclusions retain their original numerator and positive-reference hypotheses. Scalar closure would remove their remaining scalar denominator obstruction, but capacity recovery alone is not a new path-valued local law or a forward essential-likelihood theorem. No conditional law is defined on a zero reference class.

The source-height argument also remains distinct from a mixing local limit theorem tested against fixed intervals. In particular Definition 2.2 and Remark 2.5 of [3] concern continuous compactly supported additive tests and suitable continuity sets. Those statements do not bound (81.9). The present proof uses the already established positive physical collar and the BV current on the same Lorentz record, rather than differentiating an interval theorem.

Part 34. Angular overlap and the complete source identity

82. Finite-scale angular overlap on the original source

A large radial supremum-to-average ratio does not distinguish an angular birth from an angular loss. The former needs no compensation when the comparison collar is farther from the critical value. We separate these two effects without imposing a bound on that ratio. The separation is a positive weighting of the original source, not a change of trajectory or a claim that two angular sets are nested.

Fix $0 < \chi \le 1/10$ and $0 < \varepsilon \le \chi/4$. Retain the physical charts, the exact label $\lambda = (n, k)$, and the common coarea versions of Section 85. Write

$$L_z(u) = \ell_{z,\lambda}(\sqrt{2u}), \qquad b_z(t_z + u) = b_{z,\lambda}^{\varepsilon,\mathrm{clr}}(t_z + u), \qquad 0 < u < h_\chi.$$

The label is suppressed only in these two abbreviations. All angular fractions and all comparisons below refer to this same exact label. Let $I = (a, b)$ and $J = (c, d)$, where $0 \le a < b \le h_\chi$ and

$0 \le c < d < h_\chi$. Both intervals are fixed before the collision limit. Put $|J| = d - c$ and define

$$a_{z,J}(u) = \frac{1}{|J|}\int_J \min\{1, L_z(v)/L_z(u)\}\,dv \quad \text{if } L_z(u) > 0, \tag{82.1}$$

$$\Delta_{z,J}(u) = \frac{1}{|J|}\int_J (L_z(u) - L_z(v))_+\,dv. \tag{82.2}$$

Set $a_{z,J}(u) = 1$ when $L_z(u) = 0$. This convention changes no source: $b_z(t_z + u) = 0$ at almost every such level. In particular no clearance guard, positive center value, or collar mass is a denominator.

Lemma 82.1 (Maximal scalar overlap)**.** *The weights are measurable and lie in* $[0, 1]$*. Almost everywhere,*

$$L_z(u)a_{z,J}(u) = \frac{1}{|J|}\int_J \min\{L_z(u), L_z(v)\}\,dv, \qquad L_z(u)(1 - a_{z,J}(u)) = \Delta_{z,J}(u). \tag{82.3}$$

For each u, v *with* $L_z(u) > 0$*, the integrand in* (82.1) *is the largest* $a \in [0, 1]$ *satisfying* $L_z(u)a \le L_z(v)$*.*

Proof. The last constraint is equivalent to $a \le L_z(v)/L_z(u)$. Multiplication by $L_z(u)$ and the identity $x - \min(x, y) = (x - y)_+$ prove both equalities, also at $L_z(u) = 0$. Measurability follows from parameter integration of bounded measurable functions. These are statements about angular capacities; they do not assert the existence of a physical orbit joining the two levels. □

Define two positive source densities on the same physical chart by

$$s_z^{I,J}(t_z + u) = \mathbf{1}_I(u)a_{z,J}(u)b_z(t_z + u), \qquad d_z^{I,J}(t_z + u) = \mathbf{1}_I(u)(1 - a_{z,J}(u))b_z(t_z + u). \tag{82.4}$$

Thus $s_z^{I,J} + d_z^{I,J} = \mathbf{1}_I b_z$. This is a partition of weights, not a decomposition into disjoint events. Each weight is a measurable insertion into the unchanged physical probability space. Neither source is normalized again.

For positive source families use $\mathcal{H}(f) = \limsup_m \sup_{R,\lambda} m^2 \|f_{m;\lambda,R}\|_\infty$. Write

$$W_\delta(b) = \min\{1, \delta + C_g\varepsilon^{-1}\sqrt{2b}\}, \qquad E_\chi(b, d) = \exp\{K_\chi(\sqrt{2b} + \sqrt{2d})\}.$$

Here $g_z = D_z^\varepsilon(0)$ is allowed to vanish.

Theorem 82.2 (Height without a radial-ratio cutoff)**.** *Let* $s^{I,J;\le\delta}$ *be the sum of* (82.4) *over any subfamily of the selected physical charts with* $g_z \le \delta$*, where* $0 \le \delta \le 1$*. Then*

$$\mathcal{H}(s^{I,J;\le\delta}) \le C\frac{d}{d - c}(b - a + d)E_\chi(b, d)W_\delta(b). \tag{82.5}$$

No bound on a birth order, an angular persistence certificate, or a radial moment is required. On each chart the complementary density satisfies

$$d_z^{I,J}(t_z + u) \le \mathbf{1}_I(u)e^{K_\chi\sqrt{2b}}J_z \min\{1, g_z + C_g\varepsilon^{-1}\sqrt{2b}\}\Delta_{z,J}(u). \tag{82.6}$$

All source identities and inequalities hold on common coarea versions.

Proof. The physical profile and absolute guard comparison give $b_z(t_z + u) \le e^{K_\chi\sqrt{2b}}J_z W_\delta(b)L_z(u)$ for almost every $u \in I$ on the indicated subfamily. The first identity in (82.3) bounds its product with $a_{z,J}(u)$ by

$$e^{K_\chi\sqrt{2b}}J_z W_\delta(b)\frac{1}{d - c}\int_c^d L_z(v)\,dv \le E_\chi(b, d)W_\delta(b)\frac{d}{d - c}\widehat{q}_{z,d;\lambda}.$$

The last step uses the lower physical-density bound on $(0, d)$; the collar average is precisely the unguarded physical average of (85.5). At a roof t, contributing centers lie in $[t - b, t - a]$. Sum

over those centers and apply Theorem 85.2 with interval length $b-a$ and collar width d. This proves (82.5). The comparison is made to one positive event after summing the physical source. In particular it incurs neither a word count nor an exact-label count and makes no regularity assumption on the selecting insertion. The second identity in (82.3) proves (82.6). The proof uses only bounded measurable angular profiles. □

Corollary 82.3 (All-order and later-birth height without angular loss)**.** *Fix $H>0$ with $2H<h_\chi$. Consider any subfamily for which*

$$L_z(u)\le L_z(v)\quad \text{for almost every } (u,v)\in(0,H)\times(H,2H). \tag{82.7}$$

Its entire original clearance source on $0<F_z-t_z<H$, with $g_z\le\delta$, has height at most

$$6CH\exp\{(2+\sqrt{2})K_\chi\sqrt{H}\}\min\{1,\delta+C_g\varepsilon^{-1}\sqrt{2H}\}. \tag{82.8}$$

At $\delta=0$ this is $6\sqrt{2}CC_g\varepsilon^{-1}H^{3/2}\exp\{(2+\sqrt{2})K_\chi\sqrt{H}\}$. Nondecreasing angular occupation on $(0,2H)$ is sufficient for (82.7)*; angular sets themselves need not be nested. No analytic germ at the center is required.*

Proof. Under (82.7), Fubini gives $\Delta_{z,(H,2H)}=0$ almost everywhere on $(0,H)$. Consequently $s_z^{(0,H),(H,2H)}=b_z$ there. In (82.5) put $(a,b,c,d)=(0,H,H,2H)$. The factor $d(b-a+d)/(d-c)$ equals $6H$. The guard estimate proves the zero-center assertion, including infinitely flat guards. □

The no-loss condition is an exact finite-scale condition on the physical angular sets, not a claim that every Lorentz word satisfies it. It can hold even when the radial ratios are unbounded on a family. For example, the scalar profiles $L(u)=(u/(2H))^p$ on $(0,2H)$ have ratio $p+1$ on that whole interval and satisfy (82.7) for every $p\ge0$. A profile which is zero up to a positive radius and then increases is also included. These examples explain the distinction from a radial-moment hypothesis; they do not assert a Lorentz realization of every profile. An angular spike which subsequently disappears contributes to (82.2), rather than being silently controlled by the corollary.

83. The signed radial current of the physical angular set

The loss in (82.2) can be bounded by the negative variation of the actual angular occupation. For circular scatterers this variation exists at every fixed word on every compact radial interval of the selected chart. We prove this fact and identify its signed geometric source. No count-uniform variation estimate is inferred from finite-word definability.

Fix a selected chart, an exact label $\lambda=(n,k)$ and $0<S<h_\chi$. In the cylinder $(0,S)\times\mathbb{S}^1$ set

$$E_{z,\lambda}=\{(u,\phi):B_{z,\lambda}(\sqrt{2u}\,\omega_\phi)=1\},\qquad \pi(u,\phi)=u.$$

All physical and section decisions are made before this set is formed. We take distributional derivatives in the open cylinder; no artificial boundary mass is inserted at $u=0,S$ or at an angular coordinate cut.

Theorem 83.1 (Finite-word physical boundary current)**.** *The indicator $\mathbf{1}_{E_{z,\lambda}}$ has bounded variation on the cylinder, and $L_z\in BV(0,S)$. As finite signed Radon measures,*

$$DL_z=\frac{1}{2\pi}\pi_\#D_u\mathbf{1}_{E_{z,\lambda}}=-\frac{1}{2\pi}\pi_\#\big(n_u\,\mathcal{H}^1|_{\partial^*E_{z,\lambda}}\big). \tag{83.1}$$

Here n is the measure-theoretic outward normal. In particular

$$(DL_z)^-\le\frac{1}{2\pi}\pi_\#(D_u\mathbf{1}_{E_{z,\lambda}})^-. \tag{83.2}$$

Radial jumps and tangencies are included. There is no division by a boundary gradient. Finiteness is asserted for each fixed word and label, not uniformly after taking the collision-count limit.

Proof. On a closed annulus inside the Morse disk the selected contact graph and the fixed Morse map are real analytic. Indeed the selected incidences remain positive and the internal contact derivative remains invertible. The analytic implicit function theorem applies at every point, and uniqueness identifies its local solutions. The explicit matrix-square-root Morse map is analytic on the positive definite cone. Compactness permits a finite analytic coordinate cover.

On this cover the original first-hit decisions have a finite Boolean description by restricted analytic tests. Competing disks are kept separately. If the closest point of a segment changes, divide into its interior-foot and two endpoint cases before writing the distance inequality. Selected flight lengths stay positive. Section membership is tested in the original finitely many analytic rectangles. Occupation, terminal membership and the fixed exact label are finite Boolean combinations of these same tests. Passing to angular coordinates on a finite cover preserves restricted analyticity. Thus $E_{z,\lambda}$ is definable in the restricted analytic structure on this compact annulus.

By cell decomposition, its angular fibers are finite unions of intervals with definable endpoints, with finitely many changes of radial cell. Consequently their lengths divided by 2π define a bounded definable function of u. The monotonicity theorem gives finitely many monotone pieces, so this function has finite variation. These uses of cell decomposition and monotonicity are the standard ones in [9]. They give no bound independent of the defining word. The boundary curves also have finite length: divide each bounded definable curve into finitely many arcs on which both coordinates are monotone; its length is bounded by the sum of their coordinate variations. This proves finite perimeter on the annulus.

Near $u = 0$, Lemma 88.2 gives finitely many analytic angular boundary graphs $\phi = \phi_j(r)$ with a fixed cyclic order, where $u = r^2/2$. Their lengths are finite, since both $r \mapsto r^2/2$ and the analytic ϕ_j have integrable derivatives on the small closed interval. The angular fraction is analytic in r by Theorem 88.3, and therefore has finite variation after this monotone change of variable. Joining this neighborhood to one compact annulus proves the two bounded-variation assertions on $(0, S)$. Values on boundary curves may be changed without changing the source or the distributions.

For $\eta \in C_c^1(0, S)$, Fubini gives

$$-\int_0^S L_z(u)\eta'(u)\,du = -\frac{1}{2\pi}\int_{E_{z,\lambda}} \eta'(u)\,du\,d\phi = \frac{1}{2\pi}\int \eta \circ \pi\, dD_u \mathbf{1}_{E_{z,\lambda}}.$$

This proves the first equality in (83.1). The finite-perimeter derivative formula, with the stated outward normal convention, gives the second; see [13]. For a signed measure $\mu = \mu^+ - \mu^-$, the negative part of $\pi_\#\mu$ is at most $\pi_\#\mu^-$. Applying this elementary measure inequality proves (83.2). No regular-level parametrization or transversality divisor was used. □

Attribution to primitive physical decisions. There is a finite ordered list of the original scalar tests at each word. Away from their zero sets all signs, hence the actual Boolean indicator, are locally constant. The reduced boundary in (83.1) is therefore contained, up to $\mathcal{H}^1$-null sets, in the union of the genuine decision interfaces. Identically vanishing tests have no interface and cause no extra current. Assign each boundary point to the first interface in this list which contains it. This gives a disjoint measurable partition of the reduced boundary, including coincident interfaces. The current assigned there is the jump of the actual one-hot indicator, not a sum of jumps of all tests vanishing at that point. In particular births and losses are combined with their signs before taking a negative part after radial projection. The original first-clearance witness and its

next-collision mark remain in b_z; this additional boundary attribution does not replace either of them.

Proposition 83.2 (One-sided radial loss formula)**.** *Let $2H < h_\chi$ and choose S with $2H < S < h_\chi$. For almost every $0 < u < H$,*

$$\Delta_{z,(H,2H)}(u) \le \int_{(u,2H)} \kappa_H(s)\, d(DL_z)^-(s), \qquad \kappa_H(s) = \begin{cases} 1, & 0 < s \le H, \\ (2H - s)/H, & H < s < 2H. \end{cases} \tag{83.3}$$

The right side retains atomic radial deaths. In particular $DL_z \ge 0$ on $(0, 2H)$ implies that the complete inner source satisfies Corollary 82.3.

Proof. Use the precise representative of the BV function. At its continuity points $u < v$ the fundamental theorem for BV gives $L_z(u) - L_z(v) \le (DL_z)^-((u, v))$. The exceptional jump points are countable and do not affect the almost-everywhere assertion or the average in v. Average over $H < v < 2H$ and apply Tonelli to the nonnegative variation measure. A point $s \le H$ is crossed by the entire averaging interval; a point $H < s < 2H$ is crossed by a fraction $(2H - s)/H$. This proves (83.3). A BV function with nonnegative distributional derivative has a nondecreasing representative, giving the last assertion. □

For example, a downward jump of size β at $s_0 \in (H, 2H)$ contributes $\beta(2H - s_0)/H$ to the right side for $u < H$. Replacing the distributional derivative by its almost-everywhere classical derivative would lose this contribution. Conversely, upward jumps incur no negative-current cost. This is why absolute boundary variation is not used in (83.3).

Combining (82.6) and (83.3) identifies a sufficient physically weighted estimate for the remaining inner loss. Specifically its height is at most

$$e^{K_\chi \sqrt{2H}} W_\delta(H) \limsup_m \sup_{R,\lambda,t} m^2 \sum_{z:\, t_z \in [t-H,t]} J_z \int_{(0,2H)} \kappa_H(s)\, d(DL_z)^-(s).$$

This expression may be infinite. Fixed-word finite perimeter proves that its summands are defined; it does not establish the ordered weighted bound. The exact loss (82.2), rather than this sufficient upper bound, is used in the full source identity below.

84. The whole selected chart and the pointwise source identity

We now apply the overlap comparison to the entire selected Morse disk, not only to an inner germ collar. This accounts for its outer annuli on the same footing as its center. The genuinely uncovered physical source remains separate.

For each fixed χ, ε choose

$$I_\chi = (0, h_\chi), \qquad J_\chi = (0, h_\chi/2), \qquad s_{70} = \sum_z s_z^{I_\chi, J_\chi}, \quad d_{70} = \sum_z d_z^{I_\chi, J_\chi}.$$

The sums are over all selected χ-incident physical charts with the prescribed exact label. Write $e_{70} = b^{\varepsilon,\chi,\mathrm{outside}}$ for the original positive complement in (95.8).

Theorem 84.1 (Complete allocation of the selected-chart source)**.** *On the original probability space, and on the corresponding common coarea versions,*

$$b^{\varepsilon,\mathrm{clr}} = s_{70} + d_{70} + e_{70}, \qquad s_{70}, d_{70}, e_{70} \ge 0. \tag{84.1}$$

The selected-chart and outside events are disjoint; within each chart s_{70} and d_{70} are complementary positive weights. Moreover

$$\mathcal{H}(s_{70}) \le \frac{3}{2} C h_\chi \exp\{(1 + 1/\sqrt{2}) K_\chi r_\chi\} \le C' \chi^6. \tag{84.2}$$

The constants are independent of the exact label and the collision count. Every portion of every selected disk, including all positive- radius births and all its outer annuli, is allocated in (84.1)*. Neither d_{70} nor e_{70} is assigned a small height by this assertion.*

Proof. Equation (82.4), summed over the disjoint physical chart restrictions, is precisely the chart source. Add the unchanged outside source to obtain (84.1). Apply Theorem 82.2 with $(a,b,c,d) = (0, h_\chi, 0, h_\chi/2)$ and $\delta = 1$. The ratio $d/(d-c)$ is one, the center interval plus the collar width has length $3h_\chi/2$, and the distortion exponent is $(1 + 1/\sqrt{2})K_\chi r_\chi$. The source interval is open, so $b = h_\chi$ only uses estimates strictly within the Morse disk. The receiving width is strictly smaller than h_χ, as required by the inherited collar theorem. Finally $h_\chi = r_0^2\chi^6/2$ and $K_\chi r_\chi = Cr_0\chi$. This proves the uniform last bound. No exhaustion in the collision count or interchange with an infinite sum of annuli occurs. □

The complement e_{70} retains failed taper, selected grazing, noncritical roof-rank pieces, and every state outside the specified physical disks. Unlike e_{69} it does not contain annuli cut out of these disks at an additional small radial cutoff. Those annuli are instead divided, with their original weight, between s_{70} and d_{70}. They have not all been declared small: a late birth with no occupation in J_χ is allocated wholly to d_{70}. This example also explains why the complete radial tail is not proved merely by introducing the overlap decomposition.

Corollary 84.2 (An ordered whole-chart contribution to raw inversion)**.** *For every sufficiently large fixed reconstruction band B, put*

$$\varepsilon(B) = A_0 B^{-1/12}, \qquad \chi(B) = \sqrt{\varepsilon(B)}.$$

These choices are admissible for all sufficiently large B. The unchanged central pointwise error satisfies

$$\begin{aligned}\mathcal{E}_M \le C_M B^{-1/192} + C' B^{-1/4} + \mathcal{H}(b^{\varepsilon(B),\mathrm{inc}}) \\ + \mathcal{H}(d_{70}) + \mathcal{H}(e_{70}).\end{aligned} \tag{84.3}$$

Here s_{70}, d_{70}, e_{70} are formed at the fixed choices $\chi(B), \varepsilon(B)$ before taking the collision limsup. No cutoff in this statement depends on m.

Proof. For sufficiently small ε, one has $4\varepsilon \le \sqrt{\varepsilon} \le 1/10$. Thus all chart and incidence assumptions hold. Theorem 181.4, the original first-incidence/first-clearance split, and (84.1) give the asserted inequality with $\mathcal{H}(s_{70})$ in place of $C'B^{-1/4}$. By (84.2), that height is at most $C'\chi(B)^6 = C'\varepsilon(B)^3 = C'A_0^3 B^{-1/4}$. Absorb the fixed A_0^3 into the constant. Only after this fixed-band collision limit may B tend to infinity. □

The unrestricted two-sided arithmetic theorem would follow from vanishing of the last three heights in (84.3). These estimates are not consequences of finite-word BV, of the radial-moment implication, or of the finite-count bound $CA^m\varepsilon$ for first incidence. In particular no assertion of a finite uniform radial moment or a vanishing full signed-current tail is made here. The scalar endpoint, its exact arithmetic zero classes, and the original source remain unchanged. Unrestricted same-roof bridges, forward essential likelihood and pointwise roof-conditioned path laws still require that endpoint.

84.1. **Comparison with existing local-limit statements.** The local Lorentz-process theorem of [2] already gives a cell-index local limit. Theorem 2.2 and Remark 2.3 of [12] give a two-dimensional cell-index local limit with admissible endpoint observables, including piecewise Hölder endpoints under the stated uniformity conditions. Inserting a free-flight observable as an endpoint factor in that theorem is different from simultaneously resolving the accumulated roof,

exact section occupation and the present density-level boundary source. We do not infer the latter from an endpoint insertion.

Definition 2.2 of [3] formulates its mixing local limit using compactly supported continuous tests in the additive coordinate; Remark 2.5 permits appropriate continuity-set indicators. Its Proposition 6.3 applies the suspension theory to finite-horizon Sinai billiards under the joint minimality and arithmetic-group hypotheses. Those statements supply neither an essential-supremum estimate for $\Delta_{z,J}$ nor the complete three-height estimate in (84.3). Differentiating an interval law would require precisely an additional density-height argument. The present overlap and signed-current results address this source question; they are not claims of a new general suspension mechanism or a replacement for the inherited anisotropic local-limit inputs.

Part 35. Positive-source comparison and raw height

85. A positive comparison trace on the unchanged physical record

The division by the center value of a clearance guard is unnecessary for an upper height estimate. The appropriate comparison measure is the original physical measure on the selected itinerary, with the clearance multiplier omitted only in that upper comparison. In particular, omitting this multiplier does not omit a first-hit test, a section decision, or an exact integer label.

Fix $0 < \chi \le 1/10$ and $0 < \varepsilon \le \chi/4$. Retain the selected-contact chart, r_χ, $h_\chi = r_\chi^2/2$, K_χ, J_z, and the one-hot physical indicator $B_{z,\lambda}$, where $\lambda = (n,k)$, from Sections 95 and 88. Write

$$\widehat{b}_{z,\lambda}(t_z+u) = \int_0^{2\pi} \widetilde{w}_z(\sqrt{2u}\,\omega_\phi) B_{z,\lambda}(\sqrt{2u}\,\omega_\phi)\, d\phi, \qquad 0 < u < h_\chi. \tag{85.1}$$

This is an unnormalized density of an actual physical source. The normalization in $\widetilde{w}_z$ already includes c_*^{-1}. Nonphysical points on the selected continuation still have weight zero. The density is not asserted to equal the first-clearance source.

Lemma 85.1 (Physical comparison and normalization)**.** *On a common coarea representative, for almost every $0 < u < h_\chi$, with $r = \sqrt{2u}$,*

$$0 \le b_{z,\lambda}^{\varepsilon,\mathrm{clr}}(t_z+u) \le \widehat{b}_{z,\lambda}(t_z+u), \tag{85.2}$$

$$e^{-K_\chi r} J_z \ell_{z,\lambda}(r) \le \widehat{b}_{z,\lambda}(t_z+u) \le e^{K_\chi r} J_z \ell_{z,\lambda}(r). \tag{85.3}$$

More precisely, if $g_z = D_z^\varepsilon(0)$ and $L_g = C_g \varepsilon^{-1}$, then

$$b_{z,\lambda}^{\varepsilon,\mathrm{clr}}(t_z+u) \le \min\{1, g_z + L_g\sqrt{2u}\}\, \widehat{b}_{z,\lambda}(t_z+u). \tag{85.4}$$

The assertions also hold when $g_z = 0$ and when the guard is infinitely flat at the center. They use no analytic property of the guard.

Proof. The exact coarea identity (95.7) contains one additional factor D_z^ε relative to (85.1). Its values lie in $[0,1]$. The relative physical-density bound used in Theorem 95.2, and $\int B_{z,\lambda}\, d\phi = 2\pi \ell_{z,\lambda}$, prove the first two assertions. In polar coarea the arclength factor r cancels the derivative of $r^2/2$; no radial power is introduced. Finally (95.2) gives $D_z^\varepsilon(r\omega) \le \min\{1, g_z + L_g r\}$ pointwise. Multiply by the nonnegative physical indicator before integrating. No division by g_z occurs. □

For any subfamily $\mathcal{Z}$ of the disjoint physical charts put

$$\widehat{q}_{z,s;\lambda} = \frac{1}{s}\int_0^s \widehat{b}_{z,\lambda}(t_z+u)\, du, \qquad \widehat{\mathcal{Q}}_{m,s;\lambda,R} = \sum_{z\in\mathcal{Z}} \widehat{q}_{z,s;\lambda}\delta_{t_z}. \tag{85.5}$$

A subfamily may depend on m, R, λ and on parameters fixed before the collision limit. No regularity of its selection is required: the following estimate is a positive comparison with a single physical event, not a spectral estimate on the selecting multiplier.

Theorem 85.2 (Ordered unguarded physical collar trace)**.** *For every fixed $0 < s < h_\chi$ and every fixed interval length $w > 0$,*

$$\limsup_{m\to\infty} \sup_{R,\lambda,a} m^2 \widehat{\mathcal{Q}}_{m,s;\lambda,R}([a, a+w]) \le C(w+s). \tag{85.6}$$

The constant is independent of the chart subfamily, the exact labels, s, χ, ε and the number of coincident critical values. The estimate remains true when the atoms are multiplied by arbitrary numbers in $[0,1]$.

Proof. Each physical chart with $t_z \in [a, a+w]$, restricted to $0 < F_z - t_z < s$, has actual roof in $[a, a+w+s]$. Both endpoint momenta have modulus at most $C_0\sqrt{s}$ by Theorem 94.3. Different physical itineraries are disjoint; the uniqueness of the normal critical point in each selected word is the one used in (95.8). Consequently

$$s\widehat{\mathcal{Q}}_{m,s;\lambda,R}([a, a+w]) \le \nu_R^*(E_{n,k,m,R}^{(C_0\sqrt{s})}(a; [0, w+s])).$$

The initial, occupation, terminal and displacement tests on the left are precisely those on the right. The comparison adds physical points but never assigns measure to nonphysical continuations. The choice of r_0 already made in Theorem 96.1 ensures $C_0\sqrt{s} < 1/4$. Apply Lemma 170.1 at these fixed widths. Its bound is $Cs(w+s)$ after multiplication by m^2 and the ordered limsup. Divide by s. Multipliers in $[0,1]$ only decrease the positive trace. The same argument bounds every selected subfamily by the same event and therefore introduces no counting multiplier. □

The trace $\widehat{\mathcal{Q}}$ differs from the guarded trace $\mathcal{Q}$ in (96.1). It also differs from the full reversible atomic measure: its atoms are averages of the actual, exactly labelled physical source over a positive-width collar. Neither the physical angular indicator nor that width has been removed.

86. Absolute guard control and zero-center caustic height

We separate the angular certificate from the clearance guard. This removes the relative guard condition, including its singularity at zero center value, from the all-order collar theorem.

For a nonzero angular germ retain $r_{z,\lambda}^*$ and $k_{z,\lambda}^{\mathrm{a}}$ from (88.3), and define

$$\mathfrak{p}_{z,\lambda}^{\mathrm{a}} = \max\{(r_{z,\lambda}^*)^{-1}, 2k_{z,\lambda}^{\mathrm{a}}\}. \tag{86.1}$$

For an identically zero germ choose a certified zero radius $0 < r_{z,\lambda}^{\mathrm{zero}} < r_\chi$ and set $\mathfrak{p}_{z,\lambda}^{\mathrm{a}} = (r_{z,\lambda}^{\mathrm{zero}})^{-1}$. This latter convention asserts zero source only within the certified disk. It does not assign a finite birth order to a zero germ. Let $\mathcal{A}_{m,R,\lambda}(K)$ consist of all selected charts with $\mathfrak{p}_{z,\lambda}^{\mathrm{a}} \le K$. This family does not depend on H or on the value of the center guard. For a positive center guard, the old persistence number satisfies $\mathfrak{p}_{z,\lambda}^{\mathrm{a}} \le \mathfrak{p}_{z,\lambda}$.

Lemma 86.1 (An annular comparison without a guard denominator)**.** *Fix $2H < h_\chi$ and $2K\sqrt{H} < 1$. For every member of $\mathcal{A}_{m,R,\lambda}(K)$,*

$$\operatorname*{ess\,sup}_{0<u<H} \widehat{b}_{z,\lambda}(t_z + u) \le 2F_{\chi,K}(H)\widehat{q}_{z,2H;\lambda}, \tag{86.2}$$

where $F_{\chi,K}(H)$ is the factor in (89.7)*. Its constant is independent of every finite angular birth order and of the center guard.*

Proof. For a nonzero germ, (88.4) and (86.1) give $(1-Kr)\eta r^d \le \ell(r) \le (1+Kr)\eta r^d$ for $r < K^{-1}$. Insert this estimate in (85.3). On $H < v < 2H$ the lower profile is at least

$$e^{-2K_\chi\sqrt{H}}(1 - 2K\sqrt{H})J_z\eta(2H)^{d/2}.$$

On $0 < u < H$ its upper counterpart is at most $e^{\sqrt{2}K_\chi\sqrt{H}}(1+\sqrt{2}K\sqrt{H})J_z\eta(2H)^{d/2}$. Divide these two bounds and compare the positive outer-annulus average to twice the full $2H$-collar average.

This proves the assertion with exactly the indicated F. Only monotonicity of r^d, not a bound on d, has been used. For a zero germ, $2\sqrt{H} < K^{-1}$ lies within its certified zero disk, so both source integrals vanish. The guard has not entered either case. □

For $0 \le \delta \le 1$ let

$$b^{\mathrm{a,K,H};\le\delta}_{m;\lambda,R}(t) = \sum_{\substack{z\in\mathcal{A}_{m,R,\lambda}(K)\\ g_z\le\delta}} \mathbf{1}_{\{0<t-t_z<H\}} b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t). \tag{86.3}$$

For $\delta = 1$ write simply $b^{\mathrm{a,K,H}}$. These are restrictions of the original first-clearance source, not of the comparison source. Use the ordered height $\mathcal{H}$ defined before (90.9).

Theorem 86.2 (Absolute-guard height on every angular stratum)**.** *Fix $\chi, \varepsilon, K, H, \delta$ before the collision limit, with $2H < h_\chi$ and $2K\sqrt{H} < 1$. Then*

$$\mathcal{H}(b^{\mathrm{a,K,H};\le\delta}) \le 6CHF_{\chi,K}(H)\min\{1, \delta + L_g\sqrt{2H}\}. \tag{86.4}$$

In particular

$$\mathcal{H}(b^{\mathrm{a,K,H}}) \le 6CHF_{\chi,K}(H), \tag{86.5}$$

$$\mathcal{H}(b^{\mathrm{a,K,H};\le 0}) \le 6\sqrt{2}CC_g\varepsilon^{-1}H^{3/2}F_{\chi,K}(H). \tag{86.6}$$

Every finite angular birth order is included. Zero center guards, including infinitely flat ones, and arbitrarily small positive center guards require no further cutoff. A bounded measurable insertion costs only its supremum norm in these bounds.

Proof. At a fixed roof t, contributing centers lie in $[t-H, t]$. Lemma 85.1, followed by Lemma 86.1, gives almost everywhere

$$b^{\mathrm{a,K,H};\le\delta}_{m;\lambda,R}(t) \le 2F_{\chi,K}(H)\min\{1, \delta + L_g\sqrt{2H}\}\,\widehat{\mathcal{Q}}_{m,2H;\lambda,R}([t-H, t]),$$

where the trace is restricted to the same chart subfamily. Apply Theorem 85.2 with interval length H and collar width $2H$. Its bound is $3CH$, proving (86.4). The choices $\delta = 1$ and $\delta = 0$ give the two displayed consequences. The common coarea representative excludes a null set at each finite count; finite unions over words and labels remain null. Positivity proves the insertion assertion before pushforward. All birth orders are summed at fixed count; no infinite scale sum is interchanged with a limsup. □

Corollary 86.3 (Fixed-stratum shrinking collars)**.** *At fixed χ, ε, K, for every deterministic sequence $H_m \downarrow 0$,*

$$\sup_{R,\lambda} m^2 \|b^{\mathrm{a,K,H_m}}_{m;\lambda,R}\|_\infty \longrightarrow 0.$$

No quantitative rate in m is asserted.

Proof. For any fixed admissible H, eventually $H_m < H$. The chart family is independent of the collar width, so the original positive restrictions satisfy $b^{\mathrm{a,K,H_m}} \le b^{\mathrm{a,K,H}}$. Apply (86.5) at that fixed H and then let $H \downarrow 0$. This is a monotone comparison with fixed-width estimates, not a spectral evaluation at H_m or a choice $K = K_m$. □

The original relative-guard persistence theorem remains intact. Its controlled source is contained in the present one at the same K, H. The new theorem removes the guard part of the cutoff; it neither proves tightness of $\mathfrak{p}^{\mathrm{a}}$ nor identifies it with the old angular condition number $\mathfrak{a}$. Source outside the certified angular disks has not been inferred small. We next give a separate comparison which can be evaluated on that source without using an angular germ at all.

87. Finite-scale radial comparison and the complete positive remainder

A comparison on an entire fixed radial interval need not use the analytic germ at its center. We formulate it directly in terms of the exact physical angular sets. This includes portions beyond a certified germ disk and labels which are absent near the center but appear farther out. The resulting finite-scale condition and its weighted tail are kept distinct from the earlier persistence number.

Fix $0 < S < h_\chi$ and an interval $I = (a, b)$ with $0 \le a < b \le S$. Put

$$L_{z,\lambda}(u) = \ell_{z,\lambda}(\sqrt{2u}), \qquad A_{z,\lambda}(S) = \frac{1}{S}\int_0^S L_{z,\lambda}(u)\,du, \tag{87.1}$$

$$U_{z,\lambda}(I) = \operatorname*{ess\,sup}_{u\in I} L_{z,\lambda}(u), \qquad V_{z,\lambda}(I,S) = \frac{U_{z,\lambda}(I)}{A_{z,\lambda}(S)}. \tag{87.2}$$

Set $V = 0$ if $A = 0$. In that case $L = 0$ almost everywhere on $(0, S)$, hence on I, and there is no source on this radial interval. These quantities are defined on the original physical indicator, not on a Taylor polynomial. They have no guard denominator.

Lemma 87.1 (Finite-scale geometric certificate)**.** *At every fixed chart and exact label, $V(I,S)$ is finite and nonnegative. If a verified computation of the physical angular sets gives $U(I) \le u_+$ and $A(S) \ge a_- > 0$, then $V(I,S) \le u_+/a_-$. On the common coarea representative,*

$$\operatorname*{ess\,sup}_{u\in I} b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t_z + u) \le G_\chi(b,S)V_{z,\lambda}(I,S)\min\{1, g_z + L_g\sqrt{2b}\}\widehat{q}_{z,S;\lambda}, \tag{87.3}$$

where $G_\chi(b,S) = \exp\{K_\chi(\sqrt{2b} + \sqrt{2S})\}$. No analyticity or fixed radial ordering of the physical angular boundaries on $(0,S)$ is assumed.

Proof. The measurable angular fraction lies in $[0,1]$. If its average is positive the quotient is finite; if the average is zero the preceding zero convention is justified by nonnegativity and Fubini's theorem. By (85.3),

$$\widehat{q}_{z,S;\lambda} \ge e^{-K_\chi\sqrt{2S}} J_z A_{z,\lambda}(S), \qquad \operatorname*{ess\,sup}_{u\in I} \widehat{b}_{z,\lambda}(t_z + u) \le e^{K_\chi\sqrt{2b}} J_z U_{z,\lambda}(I).$$

Divide when $A > 0$ and apply (85.4). The zero case has already been settled. The certificate assertion is the same quotient inequality. It specifies what must be verified on the physical sets; no numerical certificate for all long Lorentz words is asserted. □

Let $b^{I,S,V\le T;\le\delta}$ be the original clearance source restricted to those chart–label pairs with $V(I,S) \le T$ and $g_z \le \delta$, and to roof offsets in I. The selection is made at the fixed scale S before the collision limit.

Theorem 87.2 (Height on a complete finite radial interval)**.** *For fixed admissible $\chi, \varepsilon, I, S, T$ and $0 \le \delta \le 1$,*

$$\mathcal{H}(b^{I,S,V\le T;\le\delta}) \le CTG_\chi(b,S)\min\{1, \delta + L_g\sqrt{2b}\}(b - a + S). \tag{87.4}$$

In particular, for $I = (0,S)$ the entire selected radius $0 < F_z - t_z < S$ on this stratum satisfies

$$\mathcal{H}(b^{(0,S),S,V\le T}) \le 2CTSe^{2K_\chi\sqrt{2S}}. \tag{87.5}$$

This includes source beyond an angular-germ disk, positive-radius births of zero germs, and zero-center guards, when they belong to the stated finite-scale stratum.

Proof. At roof t, the centers contributing to offsets in (a,b) lie in $[t-b, t-a]$, an interval of length $b - a$. Sum (87.3) over these centers. Bound V by T and the guard envelope by its displayed common bound. Apply Theorem 85.2 with collar width S and interval length $b - a$. This proves

the first statement. The choice $a = 0$, $b = S$, $\delta = 1$ proves the second. The averaging collar remains positive-width throughout. A fixed finite disjoint partition into radial intervals may be treated in the same way and its finitely many estimates added. No countable decomposition or exchange of an infinite sum with a collision limsup is used. □

87.1. **A physically weighted moment sufficient for the radial tail.** For the whole radial interval write $V_z(S) = V_{z,\lambda}((0,S),S)$. For $\alpha > 0$ define the positive trace moment

$$\mathcal{M}_\alpha(S) = \limsup_{m\to\infty} \sup_{R,\lambda,t} m^2 \sum_{z:t_z\in[t-S,t]} V_z(S)^{1+\alpha}\widehat{q}_{z,S;\lambda}. \tag{87.6}$$

The sum uses the actual physical collar mass with the original exact label, divided by S. It is not an unweighted word count or a sum of full reversible coefficients. The moment is allowed to be infinite.

Proposition 87.3 (Weighted radial-tail transfer)**.** *The original clearance source on $0 < F_z - t_z < S$ with $V_z(S) > T$ satisfies, for every $T > 0$,*

$$\mathcal{H}(b^{(0,S),S,V>T}) \le e^{2K_\chi\sqrt{2S}}T^{-\alpha}\mathcal{M}_\alpha(S). \tag{87.7}$$

Thus a finite bound on this physically weighted moment gives ordered tightness of this radial cutoff. No such uniform moment bound is asserted for the Lorentz family by the present proposition.

Proof. Use (87.3) with $I = (0,S)$ and bound the guard envelope by one. On $V > T$, $V \le T^{-\alpha}V^{1+\alpha}$. Sum at the prescribed roof, whose contributing centers are in $[t-S,t]$, and take the stated ordered limsup. The argument is valid in the extended nonnegative reals; it asserts a finite tail estimate only when the displayed moment is finite. □

87.2. **One disjoint source partition.** Fix $0 < H < S < h_\chi$, $2H < h_\chi$, and $2K\sqrt{H} < 1$. First take the actual source $b^{\mathrm{a,K,H}}$ of (86.3). Among all remaining physical points in the selected charts with $0 < F_z - t_z < S$, take those whose chart–label pair has $V_z(S) \le T$ and call their source o_{69}. Call the rest within that radius t_{69}. Finally let e_{69} contain every point not yet assigned: the original outside source and all portions of the selected charts with $F_z - t_z \ge S$. Then, as densities of disjoint positive restrictions,

$$b^{\varepsilon,\mathrm{clr}} = b^{\mathrm{a,K,H}} + o_{69} + t_{69} + e_{69}. \tag{87.8}$$

The decomposition preserves the first-clearance witness, the next-collision mark, the original occupation and terminal convention, and every arithmetic zero class. In particular zero-center source is not discarded. Outside an inner angular certificate it is assigned by the finite-scale partition just described.

The first two restrictions have unconditional bounds

$$\mathcal{H}(b^{\mathrm{a,K,H}}) \le 6CHF_{\chi,K}(H), \qquad \mathcal{H}(o_{69}) \le 2CTSe^{2K_\chi\sqrt{2S}}.$$

The second follows by positive domination by the full $V \le T$ source, even though an overlap was removed. Similarly t_{69} is dominated by the full $V > T$ source in Proposition 87.3. Consequently the unchanged pointwise error satisfies, for every fixed reconstruction band B and its width $\varepsilon(B) = A_0B^{-1/12}$,

$$\begin{aligned}\mathcal{E}_M \le{}& C_MB^{-1/192} + \mathcal{H}(b^{\varepsilon(B),\mathrm{inc}}) + 6CHF_{\chi,K}(H)\\ &+ 2CTSe^{2K_\chi\sqrt{2S}} + \mathcal{H}(t_{69}) + \mathcal{H}(e_{69}).\end{aligned} \tag{87.9}$$

A proved bound on $\mathcal{M}_\alpha(S)$ may replace the fifth term by (87.7); it may not be silently assumed. First fix B, χ, K, H, S, T, then take the collision limsup. Any subsequent choices may depend on B, but not on m.

The finite-scale theorem controls the entire chosen radial interval on its stated stratum, rather than treating its outer part as controlled by a shrinking inner collar. It does not estimate the high-V tail without the moment, nor the exterior e_{69}. The latter still contains failed taper, selected grazing, noncritical rank, uncovered states, and all omitted outer annuli. The complete first-incidence height remains a separate positive term. Neither its exponential finite-count estimate in Section 97, nor fixed-count finiteness of V, supplies the missing ordered estimate.

The original unrestricted two-sided pointwise theorem and its same-roof bridge, essential-likelihood and conditioned-path consequences retain precisely these complete-source obligations. The new route is an enlargement of the controlled Lorentz source, not a replacement of that endpoint by a different model or topic.

Part 36. The complete arithmetic record

88. Analytic angular germs and nonlinear birth of exact labels

A zero tangent fraction does not imply a zero physical source. We resolve this distinction for the original circular table. The exact angular sets have an analytic one-sided germ, even when several active boundaries have the same tangent. A nonzero germ consequently has a finite order of birth. That order is not assumed bounded as the word length increases.

Throughout Sections 88–90, fix $0 < \chi \le 1/10$ and $0 < \varepsilon \le \chi/4$. Use the selected-contact charts of Theorem 94.3, their original physical restrictions, and the exact label $\lambda = (n, k)$. The radius r_χ, roof width $h_\chi = r_\chi^2/2$, density constant K_χ, and coefficient J_z retain their previous meanings. The selected orbit is continued only to define coordinates and masks; no continued nonphysical point is assigned physical measure.

88.1. **Analyticity of the primitive tests.**

Lemma 88.1 (Analytic physical decision germs)**.** *At a fixed normal critical word with positive selected incidences, the complete finite scalar-margin description of Lemma 105.2 can be chosen real analytic in a neighborhood of the origin in the Morse chart. Every active margin has nonzero differential. The inactive signs remain fixed on a smaller positive radius, which is not asserted uniform in the word.*

Proof. For the circular table, the selected contact equations are analytic functions of contact arclengths and the two endpoint variables. The flight lengths are positive. Their internal derivative is invertible by Lemma 94.1, so the analytic implicit function theorem gives the selected contact graph at this fixed word. The coordinate construction in the proof of Theorem 94.3 is analytic as well: its matrix $Q_z(x)$ is analytic and positive definite, matrix square root is analytic on the positive definite cone, and the map $x \mapsto Q_z(x)^{1/2}x$ has an analytic local inverse. We keep this specific Morse map.

The original section is a finite union of rectangles in the analytic (α, p) charts; see (128.1). Its edge tests, composed with each regular selected branch, are analytic. On a near-clearance envelope the perpendicular foot lies strictly inside the flight and the signed perpendicular distance is nonzero at the center. Its sign can therefore be fixed before taking its absolute value. The resulting clearance margin is analytic. The completeness and the nonzero differentials are those of Lemma 105.2; all other first-hit tests are retained as strict inactive tests on the chosen disk. Taking a minimum is needed for the smooth guard, but not for the scalar sign list: its individual candidates are kept separately. The explicit inactive-sign argument in (91.1) gives the last assertion. No claim of a count-uniform analytic radius or derivative bound is made. □

Let $B_{z,\lambda}(y)$ be the complete physical and exact-label indicator in (95.1). For $r > 0$ put

$$\ell_{z,\lambda}(r) = \frac{1}{2\pi} \int_0^{2\pi} B_{z,\lambda}(r\omega_\phi)\, d\phi. \tag{88.1}$$

This is the unweighted angular fraction. It is distinct from the clearance-guard-weighted angular fraction in the next section and from the roof density. The initial section test, occupation at $0, \dots, m-1$, terminal membership at m, and displacement are all evaluated before a label is assigned.

Lemma 88.2 (Ordering the analytic angular boundaries)**.** *For each fixed word there is $r_z > 0$ on which every active scalar margin has exactly two angular zero germs. They are real analytic functions of r at zero. After coincident germs have been identified, their cyclic order is fixed for $0 < r < r_z$. The label on each open arc between consecutive germs is independent of r.*

Proof. For an active analytic margin f, the function

$$F_f(r, \phi) = \begin{cases} f(r\omega_\phi)/r, & r \neq 0, \\ Df(0) \cdot \omega_\phi, & r = 0 \end{cases}$$

is analytic. At either zero ϕ_0 of its value at $r = 0$,

$$\partial_\phi F_f(0, \phi_0) = Df(0) \cdot \omega'_{\phi_0} \neq 0.$$

The analytic implicit function theorem supplies one root near each of these two angles. On the compact complement of their neighborhoods, $F_f(0, \cdot)$ stays away from zero, so there are no further roots for small r. Choose an angular cut avoiding every zero at $r = 0$. All roots then have analytic lifts to one interval of length 2π.

The difference of two such lifts is either the zero analytic germ or has a first nonzero Taylor coefficient. In the latter case it has a fixed sign for all sufficiently small positive r. There are finitely many pairs. Shrinking the radius once fixes their order simultaneously. Identifying a coincident root does not remove any primitive test: all of the sign changes attached to that root are retained. On each open moving arc no active test vanishes, and all inactive signs are fixed. The full input sign vector is therefore constant there as r varies. The original one-hot visit rule assigns the same label, or no label, to that arc. Boundary conventions concern only finitely many angles at each positive radius. □

Theorem 88.3 (Complete finite-order classification of angular germs)**.** *For every fixed selected word and exact label, the function $\ell_{z,\lambda}$ has a real analytic extension from a sufficiently small right neighborhood of zero. Its value at zero is the tangent fraction $\theta_{z,\lambda}$. Precisely one of the following holds:*

$$\ell_{z,\lambda} \equiv 0 \quad \textit{near zero}, \qquad \textit{or} \qquad \ell_{z,\lambda}(r) = \eta_{z,\lambda} r^{d_{z,\lambda}} + O_z(r^{d_{z,\lambda}+1}), \tag{88.2}$$

where $d_{z,\lambda} \in \{0, 1, 2, \dots\}$ and $\eta_{z,\lambda} > 0$. If $\theta_{z,\lambda} > 0$, then $d_{z,\lambda} = 0$ and $\eta_{z,\lambda} = \theta_{z,\lambda}$. If $\theta_{z,\lambda} = 0$ and the nonlinear germ is nonzero, then $d_{z,\lambda} \geq 1$. Every such nonlinear germ is retained.

Proof. Insert the fixed angular cut as two constant endpoints in the ordered list of Lemma 88.2. The angular fraction of each label is the sum, divided by 2π, of the lengths of those open arcs carrying that label. Each length is a difference of analytic root lifts, or of a root lift and a constant endpoint. Hence this finite sum is analytic at zero. Every summand is nonnegative for positive r, including arcs whose endpoints coincide at zero. Its constant term is the tangent angular fraction: away from the finite set of tangent zero directions, all signs converge to their tangent signs, with the inactive signs fixed. This also follows directly by dominated convergence in (88.1).

An analytic scalar germ is either identically zero or has a first nonzero Taylor coefficient. Since the angular fraction is nonnegative for positive r, that coefficient is positive. These observations

prove all assertions. In the zero-germ case the corresponding physical angular source is zero on this smaller disk; no assertion is made about the rest of the original chart. □

88.2. **A recorded radius and a relative Taylor remainder.** For a nonzero germ fix $0 < r^*_{z,\lambda} < r_\chi$ within all the preceding analytic, ordering and inactive-sign radii. Put $d = d_{z,\lambda}$, $\eta = \eta_{z,\lambda}$ and

$$L_{z,\lambda} = \frac{1}{(d+1)!} \sup_{0\le r\le r^*_{z,\lambda}} |\ell^{(d+1)}_{z,\lambda}(r)|, \qquad k^{\mathrm{a}}_{z,\lambda} = L_{z,\lambda}/\eta. \tag{88.3}$$

Taylor's theorem, with the lower coefficients zero, gives

$$|\ell_{z,\lambda}(r) - \eta r^d| \le k^{\mathrm{a}}_{z,\lambda}\, r\, \eta r^d \quad (0 < r < r^*_{z,\lambda}). \tag{88.4}$$

This is relative to the first nonzero coefficient, not to a vanished tangent fraction.

The data have a finite certificate at each nonzero germ: the primitive analytic formulas, the chosen radius, the cyclic ordering of the root germs, their first separating jets, the positive coefficient η, and the derivative remainder in (88.3). An asserted equality of two root germs requires an analytic identity; it is not certified by an arbitrary finite list of vanishing derivatives. For a nonidentical pair, its first separating coefficient and a remainder bound certify the order on a sufficiently small radius. Root derivatives are obtained by differentiating $F_f(r, \phi_f(r)) = 0$; the divisor at zero is the displayed nonzero angular derivative. No lower bound on that divisor, no uniform maximum separating order, and no uniform radius is hidden in the certificate. The radius and remainder are recorded separately and will enter the physical cutoff.

Remark 88.4 (Arbitrarily high birth order is locally possible). The analytic sign conditions $x > 0$, $y > 0$, and $y < x^{d+1}$ have nonzero active gradients for every integer $d \ge 1$. On the circle of radius r, their angular width is the small positive solution $\alpha(r)$ of $\sin\alpha = r^d \cos^{d+1}\alpha$. Thus $\alpha(r)/(2\pi) = r^d/(2\pi) + O(r^{d+1})$ and its tangent fraction is zero. This local example illustrates the classification; it is not claimed to construct a Lorentz itinerary of every possible order. The following height theorem does not require such a realization or a bound on the order.

89. Relative radial persistence of the original clearance source

We now restore the smooth physical guard. Analyticity is used only for the finite sign geometry, not asserted for the guard. In particular an infinitely flat guard at a zero center value is not assigned a finite birth order.

Let D^ε_z be the original first-clearance multiplier of Section 95. It satisfies $0 \le D^\varepsilon_z \le 1$ and the Lipschitz estimate (95.2). Write

$$g_z = D^\varepsilon_z(0), \qquad L_g = C_g\varepsilon^{-1}, \qquad k^{\mathrm{g}}_z = L_g/g_z \quad \text{when } g_z > 0. \tag{89.1}$$

At a competing-clearance seam $g_z = 1$. A non-seam critical center with $g_z > 0$ is allowed as well. Define the exact weighted fraction

$$a^\varepsilon_{z,\lambda}(r) = \frac{1}{2\pi}\int_0^{2\pi} B_{z,\lambda}(r\omega_\phi) D^\varepsilon_z(r\omega_\phi)\, d\phi. \tag{89.2}$$

Thus this is $A^\varepsilon_{z,n,k}(r^2/2)/(2\pi)$, not a new source or a substituted mask.

For a nonzero angular germ and $g_z > 0$ set

$$\begin{aligned} c_{z,\lambda} &= g_z\eta_{z,\lambda}, \qquad \gamma_{z,\lambda} = J_z c_{z,\lambda}, \\ \mathfrak{p}_{z,\lambda} &= \max\{(r^*_{z,\lambda})^{-1}, 2(k^{\mathrm{a}}_{z,\lambda} + k^{\mathrm{g}}_z)\}. \end{aligned} \tag{89.3}$$

All these numbers are finite at the fixed word. A change of positive scale of a primitive scalar margin changes none of its angular sets. The profile certificate is recorded on those exact sets. No finite value of $\mathfrak{p}$ is assigned when $g_z = 0$ or the angular germ is identically zero.

Theorem 89.1 (Relative radial profile at every finite birth order)**.** *For* $\mathfrak{p}_{z,\lambda} \le K$ *and* $0 < r < K^{-1}$,

$$(1 - Kr)c_{z,\lambda} r^{d_{z,\lambda}} \le a^{\varepsilon}_{z,\lambda}(r) \le (1 + Kr)c_{z,\lambda} r^{d_{z,\lambda}}. \tag{89.4}$$

Consequently, with $u = r^2/2$, *on the common coarea representative,*

$$e^{-K_\chi r}(1 - Kr)\gamma_{z,\lambda} r^{d_{z,\lambda}} \le b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t_z + u) \le e^{K_\chi r}(1 + Kr)\gamma_{z,\lambda} r^{d_{z,\lambda}}. \tag{89.5}$$

The constants do not depend on $d_{z,\lambda}$. *For a seam and* $d_{z,\lambda} = 0$, $\gamma_{z,\lambda} = \beta_{z,\lambda}$. *For* $d_{z,\lambda} > 0$, γ *is a leading radial coefficient, not a nonzero zero-width atom.*

Proof. On the circle of radius r, the Lipschitz estimate gives $g_z(1 - k^{\mathrm{g}}_z r) \le D^{\varepsilon}_z \le g_z(1 + k^{\mathrm{g}}_z r)$. Multiply by the nonnegative exact indicator and integrate. Combine this with (88.4). Put $x = k^{\mathrm{a}}_{z,\lambda} r$ and $y = k^{\mathrm{g}}_z r$. Since $x + y \le Kr/2 < 1/2$, the lower product is at least $1 - x - y \ge 1 - Kr$. For the upper product, $x + y + xy \le 2(x + y) \le Kr$. This proves (89.4). Multiplication by the unchanged positive density distortion in (95.5) proves (89.5). There is no summation over nonphysical labels or words in this argument. The last two statements follow from the values of g_z, d and η. □

89.1. **An outer annulus controls every inner level.** For a fixed $H > 0$ with $2H < h_\chi$ and $2K\sqrt{H} < 1$ define

$$q_{z,2H} = \frac{1}{2H} \int_0^{2H} b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t_z + v)\, dv,$$

$$\overline{q}_{z,H} = \frac{1}{H} \int_H^{2H} b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t_z + v)\, dv, \tag{89.6}$$

$$F_{\chi,K}(H) = e^{(2+\sqrt{2})K_\chi \sqrt{H}} \frac{1 + \sqrt{2}K\sqrt{H}}{1 - 2K\sqrt{H}}. \tag{89.7}$$

All coefficients use the original source at its actual roof value.

Proposition 89.2 (Order-free positive annular comparison)**.** *For every selected pair with* $\mathfrak{p}_{z,\lambda} \le K$,

$$\gamma_{z,\lambda}(2H)^{d_{z,\lambda}/2} \le \frac{e^{2K_\chi\sqrt{H}}}{1 - 2K\sqrt{H}} \overline{q}_{z,H} \le \frac{2e^{2K_\chi\sqrt{H}}}{1 - 2K\sqrt{H}} q_{z,2H}, \tag{89.8}$$

$$\operatorname*{ess\,sup}_{0<u<H} b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t_z + u) \le F_{\chi,K}(H)\overline{q}_{z,H} \le 2F_{\chi,K}(H) q_{z,2H}. \tag{89.9}$$

In particular there is no cost growing with the finite birth order.

Proof. For $H < v < 2H$, the radial variable lies between $\sqrt{2H}$ and $2\sqrt{H}$. Since $d \ge 0$, the lower bound in (89.5) is at least

$$e^{-2K_\chi\sqrt{H}}(1 - 2K\sqrt{H})\gamma_{z,\lambda}(2H)^{d/2}.$$

Average over an interval of length H. Positivity gives $\overline{q}_{z,H} \le 2q_{z,2H}$ and proves (89.8). For $0 < u < H$ the upper profile is at most $e^{\sqrt{2}K_\chi\sqrt{H}}(1 + \sqrt{2}K\sqrt{H})\gamma_{z,\lambda}(2H)^{d/2}$. Combine the two inequalities. Only the monotonicity of r^d for $d \ge 0$ was used. In particular neither a factor 2^d nor a maximum allowed value of d is introduced. □

The two excluded zero cases. If the angular germ is identically zero, the angular source vanishes on its certified smaller disk, but source in the rest of the chart is retained. If $g_z = 0$, the guard can be nonzero off the center and need not be analytic. Neither a division by g_z nor a finite-order classification of that guarded source is allowed. Such source will appear in the positive complement. These exclusions do not remove a zero-tangent-fraction label whose angular germ is nonzero and whose center guard is positive; all its finite orders are included above.

90. Ordered nonlinear caustic height and the complete source

We apply the annular comparison before the collision limit. This controls angular births directly at the density level. It neither differentiates an interval local limit nor passes from a finite-count analytic radius to a uniform radius without a cutoff.

For each fixed K, χ, ε, let

$$\mathcal{P}_{m,R,\lambda}(K) = \{z : g_z > 0,\ \ell_{z,\lambda} \not\equiv 0 \text{ as a germ},\ \mathfrak{p}_{z,\lambda} \le K\}. \tag{90.1}$$

The certificate and this selection do not depend on H. There is no cutoff on the birth order. Use the same disjoint physical itinerary restrictions as in (95.8), and put

$$\mathcal{Q}^{\mathrm{p},K}_{m,2H;\lambda,R} = \sum_{z\in\mathcal{P}_{m,R,\lambda}(K)} q_{z,2H}\delta_{t_z},$$

$$\mathcal{C}^{K}_{m,H;\lambda,R} = \sum_{z\in\mathcal{P}_{m,R,\lambda}(K)} \gamma_{z,\lambda}(2H)^{d_{z,\lambda}/2}\delta_{t_z}, \tag{90.2}$$

$$b^{\mathrm{p},K,H}_{m;\lambda,R}(t) = \sum_{z\in\mathcal{P}_{m,R,\lambda}(K)} \mathbf{1}_{\{0<t-t_z<H\}} b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t). \tag{90.3}$$

The second measure is scale weighted. Except on its order-zero part, it is not the zero-width trace $\mathcal{B}$. The last density is a restriction of the original source, not of its leading approximation.

Theorem 90.1 (All-order original-source collar height)**.** *Fix K, χ, ε, H before taking $m \to \infty$, with $2H < h_\chi$ and $2K\sqrt{H} < 1$. For every $h > 0$,*

$$\limsup_{m\to\infty} \sup_{R,\lambda,t} m^2 \mathcal{C}^{K}_{m,H;\lambda,R}([t-h, t+h]) \le \frac{2Ce^{2K\chi\sqrt{H}}}{1-2K\sqrt{H}}(2h+2H), \tag{90.4}$$

$$\limsup_{m\to\infty} \sup_{R,\lambda} m^2 \|b^{\mathrm{p},K,H}_{m;\lambda,R}\|_\infty \le 6CHF_{\chi,K}(H). \tag{90.5}$$

The constant C is the unchanged physical collar constant. It is independent of the number of words, labels and active margins, of coincident critical values, and of every birth order. In particular,

$$\lim_{H\downarrow 0} \limsup_{m\to\infty} \sup_{R,\lambda} m^2 \|b^{\mathrm{p},K,H}_{m;\lambda,R}\|_\infty = 0. \tag{90.6}$$

This includes zero-tangent-fraction nonlinear sectors of every finite order and non-seam critical centers with positive center guard, when they belong to the stated stratum. A bounded measurable insertion is controlled by its supremum norm times the same positive height bound.

Proof. For an interval I, sum (89.8) over its unchanged critical centers. This is a positive comparison with $\mathcal{Q}^{\mathrm{p},K}_{m,2H}(I)$. Theorem 96.1 applies to this subfamily, at the fixed collar width $2H$. Its bound is $C(2h+2H)$, which proves (90.4).

At roof t, each summand of (90.3) has its critical center in $[t-H, t]$. By (89.9),

$$b^{\mathrm{p},K,H}_{m;\lambda,R}(t) \le 2F_{\chi,K}(H)\mathcal{Q}^{\mathrm{p},K}_{m,2H;\lambda,R}([t-H,t]) \quad \text{for almost every } t. \tag{90.7}$$

The common coarea representative excludes only a finite-count null roof set; taking the finite union over words and exact labels does not enlarge it to positive measure. Apply the same collar theorem to an interval of length H, now centered at $t - H/2$. Its normalized limsup is at most $C(H+2H) = 3CH$. This proves (90.5). At fixed K, χ, the factor $F_{\chi,K}(H)$ tends to one as $H \downarrow 0$, proving (90.6). Variation domination at the original positive source gives the insertion assertion.

The argument sums all birth orders at each fixed count before taking any limit. It uses one common outer annulus, not a countable sum of dyadic roof scales. Thus no interchange of an infinite scale sum and the collision limsup is needed. Coincident centers add with their positive coefficients throughout. □

90.1. **Relation to the angular-condition tail.** On a seam, the order-zero part has exactly the physical coefficient $J_z\theta_{z,\lambda}$ of Section 92. The positive-order part instead has zero such coefficient, although its nonlinear source is covered by Theorem 90.1. This is precisely the part which an estimate on $E(K)$ alone could not control.

The persistence cutoff $\mathfrak{p}$ is not the angular cutoff $\mathfrak{a}$ in (91.3). The original estimate $E(K) \to 0$ has not been derived from the analytic classification. At fixed m, R, every positive-center-guard nonzero angular germ has a finite persistence number. Neither this statement nor the order-independent constant in (90.5) implies tightness after the supremum in R, λ and the collision limsup. In particular there is no choice $K = K_m$ in the theorem.

90.2. **The full positive source and one canonical order of limits.** At fixed admissible parameters the original first-clearance source has an exact disjoint restriction and its positive complement:

$$b^{\varepsilon,\mathrm{clr}} = b^{\mathrm{p},K,H} + r_{68}^{\varepsilon,\chi,K,H}, \qquad r_{68}^{\varepsilon,\chi,K,H} \geq 0. \tag{90.8}$$

The complement retains the entire outside source in (95.8): failed incidence taper, selected grazing, noncritical components and all uncovered states. Within the selected charts it also retains zero-center-guard source, source outside a certified angular-germ disk, large persistence numbers and the portions outside the chosen inner collar. A zero angular germ can delete source only inside its certified zero disk; its possible outer source is not deleted. In particular, zero tangent fraction alone is no longer an exclusion from the controlled term.

Write $\mathcal{H}(f) = \limsup_m \sup_{R,n,k} m^2 \|f_{m;n,k,R}\|_\infty$. Then

$$\mathcal{H}(b^{\varepsilon,\mathrm{clr}}) \leq 6CHF_{\chi,K}(H) + \mathcal{H}(r_{68}^{\varepsilon,\chi,K,H}). \tag{90.9}$$

The complete first-incidence source remains separate. Its finite-count inverse-incidence gain in Section 97 has an exponential count constant and is not substituted for its ordered height.

For clarity let $\mathcal{E}_M$ denote the normalized central-window pointwise error limsup in the original raw-density problem. The exact positive-error identity of Theorem 181.4 and (90.9) give, for each fixed reconstruction band B and every admissible fixed χ, K, H,

$$\mathcal{E}_M \leq C_M B^{-1/192} + \mathcal{H}(b^{\varepsilon(B),\mathrm{inc}}) + 6CHF_{\chi,K}(H) + \mathcal{H}(r_{68}^{\varepsilon(B),\chi,K,H}), \qquad \varepsilon(B) = A_0 B^{-1/12}. \tag{90.10}$$

First fix B and its physical width, then choose admissible χ, K, H, and only then take the collision limsup. Subsequent outer choices of these parameters may depend on B, but never on m. To conclude $\mathcal{E}_M = 0$, the complete incidence height must tend to zero with $B \to \infty$, and there must be fixed admissible choices for which the last two terms tend to zero as well. Only the explicit inner-collar term is estimated here. No decrease of H is asserted to make its complementary source small.

The finite arithmetic transition kernel, all zero arithmetic classes, and the original unrestricted pointwise target are unchanged. The same-roof bridge, essential-likelihood and conditioned-path implications retain their full positive-source hypotheses. The local-limit inputs [2, 3, 12] are not used as density differentiation rules or as an unproved estimate on the complementary height.

The new proof route is the analytic classification, relative radial persistence, and the single positive annular comparison above.

91. Relative angular stability of the exact physical decisions

We retain the original circular Lorentz table, section and exact labels. Fix χ, ε as in Section 95, and restrict attention to its normal critical words with a competing-clearance seam. The selected incidences satisfy the stated taper; selected grazing is not included. Write $\lambda = (n, k)$, $\beta_{z,\lambda} = J_z \theta_{z,\lambda}$, and $a_{z,\lambda}(u) = A^{\varepsilon}_{z,n,k}(u)/(2\pi)$. The new estimate is relative to $\beta_{z,\lambda}$, not to the full reversible coefficient J_z. This distinction permits a positive absorption argument without a concentration theorem for the full reversible trace.

91.1. **A budget from the primitive scalar margins.** At each fixed word choose a closed disk $|y| \le \bar{r}_z$, contained strictly inside the Morse chart and inside the saturated disk of Lemma 95.1. Reduce it, if necessary, so that the finite Boolean description of Lemma 105.2 is valid throughout it. This radius is part of the data; it is not asserted to be uniform in the word. In particular no competing disk, section edge or inactive first-hit condition may be omitted unless its constant sign has been verified on this disk.

Let $\mathscr{F}_z$ be the resulting finite list of C^2 scalar margins in Morse coordinates, including all physical and section decisions. For $f \in \mathscr{F}_z$ put

$$b_f = f(0), \qquad a_f = Df(0), \qquad A_f = |a_f|, \qquad L_f = \tfrac{1}{2} \sup_{|y| \le \bar{r}_z} \|D^2 f(y)\|.$$

Every active margin, meaning $b_f = 0$, has $A_f > 0$ by the cited lemma. Set

$$\rho_z = \min \left\{ \bar{r}_z, \min_{b_f \neq 0} \frac{|b_f|}{2(A_f + L_f \bar{r}_z)} \right\}, \tag{91.1}$$

$$\kappa_z = \sum_{b_f = 0} \frac{L_f}{A_f}. \tag{91.2}$$

An empty minimum or a quotient with zero denominator in (91.1) is $+\infty$. Thus $\rho_z > 0$ and $\kappa_z < \infty$ at every fixed word. These budgets are unchanged by a positive constant rescaling of an individual margin. The sum is over the actual scalar tests, not over words or output labels. It is not claimed to have a uniform bound for all collision counts.

For $\theta_{z,\lambda} > 0$ define the angular condition number

$$\mathfrak{a}_{z,\lambda} = \max\{\rho_z^{-1}, \kappa_z/\theta_{z,\lambda}\}. \tag{91.3}$$

Labels with $\theta_{z,\lambda} = 0$ have zero atomic coefficient and are not assigned a finite condition number. Their nonlinear physical source need not vanish and will remain in the positive complement. Exact affine active margins have $\kappa_z = 0$; in that case arbitrarily small positive angular fractions cause no loss in (91.3).

Lemma 91.1 (Homogeneous threshold estimate)**.** *If $f(0) = 0$, $|Df(0)| = A > 0$ and $|f(r\omega) - rDf(0) \cdot \omega| \le Lr^2$, the normalized angular measure of the sign-disagreement set is at most $\min(1, Lr/A)$. No angle between different margin gradients enters this estimate. If $b_f \neq 0$ and $r < \rho_z$, the sign of $f(r\omega)$ is its constant sign at zero for every ω.*

Proof. Sign disagreement for an active margin implies $|Df(0) \cdot \omega| \le Lr$. After rotation the angular fraction is at most $2\pi^{-1} \arcsin(\min(1, Lr/A))$, which is at most $\min(1, Lr/A)$: convexity places $\arcsin x$ below its chord $\pi x/2$ on $[0, 1]$. For an inactive margin, Taylor's estimate gives

$$|f(r\omega) - b_f| \le A_f r + L_f r^2 \le r(A_f + L_f \bar{r}_z) < |b_f|/2.$$

This proves the second assertion. Strict or half-open conventions alter only finitely many tangent directions in the active comparison. □

Theorem 91.2 (Relative angular error with exact labels)**.** *For $r = \sqrt{2u} < \rho_z$, the exact physical angular fractions satisfy*

$$|a_{z,\lambda}(u) - \theta_{z,\lambda}| \le \min(1, \kappa_z r), \tag{91.4}$$

$$(\theta_{z,\lambda} - a_{z,\lambda}(u))_+ \le \min(\theta_{z,\lambda}, \kappa_z r), \tag{91.5}$$

$$\sum_{\lambda} |a_{z,\lambda}(u) - \theta_{z,\lambda}| \le 2\min(1, \kappa_z r). \tag{91.6}$$

Consequently, for every fixed $K > 0$, whenever $\mathfrak{a}_{z,\lambda} \le K$ and $r < K^{-1}$,

$$(1 - Kr)\theta_{z,\lambda} \le a_{z,\lambda}(u) \le (1 + Kr)\theta_{z,\lambda}. \tag{91.7}$$

These estimates concern the original nonlinear masks. The tangent masks supply the reference fraction only; they are not substituted for the physical source.

Proof. The inactive signs are fixed by Lemma 91.1. The union of active sign-disagreement sets has normalized angular measure at most $\min(1, \kappa_z r)$. Outside it the entire Boolean input vector agrees with its tangent vector. In particular the first-hit choice, the initial section decision, all occupation bits at $0, \dots, m-1$, and the separate terminal decision at m agree. The original visit rule therefore gives the same exact output label. On the disk under consideration the first-clearance multiplier equals one wherever a physical label is assigned. Its image mark is still $j + 1$. Thus $a_{z,\lambda}$ is precisely the normalized measure of this nonlinear output set.

Integrating the two indicators proves (91.4); their nonnegativity improves the one-sided estimate to (91.5). At each angle both output vectors are zero or a single unit vector. Their ℓ^1 distance is at most two, proving (91.6) without a label-cardinality factor. Finally $r < K^{-1} \le \rho_z$ and $\kappa_z \le K\theta_{z,\lambda}$ give (91.7). □

The role of the offsets. The linear-in-r estimate uses homogeneous active thresholds only. Inactive thresholds are kept fixed by the explicit radius (91.1); near-active offsets can make this radius small. It would be incorrect to apply the homogeneous estimate uniformly to arbitrary translated thresholds. The affine $u^{1/4}$ bound of Section 105 retains its original, different scope. No lower bound on an individual gradient, no separation of tangent directions and no collision-count bound on $\mathfrak{a}$ has been introduced silently.

92. Zero-width trace concentration by positive absorption

Keep χ, ε fixed, and use the same disjoint physical critical charts as in Section 96, restricted to clearance seams. For a fixed $K > 0$ define the exact-label stratum

$$\mathcal{Z}_{m,R,\lambda}(K) = \{z : \theta_{z,\lambda} > 0,\ \mathfrak{a}_{z,\lambda} \le K\}.$$

Selection depends only on the center data and the exact label, not on the collar width. Define

$$\mathcal{B}^K_{m;\lambda,R} = \sum_{z \in \mathcal{Z}_{m,R,\lambda}(K)} \beta_{z,\lambda}\delta_{t_z},$$

$$\mathcal{Q}^K_{m,s;\lambda,R} = \sum_{z \in \mathcal{Z}_{m,R,\lambda}(K)} \left(s^{-1} \int_0^s b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t_z + u)\, du \right) \delta_{t_z}. \tag{92.1}$$

Let $\mathcal{D}^K_{m,s}$ denote the same restriction of (96.7). Subscripts suppressed below are unchanged. Theorem 96.1 applies to these subfamilies: selection only removes positive source pieces. No extra regularity of the selected family is needed for that comparison.

Theorem 92.1 (An ordered zero-width bound on finite-jet strata)**.** *For every fixed K, χ, ε and $h > 0$,*

$$\limsup_{m\to\infty} \sup_{R,\lambda,t} m^2 \mathcal{B}^K_{m;\lambda,R}([t-h, t+h]) \le 2Ch, \tag{92.2}$$

where C is the collar constant, independent of K. Moreover, for $s < h_\chi$ with $K\sqrt{2s} < 1$,

$$\mathcal{B}^K_m(I) \le \frac{e^{K_\chi\sqrt{2s}}}{1 - K\sqrt{2s}} \mathcal{Q}^K_{m,s}(I), \tag{92.3}$$

$$\mathcal{D}^K_{m,s}(I) \le \tfrac{2}{3} K\sqrt{2s}\, \mathcal{B}^K_m(I). \tag{92.4}$$

In particular

$$\limsup_{m\to\infty} \sup_{R,\lambda,t} m^2 \mathcal{D}^K_{m,s}([t-1, t+1]) \le \tfrac{4}{3} CK\sqrt{2s}. \tag{92.5}$$

The statements require no concentration estimate for $\sum_z J_z \delta_{t_z}$.

Proof. For $0 < u < s$, Theorem 91.2 and the lower multiplicative density estimate give

$$b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t_z + u) \ge e^{-K_\chi\sqrt{2s}}(1 - K\sqrt{2s})\beta_{z,\lambda}.$$

Average over u, then sum at the unchanged critical values in I. This proves (92.3). All coefficients are positive; coincident critical values simply add. Alternatively the same inequality absorbs a relative angular-loss term into the left hand side of (96.8).

Fix $s, h, K, \chi, \varepsilon$ before taking the collision limsup. The positive collar comparison gives

$$\limsup_{m} \sup_{R,\lambda,t} m^2 \mathcal{B}^K_m([t-h, t+h]) \le C \frac{e^{K_\chi\sqrt{2s}}}{1 - K\sqrt{2s}} (2h + s).$$

The left side does not depend on s. Let $s \downarrow 0$ now, with K, h, χ, ε still fixed, to obtain (92.2). This does not interchange m with the fixed-count limit of $\mathcal{Q}_{m,s}$.

The one-sided angular estimate is at most $K\sqrt{2u}\theta_{z,\lambda}$ on the stratum. Since $s^{-1}\int_0^s \sqrt{2u}\, du = \frac{2}{3}\sqrt{2s}$, it proves (92.4). Apply (92.2) with $h = 1$ to obtain (92.5). □

92.1. **A pointwise height estimate for the original caustic collars.** For $H > 0$ define a positive density on the original source by

$$b^{K,H}_{m;\lambda,R}(t) = \sum_{z \in \mathcal{Z}_{m,R,\lambda}(K)} \mathbf{1}_{\{0<t-t_z<H\}}\, b^{\varepsilon,\mathrm{clr}}_{z,\lambda}(t). \tag{92.6}$$

The width H is a roof width, whereas the chart radius is $\sqrt{2H}$.

Corollary 92.2 (Ordered essential height on each stratum)**.** *If $H < h_\chi$ and $K\sqrt{2H} < 1$, then*

$$\limsup_{m\to\infty} \sup_{R,\lambda} m^2 \|b^{K,H}_{m;\lambda,R}\|_\infty \le CHe^{K_\chi\sqrt{2H}}(1 + K\sqrt{2H}). \tag{92.7}$$

Thus this original-source essential height tends to zero as $H \downarrow 0$ after the collision limit, with K, χ, ε fixed. A bounded insertion is dominated by its supremum norm times the same right hand side.

Proof. The upper multiplicative profile and (91.7) bound each summand, on the common coarea representative, by $e^{K_\chi\sqrt{2H}}(1 + K\sqrt{2H})\beta_{z,\lambda}$. Contributing centers lie in $[t - H, t]$. Apply (92.2) with center $t - H/2$ and radius $H/2$. At each finite count the union of the exceptional roof sets is null, so this gives an essential-supremum estimate for the summed density. Source domination proves the insertion statement. □

What is uniform. The constant in (92.2) is independent of the cutoff K, but the collision-count convergence is not asserted to be uniform in K. Every fixed word with positive angular fraction belongs to some finite stratum; this fixed-count exhaustion does not permit $K \to \infty$ before $m \to \infty$. The next section keeps the missing tail as a positive measure. Likewise, no height for an annulus or for a zero-angular-fraction source follows from (92.7).

93. Angular tails and the complete positive source

The preceding result replaces a fixed-word dominated-convergence argument by an ordered zero-width theorem on every finite-jet stratum. Passing to all words still requires a dynamical estimate. We state it in the original physical coefficient and do not replace it by an unweighted word count.

93.1. **A positive tail sufficient for the full angular loss.** Define

$$\mathcal{E}^K_{m;\lambda,R} = \sum_{\substack{z:\theta_{z,\lambda}>0 \\ \mathfrak{a}_{z,\lambda}>K}} \beta_{z,\lambda}\delta_{t_z}, \qquad \mathcal{B}_m = \mathcal{B}^K_m + \mathcal{E}^K_m. \tag{93.1}$$

Let

$$E(K) = \limsup_{m\to\infty} \sup_{R,\lambda,t} m^2 \mathcal{E}^K_m([t-1,t+1]) \in [0,\infty]. \tag{93.2}$$

These are the physical weights $J_z\theta_{z,\lambda}$, with the original half-word labels. They are not the full reversible trace.

Proposition 93.1 (Quantified angular exhaustion)**.** *For fixed K and $s < h_\chi$ with $K\sqrt{2s} < 1$,*

$$\limsup_{m} \sup_{R,\lambda,t} m^2 \mathcal{D}_{m,s}([t-1,t+1]) \le \tfrac{4}{3}CK\sqrt{2s} + E(K). \tag{93.3}$$

Consequently the additional estimate

$$\lim_{K\to\infty} E(K) = 0 \tag{93.4}$$

implies the full angular-loss conclusion (96.9) *for this seam-chart family, and*

$$\limsup_{m} \sup_{R,\lambda,t} m^2 \mathcal{B}_m([t-h,t+h]) \le 2Ch \quad (0 < h \le 1).$$

Condition (93.4) *is a sufficient physical weighted-tightness input, not an assertion proved for the full Lorentz family in this article.*

Proof. On the complement of the stratum the loss of each label is at most its tangent fraction, so $\mathcal{D}_{m,s} \le \mathcal{D}^K_{m,s} + \mathcal{E}^K_m$. Labels with zero tangent fraction contribute zero to the one-sided loss even when their nonlinear profile is positive. Apply (92.5) to obtain (93.3). First let $s \downarrow 0$ at fixed K, and only then let $K \to \infty$. The nonnegative left side therefore tends to zero under (93.4). For the atomic bound use (93.1), the stratum bound, and $[t-h,t+h] \subset [t-1,t+1]$; then let $K \to \infty$. No count-dependent cutoff is inserted into the local law. □

Corollary 93.2 (A moment sufficient for weighted tightness)**.** *If for some $p > 0$ the independently verified bound*

$$M_p := \limsup_{m} \sup_{R,\lambda,t} m^2 \sum_{z:\,|t_z-t|\le 1} \beta_{z,\lambda}\mathfrak{a}^p_{z,\lambda} < \infty$$

holds, with zero coefficients omitted, then $E(K) \le M_p K^{-p}$. For this hypothesis only, the ordered angular loss in (93.3) *is $O(s^{p/(2p+2)})$ as $s \downarrow 0$. No such moment bound or polynomial rate is asserted for the unrestricted Lorentz source.*

Proof. On $\mathfrak{a} > K$, the indicator is at most $\mathfrak{a}^p K^{-p}$. Positivity gives the tail bound. In (93.3) choose $K = s^{-1/(2p+2)}$. For each fixed s this is a fixed cutoff before the collision limit, and $K\sqrt{2s} \to 0$. Both terms have the stated power of s. □

93.2. **The pointwise target retains every positive component.** For K, H in Corollary 92.2, the source defining $b^{K,H}$ is a disjoint restriction of the unchanged complete first-clearance source. Thus there is the exact positive partition

$$b^{\varepsilon,\mathrm{clr}} = b^{K,H} + r^{\varepsilon,\chi,K,H}, \qquad r^{\varepsilon,\chi,K,H} \geq 0. \tag{93.5}$$

The second term includes the original outside source, tapered-incidence violations, selected grazing, noncritical rank components, non-seam critical centers, labels of zero tangent fraction, the large-condition strata and all omitted chart annuli. This is a source partition, not a cancellation or an equality inferred from small mass.

Write $\mathcal{H}(f) = \limsup_m \sup_{R,n,k} m^2 \|f_{m;n,k,R}\|_\infty$. For every fixed admissible ε, χ, K, H, positivity gives

$$\mathcal{H}(b^{\varepsilon,\mathrm{clr}}) \leq CHe^{K\chi\sqrt{2H}}(1 + K\sqrt{2H}) + \mathcal{H}(r^{\varepsilon,\chi,K,H}). \tag{93.6}$$

In particular, at each fixed ε, an arbitrarily small bound on both terms, obtained by fixed choices of χ, K, H before the collision limit, would close the clearance height at that width. For the unrestricted theorem one needs the corresponding bounds tending to zero as $\varepsilon \downarrow 0$, as well as the complete incidence height. A bound on $E(K)$ alone would not give the height of r.

The inherited positive-error representation (Theorem 181.4) remains

$$\limsup_m \sup_{R,n,k} \|P - G - m^2(b^{\varepsilon(B),\mathrm{inc}} + b^{\varepsilon(B),\mathrm{clr}})\|_\infty \leq CB^{-1/192}, \qquad \varepsilon(B) = A_0 B^{-1/12}.$$

Here B is the fixed reconstruction band, unrelated to the atomic measure $\mathcal{B}$. All new cutoffs are chosen with this band fixed before the collision limit. The complete incidence estimate and the height of the positive complement in (93.5) are not supplied by the angular calculation. The unrestricted two-sided pointwise arithmetic law and its same-roof consequences therefore remain the unchanged target, not conclusions inferred from the stratum theorem. Its finite arithmetic transition factor and zero classes are retained.

93.3. **Relation to local limit results.** Szász–Varjú [2, Theorem 1.1 and the following remark] prove a mixed local limit in weak topology for a minimal nondegenerate bounded Hölder observable over a Young system. The stated weak conclusion is distinct from a local limit for densities. Demers–Pène–Zhang [12] treat the cell index of randomly deforming billiards with piecewise Hölder endpoint observables. Dolgopyat–Nándori [3] formulate local limit and mixing conditions for suspensions, including finite-horizon Sinai billiard applications.

The comparison needed here is more specific than a change of observable: Section 96 uses the already established exact-occupation and displacement local law with two normal endpoint strips of width $O(\sqrt{s})$ and roof interval of length $2h + s$. It keeps s fixed before the collision limit. The new step in Section 92 is a relative geometric absorption which yields a zero-width trace and then an essential-height bound on specified strata. These are not obtained by differentiating a fixed-set local limit. Nor do the cited local-limit statements, without an additional hypothesis verification, estimate the weighted condition-number tail (93.4) or the full positive remainder (93.5). We make no claim of a new general local limit theorem for all singular hyperbolic systems.

94. Uniform contact continuation across physical clearance seams

The protected collars of Section 169 keep both the selected contact equations and every physical decision fixed. For a clearance caustic the latter requirement is inappropriate: the physical source occupies only part of a smooth selected-contact chart. We separate the two requirements. The

chart below continues selected reflections, whereas all first-hit and section restrictions will remain as indicators in the original source integral.

Set $q = 47/53$, $\sigma = \sqrt{q}$, $d_j = \min(j, m-j)$, and $e_j = q^{d_j/4}$. A selected normal-to-normal word is χ-*incident* if it is in the physical-side closure, its selected contacts have positive incidence, and

$$c_j(z) \geq \chi e_j \quad (0 \leq j \leq m), \qquad 0 < \chi \leq 1/10. \tag{94.1}$$

There is no clearance or section-margin condition in this definition. In particular several competing disks and section edges may meet at z. All constants in this section depend only on the fixed one-flight geometry and contact charts, not on m, R, z, or their exact labels.

94.1. **An incidence-normalized inverse.** Use the oriented contact arclengths and the matrix A of (169.2). Write $D_c = \operatorname{diag}(c_j)$. The following refinement of the inherited Neumann estimate is useful.

Lemma 94.1 (Weighted contact inverse)**.** *For every positive selected incidence,*

$$|(A^{-1})_{ij}| \leq C \min(c_i, c_j) q^{|i-j|}, \qquad |(D_c^{-1} A^{-1})_{ij}| \leq C q^{|i-j|}. \tag{94.2}$$

Consequently $N = AD_c$, the internal Hessian with row j divided by c_j, has an inverse bounded independently of m in the norm

$$\|v\|_\sigma = \max_{1 \leq j < m} \sigma^{-d_j} |v_j|.$$

The assertions about internal matrices are vacuous for $m = 1$.

Proof. Write $A = D(I - K)$, where D is its diagonal. The row sums of K are at most q, its entries are nonnegative, and K connects only neighboring indices. Thus $(K^r)_{ij} = 0$ for $r < |i-j|$ and $|(K^r)_{ij}| \leq q^r$. Moreover $D_{jj}^{-1} \leq R c_j/2$. The Neumann series gives $|(A^{-1})_{ij}| \leq (R_+/2)(1-q)^{-1} c_j q^{|i-j|}$. Symmetry gives the same inequality with c_i. Division by c_i proves the second assertion. Since $|d_i - d_j| \leq |i-j|$,

$$\sup_i \sum_j q^{|i-j|} \sigma^{d_j - d_i} \leq 1 + 2 \sum_{r \geq 1} (q/\sigma)^r < \infty.$$

This is the asserted weighted inverse bound. No product of inverse incidences remains in it. □

Theorem 94.2 (Count-uniform selected-contact jets)**.** *There are $a_0 > 0$ and $C_1, C_2, C_3 < \infty$ such that every χ-incident word has a unique selected-contact continuation on $|u|, |v| < a_0 \chi^3$ about its normal endpoints. On this square all selected incidences are at least $\chi e_j/2$. If $y_j(u, v)$ is internal arclength displacement from the original contact, then*

$$\|D^r y_j\| \leq C_r \chi^{1-r} \sigma^{d_j}, \qquad 1 \leq r \leq 3. \tag{94.3}$$

The same bounds, with fixed changes of constants and neighboring indices, apply to the selected positions, velocities and flight data. The continuation need not satisfy the first-hit or section tests. Only its restriction satisfying those tests represents physical source.

Proof. Let E_j be the internal stationary equation for the polygonal length. Freeze $c_j^0 = c_j(z)$ and put $G_j = E_j/c_j^0$. At the base point $D_y G = D_c^{-1}(D_c A D_c) = AD_c = N$. Its inverse has the weighted bound of Lemma 94.1. The endpoint forcing has only its first and last rows nonzero. Their entries have modulus $c_0 t_0$ and $c_m t_{m-1}$ after the indicated row division, and therefore have bounded weighted norm.

Each E_j depends on three adjacent contact positions. Its derivatives of any fixed order are bounded on fixed local arclength charts: the formulas divide by neighboring flight lengths, which are at least $\ell_0 = 3/50$, and not by incidence. Tangent signs are fixed from the base contact word.

Put $\omega_j = \sigma^{d_j}$, with endpoint weights one. Neighboring weights differ by at most σ^{-1}. For $r \ge 2$ a derivative of order r, with arguments of weighted norm at most one, therefore satisfies

$$\omega_j^{-1}|D^r G_j[v_1,\dots,v_r]| \le C_r \frac{\omega_j^{r-1}}{c_j^0} \le C_r \chi^{-1} q^{((r-1)/2-1/4)d_j} \le C_r \chi^{-1}. \tag{94.4}$$

The same estimate holds for mixed endpoint derivatives, whose support is confined to the endpoint rows. These are estimates for the scaled norms in varying finite dimensions; their constants do not count rows.

Apply the contraction map obtained by multiplying the stationary equations by the fixed inverse N^{-1}. On a weighted ball of radius $a\chi$ and endpoint square of radius $a\chi$, the derivative of its nonlinear part is bounded by Ca. Taking a small gives a contraction into a slightly larger fixed multiple of the endpoint radius. This proves a unique smooth solution and a uniformly bounded inverse linearization there. In particular its first derivative has weighted norm at most C. Differentiating the equation twice gives a second derivative bounded by $C\chi^{-1}$; differentiating a third time gives $C(\chi^{-1}+\chi^{-2}) \le C'\chi^{-2}$, by (94.4). This proves (94.3), on the smaller displayed square.

Positions and unit flight directions are Lipschitz functions of adjacent arclengths with uniform constants. Thus incidence changes by at most $C(|u|+|v|)\omega_j$. Relative to c_j^0 this is at most $C\chi^{-1}(|u|+|v|)q^{d_j/4}$. Reduction of a_0 keeps it below $1/2$. Endpoint incidences likewise stay above $1/2$. The local stationary solution thus retains the selected positive reflection signs. The constants were chosen inside fixed contact charts. Uniqueness on overlaps gives the whole square. This argument used neither a lower bound on a competing-hit margin nor a fixed section membership. On points satisfying the original first-hit tests it agrees with the original billiard word by endpoint injectivity in Lemma 167.1. □

94.2. **Uniform Morse coordinates and relative density.** Let F_z be the selected reduced action and let $w_z = |\partial_u \partial_v F_z|/(4\pi R c_*)$, where $c_* = 91/(10000\pi)$. This is a smooth coefficient on the continuation; it is the density of the original source only after imposing its physical and section indicators.

Theorem 94.3 (A uniform quadratic chart at the seam)**.** *There are $r_0, C > 0$ such that, putting*

$$r_\chi = r_0\chi^3, \qquad h_\chi = r_\chi^2/2, \qquad K_\chi = C\chi^{-2}, \tag{94.5}$$

every χ-incident word has a selected-contact coordinate map $x = \Phi_z(y)$ on $|y| < r_\chi$ satisfying

$$\begin{aligned} F_z(\Phi_z(y)) &= t_z + |y|^2/2, & |p_0| + |p_m| &\le C|y|, \\ \widetilde{w}_z(y) &= w_z(\Phi_z(y))|\det D\Phi_z(y)|, & \widetilde{w}_z(0) &= J_z/(2\pi), \\ \left|\log \frac{2\pi\widetilde{w}_z(y)}{J_z}\right| &\le K_\chi|y|. \end{aligned} \tag{94.6}$$

Here J_z is the unchanged reversible coefficient of Theorem 103.2, interpreted from its physical side. The constants are uniform in the word length and in every positive incidence satisfying (94.1)*. The disk is a coordinate disk, not an assertion of a full physical disk.*

Proof. The selected contact second variation has the same positive internal block throughout the continuation. Its endpoint Schur complement therefore gives

$$\lambda I \le \nabla^2 F_z \le \Lambda I, \quad \lambda = 1/(2R_+), \quad \Lambda = 2(1/R_- + 1/\ell_0),$$

after reducing the endpoint square. First variation expresses DF_z using only the two endpoint flight directions. The first three contact jets consequently bound $D^r F_z$, $2 \le r \le 4$, by $C\chi^{-2}$. The value $F_z(0)$, which grows with m, is not included in this derivative estimate.

The determinant formula (97.2) is valid on the selected continuation. In its logarithmic derivative the flight terms have summable bounds $C\omega_j$. The diagonal potential terms of $\operatorname{tr}(A^{-1}dA)$ are bounded, using the refined diagonal inverse, by $C|dc_j|/c_j$. Their sum is at most

$$C\chi^{-1}\sum_{j=1}^{m-1} q^{d_j/4}|(du,dv)| \le C'\chi^{-1}|(du,dv)|.$$

Off-diagonal terms and the two endpoint cosines have bounded sums. It follows that $|D\log w_z| \le C\chi^{-1}$. This is a relative bound and does not require a lower bound for $w_z(0)$ as m grows.

For completeness use the explicit quadratic construction

$$Q_z(x) = 2\int_0^1 (1-s)\nabla^2 F_z(sx)\,ds, \qquad y = Q_z(x)^{1/2}x.$$

It has derivative $(\nabla^2 F_z(0))^{1/2}$ at zero. Matrix square root and inversion have bounded derivatives on matrices with spectrum in $[\lambda,\Lambda]$. The preceding fourth-derivative bound gives $\|D^2(y(x))\| \le C\chi^{-2}$ on the endpoint square. On a square of radius $a_0\chi^3$ its first derivative is uniformly close to its invertible value at zero, after reducing a_0. The quantitative inverse function argument supplies a ball $|y| < r_0\chi^3$ in its image, with $D\Phi_z$ bounded and $D^2\Phi_z$ bounded by $C\chi^{-2}$. The coordinate identity is exact. Differentiating $\log\widetilde{w}_z = \log(w_z\circ\Phi_z) + \log|\det D\Phi_z|$ gives a bound $C\chi^{-2}$. Integration from zero proves the last line of (94.6). At zero its value is the original Morse coefficient $J_z/(2\pi)$. The momentum estimate follows from first variation, the upper Hessian bound and $|\Phi_z(y)| \le C|y|$. □

Coverage provided by the theorem. Every normal critical point in the physical-side closure satisfying (94.1) has this same chart budget. No nonzero clearance margin, section margin, individual boundary-gradient lower bound, or angle between boundary gradients enters the budget. Selected grazing and points violating the tapered incidence envelope are not included. The next section keeps their source, and every noncritical or uncovered source component, in an explicit positive complement. No physical probability is assigned to the continued graph.

95. A count-uniform angular formula for the original boundary source

We use the uniform selected-contact chart without linearizing a physical boundary. Thus a small boundary gradient does not enter an error constant. The resulting angular integral keeps the exact nonlinear first-hit tests, the smooth physical guard and every section decision. It is not a closed evaluation of all those angular sets.

Fix a χ-incident critical word and the chart of Theorem 94.3. Write $x_z(y)$ for its selected initial state. Define $B_{z,n,k}(y)$ to be zero unless this selected word is the actual physical word and both original section endpoint tests hold. On that physical set impose its exact original return label (n,k,m). Occupation uses collisions $0,\dots,m-1$ and terminal membership is checked at m. Consequently

$$0 \le \sum_{n=1}^{m}\sum_{k} B_{z,n,k}(y) \le 1. \tag{95.1}$$

These tests are made on the actual selected positions and velocities, not on signs of first-order approximations. A point on the continuation failing a physical test receives zero weight.

95.1. **The first-clearance multiplier on the common chart.** For $0 < \varepsilon \le \chi/4$ all selected incidence guard factors are one on this chart. Let $D_z^{\varepsilon}(y)$ be the sum of the original first-clearance multipliers there. On the physical set it is exactly $1 - H_{m,R}^{\varepsilon}$. It may be evaluated as a function of the selected contacts off that set, solely for the coordinate formula. It is always multiplied by $B_{z,n,k}$ when defining a source.

Lemma 95.1 (Uniform control of the actual clearance guard)**.** *There is C_g independent of $m, R, z, \chi, \varepsilon$ such that*

$$|D_z^\varepsilon(y) - D_z^\varepsilon(y')| \le C_g \varepsilon^{-1} |y - y'| \quad (|y|, |y'| < r_\chi). \tag{95.2}$$

If at least one original competing-disk clearance vanishes at z, then on the physical part of $|y| < r_{\chi,\varepsilon}$,

$$D_z^\varepsilon(y) = 1, \qquad r_{\chi,\varepsilon} = \min(r_\chi, \varepsilon/C_g), \tag{95.3}$$

after a fixed enlargement of C_g. This permits simultaneous seams, arbitrarily small nonzero margins and section coincidences.

Proof. Theorem 94.2 and the bounded derivative of Φ_z give a Lipschitz bound $Cq^{d'_j/2}$ for every flight-j clearance as a function of y, where $d'_j = \min(j, m-1-j)$. Distance from a segment to a fixed competing disk is Lipschitz in its two endpoints, also when its closest point changes. A fixed finite candidate list per flight suffices on the continuation, by the same uniform bound on selected flight lengths. Taking a minimum preserves this Lipschitz bound. Equivalently, on a near-clearance envelope one uses the inherited signed margin at image collision $j+1$.

The original flight guard has width $\varepsilon e'_j = \varepsilon q^{d'_j/4}$. Its Lipschitz bound is therefore $C\varepsilon^{-1} q^{d'_j/4}$. For factors in $[0,1]$, the difference of products is at most the sum of their differences. The sum over j is bounded by a geometric series independent of m. This proves (95.2). There is no factor for the number of competing disks, because the minimum is Lipschitz with the maximum of the individual constants.

If a specified competing clearance is zero at z, its absolute value on the disk of radius ε/C_g is at most $\varepsilon e'_j$. On the physical part it is nonnegative, so that flight guard is zero. Every incidence factor remains one because $c_j \ge \chi e_j/2 \ge 2\varepsilon e_j$. Hence the unchanged first-defect telescoping sum consists entirely of clearance pieces and equals one. No earlier clearance test or later incidence has been deleted from its definition. □

Set

$$A^\varepsilon_{z,n,k}(h) = \int_0^{2\pi} B_{z,n,k}(\sqrt{2h}\,\omega_\theta) D_z^\varepsilon(\sqrt{2h}\,\omega_\theta)\, d\theta, \qquad 0 < h < h_\chi. \tag{95.4}$$

Then $0 \le \sum_{n,k} A^\varepsilon_{z,n,k}(h) \le 2\pi$. The selected-coordinate source coefficient is not a probability; the physical measure is supplied only by the indicators and the original normalization c_*^{-1}.

Theorem 95.2 (Uniform nonlinear physical profile)**.** *Let $b^{\varepsilon,\mathrm{clr}}_{z,n,k}$ be the roof density of the original first-clearance source restricted to this Morse disk. For every m, every χ-incident critical word, every $0 < \varepsilon \le \chi/4$, and almost every $0 < h < h_\chi$,*

$$e^{-K_\chi\sqrt{2h}} \frac{J_z}{2\pi} A^\varepsilon_{z,n,k}(h) \le b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z + h) \le e^{K_\chi\sqrt{2h}} \frac{J_z}{2\pi} A^\varepsilon_{z,n,k}(h). \tag{95.5}$$

In particular

$$\begin{aligned} \sum_{n,k} \left| b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z + h) - \frac{J_z}{2\pi} A^\varepsilon_{z,n,k}(h) \right| &\le (e^{K_\chi\sqrt{2h}} - 1) \frac{J_z}{2\pi} \sum_{n,k} A^\varepsilon_{z,n,k}(h) \\ &\le C K_\chi J_z \sqrt{h}. \end{aligned} \tag{95.6}$$

The first bound is relative to the retained physical angular weight, not to a sum of full-word coefficients on nonphysical directions. There is no lower bound on any boundary gradient in these statements. On a clearance seam, for $h < r^2_{\chi,\varepsilon}/2$, the angular integral is simply the measure of the exact physical and exact-label angular set, by (95.3)*.*

Proof. On points where $B_{z,n,k} = 1$, the selected initial-state map is the original orbit map and its Jacobian density is w_z. Thus change of variables, followed by polar coarea, gives exactly

$$b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z + h) = \int_0^{2\pi} \widetilde{w}_z(\sqrt{2h}\,\omega_\theta) B_{z,n,k}(\sqrt{2h}\,\omega_\theta) D^{\varepsilon}_z(\sqrt{2h}\,\omega_\theta)\, d\theta. \tag{95.7}$$

The factor r in polar area cancels the derivative of $r^2/2$. Insert the relative density bound of Theorem 94.3. Positivity gives (95.5). Summing absolute errors uses (95.1), not the number of labels. The budget makes $K_\chi r_\chi$ bounded by a fixed constant, so $e^{K_\chi\sqrt{2h}} - 1 \le CK_\chi\sqrt{h}$. This proves the last line. The saturation assertion was proved on the physical source in Lemma 95.1. All identities use common coarea representatives, outside a null roof set at each finite count. □

95.2. **Positive coverage without deleting its complement.** For fixed m, R, χ, take all the preceding critical words and restrict each chart to its own physical itinerary. These restrictions are disjoint: different words give disjoint regular itinerary events, and a selected word has at most one normal critical point. The latter also holds in a physical-side closure with positive selected incidences: its contact gradient is strictly inward at every disk, so the convex minimization argument of Lemma 167.1, now including the two normal endpoints, gives uniqueness.

Let $b^{\varepsilon,\chi,\mathrm{chart}}_{n,k,m,R}$ be the sum of the original first-clearance restrictions to these physical chart pieces. Then

$$b^{\varepsilon,\mathrm{clr}}_{n,k,m,R} = b^{\varepsilon,\chi,\mathrm{chart}}_{n,k,m,R} + b^{\varepsilon,\chi,\mathrm{outside}}_{n,k,m,R}, \qquad b^{\varepsilon,\chi,\mathrm{outside}}_{n,k,m,R} \ge 0. \tag{95.8}$$

The outside term includes critical points below the tapered incidence envelope, selected-grazing neighborhoods, noncritical roof-rank components, and every physical state not in the specified disks. If smaller disks are used, the removed annuli also belong to this term. Bounded measurable insertions are dominated in variation by their supremum times this same positive partition.

Theorem 95.2 supplies count-uniform chart, guard and relative-density constants at a fixed physical width. It avoids the local affine-gradient denominators of (105.8) by retaining the nonlinear angular sets. It does not bound their concentration as the collision count grows, and it does not assign a height estimate to the outside term.

96. The physical collar trace and its dynamical bound

A full reversible determinant and a physically observable caustic coefficient are different at a seam. The former has weight J_z; the latter also contains the physical angular fraction and the exact label decision. We construct a positive collar trace on the original source, prove its ordered dynamical bound, and state explicitly what is still needed to pass from this trace to zero-width coefficients.

Fix χ, ε as in the preceding section. Let $\mathcal{Z}$ be any subfamily of its critical charts, each counted once. Exact labels below are imposed on the source points inside a chart; they need not be constant on the chart or defined by a boundary convention at its center. For $0 < s < h_\chi$ put

$$\begin{aligned} M_{z,n,k}(s) &= \int_0^s b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z + u)\, du, \\ \mathcal{Q}^{\chi,\varepsilon}_{m,s;n,k,R} &= \sum_{z\in\mathcal{Z}} s^{-1} M_{z,n,k}(s)\delta_{t_z}. \end{aligned} \tag{96.1}$$

This is a positive, unnormalized trace obtained from a restriction of the original Lorentz source. Its atom is placed at the chart's critical value only as bookkeeping; $M_{z,n,k}(s)$ integrates the actual roof $L_{m,R}$ and the unchanged return event.

Theorem 96.1 (An ordered bound for the physical collar trace)**.** *For every fixed* $0 < s < h_\chi$ *and* $h > 0$*,*

$$\limsup_{m\to\infty} \sup_{R,n,k,t} m^2 \mathcal{Q}^{\chi,\varepsilon}_{m,s;n,k,R}([t-h, t+h]) \le C(2h+s). \tag{96.2}$$

The constant is independent of χ, ε, h, s *in these ranges. In particular*

$$\lim_{s\downarrow 0} \limsup_{m\to\infty} \sup_{R,n,k,t} m^2 \mathcal{Q}^{\chi,\varepsilon}_{m,s;n,k,R}([t-s, t+s]) = 0. \tag{96.3}$$

Both statements permit clearance seams, exact-label changes and coincident critical values. They concern the stated finite-width trace, not the full reversible atomic measure of (103.10)*.*

Proof. On a chart with $t_z \in [t-h, t+h]$, the part $0 < F_z - t_z < s$ has actual roof in $[t-h, t+h+s]$ and both endpoint momenta bounded by $C_0\sqrt{s}$, by Theorem 94.3. The physical chart pieces are disjoint. Their clearance multipliers are at most one, and their initial and terminal section restrictions are exact. Hence

$$s\, \mathcal{Q}^{\chi,\varepsilon}_{m,s;n,k,R}([t-h, t+h]) \le \nu^*_R(E^{(C_0\sqrt{s})}_{n,k,m,R}(t; [-h, h+s])). \tag{96.4}$$

Choose the fixed r_0 in (94.5) small enough that $C_0\sqrt{h_\chi} < 1/4$. For fixed s, h, the inherited Lemma 170.1 bounds the limsup of the right side times m^2 by $Cs(2h+s)$. Division by s proves the first assertion. Setting $h = s$ and then taking $s \downarrow 0$ proves the second. No spectral estimate is evaluated at a width depending on m. The label supremum is outside a positive source comparison; it does not average either integer label. No word-count factor or separation of critical values occurs. □

96.1. **The zero-width observable coefficient.** Suppose now that $\mathcal{Z}$ consists of clearance seams and take $s < r^2_{\chi,\varepsilon}/2$. At each fixed finite word the active physical and section scalar margins have nonzero individual gradients in the regular selected chart, as in Lemma 105.2. No uniform gradient lower bound is imposed. Let $\theta_{z,n,k} \in [0, 1]$ be the angular fraction assigned to the original label by the signs of their linear parts; nonzero margins retain their constant signs. Then $\sum_{n,k} \theta_{z,n,k} \le 1$. Put

$$\beta_{z,n,k} = J_z \theta_{z,n,k}, \qquad \mathcal{B}^{\chi,\varepsilon}_{m;n,k,R} = \sum_{z\in\mathcal{Z}} \beta_{z,n,k} \delta_{t_z}. \tag{96.5}$$

A full physical circle has fraction one, a single strict seam one half, and a compatible quadrant one quarter. The label is always the original half-word label, never an index read from the doubled orbit.

Proposition 96.2 (Exact fixed-count source realization)**.** *For each fixed* m, R, z, n, k*,*

$$\beta_{z,n,k} = \lim_{u\downarrow 0} b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z + u) = \lim_{s\downarrow 0} s^{-1} M_{z,n,k}(s). \tag{96.6}$$

These limits use the coarea representative. Thus $\mathcal{Q}_{m,s}$ *converges to* $\mathcal{B}_m$ *against every bounded critical-value test at this fixed finite count. In general neither trace equals the unrestricted sum of full coefficients* J_z*.*

Proof. On the saturated physical disk $D^\varepsilon_z = 1$. For a fixed finite list of active scalar margins, Taylor expansion shows that their signs converge on every angular direction except the finite set where a first derivative vanishes. Initial, occupation and terminal decisions are evaluated separately by the same visit rule. Dominated convergence therefore gives $A^\varepsilon_{z,n,k}(u)/(2\pi) \to \theta_{z,n,k}$. Equation (95.5) gives the first limit; its Cesaro average gives the second. There are finitely many selected critical words at fixed count, so summation commutes with this fixed-count limit. No assertion about an interchange with $m \to \infty$ is used. □

96.2. **A quantified separation of the two limits.** Write $a_{z,n,k}(u) = A^{\varepsilon}_{z,n,k}(u)/(2\pi)$ and define for an interval I the nonnegative angular loss

$$\mathcal{D}_{m,s;n,k,R}(I) = \sum_{z:t_z\in I} \frac{J_z}{s} \int_0^s (\theta_{z,n,k} - a_{z,n,k}(u))_+ \, du. \tag{96.7}$$

It measures loss of the actual physical and exact-label angular set, not an error in the normalizing density or a discarded source mass.

Proposition 96.3 (A positive atomic-to-collar comparison)**.** *For every interval I and $0 < s < r^2_{\chi,\varepsilon}/2$,*

$$\mathcal{B}_{m;n,k,R}(I) \le e^{K_\chi\sqrt{2s}}\mathcal{Q}_{m,s;n,k,R}(I) + \mathcal{D}_{m,s;n,k,R}(I). \tag{96.8}$$

Consequently, if the further physical estimate

$$\lim_{s\downarrow 0} \limsup_{m\to\infty} \sup_{R,n,k,t} m^2 \mathcal{D}_{m,s;n,k,R}([t-1,t+1]) = 0 \tag{96.9}$$

holds, then

$$\lim_{h\downarrow 0} \limsup_{m\to\infty} \sup_{R,n,k,t} m^2 \mathcal{B}_{m;n,k,R}([t-h,t+h]) = 0. \tag{96.10}$$

The additional estimate (96.9) *is not proved here and is not implied by the count-uniform selected-contact jets. The implication concerns the physically weighted trace $\mathcal{B}$, not the larger full reversible trace.*

Proof. For each z, average the elementary inequality $\theta \le a(u) + (\theta - a(u))_+$ over $0 < u < s$ and multiply by J_z. The lower bound in (95.5) bounds $J_z a(u)$ by $e^{K_\chi\sqrt{2s}} b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z + u)$. Sum to prove (96.8). For fixed $s, h < 1$, Theorem 96.1 then bounds the normalized limsup by

$$Ce^{K_\chi\sqrt{2s}}(2h+s) + \limsup_{m} \sup_{R,n,k,t} m^2 \mathcal{D}_{m,s;n,k,R}([t-1,t+1]).$$

First let $h \downarrow 0$ and then $s \downarrow 0$, with χ, ε fixed throughout. This proves the stated implication. □

The unchanged pointwise endpoint. The new uniform charts and finite-width trace estimate do not bound the complete first-incidence source or the positive outside source in (95.8). They also do not prove (96.9), full reversible-weight concentration, or a uniform pointwise bound for the nonlinear angular profiles. The complete ordered incidence and clearance heights remain the inputs of the original positive raw-error identity. The unrestricted two-sided arithmetic raw-density law, its same-roof conditional consequences and forward essential likelihood are not inferred from a fixed-count limit or a finite-width trace. The finite arithmetic factor and its zero classes remain unchanged.

97. Inverse-incidence coarea on the original Lorentz source

We return to the physical boundary sources. The contact determinant contains a gain which is lost in the unweighted finite-count height bound: it cancels one inverse incidence at every contact, simultaneously. This gives a density estimate for an unbounded observable of the actual orbit. The estimate is uniform in the table and in the incidence threshold, but its constant is exponential in the collision count. We keep that dependence explicit.

Throughout this section the table is circular, $R \in [9/20, 47/100]$. Write $\ell_0 = 3/50$, $c_* = 91/(10000\pi)$ and $c_j(x) = \cos\varphi(T^j_R x)$. On a regular m-flight orbit put

$$\mathcal{I}_{m,R}(x) = \prod_{j=0}^{m} c_j(x)^{-1}, \qquad L_{m,R}(x) = \sum_{j=0}^{m-1} \tau_R(T^j_R x). \tag{97.1}$$

Values on singular orbits may be set to zero. These orbits form a collision-null set at each finite count. There is no added random variable or modified flight in (97.1).

97.1. **Cancellation in the endpoint density.** For a fixed regular word, use oriented endpoint arclengths (u, v) and its reduced action $F(u, v)$. Put $t_j = 1/\ell_j$. The internal contact matrix is

$$A_{jj} = t_{j-1} + t_j + \frac{2}{Rc_j} \quad (1 \le j < m),$$
$$A_{j,j+1} = A_{j+1,j} = -t_j \qquad (1 \le j < m-1).$$

An empty determinant, at $m = 1$, is one. This is the matrix in (169.2), evaluated at the original contacts.

Lemma 97.1 (A determinant gain at every incident contact)**.** *On the full regular endpoint image,*

$$|F_{uv}| = \frac{c_0 c_m \prod_{j=0}^{m-1} t_j}{\det A}, \qquad \mathcal{I}_{m,R}|F_{uv}| \le \left(\frac{R}{2}\right)^{m-1} \prod_{j=0}^{m-1} t_j. \tag{97.2}$$

For $S \subset \{1, \dots, m-1\}$, let $A^{(S)}$ denote the same matrix with c_j replaced by one for $j \in S$, solely in its indicated diagonal potentials. Then

$$\frac{\det A^{(S)}}{\det A} \le \prod_{j \in S} \frac{53 c_j}{6 + 47 c_j}. \tag{97.3}$$

The matrix $A^{(S)}$ is an algebraic comparison, not the Hessian of an asserted replacement trajectory.

Proof. The first variation gives $p_0 = -F_u$. Hence the absolute Jacobian of $(u, p_0) \mapsto (u, v)$ is $|F_{uv}|$ in the inverse direction. The off-diagonal entries of the full contact Hessian have magnitudes $c_j c_{j+1} t_j$. Its internal block is $D_c A D_c$, where $D_c = \operatorname{diag}(c_1, \dots, c_{m-1})$. Taking the endpoint Schur complement and using the corner cofactor of a tridiagonal matrix gives the first identity in (97.2). The factors $\prod_{j=1}^{m-1} c_j^2$ cancel. This is the identity (169.7); its algebra needs positive incidence, not a protected collar or a critical endpoint. The one-flight case is the direct mixed derivative of the segment length.

Write $A = L + D$, where $D_{jj} = 2/(Rc_j)$ and L is the weighted Dirichlet path Laplacian, with diagonal $t_{j-1} + t_j$ and neighboring entries $-t_j$. Its quadratic form is a sum of nonnegative squares, including the two boundary terms. Thus $L \ge 0$, and

$$\det A = \det D \, \det(I + D^{-1/2} L D^{-1/2}) \ge \prod_{j=1}^{m-1} \frac{2}{Rc_j}.$$

Substitution proves the second identity's inequality, including both endpoint incidence factors.

For the refined assertion change one selected diagonal at a time. Before restoring c_j from one to its physical value, its Schur complement s_j is positive and no larger than $t_{j-1} + t_j + 2/R \le 2/\ell_0 + 2/R$. Restoring it adds $(2/R)(c_j^{-1} - 1)$ to this complement. The corresponding inverse determinant ratio is at most

$$\frac{(1 + R/\ell_0)c_j}{1 + (R/\ell_0)c_j} \le \frac{53 c_j}{6 + 47 c_j}.$$

The last expression uses $R/\ell_0 \le 47/6$; the ratio is increasing in that parameter for $0 < c_j \le 1$. All intermediate matrices remain positive definite. Multiplication of the ratios proves (97.3), without a factor depending on the order or number of selected contacts. □

97.2. **An unbounded physical insertion at fixed roof.**

Lemma 97.2 (Endpoint level integrals)**.** *On each of a fixed finite collection of rational endpoint charts, let f be the reduced action of a regular m-flight word. There are $C > 0$ and $A_0 > 1$ such that, for almost every t,*

$$\int_{\{f=t\}} \frac{d\mathcal{H}^1}{|\nabla f|} \le C A_0^m. \tag{97.4}$$

The integral includes the entire open physical endpoint image in the chart. In particular it does not require that image to be convex.

Proof. Take charts $\alpha = \alpha_* + 2\arctan x$, $|x| < 1$, at both endpoints and use the disjoint refinements of Lemma 167.2. Their derivatives lie in $[1, 2]$. The first variation is $\nabla f = R(-a(x)p_0, a(y)p_m)$. On the region where either $|p_0|$ or $|p_m|$ is at least $\delta_0 = 1/20$, its magnitude is at least $R_-\delta_0$.

The level-length argument of that lemma also applies to these endpoint charts: impose an endpoint coordinate line, rather than an initial coordinate line, on the same auxiliary collision graph. It still has $O(m + 1)$ variables and bounded-degree polynomial sign conditions. The coefficient-independent bound (167.4) bounds each finite projection fiber by CA_0^m. Integrating coordinate multiplicities bounds the total endpoint level length by CA_0^m. Thus the asserted integral over the non-normal region has this bound.

On the complementary region the rational-coordinate Hessian satisfies $\nabla^2 f \ge \lambda_0 I$ with $\lambda_0 = R_-/2$, by (167.3). At a regular level its signed curvature with normal $\nabla f/|\nabla f|$ is $k_f = T^{\mathsf{T}}\nabla^2 f T/|\nabla f|$, and consequently $|\nabla f|^{-1} \le |k_f|/\lambda_0$. The total absolute curvature bound of Lemma 167.2 gives (97.4) on this part. It includes every connected portion, even if the level meets a physical seam in its closure. Compact regular exhaustions and monotone convergence retain all such portions. Critical roof values and exceptional coarea levels form a null set at each word; no positive-measure roof set is removed. □

Theorem 97.3 (Complete inverse-incidence roof density)**.** *The measure $\mathcal{I}_{m,R}\nu$ is finite. Its pushforward by $L_{m,R}$ has a density $\rho^{\mathcal{I}}_{m,R}$ satisfying*

$$\|\rho^{\mathcal{I}}_{m,R}\|_\infty \le CA^m \qquad (m \ge 1,\ R \in [9/20, 47/100]) \tag{97.5}$$

for fixed $C > 0$, $A > 1$. The same bound holds for every nonnegative weight $\prod_{j=0}^m c_j^{-\beta_j}$ with $0 \le \beta_j \le 1$. For complex insertions $w_{n,R}$ of supremum at most M, the exact actual-return coefficients obey

$$\operatorname*{ess\,sup}_t \sum_{n=1}^m \sum_{k\in\mathbb{Z}^2} |p^{w\mathcal{I}}_{n,R}(k, m, t)| \le \frac{CMA^m}{c_*}. \tag{97.6}$$

Here the insertion on $\{N_{n,R} = m\}$ is the product along that very m-collision orbit. Occupation is counted at $0, \dots, m-1$, and terminal membership at m is retained separately.

Proof. Under the full collision probability the endpoint density is $|F_{uv}|/(4\pi R)$. On a rational endpoint chart the further Jacobian is $R^2 a(x)a(y)$, bounded uniformly above. By Lemma 97.1, multiplication by $\mathcal{I}_{m,R}$ leaves a chart density bounded by a constant times

$$\ell_0^{-1} \max\{1, R_+/(2\ell_0)\}^{m-1}.$$

The bound holds on the full regular word, without a lower cutoff on any c_j.

The finite horizon gives a fixed finite alphabet of next centers. There are at most D^m words and a fixed number of endpoint charts per word. Endpoint injectivity is Lemma 167.1. The bounded chart density therefore first proves finiteness of the weighted source, by integration over the bounded endpoint boxes and summation over words. Coarea on those same boxes,

followed by Lemma 97.2, gives (97.5). Alternatively one may truncate the insertion and pass monotonically to its full value; every bound is independent of the truncation.

Since $c_j \le 1$, the fractional-power products are dominated by $\mathcal{I}_{m,R}$. For fixed m the original section events $\Pi_{n,k,m,R}$ are pairwise disjoint as n,k vary. Their laws are restrictions of ν/c_*, and on each such event $T_{n,R} = L_{m,R}$. Positive domination before pushforward proves (97.6), including arbitrary bounded measurable $w_{n,R}$. This is a sum of variation densities, not a probability total-variation distance; no factor $1/2$ occurs. □

Corollary 97.4 (The original first-incidence source has a height gain)**.** *Let $b^{\varepsilon,\mathrm{inc},w}_{n,k,m,R}$ be exactly the first-physical-defect incidence source in* (177.3)*. For all sufficiently small $\varepsilon > 0$, simultaneously for all finite $m \ge 2$,*

$$\operatorname*{ess\,sup}_t \sum_{n=1}^{m} \sum_{k\in\mathbb{Z}^2} |b^{\varepsilon,\mathrm{inc},w}_{n,k,m,R}(t)| \le CMA^m\varepsilon. \tag{97.7}$$

More generally, restricting the original source to $c_j < s_j$ at each contact in a prescribed set $S \subset \{0,\dots,m\}$ costs a factor $\prod_{j\in S} s_j$ in the same height bound. All constants include the fixed section normalization once.

Proof. The smooth physical guard is between zero and one. A first incidence factor below one, including its transition region, has $c_j < 2\varepsilon q_0^{d_j/4}$, where $q_0 = 47/53$ and $d_j = \min(j, m-j)$. Pointwise on the unchanged orbit,

$$(1 - H^{\varepsilon}_{m,R})\mathbf{1}_{\{\text{first type is incidence}\}} \le \sum_{j=0}^{m} \mathbf{1}_{\{c_j < 2\varepsilon q_0^{d_j/4}\}} \le 2\varepsilon \sum_{j=0}^{m} q_0^{d_j/4}\mathcal{I}_{m,R}.$$

The geometric sum is at most $2/(1-q_0^{1/4})$. Apply Theorem 97.3 and source domination. For simultaneous contact conditions use $\prod_{j\in S}\mathbf{1}_{\{c_j<s_j\}} \le (\prod_{j\in S} s_j)\prod_{j\in S} c_j^{-1} \le (\prod_{j\in S} s_j)\mathcal{I}_{m,R}$. The first-defect source is only enlarged for a positive upper bound; its actual return index, collision count and displacement are not changed. □

Collision-count dependence. Equation (97.7) is a density estimate, not the mass estimate $O(\varepsilon^2)$ of the incidence layer. It allows an arbitrary count-dependent ε_m. In particular, its normalized height tends to zero if $m^2A^m\varepsilon_m \to 0$. For a central exact label whose canonical reference $G(u)$ is at least $d > 0$, the pointwise lower law (181.8) gives $P(u) \ge d/2$ for large m. Division at the same roof then bounds the original conditional incidence source by $C_d m^2 A^m \varepsilon$ almost everywhere there. A single source disintegration supplies these inequalities for all bounded tests.

This exponential factor is not a central-scale height estimate for a fixed ε. In particular it does not establish the ordered limit in Corollary 181.5. The missing step is uniform control of the positive exact-label level sum as m grows, not integrability of the inverse incidence at a fixed count. No change of the finite arithmetic coefficient is involved.

98. Clearance coarea and its rank-deficient positive remainder

A small competing-hit clearance does not supply the incidence factor in the preceding determinant. Its density instead admits a direct estimate where the total roof and the image-side clearance coordinate are transverse. We prove that estimate on every original physical word and retain the rank-deficient source as a positive measure. In particular no continuation through a competing-hit seam is used.

Use the exact first-physical-defect partition of Section 177. If its first type is clearance, let j be the corresponding flight. Select a responsible nonincident disk by a fixed ordering of the finitely many candidates, with the inherited flight and type ordering left unchanged. The responsible

perpendicular foot lies inside the physical segment, by Lemma 178.1. In coordinates based at the actual terminal collision of this flight, its label is $d \in \mathcal{D}$. Set, on the corresponding regular word,

$$Z_{j,d,R}(x) = Z_d(T_R^{j+1}x), \qquad \mathcal{J}_{j,d,m,R}(x) = \det D_{(\alpha,p)}(L_{m,R}, Z_{j,d,R})(x). \tag{98.1}$$

The definition of Z_d is (178.1). The mark is $j+1$, not j, also for the last flight. The first-defect transition region has

$$R \le |Z_{j,d,R}| \le R + s_j, \qquad s_j = 2\varepsilon q_0^{\min(j,m-1-j)/4}, \quad q_0 = 47/53.$$

We assume $2\varepsilon < s_0$. The additional candidate selection only partitions the original positive clearance source; it changes no event.

For a fixed $\eta > 0$, split that source according as the absolute determinant in (98.1), at its selected witness, is at least η or is less than η. Denote the resulting exact roof densities by $b^{\varepsilon,\mathrm{tr},\eta}$ and $b^{\varepsilon,\tan,\eta}$, respectively. Thus

$$b^{\varepsilon,\mathrm{clr},1} = b^{\varepsilon,\mathrm{tr},\eta,1} + b^{\varepsilon,\tan,\eta,1}, \qquad b^{\varepsilon,\mathrm{tr},\eta,1}, b^{\varepsilon,\tan,\eta,1} \ge 0. \tag{98.2}$$

The name "tan" includes small determinants, not only exact tangencies.

98.1. **Multiplicity of the physical two-coordinate map.**

Lemma 98.1 (Uniform finite-word two-coordinate multiplicity)**.** *There are $C > 0$ and $D > 1$ such that, for every m, R, j, d, the number of regular preimages of any (t, z) under $(L_{m,R}, Z_{j,d,R})$ at which its determinant is nonzero is at most CD^m, after all physical words and a fixed disjoint angular chart cover have been included.*

Proof. Retain the auxiliary collision graph used in Lemma 167.2: contact positions, normals, unit velocities, flight lengths and the physical first-hit inequalities. For each fixed center word it has $O(m+1)$ variables and polynomial sign conditions of fixed degree. The equations $L_{m,R} = t$ and $Z_{j,d,R} = z$ add only a fixed number of conditions. Indeed Z_d is a scalar product of the image collision position and velocity; one may use those variables directly, without differentiating a root or eliminating the intermediate contacts. Rational chart denominators have a fixed positive sign.

At a regular preimage with nonzero determinant the inverse function theorem makes that preimage isolated in the fiber. The auxiliary variables are uniquely determined by the physical initial state and word, so it gives an isolated component of the same graph fiber. The coefficient-independent bound (167.4) is CD_0^m for its number of components. This also bounds the isolated components when the fiber has other, positive-dimensional components. There are at most D_1^m physical center words and a fixed number of charts. Their sum proves the assertion. All constants are uniform in R, t, z and the chosen mark. Singular or repeated-hit continuations are not inserted into the graph. □

Theorem 98.2 (Height of the transverse physical clearance source)**.** *There are constants $C > 0$ and $A > 1$ such that, for every $m \ge 2$, $0 < 2\varepsilon < s_0$, $\eta > 0$, and insertions with $|w| \le M$,*

$$\operatorname*{ess\,sup}_t \sum_{n=1}^{m} \sum_{k\in\mathbb{Z}^2} |b^{\varepsilon,\mathrm{tr},\eta,w}_{n,k,m,R}(t)| \le CMA^m \frac{\varepsilon}{\eta}. \tag{98.3}$$

The constants are independent of both thresholds, which may depend on m. Every trajectory, label, first defect and next-collision mark in this formula is the original one.

Proof. First fix the flight and the candidate disk. On the region where $|\mathcal{J}_{j,d,m,R}| \ge \eta$, apply the two-dimensional area formula to (98.1). The full collision density in (α, p) is $1/(4\pi)$; the section restriction multiplies it by at most $1/c_*$. The density at (t, z) is bounded by

$$\frac{1}{4\pi c_*} \sum_{x:(L_m, Z_{j,d,R})(x)=(t,z),\, |\mathcal{J}(x)|\ge\eta} |\mathcal{J}(x)|^{-1} \le \frac{CD^m}{c_*\eta}.$$

One may first exhaust compact regular subsets on which this is an ordinary finite-to-one change of variables, and then pass by monotone convergence. Lemma 98.1 controls the full multiplicity throughout. The first-defect weight $1 - H^{\varepsilon}$, the section event and every exact-label restriction are all between zero and one, so deleting them here is only a positive upper comparison on the same physical map.

Integrate z over $[R, R+s_j] \cup [-R-s_j, -R]$, whose total length is $2s_j$. Summing over candidates costs a fixed factor. Finally $\sum_{j=0}^{m-1} s_j \le 4\varepsilon/(1-q_0^{1/4})$, independently of m. This proves (98.3) for the positive source. For fixed m, the original events $\Pi_{n,k,m,R}$ are disjoint. Positive domination proves the sum of absolute inserted densities in the statement, without multiplying by the number of exact targets. Fubini and the area formula remove only Lebesgue-null roof values. No small interval of roofs is discarded. □

98.2. **An exact location of the uncontrolled clearance term.** Use $P = m^2 p$, $G = \mathcal{L}_{m,R}$ and $\varepsilon(B) = A_0 B^{-1/12}$ from Theorem 181.4. Define the actual finite-count quantity

$$e_{B,m} = \sup_{R,n,k} \|P - G - m^2 b^{\varepsilon(B),1}\|_{\infty}.$$

For each sufficiently large fixed B its collision limsup is at most $CB^{-1/192}$. The two preceding theorems and the positive source identities give the finite inequality

$$\sup_{R,n,k} \|P - G - m^2 b^{\varepsilon(B),\tan,\eta,1}\|_{\infty} \le e_{B,m} + Cm^2 A^m \varepsilon(B)(1+\eta^{-1}). \tag{98.4}$$

Here A is enlarged to work for both geometric bounds. The identity behind this inequality retains the complete incidence source, transverse clearance source, and rank-deficient clearance source, with positive signs. It is not an interchange of the collision and band limits.

In particular a normal-to-normal critical point of a regular word has $dL_m = 0$, and hence $\mathcal{J}_{j,d,m,R} = 0$ for every candidate. A clearance defect meeting such a point belongs to the retained term, not to a region on which (98.3) can be used. The intrinsic critical jump and all arithmetic phases remain in the original raw inversion problem.

Relation to the height criterion. These results verify two new density-level facts about the Lorentz source itself: simultaneous inverse-incidence cancellation, and clearance coarea at a quantified nonzero rank. They are not transfers from the Markov realization and do not require a monotone scalar coordinate on every Lorentz word. They use the endpoint action and coefficient-independent collision-graph complexity already established for this family. The elementary change-of-variables mechanism is separate from the dynamical local-limit inputs of [2, 12, 3].

At a fixed protection width the factor $m^2 A^m$ in (98.4) need not be small. Choosing B exponentially in m to offset it would require a reconstruction estimate not supplied by a fixed-band theorem. Thus neither this inequality nor the incidence gain alone proves either ordered essential-height limit in Corollary 181.5. A central-scale bound for the positive exact-label coarea sums, and control of the retained rank-deficient clearance source, are still required. The unrestricted pointwise conclusions are not invoked before these steps.

99. Finite-type clearance on the actual roof fibers

The determinant cutoff of Section 98 leaves points at which the clearance is tangent to a regular roof fiber. A higher derivative along that fiber can still give a positive height gain. We prove this on the original physical words, including their exact labels. The estimate applies to the finite-type part of the first-clearance source; no finite-type assertion is imposed on its complement.

99.1. **A one-dimensional estimate.**

Lemma 99.1 (A derivative sublevel bound)**.** *For each integer $r \geq 1$ there is C_r with the following property. If $V \subset \mathbb{R}$ is an interval, $f \in C^r(V)$, and $|f^{(r)}| \geq b > 0$ on V, then, for an interval H of length s,*

$$|\{v \in V : f(v) \in H\}| \leq C_r (s/b)^{1/r}. \tag{99.1}$$

The interval V need not include its endpoints.

Proof. Exhaust by compact subintervals. Let E be the sublevel set in one such interval and write $l = |E|$. If $l > 0$, a greedy selection gives $r+1$ points of E separated by at least $l/(4(r+1))$: the union of r neighborhoods of this radius has length less than l. Write them in increasing order as $v_0, \dots, v_r$, and subtract one fixed endpoint of H from f. The resulting values lie in $[0, s]$. The Lagrange formula for their divided difference bounds its absolute value by $(r+1)s(4(r+1)/l)^r$. Repeated Rolle's theorem identifies this difference with $f^{(r)}(\xi)/r!$ for some ξ between the extreme points. Hence $b \leq r!(r+1)s(4(r+1)/l)^r$. This proves the estimate with a constant depending only on r. Approximation of E from within and monotone convergence remove the compact exhaustion. For $s = 0$, apply the result to decreasing positive lengths. □

99.2. **The derivative on a physical fiber.** Retain the first-clearance witness (j, d) and its ordering from Section 98. Recenter at the actual image collision $y = T_R^{j+1} x$. In a fixed rational angular chart write

$$\alpha = \alpha_* + 2 \arctan x_1, \qquad p = x_2, \qquad |x_1| < 1.$$

Use a fixed finite cover and assign overlaps in a fixed order when a single source partition is needed. The invariant density in these coordinates is at most $1/(2\pi)$. On a regular, actual two-sided word through y put

$$F(y) = L_{m,R}(T_R^{-(j+1)} y), \qquad Z(y) = Z_d(y). \tag{99.2}$$

The function F sums the original m flights, including those before the mark. It is not a roof for a nearby continued word. For $i = 1, 2$, write $i' = 3 - i$ and, where $F_i = \partial_i F \neq 0$, set

$$\mathscr{D}_i = \partial_{i'} - \frac{F_{i'}}{F_i} \partial_i. \tag{99.3}$$

On a graph of $F = t$, parametrized by $x_{i'}$, this is its ordinary derivative. In $\mathscr{D}_i^r Z$ all coefficients of the vector field are differentiated as well.

For $a, b > 0$, define the order-r carrier at the selected witness by

$$\mathcal{T}_r(a, b) = \bigcup_{\text{assigned charts, } i=1,2} \{|F_i| \geq a,\ |\mathscr{D}_i^r Z| \geq b\}. \tag{99.4}$$

All functions in this definition are evaluated on the actual word of the source point. This definition includes nonzero first derivatives when $r = 1$. When $r = 2$, it also permits $\det D(F, Z) = 0$ provided the second derivative and the chosen roof derivative have the stated lower bounds.

Lemma 99.2 (Finite-order physical fiber complexity)**.** *For each fixed $r \geq 1$ there are $C_r > 0$, $A_r > 1$ such that the parts of $F = t$ satisfying either pivot condition in* (99.4) *can be covered, apart from finitely many points, by at most $C_r A_r^m$ graph intervals on which that condition holds throughout. The bound includes all original center words and charts and is independent of R, t, a, b, j, d.*

Proof. We use the auxiliary physical graph in Lemma 167.2, without eliminating intermediate collisions. The initial data are now at the mark. Forward and backward flight equations determine each of the m contacts and velocities; all selected incidence cosines are positive on a regular word. For each flight, implicit differentiation of its quadratic contact equation determines the first jet

uniquely. Its coefficient for the flight-time derivative is the nonzero incoming normal component. The reflection equation is polynomial in the normal and velocity. Differentiating these equations through order r determines each successive jet linearly in its highest-order unknown, using already determined lower jets. The same construction applies to the reversed flights. It introduces at most $C_r(m+1)$ variables and equations of degree bounded solely in terms of r. Reciprocal variables with their nonzero sign conditions record denominators; no inverse is taken at a grazing contact.

Sum the time jets to obtain the jets of F, and use the scalar-product formula for Z_d to obtain its jets. Add $F = t$, the pivot inequalities, and the rational expression for $\mathscr{D}_i^r Z$ with $|F_i| \ge a$. The number of polynomial sign conditions is still at most $C_r(m+1)$ and their degrees remain bounded in terms of r. The auxiliary jets are unique, so their graph projects homeomorphically onto the regular source set. The component estimate (167.4), namely

$$d(2d-1)^{v-1} \sum_{h=0}^{v} 4^h \binom{s}{h} \le d(2d-1)^{v-1} 5^s,$$

from [14, Section 3.2, equation (3.3)], is therefore exponential in m, not exponential in $m \log m$. Here $v, s = O_r(m+1)$ and d is fixed after r. The bound is independent of polynomial coefficients, so the thresholds and roof value do not change it.

To make the graph-interval conclusion explicit, a one-dimensional component in the pivot region is a regular level curve. Its tangent can be oriented to have positive $x_{i'}$ component, because F_i has a fixed nonzero sign. The projection to $x_{i'}$ is consequently strictly increasing along that oriented component. It has no closed component or turning point. Subdivision at endpoints and at the finitely many zero-dimensional strata gives graphs over intervals. Their number is bounded by applying the same component estimate to the sign strata and their frontier conditions. Equivalently these frontiers are obtained by adjoining equality in the finitely many pivot, jet and physical inequalities already present. A one-dimensional regular level has no branching in the pivot region. This subdivision thus keeps the exponential bound. The number of candidate center words is itself at most D^m, and the chart cover is fixed.

Only the regular physical graph and its interval pieces are used. A physical seam may end a piece; it is not filled in by a continuation of the orbit. The section and first-defect indicators can restrict a piece further, but are not needed in this positive upper cover. □

Theorem 99.3 (Positive finite-type clearance height)**.** *Restrict the original positive first-clearance source to $\mathcal{T}_r(a,b)$, and denote its exact roof density with insertion w by $b_{n,k,m,R}^{\varepsilon,\mathrm{ft}_r(a,b),w}$. For each fixed $r \ge 1$ there are $C_r > 0$, $A_r > 1$ such that*

$$\operatorname*{ess\,sup}_t \sum_{n=1}^{m} \sum_{k\in\mathbb{Z}^2} |b_{n,k,m,R}^{\varepsilon,\mathrm{ft}_r(a,b),w}(t)| \le C_r M A_r^m a^{-1} (\varepsilon/b)^{1/r} \tag{99.5}$$

for $m \ge 2$, $0 < 2\varepsilon < s_0$, $a, b > 0$ and $|w| \le M$. The thresholds may depend on m; the constants do not. The source outside the carrier is not discarded or assigned this estimate.

Proof. Change the source origin to $T_R^{j+1}x$. Invariance keeps its density bounded by $1/c_*$ times the collision density, even with the original section and exact-return conditions retained. On one pivot graph, let $X_i(t,v)$ have its i'th coordinate equal to v and its ith coordinate given by the implicit graph. Ordinary coarea gives

$$\int_{F=t} \frac{r(y)}{|\nabla F|}\, d\mathcal{H}^1(y) = \int \frac{r(X_i(t,v))}{|F_i(X_i(t,v))|}\, dv, \qquad 0 \le r \le \frac{1}{2\pi c_*}.$$

It may first be applied on compact regular subsets and then exhausted by positivity. At the selected witness the original clearance source is supported in the two intervals $Z \in [R, R+s_j] \cup$

$[-R-s_j,-R]$, with $s_j=2\varepsilon q_0^{\min(j,m-1-j)/4}$. Lemma 99.1, applied along each graph in Lemma 99.2, bounds the length contributing to this integral by $C_r(s_j/b)^{1/r}$. The pivot denominator costs a^{-1}. Sum over the two pivots, the fixed charts and candidate disks, and all physical words. Finally

$$\sum_{j=0}^{m-1} s_j^{1/r} \le \frac{2(2\varepsilon)^{1/r}}{1-q_0^{1/(4r)}}.$$

This proves the positive estimate. For fixed m the events $\Pi_{n,k,m,R}$ are disjoint. Domination before pushforward therefore proves the sum of absolute inserted densities without a factor for the number of targets. Only a Lebesgue-null set of roofs is exceptional. □

Corollary 99.4 (Fold clearance and the retained positive source)**.** *On $\mathcal{T}_2(a,b)$ the bound is $CMA_2^m a^{-1}\sqrt{\varepsilon/b}$, including points with vanishing first clearance derivative. For $1\le r\le r_*$ and thresholds b_r, assign each point of $\bigcup_{r\le r_*}\mathcal{T}_r(a,b_r)$ to its first qualifying order, and retain the rest as $b^{\varepsilon,\mathrm{rest},1}$. Then*

$$b^{\varepsilon,\mathrm{clr},1}=\sum_{r=1}^{r_*} b^{\varepsilon,[r],1}+b^{\varepsilon,\mathrm{rest},1}, \qquad b^{\varepsilon,[r],1},b^{\varepsilon,\mathrm{rest},1}\ge 0.$$

With $P,G,e_{B,m}$ as in Section 98, $\varepsilon=\varepsilon(B)$, and supremum over all R,n,k,

$$\|P-G-m^2b^{\varepsilon,\mathrm{rest},1}\|_\infty \le e_{B,m}+Cm^2A^m\varepsilon+m^2\sum_{r=1}^{r_*} C_rA_r^m a^{-1}(\varepsilon/b_r)^{1/r}. \tag{99.6}$$

Proof. Use the positive partition, Theorem 99.3, and the first-incidence bound of Corollary 97.4 in the exact positive raw-error representation. The Gaussian reference G is unchanged, including its finite arithmetic factor and zero classes. □

The estimate covers physical finite-type strata rather than assuming that every clearance point has finite type with a uniform constant. In particular, a critical point of the total roof need not lie in any pivot carrier. The factors A_r^m also remain explicit. Thus the corollary is a positive source reduction at finite collision count; it is not the ordered central-scale clearance-height limit.

100. Physical clearance caustics and positive source localization

We now locate the remaining loss of transversality by a geometric set in parameter–roof space. For each fixed collision count and a positive incidence threshold, its fiber at a fixed radius is finite. Away from this set a semialgebraic separation inequality gives a quantitative coarea denominator. The exact positive source over a neighborhood of the set remains in the raw-error formula.

100.1. **The physical-side compact graph.** Fix $m\ge 2$ and $0<\chi<1$. In every true center word, recenter at a possible clearance witness $y=T_R^{j+1}x$, as in (99.2), and require

$$c_i(x)\ge\chi \quad (0\le i\le m). \tag{100.1}$$

Take the closure of this regular physical graph with its parameter $R\in I$, rather than replacing its strict first-hit inequalities by an arbitrary nonphysical word. Retain the compact one-flight envelope

$$R\le|Z_d(y)|\le R+s_0, \qquad L_d(y)\ge\ell_*>0,$$

where ℓ_* is the constant in Lemma 178.1. Let $\mathcal{K}_{m,\chi}$ be the finite disjoint union of these closures over words, marks, candidates and compact rational chart pieces. A fixed finite chart cover can be chosen with interiors covering every collision angle. The expression $\mathcal{J}=\det D(F,Z)$ below is computed in its particular marked chart. Its zero set is independent of that choice.

The positive incidence threshold makes the chosen contact roots simple, including on these closures. The implicit contact equations therefore supply continuous jets there. These are limits of physical-side jets; the construction does not continue a source trajectory across a competing-hit seam. Uniform upper and lower flight lengths, bounded contact coordinates and velocities make $\mathcal{K}_{m,\chi}$ compact. It is semialgebraic by the auxiliary graph construction and closure.

Define the clearance caustic by

$$\mathcal{C}_{m,\chi} = \{(R, F(y)) : \ y \in \mathcal{K}_{m,\chi},\ |Z(y)| = R,\ \mathcal{J}(y) = 0\}. \tag{100.2}$$

Different word copies are allowed to have the same image. In particular all roof-critical seam contacts are retained in this definition.

Theorem 100.1 (Finite physical caustic fibers)**.** *The set $\mathcal{C}_{m,\chi}$ is compact and semialgebraic. There are $C > 0$, $A > 1$, independent of m, χ, R, such that*

$$\#\{t : (R, t) \in \mathcal{C}_{m,\chi}\} \le CA^m \qquad (R \in I). \tag{100.3}$$

Thus it is a finite union of parameter-dependent semialgebraic arcs and points and has no interval in a vertical fiber.

Proof. The compact image theorem and Tarski–Seidenberg give the first claim. For fixed R, the one-flight formulas of (178.1) give $\partial_\varphi Z_d = L_d \ge \ell_*$ on the relevant envelope. Since $c = \cos\varphi \ge \chi$, this is a nonzero derivative also in the marked (α, p) charts. Hence $dZ \ne 0$ there. On $Z = R$ or $Z = -R$, the condition $\mathcal{J} = 0$ says that dF is proportional to dZ. Every connected semialgebraic component of this set is joined by piecewise C^1 paths; this follows from cell decomposition, or [22, Sections 2.3.4 and 3.2]. Along each piece of a path, $dZ = 0$ and therefore $dF = 0$. Continuity at the finitely many joining points shows that F is constant on the whole component. Isolated points also contribute only one value.

It remains to bound the number of components before projection. First jets are included exactly as in Lemma 99.2; all implicit coefficients are nonzero under (100.1). To avoid a complexity claim about eliminating a closure operation, use the following larger graph only for counting values. Keep the contact equations, positive incidence cap, root-selection signs and jet equations; replace the physical first-hit inequalities by their weak versions and adjoin the seam and rank equations. This graph has $O(m + 1)$ variables and fixed-degree sign conditions and contains the capped physical-side closure. Its selected contact roots are still simple, so its jets are the unique implicit jets of those equations. On every connected component of this larger critical graph the same $dZ \ne 0$, $dF \wedge dZ = 0$ argument makes F constant. Thus a subset of such a component cannot create additional roof values, even if the subset is disconnected. The bound (167.4) gives CD^m components of the larger graph, uniformly in the coefficient χ and the parameter R. No measure or orbit is continued on this counting overcover. Summing the exponentially many words and possible marks changes only the exponential constant. Since each component contributes at most one roof value, this proves (100.3). Semialgebraic cell decomposition of the image, together with the finite fibers, gives the last assertion. □

100.2. **A denominator off the parameter–roof caustic.** For $v \in \mathcal{K}_{m,\chi}$ put

$$d(v) = \min\{1, \operatorname{dist}((R, F(v)), \mathcal{C}_{m,\chi})\}, \qquad f(v) = \big||Z(v)| - R\big| + |\mathcal{J}(v)|.$$

If the caustic is empty, define $d = 1$. The distance is taken in the two-dimensional parameter–roof space, not to the fiber at the same radius. This convention retains contacts at a parameter transition.

Lemma 100.2 (Physical-side semialgebraic separation)**.** *For every fixed m, χ there are $C_{m,\chi} \ge 1$ and an integer $N_{m,\chi} \ge 1$ such that*

$$d(v)^{N_{m,\chi}} \le C_{m,\chi} f(v) \qquad (v \in \mathcal{K}_{m,\chi}). \tag{100.4}$$

No bound on the growth of these constants as $m \to \infty$ or $\chi \to 0$ is asserted.

Proof. The jets, f and d are continuous semialgebraic functions on the compact finite union of graph pieces. Semialgebraicity of distance follows from its defining infimum and Tarski–Seidenberg. If $f = 0$, its parameter–roof image belongs to (100.2), so $d = 0$. The compact semialgebraic Łojasiewicz inequality [22, Theorem 2.12] gives (100.4) on each piece; finite maxima of constants and exponents combine them, since $0 \le d \le 1$. If the caustic is empty, f has a positive minimum on the compact graph, and the conclusion follows directly. □

At an original first-clearance point $f \le 2\varepsilon + |\mathcal{J}|$. Consequently, for $0 < h \le 1$,

$$\varepsilon \le \frac{h^{N_{m,\chi}}}{4C_{m,\chi}}, \quad d \ge h \quad \Longrightarrow \quad |\mathcal{J}| \ge \frac{h^{N_{m,\chi}}}{2C_{m,\chi}}. \tag{100.5}$$

This is a bound on the actual determinant, uniform in the radius for the fixed m, χ, h.

Define $b^{\varepsilon,\mathrm{cau}(\chi,h),w}$ to be the original first-clearance source restricted to all $c_i \ge \chi$ and $d < h$. The distance uses the selected witness chart through the common parameter–roof image. The source with some $c_i < \chi$ is kept as a separate positive source, even if the first physical defect was clearance and the small incidence occurs later.

Theorem 100.3 (Positive caustic localization at exact labels)**.** *There are $C > 0$, $A > 1$ such that, under the threshold condition in* (100.5)*, $0 < 2\varepsilon < s_0$ and $|w| \le M$,*

$$\begin{aligned} \operatorname*{ess\,sup}_t \sum_{n=1}^{m} \sum_{k\in\mathbb{Z}^2} \left| b^{\varepsilon,\mathrm{clr},w}_{n,k,m,R}(t) - b^{\varepsilon,\mathrm{cau}(\chi,h),w}_{n,k,m,R}(t) \right| \\ \le CMA^m \left((m+1)\chi + C_{m,\chi}\varepsilon h^{-N_{m,\chi}} \right). \end{aligned} \tag{100.6}$$

The caustic term is a restriction of the original positive source when $w = 1$. This is an identity-based localization of that source on all roofs, not a deletion of a set of roof values.

Proof. Split the original clearance source into: some $c_i < \chi$; all $c_i \ge \chi$ and $d < h$; and all $c_i \ge \chi$ and $d \ge h$. On a regular physical word $0 < c_i \le 1$, so

$$\mathbf{1}_{\{\min_i c_i < \chi\}} \le \sum_{i=0}^{m} \chi c_i^{-1} \le (m+1)\chi\, \mathcal{I}_{m,R}.$$

Theorem 97.3 bounds the height of the first part by $CMA^m(m+1)\chi$. This comparison does not change the first defect's type or erase its later history. For the third part, (100.5) gives a lower bound for the marked-chart determinant. Invariance under recentering and the bounded rational-chart density let us apply the same area formula and physical multiplicity proof as Theorem 98.2, now in that chart. The total length of the signed clearance bands, summed over marks, is $O(\varepsilon)$, so its height is at most $CMA^m C_{m,\chi}\varepsilon h^{-N_{m,\chi}}$. Their sum is the left side for $w = 1$. Domination and the disjoint exact-label partition prove the inserted statement. The retained middle part is precisely the caustic source in the theorem. □

Corollary 100.4 (The raw-error formula with the caustic source)**.** *For $\varepsilon = \varepsilon(B)$ satisfying* (100.5)*, uniformly over R, n, k,*

$$\|P - G - m^2 b^{\varepsilon,\mathrm{cau}(\chi,h),1}\|_\infty \le e_{B,m} + Cm^2A^m \left(\varepsilon + (m+1)\chi + C_{m,\chi}\varepsilon h^{-N_{m,\chi}} \right). \tag{100.7}$$

Proof. Use Theorem 181.4, the original incidence height gain and Theorem 100.3. The canonical finite arithmetic reference G is not modified. □

100.3. **Size, but not height, of the retained caustic source.**

Lemma 100.5 (Uniform vertical measure of a caustic neighborhood)**.** *For fixed m, χ there are $D_{m,\chi} < \infty$ and $\beta_{m,\chi} > 0$ such that*

$$\sup_{R\in I} |\{t : \operatorname{dist}((R,t), \mathcal{C}_{m,\chi}) < h\}| \le D_{m,\chi} h^{\beta_{m,\chi}}, \qquad 0 < h \le 1. \tag{100.8}$$

For the empty caustic the left side is zero.

Proof. First the left side tends uniformly to zero as $h \downarrow 0$. Otherwise choose $R_l \to R_*$ and $h_l \downarrow 0$ violating that assertion. All roof values lie in a fixed compact interval because m is fixed. By compactness, for every neighborhood V of the finite fiber $\mathcal{C}_{m,\chi}(R_*)$, the corresponding vertical sections are eventually contained in V. A finite union of arbitrarily short intervals about that fiber contradicts the violation.

There is also a power modulus. Cell decomposition of the semialgebraic neighborhood family expresses each vertical section as a finite union of intervals with semialgebraic endpoints, after partitioning (R, h). Its length is therefore a semialgebraic function; no assertion that arbitrary integrals of semialgebraic functions are semialgebraic is being used. Taking its supremum over $R \in I$ preserves semialgebraicity by quantifier elimination. The resulting bounded one-variable function tends to zero. Its graph, on a small interval, either vanishes or is a branch of a polynomial relation. The one-variable growth argument in the proof of [22, Theorem 2.12] bounds it by a positive power of h. Increasing the constant treats the rest of $0 < h \le 1$. □

Corollary 100.6 (A positive mass bound for the geometric remainder)**.** *For every fixed bounded roof interval J, the uninserted caustic source satisfies*

$$\int_J m^2 b^{\varepsilon,\mathrm{cau}(\chi,h),1}_{n,k,m,R}(t)\, dt \le C(1+|J|)^{144/145} \big(D_{m,\chi} h^{\beta_{m,\chi}}\big)^{1/145}. \tag{100.9}$$

Here C is the constant in the inherited local weak-integrability estimate; the new constants have the dependence just stated.

Proof. The source density is at most $P = m^2 p$ after normalization and is supported in the neighborhood in (100.8). The layer-cake consequence of the inherited weak $L^{145/144}$ bound gives, for a measurable $E \subset J$,

$$\int_E P \le C(1+|J|)^{144/145} |E|^{1/145}.$$

This is Lemma 187.2. Apply it to the indicated neighborhood intersected with J and then use Lemma 100.5. □

The last corollary is a mass estimate for an explicitly retained physical source, not an essential-height estimate for it. Neither the power modulus nor the separation exponent has been controlled as the collision count grows. In particular the ordered limits required in Corollary 181.5 do not follow by choosing ε exponentially small in m: the fixed-band spectral error has a different order of quantifiers. The present results isolate finite-type clearance and physical caustics without replacing the original next-collision source or assigning conditional laws to arithmetic zero classes.

101. Effective buffered caustics of the physical collision graph

We quantify the dependence on the collision count and on the incidence cap in the geometric localization of Section 100. The reference set used here is a specified algebraic overcover of the physical caustic. It carries no measure. The source localized near it will remain a restriction of the original orbit source. A factor two between the source cap and the reference cap is needed when the cap varies.

101.1. **A finite-format reference graph.** Choose an integer T_0 larger than the uniform flight horizon. Use compact rational angular charts with algebraic frames and introduce $c_i > 0$ by $c_i^2 + p_i^2 = 1$. For each word, mark $j+1$, candidate disk and chart, let $\widehat{\mathcal{K}}_{m,\eta}$ be the weak-first-hit graph used in the counting argument of Theorem 100.1. More precisely, retain the contact and reflection equations, the chosen entry-root signs, all first-jet equations, $\ell_0 \le \ell_i \le T_0$, and $c_i \ge \eta$ for $0 \le i \le m$, and replace only the strict first-hit inequalities by their weak versions. Retain also the compact selected-witness envelope

$$R \le |Z| \le R + s_0, \qquad L_d \ge 1/5.$$

This envelope contains the original one in module 134, since $\ell_* > 1/5$. Take the finite disjoint union over the indicated labels. The full roof F, the clearance Z and $\mathcal{J} = \det D(F, Z)$ have exactly the marked-chart meanings in (99.2). In particular F contains the entire backward and forward parts of the original m flights.

Define

$$\widehat{\mathcal{C}}_{m,\eta} = \{(R, F(X)) : X \in \widehat{\mathcal{K}}_{m,\eta}, \quad Z(X)^2 = R^2, \quad \mathcal{J}(X) = 0\}. \tag{101.1}$$

The physical caustic $\mathcal{C}_{m,\eta}$ is contained in this set. Equality is neither needed nor asserted. In contrast with the earlier use of the weak graph solely for counting, its image now serves as a larger geometric reference set. No trajectory or probability is put on its additional points.

Lemma 101.1 (Format and finite fibers of the reference graph)**.** *For each $m \ge 2$ and $0 < \eta \le 1$ the graph and its critical image are compact. At a fixed radius the image has at most CA^m points. For fixed R, η and $0 \le h \le 1$, the compact set*

$$V_{m,\eta}(R, h) = \{t : (R', t) \in \widehat{\mathcal{C}}_{m,\eta},\ |R' - R| \le h\} \tag{101.2}$$

is a union of at most CA^m closed intervals and points.

There are fixed integers $a, d \ge 2$ for which the following format bounds hold. One word graph uses at most $a(m+1)$ real variables and polynomial atoms of degree at most d. The union uses at most $\exp(a(m+1))$ atoms after Boolean expansion, with the same variable blocks. Coefficient bit lengths, after clearing rational denominators, are at most $a(m+1)^2$. The only fixed algebraic coefficient needed is $\sqrt{3}$, which may be replaced by one positive variable satisfying $z^2 = 3$. The variables R, η, h are not treated as coefficients.

Proof. This is the auxiliary physical graph of Sections 97 and 100, with first jets as in Lemma 99.2. Intermediate contacts are retained, not eliminated. Each new collision adds a fixed number of positions, velocities, flight lengths and first-derivative variables. Introducing a reciprocal for each nonzero implicit coefficient keeps the polynomial degree bounded. Under a positive cap these coefficients do not vanish. Their reciprocals and the unique first jets are bounded on each capped compact parameter set. Thus the weak graph is closed and bounded. These bounds may depend on m, η; no norm bound on the jets is being claimed independent of them.

The next-center alphabet is fixed by T_0. In local coordinates its coefficients lie in $\mathbb{Q}(\sqrt{3})$ and have a fixed height bound. Alternatively, absolute lifted centers have size $O(m+1)$, which is still within the stated bit bound. Rational angular frames may be chosen among the coordinate-axis frames; positive denominators are retained as graph variables. Neither the moving section nor any exact-label indicator is encoded here. There are exponentially many words and only polynomially many marks. Their Boolean disjunction reuses the same continuous variables: it does not introduce one quantified block for each word. This proves the format statements.

For a fixed R, $dZ \ne 0$ on the envelope: in the collision coordinates $\partial_\varphi Z = L_d \ge 1/5$. On the seam and rank locus, $dF \wedge dZ = 0$. The implicit simple-root equations define the same smooth local jets even where a weak first-hit inequality becomes an equality. On every smooth stratum of a path in that locus, $dZ = 0$ and hence $dF = 0$ along the path. Semialgebraic

connected components are piecewise smooth path connected. Continuity of F at successive stratum endpoints makes it constant on the entire component. This includes zero-dimensional strata and junctions; no differentiability across a change of selected word is used, since word copies are disjoint.

For each word, the sign-component estimate (167.4) applies to $O(m+1)$ variables and atoms of fixed degree, also after adding the two horizontal-strip inequalities in (101.2). It gives CD^m components. Summing words and marks gives CA^m. For a fixed radius each critical component contributes one value. In the horizontal strip its roof projection is a compact connected subset of the real line, and thus an interval or point. These observations prove both counting assertions. □

101.2. **An effective one-variable lower modulus.** We record the elementary scalar step used to keep the cap dependence quantitative. Its algebraic input is effective quantifier elimination, not a billiard mixing estimate.

Lemma 101.2 (A positive monotone modulus of controlled format)**.** *Suppose $a_m : (0,1] \to (0,1]$ is positive and nondecreasing. Suppose its graph has a first-order description over the reals with at most twelve quantifier blocks, at most $a(m+1)$ variables in each block, and at most a fixed number of free variables. Assume the degree, atom-count and integer coefficient bounds of Lemma 101.1. Then, for a fixed $C_0 \geq 1$ depending only on this format,*

$$E_m = \left\lceil 2^{C_0(m+1)^{16}} \right\rceil, \qquad K_m = 2^{E_m} \tag{101.3}$$

satisfy

$$a_m(\delta) \geq K_m^{-1} \delta^{E_m}, \qquad 0 < \delta \leq 1. \tag{101.4}$$

The same C_0 can cover any fixed finite list of such descriptions.

Proof. The quantifier-elimination bound in [23, Theorem 4] gives a quantifier-free description of the graph by integer polynomials whose degrees D and coefficient bit lengths H satisfy

$$D, H \leq 2^{C_1(m+1)^{12}}.$$

Indeed the degree bound is $d^{O(k_1)\cdots O(k_\omega)}$, and the bit bound is the input bit length times that quantity, with $\omega \leq 12$ and $k_i = O(m+1)$. The number of atoms can be large, but does not enter these two bounds. Adjoining the fixed algebraic coefficient can be merged into an existential block; increasing the fixed block budget to twelve includes it.

Here is a direct passage from these bounds to a lower modulus; it requires neither continuity of a_m nor a quantitative Puiseux theorem. At every graph point (δ, u) at least one nonzero defining polynomial $P(\delta, u)$ vanishes. Otherwise the truth values of all its signs are constant on a neighborhood, contradicting that the formula defines the graph of a single-valued function. Remove the largest power of u dividing this polynomial and write $P = P_0(\delta) + \sum_{j=1}^{D} P_j(\delta) u^j$, with $P_0 \neq 0$. For

$$0 < \delta \leq \delta_0 = [4(D+1)2^H]^{-1}$$

the first nonzero integer coefficient of P_0 gives $|P_0(\delta)| \geq \delta^D/2$. The sum of the absolute values of all higher terms is at most $D(D+1)2^H u$ for $0 < u \leq 1$. Consequently every such positive root satisfies

$$u \geq \frac{\delta^D}{2D(D+1)2^H}.$$

A polynomial independent of u cannot vanish in this range, by the same leading-coefficient estimate. These estimates hold for all the polynomials at once, without multiplying their number.

For $\delta \geq \delta_0$, use monotonicity and the estimate at $\delta_0/2$. It follows that the whole interval has a lower bound $c\delta^D$, where $\log_2(c^{-1}) \leq C_2 D(H + \log(2+D))$. Increasing C_0 in (101.3) dominates

this last quantity and D, and proves (101.4). All constants are controlled by the displayed finite format, not by an unrecorded real coefficient of a fixed-cap distance function. □

101.3. **Buffered separation, uniform in the cap.** For $X \in \widehat{\mathcal{K}}_{m,\chi}$, $0 < \chi \le 1$, put

$$d_{m,\chi}(X) = \min\{1, \operatorname{dist}((R, F(X)), \widehat{\mathcal{C}}_{m,\chi/2})\}, \qquad f(X) = \big||Z(X)| - R\big| + |\mathcal{J}(X)|. \tag{101.5}$$

The distance is in parameter–roof space. It is one if the reference set is empty. All inequalities below apply in particular to the original physical graph capped at χ.

Theorem 101.3 (Effective separation with a buffered incidence cap)**.** *After one fixed increase of C_0 in* (101.3)*,*

$$(\chi d_{m,\chi}(X))^{E_m} \le K_m f(X) \quad (m \ge 2,\ 0 < \chi \le 1,\ X \in \widehat{\mathcal{K}}_{m,\chi}). \tag{101.6}$$

Thus an original first-clearance point satisfies

$$4K_m\varepsilon \le (\chi h)^{E_m}, \quad d_{m,\chi} \ge h \quad \Longrightarrow \quad |\mathcal{J}| \ge \frac{(\chi h)^{E_m}}{2K_m}. \tag{101.7}$$

Proof. For $0 < \delta \le 1$, let $a_m(\delta)$ be the infimum of the set containing 1 and every $\min(1, f(X))$ such that

$$\delta \le \chi, h \le 1, \quad X \in \widehat{\mathcal{K}}_{m,\chi}, \quad \operatorname{dist}((R, F(X)), \widehat{\mathcal{C}}_{m,\chi/2}) \ge h.$$

It is nondecreasing. It is strictly positive. To prove the latter, suppose for a fixed $\delta > 0$ that $f(X_l) \to 0$ along eligible points. Pass to a subsequence in one word/chart copy with $\chi_l \to \chi_*$ and $X_l \to X_*$. The cap δ bounds the simple-root jets, so such a subsequence exists. The limit is a critical seam point with $c_i(X_*) \ge \chi_*$ for every i. For all sufficiently large l it belongs to the reference graph with cap $\chi_l/2$. Its image approaches $(R_l, F(X_l))$, contradicting the distance lower bound δ. The buffer is used precisely at this step; continuity of an unbuffered cap-dependent distance is not assumed.

The graph of this infimum has controlled format. Distance at least h is the universal statement that every critical graph point Y at cap $\chi/2$ has squared image distance at least h^2. The empty-reference case is included by that statement. The condition that u is the infimum is the conjunction of: $u \le 1$ and u is a lower bound for all eligible points; and, for every $r > 0$, either $1 < u + r$ or there is an eligible point with $\min(1, f) < u + r$. The first conjunct has prefix $\forall X\, \exists Y$ after negating eligibility; the second has prefix $\forall r\, \exists X\, \forall Y$. Caps, thresholds and the absolute-value graph variables are included in the corresponding blocks. Renaming bound variables and adjoining the algebraic frame constant uses fewer than twelve blocks of size $O(m+1)$. All atoms have the format in Lemma 101.1.

Apply Lemma 101.2. At a point with $d > 0$, take $h = d$ and $\delta = \min(\chi, d) \ge \chi d$. The definition of a_m yields (101.6); the case $d = 0$ is immediate. On the original source the seam error is at most 2ε. Subtract it from the lower bound for f to obtain (101.7). □

101.4. **A collision-count bound for the tube modulus.**

Theorem 101.4 (Effective vertical tube length)**.** *The same budgets, after increasing C_0 once, satisfy*

$$\sup_{R \in I} |\{t : \operatorname{dist}((R, t), \widehat{\mathcal{C}}_{m,\chi/2}) < h\}| \le K_m \chi^{-1} h^{1/E_m} \quad (0 < \chi, h \le 1). \tag{101.8}$$

An empty reference has zero tube length. In particular both the power and its constant are now bounded in the collision count and cap.

Proof. Define $b_m(\delta)$ to be the infimum of $\{1\}$ and of the values $h \in [0,1]$ for which there exist $\delta \le \chi \le 1$, $R \in I$ and $u \in [0, mT_0]$ with

$$[u, u+\delta] \subset V_{m,\chi/2}(R,h).$$

It is nondecreasing, since a longer interval contains a shorter one and the allowed caps decrease when δ decreases. It is positive. Indeed, if $h_l \to 0$ for a fixed $\delta > 0$, pass to limits in R_l, χ_l, u_l. For each point of $[u_*, u_* + \delta]$, compactness of the critical graph with cap at least $\delta/2$ places that point in the limiting fixed-radius critical fiber at cap $\chi_*/2$. This contradicts its finiteness in Lemma 101.1.

The interval inclusion has the literal formula

$$\forall v \in [u, u+\delta]\ \exists Y: \quad Y \text{ is a critical graph point at cap } \chi/2, \quad |R_Y - R| \le h, \quad F(Y) = v.$$

Using the lower-bound and approximation descriptions of an infimum as above gives at most nine quantifier blocks, including the fixed algebraic coefficient, each with $O(m+1)$ variables. This avoids forming an integral of a semialgebraic function or quantifying one variable for every interval endpoint. Lemma 101.2 therefore gives $b_m(\delta) \ge K^{-1}\delta^E$ with preliminary budgets of the form (101.3).

Let an interval component of $V_{m,\chi/2}(R,h)$ have length $l > 0$. It lies in $[0, mT_0]$. With $\delta = \min(l, \chi, 1)$ the preceding bound gives $h \ge K^{-1}\delta^E$. Since $\delta \ge \chi l/(1+mT_0)$,

$$l \le (1+mT_0)\chi^{-1}(Kh)^{1/E}.$$

There are at most CA^m interval or point components. The vertical section of the Euclidean h-tube is contained in the h-neighborhood of this set, whose length is at most the sum of these component lengths plus $2CA^m h$. Absorb the finite factors, $K^{1/E}$ and $1 + mT_0$ by a fixed increase of C_0. Increasing E_m weakens the power bound for $h \le 1$, so this change proves (101.8) simultaneously with the separation bound. □

The quantitative input in this section is the standard real-algebraic elimination bound, with its coefficient-height control. Its physical content is the simple-root graph, the seam/rank finite-fiber argument and the buffered-cap compactness. The estimates do not independently certify those inherited continuum inputs. They give explicit, very large finite-count moduli, not a spectral estimate at a growing band.

102. Quantitative positive localization on the unchanged source

We apply the preceding moduli before pushforward. This preserves the first-defect order, the complete incidence history, the image witness $j+1$, and every exact label. The reference overcover defines a region; all measure in that region still comes from the original Lorentz orbit.

For $0 < \chi, h \le 1$ let $\widehat{b}^{\varepsilon,\mathrm{cau}(\chi,h),w}_{n,k,m,R}$ be the first-clearance density of Section 177, restricted, before pushforward by the full roof, to

$$c_i \ge \chi \quad (0 \le i \le m), \qquad d_{m,\chi} < h. \tag{102.1}$$

The selected witness is evaluated at $T_R^{j+1}x$, with its original candidate disk and chart. The distance is (101.5), using the cap $\chi/2$ for the reference set only. For $w = 1$ this is a positive restriction of the same source, not a signed cancellation or a probability on $\widehat{\mathcal{K}}$.

Theorem 102.1 (Effective caustic localization at exact arithmetic labels)**.** *Let $m \ge 2$, $|w| \le M$, $0 < 2\varepsilon < s_0$ and $0 < \chi, h \le 1$. Assume*

$$4K_m\varepsilon \le (\chi h)^{E_m}. \tag{102.2}$$

There are $C > 0$, $A > 1$, independent of these parameters, such that

$$\operatorname*{ess\,sup}_t \sum_{n=1}^{m} \sum_{k\in\mathbb{Z}^2} \left| b^{\varepsilon,\mathrm{clr},w}_{n,k,m,R}(t) - \widehat{b}^{\varepsilon,\mathrm{cau}(\chi,h),w}_{n,k,m,R}(t) \right| \le CMA^m \left((m+1)\chi + K_m \varepsilon (\chi h)^{-E_m} \right). \tag{102.3}$$

For every bounded interval J the retained uninserted density satisfies

$$\int_J m^2 \widehat{b}^{\varepsilon,\mathrm{cau}(\chi,h),1}_{n,k,m,R}(t)\, dt \le C(1+|J|)^{144/145} \left(K_m \chi^{-1} h^{1/E_m} \right)^{1/145}. \tag{102.4}$$

The first assertion is essential-height control outside the retained source. The second is mass control of that source, not its height.

Proof. Partition the actual first-clearance source into three positive pieces: there is some $c_i < \chi$; all $c_i \ge \chi$ and $d < h$; all $c_i \ge \chi$ and $d \ge h$. This is a partition on the original trajectory, including incidences later than its first clearance defect. On every regular trajectory,

$$\mathbf{1}_{\{\min_i c_i < \chi\}} \le \sum_{i=0}^{m} \chi c_i^{-1} \le (m+1)\chi \mathcal{I}_{m,R}.$$

Theorem 97.3 pays the first piece by $CMA^m(m+1)\chi$ in height. The middle piece is exactly (102.1).

On the last piece Theorem 101.3 and (102.2) give $|\mathcal{J}| \ge (\chi h)^{E_m}/(2K_m)$. Apply the physical area formula and multiplicity bound of Theorem 98.2 in the selected chart. The marked source has the bounded rational-coordinate density inherited from ν^*_R, and the total length of the signed clearance bands, summed over the geometrically weighted mark depths, is $O(\varepsilon)$. The resulting height is at most $CMA^m K_m \varepsilon (\chi h)^{-E_m}$. Only for this positive upper comparison are exact-label restrictions dropped. Restoring their disjoint partition at fixed m gives the sum in (102.3), with no target-count multiplier. The source normalization contributes c_*^{-1} once, absorbed in C. For complex w, dominate its variation by M times the same positive source.

The normalized retained density is bounded above by the complete normalized density $P = m^2 p$. Its roof support is contained in the tube of Theorem 101.4. The layer-cake estimate of Lemma 187.2, on a measurable $E \subset J$, is

$$\int_E P \le C(1+|J|)^{144/145} |E|^{1/145}.$$

Insert the new tube bound to obtain (102.4). Coarea identities hold outside their usual Lebesgue-null roof sets; no regular conditional version on an arithmetic zero class is used. □

Corollary 102.2 (The raw-error formula with explicit cap and count budgets)**.** *Use the original reconstruction width $\varepsilon = \varepsilon(B)$. Under* (102.2)*, uniformly in R, n, k,*

$$\|P - G - m^2 \widehat{b}^{\varepsilon,\mathrm{cau}(\chi,h),1}\|_\infty \le e_{B,m} + Cm^2 A^m \left(\varepsilon + (m+1)\chi + K_m \varepsilon (\chi h)^{-E_m} \right). \tag{102.5}$$

The reference G is the unchanged signed canonical arithmetic kernel.

Proof. Substitute (102.3) and the incidence-source bound of Corollary 97.4 into the positive raw-error representation of Theorem 181.4. The retained source is not discarded. No normalization of G occurs in this operation. In probability comparisons elsewhere, only its positive part is normalized and probability TV is half the variation mass; the sum in (102.3) is variation mass itself. □

What the quantitative statement changes. For the buffered reference the fixed-count quantities in the earlier separation and tube lemmas can now be replaced by

$$N = E_m, \qquad C_{m,\chi} = K_m \chi^{-E_m}, \qquad D_{m,\chi} = K_m \chi^{-1}, \qquad \beta = E_m^{-1}.$$

These replacements concern the explicitly enlarged reference (101.1) at cap $\chi/2$, not an asserted equality with $\mathcal{C}_{m,\chi}$. The new estimate is uniform in the real cap; it is stronger in this respect than a separate finiteness statement for each χ. Its exponent and prefactor follow from a fixed quantifier-block budget and algebraic coefficient bounds. The finite word geometry is not treated as a system of independent returns.

The ordered height problem. The budgets E_m and K_m are large. In particular they do not imply

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k,\,\text{central } t} m^2 b^{\varepsilon(B),\mathrm{inc}}_{n,k,m,R}(t) = 0$$

or its complete clearance analogue. In the required order, B and $\varepsilon(B)$ are fixed before the collision limit. Making the finite-count error small by choosing B after m does not prove this statement. Moreover (102.4) does not control the height inside the tube. The two ordered heights in Corollary 181.5, and hence its unrestricted pointwise and exact-roof consequences, remain the next analytical requirements on this same source.

A direct proof route for the physical source. The dependencies of the six geometric modules are as follows. Each row separates the input from its conclusion; no height is inferred from mass alone.

Module	Input on the actual orbit	Conclusion used downstream
131	Endpoint contact Hessian and coarea	Inverse-incidence density; incidence height
132	Marked (F, Z) and multiplicity	Transverse clearance height
133	Full tangential jets on a roof fiber	Finite-type carriers, including folds
134	Capped seam/rank graph	Finite caustic fibers and positive localization
135	134; fixed-format elimination	Buffered separation and tube budgets
136	131, 132, 135; positive raw error	Quantitative localization with its positive remainder

The determinant, physical graph and jet constructions remain subject to independent specialist verification. Quantifier elimination supplies moduli once those inputs have been established; source hashes and finite fixtures do not verify the continuum geometry.

103. Reversible critical weights on the original collision record

The positive source retained near a clearance caustic is sensitive to critical coefficients, not just to the size of a roof neighborhood. We first identify the coefficient of a normal-to-normal word with a reversible linearization. The identity holds at every regular word, without a lower bound on its interior incidences. At a simple competing-hit boundary it has a physical-side interpretation, which must be distinguished from the derivative of the singular billiard map.

Determinant identities relating action Hessians and monodromy are classical; the discrete and reversible theory is developed in [24]. The purpose here is the exact source normalization, the half-word labels, and the square-root weight needed by the roof density. No periodic-orbit equidistribution theorem is inferred from the identity.

103.1. **Normal endpoints and reversible closure.** Fix a regular word of $m \geq 1$ flights contributing to the original section source. Its initial and terminal section tests and its exact half-word labels are retained. Use canonical endpoint arclengths u, v, with tangential momenta

p_0, p_m, and let $F(u, v)$ be its reduced action. Suppose z is a critical point of F. The first variation gives $p_0 = p_m = 0$. Write

$$H_z = \nabla^2 F(z) = \begin{pmatrix} h_{00} & h_{01} \\ h_{01} & h_{11} \end{pmatrix}, \qquad M_z = DT_R^m(z) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{103.1}$$

The matrix entry c in this display is not the section mass; the latter is denoted throughout this section by $c_* = 91/(10000\pi)$. Matrices use arclength and momentum, not angle and momentum with an unrecorded scale factor.

Lemma 103.1 (The half-word and its reflected closure)**.** *The point z is fixed by T_R^{2m} on the collision quotient, and*

$$P_z := DT_R^{2m}(z) = SM_z^{-1}SM_z, \qquad S = \operatorname{diag}(1, -1). \tag{103.2}$$

Moreover

$$b = -h_{01}^{-1}, \quad a = -h_{00}/h_{01}, \quad d = -h_{11}/h_{01}, \quad c = -\det H_z/h_{01}, \qquad bc = \det H_z/h_{01}^2 > 0, \tag{103.3}$$

and

$$P_z = \begin{pmatrix} ad + bc & 2bd \\ 2ac & ad + bc \end{pmatrix}, \qquad \det(I - P_z) = -4bc. \tag{103.4}$$

The period need not be primitive. There is no division by m or $2m$ in these assertions.

Proof. The collision reversal is $\iota(u, p) = (u, -p)$ and satisfies $\iota T_R \iota = T_R^{-1}$ on regular orbits. Both z and $z_m = T_R^m z$ are fixed by ι. Thus $T_R^m z_m = \iota T_R^{-m} \iota z_m = z$; geometrically the second half retraces the first. Differentiating the reversal identity at z_m gives $DT_R^m(z_m) = SM_z^{-1}S$, proving (103.2).

Differentiate $p_0 = -F_u$ and $p_m = F_v$ and solve the first equation for dv in terms of du, dp_0. This gives every entry in (103.3). In particular $\det M_z = 1$. The endpoint injectivity and nonzero mixed derivative are those of Lemma 167.1 and (97.2). Positivity of H_z gives $bc > 0$. Multiplication, using $ad - bc = 1$, gives (103.4). □

The closure is used only to calculate a weight. The displacement, return index and roof of the original source are still those of the first m flights. In particular the section is not assumed invariant under reversal. Neither its occupation count nor its arithmetic residue is read from the doubled orbit. Two starting points on one closed orbit are not identified if they label distinct original source states.

Theorem 103.2 (The intrinsic coefficient as a reversible weight)**.** *Let J_z be the full-word Morse coefficient of the common density ν/c_*, before any active one-sided restriction. At a strict section point this is* (169.6) *for the original source; boundary fractions will be imposed in the next two sections. The eigenvalues of P_z are $\lambda_z, \lambda_z^{-1}$ with $\lambda_z > 1$, and*

$$J_z = \frac{1}{Rc_*\sqrt{|\det(I - P_z)|}} = \frac{1}{Rc_*}\frac{\lambda_z^{-1/2}}{1 - \lambda_z^{-1}}. \tag{103.5}$$

There are fixed $C < \infty$ and $q_0 = 47/53 < 1$ such that

$$J_z \le C q_0^{m-2} \quad (m \ge 2), \tag{103.6}$$

uniformly in R and in all positive interior incidences. The square-root determinant in (103.5) *cannot be replaced by the ordinary inverse determinant.*

Proof. The original endpoint density at z is $w_z = |h_{01}|/(4\pi R c_*)$. Hence the two-dimensional Morse coefficient is

$$J_z = \frac{2\pi w_z}{\sqrt{\det H_z}} = \frac{1}{2Rc_*\sqrt{bc}}.$$

Equation (103.4) gives the first formula. Since $\operatorname{tr} P_z = 2 + 4bc > 2$ and $\det P_z = 1$, its eigenvalues are positive reciprocal numbers, one greater than one. Now $|\det(I - P_z)| = (\lambda_z - 1)^2/\lambda_z$, which proves the second formula and fixes its square root.

For the count bound use the internal matrix A in (169.2). The endpoint Schur complement gives, for $m \geq 2$ and normal endpoints,

$$|h_{01}| = t_0 t_{m-1} |(A^{-1})_{1,m-1}|.$$

This includes the one-entry matrix when $m = 2$. The Neumann estimate (169.3), valid for every positive incidence, bounds this by Cq_0^{m-2}. Also $H_z \geq R^{-1} I$ in arclength coordinates by the endpoint second variation. Thus $\sqrt{\det H_z} \geq R^{-1}$ and the formula for J_z proves (103.6). This proof does not bound a sum over words by its largest summand. □

103.2. **A normal-section crossing measure.** There are constants $0 < \lambda_* \leq \Lambda_* < \infty$, independent of the word and of all its positive incidences, such that at a normal critical point

$$\lambda_* I \leq H_z \leq \Lambda_* I. \tag{103.7}$$

One may take $\lambda_* = 1/R_+$ and $\Lambda_* = 2(1/R_- + 1/\ell_0)$. The lower bound was just proved. The internal Schur complement is bounded above by the endpoint principal block. Its entries are bounded by endpoint curvature and $1/\ell_0$; the stated choice also includes $m = 1$.

Proposition 103.3 (Crossing weights and fixed-count traces)**.** *Set $W_z = |\partial_u p_m(u,0)|_z^{-1} = |c|^{-1}$. Then*

$$J_z = \frac{\sqrt{\det H_z}}{2Rc_*} W_z, \qquad \frac{\lambda_*}{2R_+ c_*} W_z \leq J_z \leq \frac{\Lambda_*}{2R_- c_*} W_z. \tag{103.8}$$

For a regular critical point whose physical and section inequalities are all strict, take an isolating source neighborhood U_z. For any continuous roof test ψ,

$$\lim_{d \downarrow 0} \frac{1}{4d^2} \int_{U_z} \mathbf{1}_{\{|p_0|<d,\ |p_m|<d\}} \psi(L_{m,R})\, d\nu_R^* = \frac{W_z}{4\pi R c_*} \psi(t_z), \qquad t_z = F(z). \tag{103.9}$$

The collision count in this limit is fixed.

Proof. The ratio $c/b = \det H_z$ in (103.3) proves the equality in (103.8); then apply (103.7). The map $(u, p_0) \mapsto (p_0, p_m)$ has absolute determinant $|c|$ at z. It is a local diffeomorphism. Its inverse sends the shrinking square $(-d, d)^2$ to U_z for all sufficiently small d. The source density is $1/(4\pi R c_*)$ in (u, p_0). Change variables, divide by the square's area, and use continuity of the integrand. □

For a finite family with the same original labels define

$$\mathfrak{T}_{n,k,m,R} = \sum_z \frac{\lambda_z^{-1/2}}{1 - \lambda_z^{-1}} \delta_{t_z}. \qquad \sum_z J_z \delta_{t_z} = (Rc_*)^{-1} \mathfrak{T}_{n,k,m,R}. \tag{103.10}$$

This is a positive atomic measure, not a probability or a spectral trace of an operator already constructed in the paper. Equal roof values are added. The pointwise identity gives equality of its weighted sums against every bounded test, so it does not require separation of critical values.

103.3. **Physical-side values at a clearance seam.**

Proposition 103.4 (The boundary interpretation)**.** *Suppose a selected collision word has a physical-side limit z at a competing-hit seam, with every selected contact incidence positive and both endpoint momenta zero. Its simple contact equations define local smooth F, H_z, M_z, and w_z as limits of that branch. Define $P_z = SM_z^{-1}SM_z$ algebraically. Equations* (103.4)*,* (103.5)*,* (103.6) *and* (103.8) *remain valid for these branch data. They do not assert that DT_R^{2m} exists at the singular state.*

Proof. Positive selected incidence makes each contact root simple. The implicit contact graph and its endpoint map therefore extend smoothly locally, whether or not the strict competing-hit inequality survives on both sides. The Hessian, mixed derivative and symplectic generating identities persist by continuity from the physical side. The positive normal Hessian and the nonzero mixed derivative persist at this finite word, by the explicit contact determinant. Apply the same algebra. The extension is used to compute derivatives only; no source is placed on a nonphysical continuation. At the seam the actual collision map can change its selected first hit, and its full derivative is not used. □

Relation to the ordered height problem. The exact formula identifies a concrete positive weight to be summed in the original labels. It does not prove the required long-count concentration estimate for that sum. In particular the limit $d \downarrow 0$ in (103.9) cannot be interchanged with the collision limit in Lemma 170.1. That interchange would require a uniform transverse-trace estimate not supplied by the fixed-width local theorem. Likewise individual exponential decay in (103.6) is not a bound on an exponentially large critical cluster. The next section uses the exact weights to calculate, rather than discard, a simple roof-critical clearance source.

104. A two-scale density profile inside a simple clearance caustic

We now compute the density inside a part of the retained caustic, including a roof-critical point where no roof pivot is available. The conclusion is local on a simple competing-hit seam. It is not a uniform classification of the entire caustic, and it does not convert a tube-mass estimate into height control.

104.1. **A simple physical-side critical contact.** Fix m, R, an original center word, and a physical-side limiting critical point z of its reduced action. Assume every selected contact has positive incidence. On an endpoint neighborhood U, assume the only active physical boundary is one competing disk for flight j, with its perpendicular foot strictly inside that flight. All other first-hit inequalities and all section decisions are strict. Use the same responsible disk and the image mark $j+1$ as in Section 178. The local signed margin is

$$g(x) = |Z_d(T_R^{j+1}x)| - R, \qquad g(z) = 0. \tag{104.1}$$

After shrinking U, the original word occupies exactly $g > 0$ in this neighborhood and its exact half-word labels are constant there. The equality side has zero source measure. The local continuation of the selected contact equations merely supplies coordinates for this one-sided source.

The derivative $dg(z)$ is nonzero. Indeed $\partial_\varphi Z_d = L_d \geq \ell_* > 0$ on the selected envelope, by (178.4); the regular recentering and endpoint maps are local diffeomorphisms. Define

$$H = \nabla^2 F(z), \quad t_z = F(z), \quad a_z = dg(z)H^{-1/2}, \quad \kappa_z = |a_z| > 0, \quad J_z = \frac{2\pi w_z}{\sqrt{\det H}}. \tag{104.2}$$

Here $w_z = |F_{uv}(z)|/(4\pi R c_*)$ is the density of the original section source in endpoint arclengths. Thus J_z is given by Proposition 103.4. At a boundary point it is the full local quadratic coefficient; the physical one-sided coefficient will be one half of it.

Lemma 104.1 (Uniform angular threshold comparison)**.** *For $a \in \mathbb{R}^2 \setminus \{0\}$ put $\omega_\theta = (\cos\theta, \sin\theta)$. Suppose q_r on the circle satisfies $\|q_r - a \cdot \omega\|_\infty \le Lr$, $0 < r < r_0$. Uniformly in $\lambda \ge 0$,*

$$\big|\big|\{0 < q_r < \lambda\}\big| - 2\arcsin(\min\{1, \lambda/|a|\})\big| \le C_a\sqrt{Lr}, \tag{104.3}$$

with an enlarged constant when Lr is not small. Bars on sets mean angular Lebesgue measure. No differentiability of q_r is required.

Proof. Rotate a to $(|a|, 0)$. The angular measure of $\{0 < |a|\cos\theta < \lambda\}$ is the displayed arcsine expression. The symmetric difference of the two threshold sets lies in

$$\{|a|\cos\theta| \le Lr\} \ \cup \ \{\big||a|\cos\theta - \lambda\big| \le Lr\}.$$

For every real b and $0 < \eta \le 1$, the set $\{|\cos\theta - b| \le \eta\}$ has measure at most $C\sqrt{\eta}$. To see this, split into the arcs within distance $2\sqrt{\eta}$ of 0 and π, whose total length is $O(\sqrt{\eta})$, and their complement, where $|\sin\theta| \ge c\sqrt{\eta}$. On each monotone complementary arc the inverse function theorem bounds the length by $C\eta/\sqrt{\eta}$. If the specified level is outside $[-1, 1]$, the same bound or the empty set applies. Rescale by $|a|$ and bound the two pieces of the symmetric difference. This also proves uniformity at the transition value $\lambda = |a|$, where a linear error bound would not follow from this argument. □

Let $b_{z,s}$ denote the roof density of the original local source with the additional restriction $0 < g < s$. Its neighborhood is chosen below as a full small Morse disk, restricted to the physical side. This density is not renormalized.

Theorem 104.2 (The simple seam–minimum profile)**.** *There are $h_z > 0$, $s_z > 0$ and $C_z < \infty$ such that, for every $0 < s < s_z$ and almost every $0 < h < h_z$,*

$$\left| b_{z,s}(t_z + h) - \frac{J_z}{\pi}\arcsin\!\left(\min\left\{1, \frac{s}{\kappa_z\sqrt{2h}}\right\}\right)\right| \le C_z J_z h^{1/4}. \tag{104.4}$$

The error constant does not depend on $s/h^{1/2}$, and so includes both sides of the transition $h = s^2/(2\kappa_z^2)$. For every fixed $s > 0$ in this range,

$$\lim_{h\downarrow 0} b_{z,s}(t_z + h) = J_z/2 \tag{104.5}$$

for the representative defined by coarea. The restricted density vanishes below t_z within the chosen neighborhood. The constants are local branch constants; no count-uniform bounds for h_z, s_z, C_z are asserted.

Proof. Translate z to the origin in endpoint arclengths. The local action is smooth with $dF(0) = 0$ and positive Hessian H. A direct Morse coordinate construction is useful here. Write

$$F(x) - t_z = \tfrac{1}{2}x^{\mathsf{T}}Q(x)x, \qquad Q(x) = 2\int_0^1 (1-t)\nabla^2 F(tx)\,dt.$$

The positive square root of $Q(x)$ is smooth near zero. The map $y = Q(x)^{1/2}x$ has derivative $H^{1/2}$ at zero, so it is a local diffeomorphism with inverse $x = \Phi(y)$. It gives exactly $F(\Phi(y)) = t_z + |y|^2/2$. Choose a ball $|y| < r_z$ whose inverse lies in U, and put $h_z = r_z^2/2$, reducing it when necessary. This is the local neighborhood in the definition of $b_{z,s}$.

In these coordinates the original density is $\widetilde{w}(y) = w(\Phi(y))|\det D\Phi(y)|$, with $\widetilde{w}(0) = w_z/\sqrt{\det H} = J_z/(2\pi)$. Taylor's formula gives, uniformly in θ,

$$g(\Phi(r\omega_\theta))/r = a_z \cdot \omega_\theta + O_z(r), \qquad \widetilde{w}(r\omega_\theta) = J_z/(2\pi) + O_z(J_z r).$$

The constants are relative to the fixed positive J_z; this places no lower bound on J_z uniformly in the count. Polar coarea, including the Jacobian $r\,dr\,d\theta$, gives the exact identity, with $r=\sqrt{2h}$,

$$b_{z,s}(t_z+h)=\int_0^{2\pi}\widetilde{w}(r\omega_\theta)\mathbf{1}_{\{0<g(\Phi(r\omega_\theta))<s\}}\,d\theta. \tag{104.6}$$

There is no additional factor r^{-1}: it cancels against the radial Jacobian under $h=r^2/2$. Apply Lemma 104.1 with $\lambda=s/r$. Its error is $O_z(r^{1/2})$, and the weight error is $O_z(J_zr)$. Since $r^{1/2}=2^{1/4}h^{1/4}$, this proves (104.4). The angular formula is an actual density representative. At fixed s, its leading angular measure tends to π, proving (104.5). Positivity of the quadratic normal form gives the assertion below t_z. □

104.2. **The unchanged smooth first-clearance source.** Let $q_0=47/53$ and $e'_j=q_0^{\min(j,m-1-j)/4}$. On the fixed local neighborhood all the other physical margins are strictly positive. For each fixed finite word, by reducing both the neighborhood and $\varepsilon_z>0$, all their factors in $H^{\varepsilon}_{m,R}$ are therefore one when $0<\varepsilon<\varepsilon_z$. The sole clearance factor equals zero for $g\le\varepsilon e'_j$ and one for $g\ge 2\varepsilon e'_j$, by the inherited guard convention. The selected factor is consequently the first physical defect wherever the local defect source is nonzero. No earlier factor or later incidence is suppressed by a change of definition.

Corollary 104.3 (Height of the original local defect)**.** *Put $s=\varepsilon e'_j$. Let $b_z^{\varepsilon,\mathrm{clr}}$ be the density of the original first-clearance source restricted to the local Morse disk above. For $0<\varepsilon<\varepsilon_z$ and $0<h<h_z$, almost everywhere,*

$$\begin{aligned}\frac{J_z}{\pi}\operatorname{arcsinmin}\left\{1,\frac{s}{\kappa_z\sqrt{2h}}\right\}-C_zJ_zh^{1/4}&\le b_z^{\varepsilon,\mathrm{clr}}(t_z+h)\\&\le\frac{J_z}{\pi}\operatorname{arcsinmin}\left\{1,\frac{2s}{\kappa_z\sqrt{2h}}\right\}+C_zJ_zh^{1/4}.\end{aligned} \tag{104.7}$$

In particular the one-sided limit is $J_z/2$. On $h\ge 8s^2/\kappa_z^2$, the explicit leading upper term is at most $CJ_zs/(\kappa_z\sqrt{h})$. For a bounded measurable insertion $|w|\le M$, the variation density is bounded by M times the same positive local source.

Proof. On the original physical half-neighborhood,

$$\mathbf{1}_{\{0<g<s\}}\le 1-H^{\varepsilon}_{m,R}\le\mathbf{1}_{\{0<g<2s\}}.$$

Apply positive pushforward and Theorem 104.2 to the two bounds; this proves the inequalities and their common one-sided limit. If $h\ge 8s^2/\kappa_z^2$, the upper arcsine argument is at most $1/2$, and $\arcsin x\le 2x$ there. Variation domination is performed on the original source before pushforward. □

104.3. **A positive weighted-cluster bound.** For fixed exact half-word labels (n,k,m,R), let $\mathcal{S}$ be a finite collection of the simple seam–minimum germs just described. Use pairwise disjoint physical source neighborhoods; distinct word copies are already disjoint. Each germ is counted once, with its original responsible disk. Choose $0<r_z<h_z$ for each $z\in\mathcal{S}$. Restrict its source further to $0<F-t_z<r_z$. For $\varepsilon<\min_z\varepsilon_z$, let $b^{\varepsilon,\mathcal{S}}$ be the sum of these restrictions. The complete source then has the exact positive partition

$$b^{\varepsilon,\mathrm{clr}}=b^{\varepsilon,\mathcal{S}}+b^{\varepsilon,\text{outside }\mathcal{S}},\qquad b^{\varepsilon,\text{outside }\mathcal{S}}\ge 0. \tag{104.8}$$

The second term includes every caustic point not assigned to these neighborhoods, rather than only the transverse points.

Theorem 104.4 (Critical-cluster control inside the selected caustic)**.** *For almost every t,*

$$b^{\varepsilon,\mathcal{S}}(t) \le \sum_{z\in\mathcal{S}} J_z \mathbf{1}_{\{0<t-t_z<r_z\}} \left[\frac{1}{\pi} \operatorname{arcsinmin}\left\{1, \frac{2\varepsilon e'_{j_z}}{\kappa_z\sqrt{2(t-t_z)}}\right\} + C_z(t-t_z)^{1/4}\right]. \tag{104.9}$$

Consequently, if $r=\max_z r_z$ and $\delta=\max_z C_z r_z^{1/4}$, the positive atomic measure of (103.10)*, now formed from the specified physical-side branch data, satisfies*

$$b^{\varepsilon,\mathcal{S}}(t) \le \frac{1/2+\delta}{Rc_*}\mathfrak{T}_{n,k,m,R}([t-r,t]). \tag{104.10}$$

These estimates retain coincident critical values and contain no factor equal to the number of words or exact labels.

Proof. Sum the upper bound of Corollary 104.3 on the disjoint source neighborhoods. This proves (104.9). The arcsine divided by π is at most $1/2$, and the other term is at most δ. Every contributing critical value lies in $[t-r,t]$. Substitute (103.10) to obtain (104.10). Addition of positive measures at an equal value is exactly what the coarea sum requires. □

Corollary 104.5 (An ordered transfer for a verified simple-seam family)**.** *For each fixed reconstruction band B, suppose a chosen family of the preceding neighborhoods is valid for $\varepsilon(B)=A_0B^{-1/12}$ at every sufficiently large count, uniformly in the original exact labels and radius. Suppose its radii satisfy $\sup_z r_z \le r_m \downarrow 0$ and $\sup_z C_z r_z^{1/4}\to 0$, uniformly. If its positive reversible measures satisfy*

$$\lim_{h\downarrow 0}\limsup_{m\to\infty}\sup_{R,n,k,t} m^2\mathfrak{T}_{B,n,k,m,R}([t-h,t+h])=0, \tag{104.11}$$

then $\sup_{R,n,k} m^2\|b^{\varepsilon(B),\mathcal{S}}\|_\infty\to 0$ for that fixed B. The conclusion is also uniform over all insertions of a fixed bounded modulus.

Proof. For fixed $h>0$, eventually $r_m<h$. Equation (104.10) bounds the collision limsup by a fixed multiple of that in (104.11). Only after taking the collision limsup let $h\downarrow 0$. Positive source domination gives the insertion assertion. The band has remained fixed throughout. □

The hypotheses of this last corollary are stated as inputs, not as properties verified here for all Lorentz caustics. In particular, a fixed B does not automatically satisfy the local bound $\varepsilon(B)<\varepsilon_z$ uniformly as m increases. The trace concentration and this neighborhood coverage both require additional dynamical estimates. Multiple competing seams, vanishing selected incidence, noncritical-roof tangencies, and section-boundary junctions remain in the positive complement of (104.8). The first-incidence source is unchanged. Substituting this partition into Theorem 181.4 preserves P, the signed canonical arithmetic reference G, and both exact labels. It does not yet remove that complement or establish the full pointwise law.

Proof route and comparison with the preceding estimates. The direct dependency route, within the compiled manuscript, is

endpoint action and contact determinant (§§167, 97)

⇓

reversible weight (Theorem 103.2)

⇓

physical-side Morse coarea (Theorem 104.2)

⇓

exact-label positive cluster (Theorem 104.4).

The geometry of the image witness is supplied independently by Lemma 178.1. Modules 131–136, including their finite-type and buffered caustic estimates, remain intact. The new profile controls a stated roof-critical germ within the tube, where a pivot denominator is unavailable; it does not improve their full-source count budgets by relabelling them. The arcsine profile and the half coefficient are elementary consequences of quadratic coarea. Their role here is to identify the actual physical coefficient and its positive trace sum, rather than to claim a new general Morse or Hill theorem.

105. Affine angular profiles with every exact section decision

The preceding simple profile can be extended without assuming that only one clearance seam is active, or that the critical point avoids a section boundary. Retaining affine, rather than only homogeneous, first-order margins is important: it permits the physical guard width to be fixed while a small positive margin at the center varies. We obtain a local formula for the entire vector of exact-label densities. All estimates in this section are at a fixed word and finite count.

105.1. **A width-uniform comparison for smooth guards.** Write Θ for the original nondecreasing smooth guard, zero on $(-\infty, 1]$ and one on $[2, \infty)$. Let $\omega = \omega_\theta$.

Lemma 105.1 (Affine thresholds and bounded-variation guards)**.** *Suppose*

$$f(r\omega) = \beta + ra \cdot \omega + E_r(\theta), \qquad a \neq 0, \quad |E_r| \le Lr^2.$$

Put $f_r^{\mathrm{a}}(\theta) = \beta + ra \cdot \omega$. *Uniformly in* $\beta \in \mathbb{R}$ *and every* $s > 0$,

$$\int_0^{2\pi} |\Theta(f(r\omega)/s) - \Theta(f_r^{\mathrm{a}}(\theta)/s)|\, d\theta \le C_a\sqrt{Lr},$$

$$|\{\theta : \mathbf{1}_{f(r\omega)>0} \neq \mathbf{1}_{f_r^{\mathrm{a}}(\theta)>0}\}| \le C_a\sqrt{Lr}. \tag{105.1}$$

The same bound, summed over the scalar inputs, holds for finite products and minima of such guard values. It also bounds the measure on which a finite Boolean formula in the sign bits changes.

Proof. For every real threshold t, disagreement between $f > t$ and $f_r^{\mathrm{a}} > t$ implies $|a\cdot\omega - (t-\beta)/r| \le Lr$. The circle estimate in the proof of Lemma 104.1 is uniform in this translated threshold. This proves the sign assertion and a uniform bound for all threshold disagreements. Since Θ has total variation one, its Stieltjes representation is $\Theta(x) = \int \mathbf{1}_{\{x>v\}}\, d\Theta(v)$, with immaterial endpoint conventions. Tonelli and the threshold bound give the first inequality, including when s/r tends to zero or infinity. For factors in $[0, 1]$, the difference of products, or of minima, is at most the sum of the individual differences. A Boolean formula can change only if some input bit changes. Apply the union bound to these input events, not to the number of output labels. □

105.2. **The finite physical margin description.** Let z be a normal-to-normal critical point in the physical-side closure of a selected word. Assume every selected incidence is positive; there may be several simultaneous competing-hit seams and arbitrary section-boundary coincidences. Use the smooth endpoint continuation only to choose the Morse coordinates y of the preceding section. Choose $\chi_z > 0$ and a sufficiently small Morse disk on which all selected incidences exceed χ_z. Fix

$$0 < \varepsilon < \min\{\chi_z/4, s_0/8\}. \tag{105.2}$$

Every incidence guard is then one on the whole disk. All physical defect source in this disk is therefore the sum of the original first-clearance pieces, with their original ordering unchanged.

There is a finite list g_i of smooth near-clearance margins, grouped by their original flight $j(i)$, and a finite list q_l of smooth section-edge margins, with the following properties. A margin g_i is the distance to its selected nonincident disk minus R, read at image collision $j(i) + 1$, with the

perpendicular foot inside that flight. Candidates which are uniformly too far to affect a guard in (105.2) are omitted. Each flight guard is then

$$U_j(y) = \min_{i:j(i)=j} \Theta(g_i(\Phi(y))/(\varepsilon e_j')), \qquad \min \varnothing = 1. \tag{105.3}$$

The full physical guard on the disk is $\prod_j U_j$. The margins q_l are the rectangle edge functions for each original section membership test, composed with its actual selected collision branch. The section decisions at $0, \dots, m-1$ define the occupation, and terminal membership at m is imposed separately.

Lemma 105.2 (Nonzero scalar gradients and exact Boolean labels)**.** *The lists may be chosen so that $D(g_i \circ \Phi)(0) \neq 0$ and $D(q_l \circ \Phi)(0) \neq 0$ for every retained margin. The physical word and the original events $\Pi_{n,k,m,R}$ are described on the disk by a finite Boolean formula in their signs, with the fixed inactive conditions retained as constants. Let its indicator be $B_{n,k}$. At every input sign pattern these indicators can be defined by the original visit-count rule so that*

$$\sum_{n=1}^{m} \sum_{k} B_{n,k} \leq 1. \tag{105.4}$$

The initial and terminal section tests are included in $B_{n,k}$; no normalization is encoded in it.

Proof. A disk whose clearance can be below s_0 has its foot strictly inside the flight, since the endpoint distances to a nonincident disk are at least $\ell_0 > s_0$. The finite-horizon candidate list is fixed at each flight. On its near-clearance envelope, $\partial_\varphi Z_d = L_d \geq \ell_* > 0$ by Lemma 178.1. Its sign is fixed locally when $|Z_d|$ is near $R > 0$. Recentring at the image collision and passing to endpoint and Morse coordinates are local diffeomorphisms. This proves the nonzero derivative for every needed g_i. Candidates with clearance at least s_0 at z remain above $s_0/2$ after reducing the disk, and are inactive for (105.2). Candidates with a foot outside the segment likewise cannot enter the near-clearance range on a sufficiently small disk. These observations justify the finite list and (105.3); monotonicity of Θ allows it to commute with the minimum of the actual clearance distances.

In the fixed section charts an edge function has nonzero coordinate gradient. Its composition with a regular collision branch still has nonzero gradient. Rectangle intersections and unions, followed by the finite visit-count rule, give the claimed Boolean description. First-hit exclusions on the selected simple-root word are exactly the relevant positive clearance signs near the disk; other exclusions are strict. All quantities are evaluated on that selected branch before imposing its physical-side inequalities.

For any sign pattern, first evaluate each collision's section bit by the same rectangle formula. If either endpoint is outside the section, assign no exact return label. Otherwise the sum of bits at $0, \dots, m-1$ determines a unique n; the selected word determines k. Assign precisely that label when the physical sign conditions also hold. This construction is meaningful even for an affine sign pattern not realized by an actual trajectory, and proves (105.4). It assigns no measure to such a pattern. □

Put $r = \sqrt{2h}$ and retain the constant terms in

$$\begin{aligned} G_{i,r}(\theta) &= g_i(z) + rD(g_i \circ \Phi)(0) \cdot \omega_\theta, \\ Q_{l,r}(\theta) &= q_l(z) + rD(q_l \circ \Phi)(0) \cdot \omega_\theta. \end{aligned} \tag{105.5}$$

Let $B^{\mathrm{a}}_{n,k}(r,\theta)$ be the same Boolean formula evaluated on these affine signs. Define

$$U^{\mathrm{a}}_j(\varepsilon,r,\theta)=\min_{i:j(i)=j}\Theta(G_{i,r}(\theta)/(\varepsilon e'_j)),$$

$$\mathcal{A}_{z,n,k}(\varepsilon,h)=\int_0^{2\pi}B^{\mathrm{a}}_{n,k}(r,\theta)\left(1-\prod_j U^{\mathrm{a}}_j(\varepsilon,r,\theta)\right)d\theta. \tag{105.6}$$

In particular $0\le\sum_{n,k}\mathcal{A}_{z,n,k}\le 2\pi$. These are explicit one-dimensional angular integrals of the physical first jets and the inherited guard. They are not billiard probabilities.

Theorem 105.3 (The full local exact-label caustic profile)**.** *Let $b^{\varepsilon,\mathrm{clr}}_{z,n,k}$ be the original first-clearance source restricted to the Morse disk at z, with its exact labels. For suitable $h_z>0$ and $C_z<\infty$, uniformly for* (105.2) *and almost every $0<h<h_z$,*

$$\sum_{n=1}^{m}\sum_k\left|b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z+h)-\frac{J_z}{2\pi}\mathcal{A}_{z,n,k}(\varepsilon,h)\right|\le C_zJ_zh^{1/4}. \tag{105.7}$$

No clearance or section margin at z is required to be positive. There is no uniform lower bound assumed on distinct margin gradients or on angles between them. The theorem requires their individual nonvanishing and keeps the resulting local constant C_z. There is no factor for the number of output labels in this estimate. More explicitly, take the Morse radius at most one, put $W_z=\|D\widetilde{w}\|_\infty/\widetilde{w}(0)$, and for every scalar margin f in the two lists put $A_f=|D(f\circ\Phi)(0)|>0$ and $L_f=\frac12\|D^2(f\circ\Phi)\|_\infty$ on the disk. An absolute constant C permits the choice

$$C_z=C\left(1+W_z+\sum_f\sqrt{L_f/A_f}\right). \tag{105.8}$$

Thus the error uses explicitly stated finite-jet data, not an unrecorded separation exponent.

Proof. Use polar coarea as in (104.6), with the exact integrand

$$\widetilde{w}(r\omega)B_{n,k}(\Phi(r\omega))\left(1-\prod_j U_j(r\omega)\right).$$

Taylor remainder bounds for the finite scalar lists are $O_z(r^2)$. Lemma 105.1, summed over these lists, bounds the integral of the guard-product difference by $C_zr^{1/2}$, uniformly in every width $\varepsilon e'_j$. It also bounds by $C_zr^{1/2}$ the angular set on which any Boolean input sign differs from its affine counterpart. Outside this set every exact-label indicator agrees. On this set both the original indicator vector and the affine one are either zero or a single unit vector by (105.4). Their ℓ^1 distance is therefore at most two, regardless of the number of labels. This bounds the summed indicator error by $2C_zr^{1/2}$.

Finally, $\widetilde{w}(r\omega)=J_z/(2\pi)+O_z(J_zr)$. The summed output indicators are at most one, so its contribution to the summed density error is $O_z(J_zr)$, not that quantity times the number of labels. Combine the three estimates and use $r^{1/2}=2^{1/4}h^{1/4}$. Every source restriction used in this calculation is an original physical and section restriction. The affine formulas approximate its angular integral; they do not alter its itinerary or its pushforward measure. The circle threshold estimate costs $C\sqrt{L_f/A_f}\,r^{1/2}$ for each scalar input. The relative weight error costs CW_zr. Summing and using $r\le1$ gives (105.8) after an absolute enlargement of C. □

Corollary 105.4 (Angular trace at a multiple seam)**.** *Suppose at least one clearance margin vanishes at z and keep ε fixed in* (105.2)*. Replace each zero margin in the Boolean label formula*

by the sign of its linear part, and each nonzero margin by its constant sign. Let $\mathcal{C}_{z,n,k}$ be the resulting angular sets. Then

$$\lim_{h\downarrow 0} b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z+h) = \frac{J_z}{2\pi}|\mathcal{C}_{z,n,k}|, \qquad \sum_{n,k}|\mathcal{C}_{z,n,k}| \le 2\pi. \tag{105.9}$$

The sets may have zero angular measure; no transverse intersection between different active boundaries is assumed. For a sole active clearance seam with strict section decisions this recovers $J_z/2$.

Proof. Each nonzero linear form vanishes at only two angular directions. Off their finite union the affine signs converge to the stated signs. On an allowed physical direction an active clearance margin is $O(r)$; its guard tends to zero at fixed ε. Hence $1-\prod_j U_j^{\mathrm{a}}$ tends to one on every allowed direction. Dominated convergence in (105.6) and (105.7) prove the formula. The angular partition bound follows from the same unique occupation rule. Coincident linear forms can make some angular sets empty; the proof never divides by a determinant formed from two such gradients. □

Uniformity, source coverage and the next dynamical estimate. The formula removes the single-seam and strict-section restrictions from the local roof-critical calculation. It also avoids imposing ε below each nonzero clearance margin: the offsets $g_i(z)$ in (105.5) retain such margins exactly at first order. On a compact family of these coordinate neighborhoods, with common derivative bounds and positive lower bounds for the stated individual gradients and selected incidences, h_z and C_z can be chosen uniformly by the proof. No such bounds are asserted for all words as the collision count grows.

For disjoint original-source neighborhoods, the vector estimates may be summed with their actual positive weights J_z. In particular a local roof density is bounded by $(1+o_z(1))J_z$, and each J_z is the reversible square-root weight of (103.5). Thus the remaining long-count task is a weighted critical-cluster estimate together with uniform coverage of the neighborhoods, not a conversion of their small mass into a height estimate. This observation is a reduction for the described roof-critical source, not a proof of either ordered physical height limit. Regions with selected grazing contacts or noncritical-roof rank degeneracy remain in the original positive clearance complement. The complete incidence source, the canonical arithmetic kernel and the fixed-band order of limits are unchanged.

106. Physical pressure domination and complex Gaussian estimates

We give the scalar details used in the pressure-contact argument before turning to a correlated realization with a continuous roof. This section uses no collision norm. In particular, the pressure comparison is an inequality between original scalar integrals, not an order relation on an anisotropic space.

Lemma 106.1 (Taking roots of a visible physical branch)**.** *Let X be a centered bounded observable of a probability-preserving system. Suppose, at a fixed real tilt t,*

$$1 \le \int e^{-t\cdot S_m X}\, d\nu \le C_0 e^{m\mathscr{P}(t)}.$$

Suppose bounded physical endpoint tests a, d give, at $z = a_0 + it$,

$$I_m(z) = A(z)e^{m\psi(z)} + E_m(z), \qquad |A(z)| \ge a_* > 0, \qquad |E_m(z)| \le C_1\rho^m, \quad \rho < 1,$$

where $I_m(z) = \int a e^{iz\cdot S_m X}(d\circ T^m)d\nu$ and the logarithm is for this centered *observable. Then*

$$\mathscr{P}(t) \ge 0, \qquad \operatorname{Re}\psi(a_0+it) \le \mathscr{P}(t). \tag{106.1}$$

The conclusion is uniform on any finite cover with common bounds and a common tilt ball. The tests may change between members of that cover.

Proof. The first inequality in the hypothesis follows from centering and Jensen whenever only the upper bound is given. Taking logarithms, dividing by m, and passing to the limit gives $\mathscr{P}(t) \ge 0$. Physical domination and the reverse triangle inequality give

$$a_* e^{m \operatorname{Re}\psi(z)} \le C_0 \|a\|_\infty \|d\|_\infty e^{m\mathscr{P}(t)} + C_1 \rho^m.$$

For nonnegative u_m, v_m, $(u_m + v_m)^{1/m}$ has limsup at most the larger of their root limsups, since $u_m + v_m \le 2\max(u_m, v_m)$. Thus $e^{\operatorname{Re}\psi(z)} \le \max(e^{\mathscr{P}(t)}, \rho) = e^{\mathscr{P}(t)}$. The lower amplitude bound is used before taking roots; the observation amplitude of another pairing need not be nonzero. □

Lemma 106.2 (Finite witnesses near a compact zero set)**.** *Let finitely many compact branch supports carry continuous nonnegative dampings κ_j. Suppose every point of their zero sets has an open neighborhood on which a physical expansion as above holds. Then some $\delta_0 > 0$ has the following property: every retained branch point with $\kappa_j \le \delta_0$ is in one of finitely many such neighborhoods. All constants and a smaller positive tilt radius can be chosen in common.*

Proof. Choose a finite subcover of the zero set. Its complement in the compact disjoint union of branch supports is compact and contains no zero, so κ has a positive minimum there, unless the complement is empty. Choose δ_0 below that minimum. Take the maximum of finitely many upper bounds, the minimum of the positive amplitude lower bounds, and the minimum of smaller tilt radii. No continuous global choice of witnesses is required. □

Lemma 106.3 (Accretive Gaussian differentiation)**.** *Let $\mathcal{H}$ be a compact family of complex symmetric matrices satisfying $\operatorname{Re} H \ge \lambda I$ for some $\lambda > 0$. The Gaussian*

$$G_H(Z) = (2\pi)^{-d/2} (\det H)^{-1/2} e^{-Z^{\mathsf{T}} H^{-1} Z/2}$$

uses the determinant square root continued from the real positive matrices in the convex accretive domain. There are $C, b > 0$ such that, for real Z and complex symmetric E,

$$|G_H(Z)| + |\nabla_Z G_H(Z)| + \sup_{\|E\| \le 1} |D_H G_H(Z)[E]| \le C e^{-b|Z|^2}. \tag{106.2}$$

The same assertion holds on all line segments from $\mathcal{H}$ to any compact family of real positive matrices with a common lower bound.

Proof. Symmetry identifies the adjoint with the entrywise conjugate. The Hermitian identity is

$$\operatorname{Re}(H^{-1}) = (H^*)^{-1} (\operatorname{Re} H) H^{-1} = \overline{H}^{-1} (\operatorname{Re} H) H^{-1}. \tag{106.3}$$

Indeed multiplying $(H + H^*)/2$ on these two sides gives $((H^*)^{-1} + H^{-1})/2$. If $\|H\| \le M$, its quadratic form is at least $\lambda M^{-2} |Z|^2$. In particular every matrix in the accretive domain is invertible. The domain is convex, so the analytic logarithm of its nonvanishing determinant is fixed by its real-positive value; this defines the indicated square root unambiguously.

Differentiating gives

$$\nabla_Z G_H = -H^{-1} Z G_H, \qquad D_H G_H[E] = \tfrac{1}{2} G_H \{ Z^{\mathsf{T}} H^{-1} E H^{-1} Z - \operatorname{tr}(H^{-1} E) \}.$$

The inverse and determinant factors are uniformly bounded on the compact family. Absorb $1 + |Z| + |Z|^2$ by reducing the Gaussian exponent. Line segments retain the positive real-part bound and form a compact set, so the same argument applies there. □

The explicit diagonal and the two covariance moduli. In Section 107, h_j is the norm of the real quadratic pressure gap, whereas $\omega(\delta)$ is the supremum of the full complex Hessian mismatch. The latter controls the matrix segment in Lemma 106.3. Taking

$$\delta_m = \min\{\delta_0/2, m^{-1/2}\}$$

in (107.9) makes its right side tend to zero: $\omega(\delta_m) \to 0$, $m^{-1/6} \to 0$, and eventually $e^{-m\delta_m/4} = e^{-\sqrt{m}/4}$. This is an explicit diagonal in a proved scalar inequality; it is not a substitution of a growing band in a fixed-band spectral theorem. No rate is assigned to ω.

Source normalization and exact labels. The first $1/c$ in Φ is the restricted collision-source normalization. The second, in $\alpha = \Phi/c$, is the product of the count prefactor and the covariance determinant change: $m^{-2} \sim c^2 n^{-2}$ and $g_{\Omega_R}(\sqrt{c}L_R V) = c^{-3} g_{D_R}(V)$. A torus-frequency lift changes the phase by 2π times an exact integer label, hence leaves it unchanged. None of these operations erases a zero arithmetic class. The reference is signed until its positive part is explicitly normalized. In Proposition 108.2, the true density is extended by zero beyond its roof support and the local integral is uniform over every translated unit interval. That argument controls negative reference heights, not positive physical-source heights.

107. Flat pressure contact at a disappearing arithmetic peak

The drift bound of Theorem 114.2 makes the moving Gaussian centers tight. It does not make them coincide on the diffusive scale. There is a further piece of information at an exact resonance: its covariance is the physical covariance. Keeping this second-order contact in the pressure comparison gives a strictly stronger conclusion. The argument is a scalar spectral principle; it does not depend on a billiard norm once its physical inputs have been verified.

107.1. **A visible-branch principle.** Let Λ be a compact parameter space and let $(M_\lambda, \nu_\lambda, T_\lambda)$ be probability-preserving systems. Let $X_\lambda : M_\lambda \to \mathbb{R}^d$ be centered bounded observables, with a common supremum bound. Write $S_m X_\lambda$ for their actual orbit sums. The systems may be invertible or noninvertible. The following hypotheses concern finitely many compact branch supports; all neighborhoods and bounds may first be chosen locally and then made uniform by a finite cover.

Pressure. On a real ball $|t| < t_0$ there is a real C^3 function $\mathscr{P}_\lambda$ with uniform derivatives through order three, such that

$$\int e^{-t\cdot S_m X_\lambda}\, d\nu_\lambda \le C e^{m\mathscr{P}_\lambda(t)}, \qquad \mathscr{P}_\lambda(0) = 0, \quad D\mathscr{P}_\lambda(0) = 0, \quad D^2\mathscr{P}_\lambda(0) = \Sigma_\lambda. \tag{107.1}$$

The real symmetric matrices Σ_λ are continuous and uniformly positive definite. This hypothesis is an exponential moment of the original probability, not an order relation in a distribution space.

Branches and physical visibility. For each branch there is a real frequency $a_j(\lambda)$ and a centered holomorphic logarithm $\psi_{j,\lambda}$ near that frequency. Its first three frequency derivatives are uniformly bounded and continuous in the parameter. At the real maximum put

$$\operatorname{Re}\psi_j(a_j) = -\kappa_j \le 0, \quad D\psi_j(a_j) = iv_j, \quad -D^2\psi_j(a_j) = H_j, \qquad v_j \in \mathbb{R}^d. \tag{107.2}$$

The peak data a_j, κ_j, v_j, H_j are continuous on their compact supports. The complex symmetric H_j are bounded and have uniformly positive real part. Near every point with $\kappa_j = 0$ there are bounded physical endpoint tests a, d for which, on a fixed complex neighborhood,

$$\begin{gathered}\int a(x) e^{iz\cdot S_m X_\lambda(x)} d(T_\lambda^m x)\, d\nu_\lambda(x) = A_j^{a,d}(\lambda, z) e^{m\psi_j(z)} + E_{j,m}(\lambda, z), \\ |A_j^{a,d}| \ge a_* > 0, \qquad |E_{j,m}| \le C\rho^m.\end{gathered} \tag{107.3}$$

where $\rho < 1$. The tests may differ between neighborhoods. They need not equal the endpoint tests defining a particular observation. In particular its observation amplitude is allowed to vanish.

Second-order contact. On the compact zero-damping locus one has

$$H_j(\lambda) = \Sigma_\lambda. \tag{107.4}$$

This is a separate spectral hypothesis. It is not inferred from continuity of v_j or from nonnegativity of the pressure alone. Set

$$h_j = \|\Sigma_\lambda - \operatorname{Re} H_j\|, \qquad \omega(\delta) = \sup_{j,\lambda:\, \kappa_j \le \delta} \|H_j - \Sigma_\lambda\|,$$

where a supremum over an empty set is zero. Continuity, compactness and (107.4) imply $\omega(\delta) \to 0$ as $\delta \downarrow 0$. No rate in δ is assumed.

Theorem 107.1 (Drift at a flat pressure contact)**.** *Under these hypotheses there are uniform constants $C, \kappa_0 > 0$ such that, for $0 < \kappa_j \le \kappa_0$,*

$$|v_j| \le C(\sqrt{\kappa_j h_j} + \kappa_j^{2/3}), \qquad |v_j|^2 \le C\kappa_j(h_j + \kappa_j^{1/3}). \tag{107.5}$$

At $\kappa_j = 0$ one has $v_j = 0$. In particular

$$\lim_{\delta \downarrow 0} \sup_{0 < \kappa_j \le \delta} \frac{|v_j|^2}{\kappa_j} = 0. \tag{107.6}$$

For each fixed $L < \infty$, $\sup_{m\kappa_j \le L} \sqrt{m}\,|v_j| \to 0$ as $m \to \infty$. The conclusions remain valid when a section or observation amplitude vanishes at the resonance.

Proof. Take the physical witnesses in (107.3). For $z = a_j + it$ the modulus of their orbit pairing is at most $\|a\|_\infty \|d\|_\infty \int e^{-t\cdot S_m X_\lambda} d\nu_\lambda$. Jensen's inequality in (107.1) gives $\mathscr{P}_\lambda(t) \ge 0$ by taking mth roots. The same operation in the witness expansion therefore proves

$$F_j(t) := \mathscr{P}_\lambda(t) - \operatorname{Re} \psi_j(a_j + it) \ge 0.$$

The exponentially small remainder disappears in this step. The amplitude lower bound is used before taking roots. Compactness of the zero-damping locus gives finitely many witness neighborhoods with one smaller tilt radius and common constants.

The frequency convention in (107.2) gives

$$F_j(t) = \kappa_j + v_j \cdot t + \tfrac{1}{2} t^{\mathsf{T}} (\Sigma_\lambda - \operatorname{Re} H_j) t + O(|t|^3). \tag{107.7}$$

In particular the quadratic term is a *difference* of covariances. At zero damping, nonnegativity of F_j and $F_j(0) = 0$ give $v_j = 0$.

For positive damping and $v_j \ne 0$, insert $t = -r v_j/|v_j|$ into (107.7). It follows that

$$|v_j| \le \kappa_j/r + \tfrac{1}{2} h_j r + C_3 r^2. \tag{107.8}$$

Choose

$$r = \min\{\kappa_j^{1/3}, \sqrt{\kappa_j/h_j}\},$$

with the second quantity interpreted as infinity when $h_j = 0$. For small κ_j this lies in the fixed tilt ball. If $h_j \le \kappa_j^{1/3}$, all three terms of (107.8) are bounded by $C\kappa_j^{2/3}$. Otherwise $r = \sqrt{\kappa_j/h_j}$, and $r^2 = \kappa_j/h_j \le \sqrt{\kappa_j h_j}$; all three terms are bounded by $C\sqrt{\kappa_j h_j}$. This proves the first bound, also when $v_j = 0$. Squaring proves the second.

All sufficiently small positive dampings lie in the finitely many witness neighborhoods: otherwise a compact subsequence would limit to an uncovered zero. Since $h_j \le \omega(\delta)$ when $\kappa_j \le \delta$, (107.6) follows. Finally

$$m|v_j|^2 \le CL\{\omega(L/m) + (L/m)^{1/3}\} \quad \text{when } m\kappa_j \le L,$$

which proves the diffusive assertion. The observation amplitude has not entered any part of the proof. □

107.2. **A common Gaussian shape.** For accretive complex symmetric H define

$$G_H^{(d)}(Z) = (2\pi)^{-d/2}(\det H)^{-1/2}e^{-Z^{\mathsf T}H^{-1}Z/2}, \qquad Z \in \mathbb{R}^d.$$

The determinant branch is the one continued from real positive matrices on the convex accretive domain. Suppose the coefficients $q_{j,m}(\lambda, y)$ obey $|q_{j,m}| \le Ce^{-m\kappa_j}$. Define

$$K_m(\lambda, y, Z) = \sum_j q_{j,m}(\lambda, y)G_{H_j}^{(d)}(Z - \sqrt{m}\, v_j), \qquad S_m(\lambda, y) = \sum_j q_{j,m}(\lambda, y).$$

This $S_m(\lambda, y)$ is a scalar coefficient, not an orbit sum.

Corollary 107.2 (Uniform Gaussian shape through a resonance change)**.** *There are $a, C, \delta_0 > 0$ such that, for $m \ge 2$ and $0 < \delta < \delta_0$,*

$$\sup_{\lambda,y,Z} e^{a|Z|^2}|K_m(\lambda, y, Z) - S_m(\lambda, y)g_{\Sigma_\lambda}(Z)| \le C\{\omega(\delta) + \sqrt{\omega(\delta)} + m^{-1/6} + e^{-m\delta/4}\}. \tag{107.9}$$

In particular its left side tends to zero. This is a statement about the evaluated peak sum; it is not a density theorem for an arbitrary physical source realizing the spectral hypotheses.

Proof. The weaker bound $|v_j|^2 \le C_D\kappa_j$ holds on all compact supports. Near zero it follows from Theorem 107.1 and boundedness of h_j; away from zero use the positive minimum of κ_j and boundedness of the drift.

On the compact accretive matrix family and its line segments to Σ_λ, Gaussian differentiation and completion of the square give uniform bounds

$$|\nabla_Z G_H^{(d)}(Z)| + \|D_H G_H^{(d)}(Z)\| \le Ce^{-b|Z|^2}$$

for some $b > 0$, after absorbing the polynomial factors. The same bound holds for the Gaussian itself. Along the segment in the matrix and in the shift $d_j = \sqrt{m}\, v_j$, the fundamental theorem of calculus gives

$$|G_{H_j}^{(d)}(Z - d_j) - g_{\Sigma_\lambda}(Z)| \le C(\|H_j - \Sigma_\lambda\| + |d_j|) \sup_{0\le s\le 1} e^{-b|Z-sd_j|^2}.$$

Decrease b so that $bC_D \le 1/2$. Using $|Z - sd_j|^2 \ge |Z|^2/2 - |d_j|^2$ shows that the product with $e^{-m\kappa_j}$ is bounded by

$$Ce^{-a|Z|^2}e^{-m\kappa_j/2}(\|H_j - \Sigma_\lambda\| + \sqrt{m}\,|v_j|). \tag{107.10}$$

For $\kappa_j \le \delta$, use (107.5). The term $\sqrt{m\kappa_j h_j}e^{-m\kappa_j/2}$ is at most $C\sqrt{\omega(\delta)}$. The cubic-remainder term is bounded by

$$\sqrt{m}\,\kappa_j^{2/3}e^{-m\kappa_j/2} = m^{-1/6}(m\kappa_j)^{2/3}e^{-m\kappa_j/2} \le Cm^{-1/6}.$$

The Hessian difference costs $C\omega(\delta)$. For $\kappa_j > \delta$, bounded Hessians and $\sqrt{m}|v_j| \le C\sqrt{m\kappa_j}$ bound (107.10) by $Ce^{-a|Z|^2}e^{-m\delta/4}$. Sum the finite family. For instance, $\delta = m^{-1/2}$ gives convergence. The unknown modulus ω prevents this choice from being advertised as a polynomial rate. □

107.3. **Verification on the occupation spectrum.**

Corollary 107.3 (Flat contact for the original Lorentz record)**.** *For the actual occupation peaks of* (160.1), (107.5) *holds with* $v_j = \mu_j - \mu_R$ *and* $\Sigma_R = \Omega_R$*. All constants are uniform for* $R \in [0.45, 0.47]$*, including changes of arithmetic type. No section residue is required to be nonzero.*

Proof. Use $X_R = \mathsf{f}_R - \mu_R$, including the actual occupation η_R, not the constant collision step. Lemma 114.1 supplies the pressure. Differentiation at zero of its full occupation splitting gives $D^2\mathscr{P}_R(0) = \Omega_R$, by the same Green–Kubo identification used in Section 151. Its third derivatives are uniformly bounded by strong frequency analyticity on a fixed ball. Theorem 114.2 constructs exactly the nonzero full-collision witness neighborhoods in (107.3), also where $B_j = 0$. Lemma 160.1 supplies scalar derivative continuity, and Lemma 160.2 supplies $H_j = \Omega_R$ on $\kappa_j = 0$ and the compact accretive bounds. These verify each hypothesis separately. Apply Theorem 107.1. No parameter derivative, new anisotropic space, or growing-band estimate is used. □

107.4. **An explicit singular hyperbolic realization.** The principle is not restricted to billiard collision operators. Here is a deterministic example with a finite resonance transition, for which all physical scalar inputs can be verified directly. On the three strips $i/3 \le x < (i+1)/3$, $i = 0, 1, 2$, of the unit square, define the baker transformation

$$\mathcal{B}(x, y) = (3x - i, (y + i)/3), \qquad f_\epsilon(x, y) = \begin{cases} 0 & i = 0, \\ 1 & i = 1, \\ 2 + \epsilon & i = 2. \end{cases} \tag{107.11}$$

Boundary conventions affect a null set only. The map preserves Lebesgue probability: each strip is mapped bijectively, with Jacobian one, to a disjoint horizontal strip. Its derivative on each piece is $\operatorname{diag}(3, 1/3)$. It is an invertible singular hyperbolic map, with a geometry different from a dispersing billiard.

Proposition 107.4 (A damped arithmetic peak for the baker map)**.** *For all sufficiently small real* ϵ*, the centered observable* $f_\epsilon - \mu_\epsilon$*,* $\mu_\epsilon = 1 + \epsilon/3$*, satisfies the visible-branch and pressure hypotheses near the resonance* 2π*. Its physical covariance is* $\Sigma_\epsilon = 2/3 + 2\epsilon/3 + 2\epsilon^2/9$*. The unique nearby real maximum* a_ϵ *has data*

$$a_\epsilon = 2\pi - \pi\epsilon + \tfrac{\pi}{3}\epsilon^2 + O(\epsilon^3), \tag{107.12}$$

$$\kappa_\epsilon = \tfrac{\pi^2}{9}\epsilon^2 + O(\epsilon^3), \qquad v_\epsilon = \tfrac{\pi^2}{18}\epsilon^3 + O(\epsilon^4), \tag{107.13}$$

$$H_\epsilon - \Sigma_\epsilon = \tfrac{2\pi i}{9}\epsilon + O(\epsilon^2). \tag{107.14}$$

In particular $\epsilon_m \to 0$ *with* $m\epsilon_m^2 \to t < \infty$ *gives* $m\kappa_{\epsilon_m} \to \pi^2 t/9$ *but* $\sqrt{m}\, v_{\epsilon_m} \to 0$*. The peak can retain a nonzero limiting damping factor without retaining a displaced Gaussian center.*

Proof. Prescribing the first m strip digits fixes an interval of x of length 3^{-m} and leaves y arbitrary. Thus these digits are independent and uniform under the invariant probability, as an exact property of this deterministic map. Put

$$\varphi_\epsilon(z) = \tfrac{1}{3}(1 + e^{iz} + e^{i(2+\epsilon)z}).$$

Its physical characteristic moment is exactly $\varphi_\epsilon(z)^m$. The centered branch is $e^{-iz\mu_\epsilon}\varphi_\epsilon(z)$; the witnesses are $a = d = 1$, their amplitude is one, and the remainder is zero. The centered real exponential moment therefore has pressure

$$\mathscr{P}_\epsilon(t) = \log(e^{t\mu_\epsilon}\varphi_\epsilon(it)).$$

Its first derivative vanishes and its Hessian is the stated variance. The same identity proves the moment hypothesis at every finite count. All complex derivatives are bounded on fixed neighborhoods of zero and 2π for small ϵ.

At $\epsilon = 0$ the logarithmic real curvature at 2π is $-2/3$. The analytic implicit function theorem gives a unique analytic nearby maximum. The triangle equality $|\varphi_\epsilon(a)| = 1$ requires $e^{ia} = e^{ia(2+\epsilon)} = 1$. On a small neighborhood of $(\epsilon, a) = (0, 2\pi)$ this holds only at that point. There the centered branch is its origin branch multiplied by a constant phase; hence $H_0 = \Sigma_0$. This verifies the contact condition without assuming it throughout the transition.

We give the expansion calculation. Write $a = 2\pi + s$ and let F take the values $0, 1, 2 + \epsilon$ with equal probabilities. The three residual phases are $\Theta = (0, s, 2s+2\pi\epsilon+\epsilon s)$. For $s = A\epsilon+B\epsilon^2+O(\epsilon^3)$, the first two coefficients of $\mathrm{Re}(\varphi'/\varphi)$ are

$$-\tfrac{2}{3}(A + \pi), \qquad -\tfrac{2}{3}A - \tfrac{2}{3}B - \tfrac{4}{9}\pi.$$

Their vanishing gives $A = -\pi$, $B = \pi/3$ and (107.12). The real logarithm to second order is $-\frac{1}{2}\operatorname{Var}\Theta$, giving $\kappa_\epsilon = \pi^2\epsilon^2/9 + O(\epsilon^3)$. For the imaginary logarithmic derivative, expansion of the finite exponential sum gives

$$\mathrm{Im}(\varphi'/\varphi) - \mathbb{E}F = -\tfrac{1}{2}\mathbb{E}[(\Theta - \mathbb{E}\Theta)^2(F - \mathbb{E}F)] + O(\epsilon^4).$$

At the maximum the expectation on the right is $-\pi^2\epsilon^3/9 + O(\epsilon^4)$; its second-order term is zero. Finally $-(\log\varphi)''$ equals $\operatorname{Var}F + i\mathbb{E}[(F - \mathbb{E}F)^2(\Theta - \mathbb{E}\Theta)] + O(\epsilon^2)$, and the expectation is $2\pi\epsilon/9 + O(\epsilon^2)$. These identities prove (107.13)–(107.14). Analyticity of the implicit maximum bounds the displayed remainders. The limiting assertions follow by substitution. □

Scope of the second realization. This example verifies the same pressure/contact theorem in a piecewise linear hyperbolic map by exact physical pairings, independently of the billiard imports. Its additive observable has finite support, so no continuous roof-density theorem is claimed for it. It is not substituted for the original Lorentz record or its incidence and clearance sources.

108. One physical Gaussian through changes of arithmetic type

We apply flat pressure contact to the same complete raw return law. The finite phase and damping sum remains necessary, but the moving means and complex Hessians can now be removed from its Gaussian factor uniformly in the radius. The resulting formula has the physical return covariance even when the arithmetic type changes with the observation count.

Keep all peak data and partition functions of (160.1)–(160.5). For the real target $y = (k_1, k_2, u, n)$ define

$$\Phi_{m,R}(y) = \frac{1}{c}\,\mathrm{Re}\sum_j e^{-ia_j(R)\cdot y} z_j(R)^m B_j(R), \tag{108.1}$$

$$\mathcal{L}^\circ_{m,R}(y) = \Phi_{m,R}(y) g_{\Omega_R}\left(\frac{y - m\mu_R}{\sqrt{m}}\right), \tag{108.2}$$

$$\alpha_{m,R}(k, n, u) = c^{-1}\Phi_{m,R}(k_1, k_2, u, n). \tag{108.3}$$

The factor $c = 91/(10000\pi)$ is fixed. Each scalar sum is real and uniformly bounded, but it need not be nonnegative at finite count. Every retained branch occurs with its original amplitude, damping, phase and partition. The roof phase $e^{-i(a_j)_3 u}$ is not discarded. Different real lifts of torus frequencies give the same expression at the exact integer labels.

108.1. **Uniform removal of moving Gaussian data.**

Theorem 108.1 (Common Gaussian shape of the arithmetic transition)**.** *There are $a > 0$ and a deterministic $\theta_m \to 0$ such that*

$$\sup_{R,y} e^{a|y-m\mu_R|^2/m} |\mathcal{L}_{m,R}(y) - \mathcal{L}^{\circ}_{m,R}(y)| \le \theta_m. \tag{108.4}$$

The supremum may be taken over every real $y \in \mathbb{R}^4$ for a fixed choice of frequency lifts; in the physical law its first, second and fourth coordinates have their exact integer values. One admissible bound, for all sufficiently large m, is

$$\theta_m = C\{\omega_*(m^{-1/2}) + \sqrt{\omega_*(m^{-1/2})} + m^{-1/6} + e^{-\sqrt{m}/4}\}, \tag{108.5}$$

where $\omega_*(\delta) = \sup_{j,R:\,\kappa_j \le \delta} \|H_j - \Omega_R\|$. The modulus is uniform over the whole radius family. In particular no polynomial rate in m is asserted.

Proof. In Corollary 107.2 take $d = 4$, $Z = (y - m\mu_R)/\sqrt{m}$, $v_j = \mu_j - \mu_R$ and

$$q_{j,m}(R, y) = c^{-1} e^{-ia_j(R)\cdot y} z_j(R)^m B_j(R).$$

The B_j are uniformly bounded and $|z_j|^m = e^{-m\kappa_j}$. Corollary 107.3 verifies the hypotheses. Taking real parts gives (108.4), and $\delta = m^{-1/2}$ gives (108.5). This uses only the fixed peak charts. It substitutes no growing roof band into a spectral estimate. □

Proposition 108.2 (Slow roof variation and asymptotic reference positivity)**.** *For the same exact integer labels $1 \le n \le m$, $k \in \mathbb{Z}^2$,*

$$\sup_{R,n,k,u} (\mathcal{L}^{\circ}_{m,R}(k_1, k_2, u, n))_- \to 0. \tag{108.6}$$

Consequently, for every fixed $M < \infty$,

$$\sup_{R,n,k,u:\,|y-m\mu_R| \le M\sqrt{m}} (\Phi_{m,R}(y))_- \to 0.$$

These assertions concern the reference, not an upper bound for the positive original density. The finite reference is still treated as signed.

Proof. Put $b_j = (a_j)_3$. At every zero of κ_j one has $b_j = 0$ by Lemma 160.2. Continuity and compactness therefore give

$$\beta(\delta) := \sup_{j,R:\,\kappa_j \le \delta} |b_j| \to 0.$$

Differentiating the finite sum, and using uniform Gaussian derivative bounds, shows

$$\sup_{R,y} |\partial_u \mathcal{L}^{\circ}_{m,R}(y)| \le C \left(m^{-1/2} + \sum_j |b_j| e^{-m\kappa_j} \right) =: d_m \to 0. \tag{108.7}$$

For example the sum is bounded by $C\{\beta(m^{-1/2}) + e^{-\sqrt{m}}\}$. The coefficients outside the Gaussian do not have to be independent of the roof for this argument.

Let $e_m \to 0$ bound the inherited complete local variation error on every unit roof interval from Theorem 180.1. For $P = m^2 p \ge 0$ and $G^{\circ} = \mathcal{L}^{\circ}$, the new kernel comparison gives

$$\int_u^{u+1} |P(v) - G^{\circ}(v)|\, dv \le e_m + \theta_m.$$

If $G^{\circ}(u) = -h < 0$, then $G^{\circ}(v) \le -h + d_m$ for $u \le v \le u + 1$. When $h > d_m$, positivity of P bounds the last integral below by $h - d_m$. Hence $(G^{\circ}(u))_- \le e_m + \theta_m + d_m$ at every roof; the case $h \le d_m$ is immediate. The density is extended by zero outside its physical roof support, as

in the local theorem. Finally the Gaussian g_{Ω_R} has a positive uniform lower bound on $|Z| \le M$, proving the assertion for Φ. No estimate of the positive part of $P - G^\circ$ is inferred. □

108.2. **The fixed-return covariance on the entire record space.** Retain $\mathcal{O}_n$, $d\zeta$ and $V_{n,R}$ from Section 114. Define the signed density

$$q^\circ_{n,R}(k,m,u) = n^{-2}\alpha_{m,R}(k,n,u)g_{D_R}(V_{n,R}(k,m,u)), \quad (k,m,u) \in \mathcal{O}_n. \tag{108.8}$$

All collision counts $m \ge n$, displacements k and positive roofs are present. Define $Z^\circ_{n,R} = \int (q^\circ_{n,R})_+ d\zeta$ and, when this number is positive, let $\mathsf{A}^\circ_{n,R}$ be the probability with density $(q^\circ_{n,R})_+/Z^\circ_{n,R}$.

Theorem 108.3 (The uniform arithmetic Gaussian law of the full record)**.** *For the original actual return record, uniformly in the radius,*

$$E^\circ_n := \sup_R \sum_{m=n}^{\infty} \sum_{k\in\mathbb{Z}^2} \int_0^\infty |p_{n,R}(k,m,u) - q^\circ_{n,R}(k,m,u)|\, du \longrightarrow 0. \tag{108.9}$$

Its reference satisfies $Z^\circ_{n,R} \to 1$ uniformly and

$$d_{\mathrm{TV}}(\mathsf{P}_{n,R}, \mathsf{A}^\circ_{n,R}) \le E^\circ_n.$$

The negative mass of q° tends to zero. Here d_{TV} for probabilities is one half of variation mass. At each fixed radius the factor $\alpha_{m,R}$ may be replaced by $\mathfrak{a}_R(k,n,m)$, including its zero classes. Uniformly through arithmetic changes the finite damped factor is retained.

Proof. First compare the original transition density $\gamma_{n,R} = m^{-2}\mathcal{L}_{m,R}$ with $m^{-2}\mathcal{L}^\circ_{m,R}$. Equation (108.4) and the Gaussian count sum (114.7) give directly

$$\int_{\mathcal{O}_n} m^{-2}|\mathcal{L}_{m,R} - \mathcal{L}^\circ_{m,R}|\, d\zeta \le C \sup_{m\ge n} \theta_m \to 0. \tag{108.10}$$

Both kernels have uniform reference tails by the same estimate.

We next change from the collision scale to the return scale. On the fixed set $|V_{n,R}| \le M$ one has $m = n/c + O_M(\sqrt{n})$. The linear relations in (151.7) give

$$\frac{y - m\mu_R}{\sqrt{m}} = \sqrt{n/m}\, L_R V_{n,R}, \qquad \Omega_R = cL_R D_R L_R^{\mathsf{T}}, \qquad \det L_R = c.$$

Thus $g_{\Omega_R}(\sqrt{c}L_R V) = c^{-3} g_{D_R}(V)$. Together with $n^2/m^2 \to c^2$, uniformly on this set, this proves

$$\sup_{R,\,|V_{n,R}|\le M} n^2 |m^{-2}\mathcal{L}^\circ_{m,R}(y) - q^\circ_{n,R}(k,m,u)| \to 0. \tag{108.11}$$

The remaining factor is $\Phi/c = \alpha$, precisely as in (108.3); the section normalization is not applied a second time.

Here is the count bookkeeping for integration. On $|V| \le M$ there are $O_M(n)$ displacement labels, $O_M(\sqrt{n})$ collision counts, and $O_M(\sqrt{n})$ unit roof intervals. Therefore the bound in (108.11) integrates with the factor

$$\underbrace{O_M(n)\, O_M(\sqrt{n})\, O_M(\sqrt{n})}_{O_M(n^2)\ \text{unit cells}} \underbrace{O_M(m^{-2})}_{O_M(n^{-2})}\, o(1) = o(1). \tag{108.12}$$

Equivalently use its displayed n^{-2} normalization. All constants here are for a fixed M; no count index has been averaged.

Since $|\alpha| \le C$ and D_R is uniformly elliptic, direct Gaussian summation on the three scaled lattice coordinates and integration on the roof give

$$\int_{\mathcal{O}_n} |q^\circ_{n,R}|\, d\zeta \le C, \qquad \int_{|V_{n,R}|>M} |q^\circ_{n,R}|\, d\zeta \le Ce^{-aM^2}.$$

For example sum $e^{-ak_i^2/n}$ and $e^{-a(m-n/c)^2/n}$ separately, each at cost $C\sqrt{n}$, and integrate the roof Gaussian at the same cost. The prefactor is n^{-2}. Spend half the exponent for the tail. The old reference has the separate tail bound (114.6). Let $n \to \infty$ first and then $M \to \infty$. Equations (108.10)–(108.11) prove $\sup_R \int |\gamma - q^\circ| d\zeta \to 0$. Theorem 114.4 now proves (108.9).

For $p \geq 0$ one has $|p - q_+^\circ| \leq |p - q^\circ|$, so $|Z^\circ - 1| \leq E_n^\circ$ and normalization costs at most a second copy of this error in variation mass. Division by two proves the probability bound. On $q^\circ < 0$ its negative part is at most $|p - q^\circ|$, proving the negative-mass assertion.

At fixed R, all peaks with positive damping decay exponentially in m; the remaining peaks are actual resonances with zero roof coordinate. Their original phase sum, divided by c^2, is exactly $\mathfrak{a}_R(k, n, m)$, by the fixed-radius specialization in Section 160. The difference of the two bounded coefficients tends to zero uniformly in the labels and roof. The same Gaussian summation proves the fixed-radius assertion on all of $\mathcal{O}_n$. It does not replace a zero residue by a positive conditional law. □

108.3. **The same coupled paths and exact observation event.** Let H_R and the observation-variation/path-BL norm have exactly the definitions in Section 115. The pair consists of the collision bridge and the actual-return bridge from the *same* trajectory; its reference is the graph law $\mathrm{Law}(\mathbb{B}_{\Omega_R}, c^{-1/2} L_R^{-1} \mathbb{B}_{\Omega_R})$.

Corollary 108.4 (Canonical reference for the coupled record and paths)**.** *With $\mathsf{A}_{n,R}^\circ$ in place of $\mathsf{A}_{n,R}$,*

$$\Delta_n^\circ := \sup_R \left\| \mathrm{Law}_{\nu_R^*}(J_{n,R}, \mathcal{B}_{N_{n,R},R}, \mathcal{Y}_{n,R}) - \mathsf{A}_{n,R}^\circ \otimes \mathsf{H}_R \right\|_{\mathrm{obs,BL}} \to 0. \tag{108.13}$$

The two bridges inside H_R are not independent. For any record-only Markov observation kernel and any event in its complete output with reference probability at least $r_n > \Delta_n^\circ$, the conditional output probability-TV error is at most $\Delta_n^\circ/(r_n - \Delta_n^\circ)$, and the joint path-dual error is at most $2\Delta_n^\circ/(r_n - \Delta_n^\circ)$. Thus convergence after selection requires $\Delta_n^\circ/r_n \to 0$. A path-reading selector is not covered.

Proof. The preceding theorem and Theorem 114.4 give $d_{\mathrm{TV}}(\mathsf{A}_{n,R}, \mathsf{A}_{n,R}^\circ) \to 0$ uniformly. Tensoring a signed observation measure with the probability H_R gives mixed norm at most its variation mass. Apply Theorem 115.2 and the triangle inequality to obtain (108.13). Pullback through a record-only kernel preserves the path test ball, and restriction to the full output event does not increase the unnormalized error. The difference of event masses is at most Δ_n°; normalization costs at most twice that error divided by the true mass, which is at least $r_n - \Delta_n^\circ$. On the scalar marginal divide by two for probability TV. This reproduces the stated constants without changing the event or inferring a convergence rate for a microscopic selection. □

The original pointwise endpoint. The common Gaussian shape removes moving centers and complex covariances from the reference, not positive incidence or clearance from the source. The one-sided raw identity still has those two positive source densities at the exact labels. Their essential heights are not bounded by Theorem 108.3 or by the reference positivity proposition. In particular neither unrestricted same-roof bridges nor forward essential likelihoods are asserted here. The original pointwise criterion, arithmetic zero classes, half-open occupation at $0, \ldots, m-1$ and clearance at collision $j+1$ remain unchanged.

109. A correlated singular realization with a continuous roof

The finite-digit example of Section 107 does not test a continuous density or correlated symbols. We now verify the same pressure mechanism on a two-state Markov baker transformation. Its stable coordinate gives an actual continuous roof and, in the next section, a pointwise density theorem. The comparison is with finite-state Markov additive local limits [20]; the additional

calculation here keeps the continuous deterministic coboundary and its boundary sources inside the coarea formula. It is not a claim of a new general Markov local limit theorem.

Put

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/3 & 2/3 \end{pmatrix}, \qquad \pi = (4/7, 3/7), \qquad \rho_0 = 3/4. \tag{109.1}$$

The state space is the disjoint union of two unit squares, with $\mu(\{i\} \times dx\, dy) = \pi_i dx\, dy$. Let $a_{i0} = 0$, $a_{i1} = P_{i0}$, $b_{j0} = 0$, and $b_{j1} = P_{j0}$. On $a_{ij} \le x < a_{ij} + P_{ij}$ define

$$F(i, x, y) = \left(j, \frac{x - a_{ij}}{P_{ij}}, b_{ji} + P_{ji} y \right). \tag{109.2}$$

Values on partition boundaries are immaterial. Detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$ holds. Consequently the images partition each target square horizontally and the Jacobian P_{ji}/P_{ij} preserves μ. The map is invertible modulo boundaries, expands the first coordinate by at least $4/3$ and contracts the second by at most $3/4$. Its cuts are genuine vertical and horizontal singularity lines. The state sequence I_r has transition matrix P. In particular

$$\mathrm{Cov}_\mu(I_0, I_r) = \frac{12}{49}(5/12)^r. \tag{109.3}$$

Thus these symbols are not independent. This statement concerns the specified coding, not absence of a Bernoulli isomorphism.

For a small real ϵ set

$$\begin{aligned} \xi_\epsilon(i, j) &= j + \epsilon i j, \qquad \bar{\xi}_\epsilon = 3/7 + 2\epsilon/7, \\ \tau_\epsilon(i, x, y) &= 3 + \xi_\epsilon(i, j) + y(F(i, x, y)) - y. \end{aligned} \tag{109.4}$$

For $|\epsilon| < 1/4$ this is a bounded positive roof, piecewise affine in y with a nonzero derivative. No independent smoothing variable is added. Along the original deterministic trajectory,

$$S_m \tau_\epsilon = m(3 + \bar{\xi}_\epsilon) + \sum_{r=0}^{m-1} (\xi_\epsilon(I_r, I_{r+1}) - \bar{\xi}_\epsilon) + y_m - y_0. \tag{109.5}$$

109.1. **An infinite-dimensional physical operator.** Let $\mathcal{B}$ be pairs of complex Lipschitz functions on $[0, 1]$, with norm $\|h\|_\infty + \max_i \mathrm{Lip}(h_i)$. Define

$$\begin{aligned} (\mathcal{P}_{\epsilon,z} h)_i(y) &= \sum_j P_{ij} e^{iz(\xi_\epsilon(i,j) - \bar{\xi}_\epsilon)} h_j(b_{ji} + P_{ji} y), \\ \mathcal{Q}_{\epsilon,z} &= M_{e^{-izy}} \mathcal{P}_{\epsilon,z} M_{e^{izy}}. \end{aligned} \tag{109.6}$$

Conditional integration over the future x coordinate proves the exact scalar identity, for bounded initial tests $a(i, y)$ and Lipschitz terminal tests $d(i, y)$,

$$\int a e^{iz(S_m \tau_\epsilon - m(3 + \bar{\xi}_\epsilon))} (d \circ F^m) d\mu = \sum_i \pi_i \int_0^1 a_i(y) (\mathcal{Q}_{\epsilon,z}^m d)_i(y)\, dy. \tag{109.7}$$

This is a physical conditional-expectation operator for the stated endpoint class, not a claim that it is the complete billiard operator.

Lemma 109.1 (Constants, contraction and the visible eigenvalue)**.** *Near $(\epsilon, z) = (0, 0)$ and $(0, 2\pi)$, the family $\mathcal{Q}_{\epsilon,z}$ is strongly analytic in z and has a simple leading eigenvalue, an analytic rank-one projection, and a complementary power bound $C\rho_1^m$ with a common $\rho_1 < 1$. The leading eigenvalue is the one from the matrix*

$$D_\epsilon(z)_{ij} = P_{ij} e^{iz(\xi_\epsilon(i,j) - \bar{\xi}_\epsilon)}. \tag{109.8}$$

All constants are uniform for ϵ in a sufficiently small closed interval. The continuous roof has physical witnesses of nonzero amplitude near its resonance.

Proof. On compact complex sets the exponential and its derivatives are bounded multipliers of $\mathcal{B}$, proving strong analyticity. The subspace of functions constant in y is invariant for $\mathcal{P}$, with matrix $D_\epsilon(z)$. On its quotient the Lipschitz seminorm is a norm, and

$$\operatorname{Lip}(\mathcal{P}_{\epsilon,z}^m h) \le (\rho_0 e^{C|\operatorname{Im} z|})^m \operatorname{Lip}(h).$$

Choose the complex neighborhoods so that the base is less than one. Splitting a function into its values at $y = 0$ and a function vanishing there writes the operator in upper triangular blocks. Its quotient block has the displayed spectral-radius bound. Outside that disk its spectrum is exactly that of $D_\epsilon(z)$, by the block resolvent formula. At $(0, 0)$ the two matrix eigenvalues are $1, 5/12$; at $(0, 2\pi)$ they are these numbers multiplied by $e^{-6\pi i/7}$. Small fixed resolvent contours and the Neumann series therefore yield the stated splitting and uniform complementary powers. Conjugation by e^{izy} preserves them on the fixed neighborhoods.

At $\epsilon = 0, z = 2\pi$ choose the physical tests $a(i, y) = e^{2\pi iy}$ and $d(i, y) = e^{-2\pi iy}$. The two endpoint factors cancel $y_m - y_0$ in (109.5); the pairing is exactly $e^{-6\pi im/7}$. Its leading amplitude is one. Analyticity of the scalar pairing and projection keeps this amplitude bounded away from zero on a smaller neighborhood. These tests are fixed, not changed with the complex tilt. The remainder comes from the actual operator splitting just proved. □

109.2. **Pressure and a genuinely damped peak.** Let $\Lambda(s, t)$ be the Perron eigenvalue near one of

$$\begin{pmatrix} 3/4 & (1/4)e^s \\ 1/3 & (2/3)e^{s+t} \end{pmatrix}, \qquad C(s, t) = \log \Lambda(s, t).$$

Differentiating its characteristic polynomial

$$\Lambda^2 - (3/4 + (2/3)e^{s+t})\Lambda + (1/2)e^{s+t} - (1/12)e^s = 0 \tag{109.9}$$

gives

$$DC(0, 0) = (3/7, 2/7), \qquad D^2C(0, 0) = \frac{1}{343}\begin{pmatrix} 204 & 192 \\ 192 & 198 \end{pmatrix}. \tag{109.10}$$

Its determinant is $72/2401 > 0$. In particular the centered observable ξ_ϵ has asymptotic variance

$$\sigma_\epsilon^2 = (204 + 384\epsilon + 198\epsilon^2)/343.$$

The same variance belongs to the continuous roof, since the difference of the two centered sums in (109.5) is bounded and the symbolic variance is $O(m)$.

Theorem 109.2 (Pressure contact for the correlated continuous roof)**.** *The continuous roof in* (109.4) *satisfies all the physical pressure, visibility and contact hypotheses of Theorem 107.1 near the peak at 2π. Its real maximizer, damping and centered drift satisfy*

$$\begin{aligned} a_\epsilon &= 2\pi - \frac{32\pi}{17}\epsilon + O(\epsilon^2), \\ \kappa_\epsilon &= \frac{12\pi^2}{119}\epsilon^2 + O(\epsilon^3), \\ v_\epsilon &= -\frac{3456\pi^2}{99127}\epsilon^2 + O(\epsilon^3). \end{aligned} \tag{109.11}$$

Here $D_z\psi_\epsilon(a_\epsilon) = iv_\epsilon$ for the logarithm of the centered *pairing. Thus $v_\epsilon^2 = o(\kappa_\epsilon)$, and the peak has positive damping for all sufficiently small nonzero ϵ. This realization includes a continuous deterministic roof; it is not an independent finite-digit exponential sum.*

Proof. For real t the matrix $D_\epsilon(it)$ is positive. Its simple Perron eigenvalue gives a real analytic pressure $\mathscr{P}_\epsilon(t)$ with zero value and gradient at zero and Hessian σ_ϵ^2. Uniform Perron bounds give the exponential-moment upper estimate for the symbolic sum. The bounded factor $e^{-t(y_m-y_0)}$ in the original roof changes only its prefactor by $e^{|t|}$, not its pressure. This checks the pressure on the actual probability. Visibility was proved in Lemma 109.1.

At $\epsilon = 0$ the identity $D_0(2\pi + w) = e^{-6\pi i/7} D_0(w)$ holds for all complex w. Consequently the Hessian at resonance is exactly the physical variance. The real logarithm has a strict quadratic maximum there, so the implicit function theorem gives the real maximizer a_ϵ.

Here are the coefficient checks without a numerical fit. Put $z = 2\pi + q$. Up to the fixed logarithm branch, the centered logarithm is

$$\psi_\epsilon(z) = -iz(3/7 + 2\epsilon/7) + C(iq, iz\epsilon).$$

The linear terms reduce to $-6\pi i/7$. Minimizing the quadratic form in (109.10) gives $q = -2\pi(192/204)\epsilon + O(\epsilon^2)$ and the residual variance $\Sigma_{22} - \Sigma_{12}^2/\Sigma_{11} = 6/119$. This proves the first two expansions. The three derivatives needed for the drift are

$$C_{sss}(0) = 4908/16807, \quad C_{sst}(0) = 9600/16807, \quad C_{stt}(0) = 14016/16807.$$

Inserting $w = (-32\pi/17, 2\pi)$ into the derivative of the cubic term $-iD^3C(0)[(q, z\epsilon)]^3/6$ gives $-D^3C(0)[(1,0), w, w]/2 = -3456\pi^2/99127$. The quadratic term contributes only to the real derivative, which is zero at the real maximum. The remaining terms are $O(\epsilon^3)$. These derivatives follow directly from (109.9). Uniform analyticity justifies the remainders. After reducing the parameter interval, the positive quadratic damping shows that its only zero is $\epsilon = 0$, where exact contact has been checked. All compact-support hypotheses now hold. □

110. Pointwise roof inversion by stable coarea and cylinder bounds

We prove a density theorem, and not just a scalar spectral expansion, for the preceding Markov baker map. In this section the roof is τ_0; the uniform pressure-contact assertion for the perturbed roofs τ_ϵ is separate. Let p_m^{M} be the density of $S_m\tau_0$ under the original probability μ, and put

$$\bar\tau = 24/7, \qquad \sigma^2 = 204/343, \qquad z_m(t) = (t - m\bar\tau)/\sqrt{m}.$$

All boundary lines of the baker partition remain in the physical coding. No noise, roof convolution, or exceptional roof set is introduced.

110.1. **A lattice estimate with a stable cylinder.** Write $A_m = \sum_{r=1}^m I_r$. For a full word $w = (i_0, \dots, i_m)$ let

$$w_\mu = \pi_{i_0} \prod_{r=0}^{m-1} P_{i_r i_{r+1}}, \qquad y_m = b_w + a_w y_0, \qquad 0 < a_w \le \rho_0^m.$$

Conditional on the word, y_0 is uniform on $[0,1]$; it is independent of the future x cylinder. Here b_w is a scalar endpoint of a stable cylinder, not a physical remainder density. Put $b_m = b_w$ on that word.

Lemma 110.1 (Exact-state lattice local limit)**.** *For every pair of states i, j, uniformly in $k \in \mathbb{Z}$ and $m \ge 1$,*

$$\left|\sqrt{m}\,\Pr_i(A_m = k, I_m = j) - \pi_j g_{\sigma^2}\left(\frac{k - 3m/7}{\sqrt{m}}\right)\right| \le Cm^{-1/2}, \tag{110.1}$$

$$\sqrt{m}\,\Pr_i(A_m = k, I_m = j) \le C.$$

The same estimates hold after averaging the initial state by π.

Proof. Use the matrix $P_{ij}e^{izj}$ and Fourier inversion on $[-\pi,\pi]$. At zero its leading eigenvalue is simple. The centered logarithm has quadratic term $-\sigma^2 z^2/2$ by (109.10), and a bounded cubic remainder; the projection entry equals $\pi_j + O(|z|)$. There is no other unit-modulus eigenvalue. Indeed equality in the triangle inequality for an eigenvector, followed by irreducibility, forces all its coordinate moduli to agree and every transition phase to agree with its eigen-equation. The 00 loop gives eigenvalue one; the 11 loop then gives $e^{iz} = 1$. The only such point on the torus is zero. Compactness gives exponential decay away from zero.

On a small fixed interval the leading modulus is at most e^{-cz^2}. The projection error and the logarithmic cubic error in the inverse integrate to C/m: use $|z|e^{-cmz^2}$ and $m|z|^3e^{-cmz^2}$, respectively, after reducing the interval so that the real quadratic damping absorbs the cubic remainder. The omitted Gaussian tail and complementary terms are exponential. Multiplication by $\sqrt{m}$ proves (110.1); the bound is uniform in k because its Fourier phase has modulus one. Increase C for finitely many small m. □

Lemma 110.2 (Stable interval probabilities at an exact lattice label)**.** *For every interval $J \subset \mathbb{R}$, every $k \in \mathbb{Z}$ and $1 \le L \le m/2$,*

$$\left|\sqrt{m}\,\mu\{A_m = k, b_m \in J\} - g_{\sigma^2}\left(\frac{k-3m/7}{\sqrt{m}}\right)|J\cap[0,1]|\right| \le C\left(\frac{1+L}{\sqrt{m}} + \rho_0^L\right), \tag{110.2}$$

$$\sqrt{m}\,\mu\{A_m = k, b_m \in J\} \le C(|J\cap[0,1]| + \rho_0^L). \tag{110.3}$$

Open, closed and half-open intervals obey the same bounds.

Proof. For each terminal state j, the last L transitions partition its stable unit interval into cylinders I_w of lengths $a_w = \prod P_{i_{r+1}i_r} \le \rho_0^L$. Their interiors are disjoint. A full word with that suffix has $b_m \in \overline{I}_w$. If $i = i_{m-L}$ and $s_w = \sum_{r=m-L+1}^{m} i_r$, the probability of this suffix and $A_m = k$ is exactly

$$\Pr_\pi\{A_{m-L} = k - s_w, I_{m-L} = i\}\prod_{r=m-L}^{m-1} P_{i_r i_{r+1}}.$$

Detailed balance identifies π_i times the last product with $\pi_j|I_w|$. Apply Lemma 110.1 to the prefix. Changing its Gaussian scale and center to $(m, 3m/7)$ costs $C(1+L)/\sqrt{m}$ after normalization, uniformly in k, since $|s_w| \le L$ and both g' and zg' are bounded. Summing over any collection of suffixes does not multiply this error by their number: the sum of their transition products is at most two, the number of possible initial states. Consequently the normalized mass of every such collection differs from its total $\pi_j|I_w|$ times the full Gaussian by at most $C(1+L)/\sqrt{m}$.

Approximate $J \cap [0,1]$ from inside and outside by complete stable cylinders, separately for each terminal state. At most two boundary cylinders per state are needed, with total length at most $C\rho_0^L$. This proves (110.2). For the upper bound, use just the uniform upper estimate for the prefix, paying $\pi_i^{-1} \le 7/3$; the resulting mass is at most a constant times the total cylinder length. There is no local-limit error in this positive upper bound. Boundary atoms are contained in the same boundary cylinders. This also proves all the endpoint conventions claimed above. □

110.2. **A pointwise density, including every word.** The exact conditional roof on a word with $A_m = k$ is

$$S_m\tau_0 = 3m + k + b_w - (1-a_w)y_0.$$

The derivative in the original stable variable is bounded away from zero by $1 - \rho_0^m$. Thus the complete density, for almost every t, is the finite positive coarea sum

$$p_m^{\mathrm{M}}(t) = \sum_w \frac{w_\mu}{1-a_w}\mathbf{1}_{[b_w+a_w-1,b_w]}(t - 3m - A_m(w)). \tag{110.4}$$

In particular its integral is one. There are no omitted words near any cut, and no derivative of a long inverse density is being assumed.

Theorem 110.3 (A continuous-roof pointwise local limit)**.** *For $m \ge 2$,*

$$\operatorname*{ess\,sup}_{t\in\mathbb{R}} \left| \sqrt{m}\, p_m^{\mathrm{M}}(t) - g_{\sigma^2}(z_m(t)) \right| \le C \frac{\log(2+m)}{\sqrt{m}}. \tag{110.5}$$

Moreover $\sup_m \sqrt{m}\|p_m^{\mathrm{M}}\|_\infty < \infty$. For bounded Lipschitz functions f, d on $[0,1]$, the actual weighted roof density $p_m^{f,d}$, with insertion $f(y_0)d(y_m)$, satisfies

$$\operatorname*{ess\,sup}_{t\in\mathbb{R}} \left| \sqrt{m}\, p_m^{f,d}(t) - g_{\sigma^2}(z_m(t))\Theta_{f,d}(t-3m) \right| \le C\|f\|_{\mathrm{Lip}}\|d\|_{\mathrm{Lip}} \frac{\log(2+m)}{\sqrt{m}}, \tag{110.6}$$

where the arithmetic endpoint factor is

$$\Theta_{f,d}(r) = \sum_{k\in\mathbb{Z}} \int_{[0,1]\cap[r-k,r-k+1]} f(v-r+k)d(v)\,dv.$$

It is one-periodic and is not replaced by the product of endpoint means. For $f = d = 1$, it is identically one.

Proof. For a fixed integer k put $\delta = t - 3m - k$. Since $0 < a_w \le \rho_0^m$ and $0 \le b_w \le 1 - a_w$, the contribution of words with $A_m = k$ in (110.4) is between

$$\mu\{A_m = k, b_m \in [\delta, \delta + 1 - \rho_0^m]\} \quad \text{and} \quad (1-\rho_0^m)^{-1}\mu\{A_m = k, b_m \in [\delta, \delta+1]\},$$

up to the immaterial finite set of roof endpoints. Choose $L = \lceil \log(m)/(2\log(4/3)) \rceil$ for large m, so that $L \le m/2$ and $\rho_0^L \le m^{-1/2}$. Lemma 110.2 identifies the normalized contribution with

$$g_{\sigma^2}\left(\frac{k - 3m/7}{\sqrt{m}}\right)(1 - |\delta|)_+ + O(\log(2+m)/\sqrt{m}).$$

Only at most three integers have $|\delta| \le 1$. On this set their Gaussian argument differs from $z_m(t)$ by at most $m^{-1/2}$. The exact partition identity $\sum_{k\in\mathbb{Z}}(1 - |r-k|)_+ = 1$ proves (110.5), uniformly in t. The all-count height bound follows directly from (110.4), $1 - a_w \ge 1 - \rho_0$, and the positive lattice upper bound, again summing at most three integer labels.

For the weighted assertion, at the coarea point

$$y_0 = (b_w - \delta)/(1 - a_w), \qquad y_m = b_w + a_w y_0.$$

On its support these differ from $b_w - \delta$ and b_w, respectively, by at most $C\rho_0^m$. Lipschitz continuity changes the normalized weighted sum by at most $C\|f\|_{\mathrm{Lip}}\|d\|_{\mathrm{Lip}}\rho_0^m$. The remaining function of b_w is $f(b_w - \delta)d(b_w)\mathbf{1}_{[\delta,\delta+1]}(b_w)$, with a uniform bounded-variation norm. The interval estimate (110.2) controls its integral by integration by parts: a signed measure whose mass on every interval is at most e acts on such a function with absolute value at most $e(\|h\|_\infty + \mathrm{TV}(h))$. The two shifted interval boundaries are handled by (110.3) and the same L. The limiting integral is the one displayed in $\Theta_{f,d}$. Sum the at most three labels and make the same Gaussian-center change. Complex weights follow by the identical variation estimate. No arbitrary path-dependent insertion is assigned this amplitude. □

110.3. **Positive essential heights of boundary sources.** Let U_s be an initial vertical strip family: in each square it is the union of intervals of x-distance less than s from $0, 1, a_{i1}$. Let V_s be the analogous terminal stable strip about $y = 0, 1, b_{j1}$. They include the forward and backward partition cuts. Let $p_{m,s}^{\mathrm{vert}}$ and $p_{m,s}^{\mathrm{stab}}$ be the roof densities of the positive, unnormalized sources $\mathbf{1}_{U_s}\mu$ and $\mathbf{1}_{V_s} \circ F^m\,\mu$, respectively. Define also $p_{m,s}^{\mathrm{age}}$ by the source $\mathbf{1}_{\{y_0 < s \text{ or } y_0 > 1-s\}}\mu$.

Theorem 110.4 (Actual boundary-source height estimates)**.** *For $0 < s < 1/8$ and $m \geq 2$,*

$$\sqrt{m}(\|p^{\mathrm{vert}}_{m,s}\|_\infty + \|p^{\mathrm{stab}}_{m,s}\|_\infty + \|p^{\mathrm{age}}_{m,s}\|_\infty) \leq C(s + \rho_0^{\lfloor m/2 \rfloor}). \tag{110.7}$$

In particular the ordered limit of each essential height is zero as $m \to \infty$ and then $s \downarrow 0$. A union of these sources and any bounded insertion supported on it have the same domination bound, with its supremum norm included.

Proof. Take $L = \lfloor m/2 \rfloor$. Cover the initial vertical strips from outside by full forward cylinders of length L. Their mesh is at most ρ_0^L, so their total stationary mass is at most $C(s + \rho_0^L)$. For each such prefix, the remaining $m - L$ transitions have exact-count probability at most $C/\sqrt{m}$, by Lemma 110.1. The conditional coarea density on a complete word is at most $(1 - \rho_0)^{-1}$ and a prescribed roof uses at most three count values. Sum the unchanged positive prefix weights. This proves the vertical estimate without requiring the strip to be a union of exact cylinders.

For the terminal stable strips, $|y_m - b_m| \leq \rho_0^m$, uniformly at every coarea point. Enlarge their finitely many intervals by this amount and apply the positive bound (110.3) to each of the at most three relevant counts. For the initial age strips, the coarea identity makes $y_0 < s$ imply $b_m \in [\delta, \delta + s]$, and $y_0 > 1 - s$ imply $b_m \in [\delta + 1 - s - \rho_0^m, \delta + 1]$. The same interval bound proves the assertion. Constants absorb rounding and $\rho_0^m \leq \rho_0^L$. Source domination proves the weighted and union claims. This proof bounds essential heights directly; it does not infer them from the small masses of the strips. □

What the realization tests, and what it does not transfer. The map has correlated state observations, a nontrivial complementary operator, a continuous roof, and exponentially many singular word boundaries. Its density calculation remains on the original deterministic source. The decisive additional input for its height theorem is the uniform stable coarea derivative $1 - a_w \geq 1 - \rho_0$, together with positive local bounds for cylinders of length up to $m/2$. These inputs are proved here, not assumed as consequences of pressure contact. The roof is nevertheless an explicitly solvable coboundary extension of a finite-state additive record; this is not a claim that all singular hyperbolic density problems reduce to it.

For the Lorentz return record, the incidence and competing-hit sources in Corollary 181.5 do not yet have this cylinder/coarea comparison with a uniform derivative and preserved exact labels. An orbit is not continued across either physical seam by this example. Their two essential-height estimates and the original unrestricted four-coordinate pointwise theorem therefore remain separate requirements. The same-roof Lorentz consequences are not promoted by a pointwise theorem for the Markov model.

111. A positive coarea criterion with exact labels

Pressure contact concerns a characteristic branch. A small positive source requires a separate estimate at the density level. We isolate such an estimate before applying it to the perturbed continuous roof. All labels in the following proposition are retained under pushforward. The criterion does not identify a mass estimate with a height estimate.

Fix a time m, a parameter λ, and an exact auxiliary label ι. Suppose the original source with label ι is disintegrated into countably many words w, with masses $c_w \geq 0$, a scalar coordinate $y \in [0, 1]$, and conditional densities $r_w(y) \leq D$. On a word let $q_w \in \mathbb{Z}^d$ be its further exact label and let its scalar observation be

$$T_w(y) = \ell_\lambda(q_w) + u_w(y), \qquad |u_w| \leq H, \qquad |u_w'| \geq \eta > 0. \tag{111.1}$$

Assume u_w is continuously differentiable and monotone on its word. An additive deterministic time shift may be included in ℓ_λ. All constants below are independent of m, λ, ι.

Proposition 111.1 (Positive source height from labelled coarea)**.** *Let $a_m > 0$ and $\delta_m \downarrow 0$. The following two sufficient criteria bound the height of a positive source at its original labels.*

First, suppose an event E_s is covered by a collection $\mathcal{C}_s$ of word cylinders and

$$\sum_{C\in\mathcal{C}_s} c(C) \le K(s+\delta_m),$$
$$\sum_{\substack{w\subset C\\ |\ell_\lambda(q_w)-t|\le H}} c_w \le K a_m c(C) \quad (t\in\mathbb{R}), \tag{111.2}$$

where $c(C)=\sum_{w\subset C} c_w$. Overlap is allowed in the cover. Then its unnormalized density $p^{E_s}_{m,\lambda,\iota}$ satisfies

$$\|p^{E_s}_{m,\lambda,\iota}\|_\infty \le D\eta^{-1}K^2 a_m(s+\delta_m). \tag{111.3}$$

Alternatively suppose each word has a statistic $b_w\in[0,1]$, and nonnegative envelopes $e_m(q)$ obey

$$\sum_{w:q_w=q,\ b_w\in J} c_w \le K e_m(q)(|J\cap[0,1]|+\delta_m),$$
$$\sup_t \sum_{q:|\ell_\lambda(q)-t|\le H} e_m(q) \le K a_m \tag{111.4}$$

for every interval J. If $T_w(y)=t$ and $(w,y)\in E_s$ imply that b_w lies in a union of at most K intervals of total length at most $K(s+\delta_m)$, depending only on (q_w,t,λ,ι), then (111.3) *holds with a changed constant. Every bounded insertion supported on E_s is controlled by the same bound times its supremum norm. A finite union of such sources is controlled by the sum of the bounds.*

Proof. For almost every t the contribution of word w is

$$c_w \frac{r_w(y_w(t))}{|u_w'(y_w(t))|}\mathbf{1}_{\{t\in T_w([0,1])\}},$$

where $y_w(t)$ is its unique inverse. The change-of-variables formula on each word, followed by Tonelli, proves that the sum is the density of the original pushforward, including all its positive components. It is bounded above by $D\eta^{-1}$ times the sum of the word masses which can meet the roof value. Such a word necessarily satisfies $|\ell_\lambda(q_w)-t|\le H$.

For the first criterion, enlarge the source to its cylinder cover, apply the second line of (111.2) on each cylinder, and then sum its first line. For the second, group the same positive coarea sum by q_w, use the endpoint intervals and the first line of (111.4), and sum the second line. No signed error is used in either argument. Domination at the source proves the insertion and union assertions. Countably many null sets from the wordwise changes of variables have zero Lebesgue measure; no exceptional roof set of positive measure is deleted. □

Why the exact-label hypotheses matter. The cylinder bound in (111.2) concerns the remaining local constraint *after* a cylinder has been fixed. Its conclusion is not supplied by a bound on $c(C)$ alone. Similarly, (111.4) is an exact-label estimate for intervals depending on the roof value, not an unconditioned marginal bound for b_w. Uniformity in an auxiliary exact label must be proved, not acquired by summing that label away.

For the Lorentz source in Corollary 181.5, a version of these hypotheses at scale $a_m=m^{-2}$, with the original (n,k,m,R) fixed and with the complete incidence and next-collision clearance sources, would yield the two required heights. The results on local variation in Sections 181 and 110 do not provide those hypotheses. In particular, (111.1) is not asserted at Lorentz grazing or at a change of the first physical hit. The application below proves every input on its stated original deterministic source.

112. Pointwise inversion throughout the perturbed roof family

We retain the map, probability and roof of (109.2)–(109.4). Fix $0 < \epsilon_0 < 1/4$. The roof remains positive for $|\epsilon| \le \epsilon_0$. Define the two original integer counts

$$A_m = \sum_{r=1}^{m} I_r, \qquad B_m = \sum_{r=1}^{m} I_{r-1} I_r, \qquad Q_m = (A_m, B_m).$$

Write

$$\mu_* = (3/7, 2/7), \quad \Sigma_* = \frac{1}{343}\begin{pmatrix} 204 & 192 \\ 192 & 198 \end{pmatrix}, \quad \bar{\tau}_\epsilon = \frac{24 + 2\epsilon}{7}, \quad \sigma_\epsilon^2 = \frac{204 + 384\epsilon + 198\epsilon^2}{343}. \tag{112.1}$$

The covariance is positive definite by (109.10). The new proof first keeps both integer counts and only then projects onto $A_m + \epsilon B_m$. In particular it does not require a common lattice span for that scalar projection.

112.1. **A two-count local estimate with a positive envelope.**

Lemma 112.1 (Joint exact-state estimates)**.** *For $i, j \in \{0, 1\}$, $q \in \mathbb{Z}^2$ and $m \ge 1$,*

$$\left| m \Pr_i(Q_m = q, I_m = j) - \pi_j g_{\Sigma_*}((q - m\mu_*)/\sqrt{m}) \right| \le C m^{-1/2}, \tag{112.2}$$

$$\Pr_i(Q_m = q, I_m = j) \le \frac{C}{m} \exp(-c|q - m\mu_*|^2/m). \tag{112.3}$$

Consequently, for any fixed $H < \infty$,

$$\sup_{t,\ |\epsilon| \le \epsilon_0} \Pr_i\{|A_m + \epsilon B_m - t| \le H, I_m = j\} \le C_H m^{-1/2}. \tag{112.4}$$

The stationary initial law satisfies the same estimates after summing with the original weights π_i.

Proof. The two-variable Fourier matrix is $M(z)_{ij} = P_{ij} \exp(i(z_1 j + z_2 ij))$. If a unit-modulus eigenvalue exists, equality in the triangle inequality and positivity of all four transition probabilities force $e^{i(z_1 j + z_2 ij)} v_j = \lambda v_i$, with $|v_0| = |v_1| > 0$. The 00 loop gives $\lambda = 1$. The 10 edge gives $v_0 = v_1$, the 01 edge gives $e^{iz_1} = 1$, and the 11 loop gives $e^{iz_2} = 1$. Thus the joint lattice is the full $\mathbb{Z}^2$, including for each specified pair of endpoint states.

Near zero the analytic simple eigenvalue has centered logarithm $-z^{\mathsf{T}} \Sigma_* z/2 + O(|z|^3)$; its projection entry is $\pi_j + O(|z|)$. The characteristic polynomial in (109.9) supplies the mean and Hessian. The real part of this logarithm is at most $-c|z|^2$ on a small ball. Outside it compactness and the preceding phase argument give exponential decay. In the two-dimensional inverse, the projection error integrates to $O(m^{-3/2})$; the cubic error is bounded by $Cm \int |z|^3 e^{-cm|z|^2} dz = O(m^{-3/2})$. Gaussian tails and the complementary power have smaller errors. The inversion phase has modulus one for every q, proving (112.2) uniformly in the exact label.

For completeness the upper estimate has no local-limit error. For a small real tilt $\theta \in \mathbb{R}^2$ use the positive matrix $M(-i\theta)$. Let $C(\theta)$ be its logarithmic Perron root. Its Doob transform is a positive stochastic matrix with exactly the same four edges; its endpoint conjugating factors are uniformly bounded. The same phase argument, and the positive Hessian of C on a sufficiently small closed tilt ball, give

$$\Pr_i(Q_m = q, I_m = j) \le \frac{C}{m} \exp\{mC(\theta) - \theta \cdot q\}.$$

Indeed the absolute Fourier integral of the tilted matrix powers is at most $Ce^{mC(\theta)}/m$: use the uniform quadratic decrease near zero and the uniform gap on its compact complement. This is a bound on a positive coefficient, not subtraction of a signed Gaussian error. For a feasible count put $v = q - m\mu_*$ and choose $\theta = av/m$, where $a > 0$ is small and fixed. Feasible counts satisfy

$|v| \le C_0 m$, so the tilt lies in the chosen ball. Taylor's inequality $C(\theta) \le \mu_* \cdot \theta + C_1|\theta|^2$ gives an exponent at most $-(a - C_1 a^2)|v|^2/m$. Infeasible counts have zero probability. This proves (112.3).

At fixed $B_m = l$ the strip in (112.4) contains at most $2H + 2$ possible integers A_m. Discarding the first coordinate in the positive Gaussian envelope and summing $\exp(-c(l - 2m/7)^2/m)$ gives $C\sqrt{m}$. This proves the last bound, without a rationality or a Diophantine restriction on ϵ. □

On a full word retain the notation $y_m = b_w + a_w y_0$ and $b_m = b_w$; $0 < a_w \le \rho_0^m$, $0 \le b_w \le 1 - a_w$, where $\rho_0 = 3/4$. This b_m is the full-word endpoint, not the left endpoint of a suffix cylinder.

Lemma 112.2 (Labelled stable intervals)**.** *For every interval J, $q \in \mathbb{Z}^2$ and $1 \le L \le m/2$,*

$$|m\mu\{Q_m = q, b_m \in J\} - g_{\Sigma_*}((q - m\mu_*)/\sqrt{m})|J \cap [0,1]|| \le C\left(\frac{1+L}{\sqrt{m}} + \rho_0^L\right). \tag{112.5}$$

If in addition $L \le \sqrt{m}$, the positive bound is

$$\mu\{Q_m = q, b_m \in J\} \le \frac{C}{m} e^{-c|q - m\mu_*|^2/m}(|J \cap [0,1]| + \rho_0^L). \tag{112.6}$$

All endpoint conventions for J satisfy these estimates.

Proof. A suffix of L transitions has reward $q_w \in [0, L]^2$, forward weight $P(w)$, and a terminal stable cylinder of length at most ρ_0^L. If its initial and terminal states are i, j, detailed balance gives $\pi_i P(w) = \pi_j |I_w|$. Its mass at $Q_m = q$ is exactly

$$\Pr_\pi(Q_{m-L} = q - q_w, I_{m-L} = i)P(w).$$

Apply (112.2) to the prefix. Changing its mean and scale to those at time m costs $C(1 + L)/\sqrt{m}$ after multiplication by m, uniformly in q; the Gaussian and its first derivatives, including their linear coordinate multiples, are bounded. The errors sum with $P(w)$, not with the number of cylinders. For all suffixes ending at either state, $\sum_w P(w) \le 2$, since this sum runs over the two possible starting states. Detailed balance gives the leading stable length. Inner and outer covers of J require at most two boundary cylinders per terminal state. This proves (112.5).

For the positive statement use (112.3) instead. Since $|q_w - L\mu_*| \le CL$, for $L \le \sqrt{m}$ its exponential factor is at most $Ce^{-c'|q - m\mu_*|^2/m}$. This follows from $|v - d|^2 \ge |v|^2/2 - |d|^2$ and $m - L \ge m/2$. Sum the outer cylinder cover, using $\pi_i \ge 3/7$. It has weighted length at most $C(|J \cap [0,1]| + \rho_0^L)$. This also controls boundary atoms and proves (112.6). □

112.2. **The original continuous density for every small perturbation.** Let $p_{m,\epsilon}^{f,d}$ denote the density of $S_m \tau_\epsilon$ with insertion $f(y_0)d(y_m)$, where f, d are bounded Lipschitz functions on $[0, 1]$. Put $p_{m,\epsilon} = p_{m,\epsilon}^{1,1}$ and $\|f\|_{\mathrm{Lip}} = \|f\|_\infty + \mathrm{Lip}(f)$. For $r = t - 3m$ and $\delta_w = r - A_m(w) - \epsilon B_m(w)$ the exact coarea formula is

$$p_{m,\epsilon}^{f,d}(t) = \sum_w \frac{w_\mu}{1 - a_w} f\left(\frac{b_w - \delta_w}{1 - a_w}\right) d\left(b_w + a_w \frac{b_w - \delta_w}{1 - a_w}\right) \mathbf{1}_{[b_w + a_w - 1, b_w]}(\delta_w). \tag{112.7}$$

The functions are evaluated only on the displayed support. The stable derivative $1 - a_w \ge 1 - \rho_0$ is independent of ϵ and the word. Conditional uniformity of y_0, not independent roof noise, proves this formula. In particular the unweighted integral is one.

Retain the one-periodic endpoint overlap $\Theta_{f,d}$ from Theorem 110.3 and define the convergent sum

$$\mathcal{R}_{m,\epsilon}^{f,d}(t) = \frac{1}{\sqrt{m}} \sum_{l \in \mathbb{Z}} g_{\Sigma_*}\left(\frac{r - \epsilon l - 3m/7}{\sqrt{m}}, \frac{l - 2m/7}{\sqrt{m}}\right) \Theta_{f,d}(r - \epsilon l). \tag{112.8}$$

This profile retains the actual projected arithmetic for rational, irrational and count-dependent perturbations.

Theorem 112.3 (Uniform perturbed pointwise density)**.** *For every fixed* $0 < \epsilon_0 < 1/4$ *and* $m \geq 2$*,*

$$\sup_{|\epsilon|\leq\epsilon_0} \operatorname*{ess\,sup}_{t\in\mathbb{R}} \left|\sqrt{m}\, p^{f,d}_{m,\epsilon}(t) - \mathcal{R}^{f,d}_{m,\epsilon}(t)\right| \leq C\|f\|_{\mathrm{Lip}}\|d\|_{\mathrm{Lip}} \frac{\log(2+m)^{3/2}}{\sqrt{m}}, \tag{112.9}$$

$$\sup_{|\epsilon|\leq\epsilon_0} \operatorname*{ess\,sup}_{t\in\mathbb{R}} \left|\sqrt{m}\, p_{m,\epsilon}(t) - g_{\sigma^2_\epsilon}\left(\frac{t - m\bar{\tau}_\epsilon}{\sqrt{m}}\right)\right| \leq C\frac{\log(2+m)^{3/2}}{\sqrt{m}}. \tag{112.10}$$

Moreover $\sup_{m,|\epsilon|\leq\epsilon_0} \sqrt{m}\|p_{m,\epsilon}\|_\infty < \infty$*. The endpoint-weighted arithmetic profile is not replaced by a product of means.*

Proof. The height upper bound follows at once from the exact coarea formula and (112.4), with $H = 1$. For each fixed integer pair (k, l) set $\delta = r - k - \epsilon l$. Its unweighted coarea contribution is trapped between

$$\mu\{Q_m = (k,l), b_m \in [\delta, \delta + 1 - \rho_0^m]\} \quad \text{and} \quad (1-\rho_0^m)^{-1}\mu\{Q_m = (k,l), b_m \in [\delta, \delta+1]\}.$$

At a coarea point the two endpoints differ from $b_m - \delta, b_m$ by $O(\rho_0^m)$. The weighted version follows by applying (112.5) to $f(v-\delta)d(v)\mathbf{1}_{[\delta,\delta+1]}(v)$ through its bounded variation. The supremum and variation of this function are bounded by $C\|f\|_{\mathrm{Lip}}\|d\|_{\mathrm{Lip}}$, uniformly in δ. The changed interval boundaries are paid by (112.6). Thus, after multiplication by m, the contribution equals

$$g_{\Sigma_*}((k - 3m/7, l - 2m/7)/\sqrt{m}) \int_{[0,1]\cap[\delta,\delta+1]} f(v-\delta)d(v)\, dv + O\left(\|f\|_{\mathrm{Lip}}\|d\|_{\mathrm{Lip}}\left(\frac{1+L}{\sqrt{m}} + \rho_0^L\right)\right).$$

Choose $L = \lceil \log(m)/(2\log(4/3))\rceil$ for sufficiently large m, so that $L \leq \min(m/2, \sqrt{m})$ and $\rho_0^L \leq m^{-1/2}$.

It is important not to sum this error over m possible values of l. First restrict to $|l - 2m/7| \leq K\sqrt{m\log(2+m)}$. For each l at most three integers k have $|\delta| \leq 1$. The positive envelope (112.3) bounds the omitted coarea contribution, after normalization by $\sqrt{m}$, by Cm^{-2} when K is fixed sufficiently large. The same estimate holds for the omitted Gaussian sum. This follows by summing the Gaussian tail in the second coordinate; it is uniform in t and ϵ. There are $O(\sqrt{m\log(2+m)})$ retained pairs. Their total normalized error is $O(\log(2+m)^{3/2}/\sqrt{m})$.

On each retained pair replace k in the Gaussian by $r - \epsilon l$. The change of the normalized first coordinate is at most $m^{-1/2}$. Its derivative is bounded by a summable Gaussian envelope in the second coordinate, giving total error $O(m^{-1/2})$ after the same normalization. Summing the integrals over k gives exactly $\Theta_{f,d}(r - \epsilon l)$. This proves (112.9). Complex weights use variation and the positive envelope for domination, not an order comparison of complex densities.

For $f = d = 1$ the triangle partition identity gives $\Theta_{1,1} = 1$. Put $z = (t - m\bar{\tau}_\epsilon)/\sqrt{m}$ and $y_l = (l - 2m/7)/\sqrt{m}$. The remaining reference is the mesh $m^{-1/2}$ Riemann sum of $F_{\epsilon,z}(y) = g_{\Sigma_*}(z - \epsilon y, y)$. Uniformly in z and $|\epsilon| \leq \epsilon_0$,

$$\int_{\mathbb{R}} F_{\epsilon,z}(y)\, dy = g_{\sigma^2_\epsilon}(z), \qquad \int_{\mathbb{R}} |F'_{\epsilon,z}(y)|\, dy \leq C.$$

These follow by completing the Gaussian square; the conditional variance is $\det\Sigma_*/\sigma^2_\epsilon$, bounded above and below. The elementary interval-by-interval Riemann error is at most $m^{-1/2}\int|F'|$. This proves (112.10). Finitely many small m are absorbed using the uniform height bound. No growing-band spectral constant or changing scalar lattice span has entered the proof. □

113. Positive heights and the critical arithmetic transition

The same perturbed roof now admits positive boundary estimates and an explicit density profile in the moving-peak regime. The source, coded states and stable endpoint difference are

those already used for Theorem 109.2; no perturbation is reset to zero before the density is evaluated.

113.1. **Three original boundary sources, uniformly in the parameter.** Keep U_s, V_s and the initial-age strips of Theorem 110.4. Let $p^{\mathrm{vert}}_{m,\epsilon,s}, p^{\mathrm{stab}}_{m,\epsilon,s}$ and $p^{\mathrm{age}}_{m,\epsilon,s}$ denote their unnormalized positive $S_m\tau_\epsilon$ densities. Put $L_m = \min\{\lfloor m/2\rfloor, \lfloor\sqrt{m}\rfloor\}$.

Theorem 113.1 (Perturbed positive essential heights)**.** *For $m \geq 2$, $0 < s < 1/8$, and any fixed $0 < \epsilon_0 < 1/4$,*

$$\sup_{|\epsilon|\leq\epsilon_0} \sqrt{m}\left(\|p^{\mathrm{vert}}_{m,\epsilon,s}\|_\infty + \|p^{\mathrm{stab}}_{m,\epsilon,s}\|_\infty + \|p^{\mathrm{age}}_{m,\epsilon,s}\|_\infty\right) \leq C(s + \rho_0^{L_m}). \tag{113.1}$$

The vertical and terminal stable terms separately satisfy the stronger bound with $\rho_0^{\lfloor m/2\rfloor}$. The same domination applies to bounded insertions supported on the union, with their supremum norm. Consequently the conditional probability of this union, given the same *roof value t with $|t - m\bar{\tau}_\epsilon| \leq M\sqrt{m}$, is at most $C_M(s + \rho_0^{L_m})$ for almost every such t and all sufficiently large m, uniformly in ϵ.*

Proof. The coarea derivative is $1 - a_w \geq 1 - \rho_0$, and every contributing word satisfies $|A_m + \epsilon B_m - (t - 3m)| \leq 1$. For a fixed prefix of depth $L = \lfloor m/2\rfloor$, subtract its exact two rewards. The remaining strip probability is at most $C/\sqrt{m}$, by (112.4), uniformly in the prefix, its terminal state and the shifted roof value. The outer forward cylinder cover of U_s has stationary mass at most $C(s + \rho_0^L)$. Apply the first criterion of Proposition 111.1. This uses the residual local constraint after fixing the prefix.

For V_s cover its terminal stable intervals by suffix cylinders of the same depth. Their total stationary mass is at most $C(s + \rho_0^L)$; terminal state weights are included. After a suffix with initial state i is fixed, its exact rewards are subtracted and the prefix strip probability ending at i is at most $C/\sqrt{m}$. Multiplication by the suffix transition product and $\pi_i^{-1} \leq 7/3$ gives the required conditional cylinder estimate. Equivalently, detailed balance converts these products into the actual stable-cylinder masses. The terminal coordinate belongs to its suffix cylinder exactly, so this cover includes all coarea endpoints. The first criterion applies again.

For the initial-age source use the second criterion with $q = (k, l)$, $b_w = b_m$, $a_m = m^{-1/2}$, and the envelope $Cm^{-1}e^{-c|q-m\mu_*|^2/m}$ from (112.6) at depth L_m. At a coarea point set $\delta = t - 3m - k - \epsilon l$. The implications are

$$\begin{aligned} y_0 < s &\implies b_m \in [\delta, \delta + s], \\ y_0 > 1 - s &\implies b_m \in [\delta + 1 - s - \rho_0^m, \delta + 1]. \end{aligned}$$

Both intervals are retained. Summing their positive interval bounds requires only the labels with $|\delta| \leq 1$. For each l there are at most three k, and the second-coordinate Gaussian sum is $O(\sqrt{m})$, exactly as in (112.4). This proves the age bound without summing a signed local-limit error.

Domination before pushforward proves the union and insertion statements. Finally, (112.10) gives $p_{m,\epsilon}(t) \geq c_M/\sqrt{m}$ on the stated central set for large m, because the variances have common positive lower and upper bounds. Divide the unchanged source density by this same-roof density. Regular conditional probabilities are thereby specified for almost every roof value. No roof set of positive measure is discarded. □

113.2. **A visible transition at the pressure scale.** The endpoint overlap is continuous and one-periodic. More precisely,

$$\|\Theta_{f,d}\|_\infty + \mathrm{Lip}_{\mathbb{R}/\mathbb{Z}}(\Theta_{f,d}) \leq C\|f\|_{\mathrm{Lip}}\|d\|_{\mathrm{Lip}}. \tag{113.2}$$

To verify this, extend f periodically, allowing its one endpoint jump. This extension has bounded variation on the circle. The overlap is its circular correlation with the bounded periodic extension of d. Translation in L^1 of a function of bounded variation is bounded by the shift times its variation. This proves the Lipschitz estimate; no equality of the values at $0, 1$ is assumed.

Set

$$c_m = \epsilon\sqrt{m}, \qquad z = \frac{t - m\bar{\tau}_\epsilon}{\sqrt{m}}, \qquad r_\epsilon = t - 3m - \frac{2m\epsilon}{7}, \qquad \beta_* = \frac{16}{17}, \quad v_* = \frac{6}{119}. \tag{113.3}$$

The last two constants are the conditional mean coefficient and variance of the second coordinate of the centered Σ_* Gaussian given its first coordinate.

Theorem 113.2 (The moving arithmetic profile on its critical scale)**.** *For every $C_0 < \infty$, uniformly for $|c_m| \le C_0$, $|\epsilon| \le \epsilon_0$ and almost every $t \in \mathbb{R}$,*

$$\sqrt{m}\, p^{f,d}_{m,\epsilon}(t) = g_{204/343}(z) \int_{\mathbb{R}} \Theta_{f,d}(r_\epsilon - \beta_* c_m z - c_m v) g_{v_*}(v)\, dv + O_{C_0}\left(\|f\|_{\mathrm{Lip}} \|d\|_{\mathrm{Lip}} \frac{\log(2+m)^{3/2}}{\sqrt{m}} \right). \tag{113.4}$$

For the fixed physical witness functions $f(y) = e^{2\pi i y}$ and $d(y) = e^{-2\pi i y}$, the main term is

$$g_{204/343}(z) e^{-2\pi i (r_\epsilon - (16/17) c_m z)} e^{-12\pi^2 c_m^2/119}. \tag{113.5}$$

Thus the moving damped peak and the pointwise density are evaluated on the same nontrivial parameter family, together with the uniform positive height estimates of Theorem 113.1.

Proof. Write the sum in (112.8) on the grid $y_l = (l - 2m/7)/\sqrt{m}$. Its summand is

$$g_{\Sigma_*}(z - \epsilon y_l, y_l) \Theta_{f,d}(r_\epsilon - c_m y_l).$$

By (113.2) and Gaussian differentiation, its variation as a function of y is uniformly bounded by $C_{C_0} \|f\|_{\mathrm{Lip}} \|d\|_{\mathrm{Lip}}$, independently of z and r_ϵ. The Riemann error is therefore $O_{C_0}(m^{-1/2})$ with the same norm factor. Replacing $z - \epsilon y$ by z in the integral costs $O(|\epsilon|)$ uniformly in z: the integral of $|y|\, |\partial_1 g_{\Sigma_*}(z - \theta\epsilon y, y)|$ is uniformly bounded for $0 \le \theta \le 1$. Since $|\epsilon| \le C_0/\sqrt{m}$, this is another allowed error. Gaussian conditioning now gives

$$g_{\Sigma_*}(z, y) = g_{204/343}(z) g_{6/119}(y - (16/17)z).$$

Combine this identity with (112.9) to obtain (113.4).

For the displayed witness pair the original overlap equals $\Theta_{f,d}(r) = e^{-2\pi i r}$. Its Gaussian integral is $e^{-2\pi i (r_\epsilon - \beta_* c_m z)} e^{-2\pi^2 v_* c_m^2}$, which is (113.5). The damping coefficient is exactly the one in (109.11): $m\kappa_\epsilon = (12\pi^2/119) c_m^2 + O_{C_0}(m^{-1/2})$. This agreement is obtained from the original coarea density, not by assuming that a spectral expansion already supplies a pointwise inverse. □

Relation to the original raw inversion problem. The scalar roof projection may change from lattice to nonlattice as ϵ varies. The proof avoids any nonuniform scalar arithmetic estimate by retaining the full integer pair until the positive coarea sum is performed. The source-height criterion isolates the two inputs which pressure alone cannot supply: a wordwise derivative and positive local bounds after conditioning on the relevant cylinders. Finite-state Markov additive local-limit methods are classical [20, 21]; uniformity over compact transition families is also available there. The specific additions here are the joint-label projection across an arithmetic transition, its exact deterministic endpoint overlap, and the three original positive source heights in that same family.

This verifies the new criterion in a correlated perturbed realization, not in several unrelated non-Markov systems. In the triangular Lorentz problem the labels n, k, m and the next-collision clearance convention must still be retained when verifying the criterion. Neither the stable derivative in (112.7) nor the two-count positive cylinder estimate is a theorem about those physical sources. The two Lorentz heights in Corollary 181.5, and the unrestricted same-roof and essential-likelihood consequences which require them, are not asserted here. All earlier exact-return statements and their arithmetic zero classes remain in force.

114. Pressure control of moving peaks and the untruncated raw law

A bounded central region contains only a finite fraction of the exact return labels. Passing from its local variation law to the whole return measure therefore requires tightness of the arithmetic reference itself. Continuity of a moving spectral drift is not sufficient: its displacement is multiplied by the square root of the collision count. We obtain the required bound from a positive exponential moment of the same collision observable. No estimate at a growing frequency band is used.

Retain the peak data $a_j, z_j, \kappa_j, \mu_j, H_j, B_j$ and the exact kernel $\mathcal{L}_{m,R}$ of (160.5). Write $\mu_R = (0, 0, \bar{\tau}_R, c)$, $c = 91/(10000\pi)$, and $X_R = \mathsf{f}_R - \mu_R$. All constants below are uniform on the original circular parameter family. The finitely many peak charts and their partition are unchanged.

114.1. **A real exponential tilt dominates each visible branch.**

Lemma 114.1 (Uniform centered pressure)**.** *For some $t_0, C > 0$ there is a real function $\mathscr{P}_R(t)$, $|t| < t_0$, such that*

$$0 \le \mathscr{P}_R(t) \le C|t|^2, \qquad \int \exp\left(-t \cdot \sum_{l=0}^{m-1} X_R \circ T_R^l\right) d\nu \le C e^{m\mathscr{P}_R(t)} \quad (m \ge 1). \tag{114.1}$$

It is the logarithm of the positive simple eigenvalue of the centered occupation twist at frequency it near zero.

Proof. Center the analytic full occupation operator by multiplying it by $e^{-iz\cdot\mu_R}$. The near-zero splitting and its physical pairing in Section 151, together with the full occupation realization of Section 157, give, on a common small complex ball,

$$\int e^{-t\cdot S_m X_R}\, d\nu = \lambda_{0,R}(it)^m A_{0,R}(it) + O(\rho^m), \qquad \rho < 1.$$

Here $A_{0,R}(0) = 1$, so the amplitude is positive and bounded away from zero for real small t. Conjugation and simplicity make the eigenvalue real; closeness to one makes it positive and larger than a fixed number exceeding ρ. Jensen's inequality and $\int X_R d\nu = 0$ show that the left side is at least one for every m. Taking mth roots in the splitting proves $\lambda_{0,R}(it) \ge 1$. Put $\mathscr{P}_R(t) = \log \lambda_{0,R}(it)$. Centering removes its linear term, and the uniform analytic second-derivative bound gives $\mathscr{P}_R(t) \le C|t|^2$. The displayed splitting, with the complement absorbed using $\lambda_{0,R}(it) \ge 1$, proves the moment bound. Only a fixed complex neighborhood of zero is involved. □

Theorem 114.2 (Drift is controlled by spectral damping)**.** *There is $C_D < \infty$ such that every retained peak satisfies*

$$|\mu_j(R) - \mu_R|^2 \le C_D \kappa_j(R). \tag{114.2}$$

In particular a peak with $m\kappa_j = O(1)$ has a bounded displacement $\sqrt{m}(\mu_j - \mu_R)$. The estimate concerns the existing scalar branches, including through changes of arithmetic type.

Proof. We first work near an actual resonance on a given peak chart. Its rank-one peripheral projection is nonzero and has the physical representation established in Section 159. Choose bounded smooth initial and terminal functions a, d whose pairing with this projection is nonzero. Such functions exist by the physical representation and density of smooth functions in $L^2(\nu)$. These are proof tests on the full collision space, not section weights. In particular their pairing is not required to have the possibly vanishing section amplitude B_j.

The scalar-continuity proof of Lemma 160.1 applies also to this pair: finite-time pairings are continuous, and multiplication by the inverse leading eigenvalue makes their limit uniform on a smaller complex neighborhood. Hence its amplitude $A_j^{a,d}(R,z)$ is bounded away from zero there. After centering the branch, write

$$\widetilde{\lambda}_j(R,z) = e^{-iz\cdot\mu_R}\lambda_j(R,z).$$

Shrink the imaginary neighborhood so that the centered complementary powers are bounded by $C\rho_1^m$, $\rho_1 < 1$. The exact pairing, valid also at complex frequencies for this bounded observable, gives

$$\begin{aligned} |A_j^{a,d}(R,z)|\,|\widetilde{\lambda}_j(R,z)|^m &\le \left|\int a(x)e^{iz\cdot S_m X_R(x)} d(T_R^m x)\, d\nu(x)\right| + C\rho_1^m \\ &\le \|a\|_\infty \|d\|_\infty \int e^{-t\cdot S_m X_R}\, d\nu + C\rho_1^m, \qquad z = a_j(R) + it. \end{aligned}$$

Lemma 114.1 and mth roots imply the spectral domination

$$\operatorname{Re}\log\widetilde{\lambda}_j(R, a_j + it) \le \mathscr{P}_R(t). \tag{114.3}$$

No order structure on the anisotropic space is presumed: domination was established on physical scalar pairings with a nonzero amplitude.

At the real maximum the centered logarithmic gradient is iv_j, where $v_j = \mu_j - \mu_R$ is real. The uniform complex Taylor estimate and (114.3) give

$$-\kappa_j - v_j\cdot t \le C_1|t|^2 \qquad (|t| < t_1).$$

All drifts are bounded on the compact chart supports. Enlarge C_1 so that $|v_j|/(2C_1) < t_1$ and substitute $t = -v_j/(2C_1)$. This yields $|v_j|^2 \le 4C_1\kappa_j$.

For completeness, the local tests just chosen need not norm a whole chart. The zero set of κ_j on its compact parameter support is compact. Cover it by finitely many of the neighborhoods above. On the remaining compact set κ_j has a positive minimum, while v_j is bounded; their quotient therefore satisfies the same conclusion with a larger constant. If the zero set is empty this last argument suffices. A maximum over the finite peak family proves (114.2) without replacing its partition. □

114.2. **A Gaussian envelope centered at the physical mean.**

Lemma 114.3 (Complex Gaussian and label tails)**.** *There are $C, a > 0$ such that for every $m \ge 1$ and every real target $y = (k_1, k_2, u, n)$,*

$$|\mathcal{L}_{m,R}(y)| + |\mathcal{K}_{m,R}(y)| \le C\exp\left(-a\frac{|y - m\mu_R|^2}{m}\right). \tag{114.4}$$

For fixed return count $n \ge 2$ define

$$V_{n,R}(k,m,u) = \frac{(k_1, k_2, m - n/c, u - n\bar{\tau}_R/c)}{\sqrt{n}}, \quad \gamma_{n,R}(k,m,u) = m^{-2}\mathcal{L}_{m,R}(k_1,k_2,u,n).$$

With $m \geq n$, $k \in \mathbb{Z}^2$ and $u > 0$, one has

$$\sum_{m,k} \int_0^\infty |\gamma_{n,R}|\, du \leq C, \tag{114.5}$$

$$\sum_{m,k} \int_{\{|V_{n,R}|>M,\ u>0\}} |\gamma_{n,R}|\, du \leq Ce^{-aM^2} + Ce^{-an} \quad (M > 0). \tag{114.6}$$

The constants may change between these displays.

Proof. On the compact matrix family in (160.1), $\operatorname{Re} H \geq a_0 I$ and $\|H\| \leq A$. The Gaussian integral in (160.4), by analytic continuation of completion of the square on $\operatorname{Re} H > 0$, equals $(2\pi)^{-2}(\det H)^{-1/2} \exp(-z^{\mathsf{T}} H^{-1} z/2)$. The determinant branch is continued from positive real matrices. For real z, put $w = H^{-1}z$. Then

$$\operatorname{Re}(z^{\mathsf{T}} H^{-1} z) = \operatorname{Re}(w^* H^* w) \geq a_0 |w|^2 \geq (a_0/A^2)|z|^2.$$

Also every singular value of H is at least a_0. Consequently $|G_H(z)| \leq Ce^{-a_1|z|^2}$ uniformly.

Set $Z = (y - m\mu_R)/\sqrt{m}$ and $d_j = \sqrt{m}(\mu_j - \mu_R)$. The modulus of the jth transition summand is at most $C \exp(-m\kappa_j - a_1|Z - d_j|^2)$. By Theorem 114.2, $|d_j|^2 \leq C_D m\kappa_j$. The inequality $|Z|^2 \leq 2|Z - d_j|^2 + 2C_D m\kappa_j$ proves (114.4) after summing the finitely many bounded amplitudes, including the fixed factor $1/c$.

Summing the two displacement Gaussians and integrating the roof Gaussian bounds the contribution at count m by

$$Cm^{-1/2} \exp\left(-a\frac{(n - cm)^2}{m}\right). \tag{114.7}$$

Indeed $\sum_{k\in\mathbb{Z}^2} e^{-a|k|^2/m} \leq Cm$ and the roof integral over the whole real line is $C\sqrt{m}$. For $n/(2c) \leq m \leq 2n/c$, this sum is bounded by $Cn^{-1/2} \sum_m e^{-a'(m-n/c)^2/n} \leq C$. For $n \leq m < n/(2c)$ its total is $C\sqrt{n}e^{-a'n}$, and for $m > 2n/c$ it is bounded by $C \sum_{m>2n/c} e^{-a'm}$. Absorb the polynomial in an exponential. This proves (114.5).

On the middle range $m \asymp n$, the linear identities $m - n/c = -(n - cm)/c$ and $u - n\bar{\tau}_R/c = (u - m\bar{\tau}_R) + \bar{\tau}_R(m - n/c)$ give $|V_{n,R}| \leq C|Z|$. On $|V_{n,R}| > M$, spend half the Gaussian exponent to obtain $Ce^{-a'M^2}$, and sum the remaining half as above. The two outer count ranges have total $Ce^{-a'n}$ whether or not the target restriction is imposed. This proves (114.6). No central cutoff occurs in $\gamma_{n,R}$ itself. □

114.3. **The complete fixed-return mixed measure.** Let $\mathcal{O}_n = \mathbb{Z}^2 \times \{n, n+1, \ldots\} \times (0, \infty)$, with counting measure in its first three coordinates and Lebesgue measure in its last coordinate; denote this product measure by $d\zeta$. The law $\mathsf{P}_{n,R}$ of the original $J_{n,R}$ on this space has density $p_{n,R}(k, m, u)$. It is a probability, not the mass of a fixed-collision section intersection. Put

$$Z^{\mathrm{ref}}_{n,R} = \int_{\mathcal{O}_n} (\gamma_{n,R})_+\, d\zeta, \qquad d\mathsf{A}_{n,R} = \frac{(\gamma_{n,R})_+}{Z^{\mathrm{ref}}_{n,R}} d\zeta. \tag{114.8}$$

The second definition is used when its denominator is positive.

Theorem 114.4 (Untruncated arithmetic raw law in total variation)**.** *For the same original return index n,*

$$E_n := \sup_R \int_{\mathcal{O}_n} |p_{n,R}(k, m, u) - \gamma_{n,R}(k, m, u)|\, d\zeta \longrightarrow 0. \tag{114.9}$$

Consequently $Z^{\mathrm{ref}}_{n,R} \to 1$ uniformly in R, the reference in (114.8) *is eventually defined for every radius, and*

$$\sup_R d_{\mathrm{TV}}(\mathsf{P}_{n,R}, \mathsf{A}_{n,R}) \leq E_n \to 0. \tag{114.10}$$

Here d_{TV} is one half of the variation mass. Every displacement, collision count and positive roof is included; no central restriction, source guard, roof smoothing or average over n remains in the result. The negative transition mass also tends uniformly to zero.

Proof. Let e_m be the unit-window supremum in Theorem 180.1. It is finite and tends to zero. Replace it by its decreasing tail supremum when convenient. On $|V_{n,R}| \le M$, for a fixed M, one has $m = n/c + O_M(\sqrt{n})$. There are $O_M(n^{3/2})$ discrete label triples (k, m), and each roof interval is covered by $O_M(\sqrt{n})$ unit intervals. The local theorem, with the signed kernel still retained, therefore gives

$$\int_{\{|V_{n,R}|\le M\}} |p_{n,R} - \gamma_{n,R}|\, d\zeta \le C_M \sup_{m\ge n} e_m \longrightarrow 0. \tag{114.11}$$

These sums are taken at a fixed return index. They are the total variation norm of its joint measure, not an average of different return laws.

The fourth-moment estimate (147.6), with mark zero, yields $\int |J_{n,R} - n\bar{G}_R|^4 d\nu_R^* \le Cn^2$. Thus the original mass of $|V_{n,R}| > M$ is at most CM^{-4}. Lemma 114.3 supplies the reference tail. Combining these bounds gives, for each fixed M and all sufficiently large n,

$$E_n \le C_M \sup_{m\ge n} e_m + CM^{-4} + Ce^{-aM^2} + Ce^{-an}.$$

First let $n \to \infty$ and then $M \to \infty$. This proves (114.9). The preliminary central cutoff was only in the proof and has disappeared from both measures.

Since $p_{n,R} \ge 0$, replacing γ by γ_+ decreases its pointwise error against p. Therefore $|Z_{n,R}^{\mathrm{ref}} - 1| \le E_n$, and

$$\int \left| p_{n,R} - \frac{\gamma_+}{Z_{n,R}^{\mathrm{ref}}} \right| d\zeta \le \int |p_{n,R} - \gamma_+| d\zeta + |Z_{n,R}^{\mathrm{ref}} - 1| \le 2E_n.$$

This proves (114.10). On $\gamma < 0$ its negative mass is bounded by $|p - \gamma|$, giving the last assertion. No pointwise arithmetic floor is required to normalize this whole law. □

Corollary 114.5 (The fixed-radius residue is retained globally)**.** *At each fixed radius define on $\mathcal{O}_n$*

$$q_{n,R}^0(k, m, u) = n^{-2}\mathfrak{a}_R(k, n, m) g_{D_R}(V_{n,R}(k, m, u)).$$

Its mass tends to one, and the probability obtained by dividing it by its mass is asymptotic in total variation to $\mathsf{P}_{n,R}$. The coefficient $\mathfrak{a}_R$ is the actual finite residue, including its zero classes; it has not been replaced by one. This specialization is for fixed R, whereas (114.10) *is radius-uniform.*

Proof. On a fixed central region use the fixed-radius specialization in Theorem 160.3. With L_R as in (151.7), the centered variables satisfy $Z = \sqrt{n/m}\, L_R V_{n,R}$ and $m/n \to c^{-1}$ there. The identity $\Omega_R = cL_R D_R L_R^{\mathsf{T}}$ and $\det L_R = c$ give $g_{\Omega_R}(\sqrt{c} L_R v) = c^{-3} g_{D_R}(v)$. Consequently $cm^{-2} g_{\Omega_R}(Z) = n^{-2} g_{D_R}(V) + o(n^{-2})$ uniformly on that region. The finite-radius residue is nonnegative and bounded. Its Gaussian expression q^0 therefore has a Gaussian tail in V after summing the scaled lattice and integrating the roof, by the same elementary sums as above. Combine the central comparison with that tail and the true fourth-moment tail, then normalize exactly as in Theorem 114.4. □

The positive-height problem. The new input is (114.2); mere continuity at $\kappa_j = 0$ would not give the Gaussian tail used in the proof. The conclusion integrates the absolute error on the entire exact mixed record. It does not estimate the essential height of either physical component in Corollary 181.5. Those two estimates still govern the original unrestricted pointwise arithmetic local theorem. In particular the whole-law normalization above does not assign a nonzero probability to a single roof value or a positive likelihood to a zero arithmetic class.

115. The full record and its coupled collision and return bridges

The collision and actual-return bridges are obtained from the same trajectory. Their joint limit is consequently a graph coupling, not two independent Gaussian bridges. We prove this joint assertion in a norm which is total variation in the entire observed record and bounded-Lipschitz dual in the two paths. This also specifies exactly which adaptive observations and subsequent selections preserve the conclusion.

Let $\mathcal{C} = C([0,1], \mathbb{R}^4)$ and give $\mathcal{C}^2$ the metric $d_2((x,y),(x',y')) = \max\{\|x - x'\|_\infty, \|y - y'\|_\infty\}$. The unit test class $\mathcal{F}_2$ consists of real functions with supremum norm and Lipschitz constant at most one. Its dual norm is BL_2^*. Put

$$A_R = c^{-1/2} L_R^{-1}, \qquad \mathsf{H}_R = \mathrm{Law}(\mathbb{B}_{\Omega_R}, A_R \mathbb{B}_{\Omega_R}). \tag{115.1}$$

The second marginal has covariance D_R, by (151.7); it is not independent of the first. All these linear maps are uniformly bounded and continuous in R.

On the original source ν_R^*, define

$$O_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R}), \qquad Z_{n,R}^{\mathrm{path}} = (\mathcal{B}_{N_{n,R},R}, \mathcal{Y}_{n,R}),$$

using the paths tied to their own terminals in (161.1)–(161.2). Let $\mathsf{T}_{n,R}^z$ be the conditional law of this pair at $O_{n,R} = z$. A single regular disintegration is used on the standard Borel spaces. The discrete labels are countable. As in Section 190, a countable norming class for $\mathcal{F}_2$ fixes all roof versions off one null set for each set of labels. Tests measurable in z are interpreted through this same kernel.

115.1. **The joint, rather than marginal, local numerator.**

Lemma 115.1 (A coupled path law on central roof windows)**.** *For each fixed $M, h < \infty$, $h > 0$, uniformly in the original exact labels with $|Z(t)| \le M$,*

$$\frac{1}{h} \int_t^{t+h} \left\| m^2 p_{n,R}(k,m,u) \mathsf{T}_{n,R}^{k,m,u} - G_{m,R}^{n,k}(u) \mathsf{H}_R \right\|_{\mathrm{BL}_2^*} du \longrightarrow 0 \quad (m \to \infty). \tag{115.2}$$

There is no positive-denominator assumption.

Proof. Under the finite positive source $\lambda_m = (m^2/h) \mathbf{1}_{\{K_n = k, N_n = m, T_n \in [t, t+h]\}} \nu_R^*$, Lemma 191.1 proves the actual clock coupling

$$\int \min\{2, \|\mathcal{Y}_{n,R} - A_{m,n,R} \mathcal{B}_{m,R}\|_\infty\} \, d\lambda_m \to 0, \qquad A_{m,n,R} = \sqrt{m/n}\, L_R^{-1}.$$

This is (191.4) on the joint trajectory, not an inference from convergence of two marginal laws. On the stated central targets, $n/m = c + O_M(m^{-1/2})$ and hence $A_{m,n,R} \to A_R$ uniformly. The collision-path measures under λ_m are uniformly tight and have bounded mass, as proved there. Splitting at $\|\mathcal{B}_{m,R}\|_\infty \le K$, then taking $m \to \infty$ followed by $K \to \infty$, permits $A_{m,n,R}$ to be replaced by A_R in the last display. This argument also covers a source whose total mass tends to zero.

The graph map $x \mapsto (x, A_R x)$ is uniformly Lipschitz. Pushing the collision path numerator of Theorem 190.3 through this map costs at most $\max\{1, \|A_R\|\}$ in its dual norm. For every test in $\mathcal{F}_2$, the difference between the original pair and this graph is bounded by the truncated clock discrepancy above. Joint disintegration and the countable norming class allow the supremum over path tests before roof integration. These two bounds prove (115.2). The roof interval enlarges the central target only to $M + h/\sqrt{m}$, as in the inherited clock lemma. □

115.2. **No label or roof cutoff in the joint limit.** For two finite measures on $\mathcal{O}_n \times \mathcal{C}^2$ define

$$\|\sigma\|_{\mathrm{obs,BL}} = \sup_{\phi} \left| \int \phi(z,x,y)\, d\sigma \right|,$$

where ϕ is jointly measurable and $\phi(z,\cdot,\cdot) \in \mathcal{F}_2$ for every z. For a disintegrated signed measure this equals the integral of the fiber dual norms: measurable selection of a test from the countable norming class gives the lower inequality, and integration gives the upper one. This norm is variation mass on the observation marginal, not probability total variation; it is not path-space total variation.

Theorem 115.2 (Untruncated mixed law with coupled bridges)**.** *Uniformly in R at each fixed actual return index n,*

$$\epsilon_n := \sup_R \int_{\mathcal{O}_n} \left\| p_{n,R}(z)\mathsf{T}^z_{n,R} - \gamma_{n,R}(z)\mathsf{H}_R \right\|_{\mathrm{BL}_2^*} d\zeta \longrightarrow 0 \quad (n \to \infty). \tag{115.3}$$

Consequently, with the normalized positive arithmetic law $\mathsf{A}_{n,R}$ of (114.8)*,*

$$\Delta_n := \sup_R \left\| \mathrm{Law}_{\nu^*_R}(O_{n,R}, Z^{\mathrm{path}}_{n,R}) - \mathsf{A}_{n,R} \otimes \mathsf{H}_R \right\|_{\mathrm{obs,BL}} \le 2\epsilon_n \longrightarrow 0. \tag{115.4}$$

In particular the conditional pair bridge at its own complete observed record satisfies

$$\sup_R \int d_{\mathrm{BL}_2}(\mathsf{T}^z_{n,R}, \mathsf{H}_R)\, d\mathsf{P}_{n,R}(z) \le 2\epsilon_n \to 0. \tag{115.5}$$

All three statements are on the original unprotected source.

Proof. On $|V_{n,R}| \le M$, the discrete-label and unit-window count is the same $O_M(n^2)$ count as in (114.11). The collision central coordinates are bounded by a constant depending on M, since $m \asymp n$. Thus Lemma 115.1, including the fixed enlargement across each unit window, makes the integral of the joint dual error on this region tend uniformly to zero.

Outside that region its integrand is at most $p_{n,R} + |\gamma_{n,R}|$: positive finite path measures have dual norm equal to mass. The true fourth-moment tail and (114.6) bound its total by $CM^{-4} + Ce^{-aM^2} + Ce^{-an}$. First take the return limit and then the central radius to infinity. This proves (115.3) on the entire observation space.

For $p \ge 0$, a probability Q and a probability H, replacing a negative scalar g by zero decreases $\|pQ - gH\|_{\mathrm{BL}_2^*}$: the old measure is then positive with mass $p + |g|$, and the new measure has mass p. Thus positive-part replacement never enlarges the error in (115.3). The constant test one gives $|Z^{\mathrm{ref}}_{n,R} - 1| \le \epsilon_n$. Normalization adds exactly that mass difference in the mixed norm, proving (115.4).

Finally, pointwise in the common disintegration,

$$p\|Q - H\|_{\mathrm{BL}_2^*} \le \|pQ - gH\|_{\mathrm{BL}_2^*} + |p - g| \le 2\|pQ - gH\|_{\mathrm{BL}_2^*}.$$

Integrate with $g = \gamma_{n,R}$ to obtain (115.5). Neither a pointwise floor nor division by a zero fiber mass has been used; an arbitrary version on $p = 0$ has zero weight in this integral. □

115.3. **Observations, randomization and the exact selection event.** A Markov kernel $\mathscr{K}_{n,R}(z, do)$ from $\mathcal{O}_n$ to a standard Borel output space represents any observation of the complete record with independent randomization. It may, for example, output a label, a roof interval, an acceptance decision and the random seed. The output law is formed with this very kernel under both the true law and the arithmetic reference. A procedure observing the path itself is not a kernel of the record alone.

Theorem 115.3 (Observation and postselection with a mass budget)**.** *Apply any such kernel to the two probability measures in* (115.4)*, leaving the paths unchanged. Their output–path mixed norm is at most Δ_n, uniformly over all these kernels. Let A be a measurable output event and suppose its reference probability is at least $r_n > \Delta_n$. Then its true probability is at least $r_n - \Delta_n$, and the conditional laws obey*

$$d_{\mathrm{TV}}(\textit{true output} \mid A, \textit{reference output} \mid A) \le \frac{\Delta_n}{r_n - \Delta_n}, \tag{115.6}$$

$$\|\textit{true output–path law} \mid A - \textit{reference output law} \mid A \otimes \mathsf{H}_R\|_{\mathrm{obs,BL}} \le \frac{2\Delta_n}{r_n - \Delta_n}. \tag{115.7}$$

The true conditional mean of the same-output pair-bridge distance from H_R is bounded by the right side of (115.7)*. Hence all these errors vanish for any selection family with $\Delta_n/r_n \to 0$.*

Proof. An admissible output–path test pulls back to $\int \phi(o, x, y)\mathscr{K}_{n,R}(z, do)$, which is still bounded by one and one-Lipschitz in the two paths, measurably in z. This proves contraction of the mixed norm, without any regularity of the observation kernel in the record coordinates. Under the reference, the graph bridge remains independent of the output because the kernel does not read the paths.

Write α, β for the true and reference probabilities of A. Their difference is at most Δ_n, since the indicator of A is an admissible constant path test. Restrict both joint laws to A. Their mixed-norm error is at most Δ_n. Adding and subtracting the reference restriction divided by α shows that normalization costs at most $(\Delta_n + |\alpha - \beta|)/\alpha \le 2\Delta_n/(r_n - \Delta_n)$. On output marginals this is an L^1 variation bound, so division by two gives (115.6).

Disintegrate the output–path law. At each output use the last pointwise inequality in the proof of Theorem 115.2, now with the reference output density as g with respect to a common dominating output measure. Its integral on A is at most $2\Delta_n$. Division by $\alpha \ge r_n - \Delta_n$ proves the conditional-mean assertion. The reference output measure can be discrete, continuous or mixed; no roof representative is needed in this argument. □

Interpretation of selection. The event in Theorem 115.3 is the full output selection event. For an interval chosen from the observed record, conditioning on its reported value is not silently replaced by conditioning only on membership in that interval. Both laws use the same kernel, including all information carried by selection. Arbitrary path-dependent acceptance need not preserve a Gaussian bridge amplitude and is not asserted to do so. Nor does the qualitative sequence Δ_n provide a rate for a prescribed microscopic singleton whose mass is much smaller than Δ_n.

Proof route and the original endpoint. The route to (115.4) consists of four inputs: centered positive pressure and the physical visible-branch pairing; the resulting uniform Gaussian envelope for the full transition sum; the inherited complete-source scalar and path local-variation laws; and the actual joint clock coupling. The scalar fourth moment pays the discarded tails in the proof. All cutoffs disappear from the final probability measures. The two paths remain coupled, with the second one obtained from the first by A_R in the limit.

This integrated mixed norm does not control a density spike at an individual roof, even though it controls all bounded observations of the full record. The incidence and clearance essential-height conditions and unrestricted pointwise bridges remain the same targets as in Corollary 181.5. The arithmetic transition kernel is permanent in the uniform theorem; an unmodulated specialization requires the separate concrete residue criterion.

Part 37. Action, arithmetic and the physical local law

116. The local theorems and their proof routes

There are two local questions in this article. The stationary physical question is a lattice singleton in three coordinates, evaluated at a deterministic physical time. The actual-return question is a four-coordinate mixed density with three counting coordinates and a pointwise roof coordinate. They have different regularizing mechanisms. The first uses the actual stationary flight age; the second requires the extracted raw return correction. We keep both records, rather than identify one with the other.

116.1. **Comparison at the level of theorem statements.** The Lorentz-process local theorem of Szász–Varjú [2] concerns the periodic cell process and its recurrence, using a Young-tower local limit framework. Its existence already prevents interpreting a Gaussian cell local limit at a fixed table as a new phenomenon here. The endpoint version in [12, Theorem 2.2 and Remark 2.3] gives a mixing local limit for the two-dimensional cell index with piecewise Hölder endpoint observables, including in randomly deformed configurations. Thus endpoint factors and geometric perturbations are not by themselves new local-limit principles. That theorem is not, as stated, the joint three-frequency displacement–roof estimate used in this article. The new work required for that use is the accumulated roof action in all three norms and its uniform compact-frequency arithmetic.

The suspension theory of Dolgopyat–Nándori [3, Definition 2.2, Theorem 3.7 and Proposition 6.3] starts from a mixing local limit and the appropriate closed arithmetic group, and includes finite-horizon Sinai billiards among its applications. At a fixed parameter, the physical conclusion here belongs to that framework once the joint observable, its endpoint coboundaries and its minimality hypotheses have been verified. The scalar billiard application there is not a literal statement of our compact-family vector theorem, but neither is the suspension step claimed as a new mechanism. Our proof supplies the needed collision input directly and tracks its uniformity through the actual age integration.

The collision count is also not a new arithmetic group merely because it is recorded at a continuous time. If x_s is the last outgoing collision state at time s, the starting age is a, the ending state is $y = T^m x$, and the ending age is $v = a + t - S_m$, then exactly

$$\int_0^t \frac{ds}{\tau(x_s)} = m - \frac{a}{\tau(x)} + \frac{v}{\tau(y)}. \tag{116.1}$$

The incomplete first and last flights prove this identity, with the same formula when $m = 0$. Thus the count is an additive suspension reward up to a bounded endpoint coboundary. The arithmetic to check is the joint lattice/roof group with these endpoints. Our finite-cover argument performs that check without a rotation of the obstacle.

There is no claim that a singleton in a discrete lattice coordinate is a stronger topology than a mixing lattice local limit: a singleton indicator is continuous on the lattice. Nor is a fixed roof interval silently replaced by a pointwise roof density. The positive age overlap turns the physical event into an admissible interval-type endpoint test. That regularization is absent for the raw actual-return density. The source total-variation conclusion has a different purpose: after a positive microscopic denominator is proved, it controls every bounded test on the unchanged conditioned trajectory. It does not supply an arbitrary selector's Gaussian amplitude.

The contribution developed in this part is consequently a specific combination of verifiable statements: an action-weighted norm argument for the unsmoothed roof, a rotation-free finite-cover exclusion of joint phases, a compact-family collision-to-physical local theorem, and a quantitative source comparison after normalization. The ellipse family verifies these ingredients

beyond the circular parameter line. No claim of priority for general local inversion, abstract suspension local limits, or perturbations of dispersing tables is needed for these statements.

116.2. **A short route through the article.** The leading physical theorem can be checked in the following order. Section 117 specifies the exact norms, partition conditions, Jacobian sums, action estimates and physical peripheral representation. Section 119 excludes the joint phase by contact holonomy and finite-cover mixing. Section 120 derives the compact-family local law, verifies the ellipse geometry, and proves the normalized source comparison. Its only imported dynamical inputs are the unweighted finite-horizon collision framework of [8, 4], the piecewise multiplier criterion of [12], and the product/mixing construction of [6].

Sections 122–124 retain the detailed circular collision, endpoint-measure and age-overlap proof. The exact physical identity can also be checked directly in Section 226; the positive likelihood there is used in Corollary 120.4. The signed-complement conclusion additionally uses the exact Gaussian-plus-complement identity in Section 227; it is a consequence of the independently evaluated positive probability, not a uniform spectral estimate at a growing band.

Part II retains the full actual-return analysis. Its independent unconditional results include the moving-section geometry, exponential return bounds, Gaussian and functional laws, complete joint nondegeneracy and marked moments. Its raw inversion route consists of the critical-edge calculation, finite-count extraction, coherent cutoff transport, and the remaining common pointwise correction and return complement. In particular Theorem 143.1 keeps its original four-coordinate object and its exact analytic hypotheses. The new three-coordinate theorem is not substituted into that statement. The A–X synopsis remains compiled in Appendix E. The original sources and their proof routes remain available even where the direct physical argument bypasses them.

117. Action weights in the three collision norms

This section gives the norm calculation underlying Proposition 122.3. It also specifies the geometric inputs when the scatterer is not a circle. The time parameter is fixed throughout an orbit; a family of tables is not a randomly changing table. We work on a local common space for an unforced finite-horizon periodic billiard with strictly convex C^3 scatterers. Constants are uniform on such a local family with common horizon, separation, curvature and boundary bounds. A finite cover will suffice for a compact family.

117.1. **Norms, partitions and exponents.** We use densities relative to the invariant collision probability. Thus a smooth function h represents $h\nu$, and on densities $\mathcal{L}h = h \circ T^{-1}$. This convention is important: the Jacobian below is a stable-curve Jacobian, not an additional volume Jacobian. Let $\mathcal{W}^s$ be the homogeneous stable curves of length at most δ_0, with the cone and curvature bounds in [4, Section 3.3]. Locally write $W = \{(r, \varphi_W(r)) : r \in J_W\}$ and let G_W be this graph parametrization. Curves in different homogeneity strips have infinite distance; within one strip put

$$d_s(W_1, W_2) = |J_{W_1} \triangle J_{W_2}| + \|\varphi_{W_1} - \varphi_{W_2}\|_{C^1(J_{W_1} \cap J_{W_2})}.$$

The spaces $C^a(W)$ below are the closures of smooth functions in the a-Hölder norm. For smooth h set

$$|h|_w = \sup_{W\in\mathcal{W}^s}\sup_{\|\psi\|_{C^p(W)}\le 1}\left|\int_W h\psi\, dm_W\right|, \tag{117.1}$$

$$\|h\|_s = \sup_{W\in\mathcal{W}^s}\sup_{\|\psi\|_{C^q(W)}\le |W|^{-\varsigma}}\left|\int_W h\psi\, dm_W\right|, \tag{117.2}$$

$$\|h\|_u = \sup_{0<\epsilon\le\epsilon_0}\sup_{d_s(W_1,W_2)\le\epsilon}\sup_{\substack{\|\psi_i\|_{C^p(W_i)}\le 1\\ \psi_1\circ G_{W_1}=\psi_2\circ G_{W_2}}}\epsilon^{-\gamma}\left|\int_{W_1} h\psi_1\, dm_{W_1}-\int_{W_2} h\psi_2\, dm_{W_2}\right|. \tag{117.3}$$

Equality of the graph tests is imposed on the common interval. The strong and weak spaces are the completions of C^1 densities in $\|h\|_s + c_0\|h\|_u$ and $|h|_w$, respectively. Changing $c_0 > 0$ changes the norm, not its completion. These are the conventions in [12, Section 3.1].

The following strict choices are compatible with all the estimates used here:

$$p = \frac{1}{12},\qquad q = \frac{1}{24},\qquad \varsigma = \frac{1}{8},\qquad \zeta = \frac{1}{4},\qquad 0<\gamma\le\frac{1}{100}. \tag{117.4}$$

If necessary decrease γ to meet the local geometric restriction. For the usual homogeneity strips the inverse-Jacobian summability threshold can be chosen as $\zeta_0 = 2/3$. In particular

$$\varsigma < 1-\zeta_0,\quad \gamma < \min\{\varsigma, p-q, 1/3-q, 1-q\},\quad \max\{p,\gamma/(1-\gamma)\} < \zeta < 1/2.$$

The threshold statement uses only the standard quadratic homogeneity strips and the one-step summability in [4, Section 3.1]. The first two restrictions are the norm restrictions; the last is the piecewise multiplier restriction. No equality at a Hölder endpoint is used.

Here is the precise partition condition on a preceding-flight record $\check f = f\circ T^{-1}$. In each homogeneity strip its regularity partition has at most N_0 simply connected pieces. A stable curve intersects any piece in at most K_0 intervals, and

$$m_W\{x\in W : d(x,\partial Z)<\epsilon\} \le C_0\epsilon \tag{117.5}$$

for each partition piece Z. The constants are common to all strips and local parameters. On each piece $\check f$ has a common $C^{1/2}$ bound. A finite refinement by coordinate cuts is permitted, but a number of cuts growing with an orbit length is not part of this condition. The circular root calculation in Lemma 122.1 verifies these conditions; Lemma 120.2 will verify them for ellipses.

Lemma 117.1 (The precise multiplier consequence)**.** *Under this partition condition, $M_{\exp(iz\cdot\check f)}$ is an entire strong-space multiplier. For a compact complex set K and any multi-index β,*

$$\sup_{z\in K}\|\partial_z^\beta M_{\exp(iz\cdot\check f)}\|_{\mathcal{B}\to\mathcal{B}} \le C_{K,\beta}. \tag{117.6}$$

The same bound holds on the weak space.

Proof. The conditions just listed are precisely the bounded-complexity conditions of [12, Lemma 3.3]; its multiplier conclusion applies because of (117.4). Products on each piece obey $\|uv\|_{C^\varsigma} \le \|u\|_{C^\varsigma}\|v\|_{C^\varsigma}$. If the components of $\check f$ have norm at most F, the exponential series in that algebra is bounded by $\exp(CF|z|)$. Its derivative is multiplication by $(i\check f)^\beta\exp(iz\cdot\check f)$ and has norm at most $(CF)^{|\beta|}\exp(CF|z|)$. The fixed partition has not changed. Applying the multiplier bound to the series and its locally uniformly convergent derivatives proves the strong assertion and analyticity. For the weak norm one splits the stable test at the same finite boundaries and uses its C^p bound; alternatively the weak part of the same multiplier estimate gives the assertion. The local boundary reparametrizations have uniformly bounded derivatives and inverses, so the partition and Hölder bounds survive them. □

117.2. **The geometric sums, before inserting a weight.** Write $\mathcal{G}_m(W)$ for the homogeneous inverse pieces, subdivided at the reference length as in [4, Section 3.2], and let $\mathcal{I}_m(W)$ be the pieces without a long ancestor. For $U \in \mathcal{G}_m(W)$ set $j_U = \|J_U T^m\|_\infty$. The following are the unweighted geometric inputs, recorded with their length factors:

$$\sum_{U\in\mathcal{G}_m(W)} j_U \le C_2, \qquad \sum_{U\in\mathcal{G}_m(W)} (|U|/|W|)^a j_U \le C_2^{1-a} \qquad (0 \le a \le 1), \tag{117.7}$$

$$\sum_{U\in\mathcal{I}_m(W)} j_U \le C_1\theta_*^m, \qquad \sum_{U\in\mathcal{G}_m(W)} j_U^{1-\varsigma} \le C_3^m, \tag{117.8}$$

where $\theta_* < 1$. Precisely, the never-long sum is [4, Lemma 3.2(a)], the total sum is its part (b), the length-weighted sum is part (c), and the inverse-Jacobian power sum is part (d), on page 12 of the pinned arXiv:1210.1261v1 text. The last inequality uses $1 - \varsigma > \zeta_0$. Bounded distortion gives $\|J_U T^m / j_U\|_{C^p(U)} \le C$. Jensen's inequality and $\sum_U j_U|U| \le C|W|$ also give

$$\sum_{U\in\mathcal{I}_m(W)} (|U|/|W|)^\varsigma j_U \le C\theta_*^{(1-\varsigma)m}. \tag{117.9}$$

We retain these sums as geometric theorems, not consequences of a finite-word computation. All their constants are independent of a Fourier frequency.

The symplectic input needed to insert the roof is

$$d\tau = T^*\vartheta - \vartheta, \qquad \vartheta = \sin\varphi\, dr. \tag{117.10}$$

It follows by differentiating the length of a regular free flight at its two endpoints. The tangential component of the velocity is unchanged by reflection. Thus the derivative at the terminal endpoint is the outgoing tangential momentum there, and the derivative at the initial endpoint is its negative. This proves the formula also for noncircular scatterers.

For real (u, b) with $|b| \le B$ put $w_m = \exp\{i(u \cdot K_m + b(S_m - m\bar\tau))\}$. On a regular branch K_m is constant. For an arc γ joining x to y and regular through time m, integration of (117.10) gives

$$S_m(y) - S_m(x) = \int_{T^m\gamma} \vartheta - \int_\gamma \vartheta. \tag{117.11}$$

Uniform boundedness of ϑ and stable contraction therefore bound $\|w_m|_U\|_{C^1}$ by C_B, independently of m.

For completeness the matching conditions used here are the following. The untrimmed inverse graphs are joined by the unstable-cone foliation of [4, Lemma 3.3 and Section 4.3]. Every intermediate connector and both graphs lie in one regular branch and one homogeneity strip; a connector cut by a singularity is placed in the unmatched family. The construction gives

$$|\gamma_x| \le C\Lambda^{-m}\epsilon, \quad |T^m\gamma_x| \le C\epsilon, \quad |T^i\gamma_x| \le 2\max_{j=1,2}|T^iU_j| \quad (0 \le i \le m).$$

In particular the two endpoints have the same lattice itinerary. These are the hypotheses of the matched Jacobian estimate, rather than an additional claim about all nearby orbits. Trimming two such graphs to the common base interval moves an endpoint by at most $C\epsilon$ on a stable graph, and its image by at most $C\Lambda^{-m}\epsilon$. Equation (117.11) consequently gives

$$\|w_m^{(1)} - w_m^{(2)}\|_\infty \le C_B\epsilon, \qquad \|w_m^{(1)} - w_m^{(2)}\|_{C^q} \le C_B\epsilon^{1-q}. \tag{117.12}$$

The latter follows by interpolation with the individual Lipschitz bounds in common graph coordinates. It is valid at every length m. No forward flight singularity or homogeneity boundary has been removed: all such cuts are already in the inverse-curve partition. The weight adds none because its only discrete label is constant on a regular collision branch.

117.3. **The weighted inequalities with all length factors.**

Proposition 117.2 (Action-weighted collision inequalities)**.** *For each fixed $B < \infty$, the exact operator $Q_\omega = M_{\exp(i\omega\cdot \check f^{c})}\mathcal{L}$ satisfies*

$$|Q_\omega^m h|_w \le C_B |h|_w, \tag{117.13}$$

$$\|Q_\omega^m h\|_s \le C_B(\Lambda^{-qm} + \theta_*^{(1-\varsigma)m})\|h\|_s + C_B|h|_w, \tag{117.14}$$

$$\|Q_\omega^m h\|_u \le C_B\Lambda^{-\gamma m}\|h\|_u + C_B C_3^m\|h\|_s. \tag{117.15}$$

Here C_3 may be enlarged. The exponents and geometric constants are independent of B; the multiplier constants are not.

Proof. It suffices first to use smooth densities. A pairing on a terminal curve is the sum of

$$\int_U h\, w_m J_U T^m(\psi\circ T^m)\, dm_U.$$

The normalized Jacobian and w_m have uniformly bounded C^p norms, where the latter bound is C_B. The first sum in (117.7) proves (117.13).

For the stable norm let a_U be the arclength average of $\psi\circ T^m$ on U. Then

$$\|\psi\circ T^m - a_U\|_{C^q(U)} \le C\Lambda^{-qm}|W|^{-\varsigma}.$$

The contribution of these differences is bounded by

$$C_B\Lambda^{-qm}\|h\|_s \sum_U (|U|/|W|)^\varsigma j_U \le C_B\Lambda^{-qm}\|h\|_s.$$

On never-long pieces the average contribution is bounded by $C_B\|h\|_s$ times (117.9). For the remaining pieces group by the most recent long ancestor $V \in \mathcal{G}_k(W)$, with $|V| \ge \delta_0/3$. Descendants with no later long ancestor have Jacobian sum at most $C\theta_*^{m-k}$ relative to V. The smooth average, weight and normalized Jacobian are paid by the weak norm, exactly as for the average term in [4, equations (4.9)–(4.11)]. Keeping the length normalization, the resulting bound is at most

$$C_B\delta_0^{-\varsigma}|h|_w \sum_{k=0}^{m}\theta_*^{m-k}\sum_{V\in\mathcal{G}_k(W)}(|V|/|W|)^\varsigma j_V \le C_B|h|_w.$$

The $k=0$ term is included only if the root is long. This proves (117.14). In particular the noncontracting derivative of w_m has not been multiplied by $\|h\|_s$ in its leading term.

We spell out which unstable errors the last geometric sum pays. An unmatched piece V has $|T^m V| \le C\epsilon$; bounded distortion implies $j_V|V| \le C\epsilon$. Its stable-norm contribution is at most $C_B j_V |V|^\varsigma\|h\|_s$. Hence all unmatched pieces cost at most

$$C_B\epsilon^\varsigma\|h\|_s\sum_V j_V^{1-\varsigma} \le C_B\epsilon^\varsigma C_3^m\|h\|_s.$$

After division by ϵ^γ this is admissible. On a matched pair the old graph/test comparison, with one weight attached, has the leading term $C_B\epsilon^\gamma\Lambda^{-\gamma m}\|h\|_u$. The normalized Jacobian and graph reparametrization errors have C^q size bounded by $C(\epsilon^{p-q}+\epsilon^{1/3-q})$ in the matching calculation of [4, Section 4.3]. Their stable-norm sums, with each Jacobian factored out, are bounded by $CC_3^m\|h\|_s$. The additional difference of the two weights has C^q size $C_B\epsilon^{1-q}$ by (117.12), and the same Jacobian/length sum bounds its stable contribution. Each of $\varsigma, p-q, 1/3-q, 1-q$ exceeds γ. Dividing the total difference by ϵ^γ proves (117.15). This calculation retains the exponential stable cross coefficient. Completion of smooth densities gives the result. □

Fix B before making the following choices. Choose N so that both leading coefficients in (117.14) and (117.15) are less than $1/4$. Then choose $c_B > 0$ so that $c_B C_B C_3^N < 1/4$. In the equivalent norm $\|h\|_B = \|h\|_s + c_B\|h\|_u$,

$$\|Q_\omega^N h\|_B \le \tfrac{1}{2}\|h\|_B + C'_B|h|_w.$$

Iteration and (117.13), followed by the finitely many remaining powers, prove power boundedness. Compact embedding of the strong unit ball in the weak space gives a common essential radius less than one for this fixed band. No estimate for the growth of N, c_B^{-1} or the spectral constants as $B \to \infty$ is implied.

117.4. **From a peripheral distribution to a physical phase.** We isolate a small functional-analytic detail. The weak completion alone must not be identified with a faithful space of global functionals without checking that identification.

Lemma 117.3 (Faithfulness on the strong completion)**.** *If $h \in \mathcal{B}$ vanishes on all smooth global tests, then $h = 0$.*

Proof. Fix a stable graph compactly inside one homogeneity strip and a smooth test supported in its interior. Its small transverse translates are admissible stable graphs after choosing the cone and curvature bounds with a strict margin. Give them the same graph-coordinate test. The unstable norm bounds the difference of their pairings by $C\|h\|_u|v|^\gamma$, where v is the transverse translation. Average these pairings against a smooth transverse kernel of width ϵ. On smooth densities the average is a global test pairing: the graph arclength Jacobian is included in the test and the collision density is divided out. This density is smooth and bounded below on the compact strip under consideration. The resulting test is C^1; smooth approximation and the weak pairing bound permit its use for $h \in \mathcal{B}$. By the hypothesis the average is zero. Letting $\epsilon \downarrow 0$ makes the original graph pairing zero.

General little-Hölder tests are limits of smooth tests. Trimming endpoints has error at most $C\|h\|_s \ell^{\varsigma - q}$ on a trimmed interval of length ℓ. Curves on a boundary of the admissible family can first be approximated by ones with strict cone and curvature margins, using the unstable comparison and the same trimming estimate. We now conclude in the strong norm, without using an injective inclusion. For a fixed admissible W, its pairing is continuous in the little $C^q(W)$ norm by (117.2). Approximate each such test in that norm by smooth tests, after the endpoint and curve approximations just described. The pairings therefore vanish for every C^q stable test, not merely for the C^p tests of the weak norm. Their normalized supremum is zero, so $\|h\|_s = 0$. Every difference in (117.3) is now a difference of two zero pairings. For each fixed pair and each fixed $\epsilon > 0$ its normalized value is zero. Taking the defining supremum gives $\|h\|_u = 0$.

These equalities also hold for the completed, rather than just the smooth, norm. Indeed the defining normalized stable pairings and normalized pair differences embed smooth densities isometrically in the product of two spaces of bounded scalar families. A strong Cauchy sequence converges uniformly in those two families. Thus their suprema give the completed seminorms as well; no interchange of an uncontrolled pointwise limit with a supremum is involved. It follows that $\|h\|_s + c_0\|h\|_u = 0$ and hence $h = 0$ in the completion. This argument uses transverse regularity of the strong space and asserts no faithfulness for arbitrary weak-completion vectors. The independent inclusions recorded in [12, Lemma 3.2(c), footnote 7] are consistent with, but are not used to prove, this conclusion. □

Lemma 117.4 (Bounded physical representatives of peripheral vectors)**.** *Let Q_ω be power bounded and quasi-compact as above. If $|\lambda| = 1$ is an eigenvalue, a nonzero peripheral vector is represented by $q\nu$ with $q \in L^\infty(\nu)$. After multiplication by a constant, $|q| = 1$ and*

$$q \circ T = \lambda^{-1} e^{i\omega \cdot f^c} q \quad \textit{almost everywhere.} \tag{117.16}$$

Proof. Power boundedness excludes Jordan blocks at modulus one. The Cesaro averages $N^{-1}\sum_{j<N}\lambda^{-j}Q_\omega^j$ converge in the strong space to the spectral projection P_λ. Since C^1 densities are dense and its range is finite dimensional, $P_\lambda g \neq 0$ for some smooth bounded density g (not necessarily positive). On physical densities every summand has modulus at most $\|g\|_\infty$. Thus for smooth global ϕ,

$$|(P_\lambda g)(\phi)| \le \|g\|_\infty \|\phi\|_{L^1(\nu)}.$$

Extension by density in L^1 supplies $q \in L^\infty(\nu)$. It is nonzero by Lemma 117.3.

To justify the phase equation without testing an abstract distribution on a discontinuous pullback, regard the same averages as bounded physical densities. Their weak-star limits are precisely q, because smooth global tests are dense in L^1. Weighted composition by T^{-1} is weak-star continuous on L^∞, with its usual L^1 predual. The difference between its action on the average and λ times the average has supremum norm at most $2\|g\|_\infty/N$. Taking the weak-star limit proves the physical eigen-equation and hence (117.16). Ergodicity makes $|q|$ a positive constant. Division by that constant finishes the proof. □

The arithmetic result in the next section now applies to an actual measurable function, not an anisotropic distribution. Exclusion of nonzero peripheral frequencies, local strong-to-weak continuity, and the fixed-band inequalities yield a uniform spectral gap on each compact set avoiding zero. For parameter continuity the one-step comparison uses the same matched and unmatched curves as [4, Section 5]. On matched pieces the preceding label agrees and the explicit root is jointly 1/2-Hölder in state and parameter; interpolation gives convergence in C^q. The unmatched short pieces are bounded by the same small-length sum. This explains exactly where the parameter-uniform preceding-root condition is used. No continuity of a long singular itinerary is required.

118. Finite-cover mixing and the hypotheses of the action principle

The arithmetic argument uses an ergodic assertion on actual finite covers, in addition to local collision geometry. We give its precise source and keep its constants distinct from the parameter-uniform constants used on the base table.

Proposition 118.1 (Mixing of the physical finite covers)**.** *Fix a positive integer p. On the torus $\mathbb{R}^2/(p\Lambda)$, the collision map of each circular or oriented-ellipse table considered in this article is mixing for normalized collision measure. The same holds for the support-function tables in Proposition 121.1. For each fixed p, the correlation-decay theorem applies to the corresponding compact family. No bound uniform as $p \to \infty$ is asserted or needed.*

Proof. The external input is [18, Theorem 3], in the precise configuration class $\widetilde{\mathcal{K}}$ defined immediately before its Theorem 2 (arXiv:1210.0011v4, pages 6–7). For a fixed finite collection of $C^{3+\mathrm{Lip}}$ strictly dispersing obstacles, with bounded boundary derivatives, curvatures bounded away from zero and infinity, positive minimal flight length, and a common (t_0, φ_0) horizon, it gives

$$\left| \int f \circ T^m \, g \, d\nu - \int f \, d\nu \int g \, d\nu \right| \le C_{f,g}\theta^m, \qquad \theta < 1, \tag{118.1}$$

for Hölder f and 1/6-Hölder g. This is a theorem for the collision map, not just its suspension flow. Its proof is given in Section 8.2 of that paper. The constant-configuration specialization of its memory-loss theorem is an equivalent route to (118.1).

We check the geometric hypotheses on each cover. Its planar lift is exactly the original obstacle array. There are finitely many embedded, pairwise disjoint obstacles on the cover. The circle and ellipse boundaries are analytic; on the compact ellipse parameter set their arclength derivatives through order four and their positive curvatures have common bounds. The gap is at least 3/50.

Every obstacle contains the radius-9/20 disk. A line avoiding the interiors of those disks would be a corridor in the smallest circular table, which has none. Compactness of initial position and direction then gives a bounded length containing a strictly transverse hit, uniformly in the starting state. More explicitly, choose a transverse hit for each state, keep that hit in a neighborhood of the state, and take a finite cover. The largest hit time and the smallest positive incidence angle in this finite cover give t_0 and φ_0. Alternatively, first obtaining a common positive depth of entry into a disk gives the same bounds. Containment and the common curvature bounds give such a transverse hit for every member of the compact family. Thus this is the (t_0, φ_0) horizon required in [18], not merely an upper bound for almost-everywhere free flights.

The theorem and its local Euclidean collision proof apply on a fixed flat torus. One can represent the triangular cover by the rectangular Euclidean cover with periods $(p, 0)$ and $(0, p\sqrt{3})$ and $2p^2$ obstacles, then pass to its deck quotient. The metric is the inherited Euclidean metric; no nonisometric affine conjugacy of specular reflection is performed. Changing the fixed periods only changes the finite configuration space and the constants in the compactness and coupling argument. The normalized invariant measure is still proportional to $\cos\varphi \, dr \, d\varphi$.

Equation (118.1) proves mixing first for smooth functions. To extend it to $f, g \in L^2$, approximate each in L^2 by smooth functions. Invariance and Cauchy–Schwarz bound the replacement error uniformly in m by

$$\|f - f_j\|_2 \|g\|_2 + \|f_j\|_2 \|g - g_j\|_2.$$

Take $m \to \infty$ before $j \to \infty$. The resulting mixing excludes all unit-modulus eigenvalues except 1, and also implies ergodicity. This verifies exactly the two alternatives in the finite-character proof: mixing is used when the constant-step character is nonzero; ergodicity suffices when it is zero. Connectedness of the free region is only one table hypothesis, not a proof of either conclusion. □

Separated inputs and their uses. The hypotheses of Theorem 120.1 are now stated separately in Section 120. Local common spaces and unweighted geometric estimates yield the collision splitting. The fixed preceding-flight partition yields the analytic multiplier. The action identity and regular matching yield the three weighted inequalities. Finite-cover mixing excludes physical peripheral phases. Strong-to-weak parameter continuity gives spectral uniformity on a fixed compact frequency set. Observation-cell regularity supplies the varying endpoint tests. Continuity and positive bounds for the mean roof give the stationary normalization. The arithmetic step fixes a cover before applying its mixing theorem; compactness on the base parameter family is used only after pointwise peripheral exclusion. No summation of estimates over the covers occurs.

Pinned norm references. The four unweighted sums in (117.7)–(117.8) are [4, Lemma 3.2(a)–(d), p. 12], respectively with the assignment made just after those displays. The matched graph and stable-Jacobian estimates are [4, Lemma 3.3, pp. 13–14]; its three hypotheses are regular intermediate stable curves in a common homogeneity strip, a regular unstable-cone matching foliation, and the intermediate connector-length bound. The weak, stable and unstable calculations are [4, Proposition 4.1, equations (4.1)–(4.3), (4.7)–(4.11), (4.14), (4.18)]. These page and equation numbers refer to arXiv:1210.1261v1. The piecewise multiplier used here is [12, Lemma 3.3(b), p. 13], with its partition conditions (1)–(2) and the strict inequality $\zeta > \max\{p, \gamma/(1-\gamma)\}$. The latter numbering refers to arXiv:1902.06850v1, not to a differently paginated journal copy. These are imported continuum estimates; source hashes and finite diagnostics do not prove them.

119. Finite-cover arithmetic without rotational symmetry

The circular proof of measurable phase rigidity used rotations to eliminate a constant collision phase. The constant can instead be included in the integer cocycle. This gives an arithmetic argument which is unchanged by a noncircular deformation of the scatterer.

Let T be an invertible ergodic collision map on a periodic planar billiard, $\kappa : M \to \mathbb{Z}^2$ its lattice increment, and τ its flight time. We use the regular local stable–unstable product rectangles of the finite-horizon dispersing theory, with absolute continuity of both conditional measures, as in [6, Section 8]. The action identity is (117.10). Assume that on every finite lattice cover $p\Lambda$, p prime, the collision map is mixing. These are geometric and ergodic hypotheses; no twisted spectrum or local limit theorem is assumed.

Lemma 119.1 (A finite character separates a proper subgroup)**.** *If L is a proper subgroup of $\mathbb{Z}^d$, there are a prime p and a nonzero homomorphism $\chi : \mathbb{Z}^d \to \mathbb{Z}/p\mathbb{Z}$ which vanishes on L.*

Proof. The quotient $\mathbb{Z}^d/L$ is a nonzero finitely generated abelian group. If it has a nonzero free summand, reduction of one such coordinate modulo any prime gives a quotient of order p. Otherwise one of its nonzero finite cyclic summands has order divisible by a prime p; reduction of that summand gives the required quotient. Compose with the quotient map from $\mathbb{Z}^d$. This also covers subgroups of deficient rank without assuming they have finite index. □

Theorem 119.2 (Complete physical phase rigidity)**.** *Suppose a measurable circle-valued function q satisfies*

$$q \circ T = e^{i(u\cdot\kappa+s+b\tau)}q, \qquad u \in \mathbb{R}^2,\ s, b \in \mathbb{R}. \tag{119.1}$$

Then $b = 0$, $s \in 2\pi\mathbb{Z}$, $u \in (2\pi\mathbb{Z})^2$, and q is almost everywhere constant. No rotational or reflection symmetry of the table is required.

Proof. We first recall the measurable holonomy argument, including the step which removes the roof frequency. Choose a positive-measure compact regular product rectangle P away from grazing. Conditional pair measures on a common local stable or unstable leaf have both marginals dominated by $C_P\nu$. Approximate q in $L^1(\nu)$ by smooth bounded functions q_ϵ. Stable contraction and invariance give

$$\int |q(T^n y) - q(T^n x)|\, d\pi^s(x, y) \longrightarrow 0.$$

Indeed the replacement of the two occurrences of q costs at most $2C_P\|q - q_\epsilon\|_1$, uniformly in n, and the difference of q_ϵ tends to zero along the contracting leaves. Take n to infinity before removing the approximation. Along these leaves the lattice itinerary is identical, so iteration of (119.1) and the action identity give, almost everywhere on stable pairs,

$$q(y)/q(x) = \exp\left(ib \int_{[x,y]^s} \vartheta\right).$$

Backward iteration gives the same formula on unstable pairs. The pair disintegration and Fubini's theorem permit their multiplication around almost every small stable–unstable rectangle Q in P. Consequently $\exp(ib\int_Q d\vartheta) = 1$. Since $d\vartheta = \cos\varphi\, d\varphi \wedge dr$ is nonvanishing on this rectangle, nondegenerate rectangles of arbitrarily small positive area contradict this equality unless $b = 0$. This is the argument of Section 149, using the general one-form and not a circular parametrization.

For $b = 0$ both leaf ratios are one. Product absolute continuity and Fubini imply $q = c$ almost everywhere on P for some $|c| = 1$. Discarding the invariant null set on which an iterate of the phase equation fails, the orbit saturation of P has full measure. Hence q/c takes values in the countable group

$$\{\exp(i(u \cdot k + sn)) : (k, n) \in \mathbb{Z}^2 \times \mathbb{Z}\}.$$

Define its kernel $L = \{(k, n) : u \cdot k + sn \in 2\pi\mathbb{Z}\}$. Enumeration of the countable quotient defines a measurable function $h : M \to \mathbb{Z}^3/L$ by $q/c = \exp(i(u, s) \cdot h)$. Uniqueness in the quotient gives the exact measurable relation

$$h \circ T = h + [(\kappa, 1)]. \tag{119.2}$$

If L were proper, Lemma 119.1 would give $\chi(k, n) = a \cdot k + c_0 n \pmod p$, with $(a, c_0) \neq 0$. It descends to $\mathbb{Z}^3/L$. On the actual p-lattice cover, written as $(x, l) \in M \times (\mathbb{Z}/p\mathbb{Z})^2$, the collision map is $\widetilde{T}(x, l) = (Tx, l + \kappa(x))$. Put

$$H(x, l) = \exp\left(\frac{2\pi i}{p}(a \cdot l - \chi(h(x)))\right).$$

Equation (119.2) yields $H \circ \widetilde{T} = e^{-2\pi i c_0/p} H$. If $c_0 \neq 0$, this is a nontrivial unit-modulus eigenvalue, excluded by mixing of the finite cover. If $c_0 = 0$, then $a \neq 0$ and the sheet average of H is zero, so this is a nonconstant invariant function, excluded by ergodicity. Both cases are impossible. Thus $L = \mathbb{Z}^3$, which gives the asserted values of u, s. Finally (119.1) reduces to invariance of q. □

Corollary 119.3 (Nondegeneracy from the same arithmetic)**.** *Suppose the bounded collision observable $f^{\mathrm{c}} = (\kappa_1, \kappa_2, \tau - \bar{\tau})$ has zero mean and exponentially summable covariances. Its Green–Kubo matrix Σ is positive definite. For a compact parameter family with common correlation bounds and continuous finite-lag covariances, the matrices are continuous and have a common positive lower eigenvalue.*

Proof. Let $v \cdot \Sigma v = 0$ and put $g = v \cdot f^{\mathrm{c}}$. Exponential summability gives $\sum_{k \geq 1} k |\int g(g \circ T^k) d\nu| < \infty$. The finite covariance expansion then bounds $\|S_m g\|_2$ uniformly in m. The averages $b_M = M^{-1} \sum_{m=1}^{M} S_m g$ are bounded in L^2 and satisfy $(I - U)b_M = g - M^{-1} \sum_{m=1}^{M} U^m g \to g$, where $Ug = g \circ T$. A weakly convergent subsequence supplies $g = b - b \circ T$ with a real $b \in L^2$. For every real t, exponentiating $-itb$ gives (119.1) with frequencies (tv_1, tv_2), $-tv_3\bar{\tau}$, and tv_3. Theorem 119.2 forces $v_3 = 0$ and $t(v_1, v_2) \in (2\pi\mathbb{Z})^2$ for every t, hence $v = 0$. Uniform covariance convergence and finite-lag continuity give continuity of Σ. Compactness supplies the positive minimum eigenvalue. No common measurable representative of the transfer functions, or uniform size of the local product rectangle, is needed for this last compactness argument. □

Remark 119.4 (The section occupation is not inserted into the twist)*.* The integer collision step in (119.2) is the constant 1, not the discontinuous occupation $\mathbf{1}_{Y^*}$. The theorem strengthens the arithmetic used for the physical collision–roof spectrum. It does not provide a Lasota–Yorke estimate for the accumulated section occupation or for unbounded induced return twists. The common pointwise return correction and the return frequency complement in Theorem 143.1 remain distinct analytic requirements for the original four-coordinate problem.

120. A compact-family local principle and noncircular tables

We formulate the consequences of the action and finite-cover arguments before specializing them. The hypotheses concern collision geometry, not a local limit theorem for an induced map. The original circular family is contained in the explicit family at the end of the section.

120.1. **Admissible compact families.** Let $\mathcal{K}$ be a compact parameter space. For $\lambda \in \mathcal{K}$ place translates of a strictly convex scatterer O_λ at the points of a fixed planar lattice Λ, with one scatterer per fundamental cell. Use Euclidean billiard reflection and unit speed; lattice coordinates, rather than physical Cartesian coordinates, are used for displacement. Denote the perimeter by L_λ and normalize its arclength coordinate by $\alpha = 2\pi r/L_\lambda$. The collision probability is then the common measure

$$d\nu = (4\pi)^{-1} \cos\varphi \, d\alpha \, d\varphi.$$

We impose the following geometric conditions.
Uniform table geometry. The boundary parametrizations form a continuous C^3 family with positive lower and upper curvature bounds, a positive separation between distinct scatterers, and a common finite horizon. The free region in every finite lattice cover is connected. This last condition is topological; it is not used as a synonym for mixing.
Local spaces and unweighted geometric estimates. On a finite parameter cover there are local common collision spaces with the norms of Section 117, faithful strong completion, compact strong-to-weak embedding and the unweighted spectral splitting. The stable contraction, distortion, normalized Jacobian and length sums, and product absolute continuity are those specified in that section. Constants are uniform on the finite cover; one global Banach space over all parameters is not required.
Action and matching. The identity $d\tau = T^*\vartheta - \vartheta$ holds on regular branches, and the matched connectors remain regular at every intermediate iterate as in (117.11) and (117.12). The unweighted matching partition contains the required singularity and homogeneity cuts.
Finite-cover dynamics. For each prime p, the collision map of the actual $p\Lambda$ cover is mixing for its normalized collision measure. This is an independent hypothesis in the general principle. No mixing rate uniform in p is required. Proposition 118.1 verifies it for the circular and oriented-ellipse applications by an explicit correlation-decay theorem, not from connectedness alone.
Parameter continuity and normalization. The collision operators are strongly-to-weakly continuous in the local spaces. The preceding-flight comparison below supplies that continuity for the twists. The mean roof is continuous, bounded above, and bounded away from zero. Finite-lag covariance integrands converge almost everywhere and have common integrable bounds. Together with uniform summable correlation tails, this is the covariance continuity used in the compactness argument.
Preceding-flight regularity. The preceding displacement and roof satisfy the partition conditions (117.5), with a common number of pieces and a uniform piecewise $C^{1/2}$ norm. On matched one-step branches the roof is jointly 1/2-Hölder in the parameter and state in local coordinates. Finite itineraries and these functions converge outside the collision singularities as the parameter varies. No estimate on the complexity of an arbitrary long forward occupation word is part of this condition.
Regular observation geometry. Fix a convex polygonal fundamental cell. The labels of the cells met by a single flight belong to a fixed finite set. Write $I_{\lambda,d}(x) \subset [0, \tau_\lambda(x))$ for the ages at which a flight from collision state x lies in cell d. The endpoints of these intervals vary continuously, locally uniformly outside closed neighborhoods of a finite union of null curves and boundary pieces. For every convergent parameter sequence the neighborhoods can be chosen with arbitrarily small limiting collision measure and with null boundary. Include grazing, singular flights, cell tangencies and vertex crossings among the exceptional sets. Empty intervals are allowed. This is a condition on one flight, not on the long orbit.

Set $f_\lambda^{\mathrm{c}} = (\kappa_\lambda, \tau_\lambda - \bar\tau_\lambda)$, where $\bar\tau_\lambda = \int \tau_\lambda d\nu$. Time reversal gives $\int \kappa_\lambda d\nu = 0$. Let Σ_λ be its collision Green–Kubo matrix and put

$$D_\lambda^{\mathrm{clk}} = \mathrm{diag}(1, 1, -1/\bar\tau_\lambda), \qquad \mathcal{V}_\lambda = \bar\tau_\lambda^{-1} D_\lambda^{\mathrm{clk}} \Sigma_\lambda (D_\lambda^{\mathrm{clk}})^{\mathsf{T}}. \tag{120.1}$$

The matrix D^{clk} is a clock-change map, not the four-coordinate return covariance D_R used elsewhere in the article.

Theorem 120.1 (Uniform local principle for an action family)**.** *For a family satisfying these conditions, Σ_λ and $\mathcal{V}_\lambda$ exist, are continuous and uniformly positive definite. Under the stationary*

suspension probabilities $d\mu_\lambda(x,a)=\bar\tau_\lambda^{-1}d\nu(x)\,da$, *for every* $M<\infty$,

$$\sup_{\substack{\lambda\in\mathcal K,\ k\in\mathbb Z^2,\ m\ge0\\ |\xi_{t,\lambda}(k,m)|\le M}}\Big|t^{3/2}\mu_\lambda\{W_\lambda(t)=k,C_\lambda(t)=m\}-g_{\mathcal V_\lambda}(\xi_{t,\lambda}(k,m))\Big|\longrightarrow0, \tag{120.2}$$

where $\xi_{t,\lambda}(k,m)=(k/\sqrt t,(m-t/\bar\tau_\lambda)/\sqrt t)$. *The same formula with bounded regular initial and terminal flight-state selectors has Gaussian amplitude equal to the product of their stationary means. Here regular means jointly continuous bounded families, or bounded families continuous off the closed exceptional neighborhoods specified above, with uniform convergence on their complements. Indicator selectors must have null limiting boundaries. A uniform positive selected denominator is asserted only for nonnegative selectors whose product of stationary means is bounded below by a positive number.*

Proof. We give the entire logical route and the order of its constants. The local unweighted spectral splitting and Lemma 117.1 realize

$$Q_{\lambda,z}=M_{\exp(iz\cdot(f_\lambda^{c}\circ T_\lambda^{-1}))}\mathcal L_\lambda.$$

The covariance series is exponentially summable: replace both factors by their preceding-flight versions using invariance, and pair the unweighted splitting between the two bounded strong multipliers. For example each centered scalar component v satisfies

$$\int v(v\circ T^j)d\nu=\ell\big(M_{v\circ T^{-1}}\mathcal N^jM_{v\circ T^{-1}}\nu\big).$$

Finite-lag continuity follows by dominated convergence outside the finitely many singularity preimages. A finite parameter cover makes the correlation tails uniform. Corollary 119.3 now gives continuous uniform ellipticity. This argument does not require a uniform size for the product rectangles on all finite covers.

For a fixed B, Proposition 117.2 gives local uniform power boundedness and quasi-compactness of $Q_{\lambda,\omega}$. The parameter comparison at the end of Section 117 gives strong-to-weak continuity. Lemma 117.4 and Theorem 119.2 exclude every nonzero peripheral frequency in $\mathbb T^2\times[-B,B]$. Near zero, analytic perturbation of the simple eigenvalue and differentiation of its transfer pairing give

$$\log\lambda_\lambda^{\mathrm{sp}}(z)=-\tfrac12z^{\mathsf T}\Sigma_\lambda z+O(|z|^3),\qquad\Pi_\lambda(0)=\nu\ell. \tag{120.3}$$

The linear term vanishes by centering, and the second derivative is exactly the covariance sum just established. Resolvent contours on a finite parameter cover give uniform errors near zero. On any fixed compact frequency set avoiding zero, the strong/weak perturbation argument and a finite covering give a common spectral radius less than one, as in Theorem 122.4.

Here are the measure-theoretic consequences, with no growing-band substitution. For a time test q with integrable compactly supported Fourier transform, torus orthogonality gives the endpoint pairing

$$\begin{aligned}&\int a(x)d(T_\lambda^mx)\mathbf1_{\{K_m=k\}}q(S_m-t)d\nu(x)\\&=\frac{1}{(2\pi)^3}\int_{\mathbb T^2\times\mathbb R}e^{-i[u\cdot k+b(t-m\bar\tau_\lambda)]}\widehat q(-b)\ell(M_dQ_{\lambda,(u,b)}^m(a\nu))\,du\,db.\end{aligned}$$

The minus sign in $\widehat q(-b)$ follows from the argument S_m-t. For fixed support the complement of zero is exponentially small; rescaling $\sqrt m(u,b)$ near zero and using (120.3) gives the Gaussian with endpoint amplitude $\nu(a)\nu(d)\int q$. For an interval J, positive upper and lower Fourier envelopes of band B have masses $|J|\pm2\pi/B$. With nonnegative smooth endpoints multiply their pointwise inequalities by the nonnegative source measure. First take $m\to\infty$ with B fixed, then let $B\to\infty$. This yields the unsmoothed interval limit; one fixed majorant also bounds all local

masses by $Cm^{-3/2}$. Approximation and decomposition into real and imaginary parts extend the endpoint class to continuous complex functions. This is the argument of Theorem 123.3, now with uniform local inputs on $\mathcal{K}$.

In particular, along $\lambda_j \to \lambda_0$ and $(k_j, t_j - m_j\bar{\tau}_{\lambda_j})/\sqrt{m_j} \to \zeta$, the positive endpoint measures converge vaguely on bounded time slabs to

$$m_j^{3/2}(x, S_{m_j} - t_j, T_{\lambda_j}^{m_j} x)_*(\mathbf{1}_{\{K_{m_j}=k_j\}}\nu) \;\rightharpoonup\; g_{\Sigma_{\lambda_0}}(\zeta)\nu \otimes ds \otimes \nu. \tag{120.4}$$

For a varying overlap test use a closed exceptional neighborhood of small limiting measure and null boundary. Portmanteau bounds the source mass of this closed set; the local mass bound and uniform convergence on the complement control the other part. This is why small measure of an arbitrary open set, or pointwise convergence of tests alone, is not the regularity condition.

The exact physical identity is a finite sum over the two cell offsets. Its (d, e) term uses the collision label $k + d - e$ and the nonnegative test

$$F_\lambda^{d,e}(x, s, y) = \int \mathbf{1}_{I_{\lambda,d}(x)}(a)\mathbf{1}_{I_{\lambda,e}(y)}(a - s)\, da.$$

Indeed $s = S_m - t$ and the final age is $a - s$. The original count event is exactly $0 \le a - s < \tau_\lambda(y)$, so no boundary flight is replaced by a completed-return surrogate. The observation geometry makes these bounded, compactly supported tests admissible in (120.4). Their integrated amplitude is

$$\sum_{d,e} \int F_\lambda^{d,e}\, d\nu(x)\, ds\, d\nu(y) = \left(\int \tau_\lambda d\nu\right)^2 = \bar{\tau}_\lambda^2.$$

The source normalization is $1/\bar{\tau}_\lambda$; hence on the collision scale the main term is $\bar{\tau}_\lambda g_{\Sigma_\lambda}$. The finite label offsets change the central coordinate by $O(m^{-1/2})$. The physical clock conversion has

$$\det \mathcal{V}_\lambda = \bar{\tau}_\lambda^{-5} \det \Sigma_\lambda, \qquad m/t \longrightarrow \bar{\tau}_\lambda^{-1}.$$

This converts the collision-scale expression exactly to (120.2). Sequential compactness in λ and the central coordinates proves its stated uniformity.

For selectors insert $\chi_\lambda(x, a)\psi_\lambda(y, a - s)$ before integrating over a. Fubini gives amplitude $\bar{\tau}_\lambda^2 \mu_\lambda(\chi_\lambda)\mu_\lambda(\psi_\lambda)$. The same local-measure argument applies under the stated regularity. Uniform ellipticity and the positive product-of-means condition give the claimed selected lower bound. □

120.2. **An explicit three-parameter noncircular family.** Keep the triangular lattice of the original problem and set

$$O_{a,b,\theta} = \mathrm{Rot}_\theta \,\mathrm{diag}(a, b)\{z : |z| < 1\}, \qquad (a, b, \theta) \in [9/20, 47/100]^2 \times (\mathbb{R}/2\pi\mathbb{Z}). \tag{120.5}$$

The case $a = b = R$ is exactly the original disk table; $a \neq b$ in general removes its sixfold rotational symmetry. Rotation here is a parameter of the obstacle, not an affine conjugacy of reflection.

Lemma 120.2 (Uniform checks for the ellipse family)**.** *The family* (120.5) *satisfies all the separated geometric and dynamical conditions above. Its mean roof is*

$$\bar{\tau}_{a,b,\theta} = \frac{\pi(\sqrt{3}/2 - \pi ab)}{L_{a,b}}, \qquad 2\pi(9/20) \le L_{a,b} \le 2\pi(47/100). \tag{120.6}$$

All collision probabilities are the same in normalized arclength coordinates.

Proof. Every ellipse contains the disk of radius $9/20$ and is contained in the disk of radius $47/100$. Different obstacles are therefore separated by at least $3/50$. The smallest circular table has finite

horizon, and containment transfers its upper flight bound to every ellipse. The curvature in the usual ellipse parameter v is

$$\frac{ab}{(a^2\sin^2 v + b^2\cos^2 v)^{3/2}}.$$

It lies between $(9/20)^2/(47/100)^3$ and $(47/100)^2/(9/20)^3$. Its boundary derivatives and normalized arclength changes have common C^3 bounds, and the analytic family also has the uniform fourth-derivative bounds required by [18]. The positive gaps give a connected free region in every finite cover. Mixing is a separate consequence of Proposition 118.1, with its precise collision-map correlation theorem and transverse-horizon check; it is not deduced from connectedness. The local product construction is the unweighted one in [6, Section 8]. A rectangular Euclidean cover may be used without an affine change of the reflection law. The mean-roof formula is the collision suspension identity $\bar\tau = \pi\,\mathrm{area(free\ cell)}/L$. The area is $\sqrt{3}/2 - \pi ab$; the perimeter bounds follow from the speed of the ellipse parametrization. This proves (120.6) and the common probability normalization.

We verify the preceding-flight condition directly. Let z be the current boundary point, w the unit backwards incoming velocity and d a candidate preceding center. Put $A = \mathrm{Rot}_\theta\,\mathrm{diag}(a^{-2}, b^{-2})\,\mathrm{Rot}_\theta^{\mathsf T}$. The candidate entry length is

$$\ell_d = \frac{-B_0 - \sqrt{B_0^2 - A_0 C_0}}{A_0}, \quad A_0 = w^{\mathsf T} A w, \quad B_0 = w^{\mathsf T} A (z-d), \quad C_0 = (z-d)^{\mathsf T} A (z-d) - 1. \tag{120.7}$$

For an obstacle other than the current one, $C_0 > 0$ and $A_0 > 0$. A positive entry requires $B_0 < 0$ and $B_0^2 - A_0 C_0 \geq 0$. The displayed minus-square-root solution is the first intersection; the plus solution is the exit, and is not selected as the entry. For the current obstacle $C_0 = 0$ and the backward outgoing ray has $B_0 > 0$ off grazing, so its roots are zero and a negative value. The zero root is discarded. Finite horizon bounds all possible d in one fixed finite set. The coefficients are smooth and uniformly bounded in state and parameter, with $A_0 \geq (47/100)^{-2}$. Thus the entry root is uniformly 1/2-Hölder up to tangency, by the square-root inequality. Its preceding lattice label is $-d$.

For clarity, finiteness of candidates by itself is not the multiplier partition check. In rational trigonometric charts for the boundary, velocity and orientation, the entry inequalities, tangency equation and comparisons of the finitely many roots are semialgebraic of fixed complexity. Uniform cell decomposition splits their boundaries into a bounded number of monotone arcs, uniformly in (a, b, θ). Distinct first-hit regions can meet only on a backward singularity: a tie of two regular first entries would hit two separated obstacles at the same point. Backward singularity arcs are unstable-cone arcs, with common transversality to stable curves. A stable graph consequently crosses each such monotone arc at most once after the finite chart refinement. The length of its part within distance ϵ of an arc is at most $C\epsilon$, by the slope separation. The coordinate cuts have the same property. Intersecting this fixed finite partition with any homogeneity strip preserves a common component and intersection bound. Normalized arclength is a uniformly smooth monotone coordinate change and preserves these properties. This proves both (117.5) and the uniform partition count, not merely piecewise continuity of the root.

Finally, the bounded straight segment of a single flight intersects each translated convex cell in an interval, and meets only a fixed finite number of cells. Away from a tangent or a vertex crossing its entry and exit ages are continuous functions of its endpoints and direction. Changes of collision root, grazing, vertex crossings and flights lying in a cell edge are finite unions of semialgebraic curves or boundary pieces in collision coordinates. Each is null for ν; for a flight along a cell edge the fixed direction condition already has zero two-dimensional measure. On compact complements of their closed neighborhoods all interval endpoints vary uniformly. Choose

neighborhood radii at continuity points of their measure; their boundaries are then null and their measures tend to zero. Along any converging parameter sequence the corresponding regular roots persist on this compact complement. This gives the stated varying-test condition, including the appearance and disappearance of empty cell intervals. The same construction handles the products of the two endpoint spaces and a bounded age slab. □

Corollary 120.3 (The microscopic law for oriented ellipses)**.** *The full conclusions of Theorem 120.1 hold uniformly on* (120.5)*. In particular they hold along arbitrary sequences of eccentricities and orientations converging to a circle, as well as within a noncircular family. The original circular stationary theorem is its restriction to $a = b$.*

Proof. Apply Lemma 120.2 and the theorem. In particular finite-cover arithmetic, rather than a rotation of a circle, is what proves nondegeneracy and peripheral exclusion here. □

120.3. **Conditioning on the unchanged trajectory.**

Corollary 120.4 (Uniform posterior approximation in the same family)**.** *On a fixed central compact set the microscopic event in* (120.2) *has probability at least $c_M t^{-3/2}$ for all sufficiently large t, uniformly in the parameter. For the positive finite-band likelihood of Theorem 226.4, formed with the same physical flight-age indicator, the normalized source total-variation error is at most $C_M m^{3/2}/B$. In particular $B = m^{P+3/2}$ gives $O(m^{-P})$ for every fixed $P > 0$, against arbitrary bounded measurable tests on the original trajectory space. This assertion includes the ellipse family and involves no growing-band spectral estimate.*

Proof. Continuity and uniform ellipticity bound the Gaussian away from zero on the compact set; the local theorem gives the denominator. The positive smoothing kernel used in Theorem 226.4 has a finite first moment of order B^{-1}. Translation by v changes the indicator of one age interval in $L^1(da)$ by at most $2|v|$. There are only uniformly many initial and terminal cell intervals, and $\bar\tau$ is bounded below. The identical source calculation therefore gives unnormalized error at most C/B throughout the present family. If finite positive measures α, β have masses $a, b > 0$, then $\|\alpha/a - \beta/b\|_{\mathrm{TV}} \le 2\|\alpha - \beta\|_{\mathrm{TV}}/a$, with total variation taken as the norm of a signed measure. Use the proved microscopic mass and $t \asymp m$. The local theorem was obtained with a fixed band first; the later choice of B concerns only the explicit smoothing error. It provides a posterior comparison, not a Gaussian amplitude for an arbitrary path selector. □

121. A nonelliptic compact family on the same lattice

The action principle is not confined to quadrics or to centrally symmetric obstacles. The following family provides a separate verification on the same triangular lattice, with no change in the Euclidean reflection law or in the observation cell.

Let $n_\psi = (\cos\psi, \sin\psi)$, $t_\psi = (-\sin\psi, \cos\psi)$. For parameters $\lambda = (R, e, f, \theta, \phi)$ set

$$h_\lambda(\psi) = R + e\cos(3(\psi - \theta)) + f\cos(4(\psi - \phi)),$$
$$R \in [91/200, 93/200], \qquad |e| + |f| \le 1/250, \qquad \theta, \phi \in \mathbb{R}/2\pi\mathbb{Z}. \tag{121.1}$$

Define the obstacle by its support function h_λ, equivalently by the boundary parametrization $q_\lambda(\psi) = h_\lambda(\psi) n_\psi + h'_\lambda(\psi) t_\psi$. Place its identical translates at every lattice point.

Proposition 121.1 (Verification of the nonelliptic family)**.** *The family* (121.1) *satisfies the separated hypotheses of Section 120. Its stationary microscopic local law, regular endpoint-selected law, and normalized positive-likelihood comparison hold uniformly in λ. The local laws are qualitative $o(1)$ statements. The obstacle perimeter and area are*

$$L_\lambda = 2\pi R, \qquad |O_\lambda| = \pi R^2 - 4\pi e^2 - \tfrac{15}{2}\pi f^2, \qquad \bar\tau_\lambda = \frac{\pi(\sqrt{3}/2 - |O_\lambda|)}{L_\lambda}. \tag{121.2}$$

The physical covariance is continuous and uniformly positive definite. If $e \neq 0$, the obstacle is not centrally symmetric and hence is not an ellipse.

Proof. The support function and curvature radius satisfy

$$\frac{451}{1000} \leq h_\lambda \leq \frac{469}{1000}, \qquad \frac{79}{200} \leq h_\lambda + h''_\lambda \leq \frac{21}{40}. \tag{121.3}$$

The second bound follows from $h + h'' = R - 8e\cos(3(\psi - \theta)) - 15f\cos(4(\psi - \phi))$. In particular $q'_\lambda = (h + h'')t_\psi$ never vanishes. Positive curvature makes this a regular strictly convex closed boundary with the stated support function. Its curvature lies between 40/21 and 200/79. All boundary derivatives of any fixed order are uniformly bounded. The normalized arclength change is smooth with uniformly bounded derivatives and inverse derivatives of every fixed order.

The support inequalities imply $B(0, 451/1000) \subset O_\lambda \subset B(0, 469/1000)$. Different obstacles thus have a gap at least 31/500. Containment of the radius-0.45 disks transfers their uniform finite horizon to the present tables. The transverse finite-horizon condition and the finite-cover mixing required in the arithmetic follow from Proposition 118.1. The unweighted common spaces and normalized Jacobian estimates follow locally from the uniform smooth dispersing geometry and its compact parameter cover. The one-flight action identity uses arclength and is unchanged by the support parametrization.

Here are the additional partition checks; smooth convexity alone is not substituted for them. There are finitely many possible neighboring obstacles within the uniform horizon. On bounded rational trigonometric charts the coordinates of q_λ, the velocity, and the intersection equation with each neighboring translated boundary are semialgebraic. The coefficients $e, f, \cos(j\theta), \sin(j\theta)$ and their ϕ counterparts range in a fixed compact algebraic parameter set. Selecting the least admissible positive intersection uses only finitely many comparisons. Uniform cell decomposition therefore gives finitely many monotone arcs of the one-step backward singularity set, uniformly in the parameters. This decomposition is made in the trigonometric charts, before normalized arclength is used.

On a fixed candidate region the entry time is uniformly one-half Hölder up to its tangency boundary. Away from tangency this is the implicit function theorem with a nonzero transverse derivative. At tangency, a defining function for the boundary has a nonzero second derivative along the tangent flight direction, bounded below by the positive curvature bound. The parameter-dependent quadratic normal form at this nondegenerate contact writes the two nearby intersection times as a smooth center plus or minus a smooth bounded factor times the square root of a nonnegative parameter function. A finite cover of the compact tangency set makes these bounds uniform. The square-root inequality gives the asserted exponent. The first admissible root is the entry root; a zero root at the current obstacle is discarded. The displacement label is constant on each candidate region. Strict convexity precludes a higher-order tangency in this one-flight calculation.

Backward singularity arcs lie in the unstable cone, uniformly transverse to stable test curves. Their finitely many monotone pieces therefore have uniformly bounded stable intersection counts and $O(\epsilon)$ stable length in boundary neighborhoods. The smooth monotone arclength change preserves these estimates. This verifies the preceding-flight multiplier assumptions without a long-itinerary partition. Regular matched connectors and their intermediate singularity cuts are those of the unweighted collision construction, so the action proof introduces no new matching partition.

For the fixed convex polygonal observation cell, a flight has finitely many cell intervals. Exclude grazing and collision singularities, cell-edge tangencies and vertex crossings. These are finitely many semialgebraic curves and boundary pieces with zero collision measure; line coincidence with an edge is included among the exceptional configurations. On compact sets separated from them

the intervals and their endpoints vary uniformly with the parameters. Closed neighborhoods of the limiting exceptional sets can be chosen with arbitrarily small measure and null boundary, by taking continuity radii. These are the closed exceptional neighborhoods required in the corrected varying-test statement. Thus the observation hypothesis, not just pointwise convergence away from an open set, is verified.

All covariance and normalization continuity statements now follow from the compact-family principle. Its finite-cover arithmetic gives uniform positivity by compactness. To compute the normalizations, integrate $h + h''$ for the perimeter and use $|O| = \frac{1}{2}\int_0^{2\pi}(h^2 - (h')^2)\,d\psi$ for the area. Orthogonality of the third and fourth harmonics gives (121.2); the mean roof is the same invariant collision/flow normalization as in the ellipse application.

Finally a centrally symmetric body's support function, after a translation, is π-periodic; the translation contributes only a first harmonic. The nonzero third harmonic in (121.1) cannot be removed by such a translation. For $e \neq 0$ the obstacle is therefore not centrally symmetric. Parameter redundancy when an amplitude vanishes does not singularize the geometry or the covariance. □

The class of selected conclusions. Regular endpoint selectors always mean the null-boundary and closed exceptional-neighborhood class of the compact-family theorem. The product of their stationary means must be uniformly positive for a selected denominator. The arbitrary bounded trajectory-test conclusion concerns only the comparison of posterior measures, not a Gaussian amplitude for an arbitrary path selector. We use $\|\mu - \nu\|_{\mathrm{TV}} = \sup_{|F|\le 1} |\mu(F) - \nu(F)|$; for probabilities the often used distance $\sup_A |\mu(A) - \nu(A)|$ is one half of this norm. All comparison bounds hold in this explicitly chosen, larger convention.

122. The unsmoothed collision spectrum on a bounded frequency set

The flight-age inversion reduces the original physical observation to three collision frequencies. We now estimate these frequencies without smoothing the collision record. The relevant multiplier is the record of the *preceding* flight, on the image side of the transfer operator. Its discontinuities are transverse to the stable test curves. For a bounded set of real roof frequencies the action identity controls the accumulated multiplier on every homogeneous inverse branch. This allows the collision-space compactness argument to be applied to the actual displacement–time twist. No induced operator is introduced.

Put

$$f_R^{\mathrm{c}} = (\kappa_{R,1}, \kappa_{R,2}, \tau_R - \bar{\tau}_R), \qquad \check{f}_R^{\mathrm{c}} = f_R^{\mathrm{c}} \circ T_R^{-1}, \qquad Q_{R,z} = M_{\exp(iz\cdot \check{f}_R^{\mathrm{c}})}\mathcal{L}_R.$$

The last expression will first be defined on smooth densities and then on the collision spaces. Real frequencies are denoted by $\omega = (u, b) \in \mathbb{T}^2 \times \mathbb{R}$. Near zero, $z \in \mathbb{C}^3$ is an ordinary coordinate vector. We use the covariance Σ_R in (227.1); equivalently it is the collision Green–Kubo covariance of f_R^{c}.

122.1. **The image-side multiplier.**

Lemma 122.1 (A bounded analytic realization of the exact collision twist)**.** *The local common collision spaces can be chosen so that $Q_{R,z}$ is analytic in z in the strong operator norm. On each compact complex frequency set its norm and all fixed-order derivatives are bounded uniformly in R. For smooth functions a, d and every $m \ge 0$,*

$$\ell\big(M_d Q_{R,z}^m (a\nu)\big) = \int_M a(x) \exp\left(iz \cdot \sum_{j=0}^{m-1} f_R^{\mathrm{c}}(T_R^j x) \right) d(T_R^m x)\, d\nu(x). \tag{122.1}$$

In particular the earliest density occurs on the right of the operator product and the terminal multiplier on the left.

Proof. We specify the admissible discontinuities. In the collision coordinates (α, φ), the velocity obtained by following the incoming flight backwards is $w = \cos\varphi\, n_\alpha - \sin\varphi\, t_\alpha$. Its length to a candidate preceding disk with center d is the entry root

$$(d - Rn_\alpha) \cdot w - \sqrt{R^2 - |d - Rn_\alpha|^2 + ((d - Rn_\alpha) \cdot w)^2}.$$

Finite horizon gives a fixed finite list of candidates. On a region where one regular candidate is selected, this root is uniformly 1/2-Hölder, up to its boundary. Indeed the quantities outside and under the square root are uniformly Lipschitz and bounded, and $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for $x, y \ge 0$. The preceding displacement is the constant label $-d$ on that region. The assertion is made in (α, φ), not by an inverse change of variables at grazing.

The boundaries between preceding itineraries belong to the one-step backward singularity set. They are unstable curves, uniformly transverse to the stable cone. Splitting into their finitely many monotone pieces gives a partition with a uniformly bounded number of components in each homogeneity strip. Each stable curve meets a partition element in a uniformly bounded number of intervals; the length within distance ϵ of its boundary is $O(\epsilon)$. These are the two partition conditions of [12, Lemma 3.3]. They concern the preceding label, not the forward label.

Choose the weak stable-test exponent p with $0 < p < 1/6$, a strong stable exponent $0 < q < p$, and an unstable comparison exponent $0 < \gamma < \min\{p - q, 1/12\}$, decreasing it also to satisfy the geometric restrictions of the spaces. Choose the multiplier exponent ζ strictly between $\max\{p, \gamma/(1 - \gamma)\}$ and $1/2$. The strict inequality at $1/2$ places the preceding observables in the closure used in the Hölder test spaces. The piecewise multiplier bound of [12, Lemma 3.3(b)] therefore applies. This is the same image-side placement as in its Lemma 3.9, now including the entry-root coordinate. The exponential series and all its fixed derivatives converge in this multiplier norm on compact complex sets. This proves boundedness and analyticity.

For one step, the convention $\mathcal{L}h(\psi) = h(\psi \circ T)$ gives

$$Q_{R,z}h(\psi) = h(e^{iz \cdot f_R^{\mathrm{c}}} \psi \circ T_R).$$

Iteration proves (122.1) on smooth densities. Continuity gives the asserted realization. The local boundary reparametrizations and the deck-invariant restriction in Lemma 133.1 preserve these bounds. □

122.2. **Accumulated phases in the collision norms.** We record the geometric estimate needed for real frequencies away from zero. It is not a consequence of analyticity near zero.

Lemma 122.2 (Action bounds on homogeneous and matched curves)**.** *For $|b| \le B$, let*

$$w_{m,\omega}(x) = \exp\{i(u \cdot K_{m,R}^{\mathrm{c}}(x) + bS_{m,R}(x) - bm\bar{\tau}_R)\}.$$

On every homogeneous component of $T_R^{-m}W$, $W \in \mathcal{W}^s$, this weight has modulus one and uniformly bounded Lipschitz seminorm, with bound C_B independent of m, R, u. For the matched inverse pieces used in the unstable comparison of the collision norm, parametrized over the same interval and arising from terminal curves at distance at most ϵ, their two weights satisfy

$$\|w_{m,\omega}^{(1)} - w_{m,\omega}^{(2)}\|_{C^q} \le C_B \epsilon^{1-q}. \tag{122.2}$$

The unmatched pieces are exactly the geometric unmatched pieces; the weight creates no additional partition boundary.

Proof. Write $\vartheta_R = R\sin\varphi\, d\alpha$. The one-flight action identity, proved in Section 149, is $d\tau_R = T_R^*\vartheta_R - \vartheta_R$. On any connected arc γ regular through m collisions it telescopes to

$$S_{m,R}(y) - S_{m,R}(x) = \int_{T_R^m\gamma} \vartheta_R - \int_\gamma \vartheta_R. \tag{122.3}$$

Here x, y are the endpoints of γ; the lattice sum is constant on the corresponding regular branch. The form is uniformly bounded. A homogeneous inverse stable component is a graph with bounded slope, and its image under T_R^m contracts by at most $C\Lambda^{-m}$ in arc length. The latter is the stable-curve estimate used in [4, equation (4.6)]. Apply (122.3) to each subarc. Its two lengths are bounded by a constant times the distance of its endpoints in the graph coordinate. This proves the first assertion, uniformly in m.

We give the matching check because a bound on the variation along only one curve would not suffice. In the construction of [4, Section 4.3], the untrimmed matched inverse pieces are joined by unstable-cone connectors γ_x such that

$$|\gamma_x| \le C\Lambda^{-m}\epsilon, \qquad |T_R^m\gamma_x| \le C\epsilon.$$

Every intermediate image of such a connector is regular; otherwise the piece is assigned to the unmatched part. The two endpoints consequently have identical lattice itineraries. Formula (122.3) bounds their roof-sum difference by $C\epsilon$. Trimming to a common graph interval moves the paired point along one inverse stable piece by at most $C\epsilon$; its image moves by at most $C\Lambda^{-m}\epsilon$. The first part of the proof bounds this additional change also by $C\epsilon$. Thus the two weights in common graph coordinates differ in supremum norm by $C_B\epsilon$. Their individual Lipschitz constants in those coordinates are bounded by C_B. Interpolation between the supremum and Lipschitz bounds gives (122.2). All connectors and all trimming here are those of the existing collision partition, so no new singular boundary or long-word count is introduced. □

Proposition 122.3 (Bounded-band Lasota–Yorke estimates)**.** *For every $B < \infty$, on a finite cover by local common collision spaces there are $C_B < \infty$, $0 < \theta < 1 < \Lambda$, and $C_3 \ge 1$ such that, for all real $(u, b) \in \mathbb{T}^2 \times [-B, B]$ and $m \ge 0$,*

$$|Q_{R,\omega}^m h|_w \le C_B|h|_w, \tag{122.4}$$

$$\|Q_{R,\omega}^m h\|_s \le C_B(\theta^m + \Lambda^{-qm})\|h\|_s + C_B|h|_w, \tag{122.5}$$

$$\|Q_{R,\omega}^m h\|_u \le C_B\Lambda^{-\gamma m}\|h\|_u + C_BC_3^m\|h\|_s. \tag{122.6}$$

An equivalent strong norm, chosen after B, makes this a uniformly power-bounded, quasi-compact family with essential spectral radius strictly less than one. On each local parameter neighborhood, $Q_{R,\omega}$ is continuous from the strong space to the weak space as (R, ω) varies, uniformly on this frequency set.

Proof. We check the changes to the norm estimates rather than importing a local limit theorem for a different record. The collision norms and the inverse-curve decomposition are those in [4, Section 3.3]; its Proposition 4.1 gives the unweighted estimates with precisely the three forms above. A density paired against a terminal stable test ψ has, on each inverse piece U, the integral

$$\int_U h\, w_{m,\omega}\, J_U T_R^m\, (\psi \circ T_R^m)\, dm_U. \tag{122.7}$$

The collision probability is invariant, so there is no exponential volume-Jacobian factor. We use the same density convention as the unweighted norms.

For the weak estimate, Lemma 122.2 bounds the C^p norm of the added weight uniformly. The pulled-back test and the normalized stable Jacobian have the same C^p bounds as before. The sum of the stable Jacobian suprema over inverse pieces is bounded uniformly in m [4, Lemma 3.2(b)]. This proves (122.4).

For the strong stable estimate, on U subtract the average of $\psi \circ T_R^m$, *not* the average of the whole weighted test. The C^q norm of the resulting difference is at most $C\Lambda^{-qm}|\psi|_{C^q}$, by stable contraction. Multiplying this difference by the uniformly Lipschitz weight keeps that decaying factor. The average term times the weight and normalized Jacobian is a C^p test and is paid by the weak norm on pieces which have grown to the fixed reference length. Pieces not yet reaching that length retain the usual geometric growth-lemma factor $\theta^m\|h\|_s$. The associated Jacobian sums, including their length normalization, are unchanged, because $|w_{m,\omega}| = 1$. These are exactly the two sums in [4, equations (4.7)–(4.11)]. This proves (122.5); a nondecaying derivative of the phase has been put in the weak term, not in the leading strong coefficient.

For the unstable estimate, retain the matched and unmatched pieces of that proof. Unmatched pieces use the same stable norm estimate, since the extra weight has bounded C^q norm. On matched pieces split the test difference into the old Jacobian/test difference multiplied by one weight and the difference of the weights multiplied by the second Jacobian/test. The first term gives the old leading factor $C_B\Lambda^{-\gamma m}\|h\|_u$ and its stable-norm error. For the second term (122.2), divided by ϵ^γ, is bounded, since $1-q > \gamma$. The Jacobian and length sums bound it by $C_BC_3^m\|h\|_s$, as for the stable-norm error in [4, equation (4.3)]. We retain this exponential cross coefficient; it is not asserted uniform in m.

Choose an integer N so that the coefficients of $\|h\|_s$ and $\|h\|_u$ in their respective first terms are less than $1/4$. Then choose $c_B > 0$ with $c_BC_BC_3^N < 1/4$ and use $\|h\|_B = \|h\|_s + c_B\|h\|_u$. It follows that

$$\|Q_{R,\omega}^N h\|_B \le \tfrac{1}{2}\|h\|_B + C_B'|h|_w.$$

Iterate this inequality using (122.4), and treat the finitely many remaining powers. This proves uniform power boundedness. The compact embedding of the strong unit ball into the weak space, [12, Lemma 3.2(d)], gives quasi-compactness with a uniform essential-radius bound on the stated set. The equivalent norm does not change eigenvalues or spectral projections.

Finally apply the one-step matched-curve comparison in [4, Section 5 and Theorem 2.6] to two nearby radii. On unmatched pieces the old small-length estimate is multiplied by a bounded C^p norm. On a matched pair the preceding disk label is the same; the explicit preceding-flight root is uniformly 1/2-Hölder in (R, α, φ). Its difference tends uniformly to zero in supremum norm, and interpolation with its uniform $C^{1/2}$ bound makes the difference tend to zero in C^q. Thus the added weighted matched term tends to zero in the strong-to-weak operator norm. The same assertion in frequency follows from Lemma 122.1. The prescribed boundary-length reparametrizations are already part of this comparison. This proves the asserted local uniform continuity. □

122.3. **Physical exclusion of the peripheral spectrum.**

Theorem 122.4 (The full compact-frequency collision estimate)**.** *For every $B < \infty$ and $\epsilon > 0$ there are $C_{B,\epsilon} < \infty$ and $\rho_{B,\epsilon} < 1$ such that*

$$\|Q_{R,(u,b)}^m\| \le C_{B,\epsilon}\rho_{B,\epsilon}^m, \quad |b| \le B, \quad |(u,b)| \ge \epsilon, \quad u \in [-\pi,\pi]^2. \tag{122.8}$$

For a fixed neighborhood of zero, uniform in R, the simple eigenvalue and projection satisfy

$$Q_{R,z}^m = \lambda_R(z)^m\Pi_R(z) + N_R(z)^m, \qquad \|N_R(z)^m\| \le C\rho^m, \tag{122.9}$$

$$\log\lambda_R(z) = -\tfrac{1}{2}z^{\mathsf{T}}\Sigma_R z + O(|z|^3), \qquad \|\Pi_R(z) - \Pi_R(0)\| \le C|z|. \tag{122.10}$$

Here $\Pi_R(0)h = \ell(h)\nu$. For real z in a smaller neighborhood, $|\lambda_R(z)| \le e^{-c|z|^2}$ with $c > 0$ uniform in R.

Proof. Fix a real frequency. Quasi-compactness and power boundedness from Proposition 122.3 leave only finite-dimensional, semisimple eigenspaces on the unit circle. We show that a vector

in such an eigenspace is an actual measurable phase. If E_λ is its spectral projection, then on the strong space

$$E_\lambda = \lim_{M\to\infty} \frac{1}{M} \sum_{j=0}^{M-1} \lambda^{-j} Q_{R,\omega}^{j}.$$

Smooth densities are dense. Their images under E_λ form a dense linear subspace of the finite-dimensional range, hence its whole range. Thus any vector there is $E_\lambda(a\nu)$ for some smooth a. For every smooth test ψ, the exact pairing and invariance give

$$|Q_{R,\omega}^{j}(a\nu)(\psi)| \le \|a\|_\infty \int |\psi \circ T_R^{j}|\, d\nu = \|a\|_\infty \int |\psi|\, d\nu.$$

The same bound holds in the Cesaro limit. The vector is therefore $q\nu$ for a bounded measurable q. Invertibility and the eigenvector equation imply that $|q|$ is invariant; ergodicity makes it constant. For a nonzero vector normalize it to one. Then

$$q \circ T_R = \lambda^{-1} e^{i(u\cdot\kappa_R + b\tau_R - b\bar\tau_R)} q.$$

Apply Theorem 194.2, with constant phase $-\arg\lambda - b\bar\tau_R$. It gives $u = 0$ on the torus, $b = 0$, and $\lambda = 1$. This argument uses the already proved physical phase rigidity, not a choice of representative at isolated periodic points.

For every nonzero frequency the spectral radius is consequently less than one. The uniform Lasota–Yorke estimates and strong-to-weak continuity permit the spectral stability theorem [4, Theorem 2.1] to be applied locally in (R, u, b). A finite cover of the compact set in (122.8) gives the displayed complementary power estimate. Compactness is used only *after* the common-space inequalities and the physical peripheral exclusion have been proved.

Near zero, strong analyticity and the uniform unperturbed collision gap give (122.9). The first derivative of the eigenvalue vanishes because $\int f_R^{\mathrm{c}}\, d\nu = 0$. Twice differentiating (122.1) with $a = d = 1$, dividing by m, and using the spectral splitting identifies the Hessian of $\log\lambda_R$ as minus the limiting collision covariance. The exponentially summable covariance formula of Theorem 138.4, applied to $f_R^{\mathrm{c}} = P_R h_R$, identifies it with Σ_R. Cauchy's estimate on a fixed smaller complex neighborhood gives the uniform cubic remainder. Its strict positive lower eigenvalue bound then proves the final assertion. □

Remark 122.5. The constants in (122.8) may depend on B without any specified growth bound. We will first let m tend to infinity on each fixed frequency set and then enlarge that set by positive approximation. Substituting a growing B_m directly into (122.8) is not part of the proof. The operator also contains no section indicator and proves no corresponding spectral assertion for the unbounded four-coordinate return twist.

123. A joint local limit with both collision endpoints

Compact-frequency estimates suffice for local intervals if the passage to an unsmoothed interval preserves order. This section makes that passage before inserting the exact flight-age overlap. In particular we shall not substitute a polynomially increasing frequency cutoff into an estimate whose constants were proved only on compact sets.

For $k \in \mathbb{Z}^2$, $t \in \mathbb{R}$ write

$$\zeta_{m,R}(k,t) = \left(\frac{k}{\sqrt{m}}, \frac{t - m\bar\tau_R}{\sqrt{m}}\right), \qquad S_{m,R} = \sum_{j=0}^{m-1} \tau_R \circ T_R^{j}.$$

All compact central sets below are fixed independently of m and R. The density g_Σ is the normalized centered Gaussian in the integer cell coordinates and the time coordinate.

123.1. **A fixed Fourier support and an order-preserving limit.**

Lemma 123.1 (Endpoint insertions with a band-limited time test)**.** *Let a, d be smooth collision functions. Suppose $q \in L^1(\mathbb{R})$ has integrable Fourier transform of compact support, with the convention $\widehat{q}(b) = \int e^{ibs} q(s)\, ds$. Then*

$$\begin{aligned} m^{3/2} \int a(x) d(T_R^m x) \mathbf{1}_{\{K^{\mathrm{c}}_{m,R}(x)=k\}} q(S_{m,R}(x) - t)\, d\nu(x) \\ = g_{\Sigma_R}(\zeta_{m,R}(k,t)) \left(\int a\, d\nu \right) \left(\int d\, d\nu \right) \int q(s)\, ds + o(1). \end{aligned} \tag{123.1}$$

The error is uniform in R, k, t with $\zeta_{m,R}(k,t)$ in a fixed compact set. For fixed a, d, q the left side is bounded uniformly in all $R, k, t, m \geq 1$.

Proof. Torus orthogonality and Fourier inversion express the integral as

$$\frac{1}{(2\pi)^3} \int_{[-\pi,\pi]^2} \int_{\mathbb{R}} e^{-i[u\cdot k + b(t - m\bar{\tau}_R)]} \widehat{q}(-b) \ell\big(M_d Q^m_{R,(u,b)}(a\nu)\big)\, db\, du. \tag{123.2}$$

The sign $-b$ in $\widehat{q}$ results from evaluating $q(S_m - t)$. The equality follows directly from (122.1), without a change of the endpoint functions or of the record.

The time frequency belongs to a fixed bounded interval. Outside a fixed neighborhood of zero, (122.8) makes the integral exponentially small, even after multiplication by $m^{3/2}$. Inside that neighborhood use (122.9) and put $v = \sqrt{m}(u, b)$. The principal amplitude tends to $\nu(a)\nu(d)$; its value at zero follows from $\Pi_R(0) = \nu\ell$. Also $\widehat{q}(-v_3/\sqrt{m}) \to \int q$, uniformly on compact v sets. The absolute bound $Ce^{-c|v|^2}$ for the principal term, and the exponentially small complementary term on its fixed physical domain, justify dominated convergence. The uniform expansion (122.10) identifies the limit as the Gaussian Fourier inverse. The oscillatory factor has modulus one; convergence is uniform on central compact sets, including along arbitrary converging parameter sequences. The same absolute estimates bound the normalized inverse for all k, t. Enlarge the constant for the finitely many small values of m. □

Lemma 123.2 (Fourier envelopes for an interval)**.** *For a bounded interval J and each $B > 0$ there are real integrable functions $q^-_{J,B}, q^+_{J,B}$ whose Fourier transforms are continuous and supported in $[-B, B]$, such that*

$$q^-_{J,B} \leq \mathbf{1}_J \leq q^+_{J,B}, \qquad \int q^\pm_{J,B} = |J| \pm \frac{2\pi}{B}. \tag{123.3}$$

Endpoint conventions do not affect the inequalities if the lower bound is taken for the open interval and the upper bound for its closure.

Proof. Use the Beurling–Selberg majorant and minorant for the interval; we recall the normalization from [17, Introduction]. The Beurling functions H, K satisfy $|\operatorname{sgn} x - H(x)| \leq K(x)$, where $K(x) = (\sin(\pi x)/(\pi x))^2$, $\int K = 1$, and H is odd. Their exponential type is 2π. For J with endpoints $c < d$ and $\delta = B/(2\pi)$ take

$$q^\pm_{J,B}(s) = \tfrac{1}{2}\{H(\delta(s-c)) - H(\delta(s-d))\} \pm \tfrac{1}{2}\{K(\delta(s-c)) + K(\delta(s-d))\}.$$

The sign inequality gives the pointwise bounds. At an endpoint it gives an upper value at least one and a lower value at most zero, so open, closed and half-open intervals are all accommodated. The symmetric integral of the difference of the two H terms is $d - c$; the two K terms contribute $1/\delta$. The standard exponential-type Fourier support statement gives $[-B, B]$ in the convention adopted here. The functions are integrable; hence their transforms are continuous, and compact support also makes those transforms integrable. □

Theorem 123.3 (The mixed collision local limit with two endpoints)**.** *For continuous collision functions a, d and any fixed bounded interval J,*

$$\begin{aligned} m^{3/2} \int a(x) d(T_R^m x) \mathbf{1}_{\{K^{\mathrm{c}}_{m,R}=k,\ S_{m,R}-t\in J\}}\, d\nu(x) \\ = |J|\, g_{\Sigma_R}(\zeta_{m,R}(k,t))\nu(a)\nu(d) + o(1), \end{aligned} \tag{123.4}$$

uniformly in R and on central compact sets. For each fixed bounded interval J, independently of $k, t, R, m \geq 1$,

$$m^{3/2}\nu\{K^{\mathrm{c}}_{m,R} = k,\ S_{m,R} - t \in J\} \leq C_J. \tag{123.5}$$

Proof. First let a, d be nonnegative and smooth. Multiply the inequalities (123.3) by the nonnegative source weight $a(d \circ T_R^m)\mathbf{1}_{\{K^{\mathrm{c}}_m=k\}}$. For each fixed B, Lemma 123.1 evaluates both bounds. The difference of their limiting normalized values is $4\pi B^{-1} g_{\Sigma_R}(\zeta)\nu(a)\nu(d)$, uniformly bounded on the parameter interval. First take $m \to \infty$, then $B \to \infty$. This proves the interval formula. In particular no growing-cutoff spectral bound is used. The upper envelope for one fixed B, with $a = d = 1$, gives (123.5).

Approximate nonnegative continuous a, d uniformly by nonnegative smooth functions on the compactified collision cylinder. The change in the normalized integral is bounded by a constant times the approximation error, by (123.5). This proves the result for continuous functions. Adding constants and separating real and imaginary parts then gives the stated complex version. Uniform convergence on central compact sets follows from the uniform envelopes and the uniform fixed-band estimate. □

123.2. **Testing an endpoint-dependent overlap.** Let $\overline{M}$ be the compact collision cylinder, including the null grazing boundary. Define positive, locally finite measures on $\overline{M} \times \mathbb{R} \times \overline{M}$ by

$$\rho^{k,t}_{m,R} = m^{3/2}(x, S_{m,R}(x) - t, T_R^m x)_*(\mathbf{1}_{\{K^{\mathrm{c}}_{m,R}=k\}}\nu). \tag{123.6}$$

Corollary 123.4 (Local endpoint measures)**.** *If $R_m \to R_0$ and $\zeta_{m,R_m}(k_m, t_m) \to \zeta_0$, then*

$$\rho^{k_m,t_m}_{m,R_m} \longrightarrow g_{\Sigma_{R_0}}(\zeta_0)\, \nu(dx)\, ds\, \nu(dy) \tag{123.7}$$

vaguely. This convergence also evaluates bounded, compactly s-supported test functions continuous outside a set of zero measure for the measure on the right.

It evaluates a varying test F_{R_m} as well, provided there are open neighborhoods O_η whose closures have limiting measure tending to zero as $\eta \downarrow 0$ and whose boundaries have limiting measure zero, such that $F_{R_m} \to F_{R_0}$ uniformly outside O_η. The tests must have a common bound and a common compact s support. A small measure for the open set alone is not sufficient.

Proof. Products of continuous endpoint functions and interval indicators are evaluated by Theorem 123.3. Their linear combinations approximate continuous compactly supported tests, using uniform continuity and a finite interval partition in s. The bound (123.5) supplies uniform local masses. This proves vague convergence. The limiting measure assigns zero mass to every fixed time-slice boundary. Continuous upper and lower approximations, or the Portmanteau theorem on a compact time slab, then give the assertion about a fixed almost-everywhere continuous test.

For the varying assertion use the exceptional neighborhoods with small closed measure and null boundary required in the statement. Portmanteau bounds the limsup of the source measure of that closure by its limiting measure. The common supremum bound controls both tests there. On its complement uniform convergence controls their difference, by the local mass bound. Take $m \to \infty$ and then $\eta \downarrow 0$. Pointwise convergence of the tests alone would not justify this step. □

124. The original stationary microscopic local limit

We now evaluate the complete singleton physical probability. The initial and terminal cell offsets in (226.3) remain present throughout the argument. The overlap kernel is tested against the local endpoint measure, rather than estimated through a growing Fourier norm. This proves the disappearance of the signed complement in (227.6).

Fix once and for all the convex polygonal fundamental cell $\mathcal{P}$ of Section 226. The finite set $\mathcal{D}$ and $J_* = |\mathcal{D}|$ depend only on this cell, the lattice and the uniform horizon. They are not enlarged with m, t or with a frequency cutoff.

Lemma 124.1 (Admissibility of the actual overlap tests)**.** *For $d, e \in \mathcal{D}$ the functions*

$$F_R^{d,e}(x, s, y) = H_{I_{R,d}(x), I_{R,e}(y)}(-s)$$

have a common bound and support $|s| \le \tau_+$. They satisfy the fixed and parameter-varying test conditions of Corollary 123.4. Moreover

$$\sum_{d,e} \int F_R^{d,e}(x, s, y)\, d\nu(x)\, ds\, d\nu(y) = \bar{\tau}_R^2. \tag{124.1}$$

Proof. An overlap of two flight intervals is bounded by τ_+ and vanishes outside $[-\tau_+, \tau_+]$. At a regular collision state, excluding flights parallel to or passing through a vertex of a cell edge, the finite interval endpoints are given by intersections of a straight line with finitely many fixed lines, truncated by the regular entry root. They depend continuously on the parameter and collision state. Taking maxima and minima and allowing empty intervals preserves continuity of the intervals in symmetric-difference length. Since

$$|H_{I,J}(s) - H_{I',J'}(s)| \le |I \triangle I'| + |J \triangle J'|,$$

this implies uniform-in-s continuity of the overlap.

The exceptional sets just described, together with grazing and the one-flight singularity set, have zero collision measure: they are finitely many piecewise analytic curves and boundary pieces, with any coincident edge segment assigned the fixed cell convention. A flight contained in an edge has a prescribed position-direction constraint and also has zero collision measure. On every compact set separated from these sets at R_0, the selected roots and all transverse crossings persist uniformly for R near R_0. Thus the interval endpoints, and hence the overlap tests, converge uniformly there. Open neighborhoods of the exceptional sets can be chosen with arbitrarily small ν measure and null boundary. The union of their initial and terminal lifts to a compact s slab has arbitrarily small product measure. This proves the varying-test condition; it does not rely on an unquantified exchange of two pointwise limits.

Finally $\int_{\mathbb{R}} H_{I,J}(s)\, ds = |I||J|$. Sum over the two finite flight partitions to obtain $\tau_R(x)\tau_R(y)$ and integrate the independent endpoint measures. This gives (124.1). □

Theorem 124.2 (Uniform local limit for the original stationary event)**.** *For every $M < \infty$, uniformly for $R \in I$, $k \in \mathbb{Z}^2$, $m \ge 1$ and $t > 0$ in the indicated central sets,*

$$m^{3/2}\mathbb{P}_R\{W_R(t) = k, C_R(t) = m\} = \bar{\tau}_R g_{\Sigma_R}(\zeta_{m,R}(k,t)) + o(1), \qquad |\zeta_{m,R}(k,t)| \le M, \tag{124.2}$$

$$t^{3/2}\mathbb{P}_R\{W_R(t) = k, C_R(t) = m\} = g_{\mathcal{V}_R}(\xi_{t,R}(k,m)) + o(1), \qquad |\xi_{t,R}(k,m)| \le M, \tag{124.3}$$

where

$$\xi_{t,R}(k,m) = \left(\frac{k}{\sqrt{t}}, \frac{m - t/\bar{\tau}_R}{\sqrt{t}}\right), \qquad \mathcal{V}_R = \bar{\tau}_R^{-1} D_R^{\mathrm{clk}} \Sigma_R (D_R^{\mathrm{clk}})^{\mathsf{T}}, \quad D_R^{\mathrm{clk}} = \mathrm{diag}(1, 1, -1/\bar{\tau}_R).$$

The first error tends to zero as $m \to \infty$ and the second as $t \to \infty$. In particular there is $c_M > 0$ such that, for all sufficiently large t in the second central set,

$$\mathbb{P}_R\{W_R(t) = k, C_R(t) = m\} \geq c_M t^{-3/2}. \tag{124.4}$$

Proof. Use the exact overlap formula (226.5). Its (d, e) term has the exact collision label $k+d-e$, not k. Multiplying that term by $m^{3/2}$ expresses it as $\bar{\tau}_R^{-1}\rho_{m,R}^{k+d-e,t}(F_R^{d,e})$. The offsets belong to a fixed finite set, so $\zeta_{m,R}(k+d-e,t) - \zeta_{m,R}(k,t) = O(m^{-1/2})$.

For any sequence in the first central set choose a subsequence with $R \to R_0$ and $\zeta \to \zeta_0$. Apply Corollary 123.4 and Lemma 124.1 to every one of the finitely many pairs. The sum of their limiting integrals is $\bar{\tau}_{R_0}^{-1} g_{\Sigma_{R_0}}(\zeta_0)\bar{\tau}_{R_0}^2$. This proves the first limit by the sequential criterion for uniformity. Both cell corrections and the initial stationary age were retained in every term.

In the physical central set $t/m = \bar{\tau}_R + O(t^{-1/2})$, uniformly, and the two central sets are comparable. The matrix identity (227.9) gives $\det \mathcal{V}_R = \det \Sigma_R / \bar{\tau}_R^5$. Equivalently, for $A_R = \sqrt{\bar{\tau}_R}\,\mathrm{diag}(1, 1, -\bar{\tau}_R)$,

$$A_R \mathcal{V}_R A_R^{\mathsf{T}} = \Sigma_R, \qquad |\det A_R| = \bar{\tau}_R^{5/2}, \qquad \zeta_{m,R} = A_R \xi_{t,R} + O(t^{-1/2}).$$

Gaussian change of variables converts the first formula into the second. Uniform ellipticity, continuity and compactness imply $\inf_{R, |\xi| \leq M} g_{\mathcal{V}_R}(\xi) > 0$. The uniform error then proves (124.4). □

Corollary 124.3 (The full signed middle complement vanishes)**.** *For every fixed $P > 0$ take the explicit bandwidth $B_m = m^{P+3/2}$ and the physical central radius $r_m = 2m^{-99/200}$. For the* whole *complement $\mathfrak{M}_{m,R,B_m}$ defined in* (227.6)*,*

$$\sup_{\substack{R \in I,\ k \in \mathbb{Z}^2,\ t > 0 \\ |\xi_{t,R}(k,m)| \leq M}} t^{3/2} |\mathfrak{M}_{m,R,B_m}(k,t)| \longrightarrow 0. \tag{124.5}$$

In particular $B_m = m^{5/2}$ is one admissible polynomial bandwidth.

Proof. Theorem 227.3 gives the exact Gaussian-plus-complement identity with central error $O(m^{-11/700}\sqrt{\log(2+m)})$ and normalized far error $O(m^{-P})$. Theorem 124.2 independently evaluates its full probability, with the same Gaussian and normalization. Subtract the two formulas. This proves (124.5) on precisely the stated domain.

The mechanism is fixed-band spectral decay followed by positive interval envelopes and the exact endpoint overlap. The complement contains the entire noncentral spatial torus and every roof frequency between the central ball and B_m; none has been omitted. This proves a signed inverse estimate, not an absolute integral estimate for the modulus of its transform. No uniform-in-B spectral constant, or unproved cancellation assumption, is used in the subtraction. □

124.1. **Endpoint selectors and normalization on the same trajectory.** Let χ_R, ψ_R be bounded functions of a collision state and its age in the current flight. For definiteness assume they are restrictions of a jointly continuous bounded family on $I \times \overline{M} \times [0, \tau_+]$. The same statements hold for indicators of fixed endpoint sets whose boundaries have zero stationary measure, with the uniform exceptional-neighborhood condition used in Corollary 123.4. Write

$$\mu_R(\chi_R) = \bar{\tau}_R^{-1} \int_M \int_0^{\tau_R(x)} \chi_R(x,a)\, da\, d\nu(x).$$

On a trajectory starting at (x, a) let (y_t, v_t) be its current collision state and flight age at time t.

Theorem 124.4 (A microscopic local law with endpoint selectors)**.** *For the endpoint class just specified, uniformly on fixed physical central compact sets,*

$$t^{3/2}\mathbb{E}_R[\chi_R(x,a)\psi_R(y_t,v_t)\mathbf{1}_{\{W_R(t)=k,C_R(t)=m\}}] = \mu_R(\chi_R)\mu_R(\psi_R)g_{\mathcal{V}_R}(\xi_{t,R}(k,m)) + o(1). \quad (124.6)$$

Thus the conditional distribution of the initial and terminal flight states, tested against this endpoint class, tends to the product of the two stationary flight-state laws. For nonnegative selectors with $\inf_R \mu_R(\chi_R)\mu_R(\psi_R) > 0$*, their selected microscopic mass has a uniform positive lower bound of order* $t^{-3/2}$*.*

Proof. In the (d,e) term of the exact source formula replace the overlap by

$$H_R^{d,e;\chi,\psi}(x,y,s) = \int_{I_{R,d}(x)} \chi_R(x,a)\mathbf{1}_{I_{R,e}(y)}(s+a)\psi_R(y,s+a)\,da.$$

The same common support and bounds hold. Continuity of the selectors, the interval symmetric-difference estimate and dominated integration give the admissibility conditions of Lemma 124.1. For the stated indicator extension, apply the null-boundary approximation on the product of endpoint state–age spaces before integrating the ages. Fubini makes the exceptional overlap tests null for $\nu(dx)ds\nu(dy)$, and the specified neighborhood condition gives parameter uniformity.

Integration in s changes variables $v = s + a$ and factors the amplitude:

$$\sum_{d,e}\int H_R^{d,e;\chi,\psi}(x,y,s)\,d\nu(x)\,ds\,d\nu(y)$$
$$= \left(\int_M \int_0^{\tau_R(x)} \chi_R(x,a)\,da\,d\nu(x)\right)\left(\int_M \int_0^{\tau_R(y)} \psi_R(y,v)\,dv\,d\nu(y)\right).$$

Repeat the preceding local-measure proof and the same covariance change of variables. This gives (124.6). Division by (124.4) proves the conditional assertion. Positivity of the two stationary means and the Gaussian gives the selected denominator bound. □

Corollary 124.5 (Normalized polynomial-band reconstruction)**.** *Let* $A = \{W_R(t) = k, C_R(t) = m\}$ *be in a fixed physical central compact set, and let* $\lambda_{A,B}$ *be the positive likelihood in* (226.13)*. For* $B_m = m^{P+3/2}$*,*

$$\left\| \frac{\mathbf{1}_A\,d\mathbb{P}_R}{\mathbb{P}_R(A)} - \frac{\lambda_{A,B_m}\,d\mathbb{P}_R}{\mathbb{E}_R\lambda_{A,B_m}} \right\|_{\mathrm{TV}} = O(m^{-P}). \quad (124.7)$$

The comparison is on the original trajectory space and is uniform against every bounded measurable trajectory selector. It uses no event replacement. The same comparison holds after multiplication by fixed nonnegative endpoint selectors from Theorem 124.4 with uniformly positive selected mass on the normalized scale.

Proof. Theorem 226.4 bounds the unnormalized source error by C/B. For two finite positive measures α, β with masses $p, q > 0$,

$$\|\alpha/p - \beta/q\|_{\mathrm{TV}} \le \frac{\|\alpha-\beta\|_{\mathrm{TV}} + |q-p|}{p} \le \frac{2\|\alpha-\beta\|_{\mathrm{TV}}}{p}.$$

Use the newly proved denominator $p \ge c_M t^{-3/2}$, and note that t/m stays in a fixed positive compact interval. For large m, the source error is smaller than $p/2$, so $q > 0$. This proves (124.7). Multiplication by a bounded nonnegative endpoint weight contracts the unnormalized error up to its supremum norm; the selected denominator is provided by the preceding theorem. Total variation controls every bounded test on the unchanged trajectory space, without any regularity of that additional test. □

The two local problems. Theorems 124.2 and 124.4 prove local statements for the original stationary event at prescribed physical collision count. They do not identify a Gaussian bridge for arbitrary long path selectors. The arbitrary-selector assertion in Corollary 124.5 is a comparison of two posterior measures, not a formula for the selected Gaussian amplitude. Nor does the positive age-overlap mechanism create a pointwise density estimate for the four-coordinate return record $J_{n,R}$. The latter's common raw correction and return-frequency complement retain their separate roles in Theorem 143.1. The original raw return problem and all its geometric and probabilistic inputs remain in the article.

Part 38. The actual-return record and raw inversion

The retained return-window and bridge route. The new local constraint is proved in Sections 151 and 152. It uses the exact section geometry, the inherited compensated covariance and its nondegeneracy, and the compact collision spectrum of the preceding part. The return index is resolved on its diffusive scale; single-index inversion and pointwise roof inversion remain distinct from that theorem. Sections 153 and 154 then replace the diffusive window by any diverging integer block, prove a single-index upper bound, and locate the exact inverse on a fixed occupation torus. The new Sections 157–159 then give a full fixed-band spectral reduction and evaluate the fixed-radius interval law with its arithmetic coefficient. Sections 160–162 then evaluate the uniform transition kernel and prove the exact-event return bridge. The complete inherited arithmetic, raw edge and return-frequency analysis continues on the same physical record.

125. The mechanical question and its interfaces

Let $a = (1, 0)$, $b = (1/2, \sqrt{3}/2)$, $\Lambda = \mathbb{Z}a + \mathbb{Z}b$, and

$$I = [9/20, 47/100], \qquad Q_R = (\mathbb{R}^2/\Lambda) \setminus \overline{B}(0, R).$$

The speed is one and every reflection is specular. Three indices will be kept distinct: a physical collision count N, an induced return count n, and the number of experimental bits. An induced record has values in $\mathbb{Z}^2 \times \mathbb{Z} \times \mathbb{R}$, with counting measure in the first three coordinates and Lebesgue measure in the last. A four-dimensional Lebesgue density would be the wrong object.

The geometric inverse of the preceding article identifies the table from two fixed collision fields [1]. It supplies neither independent flights nor an induced transfer-operator estimate. Conversely, the stationary law of the equilibrium billiard is not the unknown preparation law of that inverse experiment. The distinction permits geometric estimates to be used for equilibrium statistics without first estimating the entire preparation law.

The joint arithmetic result is Theorem 128.3. It concerns the complete periodic phase of an explicitly enlarged physical first-return section. Lemma 128.1 constructs its infinite period family, and Theorem 128.2 proves the strict excess inequalities. These results go beyond the zero-roof determinant by ruling out every nonzero roof frequency, without a numerical irrationality claim. The real-rank and compact-frequency consequences are stated separately from measurable cohomology regularity and operator contraction.

The original physical record, winding geometry, critical-edge and conditional inversion arguments remain in the article. Theorem 144.1 refines the inversion requirement by using local edge errors instead of small total edge mass. Propositions 131.1 and 131.2 give exact normalizations and stationary endpoint control for the enlarged section. The full raw local limit is not inferred from the arithmetic theorem: its spectral, residual and weighted-insertion requirements are listed in the final section. All probability laws here refer to actual specular orbits, not independent flight surrogates.

The relevant prior local-limit theory includes the Lorentz-process theorem of Szász–Varjú [2] and the suspension framework of Dolgopyat–Nándori [3]. Definition 2.2 and Proposition 6.3 of the latter concern a mixing local limit tested against fixed compactly supported functions under minimality and group assumptions. They do not state absolute integrability of every raw finite-time characteristic function, or the particular geometrically uniform density theorem sought here. Demers–Zhang [4] give a perturbation framework for Lorentz-gas maps; its spectral continuity is not automatically twice differentiability in a moving geometric parameter. We use the corrected version of the flow-mixing work of Baladi–Demers–Liverani [5]. No novelty is claimed for Fourier inversion, the Morse lemma, or the suspension identity as general methods.

Theorem 132.2 compares the actual moving return events in density L^1 and first moment, at every fixed return count. Corollary 132.4 then supplies uniform stationary endpoint control over the whole radius interval. The proof uses finite-itinerary stability, coarea and exact Kac means, not a hypothetical common spectral space. These finite-record moduli do not purport to supply the quantitative long-time estimates in the local-limit interface.

Theorem 129.2 strengthens strict increase to uniform two-sided exponential bounds, obtained from the exact nonlinear half action. Theorem 129.3 uses their upper and lower bounds together: the former selects periods of logarithmic length, and the latter supplies a polynomial discrepancy rather than mere nonvanishing. Corollary 129.4 states precisely the continuous-phase consequence. It does not assume the measurable regularity needed for an operator application.

At the level of the clock, Theorems 137.2 and 137.4 turn fixed-record continuity and exact means into a uniform functional first-order law. This includes the maximum unfinished return over the observation window at scale t, whereas the retained stationary endpoint result is at scale $\sqrt{t}$ for finitely many deterministic endpoints. The distinction of scales is essential. Corollary 137.5 gives a geometric error term for the physical rate without assuming independence of the radius estimate and the observed orbit. No originality is claimed for the general ergodic, subadditive or renewal inversion arguments; the point is their verified application to the actual moving return family.

126. Physical records and an exact marked identity

Write an outgoing section point as

$$q = Rn_\alpha, \qquad v = \sqrt{1-p^2}\, n_\alpha + p\, t_\alpha, \qquad (\alpha, p) \in \mathbb{T} \times (-1, 1).$$

The collision map is T_R, its next physical flight is τ_R, and

$$\mathrm{d}\nu = \frac{\mathrm{d}\alpha\, \mathrm{d}p}{4\pi}, \qquad \bar{\tau}_R = \frac{\sqrt{3}/2 - \pi R^2}{2R}, \qquad \mathrm{d}\mu_R = \bar{\tau}_R^{-1}\, \mathrm{d}\nu\, \mathrm{d}s \quad (0 \le s < \tau_R). \tag{126.1}$$

These formulas follow by integrating the boundary flux $\cos\phi\, \mathrm{d}s_{\partial Q}\, \mathrm{d}\phi$: total section flux is $4\pi R$ and phase volume is $2\pi(\sqrt{3}/2 - \pi R^2)$. Reflection preserves this flux. Thus T_R preserves ν.

Each next disk has a lattice label $d \neq 0$. With $D = d - q$ and $L = D \cdot v$, its candidate entry time is

$$\tau_d = L - \sqrt{R^2 - |D|^2 + L^2}. \tag{126.2}$$

The physical branch is the positive entry root with the smallest time, not an arbitrarily selected root. Its endpoint normal is $n' = (q + \tau_d v - d)/R$, and its outgoing velocity is $v' = v - 2(v \cdot n')n'$. These expressions define the label, point, both velocities, and time on every regular branch. On a compact finite itinerary separated from tangencies and competing roots they depend analytically on R and the initial coordinates, by the implicit function theorem. This is a finite-itinerary assertion, not a common norm for all iterates.

Distinct disks have distance at least $g_R = 1 - 2R \ge 3/50$. A straight outgoing flight cannot return to its initial convex disk. In particular there is no finite-time collision accumulation. This

family has uniform finite horizon: a rational lattice direction has row spacing at most $\sqrt{3}/2$, so every line in that direction meets an open disk when $R > \sqrt{3}/4$; an irrational line is dense on the torus. Compactness then excludes arbitrarily long free segments at the smallest radius. Increasing the radius preserves this upper bound. Tangencies and their finite preimages have section measure zero.

Put $S_j\tau(y) = \sum_{i=0}^{j-1} \tau(T_R^i y)$. For y interpreted as the first future outgoing collision and u as its arrival time, the first k future collisions have times $u + S_j\tau(y)$, $0 \le j < k$. Their points, lattice labels, and incoming and outgoing velocities are obtained by iterating (126.2). The final incomplete flight has duration $t - u - S_{k-1}\tau(y)$.

Proposition 126.1 (Marked stationary record)**.** *For every bounded measurable functional Ψ of this record and its terminal flight,*

$$\mathbb{E}_{\mu_R}[\Psi; N_t = k] = \frac{1}{\bar{\tau}_R} \int \mathrm{d}\nu(y) \int_0^{\tau(T_R^{-1}y)} \mathbf{1}_{\{u+S_{k-1}\tau(y) \le t < u+S_k\tau(y)\}} \cdot \Psi(\mathcal{R}_{k,t}(y,u))\,\mathrm{d}u, \qquad k \ge 1. \tag{126.3}$$

The initial phase point is $\Phi_R^{-u}y$, so the entire initial residual flight is retained as well. For $t = 11/100$ there are at most two future collisions, and (126.3) *with $k = 1, 2$, together with the zero-collision case, gives the full marked law without an independence assumption.*

Proof. Under (126.1), set $y = T_R x$ and $u = \tau(x) - s$. Invariance of ν gives the displayed domain for u with Jacobian one. The kth and $(k+1)$st future collision times are exactly the two endpoints in the indicator. This proves the identity for arbitrary bounded measurable Ψ, first for indicators and then by integration. The event of equality at an endpoint has zero suspension measure. Three collisions require at least two complete intercollision flights after the first event; their total duration is at least $6/50 > 11/100$. □

127. Explicit periodic geometry and a genuine inducing set

Two elementary cycles have zero winding. The normal nearest-neighbor orbit has two flights, total length

$$L_2 = 2(1 - 2R), \qquad L_3 = 3(1 - \sqrt{3}R) \tag{127.1}$$

is the length of the inner equilateral three-cycle on centers $0, a, b$. Its contact angles are $\pi/6, 5\pi/6, 3\pi/2$ and its outgoing p coordinates are all $-1/2$. The reflection cosines are respectively 1 and $\sqrt{3}/2$. The equilateral segments lie inside their elementary triangle and outside the three vertex disks; any other lattice center is at distance at least $\sqrt{3}/2$ from that triangle. Hence all other disks are strictly avoided.

Lemma 127.1 (A winding four-cycle)**.** *For every $R \in I$ there is an analytic angle $\theta_R \in (0.92, 0.967)$ satisfying*

$$f_R(\theta) = 2R\sin\theta - \sin\theta\cos\theta + \frac{\sqrt{3}}{2}\cos(2\theta) = 0. \tag{127.2}$$

The itinerary with centers $0, a, a-b, 2a-b, a$ and normals at angles

$$-\theta_R, \quad \pi + \theta_R, \quad \theta_R, \quad \pi - \theta_R$$

is a physical four-collision periodic orbit modulo Λ, with winding a. All its incidence cosines exceed $1/2$ and its clearance from every third disk exceeds $1/200$. Rotation through $\pi/3$ gives another cycle with winding b and the same length.

Proof. Put $c = \cos\theta$, $s = \sin\theta$. Its first four contact points and the translated fifth point are

$$q_0 = (Rc, -Rs), \qquad q_1 = (1 - Rc, -Rs),$$
$$q_2 = (1/2 + Rc, -\sqrt{3}/2 + Rs), \qquad q_3 = (3/2 - Rc, -\sqrt{3}/2 + Rs),$$
$$q_4 = q_0 + a.$$

Two segments are horizontal. Write $d = 2Rc - 1/2$, $h = \sqrt{3}/2 - 2Rs$. The other two are $(d, -h)$ and (d, h). The scalar equation $d\sin 2\theta + h\cos 2\theta = 0$ is precisely (127.2). Since $d, h > 0$, it makes the diagonal direction the reflection of the horizontal direction at the intervening normal. The remaining reflection equations follow by symmetry, with cosine c at every contact. The total length is

$$L_4 = 2(1 - 2Rc) + 2\sqrt{d^2 + h^2}. \tag{127.3}$$

Here are reproducible rational bounds for the inequalities and absence of occlusion. Divide I into 32 closed intervals of width $1/1600$. On each, bisect the endpoint equations (127.2) between $9/10$ and 1 to width at most 10^{-10}. The interval derivative

$$2R\cos\theta - \cos 2\theta - \sqrt{3}\sin 2\theta$$

is less than $-1/2$. The endpoint signs and positivity of $\partial_R f$ show that the two endpoint root brackets enclose the root for the entire parameter interval. The resulting angles lie in $(0.92, 0.967)$; $d, h > 0$ and $c > 1/2$ on every interval.

For clarity, the arithmetic of this finite certificate uses no floating root solver. Square roots are enclosed by integer square roots at denominator 10^{22}. For sine and cosine use the Taylor polynomials through degrees 25 and 24 at the rational midpoint, with errors at most $|x|^{26}/26!$ and $|x|^{25}/25!$, respectively; add the interval radius, since both functions are 1-Lipschitz. All interval operations are rational.

For each segment check centers $ia + jb$, $|i|, |j| \le 4$, excluding its two endpoints. The distance from the rational midpoint segment is computed by clamping the rational orthogonal projection to $[0, 1]$. Subtract the endpoint and center enclosure radii and the largest obstacle radius. Each lower bound exceeds $1/200$. Every contact point has norm less than 3; a center outside the index box has norm at least $5\sqrt{3}/2$, so it too has distance greater than the obstacle radius from every segment. The complete deterministic calculation is specified in `tools/certify_winding.py`; it obtains lower bounds greater than 0.00925 for clearance and 0.569 for incidence, and a derivative upper bound below -0.731. The signs and strict margins prove the assertions on the full intervals, not only at sampled radii. The implicit function theorem gives analytic dependence and uniqueness in the printed root interval. □

Choose $\epsilon = 1/200$. In (α, p) coordinates let Y_R be the union of the open squares of half-side ϵ centered at

$$(0, 0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R). \tag{127.4}$$

Angles are taken modulo 2π. The squares are disjoint and $\nu(Y_R) = 4\epsilon^2/\pi$. Let r_R be the first return time to Y_R and $F_R = T_R^{r_R}$ on its recurrent full-measure domain, with invariant probability $\nu_R^Y = \nu|_{Y_R}/\nu(Y_R)$. Define

$$\kappa_R = \sum_{j<r_R} \kappa_{\mathrm{coll}} \circ T_R^j, \qquad \tau_R^Y = \sum_{j<r_R} \tau_R \circ T_R^j. \tag{127.5}$$

Poincaré recurrence and invariance give this induced probability; they do not give an exponential return tail, a Markov partition, or a differentiable family of spectral projectors.

Proposition 127.2 (Full augmented lattice at zero roof frequency)**.** *The four cycles above give fixed points of F_R with data $(\kappa_1, \kappa_2, r, \text{return period})$ equal to the rows of*

$$V = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 1 \\ 1 & 0 & 4 & 1 \\ 0 & 1 & 4 & 1 \end{pmatrix}, \qquad |\det V| = 1. \tag{127.6}$$

Consequently a circle-valued coboundary identity $e^{i(u\cdot\kappa+sr)} = e^{ic} g/(g \circ F_R)$ that can be evaluated at these fixed points forces $u = s = c = 0$ modulo 2π.

Proof. Only one contact of each cycle belongs to (127.4). For the four-cycle the successive p coordinates are $s, s, -s, -s$. The same is true after rotation. The two positive-p unselected contacts are separated from the selected centers; the smallest relevant angular separation is $2\pi/3 - 2\theta_R > 0.16$. All negative-$p$ contacts are separated in p from the selected winding contacts, and also from $0, -1/2$. The remaining contacts of the two- and three-cycles are separated by their displayed angles. Thus the physical first return counts are $2, 3, 4, 4$, not freely assigned labels. The four selected contacts are interior to their squares, so the same finite return branches exist in neighborhoods. Direct subtraction of the first two rows gives determinant of magnitude one. Evaluation first gives $2s = c = 3s$, hence $s = c = 0$; the last two rows then give $u_1 = u_2 = 0$. □

The qualification about evaluation is essential. A merely measurable transfer function is not automatically defined at a periodic point. A spectral application requires its own regularity or Livšic argument. Moreover, adding a roof twist $b\tau_R^Y$ to this proposition leaves bL_j in every periodic equation. The four rows do not eliminate it. Joint roof nonarithmeticity and returned temporal nonintegrability remain distinct requirements. As one smaller unconditional conclusion, (127.1) excludes a continuous cohomology $\tau_R = c + g - g \circ T_R$: the two periodic averages are $1 - 2R$ and $1 - \sqrt{3}R$, which differ by $(2 - \sqrt{3})R > 0$.

128. A physical period family and the complete periodic annihilator

The four records in Proposition 127.2 settle the integer coordinates at zero roof frequency. We now add a family of actual periodic trajectories that settles every roof frequency. A slightly larger section is specified explicitly; the original section and every assertion about it remain as stated in the preceding sections.

128.1. **The enlarged section.** Put

$$U = \{(\alpha, p) : |\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* := \nu(Y_R^*) = \frac{91}{10000\pi}. \tag{128.1}$$

The rectangle U is disjoint from the four squares defining Y_R. At the square with the same angular center the p coordinate is near $-1/2$; the other three squares are separated in angle or in p. The area of U is $4(3/50)(3/20)$, and the section density is $1/(4\pi)$, proving the value of c_*. Write $F_R^* = T_R^{r_R^*}$ for the first return to Y_R^*, and let κ_R^* and τ_R^* be the sums of the physical displacement and flight length before that return. Its invariant probability is $\nu_R^* = \nu|_{Y_R^*}/c_*$. An assertion about F_R does not become an assertion about F_R^* merely by changing notation.

Rotate coordinates through $-\pi/6$ and put

$$D = \sqrt{3}, \qquad O = (0,0), \quad C = (D,0), \quad A = (D/2, -1/2), \quad B = (D/2, 1/2),$$
$$d = \frac{1}{2} - R, \qquad g = D - 2R.$$

The rotated lattice consists of $(Dk/2, l/2)$ with integers k, l of the same parity. The normal orbit between O and C is unobstructed: the two intermediate centers A, B have distance $1/2$ from its supporting line, and all other centers are farther away. It has two physical flights of total length $2g$, zero winding, and exactly one visit to Y_R^* per period. Its selected state is $(\pi/6, 0)$ in the unrotated coordinates. The four cycles of Proposition 127.2 still have exactly one visit per period. Indeed, the equilateral cycle has $p = -1/2$, the two-cycle has angles $0, \pi$, and the winding cycles have $|p| = \sin\theta_R > 0.79$. None gains a visit to U.

128.2. **A convex optical action.** For $-d \le y, z \le 0$ define

$$\begin{aligned} x_R(y) &= \sqrt{R^2 - y^2}, \qquad u_R(y) = R - x_R(y), \\ e_R(y) &= \sqrt{(D/2 - x_R(y))^2 + (d + y)^2}, \\ \ell_R(y, z) &= \sqrt{(D - x_R(y) - x_R(z))^2 + (z - y)^2}. \end{aligned} \tag{128.2}$$

The function e_R is the length from the top contact $q_A = (D/2, -d)$ of the disk at A to a contact at height y on the facing arc of the disk at O or C. The function ℓ_R is a flight length between the two facing arcs. For $m \ge 1$ let

$$\mathcal{A}_{m,R}(y_1, \dots, y_{2m}) = e_R(y_1) + \sum_{j=1}^{2m-1} \ell_R(y_j, y_{j+1}) + e_R(y_{2m}), \qquad L_m(R) = \min_{[-d,0]^{2m}} \mathcal{A}_{m,R}. \tag{128.3}$$

Lemma 128.1 (A uniformly regular physical family)**.** *For each $m \ge 1$ and $R \in I$, the minimum in* (128.3) *is unique. Its coordinates satisfy*

$$-\frac{2d}{5} < y_j < 0, \qquad y_j = y_{2m+1-j}. \tag{128.4}$$

The contacts with these heights form a genuine specular periodic orbit with center word $A, (O, C)^m, A$. Every flight avoids all third disks. The orbit has zero winding, $2m + 1$ physical collisions, and exactly m returns to Y_R^*. The period length $L_m(R)$ is analytic in R for each fixed m.

Proof. We supply the variational, mechanical and return checks separately. On the whole box, $x_R(y) > 11/25$, $x_R(y) \le 47/100$, and $1.73 < D < 1.74$. The positive horizontal quantities

$$a(y) = D/2 - x_R(y), \qquad h(y, z) = D - x_R(y) - x_R(z)$$

satisfy $a > 79/200$ and $h > 79/100$. The second derivative of $-x_R(y)$ is $R^2/x_R(y)^3 > 0$. The Hessian formula for a Euclidean norm shows that the Hessian of e_R or ℓ_R is the sum of a positive semidefinite term and the Hessian of its horizontal component multiplied by its positive horizontal direction cosine. The latter cosine exceeds $9/10$ on this box. Thus each segment length is strictly convex in its variables; in fact the Hessian of $\mathcal{A}_{m,R}$ dominates the identity. One may check the last numerical bound from $(9/10)(9/20)^2/(47/100)^3 > 1$. Compactness and strict convexity give a unique minimizer.

No coordinate equals $-d$. For such a coordinate y and an incident interior segment, the derivative is

$$\partial_y \ell_R(y, z) = \frac{h(y, z) y / x_R(y) + y - z}{\ell_R(y, z)} < 0,$$

since $z \ge y = -d$. For an end segment, $e_R'(-d) = -a(-d)d/(x_R(-d)e_R(-d)) < 0$. Increasing that coordinate therefore decreases the action. At an end coordinate equal to zero, $e_R'(0) = d/e_R(0) > 0$, and all other incident derivatives are nonnegative. At an interior zero adjacent to a negative coordinate, its derivative is positive. If a block of zero coordinates had no such neighbor it would reach an endpoint. A variation into the box excludes all these cases. Every coordinate

is consequently in $(-d, 0)$. Reflection in the line $x = D/2$ reverses the center word and preserves the action, so uniqueness gives the symmetry in (128.4).

Here is a uniform improvement of the height bound. A smallest coordinate $y = -w$ cannot be interior, because both incident derivatives there are negative. Take it to be y_1, using reversal if needed, and write $z = y_2$. Stationarity gives

$$d - w = \left(a + e_R(y)\frac{h}{\ell_R(y,z)}\right)\frac{w}{x_R(y)} + \frac{e_R(y)}{\ell_R(y,z)}(w + z).$$

The last term is nonnegative since $z \geq -w$. Moreover,

$$\frac{a + e_R(y)h/\ell_R(y,z)}{x_R(y)} > \frac{(19/10)(79/200)}{47/100} = \frac{1501}{940} > \frac{3}{2}.$$

It follows that $w < 2d/5$. This bounds every coordinate.

Interior stationarity of the optical length is precisely equality of tangential incoming and outgoing velocities at each contact. The outgoing directions point into the exterior normal half-plane. For example, at a contact on O, its outward normal is $(x_R(y), y)/R$. Its dot product with a direction to C has numerator at least $x_R(y)h - |y||z - y| > 0$; for a direction to q_A both the horizontal contribution and $y(-d - y)$ are positive. The checks at C are reflections of these. The symmetry of the two end contacts implies the reflection law at q_A, whose normal is vertical. There $e_R(y_1) < 11/25$ and $d + y_1 > 3d/5 \geq 9/500$, so the normal incidence cosine exceeds $9/220 > 1/25$. The other contacts are uniformly transverse by the preceding horizontal-component bounds. Thus the stationary polygon is specular rather than the straight-through solution of the stationarity equation.

The long flights lie in $-2d/5 < y < 0$, so their distance from the disk at A is at least $3d/5 \geq 9/500$, and from the disk at B at least $d \geq 3/100$. End flights join q_A to the appropriate facing arc and leave both endpoint disks in their exterior supporting half-planes. Their third disk O or C has horizontal separation at least $D/2 > R$, and B is vertically separated. All segments lie in the rectangle

$$11/25 < x < D - 11/25, \qquad -d \leq y \leq 0.$$

For the remaining lattice centers, the possibilities $k = 0, 2$ with nonzero l have vertical separation at least $1 - d \geq 0.95$; $k = 1$, $|l| \geq 3$ have separation at least $3/2 - d$; and $k \notin \{0, 1, 2\}$ have horizontal separation exceeding 1.3. This enumerates the whole lattice and proves nonocclusion, not just a check of a finite list of nearby obstacles.

At an outgoing contact on O, the normal angle differs from $\pi/6$ by at most $|y|/x_R(y) < 1/22 < 3/50$. The outgoing velocity points toward C and its angular deviation from the horizontal is at most $|z - y|/h < 2/79$. Thus

$$|p| \leq \frac{1}{22} + \frac{2}{79} < \frac{3}{20}.$$

All m contacts on O belong to U. Contacts on C have angles near $7\pi/6$, and q_A has angle $2\pi/3$; none belongs to Y_R^*. The return counts are 2 between successive contacts on O, with one count 3 through C, A, O. Their sum is $2m + 1$, and there are m induced returns. The orbit is closed in the plane, so its winding is zero. All selected contacts and unselected contacts have strict membership margins, giving actual regular return branches. Finally the positive Hessian and the interior minimum allow the analytic implicit function theorem. This proves analyticity for each fixed m; no bound for high parameter derivatives uniform in m is being inferred. □

128.3. **Strictly increasing bounded excess lengths.** The following argument supplies an infinite sequence of period relations. It does not infer nonarithmeticity from finitely many decimal lengths.

Theorem 128.2 (Strict period excess)**.** *For the family of Lemma 128.1, define* $E_m(R) = L_m(R) - 2mg$*. Then, for every* $R \in I$*,*

$$0 < E_m(R) < E_{m+1}(R) \le \frac{2d^2}{g} < \frac{1}{100}. \tag{128.5}$$

In particular $E_{m+1}(R) - E_m(R) > 0$ *and these differences tend to zero. The convergence of* E_m *is used pointwise in* R*; the subsequent compact-frequency assertion supplies its own uniformity argument.*

Proof. Suppress R and set $K(y,z) = \ell(y,z) - g$ on $[-d,0]^2$. The horizontal component of the flight gives

$$K(y,z) \ge u(y) + u(z), \tag{128.6}$$

with equality exactly when $y = z$. Define continuous value functions

$$V_0(y) = u(y), \qquad V_{j+1}(y) = \min_{z\in[-d,0]}\{K(y,z) + V_j(z)\}.$$

They are nonnegative and vanish at zero. For $j \ge 1$, choosing $z = 0$ gives $V_j(y) \le K(y,0) = O(y^2)$ near zero; V_0 has the same bound. The implied bound is independent of j. Equation (128.6) implies $V_1(y) > V_0(y)$ for $y < 0$: equality at a minimizing z would require both $z = 0$ and equality in (128.6), hence $y = 0$.

For any $y < 0$, zero is not a minimizer of $K(y,z) + V_j(z)$. Indeed,

$$\partial_z K(y,0) = \frac{-y}{\ell(y,0)} > 0,$$

whereas $0 \le V_j(z) = O(z^2)$. A sufficiently small negative z therefore reduces the value. If $V_j > V_{j-1}$ on $[-d,0)$, evaluate at a minimizer $z_* < 0$ for $V_{j+1}(y)$. It follows that

$$V_{j+1}(y) = K(y,z_*) + V_j(z_*) > K(y,z_*) + V_{j-1}(z_*) \ge V_j(y).$$

Induction proves the strict inequality for every j and $y < 0$.

The minimizing path is symmetric, and its middle edge has length $\ell(y_m, y_m) = g + 2u(y_m)$. Splitting the action at that edge gives the exact identity

$$E_m = 2\min_{y\in[-d,0]}\{e(y) - g/2 + V_{m-1}(y)\}. \tag{128.7}$$

The outer minimizer cannot be zero, since $e'(0) = d/e(0) > 0$ and $V_{m-1}(y) = O(y^2)$. Evaluating the formula for E_{m+1} at its negative minimizer, and using strict increase of V_j there, proves $E_{m+1} > E_m$. Since $e(y) \ge D/2 - x_R(y) \ge g/2$, equality $E_m = 0$ would force $y = 0$ and then $d = 0$, which is excluded. Taking all heights zero gives

$$E_m \le 2\sqrt{(g/2)^2 + d^2} - g = \frac{2d^2}{\sqrt{(g/2)^2 + d^2} + g/2} \le \frac{2d^2}{g}.$$

Here $d \le 1/20$ and $g > 79/100$, proving the last bound in (128.5). A bounded increasing sequence has summable positive increments, which must tend to zero. □

128.4. **Joint arithmetic and its exact scope.** A regular periodic return orbit has a total record (K, N, T, h): winding $K \in \mathbb{Z}^2$, physical collision count N, flight length T, and induced period h. Define its phase at $(u, s, \beta, c) \in \mathbb{T}^2 \times \mathbb{T} \times \mathbb{R} \times \mathbb{T}$ to be $u \cdot K + sN + \beta T - ch$, with $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$.

Theorem 128.3 (Trivial joint periodic annihilator)**.** *For every $R \in I$, if*

$$\exp i(u \cdot K + sN + \beta T - ch) = 1 \tag{128.8}$$

for every regular periodic orbit of F_R^*, then $\beta = 0$ and $u = s = c = 0$ on their tori. It suffices to test the four cycles of Proposition 127.2, the long normal two-cycle, and the family of Lemma 128.1.

Proof. The long normal cycle gives $\exp i(2s + 2\beta g - c) = 1$. The mth excursion gives $\exp i((2m + 1)s + \beta L_m - mc) = 1$. Dividing the latter by the mth power of the former yields

$$\exp i(s + \beta E_m) = 1.$$

Taking consecutive quotients shows that $\beta(E_{m+1} - E_m) \in 2\pi\mathbb{Z}$ for all m. If $\beta \neq 0$, Theorem 128.2 makes these quantities nonzero and strictly smaller than 2π in absolute value for large m, a contradiction. Hence $\beta = 0$. The four unchanged first-return records in (127.6) now give $u = s = c = 0$. □

Corollary 128.4 (Finite uniform separation on compact frequency sets)**.** *For every $B < \infty$ and $\varepsilon > 0$, there are a finite list of these physical periodic words and $\eta > 0$ such that, uniformly in $R \in I$ and in $|\beta| \leq B$ at distance at least ε from the trivial phase, at least one word has $|\exp i(u \cdot K + sN + \beta T - ch) - 1| \geq \eta$.*

Proof. Each word has fixed integer record and continuous length on I. Theorem 128.3 gives a nonzero discrepancy at every point of the compact parameter–frequency set in question. The open sets on which a given word has positive discrepancy cover that set. Take a finite subcover and minimize the maximum of its discrepancies. This minimum is positive and supplies η. □

Corollary 128.5 (Full real period rank)**.** *The real vectors (K_1, K_2, N, h, T) of five fixed cycles span $\mathbb{R}^5$. The absolute determinant is $2(\sqrt{3} - 1)$, independently of R. Consequently no nonzero real linear combination of the four induced record coordinates is cohomologous to a constant by a transfer function for which periodic evaluation is valid.*

Proof. Append lengths L_2, L_3, L_4, L_4 to the four rows in (127.6), and append $(0, 0, 2, 1, 2g)$ as the fifth row. Subtracting the first row from the fifth leaves only the last entry $2g - L_2 = 2(\sqrt{3} - 1)$. The remaining determinant is that of V, of absolute value one. Evaluation of a real cohomology on these cycles therefore forces all coefficients, including the constant, to vanish. □

The preceding theorems are statements about real billiard orbits, not merely formal integer records. They also exclude a circle-valued coboundary with a continuous transfer function on the regular return branches: an almost-everywhere identity then extends to each regular periodic point by continuity and full support of the section measure. A transfer function assumed only measurable requires a separate regularity theorem. Similarly, a covariance conclusion requires a proved variance–coboundary criterion in the relevant function space. If that criterion and a continuous covariance family are established, Corollary 128.5 gives positive definiteness and, on I, a uniform positive lower eigenvalue. Existence and continuity of that covariance are not consequences of the determinant alone. Compact periodic separation is not a quantitative operator contraction or a returned temporal nonintegrability estimate.

129. Quantitative separation by logarithmic-length periods

The complete periodic annihilator was determined in Theorem 128.3. We now estimate the period increments uniformly in both the radius and the number of collisions. This gives an explicit separation at large roof frequencies, using three actual periodic words. Throughout this section $g = \sqrt{3} - 2R$, $d = 1/2 - R$, and e, ℓ and u are the functions in+(128.2). The coordinates below are heights, not arc coordinates.

Define the half action on $[-d,0]^m$ by

$$W_{m,R}(y) = e_R(y_1) - g/2 + \sum_{j=1}^{m-1} \left(\ell_R(y_j, y_{j+1}) - g\right) + u_R(y_m). \tag{129.1}$$

Reflection symmetry and uniqueness of the full minimizer imply $E_m(R) = 2\min W_{m,R}$. Denote its minimizing coordinates by $y_1^{(m,R)}, \dots, y_m^{(m,R)}$.

Lemma 129.1 (Uniform Hessian and endpoint estimates)**.** *On the entire height box, the Hessian $H = \nabla^2 W_{m,R}$ is tridiagonal and satisfies*

$$3I_m \le H \le 12 I_m, \qquad H_{jj} \le 10, \qquad \frac{1}{2} \le -H_{j,j+1} \le 2 \quad (1 \le j < m). \tag{129.2}$$

The last condition is empty when $m = 1$. For all $1 \le j \le m$,

$$\frac{3}{440}\, 20^{-(j-1)} \le -y_j^{(m,R)} \le \frac{1}{21}\left(\frac{7}{10}\right)^{j-1}. \tag{129.3}$$

All constants are independent of m and $R \in I$.

Proof. Write $x(y) = \sqrt{R^2 - y^2}$, $a = D/2 - x(y)$, $h = D - x(y) - x(z)$, $v = z - y$, with $D = \sqrt{3}$. The elementary bounds on the whole box are

$$x > 11/25, \quad |x'| \le 5/44, \quad 79/200 < a < 43/100,$$
$$79/100 < h < 86/100, \quad e < 11/25, \quad \ell < 9/10.$$

Also $u'' = R^2/x^3 \ge 1/R \ge 100/47$ and $u'' \le 55225/21296 < 3$. For the norm of a vector-valued function, the Hessian is its transverse positive semidefinite quadratic form plus the Hessian of the vector dotted with its unit direction. Consequently

$$e'' \ge (a/e)u'' > \frac{79}{88}\frac{100}{47} > \frac{3}{2}, \qquad \nabla^2 \ell \ge (h/\ell)\,\mathrm{diag}(u''(y), u''(z)) > \frac{3}{2} I_2.$$

Each variable of W_m occurs in two such terms, with $u'' > 3/2$ at the terminal endpoint. This proves the lower bound $H \ge 3I_m$, including $m = 1$.

The same norm formula gives

$$\ell_{yy}, \ell_{zz} \le \frac{1 + (5/44)^2}{79/100} + \frac{55225}{21296} < 4, \qquad e'' \le \frac{1 + (5/44)^2}{79/200} + \frac{55225}{21296} < 6.$$

An explicit mixed derivative is

$$-\ell_{yz} = \frac{(h + vy/x(y))(h - vz/x(z))}{\ell^3}. \tag{129.4}$$

Each numerator factor lies between $78/100$ and $87/100$. Using $79/100 < \ell < 9/10$ proves $1/2 < -\ell_{yz} < 2$. The diagonal bounds for W_m are therefore 10 at its first endpoint, 8 in its interior, and 7 at its last endpoint; when $m = 1$ the bound is 9. Absolute row sums are at most 12. The symmetric matrix bound $H \le 12 I_m$ follows, proving (129.2).

At zero the gradient is $a_0 e_1$, where $a_0 = d/e_R(0)$ satisfies $3/44 \le a_0 \le 1/7$. Integrate the Hessian on the segment from zero to the minimizer and write the resulting matrix as A. Stationarity gives $A y^{(m,R)} = -a_0 e_1$. The matrix $B = 10 I_m - A$ is entrywise nonnegative and tridiagonal. Its eigenvalues lie in $[-2, 7]$, so $\|B/10\|_2 \le 7/10$. Hence

$$A^{-1} = \frac{1}{10} \sum_{k \ge 0} (B/10)^k.$$

For the $(j,1)$ entry every power before $j-1$ vanishes. The nearest-neighbor product at power $j-1$ is at least $20^{-(j-1)}$. Entrywise nonnegativity and the norm bound yield

$$\frac{1}{10}20^{-(j-1)} \le (A^{-1})_{j1} \le \frac{1}{10}\sum_{k\ge j-1}(7/10)^k = \frac{1}{3}(7/10)^{j-1}.$$

Multiplication by the two bounds on a_0 proves (129.3). The matrix is an averaged Hessian of the actual nonlinear action; no quadratic approximation of the minimizing orbit has been made. □

Theorem 129.2 (Two-sided excess increments)**.** *For every $R \in I$ and $m \ge 1$,*

$$\frac{1}{10000}\,400^{-m} \le E_{m+1}(R) - E_m(R) \le \frac{1}{300}\left(\frac{49}{100}\right)^{m-1}. \tag{129.5}$$

In particular E_m converges uniformly on I to a continuous function E_∞, with

$$0 < E_\infty(R) - E_m(R) \le \frac{1}{153}\left(\frac{49}{100}\right)^{m-1}. \tag{129.6}$$

Proof. For a minimizer y of W_m, append the coordinate zero. With $K(y,z) = \ell(y,z) - g$, the resulting change in half action is

$$K(y_m,0) - u(y_m) = \sqrt{(g+u(y_m))^2 + y_m^2} - (g+u(y_m)) \le \frac{y_m^2}{2g}.$$

It follows that $E_{m+1} - E_m \le y_m^2/g$. The upper bound in (129.3) and $g > 79/100$ give the upper coefficient $100/34839 < 1/300$ in (129.5).

Conversely, truncate a minimizer z of W_{m+1} after its first m coordinates. The removed contribution is

$$K(z_m, z_{m+1}) + u(z_{m+1}) - u(z_m) \ge 2u(z_{m+1}),$$

by (128.6). Thus

$$E_{m+1} - E_m \ge 4u(z_{m+1}) \ge \frac{2z_{m+1}^2}{R} \ge \frac{9}{45496}\,400^{-m} > \frac{1}{10000}\,400^{-m}.$$

Here $u(z) = z^2/(R + x(z)) \ge z^2/(2R)$ and (129.3) were used. Both comparisons concern minima of the exact actions. Summing the geometric upper bound proves (129.6). Each E_m is continuous by Lemma 128.1; the uniform limit is continuous. □

Let P_0 denote the long normal orbit and P_m the mth excursion. For a periodic word P write

$$Z_P(u,s,\beta,c;R) = \exp i\big(u\cdot K_P + sN_P + \beta T_P - ch_P\big).$$

Put $a = \log(100/49)$ and, for $b \ge 1$, define

$$m(b) = 1 + \left\lceil \frac{\log b}{a} \right\rceil, \qquad \gamma = \frac{\log 400}{a} - 1. \tag{129.7}$$

Theorem 129.3 (Explicit large-roof phase separation)**.** *For $|\beta| = b \ge 1$, all $R \in I$, and arbitrary torus frequencies u, s, c, the three true periods $P_0, P_{m(b)}, P_{m(b)+1}$ satisfy*

$$\max_P |Z_P - 1| \ge \frac{b\,400^{-m(b)}}{15000\pi} \ge \frac{b^{-\gamma}}{2400000000\pi}. \tag{129.8}$$

The exponent is $\gamma = \log 400/\log(100/49) - 1$. For a whole band $1 \le |\beta| \le B$, the fixed three periods $P_0, P_{m(B)}, P_{m(B)+1}$ suffice, with lower bound $400^{-m(B)}/(15000\pi)$. Their longest physical word has $2m(B)+3$ collisions. All assertions are uniform in the radius.

Proof. The actual counts and lengths give the exact identity

$$Z_{P_{m+1}}\overline{Z_{P_m}Z_{P_0}} = \exp\bigl(i\beta(E_{m+1}-E_m)\bigr). \tag{129.9}$$

The winding, physical count and induced period cancel. With $m = m(b)$, (129.5) implies

$$\frac{b\,400^{-m}}{10000} \le b(E_{m+1}-E_m) \le \frac{1}{300}.$$

For $|v| \le \pi$, $|e^{iv}-1| \ge 2|v|/\pi$. The distance from one of a product of three unit complex numbers is at most the sum of their individual distances from one. Applying these facts to (129.9) proves the first bound in (129.8). Since $m(b) \le 2 + \log b/a$, the second bound follows. For the band assertion use $m(B)$; the upper phase bound still holds, while $b \ge 1$ supplies the printed lower bound. The number of collisions was proved in Lemma 128.1. □

Corollary 129.4 (An approximate-phase lower bound)**.** *Let $q : Y_R^* \to \mathbb{C}$ have modulus one and be continuous in neighborhoods of all states in the three cycles used above. Suppose q is measurable elsewhere and let*

$$D(q) = \operatorname*{ess\,sup}_{x\in Y_R^*} \left| e^{i(u\cdot\kappa_R^*(x)+sr_R^*(x)+\beta\tau_R^*(x))}q(F_R^*x) - e^{ic}q(x)\right|.$$

For $b = |\beta| \ge 1$ and $m = m(b)$,

$$D(q) \ge \frac{b\,400^{-m}}{10000\pi(m+1)}. \tag{129.10}$$

Proof. The return branches through these cycles are regular, and their section measure has full support. Continuity therefore extends the essential bound to every state of each selected cycle. Multiplication around a cycle of induced period h telescopes the unit phases and gives $|Z_P - 1| \le hD(q)$. In (129.9) the sum of the three periods is $1 + m + (m+1) = 2(m+1)$. The lower sine bound in the preceding proof therefore gives $2b(E_{m+1}-E_m)/\pi \le 2(m+1)D(q)$. Apply the lower bound in (129.5). □

The estimates are quantitative statements about the specified mechanical return map, and in particular about continuous circle-valued approximate phases. They do not assert regularity of a merely measurable transfer function, a lower bound for a spectral eigenfunction on these orbits, or a transfer-operator resolvent estimate. Those are distinct analytical steps when using (129.10) in a high-frequency argument. The raw characteristic-function tail is not replaced by the period discrepancy.

130. Uniform phase separation and positive-measure coercivity

We strengthen the frequency estimate by comparing consecutive exact minimizers. The choice of a period will depend on the frequency and possibly on the radius. This dependence is essential: a fixed triple of periods is not asserted to separate the entire unbounded frequency axis by a positive constant.

Retain $K_R(y,z) = \ell_R(y,z) - g$ and $W_{m,R}$ from (129.1). Write

$$a_m(R) = -y_m^{(m,R)} > 0, \qquad \Delta_m(R) = E_{m+1}(R) - E_m(R) > 0.$$

The symbol a_m denotes a terminal height; it is unrelated to the logarithmic constant in (129.7).

130.1. **Comparison of successive exact minima.**

Lemma 130.1 (The terminal value function)**.** *For $y \in [-d, 0]$ put*

$$V_R(y) = \min_{z\in[-d,0]} \{K_R(y, z) + u_R(z)\}, \qquad \delta_R(y) = V_R(y) - u_R(y).$$

The minimizer $f_R(y)$ is unique. It belongs to $(-d, 0)$ for $y < 0$ and equals zero for $y = 0$. The functions are twice continuously differentiable, with one-sided derivatives at the endpoints, and

$$f_R'(y) \geq \frac{1}{14}, \qquad |f_R(y)| \geq \frac{|y|}{14}, \qquad |\delta_R''(y)| \leq 2, \qquad |\delta_R'(y)| \leq 2|y|. \tag{130.1}$$

These estimates hold on the entire height interval, uniformly in R.

Proof. The second derivative in z is $K_{zz} + u'' > 0$. At $z = -d$,

$$K_z(y, -d) = \frac{h(y, -d)(-d)/x_R(-d) - d - y}{\ell_R(y, -d)} < 0, \qquad u_R'(-d) < 0.$$

At $z = 0$ and $y < 0$ the derivative is $-y/\ell_R(y, 0) > 0$. Consequently the minimizer is interior and unique. For $y = 0$, (128.6) gives the unique minimizer zero. The implicit function theorem applies to the stationarity equation, including its smooth extension near the endpoints, and gives

$$f_R'(y) = \frac{-K_{yz}(y, f_R(y))}{K_{zz}(y, f_R(y)) + u_R''(f_R(y))} \geq \frac{1/2}{4 + 3} = \frac{1}{14}.$$

Here the individual segment bounds established in the proof of Lemma 129.1 were used. Integration from y to zero proves the second inequality of (130.1).

The envelope formula is

$$V_R''(y) = K_{yy}(y, f_R(y)) - \frac{K_{yz}(y, f_R(y))^2}{K_{zz}(y, f_R(y)) + u_R''(f_R(y))}.$$

The Hessian of $K_R(y, z) + u_R(z)$ dominates $(3/2)I_2$. Its Schur complement therefore is at least $3/2$: evaluate its quadratic form on $(1, t)$ and minimize over t. Also $V_R'' \leq K_{yy} \leq 4$. Since $100/47 \leq u_R'' < 3$, subtraction shows $|\delta_R''| \leq 2$. The first derivatives of K_R and u_R vanish at the origin, so $\delta_R'(0) = 0$. Integration proves the last inequality. All estimates concern the nonlinear action itself. □

Proposition 130.2 (Consecutive terminal heights)**.** *For every $m \geq 1$ and $R \in I$,*

$$a_{m+1}(R) \geq \frac{3}{70} a_m(R). \tag{130.2}$$

Proof. Let y minimize $W_{m,R}$ and let (X, z) minimize $W_{m+1,R}$, where X consists of the first m coordinates. Eliminating the last coordinate gives exactly

$$\min_z W_{m+1,R}(X, z) = W_{m,R}(X) + \delta_R(X_m), \qquad z = f_R(X_m).$$

All minimizing coordinates are interior by Lemma 128.1. Stationarity and the fundamental theorem of calculus thus yield

$$A(X - y) = -\delta_R'(X_m) e_m, \qquad A = \int_0^1 \nabla^2 W_{m,R}(y + t(X - y))\, \mathrm{d}t \geq 3I_m.$$

The segment stays in the height box. In particular $0 < (A^{-1})_{mm} \leq 1/3$. Taking the last coordinate and applying Lemma 130.1, we obtain

$$|y_m| \leq |X_m| + \frac{1}{3}|\delta_R'(X_m)| \leq \frac{5}{3}|X_m|.$$

Finally $|z| = |f_R(X_m)| \geq |X_m|/14$. These two inequalities give (130.2). No monotonicity of the terminal heights, and no quadratic replacement of $W_{m,R}$, is assumed. □

Corollary 130.3 (A lower ratio for the excess increments)**.** *For every $m \geq 1$ and $R \in I$,*

$$\Delta_{m+1}(R) \geq 10^{-5}\Delta_m(R). \tag{130.3}$$

Proof. The append and truncate comparisons in the proof of Theorem 129.2 give

$$\Delta_m \leq \frac{a_m^2}{g}, \qquad \Delta_{m+1} \geq \frac{2}{R}a_{m+2}^2.$$

Applying (130.2) twice, and using $g > 79/100$ and $R \leq 47/100$, yields

$$\Delta_{m+1} \geq \frac{2g}{R}\left(\frac{3}{70}\right)^4\Delta_m > \frac{158}{47}\left(\frac{3}{70}\right)^4\Delta_m > \frac{\Delta_m}{100000}.$$

This is a comparison between adjacent increments, stronger for the present purpose than separate exponential upper and lower bounds. □

130.2. **A frequency-dependent triple and a fixed finite menu.** Put

$$\eta_0 = \frac{1}{4000000}, \qquad \sigma_0 = \frac{1}{6000000\pi}, \qquad M(b) = m(b) + 1, \tag{130.4}$$

where $m(b)$ is defined in (129.7).

Theorem 130.4 (Uniform positive phase separation)**.** *For every $R \in I$ and $b \geq 1$ the integer*

$$j(R, b) = \min\{j \geq 1 : b\Delta_j(R) \leq 1/2\} \tag{130.5}$$

is well defined and satisfies $j(R, b) \leq m(b)$. If $|\beta| = b$, then, for arbitrary u, s, c,

$$\max_{P \in \{P_0, P_{j(R,b)}, P_{j(R,b)+1}\}} |Z_P(u, s, \beta, c; R) - 1| \geq \sigma_0. \tag{130.6}$$

The longest of these physical words has at most $2m(b) + 3$ collisions. For the entire band $1 \leq |\beta| \leq B$, the fixed menu

$$\mathcal{P}_B(R) = \{P_0(R), P_1(R), \ldots, P_{m(B)+1}(R)\} \tag{130.7}$$

satisfies the same lower bound for its maximum, uniformly in R. It contains $m(B) + 2$ periods.

Proof. The upper bound of (129.5) gives $b\Delta_{m(b)} \leq 1/300$. Thus the defining set in (130.5) is nonempty. If $j(R, b) = 1$, its lower bound gives $b\Delta_1 \geq \Delta_1 \geq 1/4000000 = \eta_0$. If $j(R, b) > 1$, minimality and (130.3) give

$$b\Delta_{j(R,b)} > 10^{-5}/2 \geq \eta_0.$$

In either case

$$\eta_0 \leq b\Delta_{j(R,b)} \leq 1/2. \tag{130.8}$$

The exact three-period identity (129.9) therefore has distance at least $2\eta_0/\pi$ from one. The distance of a product of three unit complex numbers from one is at most the sum of their three distances. This proves (130.6), since $2\eta_0/(3\pi) = \sigma_0$. The physical and induced counts are those of Lemma 128.1; hence the stated word length. Finally $m(b) \leq m(B)$ on the band, so the selected triple is contained in (130.7).

The index selection uses a first crossing, not monotonicity of Δ_j. It need not be continuous in R or b. The menu has fixed indices and consists of analytically varying true periods, which is the asserted parameter-uniform conclusion. □

Corollary 130.5 (Logarithmic uniform-norm coercivity)**.** *Let q be circle-valued and measurable on Y_R^*, and continuous in neighborhoods of every state of the triple in* (130.6)*. With $D(q)$ as in Corollary 129.4, for $b = |\beta| \geq 1$,*

$$D(q) \geq \frac{1}{4000000\pi(j(R, b) + 1)} \geq \frac{1}{4000000\pi M(b)}. \tag{130.9}$$

Proof. Continuity and full support extend the essential bound to the selected regular periodic states. Telescoping around each period bounds its phase discrepancy by its induced length times $D(q)$. The three induced lengths sum to $2(j(R,b)+1)$. Apply the product identity and (130.8) to obtain

$$2\eta_0/\pi \le 2(j(R,b)+1)D(q).$$

This proves (130.9). The modulus-one condition is used in telescoping and is not a consequence of an arbitrary operator normalization. □

130.3. **Regular tubes of the actual return map.** We use the Euclidean metric in local (α, p) coordinates. The following uniformity concerns the specified periodic family, not all branches of the induced map.

Lemma 130.6 (Uniform return neighborhoods of the period family)**.** *There are constants $0 < \rho_0 < 1$, $L \ge 2$ and $B_0 \ge 1$, independent of R and m, with the following properties. For every selected state x_P of P_0 or any P_m, the ball $B(x_P, \rho_0)$ lies in Y_R^* and in one regular first-return branch. On this ball r_R^* and κ_R^* are constant, $r_R^* \in \{2,3\}$, and*

$$\mathrm{Lip}(F_R^*) \le L, \qquad \mathrm{Lip}(\tau_R^*) \le B_0.$$

If P has induced period h and $0 < \rho \le \rho_0 L^{-h}$, every $x \in B(x_P, \rho)$ follows the same h return branches as x_P, and, writing $F = F_R^*$,

$$d(F^k x, F^k x_P) \le L^k \rho \qquad (0 \le k \le h), \tag{130.10}$$

$$|T_{h,R}(x) - T_{h,R}(x_P)| \le 2B_0 L^h \rho, \tag{130.11}$$

$$\nu_R^*(B(x_P,\rho)) = \rho^2/(4c_*). \tag{130.12}$$

Proof. We verify that the family stays a uniform distance from every local singularity relevant to a single return. The estimates in Lemma 128.1 hold for all its contact heights: $-2d/5 < y < 0$, the long-flight third-disk clearances are at least $9/500$, and the incidence cosine at A exceeds $1/25$. At the facing contacts on O, C it is bounded below by a positive constant from the horizontal-component inequalities in that proof. End flights have a uniform positive distance from every third disk: the opposite facing center is horizontally separated by $D/2$, B is vertically separated, and the remaining lattice centers obey the enumerated uniform separation bounds. All flight lengths are bounded above and below by positive constants.

Return membership has uniform margins as well. The selected contacts on O satisfy

$$|\alpha - \pi/6| \le 1/22 < 3/50, \qquad |p| \le 1/22 + 2/79 < 3/20.$$

The unselected contacts on C and A have angles near $7\pi/6$ and equal to $2\pi/3$, respectively, uniformly separated from U and the four squares in Y_R. The long normal orbit has the same strict margins. Each selected-to-selected path consequently contains two physical flights, or three through C, A, O.

For clarity, compactness is applied only to these bounded-length paths. The parameters R and their contact coordinates range in a compact box. Every limiting path satisfies the non-grazing, nonocclusion and return-membership margins just established. In a neighborhood of such a path the selected entry roots are simple; the physical collision map and flight time are analytic in the initial state and parameter. There are only the finitely many center itineraries just listed, up to lattice translation. A finite cover of their compact closures therefore gives a common neighborhood radius and uniform first-derivative bounds for the actual two- or three-flight first returns. Decrease the radius so its coordinate balls lie in Y_R^* and contain no change of lattice labels; enlarge the derivative bounds to $L \ge 2$, $B_0 \ge 1$. This proves the first part without taking a derivative of an arbitrarily long return.

Starting in a ball of radius at most $\rho_0 L^{-h}$, induction gives the first inequality in (130.10) and preserves the same return decisions at each step. Summing the roof variations gives $B_0\rho\sum_{k<h}L^k \le 2B_0L^h\rho$. Finally $\mathrm{d}\nu_R^* = (4\pi c_*)^{-1}\,\mathrm{d}\alpha\,\mathrm{d}p$ on the ball, whose Euclidean area is $\pi\rho^2$. This proves the last equality. □

Theorem 130.7 (Positive-measure coercivity for regular phases)**.** *Fix $0<\alpha\le 1$, $H\ge 1$ and $1\le p<\infty$. Suppose q is measurable and circle-valued on Y_R^* and has an α-Hölder representative, of constant at most H, on the balls of radius ρ_0 around all states of the selected triple. Define*

$$\mathcal{D}_q(x) = e^{i(u\cdot\kappa_R^*(x)+sr_R^*(x)+\beta\tau_R^*(x))}q(F_R^*x) - e^{ic}q(x).$$

For $b=|\beta|\ge 1$, put

$$\rho(b,H) = L^{-M(b)}\min\left\{\frac{\rho_0}{2}, \frac{\sigma_0}{8bB_0}, \left(\frac{\sigma_0}{8H}\right)^{1/\alpha}\right\}. \tag{130.13}$$

Then the actual invariant return probability satisfies

$$\|\mathcal{D}_q\|_{L^p(\nu_R^*)} \ge \frac{\sigma_0}{2M(b)}\left(\frac{\rho(b,H)^2}{4c_*}\right)^{1/p}. \tag{130.14}$$

All geometric constants are uniform over $R\in I$. No independence of the returns is assumed.

Proof. Select from the triple a period P with $|Z_P-1|\ge\sigma_0$. Let its selected state be x_P and its induced period be $h\le M(b)$. Write $\varphi = u\cdot\kappa_R^* + sr_R^* + \beta\tau_R^*$, $F=F_R^*$ and $S_h\varphi=\sum_{k<h}\varphi\circ F^k$. The lattice and count terms are constant along the corresponding branches in the tube. Thus for $x\in B(x_P,\rho(b,H))$, Lemma 130.6 gives

$$|S_h\varphi(x) - S_h\varphi(x_P)| \le 2bB_0L^h\rho(b,H).$$

At x_P, the modulus of $e^{iS_h\varphi(x_P)}q(F^hx_P) - e^{ich}q(x_P)$ is $|Z_P-1|$. Since $F^hx_P=x_P$, the Hölder bounds at the two endpoints and (130.13) imply, throughout the smaller ball,

$$\begin{aligned}\left|e^{iS_h\varphi(x)}q(F^hx) - e^{ich}q(x)\right| &\ge \sigma_0 - 2bB_0L^h\rho(b,H) - 2H(L^h\rho(b,H))^\alpha\\ &\ge \sigma_0/2.\end{aligned}$$

Choose the indicated representative on these balls; changes on null sets do not affect the following integrals. Telescoping h times bounds the left side by $\sum_{k<h}|\mathcal{D}_q(F^kx)|$ almost everywhere. Minkowski's inequality on the smaller ball and invariance of ν_R^* give

$$\frac{\sigma_0}{2}\nu_R^*(B(x_P,\rho(b,H)))^{1/p} \le \sum_{k<h}\|\mathcal{D}_q\circ F^k\|_{L^p(B(x_P,\rho(b,H)),\nu_R^*)} \le h\|\mathcal{D}_q\|_{L^p(\nu_R^*)}.$$

Use the exact ball mass and $h\le M(b)$ to conclude. The estimate uses measure preservation, not disjointness or probabilistic independence of the iterates of the ball. □

Corollary 130.8 (A polynomial budget for the phase regularity)**.** *Fix α, p, H_0 as above and $\zeta\ge 0$. If the Hölder constants satisfy $H\le H_0b^\zeta$, set*

$$\kappa = \frac{\log L}{\log(100/49)} + \max\{1,\zeta/\alpha\}. \tag{130.15}$$

There is $C>0$, depending only on these fixed constants and the mechanical family, such that

$$\|\mathcal{D}_q\|_{L^p(\nu_R^*)} \ge C\frac{b^{-2\kappa/p}}{1+\log b} \qquad (b\ge 1). \tag{130.16}$$

Proof. Writing $a = \log(100/49)$, we have $M(b) \le 3 + \log b/a$ and hence $L^{-M(b)} \ge L^{-3} b^{-\log L/a}$. The minimum in (130.13) is at least

$$\min\left\{\frac{\rho_0}{2}, \frac{\sigma_0}{8B_0}, \left(\frac{\sigma_0}{8H_0}\right)^{1/\alpha}\right\} b^{-\max\{1,\zeta/\alpha\}}.$$

Substitution into (130.14) proves the claim with a positive constant independent of R and b. □

130.4. **The operator interface.** Theorems 130.4 and 130.7 have different hypotheses. The former is an unconditional estimate for the specified mechanical periods. The latter is an estimate for a stated regularity class of circle-valued phases, in the actual invariant measure. In particular it supplies a positive-measure obstruction rather than a conclusion obtained by evaluating an unspecified measurable representative at isolated periodic points.

To apply (130.16) to a transfer operator, one must construct from its approximate spectral vectors phases in that class, bound the Hölder constant, and compare their physical L^p defect with the operator defect. Such a comparison would contradict an error smaller than the right side of (130.16). It is not proved merely by writing an eigenvalue equation. In particular a division by the modulus of a spectral vector requires a separate lower-modulus estimate. General Livšic regularity results for Markov systems [7] concern precise dynamical and cohomological hypotheses; they do not supply a uniform quantitative regularity bound for these approximate phases without checking those hypotheses and the defect comparison. No such comparison is assumed here for the full induced operator. The common operator realization and the all-branch raw residual estimates in the final section remain the requirements for the original joint density theorem.

131. Exact return means and stationary unfinished flights

The enlarged section gives explicit normalization constants without any local-limit assumption. We use ergodicity of the finite-horizon periodic dispersing collision map, as supplied by the standard billiard framework [6]. No spectral conclusion for the new first-return map is imported from that fact.

Proposition 131.1 (Exact induced means)**.** *For the actual return records to Y_R^*,*

$$\int (\kappa_{R,1}^*, \kappa_{R,2}^*, r_R^*, \tau_R^*)\, \mathrm{d}\nu_R^* = \left(0, 0, \frac{1}{c_*}, \frac{\bar\tau_R}{c_*}\right), \quad \bar\tau_R = \frac{\sqrt{3}}{4R} - \frac{\pi R}{2}. \tag{131.1}$$

The mean roof satisfies

$$\left|\frac{\mathrm{d}}{\mathrm{d}R}\int \tau_R^*\, \mathrm{d}\nu_R^*\right| < \frac{15}{4c_*} \qquad (R \in I). \tag{131.2}$$

The equilibrium physical collision rate remains $1/\bar\tau_R$.

Proof. For any integrable physical collision observable f, the disjoint return-tower levels partition the collision section up to null sets. Invariance, first for nonnegative functions and then for positive and negative parts, gives the Kac identity

$$\int_{Y_R^*} \sum_{j=0}^{r_R^*(x)-1} f(T_R^j x)\, \mathrm{d}\nu(x) = \int f\, \mathrm{d}\nu.$$

Taking $f = 1$ and dividing by c_* gives the mean return count. Taking $f = \tau_R$ gives the mean induced roof. The physical suspension normalization proved earlier gives the displayed value of $\bar\tau_R$. The physical displacement is integrable by the finite-horizon bound. Spatial inversion of the entire lattice maps its value to its negative and preserves ν, so its mean is zero. This proves (131.1).

The section volume c_* is independent of R. Differentiating the explicit expression for $\bar\tau_R$ yields $-\sqrt{3}/(4R^2) - \pi/2$. The bounds $\sqrt{3} < 7/4$, $\pi < 22/7$ and $R \geq 9/20$ show its absolute value is less than $15/4$. Finally the ratio of mean physical collisions per return to mean physical time per return is $1/\bar\tau_R$. □

Proposition 131.2 (The complete induced suspension)**.** *The equilibrium flow is represented by the suspension over F_R^* with roof τ_R^* and probability*

$$\frac{\nu_R^*(\mathrm{d}x)\,\mathrm{d}s}{\int \tau_R^*\,\mathrm{d}\nu_R^*}, \qquad 0 \leq s < \tau_R^*(x). \tag{131.3}$$

If A_R and B_R denote respectively the age since the last visit and the time to the next visit to Y_R^* under this stationary law, then, for $q \geq 0$,

$$\Pr(A_R > q) = \Pr(B_R > q) = \frac{\int (\tau_R^* - q)_+\,\mathrm{d}\nu_R^*}{\int \tau_R^*\,\mathrm{d}\nu_R^*}. \tag{131.4}$$

At either one fixed endpoint of a time interval of length t, the incomplete induced flight, its physical collision count and its lattice displacement divided by $\sqrt{t}$ converge to zero in probability as $t \to \infty$, for each fixed R.

Proof. In the physical suspension over T_R, group the consecutive single-flight intervals before each first return to Y_R^*. The tower identity and the normalization $c_* \int \tau_R^*\,\mathrm{d}\nu_R^* = \bar\tau_R$ convert the original suspension measure exactly to (131.3). This rearranges the actual dependent path; it introduces no new launch law and no independent flight variables. The marked formula of Proposition 126.1 can consequently be grouped into these complete return records while keeping both endpoint pieces.

For a given roof τ, the set of $s \in [0, \tau)$ with age $s > q$ has length $(\tau - q)_+$. The set with remaining time $\tau - s > q$ has the same length. Integration proves (131.4). Its right side tends to zero by the integrability in Proposition 131.1. Each endpoint roof is finite almost surely under the length-biased stationary law; stationarity gives the same marginal at the other fixed endpoint. Thus its ratio to $\sqrt{t}$ tends to zero in probability.

Every physical flight has length at least $1 - 2R \geq 3/50$. The number of collisions in an incomplete interval of length s is at most $1 + s/(1 - 2R)$. Its displacement between fundamental cells is bounded by a constant plus a fixed multiple of s, since the speed is one and the lattice cell has bounded diameter. These bounds prove the same endpoint statement for the two marked rewards. □

A fixed-endpoint assertion is not a bound on the maximum residual over all times in $[0, t]$. Nor does pointwise integrability in R imply a uniform residual estimate over I. Such uniform or process-level claims require uniform integrability or a quantitative tail for the induced roof. The functional limit assumptions in Proposition 228.2 therefore remain explicit. The exact mean and the modulus (131.2), by contrast, are already available for geometric error propagation. They do not require recovery of the entire preparation distribution in A2-GEOM.

132. Stability of the actual return records

We compare the first returns themselves, including their changing domains. The comparison is made on the common collision space $M = \mathbb{T} \times (-1, 1)$ with the probability ν in (126.1). It is not a comparison between independently chosen inducing schemes. The squares defining Y_R move analytically with R, and U is fixed. Consequently Y_R^* has constant measure c_*, and its boundary is a finite union of smooth arcs, continuously varying with R.

We use only ergodicity, not a parameter-uniform mixing rate, from the finite-horizon Sinai theory. In the present family the obstacles are smooth, strictly dispersing and disjoint, and the

horizon is finite. The flow-mixing theorem [5, Corollary 1.3 and correction note] therefore implies ergodicity of the collision map: the suspension of a nontrivial invariant collision set would be a nontrivial invariant flow set. Thus every Y_R^* has full saturation. This explicitly supplies the full-tower hypothesis in the Kac identity used above.

Write $r = r_R^*$, $F = F_R^*$ and $\tau^* = \tau_R^*$ in sums indexed by induced returns. For $n \geq 1$ define the actual cumulative record

$$J_{n,R} = \left(\sum_{j<n} \kappa_R^* \circ F^j, \sum_{j<n} r_R^* \circ F^j, \sum_{j<n} \tau_R^* \circ F^j \right) = (K_{n,R}, N_{n,R}, T_{n,R})$$

on Y_R^*. Its values lie in $\mathbb{Z}^3 \times \mathbb{R}$; in particular the last coordinate is one cumulative time, not n independent roof coordinates. A tilde denotes extension by zero to M. Let $p_{n,R}$ denote the raw mixed density of $J_{n,R}$ under ν_R^*, whose existence is included in Theorem 132.2 below. Norms of densities use counting measure times Lebesgue measure, denoted by $\mathfrak{m}$.

Lemma 132.1 (Almost-everywhere stability of a finite return record)**.** *Fix $R_0 \in I$ and a finite n. Outside a ν-null subset of M, $\widetilde{J}_{n,R}(x) \to \widetilde{J}_{n,R_0}(x)$ as $R \to R_0$. For $x \in Y_{R_0}^*$ outside that null set, the physical word, every intermediate return decision, $K_{n,R}(x)$ and $N_{n,R}(x)$ are constant for all sufficiently close R. On a neighborhood of such a point the cumulative time is the physical analytic length of that fixed word. The neighborhood and parameter tolerance may depend on x and n.*

Proof. Remove the boundary of $Y_{R_0}^*$, all singular initial states and all preimages under T_{R_0} of that boundary. These sets are null: the boundary has zero area, the map preserves ν, and the billiard singularities and their iterates have zero flux measure. Remove also the null set on which a required return is infinite. For a remaining $x \in Y_{R_0}^*$, the first n returns contain finitely many regular physical collisions. Every tested state lies strictly inside or strictly outside the return set. The finite-itinerary analyticity following (126.2), and the continuous motion of the return boundary, preserve all these decisions for nearby (R, x). They also preserve the selected entry roots and lattice labels. The final physical count and displacement are therefore unchanged, while the sum of flight lengths varies analytically. A remaining point outside $Y_{R_0}^*$ stays outside Y_R^* for nearby R, and its zero extension stays zero. These statements prove the asserted convergence. No common neighborhood for infinitely many words is used. □

Theorem 132.2 (Finite-record stability in density and first moment)**.** *For every fixed $n \geq 1$, the maps*

$$R \longmapsto \widetilde{J}_{n,R} \in L^1(\nu; \mathbb{R}^4), \qquad R \longmapsto p_{n,R} \in L^1(\mathfrak{m}) \tag{132.1}$$

are continuous. Both maps are uniformly continuous on I. Moreover $\{\widetilde{N}_{n,R} : R \in I\}$ and $\{\widetilde{T}_{n,R} : R \in I\}$ are uniformly integrable. If $\eta_{n,R}$ is the record law, then

$$W_1(\eta_{n,R}, \eta_{n,S}) \leq c_*^{-1} \|\widetilde{J}_{n,R} - \widetilde{J}_{n,S}\|_{L^1(\nu)}, \tag{132.2}$$

where W_1 uses the Euclidean distance on $\mathbb{Z}^3 \times \mathbb{R}$. In particular, for the original, unmodified characteristic functions,

$$\begin{aligned} |\Phi_{n,R}(\omega) - \Phi_{n,S}(\omega)| &\leq \|p_{n,R} - p_{n,S}\|_{L^1(\mathfrak{m})}, \\ |\Phi_{n,R}(\omega) - \Phi_{n,S}(\omega)| &\leq c_*^{-1} |\omega| \, \|\widetilde{J}_{n,R} - \widetilde{J}_{n,S}\|_{L^1(\nu)}. \end{aligned} \tag{132.3}$$

No modulus uniform in n, and no frequency-integrated estimate, is asserted by these inequalities.

Proof. The Kac identity and invariance of F_R^* give, exactly,

$$\int_M \widetilde{N}_{n,R} \, \mathrm{d}\nu = n, \qquad \int_M \widetilde{T}_{n,R} \, \mathrm{d}\nu = n\bar{\tau}_R. \tag{132.4}$$

For nonnegative functions $h_j \to h$ almost everywhere with $\int h_j \to \int h$, the identity

$$\|h_j - h\|_1 = \int h_j + \int h - 2\int \min(h_j, h)$$

and dominated convergence for $\min(h_j, h) \le h$ prove L^1 convergence. Apply this to the two nonnegative coordinates, using Lemma 132.1, (132.4) and continuity of $\bar{\tau}_R$. Uniform finite horizon bounds every single-flight lattice label by a fixed constant. Thus $|\widetilde{K}_{n,R}| \le C\widetilde{N}_{n,R}$. The already established L^1 convergence of the counts makes these bounds uniformly integrable along each convergent parameter sequence. Almost-everywhere convergence and truncation then give L^1 convergence of the displacement as well. This proves the first continuity in (132.1).

A continuous image of the compact interval I is compact in L^1. For completeness, an L^1-compact nonnegative family is uniformly integrable. Cover it by finitely many L^1 balls of radius ϵ with centers $h_1, \dots, h_q$. For a member h in the ball about h_i,

$$\int_{\{h>A\}} h \le 2\|h - h_i\|_1 + 2\int_{\{h_i > A/2\}} h_i.$$

Choose A for the finite set of centers and then let ϵ decrease. This proves the stated uniform integrability without a tail assumption or an exponential return estimate.

We next prove existence and L^1 continuity of the density. Partition Y_R^*, up to null sets, by total physical count and ordered lattice word. The argument of Corollary 142.4 applies to this actual first-return event as well: on each regular word the cumulative length has at most one critical point by Proposition 142.3; elsewhere it is a submersion. Coarea on a countable exhaustion and then summation give absolute continuity. This argument recomputes the return event for Y_R^*; it does not transfer a density normalization from Y_R.

Fix R_0 and $\epsilon > 0$. After deleting the null sets just described, choose finitely many smooth nonnegative cutoffs ψ_i on M, with $\sum_i \psi_i \le 1$, whose supports lie compactly in regular first-n-return patches for R_0, such that

$$c_*^{-1}\int \sum_i \psi_i \, d\nu > 1 - \epsilon.$$

Refine the patches so that on each one a coordinate derivative of the cumulative length is nonzero. Compact support and the finite-return lemma preserve the same word, all return decisions and this derivative for R close to R_0. On each patch the map consisting of the other initial coordinate and the cumulative length is a local diffeomorphism. It and its inverse depend continuously in C^1 on R on slightly larger compact patches. Changing variables and integrating the other coordinate therefore shows that the pushforward of $c_*^{-1}\psi_i\nu$ has a density continuous in L^1 with R. One can see this directly by extending the transformed smooth weight by zero off its coordinate patch: the compactly supported cutoff vanishes near the patch boundary, the Jacobian is bounded away from zero, and dominated convergence applies on a fixed bounded coordinate box. The lattice coordinates are constant on the patch.

The remainder is a positive measure. Its mass is $1 - c_*^{-1}\int \sum_i \psi_i \, d\nu < \epsilon$, for all these R, because the cutoff supports stay inside Y_R^* and $\nu(Y_R^*) = c_*$. Consequently

$$\|p_{n,R} - p_{n,R_0}\|_1 \le \sum_i \|p_{i,R} - p_{i,R_0}\|_1 + 2\epsilon.$$

Let $R \to R_0$ and then $\epsilon \to 0$. This proves the second continuity in (132.1). Compactness of I gives uniform continuity of both maps.

For a 1-Lipschitz function f, subtract the constant $f(0)$. Then

$$\int f \, d\eta_{n,R} = c_*^{-1}\int_M f(\widetilde{J}_{n,R}) \, d\nu.$$

The same identity holds for S. Subtraction and the Lipschitz bound, followed by the dual characterization of W_1, give (132.2). All first moments are finite by (132.4). Direct integration against the densities gives the first bound in (132.3). For the second, use the zero extensions and $|e^{ia} - e^{ib}| \le |a - b|$; the off-section constants cancel because both section measures equal c_*. These are estimates of the raw transforms, with no convolution or change of observable. □

Proposition 132.3 (Weighted records and a positive denominator)**.** *Fix n. Suppose measurable insertions $w_R : M \to \mathbb{R}$ satisfy $|w_R| \le M_0$ and, for every R_0, $w_R(x) \to w_{R_0}(x)$ for ν-almost every x as $R \to R_0$. The w_R-weighted record densities $p^w_{n,R}$ are continuous in $L^1(\mathfrak{m})$. Set*

$$d_n(\delta) = \sup_{|R-S|\le\delta} \|p_{n,R} - p_{n,S}\|_1, \qquad d^w_n(\delta) = \sup_{|R-S|\le\delta} \|p^w_{n,R} - p^w_{n,S}\|_1.$$

These moduli tend to zero. For any Borel observation event B in the record space, if $\eta_{n,R}(B) \ge b > 0$ and $d_n(\delta) < b$, then, whenever $|R - S| \le \delta$,

$$\left| \frac{\int_B p^w_{n,R}\,\mathrm{d}\mathfrak{m}}{\eta_{n,R}(B)} - \frac{\int_B p^w_{n,S}\,\mathrm{d}\mathfrak{m}}{\eta_{n,S}(B)} \right| \le \frac{d^w_n(\delta) + M_0 d_n(\delta)}{b - d_n(\delta)}. \tag{132.5}$$

Proof. Use the compact regular patches in the preceding proof. Replacing w_R by w_{R_0} costs at most $c_*^{-1} \int |w_R - w_{R_0}|\,\mathrm{d}\nu \to 0$, by dominated convergence and contraction of total variation under pushforward. On each fixed patch approximate the bounded measurable weight w_{R_0} in L^1 by a smooth one. The approximation error in its pushforward is at most the same L^1 error, uniformly in R. For the smooth weight, the coordinate-change proof above applies. The leftover measure has total variation at most $M_0\epsilon$. This proves weighted density continuity, and compactness gives the modulus.

Write $a_R = \int_B p^w_{n,R}\,\mathrm{d}\mathfrak{m}$ and $b_R = \eta_{n,R}(B)$. Then $|a_R - a_S| \le d^w_n(\delta)$, $|b_R - b_S| \le d_n(\delta)$ and $|a_R| \le M_0 b_R$. Since $b_S \ge b - d_n(\delta) > 0$, subtraction of the two ratios gives (132.5). □

The event B may fix the lattice coordinates and restrict the roof to an interval. The denominator in (132.5) is an actual event probability. The estimate does not remove its deterioration for small events. In particular density L^1 continuity alone is not the shrinking-interval LLT of Proposition 228.1; that proposition retains its stronger pointwise density hypotheses. An error bound $|\widehat{R} - R| \le \delta$ can be inserted into the displayed moduli, but no numerical sampling budget follows from a noneffective modulus.

Corollary 132.4 (Uniform stationary endpoint control)**.** *Put $H_R = \widetilde{T}_{1,R}$ and $b_* = \min_{R\in I} \bar\tau_R > 0$. The stationary induced suspensions, regarded as probabilities on $M \times \mathbb{R}_+$, have densities*

$$h_R(x, s) = \bar\tau_R^{-1} \mathbf{1}_{\{0\le s<H_R(x)\}}$$

with respect to $\nu(\,\mathrm{d}x)\,\mathrm{d}s$, and

$$\|h_R - h_S\|_1 \le b_*^{-1}(\|H_R - H_S\|_1 + |\bar\tau_R - \bar\tau_S|). \tag{132.6}$$

Furthermore

$$\Psi(q) := b_*^{-1} \sup_{R\in I} \int_M H_R \mathbf{1}_{\{H_R>q\}}\,\mathrm{d}\nu \longrightarrow 0 \quad (q \to \infty). \tag{132.7}$$

Under the stationary flow, at either deterministic endpoint of an interval of length t, the incomplete induced time, physical collision count and lattice displacement are $o_P(\sqrt{t})$, uniformly over $R \in I$. The same is true jointly at any fixed finite number of deterministic endpoints.

Proof. The normalization in (131.3) and $c_* \int \tau^*_R\,\mathrm{d}\nu^*_R = \bar\tau_R$ give h_R exactly. For fixed x, the symmetric difference of the intervals $[0, H_R(x))$ and $[0, H_S(x))$ has length $|H_R(x) - H_S(x)|$. Integrating, and then comparing the two normalizing constants, proves (132.6). The uniform integrability in Theorem 132.2 proves (132.7).

The full roof containing a stationary endpoint has tail

$$\Pr_R\{\text{containing roof} > q\} = \bar{\tau}_R^{-1} \int_M H_R \mathbf{1}_{\{H_R > q\}} \, \mathrm{d}\nu \le \Psi(q).$$

This is the length-biased law; replacing it by $\nu_R^*(\tau_R^* > q)$ would be incorrect. Both endpoint pieces are at most this full roof. The collision count of an incomplete piece of length s is at most $1 + (50/3)s$, and its lattice displacement is bounded by $C(1+s)$, by the physical estimates in Proposition 131.2. Hence there are fixed constants C_0, C_1 such that each marked endpoint norm is bounded by $C_0 + C_1 H$ when its containing roof has length H. Its probability of exceeding $\epsilon\sqrt{t}$ is at most $\Psi((\epsilon\sqrt{t} - C_0)/C_1)$ for large t, uniformly in R. Stationarity applies at every deterministic endpoint, including an endpoint depending deterministically on t. A union bound proves the finite-endpoint assertion; independence is not used. □

This removes the additional uniform-integrability premise for fixed stationary endpoints in this family. It is not a bound on a supremum over a growing time interval or at an arbitrary stopping time. The functional clock theorem still needs its printed process hypotheses. Likewise (132.3) is a uniform comparison of raw characteristic functions, not an integrable tail for any one of them. The high-frequency and central all-branch remainder problem is unchanged by an L^1 comparison of probability densities.

133. Exponential tails for the genuine return record

We now use the spectral theory of the collision map, in addition to the geometric and ergodic inputs used above. The operator in this section is not the operator of the induced map. A smooth killing weight on the collision space gives a scalar estimate for avoiding the actual return section. In particular no multiplication by its discontinuous indicator on an anisotropic space will be needed.

133.1. **The collision-space input.** We use the strong and weak distribution spaces of Demers–Zhang [4, Section 3.3]. Their properties needed here are the local uniform spectral gap and power bounds for unforced dispersing collision maps, including small changes of the scatterers and their boundary lengths [4, Theorems 2.1–2.6 and Remark 2.7(b)]. The spectral gap for an individual unforced Lorentz map is the input from [8] used there. Here is the precise consequence, together with the checks for this family.

Lemma 133.1 (Local uniform collision spaces)**.** *There is a finite cover of I by parameter neighborhoods I_j. On each I_j there is a Banach space $\mathcal{B}_j$ of distributions on a common collision space, a continuous mass functional ℓ_j, and transfer operators $\mathcal{L}_R$, such that*

$$\mathcal{L}_R = \Pi_R + \mathcal{N}_R, \quad \Pi_R h = \ell_j(h)\nu, \quad \Pi_R \mathcal{N}_R = \mathcal{N}_R \Pi_R = 0, \quad \|\mathcal{N}_R^k\|_{\mathcal{B}_j} \le C\vartheta^k \quad (k \ge 0), \tag{133.1}$$

where $0 < \vartheta < 1$ and $C < \infty$ can be chosen for all the finitely many neighborhoods. The norms of ν, ℓ_j and $\mathcal{L}_R$ are uniformly bounded. Multiplication by a smooth function g satisfies

$$\|gh\|_{\mathcal{B}_j} \le C\|g\|_{C^2}\|h\|_{\mathcal{B}_j}. \tag{133.2}$$

The transfer convention is $(\mathcal{L}_R h)(f) = h(f \circ T_R)$. No assertion about the unbounded induced twists is included in this lemma.

Proof. For the collision-space construction use coordinates (α, φ), where $p = \sin\varphi$. The invariant probability is the same for all parameters: $\mathrm{d}\nu = (4\pi)^{-1}\cos\varphi \, \mathrm{d}\alpha \, \mathrm{d}\varphi$. We do not conjugate a distribution space by the inverse of $p = \sin\varphi$ at grazing. At a fixed R_0, use the boundary coordinate $r_0 = R_0\alpha$. The parametrization of the nearby circle is then $r_0 \mapsto R n_{r_0/R_0}$. It is a C^3-bounded deformation, C^2-close to the original parametrization, and its constant speed R/R_0

tends to one. This is precisely the boundary-length reparametrization permitted in [4, Remark 2.7(b)].

The physical gap between different disks is at least $3/50$. The curvatures belong to $[100/47, 20/9]$, and the parametrized boundaries have uniform C^3 bounds. The finite horizon is uniform on I: a line with no disk intersection at the smallest radius would be an infinite free line in that table, which has no corridor; compactness of position and direction then bounds the free length, and increasing the radius cannot increase it. If a rectangular fundamental domain is desired, use the two-scatterer cover with periods $(1, 0)$ and $(0, \sqrt{3})$ and disk centers $0, b$. Its deck translation commutes with the collision map. Restricting to the deck-invariant distributions gives the original table. Thus the local billiard estimates are applied without an affine distortion of specular reflection.

The uniform geometric hypotheses of [4, Theorem 2.5] and the perturbation hypothesis of its Theorem 2.6 are satisfied. The unforced collision map at each R_0 has the simple eigenvalue one and a spectral gap. The strong/weak perturbation theorem, specifically the complementary power bound in [4, Theorem 2.1(3)], gives (133.1) on a neighborhood of R_0. The eigenprojection has the displayed form because the invariant probability is ν and mass is preserved. A finite cover of the compact parameter interval makes the constants uniform. This uses local common spaces and does not assert that all arbitrary inducing schemes share a Banach space.

For completeness, smooth multiplication follows directly from the stable-test and unstable-comparison norms defining these spaces. On a single admissible stable curve, multiplying its test function by g changes its Hölder norm by at most $C\|g\|_{C^1}$. For two matched curves at distance ϵ, split the difference of the weighted tests into the old test difference and the difference of the two restrictions of g. After graph matching, the latter has C^q norm at most $C\|g\|_{C^2}\epsilon^{1-q}$ by interpolation between its C^0 and C^1 bounds. The defining exponents satisfy $\beta < 1 - q$, so division by ϵ^β is bounded for $\epsilon \le 1$. The stable part of the norm controls this second term and the unstable part controls the first. The weak norm has the same single-curve multiplication bound. Density of the defining smooth distributions extends these bounds to $\mathcal{B}_j$, proving (133.2). The coordinate changes above and the finite cover have uniformly bounded derivatives. □

133.2. **A smooth avoidance estimate.** Choose once and for all $\chi \in C^\infty(M)$ such that

$$0 \le \chi \le 1, \quad \operatorname{supp}\chi \Subset \{|\alpha| < 1/400,\ |p| < 1/400\}, \quad a_\chi := \int \chi \, \mathrm{d}\nu > 0. \tag{133.3}$$

The indicated rectangle is inside the square of Y_R centered at $(0, 0)$ for every R; hence it is inside Y_R^*. Its support is away from grazing, so it is smooth also in the collision coordinates used in Lemma 133.1. This is a proof weight, not an extra field in the collision record.

Proposition 133.2 (Uniform soft-killing estimate)**.** *There exist $z_0, c, C > 0$ such that for every $R \in I$ and $k \ge 0$,*

$$\int_M \exp\left(-z_0 \sum_{j=0}^{k-1} \chi \circ T_R^j\right) \mathrm{d}\nu \le Ce^{-ck}. \tag{133.4}$$

Proof. On a local collision space set $\mathcal{L}_{R,z} = \mathcal{L}_R M_{e^{-z\chi}}$ for complex z. The multiplier bound proves analyticity in the strong operator norm and, uniformly in R, $\|\mathcal{L}_{R,z} - \mathcal{L}_R\| \le C|z|$ near zero. Equation (133.1) bounds the unperturbed resolvents uniformly on a circle around one and on a second circle $|w| = \vartheta_1$ with $\vartheta < \vartheta_1 < 1$, chosen disjointly. A uniformly convergent resolvent Neumann series therefore defines a rank-one analytic projection and a single eigenvalue $\lambda_R(z)$ near one for all $|z| < z_*$. The complementary powers are bounded by $C\vartheta_1^k$, after decreasing z_* if necessary.

Since $\ell_j(\nu) = 1$, differentiation at zero gives

$$\lambda_R(0) = 1, \qquad \lambda_R'(0) = -\ell_j(\mathcal{L}_R(\chi\nu)) = -a_\chi.$$

The fixed complex disk and the uniform resolvent bounds give a uniform bound on the second derivative of λ_R on a smaller disk, by Cauchy's formula. Thus $|\lambda_R(z) - 1 + a_\chi z| \le C_2|z|^2$ uniformly in R. For real small z the eigenvalue is real, by conjugation and uniqueness, and positive, by its closeness to one. Choose a fixed $z_0 > 0$ such that $C_2 z_0 \le a_\chi/2$ and $z_0 < z_*$. Then $0 < \lambda_R(z_0) \le 1 - a_\chi z_0/2$. The spectral decomposition gives $\|\mathcal{L}_{R,z_0}^k\| \le Ce^{-ck}$. Only finitely many collision spaces occur, so the same z_0, c, C work on all of I.

Iteration of the transfer identity, first for densities and then as bounded functionals, yields

$$\ell_j(\mathcal{L}_{R,z_0}^k \nu) = \int \prod_{j=0}^{k-1} e^{-z_0 \chi(T_R^j x)} \, \mathrm{d}\nu(x).$$

The norms of ℓ_j and ν are bounded. This proves the asserted scalar estimate. In particular the argument does not apply a local large-deviation formula outside its stated neighborhood of the mean. □

Theorem 133.3 (Exponential moments of the actual return block)**.** *Let $Q_R = r_R^* + \tau_R^* + |\kappa_R^*|$ on the genuine first-return section. There are constants $c_1, a_1, C_1 > 0$, independent of $R \in I$, such that*

$$\nu_R^*(r_R^* > k) \le C_1 e^{-c_1 k} \quad (k \ge 0), \qquad \int e^{a_1 Q_R} \, \mathrm{d}\nu_R^* \le C_1. \tag{133.5}$$

For $Q_{n,R} = \sum_{j<n} Q_R \circ (F_R^*)^j$ and every $n \ge 1$,

$$\int \exp(a_1 Q_{n,R}/n) \, \mathrm{d}\nu_R^* \le C_1, \qquad \nu_R^*(N_{n,R} > L) \le C_1 e^{-a_1 L/n}. \tag{133.6}$$

These are estimates for the original deterministic mechanical records.

Proof. If $x \in Y_R^*$ and $r_R^*(x) > k$, none of $T_R x, \dots, T_R^k x$ belongs to Y_R^*. Every term of $\sum_{j=1}^k \chi \circ T_R^j$ is therefore zero. Positivity and invariance of ν imply

$$\nu_R^*(r_R^* > k) \le c_*^{-1} \int_M e^{-z_0 \sum_{j=1}^k \chi \circ T_R^j} \, \mathrm{d}\nu \le c_*^{-1} C e^{-ck}.$$

This comparison uses neither the regularity of $\mathbf{1}_{Y_R^*}$ as a multiplier nor an independent-return construction.

Uniform finite horizon and the uniform bound on a single-flight lattice step give $Q_R \le C_0 r_R^*$ for a fixed C_0. Summation of the exponential tail of the positive integer r_R^* gives a uniform exponential moment of Q_R for $a_1 > 0$ sufficiently small. Enlarge C_1 to include the first bound. Finally Hölder's inequality with n equal exponents and invariance of F_R^* give

$$\int \prod_{j<n} \exp(a_1 Q_R \circ (F_R^*)^j/n) \, \mathrm{d}\nu_R^* \le \prod_{j<n} \left(\int e^{a_1 Q_R \circ (F_R^*)^j} \, \mathrm{d}\nu_R^* \right)^{1/n} \le C_1.$$

Since $N_{n,R} \le Q_{n,R}$, exponential Markov inequality proves the last assertion. No independence is used in this step either. □

Corollary 133.4 (All finite moments under parameter variation)**.** *For fixed n and $1 \le p < \infty$, the map $R \mapsto \widetilde{J}_{n,R}$ is continuous into $L^p(\nu; \mathbb{R}^4)$. Every polynomial moment of a fixed finite list of return records is finite and continuous in R. In particular every finite-lag covariance of the induced record is continuous.*

Proof. Almost-everywhere convergence is Lemma 132.1. Equation (133.6), the fixed section mass and $|J_{n,R}| \le Q_{n,R}$ make all fixed powers uniformly integrable. Truncation followed by dominated convergence proves L^p convergence along every convergent parameter sequence. A finite list of records can be expressed as differences of cumulative records. Hölder's inequality and the same exponential moment bound give uniform integrability of each polynomial in that list. Truncation proves its continuity. This proves finite-lag, not yet infinite Green–Kubo, covariance continuity. □

134. Exponential control of cumulative collision counts

The avoidance estimate controls more than a single return. It also controls the time of the nth visit to the section, without separating the successive return blocks. This observation improves the truncation budget in (230.3) and supplies exponential moments on a fixed complex neighborhood for the entire physical record.

Retain z_0, c, C from Proposition 133.2. Write $N_{n,R} = \sum_{j<n} r_R^* \circ (F_R^*)^j$, and enlarge a fixed constant $D \ge 1$, if necessary, so that

$$|J_{n,R}|_1 \le D N_{n,R}, \qquad Q_{n,R} \le D N_{n,R}. \tag{134.1}$$

Such a constant exists because each physical flight has uniformly bounded length and lattice displacement. All probabilities and expectations in this section are with respect to ν_R^*.

Theorem 134.1 (The time of the nth actual return)**.** *There are constants $A \ge 1$, $a, c > 0$, independent of n, L and R, such that*

$$\nu_R^*\{N_{n,R} > L\} \le A \exp(an - cL) \qquad (n \ge 1,\ L \ge 0). \tag{134.2}$$

One may take $a = z_0$ and the exponent c of the soft-killing estimate. Put $v = a/c$. For $0 < s < c$ there is a constant $B_s < \infty$, uniform in n, R, for which

$$\int e^{s(N_{n,R} - vn)_+} \, \mathrm{d}\nu_R^* \le B_s, \qquad \int e^{sN_{n,R}} \, \mathrm{d}\nu_R^* \le B_s e^{svn}. \tag{134.3}$$

Consequently, for every $0 < \lambda < c/D$,

$$\int e^{\lambda Q_{n,R}} \, \mathrm{d}\nu_R^* \le B_{\lambda D} e^{\lambda D v n}. \tag{134.4}$$

The constants B_s are locally bounded for $0 \le s < c$, with $B_0 = 1$. The number v is an upper envelope, not the Kac mean c_*^{-1}.

Proof. For an integer $L \ge 0$ let

$$H_{L,R}(x) = \sum_{j=1}^{L} \mathbf{1}_{Y_R^*}(T_R^j x).$$

Since $N_{n,R}$ is the physical collision time of the nth return, $N_{n,R} > L$ if and only if $H_{L,R} < n$ on Y_R^*. The fixed smooth weight satisfies $0 \le \chi \le \mathbf{1}_{Y_R^*}$. Thus, on this event, $\sum_{j=1}^{L} \chi(T_R^j x) \le n - 1$. In particular, pointwise on Y_R^*,

$$\mathbf{1}_{\{N_{n,R} > L\}} \le e^{z_0(n-1)} \exp\left(-z_0 \sum_{j=1}^{L} \chi(T_R^j x) \right).$$

Integrate on Y_R^*, enlarge the domain to M, and use invariance to shift the sum by one collision. Equation (133.4) gives

$$\nu_R^*\{N_{n,R} > L\} \le c_*^{-1} C e^{z_0(n-1) - cL}.$$

For real L replace it by $\lfloor L \rfloor$ and enlarge the constant by e^c. This proves (134.2). It uses a single orbit's visit count, not a product of probabilities of return blocks.

It follows that, for $u \geq 0$, $\nu_R^*\{N_{n,R} > vn + u\} \leq Ae^{-cu}$. For a nonnegative random variable X, Tonelli's theorem gives

$$\int e^{sX} \, \mathrm{d}\nu_R^* = 1 + s \int_0^\infty e^{su} \nu_R^*\{X > u\} \, \mathrm{d}u.$$

Apply this with $X = (N_{n,R} - vn)_+$ to obtain $B_s = 1 + As/(c - s)$. Since $N_{n,R} \leq vn + X$, the second bound in (134.3) follows. Finally apply (134.1) with $s = \lambda D$. □

We now apply the tail to the actual record measure. For a measurable insertion $w_{n,R}$ define the finite signed or complex measure

$$\mu_{n,R}^w = (J_{n,R})_*(w_{n,R}\nu_R^*), \qquad \mu_{n,R}^{w,L} = (J_{n,R})_*(w_{n,R}\mathbf{1}_{\{N_{n,R} \leq L\}}\nu_R^*).$$

The transform uses the convention $\widehat{\mu}(\omega) = \int e^{i\omega \cdot j} \, \mathrm{d}\mu(j)$, with $\omega = (u_1, u_2, s, b) \in \mathbb{T}^3 \times \mathbb{R}$.

Proposition 134.2 (Weighted finite-band truncation)**.** *Fix a nonnegative integer d and suppose $|w_{n,R}| \leq M(1 + Q_{n,R})^d$. For every multi-index γ there is a constant $C_{d,\gamma}$, independent of n, R, L, w, such that*

$$\|\mu_{n,R}^w - \mu_{n,R}^{w,L}\|_{\mathrm{TV}} \leq C_{d,0} M (1 + L)^d e^{an - cL}, \tag{134.5}$$

$$\sup_{\omega \in \mathbb{T}^3 \times \mathbb{R}} \left| \partial_\omega^\gamma (\widehat{\mu}_{n,R}^w - \widehat{\mu}_{n,R}^{w,L})(\omega) \right| \leq C_{d,\gamma} M (1 + L)^{d + |\gamma|} e^{an - cL}.$$

For a bounded insertion ($d = 0$), a frequency region of finite volume $V > 0$, and $\varepsilon > 0$, the sufficient integer cutoff for integrated transform error at most ε is

$$L \geq \left\lceil \frac{a}{c} n + \frac{1}{c} \max\left\{0, \log \frac{C_{0,0} M V}{\varepsilon}\right\} \right\rceil. \tag{134.6}$$

If $V_n \leq V_0 e^{\kappa n}$ and the required error is $e^{-\delta n}$, then a cutoff linear in n suffices. This remains true for each fixed d, γ, with any fixed positive slack in the exponential budget.

Proof. For an integer $m \geq 0$, integration of the tail gives

$$\begin{aligned} &\int (1 + N_{n,R})^m \mathbf{1}_{\{N_{n,R} > L\}} \, \mathrm{d}\nu_R^* \\ &\quad = (1 + L)^m \nu_R^*\{N_{n,R} > L\} + m \int_L^\infty (1 + t)^{m-1} \nu_R^*\{N_{n,R} > t\} \, \mathrm{d}t \\ &\quad \leq C_m (1 + L)^m e^{an - cL}. \end{aligned}$$

The integral term is zero when $m = 0$. For $m \geq 1$, the last inequality follows by writing $t = L + u$ and using $1 + L + u \leq (1 + L)(1 + u)$. Pushforward does not increase total variation. For the derivative estimate differentiate under the integral: its absolute integrand is at most $M(1 + Q_{n,R})^d |J_{n,R}|_1^{|\gamma|} \mathbf{1}_{\{N_{n,R} > L\}}$. Equation (134.1) and the preceding tail integral prove both estimates.

Multiplication by V bounds the integral of the transform error. Solving $C_{0,0} M V e^{an - cL} \leq \varepsilon$ gives (134.6). More generally choose a slope $\ell > (a + \kappa + \delta)/c$. Then, for $L_n = \lceil \ell n \rceil$,

$$V_n (1 + L_n)^{d + |\gamma|} e^{an - cL_n} \leq C (1 + n)^{d + |\gamma|} e^{-(\delta + \epsilon_0)n}$$

for some $\epsilon_0 > 0$. A fixed additive increase of the cutoffs absorbs the remaining constant and polynomial, proving the last assertion. □

Remark 134.3. This improves the sufficient cutoff of order $n\log(V/\varepsilon)$ obtained from (230.3) to order $n+\log(V/\varepsilon)$. On an exponentially growing frequency band the former budget is quadratic in n, whereas the new one is linear. Neither estimate bounds the number of critical words, the variation of an inverse coarea Jacobian, or a pointwise raw density. In particular the all-branch derivative sum in Lemma 230.2 remains a separate requirement. The truncation removes only actual high-count records in an error estimate; the limiting datum is still the full unmodified physical record.

135. A common realization and the physical renewal identity

The preceding moment estimate gives a common realization of the unbounded record twists on a scale of Lebesgue spaces. It is useful to construct this realization before seeking an anisotropic spectral estimate: the collision and induced operators, their damping variables, and their chronological composition can then be identified without regularity assumptions on an indicator multiplier. The renewal algebra is classical; compare [10]. The assertion here is its exact realization for the moving physical first-return record.

135.1. **First-return operators on the common collision space.** Use $M=\mathbb{T}\times(-1,1)$ and $\mathrm{d}\nu=(4\pi)^{-1}\,\mathrm{d}\alpha\,\mathrm{d}p$. Let P_R be multiplication by $\mathbf{1}_{Y_R^*}$ and let $E_R=I-P_R$. Recall locally that

$$c_*=\nu(Y_R^*)=\frac{91}{10000\pi},\qquad \mathrm{d}\nu_R^*=c_*^{-1}\mathbf{1}_{Y_R^*}\,\mathrm{d}\nu.$$

On $L^1(M,\nu)$, define

$$(\mathcal{A}_{R,\omega}h)(y)=e^{i\omega\cdot f_R(T_R^{-1}y)}h(T_R^{-1}y),\qquad f_R=(\kappa_{\mathrm{coll},1},\kappa_{\mathrm{coll},2},1,\tau_R),$$

for real $\omega=(u_1,u_2,s,b)$. Collision singularities are discarded as null sets. These are isometries. The operators P_R,E_R are contractions on every L^p, although no such assertion about their action on anisotropic distribution spaces is needed or made here.

Theorem 135.1 (Exact damped renewal realization)**.** *For $m\ge 1$ set*

$$\mathcal{R}_{m,R,\omega}=P_R\mathcal{A}_{R,\omega}(E_R\mathcal{A}_{R,\omega})^{m-1}P_R,\qquad \mathcal{R}_R(z,\omega)=\sum_{m\ge1}z^m\mathcal{R}_{m,R,\omega}. \tag{135.1}$$

For $|z|<1$ the series converges in operator norm on $L^1(M,\nu)$ and

$$\|\mathcal{R}_R(z,\omega)h\|_1\le|z|\|P_Rh\|_1. \tag{135.2}$$

On the closed subspace P_RL^1, with identity P_R, one has

$$P_R(I-z\mathcal{A}_{R,\omega})^{-1}P_R=(I-\mathcal{R}_R(z,\omega))^{-1}. \tag{135.3}$$

Moreover,

$$\begin{aligned}\mathcal{R}_R(z,\omega)=zP_R\mathcal{A}P_R\\ +z^2P_R\mathcal{A}E_R(I-zE_R\mathcal{A}E_R)^{-1}E_R\mathcal{A}P_R,\end{aligned} \tag{135.4}$$

where $\mathcal{A}=\mathcal{A}_{R,\omega}$ and the middle inverse acts on E_RL^1.

For $|z|\le1$, the strongly defined boundary series represents the actual first-return twist: for $y\in Y_R^*$ and $x=(F_R^*)^{-1}y$,

$$(\mathcal{R}_R(z,\omega)h)(y)=z^{r_R^*(x)}e^{i\omega\cdot G_R(x)}h(x). \tag{135.5}$$

It is zero off Y_R^*. In particular, for every $n\ge1$,

$$c_*^{-1}\int_M\mathcal{R}_R(z,\omega)^n\mathbf{1}_{Y_R^*}\,\mathrm{d}\nu=\int_{Y_R^*}z^{N_{n,R}}e^{i\omega\cdot J_{n,R}}\,\mathrm{d}\nu_R^*. \tag{135.6}$$

Proof. The collision map is invertible and preserves ν. Poincaré recurrence in both time directions makes F_R^* invertible and measure-preserving on Y_R^*, modulo null sets. In the product (135.1), the initial and final states lie in Y_R^* and the $m-1$ intervening states lie outside it. Thus its initial domain is exactly $\{r_R^* = m\}$. Its weight is the exponential of the sum of the m actual collision records, hence $e^{i\omega\cdot G_R}$. These initial domains partition Y_R^* up to a null set. Their images under the corresponding return branches also partition Y_R^*, by invertibility of the induced map. Consequently

$$\sum_{m\ge 1} |z|^m \|\mathcal{R}_{m,R,\omega} h\|_1 = \int_{Y_R^*} |z|^{r_R^*} |h|\, \mathrm{d}\nu \le |z| \|P_R h\|_1 \quad (|z| \le 1).$$

Each summand has operator norm at most one, so for $|z| < 1$ the operator-norm series converges geometrically. At the boundary the last display proves strong convergence for each h and gives (135.5). It does not assert operator-norm convergence of the boundary series.

Let $\mathcal{U}_0 = P_R$ and $\mathcal{U}_m = P_R \mathcal{A}^m P_R$ for $m \ge 1$. Partition a trajectory with endpoints in Y_R^* according to its first return time. Chronological composition, with the first segment on the right, gives

$$\mathcal{U}_m = \sum_{r=1}^{m} \mathcal{U}_{m-r} \mathcal{R}_{r,R,\omega}.$$

All generating series converge in operator norm for $|z| < 1$. Summing the identity gives $\mathcal{U}(z) = P_R + \mathcal{U}(z)\mathcal{R}_R(z,\omega)$. The inverse of $I - \mathcal{R}_R$ exists by (135.2), and $\mathcal{U}(z) = P_R(I - z\mathcal{A})^{-1} P_R$, proving (135.3). Separating $m = 1$ from the series and summing the intervening outside-section iterates proves (135.4). All these inverses are used only in $|z| < 1$.

Finally iterate (135.5) and change variables by $(F_R^*)^n$. Its collision counts and record sums are $N_{n,R}$ and $J_{n,R}$, respectively. Division by c_* yields (135.6). No independence of successive excursions enters the identity. □

135.2. **A fixed complex neighborhood for the unbounded twists.** To cross the damping boundary at $z = 1$, use $z = e^w$. Define the operator directly by the actual nth return, rather than by an unproved continuation of a resolvent. For $y \in Y_R^*$ let $x = (F_R^*)^{-n} y$ and set

$$(\mathcal{T}_{n,R}(w,\omega)h)(y) = e^{wN_{n,R}(x) + i\omega\cdot J_{n,R}(x)} h(x), \tag{135.7}$$

with zero extension to $M \setminus Y_R^*$. Here $w \in \mathbb{C}$ and $\omega \in \mathbb{C}^4$. Holomorphy is stated on this universal cover. The formulas are invariant under real shifts by $2\pi\mathbb{Z}^3$ in the first three frequency coordinates. Local complex charts about real torus frequencies are restrictions of the same formulas; no additional global complex-torus construction is being asserted. The symbols $J_{n,R}$ and G_R denote four real coordinates, with the displacement contributing two coordinates.

Theorem 135.2 (Holomorphic realization on a common Lebesgue scale)**.** *Fix $1 \le q < p \le \infty$ and put $t = (q^{-1} - p^{-1})^{-1}$. Let*

$$\sigma(w,\omega) = |\operatorname{Re} w| + D|\operatorname{Im}\omega|_\infty, \qquad \Omega_{p,q} = \{(w,\omega) : t\sigma(w,\omega) < c\}.$$

For every $n \ge 1$, (135.7) defines a holomorphic map from $\Omega_{p,q}$ to $\mathcal{L}(L^p(M,\nu), L^q(M,\nu))$. Its norm satisfies

$$\|\mathcal{T}_{n,R}(w,\omega)\|_{p\to q} \le c_*^{1/t} B_{t\sigma}^{1/t} e^{\sigma v n}. \tag{135.8}$$

On every compact subset of $\Omega_{p,q}$, all fixed-order derivatives in (w,ω) have bounds $C_k e^{C_k n}$ uniform in R. At every fixed $(w,\omega) \in \Omega_{p,q}$ and every $h \in L^p(M,\nu)$,

$$R \longmapsto \mathcal{T}_{n,R}(w,\omega)h \quad \text{is continuous in } L^q(M,\nu). \tag{135.9}$$

For $\operatorname{Re} w < 0$ *and real* ω*, this operator agrees with* $\mathcal{R}_R(e^w, \omega)^n$ *wherever the latter is considered as an* L^1 *operator. The resulting inserted record pairings extend+holomorphically across* $w = 0$ *on this fixed neighborhood.*

Proof. Change variables $y = (F_R^*)^n x$ in the qth power of the norm. The absolute weight is at most $e^{\sigma N_{n,R}(x)}$, by (134.1). Hölder's inequality with $1/q = 1/p + 1/t$ and Theorem 134.1 give

$$\|\mathcal{T}_{n,R}(w,\omega)h\|_q \le \left(\int_{Y_R^*} e^{t\sigma N_{n,R}}\,\mathrm{d}\nu\right)^{1/t} \|h\|_p \le c_*^{1/t} B_{t\sigma}^{1/t} e^{\sigma v n}\|h\|_p.$$

For $\sigma = 0$ use $B_0 = 1$. This proves the operator bound, including $p = \infty$.

On a compact subset of the stated domain choose a slightly larger compact neighborhood for which $t\sigma' < c$. A derivative of order k in (w, ω) inserts a monomial in $N_{n,R}$ and the four coordinates of $J_{n,R}$, bounded by $C_k N_{n,R}^k$. Every such polynomial is absorbed by $e^{(\sigma'-\sigma)N_{n,R}}$. Hölder's inequality and (134.3) therefore bound the derivative integrands and the Taylor remainders in operator norm. The exponential power series on a sufficiently small polydisc converges in that norm. This proves joint holomorphy, not just weak holomorphy of individual pairings. The same estimates on the larger neighborhood, or Cauchy's inequalities there, give the claimed derivative bounds.

We give the parameter-continuity argument because the underlying return domains move. Fix R_0. Remove $\partial Y_{R_0}^*$, the forward and backward collision singularities, and all finite collision iterates of $\partial Y_{R_0}^*$. Each is a countable union of null curves, up to singular endpoints. Backward recurrence holds almost everywhere. At a remaining point of $Y_{R_0}^*$ the preceding n returns use finitely many nonsingular physical collisions; none is on a section boundary. The implicit collision-root argument of Theorem 132.2, applied backwards, preserves this finite itinerary for R sufficiently near R_0. The backward starting point and all flight lengths vary continuously; the lattice labels and collision counts are fixed. At a remaining point outside $Y_{R_0}^*$ the section indicator stays zero. Thus (135.7) converges pointwise almost everywhere when h is bounded and continuous.

Choose $q' > q$ sufficiently close to q so that $q' < p$ and $t'\sigma < c$, where $t' = (1/q' - 1/p)^{-1}$. The preceding bound in $L^{q'}$ is uniform in nearby R. It gives uniform integrability of the qth powers, so pointwise convergence implies convergence in L^q. For $p < \infty$, approximate an arbitrary h in L^p by bounded smooth functions and use the uniform $p \to q$ bound. For $p = \infty$, choose a finite $p_0 > q$ so large that $(1/q - 1/p_0)^{-1}\sigma < c$. Approximation in L^{p_0} and its uniform $p_0 \to q$ bound give the same conclusion for $h \in L^\infty$. This avoids assuming norm density of smooth functions in L^∞.

In the damped region the two operators have the same pointwise formula, by Theorem 135.1. For example, for $h \in L^p$ and $b \in L^{q^\dagger}$, where $q^\dagger$ is the conjugate exponent of q,

$$c_*^{-1} \int b\,\mathcal{T}_{n,R}(w,\omega)h\,\mathrm{d}\nu = \int_{Y_R^*} h(x) b((F_R^*)^n x) e^{wN_{n,R}(x) + i\omega\cdot J_{n,R}(x)}\,\mathrm{d}\nu_R^*.$$

This continuous pairing is holomorphic on $\Omega_{p,q}$. It agrees with the damped renewal pairing there and hence is the asserted extension across $w = 0$. □

Compatibility of the realizations. For real ω and $|z| < 1$, the constructions fit the diagram

$$\begin{array}{ccc} \mathcal{A}_{R,\omega} & \longmapsto & \mathcal{R}_R(z,\omega) \\ \big\downarrow{\scriptstyle P_R(I-z\mathcal{A})^{-1}P_R} & & \big\downarrow{\scriptstyle (I-\mathcal{R})^{-1}} \\ \mathcal{U}_R(z,\omega) & = & \mathcal{U}_R(z,\omega). \end{array}$$

The upper arrow is the first-return sum (135.4); the bottom equality is (135.3). On the common domain $z = e^w$, $\operatorname{Re} w < 0$, $\omega \in \mathbb{R}^4$, the second compatibility is

$$\mathcal{R}_R(e^w,\omega)^n = \mathcal{T}_{n,R}(w,\omega), \qquad c_*^{-1}\langle b, \mathcal{T}_{n,R}h\rangle_\nu = \int h\,(b\circ(F_R^*)^n)e^{wN_{n,R}+i\omega\cdot J_{n,R}}\,\mathrm{d}\nu_R^*.$$

The first identity is interpreted on $L^p \subset L^1$, with values in L^q, when the scale operator is used; the second has the insertion exponents of the theorem. Any anisotropic realization must reproduce these pairings on its admissible test class. Such an additional realization is not an arrow already established by this diagram.

Remark 135.3 (Relation to the spectral and raw-density estimates). The operator identities concern the genuine four-coordinate record, and all their spaces live on the same collision probability space. They specify a common realization with fixed exponential integrability and strong parameter continuity. They do not assert operator-norm continuity in R, quasi-compactness on L^p, or a holomorphic family of endomorphisms of a single anisotropic space. At real frequencies the induced L^1 operator is an isometry on its section, so its construction alone gives no spectral gap. Likewise, maps $L^p \to L^q$ with $q < p$ cannot simply be iterated on one space outside the damped region; the n-block operator above was defined directly.

The next analytical step is to establish an anisotropic realization compatible with (135.6) and (135.3), and to prove its phase reconstruction and raw-residual bounds. The collision compensation in Sections 138–140 will establish the actual Gaussian covariance and functional limits by a different route, without identifying an induced spectral family. The criteria in Theorem 141.2, Proposition 230.1, and Lemma 230.2 continue to state these requirements separately. In particular the common Lebesgue realization is not used as a substitute for any of those estimates in the raw local limit.

136. Common charts for moving return domains

We give a local change-of-variables estimate and then apply it to the actual return domains. This makes explicit the domination used in Theorem 132.2. Critical neighborhoods are discarded in the initial collision space, not by assuming a uniform density bound near their images.

Lemma 136.1 (A common submersion chart)**.** *Let $U = U_1 \times U_2 \Subset \mathbb{R}^2$ be a bounded open rectangle and let $f(R,u,v)$ be C^2 on a neighborhood of $J \times \overline{U}$, where J is a compact parameter interval. Suppose $\partial_v f \ge \eta > 0$ there. For $a \in C_c^1(U)$, let g_R be the density of the pushforward of $a(u,v)\,\mathrm{d}u\,\mathrm{d}v$ by f_R. Then*

$$\|g_R - g_S\|_{L^1(\mathbb{R})} \le C_a|R-S| \qquad (R,S \in J). \tag{136.1}$$

The constant depends only on U, a common bounded range of f, η, the C^2 bound for f, and the C^1 bound and support of a. The same assertion holds if $\partial_v f \le -\eta$.

Proof. For fixed R,u the map $v \mapsto f(R,u,v)$ is one-to-one. Write its inverse as $v_R(u,t)$. The pushforward under $(u,v) \mapsto (u, f_R(u,v))$ has density

$$A_R(u,t) = \frac{a(u,v_R(u,t))}{\partial_v f(R,u,v_R(u,t))}$$

on its image, extended by zero elsewhere. All images lie in one bounded (u,t) box. Since a vanishes on a fixed neighborhood of ∂U, this zero extension introduces no boundary jump as R varies. Wherever a or its derivatives are nonzero, the inverse is defined on a common parameter neighborhood. If M bounds the relevant derivatives of f, implicit differentiation gives

$$|\partial_R v_R| \le M/\eta, \qquad |\partial_R A_R| \le \|a_v\|_\infty M/\eta^2 + \|a\|_\infty(M/\eta^2 + M^2/\eta^3).$$

At points outside the inverse chart both sides are interpreted by the zero extension, which vanishes on an open neighborhood of its moving edge. The same bound therefore gives an L^1 majorant on the fixed box. Integrating in R proves $\|A_R - A_S\|_1 \le C_a|R-S|$. Since $g_R(t) = \int A_R(u,t)\,\mathrm{d}u$, integration in u proves (136.1). Reversing the v coordinate handles the negative derivative. The result also applies on a constant lattice component of a mixed density. □

Proposition 136.2 (Finite common patches with an explicit remainder)**.** *Fix $n \ge 1$, $R_0 \in I$, and $\epsilon > 0$. There are finitely many nonnegative smooth cutoffs ψ_i on the common collision space, a parameter neighborhood J of R_0, and $C_{\epsilon,n,R_0} < \infty$ such that $\sum_i \psi_i \le 1$, every support lies in one common regular first-n-return patch for all $R \in J$, and*

$$c_*^{-1} \int \sum_i \psi_i \,\mathrm{d}\nu > 1 - \epsilon. \tag{136.2}$$

The lattice labels and all intermediate return decisions are fixed on each patch. If $p^i_{n,R}$ is the pushforward density of $c_*^{-1}\psi_i\nu$, then

$$\sum_i \|p^i_{n,R} - p^i_{n,S}\|_1 \le C_{\epsilon,n,R_0}|R-S|, \tag{136.3}$$

$$\|p_{n,R} - p_{n,S}\|_1 \le 2\epsilon + C_{\epsilon,n,R_0}|R-S| \qquad (R,S \in J).$$

Before choosing these patches one may discard $N_{n,R_0} > L$ at cost at most $C_1 e^{-a_1 L/n}$, with the constants of Theorem 133.3.

Proof. The exponential moment bound proves the last assertion. Choose L so that this cost is less than $\epsilon/3$. On the remainder remove the initial boundary of the section, all grazing and competing-root singularities encountered before collision L, and the preimages of the moving return boundary at R_0 through those collisions. These sets have zero collision measure. On every remaining regular word also remove the critical points of its cumulative flight length. Proposition 142.3 shows that there is at most one such point on a regular optical word; the submersion complement has full measure. There are only finitely many possible physical lattice words of length at most L, since each flight label belongs to a fixed finite set. No estimate for the number of unbounded-length words is used here.

Inner regularity of the collision measure and a countable exhaustion of the regular submersion patches give a compact subset of mass greater than $1-\epsilon$ in the normalized section. At every point of this compact subset, choose a small coordinate rectangle on which one coordinate derivative of the cumulative time is bounded away from zero. The finite-itinerary lemma preserves the entire physical word and every return decision on a slightly larger rectangle for R close to R_0. Select finitely many such rectangles and a subordinate family of smooth cutoffs with sum at most one, equal to one on the compact subset. Their supports have positive distance from all the excluded sets at R_0. Finiteness gives a common parameter neighborhood J on which these distances, return decisions and derivative bounds persist. The time functions and their first two derivatives are uniformly bounded on these finitely many compact rectangles. Lemma 136.1 applies with $a = \psi_i/(4\pi c_*)$ in (α, p) coordinates. Summing its bounds proves the first line of (136.3).

For every $R \in J$ the remainder of the original probability is a positive measure. Its mass is exactly $1 - c_*^{-1}\int \sum_i \psi_i \,\mathrm{d}\nu < \epsilon$, because the section measure is the fixed number c_*. The raw densities exist by the submersion/coarea argument of Theorem 132.2. The two remainder densities have total L^1 norm less than 2ϵ. This gives the second line. Nothing has been assumed about the size of a density at a critical value or a sharp image boundary. □

The constants in (136.3) expose the order of choices: physical-count truncation, deletion of small initial-state neighborhoods, compact common charts, and finally parameter tolerance.

They are not asserted to remain bounded as n or ϵ^{-1} grows. A quantitative global density modulus requires quantitative control of those additional choices. Bounded measurable insertions are handled by L^1 approximation on the same patches, as in Proposition 132.3; no artificial observation coordinate is introduced.

137. A uniform functional law for the mechanical clock

Finite-record continuity also has a long-time consequence that does not require a mixing rate. We prove a parameter-uniform functional law at the first-order scale, including the maximum unfinished return over an entire observation interval. The $\sqrt{t}$-scale process theorem in Proposition 228.2 remains a separate statement.

Set

$$G_R = (\kappa_R^*, r_R^*, \tau_R^*), \qquad \bar{G}_R = (0, 0, c_*^{-1}, \bar{\tau}_R/c_*), \qquad J_{0,R} = 0.$$

All probabilities on the return section use ν_R^*, not an independent sequence of records. Write $\bar{r} = c_*^{-1}$ and $\bar{\tau}_R^* = \bar{\tau}_R/c_*$. Both $\bar{\tau}_R^*$ and its reciprocal are bounded on I.

Lemma 137.1 (A compact subadditive argument)**.** *Let $a_n : I \to [0,\infty)$ be continuous, $a_0 = 0$, and $a_{n+m}(R) \le a_n(R) + a_m(R)$. Suppose $\inf_{n\ge1} a_n(R)/n = 0$ for every R. Then*

$$\lim_{n\to\infty} \sup_{R\in I} \frac{a_n(R)}{n} = 0.$$

More precisely, for every $\epsilon > 0$ there is a finite integer M such that, with $C = \sup_R a_1(R)$,

$$\sup_R a_n(R)/n \le \epsilon + CM/n \qquad (n \ge 1). \tag{137.1}$$

Proof. For each R choose a block length m_R with $a_{m_R}(R)/m_R < \epsilon$. Continuity preserves this strict inequality on a neighborhood of R. Take a finite subcover and let M be the largest of its block lengths. For an arbitrary parameter choose one of the covering neighborhoods, with block length $m \le M$, and write $n = qm + r$, $0 \le r < m$. Subadditivity and $a_r \le r a_1$ give $a_n \le q\epsilon m + Cr \le \epsilon n + CM$. This proves the estimate and then the limit. The neighborhoods need not have a common block length; replacing them by an unproved common mixing time would not be justified. □

Theorem 137.2 (Uniform first-order return process)**.** *The actual induced records satisfy*

$$\lim_{n\to\infty} \sup_{R\in I} \int \left| J_{n,R}/n - \bar{G}_R \right| \, \mathrm{d}\nu_R^* = 0. \tag{137.2}$$

For every fixed $A < \infty$ and $\epsilon > 0$,

$$\lim_{n\to\infty} \sup_{R\in I} \nu_R^* \left\{ \sup_{0\le u\le A} \left| n^{-1} J_{\lfloor nu \rfloor, R} - u\bar{G}_R \right| > \epsilon \right\} = 0. \tag{137.3}$$

Proof. The ergodic collision map in Section 132 induces an ergodic map on its positive-measure return section. Indeed, the complete tower over an invariant subset of the induced map is an invariant collision set. The two towers over that subset and its complement partition the full collision space modulo null sets; collision ergodicity forces one of the two to have zero measure. Thus the integrable vector G_R obeys the L^1 ergodic theorem for each fixed R, with the exact mean in Proposition 131.1.

Put $a_n(R) = \int |J_{n,R} - n\bar{G}_R| \, \mathrm{d}\nu_R^*$. Invariance of F_R^* and the cocycle identity make a_n subadditive. It is continuous: on the common collision space it equals

$$c_*^{-1} \int_M \left| \widetilde{J}_{n,R} - n\bar{G}_R \mathbf{1}_{Y_R^*} \right| \, \mathrm{d}\nu.$$

Theorem 132.2 gives the first term's L^1 continuity, the section indicators are continuous in L^1 because their boundaries move continuously and have zero measure, and the mean is continuous by its explicit formula. The pointwise ergodic theorem gives $a_n(R)/n \to 0$. Lemma 137.1 proves (137.2).

For the functional conclusion, first test a fixed finite mesh of $[0, A]$. Equation (137.2) and Markov's inequality give convergence in probability at all its mesh points, uniformly in R. The count and roof coordinates of J_n are nondecreasing. Between mesh points their normalized deviations are therefore bounded by the endpoint deviations plus a uniformly bounded mean times the mesh width, with an $O(n^{-1})$ integer-part error. For the displacement coordinate, finite horizon gives a constant C_κ such that for every $l \geq j$,

$$|K_{l,R} - K_{j,R}| \leq C_\kappa (N_{l,R} - N_{j,R}).$$

Consequently its oscillation on a mesh interval is controlled by the count increment on that interval. Its mean is zero. On the event where all mesh errors are small, every coordinate has a uniformly small error between mesh points as well. Let n grow, then let the mesh width and the mesh error tolerance decrease. This proves (137.3) without an independence or maximal-martingale assumption. □

Lemma 137.3 (Sublinear maximum return blocks)**.** *Put* $Q_R = r_R^* + \tau_R^* + |\kappa_R^*|$ *and*

$$U(L) = \sup_R \int Q_R \mathbf{1}_{\{Q_R > L\}} \, \mathrm{d}\nu_R^*.$$

Then $U(L) \to 0$*. For* $n \geq 1$*,* $A \geq 0$*, and* $\epsilon > 0$*,*

$$\sup_R \nu_R^* \left\{ \max_{0 \leq j \leq \lfloor An \rfloor} Q_R \circ (F_R^*)^j > \epsilon n \right\} \leq \frac{A+1}{\epsilon} U(\epsilon n). \tag{137.4}$$

Both (137.3) *and the convergence to zero in* (137.4) *remain valid when the initial base point has the stationary length-biased law*

$$\mathrm{d}\rho_R = \frac{\tau_R^*}{\bar{\tau}_R^*} \, \mathrm{d}\nu_R^*. \tag{137.5}$$

Proof. Finite horizon bounds τ_R^* and $|\kappa_R^*|$ by a uniform constant times r_R^*. The uniform integrability in Theorem 132.2, and the constant section mass, therefore give $U(L) \to 0$. Invariance of F_R^*, the union bound and truncated Markov inequality give

$$\nu_R^* \{ \max_{j \leq \lfloor An \rfloor} Q_R \circ F^j > \epsilon n \} \leq (\lfloor An \rfloor + 1) \nu_R^* (Q_R > \epsilon n) \leq \frac{A+1}{\epsilon} \int Q_R \mathbf{1}_{Q_R > \epsilon n} \, \mathrm{d}\nu_R^*.$$

No independence of the events is used.

Let $d_* := \inf_R \bar{\tau}_R^* > 0$. For any event B_R determined by the forward base orbit and every $L > 0$,

$$\rho_R(B_R) \leq d_*^{-1} \left(L \nu_R^*(B_R) + \int \tau_R^* \mathbf{1}_{\{\tau_R^* > L\}} \, \mathrm{d}\nu_R^* \right). \tag{137.6}$$

If $\sup_R \nu_R^*(B_R) \to 0$, first hold L fixed and pass to that limit, then let $L \to \infty$. Uniform integrability of the roof makes the right side tend to zero uniformly. Apply this argument to the exceptional events in the two displayed convergence statements. This transfer uses a length-biased law, not the unweighted launch law. □

For the stationary physical flow, let $V_R(t)$ count future visits to Y_R^*, let $C_R(t)$ count future physical collisions, and let $W_R(t) \in \mathbb{Z}^2$ be the net fundamental-cell displacement since time zero. Changing the choice of a fixed fundamental cell changes the continuous-position comparison only by a bounded term.

Theorem 137.4 (Uniform stationary physical clock)**.** *For every $\epsilon > 0$, under the actual stationary billiard laws μ_R, uniformly for $R \in I$,*

$$\sup_{0\le u\le 1}\left|\frac{1}{t}(V_R(tu), C_R(tu), W_R(tu)) - \left(\frac{uc_*}{\bar\tau_R}, \frac{u}{\bar\tau_R}, 0\right)\right| \longrightarrow 0 \quad \text{in probability as } t\to\infty. \tag{137.7}$$

The maximum marked unfinished return encountered during $[0,t]$, including the initial return containing time zero, is $o_P(t)$ uniformly in R.

Proof. Represent the starting state by (x,s) in the induced suspension, $0 \le s < \tau_R^*(x)$. The marginal law of x is exactly ρ_R in (137.5). Write $T_{j,R}$ for the roof coordinate of $J_{j,R}$. Count future events in $(0,t]$. With this convention the following inverse identity also holds at event times:

$$V_R(tu) = \max\{j \ge 0 : T_{j,R}(x) \le s + tu\}.$$

Choose a fixed A with $Ad_* > 2$. The functional return law, transferred by Lemma 137.3, shows that $T_{\lfloor At\rfloor,R} > s + t$ with probability tending to one uniformly. Here $s/t \to 0$ uniformly because $s \le Q_R(x)$ and the same lemma applies at $j = 0$. Integer parts do not affect these statements; the return-process theorem may be applied with $n = \lceil t\rceil$ and a slightly larger fixed horizon.

On this event all inverse indices lie below At. Uniformly for $0 \le j \le At$ we have $|T_{j,R} - j\bar\tau_R^*|/t = o_P(1)$ and $\max_{j\le At} Q_R \circ F^j/t = o_P(1)$. The inequalities

$$T_{V_R(tu),R} \le s + tu < T_{V_R(tu)+1,R}$$

therefore imply

$$\sup_{u\le 1} |\bar\tau_R^* V_R(tu)/t - u| = o_P(1)$$

uniformly in R. Division by $\bar\tau_R^* \ge d_*$ yields the first component in (137.7).

The completed-return count $N_{V_R(tu),R}$ differs from the actual physical count by the initial prefix and the final incomplete return. Their sum is at most a fixed constant plus twice the maximal Q_R over these return indices. The displacement comparison has the same bound, up to a uniform multiplicative constant, since each physical lattice step is bounded. These bounds hold simultaneously for all $u \in [0,1]$. The functional return law at the random inverse indices now gives

$$C_R(tu)/t = \bar r\, V_R(tu)/t + o_P(1), \qquad W_R(tu)/t = o_P(1)$$

with uniform suprema. Since $\bar r/\bar\tau_R^* = 1/\bar\tau_R$, this proves the remaining components. Every unfinished piece in $[0,t]$ belongs to one of the same return blocks and is bounded by its marked size, proving the final assertion. □

Corollary 137.5 (Geometric error in the physical collision rate)**.** *Let $b_* := \min_{R\in I} \bar\tau_R$ and put $L_* = 15/(4b_*^2)$. There is a function $\omega_t(\epsilon) \to 0$ for every fixed $\epsilon > 0$ such that, for any I-valued radius estimate $\widehat R$ on a probability space carrying the stationary billiard at the true radius R, and any $\delta > 0$,*

$$\Pr_R\left\{\sup_{0\le u\le 1}\left|C_R(tu)/t - u/\bar\tau_{\widehat R}\right| > \epsilon + L_*\delta\right\} \le \omega_t(\epsilon) + \Pr_R\{|\widehat R - R| > \delta\}. \tag{137.8}$$

No independence of $\widehat R$ and the observed orbit is required. For the visit rate, replace the deterministic error constant by c_*L_*.

Proof. The explicit mean satisfies $|\bar\tau_R'| < 15/4$, so $|(1/\bar\tau_R)'| \le L_*$. On the event $|\widehat R - R| \le \delta$ the two proposed rates differ by at most $L_*\delta$, uniformly over $u \in [0,1]$. The triangle inequality and Theorem 137.4 prove the claim with ω_t equal to the supremum, over R, of the true rate's exceptional probability. Multiplication by c_* gives the visit-rate statement. □

The supremum statement here is at scale t. Uniform integrability alone does not turn it into an $o_P(\sqrt{t})$ bound for a growing window. Likewise (137.1) and (137.8) do not supply a numerical convergence rate for ω_t without quantitative finite-block estimates. They establish the full first-order clock comparison for this moving family. The Gaussian covariance and processes require the additional collision estimates proved in Sections 138– 140; raw density and shrinking-conditioning conclusions require the further local estimates stated there. The stationary law throughout is not substituted for an unknown experimental preparation law.

138. Bounded collision compensation and its covariance

The discontinuity of the section can be treated on the collision clock without making its indicator a multiplier on a distribution space. The argument uses bounded variation in the initial collision coordinates, smooth approximation, and the collision-space spectral splitting. It will give a covariance for the actual return record by an exact stopping identity, rather than by an assumed spectral expansion of an induced operator.

Write $\eta_R = \mathbf{1}_{Y_R^*}$ and recall

$$c_* = \frac{91}{10000\pi}, \qquad \bar{\tau}_R = \frac{\sqrt{3}/2 - \pi R^2}{2R}, \qquad \bar{G}_R = \left(0, 0, c_*^{-1}, \frac{\bar{\tau}_R}{c_*}\right). \tag{138.1}$$

Here $\bar{G}_R = \int G_R \, \mathrm{d}\nu_R^*$ is the mean of the actual induced record, by the Kac identities. On the common collision cylinder put

$$h_R = f_R - \bar{G}_R \eta_R, \qquad s_R = c_*^{-1} \eta_R, \qquad S_{m,R} u = \sum_{j=0}^{m-1} u \circ T_R^j. \tag{138.2}$$

Thus $\int h_R \, \mathrm{d}\nu = 0$ and $s_R \nu = \nu_R^*$, with the latter measure regarded as a probability on M. The four components of h_R are bounded, although those of G_R need not be.

138.1. **Variation of a single physical collision.** For a function on $\mathbb{T} \times (-1, 1)$, BV denotes bounded variation with respect to the flat coordinates (α, p): both first distributional derivatives are finite signed measures. All variation norms below include the $L^1(\nu)$ norm. They are not norms of the density of a pushed-forward record.

Lemma 138.1 (Uniform single-collision variation)**.** *There is a constant B such that*

$$\|f_R\|_\infty + \|f_R\|_{\mathrm{BV}} + \|\eta_R\|_\infty + \|\eta_R\|_{\mathrm{BV}} + \|h_R\|_\infty + \|h_R\|_{\mathrm{BV}} + \|s_R\|_\infty + \|s_R\|_{\mathrm{BV}} \le B \qquad (R \in I).$$

The norms of vector-valued functions mean the sum of the component norms.

Proof. The uniform horizon bounds all possible next-disk centers in a fixed finite subset of the triangular lattice. Cover the angular circle by finitely many bounded rational charts $t = \tan((\alpha - \alpha_0)/2)$, with uniformly bounded chart derivatives and inverse derivatives. In such a chart n_α and t_α are rational functions of t. The outgoing velocity $\sqrt{1-p^2} n_\alpha + p t_\alpha$ is semialgebraic in (t, p). For each candidate center d, the admissibility inequalities and the entry root

$$(d - R n_\alpha) \cdot v - \sqrt{R^2 - |d - R n_\alpha|^2 + ((d - R n_\alpha) \cdot v)^2}$$

are semialgebraic in (R, t, p). Taking the least admissible positive entry root and its lattice label involves only finitely many comparisons. Choose any fixed tie convention on the singular set. Thus the bounded single-collision record, extended arbitrarily by zero where no regular root is defined, is a bounded semialgebraic family on each chart. Only these one-collision functions, not an arbitrary long itinerary, are used in this assertion.

We spell out the uniformity implication. A bounded semialgebraic family $u(\lambda, x)$, with x in a bounded interval, has a uniform finite number of monotonicity intervals in x. To see this, apply

differentiable cell decomposition to its graph with the parameters λ ordered first, and refine by the sign of its fiber derivative where defined. Uniform finiteness bounds the number of remaining exceptional points and fiber intervals. On each interval the function is monotone or constant. These are the cell decomposition and monotonicity results of [9, Sections 2.1–2.2 and 6.2]. In particular the number does not depend on the parameter or on the other fixed coordinate of a slice. If $|u| \le M_0$ and there are at most b_0 intervals and exceptional points, its one-dimensional distributional variation is at most Cb_0M_0, including the jumps.

Apply this statement first with (R, p) as parameters and then with (R, t) as parameters. Integrating the two slice-variation bounds against the remaining coordinate proves the bounds for the two first distributional derivatives. Equivalently, integration by parts on almost every slice bounds the action on every compactly supported C^1 test function by its supremum norm. A fixed finite chart partition transfers these bounds to α; any jumps at the chart endpoints have uniformly bounded size and length. This proves the assertion for f_R, including the square-root behavior at grazing.

The section is a fixed finite union of coordinate rectangles with moving centers. Its perimeter and the number of its sides are uniformly bounded, so the same assertion for η_R follows directly, without requiring the centers to be semialgebraic in R. The explicit means in (138.1) are bounded on I. The remaining bounds now follow from (138.2). □

Lemma 138.2 (Mean-preserving smoothing)**.** *There are positive linear operators S_δ, $0 < \delta < 1/2$, which preserve the integral with respect to ν and satisfy*

$$\begin{aligned} \|\mathsf{S}_\delta u\|_\infty &\le \|u\|_\infty, \qquad \|u - \mathsf{S}_\delta u\|_1 \le C\delta \|u\|_{\mathrm{BV}}, \\ \|\mathsf{S}_\delta u\|_{\mathrm{BV}} &\le C\|u\|_{\mathrm{BV}}, \qquad \|\mathsf{S}_\delta u\|_{C^2(\alpha,\varphi)} \le C\delta^{-2}\|u\|_\infty. \end{aligned} \tag{138.3}$$

The last norm uses $p = \sin\varphi$, as in the collision spaces.

Proof. Reflect u evenly across $p = 1$ and $p = -1$ to a function periodic with period four in p, retaining the angular periodicity. Convolve with a nonnegative, even, smooth product kernel of mass one and scale δ, then restrict to $-1 < p < 1$. Reflection symmetry is preserved, so the integral over this half of the extended cylinder remains the original integral. The reflected function has variation bounded by a fixed multiple of the original variation; its traces match at the reflection boundaries. Translation of a bounded-variation function by a vector a changes its L^1 norm by at most $|a|$ times its variation. This follows first for smooth functions by the fundamental theorem of calculus, then for distributional derivatives by convolution and passage to the limit. Integrating this inequality against the kernel gives the L^1 error. Convolution contracts variation on the extended cylinder. The L^1 norms of the first two derivatives of the kernel are $O(\delta^{-1})$ and $O(\delta^{-2})$. These facts prove all the flat coordinate estimates. Composition with $p = \sin\varphi$ preserves the stated C^2 upper bound, since the first two derivatives of sin are bounded. No inverse coordinate change at grazing is used. □

138.2. **An exponentially summable collision covariance.**

Proposition 138.3 (Correlations of bounded-variation observables)**.** *For every $B_0 < \infty$ there are $C, \gamma > 0$, uniform in R, such that centered real functions a, b with $\|a\|_\infty + \|a\|_{\mathrm{BV}} + \|b\|_\infty + \|b\|_{\mathrm{BV}} \le B_0$ satisfy*

$$\left| \int a(b \circ T_R^k)\, \mathrm{d}\nu \right| \le Ce^{-\gamma k} \qquad (k \ge 0). \tag{138.4}$$

If also $\|a\|_1 \le C_0\delta$, then for $r = a$ and $b = a$,

$$\int |S_{m,R} r|^2\, \mathrm{d}\nu \le Cm\delta(1 + |\log\delta|). \tag{138.5}$$

The second constant may depend on C_0 and B_0, but not on m, R.

Proof. For smooth centered a, b, apply Lemma 133.1 to the distribution $a\nu$ and evaluate the result on b using smooth multiplication and the mass functional. It gives

$$\left| \int a(b \circ T_R^k) \, \mathrm{d}\nu \right| \leq C\vartheta^k \|a\|_{C^2} \|b\|_{C^2}.$$

Smooth both bounded-variation functions at scale ϵ. Invariance of ν, boundedness, and (138.3) bound the change in the correlation by $C\epsilon$. The smooth correlation is bounded by $C\vartheta^k \epsilon^{-4}$. Choose $\epsilon = \frac{1}{4}\vartheta^{k/5}$ and enlarge C for $k = 0$. This proves (138.4). In particular the constants depend only on the displayed variation and supremum bounds, not on a modulus of continuity or on a chosen smoothing scale.

For the last assertion, boundedness gives $|\int r(r \circ T_R^k) \, \mathrm{d}\nu| \leq C\delta$ at every lag, while (138.4) gives the exponential bound. Hence

$$\sum_{k \geq 0} \left| \int r(r \circ T_R^k) \, \mathrm{d}\nu \right| \leq C \sum_{k \geq 0} \min\{\delta, e^{-\gamma k}\} \leq C\delta(1 + |\log \delta|).$$

Expanding the square of the stationary sum proves (138.5). □

Put $h_{R,\delta} = \mathsf{S}_\delta h_R$. Define, on the collision clock,

$$\begin{aligned} \Gamma_R = & \int h_R \otimes h_R \, \mathrm{d}\nu \\ & + \sum_{k=1}^{\infty} \int \left[h_R \otimes (h_R \circ T_R^k) + (h_R \circ T_R^k) \otimes h_R \right] \mathrm{d}\nu. \end{aligned} \tag{138.6}$$

Use the same formula with $h_{R,\delta}$ for $\Gamma_{R,\delta}$.

Theorem 138.4 (A continuous collision Green–Kubo matrix)**.** *The series* (138.6) *and its smoothed versions converge absolutely, with exponential tails uniform in R and δ. The matrix Γ_R is continuous and positive semidefinite on I, and*

$$\Gamma_R = \lim_{m \to \infty} \frac{1}{m} \int S_{m,R} h_R \otimes S_{m,R} h_R \, \mathrm{d}\nu, \qquad \|\Gamma_{R,\delta} - \Gamma_R\| \leq C\delta(1 + |\log \delta|). \tag{138.7}$$

The first limit is uniform in R.

Proof. Lemma 138.1, smoothing, and Proposition 138.3 give the uniform exponential tails. For each fixed k, the integrand in (138.6) converges almost everywhere as $R \to R_0$. Indeed, exclude the finitely many collision singularity preimages and section boundaries encountered up to time k at R_0. On the remaining full-measure set the physical roots and finite itinerary persist and the moving rectangles have stable membership. All observables are uniformly bounded. Dominated convergence proves continuity at each fixed lag; the uniform tail then proves continuity of the infinite series. This argument establishes long-time uniformity through the correlation estimate, not through a fixed-record continuity constant.

Expansion of the square gives the finite covariance with lag weights $1 - k/m$. Absolute summability, uniformly in R, proves its limit; indeed the error is $O(m^{-1})$ by the exponential bound. Positivity follows from positivity of each finite covariance. At each lag the change made by smoothing is at most $C\delta$, using invariance and the L^1 approximation. Both the original and the smoothed lag correlations are at most $Ce^{-\gamma k}$. Summing the minimum of these two bounds proves the final estimate. □

Remark 138.5. The sum in (138.6) is for the bounded, compensated collision observable h_R under T_R. It is not the Green–Kubo sum of the unbounded observable $G_R - \bar{G}_R$ under F_R^*. Its relation to the Gaussian law of the latter will be proved by stopping the actual collision sums.

No bounded-variation estimate for a many-return coarea density, and no second derivative of that density, follows from the single-collision variation lemma.

139. A uniform Gaussian limit for the actual return record

We first obtain a quantitative characteristic-function estimate on the collision clock. Smooth approximation is used only inside the proof; the limiting statement concerns the original record and the original section probability.

139.1. **Smooth perturbation with a shrinking scale.**

Lemma 139.1 (A quantitative smooth collision expansion)**.** *On each of the collision spaces of Lemma 133.1, set*

$$\mathcal{L}_{R,\delta,z} = \mathcal{L}_R M_{\exp(iz\cdot h_{R,\delta})}, \qquad z \in \mathbb{C}^4.$$

There are $a, C > 0$ and $\rho < 1$, independent of R and $0 < \delta < 1/2$, such that for $|z| < a\delta^2$ this family has a simple analytic eigenvalue $\lambda_{R,\delta}(z)$ near one and an analytic rank-one projection $\Pi_{R,\delta}(z)$. Its complementary powers are bounded by $C\rho^m$, and, on a smaller such ball,

$$\begin{aligned} \log \lambda_{R,\delta}(z) &= -\tfrac{1}{2} z^{\mathsf{T}} \Gamma_{R,\delta} z + O(\delta^{-6}|z|^3), \\ \|\Pi_{R,\delta}(z) - \Pi_R\| &\le C\delta^{-2}|z|. \end{aligned} \tag{139.1}$$

All constants are uniform. The logarithm is the branch vanishing at $z = 0$.

Proof. The smooth multiplier estimate and the smoothing bounds give $\|\mathcal{L}_{R,\delta,z} - \mathcal{L}_R\| \le C|z|\delta^{-2}$ when $|z|\delta^{-2}$ is small. The uniform resolvent contours around one and around the complementary spectrum in the proof of Proposition 133.2 therefore apply on $|z| < a\delta^2$. The resolvent Neumann series gives the analytic projection, its difference estimate, and complementary powers on a fixed circle of radius $\rho < 1$. Decrease a so that the eigenvalue stays in a fixed disk about one not containing zero.

The eigenvalue derivative at zero is zero since $\int h_{R,\delta}\,\mathrm{d}\nu = 0$. We identify its second derivative, rather than assuming a variance formula. Fix a real direction v, put $u = v \cdot h_{R,\delta}$, and normalize a right eigenvector $e(t)$ by $\ell_j(e(t)) = 1$. Differentiating the eigenvector equation at zero yields

$$e'(0) = i \sum_{k\ge 1} \mathcal{L}_R^k(u\nu),$$

where the series converges in $\mathcal{B}_j$ by the spectral splitting. A second differentiation followed by the mass functional gives

$$\lambda''(0) = -\int u^2\,\mathrm{d}\nu - 2\sum_{k\ge 1}\int u(u\circ T_R^k)\,\mathrm{d}\nu = -v^{\mathsf{T}}\Gamma_{R,\delta} v.$$

Polarization identifies the full Hessian. Since the first derivative vanishes, the Hessian of the logarithm is the same. Cauchy's estimates for the bounded analytic logarithm on a ball of radius $a\delta^2$ bound its third derivatives by $C\delta^{-6}$ on a smaller ball. Taylor's formula proves (139.1). □

Proposition 139.2 (Uniform collision characteristic estimate)**.** *For every $K < \infty$ there is C_K such that, for $m \ge 2$ and for $\sigma_R = 1$ or $\sigma_R = s_R$,*

$$\sup_{R\in I,\ |t|\le K} \left| \int \sigma_R e^{it\cdot S_{m,R} h_R/\sqrt{m}}\,\mathrm{d}\nu - e^{-t^{\mathsf{T}}\Gamma_R t/2} \right| \le C_K m^{-1/26}\sqrt{\log(2+m)}. \tag{139.2}$$

Proof. Smooth both h_R and the initial density at scale δ. The smoothed density $\sigma_{R,\delta} = \mathsf{S}_\delta \sigma_R$ is nonnegative, has integral one, and satisfies $\|\sigma_{R,\delta}\nu\|_{\mathcal{B}_j} \le C\delta^{-2}$. For $z = t/\sqrt{m}$, the transfer pairing and the spectral decomposition of Lemma 139.1 give

$$\int \sigma_{R,\delta} e^{it\cdot S_{m,R} h_{R,\delta}/\sqrt{m}} \,\mathrm{d}\nu = \lambda_{R,\delta}(z)^m \ell_j\big(\Pi_{R,\delta}(z)\sigma_{R,\delta}\nu\big) + O(\delta^{-2}\rho^m).$$

At $z = 0$ the amplitude is one; its difference from one is $O_K(\delta^{-4}m^{-1/2})$. The logarithmic expansion gives, whenever $K/\sqrt{m} < a\delta^2$ and $\delta^{-6}m^{-1/2}$ is sufficiently small,

$$\left| \int \sigma_{R,\delta} e^{it\cdot S_{m,R} h_{R,\delta}/\sqrt{m}} \,\mathrm{d}\nu - e^{-t^{\mathsf{T}}\Gamma_{R,\delta}t/2} \right| \le C_K\big(\delta^{-6}m^{-1/2} + \delta^{-4}m^{-1/2} + \delta^{-2}\rho^m\big). \tag{139.3}$$

Here positive semidefiniteness of $\Gamma_{R,\delta}$ bounds the Gaussian factor by one. The estimate for the exponential of the Taylor remainder follows, for example, from $|e^w - 1| \le |w|e^{|w|}$.

The change of initial density costs at most $C\delta$ in the original pairing, because the exponential has modulus one. The centered residual $r_{R,\delta} = h_R - h_{R,\delta}$ has uniformly bounded supremum and variation norms and L^1 norm $O(\delta)$. Therefore Proposition 138.3, component by component, gives

$$\int \sigma_{R,\delta} |S_{m,R} r_{R,\delta}|^2 \,\mathrm{d}\nu \le Cm\delta(1 + |\log\delta|),$$

since the initial density is bounded independently of δ. The inequality $|e^{ix} - e^{iy}| \le |x - y|$ and Cauchy–Schwarz bound the change of observable by $C_K\sqrt{\delta(1 + |\log\delta|)}$. Finally, Theorem 138.4 bounds the change of the Gaussian factor by $C_K\delta(1 + |\log\delta|)$.

Take $\delta = \frac{1}{4}m^{-1/13}$. The smoothing error is $O(m^{-1/26}\sqrt{\log(2+m)})$ and the cubic spectral error is $O(m^{-1/26})$. The two smallness requirements hold for all sufficiently large m, depending only on K. The amplitude and complementary errors are smaller. Increasing C_K handles the remaining finitely many m. This proves the estimate uniformly over the finite cover of collision spaces. It does not use an induced spectral gap. □

139.2. **Stopping at the genuine return times.**

Lemma 139.3 (Exact compensation and a return-clock window)**.** *For ν_R^*-almost every initial state and every $n \ge 1$,*

$$J_{n,R} - n\bar{G}_R = S_{N_{n,R},R} h_R. \tag{139.4}$$

There is C such that for $n \ge 2$ and $2 \le a \le n$,

$$\nu_R^*\{|N_{n,R} - n/c_*| > a\} \le Cn/a^2. \tag{139.5}$$

Proof. A trajectory starting in Y_R^* has exactly n visits to this section among collision times $0, \dots, N_{n,R} - 1$. These are its initial visit and its first $n - 1$ returns. Also $J_{n,R} = S_{N_{n,R},R} f_R$. Substitution in (138.2) proves the identity with no remainder.

Let $H_{L,R} = \sum_{j=1}^{L} \eta_R \circ T_R^j$. Its stationary mean is c_*L. Proposition 138.3 applied to $\eta_R - c_*$ gives

$$\int |H_{L,R} - c_*L|^2 \,\mathrm{d}\nu \le CL.$$

The same upper bound, with a changed constant, holds under ν_R^* because its density relative to ν is at most c_*^{-1}. The exact equivalences $N_{n,R} > L \iff H_{L,R} < n$ and $N_{n,R} \le L \iff H_{L,R} \ge n$ reduce both tails to this estimate. Choose integers within distance one of $n/c_* + a$ and $n/c_* - a$. The corresponding deviations from c_*L are at least $c_*a - O(1)$, and $L \le Cn$. Chebyshev's inequality proves the assertion when a exceeds a fixed constant. For the bounded remaining range, enlarge C so that the right side is at least one. □

Lemma 139.4 (A deterministic-window maximal estimate)**.** *For $L \geq 1$, $r \geq 0$, and $R \in I$,*

$$\int \max_{0 \leq j \leq L} |S_{r+j,R} h_R - S_{r,R} h_R|^2 \, \mathrm{d}\nu \leq CL[\log(2+L)]^2. \tag{139.6}$$

The same estimate holds under ν_R^* and for backward sums on a deterministic interval contained in the nonnegative collision times.

Proof. Absolute summability of the correlations gives second moment at most $C\ell$ for the sum over every interval of ℓ collisions, by stationarity. Enlarge L to a power of two smaller than $2L$. Every prefix of that interval is a disjoint union of at most one dyadic interval at each scale. Cauchy–Schwarz bounds the square of its sum by the number of scales times the sum of the squares of these interval sums. Bound the latter by the sum over all dyadic intervals at all scales, independently of the prefix. At each scale, the expectations sum to at most CL, since the intervals partition the full interval. There are $O(\log(2+L))$ scales, proving the bound. The proof applies to each component and to intervals in reverse order. A bounded initial density gives the assertion under ν_R^* as well. □

Theorem 139.5 (Parameter-uniform Gaussian limit of the physical record)**.** *Define $D_R = c_*^{-1}\Gamma_R$. It is a continuous positive semidefinite matrix on I. For every $K < \infty$ there is C_K such that*

$$\sup_{R \in I,\ |t| \leq K} \left| \int_{Y_R^*} e^{it \cdot (J_{n,R} - n\bar{G}_R)/\sqrt{n}} \, \mathrm{d}\nu_R^* - e^{-t^{\mathsf{T}} D_R t/2} \right| \leq C_K n^{-1/26} \sqrt{\log(2+n)} \qquad (n \geq 2). \tag{139.7}$$

In particular the original, unsmoothed return record has a centered Gaussian weak limit with covariance D_R. The estimate is uniform in the physical parameter, including along convergent parameter sequences.

Proof. Put $m_n = \lfloor n/c_* \rfloor$ and $a_n = \lceil n^{3/5} \rceil$. The clock-window lemma, with a harmless change for rounding, bounds the probability of $|N_{n,R} - m_n| > a_n$ by Cn/a_n^2. On the complementary event, the difference $S_{N_{n,R},R} h_R - S_{m_n,R} h_R$ is bounded by the larger of the forward and backward maxima on intervals of length a_n about m_n. The intervals have nonnegative endpoints since $c_* < 1$. The preceding lemma and Cauchy–Schwarz consequently give

$$\begin{aligned} &\left| \int e^{it \cdot S_{N_{n,R},R} h_R/\sqrt{n}} \, \mathrm{d}\nu_R^* - \int e^{it \cdot S_{m_n,R} h_R/\sqrt{n}} \, \mathrm{d}\nu_R^* \right| \\ &\qquad \leq C_K \left(\frac{n}{a_n^2} + \sqrt{\frac{a_n}{n}} \log(2+a_n) \right) \leq C_K n^{-1/5} \log(2+n). \end{aligned} \tag{139.8}$$

The exceptional event costs at most twice its probability. No estimate of an unbounded sum on that event is used.

Apply Proposition 139.2 at time m_n and frequency $t\sqrt{m_n/n}$ with initial density s_R. This frequency stays in a fixed compact set. Since $m_n/n = c_*^{-1} + O(n^{-1})$ and Γ_R is uniformly bounded, its Gaussian differs from the one in (139.7) by $O_K(n^{-1})$. The identity (139.4) completes the estimate; the error in (139.8) is smaller than the stated one. The continuity of D_R follows from Theorem 138.4. The characteristic-function continuity theorem yields the weak limit, allowing a singular Gaussian. Along $R_n \to R_0$, the same bound and continuity give the Gaussian with covariance D_{R_0}. □

139.3. **The Gaussian kernel and measurable cohomology.**

Theorem 139.6 (The kernel is an actual L^2 coboundary)**.** *For $R \in I$ and $v \in \mathbb{R}^4$, the following three statements are equivalent:*

1. *$v^{\mathsf{T}} D_R v = 0$;*
2. *there is $b \in L^2(M, \nu)$ such that $v \cdot h_R = b - b \circ T_R$ almost everywhere;*

(3) *there is* $b_* \in L^2(Y_R^*, \nu_R^*)$ *such that* $v \cdot (G_R - \bar{G}_R) = b_* - b_* \circ F_R^*$ *almost everywhere.*

If the first condition holds, one may choose b *with* $\|b\|_2 \le C|v|$*, uniformly in* R*, and take* $b_* = b|_{Y_R^*}$.

Proof. Put $u = v \cdot h_R$ and $c_k = \int u(u \circ T_R^k)\,\mathrm{d}\nu$. By (138.4), $|c_k| \le C|v|^2 e^{-\gamma k}$. If $v^{\mathsf{T}} D_R v = 0$, then $c_0 + 2\sum_{k\ge 1} c_k = 0$. The exact variance identity gives

$$\int |S_{m,R}u|^2\,\mathrm{d}\nu = -2\sum_{k=1}^{m-1} kc_k - 2m\sum_{k\ge m} c_k \le C|v|^2 \qquad (m \ge 1).$$

Let $Uu = u \circ T_R$, a unitary operator on $L^2(\nu)$, and form $b_M = M^{-1}\sum_{m=1}^{M} S_{m,R}u$. This is a bounded sequence in L^2, with norm at most $C|v|$. Moreover

$$(I - U)b_M = u - \frac{1}{M}\sum_{m=1}^{M} U^m u \longrightarrow u \quad \text{in } L^2.$$

Here the mean ergodic theorem applies; the collision map is ergodic and u has mean zero. A weakly convergent subsequence of b_M has limit b, and bounded linearity of $I - U$ shows $(I - U)b = u$. This proves the second assertion and its norm estimate.

Discard a countable union of collision translates of the null set where this identity or a representative fails. On the remaining invariant full-measure set, telescope until the actual first return. The compensation identity for one block gives

$$v \cdot (G_R - \bar{G}_R)(x) = b(x) - b(F_R^* x).$$

The restriction belongs to $L^2(\nu_R^*)$ and has norm at most $c_*^{-1/2}\|b\|_2$. Thus the second assertion implies the third. Conversely, the third assertion telescopes for the stationary induced map, so the $L^2(\nu_R^*)$ norm of $v \cdot (J_{n,R} - n\bar{G}_R)/\sqrt{n}$ is at most $2\|b_*\|_2/\sqrt{n}$. Its characteristic function tends to one. Theorem 139.5 identifies the same limit as $\exp(-t^2 v^{\mathsf{T}} D_R v/2)$ for every real t, proving the first assertion. This also completes the equivalence without presupposing summability of induced correlations. □

Remark 139.7 (What the low-frequency theorem supplies). The matrix D_R is now established by an absolutely convergent collision Green–Kubo formula and is the covariance of the Gaussian law of the true induced record. The zero-Gaussian-variance condition has been identified with a genuine induced L^2 coboundary. A measurable representative of its transfer function has not thereby been made continuous at a selected periodic orbit. Thus the periodic rank calculation is not applied to this representative. The separate phase argument in Theorem 150.2 proves uniform positive definiteness without periodic evaluation. Neither convergence of the induced Green–Kubo series nor convergence of its normalized second moments follows from this stopping argument alone; the actual second moments and the Cesaro formula are established in Theorem 147.6 and Corollary 147.7.

The estimate (139.7) controls fixed compact sets of rescaled central frequencies. It does not give the integrated complementary-frequency bound in the raw local inversion theorem, an induced eigenvalue expansion, or the complete critical/singular branch sum. In particular it is not a raw density local limit, a weighted local-density estimate, or a functional limit under a changed exact conditioning event. These distinctions retain the original density problem while supplying its actual Gaussian weak-limit input.

140. Functional Gaussian limits and the physical clock

A central-frequency estimate alone does not give tightness of the record process. We prove it by smoothing on a slower scale, using fourth moments for the smooth sums and a maximal

second-moment estimate for the residual. The estimates remain uniform over the moving physical family.

140.1. **Fourth moments and removal of smoothing.**

Lemma 140.1 (A fourth-moment bound at a specified smoothing scale)**.** *For $m \ge 1$, $R \in I$, and $0 < \delta < 1/2$,*

$$\int |S_{m,R} h_{R,\delta}|^4 \, \mathrm{d}\nu \le C(m^2 + (m+1)\delta^{-8}). \tag{140.1}$$

For $r_{R,\delta} = h_R - h_{R,\delta}$ and $L \ge 1$,

$$\int \max_{0 \le j \le L} |S_{j,R} r_{R,\delta}|^2 \, \mathrm{d}\nu \le CL\delta(1 + |\log \delta|)[\log(2+L)]^2. \tag{140.2}$$

Both bounds hold, with changed constants, after multiplication by any nonnegative initial density bounded by a fixed constant.

Proof. For the first assertion consider one real component u of $h_{R,\delta}$ and the one-variable family in Lemma 139.1. Write $\psi(z) = \log \lambda(z)$ and $A(z) = \ell_j(\Pi(z)\nu)$. The spectral pairing for the stationary characteristic function is

$$\int e^{izS_{m,R}u} \, \mathrm{d}\nu = e^{m\psi(z)} A(z) + E_m(z).$$

On a complex disk of radius $a\delta^2$, with a decreased once, $|E_m(z)| \le C\rho^m$, $|A(z)| \le C$, and $|\psi(z)| \le C$. Cauchy's inequalities therefore give $|A^{(j)}(0)| \le C_j \delta^{-2j}$ for $j \le 4$, $|\psi^{(j)}(0)| \le C_j \delta^{-2j}$ for $j = 3, 4$, and $|E_m^{(4)}(0)| \le C\delta^{-8}\rho^m$. In addition $A(0) = 1$, $\psi'(0) = 0$, and $|\psi''(0)| \le C$, by the uniformly summable smoothed covariance. The fourth derivative of the principal term at zero is exactly

$$A^{(4)}(0) + 6m\psi''(0)A''(0) + 4m\psi'''(0)A'(0) + m\psi^{(4)}(0) + 3m^2\psi''(0)^2.$$

It is bounded by $C(m^2 + (m+1)\delta^{-8})$. Differentiating under the finite-time integral is legitimate because u is bounded. The fourth derivative is its fourth moment. Summing the component estimates proves (140.1).

For the second assertion, (138.5) bounds the second moment of every interval sum of length ℓ by $C\ell\delta(1 + |\log \delta|)$. Apply the dyadic proof of Lemma 139.4 with this constant. The final assertion follows by domination of the initial density; it does not assert stationarity under that density. □

For $A_0 < \infty$, let $X_{n,R}$ be the polygonal interpolation on $[0, A_0]$ with values

$$X_{n,R}(j/n) = n^{-1/2} S_{j,R} h_R.$$

Let $\mathcal{W}_C$ denote Brownian motion starting at zero with covariance $\mathbb{E}[\mathcal{W}_C(s)\mathcal{W}_C(t)^{\mathsf{T}}] = (s \wedge t)C$; C is allowed to be singular. We use the bounded-Lipschitz distance on probability laws on continuous path space, with its supremum norm.

Theorem 140.2 (Uniform collision functional central limit)**.** *Fix $A_0, B_0 < \infty$. Uniformly for $R \in I$ and all nonnegative probability densities σ_R satisfying $\|\sigma_R\|_\infty + \|\sigma_R\|_{\mathrm{BV}} \le B_0$,*

$$d_{\mathrm{BL}}(\mathrm{Law}_{\sigma_R \nu}(X_{n,R}), \mathrm{Law}(\mathcal{W}_{\Gamma_R})) \longrightarrow 0. \tag{140.3}$$

In particular this holds under ν and under ν_R^*.

Proof. Set $\delta_n = \frac{1}{4} n^{-1/32}$ and let $X_{n,R}^{\delta_n}$ be the polygonal process with h_R replaced by h_{R,δ_n}. Lemma 140.1 implies

$$\sup_{R,\sigma_R} \int \|X_{n,R} - X_{n,R}^{\delta_n}\|_\infty^2 \sigma_R \, \mathrm{d}\nu \le C_{A_0,B_0} \delta_n [\log(2+n)]^3 \longrightarrow 0.$$

Integer parts and interpolation change only the constant. Thus it suffices to establish the limit for the smoothed process. This scale is slower than the one optimizing the one-time estimate.

We first prove finite-dimensional convergence. Fix $0 = t_0 < t_1 < \cdots < t_q \le A_0$ and vectors $v_1, \dots, v_q$. Put $m_j = \lfloor nt_j \rfloor - \lfloor nt_{j-1} \rfloor$ and $z_j = v_j/\sqrt{n}$. Smooth the initial density at the same scale; its L^1 error is $O(\delta_n)$ and its distribution norm is $O(\delta_n^{-2})$. The characteristic function of the consecutive increments, before the harmless polygonal endpoint corrections, is

$$\ell_j(\mathcal{L}_{R,\delta_n,z_q}^{m_q} \cdots \mathcal{L}_{R,\delta_n,z_1}^{m_1}(\mathsf{S}_{\delta_n}\sigma_R)\nu).$$

The product is chronological, with the earliest segment on the right. For each fixed q, substitute the spectral decompositions of Lemma 139.1. Every term with a complementary power tends to zero uniformly, since m_j is a positive multiple of n and its remaining factors are polynomially bounded in δ_n^{-1}. The product of principal amplitudes tends to one: $\Pi(0) = \nu\ell_j$, and its error is $O_q(\delta_n^{-4}n^{-1/2})$. The cubic error in the logarithm is $O_q(\delta_n^{-6}n^{-1/2}) = o(1)$. The resulting characteristic function is therefore, uniformly in R and in the initial densities,

$$\exp\left(-\frac{1}{2}\sum_{j=1}^{q}(t_j - t_{j-1})v_j^{\mathsf{T}}\Gamma_R v_j\right) + o(1).$$

Here (138.7) removes smoothing from the covariance. This is the joint law of the independent Brownian increments, not merely a collection of marginal Gaussian laws.

For tightness put $L_n = \lceil \delta_n^{-8} \rceil$ and interpolate the smoothed sums only at collision indices $0, L_n, 2L_n, \dots$. Call this coarse polygonal process $Z_{n,R}$. An interval of $m \ge L_n$ collisions has fourth moment at most Cm^2, by (140.1) and stationarity of ν. Domination by $\sigma_R \le B_0$ gives the same bound from every deterministic starting index under the initial law. It follows at coarse grid points that the fourth moment of an increment is at most $C|t-s|^2$. For arbitrary points, split an increment into its two fractional end segments and the complete grid interval. Within one grid interval, linear interpolation gives a factor $(|t-s|n/L_n)^4$ multiplying a fourth moment $O((L_n/n)^2)$, which is bounded by $C|t-s|^2$. The same estimate follows across adjacent intervals and then across any number of intervals, with a fixed constant. Hence

$$\mathbb{E}_{\sigma_R\nu}|Z_{n,R}(t) - Z_{n,R}(s)|^4 \le C_{A_0,B_0}|t-s|^2.$$

The usual dyadic proof of the continuity and tightness criterion applies uniformly to these continuous processes. Extend the last grid interval past A_0 and restrict back to $[0, A_0]$ to avoid an endpoint convention. Finally, boundedness of h_{R,δ_n} gives deterministically

$$\|Z_{n,R} - X_{n,R}^{\delta_n}\|_\infty \le CL_n/\sqrt{n} = O(n^{-1/4}) \longrightarrow 0.$$

This proves tightness of the original processes as well.

To obtain the stated uniform distance, suppose it failed and select a violating sequence of radii and initial densities. By compactness a subsequence of the radii converges to R_0. The uniform tightness and finite-dimensional limits identify every subsequential path law as $\mathcal{W}_{\Gamma_{R_0}}$. Continuity of Γ_R, including at singular matrices, gives continuity of the Brownian law; for example couple by $\Gamma_R^{1/2}$ times one standard Brownian motion. This contradicts the violating bounded-Lipschitz distance and proves (140.3). □

140.2. **The induced record process.** Let $Y_{n,R}$ be the polygonal interpolation on $[0,1]$ of $n^{-1/2}(J_{j,R} - j\bar{G}_R)$ at times j/n.

Theorem 140.3 (Functional limit of the actual return process)**.** *Uniformly for $R \in I$,*

$$d_{\mathrm{BL}}(\mathrm{Law}_{\nu_R^*}(Y_{n,R}), \mathrm{Law}(\mathcal{W}_{D_R})) \longrightarrow 0, \qquad D_R = c_*^{-1}\Gamma_R. \tag{140.4}$$

Proof. The count coordinate of Theorem 137.2 gives

$$\sup_{0\le s\le 1} |N_{\lfloor ns\rfloor,R}/n - s/c_*| \longrightarrow 0$$

in probability, uniformly in R. Choose $A_0 > c_*^{-1} + 1$ and apply Theorem 140.2 on $[0, A_0]$ under ν_R^*. Tightness with continuous limits gives a uniform probabilistic modulus of continuity for $X_{n,R}$. Consequently replacing its random argument $N_{\lfloor ns\rfloor,R}/n$ by s/c_* changes it by $o_P(1)$ in supremum norm. The random argument remains in $[0, A_0]$ with probability tending to one uniformly; independence from $X_{n,R}$ is not required. The exact identity (139.4) identifies this composition with the step version of the centered induced process.

For its polygonal version, stationarity under F_R^* and the true one-block tail give, for every $\epsilon > 0$,

$$\nu_R^*\left\{\max_{j\le n} r_R^* \circ (F_R^*)^j > \epsilon\sqrt{n}\right\} \le C(n+1)e^{-c\epsilon\sqrt{n}} \longrightarrow 0.$$

Each centered return increment has norm at most Cr_R^*, by boundedness of h_R and exact compensation. Thus interpolation changes the process by $o_P(1)$ uniformly. Brownian time change $s \mapsto s/c_*$ has covariance $c_*^{-1}\Gamma_R$, which proves the assertion by the same compactness argument as above. □

140.3. **An unconditioned physical-time consequence.** Recall the actual stationary physical collision count $C_R(t)$ and lattice displacement $W_R(t)$ of Theorem 137.4. Define the linear map

$$B_R(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3 - x_4/\bar\tau_R), \qquad \mathcal{V}_R = \bar\tau_R^{-1} B_R \Gamma_R B_R^{\mathsf{T}}.$$

Corollary 140.4 (Physical displacement and collision fluctuations)**.** *Under the stationary laws μ_R, the processes*

$$t^{-1/2}(W_R(ts), C_R(ts) - ts/\bar\tau_R), \quad 0 \le s \le 1, \tag{140.5}$$

converge uniformly in R to Brownian motion with covariance $\mathcal{V}_R$. The convergence may be taken in the Skorokhod space, or for continuous interpolations whose uniform difference from the displayed processes is $O(t^{-1/2})$. The matrices $\mathcal{V}_R$ are continuous and positive semidefinite. No conditioning event is changed or imposed in this result.

Proof. In the physical collision suspension, the marginal of the last outgoing collision state x is $\sigma_R\nu$, where $\sigma_R = \tau_R/\bar\tau_R$. Its supremum and variation norms are uniformly bounded by Lemma 138.1 and $\min_I \bar\tau_R > 0$. The elapsed time since that collision is bounded by the uniform single-flight horizon. Thus Theorem 140.2 applies with this initial density.

The section compensation disappears under B_R:

$$B_R h_R = (\kappa_{\mathrm{coll},1}, \kappa_{\mathrm{coll},2}, 1 - \tau_R/\bar\tau_R), \qquad B_R \bar G_R = 0.$$

The collision clock satisfies $\sup_{s\le 1} |C_R(ts)/t - s/\bar\tau_R| = o_P(1)$ uniformly, by Theorem 137.4. Apply the continuous-limit random-time argument to $B_R X_{\lfloor t\rfloor,R}$ on a fixed interval larger than $1/\min_I \bar\tau_R + 1$. Rounding t affects neither the normalization nor the limit.

The initial and final partial physical flights have uniformly bounded durations. The collision sum of the lattice labels differs from $W_R(ts)$ by a uniformly bounded cell-position term. Also the sum of completed flight times differs from ts by a uniformly bounded amount. Consequently the stopped sum of $B_R h_R$ differs from the numerator in (140.5) by a uniformly bounded vector, simultaneously for all s. The jumps of the lattice and collision counts are uniformly bounded; interpolation between collision times, together with the same cell-position bound, has error $O(t^{-1/2})$. The limiting Brownian time change is $s/\bar\tau_R$, giving $\mathcal{V}_R$. Continuity follows from that of Γ_R and the explicit strictly positive mean roof. □

The functional results supply the unconditioned process input for the original clock analysis. They neither evaluate a measurable transfer function at the selected periods nor control a shrinking conditioning event. Uniform positive definiteness is supplied separately by Corollary 150.4; the weighted raw density estimates remain distinct from these path-space conclusions.

141. One-step phase coercivity and the operator criterion

The positive-measure estimate in Section 130 uses an entire periodic tube. There is a stronger estimate if one first locates a large one-step defect on the period, and only then thickens that single state. It avoids the exponentially shrinking tube of the whole period. We retain the constants ρ_0, L, B_0 of Lemma 130.6, and set $r_0 = \rho_0/(2L)$.

Theorem 141.1 (One-step thickening of the periodic defect)**.** *Fix $0 < \alpha \le 1$ and $1 \le p < \infty$. Let q be circle-valued and measurable, with an α-Hölder representative of constant $H \ge 1$ on the ρ_0 balls around the states of the triple selected in Theorem 130.4. For $b = |\beta| \ge 1$ put*

$$d_b = \frac{1}{4000000\pi M(b)}, \qquad H_b = H(1 + L^\alpha), \qquad r_b = \min\{r_0, d_b/(4bB_0), (d_b/(4H_b))^{1/\alpha}\}.$$

For the actual one-step defect $\mathcal{D}_q$ defined in Theorem 130.7,

$$\|\mathcal{D}_q\|_{L^p(\nu_R^*)} \ge \frac{d_b}{2}\left(\frac{r_b^2}{4c_*}\right)^{1/p}. \tag{141.1}$$

If $H \le H_0 b^\zeta$ with fixed $H_0 \ge 1$ and $\zeta \ge 0$, this gives

$$\|\mathcal{D}_q\|_{L^p(\nu_R^*)} \ge C\, b^{-\lambda}(1 + \log b)^{-\eta}, \qquad \lambda = \frac{2}{p}\max\{1, \zeta/\alpha\}, \quad \eta = 1 + \frac{2}{\alpha p}. \tag{141.2}$$

The constants are uniform in the radius and all torus frequencies.

Proof. Telescoping the three exact periods as in Corollary 130.5 shows that at some state x_0 of the selected triple the one-step defect has modulus at least d_b. This is initially a statement about the specified continuous representatives at finitely many states: the sum of the three induced periods is at most $2M(b)$, while their product discrepancy is at least $2/(4000000\pi)$.

On $B(x_0, r_0)$ the return has one fixed lattice label and one fixed physical count, and F_R^* maps that ball into the ρ_0 ball about the successor of x_0. For x in that ball,

$$|\mathcal{D}_q(x) - \mathcal{D}_q(x_0)| \le H_b|x - x_0|^\alpha + bB_0|x - x_0|.$$

Each term on the right is at most $d_b/4$ on the ball of radius r_b. On that ball the defect is consequently at least $d_b/2$. Its actual invariant measure is $r_b^2/(4c_*)$. Integration proves (141.1). Changes of representatives on null sets do not alter the integral.

Finally $M(b) \le 3 + \log b/\log(100/49)$ and $H_b \le Cb^\zeta$. The minimum defining r_b is bounded below by a positive constant times $b^{-\max\{1,\zeta/\alpha\}}(1 + \log b)^{-1/\alpha}$. Substitution proves (141.2). Unlike the whole-period tube estimate, its exponent contains no $\log L$ term. The regularity premise on q remains essential. □

The next theorem specifies exactly what is required to turn this physical coercivity into an operator resolvent. The hypothesis is a reconstruction of a phase from an approximate spectral vector; it is not silently identified with an eigenvalue equation.

Theorem 141.2 (A quantitative phase-reconstruction criterion)**.** *For each R, u, s, β, let $\mathcal{L}_{R,u,s,\beta}$ be a bounded operator on a Banach space $\mathcal{B}_R$. Fix the constants α, p, H_0, ζ of Theorem 141.1, and let $A \ge 1$, $v \ge 0$, $\theta > 0$. Suppose that for every $b = |\beta| \ge 1$, $|z| = 1$, and $h \in \mathcal{B}_R$ with $\|h\| = 1$*

and $e = \|(z - \mathcal{L}_{R,u,s,\beta})h\| \le 1$*, one can construct a circle-valued* q *with the local Hölder bound* $H_0 b^{\zeta}$ *and*

$$\|\mathcal{D}_q\|_{L^p(\nu_R^*)} \le A b^v e^{\theta}, \qquad e^{ic} = z. \tag{141.3}$$

Suppose also that $z - \mathcal{L}_{R,u,s,\beta}$ *is Fredholm of index zero. Then it is invertible and*

$$\|(z - \mathcal{L}_{R,u,s,\beta})^{-1}\| \le C b^{(\lambda+v)/\theta}(1 + \log b)^{\eta/\theta}, \tag{141.4}$$

with λ, η *as in* (141.2)*.*

Proof. Combining (141.2) and (141.3) yields

$$e \ge (C_0/A)^{1/\theta} b^{-(\lambda+v)/\theta}(1 + \log b)^{-\eta/\theta}$$

when $e \le 1$. Decreasing the positive constant to at most one makes the same lower bound valid when $e > 1$. Homogeneity gives a uniform lower bound for $z - \mathcal{L}$ on every vector. It is therefore injective and has closed range. Fredholm index zero makes its cokernel zero as well, so it is surjective. The lower bound gives (141.4). The Fredholm assumption cannot be omitted: an injective operator bounded below need not be onto. □

For the unbounded induced twists, the phase reconstruction (141.3), including nonvanishing modulus and quantitative regularity, is an unverified premise, not a result of Lemma 133.1. The latter treats a smooth collision-space weight near zero. It provides the return-tail theorem but is not a construction of $\mathcal{B}_R$ or of the raw high-frequency operator. The criterion separates the necessary analytical construction from the now explicit phase lower bound.
Scope of the retained edge calculation. The next section uses the original section Y_R in (127.4), not Y_R^*. Its pointwise physical critical-word calculations are retained, but an induced count or normalization is not transferred between the sections without a new return computation.

142. The exact raw density at a minimal flight length

Let $L_N(x) = S_N \tau_R(x)$ on the collision section, and put

$$g = 1 - 2R, \qquad \zeta_R = \operatorname{arcosh}(1 + g/R) > 0.$$

Near the orbit alternating normally between centers 0 and a, choose the initial arc coordinate y in the upward tangent direction and the outgoing tangential velocity p. Both vanish on the orbit. Use the same upward signed arc coordinate y_j at its successive endpoints.

Lemma 142.1 (Reduced action Hessian)**.** *The second variation of the length with all contact coordinates free is*

$$\frac{1}{2}\sum_{j=0}^{N-1}\left\{\frac{y_j^2 + y_{j+1}^2}{R} + \frac{(y_{j+1} - y_j)^2}{g}\right\}. \tag{142.1}$$

After solving the internal reflection equations, the endpoint Hessian is

$$S_N = \begin{pmatrix} A_N & -B_N \\ -B_N & A_N \end{pmatrix}, \quad A_N = \frac{\sinh \zeta_R}{g}\coth(N\zeta_R), \quad B_N = \frac{\sinh \zeta_R}{g \sinh(N\zeta_R)}. \tag{142.2}$$

In the physical initial coordinates (y, p)*,* L_N *has a nondegenerate minimum* Ng *and Hessian* H_N *satisfying*

$$\sqrt{\det H_N} = \sinh(N\zeta_R). \tag{142.3}$$

Proof. At successive facing circles the longitudinal endpoint corrections are $y_j^2/(2R)$ and $y_{j+1}^2/(2R)$. Expanding their distance gives (142.1). Its full Hessian is positive definite, since it is a sum of positive square terms including endpoint squares. The interior stationarity equations are

$$y_{j+1} - 2(1 + g/R)y_j + y_{j-1} = 0.$$

Their solution with given endpoints is

$$y_j = \frac{\sinh((N-j)\zeta_R)}{\sinh(N\zeta_R)} y_0 + \frac{\sinh(j\zeta_R)}{\sinh(N\zeta_R)} y_N.$$

Substitution into the endpoint gradients yields (142.2). The nonlinear interior equations are uniquely solvable near zero by their positive definite Hessian. Their solution is the physical reflected path by the first variation and the strict incidence at the normal orbit.

The endpoint first variation gives $p = -\partial_{y_0} L_N$ at the linearized level, so $p = -A_N y_0 + B_N y_N$. The linear map from (y_0, p) to (y_0, y_N) has determinant $1/B_N$. Thus

$$\det H_N = \frac{A_N^2 - B_N^2}{B_N^2} = \sinh^2(N\zeta_R).$$

Positivity is preserved under this nonsingular coordinate change, proving the minimum assertion. □

Theorem 142.2 (A physical edge coefficient)**.** *Let χ be a smooth nonnegative section cutoff supported in the regular alternating itinerary, equal to one near its normal initial point. The pushforward of $\chi\nu$ by L_N has, near Ng, the form*

$$f_{N,R}(Ng + s) = \mathbf{1}_{\{s \geq 0\}} a_{N,R}(s), \qquad a_{N,R}(0) = J_{N,R} := \frac{1}{2R \sinh(N\zeta_R)}. \tag{142.4}$$

The function $a_{N,R}$ is smooth on a right neighborhood of zero. For each fixed N, the construction and smooth bounds may be made uniform on I. No assertion of uniform derivative bounds as $N \to \infty$ is made.

Proof. The section density in initial arc and tangential velocity coordinates is $(4\pi R)^{-1}\,\mathrm{d}y\,\mathrm{d}p$. The Morse lemma applied to Lemma 142.1 gives $L_N = Ng + |z|^2/2$. The transformed density $w(z)$ is smooth with

$$w(0) = \frac{1}{4\pi R\sqrt{\det H_N}}.$$

In polar coordinates its pushforward density is $\mathbf{1}_{\{s\geq 0\}} \int_0^{2\pi} w(\sqrt{2s}\, n_\phi)\,\mathrm{d}\phi$ near zero. The angular integral of each odd Taylor term vanishes; Taylor's theorem at arbitrary order gives smoothness in $s \geq 0$. Its right value is $2\pi w(0)$, which is (142.4). Compactness in R and the strictly positive determinant give the fixed-N uniform statement. □

Proposition 142.3 (Regular critical words and a uniform edge bound)**.** *Fix a regular physical word with N flights between circular obstacles. Every critical point of its total flight length in the two initial section coordinates is normal at both endpoints. There is at most one such point for the fixed sequence of lattice centers. It is a nondegenerate minimum. If $S = \left(\begin{smallmatrix} A & C \\ C & D \end{smallmatrix}\right)$ is its reduced endpoint action Hessian, its section-density jump is*

$$J = \frac{|C|}{2R\sqrt{AD - C^2}}. \tag{142.5}$$

There is a fixed explicitly computable constant C_* such that, for every regular word and $N \geq 2$,

$$0 < J \leq C_* (47/53)^{N-2}. \tag{142.6}$$

No lower incidence margin for the interior collisions is needed for this coefficient bound. This is an individual-word estimate, not a bound for the sum over words.

Proof. Let $q_j(y_j)$ be arc coordinates and v_j the unit velocity of segment j. Its second length variation is

$$\frac{c_j}{R} y_j^2 + \frac{c_{j+1}}{R} y_{j+1}^2 + \frac{\{v_j^\perp \cdot (t_{j+1} y_{j+1} - t_j y_j)\}^2}{\ell_j},$$

where $c_j, c_{j+1} > 0$ are its outgoing and incoming incidence cosines. At an interior reflected contact the two cosines agree. Summing gives a positive definite tridiagonal matrix with nonzero off-diagonal entries. Its interior Schur complement S is positive definite and $C \neq 0$: eliminating the successive interior variables expresses C as the nonzero product of neighboring entries divided by positive pivots.

The endpoint first variation of the reduced action is $-p_0\,\mathrm{d}y_0 + p_N\,\mathrm{d}y_N$. Moreover $\partial_{p_0} y_N = -1/C \neq 0$. Thus the derivative of the physical length in p_0 vanishes exactly when $p_N = 0$, and its other derivative then vanishes exactly when $p_0 = 0$. At such a critical point the coordinate change gives $\det H = (AD - C^2)/C^2 > 0$. The same Morse calculation as in Theorem 142.2 proves (142.5).

There cannot be two such regular points for one word. Regard the polygonal length as a convex function of all contact positions in the product of the closed disks. At a normal endpoint its gradient is $-n$, and at an interior reflected contact it is $-2c_j n_j$. These are strictly inward normals. For a different point x_j of the same strictly convex disk, $n_j \cdot (x_j - q_j) < 0$. The convex gradient inequality therefore makes every different tuple have strictly larger action. Hence the normal-to-normal critical tuple is the unique global minimizer of that convex problem. Occlusion by a disk not in the word is excluded separately by physical admissibility, not by this minimization argument.

To prove the uniform bound, set $z_j = c_j y_j$ and choose tangent orientations successively so that the off-diagonal entries become negative. At the normal endpoints $c_0 = c_N = 1$. Write $t_j = 1/\ell_j \leq t_* = 50/3$. The interior matrix has off-diagonals $-t_j$ and diagonals

$$d_j = t_{j-1} + t_j + 2/(Rc_j) \geq t_{j-1} + t_j + v_*, \qquad v_* = 200/47.$$

Its factorization $D_0(I - K)$ has $K \geq 0$ with row sums at most $q = 2t_*/(2t_* + v_*) = 47/53$. In the Neumann series, an entry connecting interior indices 1 and $N - 1$ vanishes before power $N - 2$. Therefore

$$|C| \leq t_*^2 \frac{q^{N-2}}{v_*(1-q)}.$$

The reduced quadratic form dominates the two endpoint curvature terms, so $S \geq (100/47)I$. Inserting these estimates in (142.5), and using $R \geq 9/20$, gives (142.6) with

$$C_* = \frac{47}{90} \frac{t_*^2}{v_*(1-q)}.$$

Arbitrarily small positive interior incidence only increases the diagonal potential, so it does not weaken the estimate. □

The proposition classifies regular critical contributions and bounds each jump uniformly. Singular itinerary boundaries and the number and arrangement of critical values remain to be controlled. Multiplying (142.6) by an uncontrolled word count would not prove a summable global correction.

Corollary 142.4 (Raw finite-record absolute continuity)**.** *For every $R \in I$ and $n \geq 1$, the actual induced record $(S_n\kappa_R, S_n r_R, S_n\tau_R^Y)$ under ν_R^Y has a density with respect to counting measure on $\mathbb{Z}^3$ times Lebesgue measure. For fixed total physical count N and displacement k, it has the coarea representation, for almost every t,*

$$p_{n,R}(k, N, t) = \frac{1}{4\pi\nu(Y_R)} \sum_{\mathbf{d}} \int_{D_{n,\mathbf{d}} \cap \{L_N = t\}} \frac{\mathrm{d}\mathcal{H}^1}{|\nabla_{\alpha,p} L_N|}. \tag{142.7}$$

Here $D_{n,\mathbf{d}}$ is the actual first-return event for the ordered center word $\mathbf{d}$, total count N and displacement k. Bounded measurable insertions have the same formula with the insertion in

the numerator. Neither a uniform density bound nor a long-time local limit is asserted by this statement.

Proof. Partition the recurrent regular initial states by total collision count and ordered center word. Uniform finite horizon makes the available lattice steps finite; for fixed N there are finitely many words. On every regular branch the physical length is smooth, and Proposition 142.3 makes its critical set empty or a singleton. Away from that null set, the coarea formula applied on a countable exhaustion where $|\nabla L_N|$ is bounded below proves absolute continuity and (142.7). Restriction to the measurable first-return event does not alter absolute continuity of the initial measure. The singular initial states have zero flux measure, and summing over the countable possible N gives the full law. The same exhaustion with a bounded insertion proves its signed version. □

For the induced system (127.4), fix n and restrict its joint law to $\kappa = 0$, $r = 2n$. Every physical flight is at least g. Equality of total length with $2ng$ requires every flight to have length g, and hence a nearest-neighbor normal alternating orbit. Of the six possible normal starting states only $(0, 0)$ lies in Y_R. A neighborhood of this point makes exactly n returns of physical length two each. Compactness, or directly the equality cases of the distance bound, excludes other itineraries from a sufficiently small right neighborhood of $2ng$. Consequently this lattice component of the induced law has right density

$$\frac{J_{2n,R}}{\nu(Y_R)} > 0 \tag{142.8}$$

at its left endpoint, and zero density on its left.

Corollary 142.5 (The boundary term must be retained)**.** *For the specified inducing set, the raw joint characteristic function $\Phi_{n,R}$ is not in $L^1(\mathbb{T}^3 \times \mathbb{R})$, for any $n \geq 1$. In particular a global integrable bound for this raw function cannot be justified merely by making the normal itinerary exponentially rare.*

Proof. If it were integrable, taking its $(0, 0, 2n)$ lattice Fourier coefficient would give an integrable function of the roof frequency. Its inverse would be a continuous density of that component. A continuous representative cannot agree almost everywhere with zero to the left of $2ng$ and with a function having the positive right limit (142.8). This contradicts Theorem 142.2. □

The central local-limit problem is not decided by this corollary. The endpoint may be far outside the central window, and $J_{2n,R}$ decays exponentially. What changes is the global inversion method, not the raw observable or the intended central theorem.

Proposition 142.6 (Exact edge subtraction)**.** *Choose a smooth roof cutoff supported in the right neighborhood of Theorem 142.2, equal to one near its endpoint, and denote the resulting subdensity by f. With $J = J_{N,R}$, one has the exact identity*

$$f(Ng + s) = Je^{-s}\mathbf{1}_{\{s\geq 0\}} + q(s), \qquad \widehat{f}(b) = e^{ibNg}\left(\frac{J}{1 - ib} + \widehat{q}(b)\right), \tag{142.9}$$

where $q \in L^1(\mathbb{R})$, its distributional second derivative is a finite signed measure, and $|\widehat{q}(b)| \leq C_{N,R}(1 + b^2)^{-1}$.

Proof. Writing $f(Ng + s) = \mathbf{1}_{s\geq 0}\eta(s)a(s)$, the right value of $\eta a - Je^{-s}$ at zero is zero. Thus q is continuous across zero, piecewise smooth, and exponentially decaying at infinity. Its first derivative may jump at zero, so its second distributional derivative has one finite atom and an integrable density. Distributional integration by parts gives $b^2|\widehat{q}(b)| \leq \|D^2 q\|_{\mathrm{TV}}$; the L^1 norm gives the bound near zero. Direct integration of the exponential proves (142.9). □

This is subtraction and exact addition of an explicit density, not convolution with noise or removal of a physical event. It proves an integrable remainder for the isolated edge. Bounds for all remaining branches and edges are still needed for a long-time result.

143. A raw residual-inversion theorem

Frequency convention. The rescaled frequency is $v = \sqrt{n}\,\omega$. When the proved growing central band is used below, its radii are $2n^{1/200}$ in v and $2n^{-99/200}$ in ω. General cutoffs in the following statements retain their stated hypotheses; a Gaussian-tail estimate is not a bound for the complementary transform of the raw measure.

Let $\eta_{n,R}$ be finite measures on $\mathbb{Z}^3 \times \mathbb{R}$ with characteristic functions $\Phi_{n,R}$. Frequencies are $\omega = (u, s, b) \in [-\pi, \pi]^3 \times \mathbb{R}$, and our transform convention is $e^{i\omega\cdot z}$. Suppose there are explicit signed edge measures $E_{n,R}$ with densities $e_{n,R}$, and put $H_{n,R} = \Phi_{n,R} - \widehat{E}_{n,R}$. The following hypotheses are an interface, not consequences of the mechanical constructions above.

Theorem 143.1 (Residual criterion for the mixed density LLT)**.** *Assume the following uniformly for $R \in I$.*
(i) There are $c, C, \delta > 0$, $\rho < 1$, means $m_R \in \mathbb{R}^4$, and symmetric matrices $cI \le \Sigma_R \le CI$ such that, for $|\omega| \le \delta$,

$$\Phi_{n,R}(\omega) = e^{inm_R\cdot\omega}\{A_R(\omega)\lambda_R(\omega)^n + O(\rho^n)\},$$

where $A_R(0) = 1$, $|A_R(\omega) - 1| \le C|\omega|$, $\log\lambda_R(\omega) = -\frac{1}{2}\omega^T\Sigma_R\omega + O(|\omega|^3)$, and $|\lambda_R(\omega)| \le e^{-c|\omega|^2}$.
(ii) $H_{n,R} \in L^1(\mathbb{T}^3 \times \mathbb{R})$ and

$$\int_{|\omega|>n^{-2/5}} |H_{n,R}(\omega)|\,\mathrm{d}\omega = o(n^{-2}). \tag{143.1}$$

(iii) $\|E_{n,R}\|_{\mathrm{TV}} = o(n^{-2})$ and, for each fixed K,

$$n^2 \operatorname*{ess\,sup}_{|(z-nm_R)/\sqrt{n}|\le K} |e_{n,R}(z)| \longrightarrow 0.$$

Then $\eta_{n,R}$ has a density with respect to counting measure times Lebesgue measure, and on each such central window

$$n^2 p_{n,R}(k,m,t) = \frac{\exp(-\xi^T\Sigma_R^{-1}\xi/2)}{(2\pi)^2\sqrt{\det\Sigma_R}} + o(1), \quad \xi = \frac{(k_1,k_2,m,t) - nm_R}{\sqrt{n}}, \tag{143.2}$$

with uniform essential-supremum error. No smoothing of $\eta_{n,R}$ is used.

Proof. Fourier inversion of $H_{n,R}$ gives a continuous density of the signed measure $\eta_{n,R} - E_{n,R}$. To justify this without assuming a density, take each lattice Fourier coefficient of H. It is an integrable roof transform of the corresponding signed component. Convolve that component with a Gaussian approximate identity, use integrability to pass to the inverse-transform limit, and test against compactly supported continuous functions. Uniqueness of finite Fourier transforms identifies each component, hence the full measure. This is a proof of inversion, not a change to the datum. Adding the known density of $E_{n,R}$ gives $p_{n,R}$.

On $|\omega| \le n^{-2/5}$, replacing H by Φ costs at most a fixed volume times $\|E\|_{\mathrm{TV}} = o(n^{-2})$. Inserting (i), let $v = \sqrt{n}\,\omega$. The cubic error is uniformly $O(n^{-1/5})$ on this region; the bound on λ supplies an integrable Gaussian dominator. Uniform Taylor bounds, truncation first to $|v| \le M$, and then $M \to \infty$, give the inverse Gaussian with error $o(n^{-2})$. The inverse normalization is $(2\pi)^{-4}$ and the Gaussian integral is $(2\pi)^2/\sqrt{\det\Sigma_R}$, giving the constant in (143.2). Hypothesis (ii) bounds the remaining frequencies, and (iii) bounds the added density in the central window. All estimates are uniform in ξ in that window. □

For weighted inserted measures, the same proof applies with $A_R(0) = a_R$ and uniformly bounded amplitudes, provided the edge and tail hypotheses are checked for those measures as well. It gives a_R times the Gaussian. It does not assert uniformity for arbitrary n-dependent path functionals without such estimates.

Lemma 143.2 (An explicit frequency splice)**.** *Suppose a residual transform has a compact minor-arc bound Ce^{-cn} and, for $1 \le |b| \le e^{\kappa n}$, a bound $C(1+|b|)^A e^{-cn}$. Beyond $e^{\kappa n}$ suppose it is bounded by*

$$C_M \rho^n (1+|b|)^{-2} + C_M e^{c_M n}(1+|b|)^{-M}.$$

If one fixed integer $M > 1$ and one $\kappa > 0$ satisfy

$$\frac{c_M}{M-1} < \kappa < \frac{c}{A+1}, \tag{143.3}$$

these regions contribute exponentially small L^1 mass. Together with the Gaussian major-arc bound between $n^{-2/5}$ and δ, this proves (143.1).

Proof. The middle integral is bounded by $C\exp(-(c-(A+1)\kappa)n)$. The two outer integrals are bounded by $C_M \exp((\log\rho - \kappa)n)$ and $C_M \exp(-((M-1)\kappa - c_M)n)/(M-1)$. The torus has finite volume. The major-arc Gaussian tail is bounded by a polynomial factor times $e^{-cn^{1/5}}$, and the compact remainder is exponential. Strict positivity of the three exponent margins proves the claim. □

The inequality (143.3) must be verified with the actual derivative-growth constants. The phrases "take κ small" and "then take M large" do not imply it: for example, $c_M = 2(M-1)$ and $c/(A+1) = 1$ leave no possible κ. This checks a proposed estimate, not the truth or falsity of a central LLT. The residual formulation also requires an independently proved bound for the sum of all extracted edge terms; Proposition 142.6 treats one physical family only.

144. Local inversion with nonnegligible remote edge mass

Frequency convention. The rescaled frequency is $v = \sqrt{n}\,\omega$. When the proved growing central band is used below, its radii are $2n^{1/200}$ in v and $2n^{-99/200}$ in ω. General cutoffs in the following statements retain their stated hypotheses; a Gaussian-tail estimate is not a bound for the complementary transform of the raw measure.

Theorem 143.1 gives a sufficient global smallness condition on extracted edges. A central local limit needs less: the extracted measure must have negligible density and negligible low-frequency convolution on the central window. Its total variation need not tend to zero. The following refinement retains the raw datum and the density norm.

Fix an even real function $\chi \in C_c^\infty(\mathbb{R}^4)$ equal to one on the unit ball and supported in the ball of radius two. Let $a_n \to \infty$ with $a_n = o(n^{1/6})$, put $B_n = a_n/\sqrt{n}$, and set $\chi_n(\omega) = \chi(\omega/B_n)$. For all sufficiently large n its support fits in the fundamental torus interval in the first three coordinates. With the transform convention already fixed, its inverse kernel on $\mathbb{Z}^3 \times \mathbb{R}$ is

$$K_n(k,t) = B_n^4 \check{\chi}(B_n k, B_n t), \qquad |K_n(k,t)| \le C_M B_n^4 (1 + B_n|(k,t)|)^{-M}. \tag{144.1}$$

The check denotes the inverse transform on $\mathbb{R}^4$, including its $(2\pi)^{-4}$ factor. Convolution here and below uses counting measure in the discrete coordinates and Lebesgue measure in the roof coordinate. For fixed K, let $W_{n,R,K} = \{z : |z - nm_R| \le K\sqrt{n}\}$.

Theorem 144.1 (A local edge criterion)**.** *Assume the small-frequency hypothesis (i) of Theorem 143.1. Let $E_{n,R}$ be finite signed measures with densities $e_{n,R}$ and suppose $H_{n,R} = \Phi_{n,R} - \widehat{E}_{n,R}$*

belongs to $L^1(\mathbb{T}^3 \times \mathbb{R})$. *Assume, uniformly in* R, *that*

$$\int |1 - \chi_n(\omega)|\, |H_{n,R}(\omega)|\, \mathrm{d}\omega = o(n^{-2}). \tag{144.2}$$

For every fixed central window suppose also that

$$n^2 \operatorname*{ess\,sup}_{W_{n,R,K}} |e_{n,R}| \longrightarrow 0, \qquad n^2 \sup_{W_{n,R,K}} |K_n * E_{n,R}| \longrightarrow 0. \tag{144.3}$$

Then the raw density conclusion (143.2) *holds on that window. There is no small-total-variation hypothesis on* $E_{n,R}$.

Proof. As in the proof of Theorem 143.1, integrability of H first identifies a continuous density for $\eta - E$, and adding e gives an actual mixed density for η. Splitting this identity with χ_n gives the exact equality almost everywhere

$$p_{n,R} = K_n * \eta_{n,R} + (e_{n,R} - K_n * E_{n,R}) + \mathcal{F}^{-1}[(1 - \chi_n) H_{n,R}]. \tag{144.4}$$

All terms are defined: E is finite, K_n is bounded and integrable, and the last inverse transform is absolutely convergent. The last term is at most $(2\pi)^{-4}$ times the integral in (144.2). The middle term is controlled by (144.3).

For the first term, its Fourier integral has the factor χ_n. Insert the assumed small-frequency expansion and use $v = \sqrt{n}\,\omega$. The support has $|v| \le 2a_n$ and $nO(|\omega|^3) = O(a_n^3/\sqrt{n}) = o(1)$. The uniform quadratic bound on λ_R gives a Gaussian dominator, while the amplitude converges to one. The exponentially small spectral remainder, after integration and multiplication by n^2, is bounded by $Ca_n^4\rho^n$ and tends to zero. Truncate first at a fixed $|v|$, then let that truncation grow. This proves uniformly in R and the central variable that $n^2 K_n * \eta_{n,R}$ converges to the Gaussian in (143.2). Combining the three terms proves the assertion.

The kernel is only an analytical device in the exact decomposition (144.4). The two correction terms are retained and estimated; the result is not a local limit for a smoothed replacement of the observed law. □

Corollary 144.2 (Remote mass need not be small)**.** *Suppose* $\|E_{n,R}\|_{\mathrm{TV}} \le C$ *and, for the window under consideration, the support of* $E_{n,R}$ *is at distance at least* $\varepsilon\sqrt{n}$ *from* $W_{n,R,K}$, *for a fixed* $\varepsilon > 0$. *Then* (144.3) *holds. In particular an extracted density of bounded, nonvanishing mass can satisfy the local edge hypotheses.*

Proof. The density vanishes almost everywhere in the central window. For z in that window, (144.1) gives

$$n^2 |(K_n * E_{n,R})(z)| \le CC_M a_n^4 (1 + \varepsilon a_n)^{-M} \longrightarrow 0$$

by taking $M > 4$. The Schwartz bound is available for every fixed M because χ is smooth. These constants are kernel constants, not the derivative-growth constants of the billiard in Lemma 143.2. □

The same conclusion holds for a sum of remote extracted terms whose total variations are summable with bounded total, and a remaining nearby extracted density satisfying (144.3). This follows by linearity and the triangle inequality. Thus individual edge coefficients do not all have to be $o(n^{-2})$ merely because one proves a central limit. Nevertheless, one must still construct the full extraction and prove (144.2). Neither this corollary nor the single-word estimate in Proposition 142.3 controls the sum of all central critical words or singular boundaries. A restriction of a density to a sharp window can itself create new jump terms; it is not automatically an integrable residual.

For weighted measures the same proof gives the corresponding amplitude times the Gaussian, provided the small-frequency expansion, residual estimate and local edge errors are verified for

that weighted class. This is the precise additional input needed before applying Proposition 228.1 to a proposed family of insertions.

145. An integrated growing central band for the physical record

Frequency convention. The rescaled frequency is $v = \sqrt{n}\,\omega$. When the proved growing central band is used below, its radii are $2n^{1/200}$ in v and $2n^{-99/200}$ in ω. General cutoffs in the following statements retain their stated hypotheses; a Gaussian-tail estimate is not a bound for the complementary transform of the raw measure.

The estimate on compact rescaled frequency sets can be strengthened to an integrated estimate on an expanding band. The distinction matters for local inversion: convergence after integration controls the inverse transform uniformly in the observation variable. We prove this estimate for the actual return record, including bounded-variation initial insertions. No induced spectral realization is used.

An admissible initial insertion is a complex-valued function a_R on M, vanishing almost everywhere outside Y_R^*, such that

$$\|a_R\|_\infty + \|a_R\|_{\mathrm{BV}} \le B.$$

Its integral is denoted by α_R. The original probability has $a_R = s_R$ and $\alpha_R = 1$. Write

$$C^a_{n,R}(v) = \int_{Y_R^*} a_R(x) \exp\left(\frac{iv \cdot (J_{n,R}(x) - n\bar{G}_R)}{\sqrt{n}} \right) \mathrm{d}\nu(x), \qquad v \in \mathbb{R}^4. \tag{145.1}$$

Here v is the rescaled frequency; the physical Fourier frequency is $\omega = v/\sqrt{n}$. Constants below depend on B but not on the insertion, the radius, or n. No continuity of the insertion in R is required.

145.1. **A frequency-dependent error bound.**

Lemma 145.1 (Collision estimate with explicit frequency dependence)**.** *Put $\ell_\delta = 1 + |\log \delta|$. There are $c, C > 0$ and $\rho < 1$ such that, for $m \ge 2$, $0 < \delta < 1/2$, and real t satisfying*

$$|t| \le c\sqrt{m}\,\delta^2, \qquad \delta^{-6}|t|^3 m^{-1/2} \le c,$$

every complex function a with $\|a\|_\infty + \|a\|_{\mathrm{BV}} \le B$ satisfies

$$\begin{aligned} &\left| \int a \exp(it \cdot S_{m,R} h_R/\sqrt{m})\, \mathrm{d}\nu - \left(\int a \,\mathrm{d}\nu \right) e^{-t^{\mathsf{T}} \Gamma_R t/2} \right| \\ &\quad \le C \left\{ \delta + |t|\sqrt{\delta \ell_\delta} + |t|^2 \delta \ell_\delta + \frac{|t|\delta^{-4} + |t|^3 \delta^{-6}}{\sqrt{m}} + \delta^{-2}\rho^m \right\}. \end{aligned} \tag{145.2}$$

Proof. Set $a_\delta = \mathsf{S}_\delta a$ and $z = t/\sqrt{m}$. The operator in Lemma 139.1 gives

$$\int a_\delta e^{it \cdot S_{m,R} h_{R,\delta}/\sqrt{m}}\, \mathrm{d}\nu = \lambda_{R,\delta}(z)^m \ell_j(\Pi_{R,\delta}(z) a_\delta \nu) + O(\delta^{-2}\rho^m).$$

The embedding bound in Lemma A.4 below makes the initial-vector factor explicit. The principal amplitude at zero is $\int a \,\mathrm{d}\nu$, and its change is at most $C|t|\delta^{-4}/\sqrt{m}$. Moreover

$$m \log \lambda_{R,\delta}(t/\sqrt{m}) = -\tfrac{1}{2} t^{\mathsf{T}} \Gamma_{R,\delta} t + O(\delta^{-6}|t|^3/\sqrt{m}).$$

The stated smallness conditions put z in the analytic ball and bound the last remainder. Positive semidefiniteness of $\Gamma_{R,\delta}$ therefore bounds the principal power by a fixed constant. The inequality $|e^w - 1| \le |w| e^{|w|}$ proves the two spectral error terms in (145.2).

Mean-preserving smoothing changes the initial insertion by $O(\delta)$ in L^1. If $r = h_R - h_{R,\delta}$, then

$$\int |a_\delta|\, |S_{m,R} r|^2 \,\mathrm{d}\nu \le Cm\delta\ell_\delta,$$

by (138.5) and $|a_\delta| \le B$. Cauchy–Schwarz bounds the characteristic error from removing this smoothing by $C|t|\sqrt{\delta\ell_\delta}$. This argument uses absolute values of the insertion and so applies to signed and complex insertions as well. Finally, the segment joining the two covariance matrices consists of positive semidefinite matrices. Differentiation along that segment and (138.7) give

$$\left|e^{-t^{\mathsf T}\Gamma_{R,\delta}t/2} - e^{-t^{\mathsf T}\Gamma_R t/2}\right| \le C|t|^2\delta\ell_\delta.$$

Combining these estimates proves the lemma. □

Proposition 145.2 (Stopped estimate on finite frequency balls)**.** *For $U \ge 1$, let*

$$\mathcal{E}_n(U) = \sup_{R,a_R} \int_{|v|\le 2U} |C^a_{n,R}(v) - \alpha_R e^{-v^{\mathsf T}D_R v/2}|\,\mathrm{d}v.$$

If $U \le c\sqrt{n}\,\delta^2$ and $\delta^{-6}U^3/\sqrt{n} \le c$, with c sufficiently small, then

$$\begin{aligned}\mathcal{E}_n(U) \le C\big\{&\delta U^4 + \sqrt{\delta\ell_\delta}\,U^5 + \delta\ell_\delta U^6 + n^{-1/2}\delta^{-4}U^5 + n^{-1/2}\delta^{-6}U^7\\ &+ \delta^{-2}\rho_0^n U^4 + n^{-1/5}U^4 + n^{-1/5}\log(2+n)U^5 + n^{-1}U^6\big\},\end{aligned} \tag{145.3}$$

where $\rho_0 < 1$. The supremum is over the admissible insertions.

Proof. Put $m_n = \lfloor n/c_*\rfloor$ and $b_n = \lceil n^{3/5}\rceil$. The exact compensation identity (139.4) permits replacement of the centered record by $S_{N_{n,R},R}h_R$. The exceptional set $|N_{n,R} - m_n| > b_n$ has ν_R^*-probability $O(n/b_n^2)$ by Lemma 139.3; rounding changes only the constant. Its contribution with insertion a_R is bounded by the same order, since $|a_R|\nu \le Bc_*\nu_R^*$ on the section. On its complement use the forward and backward deterministic-window maxima from Lemma 139.4. The elementary exponential difference bound and Cauchy–Schwarz give

$$\begin{aligned}&\left|C^a_{n,R}(v) - \int a_R e^{iv\cdot S_{m_n,R}h_R/\sqrt{n}}\,\mathrm{d}\nu\right|\\ &\qquad\le C\left\{n^{-1/5} + |v|n^{-1/5}\log(2+n)\right\}.\end{aligned} \tag{145.4}$$

No second moment of an unbounded stopped sum on the exceptional set is used. In particular the estimate is valid for complex insertions.

Apply Lemma 145.1 at m_n with $t = v\sqrt{m_n/n}$. These frequencies satisfy its restrictions after changing the uniform constant c. The covariance of the deterministic sum is $(m_n/n)\Gamma_R$, which differs from $D_R = c_*^{-1}\Gamma_R$ by $O(n^{-1})$. Its Gaussian characteristic function consequently changes by at most $C|v|^2/n$. Integrating these bounds and (145.4) uses only $\int_{|v|\le 2U}|v|^j\,\mathrm{d}v = C_jU^{4+j}$, for $j \ge 0$. This proves (145.3). Finitely many small values of n can be absorbed by increasing the constants wherever the stated smallness conditions hold. □

145.2. **A polynomially expanding major arc.**

Theorem 145.3 (Integrated major arc of the actual return record)**.** *Let*

$$0 < \varepsilon < \frac{1}{134}, \qquad 10\varepsilon < \theta < \frac{1/2 - 7\varepsilon}{6}, \qquad r = \min\left\{\frac{\theta}{2} - 5\varepsilon, \frac{1}{2} - 6\theta - 7\varepsilon, \frac{1}{5} - 5\varepsilon\right\}.$$

Then $r > 0$ and

$$\sup_{R,a_R}\int_{|v|\le 2n^\varepsilon} |C^a_{n,R}(v) - \alpha_R e^{-v^{\mathsf T}D_R v/2}|\,\mathrm{d}v \le Cn^{-r}\log(2+n). \tag{145.5}$$

In particular, with $\varepsilon = 1/200$ and $\theta = 1/14$,

$$\sup_{R,a_R}\int_{|v|\le 2n^{1/200}} |C^a_{n,R}(v) - \alpha_R e^{-v^{\mathsf T}D_R v/2}|\,\mathrm{d}v \le Cn^{-3/280}\sqrt{\log(2+n)}. \tag{145.6}$$

These estimates hold for $n \geq 2$. They require only the positive semidefinite covariance already established, not positive definiteness.

Proof. Take $U = n^{\varepsilon}$ and $\delta = \frac{1}{4}n^{-\theta}$ in Proposition 145.2. Its analytic restrictions hold for all sufficiently large n: in particular $3\varepsilon + 6\theta < 1/2$ follows from $7\varepsilon + 6\theta < 1/2$, and $\varepsilon + 2\theta < 1/2$ follows as well. Constants in the radius of the analytic ball do not affect these strict inequalities.

The three potentially slow terms in (145.3) are, respectively,

$$n^{-(\theta/2-5\varepsilon)}\sqrt{\log(2+n)}, \qquad n^{-(1/2-6\theta-7\varepsilon)}, \qquad n^{-(1/5-5\varepsilon)}\log(2+n).$$

All other polynomial terms decay at least as fast as one of these; this follows by subtracting their exponents, using $\theta > 10\varepsilon$. For the rounding term also note that $1 - 6\varepsilon > 1/5 - 5\varepsilon$ in the indicated range. The exponentially decaying term is smaller than every fixed negative power. The interval for θ is nonempty precisely when $67\varepsilon < 1/2$, proving (145.5).

At the displayed explicit choice the three margins are

$$\frac{3}{280}, \qquad \frac{51}{1400}, \qquad \frac{7}{40}.$$

Only the first is minimal, and that term carries a square root of the logarithm. The strict excess of the other margins absorbs their logarithmic factors. Increasing the constant handles the finite initial range, since both characteristic functions are bounded uniformly by a constant depending on B. This proves (145.6). □

145.3. **Insertion into exact raw inversion.** Define the finite complex measure $\mu^a_{n,R}$ by pushing forward $a_R\nu$ under $J_{n,R}$, and write $\Phi^a_{n,R}$ for its transform. Take $a_n = n^{1/200}$ in the cutoff and kernel of (144.1), and use $m_R = \bar{G}_R$ in the central windows. The preceding theorem supplies a proved replacement for the spectral small-frequency hypothesis in the following inversion route.

Corollary 145.4 (Raw inversion from the proved central band)**.** *Use the uniform bound $D_R \geq d_0 I_4$ of Theorem 150.2. For an admissible family of insertions, let $E^a_{n,R}$ be finite complex measures with mixed densities $e^a_{n,R}$ and put $H^a_{n,R} = \Phi^a_{n,R} - \widehat{E}^a_{n,R}$. Assume that $H^a_{n,R} \in L^1(\mathbb{T}^3 \times \mathbb{R})$ and, uniformly in the family,*

$$\int |1 - \chi_n(\omega)|\,|H^a_{n,R}(\omega)|\,\mathrm{d}\omega = o(n^{-2}). \tag{145.7}$$

Assume also the two local edge conditions (144.3) *with e, E replaced by e^a, E^a. Then $\mu^a_{n,R}$ has a mixed density $p^a_{n,R}$, and for each fixed K,*

$$\operatorname*{ess\,sup}_{|z-n\bar{G}_R|\leq K\sqrt{n}} \left| n^2 p^a_{n,R}(z) - \frac{\alpha_R}{(2\pi)^2\sqrt{\det D_R}} \exp\left(-\tfrac{1}{2}\xi^{\mathsf{T}} D_R^{-1}\xi\right)\right| \longrightarrow 0, \quad \xi = \frac{z - n\bar{G}_R}{\sqrt{n}}, \tag{145.8}$$

uniformly in R and the admissible family. No fixed-neighborhood spectral expansion of an induced operator is assumed in this corollary.

Proof. Fourier inversion for H^a identifies a continuous density for $\mu^a - E^a$. Add e^a and use the exact decomposition (144.4), which is linear and remains valid for complex measures. After $v = \sqrt{n}\,\omega$, the difference between $n^2 K_n * \mu^a$ and the inverse Gaussian integral with cutoff $\chi(v/n^{1/200})$ is bounded, independently of z, by $(2\pi)^{-4}\|\chi\|_\infty$ times the integral in (145.6). Positive definiteness bounds the removed Gaussian tail by $C\int_{|v|>n^{1/200}} e^{-d_0|v|^2/2}\,\mathrm{d}v$, which tends to zero uniformly. This proves the main term in (145.8). The other two terms of the exact decomposition vanish on the stated scale by the assumed complementary tail and local edge conditions. These steps prove the raw density conclusion, not a conclusion only for a convolved observation. □

Remark 145.5 (Location and scope of the remaining estimates). The physical radius of the proved central band is $n^{-1/2+1/200}$. It is smaller than $n^{-2/5}$. Thus (145.7) is not a consequence of the outer-tail hypothesis in Theorem 143.1: the intervening annulus also needs an estimate. The present theorem removes the need for an assumed induced spectral expansion on the central band only. It does not assert a resolvent estimate on its complement or the full physical branch decomposition. Uniform positive definiteness is proved independently in Theorem 150.2. Both inversion routes are retained, with their distinct frequency boundaries.

The insertion class is an initial-state class with a uniform norm on the collision cylinder. Terminal insertions, multiple-time insertions, and exact-event indicators need not belong to it. In particular (145.6) does not justify a change of exact conditioning event or provide the weighted raw tail required for such a change. The count truncation and the initial-coordinate variation bounds continue to concern different objects from the derivative sums of the raw residual densities.

146. Central integrals marked at an arbitrary actual return

A weight at a terminal or intermediate return cannot simply be smoothed as a function of the initial state: composition with a long first-return iterate need not preserve a uniform variation bound. Instead we change the origin of the orbit to the marked return. The record then has a backward and a forward part, while the weight is a function of the new origin. Both parts can be estimated with the same forward collision operator. This argument does not require invariance of the section under time reversal, or mixing of the induced map.

All measures and normalizations in this section are as follows. The measure ν is the probability on the full collision cylinder; $\nu(Y_R^*) = c_*$, and $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$. For a complex function a that is zero almost everywhere outside Y_R^* put

$$M_a = \|a\|_\infty, \qquad V_a = \|a\|_{\mathrm{BV}}, \qquad \alpha_a = \int_M a \,\mathrm{d}\nu. \tag{146.1}$$

For complex functions the variation norm is the sum of the norms of the real and imaginary parts. It includes their L^1 norms and is taken in the original collision coordinates, including the section boundary. No upper bound for V_a independent of n is imposed below.

For $0 \le k \le n$ define the finite complex measure and its centered, rescaled transform by

$$\mu_{n,R}^{[k],a}(B) = \int_{Y_R^*} a((F_R^*)^k x)\mathbf{1}_{\{J_{n,R}(x)\in B\}} \,\mathrm{d}\nu(x), \tag{146.2}$$

$$C_{n,R}^{[k],a}(v) = \int_{Y_R^*} a((F_R^*)^k x) e^{iv\cdot(J_{n,R}(x)-n\bar{G}_R)/\sqrt{n}} \,\mathrm{d}\nu(x). \tag{146.3}$$

Its mass is α_a, which may be complex. The endpoints $k = 0, n$ are the initial and terminal insertions. Taking $a = s_R$ gives the original probability for every k. The physical frequency is always $\omega = v/\sqrt{n}$.

146.1. **Changing the origin to the mark.** Let $N_{j,R}^-(y)$ be the collision distance to the jth preceding section visit, so that

$$(F_R^*)^{-j} y = T_R^{-N_{j,R}^-(y)} y, \qquad N_{0,R}^- = 0.$$

The forward distance is $N_{j,R}^+ = N_{j,R}$, with $N_{0,R}^+ = 0$. These distances are finite almost everywhere, by recurrence and invertibility. For nonnegative integers r, s write

$$\mathsf{H}_{r,s,R}(y) = \sum_{j=-r}^{s-1} h_R(T_R^j y), \tag{146.4}$$

where a sum over an empty interval is zero.

Lemma 146.1 (Exact compensation about a marked return)**.** *For almost every $y \in Y_R^*$,*

$$J_{n,R}((F_R^*)^{-k}y) - n\bar{G}_R = \mathsf{H}_{N_{k,R}^-(y),N_{n-k,R}^+(y),R}(y). \tag{146.5}$$

Consequently

$$C_{n,R}^{[k],a}(v) = \int_{Y_R^*} a(y) e^{iv\cdot \mathsf{H}_{N_{k,R}^-(y),N_{n-k,R}^+(y),R}(y)/\sqrt{n}}\,\mathrm{d}\nu(y). \tag{146.6}$$

Proof. The collision interval from $(F_R^*)^{-k}y$ up to, but excluding, $(F_R^*)^{n-k}y$ has left endpoint $-N_{k,R}^-(y)$ and right endpoint $N_{n-k,R}^+(y)$ in coordinates based at y. The sum of f_R on this interval is precisely the original four-coordinate record. Its negative part contains exactly k section visits, including the left endpoint and excluding zero. Its nonnegative part contains exactly $n-k$ visits, including zero if that part is nonempty and excluding the right endpoint. Thus the sum of η_R over the entire interval is n. Substitute $h_R = f_R - \bar{G}_R\eta_R$. This proves the first identity, including $k = 0, n$. The actual first-return map preserves $\nu|_{Y_R^*}$, since it preserves its normalization ν_R^*. The change of variable $y = (F_R^*)^k x$ gives the second identity. Only additivity and invariance are used; the two parts are not asserted independent. □

Lemma 146.2 (Forward and backward return-clock windows)**.** *For $j \geq 0$, $b \geq 2$, and $m_j = \lfloor j/c_* \rfloor$, uniformly in R,*

$$\nu_R^*\{|N_{j,R}^{\pm} - m_j| > b\} \leq C\frac{j+b}{b^2}. \tag{146.7}$$

Moreover the collision maximal estimate of Lemma 139.4 holds for an interval beginning at any integer collision time, positive or negative, and in either order.

Proof. The case $j = 0$ is immediate. For either sign let $H_L^{\pm} = \sum_{i=1}^{L} \eta_R \circ T_R^{\pm i}$. Its mean under ν is c_*L. The variance is at most CL by Proposition 138.3. For the negative sign the scalar lag correlation is the same as for the positive sign by invariance. Domination $\nu_R^* \leq c_*^{-1}\nu$ gives

$$\int |H_L^{\pm} - c_*L|^2\,\mathrm{d}\nu_R^* \leq CL.$$

The equivalences $N_j^{\pm} > L \iff H_L^{\pm} < j$ and $N_j^{\pm} \leq L \iff H_L^{\pm} \geq j$ hold for the actual visits. Take integer thresholds within a fixed distance of $m_j + b$ and $m_j - b$. When the latter threshold is negative, its lower-tail event is empty. Otherwise the required deviation from c_*L is at least $c_*b - O(1)$, while $L \leq C(j+b)$. Chebyshev's inequality proves the estimate for b above a fixed constant. For the remaining bounded range enlarge C so that its right-hand side is at least one. In particular no assumption $b \leq j$ is needed.

For the maximal assertion, translate any deterministic integer interval to a nonnegative one using invariance of ν. Every subinterval still has second moment bounded by its length, from the absolutely summable collision correlations. The same dyadic prefix proof applies in the reverse order. Domination gives its version under ν_R^* as well. The assertion concerns fixed intervals, not a conditional distribution of the past given the future. □

Proposition 146.3 (Uniform stopping error about the mark)**.** *For all $0 \leq k \leq n$, $n \geq 2$, and $v \in \mathbb{R}^4$,*

$$\left| C_{n,R}^{[k],a}(v) - \int_M a(y) e^{iv\cdot \mathsf{H}_{m_k,m_{n-k},R}(y)/\sqrt{n}}\,\mathrm{d}\nu(y) \right| \leq CM_a\left(n^{-1/5} + |v|n^{-1/5}\log(2+n)\right). \tag{146.8}$$

Proof. Use (146.6) and let $b_n = \lceil n^{3/5} \rceil$. By Lemma 146.2, the union of the two events on which a clock differs from its corresponding m_j by more than b_n has probability at most $Cn^{-1/5}$ under ν_R^*. Its contribution to the difference of characteristic functions is at most $CM_a n^{-1/5}$, since $|a|\nu \le M_a c_* \nu_R^*$ on the section.

On the complement, the difference of the two collision sums is bounded by the sum of two maxima on deterministic windows of length at most $2b_n + 2$ around their respective endpoints. The preceding maximal estimate bounds its squared integral under ν by $Cb_n[\log(2 + b_n)]^2$. It remains valid when a window crosses collision time zero. Cauchy–Schwarz and $|e^{ix} - e^{iy}| \le |x - y|$ give $CM_a|v|\sqrt{b_n/n}\log(2 + b_n)$, which is the second term in (146.8). Nothing is estimated in an unbounded stopped sum on the exceptional event. Finitely many initial values of n are included by increasing the constant. □

146.2. **A collision multiplier between two spectral powers.** Set $a_\delta = \mathsf{S}_\delta a$ and $h_{R,\delta} = \mathsf{S}_\delta h_R$. The former need not be supported in the section. This is harmless:

$$\|a - a_\delta\|_1 \le C\delta V_a, \qquad \|a_\delta\|_\infty \le M_a, \qquad \|M_{a_\delta}\|_{\mathcal{B}_j \to \mathcal{B}_j} \le CM_a\delta^{-2}. \tag{146.9}$$

The last bound is Lemma A.4. In particular its constant depends on the supremum norm, not on V_a.

Lemma 146.4 (A marked deterministic collision integral)**.** *Fix $c_0, C_0 > 0$. Suppose $c_0 n \le r + s \le C_0 n$ with $r, s \ge 0$, and let $q \ge 1$ be an integer satisfying $2q \le r + s$. There are $c, C > 0$ and $\rho < 1$ such that, whenever*

$$|v|/\sqrt{n} \le c\delta^2, \qquad \delta^{-6}|v|^3/\sqrt{n} \le c,$$

the following bound holds, with $\ell_\delta = 1 + |\log\delta|$:

$$\begin{aligned}&\left|\int_M a\, e^{iv\cdot \mathsf{H}_{r,s,R}/\sqrt{n}}\,\mathrm{d}\nu - \alpha_a e^{-\frac{r+s}{2n} v^{\mathsf{T}}\Gamma_R v}\right| \\ &\quad \le C\delta V_a + CM_a\left\{|v|\sqrt{\delta\ell_\delta} + |v|^2\delta\ell_\delta + \frac{|v|\delta^{-4} + |v|^3\delta^{-6}}{\sqrt{n}} + \delta^{-2}\rho^q + \frac{|v|q}{\sqrt{n}} + \frac{|v|^2 q}{n}\right\}.\end{aligned} \tag{146.10}$$

All constants are independent of the location of the mark and of a.

Proof. First replace a by a_δ. The cost is at most $C\delta V_a$. Let $u_\delta = h_R - h_{R,\delta}$. Stationarity and (138.5) give

$$\int \left|\sum_{i=-r}^{s-1} u_\delta(T_R^i y)\right|^2 \mathrm{d}\nu(y) \le C(r + s)\delta\ell_\delta.$$

The bounded weight $|a_\delta| \le M_a$ and Cauchy–Schwarz therefore bound the cost of replacing h_R by $h_{R,\delta}$ by $CM_a|v|\sqrt{\delta\ell_\delta}$.

Put $z = v/\sqrt{n}$ and $L_z = \mathcal{L}_{R,\delta,z}$, using one of the fixed local collision spaces. Its mass functional will be denoted by ℓ to avoid confusion with block indices. Invariance and the transfer convention give the exact chronological pairing

$$\int_M a_\delta(y) e^{iz\cdot\sum_{i=-r}^{s-1} h_{R,\delta}(T_R^i y)}\,\mathrm{d}\nu(y) = \ell\big(L_z^s M_{a_\delta} L_z^r \nu\big). \tag{146.11}$$

Indeed, starting at $x = T_R^{-r} y$, the rightmost power records collisions $0, \dots, r - 1$, the multiplier is evaluated at $T_R^r x = y$, and the leftmost power records collisions $r, \dots, r + s - 1$. Thus all weights are evaluated at the prescribed states.

Suppose first that $r, s \ge q$. Write $L_z^m = \lambda(z)^m \Pi(z) + E_m(z)$ with $\|E_m(z)\| \le C\rho^m$. The hypotheses and (139.1) bound each real-frequency principal power of length at most $r + s$ by a uniform constant. Expanding (146.11), every complementary term is bounded by $CM_a\delta^{-2}(\rho^r + \rho^s)$;

the term with both complementary powers is included in this bound. For the principal amplitude use $\Pi(0) = \nu\ell$ and (146.9):

$$\ell(\Pi(0)M_{a_\delta}\Pi(0)\nu) = \alpha_a,$$
$$|\ell(\Pi(z)M_{a_\delta}\Pi(z)\nu) - \alpha_a| \le CM_a\delta^{-4}|z|.$$

The cubic remainder accumulated over $r+s$ collisions is at most $C\delta^{-6}|v|^3/\sqrt{n}$. Since the quadratic form is nonnegative, split the principal comparison as $\lambda(z)^{r+s}(A(z)-\alpha_a)+\alpha_a(\lambda(z)^{r+s}-e^{-(r+s)z^{\mathsf T}\Gamma_{R,\delta}z/2})$, where $A(z)$ is its principal amplitude. The first term uses the amplitude bound above; the second uses $|\alpha_a| \le M_a$ and $|e^w - 1| \le |w|e^{|w|}$. Thus the cubic error is multiplied only by CM_a, not by the multiplier norm. This proves the required spectral bound in the case of two long blocks.

If $r < q$, delete the first r terms from the real exponent on the left of (146.11). The cost is at most $CM_a|v|q/\sqrt{n}$, by boundedness of $h_{R,\delta}$. The remaining pairing is $\ell(L_z^s(a_\delta\nu))$. Its principal amplitude differs from α_a by at most $CM_a\delta^{-4}|z|$, by the smooth-density embedding, and its complementary term is bounded by $CM_a\delta^{-2}\rho^s$. Here $s \ge q$. If $s < q$, the remaining pairing is instead $\ell(M_{a_\delta}L_z^r\nu)$, with exactly the same bounds. At most one of r, s is less than q. Restoring the removed Gaussian time coefficient changes its value by at most $CM_a|v|^2q/n$. This handles zero blocks as well; no spectral expansion of an identity operator is used.

Finally replace $\Gamma_{R,\delta}$ by Γ_R. The segment joining these matrices is positive semidefinite, and (138.7) bounds the change by $CM_a|v|^2\delta\ell_\delta$. All the displayed errors combine to (146.10). Only forward collision operators occurred in the proof. □

146.3. **The growing band and norm growth.**

Theorem 146.5 (Integrated central band with an arbitrary return mark)**.** *There is a constant C such that for every $n \ge 2$, $R \in I$, $0 \le k \le n$, and complex a as in* (146.1)*,*

$$\int_{|v|\le 2n^{1/200}} \left|C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf T}D_Rv/2}\right| \mathrm{d}v \le C\left\{M_a n^{-3/280}\sqrt{\log(2+n)} + V_a n^{-9/175}\right\}. \tag{146.12}$$

In particular the error tends to zero uniformly for bounded M_a and $V_a = O(n^\kappa)$ with any $\kappa < 9/175$.

Proof. Take $r = m_k$ and $s = m_{n-k}$, so

$$(r+s)/n = c_*^{-1} + O(n^{-1}),$$

with an error independent of k. They satisfy the length bounds in Lemma 146.4. Choose

$$U_n = n^{1/200}, \qquad \delta_n = \tfrac{1}{4}n^{-1/14}, \qquad q_n = \left\lceil \left(1 + \frac{4}{|\log\rho|}\right)\log(2+n)\right\rceil. \tag{146.13}$$

Then $\rho^{q_n} \le (2+n)^{-4}$ and $2q_n \le r+s$ for sufficiently large n, uniformly in the mark. On $|v| \le 2U_n$, both analytic restrictions hold for large n: their positive margins are $1/2 - 2/14 - 1/200$ and $1/2 - 6/14 - 3/200$.

Apply the marked spectral estimate and Proposition 146.3. Replacing $(r+s)\Gamma_R/n$ by D_R costs at most $CM_a|v|^2/n$. Integration uses $\int_{|v|\le 2U}|v|^j\,\mathrm{d}v = C_jU^{4+j}$. The contribution involving V_a is only

$$CV_a\delta_nU_n^4 = C'V_an^{-9/175}.$$

For clarity, the negative-power margins of the remaining polynomial terms are displayed below; logarithmic factors are recorded separately.

Source	Margin	Logarithmic factor
Observable smoothing	3/280	$\sqrt{\log(2+n)}$
Covariance smoothing	29/700	$\log(2+n)$
Principal amplitude	53/280	1
Cubic remainder	51/1400	1
Clock exception	9/50	1
Clock window	7/40	$\log(2+n)$
Short collision block	19/40	$\log(2+n)$
Time coefficient and rounding	97/100	$\log(2+n)$

The complementary-power contribution is at most $CM_a n^{2/14+4/200}(2+n)^{-4}$ and is smaller than the displayed polynomial errors. The first margin in the table is strictly smaller than all the others, so it absorbs their logarithmic factors. This proves (146.12) for large n. For the remaining finite range, both transforms have modulus at most M_a, and the frequency ball has finite volume; enlargement of C proves the stated estimate for all $n \geq 2$. The last assertion follows by substitution, without changing the smoothing scale with the insertion. □

Corollary 146.6 (Conditioning on a state at a marked return)**.** *Let $E_{n,R} \subset Y_R^*$ have positive measure and a zero-extended indicator in* $\mathrm{BV}(M)$*. Set $V_{n,R} = \|\mathbf{1}_{E_{n,R}}\|_{\mathrm{BV}}$. Under the section probability, condition on the exact event*

$$A_{n,k,R} = \{x : (F_R^*)^k x \in E_{n,R}\}$$

Uniformly for $0 \leq k \leq n$,

$$\int_{|v|\leq 2n^{1/200}} \left| \mathbb{E}_{\nu_R^*}\left[e^{iv\cdot(J_{n,R}-n\bar{G}_R)/\sqrt{n}} \mid A_{n,k,R} \right] - e^{-v^{\mathsf{T}} D_R v/2} \right| \mathrm{d}v$$
$$\leq \frac{C}{\nu(E_{n,R})} \left(n^{-3/280}\sqrt{\log(2+n)} + V_{n,R} n^{-9/175} \right). \tag{146.14}$$

For example this tends to zero if, uniformly in R, $\nu_R^*(E_{n,R}) \geq cn^{-\beta}$ and $V_{n,R} \leq Cn^{\kappa}$, where $\beta < 3/280$ and $\beta + \kappa < 9/175$.

Proof. Invariance gives $\nu_R^*(A_{n,k,R}) = \nu(E_{n,R})/c_*$. The conditional characteristic function is exactly $C_{n,R}^{[k],\mathbf{1}_{E_{n,R}}}/\nu(E_{n,R})$. Apply Theorem 146.5 with $M_a = 1$. The two strict exponent inequalities absorb the logarithm. This argument uses the same event throughout and makes no comparison with another event. □

Relation to the raw density theorem. These estimates control the central integral of the actual complex measures (146.2). The exact local decomposition (144.4) also applies to these measures by linearity. More explicitly, use $D_R \geq d_0 I_4$ from Theorem 150.2; suppose they admit extracted measures $E_{n,R}^{[k],a}$ with mixed densities; and suppose the residual transform $H_{n,R}^{[k],a}$ is in $L^1(\mathbb{T}^3 \times \mathbb{R})$, satisfies the complementary-tail condition (145.7), and satisfies the local edge conditions (144.3), all uniformly in the chosen marks and insertions. If the right-hand side of (146.12) tends to zero uniformly, the proof of Corollary 145.4 gives its raw density conclusion for these marked measures. Thus its central hypothesis is now proved also for terminal and single intermediate state insertions. The other hypotheses have not been proved by the recentering argument.

The band still has physical radius $n^{-99/200}$, not $n^{-2/5}$. Multiple separated return-state factors are not one insertion at a single mark, and composing them with a long induced iterate does not give the variation norm used here. Likewise an exact final lattice or flight-time constraint is a condition on the record, not generally a collision-state set $E_{n,R}$ of the class in Corollary 146.6. Its relative error bound is consequently not a replacement for the weighted raw local limits needed

for those constraints. Nor does a two-sided orbit identity provide a pointwise representative of the zero-variance coboundary. Covariance nondegeneracy is now proved in Theorem 150.2. The complete complementary integral and full critical/singular residual estimates retain their separate roles.

147. Unsmoothed moments and the covariance of actual returns

The covariance of a weak Gaussian limit need not be the limit of the normalized second moments. We establish that identification for the actual return record, including a bounded-variation insertion at any return. The argument also gives fourth moments for the unsmoothed collision sums and improves the two-sided stopping estimate. It uses only the collision splitting, not decay of correlations for F_R^*.

Write $\|u\|_{\mathcal{V}} = \|u\|_\infty + \|u\|_{\mathrm{BV}}$. All expectations without a subscript in this section are with respect to the collision probability ν. Constants are uniform in $R \in I$. The smoothing operators preserve means and have the bounds in Lemma 138.2.

147.1. **Decoupling at a deterministic collision gap.**

Lemma 147.1 (A finite-product decoupling estimate)**.** *For $2 \le q \le 4$, integers $t_1 \le \cdots \le t_q$, and $1 \le p < q$, bounded-variation functions $u_1, \dots, u_q$ satisfy*

$$\left| \mathbb{E} \prod_{j=1}^{q} u_j \circ T_R^{t_j} - \left(\mathbb{E} \prod_{j=1}^{p} u_j \circ T_R^{t_j} \right) \left(\mathbb{E} \prod_{j=p+1}^{q} u_j \circ T_R^{t_j} \right) \right| \le C \exp\{-\gamma(t_{p+1} - t_p)\} \prod_{j=1}^{q} \|u_j\|_{\mathcal{V}}. \tag{147.1}$$

The times may be negative or repeated. The constants $C, \gamma > 0$ can be chosen for all the indicated values of q and p.

Proof. By multilinearity assume $\|u_j\|_{\mathcal{V}} \le 1$; a zero norm makes the assertion immediate. Replace every u_j by $u_{j,\epsilon} = \mathsf{S}_\epsilon u_j$. Telescoping the products, invariance of ν, and the uniform supremum bounds show that the change in the left-hand expression is at most $C_q \epsilon$, independently of all the times. In particular, no variation norm of $u_j \circ T_R^{t_j}$ is estimated.

Translate t_1 to zero. Put $d_j = t_{j+1} - t_j$ and use the transfer convention of Lemma 133.1. The smoothed joint expectation is

$$\ell\big(M_{u_{q,\epsilon}} \mathcal{L}_R^{d_{q-1}} M_{u_{q-1,\epsilon}} \cdots \mathcal{L}_R^{d_1} M_{u_{1,\epsilon}} \nu\big).$$

Replace the power across the chosen gap by $\Pi_R = \nu\ell$. The resulting expression is exactly the product of the two smoothed expectations in (147.1). The difference is bounded by $C_q \epsilon^{-2q} \vartheta^{d_p}$. Indeed each smooth multiplier costs at most $C\epsilon^{-2}$, every other power is uniformly bounded, and the complementary power across the gap costs $C\vartheta^{d_p}$. For a gap of length zero use $\mathcal{L}_R^0 - \Pi_R = I - \Pi_R$, with its uniform bound.

Choose $\epsilon = \frac{1}{4}\vartheta^{d_p/(2q+1)}$. Smoothing and operator errors are at most $C_q \vartheta^{d_p/(2q+1)}$. The finite parameter cover permits $\gamma = -\log\vartheta/9$ after changing C. Invariance justifies the initial translation also for negative times. □

Proposition 147.2 (Fourth moments without smoothing)**.** *Let u be real, centered, and $\|u\|_{\mathcal{V}} \le B$. For every deterministic integer interval of length $m \ge 1$,*

$$\mathbb{E} \left| \sum_{j=r}^{r+m-1} u \circ T_R^j \right|^4 \le C_B m^2, \qquad \mathbb{E} \max_{0 \le l \le m} \left| \sum_{j=r}^{r+l-1} u \circ T_R^j \right|^4 \le C_B m^2. \tag{147.2}$$

Both estimates hold in the reverse order of the interval and under any nonnegative initial density bounded by a fixed constant. They hold componentwise, and hence in Euclidean norm, for h_R.

Proof. Put $c_u(a) = \mathbb{E}[u(u \circ T_R^a)]$. For $a, b, c \geq 0$, apply Lemma 147.1 to four copies of u at times $0, a, a+b, a+b+c$. Splitting across the middle gap bounds

$$\left|\mathbb{E}[u(u \circ T_R^a)(u \circ T_R^{a+b})(u \circ T_R^{a+b+c})] - c_u(a)c_u(c)\right| \leq C_B e^{-\gamma b}.$$

Splitting off the first or last centered factor bounds the fourfold expectation by $C_B e^{-\gamma a}$ and $C_B e^{-\gamma c}$, respectively. Also $|c_u(a)c_u(c)| \leq C_B e^{-\gamma(a+c)}$. Taking the minimum of these three bounds gives

$$\left|\mathbb{E}[u(u \circ T_R^a)(u \circ T_R^{a+b})(u \circ T_R^{a+b+c})] - c_u(a)c_u(c)\right| \leq C_B e^{-\gamma \max\{a,b,c\}}. \tag{147.3}$$

This includes repeated times.

Expand the fourth power and order its four indices; their multiplicity is at most 24. For each initial index the sum of $e^{-\gamma(a+c)}$ over the two outer gaps is bounded, while the middle gap has at most m choices. This contributes $O(m^2)$. The sum over all three nonnegative gaps of $e^{-\gamma \max\{a,b,c\}}$ is finite, so the remaining contribution is $O(m)$. This proves the first bound.

For the second enlarge m to a power of two $M < 2m$. Every prefix is a disjoint union of at most one dyadic interval of each length. Consequently its absolute sum is bounded by the sum, over scales, of the maximum absolute interval sum at that scale. At interval length l, the fourth power of this maximum has expectation at most $C_B(M/l)l^2 = C_B M l$. Minkowski's inequality therefore gives an L^4 norm at most

$$C_B \sum_{j=0}^{\log_2 M} (M^2 2^{-j})^{1/4} \leq C_B M^{1/2}.$$

There is no logarithmic loss. Stationarity translates deterministic intervals; reversal gives suffixes of the same interval. Domination by a bounded initial density proves the remaining assertions. □

147.2. **Fourth-moment windows and two-sided stopping.** Retain $N_{j,R}^{\pm}$, $m_j = \lfloor j/c_* \rfloor$ and $\mathsf{H}_{r,s,R}$ from Section 146. Define

$$Z_{n,k,R} = \mathsf{H}_{N_{k,R}^-, N_{n-k,R}^+, R}, \qquad W_{n,k,R} = \mathsf{H}_{m_k, m_{n-k}, R}. \tag{147.4}$$

Thus $Z_{n,k,R}(y) = J_{n,R}((F_R^*)^{-k}y) - n\bar{G}_R$ almost everywhere. Zero-length backward and forward intervals are empty.

Lemma 147.3 (Fourth-moment return-clock window)**.** *For $j \geq 0$, $b \geq 2$, and either sign,*

$$\nu_R^*\{|N_{j,R}^{\pm} - m_j| > b\} \leq C\frac{(j+b)^2}{b^4}. \tag{147.5}$$

Proof. The case $j = 0$ is immediate. The centered visit count $\sum_{i=1}^{L}(\eta_R - c_*) \circ T_R^{\pm i}$ has fourth moment at most CL^2 under ν_R^*, by Proposition 147.2. The exact visit equivalences in Lemma 146.2 turn the upper and lower return-clock events into deviations of these counts. At integer thresholds within a bounded distance of $m_j \pm b$, the required deviation is at least $c_* b - O(1)$ and $L \leq C(j+b)$. Markov's inequality proves (147.5). A negative lower threshold contributes no event. Increase C for bounded b. □

Theorem 147.4 (Moment control of the actual marked stopping)**.** *Uniformly for $n \ge 2$, $0 \le k \le n$, and $R \in I$,*

$$\int (|Z_{n,k,R}|^4 + |W_{n,k,R}|^4)\,\mathrm{d}\nu_R^* \le Cn^2, \tag{147.6}$$

$$\left\| \frac{Z_{n,k,R} - W_{n,k,R}}{\sqrt{n}} \right\|_{L^2(\nu_R^*)} \le Cn^{-1/6}. \tag{147.7}$$

For every real $v \in \mathbb{R}^4$ and insertion a of (146.1)*, the stopping comparison also satisfies*

$$\left| C_{n,R}^{[k],a}(v) - \int_M a e^{iv\cdot W_{n,k,R}/\sqrt{n}}\,\mathrm{d}\nu \right| \le CM_a(1+|v|)n^{-2/9}. \tag{147.8}$$

Proof. The deterministic interval has length at most Cn, so its fourth moment follows from Proposition 147.2 and $\nu_R^* \le c_*^{-1}\nu$. For the random interval put $Q = N_{k,R}^- + N_{n-k,R}^+$. Invariance of the actual return map gives $N_{j,R}^- \overset{\text{law}}{=} N_{j,R}^+$ under ν_R^*: use $N_{j,R}^-((F_R^*)^j x) = N_{j,R}^+(x)$. Hence Theorem 134.1 and a union bound imply

$$\nu_R^*(Q > L) \le Ce^{an - cL/2}.$$

No independence of the two clocks is used. Fix $A > 2a/c + 2$. On $Q \le An$, $|Z_{n,k,R}|$ is bounded by the sum of the backward and forward collision maxima up to $\lceil An \rceil$. Their fourth moments are $O(n^2)$. On $Q > An$, boundedness of h_R gives $|Z_{n,k,R}| \le CQ$. Integrating the displayed exponential tail bounds that fourth-moment contribution by $C(1+n)^4 e^{-c'n}$ for some $c' > 0$. This proves (147.6).

Let $2 \le b \le n$ and let $\mathcal{E}_b$ be the union of the two clock-window exceptions. Lemma 147.3 gives $\nu_R^*(\mathcal{E}_b) \le Cn^2/b^4$. Off this event, the difference $Z_{n,k,R} - W_{n,k,R}$ is bounded by maxima on two deterministic collision windows of length $O(b)$, including windows that cross zero. Their second moments are $O(b)$ by the fourth-moment maximal estimate. On the exceptional event, Cauchy–Schwarz and (147.6) give

$$\int |Z_{n,k,R} - W_{n,k,R}|^2 \mathbf{1}_{\mathcal{E}_b}\,\mathrm{d}\nu_R^* \le Cn\,\nu_R^*(\mathcal{E}_b)^{1/2} \le Cn^2/b^2.$$

Thus the normalized squared L^2 error is at most $C(b/n + n/b^2)$. Taking $b = \lceil n^{2/3} \rceil$ proves (147.7).

For the characteristic comparison, the exceptional event costs only $CM_a n^2/b^4$. On its complement, the same deterministic windows and $|e^{ix} - e^{iy}| \le |x - y|$ give $CM_a|v|\sqrt{b/n}$. Choose instead $b = \lceil n^{5/9} \rceil$. Both powers are $n^{-2/9}$. The recentering identity (146.6) and $|a|\nu \le c_* M_a \nu_R^*$ complete the proof. All rounding exceptions are absorbed in C. □

147.3. **Quadratic insertions and the induced covariance.** The following estimate supplies the moment identification that is not asserted by the characteristic-function proof in Remark 139.7.

Lemma 147.5 (A centered insertion in a quadratic collision sum)**.** *Let a be a complex bounded-variation function on M, and let $\alpha = \int a\,\mathrm{d}\nu$. For all $r, s \ge 0$, with $m = r + s$,*

$$\left| \int a \mathsf{H}_{r,s,R}\,\mathrm{d}\nu \right| \le C(\|a\|_\infty + \|a\|_{\mathrm{BV}}), \tag{147.9}$$

$$\left\| \int a \mathsf{H}_{r,s,R} \otimes \mathsf{H}_{r,s,R}\,\mathrm{d}\nu - \alpha m \Gamma_R \right\| \le C(\|a\|_\infty + \|a\|_{\mathrm{BV}}). \tag{147.10}$$

The constants do not depend on the location of the origin in the interval.

Proof. Center a by subtracting α. Each component of h_R is centered. The two-factor version of Lemma 147.1 bounds $\int (a-\alpha)h_R \circ T_R^i \,\mathrm{d}\nu$ by $C\|a\|_{\mathcal{V}} e^{-\gamma|i|}$. Summation proves the first assertion.

For components u, v of h_R, sort the three times $0, i, j$ and denote their two consecutive gaps by d_1, d_2. All three factors $a-\alpha, u, v$ are centered. Splitting off the first and the last factor in (147.1) gives

$$\left| \int (a-\alpha)(u \circ T_R^i)(v \circ T_R^j) \,\mathrm{d}\nu \right| \le C\|a\|_{\mathcal{V}} e^{-\gamma \max\{d_1,d_2\}}.$$

Because $\max(d_1,d_2) \ge \max(|i|,|j|)/2$, the right side is summable over all $(i,j) \in \mathbb{Z}^2$. The centered insertion therefore changes the quadratic moment of any such interval by at most $C\|a\|_{\mathcal{V}}$. Finally the exponentially summable collision covariances give

$$\left\| \int \mathsf{H}_{r,s,R} \otimes \mathsf{H}_{r,s,R} \,\mathrm{d}\nu - m\Gamma_R \right\| \le C.$$

This is the finite covariance expansion with the lag weights $m-l$; the sum of l times the absolute lag covariances is finite. Multiplying by α proves the second assertion. Real and imaginary parts handle complex a without any positivity assumption. □

Theorem 147.6 (Moments at an arbitrary actual return mark)**.** *Put $U_{n,R} = J_{n,R} - n\bar{G}_R$. For a as in* (146.1)*, uniformly for $n \ge 2$ and $0 \le k \le n$,*

$$\left| \int_{Y_R^*} a((F_R^*)^k x) \frac{U_{n,R}(x)}{\sqrt{n}} \,\mathrm{d}\nu(x) \right| \le C\left(M_a n^{-1/6} + (M_a + V_a) n^{-1/2} \right), \tag{147.11}$$

$$\left\| \int_{Y_R^*} a((F_R^*)^k x) \frac{U_{n,R}(x) \otimes U_{n,R}(x)}{n} \,\mathrm{d}\nu(x) - \alpha_a D_R \right\| \le C\left(M_a n^{-1/6} + (M_a + V_a) n^{-1} \right). \tag{147.12}$$

These are moments of the finite complex measure of (146.2)*, using $\nu|_{Y_R^*}$, not its normalization. In particular*

$$\sup_{R \in I} \left\| \frac{1}{n} \int U_{n,R} \otimes U_{n,R} \,\mathrm{d}\nu_R^* - D_R \right\| \le C n^{-1/6}. \tag{147.13}$$

Proof. Recenter at $y = (F_R^*)^k x$. The weighted moments then involve $a(y)Z_{n,k,R}(y)$ and $a(y)Z_{n,k,R}(y) \otimes Z_{n,k,R}(y)$. Theorem 147.4 changes the normalized first moment by at most $CM_a n^{-1/6}$. For the second use

$$\|z \otimes z - w \otimes w\| \le |z-w|(|z| + |w|)$$

and Cauchy–Schwarz. Equations (147.6) and (147.7) show that its normalized change is also at most $CM_a n^{-1/6}$. The factor c_* in passing from $\nu|_{Y_R^*}$ to ν_R^* is fixed.

Apply Lemma 147.5 to $W_{n,k,R} = \mathsf{H}_{m_k, m_{n-k}, R}$. Its length satisfies $m_k + m_{n-k} = n/c_* + O(1)$ uniformly in k. Since $D_R = \Gamma_R / c_*$, division by $\sqrt{n}$ or n gives (147.11) and (147.12). For (147.13) take $a = s_R = c_*^{-1}\eta_R$, whose mass is one and whose two norms are uniformly bounded. Invariance of F_R^* and the Kac mean give $\int U_{n,R} \,\mathrm{d}\nu_R^* = 0$. Thus the last display is convergence of the actual covariance matrices. □

Corollary 147.7 (The induced Green–Kubo formula with Cesaro weights)**.** *Let $g_R = G_R - \bar{G}_R$ and $C_{j,R}^* = \int g_R \otimes (g_R \circ (F_R^*)^j) \,\mathrm{d}\nu_R^*$. Then*

$$\left\| C_{0,R}^* + \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) (C_{j,R}^* + (C_{j,R}^*)^{\mathsf{T}}) - D_R \right\| \le C n^{-1/6}. \tag{147.14}$$

Proof. The exponential one-return moment makes each term finite. Expand the second moment of $\sum_{j=0}^{n-1} g_R \circ (F_R^*)^j$ and use stationarity under F_R^*. The left matrix before subtracting D_R is exactly $n^{-1} \int U_{n,R} \otimes U_{n,R} \, \mathrm{d}\nu_R^*$. Apply (147.13). □

Corollary 147.8 (Moments under the unchanged marked-state event)**.** *Use the event $A_{n,k,R}$ and variation $V_{n,R}$ of Corollary 146.6, and put $p_{n,R} = \nu_R^*(E_{n,R}) > 0$. The conditional first moment of $U_{n,R}/\sqrt{n}$ is bounded by*

$$\frac{C}{p_{n,R}} \left(n^{-1/6} + (1 + V_{n,R}) n^{-1/2} \right).$$

Its conditional second moment about the original center $n\bar{G}_R$ differs from D_R by at most

$$\frac{C}{p_{n,R}} \left(n^{-1/6} + (1 + V_{n,R}) n^{-1} \right).$$

Consequently, if $p_{n,R} \geq c n^{-\beta}$ and $V_{n,R} \leq C n^{\kappa}$, the latter difference tends to zero when $\beta < 1/6$ and $\beta + \kappa < 1$. The conditional mean tends to zero, and the conditional covariance tends to D_R, under the stronger conditions $\beta < 1/6$ and $\beta + \kappa < 1/2$.

Proof. Take $a = c_*^{-1} \mathbf{1}_{E_{n,R}}$ in Theorem 147.6. Its mass is $p_{n,R}$. Divide the two estimates by this unchanged probability. Subtracting the tensor square of the conditional mean identifies the conditional covariance. The strict exponent inequalities prove convergence. □

Role in raw local inversion. Theorems 147.4 and 147.6 concern the unsmoothed physical record, not a replacement process. They identify its covariance by actual second moments, establish uniform integrability of its squared normalized size, and strengthen the stopping part of Theorem C. In particular all the rare marked-state events in Corollary 146.6 also satisfy the new conditional moment conclusions. Cesaro convergence in (147.14) does not assert absolute summability of the induced correlations.

Positive definiteness is a distinct assertion. The estimates above give no positive lower bound for the smallest eigenvalue of D_R and no periodic-evaluable representative of its zero-variance coboundary. Theorem 150.2 supplies positive definiteness by a separate measurable phase argument, not from these moment bounds. Likewise (147.8) is not a complementary frequency estimate. The physical central radius remains $2n^{-99/200}$, or $2n^{1/200}$ after rescaling. The annulus beyond it, the complete raw critical and singular branch sum, and the weighted tails for an exact physical observation retain their roles in Theorem 143.1. No conditioning event has been replaced.

148. Nondegeneracy of the collision-count component

Corollary 148.1 (The count component is uniformly nondegenerate)**.** *Let $e_3 = (0, 0, 1, 0)$, the collision-count direction. There is a constant $d_N > 0$ such that*

$$e_3^{\mathsf{T}} D_R e_3 \geq d_N \qquad (R \in I). \tag{148.1}$$

In particular the actual normalized variance of $N_{n,R}$ is bounded below by $d_N/2$ for all sufficiently large n, uniformly in R.

Proof. First, T_R is mixing for the collision probability. The splitting in Lemma 133.1 proves mixing for smooth densities and smooth tests. Approximation in $L^2(\nu)$ and invariance extend it to arbitrary L^2 functions: the approximation error in a correlation is bounded, uniformly in its lag, by Cauchy–Schwarz. Explicitly, if f_ϵ, g_ϵ approximate f, g in $L^2(\nu)$, then, uniformly in m,

$$\left| \int \overline{f} (g \circ T_R^m) \, \mathrm{d}\nu - \int \overline{f_\epsilon} (g_\epsilon \circ T_R^m) \, \mathrm{d}\nu \right| \leq \|f - f_\epsilon\|_2 \|g\|_2 + \|f_\epsilon\|_2 \|g - g_\epsilon\|_2.$$

The product of the two means converges under the same approximation. This implication is used for each fixed R, without a rate for the nonsmooth functions below.

Suppose $e_3^{\mathsf{T}} D_R e_3 = 0$. By Theorem 139.6, there is a real $b \in L^2(\nu)$ with

$$1 - c_*^{-1}\eta_R = b - b \circ T_R \quad \text{almost everywhere.}$$

Set $q = \exp(2\pi i c_* b)$, so $|q| = 1$. Since $2\pi c_* = 91/5000$ and η_R is integer-valued,

$$q \circ T_R = e^{-2\pi i c_*} q.$$

Here $0 < c_* < 1$, so the multiplier $\lambda = e^{-2\pi i c_*}$ is not one. Invariance gives $\int q \, \mathrm{d}\nu = \lambda \int q \, \mathrm{d}\nu$, hence $\int q \, \mathrm{d}\nu = 0$. Mixing then gives $\int \overline{q}(q \circ T_R^m) \, \mathrm{d}\nu \to 0$. But this integral is exactly λ^m, whose modulus is one. This contradiction proves strict positivity at every R. The continuity of D_R and compactness of I give the uniform constant in (148.1). Finally (147.13) identifies the actual count variance and gives the asserted lower bound for large n.

The argument uses the exact integer-valued section indicator and the nonintegral section mass. It neither chooses a pointwise representative at a periodic orbit nor, by itself, proves positive definiteness in every joint direction. The latter conclusion is proved separately in Theorem 150.2. □

149. Measurable collision phases and symplectic area

The measurable coboundary in Theorem 139.6 need not have a value at a prescribed periodic orbit. We shall instead use conditional pairs on a positive-measure hyperbolic product set. The discontinuous section contribution will be removed algebraically before this argument is applied. Throughout this section R is fixed; none of the local product constants is asserted uniform in R.

Write κ_R for the one-collision lattice label, in the basis $a = (1, 0)$, $b = (1/2, \sqrt{3}/2)$. Thus $f_R = (\kappa_{R,1}, \kappa_{R,2}, 1, \tau_R)$. All identities between measurable functions are understood almost everywhere with respect to ν. Let $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$.

149.1. **A qualitative local product input.**

Lemma 149.1 (Regular product sets for the collision map)**.** *There is a compact positive-ν-measure set $\mathcal{P}$ in a regular collision coordinate rectangle, away from grazing, with the following properties. It has product coordinates $\Psi : K^u \times K^s \longrightarrow \mathcal{P}$. Write a product point as $\Psi(u, s)$. Fixing u places points on one local homogeneous stable curve; fixing s places them on one local homogeneous unstable curve. Thus K^u labels stable leaves and K^s labels unstable leaves. These curves are C^1, have transverse tangent cones, and each stable curve meets each unstable curve exactly once.*

The probability $m = \nu(\mathcal{P})^{-1}\nu|_{\mathcal{P}}$ is equivalent, in these coordinates, to a product $m^u \otimes m^s$ of nonatomic transversal probabilities. A stable arc joining two points on the same curve remains a connected regular arc under every forward collision iterate, and its length tends to zero. The corresponding statement holds for unstable arcs under backward iterates. In particular the successive lattice labels agree along the respective arcs.

For every sufficiently small $\varepsilon > 0$, the product coordinates can be restricted to positive-measure subsets so that every four-corner stable–unstable quadrilateral lies in a coordinate box of diameter less than ε. Almost every such quadrilateral with distinct coordinates is a simple, nondegenerate Jordan domain.

Proof. We use the qualitative billiard product construction in [6, Sections 8.2–8.3]. Section 8.2B gives transverse stable and unstable cones. Homogeneous local curves and their regular iterates are described in Section 8.2D and constructed in Section 8.3, Sublemma 1. The product set chosen there has positive invariant area; this is stated at the end of its construction, before the verification of (P3)–(P5). The absolute-continuity formula (P5)(b), verified there, has positive

finite Jacobian. Apply the same local statement in the reverse time direction. Since the invariant collision measure is smooth and positive away from grazing, disintegration and the two holonomy formulas identify its null sets on the product rectangle with those of the product of the transversal measures. These measures are nonatomic, being equivalent to restricted arc length on transversals. Restriction to compact positive-measure subsets gives the stated form. No assertion that $\mathcal{P}$ contains an open set is needed.

For completeness, contraction here means contraction of the actual curves, not just their endpoints in a symbolic metric. The billiard estimates of [6, Section 8.2C] give exponential contraction of the stable arcs in its adapted metric and $\mathrm{length}(\gamma) \le C\sqrt{\mathrm{length}_{\mathrm{ad}}(\gamma)}$ for a stable or unstable curve. Thus, for fixed R and the chosen product set, the ordinary lengths of these iterated arcs are bounded by $C_{\mathcal{P}}\theta_{\mathcal{P}}^{n}$ for some $C_{\mathcal{P}} < \infty$ and $0 < \theta_{\mathcal{P}} < 1$. In particular they tend to zero. No uniform choice in R is made. Passing from (α, φ) to (α, p) uses only $p = \sin\varphi$, whose derivative is bounded; no inverse change at grazing is made. Homogeneous local curves do not cross the relevant finite-iterate collision singularities. The label κ_R is locally constant on each such physical branch, which proves the label assertion.

The hypotheses of that construction hold here: the disks are disjoint with separation at least $3/50$, their boundaries are smooth with strictly positive curvature, the free domain is connected, and the horizon is finite. These are properties of the physical Euclidean table. The same local construction works on its flat triangular torus; alternatively the rectangular two-scatterer cover in Lemma 133.1 gives the local product set in a covering chart. There is no affine change of the reflection law.

Finally choose a product support point and restrict each transversal to a sufficiently short positive-measure interval around it. Continuity of the local product map and the transverse graph cones put the connecting arcs into a small coordinate box. Unique intersections of the graph curves make the four distinct corners bound a simple Jordan domain. Nonatomicity excludes repeated transversal coordinates almost everywhere. This also proves the last assertion for product sets with gaps. □

Conditional pairs in product coordinates. To fix the measure conventions, write $\mu^u = m^u$, $\mu^s = m^s$, $\mu = \mu^u \otimes \mu^s$ and

$$\mathrm{d}m_{\mathcal{P}} = \rho(u,s)\,\mathrm{d}\mu^u(u)\,\mathrm{d}\mu^s(s), \qquad m_{\mathcal{P}} = \nu(\mathcal{P})^{-1}\nu|_{\mathcal{P}}.$$

Here $0 < \rho < \infty$ almost everywhere and $\int \rho\,\mathrm{d}\mu = 1$. Put $\rho_u(u) = \int \rho(u,s)\,\mathrm{d}\mu^s(s)$ and $\rho_s(s) = \int \rho(u,s)\,\mathrm{d}\mu^u(u)$. The stable and unstable pair probabilities are explicitly

$$\mathrm{d}\mathfrak{m}_s(u,s_0,s_1) = \frac{\rho(u,s_0)\rho(u,s_1)}{\rho_u(u)}\,\mathrm{d}\mu^u(u)\,\mathrm{d}\mu^s(s_0)\,\mathrm{d}\mu^s(s_1), \tag{149.1}$$

$$\mathrm{d}\mathfrak{m}_u(u_0,u_1,s) = \frac{\rho(u_0,s)\rho(u_1,s)}{\rho_s(s)}\,\mathrm{d}\mu^u(u_0)\,\mathrm{d}\mu^u(u_1)\,\mathrm{d}\mu^s(s).$$

The denominators are positive and finite almost everywhere. Integration of either unused coordinate shows that both marginals are $m_{\mathcal{P}}$. The displayed positive finite densities also give exactly the triple-product measure equivalences used below, without a boundedness assertion for ρ or its reciprocal.

149.2. **The physical action one-form.** On the collision cylinder define

$$\vartheta_R = Rp\,\mathrm{d}\alpha, \qquad \mathrm{d}\vartheta_R = R\,\mathrm{d}p \wedge \mathrm{d}\alpha. \tag{149.2}$$

The one-form is globally defined on the cylinder and bounded in both the flat and the original collision coordinates.

Lemma 149.2 (Exact action on a regular branch)**.** *On every regular collision branch,*

$$\mathrm{d}\tau_R = T_R^{*}\vartheta_R - \vartheta_R. \tag{149.3}$$

Consequently, if an arc γ from x to y stays on regular branches through time n, then

$$S_{n,R}\tau_R(y) - S_{n,R}\tau_R(x) = \int_{T_R^n\gamma} \vartheta_R - \int_{\gamma} \vartheta_R. \tag{149.4}$$

For backward regular arcs the corresponding identity is

$$S_{n,R}\tau_R(T_R^{-n}y) - S_{n,R}\tau_R(T_R^{-n}x) = \int_{\gamma} \vartheta_R - \int_{T_R^{-n}\gamma} \vartheta_R. \tag{149.5}$$

Proof. Lift the flight to its actual initial disk and the next disk with fixed label d. Its endpoints are $q = Rn_\alpha$ and $q' = d + Rn_{\alpha'}$. If v is its unit velocity, differentiation of its Euclidean length gives $\mathrm{d}\tau_R = v \cdot \mathrm{d}q' - v \cdot \mathrm{d}q$. The first tangential component is p, and specular reflection preserves the tangential component at arrival, which is p' in the outgoing coordinates of $T_R x$. Hence $\mathrm{d}\tau_R = Rp'\,\mathrm{d}\alpha' - Rp\,\mathrm{d}\alpha$. The lattice label is constant on the branch and contributes no derivative. This proves (149.3). Summing its pullbacks and integrating over the arc proves (149.4); applying that identity to $T_R^{-n}\gamma$ proves (149.5). □

Proposition 149.3 (A measurable physical phase has zero roof frequency)**.** *Suppose $q : M \to \mathbb{S}^1$ is measurable and, for constants $u \in \mathbb{R}^2$ and $s, t \in \mathbb{R}$,*

$$q \circ T_R = \exp\{i(u \cdot \kappa_R + s + t\tau_R)\}\, q. \tag{149.6}$$

Then $t = 0$.

Proof. Discard an invariant null set so that every finite iterate of the phase identity holds at the remaining points. Use the product set and the probability m of Lemma 149.1. Disintegrate m over its stable curves and, conditionally on the curve, draw two independent points x, y. Denote this pair probability by $\mathfrak{m}_s$. Both of its marginals are exactly m; in particular they are bounded by $\nu(\mathcal{P})^{-1}\nu$.

The following argument justifies endpoint agreement for a merely measurable q. Let q_ϵ be a smooth approximation in $L^1(\nu)$, with supremum norm at most one. Invariance of ν and the marginal bound give, for every n,

$$\begin{aligned}\int |q(T_R^n y) - q(T_R^n x)|\,\mathrm{d}\mathfrak{m}_s(x,y) \\ \le \frac{2}{\nu(\mathcal{P})}\|q - q_\epsilon\|_{L^1(\nu)} + \int |q_\epsilon(T_R^n y) - q_\epsilon(T_R^n x)|\,\mathrm{d}\mathfrak{m}_s(x,y).\end{aligned} \tag{149.7}$$

For fixed ϵ, the last term tends to zero by contraction and bounded convergence. Then let ϵ tend to zero. Thus the left side tends to zero. This uses bounded marginals, not invariance of the pair probability or a claim about every stable curve.

Let $\gamma_s(x, y)$ be the stable arc oriented from x to y. Its forward labels agree at all times. Iteration of (149.6) and (149.4) gives

$$\frac{q(T_R^n y)}{q(T_R^n x)} = \exp\left(it \left[\int_{T_R^n\gamma_s(x,y)} \vartheta_R - \int_{\gamma_s(x,y)} \vartheta_R \right] \right) \frac{q(y)}{q(x)}.$$

The first arc integral tends to zero, because ϑ_R is bounded and the arc length tends to zero. Since $|q| = 1$, the left side minus one has $L^1(\mathfrak{m}_s)$ norm tending to zero by (149.7). The right side has the pointwise limit just displayed. Fatou's lemma therefore gives

$$\frac{q(y)}{q(x)} = \exp\left(it \int_{\gamma_s(x,y)} \vartheta_R \right) \quad \text{for } \mathfrak{m}_s\text{-almost every } (x, y). \tag{149.8}$$

Disintegrating instead over unstable curves and using backward iterates in (149.7) gives, by (149.5),

$$\frac{q(y)}{q(x)} = \exp\left(it \int_{\gamma_u(x,y)} \vartheta_R \right) \quad \text{for almost every unstable pair.} \tag{149.9}$$

The sign is the same in the two formulas: the backward action in (149.5) has limiting value $\int_{\gamma_u(x,y)} \vartheta_R$.

We now use the measure-class part of the local product input. The stable and unstable pair probabilities are equivalent, in product coordinates, to the corresponding three-coordinate product measures. Fubini's theorem therefore permits (149.8) and (149.9) to be multiplied around almost every four-corner product quadrilateral Q. The oriented corner order is

$$\begin{aligned} x_{00} = \Psi(u_0, s_0) \;&\longrightarrow\; x_{01} = \Psi(u_0, s_1) \;\longrightarrow\; x_{11} = \Psi(u_1, s_1), \\ x_{11} \;&\longrightarrow\; x_{10} = \Psi(u_1, s_0) \;\longrightarrow\; x_{00}. \end{aligned}$$

The edges are stable, unstable, stable with reversed s-orientation, and unstable with reversed u-orientation. This specifies the boundary orientation used in Stokes' formula; its sign need not be fixed when only the absolute area is used. Their left sides telescope, so

$$\exp\left(it \oint_{\partial Q} \vartheta_R \right) = 1. \tag{149.10}$$

This assertion is on a positive-measure family of quadrilaterals; it does not assign values to q on a preselected closed orbit.

Suppose $t \neq 0$. Restrict to a positive-measure product subrectangle small enough that every such quadrilateral has coordinate area less than $2\pi/(R|t|)$. Almost every quadrilateral is a nondegenerate Jordan domain. Stokes' formula and (149.2) give

$$0 < \left| t \oint_{\partial Q} \vartheta_R \right| = |t| R \operatorname{area}_{\alpha,p}(Q) < 2\pi,$$

contradicting (149.10). Hence $t = 0$. □

149.3. **Rotations and the constant collision phase.** Let $S_j(\alpha, p) = (\alpha + j\pi/3, p)$, $j \in \mathbb{Z}$. In lattice coordinates rotation by $\pi/3$ is

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad A^3 = -I_2, \qquad I_2 + A^2 + A^4 = 0. \tag{149.11}$$

The actual physical symmetries give

$$T_R S_j = S_j T_R, \qquad \kappa_R \circ S_j = A^j \kappa_R, \qquad \tau_R \circ S_j = \tau_R, \qquad (S_j)_* \nu = \nu. \tag{149.12}$$

They follow by rotating both endpoints, the velocity and the selected next disk. The first admissible physical root remains the first root. No invariance of Y_R^* under these rotations is asserted.

Theorem 149.4 (Rigidity of the roof and constant parts of a phase)**.** *If a measurable circle-valued q satisfies* (149.6)*, then*

$$t = 0, \qquad s \in 2\pi\mathbb{Z}. \tag{149.13}$$

Proof. The proposition gives $t = 0$. Define

$$Q_2 = q\,(q \circ S_3), \qquad Q_3 = q\,(q \circ S_2)(q \circ S_4).$$

By (149.11) and (149.12), the lattice phases cancel exactly and

$$Q_2 \circ T_R = e^{2is} Q_2, \qquad Q_3 \circ T_R = e^{3is} Q_3.$$

Both functions have modulus one. The mixing collision map has no nontrivial circle eigenvalue, by the L^2 argument in the proof of Corollary 148.1. Therefore $e^{2is} = e^{3is} = 1$, and their quotient gives $e^{is} = 1$. □

The theorem is qualitative. It excludes exact measurable physical phases with nonzero roof frequency or a nonintegral constant phase. It is not a norm estimate for approximate spectral vectors, and supplies no decay rate for a high-frequency collision or induced resolvent.

150. Uniform nondegeneracy of the joint return record

We now combine measurable phase rigidity with the exact integer-valued section compensation. The proof treats every joint direction of the same four-coordinate record. It does not pass from the scalar count variance to a determinant by an inequality.

150.1. **A finite-cover obstruction for displacement.**

Lemma 150.1 (No real spatial collision coboundary)**.** *If $w \in \mathbb{R}^2$ and $b \in L^2(\nu;\mathbb{R})$ satisfy*

$$w \cdot \kappa_R = b - b \circ T_R, \tag{150.1}$$

then $w = 0$.

Proof. Regard w as a row vector and suppose $w \neq 0$. Composition with S_1, using (149.12), gives $(wA) \cdot \kappa_R = b \circ S_1 - (b \circ S_1) \circ T_R$. The matrix with rows $w = (w_1, w_2)$ and $wA = (w_2, -w_1 + w_2)$ has determinant

$$-(w_1^2 - w_1 w_2 + w_2^2), \qquad w_1^2 - w_1 w_2 + w_2^2 \geq \tfrac{1}{2}|w|^2 > 0. \tag{150.2}$$

Solving these two real identities gives a vector-valued $H \in L^2(\nu;\mathbb{R}^2)$ with $\kappa_R = H - H \circ T_R$.

Consider the physical billiard on the flat torus $\mathbb{R}^2/(2\Lambda)$. There are four equal scatterers, labeled by $\ell \in \Lambda/(2\Lambda) \simeq (\mathbb{Z}/2\mathbb{Z})^2$. In collision coordinates its map and invariant probability are exactly

$$\widetilde{T}_R(x,\ell) = (T_R x, \ell + \kappa_R(x) \bmod 2), \qquad \widetilde{\nu} = \nu \otimes \mathrm{unif}((\mathbb{Z}/2\mathbb{Z})^2). \tag{150.3}$$

This map is ergodic. The planar lift has precisely the original disk array, so a free flight in the cover has the original uniform length bound. The enlarged torus contains finitely many pairwise disjoint embedded disks; their complement is connected. Thus the enlarged table has smooth strictly convex scatterers, connected free space and finite horizon; the finite-horizon dispersing-billiard theorem [6, Section 8.1, Theorem 6] applies. To work only with a rectangular fundamental domain, take the physical cover with periods $(2,0)$ and $(0,2\sqrt{3})$. It has eight scatterers and satisfies the same hypotheses. Its quotient by the deck group is precisely the four-scatterer table above, so its ergodicity implies that of (150.3). Neither argument changes Euclidean length or specular reflection.

However,

$$F(x,\ell) = \exp\{i\pi(H_1(x) + \ell_1)\}$$

is invariant under (150.3): the integer increment $\kappa_{R,1}$ cancels $H_1 \circ T_R - H_1$, and reduction of ℓ_1 modulo two does not alter the exponential. The function has modulus one and integral zero by averaging over ℓ_1. It cannot be almost everywhere constant, contradicting ergodicity. This proves the lemma. No value of b or H at a periodic point is used. □

150.2. **Removing the section contribution.**

Theorem 150.2 (Uniform positive definiteness)**.** *There exist constants $0 < d_0 \le D_0 < \infty$ such that*

$$d_0 I_4 \le D_R \le D_0 I_4 \qquad (R \in I). \tag{150.4}$$

Equivalently the actual L^2 coboundaries in Theorem 139.6 have only the zero joint direction. The discontinuity of $\eta_R = \mathbf{1}_{Y_R^*}$ is allowed.

Proof. Fix R and suppose $v^{\mathsf{T}} D_R v = 0$, where $v = (w, a, b)$ with $w \in \mathbb{R}^2$. The collision-clock coboundary characterization gives a real $B \in L^2(\nu)$ such that

$$w \cdot \kappa_R + a + b\tau_R - \zeta\eta_R = B - B \circ T_R, \qquad \zeta = \frac{a + b\bar{\tau}_R}{c_*}. \tag{150.5}$$

All terms are the physical single-collision quantities. We separate whether the coefficient of the section indicator vanishes.

If $\zeta \neq 0$, set $q = \exp(2\pi i B/\zeta)$. Since η_R is integer-valued, exponentiation removes it exactly and yields

$$q \circ T_R = \exp\!\left(-\frac{2\pi i}{\zeta}(w \cdot \kappa_R + a + b\tau_R)\right) q.$$

Theorem 149.4 implies $b = 0$ and $-2\pi a/\zeta \in 2\pi\mathbb{Z}$. But now $\zeta = a/c_* \neq 0$, so $a/\zeta = c_*$. Since $0 < c_* < 1$, this is impossible. Notice that the rotations are applied only after the section indicator has disappeared. Their use requires no symmetry of the actual return section.

If $\zeta = 0$, set $q = \exp(iB)$ in (150.5). Proposition 149.3 implies $b = 0$. The definition of ζ then gives $a = 0$. The remaining identity is the real spatial coboundary $w \cdot \kappa_R = B - B \circ T_R$, and Lemma 150.1 gives $w = 0$.

Thus $v = 0$ in all cases. The symmetric matrix D_R is positive definite for each R. Theorem 138.4 gives its continuity on the compact interval I. The smallest eigenvalue is consequently bounded below by a positive number, and the largest is bounded above. This proves (150.4). Uniformity is obtained at this last step, not by assuming uniform constants for measurable phase regularity or for the product set. □

Remark 150.3 (The role of the section)*.* The algebra in (150.5) uses only that the section indicator is integer-valued and that its mean lies strictly between zero and one. The two rotation products force an integral constant phase without an arithmetic assumption on this mean. The specific section geometry is used earlier to establish the bounded collision covariance and its L^2 kernel characterization. Thus the present proof neither evaluates a discontinuous section weight on stable arcs nor replaces it by a smoother section.

Corollary 150.4 (Actual covariances and the physical-time projection)**.** *For all sufficiently large n, uniformly in R,*

$$\frac{1}{n}\int U_{n,R} \otimes U_{n,R}\, \mathrm{d}\nu_R^* \ge \frac{d_0}{2} I_4. \tag{150.6}$$

The physical-time fluctuation covariance $\mathcal{V}_R$ in Corollary 140.4 is also uniformly positive definite on its three-dimensional space.

Proof. Equation (147.13) and (150.4) give (150.6). For the physical covariance recall $\Gamma_R = c_* D_R$ and $\mathcal{V}_R = \bar{\tau}_R^{-1} B_R \Gamma_R B_R^{\mathsf{T}}$, where $B_R(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3 - x_4/\bar{\tau}_R)$. For $z \in \mathbb{R}^3$, $|B_R^{\mathsf{T}} z| \ge |z|$. Hence $z^{\mathsf{T}} \mathcal{V}_R z \ge c_* d_0 (\max_I \bar{\tau}_R)^{-1} |z|^2$. All constants are positive. □

150.3. **The Gaussian part of exact local inversion.**

Corollary 150.5 (Uniform Gaussian inversion and the remaining terms)**.** *The centered Gaussian density associated with the actual return record is*

$$g_R(\xi) = \frac{1}{(2\pi)^2\sqrt{\det D_R}} \exp\Big(-\tfrac{1}{2}\xi^{\mathsf{T}} D_R^{-1}\xi\Big). \tag{150.7}$$

It is defined for every $R \in I$. There are $C, c > 0$ such that

$$\sup_R \int_{|v|>n^{1/200}} e^{-v^{\mathsf{T}} D_R v/2}\,\mathrm{d}v \le C\exp(-cn^{1/100}). \tag{150.8}$$

In Corollary 145.4, uniform positive definiteness is therefore a proved conclusion for this physical family, not an additional hypothesis. Under its stated residual integrability, complementary-transform estimate and local-edge conditions, the raw mixed-density conclusion follows with the density (150.7)*.*

Proof. Positive definiteness gives the Gaussian inversion formula. In four dimensions, integration in polar coordinates gives, for $A > 0$,

$$\int_{|v|>A} e^{-d_0|v|^2/2}\,\mathrm{d}v = 2\pi^2\left(\frac{A^2}{d_0} + \frac{2}{d_0^2}\right)e^{-d_0A^2/2}.$$

Absorb the polynomial into a weaker exponential and put $A = n^{1/200}$. The last assertion follows from the exact decomposition and the proved central comparison in Corollary 145.4. □

The two Fourier tails in this last argument concern different objects. Equation (150.8) estimates the limiting Gaussian. It does not estimate the complementary transform of the physical raw record or of its edge-subtracted residual. The proved central band is still $|v| \le 2n^{1/200}$, equivalently $|\omega| \le 2n^{-99/200}$. The annulus beyond it, the full critical and singular edge extraction, and the weighted tails and relative event comparison for exact physical conditioning remain the analytical tasks in the raw mixed-density theorem. The covariance obstruction has been removed without changing that theorem's record or target measure.

151. The true section occupation under a local collision constraint

We return to the original circular family and the actual section Y_R^* of (128.1). The occupation of this section is not the constant collision step. Nevertheless its geometry permits a bounded multiplier at one collision, and hence a perturbation at occupation frequency of order $m^{-1/2}$. This gives a joint local and central limit without changing the section or the physical record.

Put $c = c_*$, $\eta_R = \mathbf{1}_{Y_R^*}$, and define

$$A_{m,R} = \sum_{j=0}^{m-1} \eta_R \circ T_R^j, \quad E_{m,R} = A_{m,R} - mc, \quad X_R = (\kappa_{R,1}, \kappa_{R,2}, \tau_R - \bar\tau_R, \eta_R - c). \tag{151.1}$$

The sum includes the initial collision and excludes collision m. The symbol $A_{m,R}$ is an occupation count, not a return time.

151.1. **An exact, unsmoothed occupation multiplier.**

Lemma 151.1 (The actual rectangular section is a multiplier)**.** *Multiplication by η_R is bounded on every local strong collision space, uniformly in R. Consequently, for $v \in \mathbb{C}$,*

$$M_{e^{iv(\eta_R - c)}} = e^{-ivc}\{I + (e^{iv} - 1)M_{\eta_R}\} \tag{151.2}$$

is an entire strong-space family with uniform bounds on compact sets. No smoothing of η_R occurs in this identity.

Proof. The section is the union of five coordinate rectangles, with four of them small squares, in (α, p) coordinates. Their horizontal boundaries remain horizontal after $p = \sin\varphi$, and their vertical boundaries remain vertical after the local constant arclength reparametrization. All lie in a common compact region away from grazing. Split the angular circle at a fixed meridian and partition by all rectangle endpoints in both coordinates. This gives a uniformly finite partition into open, simply connected cells on which η_R is constant. It is a partition of the collision cylinder modulo null boundaries, not a partition of a long orbit.

Admissible stable curves are graphs whose slopes are negative and bounded away from zero and infinity in these local coordinates. They therefore cross any horizontal or vertical partition side at most once. Their intersection with a rectangular cell is an interval. Their length within distance ϵ of one such side is bounded by $C\epsilon$, uniformly in the location of the side. Summing over the fixed finite number of sides gives the partition conditions (1)–(2) of [12, Lemma 3.3], uniformly on the finite parameter cover. The constant functions on the cells meet every strict piecewise-Hölder exponent requirement. Part (b) of that lemma gives $\|M_{\eta_R}\| \le C$. Formula (151.2) follows from $\eta_R^2 = \eta_R$ pointwise and proves the remaining assertions. □

Let $Q_{R,z}$ be the exact displacement–roof operator of Section 122, and set

$$\mathcal{Q}_{R,z,v} = Q_{R,z} M_{e^{iv(\eta_R - c)}}. \tag{151.3}$$

The occupation multiplier is on the source side of each collision. If $a\nu$ belongs to the strong space and M_d is bounded there, then for bounded physical representatives a, d,

$$\ell\big(M_d \mathcal{Q}_{R,z,v}^m (a\nu)\big) = \int a(x) d(T_R^m x) e^{iz\cdot(K^{\mathrm{c}}_{m,R}, S_{m,R} - m\bar{\tau}_R) + ivE_{m,R}} d\nu(x). \tag{151.4}$$

The equality follows by the transfer convention at each step. Products of the finitely many bounded multipliers are understood as physical weighted compositions first and then by continuity. This also follows from the module property of the piecewise multiplier and its dual pairing with smooth tests. In particular it applies to a, d smooth, to $a = d = \eta_R$, and to their products with smooth functions. The distinction between the source occupation and the image-side preceding roof in (151.3) fixes the endpoint count.

Proposition 151.2 (Joint expansion and a small-occupation-frequency gap)**.** *On a uniform complex neighborhood of zero, $\mathcal{Q}_{R,z,v}$ has a simple eigenvalue $\lambda_R(z,v)$ and a rank-one projection, with exponentially decaying complementary powers. Writing $\Theta = (z,v) \in \mathbb{C}^4$,*

$$\log \lambda_R(\Theta) = -\tfrac{1}{2}\Theta^{\mathsf{T}} \Omega_R \Theta + O(|\Theta|^3), \tag{151.5}$$

with constants uniform in R. The matrices Ω_R are continuous and uniformly positive definite. For each $B < \infty$ and $\epsilon > 0$ there are $v_{B,\epsilon} > 0$, $C_{B,\epsilon} < \infty$ and $\rho_{B,\epsilon} < 1$ such that

$$\|\mathcal{Q}_{R,\omega,v}^m\| \le C_{B,\epsilon} \rho_{B,\epsilon}^m \quad (|b| \le B,\ |\omega| \ge \epsilon,\ |v| < v_{B,\epsilon}), \tag{151.6}$$

where $\omega = (u, b) \in \mathbb{T}^2 \times \mathbb{R}$ is real and v may be complex. The occupation radius is chosen after B and ϵ.

Proof. Lemma 151.1 and the exact preceding-flight multiplier give strong analyticity with common bounds on a fixed complex ball. Perturb the uniform unweighted splitting by its usual resolvent contours. At zero, the first eigenvalue derivative vanishes because every component of X_R is centered. The second derivative of the logarithm is minus the asymptotic covariance of the physical sum in (151.4). One can identify it directly: differentiate that finite-time pairing twice, divide by m, and use the analytic splitting. The amplitude derivatives divided by m tend to zero and the complementary derivatives decay exponentially, by Cauchy's estimate on a smaller fixed ball.

There is an exact collision identity $X_R = L_R h_R$, where

$$L_R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -\bar{\tau}_R & 1 \\ 0 & 0 & -c & 0 \end{pmatrix}, \qquad \Omega_R = L_R \Gamma_R L_R^{\mathsf{T}} = c L_R D_R L_R^{\mathsf{T}}. \tag{151.7}$$

Here h_R is the bounded exact compensation, not an induced observable. The covariance identification and convergence follow from Theorem 138.4. Since $\det L_R = c > 0$, Theorem 150.2 implies uniform positivity and continuity of Ω_R. Uniform analyticity gives the cubic remainder in (151.5).

For the last assertion first fix B, ϵ. Theorem 122.4 bounds $\|Q_{R,\omega}^m\| \le C_0 \rho_0^m$ on this compact set. Choose $\rho_0 < \rho_1 < 1$. The series for $(w - Q_{R,\omega})^{-1}$ is uniformly bounded on $|w| = \rho_1$ and on its exterior up to a common operator-norm bound. By (151.2), $\|\mathcal{Q}_{R,\omega,v} - Q_{R,\omega}\| \le C_B |v|$ for small v. Choose $v_{B,\epsilon}$ so the resolvent Neumann series converges uniformly on that exterior. Cauchy's power formula on $|w| = \rho_1$ proves (151.6). No weighted Lasota–Yorke estimate for an entire occupation-frequency torus has been used. □

Write the positive covariance and its Schur complement as

$$\Omega_R = \begin{pmatrix} \Sigma_R & b_R \\ b_R^{\mathsf{T}} & d_R \end{pmatrix}, \qquad \beta_R = b_R^{\mathsf{T}} \Sigma_R^{-1}, \qquad \sigma_{A,R}^2 = d_R - b_R^{\mathsf{T}} \Sigma_R^{-1} b_R > 0. \tag{151.8}$$

All these matrices and scalars are continuous; their relevant positive bounds are uniform. For $z \in \mathbb{R}^3$ let $\mathcal{N}_{R,z}$ be the real normal law with mean $\beta_R z$ and variance $\sigma_{A,R}^2$.

151.2. **A local roof interval and a central occupation variable.** An endpoint family below is called multiplier-bounded if a_R, d_R^{end} are bounded physical functions, $\|a_R \nu\|_{\mathcal{B}}$ and $\|M_{d_R^{\mathrm{end}}}\|$ are bounded on the local spaces, and their supremum norms are uniformly bounded. Smooth bounded families with common C^2 bounds and $a_R = d_R^{\mathrm{end}} = \eta_R$ are examples. The endpoint notation d_R^{end} is distinct from the covariance entry d_R in (151.8).

Theorem 151.3 (Mixed local and central limit for actual occupation)**.** *Let a_R, d_R^{end} be nonnegative multiplier-bounded endpoint families, and put $\alpha_R = \nu(a_R)\nu(d_R^{\mathrm{end}})$. Fix a bounded interval $J \subset \mathbb{R}$ of positive length and a bounded continuous function $F : \mathbb{R} \to \mathbb{C}$. Uniformly for $R \in I$ and central targets with $|z_{m,R}(k,t)| \le M$, where $z_{m,R}(k,t) = (k/\sqrt{m}, (t - m\bar{\tau}_R)/\sqrt{m})$,*

$$\begin{aligned} m^{3/2} \int a_R(x) d_R^{\mathrm{end}}(T_R^m x) \mathbf{1}_{\{K_{m,R}^{\mathrm{c}} = k,\ S_{m,R} - t \in J\}} F(E_{m,R}/\sqrt{m})\, d\nu(x) \\ - \alpha_R |J| g_{\Sigma_R}(z_{m,R}(k,t)) \int F\, d\mathcal{N}_{R, z_{m,R}(k,t)} \longrightarrow 0. \end{aligned} \tag{151.9}$$

The assertion also holds for the indicator of any fixed interval in the occupation variable, including a half-line. In particular the conditional occupation law is $\mathcal{N}_{R,z}$ in the limit whenever α_R is uniformly positive. The conditioned event in this assertion is unchanged.

Proof. First replace $\mathbf{1}_J(S_m - t)$ by an integrable time test $q(S_m - t)$ with integrable Fourier transform supported in $[-B, B]$, and take $F(y) = e^{i\theta y}$. The exact Fourier expression contains $\widehat{q}(-b)$, torus orthogonality in the two displacement coordinates, and the operator $\mathcal{Q}^m_{R,(u,b),\theta/\sqrt{m}}$. Fix B and a compact set of θ before taking $m \to \infty$. Outside a fixed neighborhood of $\omega = 0$, use (151.6). Inside it, use (151.5), the principal amplitude $\alpha_R + O(|\omega| + |\theta|/\sqrt{m})$, and put $w = \sqrt{m}\,\omega$. Uniform positivity gives an integrable Gaussian bound in w for bounded θ; the complementary

power is exponentially small. Dominated convergence yields the partial inverse

$$(2\pi)^{-3}\int_{\mathbb{R}^3} e^{-iw\cdot z}e^{-\frac{1}{2}(w,\theta)^{\mathsf{T}}\Omega_R(w,\theta)}\,dw = g_{\Sigma_R}(z)\exp\{i\theta\beta_R z-\tfrac{1}{2}\theta^2\sigma_{A,R}^2\}. \tag{151.10}$$

Multiplying by $\alpha_R\int q$ proves the band-limited assertion, uniformly in central targets. The sign of the conditional mean in (151.10) follows from the inverse phase $e^{-iw\cdot z}$.

Next remove the time test. Let $q_B^-\le \mathbf{1}_J\le q_B^+$ be the positive-envelope pair used in Lemma 123.2. Their Fourier supports are fixed and $\int(q_B^+-q_B^-)=O(B^{-1})$. Although the occupation phase is complex, its modulus is one. Thus the absolute replacement error using q_B^- is bounded by the same nonnegative endpoint integral with time test $q_B^+-q_B^-$ and with occupation frequency zero. The already proved band-limited assertion at zero bounds the limsup of the scaled error by C/B. Take $m\to\infty$ for each fixed B and then $B\to\infty$. This proves characteristic-function convergence for the exact interval J, also at $\theta=0$. A singleton endpoint of J is handled by majorants of intervals of arbitrarily small length, so the choice of open or closed endpoints has no effect.

The positive finite measures in the occupation variable now have converging masses and Fourier transforms continuous at zero, with the Gaussian limits in (151.10). The continuity theorem for finite positive measures gives weak convergence. If a subsequence of masses tends to zero, boundedness of F handles it directly. Otherwise normalize by the masses and apply the probability version. This proves the assertion for bounded continuous F and for Gaussian continuity sets. Uniformity follows by contradiction: along a violating sequence take convergent subsequences of R, z, and the bounded amplitudes α_R; the Fourier proof and covariance continuity identify every limit as the displayed Gaussian finite measure. For a uniformly positive α_R, division by its exact local mass proves the conditional claim. No event replacement and no growing-band spectral estimate enters the argument. □

Remark 151.4 (What is and is not estimated). The occupation inserted in (151.3) is the exact η_R, not a surrogate chosen after a smoothing limit. The estimate uses the entire fixed displacement–roof band and an occupation frequency shrinking to zero. It does not estimate fixed nonzero occupation frequencies, or a pointwise roof density. These two additional inversions are not part of (151.9).

152. Local collision constraints and a window of actual returns

The occupation variable in the preceding section is exactly the missing integer in an actual-return description at a prescribed collision time. We now use this fact as an identity of measures. There is no random-time approximation in the identity and no replacement of a rare event.

For $m\ge 1$ define a finite measure on $\mathbb{R}$ by

$$\mathfrak{R}_{m,R}^{k,t,J}=m^{3/2}\sum_{n=1}^{m}\nu_R^*\{K_{n,R}=k,\ N_{n,R}=m,\ T_{n,R}-t\in J\}\delta_{(n-mc)/\sqrt{m}}. \tag{152.1}$$

Here J is a fixed bounded roof interval of positive length, and $c=c_*$. Every term concerns the same four-coordinate physical return record as Theorem 143.1. In particular n is a genuine number of returns, not a deterministic substitute for a collision count.

Proposition 152.1 (Exact return and occupation disintegration)**.** *For every bounded measurable* $F:\mathbb{R}\to\mathbb{C}$,

$$\int F\,d\mathfrak{R}_{m,R}^{k,t,J}=\frac{m^{3/2}}{c}\int_M \eta_R(x)\eta_R(T_R^m x)\mathbf{1}_{\{K_{m,R}^{\mathrm{c}}=k,\ S_{m,R}-t\in J\}}F\left(\frac{A_{m,R}-mc}{\sqrt{m}}\right)d\nu(x). \tag{152.2}$$

Proof. Discard the invariant null set of singular trajectories and undefined returns. Start at $x \in Y_R^*$. If $T_R^m x \in Y_R^*$, list the section visits at collision times $0 = j_0 < j_1 < \cdots < j_n = m$. There are exactly n visits in $[0, m)$, so $A_{m,R}(x) = n$ and $N_{n,R}(x) = m$. Conversely $N_{n,R}(x) = m$ implies this list and this occupation count. Thus

$$\{x \in Y_R^*,\ T_R^m x \in Y_R^*,\ A_{m,R} = n\} = \{x \in Y_R^*,\ N_{n,R} = m\}.$$

On that set additivity gives $K_{n,R} = K_{m,R}^{\mathrm{c}}$ and $T_{n,R} = S_{m,R}$. The sets are disjoint as n varies and exhaust the terminal-section event. Since at most one visit occurs at each collision, $1 \le n \le m$. Integrate and use $d\nu_R^* = c^{-1}\eta_R\, d\nu$. This proves the identity, including both endpoint conventions. □

Theorem 152.2 (A local limit in a diffusive actual-return window)**.** *Let $\Sigma_R, \beta_R, \sigma_{A,R}^2$ be as in* (151.8)*, and set $z = z_{m,R}(k,t) = (k/\sqrt{m}, (t - m\bar{\tau}_R)/\sqrt{m})$. For every fixed bounded continuous F, uniformly in $R \in I$ and $|z| \le M$,*

$$\int F\, d\mathfrak{R}_{m,R}^{k,t,J} - c|J| g_{\Sigma_R}(z) \int_{\mathbb{R}} F(y) g_{\sigma_{A,R}^2}(y - \beta_R z)\, dy \longrightarrow 0. \tag{152.3}$$

The formula holds also when F is the indicator of a fixed interval, including a half-line. For each fixed $a < b$, the unscaled probability sum in (152.1) *over $a \le (n - mc)/\sqrt{m} \le b$ is at least $c_{M,a,b,J} m^{-3/2}$ for all sufficiently large m, with $c_{M,a,b,J} > 0$ uniform in R.*

Proof. Apply Theorem 151.3 with $a_R = d_R^{\mathrm{end}} = \eta_R$. These are admissible multipliers by Lemma 151.1. Their product of means is c^2, while the exact return disintegration has the factor c^{-1}. This gives precisely the amplitude c in (152.3). The normal law has no atoms, so fixed interval endpoints are continuity sets. Continuity, uniform ellipticity and compactness give a positive lower bound for

$$c|J| g_{\Sigma_R}(z) \int_a^b g_{\sigma_{A,R}^2}(y - \beta_R z)\, dy$$

on $R \in I$, $|z| \le M$. The uniform error tends to zero, proving the last assertion. The measure is positive throughout the argument. □

152.1. **Agreement with the four-coordinate Gaussian normalization.** The new amplitude is compatible with the covariance of the original induced record, rather than a newly chosen covariance. From (151.7),

$$\det \Omega_R = c^6 \det D_R, \qquad L_R^{-1}(z_1, z_2, z_3, y) = (z_1, z_2, -y/c, z_3 - \bar{\tau}_R y/c). \tag{152.4}$$

Indeed the scalar factor c in four dimensions contributes c^4 and $(\det L_R)^2 = c^2$. The density transformation consequently gives

$$c\, g_{\Sigma_R}(z) g_{\sigma_{A,R}^2}(y - \beta_R z) = c\, g_{\Omega_R}(z, y) = c^{-2} g_{D_R}(c^{-1/2} L_R^{-1}(z, y)). \tag{152.5}$$

For $n = mc + \sqrt{m}\, y$, the centered return coordinates divided by $\sqrt{m}$ are exactly $(z_1, z_2, -y/c, z_3 - \bar{\tau}_R y/c)$. Since $n/m \to c$ on bounded y windows, the argument and prefactor on the right of (152.5) are precisely those obtained by summing a putative n^{-2} four-coordinate local law over $\sqrt{m}$ consecutive return counts. This calculation checks the normalization; it is not an inference of that pointwise law.

The remaining inversions. Theorem 152.2 fixes both displacement coordinates and the collision count exactly, keeps a roof interval of unscaled positive length, and resolves the actual return index on its diffusive scale. It supplies a local constraint for the true four-coordinate return measure, not merely for the stationary flow. To resolve one return index one must control occupation frequencies outside the shrinking neighborhood used in (151.6). To resolve a pointwise roof density one must also control the raw edge correction and the full roof-frequency complement. Neither inverse is supplied by convergence of the window measures. The common pointwise correction and the complete return complement in the original raw theorem are retained as distinct analytical requirements.

153. Four-dimensional concentration from a fixed occupation strip

The joint collision operator contains more local information than its application to a fixed diffusive occupation test. Its expansion is four-dimensional, and the compact displacement–roof estimate persists on a fixed, sufficiently small occupation strip. We exploit that strip without assuming a gap on the entire occupation torus. The conclusions are a single-index upper bound at the four-dimensional scale and a local law for windows containing any diverging number of return indices.

Retain $c, \eta_R, A_{m,R}, \Omega_R$ and $\mathcal{Q}_{R,\omega,v}$ from Section 151. Throughout this section, a_R, d_R are nonnegative multiplier-bounded endpoint families in the sense stated there. Write $\alpha_R = \nu(a_R)\nu(d_R)$ and

$$\Phi_{m,R}(\omega, v) = \ell\big(M_{d_R}\mathcal{Q}^m_{R,\omega,v}(a_R\nu)\big), \qquad \omega = (u, b) \in \mathbb{T}^2 \times \mathbb{R}.$$

The letter d_R here denotes the terminal function, not the lower-right entry of Ω_R. The physical integral defining Φ is exactly (151.4); occupation counts the times $[0, m)$.

153.1. **An integrable four-dimensional strip.**

Proposition 153.1 (The fixed-strip integral)**.** *For every $B \geq 1$ there is $0 < \sigma_B < \pi/4$ such that*

$$\int_{[-\pi,\pi]^2}\int_{-B}^{B}\int_{-\sigma_B}^{\sigma_B} |\Phi_{m,R}(u, b, v)|\, dv\, db\, du \leq C_B m^{-2} \qquad (m \geq 1), \tag{153.1}$$

uniformly in R and in any fixed bounded class of endpoints. On a fixed neighborhood of the four-dimensional origin the integrand is the sum of a principal term bounded by $Ce^{-c_0 m|(u,b,v)|^2}$ and an exponentially small complementary term. On the remainder of the displayed strip its integral is exponentially small. The width σ_B is chosen after B; no positive lower bound for σ_B as $B \to \infty$ is asserted.

Proof. Choose $r > 0$ so small that the joint spectral splitting in Proposition 151.2 holds on $|(u, b, v)| \leq 3r$. Uniform positivity of Ω_R and its cubic remainder imply, after decreasing r,

$$|\lambda_R(u, b, v)|^m \leq e^{-c_0 m|(u,b,v)|^2} \quad \text{for real } |(u, b, v)| \leq 3r.$$

The principal amplitude is $\alpha_R + O(|(u, b, v)|)$, uniformly for bounded endpoints; the complementary pairing is $O(\rho^m)$. Choose $\sigma_B < \min(r, \pi/4)$ and also smaller than the occupation radius supplied by (151.6) for $|\omega| \geq r$ and $|b| \leq B$. The region $|\omega| < r$ is covered by the joint expansion, while its complement is covered by that compact estimate. The integral of the Gaussian in four dimensions is $O(m^{-2})$; the remaining fixed-volume terms are exponentially small. The norms of the endpoint density and multiplier bound their pairings. Increasing C_B handles the bounded initial range of m. □

For $k \in \mathbb{Z}^2$, $t, \ell \in \mathbb{R}$ put

$$z = \left(\frac{k}{\sqrt{m}}, \frac{t - m\bar{\tau}_R}{\sqrt{m}}\right), \qquad y = \frac{\ell - mc}{\sqrt{m}}, \qquad Z = (z, y). \tag{153.2}$$

For integrable q, p with integrable compactly supported Fourier transforms define

$$I_{m,R}(q,p;k,t,\ell) = \int a_R(x) d_R(T_R^m x) \mathbf{1}_{\{K^{\mathrm{c}}_{m,R}=k\}} q(S_{m,R}-t) p(A_{m,R}-\ell)\, d\nu(x).$$

If their Fourier supports lie in $[-B,B]$ and $[-\sigma_B,\sigma_B]$, respectively, then the exact formula is

$$I_{m,R}(q,p;k,t,\ell) = \frac{1}{(2\pi)^4} \int e^{-i[u\cdot k + b(t-m\bar\tau_R)+v(\ell-mc)]} \widehat{q}(-b)\widehat{p}(-v)\Phi_{m,R}(u,b,v)\, du\, db\, dv. \quad (153.3)$$

The integral ranges over that strip. Both minus signs follow from the convention $\widehat{q}(b) = \int e^{ibs} q(s)\, ds$. This is Fourier inversion of two real tests; it is not full occupation-torus inversion. Fubini is justified by the bounded physical factors and the integrable Fourier transforms. Proposition 153.1 gives

$$|I_{m,R}(q,p;k,t,\ell)| \le C_B m^{-2} \|q\|_1 \|p\|_1, \quad (153.4)$$

independently of all targets.

153.2. **Two envelopes, with an order-preserving lower bound.** Use the interval envelopes of Lemma 123.2. For an interval J and band B denote them by $q^{\pm}$, and for $[-H/2,H/2]$ and band σ denote them by $p_H^{\pm}$. Put $e_B = 2\pi/B$ and $e_\sigma = 2\pi/\sigma$. In addition to their pointwise order and masses, they satisfy

$$\|q^{\pm} - \mathbf{1}_J\|_1 = e_B, \quad \|p_H^{\pm} - \mathbf{1}_{[-H/2,H/2]}\|_1 = e_\sigma, \quad \|p_H^{\pm}\|_1 \le H + e_\sigma. \quad (153.5)$$

Indeed each difference has one sign, and its integral is the prescribed mass error. In particular

$$\sup_v \left| \widehat{p_H^{\pm}}(v) - \widehat{\mathbf{1}}_{[-H/2,H/2]}(v) \right| \le e_\sigma. \quad (153.6)$$

These are the standard one-dimensional Beurling–Selberg envelopes [17, Introduction]; no multidimensional extremizer is assumed.

Lemma 153.2 (Product envelopes)**.** *If f, g are interval indicators and $q^- \le f \le q^+$, $p^- \le g \le p^+$ are their envelopes, then*

$$q^- p^+ + q^+ p^- - q^+ p^+ \le fg \le q^+ p^+. \quad (153.7)$$

The inequalities remain valid after multiplication by the nonnegative physical endpoint weight. They do not require either minorant to be nonnegative.

Proof. The majorants are nonnegative. For the lower bound observe pointwise that

$$q^+p^+ - fg = (q^+ - f)p^+ + f(p^+ - g) \le (q^+ - q^-)p^+ + q^+(p^+ - p^-).$$

Rearranging proves the result. Multiplying two possibly negative minorants instead would not prove the required lower bound. □

Theorem 153.3 (Single-occupation local concentration)**.** *For every bounded roof interval J, uniformly for all R, k, t, all integers n, and $m \ge 1$,*

$$\int a_R(x) d_R(T_R^m x) \mathbf{1}_{\{K^{\mathrm{c}}_{m,R}=k,\ A_{m,R}=n,\ S_{m,R}-t\in J\}}\, d\nu(x) \le C(1+|J|)m^{-2}. \quad (153.8)$$

The constant is independent of the position of J and depends only on the bounded endpoint class and the physical family.

Proof. First take an interval of length at most one. Fix $B = 1$ and the strip width in Proposition 153.1. A translate of a single roof majorant of length one bounds its indicator. Because $A_{m,R}$ is integer-valued,

$$\mathbf{1}_{\{A_{m,R}=n\}} = \mathbf{1}_{[-1/2,1/2]}(A_{m,R}-n) \le p_1^+(A_{m,R}-n).$$

The product of these two nonnegative majorants bounds the integrand. Its integral is bounded by (153.4), with $\|q^+\|_1 = 1+2\pi$ and $\|p_1^+\|_1 = 1+2\pi/\sigma_1$. Translation changes no Fourier absolute bound. This proves the result for every interval of length at most one, including any choice of endpoints. Cover an arbitrary bounded interval by at most $|J|+2$ such intervals and sum. No asymptotic lower bound for a singleton is used or obtained here. □

153.3. **Moving tests at an arbitrarily slow diverging width.** For $H \geq 1$ set

$$\mathcal{G}_{m,H,R}(z,y) = \frac{1}{H}\int_{-H/2}^{H/2} g_{\Omega_R}(z, y+s/\sqrt{m})\,ds. \tag{153.9}$$

This expression retains the variation of the Gaussian across the whole window, including windows of diffusive size.

Lemma 153.4 (A uniform moving-envelope comparison)**.** *Fix B, $0<\sigma\leq\sigma_B$, and a roof test q whose integrable Fourier transform is supported in $[-B,B]$. For each $H\geq 1$ let $p_H^{\pm}$ be the interval envelopes with band σ. There are $\epsilon_m\to 0$ and $C_q<\infty$, depending on the fixed bands and q but not on H,R,k,t,ℓ, such that*

$$\left|\frac{m^2}{H} I_{m,R}(q,p_H^{\pm};k,t,\ell) - \alpha_R\left(\int q\right)\mathcal{G}_{m,H,R}(z,y)\right| \leq \epsilon_m + \frac{C_q}{\sigma H}. \tag{153.10}$$

It suffices here that q is fixed; no convergence rate for ϵ_m is asserted.

Proof. Apply (153.3). Divide its Fourier weight by H and use $|\widehat{p}_H^{\pm}|/H \leq 1+e_\sigma/H$. Outside the fixed origin neighborhood, the scaled inverse is at most $C_B m^2 \rho_B^m(1+e_\sigma/H)$ and tends to zero uniformly for $H\geq 1$. Inside it put $w=\sqrt{m}(u,b,v)$. The spectral expansion, its Gaussian bound and the principal amplitude give an $L^1(dw)$ error tending uniformly to zero when the operator pairing is replaced by $\alpha_R e^{-w^{\mathsf{T}}\Omega_R w/2}$. The same holds when $\widehat{q}(-w_3/\sqrt{m})$ is replaced by $\int q$. The bounded factor $1+e_\sigma/H$ is at most $1+e_\sigma$; thus all these errors are absorbed in one ϵ_m independent of H. The inverse oscillation has modulus one, so this reasoning is uniform in all central coordinates, not just convergent target sequences.

Replace $\widehat{p}_H^{\pm}(-w_4/\sqrt{m})/H$ by $\widehat{\mathbf{1}}_{[-H/2,H/2]}(-w_4/\sqrt{m})/H$. Equation (153.6) and the integrable Gaussian bound make the error at most $C_q/(\sigma H)$. Extending the remaining Gaussian integral to $\mathbb{R}^4$ costs another uniformly vanishing error. Finally, Fourier inversion of the Gaussian, followed by integration of the interval indicator, gives exactly

$$\frac{1}{(2\pi)^4}\int_{\mathbb{R}^4} e^{-iw\cdot(z,y)} e^{-w^{\mathsf{T}}\Omega_R w/2}\frac{1}{H}\int_{-H/2}^{H/2} e^{-iw_4 s/\sqrt{m}}\,ds\,dw = \mathcal{G}_{m,H,R}(z,y).$$

The sign is consistent with testing $A_m-\ell$. The centered interval is symmetric, but no sign is discarded in this calculation. □

Theorem 153.5 (Local law in an arbitrarily slowly widening occupation block)**.** *Let J be a fixed bounded interval of positive length. Let $\mathcal{B}=\{j,j+1,\dots,j+H-1\}$ be a block of $H\geq 1$ integers with center $\ell = j+(H-1)/2$, and use* (153.2)*. For every sequence $h_m\to\infty$, uniformly over $H\geq h_m$, R, and $|Z|\leq M$,*

$$\frac{m^2}{H}\int a_R(x) d_R(T_R^m x)\mathbf{1}_{\{K_{m,R}^{c}=k,\ A_{m,R}\in\mathcal{B},\ S_{m,R}-t\in J\}}\,d\nu(x) - \alpha_R|J|\mathcal{G}_{m,H,R}(z,y) \longrightarrow 0. \tag{153.11}$$

In particular, if also $H\leq u_m\sqrt{m}$ with $u_m\to 0$, the Gaussian average can be replaced by $g_{\Omega_R}(z,y)$. There is no lower polynomial bound on h_m: every diverging width is admissible.

Proof. The half-integer endpoints of the interval $[j-1/2, j+H-1/2]$ select exactly the block $\mathcal{B}$, so its indicator is the centered interval test of length H at ℓ. First fix $B \geq 1$, then fix $\sigma \leq \sigma_B$ and choose the two pairs of envelopes. Apply Lemma 153.2. Lemma 153.4 evaluates all three terms in the lower bound and the one term in the upper bound. In the main term the upper roof mass is $|J| + e_B$; the lower combination has roof mass $(|J| - e_B) + (|J| + e_B) - (|J| + e_B) = |J| - e_B$. Uniform boundedness of α_R and g_{Ω_R} therefore gives

$$\left| \frac{m^2}{H} \int a_R (d_R \circ T_R^m) \mathbf{1}_{\{K_m^c = k, A_m \in \mathcal{B}, S_m - t \in J\}} \, d\nu - \alpha_R |J| \mathcal{G}_{m,H,R}(z,y) \right| \leq \epsilon_m(B) + \frac{C_J}{B} + \frac{C_J}{\sigma H}, \tag{153.12}$$

where $\epsilon_m(B) \to 0$ for the already fixed B, σ. Take $m \to \infty$ uniformly for $H \geq h_m$ and only afterwards $B \to \infty$. The width σ may become smaller when B changes; this causes no problem because it was fixed before the first limit. No m-dependent band is inserted into the spectral theorem.

When $H/\sqrt{m} \to 0$, uniform continuity of the Gaussian densities on central compact sets makes the average in (153.9) differ from $g_{\Omega_R}(z,y)$ by $o(1)$ uniformly. This proves the final assertion. The argument is a uniform moving-test estimate; weak convergence for fixed diffusive intervals alone would not justify it. □

154. Return-index resolution and its remaining torus complement

We apply the preceding estimates to the same actual return law as Theorem 143.1. Write

$$P_{n,m,R}(k,t;J) = \nu_R^*\{K_{n,R} = k,\ N_{n,R} = m,\ T_{n,R} - t \in J\}. \tag{154.1}$$

It is zero for $n > m$. The exact disintegration in Proposition 152.1 applies to a single n and to every block of consecutive n; no limiting step is needed for either statement.

Theorem 154.1 (Concentration and fine windows for actual returns)**.** *For all $m \geq n \geq 1$, all k, t, R and bounded intervals J,*

$$P_{n,m,R}(k,t;J) \leq C(1+|J|)m^{-2}. \tag{154.2}$$

For a fixed J of positive length, a block $\mathcal{B} = \{j, \dots, j+H-1\}$ with center $\ell = j + (H-1)/2$, and $Z = (z,y)$ as in (153.2)*, one has*

$$\frac{m^2}{H} \sum_{n \in \mathcal{B} \cap [1,m]} P_{n,m,R}(k,t;J) - c|J| \mathcal{G}_{m,H,R}(z,y) \longrightarrow 0, \tag{154.3}$$

uniformly for R, $|Z| \leq M$, and $H \geq h_m$, for any $h_m \to \infty$. If $h_m \leq H \leq u_m \sqrt{m}$ and $u_m \to 0$, this becomes

$$\frac{m^2}{H} \sum_{n \in \mathcal{B}} P_{n,m,R}(k,t;J) = c|J| g_{\Omega_R}(z,y) + o(1). \tag{154.4}$$

The block lies in $[1,m]$ for all sufficiently large m in this last central regime. Its probability is bounded below by $c_{M,J} H m^{-2}$. The same lower bound holds for $H \leq C_0 \sqrt{m}$ with a constant also depending on C_0.

Proof. Use $a_R = d_R = \eta_R$ in Theorems 153.3 and 153.5. These endpoints are admissible by Lemma 151.1. Their amplitude is c^2. The exact disintegration divides the collision integral by c, so the amplitude is c, proving every displayed formula. Since $0 < c < 1$, a block centered at $mc + O(\sqrt{m})$ and of width $O(\sqrt{m})$ is contained in $[1,m]$ eventually. On the compact set $|Z| \leq M$ and $|s|/\sqrt{m} \leq C_0/2$, uniform ellipticity gives a positive lower bound for $g_{\Omega_R}(z, y + s/\sqrt{m})$. The error in (154.3) tends uniformly to zero. □

Corollary 154.2 (The original return covariance normalization)**.** *In the subdiffusive regime of* (154.4)*, put*

$$V_{\ell,R}(k,m,t) = \ell^{-1/2}(k_1, k_2, m - \ell/c, t - \ell\bar{\tau}_R/c).$$

Then, uniformly on the same central compact sets,

$$\frac{\ell^2}{H}\sum_{n\in\mathcal{B}} P_{n,m,R}(k,t;J) = |J| g_{D_R}(V_{\ell,R}(k,m,t)) + o(1). \tag{154.5}$$

The center ℓ may be a half-integer; every summand still has an integer return index.

Proof. Equation (152.5) gives $cg_{\Omega_R}(Z) = c^{-2} g_{D_R}(c^{-1/2} L_R^{-1} Z)$. Here $L_R^{-1} Z = m^{-1/2}(k_1, k_2, m - \ell/c, t - \ell\bar{\tau}_R/c)$ exactly, and $\ell/m = c + O(m^{-1/2})$. Multiply (154.4) by ℓ^2/m^2 and use uniform continuity of g_{D_R} on compact sets. No independent covariance or clock is introduced. □

Corollary 154.3 (Regular endpoint amplitudes in a fine return window)**.** *Let a_R, d_R be bounded nonnegative functions such that $\eta_R a_R, \eta_R d_R$ form multiplier-bounded endpoints. Insert $a_R(x) d_R((F_R^*)^n x)$ in each original return probability. Then the right side of* (154.3) *is multiplied by $\nu_R^*(a_R)\nu_R^*(d_R)$. A uniform positive lower bound for the selected probability requires this product to be uniformly positive. The concentration estimate holds with a changed endpoint constant.*

Proof. On $N_{n,R} = m$ one has $(F_R^*)^n x = T_R^m x$. Apply the two collision estimates to $\eta_R a_R, \eta_R d_R$. Their product of collision means, divided by c, equals $c\nu_R^*(a_R)\nu_R^*(d_R)$. □

154.1. **Exact single-index inversion with a localized unresolved term.** The upper bound and the window asymptotic do not alone identify each single return mass. The following identity locates the remaining condition directly on the bounded collision operator, rather than on an unspecified induced spectral realization.

Let q be an integrable roof test with integrable Fourier transform supported in $[-B, B]$. Define the actual single-return functional

$$P^q_{n,m,R}(k,t) = \int_{Y_R^*} \mathbf{1}_{\{K_{n,R}=k, N_{n,R}=m\}} q(T_{n,R} - t)\, d\nu_R^*. \tag{154.6}$$

In the following formulas $\Phi^{\eta,\eta}_{m,R}$ has the two actual section endpoints. Choose a real smooth function χ_B on the occupation torus, equal to one near zero and supported in $(-\sigma_B, \sigma_B)$. It is chosen once after B. Set

$$\mathcal{R}^q_{n,m,R}(k,t) = \frac{1}{c(2\pi)^4}\int_{[-\pi,\pi]^2}\int_{-B}^{B}\int_{-\pi}^{\pi} e^{-i[u\cdot k + b(t - m\bar{\tau}_R) + v(n-mc)]}\widehat{q}(-b)(1-\chi_B(v)) \cdot \Phi^{\eta,\eta}_{m,R}(u,b,v)\, dv\, db\, du. \tag{154.7}$$

Every operator in this expression is a finite power of bounded one-step multipliers. No uniform bound on its long powers over the whole occupation torus is presupposed.

Proposition 154.4 (Exact-index Gaussian term and the fixed-torus remainder)**.** *With $Z = (k/\sqrt{m}, (t - m\bar{\tau}_R)/\sqrt{m}, (n - mc)/\sqrt{m})$,*

$$m^2 P^q_{n,m,R}(k,t) = c\left(\int q\right) g_{\Omega_R}(Z) + m^2 \mathcal{R}^q_{n,m,R}(k,t) + o(1), \tag{154.8}$$

uniformly in R and central compact sets, for each fixed q, B, χ_B. Thus for this roof test the single-index local limit is equivalent to $m^2\mathcal{R}^q_{n,m,R}(k,t) \to 0$ in the same uniform sense.

Proof. Apply the exact return disintegration, torus orthogonality for K_m^c and the integer A_m, and real Fourier inversion for $q(S_m - t)$. This gives the full integral in (154.7) with $1 - \chi_B$ replaced by one. Centering the occupation only moves a scalar phase between the inverse factor

and Φ; their product remains 2π-periodic in v. The identity is therefore valid even when mc is not an integer.

Split the integral with $1 = \chi_B + (1 - \chi_B)$. On the first piece, Proposition 153.1 and the joint expansion allow the four-dimensional rescaling $w = \sqrt{m}(u, b, v)$. The factors $\chi_B(w_4/\sqrt{m})$ and $\widehat{q}(-w_3/\sqrt{m})$ tend to one and $\int q$, respectively. The endpoint amplitude is c^2, divided by the source normalization c. Gaussian domination proves the first term and the uniform $o(1)$. The second piece is exactly the stated remainder. Subtraction proves the equivalence, with no estimate of the modulus of its integrand. □

Corollary 154.5 (A sufficient occupation-torus condition for a sharp roof interval)**.** *Fix a roof interval J. Suppose that, for every fixed $B \geq 1$ and its two interval envelopes $q^{\pm}_{J,B}$,*

$$\sup_{R,\,|Z|\leq M} m^2 |\mathcal{R}^{q^{\pm}_{J,B}}_{n,m,R}(k,t)| \longrightarrow 0.$$

Then $m^2 P_{n,m,R}(k,t;J) = c|J| g_{\Omega_R}(Z) + o(1)$ uniformly on those central compact sets. This is a sufficient condition, not an additional proved cancellation assertion.

Proof. Multiply the two roof-envelope inequalities by the unchanged positive single-index source measure. For each fixed B the proposition and the assumed cancellation give lower and upper limiting Gaussian masses $c(|J| - 2\pi/B)$ and $c(|J| + 2\pi/B)$. Take $m \to \infty$ first and $B \to \infty$ afterwards. □

Proof dependencies and relation to the raw endpoint. The new unconditional results use the existing exact occupation multiplier, joint covariance and small-occupation resolvent estimate, together with the one-dimensional interval envelopes. They add no continuum partition estimate and do not assume a gap on a full occupation torus. The fixed-strip argument is a local-limit and concentration mechanism; its present application concerns the actual five-rectangle section and its true return law. The classical Fourier envelopes are not claimed as new, and a general priority claim for mesoscopic local limits is not made.

Compared with Theorem 152.2, the window may now contain H_m consecutive return counts with $H_m \to \infty$ arbitrarily slowly, rather than a fixed positive multiple of $\sqrt{m}$. Equation (154.2) also applies to each single return index at its natural four-dimensional upper scale. Neither result sets $H_m = 1$ in the window limit: the explicit error $C_J/(\sigma H_m)$ in (153.12) would persist. The cancellation in (154.7) is the separate exact-index requirement. Pointwise roof inversion additionally requires the common raw correction and full roof complement of the original four-coordinate theorem; fixed roof intervals do not supply it. All original raw-edge and return-inversion statements are retained.

155. The measurable occupation resonance group

The occupation phase cannot be identified with a constant phase per collision. We determine the possible structure of its measurable obstructions before making any assertion about long powers on the full occupation torus. This distinction also separates an arithmetic obstruction from an estimate for the signed inverse.

All assertions in this section concern the original circular family and its actual section. Write $c = c_*$ and use the uncentered phase

$$q \circ T_R = e^{i(u\cdot\kappa_R + b\tau_R + v\eta_R + s)} q, \qquad |q| = 1 \quad \text{almost everywhere.} \tag{155.1}$$

Here $u \in \mathbb{T}^2$, $v, s \in \mathbb{T}$, $b \in \mathbb{R}$, and each torus has period 2π. A choice of representatives in the exponent is immaterial. Let $\mathcal{A}_R$ be the set of (u, b, v, s) for which (155.1) has a measurable solution. Let $\mathcal{G}_R$ be its projection onto $(u, b, v) \in \mathbb{T}^2 \times \mathbb{R} \times \mathbb{T}$. These are groups of physical measurable phases, not spectra on a postulated induced Banach space.

155.1. **A uniformly isolated group.**

Lemma 155.1 (Exclusion near the joint origin for measurable phases)**.** *There is $\delta_* > 0$, independent of R, such that* (155.1) *has no solution when*

$$0 < \operatorname{dist}((u,b,v),0) < \delta_*.$$

When $(u,b,v) = 0$, it has a solution only for $s = 0$; that solution is constant almost everywhere. No regularity of q is assumed.

Proof. In a coordinate neighborhood of zero put $\Theta = (u,b,v)$ and $X_R = (\kappa_{R,1}, \kappa_{R,2}, \tau_R - \bar{\tau}_R, \eta_R - c)$. The positive covariance and the uniform cubic expansion in Proposition 151.2 give, after reducing a uniform radius,

$$|\lambda_R(\Theta)| \le e^{-a|\Theta|^2} < 1 \qquad (0 < |\Theta| < \delta_*).$$

The complementary powers decay exponentially on the same neighborhood. Thus the exact collision pairing with any two fixed smooth endpoint functions tends to zero at each such nonzero frequency. This assertion uses the exact occupation multiplier, not smoothing of X_R.

Suppose a measurable q solves (155.1). Iteration gives the exact identity

$$\int q(x)\overline{q(T_R^m x)} e^{i\Theta\cdot S_{m,R}X_R(x)}\, d\nu(x) = e^{-im(s+b\bar{\tau}_R+vc)}. \tag{155.2}$$

Choose smooth complex q_ϵ, bounded by one, with $\|q - q_\epsilon\|_1 < \epsilon$. Such approximations follow by convolution in collision charts and truncation or by a positive smoothing of a bounded representative. Invariance and the bounded moduli show that replacing both occurrences of q changes the integral by at most 2ϵ, for every m. With ϵ fixed, the smooth pairing tends to zero by the preceding spectral estimate. The right side of (155.2) has modulus one, so $1 \le 2\epsilon$, a contradiction. The approximating norm need not be uniform in q or ϵ: the limit in m is taken first.

At $\Theta = 0$, (155.1) is the ordinary circle eigenfunction equation. Mixing of T_R excludes $s \neq 0$, and ergodicity makes an invariant q constant. The mixing input is the collision splitting, extended from smooth tests by L^2 approximation. □

Theorem 155.2 (Structure and uniform separation of occupation resonances)**.** *For each R, the phase s above a point of $\mathcal{G}_R$ is unique, and its transfer function q is unique up to a constant of modulus one. The group $\mathcal{G}_R$ is closed and uniformly discrete. There is a constant C_*, independent of R and $B \ge 0$, such that*

$$\#\{(u,b,v) \in \mathcal{G}_R : |b| \le B\} \le C_*(1+B). \tag{155.3}$$

Projection from $\mathcal{G}_R$ to its occupation coordinate v is injective. In particular its zero-roof subgroup is finite cyclic, of some order $d_R \le C_*$; its occupation coordinates are exactly the d_Rth roots of unity in angular coordinates. Every u and s in that subgroup belongs to $(2\pi/d_R)\mathbb{Z}$ modulo 2π.

There are precisely two possible forms. Either $\mathcal{G}_R$ is this finite cyclic group, or its roof projection is $\beta_R\mathbb{Z}$ for some $\beta_R > 0$ and

$$\mathcal{G}_R = \{n\gamma_R + j\gamma_R^{\mathrm{tor}} : n \in \mathbb{Z},\ 0 \le j < d_R\}. \tag{155.4}$$

Here γ_R has roof coordinate β_R and γ_R^{tor} generates the zero-roof subgroup. The occupation coordinate of γ_R, divided by 2π, is irrational. The positive roof spacing, when present, is bounded below uniformly in R. No assertion that either type of nonzero resonance actually exists is included in this classification.

Proof. Products and quotients of circle-valued solutions give addition and subtraction of phases. If two solutions have the same (u,b,v), their quotient is a collision eigenfunction with constant

phase the difference of the two s's. Mixing gives equality of those phases; ergodicity gives constancy of the quotient. Consequently $\mathcal{A}_R$ is the graph of a character s_R on $\mathcal{G}_R$.

The difference of two distinct elements of $\mathcal{G}_R$ is a nonzero element. Lemma 155.1 shows that its distance from zero is at least δ_*. Thus distinct elements are separated by a common positive distance, without any compactness assumption on their transfer functions. A convergent sequence in the group is eventually constant, proving closedness. Packing disjoint balls of radius $\delta_*/3$ into $\mathbb{T}^3 \times [-B - \delta_*, B + \delta_*]$ gives (155.3); reduce δ_* below one if needed.

If $v = 0$, the phase equation is precisely the complete physical collision phase equation. Theorem 194.2 gives $u = 0$, $b = 0$, $s = 0$. The kernel of projection to v is therefore trivial. The same argument applied to a difference proves injectivity. The zero-roof subgroup is finite by (155.3), and injects into the circle. Every finite subgroup of the circle is cyclic and consists of all dth roots for its order d. Multiplying any of its elements by d gives zero in the full group, including its unique s coordinate. This proves the rational-coordinate claim.

Let H_R be the roof projection. A bounded interval contains only finitely many of its points, by (155.3). Hence this additive subgroup of $\mathbb{R}$ is either zero or $\beta_R\mathbb{Z}$ for its least positive element. To see the latter assertion directly, choose the least positive element in any nonempty finite set of positive roof coordinates below a fixed bound. A smaller positive element anywhere would be in that set. Division with remainder by this minimum shows that every roof coordinate is its integer multiple. Choose γ_R over β_R. Subtracting $n\gamma_R$ from any element with roof $n\beta_R$ leaves the finite kernel, proving (155.4) and uniqueness of its expression. If the occupation coordinate of γ_R were a rational multiple of 2π, a positive multiple of γ_R would have zero occupation coordinate and nonzero roof coordinate. This contradicts injectivity. Finally, if $0 < \beta_R \le 1$, the elements $0, \gamma_R, \dots, \lfloor 1/\beta_R \rfloor \gamma_R$ lie in the unit roof slab. Their number is at most $2C_*$. This gives a positive uniform lower bound for β_R; the case $\beta_R > 1$ is immediate. □

155.2. **The exact arithmetic seen by an actual return.**

Proposition 155.3 (All-frequency induction of the phase equation)**.** *A phase (u, b, v, s) belongs to $\mathcal{A}_R$ if and only if there is a measurable circle-valued q_* on Y_R^* satisfying*

$$q_* \circ F_R^* = e^{i(u\cdot\kappa_R^* + b\tau_R^* + sr_R^* + v)} q_*. \tag{155.5}$$

This equivalence holds for every frequency, not only near zero, and uses the actual return tower. In particular ordinary finite lattice cover mixing does not by itself remove a nonzero v: on the section v is the return spectral phase.

Proof. Discard the invariant null set on which an iterate fails and telescope (155.1) to the true first return. The occupation sum on $[0, r_R^*)$ is exactly one. Restriction gives (155.5).

Conversely, write almost every collision state uniquely as $T_R^j y$, where $y \in Y_R^*$ and $0 \le j < r_R^*(y)$. With $\psi_R = u \cdot \kappa_R + b\tau_R + s$, define

$$q(T_R^j y) = e^{iS_{j,R}\psi_R(y) + iv\mathbf{1}_{\{j\ge 1\}}} q_*(y).$$

The genuine tower is a measurable partition and this defines a measurable circle-valued function. Its one-step equation is exact at every nontop level. At the top it is exactly (155.5). This includes $r_R^* = 1$, when the initial level is already the top. No induced mixing is used. □

What is and is not removed. The theorem replaces an unspecified continuum of possible measurable occupation obstructions by a uniformly separated group with finitely many points in each bounded roof band. It also identifies their exact return equation. It does not assert an anisotropic essential spectral radius below one on the full occupation torus. Nor does it assert that the finite collection on a fixed band is empty. Absence of a measurable phase and a power estimate for an operator are

different assertions. The signed remainder in Proposition 154.4 retains its original normalization and is not set to zero by this classification.

156. Cancellation carried by the true section source

The existence of a peripheral phase is not by itself a nonzero contribution to a prescribed endpoint pairing. For the actual section we can calculate a source cancellation on the full occupation torus. The calculation takes place in the physical L^2 space and requires neither a full-torus anisotropic power bound nor regularity of an eigenfunction.

Use the inner product $\langle f, g\rangle = \int \overline{f} g \, d\nu$. Let $f_R^{\mathrm{c}} = (\kappa_{R,1}, \kappa_{R,2}, \tau_R - \bar{\tau}_R)$, $z = e^{iv} \neq 1$, and define the physical unitary operator

$$U_{R,\omega,v} h = \big(e^{i(\omega \cdot f_R^{\mathrm{c}} + v\eta_R)} h\big) \circ T_R^{-1}. \tag{156.1}$$

It is the physical realization of $e^{ivc}\mathcal{Q}_{R,\omega,v}$. Unlike the anisotropic realization, its full L^2 operator norm is one at every power. Its inverse is $U^{-1}h = e^{-i(\omega \cdot f_R^{\mathrm{c}} + v\eta_R)} h \circ T_R$.

156.1. **A source resolvent, not a full-space resolvent bound.**

Theorem 156.1 (Section-source spectral cancellation)**.** *For $|\omega| \leq 1$, $v \notin 2\pi\mathbb{Z}$, $0 \leq r < 1$ and $0 < \epsilon \leq 2$, let $E_{R,\omega,v}(\epsilon)$ be the spectral projection of $U_{R,\omega,v}$ on $|\lambda - 1| \leq \epsilon$. Uniformly in R,*

$$\|E_{R,\omega,v}(\epsilon)\eta_R\|_2 \leq \frac{\epsilon + C|\omega|}{|e^{iv} - 1|}, \tag{156.2}$$

$$\|(I - rU_{R,\omega,v})^{-1}\eta_R\|_2 \leq \frac{2/(1+r) + C|\omega|/(1-r)}{|e^{iv} - 1|}, \tag{156.3}$$

$$\left\| \frac{1}{M} \sum_{j=0}^{M-1} U_{R,\omega,v}^{j} \eta_R \right\|_2 \leq \frac{2/M + C|\omega|}{|e^{iv} - 1|} \quad (M \geq 1). \tag{156.4}$$

At $\omega = 0$, the spectral measures of η_R and 1 satisfy the exact identity

$$d\mu_{\eta_R,v}(\lambda) = \frac{|\lambda - 1|^2}{|e^{iv} - 1|^2} \, d\mu_{1,v}(\lambda). \tag{156.5}$$

In particular the projection of η_R onto the eigenvalue 1 is zero at every nonzero occupation phase, including an occupation phase for which that eigenvalue exists.

Proof. The elementary identity $e^{-iv\eta_R} = 1 + (z^{-1} - 1)\eta_R$ and the formula for U^{-1} give

$$(z^{-1} - 1)\eta_R = (U^{-1} - I)1 + h_{R,\omega,v}, \qquad h_{R,\omega,v} = (e^{i\omega \cdot f_R^{\mathrm{c}}} - 1)U^{-1}1. \tag{156.6}$$

The bounded one-collision record implies $\|h_{R,\omega,v}\|_2 \leq C|\omega|$. This estimate is independent of v; no derivative of an occupation itinerary occurs.

Apply the spectral projection in the statement to (156.6). On its range, $U^{-1} - I$ has norm at most ϵ. The projection is a contraction and $\|1\|_2 = 1$. Division by $|z^{-1} - 1| = |z - 1|$ proves (156.2). For the resolvent, the spectral theorem gives

$$\sup_{|\lambda|=1} \frac{|\lambda^{-1} - 1|}{|1 - r\lambda|} = \frac{2}{1+r}, \qquad \|(I - rU)^{-1}\| \leq \frac{1}{1-r}.$$

Indeed $|1 - r\lambda|^2 = (1-r)^2 + r|\lambda - 1|^2$, and the ratio is increasing in $|\lambda - 1|^2 \in [0, 4]$. Apply these bounds to the two terms in (156.6). For the Cesaro bound,

$$\sum_{j=0}^{M-1} U^j (U^{-1} - I)1 = (U^{-1} - U^{M-1})1$$

has norm at most two, and the average applied to h has norm at most $\|h\|_2$. This proves (156.4). At $\omega = 0$, $h = 0$ in the exact source identity. Functional calculus then proves (156.5), including its assertion at the atom $\lambda = 1$. □

Corollary 156.2 (An invisible pure occupation eigenphase)**.** *If q is a measurable circle-valued solution of $q \circ T_R = e^{iv\eta_R} q$ with $v \notin 2\pi\mathbb{Z}$, then*

$$\int \eta_R q \, d\nu = 0, \qquad \int q \, d\nu_R^* = 0.$$

Thus its rank-one physical eigenspace has zero coefficient in the η_R–η_R pairing. No analogous vanishing is asserted for an arbitrary physical frequency or for an eigenvalue different from one.

Proof. Invariance gives $0 = \int (q \circ T_R - q) \, d\nu = (e^{iv} - 1) \int \eta_R q \, d\nu$. Division by c proves the second equality. Equivalently, Proposition 155.3 restricts q to an induced eigenfunction with eigenvalue e^{iv}, whose mean is zero by measure preservation. The physical eigenspace at eigenvalue one is at most one dimensional: the quotient of two circle solutions is invariant. The assertion about the pairing follows, also from Theorem 156.1. □

156.2. **An exact finite-time cancellation and its Abel limit.** For $m \geq 0$ put

$$F_{m,R}(v) = \int e^{ivA_{m,R}} \, d\nu, \qquad F_{0,R} = 1,$$

and for $m \geq 1$ put

$$C_{m,R}(v) = \int \eta_R(x) \eta_R(T_R^m x) e^{ivA_{m,R}(x)} \, d\nu(x). \tag{156.7}$$

By the exact return disintegration, the same function is

$$C_{m,R}(v) = c \sum_{n \geq 1} e^{ivn} \nu_R^*\{N_{n,R} = m\}. \tag{156.8}$$

Only finitely many terms occur at any given m.

Theorem 156.3 (Full-torus section cancellation in the collision clock)**.** *For every $m \geq 1$ and $v \notin 2\pi\mathbb{Z}$,*

$$C_{m,R}(v) = \frac{z}{(z-1)^2} (F_{m+1,R}(v) - 2F_{m,R}(v) + F_{m-1,R}(v)). \tag{156.9}$$

Consequently, uniformly in R and $M \geq 1$,

$$\left| \sum_{m=1}^{M} C_{m,R}(v) \right| \leq \frac{4}{|e^{iv} - 1|^2}. \tag{156.10}$$

For $1/2 \leq r < 1$ the absolutely convergent Abel sum obeys

$$\left| \sum_{m \geq 1} r^m C_{m,R}(v) - \frac{cz}{1-z} \right| \leq \frac{4(1-r)}{|e^{iv} - 1|^2}. \tag{156.11}$$

In particular these estimates are uniform on the entire occupation torus outside any fixed neighborhood of zero.

Proof. Write $\eta_j = \eta_R \circ T_R^j$. Stationarity gives

$$F_{m+1,R} - F_{m,R} = (z-1) \int \eta_0 z^{\eta_1 + \cdots + \eta_m} \, d\nu.$$

Subtract its version with m replaced by $m-1$. The result is $(z-1)^2 \int \eta_0 \eta_m z^{\eta_1 + \cdots + \eta_{m-1}} \, d\nu$. On the support of $\eta_0 \eta_m$, multiplication by z inserts the initial visit and not the terminal one. This is precisely (156.9). The convention also covers $m = 1$. Summing the second difference leaves $F_{M+1,R} - F_{M,R} - F_{1,R} + F_{0,R}$, whose modulus is at most four. This proves (156.10).

Let $F_R(r,v) = \sum_{m\ge 0} r^m F_{m,R}(v)$. Absolute convergence follows from $|F_{m,R}| \le 1$. Since $F_{1,R} = 1 + c(z-1)$, summing (156.9) gives the exact identity

$$\sum_{m\ge 1} r^m C_{m,R}(v) = \frac{z}{(z-1)^2}\left\{\frac{(1-r)^2}{r}F_R(r,v) - \frac{1-r}{r} - c(z-1)\right\}. \tag{156.12}$$

The first two terms in braces have total modulus at most $2(1-r)/r \le 4(1-r)$. The last term gives $cz/(1-z)$, proving (156.11). No mixing estimate was used: the cancellation comes from the same initial and terminal section indicator and from integer-valued occupation. □

Relation to the fixed-count inverse. These are new estimates for the unmodified section source. They show that a pure occupation eigenphase at eigenvalue one is invisible to the required endpoints, and that a whole occupation-torus complement has bounded collision-clock partial sums and an explicit Abel limit. They do not give decay of $C_{m,R}(v)$ for each fixed m. For example, on a stationary two-cycle with section indicator $(1,0)$ and $z=-1$, one has $C_{2j}(v) = (-1)^j/2$ and $C_{2j+1}(v) = 0$. Its partial sums are bounded but its coefficients do not tend to zero. This example is not the billiard and is used only to distinguish the two conclusions of the source identity.

In particular the m^2-normalized remainder of Proposition 154.4 is still a fixed-m coefficient and includes nonzero displacement–roof frequencies. Neither its cancellation nor a pointwise roof-density estimate follows by removing the collision-clock average from (156.11). The original raw-return correction, full complement and weighted microscopic targets remain unchanged.

157. Polynomial occupation cuts and the full fixed-band spectrum

We estimate the true section occupation at every real occupation frequency. The extra discontinuities are counted on an already regular inverse collision curve, not by multiplying the number of cells in a one-step partition. This distinction changes the complexity bound from an exponential bound to a polynomial one.

Throughout this section the table is circular, $R \in I$, and the section is the original Y_R^*. Put

$$\mathscr{T}_{R,\theta} = e^{i(b\bar\tau_R + vc)}\mathcal{Q}_{R,(u,b),v}, \qquad \theta = (u,b,v) \in \mathbb{T}^2 \times \mathbb{R} \times \mathbb{T}.$$

This is the uncentered operator. On physical densities its action is

$$\mathscr{T}_{R,\theta}h = \big(e^{i(u\cdot\kappa_R + b\tau_R + v\eta_R)}h\big)\circ T_R^{-1}. \tag{157.1}$$

It is defined on the local strong and weak spaces by the preceding flight and section multipliers already proved. The expression does not require the forward roof alone to be a strong-space multiplier. The torus convention is $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$; its metric is shortest angular distance, combined with Euclidean distance in the real roof coordinate. The roof band B is always fixed before any limit in the collision count.

157.1. **A refinement of the existing inverse-curve partition.** Retain $\mathcal{G}_m(W), \mathcal{I}_m(W), j_U, \varsigma$ and the four geometric sums of Section 117.

Lemma 157.1 (Linear occupation complexity on one inverse curve)**.** *There is $K < \infty$, independent of m, R, W, such that every $U \in \mathcal{G}_m(W)$ can be cut into at most $r_m = 1 + Km$ intervals on which $(\eta_R \circ T_R^j)_{0\le j<m}$ is constant. On each such interval the accumulated uncentered phase has C^p norm at most C_B, for $|b| \le B$, uniformly in u, v, m, R.*

If $\mathcal{G}_m^\eta(W)$ denotes this refinement, and j_V is the supremum of the same stable Jacobian restricted to V, then

$$\sum_{V\in\mathcal{G}_m^\eta(W)} j_V \le Cr_m, \tag{157.2}$$

$$\sum_{V\in\mathcal{G}_m^\eta(W)} (|V|/|W|)^\varsigma j_V \le Cr_m, \tag{157.3}$$

$$\sum_{\substack{U\in\mathcal{I}_m(W)\\ V\subset U}} (|V|/|W|)^\varsigma j_V \le Cr_m\theta_*^{(1-\varsigma)m}, \tag{157.4}$$

$$\sum_{V\in\mathcal{G}_m^\eta(W)} j_V^{1-\varsigma} \le Cr_mC_3^m. \tag{157.5}$$

The designation "without a long ancestor" in the third sum is that of the original collision tree; no new growth lemma for an iterated section partition is being assumed.

Proof. Every intermediate image T_R^jU is a regular stable graph. Its slope is negative and bounded away from zero and infinity. A horizontal or vertical side of one of the five section rectangles is therefore crossed at most once, after the fixed angular chart cuts. This is the same transversality used in Lemma 151.1. There are a fixed number of sides. Each of the m intermediate images contributes at most that number of cut points on U, since the map from U to its image is one-to-one. Their union has at most Km points, not K^m points.

On a component the occupation phase is constant, of modulus one. The displacement label is also constant on the old regular branch. The roof action estimate (117.11) supplies the remaining C^p bound. Normalized stable-Jacobian distortion bounds are unchanged by restriction to a subinterval.

For nonnegative lengths $l_1,\dots,l_r$ with sum l, $\sum l_i^\varsigma \le r^{1-\varsigma}l^\varsigma$. Apply this to the pieces of each U, and use $j_V \le j_U$. The first and fourth sums cost at most r_m times their old sums. The second and third cost at most $r_m^{1-\varsigma}$ times the corresponding old length sums. The weaker displayed bounds follow since $r_m \ge 1$. □

We also need compatibility with the unstable comparison; counting cuts on a single curve would not by itself establish this.

Lemma 157.2 (Occupation matching and its unmatched cost)**.** *The matched pairs of Section 117 admit a refinement with at most Cr_m pieces per old pair. On each retained pair the occupation itineraries agree. Their graph distance retains the old $C\Lambda^{-m}\epsilon$ bound, and the two accumulated weights have C^q difference at most $C_B\epsilon^{1-q}$. The additional unmatched pieces V satisfy*

$$|T_R^mV| \le C\epsilon, \qquad \sum_V j_V^{1-\varsigma} \le Cr_mC_3^m. \tag{157.6}$$

All constants are uniform in the occupation frequency.

Proof. Start with the regular connector foliation of an old matched pair. At time j, its connectors have length at most $C\Lambda^{-(m-j)}\epsilon$, by backward contraction of unstable connectors. Remove connectors meeting a section side. Transversality of that side with the two stable graphs bounds the length removed on either graph at time j by $C\Lambda^{-(m-j)}\epsilon$ per crossing. Forward stable contraction from j to m makes the corresponding terminal length at most $C\Lambda^{-2(m-j)}\epsilon \le C\epsilon$. There are at most a fixed number of crossings at each time, by the preceding lemma.

Use closed removed intervals before trimming to a common graph domain. If changing from the connector pairing to the common graph-coordinate pairing would cross an occupation cut, remove that interval as well. The old trimming moves a point by at most $C\epsilon$ along an inverse

stable graph. At time j this displacement is at most $C\Lambda^{-j}\epsilon$; its terminal image is at most $C\Lambda^{-m}\epsilon$. It has the same admissible unmatched cost. This construction has at most Cr_m endpoints per old pair and ensures agreement of the entire half-open occupation itinerary on the retained common domains. Restriction to equal graph domains preserves the old bound on the distance of the inverse graphs.

On retained pairs the occupation factors cancel exactly in the weight difference. The old roof-action estimate and interpolation give $C_B\epsilon^{1-q}$, without an extra factor growing with m. For every removed piece bounded distortion gives $j_V|V| \le C\epsilon$. The last sum in (157.6) follows by charging each restriction to its old parent and using the Cr_m multiplicity and (117.8). Thus their stable-norm contributions sum to at most $C_B r_m C_3^m \epsilon^{\varsigma}\|h\|_s$. The old collision singularities and homogeneity cuts remain in the old unmatched family throughout. No connector through such a cut is used in the argument. □

157.2. **The three estimates and their spectral consequence.**

Theorem 157.3 (Full occupation-torus inequalities)**.** *For each fixed $B < \infty$, all real θ with $|b| \le B$, and $m \ge 0$, the local common collision norms satisfy*

$$|\mathscr{T}_{R,\theta}^m h|_w \le C_B(1+m)|h|_w, \tag{157.7}$$

$$\|\mathscr{T}_{R,\theta}^m h\|_s \le C_B(1+m)(\theta_*^{(1-\varsigma)m} + \Lambda^{-qm})\|h\|_s + C_B(1+m)|h|_w, \tag{157.8}$$

$$\|\mathscr{T}_{R,\theta}^m h\|_u \le C_B(1+m)\Lambda^{-\gamma m}\|h\|_u + C_B(1+m)C_4^m\|h\|_s. \tag{157.9}$$

In a B-dependent equivalent strong norm there are $\sigma_B < 1$ and $C'_B < \infty$ such that

$$\|\mathscr{T}_{R,\theta}^m h\|_B \le C'_B\sigma_B^m\|h\|_B + C'_B(1+m)|h|_w. \tag{157.10}$$

The essential spectral radius is uniformly less than one on the full fixed band and occupation torus. The operators vary locally strong-to-weak continuously in (R,θ).

Proof. For the weak norm split the weighted curve pairing (122.7) over the intervals in Lemma 157.1. Each normalized test has C^p norm at most C_B. Equation (157.2) proves (157.7).

For the strong stable norm subtract the average a_U of the pulled-back terminal test on each *old* inverse piece U. Keep that same average on every occupation subinterval $V \subset U$. Stable contraction gives $\|\psi \circ T^m - a_U\|_{C^q(U)} \le C\Lambda^{-qm}|W|^{-\varsigma}$. Its refined sum is controlled by (157.3). The average on old never-long pieces is controlled by (157.4). For all other pieces retain their old last long ancestor. The weak payment of each descendant now has at most r_m summands. In the notation of Section 117 its total is bounded by

$$C_B r_m \delta_0^{-\varsigma}|h|_w \sum_{k=0}^{m} \theta_*^{m-k} \sum_{V\in\mathcal{G}_k(W)} (|V|/|W|)^{\varsigma} j_V \le C_B r_m |h|_w.$$

This is the last-long sum from [4, equations (4.7)–(4.11)], with its length factors unchanged and an explicitly counted multiplicity. Newly short occupation pieces are not incorrectly put in the old never-long family. This proves (157.8).

For the unstable norm use Lemma 157.2. The old matched leading comparison acquires at most Cr_m copies. The unmatched contribution is bounded there. The other errors have powers $\epsilon^{p-q}, \epsilon^{1/3-q}, \epsilon^{1-q}$ and the old Jacobian sum, now with multiplicity at most Cr_m. Each of these powers, as well as ς, exceeds γ by (117.4). Division by ϵ^{γ} proves (157.9), after enlarging C_4. This is the matched/unmatched calculation of [4, Section 4.3], not an inference from the one-step multiplier norm.

Choose $N = N(B)$ so that both leading strong coefficients at time N are below $1/4$. This is possible despite their factor $1+N$. Next choose $a_B > 0$ so that the stable cross term, multiplied by a_B, is below $1/4$. With $\|h\|_B = \|h\|_s + a_B\|h\|_u$ this gives $\|\mathscr{T}^N h\|_B \le \frac{1}{2}\|h\|_B + D_B|h|_w$. Iteration

uses the already established bound $|\mathscr{T}^{jN}h|_w \le C_B(1+jN)|h|_w$. The geometric convolution with 2^{-j} is at most $C_B(1+m)$; treating the remaining powers below N proves (157.10). Compact embedding gives the essential-radius assertion. This conclusion needs no claim that the weak powers are uniformly bounded.

For parameter continuity, the roof operator is already locally strong-to-weak continuous. The symmetric difference of the two moving sections lies in finitely many strips whose width tends to zero. Its intersection with a stable curve consists of a bounded number of intervals of length $O(\delta)$, where δ bounds the motion of the rectangle sides. The strong stable norm bounds their weak pairing by $C\delta^{\varsigma}\|h\|_s$. Thus $\|M_{\eta_R} - M_{\eta_{R_0}}\|_{\mathcal{B}\to\mathcal{B}_w} \le C\delta^{\varsigma}$. The exact affine formula (151.2), weak boundedness of one roof step, and the existing roof comparison give the assertion on the full occupation torus. Frequency continuity is strong analyticity in each local coordinate chart. Completion extends the estimates from smooth physical densities to the stated spaces. □

Theorem 157.4 (Physical peripheral spectrum on the full fixed band)**.** *For every fixed B, $\sup_{R,\theta,m}\|\mathscr{T}_{R,\theta}^m\|_B$ is finite for $|b| \le B$. At a frequency θ there is a unit-modulus eigenvalue if and only if $\theta \in \mathcal{G}_R$. It is then the single simple eigenvalue $e^{-is_R(\theta)}$, with a bounded physical eigenvector $q_\theta\nu$ satisfying* (155.1)*.*

Every compact subset of $\{(R,\theta) : |b| \le B,\ \theta \notin \mathcal{G}_R\}$ admits constants $C < \infty$ and $\rho < 1$ for which $\|\mathscr{T}_{R,\theta}^m\|_B \le C\rho^m$. This statement is about a compact subset of the joint *parameter-frequency complement, not a pointwise choice of gaps.*

Proof. The polynomial power bound in (157.10) excludes spectral radius greater than one. Quasi-compactness leaves a finite-dimensional peripheral space at each fixed parameter and frequency. We first exclude its Jordan blocks, without presupposing power boundedness. For a smooth bounded initial density and a smooth global test the physical pairing of every power is bounded independently of m. Its spectral decomposition is a finite sum of exponential polynomials plus an exponentially decaying term. A bounded exponential polynomial has no positive-degree coefficient: divide by its highest power of m, multiply by a conjugate peripheral phase, and take Cesaro means. Distinct phases have vanishing cross means. Faithfulness on the strong space, Lemma 117.3, therefore makes every positive-degree vector coefficient zero. Density of smooth sources in the strong space eliminates every nonzero nilpotent peripheral part. The powers at this fixed point are now bounded.

The physical Cesaro argument of Lemma 117.4 applies verbatim to (157.1). Each peripheral vector has a bounded physical representative. The modulus is constant by ergodicity. Quotients of two such circle-valued representatives solve an ordinary collision eigenfunction equation. Mixing makes their eigenvalues equal, and ergodicity makes their quotient constant. Thus the peripheral eigenvalue is unique and simple. Its physical equation is exactly (155.1) with eigenvalue e^{-is}.

Conversely, if a measurable phase exists and all strong eigenvalues are strictly inside the unit circle, smooth endpoint pairings decay exponentially. Approximate the measurable phase in L^1, keep that approximation fixed while $m \to \infty$, and then remove the approximation, just as in Lemma 155.1. Its unit-modulus pairing gives a contradiction. This proves the converse without assuming that an arbitrary measurable multiplier acts on the strong space.

Finally apply the strong-to-weak perturbation theorem with (157.10); the factor $1+m$ is bounded by $C_\epsilon(1+\epsilon)^m$. Near a point without peripheral spectrum all powers decay on a common contour. Near a point with peripheral spectrum the single rank-one projection persists, the complementary powers decay on a common contour, and the remaining eigenvalue has modulus at most one by the global polynomial bound. Its projection is locally uniformly bounded. A finite cover of the compact parameter-frequency band gives uniform power boundedness. The same argument on a compact joint nonresonant subset gives a uniform strictly smaller radius. It also

shows that the joint resonance locus is closed. Constants may depend on B; no growing-band estimate is asserted. □

158. Occupation holonomy, finite resonances and their endpoint residues

The new full-band estimate permits the measurable phase group to be used as an actual peripheral spectrum. We first strengthen its classification. The occupation term changes a stable or unstable holonomy by an integer. Countability of those changes, rather than continuity of an occupation itinerary, excludes every nonzero roof coordinate.

Theorem 158.1 (No roof coordinate in an occupation resonance)**.** *Every solution of* (155.1) *has* $b = 0$*. Consequently* $\mathcal{G}_R$ *is a finite cyclic group of order* $d_R \le D$*, where* $D < \infty$ *is independent of* R*. Its coordinates* (u, v, s) *belong to a fixed finite set of rational torus points, with possible denominators at most* D*. The lattice alternative in Theorem 155.2 is thereby excluded; the finite cyclic group is not thereby asserted trivial.*

Proof. Use a regular positive-measure product block and its stable and unstable conditional pair measures, as in the measurable holonomy proof of Theorem 119.2. Restrict the block if necessary so that the two marginals of each pair measure are bounded by a constant times ν, and so that leaf contraction has a common constant. This uses only the local product and absolute continuity input already used there. For a stable pair (x, y), $d(T_R^j x, T_R^j y) \le C\Lambda^{-j}$. The boundary of Y_R^* is a fixed finite collection of rectangle sides, with uniformly bounded tubular-neighborhood measure. Thus

$$\pi^s\{\eta_R(T_R^j x) \ne \eta_R(T_R^j y)\} \le C\nu\{d(\cdot, \partial Y_R^*) \le C\Lambda^{-j}\} \le C'\Lambda^{-j}.$$

Invariance of ν preserves the marginal domination at time j. The first Borel–Cantelli lemma now shows that

$$n^s(x, y) = -\sum_{j\ge 0}(\eta_R(T_R^j y) - \eta_R(T_R^j x)) \in \mathbb{Z}$$

is a finite sum almost everywhere for the stable pair measure. Backward contraction gives likewise the finite integer

$$n^u(x, y) = \sum_{j<0}(\eta_R(T_R^j y) - \eta_R(T_R^j x))$$

for unstable pairs. Independence is not used.

Approximate the measurable circle-valued q by smooth bounded functions in L^1. Marginal domination and leaf contraction show that the ratio of the two far-future values tends to one in measure on stable pairs; the far-past ratio does so on unstable pairs. Fix the approximation first, take time to infinity next, and only then remove the approximation. The displacement itinerary agrees along the corresponding regular leaves, while the constant collision phase cancels between the two endpoints. The action identity and the finite occupation sums therefore give

$$\frac{q(y)}{q(x)} = \exp\left(ib\int_{[x,y]^{s/u}} \vartheta_R + iv n^{s/u}(x, y)\right) \tag{158.1}$$

almost everywhere for the respective pair measures. The sign in n^s comes from forward telescoping; n^u uses backward telescoping.

Disintegration and Fubini permit multiplication around almost every small stable–unstable rectangle Q of this product block. They yield

$$\exp\left(ib\int_Q d\vartheta_R\right) \in \{e^{ivj} : j \in \mathbb{Z}\}. \tag{158.2}$$

The set on the right is countable, including when it is dense in the circle. The area on the left has no atoms under the product rectangle parameters. To check this last point, fix three

sides and vary the fourth leaf through a nonatomic transverse conditional parameter. The resulting rectangles are nested. Their symplectic area is continuous and strictly monotone, since $d\vartheta_R = R\cos\varphi\, d\varphi \wedge d\alpha$ has a fixed nonzero sign on the regular block. Each prescribed area therefore has at most one value of that transverse parameter. Absolute continuity of the product coordinates and Fubini give zero measure to any countable collection of such area levels. The argument also works on a positive-measure product block with gaps: restrict the nonatomic transverse parameters to that block.

If $b \neq 0$, the inverse image under $A \mapsto e^{ibA}$ of a countable set is countable. Equation (158.2) would then hold only on a null set of rectangles, a contradiction. Hence $b = 0$. The uniform cyclic and denominator conclusions now follow from Theorem 155.2. No choice of periodic representative of q has been made. □

Corollary 158.2 (Full-occupation decay away from zero roof frequency)**.** *For each $0 < \epsilon < B < \infty$ there are $C_{B,\epsilon} < \infty$ and $\rho_{B,\epsilon} < 1$ such that*

$$\|\mathscr{T}^m_{R,(u,b,v)}\|_B \le C_{B,\epsilon}\rho^m_{B,\epsilon} \quad (R \in I,\ u \in \mathbb{T}^2,\ v \in \mathbb{T},\ \epsilon \le |b| \le B).$$

Proof. This is a compact joint nonresonant set by the preceding theorem. Apply Theorem 157.4. □

158.1. **The actual source coefficient at every resonance.** Fix R and $\gamma = (u, b, v) \in \mathcal{G}_R$, and let s_γ be its unique scalar phase. Normalize $|q_\gamma| = 1$. Write Π_γ for the spectral projection of $\mathscr{T}_{R,\gamma}$ at $\zeta_\gamma = e^{-is_\gamma}$.

Proposition 158.3 (Physical projection and general endpoint residue)**.** *For every bounded physical density $a\nu \in \mathcal{B}$,*

$$\Pi_\gamma(a\nu) = q_\gamma \nu \int \overline{q_\gamma} a\, d\nu. \tag{158.3}$$

For every bounded terminal multiplier d its endpoint coefficient is

$$\ell(M_d \Pi_\gamma(a\nu)) = \nu(a\overline{q_\gamma})\nu(dq_\gamma). \tag{158.4}$$

In particular the actual section-to-section coefficient is $A_\gamma = |\nu(\eta_R q_\gamma)|^2$ at every scalar eigenvalue, not only at one.

More generally, before using $b = 0$, invariance alone gives the exact identity

$$\begin{aligned}(e^{iv} - 1)\nu(\eta_R q_\gamma) = (e^{-is_\gamma} - 1)\nu(q_\gamma) \\ - \nu((e^{i(u\cdot\kappa_R + b\tau_R)} - 1)e^{iv\eta_R} q_\gamma).\end{aligned} \tag{158.5}$$

Thus the pure-occupation cancellation holds when $u = b = s_\gamma = 0$ and $v \neq 0$. It does not annihilate a general residue with $u \neq 0$ or $s_\gamma \neq 0$.

Proof. The averages $M^{-1}\sum_{j<M} \zeta_\gamma^{-j} \mathscr{T}^j_{R,\gamma}$ converge in the strong space to Π_γ. On physical L^2 the same averages converge to the orthogonal projection onto the unitary eigenspace. That eigenspace is one-dimensional: quotients of two circle eigenfunctions are invariant and hence constant. Comparison on smooth global tests and strong-space faithfulness prove (158.3). The terminal pairing gives (158.4). This proof does not require M_{q_γ} or $M_{\overline{q}_\gamma}$ to be a strong multiplier.

Integrate (155.1), use invariance, and write $e^{iv\eta_R} = 1 + (e^{iv} - 1)\eta_R$. Separating $e^{i(u\cdot\kappa_R + b\tau_R)} - 1$ gives (158.5) with the displayed signs. The final assertions follow directly. For representatives with $|(u, b)| \le 1$ its last term has modulus at most $C|(u, b)|$; no division by $e^{iv} - 1$ is uniform as $v \to 0$. □

Proposition 158.4 (The same Gaussian curvature at each resonance)**.** *Put $\mu_R = (0, 0, \bar{\tau}_R, c)$. On a coordinate neighborhood of a resonance γ, the simple eigenvalue and projection are analytic in the displacement $\delta \in \mathbb{C}^4$, and*

$$\lambda_\gamma(\delta) = \zeta_\gamma \exp\left(i\delta \cdot \mu_R - \tfrac{1}{2}\delta^{\mathsf{T}}\Omega_R\delta + O(|\delta|^3)\right), \qquad \Pi_\gamma(\delta) = \Pi_\gamma + O(|\delta|). \tag{158.6}$$

The constants and radii may be chosen uniformly over all actual resonances and $R \in I$. In particular the centered modulus has a Gaussian upper bound for small real δ.

Proof. Use the analytic family $V_\delta = e^{-i\delta\cdot\mu_R}\mathscr{T}_{R,\gamma+\delta}$. Its strong analyticity follows from the image-side roof and the source-side occupation realization, not from a presumed bounded forward-roof multiplier. Isolate the simple eigenvalue ζ_γ by a resolvent contour. Let l_γ be its left functional with $l_\gamma(q_\gamma\nu) = 1$. On bounded physical strong vectors, (158.3) identifies it as integration against $\overline{q}_\gamma$.

For a fixed real direction w put $X = w \cdot X_R$ and differentiate V_{tw} in the scalar variable t at zero. Denote its eigenvalue by $\widetilde{\lambda}_\gamma(t) = e^{-itw\cdot\mu_R}\lambda_\gamma(tw)$. A first derivative, acting on a bounded physical strong vector h, has physical representative $i\mathscr{T}_{R,\gamma}(Xh)$. This is a description of the derivative of the entire operator family; no boundedness of M_X in isolation is needed. In particular

$$V_0' q_\gamma = i\zeta_\gamma q_\gamma(X \circ T_R^{-1}), \qquad l_\gamma(V_0' q_\gamma) = 0.$$

The complementary powers on this centered vector are summable in the strong norm. The derivative of the normalized right eigenvector is therefore

$$e_0' = i\sum_{k\geq 1} q_\gamma(X \circ T_R^{-k})$$

with strong convergence; each partial sum is a bounded physical representative. Applying l_γ to the second differentiated eigen-equation gives

$$\widetilde{\lambda}_\gamma''(0) = -\zeta_\gamma\left(\nu(X^2) + 2\sum_{k\geq 1}\nu(X(X \circ T_R^k))\right).$$

Continuity of $l_\gamma V_0'$ justifies passing to the strong sum. The physical covariance series is absolutely summable. Polarization identifies the Hessian as $-\Omega_R$ for the logarithm of the centered eigenvalue. Uniform contour analyticity bounds the cubic remainder by Cauchy's estimate and gives the projector estimate.

For uniformity, the joint resonance set is compact by Theorem 157.4 and lies in $b = 0$. All its peripheral eigenvalues are simple and separated from complementary spectrum locally. Strong-to-weak spectral stability and a finite cover give common contour bounds on the local spaces. The uniform positive lower bound for Ω_R then gives the Gaussian modulus bound after reducing the coordinate radius. This argument does not assert that the set of resonances of one radius persists at all nearby radii. □

159. Fixed collision coefficients and the arithmetic return local law

We now use long powers, rather than clock averages. There are two statements with different parameter quantifiers. The finite spectral reduction is uniform over the entire radius interval. The explicit arithmetic local law is proved at each fixed radius. We do not assume that a measurable phase, or its residue, varies continuously through a parameter at which an arithmetic component appears or disappears.

159.1. **A uniform finite spectral reduction of the exact inverse.** For a roof test $q \in L^1(\mathbb{R})$ with $\widehat{q} \in L^1(\mathbb{R})$ supported in $[-B, B]$, retain $P^q_{n,m,R}$ from (154.6).

Theorem 159.1 (Fixed-count resonant reduction, uniform in the radius)**.** *For each fixed B there are finitely many local parameter-frequency charts, rank-one spectral branches $\lambda_j(R,\theta), \Pi_j(R,\theta)$, and a subordinate partition of unity $\chi_0, \chi_1, \dots, \chi_J$ on $I \times \mathbb{T}^2 \times [-B, B] \times \mathbb{T}$, with the following properties. The support of χ_0 is a compact nonresonant set. Each other chart is centered at a genuine measurable resonance; its isolated branch is retained throughout that chart even at nearby parameters where its modulus is less than one. Put*

$$A_j(R,\theta) = \ell(M_{\eta_R} \Pi_j(R,\theta)(\eta_R \nu)).$$

There are $C_B < \infty$ and $\rho_B < 1$ such that, uniformly in R, n, m, k, t,

$$P^q_{n,m,R}(k,t) = \frac{1}{c(2\pi)^4} \sum_{j=1}^{J} \int_{\mathbb{T}^2} \int_{-B}^{B} \int_{\mathbb{T}} e^{-i(u\cdot k + bt + vn)} \widehat{q}(-b) \chi_j(R,\theta) \lambda_j(R,\theta)^m A_j(R,\theta)\, dv\, db\, du + O(C_B \|\widehat{q}\|_1 \rho_B^m). \tag{159.1}$$

At an actual resonance the amplitude and quadratic expansion are (158.4) *and* (158.6)*. The old remainder* (154.7) *has the same reduction with each integrand multiplied by $1 - \chi_B(v)$, with the same exponential error. In particular no continuous near-unit spectral remainder is left outside these finitely many branches on a fixed band.*

Proof. At each nonresonant point Theorem 157.4 gives a local decaying contour. At each resonant point its simple peripheral eigenvalue has a local rank-one continuation and an exponentially decaying complementary contour. Strong-to-weak spectral stability supplies uniform bounds in a smaller neighborhood. Cover the compact joint parameter-frequency set by finitely many such neighborhoods, and choose a partition subordinate to smaller sets with compact closure in them. Combine nonresonant members as χ_0. The remaining χ_j may be taken continuous in the parameter and smooth in the frequency coordinates. There is no need to select a globally continuous generator of a resonance group.

The exact physical pairing of each power is

$$\int \eta_R(x) \eta_R(T_R^m x) e^{i(u\cdot K^c_{m,R} + bS_{m,R} + vA_{m,R})}\, d\nu(x).$$

Insert the local spectral splitting after multiplying by its partition function. The error in this pairing is bounded by $C_B \rho_B^m$, since the source and terminal section multiplier norms are uniformly bounded. The exact return disintegration, the three torus inversions and the roof Fourier inversion now give (159.1). Its error is at most the uniform pairing error times the finite torus volume and $\|\widehat{q}\|_1$. Multiplication by the fixed $1 - \chi_B$ proves the remainder version. All statements concern the power at the specified m; neither Cesaro nor Abel averaging is used here. □

159.2. **The finite arithmetic coefficient.** Fix a radius R. By Theorem 158.1 choose the unique generator whose occupation coordinate is $2\pi/d$, where $d = d_R$. Write its other coordinates as

$$u = 2\pi a/d, \qquad s = 2\pi h/d, \qquad a \in (\mathbb{Z}/d\mathbb{Z})^2, \quad h \in \mathbb{Z}/d\mathbb{Z}.$$

Its circle-valued transfer function can be normalized so that $q_*^d = 1$: indeed the dth power is invariant and hence constant. Write

$$w_l = \nu\{x \in Y_R^* : q_*(x) = e^{2\pi i l/d}\}, \quad l \in \mathbb{Z}/d\mathbb{Z}, \qquad \sum_l w_l = c. \tag{159.2}$$

For $d = 1$ take $a = h = 0$, $q_* = 1$ and $w_0 = c$. Define

$$r_R(k,n,m) = a\cdot k + n + hm \pmod d, \qquad \mathfrak{a}_R(k,n,m) = \frac{d}{c^2}\sum_{l \bmod d} w_l w_{l+r_R(k,n,m)}. \tag{159.3}$$

Changing the normalization of q_* only permutes the labels and does not change this factor. It is nonnegative, at most d, and its average over any d consecutive return indices is one.

Theorem 159.2 (The exact-index, fixed-interval return law)**.** *Fix $R \in I$, a bounded interval J of positive length, and $M < \infty$. Then, uniformly in the integer targets k, n, m and real t with*

$$Z = (k/\sqrt{m}, (t - m\bar\tau_R)/\sqrt{m}, (n - mc)/\sqrt{m}), \qquad |Z| \le M,$$

one has

$$m^2 P_{n,m,R}(k,t;J) = c|J|\mathfrak{a}_R(k,n,m) g_{\Omega_R}(Z) + o_R(1). \tag{159.4}$$

Equivalently, for $V_n = n^{-1/2}(k_1, k_2, m - n/c, t - n\bar\tau_R/c)$,

$$n^2 P_{n,m,R}(k,t;J) = |J|\mathfrak{a}_R(k,n,m) g_{D_R}(V_n) + o_R(1). \tag{159.5}$$

If $\mathfrak{a}_R(k,n,m) = 0$, this event has probability exactly zero, for every roof interval. On its positive residue classes the central probabilities are bounded below by $c_{R,M,J} m^{-2}$ for all sufficiently large m. The asserted $o_R(1)$ is at fixed radius; a global radius-uniform arithmetic asymptotic and a pointwise roof density are not included.

Proof. Begin with a fixed integrable roof test q with integrable compactly supported Fourier transform. At this fixed radius there are only d resonances, all with $b = 0$. Choose disjoint small frequency neighborhoods of them. Outside these neighborhoods the full-band powers decay exponentially by Theorem 157.4. Inside the neighborhood of γ, use (158.6). The complementary powers decay exponentially. With $\delta = y/\sqrt{m}$, the inverse phase and the linear eigenvalue phase cancel their means and leave $e^{-iy\cdot Z}$. The constant phase is

$$e^{-i(u_\gamma\cdot k + v_\gamma n + m s_\gamma)}.$$

The projector amplitude tends to $|\nu(\eta_R q_\gamma)|^2$, and $\widehat q(-y_3/\sqrt{m}) \to \int q$. The Gaussian upper bound dominates the rescaled integrand; cubic remainders tend to zero on each compact y set. Splitting the rescaled integral into a fixed ball and its Gaussian tail proves, uniformly on the target compact set,

$$m^2 P^q_{n,m,R}(k,t) = \frac{\int q}{c} g_{\Omega_R}(Z) \sum_{\gamma\in\mathcal{G}_R} e^{-i(u_\gamma\cdot k + v_\gamma n + m s_\gamma)} |\nu(\eta_R q_\gamma)|^2 + o_R(1). \tag{159.6}$$

The argument is an estimate of a fixed coefficient, not a Tauberian passage from an average.

The transfer functions of the cyclic group can be chosen as q_*^j, $0 \le j < d$. Finite Fourier orthogonality gives

$$\sum_{j=0}^{d-1} e^{-2\pi i j r/d} \left|\sum_l w_l e^{2\pi i j l/d}\right|^2 = d\sum_l w_l w_{l+r} = c^2 \mathfrak{a}_R(k,n,m). \tag{159.7}$$

This also proves nonnegativity and the claimed residue average. Equation (159.6) is thus the desired formula with $|J|$ replaced by $\int q$.

For the sharp interval use the pointwise upper and lower Fourier envelopes already used in Theorem 123.3. Their integrals are $|J| \pm 2\pi/B$; the minorant need not be nonnegative. Multiply their inequalities by the unchanged positive exact-index source. For each fixed B take $m \to \infty$, and only then let $B \to \infty$. Since all resonances have roof coordinate zero, the same arithmetic factor occurs on both sides for every B. This proves (159.4). In the original covariance coordinates

use (152.5) and $n/m = c + O(m^{-1/2})$, exactly as in Corollary 154.2. The bounded factor $\mathfrak{a}_R \le d$ preserves the error and gives (159.5).

Finally the iterated phase equation on the exact section event gives

$$q_*(T_R^m x) = e^{2\pi i r_R(k,n,m)/d} q_*(x).$$

If the sum of products in (159.3) is zero, every permitted pair of section phase classes has a null initial or null terminal class. Invariance of T_R makes the event null. There are only finitely many positive residue factors at fixed R, so their minimum is positive. Uniform positivity of the Gaussian on the stated target compact set gives the last lower bound. □

Corollary 159.3 (Which exact-index complement remains)**.** *For every fixed radius and compactly band-limited roof test,*

$$m^2 \mathcal{R}^q_{n,m,R}(k,t) = c\left(\int q\right)(\mathfrak{a}_R(k,n,m) - 1) g_{\Omega_R}(Z) + o_R(1).$$

In particular the unmodulated fixed-interval local law follows if all nontrivial section residues vanish, and in particular if $d_R = 1$. The general theorem does not assume either statement.

Proof. Subtract Proposition 154.4 from (159.6), using (159.7). No inference from the pure occupation Abel limit enters this subtraction. □

Corollary 159.4 (Selected endpoints and the fine-window limit)**.** *For nonnegative bounded endpoints a_0, a_1 with $\eta_R a_0, \eta_R a_1$ multiplier-bounded, define $w_l^{(i)} = \nu(\eta_R a_i \mathbf{1}_{\{q_* = e^{2\pi i l/d}\}})$ for $i = 0, 1$. The fixed-radius selected version of* (159.4) *is obtained by replacing $\mathfrak{a}_R$ by*

$$\mathfrak{a}_R^{a_0,a_1}(k,n,m) = \frac{d}{c^2} \sum_l w_l^{(0)} w_{l+r_R(k,n,m)}^{(1)}.$$

Its average over d consecutive return indices is $\nu_R^*(a_0)\nu_R^*(a_1)$. Thus averaging any diverging subdiffusive block recovers the unmodulated amplitude in Corollary 154.3 at fixed R.

Proof. Use the general residue formula (158.4) in the same fixed-band proof. Its finite Fourier sum equals the displayed cross correlation of phase-class masses. The sum over all residues factors as $(\sum_l w_l^{(0)})(\sum_l w_l^{(1)})$. Within a subdiffusive block the Gaussian varies by $o(1)$, and the incomplete final residue block has relative size at most $d/H_m \to 0$. The error in the fixed-index formula is uniform over the target compact set and remains $o_R(1)$ after division by the block width. No Gaussian amplitude for an arbitrary path-dependent selector is asserted. □

The remaining raw and parameter questions. The full occupation-torus complement is now a controlled spectral problem on each fixed roof band. At a fixed radius its exact-index contribution is the finite arithmetic expression above. This does not assert that the arithmetic factor equals one, or that a pointwise selection of exact resonances gives a radius-uniform asymptotic near a change of resonance type. The uniform statement in such regions is (159.1), which retains all nearby spectral branches. Establishing triviality of the section residues, or tracking their parameter transitions, is a distinct remaining task.

A fixed positive roof interval is not a density value. None of the arguments here bounds the complete second-derivative coarea sum, removes every critical edge, or permits B to grow with m inside an unquantified spectral estimate. The common pointwise raw correction and the full roof-frequency estimate in Theorem 143.1 remain requirements of that unchanged theorem. All its inherited geometry and inversion arguments are retained.

160. A uniform local expansion through changes of arithmetic type

The fixed-radius arithmetic formula does not describe a sequence of radii on which a peripheral eigenvalue moves into the unit disk at the inverse collision scale. We retain that eigenvalue and evaluate its Fourier integral. The resulting formula is a finite sum of Gaussian terms with explicitly specified damping, phase and drift. It is uniform in the radius and does not require a continuous choice of an exact resonance group.

Write $\mathsf{f}_R = X_R + \mu_R$, where $\mu_R = (0, 0, \bar{\tau}_R, c)$ and X_R is (151.1). Thus $\mathsf{f}_R = (\kappa_{R,1}, \kappa_{R,2}, \tau_R, \eta_R)$. For a target put $y = (k_1, k_2, t, n)$, in this coordinate order, and $Z = (y - m\mu_R)/\sqrt{m}$. The probability $P_{n,m,R}(k, t; J)$ retains the exact return meaning of Theorem 159.2.

160.1. **Local spectral maxima and their scalar data.**

Lemma 160.1 (Continuous analytic scalar data)**.** *Near each point of the joint resonance locus, the simple eigenvalue $\lambda(R, \theta)$ and its logarithm are continuous in R, analytic in the complex frequency coordinates, and have uniformly bounded frequency derivatives on a smaller neighborhood. Their frequency derivatives are continuous in R there. The same assertions hold for*

$$A(R, \theta) = \ell\big(M_{\eta_R} \Pi(R, \theta)(\eta_R \nu)\big).$$

These are scalar continuity statements on local common spaces. No strong-norm differentiability in the table parameter is asserted.

Proof. Use the local rank-one continuation and complementary contour of Theorems 157.4 and 159.1. Shrink its complex frequency neighborhood so that the continued eigenvalue is nonzero and its modulus exceeds the complementary contour radius. The contour estimates and strong frequency analyticity give common bounds for the projection, the eigenvalue and all derivatives on a smaller polydisk. The strong-to-weak spectral stability already used in those theorems gives continuity of the simple eigenvalue at real frequencies. The common complex bounds make every convergent parameter sequence a normal family of holomorphic eigenvalues. Every subsequential limit agrees on a real open frequency ball, and hence on the polydisk, with the limiting eigenvalue; uniqueness follows by the one-variable identity theorem successively in each coordinate. Thus convergence is locally uniform also in complex frequencies. Cauchy's formula gives continuity of the frequency derivatives. It is not differentiation in the radius.

We justify the moving endpoint amplitude without asserting strong-norm continuity of the indicator multiplier. For a fixed integer N, put

$$F_N(R, \theta) = \int \eta_R(x) \eta_R(T_R^N x) e^{i\theta \cdot \sum_{j<N} \mathsf{f}_R(T_R^j x)} \, d\nu(x).$$

This finite pairing is continuous in R: outside the finitely many singularity preimages and section boundaries at the limiting radius, the finite itinerary and all membership decisions persist. Dominated convergence applies, also uniformly on compact complex frequency sets for this fixed N. The local splitting gives

$$\lambda(R, \theta)^{-N} F_N(R, \theta) = A(R, \theta) + O(\rho_1^N), \qquad \rho_1 < 1,$$

uniformly on a smaller neighborhood. Therefore A is continuous as a uniform limit of continuous scalar functions. Its holomorphy and common bounds follow either from this formula or from the projection pairing. Cauchy's formula again gives derivative continuity. A finite cover of the compact joint resonance locus supplies the stated uniform bounds. □

Lemma 160.2 (Real maxima of the continued branches)**.** *One may choose finitely many resonance charts and frequency-smooth partition functions χ_j, with compact supports in those*

charts, so that the spectral complement decays exponentially on each fixed roof band and the following data are defined on every parameter projection of a chart:

$$a_j(R)\in\mathbb{R}^4,\qquad z_j(R)=\lambda_j(R,a_j(R)),\qquad \kappa_j(R)=-\log|z_j(R)|\ge 0,$$
$$\mu_j(R)=\frac{1}{i}\nabla_\theta\log\lambda_j(R,a_j(R))\in\mathbb{R}^4,$$
$$H_j(R)=-\nabla_\theta^2\log\lambda_j(R,a_j(R)). \tag{160.1}$$

Here a_j is the unique real critical point of $\log|\lambda_j|$ in its chart. All these data are continuous, H_j is complex symmetric, and $\operatorname{Re}H_j\ge aI_4$ for a common $a>0$. In local real frequency coordinates,

$$|\lambda_j(R,a_j+x)|\le e^{-\kappa_j-a|x|^2}, \tag{160.2}$$
$$\log\lambda_j(R,a_j+x)=\log z_j+i\mu_j\cdot x-\tfrac12 x^{\mathsf T}H_jx+O(|x|^3). \tag{160.3}$$

If $\kappa_j(R)=0$, then $a_j(R)$ is an actual resonance. At such a parameter its roof coordinate is zero, $\mu_j(R)=\mu_R$, and $H_j(R)=\Omega_R$.

Proof. At an actual resonance, Proposition 158.4 gives zero real gradient and negative definite real Hessian for $\log\lambda$. Lemma 160.1 makes the Hessian uniformly negative on a smaller real ball and for nearby parameters. The equation for its real gradient has a unique nearby zero. For completeness this parameter version of the implicit function argument needs only continuity in R: multiply the gradient by the inverse of its Hessian at the central parameter, and use the resulting uniform contraction on a smaller closed ball. The fixed point is continuous in R and remains strictly inside that ball.

The real gradient vanishes there, so the complex gradient is purely imaginary. Taylor's formula gives (160.3); integrating the negative real Hessian along a segment gives (160.2), after reducing a. All real-frequency spectral radii are at most one by Theorem 157.4, which proves $\kappa_j\ge 0$. Equality identifies a peripheral frequency, so the last assertions follow from Theorem 158.1 and Proposition 158.4.

Choose these charts first around the compact joint resonance locus, and then cover the remaining compact set by nonresonant charts. A subordinate partition may have common bounds for each fixed number of frequency derivatives. Each peak is defined on an open parameter neighborhood of the compact parameter support of its partition function, including when that function vanishes at the peak. All actual resonances have roof coordinate zero. Thus the peak charts may be fixed in one small roof neighborhood, independently of any later fixed outer band. Enlarging that band adds only a compact exponentially decaying complement. No limit is taken with the outer band growing. □

For a complex symmetric matrix H with positive real part define

$$G_H(z)=(2\pi)^{-4}\int_{\mathbb{R}^4}e^{-iv\cdot z-v^{\mathsf T}Hv/2}\,dv,\qquad z\in\mathbb{R}^4. \tag{160.4}$$

This converges absolutely. On the matrix set in the preceding lemma, G_H and every fixed derivative in z are uniformly bounded. For real $H>0$ it is the ordinary density g_H. Put

$$b_j(R)=(a_j(R))_3,\qquad B_j(R)=\chi_j(R,a_j(R))A_j(R,a_j(R)),$$
$$\mathcal{K}_{m,R}(y)=\frac1c\sum_j e^{-ia_j(R)\cdot y}z_j(R)^mB_j(R)G_{H_j(R)}\left(\frac{y-m\mu_j(R)}{\sqrt m}\right),$$
$$\mathcal{L}_{m,R}(y)=\operatorname{Re}\mathcal{K}_{m,R}(y). \tag{160.5}$$

Only charts whose parameter supports contain R contribute. Different real lifts of the torus coordinates give the same expression, since k_1,k_2,n are integers. The data in this formula are

values and the first two frequency derivatives of finitely many scalar spectral branches. The formula contains no unevaluated Fourier integral of an operator power; (160.4) is an ordinary Gaussian integral. The peak charts are fixed once as above.

160.2. **Evaluation of the fixed coefficient.**

Theorem 160.3 (A uniform transition formula at the exact return index)**.** *For every fixed bounded interval J of positive length,*

$$\sup_{R,n,k,t}\left|m^2P_{n,m,R}(k,t;J)-|J|\mathcal{L}_{m,R}(k_1,k_2,t,n)\right|\longrightarrow 0\qquad(m\to\infty),\tag{160.6}$$

where $m\ge n\ge 1$ and the supremum is over all the other targets. The kernels are uniformly bounded and $\sup_{R,y}(-\mathcal{L}_{m,R}(y))_+\to 0$ on these targets. For a fixed integrable roof test with integrable compactly supported Fourier transform,

$$m^2P^q_{n,m,R}(k,t)=\left(\int q\right)\mathcal{K}_{m,R}(y)+o(1)\tag{160.7}$$

uniformly in the same parameters, including complex q. Moreover $\sup_{R,y}|\operatorname{Im}\mathcal{K}_{m,R}(y)|\to 0$. At each fixed radius the formula reduces, on central compact sets, to

$$\mathcal{L}_{m,R}(y)=c\mathfrak{a}_R(k,n,m)g_{\Omega_R}(Z)+o_R(1).$$

Thus the uniform formula retains a branch even when its modulus is strictly below one at the radius in question.

Proof. Use the fixed-count reduction (159.1). On chart j set $\theta=a_j+x$ and then $x=v/\sqrt{m}$. Equations (160.2)–(160.3) give its contribution, after multiplication by m^2, as

$$\frac{1}{c}e^{-ia_j\cdot y}z_j^mB_j\widehat{q}(-b_j)G_{H_j}\left(\frac{y-m\mu_j}{\sqrt{m}}\right)+o(1).\tag{160.8}$$

The error is uniform in R and in all targets. Indeed the target oscillation has modulus one. The real Gaussian bound dominates the rescaled integrand. On $|v|\le m^{1/12}$ the cubic error is bounded by $Cm^{-1/2}|v|^3$ times a fixed Gaussian; outside that ball the Gaussian tail tends to zero uniformly. Frequency variation of the projector and partition costs $O(m^{-1/2})$ in the integrated estimate. Continuity of $\widehat{q}$ pays its remaining variation. Extend the smooth partition by zero inside the larger chart; the same estimate then holds if its value at the peak is zero. Summing the finitely many charts and the exponentially decaying complement proves (160.8) with a uniform total error.

We may replace $\widehat{q}(-b_j)$ by $\widehat{q}(0)=\int q$ in this sum. Here is the uniform argument. On each compact parameter support, $\kappa_j=0$ implies $b_j=0$. For any fixed $\delta>0$, continuity and compactness give a positive minimum of κ_j on $|b_j|\ge\delta$, unless that set is empty. Its contribution is exponentially small. On $|b_j|<\delta$, uniform continuity of $\widehat{q}$ gives a bound which tends to zero with δ. First let $m\to\infty$ and then $\delta\to 0$. This proves (160.7). All prefactors and Gaussian integrals are bounded, so $|\mathcal{K}_{m,R}|\le C$. Taking one fixed real nonnegative band-limited test of integral one shows that its imaginary part tends to zero uniformly.

For $\mathbf{1}_J$ use the positive-order upper and lower band-limited envelopes from Theorem 123.3. Their integrals are $|J|\pm 2\pi/B$. The upper envelope is nonnegative; the lower need not be. Order is preserved by the original exact-index source. For fixed B, (160.7) bounds the upper and lower expressions by their masses times $\mathcal{L}_{m,R}$, with uniform errors. Their difference and $|\mathcal{L}_{m,R}|\le C$ give a limsup error at most C/B. Let $B\to\infty$ only after the collision limit. Positivity of the actual probability gives the assertion about the negative part by taking one interval of length one. This is a qualitative limit, not a polynomial local-limit rate.

At fixed R, charts whose peaks have $\kappa_j > 0$ decay exponentially. Every other peak is an actual resonance, where its drift and curvature are μ_R, Ω_R. The partition weights at each such point sum to one. Proposition 158.3 and finite Fourier orthogonality (159.7) now give the last formula. In contrast, along $R = R_m$ a factor $e^{-m\kappa_j(R_m)}$ need not vanish merely because every $\kappa_j(R_m)$ is positive. □

Corollary 160.4 (A common, slowly shrinking roof interval)**.** *There is a deterministic sequence $h_m \downarrow 0$ such that, for $J_m = [0, h_m]$,*

$$\sup_{R,n,k,t} \left| \frac{m^2}{h_m} P_{n,m,R}(k, t; J_m) - \mathcal{L}_{m,R}(y) \right| \longrightarrow 0.$$

No specified polynomial rate of decrease for h_m is asserted.

Proof. Apply (160.6) to $[0, 1/j]$ for each integer $j \geq 1$. Choose increasing integers M_j such that its absolute error is at most j^{-2} whenever $m \geq M_j$. Set $h_m = 1/j$ for $M_j \leq m < M_{j+1}$, increasing M_j if necessary. The normalized error is at most $1/j$. This diagonal uses only fixed-band statements at each fixed j; it supplies neither a density value nor a high-roof frequency estimate at a prescribed growing band. □

161. Conditional bridges for the unchanged exact return event

The scalar arithmetic coefficient can change with the radius. A path which is tied to its own endpoint has a more stable conditional limit: the drift of each long-lived spectral branch cancels before the Fourier inverse is taken. We prove this cancellation, a conditional fourth-moment bound, and the passage from the collision clock to the actual return clock. Conditioning always uses the original event.

For a positive definite matrix C, let $\mathbb{B}_C$ denote the centered continuous Gaussian bridge with covariance

$$\mathbb{E}\mathbb{B}_C(s)\mathbb{B}_C(t)^{\mathsf{T}} = (s \wedge t - st)C.$$

Put $\mathsf{S}_{j,R} = \sum_{a<j} \mathsf{f}_R \circ T_R^a$, using the coordinate order in Section 160. Define polygonal processes by their values at grid points:

$$\mathcal{B}_{m,R}(j/m) = m^{-1/2} \left(\mathsf{S}_{j,R} - \frac{j}{m} \mathsf{S}_{m,R} \right), \tag{161.1}$$

$$\mathcal{Y}_{n,R}(l/n) = n^{-1/2} \left(J_{l,R} - \frac{l}{n} J_{n,R} \right). \tag{161.2}$$

Both processes are tied to their actual, not nominal, terminal record. Fix a bounded interval J with $|J| > 0$ and write

$$E = E_{n,m,R}(k, t; J) = \{K_{n,R} = k,\ N_{n,R} = m,\ T_{n,R} - t \in J\}, \qquad p_E = \nu_R^*(E). \tag{161.3}$$

161.1. **Pinned characteristic functions before division by the mass.** Let $0 = s_0 < s_1 < \cdots < s_h = 1$ be fixed. Write $m_i = \lfloor m s_i \rfloor - \lfloor m s_{i-1} \rfloor$ and $t_i^{(m)} = m_i/m$. Given fixed vectors w_i, put $v_i^{(m)} = w_i - \sum_j t_j^{(m)} w_j$. Then $\sum_i m_i v_i^{(m)} = 0$. The exponential of

$$m^{-1/2} \sum_i v_i^{(m)} \cdot (\mathsf{S}_{\lfloor m s_i \rfloor, R} - \mathsf{S}_{\lfloor m s_{i-1} \rfloor, R})$$

is exactly the characteristic test of the corresponding increments of (161.1). These tests determine its finite-dimensional laws. Let $N_{n,m,R}^q$ be the expectation of this exponential times the exact displacement, occupation and two-section indicators, with a roof test $q(S_{m,R} - t)$, under $c^{-1}\nu$.

Lemma 161.1 (Uniform pinned factorization on a fixed roof interval)**.** *For the preceding fixed observation times and vectors, uniformly in R, n, k, t,*

$$m^2 \left[N^J_{n,m,R}(k,t) - \Phi_{m,R}\, P_{n,m,R}(k,t;J) \right] \longrightarrow 0, \tag{161.4}$$

$$\Phi_{m,R} = \exp\left(-\frac{1}{2} \sum_i t_i^{(m)} (v_i^{(m)})^{\mathsf{T}} \Omega_R v_i^{(m)} \right). \tag{161.5}$$

Here N^J denotes the test with $\mathbf{1}_J$. The factor in (161.5) *tends to the characteristic function of the corresponding increments of $\mathbb{B}_{\Omega_R}$. No lower bound for the conditioning probability is used in this lemma.*

Proof. First take an integrable band-limited roof test. Fourier inversion gives the chronological product

$$\ell \left(M_{\eta_R} \mathscr{T}^{m_h}_{R,\theta+v_h^{(m)}/\sqrt{m}} \cdots \mathscr{T}^{m_1}_{R,\theta+v_1^{(m)}/\sqrt{m}} (\eta_R \nu) \right) \tag{161.6}$$

inside the same inverse as in (159.1). All block lengths are positive multiples of m for large m. On the compact nonresonant complement, the small frequency shifts remain in common decaying neighborhoods, so this part is exponentially small. Near a peak a_j, set $\theta = a_j + y/\sqrt{m}$ and use the local splitting on every block. A term with a complementary power decays exponentially. The product of the principal projections has endpoint amplitude $A_j(R, a_j) + O((1+|y|)/\sqrt{m})$: at the common center the projection is idempotent and of rank one.

The logarithm of the principal eigenvalue product is

$$m \log z_j + i\sqrt{m}\, \mu_j \cdot y - \frac{1}{2} \sum_i t_i^{(m)} (y + v_i^{(m)})^{\mathsf{T}} H_j (y + v_i^{(m)}) + O(m^{-1/2}(1+|y|)^3).$$

The drift contribution of the $v_i^{(m)}$ is zero. Their cross term with y is zero for the same exact reason. The Gaussian integral on this chart is consequently the unmarked term from (160.8), multiplied by

$$\exp\left(-\tfrac{1}{2} \sum_i t_i^{(m)} (v_i^{(m)})^{\mathsf{T}} H_j v_i^{(m)} \right).$$

The real quadratic bounds dominate the product, since $\sum_i t_i^{(m)} |y + v_i^{(m)}|^2 = |y|^2 + \sum_i t_i^{(m)} |v_i^{(m)}|^2$. The rescaling and tail argument of Theorem 160.3 therefore applies uniformly.

On a compact parameter support, $\kappa_j = 0$ implies $H_j = \Omega_R$. It follows that

$$\sup_R e^{-m\kappa_j(R)} \| H_j(R) - \Omega_R \| \longrightarrow 0. \tag{161.7}$$

To see this, use continuity near the zero set of κ_j, and a positive minimum of κ_j on the compact complement of any such neighborhood. All Gaussian integrals and other scalar factors are bounded. Thus replacing every H_j in the extra characteristic factor by Ω_R changes the total by $o(1)$. This proves the band-limited version of (161.4), including parameter sequences which approach a change of arithmetic type.

For the sharp interval take its upper envelope q_B^+ and lower envelope q_B^-. Since the inserted path exponential has modulus one,

$$|N^J - N^{q_B^+}| \le P^{q_B^+} - P^J \le P^{q_B^+} - P^{q_B^-}.$$

The uniform scalar formula bounds the limsup of m^2 times the last difference by C/B. The same estimate pays the change in the unmarked probability. Fix B, let $m \to \infty$, and then let $B \to \infty$. This proves (161.4). Finally the displayed quadratic form is the elementary covariance formula for independent Gaussian increments tied to their sum. □

161.2. **Fourth moments under a microscopic constraint.**

Lemma 161.2 (A derivative estimate centered at a moving spectral peak)**.** *On each peak chart, for a real unit vector e, $0 \le p \le 4$ and $L \ge 1$, there are uniform $C, a > 0$ and $\rho < 1$ such that*

$$\left\| \frac{d^p}{dz^p} \left(e^{-ize\cdot\mu_j(R)} \mathscr{T}_{R,\theta+ze} \right)^L \Big|_{z=0} \right\| \le C L^{p/2} \left(e^{-aL|\theta - a_j(R)|^2} + \rho^L \right). \tag{161.8}$$

For $L = 0$ the derivative is zero when $p > 0$, and is the identity when $p = 0$. On a compact nonresonant part of a fixed band, the corresponding uncentered derivatives have a bound $C_p \rho^L$.

Proof. Put $x = \theta - a_j$. The local analytic splitting is uniform in a complex neighborhood. On a circle $|z| = d/\sqrt{L}$, with d fixed small, its principal eigenvalue after centering satisfies

$$\left| e^{-ize\cdot\mu_j} \lambda_j(R, \theta + ze) \right|^L \le e^{-L\kappa_j} \exp\{-a_0 L|x|^2 + CL|x||z| + CL|z|^2\} \le C e^{-a_1 L|x|^2}.$$

The first inequality follows from the zero real gradient at the peak, the positive real Hessian there and uniform third derivatives, reducing the chart once. Completing the square gives the second. The projection has uniformly bounded norm on this circle. Its complementary power is bounded by $C\rho_0^L e^{C\sqrt{L}}$, which is at most $C\rho^L$ for some $\rho_0 < \rho < 1$. Cauchy's derivative estimate supplies the factor $L^{p/2}$. Increasing C treats the finite initial range of L. On the nonresonant complement use a fixed complex circle small enough that the spectral contour still has radius less than one. Cauchy's estimate on that circle proves the last assertion. □

Proposition 161.3 (An unnormalized pinned fourth moment)**.** *For $0 \le r < r + l \le m$, $l \ge 1$, uniformly in all radii and targets,*

$$\int_E \left| \mathsf{S}_{r+l,R} - \mathsf{S}_{r,R} - \frac{l}{m} \mathsf{S}_{m,R} \right|^4 d\nu_R^* \le C_J l^2 m^{-2}. \tag{161.9}$$

The numerator uses the original exact return event (161.3)*. The estimate has not yet been divided by its probability.*

Proof. It suffices to prove the scalar estimate in each coordinate. Majorize $\mathbf{1}_J$ by one fixed nonnegative band-limited upper envelope. The other source factors and the fourth power are nonnegative, so this increases the integral. Put $\alpha = l/m$ and $s = m - r - l$. The scalar fourth moment inside the Fourier inverse is the fourth derivative at $z = 0$ of

$$\ell \left(M_{\eta_R} \mathscr{T}^s_{R,\theta-\alpha ze} \mathscr{T}^l_{R,\theta+(1-\alpha)ze} \mathscr{T}^r_{R,\theta-\alpha ze} (\eta_R \nu) \right).$$

The identity follows by differentiating the exact physical pairing. Every one-collision observable is bounded, so differentiation under the finite-time integral and the fixed-band inverse is legitimate.

On peak chart j, center each of the three operators at $\mu_j(R)$. The scalar factors introduced by this centering multiply to one, since $-\alpha(s + r) + (1 - \alpha)l = 0$. Apply Lemma 161.2 to each differentiated factor. The sum of the size factors in Leibniz's formula is bounded by

$$C\{\alpha(\sqrt{r} + \sqrt{s}) + (1 - \alpha)\sqrt{l}\}^4 \le C l^2.$$

Indeed $\alpha(\sqrt{r} + \sqrt{s}) \le \sqrt{2}\, l/\sqrt{m} \le \sqrt{2l}$. At least one block has length at least $m/3$. Its Gaussian factor has integral $O(m^{-2})$ in four frequencies; its complementary factor is exponentially small. The other factors are bounded by the preceding size estimate. Thus the integral over each chart is at most $C_J l^2 m^{-2}$. A zero-length factor is interpreted as in the lemma and causes no exception.

On the compact nonresonant part, derivatives of the longest block decay exponentially. The remaining fixed-order derivatives can be bounded polynomially in m, either by the lemma or by differentiating a product of uniformly bounded real-frequency powers. The resulting $C_J(1 + m)^4 \rho^{m/3}$ is at most $C_J l^2 m^{-2}$ after increasing the constant. Multiplication by the integrable Fourier transform of the fixed envelope and summation over the finite cover prove

the scalar estimate. Summing coordinates proves the vector assertion. The fixed factor c^{-1} is absorbed in C_J. □

Theorem 161.4 (A uniform conditional collision bridge)**.** *Fix $M, \epsilon > 0$ and J as above. Uniformly over the original events with*

$$|Z| \le M, \qquad m^2 p_E \ge \epsilon, \tag{161.10}$$

one has, on $C([0,1], \mathbb{R}^4)$ with its supremum norm,

$$d_{\mathrm{BL}}\left(\mathrm{Law}_{\nu_R^*(\cdot|E)}(\mathcal{B}_{m,R}), \mathrm{Law}(\mathbb{B}_{\Omega_R})\right) \longrightarrow 0. \tag{161.11}$$

Proof. Divide (161.4) by p_E, using (161.10). This proves uniform convergence of finite-dimensional characteristic functions to those of the Gaussian bridge. Polygonal interpolation has no effect on these limits, since the collision record is bounded and the total centered endpoint is bounded on the stated event class.

By Proposition 161.3, a grid increment of the normalized process has conditional fourth moment at most $C_{J,\epsilon}(l/m)^2$. Linear interpolation gives the same bound $C_{J,\epsilon}|t-s|^2$ at arbitrary times: within one grid interval its fourth moment is multiplied by the fourth power of the fractional length; an arbitrary increment is the sum of two such end pieces and a complete grid interval. The fourth-moment continuity criterion therefore gives uniform tightness in continuous path space.

For the bounded-Lipschitz formulation, take any hypothetical violating sequence. Compactness gives a convergent subsequence of its radii. Uniform tightness and the finite-dimensional limits identify every subsequential law with the Gaussian bridge at that limiting radius. Continuity of Ω_R gives continuity of these bridge laws, for example by coupling with $\Omega_R^{1/2}$ times one standard bridge. This contradicts the violating sequence. The proof uses no mixing assumption for the induced return map. □

161.3. **Returning to the genuine return clock.**

Theorem 161.5 (The microscopic bridge of the actual return record)**.** *Under the same event class* (161.10)*, uniformly in the radius and central targets,*

$$d_{\mathrm{BL}}\left(\mathrm{Law}_{\nu_R^*(\cdot|E)}(\mathcal{Y}_{n,R}), \mathrm{Law}(\mathbb{B}_{D_R})\right) \longrightarrow 0 \qquad (m \to \infty). \tag{161.12}$$

At every fixed radius the conclusion holds uniformly on central target compact sets over all positive arithmetic residue classes in Theorem 159.2; no additional denominator hypothesis is then needed. The conditioning fixes both lattice displacement coordinates and the collision count exactly, and keeps the roof in the specified fixed interval.

Proof. Let $\mathcal{X}_{m,R}$ be the polygonal path with values $m^{-1/2}\sum_{a<j} X_R(T_R^a x)$ at j/m. On E its endpoint equals $Z + O_J(m^{-1/2})$. It is consequently tight under the conditional laws, because $\mathcal{X}_{m,R}(s) = \mathcal{B}_{m,R}(s) + s\mathcal{X}_{m,R}(1)$ and Theorem 161.4 applies. In particular

$$\sup_{j \le m} |A_{j,R} - cj|/m \longrightarrow 0$$

in conditional probability, uniformly in the event class. At a genuine return, the half-open count identity is $A_{N_{l,R},R} = l$. On E every $N_{l,R}$ with $0 \le l \le n$ lies in $[0,m]$ and $N_{n,R} = m$. Since $n/m = c + O(m^{-1/2})$, the preceding bound implies

$$\max_{0 \le l \le n} \left| \frac{N_{l,R}}{m} - \frac{l}{n} \right| \longrightarrow 0 \tag{161.13}$$

in those same conditional probabilities. No comparison with a substitute conditioning event has been made.

Let L_R be the invertible matrix in (151.7). The exact compensation and $X_R = L_R h_R$ give

$$J_{l,R} - l\bar{G}_R = L_R^{-1} \sum_{a<N_{l,R}} X_R(T_R^a x).$$

Writing $\theta_l = N_{l,R}/m$, subtraction of the terminal record gives at every return grid point the exact identity

$$\mathcal{Y}_{n,R}(l/n) = \sqrt{m/n}\, L_R^{-1} \left\{ \mathcal{B}_{m,R}(\theta_l) + (\theta_l - l/n)\mathcal{X}_{m,R}(1) \right\}. \tag{161.14}$$

The second term tends to zero in supremum norm by (161.13) and the bounded endpoint. Uniform tightness with continuous limits permits replacing $\mathcal{B}_{m,R}(\theta_l)$ by $\mathcal{B}_{m,R}(l/n)$. Linear interpolation between the return grid points preserves this conclusion; it follows from the same modulus of continuity and does not require a separate tail estimate divided by a rare-event mass. Finally $m/n \to c^{-1}$ and

$$c^{-1} L_R^{-1} \Omega_R (L_R^{-1})^{\mathsf{T}} = D_R.$$

This proves (161.12) and its uniformity. For fixed R, the finitely many positive arithmetic coefficients have a positive minimum. The sharp-interval local theorem supplies (161.10) with a fixed $\epsilon_{R,M,J} > 0$, which proves the last assertion. □

Topology and the remaining raw problem. These are continuous-path limits under the unchanged exact discrete return constraints and a fixed positive roof interval. The lower mass condition is explicit in the radius-uniform assertion, and is proved on every positive fixed-radius arithmetic class. A pointwise roof conditioning law, arbitrary selected-path Gaussian amplitudes, and the common raw density correction do not follow by division by the interval length. The original pointwise theorem retains those requirements.

162. Residue cancellation, finite index packets and positive denominators

The transition formula also identifies when the unmodulated local law is uniform. It gives a finite-resolution consequence requiring no arithmetic assumption at all. This consequence concerns a specified finite packet of indices; it is not substituted for a prescribed single index in the conditional bridge theorem.

Theorem 162.1 (An exact criterion for the unmodulated interval law)**.** *Let $K \subset I$ be compact. The following assertions are equivalent.*

1. *For every $R \in K$ and every nontrivial occupation resonance, $\nu(\eta_R q_\gamma) = 0$.*
2. *For each $R \in K$, the section phase masses in* (159.2) *satisfy $w_{R,l} = c/d_R$.*
3. *For each fixed positive bounded roof interval J, uniformly for $R \in K$ and $|Z| \le M$,*

$$m^2 P_{n,m,R}(k,t;J) = c|J| g_{\Omega_R}(Z) + o(1). \tag{162.1}$$

The third assertion is the original D_R Gaussian normalization after multiplication by n^2 as in (159.5)*. It is still an interval statement, not a pointwise density theorem.*

Proof. At fixed R, Fourier coefficients of the finite vector $(w_{R,l})_{l \bmod d_R}$ are the section means of the powers of the normalized generator. Finite Fourier inversion proves equivalence of the first two assertions.

Assume the first. Choose the peak charts so that the neighborhood of the origin is disjoint from every nonzero possible resonance. Such a choice is uniform by local phase exclusion. On the origin branch, $a_j = 0$, $z_j = 1$, $\mu_j = \mu_R$, $H_j = \Omega_R$, and the source amplitude is c^2. Its partition

weights sum to one. On every other chart over K, $\kappa_j = 0$ implies $A_j(R, a_j) = 0$ by the first assertion. Scalar continuity and compactness, exactly as in (161.7), give

$$\sup_{R\in K} e^{-m\kappa_j(R)}|A_j(R, a_j(R))| \longrightarrow 0.$$

The bounded Gaussian integrals in (160.5) show that all these charts contribute $o(1)$ uniformly, even through changes of resonance type. Theorem 160.3 proves the third assertion.

Conversely fix any $R \in K$. Comparison with the fixed-radius arithmetic law shows $\mathfrak{a}_R(k, n, m) = 1$ on every residue. Every residue can be tested on a central sequence: take $k = 0$, $t = m\bar{\tau}_R$ and choose n within d_R of mc with $n + h_R^{\mathrm{ar}} m$ in the prescribed congruence class. Then $Z \to 0$, and the positive Gaussian at zero forces that residue factor to equal one. Finite Fourier inversion of (159.7) makes every nonconstant squared Fourier coefficient of $(w_{R,l})$ zero. This proves the first assertion. The argument uses endpoint residues, not triviality of the whole measurable phase group. □

Let D_0 be an integer upper bound for all the orders in Theorem 158.1, and fix once and for all

$$L = \operatorname{lcm}\{1, 2, \dots, D_0\}, \qquad p_L(v) = \frac{1}{L}\sum_{j=0}^{L-1} e^{-ijv}. \tag{162.2}$$

This is a finite constant of the mechanical family, not a function of m. No numerical value for the inherited order bound is claimed.

Theorem 162.2 (A radius-uniform local law on a fixed finite packet)**.** *For every fixed positive bounded interval J and $M < \infty$,*

$$\frac{m^2}{L}\sum_{j=0}^{L-1} P_{n+j,m,R}(k, t; J) = c|J|g_{\Omega_R}(Z) + o(1) \tag{162.3}$$

uniformly in $R \in I$ and $|Z| \le M$. For all large m, the unchanged union of these L exact-index events has probability at least $c_{M,J}Lm^{-2}$. At least one constituent exact event has probability at least $c_{M,J}m^{-2}$.

Proof. Sum the exact fixed-count inverse over $n, n + 1, \dots, n + L - 1$ and divide by L. This inserts exactly the trigonometric polynomial $p_L(v)$. On the origin branch its value is one. On a nonzero actual resonance its value is zero: the occupation coordinate has nontrivial finite order dividing L, by the injectivity of the occupation projection and the order bound. Thus, on each nonorigin peak chart,

$$\sup_R e^{-m\kappa_j(R)}|p_L((a_j(R))_4)| \longrightarrow 0.$$

This is again the compactness argument at the zero set of κ_j; it handles nearby nonperipheral branches, not only exact resonances. The proof of (160.8) with the additional smooth factor p_L consequently leaves just $c(\int q)g_{\Omega_R}(Z)$, uniformly in the radius. Positive interval envelopes then give (162.3), with the band fixed first. The averaging has been performed on the actual inverse, not on a fixed-radius formula with an unproved uniform error.

Different genuine return indices cannot share the same collision time on a single trajectory. These events are therefore disjoint. Uniform ellipticity and compactness give a positive lower bound for the right-hand Gaussian on $|Z| \le M$. The two probability assertions follow. For large m, all indices in the packet lie between one and m because $n/m \to c \in (0, 1)$. □

Corollary 162.3 (Verified classes for the conditional bridge)**.** *The radius-uniform conditional bridge theorem applies to every central exact event on a compact set satisfying the equivalent conditions of Theorem 162.1. At each fixed radius it applies on all positive arithmetic classes.*

Without either condition, Theorem 162.2 still guarantees at least one qualifying exact event in each central packet of the fixed length L.

Proof. The respective local laws give the lower mass assumption in (161.10). Invoke Theorem 161.5. The existence assertion in a packet does not identify an arbitrary preassigned index with that index, nor does it replace one conditioning event by another. □

Position in the raw inversion argument. The exact-index fixed-interval inverse now has an evaluated expression uniform through arithmetic transitions, rather than only a uniform spectral integral. The pinned return path has a conditional Gaussian bridge whenever the specified microscopic denominator has the stated lower bound. A fixed finite packet removes all possible occupation aliases without a diverging resolution scale. The single-index unmodulated interval law has the exact section-residue criterion above. None of these assertions sets the remaining section residues to zero for this family without proof. Nor does the slowly shrinking interval in Corollary 160.4 control density spikes below that scale. The common pointwise coarea correction and full roof tail in Theorem 143.1 remain the requirements for its original raw density conclusion.

163. Compatible critical edges in guarded return packets

Proposition 142.3 already classifies every regular critical word and bounds its leading jump. The purpose here is different: to sum a two-jet subtraction over all regular words below a collision cutoff, with a quantitative derivative budget, without removing the normal initial states. The conservative guard set contains grazing, first-hit singularities and section-decision boundaries, as well as the auxiliary zeros in (222.7). No separation between critical roof values is required.

Throughout this section the table is circular, $R \in I$, and the source law is the actual section probability ν_R^*. Put $c = \nu(Y_R^*) = c_*$ and $L_{m,R} = \sum_{j<m} \tau_R \circ T_R^j$. The symbol L without a subscript is an integer collision cutoff. The total variation of a signed measure is its full variation mass.

163.1. **Quantitative coordinates near a regular critical point.**

Lemma 163.1 (Critical coordinates on a guarded branch)**.** *On a regular m-collision branch write $DT_R^m = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. The critical points of $L_{m,R}$ are precisely the states with $p = p_m = 0$. At such a point its Hessian in (α, p) is*

$$\nabla^2 L_{m,R} = R \begin{pmatrix} AC & BC \\ BC & BD \end{pmatrix}, \qquad \det \nabla^2 L_{m,R} = R^2 BC > 0. \tag{163.1}$$

In particular it is positive definite. There is at most one regular critical point for a fixed ordered sequence of centers.

On a source region where the guards through time m are bounded away from their excluded sets at scale ε, the branch maps, their first ten derivatives and the regularity neighborhoods have upper and reciprocal lower bounds of the form

$$H = \exp\{C(L+1)\log(C/\varepsilon)\}, \qquad m \le L. \tag{163.2}$$

After increasing the absolute constant C, the following additional bounds hold. A relevant critical point has a Morse coordinate chart with C^6 coordinate and inverse bounds at most H, radius at least H^{-1}, and

$$L_{m,R}(x) = t_z + |z|^2/2.$$

Outside the smaller critical neighborhoods the roof gradient on the source supports is at least H^{-1}.

Proof. The action and differential identities in Lemma 224.1 give

$$\partial_\alpha L_{m,R} = R(p_m A - p), \qquad \partial_p L_{m,R} = R p_m B.$$

Every entry of $(-1)^m DT_R^m$ is positive, by (222.3); its determinant is one. In particular $B \neq 0$, which proves the critical-point characterization. Differentiating at $p = p_m = 0$ gives respectively RAC, $R(AD-1) = RBC$ and RBD. This proves (163.1). Its leading diagonal entry and its determinant are positive. The uniqueness assertion is precisely the convex endpoint-action argument of Proposition 142.3; it concerns a physical word, not a change of inducing section.

Here are the quantitative details used below. Use the explicit one-collision guard of (222.7), with a slightly enlarged regular region obtained by replacing its margins by half their values. On that region each positive selected discriminant and each incidence denominator has a power-law lower bound in ε. The finite root formulas and reflection formulas may therefore be differentiated through any fixed order. Through order ten their bounds are $(C/\varepsilon)^C$. Repeated chain rules for at most $L+1$ maps give (163.2). For clarity, one can induct on the derivative order: the highest derivative of a composition occurs linearly, and all products of lower derivatives are bounded by a fixed power of the previous exponential bound. Thus no exponent of the form C^L is introduced into $\log H$. The same argument, applied to the finitely many guard functions and their margins, gives a regular coordinate ball of radius H^{-1} about every point of the smaller guarded region. Constants in the exponent may be increased a fixed number of times below.

Each entry of a positive one-flight matrix is at least $v_0 = (3/50)/(47/100) = 6/47$. Hence each entry of its m-fold positive product is at least v_0^m. This crude lower bound suffices; no hyperbolic asymptotic is needed. The inverse of

$$(\alpha, p) \longmapsto (p, p_m), \qquad \det D(p, p_m) = -C,$$

has norm at most a fixed power of H on each regularity ball. The quantitative inverse theorem follows here by applying the contraction map to the linearized equation on a smaller ball: the inverse derivative and second derivative are bounded by powers of H, so its image contains a ball of radius H^{-C_1}. Consequently a sufficiently small value of $|(p, p_m)|$ is within H^{-C_1} of its unique zero in that regular branch. The displayed roof-gradient identities have an inverse coefficient matrix bounded by a power of H. They give the asserted lower gradient bound outside smaller critical neighborhoods after increasing C in (163.2). Apply this reasoning on the enlarged guarded region, so a zero found near a source support is still regular.

At a critical point, (163.1) has norm at most a power of H and smallest eigenvalue at least a reciprocal power of H. An explicit Morse chart can be constructed as follows. With $h = x - x_0$, set

$$Q(h) = 2\int_0^1 (1-s)\nabla^2 L_{m,R}(x_0 + sh)\, ds, \qquad z = Q(h)^{1/2} h.$$

On a ball of reciprocal-power radius this matrix remains positive definite and $L_{m,R}(x) - L_{m,R}(x_0) = |z|^2/2$. Differentiating the positive matrix square root, or its defining Sylvester equation, bounds its first six derivatives by powers of H. Its derivative at zero is the invertible positive square root of the Hessian. The same contraction-map inverse argument yields the chart and inverse bounds claimed. Relabeling a fixed power of H by increasing C completes the proof. These estimates hold for each regular branch and do not identify its boundary with a smooth branch. □

163.2. **Two jets rather than a single jump.**

Lemma 163.2 (An invariant two-jet edge subtraction)**.** *Let f have a nondegenerate minimum x_0 on a smooth two-dimensional source chart, and let $b(x)\,dx$ be a smooth compactly supported source in that chart. Choose a smooth source cutoff equal to one near x_0 and supported in a*

Morse chart $f = t_0 + |z|^2/2$. Write the resulting source density in that chart as $g(z)$, extended smoothly by zero. Its roof density is, for $s > 0$,

$$F(t_0 + s) = \int_0^{2\pi} g(\sqrt{2s}\, n_\theta)\, d\theta, \qquad F(t_0 + s) = 0 \quad (s < 0).$$

The right-hand value and derivative are

$$a_0 = 2\pi g(0) = \frac{2\pi b(x_0)}{\sqrt{\det \nabla^2 f(x_0)}}, \qquad a_1 = \pi \Delta g(0). \tag{163.3}$$

They do not depend on the Morse chart or on a cutoff equal to one in a neighborhood of x_0. With

$$e_{x_0}(t_0 + s) = \mathbf{1}_{s \geq 0} e^{-s}\{a_0 + (a_0 + a_1)s\}, \tag{163.4}$$

one has $F - e_{x_0} \in W^{2,1}(\mathbb{R})$ and

$$\widehat{e}_{x_0}(b_0) = e^{ib_0 t_0}\left(\frac{a_0}{1 - ib_0} + \frac{a_0 + a_1}{(1 - ib_0)^2}\right). \tag{163.5}$$

On charts with the bounds of Lemma 163.1, the coefficient sizes and the $W^{2,1}$ norm of the residual are bounded by a fixed power of H, times the corresponding bounded source norm.

Proof. Polar coordinates, with $s = r^2/2$ and $r\, dr = ds$, give the density. In the Taylor expansion of $g(rn_\theta)$ all odd angular monomials integrate to zero, while $\int n_\theta n_\theta^{\mathsf{T}} d\theta = \pi I_2$. This gives (163.3). The same expansion with integral remainders through order four shows that the radial average has two bounded right derivatives at zero. Away from zero, differentiating under the angular integral is legitimate. Its C^2 bound is controlled by the C^4 norm of g and the reciprocal chart radius. The cutoff makes it compactly supported in s. Changing a cutoff away from the minimum changes no sufficiently small roof level in this chart. Changing the chart changes no pushforward measure. Thus the two right jets are intrinsic to the local source contribution.

The right value of (163.4) is a_0 and its right derivative is a_1. The residual and its first derivative therefore match their zero left values. Its second distributional derivative has no atom, and its two smooth pieces have integrable second derivatives. This proves $W^{2,1}$, including the exponential part outside the original source chart. Integrating $e^{-(1-ib_0)s}$ and $se^{-(1-ib_0)s}$ gives (163.5). All bounds follow from the displayed formulas and the chart bounds. Subtracting only the leading exponential would instead leave, in general, an atom in the second derivative; the second jet removes it. □

163.3. **The actual stopped source, with its normal states retained.** For a regular initial state with $N_{n,R} = m \leq L$ set

$$G_{n,L,R,\varepsilon}(x) = \prod_{j=0}^{m} \chi_{R,\varepsilon}(T_R^j x), \tag{163.6}$$

and set it to zero on $N_{n,R} > L$ and on undefined itineraries. Unlike (222.8), this definition contains *no* factor cutting out $p = 0$. The product also stops at the actual terminal collision, rather than testing subsequent collisions. Define

$$Q_{n,R}^{L,\varepsilon} = (J_{n,R})_*(G_{n,L,R,\varepsilon}\nu_R^*), \qquad \mu_{n,R} = (J_{n,R})_*\nu_R^*.$$

Both are measures on $\mathbb{Z}^3 \times \mathbb{R}$. The label $\ell = (k_1, k_2, m)$ contains the actual collision count.

Theorem 163.3 (A compatible all-word regular critical packet)**.** *There is an exact decomposition*

$$\mu_{n,R} = e_{n,R}^{L,\varepsilon} + q_{n,R}^{L,\varepsilon} + r_{n,R}^{L,\varepsilon}, \tag{163.7}$$

where the notation also denotes mixed densities. The last density is nonnegative and

$$\sum_{\ell} \|r_{n,R}^{L,\varepsilon}(\ell,\cdot)\|_1 \le C\{e^{an-c_0L} + (L+1)\varepsilon^{1/16}\} =: \delta_{n,L,\varepsilon}. \tag{163.8}$$

Here $a, c_0 > 0$ are fixed cumulative-tail constants, and $n \le L$. The edge density is the finite sum of (163.4)*, at the actual label and roof value of* every *regular critical point meeting the guarded source. Its coefficients use the density $(4\pi c)^{-1}G_{n,L,R,\varepsilon}$ in (α, p) on that source piece. Each component of $q_{n,R}^{L,\varepsilon}$ belongs to $W^{2,1}(\mathbb{R})$, and for a fixed family constant C,*

$$\sum_{z}(|a_{0,z}| + |a_{1,z}|) + \sum_{\ell} \|q_{n,R}^{L,\varepsilon}(\ell,\cdot)\|_{W^{2,1}} \le \exp\{C(L+1)\log(C/\varepsilon)\} =: \mathcal{A}_{L,\varepsilon}. \tag{163.9}$$

There is no separate normal-strip removal, and coincident critical roof values are allowed. Regular points on auxiliary guard zeros may still lie in r; the conclusion concerns the whole guarded packet. The edge density need not be nonnegative.

For a fixed regular critical return word with strictly positive margins for every factor of (222.7) *through time m, its two coefficients stabilize as soon as $L \ge m$ and ε is small enough. Its stabilized leading coefficient is*

$$a_{0,z} = \frac{1}{2cR\sqrt{B_m C_m}}. \tag{163.10}$$

The stabilized first jet is the intrinsic right roof derivative of that same physical source contribution. This coefficient compatibility is a statement about regular words; it does not assert convergence of the infinite sum of their coefficients.

Proof. Since $0 \le G \le 1$, the difference $\mu - Q$ is positive. The cumulative return tail bounds its part on $N_n > L$. On the rest,

$$1 - \prod_{j=0}^{N_n} \chi \circ T_R^j \le \sum_{j=0}^{L} \mathbf{1}_{\{\chi \circ T_R^j \ne 1\}}.$$

Use $\nu_R^* \le c^{-1}\nu$, invariance of ν, and (222.6) to prove (163.8). Coarea absolute continuity applies to Y_R^* exactly as in Corollary 142.4: the critical set on each regular center word is at most a singleton, independent of the choice of inducing section. Bounded source restriction and the countable sum over m preserve absolute continuity. Thus $\mu - Q$ has the stated nonnegative density.

Partition $\{N_n \le L\}$ by center words and section decisions. All resulting source functions extend by zero smoothly: at the first collision or decision where the word can change, its corresponding guard vanishes on a two-sided neighborhood. This is the first-decision proof of Lemma 224.3, now through any fixed derivative order and with the product stopped at m. No pole $1/p$ is used in this argument. The one-step list of centers and the list of section decisions are finite. The number of records through L is consequently at most C_1^{L+1}; their open sets may be disconnected without affecting the zero extension. Each fixed center word has at most one regular critical point.

On each source piece apply Lemma 163.1 in a slightly enlarged guarded region. If a critical point lies near the support, use a source cutoff in its Morse chart equal to one on a smaller ball. If the original source vanishes near that point its two coefficients are zero. The remaining source is supported where $|\nabla L_m| \ge H^{-1}$. Its vector field

$$V = \frac{\nabla L_m}{|\nabla L_m|^2}$$

has the derivative bounds required for two integrations by parts. For this part, writing b_D for its zero-extended source density,

$$D_t(L_m)_*(b_D\,dx) = (L_m)_*(\operatorname{div}(b_D V)\,dx),$$
$$D_t^2(L_m)_*(b_D\,dx) = (L_m)_*(\operatorname{div}(V\operatorname{div}(b_D V))\,dx).$$

The signs are positive: for a scalar test ϕ, the distributional first derivative is $-\int b_D\phi'(L_m)$, and source integration by parts gives the first line. The second follows by iteration. There are no boundary terms because of the zero extensions and the cutoffs. The submersion coarea formula makes each right-hand measure an L^1 roof density, whose norm is bounded by the absolute source integral. These integrals and the undifferentiated mass are at most a fixed power of H.

For each critical part apply Lemma 163.2. Its two-jet edge is explicit and its residual has the same kind of bound. Sum the identities over all source pieces, grouping by ℓ. The factor C_1^{L+1} and each fixed power of H are absorbed by increasing C in (163.9). Equal roof values require only addition of their jets, not division by their distance. This proves the exact decomposition and the summed norm estimate. The real-variable cutoffs used to construct a local Morse contribution change none of its two jets.

Finally fix a regular critical word of length m with the stated guard margins. These include unselected candidate discriminants and auxiliary angular or section-coordinate factors, not only the actual first-hit discriminant. Every factor along this finite orbit is thus a positive distance from its zero set. For all small enough ε, each guard through time m is identically one on a neighborhood of the critical point. No condition is imposed after time m. The local density is then the physical constant $(4\pi c)^{-1}$, independent of $L \ge m$ and of smaller ε. Formula (163.3) and (163.1) give (163.10) and stabilization of both jets. At a true boundary or at a redundant guard zero the hypothesis fails, and no such stabilization is asserted. Any contribution excluded there remains in the exact nonnegative source r; it is not discarded from the raw law. □

163.4. **Pointwise inversion of the extracted packet.** Write $q_\ell(t)$ for the residual in (163.7), and define, for $B \ge 1$,

$$p_{n,R}^{L,\varepsilon,B}(\ell,t) = e_{n,R}^{L,\varepsilon}(\ell,t) + \frac{1}{2\pi}\int_{-B}^{B} e^{-ibt}\widehat{q}_\ell(b)\,db. \tag{163.11}$$

Every rational edge is evaluated explicitly before the residual inverse. For a jump value we choose the right trace, equivalently $\mathbf{1}_{s\ge 0} = 1$ at $s = 0$ in (163.4).

Corollary 163.4 (An explicit pointwise error and its remaining source)**.** *The extracted positive source $Q_{n,R}^{L,\varepsilon}$ has the right-trace density $e + q$, and, at every t,*

$$|(e+q)(\ell,t) - p_{n,R}^{L,\varepsilon,B}(\ell,t)| \le \frac{\mathcal{A}_{L,\varepsilon}}{\pi B}. \tag{163.12}$$

For the original raw density, at almost every (ℓ,t),

$$|p_{n,R}(\ell,t) - p_{n,R}^{L,\varepsilon,B}(\ell,t)| \le \frac{\mathcal{A}_{L,\varepsilon}}{\pi B} + r_{n,R}^{L,\varepsilon}(\ell,t). \tag{163.13}$$

For every $\lambda > 0$, the counting–Lebesgue measure of the output set where the left-hand error exceeds $\mathcal{A}_{L,\varepsilon}/(\pi B) + \lambda$ is at most $\delta_{n,L,\varepsilon}/\lambda$.

Proof. Distributional integration by parts gives $|\widehat{q}_\ell(b)| \le |b|^{-2}\|q_\ell''\|_1$ for $b \ne 0$. Together with the L^1 bound this makes the Fourier transform integrable. Its inverse agrees with the continuous $W^{2,1}$ representative. The integral of b^{-2} over $|b| > B$ is $2/B$, which proves (163.12). The exact density identity gives (163.13). Markov's inequality for the nonnegative output density r, with (163.8), gives the last assertion. □

The corollary is a pointwise reconstruction theorem for each regular cutoff packet, not a density local limit for the full uncut law. In particular small total mass of r does not imply a pointwise bound for r. Nor may the derivative budget $\exp\{C(L+1)\log(C/\varepsilon)\}$ be replaced by a polynomial in n when L is of order n. The remaining singular/decision boundary source and its high-frequency contribution are precisely the terms which must be controlled in a uniform raw local theorem. The normal critical words themselves now have compatible two-jet coefficients and a summed finite-cutoff residual estimate on the actual section.

164. The microscopic posterior of the arithmetic endpoint classes

The section residues are physical source masses. Their role can be made more explicit than a scalar correction to the local theorem: they determine a finite conditional law of the two endpoint phases. The phase-class indicators need not be strong multipliers. We instead realize the required initial densities as finite sums of physical peripheral vectors. This keeps the exact conditioning event unchanged.

Fix $R \in I$ in this section. Use the generator q_*, order $d = d_R$, phase masses w_l, and residue $r = r_R(k,n,m)$ of (159.2)–(159.3). Let $\omega = e^{2\pi i/d}$ and define $H_R(x) \in \mathbb{Z}/d\mathbb{Z}$ by $q_*(x) = \omega^{H_R(x)}$ on its invariant full-measure set. Write

$$S_r = \sum_{l \bmod d} w_l w_{l+r}, \qquad E = \{K_{n,R} = k,\ N_{n,R} = m,\ T_{n,R} - t \in J\},$$

where J is a fixed bounded interval of positive length. On E the exact phase equation gives

$$H_R((F_R^*)^n x) = H_R(x) + r \pmod d. \tag{164.1}$$

For $S_r > 0$ put

$$\pi_{R,r}(l,l') = \mathbf{1}_{\{l'=l+r\}} \frac{w_l w_{l+r}}{S_r}. \tag{164.2}$$

This is a probability, invariant under simultaneous relabeling of the phase classes. For $d = 1$ it is the point mass at $(0,0)$.

Lemma 164.1 (Strong initial densities for measurable phase classes)**.** *Each nonnegative bounded density $a_l = \eta_R \mathbf{1}_{\{H_R=l\}}$ has $a_l\nu \in \mathcal{B}$ on the local collision space at this fixed radius. Its construction does not assert that multiplication by q_* or by the phase-class indicator is bounded on $\mathcal{B}$.*

Proof. The full peripheral theorem gives a strong physical eigenvector $q_*^j\nu$ for every $0 \le j < d$. Indeed q_*^j solves the phase equation for the jth resonance; its bounded physical representative agrees up to a constant with the strong eigenvector by simplicity. The exact section multiplier is bounded on the strong space. Therefore

$$a_l\nu = \frac{1}{d}\sum_{j=0}^{d-1} \omega^{-jl} M_{\eta_R}(q_*^j\nu) \tag{164.3}$$

is a strong vector. The identity is finite Fourier inversion of the pointwise relation $q_*^d = 1$. Its physical representative is the nonnegative function a_l. Only multiplication by η_R, not by an unknown phase, has been used. □

Theorem 164.2 (Exact-index arithmetic endpoint posterior)**.** *As $m \to \infty$, uniformly over the central targets $|Z| \le M$ at this fixed radius and over $l \in \mathbb{Z}/d\mathbb{Z}$,*

$$m^2 \nu_R^*(E \cap \{H_R = l\}) = \frac{d}{c}|J| g_{\Omega_R}(Z) w_l w_{l+r} + o_R(1). \tag{164.4}$$

If $w_l w_{l+r} = 0$, the event on the left is exactly null, not merely asymptotically negligible. On every positive residue class $S_r > 0$,

$$d_{\mathrm{TV}}\Big(\mathrm{Law}_{\nu_R^*(\cdot|E)}(H_R(x), H_R((F_R^*)^n x)),\ \pi_{R,r}\Big) \longrightarrow 0. \tag{164.5}$$

Here d_{TV} on finite probability spaces is one half of the sum of the absolute mass differences. Neither assertion is made uniform through radius-dependent changes in the phase group.

Proof. For a compactly band-limited roof test, repeat the fixed-count inverse in Theorem 159.2, now with the strong initial vector $a_l\nu$ and terminal multiplier η_R. This is a legitimate endpoint pairing by Lemma 164.1. The general physical projection formula gives, at resonance j,

$$\nu(a_l \overline{q_*^j})\nu(\eta_R q_*^j) = w_l \omega^{-jl} \sum_h w_h \omega^{jh}.$$

The constant inverse phase is ω^{-jr}. Hence its complete finite residue sum is

$$\sum_{j=0}^{d-1} \omega^{-jr} w_l \omega^{-jl} \sum_h w_h \omega^{jh} = d w_l w_{l+r}. \tag{164.6}$$

All local curvatures are the same Ω_R. At fixed R there are finitely many initial vectors and resonances, so the Gaussian localization errors are uniform in l and in central targets. This proves (164.4) for the band-limited test with $|J|$ replaced by its integral. To pass to J, multiply ordered Fourier envelopes by the positive physical source $a_l(x)\eta_R(T^m x)$ and the unchanged exact lattice/occupation indicators. Fix the roof band, take $m \to \infty$, and then enlarge the band. All roof resonance coordinates vanish, so both envelopes have the same coefficient (164.6). This proves the stated interval law.

Equation (164.1) confines the endpoints to the pair $(l, l+r)$. If its initial or terminal section class is null, invariance of the collision measure and its fixed normalization on the section make the event null. For $S_r > 0$, summing (164.4) in l gives the denominator $(d/c)|J|g_{\Omega_R}(Z)S_r + o_R(1)$. It is bounded away from zero on the scaled central sets. Division proves convergence of every finite conditional mass. There are only finitely many positive S_r at this fixed radius. Summing the finitely many errors proves (164.5) uniformly over them. □

Corollary 164.3 (Phase classes and the actual Gaussian return bridge)**.** *Let $\mathcal{Y}_{n,R}$ be the return path tied to its actual endpoint in* (161.2)*. On the same fixed-radius central positive residue classes, its joint conditional law with the two phase classes converges, in bounded-Lipschitz distance on the product of the finite class space and continuous path space, to*

$$\pi_{R,r} \otimes \mathrm{Law}(\mathbb{B}_{D_R}). \tag{164.7}$$

Thus the arithmetic posterior need not be uniform, but is asymptotically independent of the pinned Gaussian fluctuations. The statement uses a fixed positive roof interval, not conditioning on a roof density value.

Proof. Insert the initial vector $a_l\nu$ in the chronological pinned characteristic calculation of Lemma 161.1. At this fixed radius the resonances and their curvatures are fixed. The block perturbations have zero weighted sum, so the drift and Fourier/bridge cross terms cancel as before. The only changed zeroth-order factor is (164.6). Complementary powers and projector errors obey the same estimates, with constants allowed to depend on the finite list of $a_l\nu$. The sharp interval error is bounded by the corresponding positive source-envelope error before any probability division. Thus the conditional finite-dimensional characteristic limit on each class with $w_l w_{l+r} > 0$ is the same Gaussian bridge.

The unnormalized fourth-moment bound of Proposition 161.3 also holds after restricting to $\{H_R = l\}$: its integrand is nonnegative and its source event has only been restricted. The new

class local law supplies its positive denominator of order m^{-2}. This proves tightness on each of the finitely many positive class pairs. The inverse occupation clock and the exact compensation in Theorem 161.5 are pathwise identities on this unchanged subevent, so the same return-clock conversion gives $\mathbb{B}_{D_R}$. Class pairs of zero weight are exactly null. Combining these conditional path limits with (164.5) proves the finite-mixture product limit. □

The arithmetic form of the raw target. The exact-index main term at fixed radius is the arithmetic expression $c\,\mathfrak{a}_R(k,n,m)g_{\Omega_R}(Z)$, or equivalently $\mathfrak{a}_R(k,n,m)g_{D_R}(V_n)$ in return normalization. The radius-uniform object is the finite transition kernel $\mathcal{L}_{m,R}$, not a pointwise choice of an exact resonance group. These are the candidate main terms for a pointwise roof-density extension of the corresponding interval theorems. The unmodulated specialization requires the exact zero-residue criterion of Theorem 162.1.

Neither the finite posterior formula nor a finite packet average proves that the concrete section phase masses equal c/d. The new critical subtraction in Section 163 works with each actual index and therefore does not average away this arithmetic. Its boundary-source error must still be controlled pointwise before the interval main term can be promoted to a raw density theorem.

165. A summable correction for the complete raw source

A collision cutoff has no effect on a component whose collision label is already fixed. We use this elementary fact before extracting the singularities. It gives one correction for all collision labels, rather than a sequence of corrections whose germs are chosen anew at every cutoff. The correction includes the decision and grazing boundaries through the complete finite graph of Section 200. Its construction is different from the exponential two-jet word sum in Section 163.

Fix $n \ge 1$ and $R \in I$. Let w be one bounded complex function on the actual initial section, admissible in the sense of Definition 200.1 for every finite cutoff, and put $M = \|w\|_\infty$. The same function w is used at every cutoff. Write $\mu^w = (J_{n,R})_*(w\nu_R^*)$. For $\ell = (k_1, k_2, m)$ set

$$b_\ell = \nu_R^*\{(K_{n,R}, N_{n,R}) = (k,m)\}, \qquad f_\ell(t) = \frac{d\mu^w(\ell,\cdot)}{dt}(t). \tag{165.1}$$

Thus $\|f_\ell\|_1 \le M b_\ell$ and $\sum_\ell b_\ell = 1$. The full variation convention is used throughout. At a fixed m only finitely many k occur, and f_ℓ is supported in $[0, m\tau_{\max}]$. Components of zero mass are identically zero.

165.1. **Intrinsic germs with an independently prescribed mass.**

Lemma 165.1 (Localized singular jets with arbitrary variation budget)**.** *Let f be a compactly supported integrable function with constructible real and imaginary parts. For every $\beta > 0$ one may select a finite power–logarithm density e such that*

$$\|e\|_1 \le \beta, \qquad q = f - e \in W^{2,1}(\mathbb{R}), \qquad \|q\|_1 \le \|f\|_1 + \beta. \tag{165.2}$$

Every one-sided term of exponent at most one in the intrinsic density germ is retained. Only its localization radius is changed to meet β. In particular this is not an approximation of a singular coefficient by a smaller coefficient.

The choices can be made once from f and β, independently of any collision cutoff. The value and first derivative of q have zero traces at every extracted point.

Proof. Use the finite analytic partition and the convergent one-sided germs of Lemma 200.4. Equivalently, take the intrinsic finite set outside which the density has a real-analytic representative, including the endpoints of its nonzero support. Null values of the original density do not

affect this set or any germ. Denote the jet at s on side σ by

$$J_{s,\sigma}(x) = \sum_{\alpha,k} c_{s,\sigma,\alpha,k} x^{\alpha} (\log x)^k, \qquad -1 < \alpha \le 1.$$

These are the germs of the *whole* component. Contributions with the same roof value have already been added. No division by a separation between critical values occurs.

Choose one fixed C^2 piecewise-polynomial function ψ on $[0,\infty)$, equal to one on $[0,1/2]$, zero on $[1,\infty)$, and between zero and one. Its first two derivatives match at the joining points. For a dyadic $d < 1$, the absolute integral of the localized term is bounded by

$$|c_{s,\sigma,\alpha,k}| \int_0^d x^{\alpha} |\log x|^k \, dx = |c_{s,\sigma,\alpha,k}| \, \mathcal{J}(\alpha+1, k, d). \tag{165.3}$$

Every quantity on the right tends to zero as $d \downarrow 0$. There are finitely many terms. Choose the first dyadic radius small enough that their total majorant is at most β, that the neighborhoods of distinct s are disjoint, and that each neighborhood lies in its germ interval. One fixes a dyadic subinterval of convergence of each intrinsic Puiseux–logarithm expansion before this choice. The positive radii of its coefficient power series, after clearing denominators, determine such intervals; real analytic continuation alone is not used as a Taylor convergence radius. None of these choices uses an external cutoff. For $f = 0$ use $e = q = 0$.

Set

$$e(t) = \sum_{s,\sigma} \mathbf{1}_{\{\sigma(t-s)>0\}} \psi(\sigma(t-s)/d) J_{s,\sigma}(\sigma(t-s)).$$

The majorants prove its mass bound. The convergent remainder has exponent strictly larger than one after the jet is removed. Hence its value and first derivative vanish at the endpoint and its second derivative is integrable. On the transition intervals the cutoff is C^2 and the density is analytic. The zero-trace argument of Proposition 200.5 proves $q \in W^{2,1}$. The final bound is the triangle inequality. Shrinking d can increase $\|q''\|_1$; no bound for that norm is inferred from β. □

165.2. **One complete correction, compatible with every count cutoff.**

Theorem 165.2 (Count-compatible extraction of all boundary germs)**.** *For every prescribed $0 < \varepsilon \le 1$ there are finite complex measures $E^{\varepsilon}, Q^{\varepsilon}$ such that*

$$\mu^{w} = E^{\varepsilon} + Q^{\varepsilon}, \qquad \|E^{\varepsilon}\|_{\mathrm{TV}} \le \varepsilon, \qquad \|Q^{\varepsilon}\|_{\mathrm{TV}} \le M + \varepsilon. \tag{165.4}$$

Their densities e_ℓ, q_ℓ satisfy

$$\|e_\ell\|_1 \le \varepsilon b_\ell, \qquad q_\ell \in W^{2,1}(\mathbb{R}), \qquad \|q_\ell\|_1 \le (M+\varepsilon) b_\ell. \tag{165.5}$$

Each e_ℓ is a finite localized power–logarithm sum for the complete actual component, including singular and section-decision boundaries. Both series over ℓ converge absolutely in total variation. There is no omitted source in (165.4)*.*

For every $L \ge n$, define $E^{\varepsilon,L}, Q^{\varepsilon,L}$ by restriction to labels $m \le L$. Then

$$\mu^{w,L} = E^{\varepsilon,L} + Q^{\varepsilon,L}, \qquad E^{\varepsilon,L'}|_{m \le L} = E^{\varepsilon,L} \quad (L' \ge L), \tag{165.6}$$

and the same restriction identity holds for Q. With the cumulative return constants $a, c_0, C > 0$,

$$\begin{aligned} \|E^{\varepsilon} - E^{\varepsilon,L}\|_{\mathrm{TV}} &\le C\varepsilon e^{an - c_0 L}, \\ \|Q^{\varepsilon} - Q^{\varepsilon,L}\|_{\mathrm{TV}} &\le C(M+\varepsilon) e^{an - c_0 L}. \end{aligned} \tag{165.7}$$

For each $0 < s < c_0$ there is $C_s < \infty$ such that

$$\sum_{\ell} e^{sm} \|e_\ell\|_1 \le \varepsilon C_s e^{s(a/c_0)n}, \qquad \sum_{\ell} e^{sm} \|q_\ell\|_1 \le (M+\varepsilon) C_s e^{s(a/c_0)n}. \tag{165.8}$$

The constants in these mass estimates are uniform in R.

Proof. For a fixed $\ell = (k, m)$ with $b_\ell > 0$, apply the complete finite collision graph at cutoff m in Lemma 200.2. The resulting component is exactly f_ℓ, not a truncated approximation to it. Indeed the event specifying $N_n = m$ contains no trajectories with a different collision count. A larger cutoff therefore gives the identical measure, identical analytic density pieces and identical intrinsic one-sided germs. This observation is made before any neighborhood or localization radius is chosen.

Apply Lemma 165.1 to f_ℓ with $\beta = \varepsilon b_\ell$. Use zero for a null component. The component estimates sum to (165.4). The measure identity follows component by component; countably many null sets are harmless. In particular grazing and decision boundaries are accounted for by their contribution to the finite sublevel integrals, rather than by a smooth extension of their inverse Jacobian. The geometric integration input is exactly [15, Theorem 1.3 and Theorem 3.11], as verified in Section 200.

Use these same component choices for every L. This proves (165.6). Summing (165.5) over $m > L$ and applying Theorem 134.1 proves (165.7). Applying the exponential-moment part of that theorem proves (165.8). The construction may vary with R, but none of these inequalities requires continuity of its germ labels or radii in R. □

Corollary 165.3 (An absolutely convergent correction transform)**.** *For real frequencies $\omega = (u_1, u_2, s, b)$,*

$$\widehat{E}^{\varepsilon}(\omega) = \sum_{\ell=(k,m)} e^{i(u\cdot k+sm)}\widehat{e}_\ell(b)$$

converges absolutely and uniformly, and the analogous assertion holds for Q^{ε}. The count-truncated transforms have the uniform errors in (165.7)*. For each fixed n, R both transforms extend holomorphically to a nonempty complex tube about the real frequencies. This tube can be chosen uniformly in R.*

Proof. The first assertion follows from $|\widehat{e}_\ell(b)| \le \|e_\ell\|_1$ and the summable majorant. A nonzero component has $|k|_1 \le C_\kappa m$. Its original roof support is inside $[0, m\tau_{\max}]$, and the localized correction has support within distance one of that interval. Thus both e_ℓ and q_ℓ are supported where $|t| \le m\tau_{\max} + 1$. On a compact sub-tube with

$$C_\kappa |\operatorname{Im} u|_\infty + |\operatorname{Im} s| + \tau_{\max}|\operatorname{Im} b| < c_0,$$

choose an intermediate $s_0 < c_0$ exceeding the left side. The modulus of each exponential integrand is bounded by a fixed constant times $e^{s_0 m}$. Equation (165.8) and its version with a slightly larger exponent dominate all fixed-order derivatives as well. Locally uniform convergence of the entire component transforms proves holomorphy. This is exponential control in the output variables, not decay as the real roof frequency tends to infinity. □

What has been summed. The series above is a count-labeled series of localized germs of the whole raw density. It includes fractional powers and logarithms from boundary sources, and converges without any separation assumption between moving critical roof values. It is not a proof that the unlocalized exponential two-jet coefficients of all critical words are absolutely summable. Nor does (165.4) bound $\|e^{\varepsilon}\|_\infty$ or $\sum_\ell \|q_\ell''\|_1$. In particular it proves a common all-boundary correction as a finite measure, not the small common pointwise remainder required by Theorem 143.1.

166. Full-source pointwise reconstruction and the arithmetic remainder

We use the correction of Section 165 without assuming that its full mixed residual transform is integrable. The order of inversion matters: take the discrete label coefficient first, and then

invert its roof transform. Each roof inverse is absolutely convergent. This gives a reconstruction of the full law, including the source left outside the guarded packet.

166.1. **Labelwise inversion with a prescribed pointwise error.** Retain e_ℓ, q_ℓ, b_ℓ and ε from Theorem 165.2. Define the exact component transform and its residual by

$$\Phi_\ell^w(b) = \int_{Y_R^*} \mathbf{1}_{\{(K_n,N_n)=(k,m)\}} w(x) e^{ibT_n(x)}\, d\nu_R^*(x), \qquad h_\ell(b) = \Phi_\ell^w(b) - \widehat{e}_\ell(b) = \widehat{q}_\ell(b). \tag{166.1}$$

The physical record has not been convolved. Its finite component transform is given by the same exact finite collision graph as its germs. Put $A_{j,\ell} = \|\partial_t^j q_\ell\|_1$ for $j = 0, 2$. The individual $A_{2,\ell}$ are finite; their unweighted sum is not assumed finite.

Theorem 166.1 (Complete raw inversion in coefficient-first order)**.** *For every component of positive reference mass,*

$$\int_{\mathbb{R}} |h_\ell(b)|\, db \le 4\sqrt{A_{0,\ell} A_{2,\ell}}, \qquad f_\ell(t) = e_\ell(t) + \frac{1}{2\pi}\int_{\mathbb{R}} e^{-ibt} h_\ell(b)\, db. \tag{166.2}$$

The equality holds almost everywhere, at every regular density point, and with the right trace at every finite-trace edge. At such an edge the point value of e_ℓ is assigned its right trace, while q_ℓ uses its continuous representative. At a divergent germ it is an equality of densities off the singular value, not an assertion of a finite density there.

For $\delta > 0$ choose, when $b_\ell > 0$,

$$\Lambda_\ell = \max\left\{1, \frac{A_{2,\ell}}{\pi\delta b_\ell}\right\}, \qquad p_\delta^w(\ell, t) = e_\ell(t) + \frac{1}{2\pi}\int_{-\Lambda_\ell}^{\Lambda_\ell} e^{-ibt} h_\ell(b)\, db. \tag{166.3}$$

For a null component put $p_\delta^w = 0$. Then

$$\sum_\ell \operatorname*{ess\,sup}_{t\in\mathbb{R}} |f_\ell(t) - p_\delta^w(\ell, t)| \le \delta. \tag{166.4}$$

For a fixed finite label set one may use the maximum of its Λ_ℓ as a single bandwidth. The bounds do not omit grazing, section decisions, normal states, or trajectories with any prescribed actual collision label.

Proof. The $W^{2,1}$ residual satisfies

$$|h_\ell(b)| \le \min\{A_{0,\ell}, A_{2,\ell}|b|^{-2}\}.$$

The integral of the right side is $4\sqrt{A_{0,\ell}A_{2,\ell}}$. If either norm is zero, the compactly supported residual is zero. Absolute Fourier inversion gives its continuous representative. The zero value and derivative traces in Lemma 165.1 make the trace conventions in the statement consistent with the intrinsic germs. Integration over $|b| > \Lambda$ gives the exact bound $A_{2,\ell}/(\pi\Lambda)$. Substitution of (166.3) bounds it by δb_ℓ, and summing $\sum b_\ell = 1$ proves (166.4). Increasing a bandwidth preserves this upper bound.

There is no unjustified interchange of two infinite integrals. By Corollary 165.3, the full residual transform has the uniformly absolutely convergent discrete expansion

$$\widehat{Q}^\varepsilon(u, s, b) = \sum_{k,m} e^{i(u\cdot k + sm)} h_{(k,m)}(b).$$

For each fixed b, its torus coefficient is therefore exactly $h_\ell(b)$. Only that coefficient is subsequently integrated in b, where (166.2) applies. Joint $L^1(\mathbb{T}^3 \times \mathbb{R})$ integrability of the transform is neither used nor obtained by this argument. □

Corollary 166.2 (A prescribed reconstruction scale at long return times)**.** *Fix $P > 0$. Choose $\varepsilon_n = \delta_n = (1+n)^{-P-3}$ in the preceding construction. Then*

$$\sup_{R\in I} n^2 \sum_{\ell} \operatorname*{ess\,sup}_{t} |p^{w}_{n,R}(\ell,t) - p^{w}_{\delta_n}(\ell,t)| \le n^2(1+n)^{-P-3}. \tag{166.5}$$

Here the preparation data and the bandwidths are allowed to depend on n, R, ℓ, w. For central return targets it suffices to retain labels $m \le L_n$, where

$$L_n = \left\lceil \lambda n + \frac{P+4}{c_0}\log(2+n) \right\rceil, \qquad \lambda > \max\{c^{-1}+1, a/c_0+1\}. \tag{166.6}$$

On these retained labels the reconstruction error contains no collision-tail density. The total variation of the omitted correction and residual is bounded by (165.7)*.*

Proof. Equation (166.4) has constant one at every R, proving the first assertion with the indicated parameter-dependent choices. A fixed return central window has $m = n/c + O(\sqrt{n})$; its labels are at most L_n eventually. Restrictions by the collision label are exact, so a discarded label cannot contribute to the density of a retained label. Finally $an - c_0 L_n \le -c_0 n - (P+4)\log(2+n)$, up to the harmless strengthening from the strict choice of λ. Apply the preceding tail bound. □

This is a prescribed error for *reconstruction*, not a prescribed rate for a Gaussian local limit. In particular the formula for Λ_ℓ retains the actual germ derivative budget; it provides no polynomial or exponential upper bound for that budget as n grows. The global mass budget for E^{ε_n} can be made smaller than n^{-2} while its pointwise height remains large. These statements must not be interchanged.

166.2. **The arithmetic main term for the unchanged singleton.** We now take $w = 1$. Fix once a real even function $\chi \in C_c^\infty((-1,1))$, with $0 \le \chi \le 1$ and $\chi = 1$ near zero. For a *fixed* $B > 0$ put

$$K_B(t) = \frac{1}{2\pi}\int e^{-ibt}\chi(b/B)\,db, \qquad \int K_B = 1.$$

The kernel is real and even, but need not be nonnegative. The band-limited test theorem applies to it without a positivity assumption. Define the full-source signed correction

$$\begin{aligned}\mathscr{D}_{B,n,R}(\ell,t) &= e_\ell(t) - (K_B * e_\ell)(t)\\ &\quad + \frac{1}{2\pi}\int_{\mathbb{R}} e^{-ibt}(1-\chi(b/B))h_\ell(b)\,db.\end{aligned} \tag{166.7}$$

This integral is absolute componentwise. All measures in this formula refer to the full actual law at the stated return index.

Theorem 166.3 (Arithmetic raw closure with an evaluated central term)**.** *At every fixed $B > 0$ there is the exact almost-everywhere identity*

$$\mathscr{D}_{B,n,R}(\ell,t) = p_{n,R}(\ell,t) - (K_B * p_{n,R}(\ell,\cdot))(t). \tag{166.8}$$

In particular the correction is independent of the germ localization radii and of ε. The same identity applies to all finite truncations once the specified label is retained.

Write $y = (k_1, k_2, t, n)$ and $Z = (y - m(0,0,\bar\tau_R, c))/\sqrt{m}$, in that order. Uniformly in R, $m \ge n \ge 1$, and $|Z| \le M$,

$$m^2 p_{n,R}(k,m,t) = \mathcal{L}_{m,R}(y) + m^2 \mathscr{D}_{B,n,R}(k,m,t) + o(1). \tag{166.9}$$

The error is in essential supremum in the roof target and ordinary supremum in the other displayed parameters. The transition kernel is exactly the evaluated finite Gaussian peak sum of

Theorem 160.3, with its arithmetic branches retained. Consequently the radius-uniform pointwise transition local law holds if and only if

$$\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t:\ |Z|\le M} |\mathscr{D}_{B,n,R}(k,m,t)| \longrightarrow 0 \tag{166.10}$$

for every M. If it holds for one fixed $B>0$, it holds for every fixed $B>0$.

At a fixed radius, the corresponding main term is $c\mathfrak{a}_R(k,n,m)g_{\Omega_R}(Z)$; equivalently in the return normalization it is $\mathfrak{a}_R(k,n,m)g_{D_R}(V_n)$. No zero-residue assertion or replacement of the prescribed singleton by a packet is needed for these statements.

Proof. The absolutely convergent component inverse gives

$$\frac{1}{2\pi}\int e^{-ibt}(1-\chi(b/B))h_\ell(b)\,db = q_\ell(t) - (K_B * q_\ell)(t).$$

Add the edge difference and use $e_\ell + q_\ell = p_{n,R}(\ell,\cdot)$. This proves (166.8). If a different extraction is used, it gives the identical right side, so the invariance is exact, not an asymptotic comparison of absolute bounds.

Because K_B is even, its convolution at t is precisely $P^{K_B}_{n,m,R}(k,t)$ with the sign convention $K_B(T_n - t)$. This is one fixed integrable test, with integrable Fourier transform supported in $[-B,B]$. Equation (160.7) and its vanishing imaginary part give

$$m^2(K_B * p_{n,R}(\ell,\cdot))(t) = \mathcal{L}_{m,R}(y) + o(1)$$

uniformly even without restricting the targets to a central compact set. Substitute (166.8). This proves (166.9) and both implications in (166.10). Comparing the identity at two fixed bands proves the last band-independence assertion. Each band is fixed before $m\to\infty$.

At fixed radius use the specialization of Theorem 160.3 and the covariance change in (159.5). On the central sets $n/m = c + O(m^{-1/2})$, so the same conversion applies to the error and to the signed correction. The arithmetic coefficient is uniformly bounded by the finite resonance order. When it is zero, the exact component measure is zero by Theorem 159.2; it is not assigned a conditional density law with a zero denominator. □

Corollary 166.4 (A finite integral certificate for the remaining error)**.** *Replace the last integral of* (166.7) *by its restriction to $|b|\le\Lambda_\ell$, where*

$$\Lambda_\ell \ge \max\{B, 1, A_{2,\ell}/(\pi\delta b_\ell)\}.$$

Call the resulting quantity $\mathscr{D}^{\mathrm{fin}}_{B,n,R}$. Then

$$\sum_\ell \operatorname*{ess\,sup}_t |\mathscr{D}_{B,n,R} - \mathscr{D}^{\mathrm{fin}}_{B,n,R}|(\ell,t) \le \delta.$$

With $\delta_n = (1+n)^{-P-3}$ this difference is negligible at the local scale on every central set. Thus the necessary and sufficient criterion may be tested on this finite, full-source inverse with an explicit tail certificate. No spectral power estimate is evaluated at the varying Λ_ℓ in this assertion.

Proof. Beyond $\Lambda_\ell \ge B$, the factor $1-\chi(b/B)$ is exactly one. The omitted integral is at most $A_{2,\ell}/(\pi\Lambda_\ell) \le \delta b_\ell$. Sum the estimate. For central targets m/n is bounded, so $m^2\delta_n \to 0$. The finite integral is left as an exact physical quantity; its smallness is not deduced from the fixed-band spectral theorem. □

The remaining analytical estimate. The complete correction now exists, with all boundary germs included, in a topology with an explicit summable mass budget. Its full-source pointwise reconstruction has a certified error after the label is fixed. What is not yet estimated is the signed finite inverse in Corollary 166.4 on the central long-time scale. The old guarded derivative bound cannot be inserted as a polynomial bound for the new germ budgets, and the small edge mass cannot be substituted for its pointwise height. A proof of (166.10), including the arithmetic transition kernel rather than an assumed unit residue, remains the precise raw-density closure step. The definition of that step is now independent of count cutoffs, auxiliary source guards and choices of singular localization.

167. Uniform height bounds for the complete collision source

The integrable power–logarithm class is larger than the class of singularities which the physical roof density can actually have. We prove a height bound before preparing any germ. The argument includes arbitrarily small positive interior incidence and every section-decision boundary. Its constants grow exponentially in the collision count; it is not a local-limit estimate.

Let $\rho_{m,R}$ be the density of $L_{m,R} = \sum_{j<m} \tau_R \circ T_R^j$ under the full collision probability ν. Put $R_- = 9/20$, $R_+ = 47/100$, $v_0 = (3/50)/R_+ = 6/47$, and $c = \nu(Y_R^*)$. Densities and inequalities between them are understood almost everywhere. We use full variation mass, without the probability-distance factor $1/2$.

167.1. **Endpoint curvature without an interior incidence margin.**

Lemma 167.1 (The reduced action on an entire regular word)**.** *Fix a regular word of m flights. The map from its initial state to its two endpoint angles (α_0, α_m) is injective and locally invertible. On its open image the reduced action F satisfies*

$$dF = R(-p_0\, d\alpha_0 + p_m\, d\alpha_m), \qquad \nabla^2 F \geq R\,\mathrm{diag}(c_0, c_m), \quad c_i = \sqrt{1-p_i^2}. \tag{167.1}$$

Writing $B_m = \partial_p \alpha_m$ on the original branch, one has

$$|B_m| \geq v_0^m, \qquad d\alpha_0\, dp_0 = |B_m|^{-1} d\alpha_0\, d\alpha_m. \tag{167.2}$$

No assertion that the endpoint image is convex is needed.

Proof. Use oriented arclength at each contact. The second variation of a segment is the sum of the two endpoint curvature terms $c_j y_j^2/R + c_{j+1} y_{j+1}^2/R$ and the nonnegative transverse square in the proof of Proposition 142.3. Summing the segments gives a positive definite contact Hessian. The internal reflection equations are stationary equations, so the Hessian of the reduced action is its internal Schur complement. Minimizing the full quadratic form over the internal coordinates retains the two endpoint curvature terms. Changing from arclength to angles gives the inequality in (167.1). The first variation gives its equality. All statements are local on a regular branch and remain valid as interior incidence tends to zero through positive values.

For injectivity, fix both endpoint positions and minimize the polygonal length over the product of the closed internal disks. At a regular reflected contact its gradient is $-2c_j n_j$. For a different point q_j' of the same strictly convex disk, $n_j \cdot (q_j' - q_j) < 0$. The convex gradient inequality therefore makes any different internal tuple have strictly larger length. Thus two regular reflected paths of this word cannot share both endpoints. For $m = 1$ their straight segment determines the initial velocity directly. This use of a convex minimization does not assert physical admissibility of an arbitrary minimizing path: the word under consideration is already physical, and other-disk exclusions are kept in its domain.

By (222.3), every entry of the positive one-flight matrix $-DT_R$ is at least v_0. Every entry of the product of m such matrices is at least v_0^m. In particular $B_m \neq 0$. The change-of-variables formula now proves (167.2), with an absolute Jacobian for either parity. □

Use finitely many rational angular charts $\alpha = \alpha_* + 2\arctan x$, $|x| < 1$, with a fixed semialgebraic disjoint refinement at their overlaps. The chart derivative $a(x)$ lies in $[1, 2]$ and $|a'(x)| \leq 4$. Set $\delta_0 = 1/20$. On the part where $|p_0|, |p_m| < \delta_0$, the endpoint-chart function $f(x, y)$ satisfies

$$\nabla^2 f \geq \lambda_0 I_2, \qquad \lambda_0 = R_-/2. \tag{167.3}$$

Indeed the congruence of the first Hessian in (167.1) is at least $R\sqrt{1-\delta_0^2}I_2$. The diagonal coordinate correction has norm at most $4R\delta_0$. Since $\sqrt{1-\delta_0^2} - 4\delta_0 > 1/2$, (167.3) follows. The collision-density weight in these coordinates is bounded by v_0^{-m}/π.

167.2. **A coefficient-independent bound on level complexity.**

Lemma 167.2 (Length and turning of physical roof levels)**.** *There are constants $C_1 > 1$ and $C_2 > 1$, depending only on the one-flight geometry and the fixed charts, with the following properties. For each fixed word and almost every roof value, its roof level in an initial (x, p) chart has total length at most $C_1 C_2^m$. On its near-normal endpoint-chart portions the total absolute curvature of the regular roof level is at most $C_1 C_2^m$. These bounds are uniform in R and require no distance from singular or decision boundaries. They count every connected portion, not only one selected inverse curve.*

Proof. Keep the finite collision graph in auxiliary variables instead of eliminating all intermediate states. Positions, unit normals, incoming and outgoing unit velocities, and flight lengths through m collisions require at most $a(m+1)$ real variables. The flight equations, circle equations and reflections have bounded degree. Positivity of flight lengths and incidence, and exclusion of an earlier hit, are finite Boolean combinations of bounded-degree sign conditions for each flight. For example, the minimum squared distance of a straight segment from a candidate center is tested at its endpoints and, when its orthogonal projection lies inside the segment, at that projection. Positive length denominators are cleared with their signs retained. The uniform finite horizon gives a fixed finite candidate list per flight. Consequently at most $a(m+1)$ polynomials of degree at most d_* describe the graph, for fixed constants a, d_*. The same format bounds hold after imposing a roof value, a coordinate line, or an endpoint-gradient direction. Chart denominators are positive.

For precision, the bound used on sign-condition components is the finite inequality in [14, Section 3.2, equation (3.3)], with $i = 0$ and ambient variety $\mathbb{R}^r$:

$$b_0 \leq d(2d-1)^{r-1} \sum_{j=0}^{r} 4^j \binom{s}{j} \leq d(2d-1)^{r-1} 5^s. \tag{167.4}$$

Adding a bounding ball changes s by one. A Boolean union has no more components than the sum over its sign conditions. Here r, s are linear in m, and d is fixed, so the bound is exponential in m. It is independent of polynomial coefficients, including R and the accumulated lattice translations. The fixed-dimension asymptotic in the main theorem of that reference is not used with a varying dimension. A projection of a connected set is connected. Thus the same component bound controls every finite projection fiber of the physical graph.

For a regular one-dimensional level, almost every horizontal or vertical line has a finite intersection with it. These intersections are bounded by (167.4), applied before projection. Integrating the two coordinate multiplicities bounds respectively $\int |dx|$ and $\int |dp|$. Since arclength is at most $|dx| + |dp|$, and each coordinate range has length two, this proves the length estimate.

Straight segments parallel to a slicing line cause an exceptional value only, and are counted by the other slicing.

For the endpoint assertion orient a regular level by its unit normal $N = \nabla f/|\nabla f|$. If T is its unit tangent, its curvature is

$$k_f = \frac{T^{\mathsf{T}} \nabla^2 f\, T}{|\nabla f|} > 0. \tag{167.5}$$

Thus its Gauss map has no constant interval on the near-normal portion. A fiber of this map is semialgebraic: by the first variation its unnormalized normal is $R(-a(x)p_0, a(y)p_m)$. Parallelism to a specified unit direction and the positive dot product are bounded-degree conditions on the same auxiliary graph. Each fiber is finite, with the component bound just proved. The one-dimensional area formula for the Gauss map gives $\int |k_f|\, ds \le 2\pi C_1 C_2^m$. Adjust C_1.

One may first apply both area formulas on compact regular exhaustions. Their right sides do not depend on the exhaustion. Monotone convergence therefore includes portions approaching grazing, chart or decision boundaries. Exceptional roof levels are discarded as permitted by the coarea formula. No boundary-neighborhood mass is converted to a density bound in this argument. □

Theorem 167.3 (Complete fixed-count height bound)**.** *There are $C > 0$ and $A > 1$, independent of m, R, such that*

$$\|\rho_{m,R}\|_\infty \le CA^m \qquad (m \ge 1, R \in [0.45, 0.47]). \tag{167.6}$$

For any family of bounded measurable insertions $w_{n,R}$, all of supremum norm at most M, the complete actual-return components obey

$$\operatorname*{ess\,sup}_t \sum_{n=1}^{m} \sum_{k \in \mathbb{Z}^2} |p^w_{n,R}(k, m, t)| \le MCA^m/c. \tag{167.7}$$

In particular a prescribed component satisfies this bound with no regularity condition on its insertion. No source guard or count cutoff changes the measure in these statements.

Proof. Work first on one regular word in an initial rational chart. If $|p_0| \ge \delta_0$, the nonstationary estimate (222.1) gives a fixed positive lower gradient bound. If $|p_0| < \delta_0$ and $|p_m| \ge \delta_0$, use instead $|\partial_p L_m| = R|p_m B_m| \ge R_- \delta_0 v_0^m$. The rational change of the initial angle has derivative at least one. The gradient is therefore at least $c_0 v_0^m$ on the union of these two regions, for a fixed $c_0 > 0$. The initial collision density is at most $1/(2\pi)$ in this chart. Coarea and the level-length estimate give a bound $C(C_2/v_0)^m$ for this part of the roof density.

On the remaining near-normal part change to the two endpoint charts. The Jacobian weight is at most v_0^{-m}/π. Equations (167.3) and (167.5) give

$$\int_{\{f=t\}} \frac{ds}{|\nabla f|} \le \lambda_0^{-1} \int_{\{f=t\}} |k_f|\, ds \le CC_2^m,$$

where both integrals are restricted to the near-normal part. This argument does not extend the endpoint domain to its convex hull. Coarea again gives the same exponential bound. Critical points on a regular word are isolated by Proposition 142.3; the critical roof levels do not affect this almost-everywhere bound. The regular exhaustions in the preceding lemma include every portion of each open physical branch. Singular source states themselves have zero collision measure.

There are at most D^m center words for a fixed finite alphabet of size D, and only a fixed number of initial and endpoint charts. Absorbing these factors in A proves (167.6). The constants are determined by the one-flight format, not by a prepared singularity neighborhood or a lower interior incidence.

Finally, on Y_R^* the events $\{K_{n,R} = k, N_{n,R} = m\}$ are disjoint as (n, k) varies with m fixed: at collision m a trajectory has a unique return index and lattice displacement, when it returns at all. Their roofs are all $L_{m,R}$. Pushforward domination and $\nu_R^* \le c^{-1}\nu$ imply the measure inequality $\sum_{n,k} |\mu_{n,R}^w(k, m, \cdot)| \le (M/c)(L_{m,R})_*\nu$. Radon–Nikodym derivatives prove (167.7). □

Interpretation. The exponential bound supplies uniform height control for the complete finite-collision source and all its bounded restrictions. It does not have the m^{-2} scale of the arithmetic local law. Its important consequence for raw extraction is that divergent density germs cannot occur in these physical components, even though they occur in the larger constructible L^1 class allowed in the earlier preparation lemma.

168. Finite jumps and pointwise-small continuous singularities

We apply the complete height bound to the original density, before any localization. It separates two different effects. A genuine jump keeps its height when its support is shortened. A positive-exponent cusp vanishes at its endpoint and can be localized with a prescribed supremum budget, without scaling its coefficient. This distinction allows an actual pointwise estimate for the entire continuous singular part of the count-compatible correction.

168.1. **The germs permitted by the physical source.**

Corollary 168.1 (No divergent physical density germ)**.** *Fix n, R and a bounded finite-record subanalytic insertion. For every actual label $\ell = (k, m)$, the convergent expansion* (200.3) *of the complete component f_ℓ has no nonzero term with negative exponent and no term $(\log x)^j$ with $j \ge 1$ and exponent zero. Its one-sided limits are finite. The component, extended by zero, belongs to* $\mathrm{BV}(\mathbb{R})$*. The same exclusion holds at singular and decision-boundary values, not only at regular critical values.*

Proof. Theorem 167.3 bounds the absolute component density. Its constructible representative is analytic on each of the finitely many regular roof intervals, so an almost-everywhere bound holds everywhere on each such interval. In a convergent power–logarithm expansion, the smallest nonzero exponent and then the largest logarithmic power at that exponent determine its magnitude near zero. A negative exponent or a nonconstant zero-exponent logarithmic polynomial would contradict boundedness. Thus near each one-sided endpoint

$$f_\ell(s \pm x) = l_{s,\pm} + \sum_{0<\alpha\le 1} \sum_j c_{s,\pm,\alpha,j} x^\alpha (\log x)^j + r_{s,\pm}(x), \tag{168.1}$$

where the sum is finite and the remainder has the derivative bounds (200.5) with exponent greater than one. Each positive-exponent term has an integrable first derivative. The remainder does too. On closed subintervals away from the finitely many endpoints the density is analytic, with bounded derivative. Adding the finitely many finite jumps proves bounded variation. Apply the argument to real and imaginary parts. It concerns the full component; separate branch singularities must first be combined as in the finite-graph lemma. □

Proposition 168.2 (Uniform finite-count variation and smoothing)**.** *For each m there is $V_m < \infty$, independent of R, such that*

$$\sum_{n=1}^{m} \sum_k \|D_t p_{n,R}(k, m, \cdot)\|_{\mathrm{TV}} \le V_m. \tag{168.2}$$

Here D_t is a distributional derivative, not the covariance D_R. The same assertion holds for any specified jointly globally subanalytic finite-record family of weights with a common bound and a fixed finite parameter description. No such uniform variation conclusion is asserted for all merely measurable bounded weights.

For the even kernel $K_B = BK_1(B\cdot)$ of Theorem 166.3,

$$\sum_{n,k} \|p_{n,R}(k,m,\cdot) - K_B * p_{n,R}(k,m,\cdot)\|_1 \le \frac{V_m}{B} \int_{\mathbb{R}} |s|\,|K_1(s)|\,ds. \tag{168.3}$$

No growth rate for V_m is asserted.

Proof. For fixed m, allow the finitely many rectangle endpoints defining the section to be independent bounded parameters. In the fixed rational charts their membership tests are semialgebraic with these endpoints as coefficients, retaining the fixed factor c^{-1} rather than renormalizing the enlarged sections; finite chart cases also cover endpoints passing a chart cut. Thus the full finite graph and cumulative component integral in (200.2) are jointly constructible in these parameters, R and t, by [15, Theorem 1.3]. Differentiation in t on a definable partition gives a definable density representative in the o-minimal real field with restricted analytic functions and exponentiation. The uniform monotonicity consequence of cell decomposition [9] bounds the number of monotonicity intervals of every real and imaginary component by a finite number depending on m and the fixed description, not on the parameters. The height bound is uniform. A function bounded by U on N monotonicity intervals, extended by zero, has variation at most $4(N+1)U$, including endpoint jumps. There are only finitely many labels and return indices at fixed m. This proves (168.2). The actual moving endpoints are a subfamily of this enlarged parameter family; no unproved definability in R of their chosen parametrization is needed. A specified jointly subanalytic weight is treated by the same parameter integral. This proof is not an effective long-time bound on the monotonicity count.

For a compactly supported BV function f, $\|f(\cdot - s) - f\|_1 \le |s|\|Df\|_{\mathrm{TV}}$, by integration of its derivative measure. Use $\int K_B = 1$, the triangle inequality and Tonelli, then sum (168.2). Scaling the kernel gives (168.3). The kernel need not be positive. □

168.2. **A count-compatible decomposition with a supremum budget.** Fix n, R, w as in Corollary 168.1, and put $b_\ell = \nu_R^*\{(K_n, N_n) = (k,m)\}$. Null components are set to zero. At a nonnull component let $J_{\ell,s} = f_\ell(s+) - f_\ell(s-)$ be each intrinsic jump.

Theorem 168.3 (Jump–cusp separation of the complete source)**.** *For every $\varepsilon, \delta > 0$ there is an exact decomposition*

$$f_\ell = j_\ell + z_\ell + q_\ell, \qquad q_\ell \in W^{2,1}(\mathbb{R}), \tag{168.4}$$

with compactly supported components, such that

$$\|j_\ell\|_1 + \|z_\ell\|_1 \le \varepsilon b_\ell, \tag{168.5}$$

$$\|z_\ell\|_\infty \le \delta b_\ell, \qquad \|j_\ell\|_\infty \le \max_s |J_{\ell,s}|. \tag{168.6}$$

The empty maximum is zero. The function j_ℓ consists only of localized step functions with coefficients $J_{\ell,s}$. The continuous function z_ℓ contains precisely the localized positive-exponent terms of (168.1) *through exponent one. All intrinsic coefficients are unchanged.*

The correction series converge absolutely in total variation and

$$\sum_\ell \|z_\ell\|_\infty \le \delta. \tag{168.7}$$

All collision-count truncations are restrictions of these same components. Their total-variation tails and exponential count moments satisfy the estimates of Theorem 165.2.

Proof. Fix the complete component before choosing any neighborhoods. Choose a C^2 cutoff θ on $[0,\infty)$, equal to one near zero, zero beyond one, taking values in $[0,1]$, with matching derivatives at the transition endpoints. At every singular value s put

$$j_{\ell,s}(t) = J_{\ell,s}\mathbf{1}_{t>s}\theta((t-s)/d_s).$$

Use disjoint two-sided neighborhoods of the distinct singular values. Subtract on each side the positive-exponent jet in (168.1), multiplied by the corresponding $\theta(x/d_s)$, to define $z_{\ell,s}$. Every such jet tends to zero at its endpoint. Its supremum on $0 < x < d_s$ and its integral there tend to zero as $d_s \downarrow 0$; the jump term's integral also tends to zero. Shrink the finitely many radii to make the sum of the integrals at most εb_ℓ and the sum of cusp suprema at most δb_ℓ. This changes no coefficient. Disjoint jump supports and $0 \le \theta \le 1$ give the second bound in (168.6).

Near s, the residual on both sides has the same limiting value $l_{s,-}$, because the right jump $l_{s,+} - l_{s,-}$ was removed. Both residual first derivatives tend to zero, and their second derivatives are integrable by the remainder estimate with exponent greater than one. The constant common trace need not be subtracted. At the cutoff transitions the values and first derivatives match. Thus there is no atom in the residual's first or second distributional derivative, and $q_\ell \in W^{2,1}$. Unlike the earlier independent one-sided subtraction, this construction has a possibly nonzero, common value trace; zero traces are not needed for Fourier inversion.

Sum the budgets over labels using $\sum b_\ell = 1$ and $\|f_\ell\|_1 \le \|w\|_\infty b_\ell$. This proves all claimed series bounds. Choose $d_s \le 1$; the supports then stay in $[-1, m\tau_{\max} + 1]$. Restricting by $m \le L$ changes no component already chosen, so the count compatibility is exact. The cumulative return tail applied to $\sum_{m>L,k} b_{(k,m)}$ proves the same variation tails; the enlarged roof support changes only a constant in the exponential-moment calculation. Selection need not be continuous in the radius for any of these conclusions. □

Corollary 168.4 (A pointwise bound for the full continuous singular part)**.** *Let $B > 0$ be fixed and let $\mathscr{D}_{B,n,R}$ be the unchanged full-source signed correction. With $h_\ell = \widehat{q_\ell}$ from* (168.4)*, it has the exact decomposition*

$$\mathscr{D}_{B,n,R}(\ell,t) = j_\ell(t) - (K_B * j_\ell)(t) + \frac{1}{2\pi}\int e^{-ibt}(1-\chi(b/B))h_\ell(b)\,db + r_\ell(t), \tag{168.8}$$

$$\sum_\ell \|r_\ell\|_\infty \le (1 + \|K_1\|_1)\delta.$$

The inverse is absolutely convergent after taking the label coefficient. For $\delta_n = (1+n)^{-P-3}$ and central labels $m \asymp n$,

$$m^2 \sum_\ell \|r_\ell\|_\infty = O(n^{-P-1}),$$

uniformly in R. This estimate includes every positive-exponent singular and decision-boundary germ of the complete component.

Proof. Set $r_\ell = z_\ell - K_B * z_\ell$. The convolution inequality and (168.7) give its bound; $\|K_B\|_1 = \|K_1\|_1$. The remaining identity follows from $f_\ell - K_B * f_\ell$ and the $W^{2,1}$ coefficient-first inversion already proved. The common nonzero trace of the residual introduces no distributional atom. Multiplication by the central $m^2 = O(n^2)$ proves the stated rate. □

Proposition 168.5 (The jump size visible to raw unsmoothing)**.** *At an intrinsic jump s and for every fixed $B > 0$,*

$$\operatorname*{ess\,sup}_{|t-s|<h} |\mathscr{D}_{B,n,R}(\ell,t)| \ge \tfrac{1}{2}|J_{\ell,s}| \qquad (h > 0). \tag{168.9}$$

Thus a central pointwise local law with a continuous roof main term requires $m^2|J_{\ell,s}| \to 0$ for jumps lying in the interior of the central target intervals.

Proof. The convolution of the compactly supported L^1 density with the Schwartz kernel is continuous. The two one-sided traces of $\mathscr{D}_B = f_\ell - K_B * f_\ell$ consequently differ by $J_{\ell,s}$. At least one trace has modulus at least half this difference. Continuity on the adjoining regular intervals gives the essential-supremum bound. Apply the same argument to the difference from a continuous main term. □

What remains on the local scale. The theorem gives a height estimate for the actual source, excludes divergent germs, and pays every continuous singular term in a summed supremum norm. It does not bound the jump part or the residual inverse in (168.8) by $o(m^{-2})$. The selected radii can still make $\|q_\ell''\|_1$ large. Their supremum budget is not a bound for those derivatives, and the finite-count variation theorem gives no usable growth rate for V_m. The main term remains $\mathcal{L}_{m,R}$, or $c\mathfrak{a}_R g_{\Omega_R}$ at fixed radius. No arithmetic residue, prescribed singleton, conditioning event or fixed-band quantifier has been changed.

169. Endpoint collars of long regular critical words

The individual edge estimate does not control a cluster of critical values. We use a physical neighborhood whose size is independent of word length and allow its interior margins to decrease exponentially with distance from the two endpoints. This is possible because endpoint influence decays faster than the permitted loss of margin.

Use arclength at the contacts, let c_j be the incidence cosine, and let ℓ_j be flight j's length. Put

$$q = 47/53, \qquad d_j = \min\{j, m-j\}, \qquad e_j = q^{d_j/4}, \qquad e'_j = q^{\min\{j,m-1-j\}/4}.$$

Fix $0 < \varepsilon < 1/10$. A regular normal-to-normal critical word, starting and ending in Y_R^*, is called ε-protected if $c_j \geq \varepsilon e_j$, its collision state at time j has distance at least εe_j from ∂Y_R^* in the fixed cylinder metric, and flight j has clearance at least $\varepsilon e'_j$ from every nonincident disk. Clearance excludes the two incident disks. The section condition applies to states outside the section as well as to its visits. A word with a fixed margin ε everywhere is protected, but the new class also allows very small interior incidence and decision margins. No positive source probability for this class is presumed.

169.1. **Decay of endpoint influence.**

Lemma 169.1 (A uniform endpoint neighborhood)**.** *There are constants $r_* > 0$ and $C < \infty$, independent of ε, m, R and of the protected word, with the following properties. Its physical reflected path extends to every endpoint pair $|u|, |v| < \rho_\varepsilon = r_*\varepsilon^3$. On this square it retains its center itinerary and every section-membership decision. Its exact displacement and return index are unchanged. Internal contacts satisfy*

$$|\partial_u y_j| \leq C\varepsilon^{-1} q^{j-1-d_j/4}, \qquad |\partial_v y_j| \leq C\varepsilon^{-1} q^{m-1-j-d_j/4}, \qquad 1 \leq j \leq m-1. \tag{169.1}$$

Every protected margin remains at least half its stated lower bound.

Proof. At an internally reflected contact choose tangent signs successively so that neighboring off-diagonal Hessian entries are negative. Put $t_j = 1/\ell_j$ and $D_c = \operatorname{diag}(c_1, \dots, c_{m-1})$. The internal action Hessian, in arclength coordinates, is exactly $H_{\mathrm{int}} = D_c A D_c$, where

$$A_{jj} = t_{j-1} + t_j + \frac{2}{Rc_j}, \qquad A_{j,j+1} = A_{j+1,j} = -t_j. \tag{169.2}$$

This is the second-variation identity of Proposition 142.3, before eliminating the internal contacts. Write $A = D_0(I - K)$, with D_0 its diagonal and $K \geq 0$. Its row sums are at most

$$q = \frac{2(50/3)}{2(50/3) + 200/47} = 47/53 < 1.$$

The Neumann series consequently gives

$$|(A^{-1})_{ij}| \leq \frac{q^{|i-j|}}{(200/47)(1-q)}, \qquad |(H_{\mathrm{int}}^{-1})_{ij}| \leq \frac{Cq^{|i-j|}}{c_i c_j} \tag{169.3}$$

for every positive internal incidence. In the endpoint derivative, one incidence factor cancels: the endpoint–internal entry has modulus $c_0c_1t_0$, hence $|\partial_u y_j| \le Cc_0t_0q^{j-1}/c_j$. The other endpoint is identical. As long as $c_j \ge \varepsilon e_j/2$, this proves (169.1) and, for endpoint differentials,

$$|dy_j| \le C\varepsilon^{-1}q^{3d_j/4}(|du| + |dv|),$$
$$|du| + |dv| + \sum_{j=1}^{m-1} |dy_j| \le C\varepsilon^{-1}(|du| + |dv|). \tag{169.4}$$

There are at most two indices at any given distance d_j. Empty internal matrices, when $m = 1$, require no inversion.

Continue the local implicit solution along (su, sv), $0 \le s \le 1$. Flight directions, incidences and collision states are uniformly Lipschitz functions of adjacent contact positions, including at a limiting zero incidence: their formulas divide by flight lengths, not by incidence cosines. All lengths are bounded below by $3/50$, and above by the fixed finite alphabet of neighboring centers. Clearance and distance to a section boundary are Lipschitz as well. Thus the change of a collision margin at index j is at most $C\varepsilon^{-1}q^{3d_j/4}(|u| + |v|)$; shifting to an adjacent index changes only a fixed factor. Divide this by its allowed lower bound $\varepsilon q^{d_j/4}$. The quotient is at most $C\varepsilon^{-2}q^{d_j/2}(|u| + |v|)$, hence less than $1/2$ for $|u|, |v| < r_*\varepsilon^3$ if r_* is small enough. The flight clearance estimate is the same with e'_j. No protected margin can first reach half its lower bound.

For each fixed finite word the internal Hessian remains invertible. Compactness of its contact-coordinate boxes and the strict half-margins continue the solution to $s = 1$. Choose r_* also below a fixed contact-chart radius. Local uniqueness and the endpoint injectivity of Lemma 167.1 make the continuations agree on overlaps and supply the entire square. Positive incidence and clearance retain physical admissibility. No collision crosses the section boundary, so every visit, including both endpoint visits, retains its original index. The choice of r_* will be reduced once more for relative distortion below. □

169.2. **Relative source distortion.** Write $F_z(u, v)$ for the reduced action in this square about critical state z, and $t_z = F_z(0, 0)$. Let $w_z(u, v)\,du\,dv$ be the original section probability in these endpoint coordinates. The subscript z labels a physical critical word, not a new probability law.

Lemma 169.2 (Distortion relative to the intrinsic edge coefficient)**.** *After a fixed reduction of r_*, there are $0 < \lambda_* \le \Lambda_* < \infty$, independent of ε, m, R, z, such that*

$$\lambda_* I_2 \le \nabla^2 F_z \le \Lambda_* I_2, \qquad e^{-1} \le w_z(u, v)/w_z(0, 0) \le e. \tag{169.5}$$

At the critical point the coefficient of the local density jump is

$$J_z = \frac{2\pi w_z(0, 0)}{\sqrt{\det \nabla^2 F_z(0, 0)}} > 0. \tag{169.6}$$

It is the coefficient for ν_R^*; the coefficient for ν is cJ_z, where $c = 91/(10000\pi)$.

Proof. The first variation is $dF_z = -p_0\,du + p_m\,dv$. Endpoint influence bounds the changes of both endpoint tangential velocities. They start at zero, so the endpoint cosines can be kept above $1/2$. The reduced Hessian is an internal Schur complement. It dominates the endpoint curvature terms and is bounded above by the endpoint principal block of the full contact Hessian. The latter has bounded entries, since $1/\ell_j \le 50/3$. Thus one may take, for example, a lower bound $1/(2R_+)$ and an upper bound $2(1/R_- + 50/3)$, enlarging constants if necessary. These estimates hold throughout the convex endpoint square, not just at its center.

Let $C_z = \partial_u \partial_v F_z$. The change of variables from (u, p_0) to (u, v) and the collision normalization give

$$w_z = \frac{|C_z|}{4\pi R c}, \qquad |C_z| = \frac{c_0 c_m \prod_{j=0}^{m-1} t_j}{\det A}. \tag{169.7}$$

The last formula follows by successive tridiagonal elimination; $\det A = 1$ for the empty matrix. The internal factors c_j^2 cancel between the product of off-diagonal entries and $\det H_{\mathrm{int}}$. In particular the formula is nonzero and may be differentiated logarithmically on the collar. It gives

$$d \log |C_z| = d \log c_0 + d \log c_m + \sum_{j=0}^{m-1} d \log t_j - \operatorname{tr}(A^{-1} dA). \tag{169.8}$$

The differentials of t_j and c_j have uniformly bounded coefficients in adjacent contact displacements. The only nonzero entries of dA are on three diagonals. Its diagonal potential derivative is bounded by $C|dc_j|/c_j^2$, and the relevant entries of A^{-1} are uniformly bounded by (169.3). Consequently (169.4) bounds the potential part of the trace by

$$C\varepsilon^{-3} \sum_{j=1}^{m-1} q^{d_j/4} (|du| + |dv|) \le C\varepsilon^{-3} (|du| + |dv|).$$

The two endpoint cosines stay above $1/2$. The remaining terms of (169.8) are bounded by $C\varepsilon^{-1}(|du| + |dv|)$. Hence $|d \log w_z| \le C\varepsilon^{-3}(|du| + |dv|)$. Integration on the square of radius $r_* \varepsilon^3$ makes the absolute logarithmic ratio at most one after reducing r_*. An absolute derivative bound for an m-fold collision map has not been used in its place.

Finally the Morse lemma at the strictly convex minimum gives (169.6), or equivalently the edge formula in Proposition 142.3, divided by c. This coefficient is not scaled when a neighborhood is chosen. □

Proposition 169.3 (A physical mass collar of fixed roof width)**.** *There are $h_* > 0$, $a_* > 0$ and $A_* < \infty$, independent of ε, m, R, z, with the following properties. Set $h_\varepsilon = h_* \varepsilon^6$. For $0 < h < h_\varepsilon$ define*

$$U_z(h) = \{(u, v) : 0 \le F_z(u, v) - t_z < h\}$$

inside the endpoint square and identify it with its original source states. This sublevel lies strictly inside the square and all its states satisfy

$$|p_0| + |p_m| \le A_* \sqrt{h}. \tag{169.9}$$

Its roof density $g_{z,h}$ is supported on $[t_z, t_z + h]$ and obeys

$$a_* J_z \le g_{z,h}(t_z + s) \le A_* J_z \quad (0 < s < h), \qquad \nu_R^*(U_z(h)) \ge a_* h J_z. \tag{169.10}$$

The inequalities hold almost everywhere. Collars belonging to different physical words are disjoint in the original source.

Proof. Strong convexity and $\nabla F_z(0) = 0$ give $\lambda_* |(u, v)|^2/2 \le F_z - t_z \le \Lambda_* |(u, v)|^2/2$. Choose $h_* < \lambda_* r_*^2/8$, so that $h_\varepsilon < \lambda_* \rho_\varepsilon^2/8$. The sublevel is then a compactly contained strictly convex disk, up to its open outer boundary. The upper Hessian bound and the first variation prove (169.9).

For $0 < s < h$, orient the smooth closed convex level by its normal. Its curvature is $k = (T^{\mathsf{T}} \nabla^2 F_z T)/|\nabla F_z|$, and its total turning is 2π. Therefore

$$\frac{2\pi}{\Lambda_*} \le \int_{\{F_z = t_z + s\}} \frac{d\mathcal{H}^1}{|\nabla F_z|} \le \frac{2\pi}{\lambda_*}.$$

Multiply by the relative bounds for w_z and use (169.6). Coarea gives both density inequalities, and integration in s gives the mass lower bound. Distinct center words are disjoint regular

itinerary events; a single word has at most one critical point. Thus no source mass is counted twice. □

170. Dynamical deconcentration of critical edges at an exact index

We now use the arithmetic local law to estimate the actual edge coefficients, rather than only reconstruct their sum. The argument compares a positive endpoint collar with a positive endpoint-selected local probability. No count of words or separation of critical values enters the comparison. The final conclusions are pointwise statements at the original exact displacement and return index.

170.1. **A two-endpoint small-source estimate.** Write $p(x)$ for the tangential collision velocity, and for a fixed bounded interval J put

$$E_{n,k,m,R}(t;J)=\{K_{n,R}=k,\ N_{n,R}=m,\ T_{n,R}-t\in J\},$$
$$E^{(d)}_{n,k,m,R}(t;J)=E_{n,k,m,R}(t;J)\cap\{|p(x)|\le d,\ |p((F_R^*)^n x)|\le d\}. \tag{170.1}$$

The terminal state is $T_R^m x$ on this unchanged exact event.

Lemma 170.1 (Local mass with two normal endpoint strips)**.** *There is $C<\infty$, independent of $0<d<1/4$ and J, such that*

$$\limsup_{m\to\infty}\sup_{\substack{R\in I,\,1\le n\le m\\ k\in\mathbb{Z}^2,\,t\in\mathbb{R}}} m^2\nu_R^*(E^{(d)}_{n,k,m,R}(t;J))\le Cd^2|J|. \tag{170.2}$$

The interval and strip width are fixed before the collision count tends to infinity. The assertion is an upper bound and does not replace the arithmetic factor by one or assert a positive denominator.

Proof. Choose a smooth $0\le b_d\le 1$ on the collision cylinder, equal to one for $|p|\le d$ and zero for $|p|\ge 2d$. Put $a_{R,d}=\eta_R b_d$. The initial vector $a_{R,d}\nu$ and terminal multiplier $a_{R,d}$ are bounded in the relevant collision norms for each fixed d, uniformly in R. Their physical masses satisfy

$$\nu(a_{R,d})\le\nu\{|p|<2d\}=2d. \tag{170.3}$$

No estimate uniform in d for their strong norms is needed.

Repeat the finite-branch fixed-band inverse of Theorem 160.3 with these two endpoints. The spectral charts, eigenvalues, real maxima and covariance bounds are unchanged; only the scalar amplitudes become

$$A_{j,d}(R,\theta)=\ell(M_{a_{R,d}}\Pi_j(R,\theta)(a_{R,d}\nu)).$$

For fixed d these amplitudes and their frequency derivatives have uniform bounds and are continuous in R. To check the latter use finite-time physical pairings with the two factors $a_{R,d}$, as in Lemma 160.1. Finite itineraries and section membership persist off a null set at the limiting radius. Normalized powers converge uniformly to the scalar projection on the local contour. Thus the same uniform-limit argument applies to these endpoints, without asserting strong-norm continuity of η_R.

Take any parameter and target sequence in the supremum and pass to $R_m\to R_0$ inside the finite chart cover. A peak branch whose limiting maximum has modulus less than one contributes zero in the limit. Every remaining limiting peak is an actual resonance. The physical projection formula there gives

$$|A_{j,d}(R_0,a_j(R_0))|=|\nu(a_{R_0,d}\bar q_j)\nu(a_{R_0,d}q_j)|\le 4d^2, \tag{170.4}$$

where $|q_j|=1$ almost everywhere. This uses all resonances, including those reached through parameter transitions. Each damping factor has modulus at most one. The complex Gaussians have a common bound from $\operatorname{Re}H_j\ge aI_4$, for every target argument. The number of peak charts

is fixed independently of the endpoints and of the outer roof band. Indeed every exact roof resonance has roof coordinate zero, as specified in Lemma 160.2.

Let q_B be a positive Fourier majorant of the interval indicator, with mass $|J| + 2\pi/B$. For each fixed B, the band inverse and the preceding peak estimate bound the limsup of its endpoint-selected expectation, multiplied by m^2, by $Cd^2(|J| + 2\pi/B)$. In this step the roof coordinates of surviving limiting peaks are zero, so their test factor is $\widehat{q}_B(0) = \int q_B$. The positive exact source permits the envelope inequality before any limit is taken. First let $m \to \infty$, then $B \to \infty$. Equation (170.1) is bounded by that selected source, which proves (170.2). The subsequence argument proves uniformity even when the arithmetic type changes with R. □

170.2. **Coalescing critical values.** Let $\mathcal{C}^{\varepsilon}_{n,k,m,R}$ be all ε-protected normal-to-normal critical words whose original source has $N_{n,R} = m$ and $K_{n,R} = k$. This is a finite set for each m. The coefficient J_z is that in (169.6). Define the positive atomic measure

$$\mathcal{J}^{\varepsilon}_{n,k,m,R} = \sum_{z \in \mathcal{C}^{\varepsilon}_{n,k,m,R}} J_z \delta_{t_z}. \tag{170.5}$$

Equal critical values are combined by addition in this measure. The definition does not change any of their intrinsic coefficients.

Theorem 170.2 (A local bound for the complete regular critical cluster)**.** *There are $C < \infty$ and $h_* > 0$, independent of ε, such that, with $h_\varepsilon = h_* \varepsilon^6$, for every fixed $0 < h < h_\varepsilon$,*

$$\limsup_{m \to \infty} \sup_{R,n,k,t} m^2 \mathcal{J}^{\varepsilon}_{n,k,m,R}([t-h, t+h]) \le Ch. \tag{170.6}$$

Here and below $1 \le n \le m$. In particular,

$$\sup_{R,n,k,t} m^2 \mathcal{J}^{\varepsilon}_{n,k,m,R}(\{t\}) \longrightarrow 0. \tag{170.7}$$

No separation of critical values, bound on the number of words, or triviality of a section residue is assumed.

Proof. For each critical value in $[t-h, t+h]$, use the actual source collar $U_z(h)$ of Proposition 169.3. These collars are disjoint and keep exactly the prescribed labels (n, k, m). Their roof values lie in $[t - h, t + 2h]$. Both endpoint tangential velocities have modulus at most $A_*\sqrt{h}$. Thus

$$a_* h \, \mathcal{J}^{\varepsilon}_{n,k,m,R}([t-h, t+h]) \le \nu^*_R(E^{(A_*\sqrt{h})}_{n,k,m,R}(t; [-h, 2h])). \tag{170.8}$$

Reduce h_ε so that the strip width is below $1/4$. For this fixed h, Lemma 170.1 bounds the limsup of the right side times m^2 by Ch^2. Division by $a_* h$ proves (170.6). An atom is bounded by every interval in that estimate. Let h tend to zero after the limsup in m to obtain (170.7). The auxiliary roof interval is used only to bound a positive source; neither exact integer label is averaged, and the conclusion concerns the original individual roof value, including coincident critical values. □

170.3. **A pointwise-small part of the original source.** For any sequence $r_m \downarrow 0$, eventually less than h_ε, restrict the original exact-label source to

$$\mathcal{U}^{\varepsilon}_{n,k,m,R}(r_m) = \bigcup_{z \in \mathcal{C}^{\varepsilon}_{n,k,m,R}} U_z(r_m).$$

Write $p^{\varepsilon,\mathrm{crit}}_{n,k,m,R}$ for its roof density. The source has not been replaced by a probability with a new normalization; it is a positive restriction of ν^*_R.

Theorem 170.3 (Uniform pointwise removal of regular critical collars)**.** *For every fixed $\varepsilon > 0$ and every $r_m \downarrow 0$,*

$$\sup_{R,n,k} m^2 \|p^{\varepsilon,\mathrm{crit}}_{n,k,m,R}\|_\infty \longrightarrow 0. \tag{170.9}$$

For arbitrary measurable insertions of modulus at most M, the same restricted-source conclusion holds with the upper bound multiplied by M. It is uniform over this whole bounded insertion class.

There is also a coefficient-level version. For each z choose a coefficient β_z with $|\beta_z| \le MJ_z$ and a function $0 \le \vartheta_z \le 1$ supported in $(0, r_m)$. Then

$$j^\varepsilon_{n,k,m,R}(t) = \sum_{z \in \mathcal{C}^\varepsilon_{n,k,m,R}} \beta_z \vartheta_z(t - t_z) \quad \textit{satisfies} \quad \sup_{R,n,k} m^2 \|j^\varepsilon_{n,k,m,R}\|_\infty \longrightarrow 0. \tag{170.10}$$

In particular this applies to localized step profiles with their original intrinsic regular critical coefficients.

Proof. The collar density upper bound in (169.10), disjointness, and positivity give

$$p^{\varepsilon,\mathrm{crit}}_{n,k,m,R}(t) \le A_* \mathcal{J}^\varepsilon_{n,k,m,R}([t - r_m, t]). \tag{170.11}$$

Fix $h > 0$ in (170.6). Eventually $r_m < h$, so that estimate bounds the limsup of the left side's essential-supremum norm, multiplied by m^2, by Ch. Now let $h \downarrow 0$. This proves (170.9) for every such sequence; no spectral constant is evaluated at r_m^{-1}. A bounded insertion is dominated in variation by M times the same positive source, so its density has the same pointwise domination. For the coefficient version use directly $|j^\varepsilon(t)| \le M\mathcal{J}^\varepsilon([t - r_m, t])$. The identical argument proves (170.10), even when many supports overlap or several critical values coincide. □

Corollary 170.4 (Removal from the complete signed correction)**.** *Let K_B be the unchanged convolution kernel in Theorem 166.3. For either the weighted critical-source density or the step profile in the preceding theorem, write f^{crit} for that component. Then*

$$\sup_{B>0} \sup_{R,n,k} m^2 \|f^{\mathrm{crit}} - K_B * f^{\mathrm{crit}}\|_\infty \longrightarrow 0. \tag{170.12}$$

For the positive-source decomposition $p = p^{\varepsilon,\mathrm{crit}} + p^{\varepsilon,\mathrm{rest}}$ this gives, in the same uniform pointwise sense,

$$\mathscr{D}_{B,n,R} = p^{\varepsilon,\mathrm{rest}} - K_B * p^{\varepsilon,\mathrm{rest}} + o(m^{-2}). \tag{170.13}$$

The coefficient and source normalizations in this identity are the original ones. Its error estimate is uniform in $B > 0$.

Proof. The scale-independent identity $\|K_B\|_1 = \|K_1\|_1$ gives $\|f - K_B * f\|_\infty \le (1 + \|K_1\|_1)\|f\|_\infty$. Apply Theorem 170.3 and use linearity. The uniformity in B here is a convolution inequality for a part of the source already proved pointwise small. It is not an estimate for transfer-operator constants on growing frequency bands. □

Boundary of the estimate. The argument estimates the entire regular critical cluster under the fixed exponentially relaxed margin envelope, rather than multiplying an individual edge bound by an uncontrolled word count. The margin is fixed in (170.6); no usable dependence of the asymptotic threshold on a margin tending to zero is asserted. Critical words violating this envelope near grazing or section decisions, other boundary contributions, and the remaining noncritical signed inverse are not covered by (170.13). The full raw target retains $\mathcal{L}_{m,R}$, and at fixed radius $c\mathfrak{a}_R g_{\Omega_R}$, as its main term. The pointwise-small positive restriction is returned to the exact source identity; it is not a replacement of the requested singleton by a roof window.

171. Positive windows and the noncritical raw inverse

The fixed-band theorem controls interval probabilities at the original integer labels. We use those positive probabilities to control a noncritical density and its first derivative. The comparison moves the two endpoint contacts, preserves every section decision, and changes the roof by a prescribed amount. The number of collision words never enters the bound. This supplies a local-scale estimate for a regular part of the signed inverse, rather than a certificate for reconstructing that part at a bandwidth chosen from an unknown derivative norm.

Throughout this section m is the collision count, n is the return index, and $1 \le n \le m$. Set

$$\Pi_{n,k,m,R} = \{K_{n,R} = k,\ N_{n,R} = m\}.$$

All source restrictions are taken under the original probability ν_R^*, without subsequent normalization. On this event the roof $T_{n,R}$ is the length of the same m-flight word. Endpoint coordinates u, v are contact arclengths. Local arclength coordinates differ by translations or changes of orientation; the gradients and divergences below use their flat metric.

171.1. **Protection away from a critical point.** Use q, d_j, e_j, e'_j of Section 169. Call an arbitrary regular m-flight source state ε-protected if its incidences, section distances and nonincident clearances satisfy the same inequalities as there. Normality of the endpoints is not required. Denote this set by $\mathsf{P}_{m,R}(\varepsilon)$, and fix a sufficiently small constant $\varepsilon_0 > 0$. All statements below use $0 < \varepsilon < \varepsilon_0$.

Lemma 171.1 (Endpoint continuation for a protected regular source)**.** *There are constants $r, C > 0$, independent of m, R, ε, such that every source state in $\mathsf{P}_{m,R}(\varepsilon)$ has an endpoint square of radius $r\varepsilon^3$ on which its physical word and every section decision persist. If F is the reduced action and $w\,du\,dv$ is the section source density on that square, then*

$$\nabla F = (-p_0, p_m), \qquad 0 < \nabla^2 F \le C I_2, \qquad |\nabla \log w| \le C\varepsilon^{-3}. \tag{171.1}$$

All margins remain at least half their original lower bounds. The lower Hessian bound may be taken as $\varepsilon I_2/(2R_+)$. In a square whose endpoint tangential velocities have modulus at most $1/2$, it may instead be taken as a fixed positive multiple of I_2. Here $R_+ = 47/100$.

Proof. The internal matrix (169.2), its row-sum bound, and the cancellation in the endpoint–internal derivative do not use normality of either endpoint. In fact their endpoint factors c_0, c_m are at most one. Thus (169.1) and (169.4) hold at every protected regular source. The continuation argument in Lemma 169.1 therefore preserves the same half-margins on the stated square. For each fixed word, endpoint injectivity glues the local continuations; there is no new multiplicity in these coordinates.

The first variation gives the gradient. The Schur-complement argument of Lemma 167.1 bounds the reduced Hessian below by the two endpoint curvature terms. The upper bound is the endpoint principal block of the full contact Hessian, since the eliminated block is positive. Flight lengths are bounded below by $3/50$ and $R \ge 9/20$, so this upper bound is independent of the internal incidences and of word length.

The cross-density formula (169.7) is also valid without normal endpoints. It retains the factor $(4\pi Rc)^{-1}$, where $c = \nu(Y_R^*)$. The internal potential terms in its logarithmic derivative are bounded by $C\varepsilon^{-3}\sum_j q^{d_j/4}$, as before. Now the two endpoint cosines are only bounded below by $\varepsilon/2$. Their differentials are at most $C\varepsilon^{-1}(|du| + |dv|)$: incidence is a Lipschitz function of adjacent contact positions, and their endpoint responses satisfy (169.4). Their logarithmic differentials therefore cost at most $C\varepsilon^{-2}(|du| + |dv|)$, still within the claimed budget. The other length terms cost at most $C\varepsilon^{-1}(|du| + |dv|)$. This proves the last estimate. Empty internal matrices require only the two endpoint terms. □

Let $\Theta : [0,\infty) \to [0,1]$ be a fixed nondecreasing smooth function which is zero on $[0,1]$, one on $[2,\infty)$, and has bounded derivative. At contact j let s_j be the distance to ∂Y_R^* in the fixed collision cylinder, on either side of that boundary. At flight j let Δ_j be its least clearance from a nonincident disk. Distances and clearances may be capped at one. Only a fixed finite collection of disks can affect that capped clearance. Define the source guard

$$\Xi_{m,R}^{\varepsilon} = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon e_j}\right) \Theta\left(\frac{s_j}{\varepsilon e_j}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon e_j'}\right). \tag{171.2}$$

Set it to zero on nonregular states. This is a restriction of the physical source, not a change of its collision record or return index.

Lemma 171.2 (A guard with a word-length-independent derivative)**.** *The guard is between zero and one, is one on $\mathsf{P}_{m,R}(2\varepsilon)$, and can be nonzero only on $\mathsf{P}_{m,R}(\varepsilon)$. On regular endpoint charts it is locally Lipschitz and*

$$|\nabla \Xi_{m,R}^{\varepsilon}| \le C\varepsilon^{-2} \quad \textit{almost everywhere.} \tag{171.3}$$

For each fixed m its zero extension vanishes near every boundary of its physical word and of every section-decision domain. No constants in (171.3) *grow with m.*

Proof. The first assertions follow directly from (171.2). Where a term in its derivative is nonzero, all the other nonzero factors impose their ε-margins. Adjacent-contact formulas and (169.1) consequently give

$$|\nabla c_j| + |\nabla s_j| \le C\varepsilon^{-1} q^{3d_j/4}.$$

A corresponding bound holds for Δ_j with the flight distance index; taking the minimum of finitely many Lipschitz clearances does not increase their largest Lipschitz constant. Dividing a differentiated factor by εe_j costs at most $C\varepsilon^{-2} q^{d_j/2}$. At most two contacts occur at each distance from the endpoints, and the geometric series is finite. This proves (171.3). At points where a factor is zero the same conclusion holds almost everywhere by the Lipschitz product rule. For fixed m each lower margin is strictly positive; hence the zero extension has no boundary jump. Distances at rectangle corners need not be smooth: their weak first derivatives suffice here. □

171.2. **A positive flow-box comparison.** For $h > 0$ define the finite-time local-window budget

$$\mathcal{Q}_m(h) = \frac{m^2}{2h} \sup_{R,n,k,t} \nu_R^*(\Pi_{n,k,m,R} \cap \{|T_{n,R} - t| < h\}). \tag{171.4}$$

Theorem 160.3 and the boundedness of its transition kernel give

$$\limsup_{m\to\infty} \mathcal{Q}_m(h) \le C_0 \quad \text{for each fixed } h > 0, \tag{171.5}$$

with C_0 independent of h. The window remains fixed in this limit. This upper bound uses all arithmetic branches and all targets; it uses neither a positive microscopic denominator nor a trivial residue.

Lemma 171.3 (Transport from a roof level to a positive window)**.** *Fix $0 < \delta < 1/10$. On the protected source with $|\nabla F| \ge \delta$, put $V = \nabla F/|\nabla F|^2$. For a fixed sufficiently small $c_1 > 0$, set*

$$h_{\varepsilon,\delta} = c_1 \varepsilon^3 \delta^2. \tag{171.6}$$

The flow of V exists for $|s| \le h_{\varepsilon,\delta}$ from every such point, changes F to $F + s$, and keeps its original exact labels. Its physical density Jacobian is between e^{-1} and e. Consequently, for every measurable subset A of that protected noncritical source, its roof density satisfies

$$\rho_A(t) \le \frac{e}{2h_{\varepsilon,\delta}} \nu_R^*(\Pi_{n,k,m,R} \cap \{|T_{n,R} - t| < h_{\varepsilon,\delta}\}) \tag{171.7}$$

almost everywhere. The inequality also holds when A is the union of such subsets over all physical words with the same exact labels.

Proof. As long as $|\nabla F| \ge \delta/2$, the flow speed is at most $2/\delta$. Its endpoint displacement is at most $2h_{\varepsilon,\delta}/\delta$. Lemma 171.1 and a first-exit argument show that this stays inside the endpoint square and keeps the gradient above $\delta/2$, after reducing c_1. Indeed $h_{\varepsilon,\delta} \le c_1\varepsilon^3\delta$ and $h_{\varepsilon,\delta} \le c_1\delta^2$. All half-margins and all section decisions are retained. The identity $VF = 1$ gives the exact roof translation.

Writing divergence relative to $w\,du\,dv$ gives

$$\operatorname{div}_w V = \operatorname{div} V + \nabla \log w \cdot V, \qquad |\operatorname{div}_w V| \le C(\delta^{-2} + \varepsilon^{-3}\delta^{-1}).$$

This follows by differentiating V and using the Hessian and log-density bounds. Multiplication by $h_{\varepsilon,\delta}$ gives at most $Cc_1(\varepsilon^3 + \delta)$. The Liouville formula for the flow therefore bounds $w(\Phi_s z)|\det D\Phi_s(z)|/w(z)$ between e^{-1} and e.

On a regular level $F = t$, the flow map from that level times $(-h_{\varepsilon,\delta}, h_{\varepsilon,\delta})$ is injective. Equality of two images first forces equality of the flow times, by the roof value, and then equality of the starting points, by uniqueness of the flow. Coarea and the density Jacobian give its source mass at least

$$2h_{\varepsilon,\delta}e^{-1}\int_{A\cap\{F=t\}} \frac{w}{|\nabla F|}\, d\mathcal{H}^1.$$

Its image lies in the positive event on the right of (171.7). Different physical words remain disjoint throughout the flow. Summing their mass inequalities thus proves the same bound for their union, without a multiplicity or word-count factor. Coarea proves the statement for almost every level. □

171.3. **A first-derivative budget for the actual regular density.** We specify the weighted class. An insertion a is admissible with bounds (M, L) if $|a| \le M$ and, on each regular endpoint chart where the $\varepsilon/2$-margins hold, it has weak endpoint gradient at most L. Bounds on a complex gradient mean the sum of the bounds for its real and imaginary parts. No derivative across a physical singularity is requested. Products $a_0(x)a_1(T_R^m x)$ of uniformly bounded Lipschitz collision-state functions are admissible with a common (M, L): the endpoint-state maps are $(u, -\partial_u F)$ and $(v, \partial_v F)$, up to the fixed arclength conversion, and their first derivatives are bounded by (171.1). This assertion does not include an arbitrary measurable path selector.

Put $H_\delta = \Theta(|\nabla F|/\delta)$ on each word, and define

$$g^{\varepsilon,\delta,a}_{n,k,m,R}(t)\,dt = (T_{n,R})_*(\mathbf{1}_{\Pi_{n,k,m,R}} a\Xi^{\varepsilon}_{m,R} H_\delta\, \nu^*_R). \tag{171.8}$$

Thus both the exact labels and the observed roof are unchanged.

Theorem 171.4 (Noncritical density and derivative at the local scale)**.** *For every fixed ε, δ and every admissible insertion, $g = g^{\varepsilon,\delta,a}_{n,k,m,R}$ belongs to $W^{1,\infty}(\mathbb{R})$. There is an absolute constant C such that, for every $m \ge 1$,*

$$\begin{aligned} m^2\|g\|_\infty &\le CM\mathcal{Q}_m(h_{\varepsilon,\delta}),\\ m^2\|g'\|_\infty &\le C\mathcal{T}_{\varepsilon,\delta}(M,L)\mathcal{Q}_m(h_{\varepsilon,\delta}),\\ \mathcal{T}_{\varepsilon,\delta}(M,L) &= L\delta^{-1} + M(\varepsilon^{-3}\delta^{-1} + \delta^{-2}). \end{aligned} \tag{171.9}$$

The bounds are uniform over R, n, k and the whole stated insertion class. In particular the two normalized norms have finite limsups, with no factor exponential in the collision count.

Proof. Let A be the support of $a\Xi^{\varepsilon}_{m,R}H_\delta$ and of its weak first derivatives, ignoring null sets. Every such source point is ε-protected and has $|\nabla F| \ge \delta$. The first assertion in (171.9) follows from domination by $M\rho_A$ and Lemma 171.3.

For a compactly supported smooth roof test ψ, integration by parts on the regular endpoint domains gives

$$\int g(t)\psi'(t)\,dt = \sum_{\text{words}} \int a\Xi^{\varepsilon} H_\delta w V \cdot \nabla(\psi \circ F)\,du\,dv$$
$$= -\sum_{\text{words}} \int \operatorname{div}(a\Xi^{\varepsilon} H_\delta w V)\psi(F)\,du\,dv. \tag{171.10}$$

The guard vanishes near each physical or section-decision boundary, and H_δ vanishes near every critical point. For fixed m there are finitely many word domains. Arclength seams cancel as coordinate seams on the contact torus; they are not source boundaries. Lipschitz approximation justifies the same formula for the stated weak derivatives. There is no omitted boundary measure.

Almost everywhere on the nonzero integrand,

$$\left|\operatorname{div}(a\Xi^{\varepsilon} H_\delta w V)\right| \le C\mathcal{T}_{\varepsilon,\delta}(M,L) w \mathbf{1}_A.$$

Indeed the insertion derivative costs L/δ, the guard derivative costs $CM\varepsilon^{-2}/\delta$, the gradient cutoff costs CM/δ^2, and the weighted divergence costs $CM(\delta^{-2} + \varepsilon^{-3}\delta^{-1})$. The second budget in (171.9) now follows from (171.10) and the same positive slice comparison. These identities identify the distributional derivative as a bounded density, proving $W^{1,\infty}$, including the zero extension outside the physical roof range. Finally use (171.5) for the fixed flow width. □

Use exactly the real even kernel K_B of Section 166. It is Schwartz, $K_B(t) = BK_1(Bt)$, and $\int K_B = 1$. Put $\kappa_1 = \int |s|\,|K_1(s)|\,ds < \infty$.

Corollary 171.5 (A common-band estimate for the noncritical inverse)**.** *For all $B > 0$,*

$$m^2 \|g - K_B * g\|_\infty \le \frac{C\kappa_1}{B}\,\mathcal{T}_{\varepsilon,\delta}(M,L)\mathcal{Q}_m(h_{\varepsilon,\delta}). \tag{171.11}$$

Consequently this signed correction tends to zero at the m^{-2} scale for every $B = B_m \to \infty$, with ε, δ fixed. No upper bound on the rate of growth of B_m is required.

Proof. The continuous representative of a $W^{1,\infty}$ function satisfies $|g(t) - g(t-s)| \le \|g'\|_\infty |s|$. Integrate this inequality against $|K_B(s)|$ and apply (171.9). The spectral input has been used only at the fixed positive window $h_{\varepsilon,\delta}$. The bandwidth in this corollary belongs to an approximate-identity inequality for a proved density derivative; it is not substituted into a fixed-band spectral theorem. □

172. The complete protected correction and collapsing margins

We combine the regular density estimate with the positive critical collars. The resulting estimate concerns the entire smoothly protected source, including its noncritical inverse. The remaining source is identified positively and has uniformly small total mass as protection is removed. Small mass will not be substituted for a pointwise density estimate; the final arithmetic criterion retains that remaining signed correction explicitly.

172.1. **Near-normal sources belong to actual critical collars.**

Lemma 172.1 (Completion to a protected normal critical word)**.** *There are $d_*, C_* > 0$, independent of m, R, ε, such that if $x \in \mathsf{P}_{m,R}(\varepsilon)$ and $|\nabla F(x)| \le 2\delta$, with $0 < \delta < d_*\varepsilon^3$, then x belongs to a collar $U_z(C_*\delta^2)$ from Proposition 169.3. Its critical center z is an $\varepsilon/2$-protected normal-to-normal word with exactly the same (n,k,m) labels. Moreover $C_*\delta^2 < h_{\varepsilon/2}$.*

Proof. The two endpoint tangential velocities at x have modulus at most 2δ. On the endpoint square in Lemma 171.1, reduce its fixed radius constant so that they stay below $1/2$. Its reduced Hessian consequently has lower and upper bounds $\lambda I_2 \le \nabla^2 F \le \Lambda I_2$ with absolute positive constants. On a ball of radius proportional to ε^3 inside that square,

$$F(x+v) - F(x) \ge -2\delta|v| + \lambda|v|^2/2.$$

For d_* sufficiently small the minimum on this closed ball is interior. Strict convexity makes it unique; call it z. The first variation gives $p_0(z) = p_m(z) = 0$, and strong monotonicity of the gradient gives $|z - x| \le 2\delta/\lambda$. Continuation preserved every half-margin and every exact label, so z is $\varepsilon/2$-protected.

After a further fixed reduction of d_*, the original point x lies in the endpoint square of radius $r_*(\varepsilon/2)^3$ about z. The continuations coincide by endpoint injectivity. Taylor's formula there gives $0 \le F(x) - F(z) \le C_*\delta^2$. Take C_* slightly larger if needed to make the upper inequality strict on the indicated support. Finally choose d_* so that $C_* d_*^2 < h_* 2^{-6}$. This proves the last assertion and the inclusion in the actual collar, without replacing the return event by a nearby index. □

Let

$$\mathcal{C}_m^{\eta}(r) = \sup_{R,n,k,t} m^2 \mathcal{J}^{\eta}_{n,k,m,R}([t-r, t+r]). \tag{172.1}$$

For fixed $\eta > 0$ and $0 < r < h_\eta$, the proved cluster bound is

$$\limsup_{m\to\infty} \mathcal{C}_m^{\eta}(r) \le Cr. \tag{172.2}$$

Keep the kernel K_B fixed as in the preceding section and put $\kappa_0 = 1 + \|K_1\|_1$. For an admissible insertion define the complete protected density and its signed correction by

$$\begin{aligned} f^{\varepsilon,a}_{n,k,m,R}(t)\,dt &= (T_{n,R})_*(\mathbf{1}_{\Pi_{n,k,m,R}} a \Xi^{\varepsilon}_{m,R}\, \nu_R^*), \\ \mathscr{D}^{\varepsilon,a}_{B,n,k,m,R} &= f^{\varepsilon,a}_{n,k,m,R} - K_B * f^{\varepsilon,a}_{n,k,m,R}. \end{aligned} \tag{172.3}$$

The cutoff is the explicit physical guard (171.2); no intrinsic edge coefficient is altered.

Theorem 172.2 (The full smoothly protected signed correction)**.** *For $0 < \delta < d_*\varepsilon^3$ and every $B > 0$, uniformly over all exact labels and all admissible insertions with bounds (M, L),*

$$\sup_{R,n,k} m^2 \|\mathscr{D}^{\varepsilon,a}_{B,n,k,m,R}\|_\infty \le C\left[\frac{\mathcal{T}_{\varepsilon,\delta}(M,L)}{B}\mathcal{Q}_m(h_{\varepsilon,\delta}) + M\mathcal{C}_m^{\varepsilon/2}(C_*\delta^2)\right]. \tag{172.4}$$

In particular there are absolute constants C, C_2 such that, for $B \ge C_2\varepsilon^{-12}$,

$$\limsup_{m\to\infty} \sup_{R,n,k,a} m^2 \|\mathscr{D}^{\varepsilon,a}_{B,n,k,m,R}\|_\infty \le C(M+L)B^{-1/2}. \tag{172.5}$$

The bandwidth and protection are fixed in this limsup. Also, for every fixed $\varepsilon > 0$ and every deterministic $B_m \to \infty$,

$$\sup_{R,n,k,a} m^2 \|\mathscr{D}^{\varepsilon,a}_{B_m,n,k,m,R}\|_\infty \longrightarrow 0. \tag{172.6}$$

No rate in m is asserted in either conclusion.

Proof. Split the guard source using H_δ from (171.8). Its high-gradient part is $g^{\varepsilon,\delta,a}$, controlled by Corollary 171.5. Its remaining part has modulus dominated by M times the positive source with $\Xi^{\varepsilon} > 0$ and $|\nabla F| \le 2\delta$. Lemma 172.1 places this latter source in the union of the $\varepsilon/2$-protected critical collars of roof width $C_*\delta^2$. Equation (170.11) therefore bounds its density in modulus by

$$MA_*\mathcal{J}^{\varepsilon/2}_{n,k,m,R}([t - C_*\delta^2, t]).$$

The operator $f \mapsto f - K_B * f$ has L^∞ norm at most κ_0, for every B. Combining the two bounds proves (172.4). It controls the actual regular inverse and the actual near-critical source, not an absolute sum of formal word jets.

For fixed ε, δ, B, take the collision limsup using (171.5) and (172.2). The result is

$$C\left[\frac{L\delta^{-1} + M(\varepsilon^{-3}\delta^{-1} + \delta^{-2})}{B} + M\delta^2\right]. \tag{172.7}$$

For $\delta = B^{-1/4}$ the geometric restriction holds if $B \geq C_2\varepsilon^{-12}$, with C_2 chosen strictly larger than d_*^{-4}. The two terms $B^{-1}\delta^{-2}$ and δ^2 equal $B^{-1/2}$. Also $\varepsilon^{-3}B^{-1}\delta^{-1} = \varepsilon^{-3}B^{-3/4} \leq CB^{-1/2}$ under the same restriction, and $LB^{-3/4} \leq LB^{-1/2}$. This proves (172.5).

For a prescribed diverging sequence B_m, keep ε, δ fixed in (172.4). The first term tends to zero because its local-window budget is eventually bounded. The second has limsup at most $CM\delta^2$. Only then let $\delta \downarrow 0$. This proves (172.6). It does not use (172.5) with $B = B_m$ inside a collision limsup. In particular it gives no polynomial rate along that sequence. □

172.2. **The positive source outside the smooth protection.** We next quantify the mass of the exact remainder, uniformly in the length of the history. Its density is not bounded by its mass.

Lemma 172.3 (One-flight boundary neighborhoods)**.** *There is $C < \infty$, uniform in R, such that for $0 < s < 1$,*

$$\nu\{c_0 < s\} \leq s^2, \qquad \nu\{\operatorname{dist}(x, \partial Y_R^*) < s\} \leq Cs, \qquad \nu\{\Delta_0 < s\} \leq Cs. \tag{172.8}$$

The last event refers to a regular physical flight and excludes its two incident disks.

Proof. In the flat coordinates (α, p) the collision probability is $(4\pi)^{-1}d\alpha\, dp$. Since $c_0 = (1-p^2)^{1/2}$, the first mass is $1 - \sqrt{1-s^2} \leq s^2$. The section is a fixed finite union of moving rectangles with uniformly bounded perimeter, which proves the second estimate, including their corners.

For the third, finite horizon restricts the relevant nonincident disks to a fixed finite set. Take first $s < s_0$ with $s_0 > 0$ sufficiently smaller than the disk separation $3/50$. A segment at clearance less than s from a nonincident disk attains its closest point in its interior: both segment endpoints lie on other disks and are at least $3/50$ from that disk. Write $r = |d - Rn_\alpha| \geq 1 - R \geq 53/100$ for the distance from the initial point to the candidate center. A regular segment misses its open disk, so its line has perpendicular distance in $[R, R+s]$. On either of the two tangent directions the derivative of this distance with respect to the direction angle has modulus

$$\sqrt{r^2 - (R+s)^2} \geq \sqrt{(53/100)^2 - (47/100 + s_0)^2} > 0.$$

Thus the allowed direction angles have total length at most Cs, uniformly in the initial angle and in the candidate. Restricting to outgoing directions or to the actual first-hit segment only decreases the set. The density in the direction angle is bounded, since $dp = \cos\varphi\, d\varphi$. Integrate in α and sum the fixed candidate set. For $s \geq s_0$ increase C and use the mass bound one. □

Proposition 172.4 (Uniform exhaustion in original source mass)**.** *The guards increase to one almost everywhere as $\varepsilon \downarrow 0$ for every fixed m, and*

$$\sup_{m \geq 1,\ R \in I} \int (1 - \Xi^\varepsilon_{m,R})\, d\nu_R^* \leq C\varepsilon. \tag{172.9}$$

Define the nonnegative exact-label remainder density by

$$b^\varepsilon_{n,k,m,R}(t)\, dt = (T_{n,R})_*(\mathbf{1}_{\Pi_{n,k,m,R}}(1 - \Xi^\varepsilon_{m,R})\, \nu_R^*). \tag{172.10}$$

It is supported by the source failing the 2ε protection conditions, and

$$\sup_{m,R}\sum_{n=1}^{m}\sum_{k}\|b^{\varepsilon}_{n,k,m,R}\|_1\le C\varepsilon. \tag{172.11}$$

For a bounded measurable insertion the same assertion holds in total variation, multiplied by its supremum bound. The total-variation norm here is the mass of the variation measure, without a factor $1/2$.

Proof. For numbers $0\le a_i\le 1$, $1-\prod_i a_i\le\sum_i(1-a_i)$. Apply this to (171.2); a summand can be nonzero only when its margin is less than twice its threshold. Invariance of ν and domination $\nu_R^*\le c^{-1}\nu$ permit Lemma 172.3 to be applied at every collision or flight. Consequently the expectation is at most

$$C\sum_{j=0}^{m}((2\varepsilon e_j)^2+2\varepsilon e_j)+C\sum_{j=0}^{m-1}2\varepsilon e'_j\le C\varepsilon.$$

At most two indices have any given distance from the ends, so all the geometric sums are bounded independently of m. No independence or mixing assertion is used here.

Every finite regular history away from section boundaries has finitely many positive margins. Monotonicity of Θ therefore gives the almost-everywhere exhaustion. The exact events $\Pi_{n,k,m,R}$ are disjoint as (n,k) varies at fixed m: collision m has at most one actual return index and displacement. Sum their restricted masses to obtain (172.11). Bounded insertions are dominated in variation by the same positive source. Absolute continuity of its roof pushforward follows also by domination by the complete density in Theorem 167.3. □

172.3. **Ordered protection scales and the arithmetic target.** Choose a fixed sufficiently large $A>0$ and, for all sufficiently large B, put

$$\varepsilon(B)=AB^{-1/12}<\varepsilon_0. \tag{172.12}$$

Take $A^{12}\ge C_2$. Then the condition of (172.5) holds for each such fixed B. It follows that the complete protected correction is at most $CB^{-1/2}$ at the normalized local scale after the collision limsup, while the complementary source has total mass at most $CB^{-1/12}$. These two estimates have different norms.

Theorem 172.5 (Localization of the full arithmetic pointwise obstruction)**.** *For every m,R,n,k and $B>0$ the unweighted full source has the exact almost-everywhere decomposition*

$$\begin{aligned}p_{n,R}(k,m,\cdot)&=f^{\varepsilon,1}_{n,k,m,R}+b^{\varepsilon}_{n,k,m,R},\\ \mathscr{D}_{B,n,R}(k,m,\cdot)&=\mathscr{D}^{\varepsilon,1}_{B,n,k,m,R}+b^{\varepsilon}_{n,k,m,R}-K_B*b^{\varepsilon}_{n,k,m,R}.\end{aligned} \tag{172.13}$$

With $\varepsilon=\varepsilon(B)$, the radius-uniform raw arithmetic local law with main term $\mathcal{L}_{m,R}$ holds if and only if, for every central compact set $|Z|\le M_0$,

$$\lim_{B\to\infty}\limsup_{m\to\infty}\sup_{R,n,k}m^2\operatorname*{ess\,sup}_{t:\ |Z|\le M_0}|b^{\varepsilon(B)}_{n,k,m,R}(t)-(K_B*b^{\varepsilon(B)}_{n,k,m,R})(t)|=0. \tag{172.14}$$

Here $Z=((k_1,k_2,t,n)-m(0,0,\bar\tau_R,c))/\sqrt{m}$, exactly as in Theorem 166.3. Its fixed-radius specialization retains $\mathfrak{a}_R$; no arithmetic factor is set to one.

Proof. The first identity is a partition of the original positive source. The second follows by linearity and (166.8); it does not depend on any choice of germ localization. For each fixed B, the full-source fixed-band inverse has main term $\mathcal{L}_{m,R}$ with error $o(1)$ after multiplication by m^2. Theorem 172.2 bounds the protected term's limsup by $CB^{-1/2}$ under (172.12). Thus (172.14), the triangle inequality, and the order $m\to\infty$ followed by $B\to\infty$ imply the full raw local law.

Conversely, that law and the fixed-band inverse make the full correction tend to zero at each fixed B. The same split bounds the limsup of the boundary correction by $CB^{-1/2}$, proving the stated criterion. The fixed-radius assertion is the unchanged specialization of the transition kernel. □

Corollary 172.6 (A count-dependent protection with no prescribed rate)**.** *For any given $B_m \to \infty$ there is a deterministic $\varepsilon_m \downarrow 0$ for which*

$$\sup_{R,n,k} m^2 \|\mathscr{D}^{\varepsilon_m,1}_{B_m,n,k,m,R}\|_\infty \to 0, \qquad \sup_R \sum_{n,k} \|b^{\varepsilon_m}_{n,k,m,R}\|_1 \to 0.$$

No rate for ε_m as a function of m is supplied.

Proof. Take any decreasing $\varepsilon_j \to 0$. For each j, (172.6) supplies an integer M_j beyond which its normalized error is at most 2^{-j}. Make these integers strictly increasing and take $\varepsilon_m = \varepsilon_j$ on $[M_j, M_{j+1})$. The mass assertion follows from (172.11). This diagonal argument does not replace the L^1 conclusion by a density bound, nor does it evaluate the full transition inverse at B_m. □

The remaining estimate. The protected noncritical inverse has now been controlled at the local scale, and the critical collars complete the same statement for the entire smooth protection. The remaining positive source in (172.10) includes near-grazing, competing-hit and section-decision histories with collapsing margins, together with their finite jumps and noncritical residuals. Its total mass tends to zero under the explicit ordered protection scale, but (172.14) is still a pointwise signed estimate. Neither small source mass nor the exponential fixed-count height bound implies it. The criterion retains precisely that estimate for the original four-coordinate record. The corresponding complete weighted boundary estimate and pointwise roof-conditioned bridge are not consequences of the regular insertion theorem above.

173. Pointwise deconcentration at endpoint decision boundaries

A section-decision boundary is not a singularity of a fixed physical collision word. A roof flow may cross such a boundary while retaining the same displacement and collision count. The return index can then change, but by no more than the number of decisions allowed to change. We use this finite enlargement only in a positive upper comparison. The density being estimated is always that of the original exact (n,k,m) event. This distinction permits a pointwise estimate for part of the collapsing-margin source, without a positive lower bound for its decision distances.

Write $c = \nu(Y^*_R)$, $q = 47/53$, and use e_j, e'_j and Θ from Sections 169 and 171. Fix an integer $J \ge 0$ and first take $m > 2J+2$. Put

$$\begin{aligned} \mathcal{E}_{m,J} &= \{0,\dots,J\} \cup \{m-J,\dots,m\}, \\ r_J &= 2J+2, \qquad \mathcal{I}_{n,J} = \{\ell \in \mathbb{Z} : |\ell - n| \le r_J\}. \end{aligned} \tag{173.1}$$

There are at most r_J occupation summands in this set of free decisions; the terminal decision at m is not an occupation summand. There is no restriction on the initial or terminal section membership in an auxiliary collision comparison below. Its measure is ν/c, which agrees with ν^*_R on the original initial section, but is not a probability on the whole cylinder.

For $0 < \eta < \varepsilon_0$ define

$$G^{\eta,J}_{m,R} = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\eta e_j}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\eta e'_j}\right) \prod_{j \notin \mathcal{E}_{m,J}} \Theta\!\left(\frac{s_j}{\eta e_j}\right). \tag{173.2}$$

Only the section factors in $\mathcal{E}_{m,J}$ have been omitted. All incidence and clearance factors remain. Let $\mathsf{P}^{\eta,J}$ denote the corresponding set of once-thresholded regular histories. For a second

parameter $0 < \varepsilon < \varepsilon_0$ put

$$E^{\varepsilon,J}_{m,R} = \prod_{j\in\mathcal{E}_{m,J}} \Theta\!\left(\frac{s_j}{\varepsilon e_j}\right),$$

$$d^{\eta,\varepsilon,J,a}_{n,k,m,R}(t)\,dt = (T_{n,R})_*(\mathbf{1}_{\Pi_{n,k,m,R}} a G^{\eta,J}_{m,R}(1 - E^{\varepsilon,J}_{m,R})\nu^*_R). \tag{173.3}$$

The insertion may be any measurable function with $|a| \le M$. Absolute continuity follows by domination by the complete physical density. The two margin parameters will be set equal only after the pointwise estimate has been proved.

173.1. **Small endpoint sets in the arithmetic local upper bound.** We require a version of the endpoint estimate without section factors. Use the uncentered collision occupation $A_{m,R}$, displacement $K^{\mathrm{c}}_{m,R}$, and roof $S_{m,R}$, so that on the original event

$$(K^{\mathrm{c}}_{m,R}, A_{m,R}, S_{m,R}) = (k, n, T_{n,R}).$$

In particular $A_{m,R}$ sums the visits at times $0, \dots, m-1$.

Lemma 173.1 (Small endpoint mass and an exact collision coefficient)**.** *Let $0 \le a_R, d_R \le 1$ be smooth endpoint functions with common collision-norm bounds and parameter continuity, for the fixed choices under consideration. Suppose $\nu(a_R) \le \alpha$ and $\nu(d_R) \le \beta$. For every fixed bounded interval I,*

$$\limsup_{m\to\infty} \sup_{R,k,\ell,t} m^2 \int a_R(x) d_R(T^m_R x) \mathbf{1}_{\{K^{\mathrm{c}}_{m,R}=k,\ A_{m,R}=\ell,\ S_{m,R}-t\in I\}}\, d\nu(x) \le C\alpha\beta|I|. \tag{173.4}$$

The constant does not depend on their fixed smooth norms. The threshold in the collision limit may depend on those norms and on I. The same conclusion follows by positive smooth upper approximation of endpoint sets.

Proof. Apply the exact full-occupation pairing and the finite-branch argument of Theorem 160.3, with endpoints a_R, d_R in place of η_R, η_R. The eigenvalues, moving maxima and quadratic matrices are unchanged. On every branch surviving along a convergent radius sequence, the limiting physical projection is an actual resonance. Its scalar amplitude has modulus

$$|\nu(d_R q_\gamma)\nu(a_R \overline{q}_\gamma)| \le \alpha\beta,$$

since $|q_\gamma| = 1$. The complex Gaussian integrals have a common upper bound for every real target, and the finite number of surviving peak charts is independent of the endpoint functions and the outer roof band. Every limiting roof resonance has roof coordinate zero. This is the same subsequence argument used in Lemma 170.1; it does not assume continuity of a generator of the arithmetic group.

For a positive Fourier majorant of $\mathbf{1}_I$, the limiting roof factor is its mass $|I| + 2\pi/B$. Positivity of the unchanged collision coefficient gives the upper inequality before taking limits. First send m to infinity with B and the endpoints fixed, and then send B to infinity. This proves (173.4). An upper approximation is taken with fixed smooth norms before either limit, and its excess mass is removed afterwards. No rate uniform in a shrinking endpoint set or roof interval is asserted. □

For a fixed constant $a_* > 0$ define the forward and backward endpoint sets

$$U^+_{R,J}(s) = \{x : \exists\, 0 \le j \le J, \operatorname{dist}(T^j_R x, \partial Y^*_R) < a_* s q^{j/4}\},$$
$$U^-_{R,J}(s) = \{y : \exists\, 0 \le j \le J, \operatorname{dist}(T^{-j}_R y, \partial Y^*_R) < a_* s q^{j/4}\}. \tag{173.5}$$

Singular finite histories may be included in these sets; they are null and are included in their smooth upper approximations.

Lemma 173.2 (Finite-block endpoint envelopes)**.** *For fixed J and $s>0$, both sets in* (173.5) *admit positive smooth upper approximations, locally in the parameter and uniformly on a finite parameter cover, with masses at most $Cs+\xi$ for arbitrary $\xi>0$. The constant C is independent of J. Consequently*

$$\limsup_{m\to\infty}\sup_{R,k,\ell,t} m^2\nu\{K^{\mathrm{c}}_{m,R}=k, A_{m,R}=\ell, S_{m,R}-t\in I, \\ x\in U^{+}_{R,J}(s)\ \text{or}\ T_R^m x\in U^{-}_{R,J}(s)\}\le Cs|I|. \tag{173.6}$$

Also the same collision event restricted by $|p_0|,|p_m|\le d$ has normalized limsup at most $Cd^2|I|$.

Proof. Invariance and the section-neighborhood estimate give

$$\nu(U^{\pm}_{R,J}(s))\le Cs\sum_{j=0}^{J} q^{j/4}\le Cs.$$

We explain why the endpoint regularity needed here follows from finite-time geometry rather than from a presumed long-pullback norm. For fixed J the finite union of forward or backward collision singularities and their preimages is null. At a fixed limiting radius, remove a closed neighborhood of this set and of grazing with mass less than ξ. On its compact complement every finite itinerary and every distance function in (173.5) persists uniformly under a small parameter change. A closed bad set with threshold a_*s is then contained in the corresponding open set with threshold $2a_*s$, together with that neighborhood. The latter union has mass at most $Cs+C\xi$, again by invariance. A smooth function equal to one on the compact bad set and on a slightly smaller singular neighborhood can be supported in a slightly larger such union. Its mass is at most $Cs+C\xi$. This also covers states close to the boundary of the compact complement. Finite parameter compactness and a subordinate continuous partition in R give common smooth norms and the required scalar continuity; the smooth norms may depend on J,s,ξ. Rescale ξ to obtain the stated excess. This argument uses a doubled distance threshold and therefore does not require a regular value for that distance.

Apply Lemma 173.1 separately to the initial and terminal upper envelopes, with the other endpoint one. The union is bounded by their sum. Remove ξ after the collision and band limits to prove (173.6). For the last assertion use smooth strip functions of mass at most $2d$ at both ends, exactly as in Lemma 170.1, but without multiplying by the section indicators. □

173.2. **Roof transport through finitely many section cuts.**

Lemma 173.3 (Continuation with free endpoint decisions)**.** *Every state in $\mathsf{P}^{\eta,J}$ has the endpoint continuation of Lemma 171.1, of radius $r\eta^3$. It preserves the center word, $K^{\mathrm{c}}_{m,R}$, m, and all section decisions outside $\mathcal{E}_{m,J}$. On a continuation starting with $A_{m,R}=n$ the occupation always belongs to $\mathcal{I}_{n,J}$. The Hessian and logarithmic-density bounds hold with η in place of ε. Moreover*

$$\max_{0\le j\le m} e_j^{-1}|\nabla s_j|\le C\eta^{-1}\quad \text{almost everywhere.} \tag{173.7}$$

The physical density is $w\,du\,dv=|\partial_u\partial_v F|(4\pi Rc)^{-1}du\,dv$, including when an endpoint crosses the section.

Proof. The endpoint response and density distortion use incidence and clearance, not section membership. The continuation proof checks a section distance only to preserve that particular decision. Omit those checks at the free indices and retain every other half-margin check. This proves the same endpoint square and all physical estimates. The collision-state response bound is $C\eta^{-1}q^{3\min(j,m-j)/4}$, up to a fixed adjacent-index factor. Distances to the finite rectangle boundary are Lipschitz, including at its corners. Division by e_j proves (173.7) without a factor

depending on J or m. A count of the free summands of A_m gives $|A_m - n| \le r_J$. The terminal visit is treated as a membership condition only and is dropped in the upper comparison.

A section cut does not change the reduced physical action or its cross Hessian. Thus these functions glue across a free decision cut. The displayed density is precisely ν/c on the continued word; no extension of ν_R^* as a probability or new section normalization is used. □

Proposition 173.4 (A pointwise bound off the normal endpoints)**.** *Let A be any measurable subset of $\Pi_{n,k,m,R} \cap \mathsf{P}^{\eta,J}$ with $|\nabla F| \ge \delta$ and $s_j < 2\varepsilon e_j$ at some $j \in \mathcal{E}_{m,J}$. Its roof density, for the original source ν_R^*, satisfies*

$$\limsup_{m\to\infty} \sup_{R,n,k,A} m^2 \|\rho_A\|_\infty \le C(J+1)\varepsilon. \tag{173.8}$$

Here $J \ge 0$, $\eta, \varepsilon > 0$ and $0 < \delta < 1/10$ are fixed before the limit, with J an integer. The constant C is independent of these choices.

Proof. Use $V = \nabla F/|\nabla F|^2$. With a fixed sufficiently small c_2, choose the fixed roof width

$$h = c_2 \min\{\eta^3\delta^2,\, \varepsilon\eta\delta\}. \tag{173.9}$$

The first term ensures the continuation, the lower gradient $\delta/2$ and the physical density Jacobian between e^{-1} and e, as in Lemma 171.3. The second term and (173.7) ensure that at least the initially bad free decision still has $s_j < 3\varepsilon e_j$ throughout the flow. The flow is allowed to cross the free section cuts; it crosses no physical-word or protected middle-decision boundary.

For almost every level $F = t$, the map from that level times $(-h, h)$ is injective. The roof identifies the flow time and uniqueness identifies the starting point. Across free decision cuts uniqueness is applied to the single physical-word vector field, not to separate cut domains. Different center words remain disjoint. Its image lies in the event

$$K^{\mathrm{c}}_{m,R} = k, \quad A_{m,R} \in \mathcal{I}_{n,J}, \quad |S_{m,R} - t| < h,$$

with $x \in U^+_{R,J}(\varepsilon)$ or $T^m_R x \in U^-_{R,J}(\varepsilon)$, taking $a_* = 3$. Weighted coarea consequently gives

$$\rho_A(t) \le \frac{e}{2hc} \sum_{\ell \in \mathcal{I}_{n,J}} \nu\{K^{\mathrm{c}}_{m,R} = k, A_{m,R} = \ell, |S_{m,R} - t| < h, \text{bad endpoint set}\}. \tag{173.10}$$

There are at most $4J + 5$ terms. Apply (173.6) to this fixed h and cancel the factor $2h$. This proves (173.8). The cancellation occurs after a fixed-window local upper bound; it is not a substitution of a shrinking window into that theorem. The left side retains the prescribed original return index. □

173.3. **Critical completion with the same finite decision allowance.**

Lemma 173.5 (Low-gradient completion and cluster bound)**.** *For fixed η, J and $0 < \delta < d_*\eta^3$, the density of any positive restriction of $\Pi_{n,k,m,R} \cap \mathsf{P}^{\eta,J} \cap \{|\nabla F| \le 2\delta\}$ satisfies*

$$\limsup_{m\to\infty} \sup_{R,n,k} m^2 \|\rho\|_\infty \le C(J+1)\delta^2. \tag{173.11}$$

The source is still unnormalized. No endpoint section membership is required at the completed critical center.

Proof. Complete each source point in its free-decision endpoint square to the unique nearby normal-to-normal critical center. The proof of Lemma 172.1 uses strong convexity and the preserved physical and middle margins, so applies unchanged after omitting the free section checks. The center is within $C\delta$ and its roof within $C\delta^2$. It is $\eta/2$-protected at every retained margin. Its middle occupation is the original one; its total occupation can differ from n by at most r_J.

Use the actual mass collars of these centers in the physical word, with source ν/c. Their intrinsic coefficients are $J_z = 2\pi w_z(z)/\sqrt{\det \nabla^2 F_z(z)}$ in this same measure. For $0 < r < h_{\eta/2}$, a collar of roof width r has mass at least $a_* r J_z$ and both endpoint tangential velocities at most $A_*\sqrt{r}$. Take each physical collar only once even if its free decisions split it into several return-index pieces. Physical endpoint injectivity and the inherited collar construction give disjoint original-source collars. The continuation keeps the middle occupation, so every point of their union has $A_m \in \mathcal{I}_{n,J}$. Here the middle count of a center is the same as that of its starting source: both occupations lie between that count and that count plus r_J. Thus their mutual difference is at most r_J, not an iterated enlargement.

For centers with critical roof in $[t-r, t+r]$, the positive collar comparison now uses the interval $[t-r, t+2r]$, the two normal endpoint strips, and at most $4J+5$ occupation coefficients. The last assertion of Lemma 173.2 bounds its normalized limsup by $C(J+1)r^2$. Divide by the collar mass factor $a_* r$ to obtain the cluster bound $C(J+1)r$ for the sum of intrinsic coefficients. The collar-density upper estimate then bounds the completed source by this cluster on a roof interval of width $C\delta^2$. This proves (173.11). All collar widths and strip widths are fixed in the collision limit. □

Theorem 173.6 (Endpoint decision boundaries are pointwise small)**.** *For fixed $J \geq 0$, $\eta, \varepsilon \in (0, \varepsilon_0)$,*

$$\limsup_{m\to\infty} \sup_{R,n,k,\,|a|\leq M} m^2 \|d^{\eta,\varepsilon,J,a}_{n,k,m,R}\|_\infty \leq CM(J+1)\varepsilon. \tag{173.12}$$

Consequently the unchanged kernel K_B satisfies

$$\limsup_{m\to\infty} \sup_{B>0} \sup_{R,n,k,\,|a|\leq M} m^2 \|d^{\eta,\varepsilon,J,a} - K_B * d^{\eta,\varepsilon,J,a}\|_\infty \leq CM(J+1)\varepsilon. \tag{173.13}$$

The supremum over insertions includes arbitrary bounded measurable path functions, by domination. No derivatives of the insertion are needed. The constants are independent of the fixed protection η; the asymptotic threshold is not asserted independent of it.

Proof. It suffices to estimate the positive source with $a = 1$. On its support some free decision has $s_j < 2\varepsilon e_j$. Split at $|\nabla F| = \delta$, applying Proposition 173.4 to the high-gradient part and Lemma 173.5 to the rest. The normalized limsup is at most $C(J+1)(\varepsilon + \delta^2)$. Send δ to zero only after that limsup, with η, ε, J fixed, to prove (173.12). Radon–Nikodym domination gives the uniform bound for $|a| \leq M$. Finally $\|K_B\|_1 = \|K_1\|_1$ and $\|f - K_B * f\|_\infty \leq (1 + \|K_1\|_1)\|f\|_\infty$ prove (173.13). The all-band statement is a convolution consequence of pointwise density control, not a spectral bound with growing frequency. □

Corollary 173.7 (Finite jumps of this boundary component)**.** *Where finite one-sided traces of the unweighted boundary density exist, the height of each of its jumps satisfies the bound* (173.12)*, with twice its constant. In particular these jump heights are $o(m^{-2})$ in the ordered limits $m \to \infty$, then $\varepsilon \downarrow 0$, for fixed J, η. This statement does not bound the sum of the absolute heights of all jumps and does not apply to the remaining grazing or clearance source.*

Proof. Both finite traces of an essentially bounded density have modulus at most its essential bound whenever they exist. Their difference is at most twice that bound. Apply Theorem 173.6. The intrinsic right-trace convention is unchanged. □

174. First bad margins after endpoint deconcentration

The endpoint estimate pays a genuine part of the pointwise boundary correction. We now separate it from the remaining incidence, clearance and interior-decision sources. The split is positive, takes place before pushforward, and preserves every original coefficient. A mass estimate for the residual is recorded only in its own norm.

Set $\eta = \varepsilon$ in (173.2) and (173.3). Then the original guard factors exactly as

$$\Xi^{\varepsilon}_{m,R} = G^{\varepsilon,J}_{m,R} E^{\varepsilon,J}_{m,R}, \qquad 1 - \Xi^{\varepsilon}_{m,R} = (1 - G^{\varepsilon,J}_{m,R}) + G^{\varepsilon,J}_{m,R}(1 - E^{\varepsilon,J}_{m,R}). \tag{174.1}$$

For $m \le 2J + 2$ take $\mathcal{E}_{m,J} = \{0, \dots, m\}$, omitting repeated indices. This defines the identities for every finite count; it does not affect any fixed-J collision limit.

List the factors of $G^{\varepsilon,J}_{m,R}$ as $w_1, \dots, w_N$, ordered by distance from the nearest endpoint and, at ties, by incidence, clearance, section distance, then increasing physical index. Flight distance means $\min(j, m - 1 - j)$. Fix this order once. Define disjoint measurable sets

$$A_i = \{w_i < 1\} \cap \bigcap_{h<i} \{w_h = 1\}. \tag{174.2}$$

They partition $\{G < 1\}$, not just the set where a factor vanishes. A first defect includes the transition region of the smooth guard. Let $\mathcal{A}_{\mathrm{inc}}, \mathcal{A}_{\mathrm{clr}}, \mathcal{A}_{\mathrm{mid}}$ be the unions corresponding to the three kinds of factor. Put

$$\begin{aligned} r^{\varepsilon,J,\mathrm{a}}_{n,k,m,R}(t)\,dt &= (T_{n,R})_*(\mathbf{1}_{\Pi_{n,k,m,R}}(1 - G^{\varepsilon,J}_{m,R})\mathbf{1}_{\mathcal{A}_{\mathrm{a}}}\nu^*_R), \qquad \mathrm{a} \in \{\mathrm{inc}, \mathrm{clr}, \mathrm{mid}\}, \\ r^{\varepsilon,J}_{n,k,m,R} &= r^{\varepsilon,J,\mathrm{inc}}_{n,k,m,R} + r^{\varepsilon,J,\mathrm{clr}}_{n,k,m,R} + r^{\varepsilon,J,\mathrm{mid}}_{n,k,m,R}. \end{aligned} \tag{174.3}$$

All these densities are nonnegative and have their original normalization. Together with f^{ε} from (172.3) and $d^{\varepsilon,\varepsilon,J}$ from (173.3), they give the exact full density

$$p_{n,R}(k, m, \cdot) = f^{\varepsilon}_{n,k,m,R} + d^{\varepsilon,\varepsilon,J}_{n,k,m,R} + r^{\varepsilon,J}_{n,k,m,R}. \tag{174.4}$$

This is not a mixture of differently conditioned probabilities.

Proposition 174.1 (First-defect masses and their quantifiers)**.** *There is an absolute constant C such that, for every m, J, R,*

$$\begin{aligned} \sum_{n,k} \|r^{\varepsilon,J,\mathrm{inc}}_{n,k,m,R}\|_1 &\le C\varepsilon^2, \\ \sum_{n,k} \|r^{\varepsilon,J,\mathrm{clr}}_{n,k,m,R}\|_1 &\le C\varepsilon, \\ \sum_{n,k} \|r^{\varepsilon,J,\mathrm{mid}}_{n,k,m,R}\|_1 &\le C\varepsilon q^{(J+1)/4}. \end{aligned} \tag{174.5}$$

The last source is zero if there are no retained middle decisions. For bounded insertions of modulus at most M, the variation-mass bounds acquire only the factor M. Variation here has no factor $1/2$. These are not pointwise density bounds.

Proof. A first incidence defect at contact j has $c_j < 2\varepsilon e_j$, a clearance defect has $\Delta_j < 2\varepsilon e'_j$, and a middle-decision defect has $s_j < 2\varepsilon e_j$ with $\min(j, m - j) > J$. Drop the earlier-good conditions in (174.2), use $1 - G \le 1$, and apply Lemma 172.3 at the indicated collision or flight. Collision invariance and $\nu^*_R \le \nu/c$ are sufficient. At most two indices have a given endpoint distance, so the three sums are bounded respectively by

$$C\varepsilon^2 \sum_{d \ge 0} q^{d/2}, \qquad C\varepsilon \sum_{d \ge 0} q^{d/4}, \qquad C\varepsilon \sum_{d > J} q^{d/4}.$$

These prove (174.5). At fixed m the original events $\Pi_{n,k,m,R}$ are disjoint, so the component mass sum is at most the original restricted-source mass. Domination gives the assertion for bounded complex insertions. No independence is used. □

174.1. **An explicit endpoint depth in the ordered bandwidth limit.** For a fixed sufficiently large A, choose

$$\varepsilon(B) = AB^{-1/12}, \qquad J(B) = \lfloor B^{1/24} \rfloor, \qquad B \geq B_0. \tag{174.6}$$

Increase B_0 so that $\varepsilon(B) < \varepsilon_0$ and choose $A^{12} \geq C_2$ from Theorem 172.2. Both $\varepsilon(B)$ and $J(B)$ are fixed before the collision limit. This is an explicit bandwidth-dependent depth, not a rate depending on m.

Theorem 174.2 (Removal of a growing endpoint decision layer)**.** *At the choices* (174.6)*,*

$$\limsup_{m\to\infty} \sup_{R,n,k} m^2 \big\| \mathscr{D}_{B,n,R}(k,m,\cdot) - [r^{\varepsilon(B),J(B)}_{n,k,m,R} - K_B * r^{\varepsilon(B),J(B)}_{n,k,m,R}] \big\|_\infty \leq CB^{-1/24}. \tag{174.7}$$

Here $\mathscr{D}_{B,n,R} = p_{n,R} - K_B * p_{n,R}$ is the same complete coefficient-first correction as in Theorem 166.3. For the residual sources the explicit uniform mass estimates are

$$\begin{aligned} \sum_{n,k} \|r^{\varepsilon(B),J(B),\mathrm{inc}}\|_1 &\leq CB^{-1/6}, \\ \sum_{n,k} \|r^{\varepsilon(B),J(B),\mathrm{clr}}\|_1 &\leq CB^{-1/12}, \\ \sum_{n,k} \|r^{\varepsilon(B),J(B),\mathrm{mid}}\|_1 &\leq CB^{-1/12} q^{(\lfloor B^{1/24} \rfloor + 1)/4}. \end{aligned} \tag{174.8}$$

The endpoint component removed in (174.7) *satisfies the same pointwise estimate, uniformly for every bounded measurable insertion. The protected weighted component retains the endpoint-derivative assumptions of Theorem 172.2.*

Proof. Apply $I - K_B*$ to (174.4). The protected part has normalized collision limsup at most $CB^{-1/2}$, since $B \geq C_2\varepsilon(B)^{-12}$. The endpoint part has bound $C(J(B)+1)\varepsilon(B)$ by (173.13). This is at most $CB^{-1/24}$. Their sum proves (174.7), with no rate in the collision count. The mass statements are direct substitutions in (174.5). The two weighted classes are exactly those in their respective source estimates and are not conflated. □

Corollary 174.3 (The remaining arithmetic correction)**.** *On every central target set used in Theorem 166.3, the full uniform arithmetic pointwise law is equivalent to*

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\text{central } t} \Big| r^{\varepsilon(B),J(B)}_{n,k,m,R}(t) - (K_B * r^{\varepsilon(B),J(B)}_{n,k,m,R})(t) \Big| = 0. \tag{174.9}$$

Its main term remains $\mathcal{L}_{m,R}$ uniformly in the radius, and $c\mathfrak{a}_R(k,n,m) g_{\Omega_R}$ at fixed radius. The three residual corrections are summed with their signs before taking the absolute value in (174.9)*. No separate vanishing assertion for each of them is needed for this equivalence or proved here.*

Proof. The triangle inequality and (174.7) show that the ordered limsup of the full correction equals zero exactly when that of the residual does. Apply the unchanged arithmetic criterion. No arithmetic residue is replaced by one, and no exact coefficient is replaced by a packet average. □

Scope of the pointwise advance. Equation (173.12) is a density estimate for an actual collapsing decision source; it does not follow from (174.5). It also controls all of that component's signed inverse and its finite trace jumps, with arbitrary bounded source insertions. The positive comparison crosses finitely many section cuts, but the conclusion holds for the original return index, roof and displacement.

Incidence and clearance margins are different: crossing them can change the physical word and collision count, so the same flow cannot be used there. Nor has a pointwise bound for the middle-decision source been inferred from its exponentially small mass in J. These three sources retain the signed correction in (174.9). The full raw-return theorem and a pointwise

roof-conditioned bridge still require that analytical estimate. A bandwidth-dependent choice in (174.6) supplies no prescribed count-dependent protection rate and no growing-band spectral bound.

175. Thin transverse strips at an arbitrary collision mark

The endpoint envelopes of Section 173 are taken at fixed depth. They do not give a bound for a decision whose depth increases with the collision count. We obtain that bound by inserting a thin strip at its actual collision time. Its strong multiplier norm is bounded independently of the strip width, while its strong-to-weak norm tends to zero. The full occupation spectrum then converts this smallness into an exact-coefficient local upper bound. No long pullback of a strip is used as a multiplier.

The stable length exponent in (117.2) is $\varsigma = 1/8$. In this section put

$$\alpha_0 = \varsigma/2 = 1/16. \tag{175.1}$$

The geometric number $q_0 = 47/53$ used below is unrelated to the stable test exponent denoted by q in the collision norms.

175.1. **Uniform multipliers and a small weak norm.** For $0 < s < s_0$, let $b_{R,s}$ be the indicator of the union of the coordinate strips of width $2s$ about the supporting lines of all sides of Y_R^*. Angular strips are interpreted in fixed circle charts. The horizontal strips lie in a fixed compact set away from grazing. We may increase their widths by any one fixed factor without changing the assertions. In particular

$$\mathbf{1}_{\{\operatorname{dist}(x,\partial Y_R^*)<s\}} \le b_{R,s}(x).$$

The strips may overlap. Their number is fixed, not proportional to a collision count. Define $b_{R,0} = 0$ modulo the null supporting lines.

Lemma 175.1 (A thin strip as a strong-to-weak multiplier)**.** *On the local common collision spaces, uniformly in R and s,*

$$\begin{aligned} \|M_{b_{R,s}} h\|_{\mathcal{B}} &\le C\|h\|_{\mathcal{B}}, \\ |M_{b_{R,s}} h|_w &\le C s^{\varsigma}\|h\|_{\mathcal{B}}. \end{aligned} \tag{175.2}$$

The first estimate does not assert smallness in the strong operator norm. Both assertions also hold for any one of the strips and for unions of a fixed bounded number of such strips.

Proof. Partition by all strip endpoints in (α, p) coordinates. These are vertical or horizontal lines; the latter remain horizontal in (α, φ). There is a common finite number of partition cells in each homogeneity strip. A homogeneous stable graph has slope bounded away from zero and infinity. It meets a cell in a bounded number of intervals and spends length at most $C\epsilon$ within distance ϵ of its boundary. These constants do not depend on the separation between two parallel partition lines. Lemma 3.3(b) of the pinned version of [12], on page 13, therefore gives the first estimate, as in Lemma 151.1.

For clarity, the uniformity as a strip collapses can also be seen in the norm calculation. In the stable norm, splitting a test on W into at most N subintervals costs at most N in the sum of $(|U|/|W|)^{\varsigma}$. In the unstable comparison, matching the strip endpoints removes intervals of length at most $C\epsilon$. If the strip width is smaller than the matching distance, its whole intersection may be assigned to that unmatched part and has the same bound. Its stable-norm cost is $C\epsilon^{\varsigma}$, which is admissible since $\gamma < \varsigma$. On the retained pieces the indicator is constant. No inverse strip width enters either bound. Membership in the strong completion and extension from smooth densities follow from the same piecewise multiplier lemma.

To prove the second assertion, start with a smooth density h and a weak test ψ on W. The intersection of W with the strip union is a union of at most N intervals U, each of length at most Cs. Restriction of a C^p test to U has uniformly bounded C^q norm. The length normalization of the strong stable norm gives

$$\left|\int_U h\psi \, dm_U\right| \le C\|h\|_s |U|^{\varsigma}.$$

Summing these finitely many intervals and taking the weak supremum proves the bound. Strong approximation and boundedness of the multiplier extend it to $h \in \mathcal{B}$. At $s = 0$ the same bound gives the zero operator. The argument is uniform in the locations of the lines, hence also in the moving radius. □

175.2. **Weak smallness on a moving spectral chart.** Write $Q = \mathscr{T}_{R,\theta}$ for the exact uncentered full-occupation operator of (157.1). Fix the finitely many peak charts in Lemma 160.2. They all lie in one fixed small roof neighborhood, independently of a later outer roof band. On a smaller chart write

$$Q^a = \lambda^a \Pi + N^a, \qquad \|N^a\| \le C\rho^a, \qquad \|Q^a\| \le C \quad (a \ge 0).$$

Here and below N^0 in this formula means $I - \Pi$. The last bound follows from $|\lambda| \le 1$, the bounded projection, and the complementary estimate. Use a common equivalent norm on this fixed finite cover. The full Doeblin–Fortet estimate gives

$$\|Q^a h\| \le C\sigma^a \|h\| + C(1+a)|h|_w, \qquad |Q^a h|_w \le C(1+a)|h|_w, \qquad \sigma < 1. \tag{175.3}$$

Shrink the peak charts so that $|\lambda| \ge r > \sigma^{1/4}$. This is possible since their central eigenvalues have modulus one. The constants in this paragraph concern the peak neighborhood only.

Lemma 175.2 (Projection and terminal-function interpolation)**.** *Uniformly on these charts,*

$$\|\Pi h\| + \sup_{a \ge 0} |\ell(Q^a h)| \le C\|h\|^{1/2} |h|_w^{1/2}. \tag{175.4}$$

Consequently, for every $a \ge 0$,

$$\|\Pi M_{b_{R,s}} Q^a \nu\| \le C s^{\alpha_0},$$
$$|\ell(Q^a M_{b_{R,s}} \Pi \nu)| \le C s^{\alpha_0}. \tag{175.5}$$

Proof. By homogeneity take $\|h\| = 1$ and put $t = |h|_w$. The case t bounded below follows from power boundedness. For $0 < t < 1/2$, choose an integer L with $\sigma^L \le t < \sigma^{L-1}$. Commutation of Q and Π gives

$$\|\Pi h\| = |\lambda|^{-L} \|\Pi Q^L h\| \le C r^{-L} (\sigma^L + (1+L)t) \le C(1 + |\log t|) t^{3/4} \le C t^{1/2}.$$

We used $-\log r/(-\log \sigma) < 1/4$. For $a \le L$, continuity of mass on the weak space and (175.3) give $|\ell(Q^a h)| \le C(1+L)t$. For $a > L$, first apply the strong bound in (175.3) at time L, then the uniform strong power bound for Q^{a-L}. This gives the same upper bound, with a changed constant. It is at most $Ct^{1/2}$. If $t = 0$, let $L \to \infty$ in these estimates: the projection is zero because $\sigma < r$, and every fixed terminal pairing is zero by the weak inequality. Restoring homogeneity proves (175.4). Apply it to $M_{b_{R,s}} Q^a \nu$ and $M_{b_{R,s}} \Pi \nu$. Their strong norms are bounded; their weak norms are at most Cs^{ς} by Lemma 175.1. This proves (175.5). □

Theorem 175.3 (A local upper bound uniform in strip width and mark)**.** *There is $C < \infty$ with the following property. For each fixed $B \ge B_0$ there are $C_B < \infty$ and $\rho_B < 1$ such that for every*

$m \ge 2$, $0 \le j \le m$, $0 < s < s_0$, radius R, discrete targets $k \in \mathbb{Z}^2$, $l \in \mathbb{Z}$, real t, and interval I of length $h > 0$,

$$m^2\nu\{K^{\mathrm{c}}_{m,R} = k, A_{m,R} = l, S_{m,R} - t \in I, b_{R,s}(T_R^j x) = 1\} \le Cs^{\alpha_0}(h + B^{-1}) + C_B(1 + Bh)m^2\rho_B^{m/2}. \tag{175.6}$$

The leading constant is independent of the outer band B. In particular s and j may depend on m; the error does not depend on them. This is an upper bound, not a Gaussian-amplitude theorem for a shrinking strip.

Proof. The exact characteristic pairing with the mark is

$$\ell(Q^{m-j}M_{b_{R,s}}Q^j\nu) = \int b_{R,s}(T_R^j x)e^{i\theta\cdot(K^{\mathrm{c}}_{m,R},S_{m,R},A_{m,R})}\,d\nu(x). \tag{175.7}$$

This follows first on physical densities from (157.1). The sharp strip is the bounded piecewise multiplier of Lemma 175.1; its physical multiplication agrees with that realization. One can check the last assertion by smooth monotone averages of parallel-strip indicators: their operator norms are uniformly bounded and their strong-to-weak differences tend to zero, while their physical integrals converge by dominated convergence. Thus no orbit-length regularity of $b_{R,s} \circ T_R^j$ is needed.

On a peak chart, if $m - j \ge m/2$, expand the left long power in (175.7) and use the first inequality of (175.5). If $j \ge m/2$, expand the right long power and use its second inequality. The unexpanded short power is uniformly bounded in the strong norm. In either case

$$|\ell(Q^{m-j}M_{b_{R,s}}Q^j\nu)| \le Cs^{\alpha_0}|\lambda|^{m/2} + C\rho^{m/2}.$$

The peak bound (160.2) integrates the first term in four real frequency coordinates to $Cs^{\alpha_0}m^{-2}$. On the nonresonant part of any fixed outer band, expand a long power using the uniform spectral gap. The other power and the strip multiplier remain bounded, giving $C_B\rho_B^{m/2}$. These assertions hold on a finite parameter and frequency cover.

Choose the positive upper envelope of $\mathbf{1}_I$ with Fourier support in $[-B, B]$, as in Section 123. It is nonnegative, has mass $h + 2\pi/B$, and its Fourier transform has supremum at most that mass. Positivity gives the upper comparison before Fourier inversion. Torus orthogonality, with the same signs as the original fixed-coefficient inverse, and the preceding integrated estimate prove (175.6). The nonpeak integral is at most a constant times $B(h + B^{-1})\rho_B^{m/2}$. All actual resonances have roof coordinate zero, so the peak cover and its leading constant are chosen once, independently of B. No arithmetic residue has been discarded. □

Corollary 175.4 (Summation over a graded family of marks)**.** *Let $\mathcal{J} \subset \{0, \dots, m\}$ and choose widths $s_j < s_0$. The left side of* (175.6)*, with the strip condition replaced by its union over $j \in \mathcal{J}$, is at most*

$$C(h + B^{-1})\sum_{j\in\mathcal{J}} s_j^{\alpha_0} + C_B(1 + Bh)|\mathcal{J}|m^2\rho_B^{m/2}.$$

This finite-count inequality may be summed before taking a limit.

Proof. Apply the union bound to the positive, unchanged coefficient and use Theorem 175.3 at every mark. The uniform error, rather than an interchange of separate limsups, justifies the assertion when the number or locations of the marks depend on m. □

176. Pointwise removal of every section-decision layer

We retain the physical incidence and clearance protection, but no longer restrict the free section decisions to a fixed endpoint depth. The uniform marked-strip estimate permits summation over all depths before taking the collision limit. This removes the interior-decision part which cannot be controlled by the fixed-history envelopes alone.

Write $d_j = \min(j, m-j)$, $e_j = q_0^{d_j/4}$ and retain e'_j for flights. For fixed $\eta, \varepsilon \in (0, \varepsilon_0)$ put

$$H^{\eta}_{m,R} = \prod_{j=0}^{m} \Theta(c_j/(\eta e_j)) \prod_{j=0}^{m-1} \Theta(\Delta_j/(\eta e'_j)),$$
$$D^{\varepsilon}_{m,R} = \prod_{j=0}^{m} \Theta(s_j/(\varepsilon e_j)). \tag{176.1}$$

The first product contains only physical margins. The original full protection is $\Xi^{\varepsilon}_{m,R} = H^{\varepsilon}_{m,R} D^{\varepsilon}_{m,R}$. In all auxiliary continuations the measure is ν/c, with $c = 91/(10000\pi)$; on the original initial section this is exactly ν^*_R.

176.1. **A positive telescoping by decision depth.** Set $D_m = \lfloor m/2 \rfloor$. For $-1 \le a \le D_m$ define

$$G_a = H^{\eta}_{m,R} \prod_{d_j > a} \Theta(s_j/(\varepsilon e_j)), \qquad G_{-1} = H^{\eta}_{m,R} D^{\varepsilon}_{m,R},$$
$$Z_a = G_a - G_{a-1} \ge 0 \qquad (0 \le a \le D_m). \tag{176.2}$$

Empty products equal one. Therefore

$$\sum_{a=0}^{D_m} Z_a = H^{\eta}_{m,R}(1 - D^{\varepsilon}_{m,R}). \tag{176.3}$$

If $Z_a > 0$, every physical margin is η-protected, every section margin at depth greater than a is ε-protected, and at least one of the at most two contacts of depth a satisfies $s_j < 2\varepsilon e_j$. Contacts of depth at most a may cross the section. At most $r_a = 2a + 2$ occupation summands can then change. The terminal membership at m is not one of the summands of A_m.

For $|w| \le M$ let $z^{a,w}_{n,k,m,R}(t)\,dt$ be the pushforward of $\mathbf{1}_{\Pi_{n,k,m,R}} w Z_a \nu^*_R$ by $T_{n,R}$. These are well-defined finite complex densities by domination by the complete fixed-count source. Their labels have not been enlarged. We use an enlargement only in a positive comparison below.

Lemma 176.1 (One geometric continuation for all decision depths)**.** *Put $\zeta = \min(\eta, \varepsilon)$. There is a common endpoint continuation of radius $r_*\zeta^3$ for every source point with $Z_a > 0$, with the decisions of depth at most a freed. The radius, Hessian and relative-density bounds are independent of a and m. For collision index j its section-distance differential obeys*

$$|\nabla s_j| \le C\zeta^{-1} q_0^{3d_j/4} \quad \text{almost everywhere.} \tag{176.4}$$

The continuation preserves the physical word, displacement, collision count and every retained decision. Its occupation lies in $\{l : |l - n| \le r_a\}$ when it starts at the original coefficient n.

Proof. Every retained margin is bounded below by ζ times its geometric weight. Apply the endpoint continuation and response proof of Lemma 173.3, using ζ for this common lower bound and omitting precisely the decisions of depth at most a. That proof checks margins separately: the relative change at depth d is bounded by $C\zeta^{-2} q_0^{d/2}(|du| + |dv|)$. The number of omitted checks does not occur in the estimate. Consequently it applies also when a depends on m, up to D_m. For overlapping endpoint layers omit duplicate contacts. Equation (176.4) is the unscaled

endpoint response before division by e_j. It remains an almost-everywhere bound at rectangle corners. The occupation assertion counts only the free terms at times $0,\dots,m-1$. The source density on the continuation is the same $|F_{uv}|/(4\pi Rc)$ as in the earlier physical continuation. □

176.2. **A summable high-gradient comparison.** Fix once, for these η,ε, a sufficiently small $\delta>0$ satisfying

$$\delta\le c_0\min\{\zeta^3,\varepsilon\zeta\}. \tag{176.5}$$

The constant c_0 below is independent of the margin parameters. This is not a count-dependent gradient scale. Use the roof vector field $V=\nabla F/|\nabla F|^2$ on $|\nabla F|\ge\delta$ and the common roof width

$$h=c_1\min\{\zeta^3\delta^2,\varepsilon\zeta\delta\}>0. \tag{176.6}$$

All these quantities are independent of the layer a.

Proposition 176.2 (Uniform layer estimate away from critical points)**.** *Let $z^{a,+}$ denote the unweighted layer density restricted by $|\nabla F|\ge\delta$. For every fixed sufficiently large auxiliary band B_*,*

$$\begin{aligned}m^2\|z^{a,+}_{n,k,m,R}\|_\infty &\le C(a+1)\varepsilon^{\alpha_0}q_0^{\alpha_0 a/4}(1+(B_*h)^{-1})\\ &\quad+C_{B_*,h}(a+1)m^2\rho_{B_*}^{m/2}.\end{aligned} \tag{176.7}$$

The constants in the first line are independent of $a,m,\eta,\varepsilon,\delta,B_*$. The constants in the second line need not be uniform in B_* or h.

Proof. The continuation, lower gradient and relative Jacobian estimates used in Proposition 173.4 hold with ζ in place of η. Along the flow,

$$e_j^{-1}|ds_j/dt|\le C\zeta^{-1}\delta^{-1}q_0^{d_j/2}.$$

At a contact of depth a which is initially bad, the choice (176.6) thus keeps its distance below $C\varepsilon q_0^{a/4}$. It suffices to test this one contact, not the entire history of length a. Its index is a or $m-a$. The flow crosses free decision cuts only; no incidence, clearance, physical-word or retained section boundary is crossed.

For almost every level $F=t$, roof time and ODE uniqueness give the same injective level-to-window map on each glued physical word as in (173.10). Thus the density is at most $C/(hc)$ times the sum, over $|l-n|\le 2a+2$ and the two possible marks j, of

$$\nu\{K^{\mathrm c}_m=k,A_m=l,|S_m-t|<h,b_{R,C\varepsilon q_0^{a/4}}(T_R^jx)=1\}.$$

There are at most $4a+5$ occupation coefficients; this allowance appears only on the upper side. Apply Theorem 175.3 at the common fixed interval of length $2h$. Its strip width and mark are allowed to depend on m. Division by h proves (176.7). This is the point where the new uniform error, absent from the old finite-history envelope, is indispensable. □

176.3. **Critical collars retain the bad decision.**

Proposition 176.3 (Uniform layer estimate near critical points)**.** *Let $z^{a,-}$ be the unweighted layer density restricted by $|\nabla F|<\delta$. There is a fixed roof width $r=C_0\delta^2>0$, independent of a,m, such that*

$$\begin{aligned}m^2\|z^{a,-}_{n,k,m,R}\|_\infty &\le C(a+1)\varepsilon^{\alpha_0}q_0^{\alpha_0 a/4}(1+(B_*r)^{-1})\\ &\quad+C_{B_*,r}(a+1)m^2\rho_{B_*}^{m/2}.\end{aligned} \tag{176.8}$$

Here B_* is fixed, as above. The continued collar source is ν/c, including when its critical center is outside the return section.

Proof. Complete a low-gradient source point to the unique physical normal-to-normal center as in Lemma 173.5. All retained margins are at least the common ζ threshold, so the center lies within $C\delta$ and its critical roof within $C\delta^2$. Choose c_0 in (176.5) so that the required collars remain inside the half-margin continuations. Their radius in endpoint coordinates is $O(\delta)$. Equation (176.4), on the completed square with half-margins, shows that an initially bad contact of depth a stays within

$$2\varepsilon q_0^{a/4} + C\delta\zeta^{-1}q_0^{3a/4} \le C'\varepsilon q_0^{a/4}$$

of its section boundary throughout the collar. This retains the small strip in the critical comparison. It is not enough to retain only the two normal endpoint strips and then sum a bound independent of a.

Use the intrinsic physical coefficient J_z of each completed center and the mass collar of roof width $r = C_0\delta^2$, increasing C_0 to cover all completed source points and reducing c_0 to keep $r < h_{\zeta/2}$. Each collar has mass at least $a_* r J_z$ in ν/c and density at most $A_* J_z$. For centers whose collars can contribute at roof t, their critical roofs lie in an interval of length Cr, and the collar union lies in a roof interval of length $C'r$. Take each physical collar once. Endpoint injectivity makes collars from different physical words disjoint. Splitting a word by free decisions does not produce extra collars.

Every collar retains its original middle occupation, and at most $2a + 2$ free summands can differ from the original n. Hence the whole collar union is included in the same occupation allowance, not a second enlargement of it. It also has one of the two marked strip conditions at a or $m - a$ just proved. Apply Theorem 175.3 to this positive union. Dividing by $a_* r$ bounds the sum of the intrinsic coefficients. The collar-density upper estimate then gives (176.8). No extra normal-strip smallness is needed here: the retained bad decision supplies the summable depth factor. □

Theorem 176.4 (All-depth pointwise decision deconcentration)**.** *For fixed $\eta, \varepsilon \in (0, \varepsilon_0)$ and $|w| \le M$, let*

$$d^{\eta,\varepsilon,w}_{n,k,m,R}(t)\,dt = (T_{n,R})_*\big(\mathbf{1}_{\Pi_{n,k,m,R}} w H^{\eta}_{m,R}(1 - D^{\varepsilon}_{m,R})\nu^*_R\big). \tag{176.9}$$

There is an absolute C such that

$$\limsup_{m\to\infty} \sup_{R,n,k,\,|w|\le M} m^2 \|d^{\eta,\varepsilon,w}_{n,k,m,R}\|_\infty \le CM\varepsilon^{1/16}. \tag{176.10}$$

More precisely, for the tail $d^{\eta,\varepsilon,w}_{>J}$ obtained by summing the layers $a > J$, $J \ge -1$, there are $C_{\eta,\varepsilon} < \infty$ and $\rho_{\eta,\varepsilon} < 1$ such that, for every $m \ge 2$ and every such J,

$$m^2 \|d^{\eta,\varepsilon,w}_{>J,n,k,m,R}\|_\infty \le CM\varepsilon^{1/16}(J+2)q_0^{(J+1)/64} + C_{\eta,\varepsilon} M m^4 \rho^{m/2}_{\eta,\varepsilon}. \tag{176.11}$$

A tail beyond D_m is zero. In particular the normalized tail tends to zero for any $J = J(m) \to \infty$, at fixed η, ε. The estimate holds for arbitrary bounded measurable source insertions.

Proof. First take $w = 1$. Choose δ, h, r as above, once for the fixed margin parameters, and then choose one auxiliary band $B_* \ge \max\{B_0, h^{-1}, r^{-1}\}$. This band is fixed before the collision limit and before summing the layers. The two layer estimates now have leading bound $C(a+1)\varepsilon^{1/16}q_0^{a/64}$ and an error $C_{\eta,\varepsilon}(a+1)m^2\rho^{m/2}_{\eta,\varepsilon}$. Summing the finite positive telescope first, use

$$\sum_{a>J}(a+1)x^a = x^{J+1}\left(\frac{J+2}{1-x} + \frac{x}{(1-x)^2}\right), \qquad x = q_0^{1/64} < 1,$$

and $\sum_{a=0}^{D_m}(a+1) \le Cm^2$. These give (176.11). In particular the summed error vanishes; no interchange of a limsup and an uncontrolled infinite depth sum is used. Setting $J = -1$ gives (176.10). Radon–Nikodym domination by M times the positive source proves the assertion for all w. Neither w nor a long pullback of a section indicator is differentiated. □

Corollary 176.5 (All roof bands and finite trace jumps)**.** *With the same sources,*

$$\limsup_{m\to\infty} \sup_{B>0,R,n,k,\,|w|\le M} m^2\|d^{\eta,\varepsilon,w} - K_B * d^{\eta,\varepsilon,w}\|_\infty \le CM\varepsilon^{1/16}. \tag{176.12}$$

The analogous bound with the first term of (176.11) *holds for a depth tail. Where finite one-sided traces exist, each jump is bounded by twice its essential-height estimate. No sum of absolute jump heights is asserted.*

Proof. Use $\|K_B\|_1 = \|K_1\|_1$ and $\|f - K_B * f\|_\infty \le (1 + \|K_1\|_1)\|f\|_\infty$. The supremum over the displayed reconstruction band B is therefore a convolution consequence, not a spectral estimate at a growing band. The auxiliary band B_* used in the preceding proof was fixed. The trace assertion follows from the essential bound. □

177. The raw correction after all decision layers have been removed

The new decomposition puts all physical defects in one source before ordering them. This is necessary: the old first-bad-middle-decision source can contain a later bad physical margin. We do not declare that old source pointwise small merely by renaming it.

At $\eta = \varepsilon$, define the physical remainder by

$$b^{\varepsilon,w}_{n,k,m,R}(t)\,dt = (T_{n,R})_*\big(\mathbf{1}_{\Pi_{n,k,m,R}} w(1 - H^\varepsilon_{m,R})\nu^*_R\big). \tag{177.1}$$

For the unweighted source the exact positive identity is

$$p_{n,R}(k,m,\cdot) = f^\varepsilon_{n,k,m,R} + d^{\varepsilon,\varepsilon}_{n,k,m,R} + b^\varepsilon_{n,k,m,R}. \tag{177.2}$$

Here f^ε is precisely the old smoothly protected source, not a new approximation to it.

Order only the incidence and clearance factors of H^ε by depth and the tie convention of Section 174. Partition $\{H^\varepsilon < 1\}$ by the first factor strictly below one, including its transition region. Group these disjoint sets by physical type to obtain $b^{\varepsilon,\mathrm{inc}}$ and $b^{\varepsilon,\mathrm{clr}}$. Then

$$b^\varepsilon = b^{\varepsilon,\mathrm{inc}} + b^{\varepsilon,\mathrm{clr}}, \qquad \sum_{n,k} \|b^{\varepsilon,\mathrm{inc}}\|_1 \le C\varepsilon^2, \quad \sum_{n,k} \|b^{\varepsilon,\mathrm{clr}}\|_1 \le C\varepsilon. \tag{177.3}$$

The proof is the invariant one-flight neighborhood sum in Proposition 174.1, with no section factors in the first-defect order. It gives only mass bounds for these two sources. All identities and domination statements retain the exact labels and the original source normalization.

Theorem 177.1 (Only physical-word defects remain in the correction)**.** *Choose $\varepsilon(B) = AB^{-1/12}$ with A sufficiently large as in Theorem 172.2. For all sufficiently large fixed B,*

$$\limsup_{m\to\infty} \sup_{R,n,k} m^2\big\|\mathscr{D}_{B,n,R}(k,m,\cdot) - [b^{\varepsilon(B)}_{n,k,m,R} - K_B * b^{\varepsilon(B)}_{n,k,m,R}]\big\|_\infty \le CB^{-1/192}. \tag{177.4}$$

Thus, on the central target sets of Theorem 166.3, the full uniform arithmetic pointwise law is equivalent to

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{\text{central } t} |b^{\varepsilon(B)}_{n,k,m,R}(t) - (K_B * b^{\varepsilon(B)}_{n,k,m,R})(t)| = 0. \tag{177.5}$$

The main term is still $\mathcal{L}_{m,R}$ uniformly through arithmetic transitions, and $c\mathfrak{a}_R(k,n,m)g_{\Omega_R}$ at a fixed radius. No zero-residue assertion is added.

Proof. Apply $I - K_B*$ to (177.2). The old protected contribution has normalized limsup $CB^{-1/2}$ when $B \ge C_2\varepsilon(B)^{-12}$; this follows by choosing $A^{12} \ge C_2$. Corollary 176.5 bounds the full decision contribution by $C\varepsilon(B)^{1/16} = CA^{1/16}B^{-1/192}$. Here the protection and the auxiliary spectral band used to prove the decision theorem are fixed for each collision limsup. Their possibly large constants do not enter the leading limsup bound. This proves (177.4). The triangle inequality

and the unchanged arithmetic reconstruction criterion prove the equivalence. The incidence and clearance corrections in (177.3) are summed with their signs before taking the absolute value. □

Exact scope. Theorem 176.4 is a pointwise theorem for every decision depth on a physically protected source, including arbitrary bounded measurable insertions. It is stronger in source scope than a fixed-depth endpoint theorem; the exponent 1/16 is not claimed optimal. In the old first-defect decomposition, a middle decision followed by a bad physical margin is now assigned to b^{ε}, not silently discarded. No uncontrolled depth tail is left in $d^{\eta,\varepsilon}$.

The two physical margins are different. Crossing a grazing or a competing-hit boundary can change the physical word and sometimes the collision count. None of the flows above crosses such a boundary. Equation (177.5) is therefore a remaining requirement, not a theorem of vanishing. The small masses in (177.3) do not control this signed pointwise correction. In particular neither the full raw density law nor the pointwise roof-conditioned bridge is asserted here. The finite-count exponential height bound of Theorem 167.3 retains exactly that finite-count scope; it is not used as a central-scale estimate.

For weighted raw inversion, the new decision bound needs only $|w| \le M$. The protected weighted estimate still needs the endpoint-derivative budget in Theorem 172.2. A common weighted theorem requires the physical remainder for that same insertion and roof value. Source domination alone gives no Gaussian amplitude for an arbitrary path selector and no missing denominator.

178. Physical boundary layers on the image collision space

We estimate the true incidence and clearance defects without continuing an orbit across a physical singularity. The clearance of flight j is read at collision $j+1$, from the preceding flight. Its thin positive upper envelope then has boundaries transverse to the stable cone. This gives a local bound for the actual defective histories, not for histories with an inserted, deleted or relabelled collision.

The table in this section is the original circular family. Write $r = R\alpha$ for arclength, $p = \sin\varphi$, and $c(x) = \cos\varphi$. The section mass remains $c = 91/(10000\pi)$ when no argument is displayed. To avoid confusion, the incidence below is denoted by c_j, as in Section 169. Choose

$$s_0 = 1/100, \qquad \ell_* = \sqrt{(53/100)^2 - (48/100)^2} > 0.$$

Enlarging a constant permits any fixed multiple of a width smaller than s_0. All multipliers use the density convention and local common spaces of Section 117.

178.1. **A one-flight geometric upper envelope.** At an image collision with its disk centered at zero, put

$$q_R = Rn_\alpha, \quad v^- = \cos\varphi\, n_\alpha - \sin\varphi\, t_\alpha, \quad (v^-)^\perp = \sin\varphi\, n_\alpha + \cos\varphi\, t_\alpha.$$

Thus v^- points backwards along the incoming flight. Let $\mathcal{D}$ be the finite set of nonzero lattice centers of norm at most $\tau_{\max} + 2R_+ + s_0$. For $d \in \mathcal{D}$ define

$$L_d = (d - q_R)\cdot v^-, \qquad Z_d = (d - q_R)\cdot (v^-)^\perp. \tag{178.1}$$

Here L_d is a geometric distance, not a reconstruction band or a collision count. Define the two indicators

$$\begin{aligned} g_{R,s}(x) &= \mathbf{1}_{\{\cos\varphi < s\}}, \\ b^-_{R,s}(x) &= \mathbf{1}_{\bigcup_{d\in\mathcal{D}}\{L_d>0,\ R\le|Z_d|\le R+s\}}(x). \end{aligned} \tag{178.2}$$

The second set is an upper envelope. It does not require that the perpendicular foot occur before the preceding collision, and it is not asserted to be the actual clearance set.

Lemma 178.1 (Transverse envelopes for actual physical defects)**.** *For every regular flight with clearance $\Delta_j < s < s_0$,*

$$b^-_{R,s}(T_R^{j+1}x) = 1. \tag{178.3}$$

On a fixed angular chart the boundary curves of the clearance envelope are uniformly C^2, and are transverse to stable graphs. Every homogeneous stable curve intersects the envelope in a uniformly bounded number of intervals, each of length at most Cs. The same conclusion holds for $g_{R,s}$, including the homogeneity strips accumulating at grazing. Partition and matching constants are independent of s and R.

Proof. The endpoints of a physical flight are on its two incident disks. Their distances from any other disk are at least $3/50$. Since $s < s_0$, a nonincident disk at clearance less than s has its nearest point strictly inside the segment. Reading this segment backwards from its terminal collision, the foot therefore has $L_d > 0$, the supporting line has distance $|Z_d| \in [R, R+s]$, and d belongs to $\mathcal{D}$. This proves (178.3). The actual previous disk can be left in the upper-envelope candidate list: an extra upper-comparison set causes no identification of labels.

On the indicated strips, $|d-q_R| \geq 1-R \geq 53/100$ and $|Z_d| \leq R+s_0 \leq 48/100$, so $L_d \geq \ell_*$. The sign condition $L_d > 0$ creates no additional boundary meeting these strips. Direct differentiation in (r, φ) gives

$$\partial_r Z_d = -\cos\varphi - L_d/R, \qquad \partial_\varphi Z_d = L_d. \tag{178.4}$$

Consequently a level $Z_d = z$ in a strip is a graph with slope

$$\frac{d\varphi}{dr} = R^{-1} + \frac{\cos\varphi}{L_d} > 0.$$

Its slope and first two derivatives have common upper bounds, since $L_d \geq \ell_*$ and $\mathcal{D}$ is finite. A stable graph has negative slope bounded away from zero and infinity. Along it,

$$\frac{dZ_d}{dr} = -\cos\varphi - L_d/R + L_d\varphi'(r) \leq -\ell_*/R_+.$$

Thus a change of Z_d by s occupies length at most Cs. The four levels $\pm R, \pm(R+s)$ and fixed angular chart cuts form an admissible partition. Their number of components in each homogeneity strip is uniformly bounded: at a horizontal strip edge the level equation is a trigonometric equation of bounded degree in α, with a fixed number of candidates. Refining by these finitely many arcs gives simply connected cells. Uniform transversality gives length $O(\delta)$ in a δ-neighborhood of every cell boundary. Neither bound depends on the separation of the parallel level curves.

For the incidence envelope, $\cos\varphi < s$ is the union of two horizontal bands of angular width $\arcsin s \leq 2s$. Stable slopes are bounded away from zero, so each intersection has length Cs. A homogeneous stable curve is already inside one homogeneity strip. Most such strips are either entirely included or entirely excluded; only the two horizontal thresholds require new cuts. The number of new cuts per strip is bounded, even when the threshold approaches $\pm\pi/2$. No inverse change from p to φ is used at grazing. Matching intersections of two stable graphs moves a transverse cut by at most a constant times their graph distance. If the thin band is smaller than that distance, its whole intersection has that same unmatched-length bound. This proves the stated uniformity. □

178.2. **Strong multiplication and a weak gain.**

Proposition 178.2 (Physical thin-layer multipliers)**.** *For $a_{R,s} = g_{R,s}$ or $b^-_{R,s}$, uniformly on each local common collision space,*

$$\|M_{a_{R,s}}h\|_{\mathcal{B}} \leq C\|h\|_{\mathcal{B}}, \qquad |M_{a_{R,s}}h|_w \leq Cs^{1/8}\|h\|_{\mathcal{B}}. \tag{178.5}$$

These assertions hold for the physical multiplication operation on the strong completion. They do not assert a small strong norm.

Proof. Use the partition in Lemma 178.1 and the piecewise multiplier criterion [12, Lemma 3.3(b), p. 13], with the pinned numbering specified in the bibliography. On each partition cell the indicator is constant. The cell-count, intersection and boundary-neighborhood hypotheses have just been verified uniformly. The fixed exponent choices of Section 117 satisfy the multiplier restrictions.

For the weak estimate, restrict a weak stable test to its at most N retained intervals U. Each has length at most Cs. The strong stable length normalization, with $\varsigma = 1/8$, gives

$$\left| \int_U h\psi \, dm_U \right| \le C\|h\|_s |U|^{\varsigma}.$$

Summation proves the second estimate. In the strong unstable comparison the unmatched lengths are $O(\delta)$, including when $s < \delta$, and cost $C\delta^{\varsigma}$. This is admissible because $\gamma < \varsigma$. On retained pieces the multipliers agree. No inverse-width derivative is paid. The multiplier theorem extends this physical product from smooth densities to the strong completion. Equivalently, smooth averages of the transverse level indicators have uniform strong bounds and converge in the strong-to-weak norm; their physical integrals converge by domination. The local changes of arclength have uniformly bounded scale factors. □

Theorem 178.3 (A marked physical-defect local upper bound)**.** *Put $\alpha_* = 1/16$. There is C such that, for each fixed auxiliary roof band $B_* \ge B_0$, there are $C_{B_*} < \infty$ and $\rho_{B_*} < 1$ for which*

$$\begin{aligned} m^2\nu\{K^{\mathrm{c}}_{m,R} = k, A_{m,R} = l, S_{m,R} \in J,\ a_{R,s}(T^j_R x) = 1\} \\ \le Cs^{\alpha_*}(|J| + B_*^{-1}) + C_{B_*}(1 + B_*|J|)m^2\rho^{m/2}_{B_*}. \end{aligned} \tag{178.6}$$

This holds for either multiplier in (178.2)*, every interval J, every exact discrete target, $m \ge 2$, $0 \le j \le m$, $0 < s < s_0$, and $R \in [0.45, 0.47]$. The leading constant is independent of B_*. The mark and width may vary with m. A defect of flight j is estimated with the clearance mark at $j + 1$, not at j.*

Proof. The exact full-occupation pairing is

$$\ell(\mathscr{T}^{m-j}_{R,\theta} M_{a_{R,s}} \mathscr{T}^{j}_{R,\theta}\nu).$$

It has the physical meaning in (175.7), now with the physical-layer multiplier. On every fixed moving-peak chart, Lemma 175.2 and (178.5) give projection and terminal amplitudes at most $Cs^{1/16}$. At least one chronological block has length $m/2$. Expanding that block, and using the strong power bound for the other one, gives $Cs^{1/16}|\lambda|^{m/2} + C\rho^{m/2}$. The four-frequency Gaussian peak bound integrates this to $Cs^{1/16}m^{-2} + C\rho^{m/2}$. The complement on a fixed outer band has an exponentially decaying long block. All occupation resonances are retained. The positive interval majorant has mass $|J| + 2\pi/B_*$ and Fourier support in $[-B_*, B_*]$. Its inversion gives (178.6), exactly as in Theorem 175.3. Constants on the complement may depend on B_*; none depends on j or s. □

Physical interpretation. This is an estimate on the union of actual singular regimes, not a transport across a grazing or competing-hit seam. Each trajectory keeps its own m collisions and all its original occupation decisions. The source at a fixed return index is bounded above by the ambient source ν/c with the same (k, n, m) coefficient. No factor of c is absorbed into a changed probability and no extra terminal visit is added to $A_{m,R} = \sum_{j=0}^{m-1} \eta_R \circ T^j_R$.

179. Local variation of the complete physical remainder

The multiplier estimate applies directly to the source left in Section 177. We sum the actual physical margins before taking a collision limit. The resulting estimate is local in the roof variable and retains every exact discrete label.

For $f \in L^1_{\mathrm{loc}}(\mathbb{R})$ and $h > 0$ define

$$\|f\|^{\mathrm{loc}}_{1,h} = \sup_{t\in\mathbb{R}} \int_t^{t+h} |f(u)|\, du. \tag{179.1}$$

For a signed or complex measure this notation refers to its total variation on the interval. No factor $1/2$ occurs in this norm. The factor $1/2$ will be used only for the distance between two probability laws.

Keep the physical guard $H^{\varepsilon}_{m,R}$ of (177.1), and put $q_0 = 47/53$, $x_0 = q_0^{1/64}$. The two depth functions are $d_j = \min(j, m-j)$ for contacts and $d'_j = \min(j, m-1-j)$ for flights. A physical factor below one has, respectively,

$$c_j < 2\varepsilon q_0^{d_j/4} \quad \text{or} \quad \Delta_j < 2\varepsilon q_0^{d'_j/4}.$$

We initially take $0 < 2\varepsilon < s_0$; increasing constants handles any fixed remaining compact range of margins.

Theorem 179.1 (A local bound on all physical defects)**.** *Let $|w| \le M$, with no regularity assumption on w. For every fixed auxiliary band $B_* \ge B_0$, every $h > 0$ and $m \ge 2$,*

$$\sup_{R,n,k} m^2 \|b^{\varepsilon,w}_{n,k,m,R}\|^{\mathrm{loc}}_{1,h} \le CM\varepsilon^{1/16}(h + B_*^{-1}) + C_{B_*} M(1 + B_* h) m^3 \rho_{B_*}^{m/2}. \tag{179.2}$$

The same estimate applies to either first-physical-defect type, and their sum is the original physical remainder. The width ε may depend on m in this finite-count estimate.

Let $b^{\varepsilon,w}_{>J}$ be the part whose first physical factor strictly below one has depth greater than J, in the physical-only order used in (177.3)*. Then*

$$\sup_{R,n,k} m^2 \|b^{\varepsilon,w}_{>J,n,k,m,R}\|^{\mathrm{loc}}_{1,h} \le CM\varepsilon^{1/16} x_0^{J+1}(h + B_*^{-1}) + C_{B_*} M(1 + B_* h) m^3 \rho_{B_*}^{m/2}. \tag{179.3}$$

An empty tail is zero. In particular, at each fixed margin and roof length,

$$\limsup_{m\to\infty} \sup_{R,n,k,|w|\le M} \frac{m^2}{h} \|b^{\varepsilon,w}_{n,k,m,R}\|^{\mathrm{loc}}_{1,h} \le CM\varepsilon^{1/16}. \tag{179.4}$$

Proof. For factors in $[0,1]$, $1 - \prod_i a_i \le \sum_i \mathbf{1}_{\{a_i<1\}}$. Thus the original positive source $1 - H^{\varepsilon}_{m,R}$ is bounded by the union sum of the contact and flight events displayed above. There are at most two contacts and two flights of any given depth. Use the incidence indicator at time j and, by (178.3), the backward-clearance indicator at time $j+1$. These marks always lie between zero and m.

On the exact return component $\Pi_{n,k,m,R}$, the collision labels are exactly $(K^{\mathrm{c}}_m, A_m) = (k, n)$ and the roof is S_m. Enlarging the source from the initial section to the whole cylinder, and dropping the terminal membership only on the upper side, costs exactly $1/c$. Apply Theorem 178.3 to every term. The leading terms sum because

$$\sum_{a\ge 0} (2\varepsilon q_0^{a/4})^{1/16} = \frac{(2\varepsilon)^{1/16}}{1 - x_0}.$$

There are $2m+1$ contact and flight terms in total. Their finite-count errors sum to $C_{B_*}(1+B_*h)m^3\rho_{B_*}^{m/2}$, before any limit is taken. This proves (179.2) for the unweighted positive source.

A first physical defect of depth greater than J is contained in the same union restricted to depths $a>J$. The leading geometric sum is multiplied by x_0^{J+1}; the error can be bounded by the same full finite sum. This proves (179.3). Grouping a disjoint first-defect partition by type gives the two individual conclusions. We have not identified first-defect pieces with independently normalized laws.

For complex w, pushforward and total variation give $|b^{\varepsilon,w}(u)|\le Mb^{\varepsilon,1}(u)$ almost everywhere. Absolute continuity is already available for the complete fixed-count source from Theorem 167.3; no derivative of w is used. Finally fix ε,h,B_*, take the collision limsup, and only then let $B_*\to\infty$. The leading constant in (179.2) is independent of B_*. This proves (179.4). The finite-count assertion allows $\varepsilon=\varepsilon_m$ because its remainder is independent of the strip widths; it says nothing about a count-dependent pointwise protected inverse. □

179.1. **The full signed correction in a local density norm.** The elementary convolution estimate needed here is

$$\|K*f\|_{1,h}^{\mathrm{loc}}\le\|K\|_1\|f\|_{1,h}^{\mathrm{loc}}. \tag{179.5}$$

Indeed, integrate the pointwise convolution upper bound over a translated interval and use Tonelli followed by the supremum in (179.1). This argument does not require that K be positive or compactly supported.

Corollary 179.2 (All reconstruction bands on the physical source)**.** *For every fixed $\varepsilon,h>0$,*

$$\limsup_{m\to\infty}\sup_{R,n,k,|w|\le M,\,B>0}\frac{m^2}{h}\|b_{n,k,m,R}^{\varepsilon,w}-K_B*b_{n,k,m,R}^{\varepsilon,w}\|_{1,h}^{\mathrm{loc}}\le CM\varepsilon^{1/16}. \tag{179.6}$$

There is also a version of (179.3) *with the same geometric tail and finite-count error for this correction.*

Proof. Apply (179.5) and $\|K_B\|_1=\|K_1\|_1$ before taking any supremum or limit. Multiply (179.2) or its depth-tail version by $1+\|K_1\|_1$, take the collision limsup, and then enlarge the auxiliary band B_*. The reconstruction band B and the auxiliary spectral band B_* are distinct throughout. □

For a protected-admissible insertion w, let (M,L) be the supremum and endpoint-gradient budget of Theorem 172.2, uniform in the parameters being considered. The original insertion $w=1$ has $(M,L)=(1,0)$. Write p^w for its full exact-label density, with no physical or section guard left in that density.

Theorem 179.3 (The complete raw correction in local variation)**.** *Choose $\varepsilon(B)=AB^{-1/12}$ with A sufficiently large as in Theorem 177.1. For every sufficiently large fixed B and every fixed $h>0$,*

$$\limsup_{m\to\infty}\sup_{R,n,k,w}\frac{m^2}{h}\|p_{n,R}^w(k,m,\cdot)-K_B*p_{n,R}^w(k,m,\cdot)\|_{1,h}^{\mathrm{loc}}\le C(M+L)B^{-1/192}. \tag{179.7}$$

The constant does not depend on h. The collision threshold and auxiliary spectral constants may depend on h and the fixed B. Every physical and section source is included on the left.

Proof. Use the exact source identity, also for the same insertion,

$$p^w=f^{\varepsilon,w}+d^{\varepsilon,\varepsilon,w}+b^{\varepsilon,w}.$$

The smoothly protected correction has essential-supremum limsup $C(M+L)B^{-1/2}$ after normalization by m^2, since $B\ge C_2\varepsilon(B)^{-12}$ once $A^{12}\ge C_2$. Its local variation norm is at most h times

that bound. The all-depth decision correction has normalized essential height $CM\varepsilon(B)^{1/16}$ by Corollary 176.5; pay it in the same way. For the physical term apply Corollary 179.2. This last term is estimated in local variation, not in essential supremum. Now $\varepsilon(B)^{1/16} = A^{1/16}B^{-1/192}$, and the first term is smaller. Adding these three estimates proves the theorem.

For each fixed B, its margin is fixed before the collision limit. The auxiliary band B_* in the physical estimate is enlarged only after that limit, with B still fixed. Only after all these estimates may B tend to infinity. No spectral estimate has been used at a band depending on m. □

Pointwise values are a separate question. Equation (179.7) estimates the integral of the absolute density error, and not only the signed error of a fixed test interval. It supplies no essential-height bound for the physical remainder. Even a bounded density of arbitrarily small local integral can have thin spikes. The physical criterion (177.5) therefore retains its original pointwise meaning and is not declared proved by this theorem.

180. Arithmetic local variation of the unmodified raw density

The local variation estimate now removes every source guard from a raw density theorem. It also allows a roof test to vary arbitrarily with the return and collision indices. The arithmetic modulation is not removed.

Theorem 180.1 (The full-source local variation law)**.** *For every fixed $h > 0$,*

$$\lim_{m\to\infty} \sup_{R,\,1\le n\le m,\,k,t} \frac{1}{h}\int_t^{t+h} \left| m^2 p_{n,R}(k,m,u) - \mathcal{L}_{m,R}(k_1,k_2,u,n)\right| du = 0. \tag{180.1}$$

This is the density of the complete original return law, with respect to counting measure in (k,m) and Lebesgue measure in u. No return index is averaged. The transition kernel is the one in (160.5)*.*

At a fixed radius, on every central compact set, it may be replaced by $c\mathfrak{a}_R(k,n,m)g_{\Omega_R}(Z(u))$, where

$$Z(u) = \frac{(k_1,k_2,u,n) - m(0,0,\bar\tau_R,c)}{\sqrt{m}}.$$

The equivalent return normalization has the same arithmetic factor and the matrix D_R, as in Theorem 166.3.

Proof. At every fixed reconstruction band B, the band-limited test theorem already gives

$$\sup_{R,n,k,u} \left| m^2 (K_B * p_{n,R}(k,m,\cdot))(u) - \mathcal{L}_{m,R}(k_1,k_2,u,n)\right| \longrightarrow 0.$$

This is the last displayed estimate in the proof of Theorem 166.3, on the full source. Integrating its absolute error on any interval of length h, and using Theorem 179.3 with $w = 1$, shows that the limsup in (180.1) is at most $CB^{-1/192}$. Let $B \to \infty$ after the collision limsup. This proves the first assertion, without selecting pointwise representatives of the densities. The fixed-radius specialization of Theorem 160.3 is uniform on a slightly larger central compact set containing the intervals under discussion. Its uniform error can be integrated, giving the second assertion. □

Corollary 180.2 (Arbitrary bounded roof selectors)**.** *The supremum in* (180.1) *may equivalently be written*

$$\sup_{R,n,k,t} \sup_{\substack{\psi:\ |\psi|\le 1\\ \operatorname{supp}\psi\subset[t,t+h]}} \frac{1}{h}\left| \int \psi(u)[m^2 p_{n,R}(k,m,u) - \mathcal{L}_{m,R}(k_1,k_2,u,n)]\,du\right|. \tag{180.2}$$

Here ψ is measurable and may depend on every displayed parameter, including m. In particular the approximation holds uniformly for all Borel subsets of a translated fixed roof interval.

Proof. The total variation norm of an absolutely continuous signed measure is its L^1 norm. The upper inequality is immediate; equality follows by the measurable sign of its density. This is a selector in the observed roof variable, not an arbitrary trajectory selector with a new Gaussian amplitude. □

180.1. **A consequence for the original mixed measure.** The preceding theorem can also be summed on a bounded central region at a fixed return index. For $M < \infty$ put

$$V_{n,R}(k,m,u) = n^{-1/2}(k_1, k_2, m - n/c, u - n\bar{\tau}_R/c), \qquad \mathcal{C}_{n,R}(M) = \{|V_{n,R}| \le M,\ m \ge n\}.$$

The positive part below is needed because a finite transition kernel need not be pointwise positive at finite count.

Corollary 180.3 (Central mixed-measure total variation)**.** *For every fixed $M < \infty$,*

$$\lim_{n\to\infty} \sup_R \sum_{m\ge n} \sum_{k\in\mathbb{Z}^2} \int_{\{u:(k,m,u)\in\mathcal{C}_{n,R}(M)\}} \left| p_{n,R}(k,m,u) - m^{-2}(\mathcal{L}_{m,R}(k_1,k_2,u,n))_+ \right| du = 0. \quad (180.3)$$

Thus this restriction of the original mixed measure is approximated in total variation, not only weakly. No assertion outside the stated central region or at a prescribed individual roof value is included.

Proof. On the central region, $m = n/c + O_M(\sqrt{n})$ and hence $m \asymp n$. There are $O_M(\sqrt{n})$ collision labels and $O_M(n)$ displacement labels. For each pair the roof set is contained in an interval of length $O_M(\sqrt{n})$, covered by $O_M(\sqrt{n})$ unit intervals. Theorem 180.1, with $h = 1$, bounds each integral by $m^{-2}e_m$, where $e_m \to 0$ uniformly in all targets. The total number of unit intervals is $O_M(n^2)$, so their sum is bounded by $C_M \sup_{m\ge a_M n} e_m \to 0$. Finally, for $p \ge 0$, $|m^2 p - \mathcal{L}_+| \le |m^2 p - \mathcal{L}|$. This justifies the positive part and proves the claim. □

180.2. **Conditional roof laws and the remaining pointwise issue.** For $I_t = [t, t+h]$, define

$$G^{n,k}_{m,R}(I_t) = \int_{I_t} (\mathcal{L}_{m,R}(k_1,k_2,u,n))_+ \, du.$$

Consider any family of these targets on which $G^{n,k}_{m,R}(I_t) \ge dh$ for a fixed $d > 0$.

Corollary 180.4 (Total variation of the exact conditional roof law)**.** *Uniformly on such a target family, the law of the original roof $T_{n,R}$ conditioned on*

$$K_{n,R} = k, \qquad N_{n,R} = m, \qquad T_{n,R} \in I_t$$

is asymptotic in total variation to the probability with density

$$\frac{\mathbf{1}_{I_t}(u)(\mathcal{L}_{m,R}(k_1,k_2,u,n))_+}{G^{n,k}_{m,R}(I_t)}. \quad (180.4)$$

The convention is $d_{\mathrm{TV}}(P,Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2}\int |dP - dQ|$. The conditioning event is unchanged. This is not a bridge conditioned at the single equation $T_{n,R} = t$.

Proof. Let $f = m^2 p \mathbf{1}_{I_t}$, $g = \mathcal{L}_+ \mathbf{1}_{I_t}$ and $E = \|f - g\|_1$. By Theorem 180.1, $E/h \to 0$ uniformly. Write $F = \int f$ and $G = \int g$. Then $|F - G| \le E$, $F \ge dh/2$ eventually, and

$$\frac{1}{2}\left\| \frac{f}{F} - \frac{g}{G} \right\|_1 \le \frac{E}{F} \le \frac{2E}{dh} \longrightarrow 0.$$

The source normalization is the original ν^*_R on both sides of the conditional probability, so no change of event or section occurs. □

Corollary 180.5 (Size of the exceptional roof set)**.** *For every fixed $h, \lambda > 0$,*

$$\sup_{R,n,k,t} \frac{1}{h} \operatorname{Leb}\{u \in [t, t+h] : |m^2 p_{n,R}(k, m, u) - \mathcal{L}_{m,R}(k_1, k_2, u, n)| > \lambda\} \longrightarrow 0.$$

For the full signed correction at fixed large B, its corresponding limsup is at most $C\lambda^{-1}B^{-1/192}$.

Proof. Apply the elementary inequality $\operatorname{Leb}\{|F| > \lambda\} \le \lambda^{-1} \int |F|$ to Theorem 180.1 and to (179.7). □

Relation to the original raw endpoint. The preceding results include the entire incidence and clearance source in a density-norm local limit and in exact-window conditional roof laws. They strengthen a fixed-test local theorem: the measurable roof selector in (180.2) is chosen after m, not fixed before the limit. They do not upgrade this norm convergence to uniform convergence of density values. Narrow physical-boundary spikes are still compatible with all the estimates above. Excluding those spikes, or estimating their signed correction at the same roof value, remains the extra assertion in (177.5). Its main term continues to carry $\mathfrak{a}_R$ at fixed radius and $\mathcal{L}_{m,R}$ uniformly. Neither residue triviality nor a single-roof conditioned path bridge follows from local total variation.

181. Positive physical remainders and a pointwise lower local law

The local variation theorem does not bound the height of the physical remainder. It does, however, bound the height of its convolution at any fixed band. Positivity then gives a one-sided pointwise theorem for the original density. More precisely, the entire density error is, in essential supremum, approximated by the positive physical remainder. This identifies which sign of concentration remains to be controlled; it does not discard that concentration.

All densities in this section are defined up to Lebesgue-null sets. Every supremum in the roof variable is an essential supremum. Set

$$P^{n,k}_{m,R}(u) = m^2 p_{n,R}(k, m, u), \qquad G^{n,k}_{m,R}(u) = \mathcal{L}_{m,R}(k_1, k_2, u, n).$$

The labels always satisfy $R \in [0.45, 0.47]$, $1 \le n \le m$ and $k \in \mathbb{Z}^2$. The source of p is the original ν^*_R. In particular the factor $1/c$ has already been included; it is not inserted again in the estimates below. The real, even reconstruction kernel is $K_B(u) = BK_1(Bu)$ from Section 166. It has integral one and is a Schwartz function; it is not presumed nonnegative.

181.1. **A fixed convolution of a locally small source.**

Lemma 181.1 (Local mass controls a fixed smooth convolution)**.** *Let $K \in W^{1,1}(\mathbb{R})$ and let f have finite $\|f\|^{\mathrm{loc}}_{1,h}$ for some $h > 0$. Then*

$$\|K * f\|_\infty \le (h^{-1}\|K\|_1 + \|K'\|_1)\|f\|^{\mathrm{loc}}_{1,h}. \tag{181.1}$$

Consequently, if a family $f_{m,\lambda}$ satisfies

$$\limsup_{m\to\infty} \sup_\lambda h^{-1}\|f_{m,\lambda}\|^{\mathrm{loc}}_{1,h} \le s \quad \text{for every fixed } h > 0,$$

with the same s, then for every fixed $K \in W^{1,1}(\mathbb{R})$,

$$\limsup_{m\to\infty} \sup_\lambda \|K * f_{m,\lambda}\|_\infty \le s\|K\|_1. \tag{181.2}$$

No analogous conclusion for $\|f_{m,\lambda}\|_\infty$ is asserted.

Proof. Take the absolutely continuous representative of K. On each interval $I_j = [jh, (j+1)h]$, the fundamental theorem of calculus gives

$$\sup_{I_j} |K| \le h^{-1} \int_{I_j} |K| + \int_{I_j} |K'|.$$

For any x, the integral of $|f(x-y)|$ over I_j is at most $\|f\|_{1,h}^{\mathrm{loc}}$. Multiply by the displayed supremum and sum over $j \in \mathbb{Z}$. The sum is finite, and Tonelli proves (181.1), including absolute convergence. At fixed h, its collision limsup is at most $s(\|K\|_1 + h\|K'\|_1)$. Now let $h \downarrow 0$, after that limsup and with K still fixed. This proves (181.2). The last operation regularizes f by the fixed kernel throughout; it does not differentiate a local-window theorem at a count-dependent scale. □

The following elementary principle separates the new logical step from its billiard inputs.

Proposition 181.2 (A positive-remainder lower principle)**.** *Let $P_{m,\lambda} \ge 0$ be densities and $G_{m,\lambda}$ real functions. For each fixed $B \ge B_0$, suppose $P = A_{B,m,\lambda} + R_{B,m,\lambda}$ with $R_{B,m,\lambda} \ge 0$, and suppose*

$$\sup_{\lambda} \|K_B * P_{m,\lambda} - G_{m,\lambda}\|_\infty \longrightarrow 0,$$
$$\limsup_m \sup_{\lambda} \|A_{B,m,\lambda} - K_B * A_{B,m,\lambda}\|_\infty \le r_B,$$
$$\limsup_m \sup_{\lambda} h^{-1} \|R_{B,m,\lambda}\|_{1,h}^{\mathrm{loc}} \le s_B \quad (h > 0 \text{ fixed}).$$

Then

$$\limsup_m \sup_{\lambda} \|P_{m,\lambda} - G_{m,\lambda} - R_{B,m,\lambda}\|_\infty \le r_B + \|K_1\|_1 s_B. \tag{181.3}$$

The same upper bound holds for the limsup of $\|(G_{m,\lambda} - P_{m,\lambda})_+\|_\infty$. If $r_B + s_B \to 0$, this last lower-law error tends to zero as $m \to \infty$.

Proof. The exact identity is

$$P - G - R = (A - K_B * A) + (K_B * P - G) - K_B * R.$$

Apply Lemma 181.1 to the last term, keeping B fixed, and use $\|K_B\|_1 = \|K_1\|_1$. This proves (181.3). Since $R \ge 0$, $(G-P)_+ \le |P-G-R|$ almost everywhere. Its limsup is therefore bounded by the same expression for every fixed B. Only now send B to infinity. Neither the regular part A nor K_B needs to be positive for this argument; the sign of the remainder is indispensable. □

181.2. **Application to the complete exact return coefficient.** Use the exact physical-only remainder $b^{\varepsilon,1}$ of Section 177. The source decomposition is

$$p = f^{\varepsilon,1} + d^{\varepsilon,\varepsilon,1} + b^{\varepsilon,1}, \qquad b^{\varepsilon,1} = b^{\varepsilon,1}_{\mathrm{inc}} + b^{\varepsilon,1}_{\mathrm{clr}} \ge 0. \tag{181.4}$$

All terms have the same original labels. The first-physical-defect partition defining the two terms on the right is disjoint before pushforward, including the transition region of every smooth guard. There is no new normalization of a component.

Lemma 181.3 (Smoothed physical remainder at the local height scale)**.** *For every fixed $B > 0$, every sufficiently small fixed $\varepsilon > 0$, and all $|w| \le M$,*

$$\limsup_{m\to\infty} \sup_{R,n,k,|w|\le M} m^2 \|K_B * b^{\varepsilon,w}_{n,k,m,R}\|_\infty \le CM\varepsilon^{1/16}\|K_1\|_1. \tag{181.5}$$

The constant is independent of the fixed reconstruction band B. This is not a supremum over count-dependent bands inside the limsup.

Proof. Equation (179.4) gives the hypothesis of Lemma 181.1, with $f_{m,\lambda} = m^2 b^{\varepsilon,w}$ and $s = CM\varepsilon^{1/16}$. Its uniformity in w is by domination of the original positive source. Apply (181.2) with this fixed K_B.

There is also a useful finite-count reading. Put $h = B^{-1}$ in (181.1) and use (179.2). With $C_K = \|K_1\|_1 + \|K_1'\|_1$, the upper bound is

$$C_K M \left[C\varepsilon^{1/16}(1 + B/B_*) + C_{B_*}(B + B_*)m^3 \rho_{B_*}^{m/2} \right]. \tag{181.6}$$

Here B and B_* are independently fixed before m tends to infinity. The auxiliary band may then be enlarged. In the sharper norm bound (181.5), the final $h \downarrow 0$ in the lemma is taken only after these limits. No orbit is continued across an incidence or competing-hit seam. □

Choose once and for all a sufficiently large constant A_0 and put $\varepsilon(B) = A_0 B^{-1/12}$, for large B, as in Theorem 177.1. Write

$$\mathfrak{b}_{B,m,R}^{n,k} = m^2 b_{n,k,m,R}^{\varepsilon(B),1}, \qquad \mathfrak{h}_{B,m,R}^{n,k} = m^2 (f_{n,k,m,R}^{\varepsilon(B),1} + d_{n,k,m,R}^{\varepsilon(B),\varepsilon(B),1}).$$

The symbol $\mathfrak{h}_{B,m,R}^{n,k}$ here denotes a controlled source density; it is not the arithmetic residue $\mathfrak{a}_R$.

Theorem 181.4 (Uniform positive-remainder representation of the error)**.** *For every sufficiently large fixed B,*

$$\limsup_{m\to\infty} \sup_{R,n,k} \|P_{m,R}^{n,k} - G_{m,R}^{n,k} - \mathfrak{b}_{B,m,R}^{n,k}\|_\infty \le CB^{-1/192}. \tag{181.7}$$

In particular the complete original density obeys the arithmetic pointwise lower local law

$$\delta_m := \sup_{R,n,k} \|(G_{m,R}^{n,k} - P_{m,R}^{n,k})_+\|_\infty \longrightarrow 0. \tag{181.8}$$

The main term is the full finite transition kernel, uniformly in R. On central compact sets at a fixed radius it can be replaced by $c\mathfrak{a}_R(k,n,m)g_{\Omega_R}(Z(u))$, with the unchanged arithmetic coefficient and collision normalization.

Proof. For each fixed B, Theorem 172.2 gives

$$\limsup_{m} \sup_{R,n,k} m^2 \|f^{\varepsilon(B),1} - K_B * f^{\varepsilon(B),1}\|_\infty \le CB^{-1/2}.$$

Indeed $B \ge C_2\varepsilon(B)^{-12}$ follows by choosing $A_0^{12} \ge C_2$. Corollary 176.5 gives for the entire decision source the bound $C\varepsilon(B)^{1/16} = CB^{-1/192}$ in the same normalized essential-supremum norm. Thus the controlled source has error $r_B \le C(B^{-1/2} + B^{-1/192})$.

The fixed-band arithmetic reconstruction on the original source, used in Theorem 180.1, gives $\sup \|K_B * P - G\|_\infty \to 0$. The physical remainder has the local variation bound $s_B \le C\varepsilon(B)^{1/16}$ of (179.4). Apply Proposition 181.2 to the exact split (181.4). This proves (181.7). Positivity of $\mathfrak{b}$ gives $\limsup_m \delta_m \le CB^{-1/192}$ for every fixed large B; let $B \to \infty$ to obtain (181.8). The fixed-radius specialization follows from the uniform central specialization in Theorem 160.3.

The three source estimates are used in their own norms. Only the convolution of the physical remainder is converted from local variation to height. Its unconvolved height has not been estimated. □

181.3. **A positive, rather than signed, remaining criterion.** For a central radius $M_0 < \infty$, put

$$\|H\|_{\infty,M_0} = \sup_{R,n,k} \operatorname*{ess\,sup}_{u:\ |Z(u)|\le M_0} |H_{R,n,k}(u)|, \qquad Z(u) = \frac{(k_1,k_2,u,n) - m(0,0,\bar\tau_R,c)}{\sqrt{m}}.$$

The essential supremum on an empty set is zero.

Corollary 181.5 (Equivalence with positive physical concentration)**.** *For every fixed M_0, the two-sided arithmetic pointwise local law on this central set is equivalent to*

$$\lim_{B\to\infty}\limsup_{m\to\infty}\|\mathfrak{b}_{B,m}\|_{\infty,M_0}=0. \tag{181.9}$$

It is also equivalent to the two corresponding positive-height conditions for $m^2 b_{\mathrm{inc}}^{\varepsilon(B),1}$ and $m^2 b_{\mathrm{clr}}^{\varepsilon(B),1}$ separately. Moreover

$$\big|\|(P_m-G_m)_+\|_{\infty,M_0}-\|\mathfrak{b}_{B,m}\|_{\infty,M_0}\big|\le\|P_m-G_m-\mathfrak{b}_{B,m}\|_{\infty}.$$

Thus a cancellation between the two signed physical corrections cannot eliminate a positive height obstruction left by (181.9)*.*

Proof. For real x and $b\ge 0$, $|x_+-b|\le|x-b|$. Apply this pointwise to $x=P_m-G_m$ and then take essential suprema to get the last inequality. Equation (181.7), restricted to the central set, proves both directions of the equivalence by the triangle inequality, with $m\to\infty$ before $B\to\infty$. The negative part already tends to zero by (181.8). Finally each nonnegative physical component is at most their sum, and the norm of that sum is at most the sum of their norms. This proves the separate equivalences. The signed criterion in (177.5) is not assumed: its equivalence with the new positive criterion follows also from (181.5). □

What remains to be bounded. The conclusion excludes downward density defects at the microscopic scale and replaces a possible signed-cancellation argument by the positive criterion (181.9). It does not prove that criterion. Positive, narrow incidence or clearance spikes are still compatible with the local variation theorem and with the new lower law. The original two-sided raw endpoint is retained with its correct arithmetic main term.

182. Microscopic lower denominators and exact-window minorization

The lower local law has a consequence not supplied by local total variation alone: a positive arithmetic reference is dominated by the original density, uniformly in the roof outside its null set. This gives lower denominators for arbitrarily small measurable roof sets. For fixed windows it yields a positive common part of the two conditional roof laws, and hence a reverse likelihood bound.

Use $P_{m,R}^{n,k}$, $G_{m,R}^{n,k}$ and δ_m from Section 181. Let $d>0$ be fixed and set $\mathcal{T}_{m,R}^{n,k}(d)=\{u: G_{m,R}^{n,k}(u)\ge d\}$. Neither this definition nor any result below assigns a positive reference to a zero arithmetic residue.

Corollary 182.1 (Lower denominators at the original discrete labels)**.** *For all sufficiently large m, uniformly in R,n,k,*

$$p_{n,R}(k,m,u)\ge\frac{d}{2m^2}\quad\text{for almost every } u\in\mathcal{T}_{m,R}^{n,k}(d). \tag{182.1}$$

For every Borel set $E\subset\mathcal{T}_{m,R}^{n,k}(d)$ of finite Lebesgue measure,

$$\begin{aligned}&\nu_R^*\{K_{n,R}=k,N_{n,R}=m,T_{n,R}\in E\}\\&\qquad\ge m^{-2}(1-\delta_m/d)\int_E G_{m,R}^{n,k}(u)\,du\ge\frac{d}{2m^2}\,\mathrm{Leb}(E).\end{aligned} \tag{182.2}$$

The set E may be chosen after m and may have arbitrarily small positive measure. No asymptotic upper bound on that event is asserted.

Proof. By (181.8), $P\ge G-\delta_m$ almost everywhere. On $\{G\ge d\}$ this is at least $(1-\delta_m/d)G$. Take m sufficiently large that $\delta_m<d/2$ and integrate on E. The measure is the unchanged original source, so the displayed probability is exactly the integral of p. □

Fix also $h > 0$ and consider all target windows $I_t = [t, t+h]$ on which $G^{n,k}_{m,R} \geq d$ almost everywhere. Define

$$e_m(h) = \sup_{R,n,k,t} h^{-1} \int_{I_t} |P^{n,k}_{m,R} - G^{n,k}_{m,R}|, \quad F = \int_{I_t} P^{n,k}_{m,R}, \quad H = \int_{I_t} G^{n,k}_{m,R}.$$

Thus $e_m(h) \to 0$ by Theorem 180.1 and $H \geq dh$. Let $\mathbb{P}^{n,k,t}_{m,R}$ be the law of the roof conditioned on the unchanged exact event

$$K_{n,R} = k, \qquad N_{n,R} = m, \qquad T_{n,R} \in I_t,$$

and let $\mathbb{Q}^{n,k,t}_{m,R}$ have density $\mathbf{1}_{I_t} G^{n,k}_{m,R}/H$. The lower bound on the reference is pointwise here; a bound only on its total mass would not suffice for the domination asserted next.

Theorem 182.2 (An asymptotically full common conditional measure)**.** *For all sufficiently large m, set*

$$\alpha_m(d,h) = \frac{1 - \delta_m/d}{1 + e_m(h)/d} \in (0,1]. \tag{182.3}$$

Uniformly on the indicated target family,

$$\mathbb{P}^{n,k,t}_{m,R} \geq \alpha_m(d,h)\mathbb{Q}^{n,k,t}_{m,R} \quad \text{as measures}, \qquad \alpha_m(d,h) \longrightarrow 1. \tag{182.4}$$

If $\alpha_m < 1$ there is a probability $\mathbb{S}^{n,k,t}_{m,R}$ on the same roof interval such that exactly

$$\mathbb{P} = \alpha_m \mathbb{Q} + (1 - \alpha_m)\mathbb{S}. \tag{182.5}$$

If $\alpha_m = 1$, the two conditional laws are equal. In particular, defining reverse likelihood divergence by

$$D_\infty(Q\|P) = \log \operatorname*{ess\,sup}_Q \frac{dQ}{dP}, \qquad D(Q\|P) = \int \log \frac{dQ}{dP}\, dQ,$$

one has

$$0 \leq D(\mathbb{Q}\|\mathbb{P}) \leq D_\infty(\mathbb{Q}\|\mathbb{P}) \leq -\log \alpha_m(d,h) \longrightarrow 0. \tag{182.6}$$

No forward likelihood bound or single-roof path bridge is included.

Proof. The lower local law gives $P \geq (1 - \delta_m/d)G$ on I_t. Also $F \leq H + he_m(h) \leq H(1 + e_m(h)/d)$. Divide the first inequality by this upper bound for F, which is positive by Corollary 182.1. This proves (182.4). The difference measure $\mathbb{P} - \alpha_m\mathbb{Q}$ is nonnegative and has mass $1 - \alpha_m$; normalize it to obtain (182.5). This residual is a decomposition of the same conditional law, not a replacement conditioning event.

Domination implies $\mathbb{Q} \ll \mathbb{P}$ and $d\mathbb{Q}/d\mathbb{P} \leq \alpha_m^{-1}$, proving the D_∞ estimate. The relative entropy is nonnegative by convexity and is at most the essential logarithmic bound. It is well defined: the negative part of $r \log r$, with $r = dQ/dP$, is integrable against the probability P, and $r \leq \alpha_m^{-1}$ bounds the positive part. Finally (182.3) tends to one, since δ_m and $e_m(h)$ tend to zero uniformly. □

Remark 182.3 (A fixed-radius arithmetic specialization). At fixed R, on a fixed central compact set and on labels with $\mathfrak{a}_R(k,n,m) \geq a_0 > 0$, the Gaussian has a positive minimum and the fixed-radius kernel approximation is uniform. It therefore supplies a $d > 0$ for all sufficiently large counts. For a parameter-uniform family the stated lower condition on $\mathcal{L}_{m,R}$ must be retained through arithmetic transitions. No claim that $\mathfrak{a}_R$ is identically one, or that all labels satisfy this lower condition, is needed.

Density versions and the remaining bridge problem. The essential bound (182.1) is a statement about a density, not a positive probability for the event $T_{n,R} = u$. That event has probability zero. At Lebesgue points it can be read using the differentiation representative of the original density. It supplies neither a value at an arbitrarily designated exceptional roof nor an asymptotic numerator for a path selector conditioned at that roof. The two-sided density estimate and a same-roof path numerator are still required for the original pointwise bridge problem.

Why the two directions of likelihood are different. For example, on $[0,1]$ take the reference density one. Let $H_j = 2^{j^2}$, $w_j = (jH_j)^{-1}$ and

$$p_j(u) = \frac{1 + H_j \mathbf{1}_{[0,w_j]}(u)}{1 + 1/j}.$$

These are probability densities, their L^1 errors tend to zero, and they dominate the reference by $(1 + 1/j)^{-1}$. Hence their reverse D_∞ tends to zero, while their essential heights diverge. The contribution of the spike to forward relative entropy is asymptotic to $j \log 2$, so forward relative entropy diverges. This example is a check on the logical scope of the conclusions, not a model for a billiard singularity. It explains why neither the minorization nor the local variation theorem closes (181.9).

183. Finite-count protection and higher integrability of the raw density

We return to the scalar physical density. A fixed-band positive upper bound is less precise than the local limit theorem, but it has a useful advantage here: its error can be retained uniformly as the protection scale decreases. We use this bound to obtain a polynomial height budget for the protected source. The thin-layer estimates then control the mass above a large density level. The complete exponential finite-count height bound closes the remaining tail integration. None of these steps continues a trajectory across a physical singularity.

Use the original exact labels and normalization

$$P(u) = m^2 p_{n,R}(k, m, u), \qquad G(u) = \mathcal{L}_{m,R}(k_1, k_2, u, n), \qquad 1 \le n \le m.$$

The section source is $\nu_R^* = \nu|_{Y_R^*}/c$. In particular P is nonnegative, whereas G is the real arithmetic transition kernel and need not be positive at every finite count. All density statements are almost everywhere. The constants below are uniform in R, n, k and in translated roof intervals.

183.1. **One auxiliary band for all protection scales.** Fix once an auxiliary roof band $B_\circ \ge B_0$. It is independent of the collision count, the protection scale, the roof interval and the density level to be introduced below.

Lemma 183.1 (Finite-count positive window bounds at one band)**.** *There are $C < \infty$ and $0 < \rho < 1$ such that, for every interval I of length $h > 0$ and $m \ge 2$,*

$$m^2 \nu_R^*\{K_{n,R} = k, N_{n,R} = m, T_{n,R} \in I\} \le C(1 + h)(1 + m^2 \rho^{m/2}). \tag{183.1}$$

For the full guard $\Xi_{m,R}^{\varepsilon}$ in (171.2)*, define the positive densities*

$$F_\varepsilon = m^2 f_{n,k,m,R}^{\varepsilon,1}, \qquad R_\varepsilon = P - F_\varepsilon \ge 0. \tag{183.2}$$

Then, simultaneously for all $0 < \varepsilon < \varepsilon_0$,

$$\int_I R_\varepsilon(u)\, du \le C(1 + h)\big(\varepsilon^{1/16} + m^3 \rho^{m/2}\big). \tag{183.3}$$

The scale ε may depend on m in these finite inequalities.

Proof. On the full collision source, the unmarked characteristic pairing is $\ell(\mathscr{T}^{m}_{R,\theta}\nu)$. The fixed-band moving-peak expansion used in Theorem 178.3, with no intermediate multiplier, bounds its four-frequency integral by $Cm^{-2}+C\rho^{m/2}$. Indeed, on each peak the amplitude is bounded and the real quadratic damping has a common positive lower bound; the Gaussian integral is $O(m^{-2})$. The fixed-band complementary powers are exponential. The finite number of moving peak charts is uniform, and no occupation resonance is removed.

Use a positive interval majorant of mass $h+2\pi/B_\circ$ and Fourier support in this one fixed band. Its Fourier supremum is at most its mass. Torus orthogonality and roof inversion therefore give $C(h+B_\circ^{-1})+C(1+B_\circ h)m^2\rho^{m/2}$ for the normalized positive probability. Dropping initial and terminal section membership on the upper side and using $\nu_R^*\le\nu/c$ proves (183.1). Constants depending on the fixed $B_\circ$ have now been absorbed into C. The argument is finite in m and uniform in the interval length; it is not an application of a fixed-window limit to a shrinking interval.

Write $q_0=47/53$, $d_j=\min(j,m-j)$ and $d'_j=\min(j,m-1-j)$. Since the factors of Ξ lie in $[0,1]$, $1-\Xi$ is bounded by the union sum of

$$c_j<2\varepsilon q_0^{d_j/4},\qquad s_j<2\varepsilon q_0^{d_j/4},\qquad \Delta_j<2\varepsilon q_0^{d'_j/4}.$$

There are $3m+2$ terms before overlaps are removed. Section strips are bounded by Theorem 175.3; incidence and clearance strips are bounded by Theorem 178.3. All use $B_\circ$. The clearance of flight j is read at collision $j+1$. The mark indices range from zero to m, while occupation still sums exactly the times $0,\dots,m-1$.

At most two contacts or flights have any prescribed depth, and $\sum_{a\ge0}q_0^{a/64}<\infty$. Thus the leading thin-layer terms sum to $C(1+h)\varepsilon^{1/16}$ and the width-independent errors to $C(1+h)m^3\rho^{m/2}$. The same enlargement by $1/c$ proves (183.3) on the original exact-label source. Take the largest of the finitely many spectral radii used above. The estimates being summed are uniform in the widths and marks, which proves the simultaneous assertion. □

Proposition 183.2 (A polynomial protection loss in height)**.** *There is a constant C_A such that for every $m\ge2$ and $0<\varepsilon<\varepsilon_0$,*

$$\sup_{R,n,k}\|F_\varepsilon\|_\infty\le C_A\varepsilon^{-9}(1+m^2\rho^{m/2})\le C'_A\varepsilon^{-9}. \tag{183.4}$$

The constants do not depend on m or ε. In particular this is a finite-count assertion, rather than a separate limsup for each fixed protection scale.

Proof. Choose an absolute $d>0$, sufficiently small for Lemma 172.1, and put

$$\delta=d\varepsilon^3,\qquad h'_\varepsilon=c_1\varepsilon^3\delta^2=c_1d^2\varepsilon^9,\qquad r_\varepsilon=C_*\delta^2=C_*d^2\varepsilon^6.$$

Reduce d so that $r_\varepsilon<h_{\varepsilon/2}$ and both roof widths are at most one. The guard is supported in $\mathsf{P}_{m,R}(\varepsilon)$.

On $|\nabla F|\ge\delta$, the positive flow-box comparison (171.7) bounds the density by the original exact-label probability in a window of length $2h'_\varepsilon$, divided by $2h'_\varepsilon/e$. Apply the finite inequality (183.1), not (171.5). The normalized height is at most $C\varepsilon^{-9}(1+m^2\rho^{m/2})$. This use of a short window pays its inverse length explicitly; it does not enlarge the auxiliary Fourier band.

For the part $|\nabla F|<\delta$, complete each point to its $\varepsilon/2$-protected normal critical center using Lemma 172.1. If the point has roof u, its critical roof lies in $[u-r_\varepsilon,u]$. The mass collars of width r_ε have density at most A_*J_z and mass at least $a_*r_\varepsilon J_z$, by Proposition 169.3. They have the same original (n,k,m) labels. Different physical collars are disjoint and a word has at most one

center. Consequently

$$m^2 \sum_{t_z \in [u-r_\varepsilon, u]} J_z \le \frac{Cm^2}{r_\varepsilon} \nu_R^*\{K_{n,R} = k, N_{n,R} = m, |T_{n,R} - u| < r_\varepsilon\}.$$

No normal-strip smallness is needed for this upper bound; removing that condition only enlarges the positive comparison. The collar density upper bound and (183.1) now give $C\varepsilon^{-6}(1 + m^2\rho^{m/2})$ for the normalized low-gradient height. It is smaller than the required budget. Every collar is charged once and the source is still ν_R^*, not a renormalized collar law. Add the two contributions. Finally $\sup_{m\ge 2} m^2\rho^{m/2} < \infty$ proves the second inequality. □

183.2. **A finite-error high-density argument.** The following measure-theoretic lemma states exactly how the exponentially small finite-count error is used. It avoids interchanging a limsup with an integral over unbounded density levels.

Lemma 183.3 (Protection, thin mass and an exponential cap)**.** *Let $P_m \ge 0$ on a finite measure space. Suppose that, for constants $K, \alpha, \kappa, \Gamma > 0$ and every $0 < \varepsilon < \varepsilon_0$,*

$$P_m = A_{m,\varepsilon} + B_{m,\varepsilon}, \quad A_{m,\varepsilon}, B_{m,\varepsilon} \ge 0, \quad \|A_{m,\varepsilon}\|_\infty \le C_A \varepsilon^{-K}, \quad \int B_{m,\varepsilon} \le C_I(\varepsilon^\alpha + e^{-\kappa m}),$$

with the same constants for all m, ε. Assume also $\|P_m\|_\infty \le C_H e^{\Gamma m}$ and a uniform bound on $\int P_m$. There are constants L_0, C_I' such that for $L \ge L_0$,

$$\int_{\{P_m > L\}} P_m \le C_I'(L^{-\alpha/K} + e^{-\kappa m}). \tag{183.5}$$

For every $0 < s < \min\{\alpha/K, \kappa/\Gamma\}$ one has $\sup_m \int P_m^{1+s} < \infty$.

Proof. Take $\varepsilon = (2C_A/L)^{1/K}$ and increase L_0 so that this scale is admissible. Then $A_{m,\varepsilon} \le L/2$. On $\{P_m > L\}$ this implies $B_{m,\varepsilon} \ge P_m/2$, which proves (183.5). The decomposition is exact at this chosen scale for each finite m.

Let $H_m = \max\{1, C_H e^{\Gamma m}\}$. The layer-cake identity, with an upper bound on the part $P_m \le 1$, gives

$$\int P_m^{1+s} \le \int P_m + s \int_1^{H_m} L^{s-1} \int_{\{P_m > L\}} P_m \, dL.$$

The bounded interval up to L_0 is harmless. The algebraic tail integrates uniformly when $s < \alpha/K$. The finite-error term is at most $C_I' e^{-\kappa m} H_m^s$, which is bounded, and tends to zero, when $s\Gamma < \kappa$. This integrates the finite error before taking a collision limit. A collection of separate asymptotic tail bounds, without this error and the cap, would not justify the same conclusion. □

Choose constants once as follows. With ρ from Lemma 183.1, put

$$\kappa = -\tfrac{1}{4} \log \rho > 0.$$

The polynomial errors are bounded by $Ce^{-\kappa m}$ after changing C. If $A > 1$ is the complete finite-count height constant in (167.6), put $\Gamma = \log(2A)$. Equation (167.7) then gives $\|P\|_\infty \le C_H e^{\Gamma m}$. Define

$$s_* = \frac{1}{2} \min\left\{\frac{1}{144}, \frac{\kappa}{\Gamma}\right\} > 0, \qquad q_* = 1 + s_*. \tag{183.6}$$

All constants arise from one fixed auxiliary band and the complete finite-count geometry. No numerical lower bound on s_* beyond this formula is asserted.

Theorem 183.4 (Uniform higher integrability of the unprotected density)**.** *For every $h > 0$ and $m \geq 2$,*

$$\sup_{R,n,k,t} \int_t^{t+h} |m^2 p_{n,R}(k,m,u)|^{q_*}\, du \leq C_*(1+h). \tag{183.7}$$

More precisely, for all $L \geq L_0$,

$$\sup_{R,n,k,t} \int_{[t,t+h]\cap\{P>L\}} P(u)\, du \leq C(1+h)(L^{-1/144} + e^{-\kappa m}). \tag{183.8}$$

The density is the complete original source, with no protection or index allowance in its definition.

Proof. On $[t, t+h]$ apply Lemma 183.3 to the exact split (183.2), with $K = 9$ and $\alpha = 1/16$. The needed protected height is Proposition 183.2; its positive remainder mass is (183.3). The total local mass is bounded by (183.1). The complete exponential cap and the strict inequalities for s_* were established just above. All interval-dependent constants are at most a fixed multiple of $1 + h$. This proves both assertions uniformly in the stated parameters. □

The role of physical boundaries. The proof uses a flow only on the fully protected source. Histories with incidence, clearance or section defects enter the positive remainder before any upper comparison. Their local mass, the polynomial protected height and the exponential full-source cap are three different inputs. The new bound does not infer height from mass and does not remove a collision at grazing. It rules out lack of uniform integrability of $P \log_+ P$, but not unbounded thin density spikes.

184. Strong local density limits and forward arithmetic likelihood

Higher integrability upgrades the complete raw local law and the positive physical remainders. It also distinguishes two questions which were both left open by the lower local law: forward relative entropy can now be controlled, although forward essential likelihood and the two-sided pointwise local theorem still require physical height estimates.

Retain $q_* = 1 + s_*$ from (183.6), and fix $1 < q < q_*$. All local integrals below concern a translated interval $I_t = [t, t+h]$ of fixed length $h > 0$. The normalized original density and signed transition kernel are P and G as in the preceding section. For this proof, the elementary interpolation inequality is

$$\int_I |v|^q \leq \left(\int_I |v|\right)^{\theta_q} \left(\int_I |v|^{q_*}\right)^{1-\theta_q}, \qquad \theta_q = \frac{q_* - q}{q_* - 1} \in (0,1). \tag{184.1}$$

It follows from Hölder's inequality applied to $|v|^q = |v|^{\theta_q}(|v|^{q_*})^{1-\theta_q}$.

184.1. **Physical remainders in a stronger local norm.**

Theorem 184.1 (Higher-integrable control of both physical sources)**.** *For the positive physical remainder of* (177.1)*, put $B_\varepsilon = m^2 b^{\varepsilon,1}_{n,k,m,R}$. Uniformly for $m \geq 2$ and $0 < \varepsilon < \varepsilon_0$,*

$$\sup_{R,n,k,t} \int_{I_t} B_\varepsilon^q \leq C_q(1+h)(\varepsilon^{1/16} + e^{-\kappa m})^{\theta_q}. \tag{184.2}$$

The same bound holds for each first-incidence and first-clearance component separately. For an arbitrary measurable insertion $|w| \leq M$ it holds with $|m^2 b^{\varepsilon,w}|^q$ and a factor M^q. Consequently

$$\limsup_{m\to\infty} \sup_{R,n,k,t,\,|w|\leq M,\,B>0} \int_{I_t} \left| m^2\big(b^{\varepsilon,w} - K_B * b^{\varepsilon,w}\big)(u)\right|^q du \leq C_q M^q (1+h)\varepsilon^{\theta_q/16}. \tag{184.3}$$

This is not an essential-height estimate.

Proof. The physical measure is a positive submeasure of the original source, so $0 \le B_\varepsilon \le P$ almost everywhere. Its local mass is bounded by (179.2), at the same fixed auxiliary band as above, by $C(1+h)(\varepsilon^{1/16} + e^{-\kappa m})$. Theorem 183.4 bounds its q_* moment. Apply (184.1). Each disjoint physical type is dominated by their sum, and Radon–Nikodym domination gives $|b^{\varepsilon,w}| \le M b^{\varepsilon,1}$.

For completeness, Minkowski's integral inequality gives

$$\sup_t \|K_B * v\|_{L^q(I_t)} \le \|K_1\|_1 \sup_t \|v\|_{L^q(I_t)}.$$

Apply this finite inequality before taking any supremum or limit. It holds also for the real sign-changing reconstruction kernel. The factor $1 + \|K_1\|_1$ is independent of B, which proves (184.3). No bandwidth depending on m has entered a spectral estimate. □

Theorem 184.2 (The complete arithmetic raw law in local L^q)**.** *For every fixed $h > 0$ and $1 < q < q_*$,*

$$\sup_{R,\, 1\le n\le m,\, k,t} \frac{1}{h} \int_{I_t} \left| m^2 p_{n,R}(k,m,u) - \mathcal{L}_{m,R}(k_1,k_2,u,n) \right|^q du \longrightarrow 0. \tag{184.4}$$

For the pinned collision-path measure $\mathbf{P}(u)$ and Gaussian bridge law W_R, the same limit holds with the integrand

$$\|\mathbf{P}(u) - G(u)\mathsf{W}_R\|^q_{\mathrm{BL}^*}. \tag{184.5}$$

For the actual-return path this conclusion holds on the central classes of Lemma 191.1. The path metric and bounded-Lipschitz convention are unchanged.

Proof. The scalar L^1 error tends to zero uniformly by Theorem 180.1. The kernel G is uniformly bounded, as in the finite moving-peak formula. Hence

$$\int_{I_t} |P - G|^{q_*} \le C_*(1+h).$$

Interpolation proves (184.4); division by the fixed h is harmless. This gives a modulus in the already proved local L^1 error, not a rate in m.

For the path statement put $v(u) = \|\mathbf{P}(u) - G(u)\mathsf{W}_R\|_{\mathrm{BL}^*}$. The constant function one belongs to the test ball, a positive path measure has this norm equal to its mass, and thus $v(u) \le P(u) + |G(u)|$. Its q_* moment is bounded by the same scalar estimate, while its L^1 norm tends to zero by Theorem 190.3. Apply the same interpolation. The countable norming class makes v measurable off the common roof-null set. For the return path use the roof-integrated transfer in Lemma 191.1; its positive path measure again has mass P. No path-space total variation is asserted. □

184.2. **Forward likelihood and relative entropy.** Fix $d > 0$ and restrict to windows I_t on which $G \ge d$ almost everywhere. This is a pointwise reference-floor condition, not merely a lower bound for its integrated mass. Put

$$F = \int_{I_t} P(u)\, du, \qquad H = \int_{I_t} G(u)\, du, \qquad d\mathbb{P} = \mathbf{1}_{I_t} \frac{P}{F}\, du, \qquad d\mathbb{Q} = \mathbf{1}_{I_t} \frac{G}{H}\, du.$$

Here $\mathbb{P}$ is precisely the roof law under the unchanged event $K_{n,R} = k$, $N_{n,R} = m$, $T_{n,R} \in I_t$. The factors m^2 cancel in its normalization. The reference G is positive only because of the stated floor on this window.

Theorem 184.3 (Forward finite-order arithmetic likelihood)**.** *Uniformly over these windows and exact targets, for every $1 < q < q_*$,*

$$\int \left| \frac{d\mathbb{P}}{d\mathbb{Q}} - 1 \right|^q d\mathbb{Q} \to 0. \tag{184.6}$$

The forward relative entropy and every fixed forward Rényi divergence of order $1 < s < q_*$ consequently tend to zero:

$$D(\mathbb{P}\|\mathbb{Q}) \to 0, \qquad D_s(\mathbb{P}\|\mathbb{Q}) = \frac{1}{s-1}\log\int\left(\frac{d\mathbb{P}}{d\mathbb{Q}}\right)^s d\mathbb{Q} \to 0. \tag{184.7}$$

Combined with Theorem 182.2, this also proves convergence of the sum of the two directed relative entropies. A forward essential likelihood bound is not asserted.

Proof. The local law gives $F - H \to 0$ uniformly, $H \geq dh$, and $F \geq dh/2$ for sufficiently large counts. Since G is uniformly bounded, F, H have uniform upper bounds on the fixed window. Thus $\mathbb{P} \ll \mathbb{Q}$, with likelihood $r(u) = HP(u)/(FG(u))$. For a constant depending only on d, h, q,

$$\int |r-1|^q\, d\mathbb{Q} = \frac{1}{H}\int_{I_t} G^{1-q}|(H/F)P - G|^q\, du \leq C_{d,h,q}\left(\int_{I_t}|P-G|^q\, du + |F-H|^q\right).$$

Theorem 184.2 proves (184.6). For $1 < s < q_*$ apply it with $q = s$. The $L^s(\mathbb{Q})$ triangle inequality gives $|\|r\|_s - 1| \leq \|r-1\|_s$, proving the Rényi assertion.

To verify the entropy step without assuming uniform boundedness of r, choose any $1 < q < q_*$; in particular $q < 2$. For $x \geq 0$,

$$0 \leq x\log x - x + 1 \leq C_q|x-1|^q, \qquad 0\log 0 = 0.$$

Taylor's formula proves this on $[1/2, 2]$ since the quadratic error is bounded by a constant times $|x-1|^q$ there. On $[0, 1/2]$ use boundedness and $|x-1| \geq 1/2$; on $[2, \infty)$ use $x\log x \leq C_q x^q$. Integrate with $x = r$ and $\int(r-1)d\mathbb{Q} = 0$. This proves $D(\mathbb{P}\|\mathbb{Q}) \leq C_q\|r-1\|_q^q \to 0$. The reverse entropy already tends to zero by conditional minorization. No estimate of $\operatorname{ess\,sup} r$ is used. □

Remark 184.4 (Arithmetic and positive reference classes). At fixed R, on a central compact set and a positive arithmetic class $\mathfrak{a}_R(k,n,m) \geq a_0 > 0$, the fixed-radius kernel specialization supplies the reference floor for sufficiently large m, as in Remark 182.3. Uniformly through changes of arithmetic type the floor on $\mathcal{L}_{m,R}$ is retained explicitly. A zero residue is not assigned a positive conditional reference. The roof interval and the discrete labels are identical on both sides of every likelihood comparison.

The remaining height question. Higher integrability is a stronger scalar conclusion than local total variation, and forward entropy does not follow from the old reverse minorization alone. Nevertheless finite L^q control does not imply an essential-supremum bound. For example, with the present fixed $q_* > 1$, normalize $1 + j\mathbf{1}_{[0,j^{-(q_*+1)}]}$ to a probability density on $[0,1]$. Its L^{q_*} norms stay bounded, its forward entropy relative to the constant density tends to zero, and its heights tend to infinity. This only distinguishes the topologies; it is not a billiard counterexample. The positive incidence and clearance heights in (181.9), and hence the two-sided pointwise raw law and unconditional uniform same-roof bridge, remain to be estimated on the actual source.

185. Cap-free control of positive boundary layers

The finite-count error in a local thin-layer estimate is not a mass which must be carried by the physical source. At small collision counts there is a second estimate: the exact global mass of the same layer. Combining these two inequalities removes the error uniformly in both variables. This gives an explicit endpoint for higher integrability, without using an exponential cap on the complete density.

We keep the original normalization

$$P(u) = m^2 p_{n,R}(k,m,u), \qquad F_\varepsilon = m^2 f^{\varepsilon,1}_{n,k,m,R}, \qquad R_\varepsilon = P - F_\varepsilon \geq 0.$$

Here $m \ge 2$, $1 \le n \le m$, and $k \in \mathbb{Z}^2$. These are the exact labels, not an occupation allowance. The source is $\nu_R^* = c^{-1}\nu|_{Y_R^*}$. All roof inequalities are understood almost everywhere and all local estimates are uniform in translated intervals I of fixed length $h > 0$.

185.1. **Two finite estimates for the same source.**

Lemma 185.1 (Global all-margin mass)**.** *For $0 < \varepsilon < \varepsilon_0 < 1$,*

$$\sup_R \sum_{n=1}^{m} \sum_{k\in\mathbb{Z}^2} \int_{\mathbb{R}} R_{\varepsilon,n,k,m,R}(u)\, du \le Cm^2\varepsilon. \tag{185.1}$$

The same bound holds for either positive physical component, and with an extra factor M in variation for an insertion of modulus at most M.

Proof. Every factor of the guard Ξ^ε is between zero and one. Its complement is bounded by the sum of the strip indicators used in Lemma 183.1. At collision j, an incidence strip of width a has collision mass at most Ca^2, and a section strip of width a has mass at most Ca. The latter follows directly from the bounded perimeter of the section rectangles in the flat collision coordinates. A clearance strip has mass at most Ca; it is read at collision $j+1$ as in Section 178. These are invariant one-flight mass estimates, not conditional independence assertions.

Use $a = 2\varepsilon q_0^{d_j/4}$ for contacts and section decisions, and $a = 2\varepsilon q_0^{d'_j/4}$ for flights, where $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$. At most two indices have any given depth. The convergent geometric sums give $\int(1-\Xi^\varepsilon)\, d\nu \le C(\varepsilon+\varepsilon^2)$. Enlarging the section source costs $1/c$ once. For fixed m, the original events $\Pi_{n,k,m,R}$ are disjoint as (n,k) varies. Consequently summing their pushforward masses and multiplying by m^2 proves (185.1). Neither the endpoint membership at m nor this summation adds a term to the occupation at $0,\dots,m-1$. The positive physical components are dominated by R_ε, since $\Xi^\varepsilon \le H^\varepsilon$. Source domination proves the weighted assertion. □

Lemma 185.2 (Removal of a finite-count error)**.** *Let $0 < \alpha < \beta$, $\kappa > 0$ and $r \ge 0$. Suppose that $U_m(\varepsilon) \ge 0$ satisfies, simultaneously for all $m \ge 2$ and $0 < \varepsilon < \varepsilon_0 < 1$,*

$$U_m(\varepsilon) \le C_1(\varepsilon^\alpha + e^{-\kappa m}), \qquad U_m(\varepsilon) \le C_2 m^r \varepsilon^\beta.$$

Then $U_m(\varepsilon) \le C\varepsilon^\alpha$ with a constant independent of both variables.

Proof. If $\kappa m \ge \alpha \log(1/\varepsilon)$, use the first inequality. Otherwise the second inequality gives

$$U_m(\varepsilon) \le C_2(\alpha/\kappa)^r [\log(1/\varepsilon)]^r \varepsilon^\beta \le C\varepsilon^\alpha,$$

because $x^r e^{-(\beta-\alpha)x}$ is bounded for $x \ge 0$. For $r = 0$ the same calculation has the factor $x^0 = 1$. Both estimates concern the same source at the same finite count; no asymptotic diagonal or exchange of limits is used. □

Theorem 185.3 (Simultaneous local thin mass without a spectral error)**.** *For every interval I of length $h > 0$, all $m \ge 2$ and $0 < \varepsilon < \varepsilon_0$,*

$$\sup_{R,n,k} \int_I R_\varepsilon(u)\, du \le C(1+h)\varepsilon^{1/16}. \tag{185.2}$$

The same bound holds for each normalized positive incidence and clearance density and for their sum. For an arbitrary bounded insertion, its variation is bounded by M times this right side. The protection scale is permitted to depend on m.

Proof. In (183.3) absorb $m^3\rho^{m/2}$ into $Ce^{-\kappa m}$ with $\kappa = -\frac{1}{4}\log\rho > 0$. Divide the local integral by $1+h$. Its second bound is (185.1), since a local mass is no larger than the global mass and $1+h \geq 1$. Apply Lemma 185.2 with $(\alpha, \beta, r) = (1/16, 1, 2)$. Domination gives the other assertions. The fixed auxiliary band used to establish the first inequality has not been altered. The dependence of the final constant on its decay rate is allowed; the exponent $1/16$ does not have that dependence. □

185.2. **A weak endpoint independent of a height cap.** We use the weak L^p distribution convention $\|f\|^p_{p,\infty;I} = \sup_{L>0} L^p|\{u \in I : |f(u)| > L\}|$. This specifies a quasi-norm, not a claim about the strong L^p integral.

Theorem 185.4 (A positive-source endpoint principle)**.** *Suppose $P_m \geq 0$ has exact positive decompositions $P_m = A_{m,\varepsilon} + B_{m,\varepsilon}$ on a finite measure space, and, uniformly for all finite m, ε,*

$$\|A_{m,\varepsilon}\|_\infty \leq C_A\varepsilon^{-K}, \quad \int B_{m,\varepsilon} \leq C_1(\varepsilon^\alpha + e^{-\kappa m}), \quad \int B_{m,\varepsilon} \leq C_2 m^r \varepsilon^\beta,$$

where $K, \kappa > 0$ and $0 < \alpha < \beta$. Suppose also that $\int P_m$ is uniformly bounded. Put $a = \alpha/K$ and $p_c = 1 + a$. Then, for a fixed L_0 and all $L \geq L_0$,

$$\int_{\{P_m > L\}} P_m \leq CL^{-a}, \qquad |\{P_m > L\}| \leq CL^{-p_c}. \tag{185.3}$$

Thus P_m is uniformly in weak L^{p_c} and in strong L^q for every $1 < q < p_c$. No complete-source height bound is assumed.

Proof. Lemma 185.2 first gives $\int B_{m,\varepsilon} \leq C\varepsilon^\alpha$. Choose $\varepsilon = (2C_A/L)^{1/K}$, with L_0 large enough that this is admissible. On $\{P_m > L\}$ one has $B_{m,\varepsilon} \geq P_m/2$. This proves the first bound in (185.3); division by L proves the second. For $1 < q < 1 + a$, the layer-cake integral

$$\int P_m^q \leq \int P_m + (q-1)\int_1^\infty L^{q-2}\int_{\{P_m > L\}} P_m \, dL$$

uses $P_m^q \leq P_m$ on $P_m < 1$. The tail is integrable because $q - 1 < a$. For smaller thresholds the uniform first moment bounds the weak quasi-norm. All integration now uses an error-free tail, so there is no exponential cap or rate ratio in the exponent range. □

For the actual record define the explicit endpoint

$$a_c = \frac{1}{144}, \qquad p_c = \frac{145}{144}. \tag{185.4}$$

Corollary 185.5 (The complete exact-label density at the weak endpoint)**.** *For all translated intervals I of length $h > 0$,*

$$\sup_{R,n,k,m} \int_{I\cap\{P>L\}} P(u)\, du \leq C(1+h)L^{-1/144} \qquad (L \geq L_0), \tag{185.5}$$

$$\sup_{R,n,k,m} \|P\|^{p_c}_{p_c,\infty;I} \leq C(1+h), \qquad \sup_{R,n,k,m} \int_I P^q\, du \leq C_q(1+h) \quad (1 < q < p_c). \tag{185.6}$$

The supremum includes every $m \geq 2$ and $1 \leq n \leq m$.

Proof. Apply Theorem 185.4 with the original guard split, $K = 9$, $\alpha = 1/16$, $\beta = 1$ and $r = 2$. Proposition 183.2 supplies the finite protected height. Lemma 183.1 and Lemma 185.1 supply the two mass bounds and uniform first moments. Equivalently use the already error-free Theorem 185.3. Divide the measure on I by $1 + h$ before applying the principle, and multiply back afterwards. The complete finite-count density exists by Theorem 167.3, but its exponential height estimate is not an input to this argument. □

The remaining pointwise estimate. The endpoint is independent of κ/Γ, unlike the sufficient interior exponent in (183.6). Constants still depend on the fixed-band estimates. Weak $L^{145/144}$ does not assert strong integrability at that endpoint, and neither assertion is an essential-height bound. In particular the positive incidence and clearance criterion in Corollary 181.5 remains the original pointwise target. No physical seam has been crossed in deriving the new estimates.

186. Endpoint-modular raw laws and likelihoods

The error-free positive-source estimate gives more than a larger open interval of powers. It yields quantitative local norms of each physical remainder at every finite count, and a logarithmically weakened endpoint law. We keep the arithmetic transition kernel in all these conclusions, including through a change of arithmetic type.

186.1. **Small mass and a weak endpoint.**

Lemma 186.1 (Interpolation from the distribution tail)**.** *Let $f \ge 0$ on a finite measure space satisfy $\int f \le \delta$ and $|\{f > L\}| \le C_0 L^{-p_c}$ for $L > 0$. For $1 < q < p_c$,*

$$\int f^q \le C_{q,p_c,C_0} \delta^{(p_c-q)/(p_c-1)}. \tag{186.1}$$

Proof. Markov's inequality supplies the other bound $|\{f > L\}| \le \delta/L$. For $\delta > 0$, split $q\int_0^\infty L^{q-1}|\{f > L\}|\,dL$ at $T = (C_0/\delta)^{1/(p_c-1)}$. The lower part is bounded by $q\delta T^{q-1}/(q-1)$ and the upper part by $qC_0T^{q-p_c}/(p_c-q)$. Both have the exponent in (186.1). If $\delta = 0$, the assertion is immediate. This proof applies to a normalized finite measure and does not require it to be a probability. □

Write $B_{\varepsilon,j} = m^2 b^{\varepsilon,j,1}_{n,k,m,R}$ for $j = \mathrm{inc}, \mathrm{clr}$, and include $j = \mathrm{phys}$ for their sum. For a complex insertion w of modulus at most M, use $B^w_{\varepsilon,j}$ for the corresponding density.

Theorem 186.2 (Finite-count norms of both physical remainders)**.** *For $1 < q < p_c$, $0 < \varepsilon < \varepsilon_0$ and every $m \ge 2$,*

$$\sup_{R,n,k,t} \int_t^{t+h} |B^w_{\varepsilon,j}(u)|^q\,du \le C_q M^q (1+h)\varepsilon^{9(p_c-q)}. \tag{186.2}$$

The same inequality, with a changed constant, holds for $B^w_{\varepsilon,j} - K_B * B^w_{\varepsilon,j}$, uniformly over all reconstruction bandwidths $B > 0$. The case $q = 1$ has the bound $CM(1+h)\varepsilon^{1/16}$.

Proof. Source domination gives $|B^w_{\varepsilon,j}| \le MB_{\varepsilon,j} \le MP$. On the measure $du/(1+h)$ restricted to the interval, Theorem 185.3 gives first moment $C\varepsilon^{1/16}$, while Corollary 185.5 gives the weak endpoint. Apply Lemma 186.1 and use the identity

$$\frac{1}{16}\frac{p_c-q}{p_c-1} = 9(p_c-q).$$

The case $M = 0$ is immediate; otherwise divide the inserted density by M. For convolution, Minkowski's integral inequality and translation of the roof interval give

$$\|K_B * B^w_{\varepsilon,j}\|_{L^q(I)} \le \int |K_B(v)|\,\|B^w_{\varepsilon,j}\|_{L^q(I-v)}\,dv.$$

The kernel has $\|K_B\|_1 = \|K_1\|_1$. This proves the all-band assertion without a spectral statement at that band. The first-moment case follows by the same argument. All exact labels and the unchanged roof value remain in these densities. □

186.2. **A logarithmically weakened endpoint.** For $\beta > 1$ and $x \geq 0$ define a convex function by

$$\Phi_\beta(x) = \int_0^x \frac{(x-t)t^{p_c-2}}{[\log(e+t)]^\beta}\, dt. \tag{186.3}$$

It has $\Phi_\beta(0) = \Phi'_\beta(0) = 0$ and $\Phi''_\beta(x) = x^{p_c-2}/[\log(e+x)]^\beta > 0$ for $x > 0$. Its behavior is comparable to x^{p_c} near zero and to $x^{p_c}/[\log(e+x)]^\beta$ at infinity, with constants depending on β. To check the latter assertion, integrate over $t \in [x/4, x/2]$ for the lower bound. For the upper bound split at $\sqrt{x}$: the first part is $O(x^{(p_c+1)/2})$ and the second at most $Cx^{p_c}/(\log x)^\beta$. These also imply the doubling bound $\Phi_\beta(Cx) \leq C'\Phi_\beta(x)$ for each fixed C.

Lemma 186.3 (Uniform endpoint modular and small-mass convergence)**.** *Suppose a family $f \geq 0$ has uniformly bounded first moments and weak L^{p_c} constants. Then its Φ_β integrals are uniformly bounded for every $\beta > 1$, and*

$$\int_{\{f>T\}} \Phi_\beta(f) \leq C_\beta[\log(e+T)]^{1-\beta} \qquad (T \geq 2). \tag{186.4}$$

If also $\int f \leq \delta \leq e^{-2}$, then

$$\int \Phi_\beta(f) \leq C_\beta \left(\delta^{1/2} + [1+\log(1/\delta)]^{1-\beta}\right). \tag{186.5}$$

Proof. The distribution formula for the truncated integral is

$$\int_{\{f>T\}} \Phi_\beta(f) = \Phi_\beta(T)|\{f > T\}| + \int_T^\infty \Phi'_\beta(L)|\{f > L\}|\, dL.$$

The same split at $\sqrt{L}$ used above gives $\Phi'_\beta(L) \leq C_\beta L^{p_c-1}/[\log(e+L)]^\beta$ for $L \geq 2$. The weak endpoint bounds the last integrand by $C_\beta/(L(\log L)^\beta)$, whose integral proves (186.4). The part $f \leq 2$ is controlled by the first moment. For the small-mass assertion put $T = \delta^{-1/(2(p_c-1))}$. On $f \leq T$ use $\Phi_\beta(f) \leq Cf^{p_c} \leq CT^{p_c-1}f$; its integral is at most $C\delta^{1/2}$. Apply (186.4) to the rest. □

Theorem 186.4 (Explicit subcritical and endpoint-modular raw laws)**.** *For every fixed $h > 0$, the complete scalar raw law holds in local L^q for every $1 < q < 145/144$. It also satisfies, for every $\beta > 1$,*

$$\sup_{R,n,k,t} \frac{1}{h}\int_t^{t+h} \Phi_\beta(|P(u) - G(u)|)\, du \longrightarrow 0. \tag{186.6}$$

Here $G(u) = \mathcal{L}_{m,R}(k_1, k_2, u, n)$, with no positivity assumption. The same conclusions hold with $|P - G|$ replaced by

$$E^{\mathrm{path}}(u) = \|\mathbf{P}(u) - G(u)\mathsf{W}_R\|_{\mathrm{BL}^*}. \tag{186.7}$$

For the actual-return path, the assertion is on the central target classes of Lemma 191.1.

Proof. The transition kernel is uniformly bounded. Hence $|P - G| \leq P + C$ has uniformly bounded weak L^{p_c} constants on translated intervals. Its local first moment tends uniformly to zero by Theorem 180.1. Apply Lemmas 186.1 and 186.3. The path error in (186.7) is at most $P + |G|$, and its first moment tends to zero by Theorem 190.3. The same proof applies; it uses the common roof versions constructed there, not a null set chosen separately for each path test. The return-path first-moment transfer is precisely Lemma 191.1. Its source remains positive, so the same scalar domination applies on the stated central class, with its prescribed enlargement across the roof window. The constants may depend on the fixed h, which is never allowed to shrink with m in this argument. □

For example, if the corresponding uniform mean error is $e_m \to 0$, (186.5) gives the explicit modulus $C_\beta(e_m^{1/2} + (1 + |\log e_m|)^{1-\beta})$ for large m. This is a modulus in the known error, not an asserted collision-count rate. The normalized positive physical remainders separately satisfy

$$\sup_{R,n,k,t,m} \frac{1}{1+h} \int_t^{t+h} \Phi_\beta(B_{\varepsilon,j}(u))\, du \le C_\beta(1 + |\log \varepsilon|)^{1-\beta}$$

for sufficiently small ε, by the same lemma. Arbitrary bounded insertions obey the corresponding bound by domination and the doubling property. This endpoint statement is still an integral estimate and does not remove their possible positive essential heights.

186.3. **The original conditional roof likelihood.**

Theorem 186.5 (Forward likelihood up to the explicit endpoint)**.** *On a fixed roof window I of length $h > 0$, assume the pointwise arithmetic floor $G \ge d > 0$. Put*

$$F = \int_I P, \qquad H = \int_I G, \qquad d\pi = P\, du/F, \quad dQ = G\, du/H, \quad r = d\pi/dQ.$$

Uniformly in the same exact targets, for all $1 < q < 145/144$ and $\beta > 1$,

$$\int |r-1|^q\, dQ \longrightarrow 0, \qquad \int \Phi_\beta(|r-1|)\, dQ \longrightarrow 0. \tag{186.8}$$

In particular $D(\pi\|Q) \to 0$ and $D_q(\pi\|Q) \to 0$ for every fixed $1 < q < 145/144$. The second integral in (186.8) *is the convex f-divergence generated by $f(x) = \Phi_\beta(|x-1|)$.*

Proof. The scalar first-moment law gives $F - H \to 0$. The floor and the upper bound on G imply $dh \le H \le Ch$ and, for all sufficiently large counts, $F \ge dh/2$. The exact likelihood satisfies

$$r - 1 = \frac{H(P-G)}{FG} + \frac{H-F}{F}.$$

The first term has the subcritical and endpoint-modular convergence of Theorem 186.4; the reference density G/H is uniformly bounded on I. For the modular use convexity in the form $\Phi_\beta(x+y) \le \frac{1}{2}\Phi_\beta(2x) + \frac{1}{2}\Phi_\beta(2y)$ and the fixed-factor doubling bound. The constant term tends to zero. This proves (186.8). Convexity and monotonicity of Φ_β show that f is convex and nonnegative with $f(1) = 0$. For relative entropy use $0 \le x\log x - x + 1 \le C_q|x-1|^q$ with $1 < q < 145/144 < 2$. For Rényi divergence, $L^q(Q)$ convergence gives $\int r^q\, dQ \to 1$. These are the unchanged conditional laws; a positive integrated reference mass without the pointwise floor would not justify the likelihood calculation. □

Proof dependencies and pointwise scope. The new route consists of the exact geometric global strip mass, one fixed-band finite-count local strip estimate, and the polynomial protected height. It uses no complete exponential height cap, no new high-frequency resolvent, and no extension through grazing or a competing hit. Thus it is not an invocation of the billiard-flow mixing result of [5] as an exact occupation–roof density theorem, nor a differentiation of the suspension interval theorem of [3]. The density and its arithmetic main term are unchanged. The positive height criterion of Corollary 181.5, and hence the uniform pointwise same-roof bridge implication, remain separate from all finite-norm conclusions proved here.

187. Maximal localization of the two positive physical sources

A bound for the mass of a boundary source does not bound its density at every roof. It can, however, control all interval averages at once, outside one explicitly estimated set of roofs. We prove that statement for the original incidence and clearance sources. The exceptional set is

chosen before a roof resolution or a bounded source insertion is specified. No physical history is removed from the density.

Throughout this section $I = [t, t + h]$ has fixed length $h > 0$. Its role is to localize a finite measure; the subintervals used below have no positive lower bound on their lengths. For a nonnegative integrable function f on I define

$$\mathcal{M}_I f(u) = \sup_{J \ni u} \frac{1}{|J|} \int_J \mathbf{1}_I(v) f(v)\, dv, \tag{187.1}$$

where J ranges over all bounded intervals of positive length in $\mathbb{R}$. The supremum is unchanged if rational endpoints are used and limits of such intervals are admitted. This defines a Borel function on $\mathbb{R}$. In particular it bounds the average over every $J \subset I$ containing u, including one-sided intervals with endpoint u. The letters B, ε and λ denote respectively a reconstruction band, a physical protection width and a density level; none denotes the interval length.

187.1. **A covering estimate and its source-weighted form.**

Lemma 187.1 (One envelope for every interval)**.** *For $f \geq 0$ in $L^1(I)$ and $\lambda > 0$, the set $O_\lambda = \{\mathcal{M}_I f > \lambda\}$ is open and*

$$|O_\lambda| \leq \frac{3}{\lambda} \int_I f. \tag{187.2}$$

If, in addition, $|\{u \in I : f(u) > L\}| \leq DL^{-p}$ for $L > 0$ and $p > 1$, then

$$|O_\lambda| \leq C_p D \lambda^{-p}. \tag{187.3}$$

Consequently, for $1 < q < p$ and $\delta = \int_I f$,

$$\int_{\mathbb{R}} (\mathcal{M}_I f)^q \leq C_{p,q} D^{(q-1)/(p-1)} \delta^{(p-q)/(p-1)}. \tag{187.4}$$

The zero function is interpreted separately if $D\delta = 0$.

Proof. Extend f by zero. An interval with average strictly greater than λ can be enlarged slightly and retain this inequality, by absolute continuity of the integral. Such enlarged open intervals cover O_λ, proving openness. Cover a compact subset by finitely many witnessing intervals. Select one of greatest length, delete all intervals intersecting it, and repeat. The selected intervals are disjoint. Every deleted interval is contained in the concentric triple of its selected interval. The compact subset therefore has measure at most three times the sum of the selected lengths, at most $3 \int f/\lambda$. Inner regularity proves (187.2). This is the elementary interval covering proof of the uncentered maximal inequality.

Write $f = f\mathbf{1}_{\{f \leq \lambda/2\}} + f\mathbf{1}_{\{f > \lambda/2\}}$. The first summand has maximal average at most $\lambda/2$. The layer-cake identity and the weak tail give

$$\int_{\{f > \lambda/2\}} f \leq D(\lambda/2)^{1-p} + D \int_{\lambda/2}^{\infty} L^{-p}\, dL \leq C_p D \lambda^{1-p}.$$

Apply (187.2) to the second summand with threshold $\lambda/2$. This proves (187.3). Finally integrate the minimum of $3\delta/\lambda$ and $C_p D \lambda^{-p}$ in the distribution formula for the qth power. Splitting at a constant multiple of $(D/\delta)^{1/(p-1)}$ proves (187.4). The integral near zero is finite since $q > 1$, even though the maximal function need not have an integrable first power on all of $\mathbb{R}$. □

Lemma 187.2 (Mass of a small roof set under the complete source)**.** *Suppose $P \geq 0$ on I satisfies $|\{P > L\}| \leq DL^{-p}$ for $L > 0$, $p > 1$. For every measurable $E \subset I$,*

$$\int_E P \leq \frac{p}{p-1} D^{1/p} |E|^{1-1/p}. \tag{187.5}$$

If $P = G + f + e$, $f \geq 0$, $\|G\|_\infty \leq C_G$ and $\|e\|_\infty \leq \eta$, there is also the distinct estimate

$$\int_E P \leq (C_G + \eta)|E| + \int_I f. \tag{187.6}$$

Proof. Integrate $\min\{|E|, DL^{-p}\}$ over $L > 0$, splitting at $(D/|E|)^{1/p}$. This proves the first estimate, with the cases of zero measure immediate. The second follows by integrating the exact positive-remainder representation. It uses a bounded controlled part; it is not a consequence of the weak endpoint alone. □

187.2. **Finite-count localization of physical heights.** Use the unchanged densities of Section 177 and put

$$P = m^2 p_{n,R}(k, m, \cdot), \qquad V_{\varepsilon,j} = m^2 b^{\varepsilon,j,1}_{n,k,m,R}, \quad V_\varepsilon = V_{\varepsilon,\mathrm{inc}} + V_{\varepsilon,\mathrm{clr}}.$$

Thus $0 \leq V_{\varepsilon,j} \leq V_\varepsilon \leq P$. The normalization is $\nu^*_R = c^{-1}\nu|_{Y^*_R}$, used once. The exact n, k, m and the half-open occupation at $0, \ldots, m-1$ are unchanged; clearance of flight j remains read at collision $j + 1$.

Theorem 187.3 (Finite-count maximal boundary control)**.** *Let $p_c = 145/144$ and $0 < \varepsilon < \varepsilon_0$. For every $m \geq 2$, every exact label, and every translated interval I, set*

$$O_{\varepsilon,\lambda} = \{u \in \mathbb{R} : \mathcal{M}_I V_\varepsilon(u) > \lambda\}. \tag{187.7}$$

With constants uniform in $m, R, n, k, t, \varepsilon, \lambda$,

$$|O_{\varepsilon,\lambda}| \leq C(1 + h)\min\{\varepsilon^{1/16}/\lambda, \lambda^{-p_c}\}, \tag{187.8}$$

$$\int_{I \cap O_{\varepsilon,\lambda}} P \leq C(1 + h)\min\{1, (\varepsilon^{1/16}/\lambda)^{1/145}\}. \tag{187.9}$$

On $I \setminus O_{\varepsilon,\lambda}$ both physical components have height at most λ almost everywhere, and

$$\sup_{J \subset I:\, u \in J} \frac{1}{|J|} \int_J V_{\varepsilon,j} \leq \lambda \quad (u \in I \setminus O_{\varepsilon,\lambda}). \tag{187.10}$$

The supremum is over all positive lengths. The same set works for every bounded source insertion: the absolute density and every interval average for $|w| \leq M$ are bounded by M times these quantities. Moreover, for $1 < q < p_c$,

$$\int_{\mathbb{R}} (\mathcal{M}_I V_\varepsilon)^q \leq C_q(1 + h)\varepsilon^{9(p_c - q)}. \tag{187.11}$$

Proof. Theorem 185.3 gives $\int_I V_\varepsilon \leq C(1 + h)\varepsilon^{1/16}$, simultaneously for every finite count and protection scale. Corollary 185.5 gives the weak L^{p_c} bound for P and hence for V_ε. Apply Lemma 187.1 to get (187.8). Lemma 187.2 then gives (187.9), since $1 - 1/p_c = 1/145$. The uniform first moment of P supplies the alternative bound by $C(1 + h)$.

Lebesgue differentiation gives $V_\varepsilon \leq \mathcal{M}_I V_\varepsilon$ almost everywhere on I. This proves the height assertion. The interval assertion is the definition of the maximal function and needs no exceptional null set depending on an interval. Positive source domination gives the inserted assertions. More precisely, fix a regular disintegration of the finite physical source over its roof. Defining every inserted density by this one kernel gives $|V^w_{\varepsilon,j}(u)| \leq MV_{\varepsilon,j}(u)$ in that common version. A separately chosen density version for each w is not being used. The last bound follows from (187.4), with $\delta = C(1 + h)\varepsilon^{1/16}$, $D = C(1 + h)$ and $(1/16)/(p_c - 1) = 9$. All these are finite inequalities; no spectral band is made to depend on a roof resolution. □

187.3. **A sharper source estimate in the ordered raw limit.** For large fixed B, take $\varepsilon(B) = A_0 B^{-1/12}$ and write $V_B = \mathfrak{b}^{n,k}_{B,m,R}$. Define

$$\eta_{B,m} = \sup_{R,n,k} \|P - G - V_B\|_\infty, \qquad G(u) = \mathcal{L}_{m,R}(k_1, k_2, u, n).$$

It is finite for each count, and $\limsup_m \eta_{B,m} \le CB^{-1/192}$ by Theorem 181.4. The parameter-uniform arithmetic kernel is bounded by a fixed C_G. Let

$$\lambda_B = B^{-1/384}, \qquad O^{n,k,I}_{B,m,R} = O_{\varepsilon(B),\lambda_B}. \tag{187.12}$$

Corollary 187.4 (Resolution-uniform localization of the raw error)**.** *For every fixed $h > 0$,*

$$\sup_{m,R,n,k,t} |O^{n,k,I}_{B,m,R}| \le C(1+h)B^{-1/384}, \tag{187.13}$$

$$\limsup_{m\to\infty} \sup_{R,n,k,t} \int_{I \cap O^{n,k,I}_{B,m,R}} P \le C(1+h)B^{-1/384}. \tag{187.14}$$

For $u \in I \setminus O^{n,k,I}_{B,m,R}$,

$$\sup_{J \subset I:\, u \in J} \frac{1}{|J|} \int_J |P - G| \le \lambda_B + \eta_{B,m}. \tag{187.15}$$

The pointwise bound $|P(u) - G(u)| \le \lambda_B + \eta_{B,m}$ also holds almost everywhere on this same complement. Consequently all these errors have ordered limsup at most $CB^{-1/384}$.

Proof. Equation (187.8) gives (187.13). Apply (187.6) with $f = V_B$ to obtain the finite estimate

$$\int_{I \cap O} P \le C(1+h)\{(C_G + \eta_{B,m})B^{-1/384} + B^{-1/192}\}.$$

Its collision limsup proves (187.14). This is sharper than the finite-count weak-endpoint estimate (187.9), because it also uses the positive-remainder raw representation. Finally $|P - G| \le V_B + \eta_{B,m}$. Take interval averages at a good anchor u and use (187.10), then use Lebesgue differentiation for the pointwise assertion. □

The part of the height problem still requiring geometry. The open set in (187.7) localizes all upward physical concentration, not just concentration at one prescribed resolution. Its length and original source mass tend to zero in the stated ordered limit. The set need not be empty. A thin positive spike can lie entirely inside it. Thus (187.15) retains the original arithmetic pointwise target but does not prove (181.9) on the excluded roofs. No incidence or competing-hit seam has been traversed in this argument.

188. Exact roof laws at all interval resolutions

We apply the same positive maximal envelope to the path-valued raw measure. It gives estimates for every subinterval containing a good roof, without prescribing how fast the subinterval shrinks. The conditioning event remains the original exact-label event with that subinterval. A statement uniform over intervals is not an assertion that conditioning on the output of an adaptive interval-selection procedure is the same event.

Let $\mathbf{P}(u)$, $\mathbf{b}_B(u)$ and W_R be the positive collision-bridge kernels and Gaussian reference of Section 190. Their masses are $P(u)$ and $V_B(u)$, respectively. All versions are fixed by its common source disintegration. Put

$$\eta^{\mathrm{path}}_{B,m} = \sup_{R,n,k} \operatorname*{ess\,sup}_u \|\mathbf{P}(u) - G(u)\mathsf{W}_R - \mathbf{b}_B(u)\|_{\mathrm{BL}^*}. \tag{188.1}$$

Theorem 190.3 gives $\limsup_m \eta_{B,m}^{\mathrm{path}} \le CB^{-1/192}$ for each fixed B. The constant test one belongs to the norming class, so the scalar discrepancy $\eta_{B,m}$ is at most $\eta_{B,m}^{\mathrm{path}}$. This is a path bounded-Lipschitz dual norm, not variation mass on path space.

188.1. **One physical envelope for a continuum of resolutions.**

Theorem 188.1 (Path and scalar control on all anchored intervals)**.** *Fix $h > 0$ and a sufficiently large band B. Use exactly the open set $O_{B,m,R}^{n,k,I}$ of* (187.12)*, and put $r_{B,m} = \lambda_B + \eta_{B,m}^{\mathrm{path}}$. For every $u \in I \setminus O_{B,m,R}^{n,k,I}$,*

$$\sup_{J \subset I:\, u \in J} \frac{1}{|J|} \int_J \|\mathbf{P}(v) - G(v)\mathsf{W}_R\|_{\mathrm{BL}^*}\, dv \le r_{B,m}. \tag{188.2}$$

Almost everywhere on this same complement the integrand at u is at most $r_{B,m}$. The supremum includes every positive interval length; the open exceptional set does not depend on J or on a path test. The estimates (187.13) *and* (187.14) *hold for this set, and*

$$\limsup_{m \to \infty} r_{B,m} \le CB^{-1/384}.$$

The same assertions hold for the scalar absolute error and for integrals against measurable roof-dependent unit bounded-Lipschitz path tests.

Proof. Positivity gives $\|\mathbf{b}_B(v)\|_{\mathrm{BL}^*} = V_B(v)$. Hence, on the common version,

$$\|\mathbf{P}(v) - G(v)\mathsf{W}_R\|_{\mathrm{BL}^*} \le V_B(v) + \eta_{B,m}^{\mathrm{path}}$$

for almost every v. Average over an interval containing a good anchor and use (187.10). The pointwise assertion follows from $V_B \le \mathcal{M}_I V_B$ almost everywhere, not from an interchange of a collision limit and differentiation. The source bounds were already proved for this very set. A measurable test whose restriction to each roof is in the unit path class is bounded by the displayed dual norm under the integral. One common countable norming class supplies the roof versions; no uncountable union of exceptional null sets is taken. □

For a subinterval J of positive length let

$$F_J = \int_J P, \quad H_J = \int_J G_+, \qquad d\pi_J = P\mathbf{1}_J\, du/F_J, \quad dQ_J = G_+\mathbf{1}_J\, du/H_J. \tag{188.3}$$

These are conditional roof laws only when their denominators are positive. We use probability total variation $d_{\mathrm{TV}}(\pi, Q) = \frac{1}{2}\int |d\pi - dQ|$. No factor $1/2$ is inserted in a variation mass of an unnormalized source.

Corollary 188.2 (Conditional laws on arbitrarily short anchored intervals)**.** *Let $u \in I \setminus O_{B,m,R}^{n,k,I}$, let $J \subset I$ contain u, and assume $H_J \ge d|J|$ for a fixed $d > 0$. If $r = r_{B,m} < d$, then*

$$F_J \ge (d - r)|J|, \qquad d_{\mathrm{TV}}(\pi_J, Q_J) \le \frac{r}{d - r}, \tag{188.4}$$

$$\int_J d_{\mathrm{BL}}(\mathsf{Q}_{m,R}^{n,k,v}, \mathsf{W}_R)\, \pi_J(dv) \le \frac{2r}{d - r}. \tag{188.5}$$

The joint roof–path law is also within $2r/(d-r)$ of $Q_J \otimes \mathsf{W}_R$ when tested by functions measurable in the roof and unit bounded-Lipschitz in the path. These estimates hold simultaneously for all such J, including intervals depending on m with no positive lower resolution.

If $G(u) \ge d$ at a good roof, its original same-roof collision bridge satisfies, almost everywhere there and for $\eta_{B,m}^{\mathrm{path}} < d$,

$$d_{\mathrm{BL}}(\mathsf{Q}_{m,R}^{n,k,u}, \mathsf{W}_R) \le \frac{2r_{B,m}}{d - \eta_{B,m}^{\mathrm{path}}}. \tag{188.6}$$

If $G \geq d$ almost everywhere on J and $r \leq d/2$, then the likelihood $r_J = d\pi_J/dQ_J$ also obeys

$$|r_J(u) - 1| \leq 4r/d \tag{188.7}$$

almost everywhere at good roofs $u \in J$. This is not an essential-supremum estimate on the whole of J.

Proof. Write $\mathbf{P}(v) = P(v)\mathsf{Q}^{n,k,v}_{m,R}$. For $G(v) < 0$, positivity implies $\|\mathbf{P}(v) - G(v)\mathsf{W}_R\|_{\mathrm{BL}^*} = P(v) + |G(v)|$. Thus replacing G by G_+ never increases this norm. In particular,

$$\int_J |P - G_+| \leq r|J|, \qquad \int_J \|\mathbf{P} - G_+\mathsf{W}_R\|_{\mathrm{BL}^*} \leq r|J|.$$

It follows that $|F_J - H_J| \leq r|J|$ and $F_J \geq (d-r)|J|$. Adding and subtracting G_+/F_J bounds the roof variation mass by $2r|J|/F_J$. Dividing by two proves (188.4). Normalizing the path-valued inequality costs at most a further $|F_J - H_J|/F_J$, proving the joint-test assertion. Also, pointwise,

$$P(v)d_{\mathrm{BL}}(\mathsf{Q}^{n,k,v}_{m,R}, \mathsf{W}_R) \leq 2\|\mathbf{P}(v) - G(v)\mathsf{W}_R\|_{\mathrm{BL}^*}.$$

Integration proves (188.5).

At a good roof the unnormalized dual error is at most r. The scalar positive-remainder identity gives $P(u) \geq G(u) - \eta^{\mathrm{path}}_{B,m}$; division proves (188.6). For the last assertion $G_+ = G$ on J, and the exact identity is

$$r_J(u) - 1 = \frac{H_J(P(u) - G(u))}{F_J G(u)} + \frac{H_J - F_J}{F_J}.$$

Here $H_J/F_J \leq 1 + r/(d-r) \leq 2$ and $|P(u) - G(u)| \leq r$ at almost every good roof. The two terms are each at most $2r/d$. The pointwise floor is used exactly here; an integrated lower bound alone would not justify the likelihood claim. □

188.2. **The original roof law sees few bad anchors.**

Corollary 188.3 (Probability of the common exceptional set)**.** *On a mother interval I with $\int_I G_+ \geq dh$, the conditional roof laws of* (188.3) *satisfy*

$$\limsup_{m\to\infty} \sup_{R,n,k,t} \{\pi_I(O^{n,k,I}_{B,m,R}) + Q_I(O^{n,k,I}_{B,m,R})\} \leq C_{h,d}B^{-1/384}. \tag{188.8}$$

There are sequences $B_m \to \infty$ and $a_m \to 0$, independent of the exact labels and translated mother interval, such that the errors in Theorem 188.1 and Corollary 188.2 tend uniformly to zero outside sets of Lebesgue, π_I and Q_I measure tending to zero. The errors are simultaneous over all anchored subintervals. No polynomial rate in m is asserted for B_m or a_m.

Proof. The inherited local L^1 law gives $\int_I P \geq dh/2$ eventually. Divide (187.14) by this bound. For the reference use its density $G_+/\int_I G_+ \leq C_G/(dh)$ and (187.13). This proves (188.8) without assuming that bad anchors are independent of the source.

For clarity, the diagonal choice uses only uniform fixed-band limits. Choose $B_j \uparrow \infty$ such that $CB_j^{-1/384} \leq 2^{-j}$. For each j, choose increasing integers M_j so that for every $m \geq M_j$ the path discrepancy at this fixed B_j and the mother normalization error satisfy their corresponding upper bounds with slack 2^{-j}. Set $B_m = B_j$ for $M_j \leq m < M_{j+1}$. The finite length bound and the finite source bound in the proof of Corollary 187.4 then give the stated vanishing exceptional measures. The all-interval inequality is already finite for each (B, m), so it requires no further diagonal over subinterval lengths. For example $a_m = C_{h,d}2^{-j}$ on these blocks bounds the errors once j is large. Constants at growing spectral bands have not been estimated or substituted into a rate. □

188.3. **All-resolution transfer for the actual return bridge.** The physical maximal set above is quantitative in the ordered band limit for the collision bridge. The actual-return clock transfer is available in an integrated norm. The following separate application makes its quantifiers explicit and does not assign it the preceding band exponent.

Let

$$E_m^{\mathrm{ret}}(v) = \|P(v)\mathsf{R}_{m,R}^{n,k,v} - G(v)\mathsf{V}_R\|_{\mathrm{BL}^*}, \qquad E_m^{\mathrm{col}}(v) = \|\mathbf{P}(v) - G(v)\mathsf{W}_R\|_{\mathrm{BL}^*}.$$

The target class is $|Z(t)| \le M_0$ with fixed h, M_0, using the prescribed enlargement $|Z(v)| \le M_0 + h/\sqrt{m}$ on I.

Theorem 188.4 (A common all-resolution set for both actual bridges)**.** *There is a deterministic $e_m \downarrow 0$ such that, uniformly in this central class, $\int_I (E_m^{\mathrm{ret}} + E_m^{\mathrm{col}}) \le (1+h)e_m$. Set $\ell_m = \sqrt{e_m} + m^{-1}$ and*

$$O_m^{\mathrm{both}} = \{\mathcal{M}_I(E_m^{\mathrm{ret}} + E_m^{\mathrm{col}}) > \ell_m\}.$$

Then

$$|O_m^{\mathrm{both}}| \le C(1+h)e_m/\ell_m, \tag{188.9}$$

$$\int_{I \cap O_m^{\mathrm{both}}} P \le C(1+h)(e_m/\ell_m)^{1/145}. \tag{188.10}$$

For every anchor outside this set and every subinterval $J \subset I$ containing it, both integrated dual errors divided by $|J|$ are at most ℓ_m. The following conditional bounds are for $\ell_m < d$, which holds eventually for each fixed $d > 0$. If $\int_J G_+ \ge d|J|$, both original conditional bridges obey the joint-test and conditional-mean estimates of Corollary 188.2, with $r = \ell_m$. At almost every good roof where $G \ge d$, their same-roof bounded-Lipschitz distances are at most $2\ell_m/(d - \ell_m)$. These conclusions are uniform over every positive subinterval length, not over arbitrary externally chosen roofs inside the exceptional set.

Proof. The first assertion follows from Lemma 191.1 and Theorem 190.3; replace the suprema of their mean errors by decreasing tail suprema. Their errors are bounded by $2(P + |G|)$ in sum, so their local integrability and common roof versions cause no difficulty. Apply (187.2) to that sum and then (187.5) to P. This proves both set bounds. All interval averages at a good anchor are at most ℓ_m by definition, and both pointwise errors have that bound almost everywhere there by differentiation. The constant path test yields $|P - G| \le \ell_m$ at such roofs. Repeat the positive-part and normalization argument of Corollary 188.2 for each bridge separately. Since $P \ge G - \ell_m$, the claimed same-roof denominator follows. No return event, clock value or source law is replaced, and no quantitative rate is assigned to e_m. □

Relation to the full pointwise endpoint. The distinction from Corollary 191.3 is the supremum over all intervals containing the same good anchor. That corollary controls the kernel at most roofs but places no simultaneous bound on every arbitrarily short surrounding interval. The covering argument supplies this uniformity directly from the unchanged positive source or the already proved integrated return numerator. It is a real-variable consequence of those dynamical inputs, not a new spectral theorem or a claim of a second singular-hyperbolic application. The local limit results of [2, 12, 3] supply the established background; none is differentiated here into an unproved density bound.

The common sets are target- and count-dependent. All densities and conditional laws in the conclusions still include the full physical source, including its values on these sets. Ruling out an upward spike inside them remains precisely the positive incidence and clearance height problem of Corollary 181.5. A vanishing bound on their measure is not a proof that they are empty.

189. A signed-measure principle for exact-label path inversion

We separate the measure-theoretic step from the collision geometry. The distinction between tight total variations and small total variation is essential. The first gives compactness of signed inverses; it does not give total-variation convergence on path space.

Lemma 189.1 (Tight signed measures on continuous path space)**.** *Let σ_j, σ be finite real signed measures on $\mathcal{C} = C([0,1], \mathbb{R}^d)$. Suppose that $\sup_j \|\sigma_j\|_{\mathrm{TV}} < \infty$, the measures $|\sigma_j|$ are uniformly tight, the total masses converge, and*

$$\int \exp\left(i \sum_{a=1}^{r} v_a \cdot x(t_a)\right) d\sigma_j(x) \longrightarrow \int \exp\left(i \sum_{a=1}^{r} v_a \cdot x(t_a)\right) d\sigma(x)$$

for all finite lists of rational times and real vectors. Then $\|\sigma_j - \sigma\|_{\mathrm{BL}^*} \to 0$, with the supremum path metric and the bounded-Lipschitz convention of Section 190. It suffices to dominate $|\sigma_j|$ by positive finite measures τ_j of bounded masses which are uniformly tight.

Proof. The Jordan parts of σ_j have bounded masses and are tight. Positive finite-measure compactness, applied to both parts along a subsequence, gives a weakly convergent signed subsubsequence. Equivalently, under the last hypothesis, apply this compactness to $\tau_j + \sigma_j$ and $\tau_j - \sigma_j$; these measures are positive because $|\sigma_j| \le \tau_j$. Cylinder characteristic functions identify every subsequential limit with σ. Indeed uniqueness of the Fourier transform identifies each finite-dimensional signed marginal. Rational evaluations generate the Borel sigma field on continuous path space; uniqueness on their cylinder sets identifies the finite signed measure. Mass convergence includes the constant cylinder and rules out an unstated loss of mass.

Given $\epsilon > 0$, choose a compact path set carrying all but ϵ of every variation and of $|\sigma|$. The restrictions of unit bounded-Lipschitz functions to this set have a finite uniform ϵ-net. Choose its representatives from the original test class, so they are defined on the whole path space. Convergence on these finitely many bounded continuous functions and the variation bound control the supremum on the compact set. The complement costs at most 2ϵ. Letting $\epsilon \downarrow 0$ proves the claim. The argument applies to every subsequence and hence to the full sequence. It proves a bounded-Lipschitz conclusion, not convergence of the Jordan parts in total variation. □

189.1. **Positive remainders shared by all tests.** The following statement applies to a compact family of reference path laws and any collection of discrete labels. The labels may vary with the integer m. Write λ for their aggregate, and let W_λ range over a compact subset of the probability measures on $\mathcal{C}$ in the weak topology. For each (m, λ) let $\mathbf{P}_{m,\lambda}(u)$ be a Borel kernel of positive finite path measures, of finite total mass after integration in u. Put $P_{m,\lambda} = \mathbf{P}_{m,\lambda}(1)$. Let $G_{m,\lambda}$ be a real scalar reference. Convolution is in u only. Suppose $K_B \in W^{1,1}(\mathbb{R})$ and $\sup_B \|K_B\|_1 \le \kappa < \infty$. All the essential norms below use common versions obtained from a countable bounded-Lipschitz norming class.

Theorem 189.2 (Exact-label path inversion with a positive remainder)**.** *For each fixed B, suppose there are positive kernels $0 \le \mathbf{R}_{B,m,\lambda} \le \mathbf{P}_{m,\lambda}$ and put $\mathbf{A}_B = \mathbf{P} - \mathbf{R}_B$. Assume the following four properties, in each of which B is fixed before $m \to \infty$.*

Scalar and cylinder inversion. *The masses of $\sigma_{B,m,\lambda,u} = (K_B * \mathbf{P}_{m,\lambda})(u)$ satisfy $\sup_{\lambda,u} |\sigma_{B,m,\lambda,u}(1) - G_{m,\lambda}(u)| \to 0$. For each rational-time characteristic cylinder ϕ,*

$$\sup_{\lambda,u} |\sigma_{B,m,\lambda,u}(\phi) - \sigma_{B,m,\lambda,u}(1)\mathsf{W}_\lambda(\phi)| \to 0.$$

Variation compactness. *The variations of the signed measures $\sigma_{B,m,\lambda,u}$ have bounded masses and are uniformly tight, including over the labels and u.*

Controlled-source correction. *For numbers $r_B \ge 0$,*

$$\limsup_m \sup_{\lambda, F\in\mathcal{F}} \|\mathbf{A}_{B,m,\lambda}(F) - K_B * \mathbf{A}_{B,m,\lambda}(F)\|_\infty \le r_B.$$

Positive local mass. *For numbers $s_B \ge 0$ independent of h, and every fixed $h > 0$,*

$$\limsup_m \sup_{\lambda,t} \frac{1}{h}\int_t^{t+h} \mathbf{R}_{B,m,\lambda}(u)(1)\,du \le s_B.$$

Then

$$\limsup_m \sup_\lambda \operatorname*{ess\,sup}_u \|\mathbf{P}_{m,\lambda}(u) - G_{m,\lambda}(u)\mathsf{W}_\lambda - \mathbf{R}_{B,m,\lambda}(u)\|_{\mathrm{BL}^*} \le r_B + \kappa s_B, \tag{189.1}$$

$$\limsup_m \sup_{\lambda,t} \frac{1}{h}\int_t^{t+h} \|\mathbf{P}_{m,\lambda}(u) - G_{m,\lambda}(u)\mathsf{W}_\lambda\|_{\mathrm{BL}^*}\,du \le r_B + (\kappa+1)s_B. \tag{189.2}$$

In particular $r_B + s_B \to 0$ gives a path-valued local law and same-roof conditional bounded-Lipschitz convergence in mean whenever $\int_{I_t}(G_{m,\lambda})_+ \ge dh$ for fixed $d, h > 0$. If, in addition, the ordered essential height of $\mathbf{R}_{B,m,\lambda}(1)$ tends to zero on a set where $G_{m,\lambda} \ge d$, the conditional path convergence is uniform in essential supremum on that set.

Proof. The first two hypotheses and Lemma 189.1 give, at fixed B,

$$\sup_{\lambda,u} \|K_B * \mathbf{P} - G\mathsf{W}_\lambda\|_{\mathrm{BL}^*} \to 0.$$

For completeness, if uniformity failed, choose a violating sequence. Compactness of the reference laws and boundedness of the signed masses allow a subsequence with limits W and a. The signed-measure lemma identifies its limit as $a\mathsf{W}$, contradicting the violation. Thus no uniformity over a growing band has been used.

For $F \in \mathcal{F}$ the exact source identity is

$$\mathbf{P}(F) - G\mathsf{W}_\lambda(F) - \mathbf{R}_B(F) = \mathbf{A}_B(F) - K_B * \mathbf{A}_B(F) + K_B * \mathbf{P}(F) - G\mathsf{W}_\lambda(F) - K_B * \mathbf{R}_B(F).$$

Since $|\mathbf{R}_B(F)| \le \mathbf{R}_B(1)$, the local mass hypothesis and Lemma 181.1 bound the last term, after the collision limsup, by $\|K_B\|_1 s_B$ uniformly in F. A countable norming class supplies one exceptional roof-null set. This proves (189.1). Integrating and using $\|\mathbf{R}_B(u)\|_{\mathrm{BL}^*} = \mathbf{R}_B(u)(1)$ proves (189.2).

Write $\mathbf{P}(u) = P(u)Q_u$. The constant function one belongs to $\mathcal{F}$, so

$$P(u)d_{\mathrm{BL}}(Q_u, \mathsf{W}_\lambda) \le 2\|\mathbf{P}(u) - G(u)\mathsf{W}_\lambda\|_{\mathrm{BL}^*}.$$

Also $|P - G_+| \le |P - G|$, since $P \ge 0$. The integrated positive reference bound therefore gives $\int_{I_t} P \ge dh/2$ eventually. Integration and division by this unchanged source mass prove the conditional mean assertion.

For the last assertion put $\mathbf{E} = \mathbf{P} - G\mathsf{W}_\lambda - \mathbf{R}_B$ and $R_B = \mathbf{R}_B(1)$. Exactly,

$$P(Q_u - \mathsf{W}_\lambda) = \mathbf{R}_B - R_B\mathsf{W}_\lambda + \mathbf{E} - \mathbf{E}(1)\mathsf{W}_\lambda.$$

Positivity gives a norm bound $2R_B + 2\|\mathbf{E}\|_{\mathrm{BL}^*}$. Moreover (189.1) gives the lower law $P \ge G - o(1)$ in essential supremum when $r_B + s_B \to 0$. Thus $P \ge d/2$ eventually on the stated set. Divide there and first take the collision limit, then the reconstruction-band limit. The required positive height is an additional hypothesis, not a consequence of positive local mass. □

Verification for the present return record. Here $\lambda = (R, n, k)$, $\mathbf{R}_B = \mathbf{b}_B$ from (190.2), and the reference is the bridge of covariance Ω_R. Lemma 190.1 verifies the insertion step on each protected physical word. Lemma 190.2 supplies cylinder inversion and a positive variation dominator. The protected correction and the all-depth decision bound give $r_B \le C(B^{-1/2} + B^{-1/192})$; the physical local-variation theorem gives $s_B \le CB^{-1/192}$. These are separate, source-pinned inputs to the abstract theorem. No hypothesis here realizes a discontinuous observable on a new Banach space, proves a physical seam estimate, or adds an independent application outside this billiard family.

190. Path-valued inversion of the original raw density

A path limit under a fixed roof window does not by itself control the conditional path law at the individual roof values inside that window. We estimate the path numerator in a norm which permits the path test to be chosen after the roof value. The estimate retains the complete physical remainder. In particular, the unresolved positive height is not replaced by a local integral.

Let $\mathcal{C} = C([0,1], \mathbb{R}^4)$ with its supremum metric, and let $\mathcal{F}$ consist of the real functions F on $\mathcal{C}$ satisfying $\|F\|_\infty \le 1$ and $\mathrm{Lip}(F) \le 1$. For a finite signed measure σ on $\mathcal{C}$ put

$$\|\sigma\|_{\mathrm{BL}^*} = \sup_{F \in \mathcal{F}} |\sigma(F)|.$$

On probability measures this is d_{BL}. On positive finite measures it is exactly the mass. This is not total variation in the path variable. Write $\mathsf{W}_R = \mathrm{Law}(\mathbb{B}_{\Omega_R})$ for the collision bridge law in Section 161.

The original exact-label source is $\mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*$, and its roof density is $p_{n,R}(k, m, u)$. Disintegrate this finite source with respect to its roof. Let $\mathsf{Q}_{m,R}^{n,k,u}$ be the resulting conditional law of $\mathcal{B}_{m,R}$, the collision path tied to its own terminal record in (161.1). For almost every roof value define finite path measures

$$\mathbf{P}_{m,R}^{n,k}(u) = m^2 p_{n,R}(k, m, u) \mathsf{Q}_{m,R}^{n,k,u}, \tag{190.1}$$

$$\mathbf{b}_{B,m,R}^{n,k}(u)(F) = m^2 b_{n,k,m,R}^{\varepsilon(B), F(\mathcal{B}_{m,R})}(u), \qquad \varepsilon(B) = A_0 B^{-1/12}. \tag{190.2}$$

The second measure is obtained by disintegrating the positive source $(1 - H_{m,R}^{\varepsilon(B)}) \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*$ jointly with the first one. Thus $0 \le \mathbf{b}_B(u) \le \mathbf{P}(u)$ as measures, and its mass is $\mathfrak{b}_B(u) = m^2 b_{n,k,m,R}^{\varepsilon(B),1}(u)$. We take both measures to be zero when their scalar density is zero. The source factor $1/c$ is already in p and b and is not inserted again. The scalar reference is $G_{m,R}^{n,k}(u)$ of Section 181.

These disintegrations exist on the standard Borel source and path spaces. Constancy of discrete prefixes in the endpoint argument below is only a statement on one protected word chart, not across a physical or section boundary. A countable class $\mathcal{F}_0 \subset \mathcal{F}$ norms BL^* on all finite measures on $\mathcal{C}$: choose a countable dense set of paths, approximate the admissible values on its finite subsets by rationals, and extend them with the same Lipschitz bound, followed by truncation to $[-1, 1]$. Diagonal approximation gives pointwise convergence to every member of $\mathcal{F}$, and bounded convergence gives the norming assertion. Consequently all roof norms below are measurable, and all estimates can be imposed off one roof-null set for each fixed set of labels.

190.1. **A uniform endpoint budget for the entire path test class.**

Lemma 190.1 (Lipschitz path tests are protected-admissible)**.** *On every regular endpoint chart with the $\varepsilon/2$-margins,*

$$\mathrm{Lip}_{\mathrm{endpoints}}(\mathcal{B}_{m,R}) \le C\varepsilon^{-1} m^{-1/2}. \tag{190.3}$$

Here the image carries the supremum metric of $\mathcal{C}$. In particular, uniformly over $F \in \mathcal{F}$, the insertion $F(\mathcal{B}_{m,R})$ has supremum at most one and weak endpoint gradient at most $C\varepsilon^{-1}m^{-1/2}$. For each fixed margin, the whole class is eventually admissible with the common budget $(M, L) = (1, 1)$ of Theorem 172.2.

Proof. Write the endpoint arclengths as a, b and the internal contact arclengths as $y_j(a, b)$. The first variation of the prefix roof is

$$dS_{j,R} = -p_0\,da + p_j\,dy_j \quad (1 \le j < m), \qquad dS_{m,R} = -p_0\,da + p_m\,db.$$

The internal reflection equations cancel all other contact terms in each prefix. Equations (169.1) and (169.4), in their regular version supplied by Lemma 171.1, give $|dy_j| \le C\varepsilon^{-1}(|da| + |db|)$ uniformly in j, m. Since $|p_j| \le 1$, every prefix roof has endpoint derivative at most $C\varepsilon^{-1}$. On the same chart every displacement prefix and every occupation prefix is constant: the physical word and every section decision persist. Only the roof coordinate of $\mathsf{S}_j - (j/m)\mathsf{S}_m$ can therefore have a nonzero derivative. After division by $\sqrt{m}$ its bound is $C\varepsilon^{-1}m^{-1/2}$. Linear interpolation takes convex combinations of consecutive grid values and does not enlarge this supremum bound. Integrate on smaller convex chart squares to obtain (190.3). Composition with a member of $\mathcal{F}$ is a scalar Lipschitz function on a two-dimensional chart; its almost-everywhere gradient has the stated bound.

This argument differentiates neither a measurable path selector nor a function across a physical or section boundary. For a reconstruction band B the margin $\varepsilon(B)$ remains fixed before m grows. The threshold for the common $(1, 1)$ budget is permitted to depend on that fixed margin. □

190.2. **Fixed-band inversion in the path norm.** For $F \in \mathcal{F}$ let p^F denote the exact-label roof density with insertion $F(\mathcal{B}_{m,R})$. The scalar density is $p = p^1$.

Lemma 190.2 (Uniform bounded-Lipschitz factorization at fixed band)**.** *For every fixed reconstruction band B,*

$$\sup_{R,n,k,u,F\in\mathcal{F}} \left| m^2(K_B * p^F)(u) - \mathsf{W}_R(F)G^{n,k}_{m,R}(u) \right| \longrightarrow 0. \tag{190.4}$$

The band is fixed in this limit. No estimate for its spectral constants at $B = B_m$ is included.

Proof. We first explain why the cylinder factorization extends to the whole class $\mathcal{F}$, rather than merely to one fixed cylinder. The Schwartz function $|K_B|$ has an integrable nonnegative band-limited upper function. For example put

$$q(s) = \left(\frac{\sin s}{s}\right)^2 + \left(\frac{\sin(s + \pi/2)}{s + \pi/2}\right)^2,$$

with the removable values filled in. The two numerators never vanish simultaneously, and $q(s) \ge c_0/(1 + s^2)$: this follows directly for $|s| \ge \pi$ and by positivity and compactness on the remaining interval. With $\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-i\xi s} f(s)\,ds$, one has

$$\widehat{q}(\xi) = \pi(1 - |\xi|/2)_+(1 + e^{i\pi\xi/2}) \in L^1(\mathbb{R}).$$

Its support is $[-2, 2]$, and inversion uses the factor $(2\pi)^{-1}$. Thus $\mathsf{q}^{[B]}(s) = C_B q(s) \ge \max\{|K_B(s)|, |K_B(-s)|\}$ for a finite constant depending on the fixed reconstruction band. The bracketed superscript denotes multiplication by C_B, not a rescaling of the Fourier support. This dependence is not used at a growing band.

At a target (R, n, k, u) consider the signed finite measure on paths

$$\sigma_m(F) = m^2 \int_{\Pi_{n,k,m,R}} F(\mathcal{B}_{m,R})K_B(u - T_{n,R})\,d\nu^*_R.$$

Its variation is dominated by the positive path measure obtained with $\mathsf{q}^{[B]}(T_{n,R}-u)$. The latter has uniformly bounded mass by the fixed-band scalar inversion. The derivative calculation in Proposition 161.3, now with the fixed nonnegative roof function $\mathsf{q}^{[B]}$, gives for its grid increments

$$m^2\int_{\Pi_{n,k,m,R}}\mathsf{q}^{[B]}(T_{n,R}-u)|\mathcal{B}_{m,R}(s)-\mathcal{B}_{m,R}(r)|^4\,d\nu_R^*\le C_B|s-r|^2. \tag{190.5}$$

Indeed its Fourier transform is integrable and compactly supported, so the same fourth derivative of the chronological three-block product is integrated. Centering at each moving peak cancels the linear drift, the size factors are $O(l^2)$ for a block of l collisions, and the four-frequency inverse costs m^{-2}. Division by m^2 from the fourth power of the path normalization, followed by the outside factor m^2, gives $C_B(l/m)^2$. The interpolation argument of that proposition gives (190.5) at all real times. The paths start at zero. The fourth-moment continuity criterion therefore gives uniform tightness of these positive finite path measures; equivalently, add mass at the zero path and divide by a common mass upper bound before applying the probability version. This also proves tightness of the variations of σ_m.

For a fixed grid of observation times and a fixed characteristic cylinder, Lemma 161.1, in its band-limited proof with K_B, gives

$$\sigma_m(\phi)-\sigma_m(1)\mathsf{W}_R(\phi)\longrightarrow 0$$

uniformly in all targets. The exact identities $\sum_i m_i v_i^{(m)}=0$ cancel both the drift and the mixed Gaussian term on every moving resonance chart. The exponentially damped curvature-continuity estimate then replaces the chart Hessian by Ω_R. All occupation resonances are retained. Polygonal observation times rather than their grid approximations change the test by $O(m^{-1/2})$ uniformly, since the one-collision record is bounded; the dominating mass bound pays that error.

To prove uniformity over $\mathcal{F}$, suppose it failed and choose a violating sequence of targets and counts. Pass to a subsequence with $R\to R_0$ and $\sigma_m(1)\to a$. Lemma 189.1, using the variation-tightness just proved, gives a weakly convergent subsequence of the signed finite measures. One can obtain it by applying compactness to the two positive measures $\tau_m+\sigma_m$ and $\tau_m-\sigma_m$, where τ_m is their positive dominating measure and $|\sigma_m|\le\tau_m$ explicitly. Cylinder characteristic functions at rational times determine finite signed measures on continuous path space. The factorization identifies the limit as $a\mathsf{W}_{R_0}$. Continuity of Ω_R gives continuity of W_R. On a compact set carrying all but an arbitrarily small amount of the variations, the restrictions of $\mathcal{F}$ have a finite uniform net by equicontinuity and boundedness. It follows that $\|\sigma_m-\sigma_m(1)\mathsf{W}_R\|_{\mathrm{BL}^*}\to 0$, contradicting the chosen sequence. Finally the scalar fixed-band reconstruction gives $\sup|\sigma_m(1)-G^{n,k}_{m,R}(u)|\to 0$. This proves (190.4). □

190.3. **The path-valued physical remainder.**

Theorem 190.3 (A common positive remainder for all path tests)**.** *For every sufficiently large fixed B,*

$$\limsup_{m\to\infty}\sup_{R,n,k}\operatorname*{ess\,sup}_{u}\left\|\mathbf{P}^{n,k}_{m,R}(u)-G^{n,k}_{m,R}(u)\mathsf{W}_R-\mathbf{b}^{n,k}_{B,m,R}(u)\right\|_{\mathrm{BL}^*}\le CB^{-1/192}. \tag{190.6}$$

The same physical incidence and clearance source is used for every test F; it is not selected anew from the test or the roof value. Consequently, for every fixed $h>0$,

$$\sup_{R,n,k,t}\frac{1}{h}\int_t^{t+h}\left\|\mathbf{P}^{n,k}_{m,R}(u)-G^{n,k}_{m,R}(u)\mathsf{W}_R\right\|_{\mathrm{BL}^*}du\longrightarrow 0. \tag{190.7}$$

Both statements concern the complete original exact-label source.

Proof. For a test $F \in \mathcal{F}$ use the exact source identity $p^F = f^{\varepsilon,F} + d^{\varepsilon,\varepsilon,F} + b^{\varepsilon,F}$. Write $a^F = f^{\varepsilon,F} + d^{\varepsilon,\varepsilon,F}$. Then exactly

$$m^2 p^F - G\mathsf{W}_R(F) - m^2 b^{\varepsilon,F} = m^2(a^F - K_B * a^F) + \{m^2 K_B * p^F - G\mathsf{W}_R(F)\} - m^2 K_B * b^{\varepsilon,F}.$$

Fix B and hence $\varepsilon = \varepsilon(B)$. Lemma 190.1 makes every insertion in this test class eventually protected-admissible with budget $(1,1)$. The protected correction is therefore $O(B^{-1/2})$ in the normalized essential norm. The complete decision correction is $O(\varepsilon^{1/16})$ in the same norm by bounded-source domination. The middle term tends to zero uniformly over the full test class by Lemma 190.2. The last term is at most $C\varepsilon^{1/16}\|K_1\|_1$ after the collision limsup, by Lemma 181.3, uniformly over these bounded insertions. Thus the limsup of the displayed error is at most $CB^{-1/192}$, uniformly over F.

Take the countable norming class $\mathcal{F}_0$ to make the exceptional roof set common to all tests. This converts the uniform scalar estimate into the essential norm in (190.6). No derivative of a roof-dependent choice of test is taken. The positive measure $\mathbf{b}_B(u)$ has norm equal to its mass. Hence integration over a translated interval and (179.4) give a limsup at most $CB^{-1/192} + C\varepsilon(B)^{1/16}$ in (190.7). Let B tend to infinity only after this collision limsup. This proves the second assertion. The local window and the auxiliary spectral band used for its physical estimate have precisely the order of limits in Theorem 179.3. □

Corollary 190.4 (Roof-dependent Lipschitz path selectors)**.** *The local bound* (190.7) *controls, uniformly, all jointly Borel functions $\Psi(u,x)$ supported in a translated fixed roof interval for which $\Psi(u,\cdot) \in \mathcal{F}$ for every u. The selector may depend on m, R, n, k and may be chosen after u. There is no regularity condition in u. This assertion is not a Gaussian-amplitude theorem for arbitrary measurable path selectors.*

Proof. At almost every u, evaluation against $\Psi(u,\cdot)$ is bounded by the path dual norm in (190.7). Integrate that bound. The countable norming construction, or bounded measurable approximation on the standard Borel product, gives the same conclusion for a measurable selector. The proof first bounds the whole test class at each roof; it does not differentiate $\Psi(T_{n,R}, \mathcal{B}_{m,R})$ as a source insertion. □

191. Conditional path laws at the same roof value

We now pass from a path-valued raw numerator to a conditional path kernel at its own roof value. Convergence is first proved in mean with respect to the unchanged conditional roof law. A second result shows that the missing scalar positive-height estimate would also supply the uniform pointwise path bridge, without a separate uncontrolled bounded-Lipschitz numerator.

Fix $h, d > 0$ and $M_0 < \infty$. Put $I_t = [t, t+h]$ and retain the central coordinate $Z(t)$ of Section 181. Our uniform target class is

$$|Z(t)| \le M_0, \qquad H_m := \int_{I_t} (G^{n,k}_{m,R}(u))_+ \, du \ge dh. \tag{191.1}$$

Only an integrated lower bound is required here. Let $F_m = \int_{I_t} P^{n,k}_{m,R}(u)\, du$. The scalar local-variation law gives $F_m \ge dh/2$ eventually. The original roof law under $E = \{K_{n,R} = k, N_{n,R} = m, T_{n,R} \in I_t\}$ and its reference are

$$\pi_m(du) = \frac{\mathbf{1}_{I_t}(u) P^{n,k}_{m,R}(u)}{F_m}\, du, \qquad \widehat{\pi}_m(du) = \frac{\mathbf{1}_{I_t}(u)(G^{n,k}_{m,R}(u))_+}{H_m}\, du.$$

Their probability total-variation distance is one half the variation mass and tends uniformly to zero by Corollary 180.4.

191.1. **Return-clock transfer in the roof-integrated path norm.** Let $\mathsf{R}^{n,k,u}_{m,R}$ denote the conditional law of the actual return bridge $\mathcal{Y}_{n,R}$ under the same source disintegration used to define $\mathsf{Q}^{n,k,u}_{m,R}$. Its reference is $\mathsf{V}_R = \mathrm{Law}(\mathbb{B}_{D_R})$.

Lemma 191.1 (Integrated transfer of the pinned clock)**.** *For every fixed h, M_0, uniformly over $|Z(t)| \le M_0$, without requiring a positive denominator,*

$$\frac{1}{h}\int_{I_t} \left\| P^{n,k}_{m,R}(u)\mathsf{R}^{n,k,u}_{m,R} - G^{n,k}_{m,R}(u)\mathsf{V}_R \right\|_{\mathrm{BL}^*} du \longrightarrow 0. \tag{191.2}$$

Proof. On E one has $n/m = c + O_{M_0}(m^{-1/2})$. Put $\mathsf{A}_{m,R} = \sqrt{m/n}\, L_R^{-1}$, where L_R is the covariance change matrix in (151.7). These matrices are uniformly bounded, and $(\mathsf{A}_{m,R})_*\mathsf{W}_R \to \mathsf{V}_R$ uniformly in bounded-Lipschitz distance, by covariance continuity and $m/n \to c^{-1}$.

Consider the finite, unnormalized source measure $\lambda_m = (m^2/h)\mathbf{1}_E \nu_R^*$. Its mass is bounded uniformly by the scalar fixed-window upper bound. The unnormalized fourth moment (161.9) gives a uniformly tight family of collision-path measures under λ_m, just as in (190.5). This argument is valid when the mass tends to zero: add mass at the zero path and normalize by a common upper bound. No rare-event denominator is introduced.

At return l put $\theta_l = N_{l,R}/m$. The exact half-open occupation convention gives $A_{N_{l,R},R} = l$ and $A_{m,R} = n$. The fourth collision-bridge coordinate consequently satisfies

$$\theta_l - l/n = -\frac{\sqrt{m}}{n}(\mathcal{B}_{m,R})_4(\theta_l). \tag{191.3}$$

Tightness makes the maximum of these displacements tend to zero in λ_m-measure. The centered terminal vector $\mathcal{X}_{m,R}(1)$ is bounded on E, because the actual roof lies in I_t and $|Z(t)| \le M_0$; explicitly, $|Z(u)| \le M_0 + h/\sqrt{m}$ on that interval. The exact pinned time-change identity (161.14), followed by the uniform modulus of continuity of a tight family of continuous paths, now gives

$$\int \min\{2, \|\mathcal{Y}_{n,R} - \mathsf{A}_{m,R}\mathcal{B}_{m,R}\|_\infty\}\, d\lambda_m \to 0. \tag{191.4}$$

Linear interpolation on the return grid is paid by the same modulus. Neither a changed roof event nor a change of return index occurs.

Use the joint source disintegration to couple the two conditional paths at each individual roof. Integrating their bounded-Lipschitz distance, with weight P/h, is bounded by (191.4). Under a linear map A, the path dual norm grows by at most $\max\{1, \|A\|\}$. Apply this to (190.7). Finally G is uniformly bounded and the reference covariance convergence pays $G[(\mathsf{A}_{m,R})_*\mathsf{W}_R - \mathsf{V}_R]$ in the integrated norm. These three estimates prove (191.2). □

Theorem 191.2 (A same-roof bridge in conditional mean)**.** *Uniformly on the class* (191.1)*,*

$$\int_{I_t} d_{\mathrm{BL}}(\mathsf{Q}^{n,k,u}_{m,R}, \mathsf{W}_R)\, \pi_m(du) \to 0, \tag{191.5}$$

$$\int_{I_t} d_{\mathrm{BL}}(\mathsf{R}^{n,k,u}_{m,R}, \mathsf{V}_R)\, \pi_m(du) \to 0. \tag{191.6}$$

Both conclusions also hold with $\widehat{\pi}_m$ in place of π_m. The conditional kernels fix the original exact n, k, m and their own roof value u. They are not the path law under a substituted window. No polynomial rate in m is asserted.

Proof. For a nonnegative scalar P, a real scalar G, and probabilities Q, W, the constant test one belongs to $\mathcal{F}$, so

$$P\|Q - W\|_{\mathrm{BL}^*} \le \|PQ - GW\|_{\mathrm{BL}^*} + |P - G| \le 2\|PQ - GW\|_{\mathrm{BL}^*}.$$

Integrate this inequality over I_t and divide by $F_m \geq dh/2$. Theorem 190.3 proves the collision assertion; Lemma 191.1 proves the return assertion. The integrands are measurable and bounded by two. Replacing the roof measure by $\widehat{\pi}_m$ changes their integrals by at most four times the probability total-variation distance, which tends to zero. This uses the actual path numerators, not merely the scalar roof approximation. □

Corollary 191.3 (Asymptotically full sets of roof values)**.** *There is a deterministic sequence $a_m \downarrow 0$, depending on h, d, M_0, such that outside target-dependent Borel sets of both π_m- and $\widehat{\pi}_m$-mass tending uniformly to zero, the sum of the two distances in* (191.5)–(191.6) *is at most a_m. The sequence is qualitative, not a prescribed shrinking roof resolution. No assertion is made at every externally chosen roof value.*

Proof. Let γ_m be the supremum over targets of the two expectations under both roof laws. It tends to zero. Replace it by its decreasing tail supremum and take, for example, $a_m = \sqrt{\sup_{j\geq m} \gamma_j} + m^{-1}$. Markov's inequality bounds each exceptional mass by $\gamma_m/a_m \to 0$. Kernel versions can first be fixed on the standard Borel spaces; all source-null modifications are immaterial. □

191.2. **The scalar height condition also controls the path numerator.** On the set $\{G^{n,k}_{m,R} \geq d\}$ the lower law ensures $P \geq d/2$ eventually. There the exact physical-defect posterior is

$$\beta^{n,k}_{B,m,R}(u) = \frac{\mathfrak{b}^{n,k}_{B,m,R}(u)}{P^{n,k}_{m,R}(u)} = \mathbb{E}_{\nu^*_R}[1 - H^{\varepsilon(B)}_{m,R} \mid \Pi_{n,k,m,R}, T_{n,R} = u] \in [0,1]. \tag{191.7}$$

This is a regular conditional expectation, not a positive probability for the equality $T_{n,R} = u$.

Proposition 191.4 (Pointwise path error charged to the same physical source)**.** *For every fixed sufficiently large B,*

$$\limsup_{m\to\infty} \sup_{R,n,k} \operatorname*{ess\,sup}_{u: G(u)\geq d} \left(d_{\mathrm{BL}}(\mathsf{Q}^{n,k,u}_{m,R}, \mathsf{W}_R) - 2\beta^{n,k}_{B,m,R}(u)\right)_+ \leq C_d B^{-1/192}. \tag{191.8}$$

On a fixed central compact set, the scalar positive-height condition (181.9) *therefore implies*

$$\sup_{R,n,k} \operatorname*{ess\,sup}_{u:\ |Z(u)|\leq M_0\,,\ G(u)\geq d} d_{\mathrm{BL}}(\mathsf{R}^{n,k,u}_{m,R}, \mathsf{V}_R) \longrightarrow 0. \tag{191.9}$$

It also implies the corresponding collision-bridge statement. This is an implication from the still-unproved height condition, not an unconditional uniform pointwise bridge theorem.

Proof. Put $\mathsf{E} = \mathbf{P} - G\mathsf{W}_R - \mathbf{b}_B$. Taking masses gives $P - G - \mathfrak{b}_B = \mathsf{E}(1)$. Consequently

$$P(\mathsf{Q} - \mathsf{W}_R) = \mathbf{b}_B - \mathfrak{b}_B\mathsf{W}_R + \mathsf{E} - \mathsf{E}(1)\mathsf{W}_R.$$

Its dual norm is at most $2\mathfrak{b}_B + 2\|\mathsf{E}\|_{\mathrm{BL}^*}$. Divide by $P \geq d/2$ and apply Theorem 190.3. This proves (191.8). If (181.9) holds, the same division shows that the conditional collision-bridge error tends uniformly to zero on the displayed positive-reference central sets, taking the collision limit before the reconstruction band tends to infinity.

For the return conclusion take any hypothetical violating sequence of such targets and nonexceptional roof values. After a subsequence $R \to R_0$. The just-proved conditional collision-path convergence gives tightness for this sequence of conditional laws. The exact identity (191.3) and the bounded central terminal vector give the same pinned clock comparison as in Lemma 191.1, now in those individual conditional laws. The deterministic linear covariance change tends to D_{R_0}. Thus the conditional return laws converge to V_{R_0}, contradicting the sequence. This proves (191.9) in essential supremum. The new numerator estimate removes the need for a separate bounded- Lipschitz path remainder condition; it does not prove the remaining scalar incidence or clearance height bounds. □

The remaining pointwise endpoint. Theorems 190.3 and 191.2 give a path-valued local density law and same-roof conditional convergence in roof probability on the full original source. Their path topology is weak, expressed by bounded-Lipschitz distance; it is not path-space total variation. The scalar two-sided essential-height theorem still requires (181.9). Its positive incidence and clearance components have not been shown small in height here. The arithmetic transition kernel remains in every uniform assertion; no residue is set equal to one. The fixed-radius positive-class specialization uses precisely the unchanged arithmetic lower bounds already established in Section 182.

192. Exponential path integrability under the exact microscopic event

The fourth-moment argument can be strengthened, but only if its constants are kept explicit in the differentiation order. We do so on fixed complex neighborhoods of the collision spectrum. This yields an exponential moment of the supremum of the entire pinned path, not an exponential moment of the roof density and not a high-frequency estimate.

192.1. **Cauchy bounds of every order.**

Lemma 192.1 (Factorial bounds at the moving peaks)**.** *On a fixed spectral band, reduce the finitely many peak charts once. There are $C, D, a > 0$ and $\rho < 1$, independent of the derivative order $q \geq 0$, the collision block length $L \geq 1$, the radius and the peak, such that, for every real unit vector e,*

$$\left\| \frac{d^q}{dz^q} \left(e^{-ize\cdot\mu_j(R)} \mathscr{T}_{R,\theta+ze} \right)^L \Big|_{z=0} \right\| \leq Cq! D^q L^{q/2} \left(e^{-aL|\theta-a_j(R)|^2} + \rho^L \right). \tag{192.1}$$

On the compact nonresonant complement the uncentered derivative is bounded by $Cq!D^q\rho^L$. At $L = 0$ the derivative is zero for $q > 0$ and the operator is the identity for $q = 0$.

Proof. The proof of Lemma 161.2 provides more than its stated fourth-order estimate. On the complex circle $|z| = d_0/\sqrt{L}$ it bounds the centered principal power by $C\exp(-aL|\theta - a_j|^2)$ and the complementary power by $C\rho_0^L \exp(C\sqrt{L}) \leq C'\rho^L$, with $\rho_0 < \rho < 1$. The projection is bounded on the same circle. The circle, the analytic neighborhood and these upper bounds do not depend on q. Cauchy's formula therefore gives exactly the factor $q!(\sqrt{L}/d_0)^q$. Taking $D \geq d_0^{-1}$ proves (192.1). A fixed complex circle on the nonresonant complement gives its asserted bound in the same way. The finitely many charts and local parameter spaces permit common constants. No claim is made about their growth with the fixed band. □

Fix a bounded roof interval J of positive length and the original event $E = E_{n,m,R}(k, t; J)$ in (161.3). The vector S_j keeps the displacement, roof and occupation order of Section 161.

Proposition 192.2 (All pinned increment moments with factorial constants)**.** *There is $C_J < \infty$, independent of the integer $r \geq 1$, such that for $0 \leq a < a + l \leq m$,*

$$\int_E \left| \mathsf{S}_{a+l,R} - \mathsf{S}_{a,R} - \frac{l}{m} \mathsf{S}_{m,R} \right|^{2r} d\nu_R^* \leq (2r)! C_J^{2r} l^r m^{-2}. \tag{192.2}$$

The estimate is uniform over all original exact labels, roof targets and radii. Neither J nor an auxiliary spectral band depends on m or on r.

Proof. Majorize $\mathbf{1}_J$ by one fixed nonnegative band-limited function with integrable Fourier transform, as in Proposition 161.3. This increases the positive moment integrand. Put $\alpha = l/m$

and $b = m - a - l$. In one coordinate, $(-1)^r$ times the derivative of order $2r$ at zero of

$$\ell\left(M_{\eta_R}\mathscr{T}^{b}_{R,\theta-\alpha z e}\mathscr{T}^{l}_{R,\theta+(1-\alpha)z e}\mathscr{T}^{a}_{R,\theta-\alpha z e}(\eta_R\nu)\right)$$

is the required Fourier-transformed moment. Differentiation under the finite-time integral is justified for every fixed order by boundedness of the single-collision record. We bound its absolute value.

On each moving peak, the centering factors cancel identically since $-\alpha(a+b)+(1-\alpha)l = 0$. By Lemma 192.1, Leibniz's formula has total size at most

$$C(2r)!D^{2r}\{\alpha(\sqrt{a}+\sqrt{b})+(1-\alpha)\sqrt{l}\}^{2r}\left(e^{-a_1 m|\theta-a_j|^2}+\rho^{m/3}\right).$$

To see the factorial dependence, the multinomial coefficient cancels the three derivative factorials, leaving $(2r)!$ times the sum of monomials in the three displayed size factors. That sum is at most their sum raised to $2r$. In every spectral term at least one block has length at least $m/3$; its Gaussian or complementary factor gives the final parenthesis. The other two blocks have bounded decay factors, including the zero-length convention. Since $\alpha(\sqrt{a}+\sqrt{b}) \le \sqrt{2}\,l/\sqrt{m} \le \sqrt{2l}$, the size is at most $(2r)!C^{2r}l^r$ times that parenthesis. The Gaussian integral is $O(m^{-2})$ in the four collision frequencies.

On the compact nonresonant part, use a single fixed complex disk for the entire three-block product. All its powers remain exponentially contracting there; its norm is at most $C\rho^m$. Cauchy's formula gives $C(2r)!D^{2r}\rho^m$, bounded by $(2r)!C_J^{2r}l^r m^{-2}$ for all $l \ge 1$. Integrate against the fixed roof majorant's Fourier transform and sum the finite charts. Passing from coordinates to the Euclidean norm costs only a factor of the form C^{2r}. The original source factor c^{-1} is fixed and is absorbed in C_J, not reintroduced as a changed probability. □

192.2. **A maximal estimate before conditioning.** For $I_t = [t, t+h]$, $h > 0$, write

$$\lambda^{n,k,t}_{m,R} = (m^2/h)\mathbf{1}_E\nu^*_R, \qquad E = \{K_{n,R} = k, N_{n,R} = m, T_{n,R} \in I_t\}.$$

Its total mass is bounded uniformly, by the fixed-window local upper bound, even on label sequences for which the mass tends to zero.

Theorem 192.3 (Exponential supremum moments of the complete paths)**.** *For every fixed $h > 0$ there are $a_h, C_h > 0$ such that*

$$\sup_{m\ge 2,R,n,k,t}\int \exp(a_h\|\mathcal{B}_{m,R}\|_\infty)\,d\lambda^{n,k,t}_{m,R} \le C_h. \tag{192.3}$$

For every fixed $M_0 < \infty$ there are $a_{h,M_0}, C_{h,M_0} > 0$ such that, on $|Z(t)| \le M_0$ and for all sufficiently large m,

$$\int \exp(a_{h,M_0}(\|\mathcal{B}_{m,R}\|_\infty + \|\mathcal{Y}_{n,R}\|_\infty))\,d\lambda^{n,k,t}_{m,R} \le C_{h,M_0}. \tag{192.4}$$

No physical or section guard is imposed in these integrals.

Proof. Normalize (192.2) by the path scale and by m^2/h. Interpolation gives, for all real $s, t \in [0,1]$ and all even $q \ge 2$,

$$\int |\mathcal{B}_{m,R}(t) - \mathcal{B}_{m,R}(s)|^q\,d\lambda \le q!C_h^q|t-s|^{q/2}.$$

Within one grid interval the fractional interpolation factor is of order $|t-s|^q m^q$, which gives this bound. For a general increment split off its two fractional end pieces and the intervening grid interval; Minkowski's inequality changes C_h, independently of q.

For even $q \geq 4$, the $L^q(\lambda)$ norm of the maximum of the 2^j dyadic increments of length 2^{-j} is at most

$$(q!)^{1/q} C_h 2^{-j(1/2-1/q)}.$$

Every continuous path starting at zero is bounded in supremum by the sum of these maxima over dyadic levels. Minkowski's inequality and $1/2 - 1/q \geq 1/4$ therefore yield

$$\int \|\mathcal{B}_{m,R}\|_\infty^q \, d\lambda \leq q! C_h'^q, \qquad q \geq 4.$$

The finite mass bound pays the smaller orders. Cauchy–Schwarz between consecutive even moments pays odd orders; the ratio $\sqrt{(q-1)!(q+1)!}/q! = \sqrt{(q+1)/q}$ is bounded. After one enlargement the factorial bound holds at every integer order. Expanding the exponential with $a_h C_h' < 1$ proves (192.3). This is an argument for finite measures and does not divide by a rare-event probability.

On the central event, $n/m = c + O_{M_0}(m^{-1/2})$ and $|\mathcal{X}_{m,R}(1)| \leq M_0 + h/\sqrt{m}$. The exact identities (161.14) and (191.3) give, at every return grid point l/n,

$$|\mathcal{Y}_{n,R}(l/n)| \leq \|\sqrt{m/n} L_R^{-1}\| \left(1 + \frac{\sqrt{m}}{n} |\mathcal{X}_{m,R}(1)|\right) \|\mathcal{B}_{m,R}\|_\infty.$$

The coefficient is uniformly bounded for large m. Linear interpolation on the return grid cannot enlarge the maximum. Reducing a_{h,M_0} in (192.3) proves (192.4). Thus the time change preserves a quantitative moment budget on the unchanged source. □

Corollary 192.4 (Exponentially weighted physical-defect mass)**.** *On fixed central windows there are $a, C > 0$ such that, for every fixed sufficiently small $\varepsilon > 0$,*

$$\limsup_m \sup_{R,n,k,t:\ |Z(t)| \leq M_0} \frac{m^2}{h} \int_E (1 - H_{m,R}^\varepsilon) e^{a(\|\mathcal{B}_{m,R}\|_\infty + \|\mathcal{Y}_{n,R}\|_\infty)} d\nu_R^* \leq C\varepsilon^{1/32}. \tag{192.5}$$

The same estimate holds for either disjoint first-physical-defect type. It concerns the original positive incidence and clearance sources, without transporting them across a physical boundary.

Proof. Use Cauchy–Schwarz in λ, noting that $(1 - H^\varepsilon)^2 \leq 1 - H^\varepsilon$. The first factor is bounded by $C\varepsilon^{1/32}$ after the collision limsup, by (179.4). Choose $2a$ below the exponential constant in (192.4) to bound the second factor. Each positive first-defect subsource is dominated by the full physical remainder, proving its assertion. This remains an integrated estimate; it supplies no essential-height estimate in u. □

193. Unbounded path observables and same-roof transport

We strengthen the path topology without changing the roof conditioning. The new exponential moment pays the tails of unbounded path tests. It does not bound the height of a density in the roof coordinate.

For a path x put $r(x) = \|x\|_\infty$. For $p \geq 1$ let $\mathcal{F}_p$ be the continuous real functions on path space with

$$|F(x)| \leq 1 + r(x)^p, \qquad |F(x) - F(y)| \leq (1 + r(x)^{p-1} + r(y)^{p-1}) \|x - y\|_\infty.$$

When $p = 1$ take $r^0 = 1$. Write $\|\sigma\|_{p,*} = \sup_{F \in \mathcal{F}_p} |\sigma(F)|$ when the variation has a finite pth moment. This is a weighted test norm, not path-space total variation. A countable norming subclass is obtained from the pointwise topology on a fixed countable dense set of paths: the displayed value and difference bounds form a separable metrizable subspace of a countable product. Continuity extends its dense approximants to all paths, and the growth bound permits dominated convergence. Thus the kernel norms used below are measurable and can be assigned a common roof-null set, as for BL*.

Lemma 193.1 (Exponential tails upgrade a local test norm)**.** *Let σ_u be a measurable signed-measure kernel and let π be a probability or a finite measure of uniformly bounded mass. Suppose*

$$e = \int \|\sigma_u\|_{\mathrm{BL}^*}\, d\pi(u), \qquad \int\int e^{ar(x)}\, d|\sigma_u|(x)\, d\pi(u) \le M$$

for fixed $a, M > 0$. For $0 < e \le 1/2$ and each fixed $p \ge 1$,

$$\int \|\sigma_u\|_{p,*}\, d\pi(u) \le C_{p,a,M}\, e(1 + |\log e|^p). \tag{193.1}$$

At $e = 0$ the left side is zero. The same estimate holds for jointly Borel functions $\Psi(u, x)$ whose path sections belong to $\mathcal{F}_p$, including a selection made after u is observed.

Proof. Choose a Lipschitz cutoff $\chi_R(x)$ which equals one on $r \le R$, vanishes on $r \ge 2R$ and has Lipschitz constant at most R^{-1}, where $R \ge 1$. For every $F \in \mathcal{F}_p$, the product $F\chi_R$ has bounded-Lipschitz norm at most $C_p R^p$, uniformly in F. The difference has absolute value at most $(1 + r^p)\mathbf{1}_{r>R} \le C_{p,a} e^{-aR/2} e^{ar}$. Consequently the left side of (193.1) is at most $C_p R^p e + C_{p,a} M e^{-aR/2}$. Take $R = \max(1, 2a^{-1}\log(1/e))$. This proves the estimate, without choosing a test before the roof value. For $e = 0$, the signed measure is zero almost everywhere, since bounded-Lipschitz tests determine finite measures. Joint Borel measurability and the countable norming class justify evaluation and integration for the stated selectors. □

Theorem 193.2 (Polynomial-growth path-valued raw local laws)**.** *For each fixed $h, M_0 > 0$ and $p \ge 1$, uniformly on $|Z(t)| \le M_0$,*

$$\frac{1}{h}\int_t^{t+h} \|P^{n,k}_{m,R}(u)\mathsf{Q}^{n,k,u}_{m,R} - G^{n,k}_{m,R}(u)\mathsf{W}_R\|_{p,*}\, du \to 0, \tag{193.2}$$

$$\frac{1}{h}\int_t^{t+h} \|P^{n,k}_{m,R}(u)\mathsf{R}^{n,k,u}_{m,R} - G^{n,k}_{m,R}(u)\mathsf{V}_R\|_{p,*}\, du \to 0. \tag{193.3}$$

No positive denominator is needed. Both conclusions are uniform for jointly Borel roof-dependent selectors in $\mathcal{F}_p$.

Proof. The bounded-Lipschitz errors tend uniformly to zero by (190.7) and (191.2). For each difference measure, its variation is bounded by the sum of its two positive terms, using $|G|$ for the reference term. The first term has uniformly bounded exponential moment on the translated roof interval by Theorem 192.3. The second does also: G is uniformly bounded, the covariance families are bounded, and Gaussian bridges have a common exponential supremum moment. For the last assertion one may repeat the dyadic argument above using the Gaussian increment moments, whose factorial bounds are uniform on a bounded covariance family. Apply Lemma 193.1 with du/h restricted to the window. Its constants are uniform in all the indicated targets.

The central enlargement is explicit: for $u \in [t, t+h]$, $|Z(u)| \le M_0 + h/\sqrt{m}$. This pays the return-clock estimate with one fixed enlarged compact set. The raw reference is the signed G, not G_+; no conditional probability has been introduced. □

193.1. **A transport estimate for conditional kernels.** For probabilities Q, W on path space define

$$W_p(Q, W)^p = \inf_{\gamma \in \mathrm{Cpl}(Q,W)} \int \|x - y\|^p_\infty\, d\gamma(x, y), \qquad p \ge 1.$$

This distance uses the original, unbounded supremum metric. We use only bounded-cost Kantorovich duality from [19, Chapters 5–6], and give the tail estimate explicitly.

Lemma 193.3 (From mean bounded-Lipschitz error to mean transport cost)**.** *Let Q_u, W_u be Borel probability kernels, let π be a probability, and suppose*

$$\int \{Q_u(e^{ar}) + W_u(e^{ar})\}\, d\pi(u) \le M.$$

Put $e = \int d_{\mathrm{BL}}(Q_u, W_u)\, d\pi(u)$. For every fixed finite $p \ge 1$ and $0 < e \le 1/2$,

$$\int W_p(Q_u, W_u)^p\, d\pi(u) \le C_{p,a,M}\, e(1 + |\log e|^p). \tag{193.4}$$

The left side is zero when $e = 0$.

Proof. The bounded metric $\bar{d}(x, y) = \min(\|x - y\|_\infty, 1)$ has transport cost at most $d_{\mathrm{BL}}(Q_u, W_u)$. Indeed every 1-Lipschitz test for $\bar{d}$ has oscillation at most one; after subtracting a constant it is in the bounded-Lipschitz class used here. Bounded-cost duality yields the assertion.

Take an arbitrarily accurate coupling for this bounded cost. On $\|x - y\|_\infty \le R$, $R \ge 1$, the pth power of the distance is bounded by $R^p \bar{d}(x, y)$. On its complement,

$$\|x - y\|_\infty^p \mathbf{1}_{\|x-y\|_\infty > R} \le C_{p,a} e^{-aR/4} e^{a\|x-y\|_\infty/2} \le \tfrac{1}{2} C_{p,a} e^{-aR/4} (e^{ar(x)} + e^{ar(y)}).$$

The second inequality uses the triangle inequality and the arithmetic– geometric mean inequality. Taking the infimum for the unbounded cost and then removing the bounded-cost approximation gives a pointwise upper bound. Integrate it in u and choose $R = \max(1, 4a^{-1} \log(1/e))$. This proves (193.4). No measurable choice of couplings is needed: the pointwise inequality is integrated after taking its infima. Wasserstein costs are Borel for Borel probability kernels on a Polish space, with value infinity allowed before the moment hypothesis is imposed. If $e = 0$, the two kernels agree almost everywhere and the conclusion follows directly. □

Theorem 193.4 (Same-roof bridges in every finite transport order)**.** *Fix the target class* (191.1)*. For every finite $p \ge 1$, uniformly in that class,*

$$\int_{I_t} W_p(\mathsf{Q}_{m,R}^{n,k,u}, \mathsf{W}_R)^p\, \pi_m(du) \to 0, \tag{193.5}$$

$$\int_{I_t} W_p(\mathsf{R}_{m,R}^{n,k,u}, \mathsf{V}_R)^p\, \pi_m(du) \to 0. \tag{193.6}$$

The original kernels retain the exact labels and their own roof value. Their exponential path moments are uniformly bounded in mean under π_m. If the stronger condition $G(u) \ge d_0 > 0$ holds almost everywhere on I_t, both conclusions also hold under $\widehat{\pi}_m$. No such reference-law transport assertion is made under the integrated reference lower bound alone.

Proof. The true event has $F_m = \int_{I_t} P \ge dh/2$ eventually. Dividing (192.4) by F_m/h gives the uniform exponential moment of the conditional path mixture under π_m, and hence of the same-roof kernels in mean. Gaussian bridge exponential moments are uniform as above. The two mean bounded-Lipschitz errors tend uniformly to zero by Theorem 191.2. Apply Lemma 193.3 separately to the two kernels. It supplies the explicit modulus in the mean bounded-Lipschitz error, but not a rate in m.

Under the additional pointwise reference floor, Theorem 182.2 gives $\pi_m \ge \alpha_m \widehat{\pi}_m$, $\alpha_m \to 1$. Thus the reference means of the nonnegative transport costs are at most α_m^{-1} times their true-roof means. This proves the last assertion without using total variation to integrate an unbounded function. Under an integrated lower bound alone the roof TV comparison only transfers bounded costs; we do not use it for these transport costs. The extra floor also ensures that kernel versions agree up to null sets for both roof measures. □

Corollary 193.5 (Conditional covariance kernels and path maxima)**.** *Let (Q^u, W, C_R) denote either $(\mathsf{Q}^u, \mathsf{W}_R, \Omega_R)$ or $(\mathsf{R}^u, \mathsf{V}_R, D_R)$, and put $\mu^u(s) = \int x(s)\, dQ^u(x)$. On the preceding target class,*

$$\int \sup_s |\mu^u(s)|^2 \, d\pi_m(u) \to 0,$$

$$\int \sup_{s,t} \|\mathrm{Cov}_{Q^u}(x(s), x(t)) - (s \wedge t - st) C_R\| \, d\pi_m(u) \to 0.$$

For every finite $p \geq 1$,

$$\int |Q^u(r^p) - W(r^p)| \, d\pi_m(u) \to 0.$$

These conclusions may also be read after jointly Borel roof-dependent choices of observation times. Under the pointwise reference floor of Theorem 193.4, they hold under $\widehat{\pi}_m$ as well.

Proof. A path coupling bounds $\sup_s |\mu^u(s)|$ by its mean path distance, and hence by $W_2(Q^u, W)$. The difference of second moment kernels, uniformly in both times, is bounded by the coupling expectation of $(r(x) + r(y))\|x - y\|_\infty$. Cauchy–Schwarz first in the coupling and then in u, the uniform second moments, and (193.5)–(193.6) prove its convergence. Subtracting $\mu^u(s) \otimes \mu^u(t)$ proves the covariance assertion. For r^p use the same coupling argument with a higher finite moment, or truncate it as in Lemma 193.1. Its growth and local Lipschitz bounds are uniform in the roof. The supremum over times already precedes roof integration, so a jointly Borel time selection costs nothing further. Domination transfers these nonnegative errors to the reference law under the stated extra floor. □

The unresolved scalar height is unchanged. All the new bounds retain the actual incidence and clearance source. They control large path amplitudes in the original microscopic event, not concentration of its scalar mass in a narrow roof set. Neither exponential path integrability nor any of (193.2)–(193.6) bounds $\|\mathfrak{b}_{B,m}\|_\infty$. The two-sided pointwise raw theorem and the unconditional uniform pointwise bridge therefore still require (181.9). The arithmetic transition kernel and its fixed-radius residue remain in every raw main term. No forward roof likelihood, entropy estimate, path-space total variation, or assertion at every externally prescribed roof is inferred here.

194. Complete physical phase arithmetic

The roof and constant parts of a physical collision phase have already been determined in Theorem 149.4. We now determine its two spatial parts. The argument uses an atom of the phase on a positive-measure product set, not a value at a periodic point. It applies to arbitrary real spatial frequencies, including irrational ones.

Write

$$\Omega = \mathbb{T}^2 \times \mathbb{T} \times \mathbb{R}, \qquad \phi_{R,\omega} = u \cdot \kappa_R + s + t\tau_R, \quad \omega = (u, s, t).$$

The three torus coordinates have period 2π. The origin of Ω therefore means $u \in 2\pi\mathbb{Z}^2$, $s \in 2\pi\mathbb{Z}$, $t = 0$.

Lemma 194.1 (Separation of a proper integer subgroup)**.** *If L is a proper subgroup of $\mathbb{Z}^d$, then there are a prime p and a nonzero $a \in (\mathbb{Z}/p\mathbb{Z})^d$ with $a \cdot L = 0$ modulo p.*

Proof. An integer subgroup has an integer basis. If its rank is less than d, choose a nonzero integer row annihilating it and a prime not dividing all coordinates of that row. If it has full rank, its index is the absolute determinant of a basis matrix. This integer exceeds one. Modulo a prime divisor of the index, the basis matrix is singular and has a nonzero left null vector. These

two constructions prove the claim without a rationality assumption on the character that might have defined L. □

Theorem 194.2 (Complete measurable physical phase rigidity)**.** *For every $R \in I$, a measurable $q : M \to \mathbb{S}^1$ satisfies*

$$q \circ T_R = e^{i\phi_{R,\omega}} q \tag{194.1}$$

if and only if $\omega = 0$ in Ω and q is constant almost everywhere.

Proof. Theorem 149.4 gives $t = 0$ and $s \in 2\pi\mathbb{Z}$. Both holonomy formulas (149.8) and (149.9) consequently have right side one. In the product coordinates of Lemma 149.1, the first says that q is almost everywhere a function only of the unstable transversal coordinate, and the second that it is almost everywhere a function only of the stable transversal coordinate. Product measure equivalence and Fubini imply that $q = c$ almost everywhere on $\mathcal{P}$, for one $c \in \mathbb{S}^1$.

The saturation $\bigcup_{j\geq 0} T_R^j \mathcal{P}$ has full measure by ergodicity and recurrence. Iterating (194.1) from $\mathcal{P}$ therefore shows that

$$q(x)/c \in \{e^{iu\cdot k} : k \in \mathbb{Z}^2\} \quad \text{for almost every } x. \tag{194.2}$$

Let $L = \{k \in \mathbb{Z}^2 : u \cdot k \in 2\pi\mathbb{Z}\}$. The map $[k] \mapsto e^{iu\cdot k}$ identifies the countable group $\mathbb{Z}^2/L$ with the displayed range. Its inverse on that countable range is measurable. Thus (194.2) defines a measurable $h : M \to \mathbb{Z}^2/L$ with

$$h(T_R x) = h(x) + [\kappa_R(x)]. \tag{194.3}$$

This is a countable-valued lift, not a real logarithm with a chosen branch. No integrability of the lift is required.

If $L \neq \mathbb{Z}^2$, apply Lemma 194.1. The resulting nonzero row a defines a homomorphism $\chi : \mathbb{Z}^2/L \to \mathbb{Z}/p\mathbb{Z}$. On the actual physical billiard modulo $p\Lambda$ the collision map is

$$\widetilde{T}_R(x, \ell) = (T_R x, \ell + \kappa_R(x) \bmod p), \qquad \ell \in (\mathbb{Z}/p\mathbb{Z})^2.$$

It preserves ν times the uniform sheet probability and is ergodic. Indeed its lift to the plane has exactly the original disk array, so its free-flight bound is unchanged. Its finitely many disjoint circular scatterers have positive curvature, and the complement of these disks in the torus is connected. Thus the finite-horizon billiard theorem [6, Section 8.1] applies. Equivalently use the rectangular Euclidean cover with periods $(p, 0)$ and $(0, p\sqrt{3})$, with $2p^2$ scatterers, and then its quotient. No affine deformation or independent sheet dynamics is introduced.

Equation (194.3) makes

$$\exp\left(\frac{2\pi i}{p}[a \cdot \ell - \chi(h(x))]\right)$$

invariant under this collision map. The brackets in this formula are interpreted modulo p. The function has modulus one and mean zero upon averaging over ℓ, since $a \neq 0$. This contradicts ergodicity. Hence $L = \mathbb{Z}^2$, or $u \in 2\pi\mathbb{Z}^2$. The original phase equation now says that q is invariant, so it is constant. The converse is immediate. □

194.1. **Finite positive-measure return witnesses.** This consequence will replace a nonquantitative appeal to absence of exact eigenfunctions in the next section. The auxiliary set below is not the return section Y_R^* of the physical record.

Lemma 194.3 (A finite generating family of product-set returns)**.** *Fix R and a positive-measure regular product set $\mathcal{P}$. Let $r_{\mathcal{P}}$ be its actual first-return time under T_R. There are finitely many pairs $(n_i, k_i) \in \mathbb{Z}_{>0} \times \mathbb{Z}^2$ such that*

$$E_i = \{x \in \mathcal{P} : r_{\mathcal{P}}(x) = n_i, S_{n_i,R}\kappa_R(x) = k_i\}, \qquad \nu(E_i) > 0, \tag{194.4}$$

and the vectors $g_i = (k_{i,1}, k_{i,2}, n_i)$ generate $\mathbb{Z}^3$ over $\mathbb{Z}$. In particular, integers b_{ji} can be chosen with $e_j = \sum_i b_{ji} g_i$ for $1 \le j \le 3$.

Proof. Let L be the subgroup generated by all positive-measure cells in (194.4). The cells form a countable partition of $\mathcal{P}$ modulo a null set. If L were proper, Lemma 194.1 would give a nontrivial finite character $\theta = (u, s) \in \mathbb{T}^3$ annihilating every cell record. For $\phi = u \cdot \kappa_R + s$ this means $e^{iS_{r_{\mathcal{P}},R}\phi} = 1$ on $\mathcal{P}$ almost everywhere.

The first-return tower over $\mathcal{P}$ represents almost every point of the invertible ergodic collision system uniquely as $T_R^j y$, where $y \in \mathcal{P}$ and $0 \le j < r_{\mathcal{P}}(y)$. Set $q(T_R^j y) = e^{iS_{j,R}\phi(y)}$ on these measurable tower levels. The preceding return identity verifies the phase equation also at the top of each level. This is an exact measurable physical phase at $(u, s, 0) \neq 0$, contradicting Theorem 194.2. Thus $L = \mathbb{Z}^3$. Writing the three standard basis vectors as integer combinations uses only finitely many cell records, which gives the asserted family. The witnesses have positive measure; they are not isolated orbits. □

195. Quantitative defects of regular collision phases

Absence of exact phases does not itself control approximate vectors. We give a quantitative version on the space of bounded functions of bounded variation. The phase is reconstructed from a complex vector; no pointwise lower bound for its modulus is assumed. The logarithmic bounds in this section are at a fixed radius and on a fixed compact frequency set. A separate compactness argument gives a parameter-uniform bound when the regularity budget is fixed.

For $\omega \in \Omega$ put

$$\begin{aligned} d_{R,\omega}(q) &= \|q \circ T_R - e^{i\phi_{R,\omega}} q\|_{L^1(\nu)}, \qquad (|q| = 1), \\ \mathcal{E}_{R,\omega}(f) &= \|f \circ T_R - e^{i\phi_{R,\omega}} f\|_{L^2(\nu)}. \end{aligned} \tag{195.1}$$

The variation is always in the original collision coordinates. In particular it is neither an anisotropic distribution norm nor a norm of a many-return pushforward density.

195.1. **Finite-time measurable holonomy with a defect.** Use the product coordinates and the pair probabilities (149.1). For a stable or unstable oriented arc from x to y define

$$H_{s/u}(x, y) = \left| \frac{q(y)}{q(x)} - \exp\left(it \int_{\gamma_{s/u}(x,y)} \vartheta_R \right) \right|.$$

These are bounded measurable functions of pairs. Null sets are taken with respect to the indicated pair probabilities.

Lemma 195.1 (A finite-time holonomy defect)**.** *Fix a regular product set $\mathcal{P}$ at radius R. There are $C_{\mathcal{P}} < \infty$ and $0 < \theta_{\mathcal{P}} < 1$ such that, for $m \ge 1$, $0 < \epsilon < 1/2$, and $q : M \to \mathbb{S}^1$ with $\|q\|_{\mathrm{BV}} \le H$, $H \ge 2$,*

$$\int H_{s/u} \, \mathrm{d}\mathfrak{m}_{s/u} \le C_{\mathcal{P}} \left[m d_{R,\omega}(q) + H\epsilon + (\epsilon^{-1} + |t|)\theta_{\mathcal{P}}^m \right]. \tag{195.2}$$

The same expression controls both signs of time. The constants do not depend on u, s, t, H, m, ϵ.

Proof. Iterating the approximate phase equation along one orbit introduces an error at most the sum of its m one-step absolute defects. The stable-pair version therefore has errors at its two endpoints. Both pair marginals equal $m_{\mathcal{P}} = \nu(\mathcal{P})^{-1} \nu|_{\mathcal{P}}$. Invariance of ν bounds the integrated sum of endpoint defects by $2m\nu(\mathcal{P})^{-1} d_{R,\omega}(q)$.

Let $q_\epsilon = \mathsf{S}_\epsilon q$. Positivity of smoothing gives $|q_\epsilon| \le 1$, and $\|q - q_\epsilon\|_1 \le CH\epsilon$, $\|q_\epsilon\|_{C^1(\alpha,p)} \le C\epsilon^{-1}$. The two endpoint approximation errors are bounded at every iterate by $2\nu(\mathcal{P})^{-1} \|q - q_\epsilon\|_1$. The actual stable arc lengths are at most $C_{\mathcal{P}} \theta_{\mathcal{P}}^m$ after m forward iterates, by the contraction estimate

in the proof of Lemma 149.1. Thus the smoothed endpoint difference costs $C_{\mathcal{P}}\epsilon^{-1}\theta_{\mathcal{P}}^{m}$. The terminal action integral in (149.4) costs at most $C_{\mathcal{P}}|t|\theta_{\mathcal{P}}^{m}$. The lattice and constant phases cancel along the stable pair exactly. These estimates prove (195.2) for stable arcs.

For an unstable pair telescope backwards. Its defect is the sum of the one-step defects at $T_R^{-j}x, T_R^{-j}y$, $1 \le j \le m$; the same marginal domination applies by invertibility and invariance. Use (149.5) and backward contraction. The holonomy sign is again positive, as in (149.9). This proves the other case. No invariant pair measure and no independent orbit assumption is used. □

We make explicit how a measure-class assertion can be used quantitatively without replacing it by an unproved bounded-density assertion. Write $\mu = \mu^u \otimes \mu^s$ and $\mathrm{d}m_{\mathcal{P}} = \rho\,\mathrm{d}\mu$ in the product coordinates. Let $\lambda_s = \mu^u \otimes \mu^s \otimes \mu^s$ and $\lambda_u = \mu^u \otimes \mu^u \otimes \mu^s$ be the reference triple probabilities, and set

$$w_s = \frac{\mathrm{d}\lambda_s}{\mathrm{d}\mathfrak{m}_s}, \qquad w_u = \frac{\mathrm{d}\lambda_u}{\mathrm{d}\mathfrak{m}_u}.$$

These derivatives are positive and finite almost everywhere. For $0 \le F \le 2$ and either pair type,

$$\int F\,\mathrm{d}\lambda_{s/u} \le K \int F\,\mathrm{d}\mathfrak{m}_{s/u} + 2\lambda_{s/u}\{w_{s/u} > K\}. \tag{195.3}$$

The last probability tends to zero as $K \to \infty$. Its value is part of the constant below; no uniform pointwise bound on ρ is needed.

Lemma 195.2 (An almost constant phase on the product set)**.** *For every $\eta > 0$ there is a finite $C_{\mathcal{P},\eta}$ such that one can choose $c \in \mathbb{S}^1$ with*

$$\int_{\mathcal{P}} |q - c|\,\mathrm{d}\nu \le \eta + C_{\mathcal{P},\eta}\big[|t| + m d_{R,\omega}(q) + H\epsilon + (\epsilon^{-1} + |t|)\theta_{\mathcal{P}}^{m}\big]. \tag{195.4}$$

This holds for the same q, H, m, ϵ as in the preceding lemma.

Proof. The action integrals on the original arcs are uniformly bounded on $\mathcal{P}$. Hence the stable and unstable pair averages of $|q(y) - q(x)|$ are bounded by the right side of (195.2) plus $C_{\mathcal{P}}|t|$.

Here are the density choices needed to pass to two arbitrary product points. Choose J so that $\int_{\{\rho > J\}} \rho\,\mathrm{d}\mu$ is less than a prescribed small number. Choose K so that both tail probabilities in (195.3) are less than that number divided by J. Joining (u, s) to (u, s') and then to (u', s'), Fubini bounds the $\mu \otimes \mu$ average of $|q(x) - q(y)|$ by the sum of the two reference triple averages. Thus there is a value $c = q(y)$ for which the μ average of $|q - c|$ is at most that sum. Finally

$$\int |q - c|\,\mathrm{d}m_{\mathcal{P}} \le J \int |q - c|\,\mathrm{d}\mu + 2\int_{\{\rho > J\}} \rho\,\mathrm{d}\mu.$$

Take the prescribed small numbers sufficiently small in terms of η and $\nu(\mathcal{P})$, and apply (195.3). The remaining multiplier is a finite function of $J, K, C_{\mathcal{P}}, \nu(\mathcal{P})$. This proves (195.4). □

195.2. **A quantitative roof obstruction.**

Proposition 195.3 (A logarithmic defect bound on a roof band)**.** *Fix R and $0 < t_0 \le T < \infty$. There is $c_{R,t_0,T} > 0$ such that*

$$d_{R,(u,s,t)}(q) \ge \frac{c_{R,t_0,T}}{1 + \log H} \tag{195.5}$$

whenever $t_0 \le |t| \le T$, $|q| = 1$, $H \ge 2$, and $\|q\|_{\mathrm{BV}} \le H$. The estimate is independent of u, s.

Proof. Restrict a regular product set so that every oriented four-corner quadrilateral has action $\mathcal{A} = \oint \vartheta_R$ satisfying $|\mathcal{A}| \le \pi/T$. By Stokes' formula and nonatomicity, $|\mathcal{A}| > 0$ for almost every quadruple under $\lambda_4 = (\mu^u)^2 \otimes (\mu^s)^2$. Choose $a_0 > 0$ such that

$$p_0 := \lambda_4\{|\mathcal{A}| \ge a_0\} > 0.$$

For this set and the entire roof band, $|e^{it\mathcal{A}} - 1| \geq 2t_0 a_0/\pi$.

Each of the four edge marginals of λ_4 is one of the reference triple measures. Choose K so that the sum of their probabilities of $w_{s/u} > K$ is less than $p_0/2$. Intersect the set $\{|\mathcal{A}| \geq a_0\}$ with all four complementary conditions; call the resulting set $\mathcal{Q}$. It has measure at least $p_0/2$, and each of its unnormalized edge marginals is bounded by $K\mathfrak{m}_s$ or $K\mathfrak{m}_u$.

Multiply the four oriented approximate holonomies. The product of q-ratios is exactly one, and the error in a product of unit complex numbers is at most the sum of its four errors. Integration over $\mathcal{Q}$ and Lemma 195.1 give

$$\frac{p_0 t_0 a_0}{\pi} \leq 4KC_{\mathcal{P}}\big[m d_{R,\omega}(q) + H\epsilon + (\epsilon^{-1} + T)\theta_{\mathcal{P}}^{m}\big]. \tag{195.6}$$

All constants have now been selected from the product set, its density tails and its action distribution, independently of q and H. Choose $\epsilon = c_1/H$, with a fixed sufficiently small $c_1 > 0$. Then choose $m = \lceil c_2 + c_3 \log H \rceil$ so that the final term in brackets is also less than one quarter of the required lower bound. The positive contraction exponent makes this possible. Equation (195.6) leaves $m d_{R,\omega}(q) \geq c_4 > 0$. This proves (195.5). □

195.3. **All compact nonzero physical frequencies.**

Theorem 195.4 (A logarithmic phase defect on a compact band)**.** *For a fixed $R \in I$ and a compact set $\mathcal{K} \subset \Omega \setminus \{0\}$ there is $c_{R,\mathcal{K}} > 0$ such that, for every $\omega \in \mathcal{K}$,*

$$|q| = 1, \quad \|q\|_{\mathrm{BV}} \leq H, \quad H \geq 2 \quad \Longrightarrow \quad d_{R,\omega}(q) \geq \frac{c_{R,\mathcal{K}}}{1 + \log H}. \tag{195.7}$$

The set includes compact nonzero spatial frequencies with $s = t = 0$.

Proof. Set

$$\rho(u,s) = |e^{iu_1} - 1| + |e^{iu_2} - 1| + |e^{is} - 1|, \qquad \rho_* := \min_{\mathcal{K}}(|t| + \rho(u,s)) > 0.$$

Let $T = 1 + \max_{\mathcal{K}} |t|$. Fix a product set $\mathcal{P}$ and the finite witnesses $(n_i, k_i), E_i, b_{ji}$ in Lemma 194.3. Write $p_i = \nu(E_i) > 0$ and $d = d_{R,\omega}(q)$. For a constant $c \in \mathbb{S}^1$, iteration of the approximate phase equation on E_i gives

$$p_i |e^{i(u\cdot k_i + s n_i)} - 1| \leq 2 \int_{\mathcal{P}} |q - c| \, \mathrm{d}\nu + n_i d + p_i |t| n_i \|\tau_R\|_{\infty}. \tag{195.8}$$

Indeed $T_R^{n_i} E_i \subset \mathcal{P}$, its contribution to the endpoint error is bounded by the integral over $\mathcal{P}$ by invariance, and the n_i one-step defects integrate to at most $n_i d$. The roof sum is bounded by $n_i \|\tau_R\|_{\infty}$.

The integer identities $e_j = \sum_i b_{ji} g_i$ and $|z^b - 1| \leq |b||z - 1|$ for $|z| = 1$, $b \in \mathbb{Z}$, show that finite constants A, B, C satisfy

$$\rho(u,s) \leq A \int_{\mathcal{P}} |q - c| \, \mathrm{d}\nu + Bd + C|t|. \tag{195.9}$$

For example one may take $A = 2\sum_{j,i} |b_{ji}|/p_i$, $B = \sum_{j,i} |b_{ji}| n_i / p_i$ and $C = \|\tau_R\|_{\infty} \sum_{j,i} |b_{ji}| n_i$. These constants include the actual positive masses of the witnesses.

Choose $\eta > 0$ with $A\eta < \rho_*/16$, and use Lemma 195.2 with this η. Choose $t_0 \in (0, \rho_*/4)$ so small that all terms proportional to $|t|$ in (195.9), after substitution of that lemma, are less than $\rho_*/16$ when $|t| \leq t_0$. Choose $\epsilon = c_5/H$, then $m = \lceil c_6 + c_7 \log H \rceil$, so that the two remaining approximation terms contribute less than $\rho_*/8$. For $|t| \leq t_0$, the left side is at least $3\rho_*/4$. The inequalities thus force $md \geq c_8 > 0$. For $|t| \geq t_0$ use Proposition 195.3 on $[t_0, T]$. Taking the smaller of the two constants proves the assertion. All selections precede the choice of q, H. □

195.4. **Reconstructing a phase without a lower-modulus assumption.**

Lemma 195.5 (A bounded-variation circle truncation)**.** *For every complex $f \in \mathrm{BV}(M)$ there is a measurable $q : M \to \mathbb{S}^1$ such that*

$$\|q\|_{\mathrm{BV}} \le C(1 + \|f\|_{\mathrm{BV}}), \qquad \|q - f\|_2 \le 3\||f| - 1\|_2 \tag{195.10}$$

whenever the latter norm is finite.

Proof. Put $a = |f|$. The BV chain inequality gives $\mathrm{TV}(a) \le C\,\mathrm{TV}(f)$. The coarea formula gives a level $\tau \in [1/4, 1/2]$ for which $E = \{a > \tau\}$ has perimeter at most $4\,\mathrm{TV}(a)$. These chain, product and coarea statements are for first distributional derivatives in the collision coordinates; see [13, Chapter 3]. Let $N_\tau(z) = z/\max\{|z|, \tau\}$ and define $q = N_\tau(f)\mathbf{1}_E + \mathbf{1}_{M\setminus E}$. The map N_τ is uniformly Lipschitz for the selected levels and has modulus at most one. The BV product inequality, including the jump on the finite-perimeter boundary of E, proves the first bound. On E, $|q - f| = |1 - a|$. On its complement $a \le 1/2$ and $|q - f| \le 1 + a \le 3(1 - a)$. This proves the second bound. In particular no value of $f/|f|$ is used at a zero of f. □

Theorem 195.6 (A logarithmic obstruction for complex approximate vectors)**.** *Fix R and a compact $\mathcal{K} \subset \Omega \setminus \{0\}$. There is $c'_{R,\mathcal{K}} > 0$ such that*

$$\|f\|_2 = 1, \quad \|f\|_\infty + \|f\|_{\mathrm{BV}} \le H, \quad H \ge 2 \quad \Longrightarrow \quad \inf_{\omega\in\mathcal{K}} \mathcal{E}_{R,\omega}(f) \ge \frac{c'_{R,\mathcal{K}}}{(1 + \log H)^2}. \tag{195.11}$$

The bound applies to complex f, with no restriction on its zero set.

Proof. Write $\varepsilon = \mathcal{E}_{R,\omega}(f)$, $a = |f|$, $\bar a = \int a\,\mathrm{d}\nu$ and $x = \|a - \bar a\|_2$. The reverse triangle inequality gives $\|a\circ T_R - a\|_2 \le \varepsilon$, hence $\|a\circ T_R^m - a\|_2 \le m\varepsilon$. The BV correlation bound of Proposition 138.3, rescaled by homogeneity, gives constants $C_0, \gamma > 0$ with

$$x^2 \le C_0 H^2 e^{-\gamma m} + x m\varepsilon. \tag{195.12}$$

Indeed add and subtract the lag-m covariance of the centered a and use Cauchy–Schwarz on the remaining term. The constants in this collision estimate can be taken uniform in R.

Given $0 < \eta < 1$, choose $m \le C[1 + \log H + |\log \eta|]$ with $C_0 H^2 e^{-\gamma m} \le \eta^2$. Equation (195.12) implies $x \le \eta + m\varepsilon$. Since $\|a\|_2 = 1$ and $\bar a \ge 0$,

$$\|a - 1\|_2^2 = 2(1 - \bar a) \le 2(1 - \bar a^2) = 2x^2.$$

Lemma 195.5 constructs q with $\|q\|_{\mathrm{BV}} \le C(1 + H)$ and $\|q - f\|_2 \le 3\sqrt{2}(\eta + m\varepsilon)$. Invariance and the unit modulus of the multiplier yield

$$d_{R,\omega}(q) \le \mathcal{E}_{R,\omega}(q) \le \varepsilon + 6\sqrt{2}(\eta + m\varepsilon). \tag{195.13}$$

This is an explicit reconstruction and defect comparison.

Let $L = 1 + \log H$. Theorem 195.4, with the bound $C(1 + H)$ for q, makes the left side at least c_9/L. Choose $\eta = c_{10}/L$ so that $6\sqrt{2}\eta < c_9/(2L)$. The selected m is at most $C_{R,\mathcal{K}}L$, since $\log L \le L$. Equation (195.13) then gives $\varepsilon \ge c'_{R,\mathcal{K}}/L^2$. No lower-modulus estimate for f was assumed; it was replaced by (195.12) and the controlled truncation. □

Corollary 195.7 (Polynomial regularity budgets on a fixed band)**.** *For fixed $R, \mathcal{K}, H_0, \zeta$ with $H_0 \ge 2$, $\zeta \ge 0$, normalized vectors with $\|f_N\|_\infty + \|f_N\|_{\mathrm{BV}} \le H_0 N^\zeta$ satisfy $\inf_{\omega\in\mathcal{K}} \mathcal{E}_{R,\omega}(f_N) \ge c/(1 + \log(2 + N))^2$. In particular no such vectors have defect $O(N^{-a})$ for any fixed $a > 0$.*

Proof. Substitute the regularity budget into (195.11). A positive power of N dominates $(1 + \log(2 + N))^2$. □

195.5. **Uniform compact separation across the physical family.**

Theorem 195.8 (Parameter-uniform separation at bounded regularity)**.** *For every $H < \infty$ and compact $\mathcal{K} \subset \Omega \setminus \{0\}$ there is $c_{H,\mathcal{K}} > 0$ such that*

$$\inf_{\substack{R\in I,\ \omega\in\mathcal{K},\ \|f\|_2=1\\ \|f\|_\infty+\|f\|_{\mathrm{BV}}\le H}} \mathcal{E}_{R,\omega}(f) \ge c_{H,\mathcal{K}}. \tag{195.14}$$

An empty admissible class causes no difficulty. The theorem asserts no uniform logarithmic dependence of this constant on H.

Proof. Suppose a sequence made the infimum zero. Pass to subsequences with $R_j \to R$, $\omega_j \to \omega \in \mathcal{K}$ and $f_j \to f$ in $L^1(\nu)$. The last subsequence exists by BV compactness. In the present coordinates it also follows directly from Lemma 138.2: the approximation is uniform in L^1, and at every fixed smoothing scale the bounded smoothed functions form a precompact family on the reflected compact cylinder. The supremum bound upgrades this convergence to L^2, preserves $\|f\|_2 = 1$, and gives $|f| \le H$ almost everywhere.

Outside the finite one-collision singularity set at R, $T_{R_j}x \to T_R x$, $\kappa_{R_j}(x) \to \kappa_R(x)$ and $\tau_{R_j}(x) \to \tau_R(x)$, by persistence of the regular physical root. Thus $e^{i\phi_{R_j,\omega_j}} \to e^{i\phi_{R,\omega}}$ almost everywhere and in every finite L^p norm. For a smooth g, invariance of the common probability gives

$$\|f_j \circ T_{R_j} - f \circ T_R\|_2 \le \|f_j - f\|_2 + 2\|f - g\|_2 + \|g \circ T_{R_j} - g \circ T_R\|_2.$$

Take $j \to \infty$, then $g \to f$ in L^2. Dominated convergence controls the last term. The multiplier term also converges in L^2. The limiting defect is therefore zero.

The identity for f makes $|f|$ invariant. Ergodicity and $\|f\|_2 = 1$ give $|f| = 1$ almost everywhere. Theorem 194.2 now forces $\omega = 0$, contradicting $\mathcal{K} \subset \Omega \setminus \{0\}$. This proves the positive infimum, without claiming continuity of product sets or their density-tail constants in R. □

Relation to the complementary integral. Theorems 195.6 and 195.8 are estimates on actual collision functions, with their stated norms. They cannot be applied to an anisotropic distribution until a map to such functions, its normalization, regularity and defect comparison have been proved. They also do not assert that these BV classes are invariant under the twisted collision operator. Absence of exact phases alone has not been substituted for either assertion.

The logarithmic estimate allows a growing regularity budget on a fixed compact nonzero band at each fixed R. The parameter-uniform theorem instead fixes that budget. No estimate of $c_{R,\mathcal{K}}$ is supplied when the roof band grows or $\mathcal{K}$ approaches the origin. Consequently the physical annulus beginning at $2n^{-99/200}$, equivalently the rescaled boundary $2n^{1/200}$, is not declared controlled by these theorems. The full complementary integral, the complete critical and singular raw residual sum and the weighted exact-event comparison retain their distinct roles in Theorem 143.1. The present estimates give a quantitative phase and function-vector step toward that same inversion problem.

196. Uniform defect bounds at frequencies approaching zero

The compact bands of Section 195 exclude the origin by a fixed amount. We now allow the physical frequency to approach zero and the variation budget to grow. The observable is the bounded compensation h_R, including its original section discontinuity. Uniform ellipticity, rather than compact-band phase rigidity, supplies the obstruction.

For $z \in \mathbb{R}^4$, $\sigma \in \mathbb{R}$, and a collision function f, put

$$\mathcal{D}_{R,z,\sigma} f = f \circ T_R - e^{i(\sigma + z\cdot h_R)} f. \tag{196.1}$$

The constant phase σ is arbitrary. All norms in this section are with respect to ν on the collision cylinder. Constants are uniform in R; none depends on σ.

196.1. **A twisted middle block with untwisted ends.**

Lemma 196.1 (Damping on the diffusive collision scale)**.** *There exist $r_0 > 0$, $A > 1$, and $C < \infty$ with the following properties. If $0 < r = |z| \le r_0$, set*

$$\delta = \tfrac{1}{4} r^{1/12}, \qquad m = \lceil A/r^2 \rceil, \qquad P_{R,z} = \mathcal{L}_R M_{\exp(-iz\cdot h_{R,\delta})}.$$

On every relevant local collision space,

$$\sup_{j\ge 0} \|P_{R,z}^j\| \le C, \qquad |\ell(P_{R,z}^m \nu)| \le \tfrac{1}{8}. \tag{196.2}$$

Moreover

$$r\|S_{m,R}(h_R - h_{R,\delta})\|_2 \le C r^{1/24} \sqrt{1 + |\log r|}. \tag{196.3}$$

Proof. By Theorem 150.2, $\Gamma_R = c_* D_R \ge c_* d_0 I_4$. The covariance smoothing estimate makes $\Gamma_{R,\delta} \ge (c_* d_0/2) I_4$ for small r. The spectral expansion of Lemma 139.1 applies since $r\delta^{-2} = O(r^{5/6})$. Its cubic error is $O(r^3\delta^{-6}) = O(r^{5/2})$. Reducing r_0 gives

$$\operatorname{Re} \log \lambda_{R,\delta}(-z) \le -c_1 r^2$$

for some uniform $c_1 > 0$. The projections and complementary powers are uniformly bounded, proving the power bound. The stationary principal amplitude is $1 + O(r\delta^{-2})$. Choose A large enough that $2e^{-c_1 A} < 1/16$, and then reduce r_0 so that the complementary term at m is less than $1/16$. This proves the second bound. Finally Proposition 138.3 gives $\|S_m(h - h_\delta)\|_2^2 \le Cm\delta(1 + |\log \delta|)$. Since $mr^2 \le A + 1$, it proves (196.3). No multiplier bound for the unsmoothed h_R is asserted. □

Theorem 196.2 (Quadratic coercivity for collision phases)**.** *There exist $a_0, c_0, r_0 > 0$ such that, if $H \ge 2$, $0 < r = |z| \le r_0$, and $r(1 + \log H) \le a_0$, then*

$$|q| = 1, \quad \|q\|_{\mathrm{BV}} \le H \quad \Longrightarrow \quad \inf_{R\in I,\ \sigma\in\mathbb{R}} \|\mathcal{D}_{R,z,\sigma} q\|_1 \ge c_0 r^2. \tag{196.4}$$

Here the infimum means the same uniform bound for every admissible q at each radius; q itself may depend on R, z, σ.

Proof. Fix the radius and write $d = \|\mathcal{D}_{R,z,\sigma} q\|_1$. Let $q_\epsilon = \mathsf{S}_\epsilon q$, where $\epsilon = b_0/H$ and $b_0 > 0$ is a sufficiently small fixed number. Then $|q_\epsilon| \le 1$, $\|q - q_\epsilon\|_1 \le C b_0$, and $\|q_\epsilon\|_{C^2} \le C\epsilon^{-2}$. Choose integers

$$l = \lceil b_1 + b_2 \log H \rceil, \qquad N = m + 2l,$$

where m is as in Lemma 196.1. The constants b_1, b_2 will be fixed before H, r, q are chosen.

Iteration of the defect and invariance give

$$\left| \int \overline{q}\, (q \circ T_R^N) e^{-iz\cdot S_{N,R} h_R}\, \mathrm{d}\nu \right| \ge 1 - Nd. \tag{196.5}$$

Indeed the corresponding unit-modulus integrand differs in L^1 from $e^{iN\sigma}$ by at most Nd. Remove the phase from the first and last l collisions. Since h_R is bounded, this changes the pairing by at most $2lr\|h_R\|_\infty$. Replacing its two endpoint factors by q_ϵ costs at most $2\|q - q_\epsilon\|_1$. Smoothing only the middle collision observable costs at most (196.3), by stationarity and the unit supremum bounds on the endpoint factors.

The resulting pairing is exactly

$$\ell\left(M_{q_\epsilon} \mathcal{L}_R^l P_{R,z}^m \mathcal{L}_R^l (\overline{q_\epsilon} \nu) \right). \tag{196.6}$$

Replace both untwisted end powers by $\Pi_R = \nu\ell$. The total error is bounded by $C\epsilon^{-4}\vartheta^l$: both smooth endpoint factors cost $C\epsilon^{-2}$, all other untwisted powers are bounded, and the middle power is bounded by (196.2). The remaining expression is

$$\left|\int q_\epsilon \,\mathrm{d}\nu\right|^2 \ell(P^m_{R,z}\nu),$$

whose modulus is at most $1/8$.

Choose b_0 to make the endpoint smoothing error less than $1/16$; then choose b_1, b_2 to make $C\epsilon^{-4}\vartheta^l < 1/16$ for every $H \geq 2$. Reducing r_0 makes the middle unsmoothing error less than $1/16$. Finally choose a_0 so that $r(1+\log H) \leq a_0$ makes $2lr\|h_R\|_\infty < 1/16$. Thus the modulus in (196.5) is at most $1/2$. Since $N \leq C/r^2$ under the same constraint, $Nd \geq 1/2$ proves the assertion. Notice that the end blocks are untwisted only inside this comparison; the original record is unchanged. □

196.2. **Complex vectors and their modulus.**

Theorem 196.3 (A uniform small-frequency function bound)**.** *There exist $a_1, c_1, r_1 > 0$ such that, with*

$$\Lambda(H,r) = 1 + \log(H/r), \qquad H \geq 2, \quad 0 < r \leq r_1,$$

$r\Lambda(H,r) \leq a_1$ implies

$$\|f\|_2 = 1, \quad \|f\|_\infty + \|f\|_{\mathrm{BV}} \leq H \quad \Longrightarrow \quad \|\mathcal{D}_{R,z,\sigma} f\|_2 \geq \frac{c_1 r^2}{\Lambda(H,r)} \quad (|z| = r). \tag{196.7}$$

The assertion is uniform in R and σ, and allows zeros of f.

Proof. Put $e = \|\mathcal{D}_{R,z,\sigma} f\|_2$, $a = |f|$ and $x = \|a - \int a \,\mathrm{d}\nu\|_2$. The modulus estimate (195.12) uses only the unit modulus of the twist and the collision BV correlation estimate. It therefore applies here with constants uniform in R:

$$x^2 \leq CH^2 e^{-\gamma j} + xje.$$

Given $\eta = br^2$ with fixed small $b > 0$, choose $j \leq C\Lambda(H,r)$ so that the first term is at most η^2. Then $x \leq \eta + je$. The circle truncation of Lemma 195.5 gives a phase q with

$$\|q\|_{\mathrm{BV}} \leq C(1+H), \qquad \|q - f\|_2 \leq 3\sqrt{2}(\eta + je).$$

Consequently

$$\|\mathcal{D}_{R,z,\sigma} q\|_1 \leq e + 6\sqrt{2}(\eta + je).$$

After decreasing a_1, r_1, the hypotheses of Theorem 196.2 hold for the budget $C(1+H)$. Choose b so that the η term is less than $c_0 r^2/2$. The remaining inequality gives $e \geq cr^2/(1+j) \geq c_1 r^2/\Lambda(H,r)$. This proves the claim without dividing by f on its zero set. □

The frequency being estimated. The twist in (196.1) contains $z \cdot h_R$, not just the physical one-collision displacement and flight time. Its sum over a genuine return is exactly $z \cdot (G_R - \bar{G}_R)$. This exact relation is used next to obtain bounds for the actual induced record. The preceding theorems bound defects of functions; neither is an integrated Fourier-tail estimate.

197. Regularizing the actual tower and excluding return defects

A function of an actual return state lifts to the collision tower with an exactly localized defect. That lift need not have bounded variation. We construct a regular approximation with a controlled cost before using Section 196. The construction concerns functions and first distributional derivatives, not anisotropic vectors or inverse-coarea densities.

197.1. **Finite collision records in the initial coordinates.**

Lemma 197.1 (A quantitative finite-record variation bound)**.** *There is $C \geq 1$ such that, for every $L \geq 1$ and $R \in I$, the following bounded functions have variation norm at most*

$$B_L = \exp\{CL \log(2 + L)\} : \tag{197.1}$$

all components of $\chi \circ T_R^j$, $h_R \circ T_R^j$, and $\eta_R \circ T_R^j$, $|j| \leq L$, where $\chi(\alpha, p) = (\cos\alpha, \sin\alpha, p)$. Their supremum norms are bounded independently of L. Null singular sets are assigned any fixed bounded semialgebraic convention. Moreover, if g is smooth on the reflected collision cylinder, then

$$\|g \circ T_R^j\|_{\mathrm{BV}} \leq C\|g\|_{C^1(\alpha,p)} B_L \qquad (|j| \leq L). \tag{197.2}$$

The constant in B_L is independent of the coefficients specifying the moving section rectangles.

Proof. We supply the finite-complexity argument, since a one-collision BV bound does not alone imply this lemma. A fixed finite set of disk labels contains every next collision, by the uniform horizon. Use normal vectors $n_j \in \mathbb{S}^1$, outgoing velocities $v_j \in \mathbb{S}^1$, and flight lengths τ_j as auxiliary variables. For a fixed label word d_j, each step is given by

$$Rn_j + \tau_j v_j - d_j = Rn_{j+1}, \qquad v_{j+1} = v_j - 2(v_j \cdot n_{j+1})n_{j+1}, \qquad p_j = v_j \cdot Jn_j, \tag{197.3}$$

where J rotates vectors by $\pi/2$. Require $\tau_j > 0$, $v_j \cdot n_j > 0$, and $v_j \cdot n_{j+1} < 0$. Require also that the open flight not enter any other candidate disk. These last conditions are finite Boolean combinations of polynomial sign conditions of bounded degree: for each center d, minimize $|Rn_j + tv_j - d|^2$ over $0 \leq t \leq \tau_j$. The minimizing value occurs at an endpoint or at $t = (d - Rn_j) \cdot v_j$; splitting into these three cases gives the required inequalities without a quantified time variable. Endpoints and tangencies are included in the declared singular convention. Thus the graph records the actual first collision, not merely some root of the flight equation.

On each bounded rational angular chart, the initial normal and outgoing velocity also have such a graph, using a positive square-root variable for $\sqrt{1 - p^2}$. A word of length at most $2L + 2$ therefore uses at most $C_2 L$ variables and $C_2 L$ polynomials, of degree bounded by a fixed d. All these variables can be confined to a bounded ball. The same representation gives negative iterates by fixing the terminal rather than the initial state; the regular billiard map is invertible. Every lifted fiber over a regular initial state is unique.

Section membership is equally finite. Each angular interval has length less than π and is described by two oriented half-plane tests on n_j, together with the two inequalities for p_j. The moving angular centers enter only as real coefficients. Taking the finite union of rectangles gives η_R; adjoining its two possible values also gives the graph of $h_R = f_R - \bar{G}_R \eta_R$. All degree and variable counts remain as above. There are at most C_3^L choices of label word and of these finite membership alternatives.

Here is the topological bound we use. The finite sign-condition bound recorded in [14, Section 3.2, equation (3.3)], with the zeroth Betti number and ambient dimension k, bounds the total number of connected components by

$$d(2d - 1)^{k-1} \sum_{i=0}^{k} \binom{s}{i} 4^i \leq (C(1 + s)d)^{Ck}.$$

The bound applies uniformly in the polynomial coefficients and to any Boolean union of the sign conditions. We use this finite inequality, not an asymptotic formula whose dimension is held fixed. Fix the second collision coordinate and a real level a. Add the condition that a chosen scalar output u exceed a. The lifted semialgebraic set has the preceding complexity. Its projection onto the remaining coordinate has at most as many connected components: every connected component projects to an interval or a point. Summing over the at most C_3^L alternatives bounds

the number of components of the slice superlevel set by $\exp\{CL\log(2+L)\}$. The same reasoning applies with the two input coordinates interchanged.

If $|u| \le M_0$, the perimeter of a one-dimensional superlevel set with at most b components is at most $2b$ in the interior. The one-dimensional coarea formula therefore bounds the variation of that slice by $4M_0 b$. Integrating both slice bounds gives the two first distributional derivative bounds, including jumps. The fixed angular chart cover has bounded derivatives and inverse derivatives; its seams add only uniformly bounded boundary terms. This proves (197.1), after increasing C. No inverse derivative near grazing has been bounded pointwise.

Finally extend g from the embedded cylinder $\chi(M) \subset \mathbb{R}^3$ to a neighborhood, with Lipschitz norm at most $C\|g\|_{C^1(\alpha,p)}$, by a fixed tubular chart and cutoff. The BV chain inequality applied to the three components of $\chi \circ T_R^j$ proves (197.2). This does not assert a BV composition bound for an arbitrary nonsmooth g; only the smoothed functions below require this estimate. □

197.2. **Exact lifting and its regular approximation.** Write $Y = Y_R^*$, $F = F_R^*$, r_R^* for its actual return time and $g_R = G_R - \bar{G}_R$. For almost every collision state let its age since the most recent section visit be j. The corresponding disjoint levels are

$$D_{j,R} = T_R^j\{y \in Y : r_R^*(y) > j\}, \qquad j \ge 0. \tag{197.4}$$

They partition a full-measure set, by recurrence and invertibility. For a measurable section function f, define its exact phase lift by

$$(\mathsf{L}_{R,z} f)(T_R^j y) = e^{iz\cdot S_{j,R}h_R(y)} f(y), \qquad 0 \le j < r_R^*(y). \tag{197.5}$$

This is a function lift, not an induced transfer operator.

Lemma 197.2 (The defect is supported at the true tower tops)**.** *For $1 \le p < \infty$, whenever the terms are finite,*

$$\|\mathsf{L}_{R,z} f\|_{L^p(\nu)}^p = c_* \int_Y r_R^* |f|^p \,\mathrm{d}\nu_R^*, \tag{197.6}$$

$$\|\mathcal{D}_{R,z,0}(\mathsf{L}_{R,z} f)\|_{L^p(\nu)}^p = c_* \int_Y |f\circ F - e^{iz\cdot g_R} f|^p \,\mathrm{d}\nu_R^*. \tag{197.7}$$

The analogous supremum bound is $\|\mathsf{L}_{R,z} f\|_\infty \le \|f\|_\infty$.

Proof. Invariance and the disjoint levels give, for every nonnegative b,

$$\int_M b \,\mathrm{d}\nu = c_* \int_Y \sum_{j=0}^{r_R^*-1} b(T_R^j y) \,\mathrm{d}\nu_R^*(y).$$

Apply this first to the modulus of the lift. At every nontop level its next value is exactly $e^{iz\cdot h_R}$ times its current value. At the last level the defect is

$$f(Fy) - e^{iz\cdot S_{r_R^*,R}h_R(y)} f(y) = f(Fy) - e^{iz\cdot g_R(y)} f(y),$$

using the one-return compensation identity. This proves (197.7). Only one top occurs per return; its measure is integrated against $c_*\nu_R^*$, not against a return-time-biased probability. □

For a section function f let f^0 be its zero extension to M, $M_f = \|f\|_\infty$ and $V_f = \|f^0\|_{\mathrm{BV}}$. Set $f_\epsilon = \mathsf{S}_\epsilon f^0$ and define

$$Q_{L,\epsilon}(x) = \sum_{j=0}^{L-1} \mathbf{1}_{D_{j,R}}(x) e^{iz\cdot\sum_{i=1}^{j} h_R(T_R^{-i}x)} f_\epsilon(T_R^{-j}x). \tag{197.8}$$

The sum for $j = 0$ is empty. The approximant is zero on ages at least L; the original lift is not truncated in any theorem statement.

Proposition 197.3 (A uniform cost for approximating the lift)**.** *For $L \geq 2$ and $0 < \epsilon < 1/2$, uniformly in R, z, f,*

$$\|Q_{L,\epsilon}\|_\infty \leq M_f, \tag{197.9}$$

$$\|\mathsf{L}_{R,z} f - Q_{L,\epsilon}\|_1 \leq C(L\epsilon V_f + M_f e^{-cL}), \tag{197.10}$$

$$\|\mathsf{L}_{R,z} f - Q_{L,\epsilon}\|_2^2 \leq C(M_f L\epsilon V_f + M_f^2 e^{-cL}), \tag{197.11}$$

$$\|Q_{L,\epsilon}\|_{\mathrm{BV}} \leq M_f \exp\{CL\log(2+L)\}(1+\epsilon^{-1}+|z|). \tag{197.12}$$

Proof. The level indicators are disjoint, proving the supremum bound. On the first L levels, invariance bounds the total L^1 change of the initial function by $L\|f^0 - f_\epsilon\|_1 \leq CL\epsilon V_f$. The remaining age probability is exactly

$$\nu\{\text{age} \geq L\} = c_* \int_Y (r_R^* - L)_+ \, \mathrm{d}\nu_R^* \leq Ce^{-cL},$$

by Theorem 133.3. This proves the L^1 bound. Both functions have supremum at most M_f; multiplying the L^1 bound by $2M_f$ gives the L^2 bound.

For variation, the exact level indicator is

$$\mathbf{1}_{D_{j,R}} = (\eta_R \circ T_R^{-j}) \prod_{i=0}^{j-1} (1 - \eta_R \circ T_R^{-i}).$$

The finite-record lemma and the BV product inequality bound its variation by $C(j+1)B_L$. The same lemma bounds the phase sum variation by CjB_L. The exponential is a Lipschitz function of this real vector sum, with Lipschitz constant at most $|z|$. Moreover $\|f_\epsilon\|_{C^1} \leq CM_f\epsilon^{-1}$, so (197.2) applies to its pullback. Apply the BV product inequality to each summand and then sum over $j < L$. Absorb the resulting fixed powers of L into B_L by increasing its constant. This proves (197.12). Neither the levels nor their images have been declared smooth; their indicator jumps are included in the BV estimates. □

197.3. **Defects for the actual unbounded return record.** For $H \geq 2$, $0 < r < 1/2$ define

$$\Lambda = 1 + \log(H/r), \qquad \mathcal{K}(H, r) = \Lambda \log(2 + \Lambda). \tag{197.13}$$

The letter $\mathcal{K}$ here denotes a scalar budget, not a compact frequency set as in Section 195.

Theorem 197.4 (Uniform return-phase coercivity near the origin)**.** *There exist $a_2, c_2, r_2 > 0$ such that, for $H \geq 2$, $0 < r = |z| \leq r_2$, and $r\mathcal{K}(H, r) \leq a_2$,*

$$|q| = 1 \text{ on } Y_R^*, \quad \|q^0\|_{\mathrm{BV}} \leq H \quad \Longrightarrow \quad \|q \circ F_R^* - e^{iz\cdot(G_R - \bar{G}_R)} q\|_{L^1(\nu_R^*)} \geq c_2 r^2. \tag{197.14}$$

All constants are independent of R, and no induced mixing hypothesis is used.

Proof. The exact lift $Q = \mathsf{L}_{R,z} q$ has modulus one. Given $\eta = br^2$ with $b > 0$ fixed and small, choose $L \leq C(1 + |\log \eta|)$ and $\epsilon = \eta/(CLH)$ so that (197.10) gives $\|Q - Q_{L,\epsilon}\|_1 \leq \eta$. The circle truncation in Lemma 195.5, applied to $Q_{L,\epsilon}$, gives a phase $\widetilde{q}$ satisfying

$$\|\widetilde{q} - Q\|_1 \leq 4\eta, \qquad 1 + \log(2 + \|\widetilde{q}\|_{\mathrm{BV}}) \leq C\mathcal{K}(H, r).$$

For the first bound, use the pointwise inequality from that lemma, $|\widetilde{q} - Q_{L,\epsilon}| \leq 3\big||Q_{L,\epsilon}| - 1\big| \leq 3|Q_{L,\epsilon} - Q|$. The second follows from (197.12); the constants include the chosen b.

After decreasing a_2, r_2, Theorem 196.2 applies to $\widetilde{q}$. Changing a function by 4η in L^1 changes its collision defect by at most 8η. Equation (197.7) therefore gives

$$c_0 r^2 \leq c_* \|q \circ F_R^* - e^{iz\cdot g_R} q\|_1 + 8br^2.$$

Choose $b < c_0/16$. This proves the assertion with a uniform $c_2 > 0$. In particular the regularity of the exact lift was not assumed. □

Theorem 197.5 (Uniform coercivity for actual return functions)**.** *There exist $a_3, c_3, r_3 > 0$ such that $0 < r = |z| \le r_3$, $H \ge 2$, and $r\mathcal{K}(H,r) \le a_3$ imply*

$$\|f\|_{L^2(\nu_R^*)} = 1, \quad M_f + V_f \le H \quad \Longrightarrow$$

$$\|f \circ F_R^* - e^{iz\cdot(G_R - \bar{G}_R)} f\|_{L^2(\nu_R^*)} \ge \frac{c_3 r^2}{\mathcal{K}(H,r)}. \tag{197.15}$$

The section function may be complex and may vanish. The conclusion is uniform in the radius and in the direction of z.

Proof. Let $Q = \mathsf{L}_{R,z} f$ and $S = \|Q\|_2$. By (197.6), $\sqrt{c_*} \le S \le H$; the upper bound also follows from $\|Q\|_\infty \le H$ and $\nu(M) = 1$. If e denotes the defect on the left of (197.15), then

$$\|\mathcal{D}_{R,z,0}(Q/S)\|_2 = \sqrt{c_*}\, e/S \le e. \tag{197.16}$$

Choose an accuracy $\eta = br^2/\mathcal{K}(H,r)$, where $0 < b < 1$ will be fixed. The bounds in Proposition 197.3 allow $L \le C[\Lambda + |\log b|]$ and $\epsilon \ge c\eta^2/(H^2 L)$ with $\|Q - Q_{L,\epsilon}\|_2 \le \sqrt{c_*}\eta$. For example choose L to make the exponential term at most $c_*\eta^2/2$, then choose ϵ equal to a sufficiently small fixed multiple of $\eta^2/(H^2 L)$. Both choices are uniform in R. Normalize $Q_{L,\epsilon}/S$ to an L^2 unit function v. For $\eta < 1/4$ this is well defined and

$$\|v - Q/S\|_2 \le 2\eta, \qquad 1 + \log(2 + \|v\|_\infty + \|v\|_{\mathrm{BV}}) + |\log r| \le C(1 + |\log b|)^2 \mathcal{K}(H,r).$$

The last bound is (197.12), using the lower bound on S, and $\log \mathcal{K}(H,r) \le C\Lambda$. The displayed dependence on b prevents a circular choice of accuracy.

Choose b so small that $4b < c_1/[2C(1+|\log b|)^2]$; such a choice exists because $b(1+|\log b|)^2 \to 0$. Fix this b once and for all. Decrease a_3, r_3 so that Theorem 196.3 applies to v under the preceding budget. That theorem and (197.16) give

$$e + 4\eta \ge \|\mathcal{D}_{R,z,0} v\|_2 \ge \frac{c_1 r^2}{C(1 + |\log b|)^2 \mathcal{K}(H,r)}.$$

The selected accuracy absorbs 4η and proves (197.15). No lower bound for $|f|$ and no mixing rate for F_R^* enter the proof. □

Corollary 197.6 (The intervening annulus with polynomial budgets)**.** *Fix $H_0 \ge 2$ and $\zeta \ge 0$. For all sufficiently large n, uniformly in R and in physical frequencies*

$$2n^{-99/200} \le |z| \le n^{-2/5}, \qquad 2n^{1/200} \le |\sqrt{n}\, z| \le n^{1/10}, \tag{197.17}$$

unit section phases with zero-extension variation at most $H_0 n^\zeta$ have defect at least $c|z|^2$. Normalized complex section functions with $M_f + V_f \le H_0 n^\zeta$ have defect at least

$$\frac{c|z|^2}{\log(2+n)\log\log(e^e + n)}. \tag{197.18}$$

In particular this last lower bound is at least $cn^{-99/100}/[\log(2+n)\log\log(e^e+n)]$ on the whole annulus.

Proof. On this annulus $\Lambda(H_0 n^\zeta, |z|) \le C\log(2+n)$, so $\mathcal{K} \le C\log(2+n)\log\log(e^e+n)$. The largest possible value of $|z|\mathcal{K}$ is bounded by $Cn^{-2/5}\log(2+n)\log\log(e^e+n)$, which tends to zero. Thus both preceding theorems apply. Squaring the lower frequency endpoint gives the final power. The interval is nonempty for all sufficiently large n, since its endpoint ratio is $n^{19/200}/2 \to \infty$. □

What the annulus estimate supplies. The norm in (197.18) is a one-step cohomological defect for the *actual unbounded return record*, with the normalization and first-variation budget stated above. The lifted defect and the regularization error have both been proved, rather than assumed to come from an induced spectral construction. No conclusion is made here for arbitrary eigenvalues of the induced operator: the multiplier in the theorem is precisely $e^{iz\cdot(G_R-\bar{G}_R)}$.

This estimate is a quantitative input on the intervening annulus, not the complementary Fourier integral itself. To deduce decay of powers from it on a distribution space still requires a construction of that space, a function reconstruction with the displayed budgets, and a bound relating the operator defect to the present physical L^2 defect. Nor does the finite-record BV lemma estimate any second derivative of a raw pushforward density. The growing roof region, the complete critical/singular extraction, and the weighted exact-event comparison remain distinct requirements of Theorem 143.1. All conditioning indicators in the inherited central and moment theorems are unchanged.

198. Peripheral return phases and a quadratic regularity budget

The spectral phase of a return operator is not a constant phase per collision. We keep these parameters separate. The physical Fourier variable is $z \in \mathbb{R}^4$, the return spectral phase is $\xi \in [-\pi, \pi]$, and the corresponding unit spectral parameter is $\lambda = e^{i\xi}$. For a section function define

$$\mathcal{E}_{R,z,\xi} f = f \circ F_R^* - e^{i(\xi + z\cdot g_R)} f, \qquad g_R = G_R - \bar{G}_R. \tag{198.1}$$

Theorem G concerns $\xi = 0$. We prove a joint small-parameter estimate, including $z = 0$, and an exact lifting identity for every ξ. We also improve the regularity constraint by using the centered second moment when removing the short collision end blocks.

198.1. **Collision end blocks in the stationary second moment.**

Lemma 198.1 (Quadratic endpoint budget)**.** *There are $a, b, r_0 > 0$, uniform in R, such that*

$$0 < r = |w| \le r_0, \quad H \ge 2, \quad r^2(1 + \log H) \le a \quad \Longrightarrow \quad \|\mathcal{D}_{R,w,\sigma} q\|_1 \ge b r^2 \tag{198.2}$$

for every $\sigma \in \mathbb{R}$ and every collision phase q with $|q| = 1$ and $\|q\|_{\mathrm{BV}} \le H$.

Proof. Use the diffusive middle block of Lemma 196.1, of length $m = \lceil A/r^2 \rceil$, and the endpoint smoothing in the proof of Theorem 196.2. Thus $\epsilon = b_0/H$, $\ell = \lceil b_1 + b_2 \log H \rceil$, and $N = m + 2\ell$. The lower pairing is at least $1 - Nd$, where $d = \|\mathcal{D}_{R,w,\sigma} q\|_1$.

The improvement concerns the removal of the phase on the first and last ℓ collisions. Since h_R is centered and its collision covariances are absolutely summable, uniformly in R,

$$\|S_{\ell,R} h_R\|_2 \le C\sqrt{\ell}.$$

All endpoint factors have modulus at most one. Invariance and $|e^{iu} - e^{iv}| \le |u - v|$ therefore bound the change in the pairing by

$$|w| \int \left| S_{\ell,R} h_R + (S_{\ell,R} h_R) \circ T_R^{m+\ell} \right| \, \mathrm{d}\nu \le C r \sqrt{\ell}. \tag{198.3}$$

No independence of the two end blocks is used. This replaces the supremum estimate $Cr\ell$ in the earlier proof.

After endpoint smoothing and middle-block smoothing, the same chronological word (196.6) remains. Its two untwisted powers may be replaced by Π_R at cost $C\epsilon^{-4}\vartheta^{\ell}$. Its principal pairing has modulus at most $1/8$. Choose b_0, b_1, b_2 exactly as in the earlier proof so that the two endpoint smoothing and mixing errors total less than $1/8$. Reduce r_0 so that the middle unsmoothing error is less than $1/16$. Finally choose a so that (198.3) is less than $1/16$. The original pairing

then has modulus at most $1/2$. The new constraint also gives $N \le C/r^2$, whence $Nd \ge 1/2$ proves (198.2). □

Lemma 198.2 (Joint small collision frequency and constant phase)**.** *Let $s = \operatorname{dist}(\sigma, 2\pi\mathbb{Z})$ and $t = (|w|^2 + s^2)^{1/2}$. There are $a, b, t_0 > 0$ such that*

$$0 < t \le t_0, \quad t^2(1 + \log H) \le a \quad \Longrightarrow \quad \|\mathcal{D}_{R,w,\sigma} q\|_1 \ge bt^2 \tag{198.4}$$

for every unit-modulus q of variation at most $H \ge 2$. All constants are uniform in R.

Proof. Write $r = |w|$ and choose a fixed large number B. If $s \le Br^2$, then $r > 0$ and $t^2 \le (1 + B^2 t_0^2) r^2$. Lemma 198.1 proves the claim, after adjusting the constants.

Suppose $s > Br^2$. Centering q in the collision correlation bound and normalizing its BV norm give

$$\left| \int \overline{q}(q \circ T_R^j) \, \mathrm{d}\nu - \left| \int q \, \mathrm{d}\nu \right|^2 \right| \le CH^2 e^{-\gamma j}.$$

Choose $\ell = \lceil b_1 + b_2 \log H \rceil$ so that this bound is at most $1/16$ for every $j \ge \ell$. Reduce t_0 below $1/4$. There is an integer

$$\ell \le j \le \ell + \lceil 2\pi/s \rceil + 1 \quad \text{such that} \quad \cos(j\sigma) \le -3/4.$$

Indeed consecutive phases, in the direction of the representative of σ of absolute value s, traverse a full circle in this many steps; a phase within s of π has the required cosine. This observation includes a negative representative.

The exact iteration of the phase defect implies

$$\left| \int \overline{q}(q \circ T_R^j) \, \mathrm{d}\nu - e^{ij\sigma} \int e^{iw \cdot S_{j,R} h_R} \, \mathrm{d}\nu \right| \le jd, \qquad d = \|\mathcal{D}_{R,w,\sigma} q\|_1.$$

The centered second moment bounds the difference of the last integral from one by $Cr\sqrt{j}$. The selected time satisfies

$$r^2 j \le C(r^2(1 + \log H) + r^2/s) \le C(a + B^{-1}).$$

First choose B large and then a small, so that $Cr\sqrt{j} \le 1/8$. The real part of the first correlation is at least $-1/16$, whereas that of the second term is at most $-3/4 + 1/8$. Hence $jd \ge 1/2$. Finally

$$jt^2 \le C(t^2(1 + \log H) + r^2/s + s) \le C(a + B^{-1} + t_0),$$

so $d \ge bt^2$. This argument also covers $w = 0$, $s > 0$; thus no pure constant-phase direction has been lost. □

Proposition 198.3 (Joint collision defects for complex functions)**.** *With t as in Lemma 198.2, put $L(H, t) = 1 + \log(H/t)$. There are uniform $a, b, t_0 > 0$ such that*

$$0 < t \le t_0, \quad t^2 L(H,t) \le a, \quad \|f\|_2 = 1, \quad \|f\|_\infty + \|f\|_{\mathrm{BV}} \le H \quad \Longrightarrow \quad \|\mathcal{D}_{R,w,\sigma} f\|_2 \ge \frac{bt^2}{L(H,t)}. \tag{198.5}$$

The function f may vanish.

Proof. Put $e = \|\mathcal{D}_{R,w,\sigma} f\|_2$ and $x = \| |f| - \int |f| \, \mathrm{d}\nu \|_2$. The modulus-variance estimate (195.12) gives $x^2 \le CH^2 e^{-\gamma j} + xje$. For $\eta = \varepsilon_0 t^2$, choose $j \le CL(H,t)$ so that the first term is at most η^2. Then $x \le \eta + je$. Lemma 195.5 provides a unit phase q with

$$\|q\|_{\mathrm{BV}} \le C(1 + H), \qquad \|q - f\|_2 \le 3\sqrt{2}(\eta + je).$$

Its L^1 collision defect is at most $e + 6\sqrt{2}(\eta + je)$. After reducing a, t_0, Lemma 198.2 applies to this phase. Choose ε_0 small enough to absorb the η term. The resulting inequality proves

(198.5). This is a controlled modulus argument, not division by an unspecified nonvanishing representative. □

198.2. **The exact spectral phase on the actual tower.** Recall $\eta_R = \mathbf{1}_{Y_R^*}$ and $e_3 = (0,0,1,0)$. For every real ξ define

$$b_{R,\xi} = \eta_R + e^{i\xi}(1-\eta_R), \qquad \mathsf{L}_{R,z,\xi} f = b_{R,\xi}\mathsf{L}_{R,z} f, \qquad \sigma = c_*\xi, \quad w = z - \sigma e_3. \tag{198.6}$$

The multiplier $b_{R,\xi}$ has modulus one and uniformly bounded variation, independently of ξ. In particular its introduction does not entail a variation estimate for a long return iterate.

Lemma 198.4 (Exact peripheral lift and its two normalizations)**.** *For every real ξ, every $z \in \mathbb{R}^4$, and $1 \le p < \infty$, whenever the terms are finite,*

$$\|\mathsf{L}_{R,z,\xi} f\|_{L^p(\nu)}^p = c_* \int_{Y_R^*} r_R^* |f|^p \, \mathrm{d}\nu_R^*, \tag{198.7}$$

$$\|\mathcal{D}_{R,w,\sigma}(\mathsf{L}_{R,z,\xi} f)\|_{L^p(\nu)}^p = c_* \|\mathcal{E}_{R,z,\xi} f\|_{L^p(\nu_R^*)}^p. \tag{198.8}$$

The defect vanishes at every nontop level of the genuine return tower. For $Q_{L,\epsilon}$ in (197.8)*, the approximant $b_{R,\xi} Q_{L,\epsilon}$ has the same L^1, L^2 and supremum error bounds as in Proposition 197.3; its BV bound is at most a fixed multiple of* (197.12)*.*

Proof. The third collision compensation coordinate is $h_{R,3} = 1 - c_*^{-1}\eta_R$. Consequently the real identity

$$\sigma + w \cdot h_R = z \cdot h_R + \xi \eta_R \tag{198.9}$$

holds, before taking exponentials. On a tower starting at $y \in Y_R^*$,

$$\sum_{i=0}^{j-1} \eta_R(T_R^i y) = \begin{cases} 0, & j = 0, \\ 1, & 1 \le j \le r_R^*(y). \end{cases}$$

Thus the lift in (198.6) accumulates exactly the phase in (198.9). At a nontop level its collision equation is exact. At the last level its defect is

$$f(F_R^* y) - e^{i(\xi + z \cdot g_R(y))} f(y).$$

This includes $r_R^*(y) = 1$: the initial level is then itself the top. The disjoint-level integration formula in Lemma 197.2 gives both identities. The norm has the height weight; the top defect does not.

Multiplication by $b_{R,\xi}$ preserves pointwise absolute errors. The BV product inequality gives $\|b_{R,\xi} Q_{L,\epsilon}\|_{\mathrm{BV}} \le C\|Q_{L,\epsilon}\|_{\mathrm{BV}} + C M_f$, which proves the last assertion. No BV regularity is assumed for the full exact lift. The construction is periodic in ξ, although the real coordinates (w, σ) used to express it change with its lift from the circle. □

198.3. **A neighborhood of the induced peripheral parameter.** For $H \ge 2$, $0 < \rho < 1/2$, write

$$\Lambda(H,\rho) = 1 + \log(H/\rho), \qquad K(H,\rho) = \Lambda(H,\rho)\log(2 + \Lambda(H,\rho)). \tag{198.10}$$

Theorem 198.5 (Joint physical and return spectral phases)**.** *There are uniform constants $a, b, \rho_0 > 0$ such that, with*

$$\xi \in [-\pi, \pi], \qquad \rho = (|z|^2 + \xi^2)^{1/2}, \qquad 0 < \rho \le \rho_0, \qquad \rho^2 K(H,\rho) \le a, \tag{198.11}$$

the following estimates hold:

$$|q| = 1 \text{ on } Y_R^*, \quad \|q^0\|_{\mathrm{BV}} \le H \quad \Longrightarrow \quad \|\mathcal{E}_{R,z,\xi} q\|_{L^1(\nu_R^*)} \ge b\rho^2, \tag{198.12}$$

$$\|f\|_{L^2(\nu_R^*)} = 1, \quad M_f + V_f \le H \quad \Longrightarrow \quad \|\mathcal{E}_{R,z,\xi} f\|_{L^2(\nu_R^*)} \ge \frac{b\rho^2}{K(H,\rho)}. \tag{198.13}$$

Zeros of f are allowed. The case $\xi = 0$ improves the sufficient budget of Theorem G from $|z|K(H,|z|) \le a$ to $|z|^2K(H,|z|) \le a$.

Proof. Put $(w,\sigma) = (z - c_*\xi e_3, c_*\xi)$. For small ρ, $\operatorname{dist}(\sigma, 2\pi\mathbb{Z}) = |\sigma|$. The linear map $(z,\xi) \mapsto (w,\sigma)$ is invertible, with fixed coefficients. Therefore there are $c_1, C_1 > 0$ such that

$$c_1\rho \le t = (|w|^2 + \sigma^2)^{1/2} \le C_1\rho. \tag{198.14}$$

In particular the line $w = 0$ causes no loss: its nonzero constant collision phase is covered by Lemma 198.2.

For a unit section phase, let $Q = \mathsf{L}_{R,z,\xi}q$. It has modulus one. Choose accuracy $\eta = \varepsilon_0\rho^2$. The approximation in Lemma 198.4 and circle truncation give a phase $\widetilde{q}$ with

$$\|\widetilde{q} - Q\|_1 \le 4\eta, \qquad 1 + \log(2 + \|\widetilde{q}\|_{\mathrm{BV}}) \le C_{\varepsilon_0}K(H,\rho).$$

Explicitly one may take $L \le C(1 + |\log\eta|)$ and $\epsilon = \eta/(CLH)$. Apply Lemma 198.2 at (w,σ) and use (198.8). Changing a function by 4η changes its L^1 collision defect by at most 8η. Choose ε_0 to absorb this error relative to $b_0c_1^2\rho^2$, then decrease a, ρ_0 so that the required quadratic budget holds. This proves (198.12).

For a normalized complex section function let $Q = \mathsf{L}_{R,z,\xi}f$ and $S = \|Q\|_2$. The exact norm identity gives $\sqrt{c_*} \le S \le H$. If $e = \|\mathcal{E}_{R,z,\xi}f\|_2$, then

$$\|\mathcal{D}_{R,w,\sigma}(Q/S)\|_2 = \sqrt{c_*}\, e/S \le e.$$

Choose now $\eta = \varepsilon_0\rho^2/K(H,\rho)$. Take $L \le C(1 + \log(H/\eta))$ and $\epsilon = \eta^2/(CH^2L)$ in the L^2 approximation and normalize the resulting approximant. It gives a collision function v with

$$\|v\|_2 = 1, \qquad \|v - Q/S\|_2 \le 2\eta,$$
$$1 + \log(2 + \|v\|_\infty + \|v\|_{\mathrm{BV}}) + |\log t| \le C(1 + |\log\varepsilon_0|)^2K(H,\rho).$$

These inequalities follow directly by taking logarithms in (197.12); $\log K(H,\rho) \le C\Lambda(H,\rho)$ and (198.14) are used here. They display the dependence on the accuracy choice. Proposition 198.3 yields

$$e + 4\eta \ge \frac{b_0c_1^2\rho^2}{C(1 + |\log\varepsilon_0|)^2K(H,\rho)}.$$

Fix ε_0 so small that the accuracy term is absorbable; this is possible since $\varepsilon_0(1 + |\log\varepsilon_0|)^2 \to 0$. Only after that choice reduce a, ρ_0 to satisfy the collision quadratic budget. This proves (198.13) with no circular choice of a regularity constant. □

Corollary 198.6 (Peripheral annuli with enlarged regularity classes)**.** *Consider the physical annulus and its rescaling*

$$2n^{-99/200} \le |z| \le n^{-2/5}, \qquad 2n^{1/200} \le |\sqrt{n}\, z| \le n^{1/10}. \tag{198.15}$$

For $H \le H_0n^{\zeta}$, there is $a_0 > 0$ such that the preceding theorem holds throughout this annulus and

$$|\xi| \le \frac{a_0}{\sqrt{\log(2+n)\log\log(e^e+n)}}$$

for all sufficiently large n. Its complex-function lower bound is at least $b(|z|^2 + \xi^2)/[\log(2 + n)\log\log(e^e + n)]$.

More generally, fix $\gamma > 0$ and $0 < \beta < \min\{4/5, 2\gamma\}$. For $H \le H_0\exp(n^\beta)$ and $|\xi| \le n^{-\gamma}$, the theorem holds uniformly on (198.15), with complex-function lower bound at least $b(|z|^2 + \xi^2)/(n^\beta\log(2+n))$.

Proof. The lower annulus endpoint gives $\rho \geq 2n^{-99/200}$. For polynomial budgets this implies $K(H,\rho) \leq C\log(2+n)\log\log(e^e+n)$. Choose a_0 small enough that the $\xi^2 K$ term meets (198.11); the $|z|^2 K$ term tends to zero. For the exponential budget, $K(H,\rho) \leq Cn^\beta \log(2+n)$, whereas $\rho^2 \leq n^{-4/5} + n^{-2\gamma}$. The strict inequalities on β make their product tend to zero. The lower bounds follow from (198.13). □

Spectral scope. The lift in Lemma 198.4 treats every return phase exactly. The quantitative conclusion, in contrast, is on the joint region (198.11); it is not a bound on the entire unit circle at a growing regularity budget. The pure spectral direction $z = 0$, $\xi \neq 0$ and the shifted-frequency cancellation line are included. Theorem 198.5 remains a physical function theorem. The next section gives a precise operator-defect comparison for finite-rank compressions, without identifying them with an anisotropic realization of the unmodified return dynamics.

199. Function reconstruction for compressed actual return operators

A distributional spectral vector does not automatically belong to a bounded-variation class. We give a setting where the reconstruction and the defect comparison can both be proved: finite-rank orthogonal compressions of the actual return operator. The multiplier uses the full unbounded return record. Only the function space is compressed. We keep the mesh dependence explicit and prove a separate finite-time comparison with the uncompressed characteristic function.

199.1. **An explicit regular function class.** For a sufficiently small mesh size $h > 0$, subdivide the five disjoint coordinate rectangles making up Y_R^* into rectangles with side lengths between $h/2$ and h. Equal subdivisions of each side have this property. Angular coordinates are taken in a chart through each rectangle; no artificial small cell is created at the angular seam. Denote the partition by $\mathcal{P}_{R,h}$, its piecewise constant space by $V_{R,h} \subset L^2(\nu_R^*)$, and its conditional expectation by $P_{R,h}$. The side lengths of the original rectangles are fixed; only their locations move with R.

Lemma 199.1 (Uniform reconstruction on the section partition)**.** *Uniformly in R and h,*

$$\dim V_{R,h} \leq Ch^{-2}, \qquad \|f\|_\infty + \|f^0\|_{\mathrm{BV}} \leq Ch^{-1}\|f\|_{L^2(\nu_R^*)} \quad (f \in V_{R,h}). \tag{199.1}$$

The map $P_{R,h}$ is an orthogonal projection, preserves constants, and contracts both L^2 and L^∞. For every bounded section function v with zero extension of bounded variation,

$$\|(I - P_{R,h})v\|_{L^1(\nu_R^*)} \leq Ch\|v^0\|_{\mathrm{BV}}, \tag{199.2}$$

$$\|(I - P_{R,h})v\|_{L^2(\nu_R^*)}^2 \leq Ch\|v\|_\infty \|v^0\|_{\mathrm{BV}}. \tag{199.3}$$

The zero extension in (199.1) *includes all jumps at the outer section boundary.*

Proof. The normalized collision density on the section is the constant $(4\pi c_*)^{-1}$ in (α, p) coordinates. Each cell has mass between ch^2 and Ch^2, and there are at most Ch^{-2} cells. If f has value a_B on cell B and norm one, then $\sum_B |a_B|^2 \leq Ch^{-2}$ and $\max_B |a_B| \leq Ch^{-1}$. Every cell side has length at most h. Summing internal jumps and outer traces bounds the first variation by

$$Ch\sum_B |a_B| \leq Ch(\#\mathcal{P}_{R,h})^{1/2}\left(\sum_B |a_B|^2\right)^{1/2} \leq Ch^{-1}.$$

Real and imaginary parts give the stated complex BV norm. The L^1 part of the norm is uniformly bounded, proving (199.1).

For a smooth v, compare its value to its cell average by integrating $|v(x) - v(y)|$ over pairs in that cell. Change the two coordinates one at a time and integrate the fundamental theorem of calculus. Since the aspect ratios are bounded, the result on a cell is at most Ch times the integral of its first derivatives. BV approximation, or the same one-dimensional slicing argument

for distributional variation, gives (199.2). Summation over cells and the fixed normalization leave the constant uniform. Finally $|(I-P)v| \le 2\|v\|_\infty$; multiplication of the L^1 bound gives (199.3). The projection and contraction properties follow directly from conditional expectation. □

Define the actual twisted Koopman operator and its compression by

$$\mathcal{U}_{R,z} f = e^{-iz\cdot g_R} f \circ F_R^*, \qquad A_{R,h,z} = P_{R,h}\mathcal{U}_{R,z}|_{V_{R,h}}. \tag{199.4}$$

The first operator is unitary on $L^2(\nu_R^*)$: F_R^* is invertible and measure preserving, and the real-frequency multiplier has modulus one. Thus $\|A_{R,h,z}\| \le 1$. In the orthonormal cell basis $e_B = \mathbf{1}_B/\sqrt{\nu_R^*(B)}$, its exact matrix entries are

$$(A_{R,h,z})_{BC} = \frac{\displaystyle\int_{B\cap (F_R^*)^{-1}C} e^{-iz\cdot g_R}\,\mathrm{d}\nu_R^*}{\sqrt{\nu_R^*(B)\nu_R^*(C)}}. \tag{199.5}$$

These integrals retain every return length; they are not truncated flight records or matrices from an independent stochastic model.

Lemma 199.2 (Compression residual versus physical defect)**.** *Let U be an isometry of a Hilbert space, P an orthogonal projection, and $A = PU|_{\operatorname{ran} P}$. If $\|f\| = 1$, $Pf = f$, $|\lambda| = 1$, and $\varepsilon = \|(A-\lambda)f\| \le 1$, then*

$$\|(U-\lambda)f\|^2 = \|(A-\lambda)f\|^2 + 1 - \|Af\|^2 \le 3\varepsilon. \tag{199.6}$$

In particular, a physical lower bound $\|(U-\lambda)f\| \ge d$ for all normalized $f \in \operatorname{ran} P$, with $0 < d \le 1$, gives

$$\|(A-\lambda)f\| \ge (d^2/3)\|f\|. \tag{199.7}$$

Proof. The orthogonal decomposition of $(U-\lambda)f$ has components $(A-\lambda)f$ and $(I-P)Uf$. The latter has squared norm $1-\|Af\|^2$. Also $\|Af\| \ge 1-\varepsilon$, so $1-\|Af\|^2 \le 2\varepsilon - \varepsilon^2$ when $\varepsilon \le 1$. This proves the displayed, slightly weaker, constant 3. If $\varepsilon > 1$, the lower bound in (199.7) is immediate. Homogeneity finishes the proof. The argument controls arbitrary approximate vectors, not only exact eigenvectors, and makes no normality assumption on the compression. □

Theorem 199.3 (Quantitative peripheral resolvents of the compression)**.** *Choose $H_h = C_0 h^{-1} \ge 2$, with C_0 large enough for Lemma 199.1. Set*

$$\rho = (|z|^2 + \xi^2)^{1/2}, \quad K_h = K(H_h, \rho), \quad \delta_h = b_0\rho^4/K_h^2.$$

There are uniform $a, \rho_0, b_0 > 0$ such that, whenever

$$\xi \in [-\pi, \pi], \quad 0 < \rho \le \rho_0, \quad \rho^2 K_h \le a, \tag{199.8}$$

the finite matrix $e^{i\xi} - A_{R,h,z}$ is invertible and

$$\|(e^{i\xi} - A_{R,h,z})^{-1}\|_{L^2\to L^2} \le \delta_h^{-1}. \tag{199.9}$$

For $u \ge 1$ the same bound holds at $ue^{i\xi}$. It holds with constant $2\delta_h^{-1}$ for $1 - \delta_h/2 \le u \le 1$. An eigenvalue $\lambda = ue^{i\xi}$ satisfying the angular and physical conditions in (199.8) *obeys*

$$1 - u \ge \delta_h. \tag{199.10}$$

The adjoint compression has the corresponding resolvent bound at $e^{-i\xi}$.

Proof. Every normalized vector of $V_{R,h}$ satisfies the actual section function budget H_h. Also

$$\|\mathcal{U}_{R,z}f - e^{i\xi}f\|_2 = \|\mathcal{E}_{R,z,\xi}f\|_2.$$

Theorem 198.5 therefore supplies $d = b\rho^2/K_h$, reduced if necessary so that $d \le 1$. Apply Lemma 199.2 and choose $b_0 = b^2/3$. Finite dimensionality turns the lower singular-value bound into invertibility and (199.9).

For a normalized f and $u \ge 1$,

$$\begin{aligned}\|(ue^{i\xi} - A)f\|^2 - \|(e^{i\xi} - A)f\|^2 \\ = (u-1)(u+1-2\,\mathrm{Re}(e^{-i\xi}\langle f, Af\rangle)) \ge 0,\end{aligned}$$

where $\langle f, g\rangle = \int \overline{f} g \, \mathrm{d}\nu_R^*$ and $|\langle f, Af\rangle| \le 1$. For the inner radial interval use the reverse triangle inequality; the lower singular value is at least $\delta_h - (1-u) \ge \delta_h/2$. If $Af = ue^{i\xi}f$, the singular-value estimate at $e^{i\xi}$ gives $1 - u \ge \delta_h$. Finally take the adjoint of the inverse to obtain the last assertion. All statements are resolvent bounds for the specified compression, not a spectral assertion about $\mathcal{U}_{R,z}$ on the full L^2 space. □

Compatibility of scales. The finite-time estimate below and the local resolvent estimate have separate domains. Proposition 204.1 shows explicitly that even their improved versions cannot be concatenated on the target annulus using a mesh that makes the displayed time-n grid error vanish. Sections 204 and 205 give a direct alternative for averaged physical coefficients, not an unstated annular power theorem.

199.2. **Finite-time consistency with the unmodified record.** To relate this operator to the raw problem, we record an error bound which is independent of a spectral convergence assertion. Use $U_{j,R} = J_{j,R} - j\bar{G}_R$ and $N_{0,R} = 0$.

Lemma 199.4 (Variation after a collision-count cutoff)**.** *For integers $1 \le j \le n \le L$ the zero extension of*

$$v_{j,L}(x) = \mathbf{1}_{\{N_{j,R}(x) \le L\}} e^{-iz\cdot U_{j,R}(x)}, \quad x \in Y_R^*,$$

satisfies, uniformly in R,

$$\|v_{j,L}\|_\infty \le 1, \qquad \|v_{j,L}^0\|_{\mathrm{BV}} \le (1+|z|)\exp\{CL\log(2+L)\}. \tag{199.11}$$

Proof. Split according to the collision time $m = N_{j,R}$, $j \le m \le L$. Starting in Y_R^*, this event says that $\eta_R \circ T_R^m = 1$ and exactly $j-1$ of the indicators $\eta_R \circ T_R^i$, $1 \le i < m$, equal one. Write it as a disjoint union of the at most 2^{m-1} binary membership patterns, multiplied by the initial section indicator. Each pattern is a product of at most $m+1$ factors from $\eta_R \circ T_R^i$ and $1 - \eta_R \circ T_R^i$. On this event exact compensation gives $U_{j,R} = S_{m,R}h_R$.

Lemma 197.1 and the BV product inequality bound the variation of each pattern by $C(m+1)B_L$. The phase $e^{-iz\cdot S_m h_R}$ has supremum one and variation at most $C(1+|z|m)B_L$, by the BV chain inequality. Sum their product bounds over $m \le L$ and all patterns. The factors 2^L and the polynomial powers of L are absorbed by increasing the constant in B_L. All section and itinerary jumps, including the initial boundary, have been counted. No variation bound for an inverse-coarea density is used. □

Proposition 199.5 (An exact telescoping comparison of powers)**.** *There are $C, C_1, a, c > 0$ such that for $n \ge 2$, $L \ge n$,*

$$\begin{aligned}\left|\int_{Y_R^*} e^{-iz\cdot U_{n,R}}\,\mathrm{d}\nu_R^* - \langle 1, A_{R,h,z}^n 1\rangle\right| \\ \le Cn\left[e^{(an-cL)/2} + \sqrt{h(1+|z|)\exp\{C_1 L\log(2+L)\}}\right].\end{aligned} \tag{199.12}$$

In particular the matrix pairing approximates the actual characteristic function at every fixed return time when the cutoff and mesh are chosen so that the right side tends to zero.

Proof. Abbreviate $\mathcal{U}_{R,z}$, $P_{R,h}$ and $A_{R,h,z}$ by U, P, A. For $f \in V_{R,h}$ the exact identity

$$A^n f - PU^n f = \sum_{j=0}^{n-1} A^{n-1-j} PU(P-I)U^j f \tag{199.13}$$

follows by telescoping. All operators on the right are contractions. For $f = 1$, $U^j 1 = e^{-iz\cdot U_{j,R}}$ under the section law. The cumulative return tail gives

$$\|U^j 1 - v_{j,L}\|_2 = \nu_R^*(N_{j,R} > L)^{1/2} \le Ce^{(an-cL)/2}.$$

For $j = 0$, $P1 = 1$ and no error occurs. For $1 \le j < n$, Lemma 199.1 applied to Lemma 199.4 gives

$$\|(I-P)U^j 1\|_2 \le Ce^{(an-cL)/2} + C\sqrt{h(1+|z|)\exp\{C_1 L\log(2+L)\}}.$$

Take the norm in (199.13) and pair with the unit vector 1. Since $P1 = 1$, the pairing of $PU^n 1$ is the unmodified characteristic function. This proves the estimate. □

The remaining power and inversion estimates. The operator (199.4) supplies an explicit regular reconstruction class, a physical defect comparison, and a local peripheral resolvent. These are proved for every mesh meeting (199.8). Passing to the uncompressed raw record additionally requires a mesh and time scale satisfying (199.12), control of the other spectral arcs, and a quantitative conversion of the resulting contour estimates into powers. The finite-time consistency statement alone supplies none of these spectral limits. The full L^2 operator is unitary; its contraction properties must not be confused with the spectral behavior sought on an anisotropic space.

The physical annulus remains (198.15): $2n^{-99/200} \le |z| \le n^{-2/5}$, or $2n^{1/200} \le |\sqrt{n}\, z| \le n^{1/10}$ after rescaling. The compact nonzero and growing roof-frequency regions, the far roof tail, and the global critical/singular raw density extraction remain the other terms of Theorem 143.1. The finite-count cutoff is used only in the proved error comparison; the matrix entries and the characteristic function in that comparison retain the full actual return record. No conditioning event or denominator is replaced.

200. Complete edge extraction at a finite physical count

The individual critical-word calculation does not describe singular itinerary boundaries. We now include all such boundaries in a finite-count extraction. The argument integrates the full finite collision graph, rather than extending an inverse-coarea Jacobian through grazing. It gives an integrable Fourier remainder for every fixed count cutoff. The constants of this remainder retain their dependence on the cutoff; they will not be treated as uniform long-time estimates.

Throughout this section $R \in I$, $n \ge 1$, and $L \ge 1$ are fixed. The section is the actual Y_R^*, the measure is its probability $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, and $J_{n,R}$ is unchanged. Write $\ell = (k, m) \in \mathbb{Z}^2 \times \mathbb{N}$ for displacement and physical collision count. Define

$$\mu_{n,R}^{w,L} = (J_{n,R})_*(w\mathbf{1}_{\{N_{n,R}\le L\}}\nu_R^*). \tag{200.1}$$

The weight is not normalized. In particular $w = 1$ gives the restricted original law, not the law conditioned on the cutoff event.

200.1. **The whole finite collision graph.**

Definition 200.1 (Finite-record subanalytic weights)**.** A bounded complex weight is admissible at (n, L, R) if its real and imaginary parts, on every finite collision chart with $N_{n,R} \le L$, are globally subanalytic functions of the initial coordinates. Values on null collision singularities may be fixed arbitrarily. Denote its supremum bound by M_w. This class includes the weight one.

Here globally subanalytic has its usual real-geometric meaning; it is not a regularity assertion about every bounded-variation function. Finite Boolean combinations of subanalytic conditions on a finite list of actual return states and partial records up to the nth return belong to this class. Indeed, after fixing the successive return counts, each state is a specified finite collision iterate. There are only finitely many such choices below L. Products of bounded subanalytic initial, intermediate and terminal weights are allowed for the same reason. No assertion about their variation uniformly in the number of observations is made.

Lemma 200.2 (Finite graph and actual mixed density)**.** *The measure* (200.1) *has a compactly supported mixed density $p^{w,L}_{n,R}(\ell, t)$. It has only finitely many nonzero lattice components. For every fixed ℓ and admissible w, the real and imaginary parts of this density have constructible representatives in the sense of Cluckers–Miller: finite sums of products of globally subanalytic functions and logarithms of positive globally subanalytic functions. Each representative is analytic off a finite set.*

Proof. Use finitely many bounded rational angular charts and disjoint measurable chart pieces. Uniform finite horizon gives a finite set of possible next-disk labels. The finite physical graph in the proof of Lemma 197.1 specifies flight equations, reflection, positive entry roots and the exclusion of an earlier collision with any candidate disk. For a fixed word of at most L flights, the graph is semialgebraic in its initial and auxiliary variables. Projection gives semialgebraic finite collision records. Return membership adds the exact rectangle tests at every collision, not just the terminal test. Angular intervals become finite rational-chart intervals; at fixed R their endpoints are fixed real coefficients. Thus the events of exactly n returns, total count $m \le L$, and displacement k are finite unions of semialgebraic chart pieces. The collision probability in a rational angular coordinate has a bounded semialgebraic density. In particular, the construction includes the limiting grazing, competing-root and return-membership boundaries through the domains of integration.

On a regular word, the total flight time is analytic. Proposition 142.3 identifies its critical points as normal-to-normal nondegenerate minima; in particular its gradient is nonzero almost everywhere. That proposition is a physical collision-word statement and does not depend on which measurable section is used to restrict the initial state. Apply coarea on an exhaustion where the gradient is bounded below, multiply by the bounded weight, and then sum the finitely many words. This gives the density, with the normalization $1/(4\pi c_*)$ in (α, p) coordinates. Singular initial states and chart boundaries have zero initial measure. Restricting to Y^*_R does not import the old section's return counts. The time support lies in $[0, L\tau_{\max}]$; the count and displacement labels also range over finite sets.

We identify the density without bounding an inverse Jacobian near a singularity. On each chart let $b(x)$ be the product of its measure density and w, and let Ω_ℓ be its exact finite-record event. For real t, the cumulative component is a sum of integrals

$$P_\ell(t) = \sum_{\text{charts}} \int \mathbf{1}_{\Omega_\ell}(x) b(x) \mathbf{1}_{\{\tau_m(x) < t\}} \, \mathrm{d}x. \tag{200.2}$$

Extend the chart integrands by zero. They are globally subanalytic and integrable for every t. The stability-under-integration theorem [15, Theorem 1.3] makes P_ℓ constructible. The absolute continuity just proved identifies its almost-everywhere derivative with $p^{w,L}_{n,R}(\ell,\cdot)$.

For completeness, only one-variable differentiation is required here. Write a constructible representation as $\sum_i a_i(t)\prod_j \log b_{ij}(t)$, with $b_{ij} > 0$. A common finite subanalytic partition makes all generators analytic on each open interval. Their derivatives are subanalytic there, and the product rule, with $(\log b)' = b'/b$, gives a constructible derivative. Define its value arbitrarily at the finitely many endpoints. This is a constructible representative of the density. The same argument treats real and imaginary parts and proves the last assertion. □

Remark 200.3 (Relation to the retained physical edge calculation). The regular critical points and their exponentially small individual jumps were already classified in Proposition 142.3. They are not new consequences of the present lemma. The added conclusion is constructible structure for the *entire* finite-count component, including all boundaries of its exact integration domain. A singular boundary may produce a fractional power or a logarithm instead of a finite jump. The extraction below allows all of these forms.

200.2. **One-sided power–logarithm jets.**

Lemma 200.4 (A finite jet removes every second-derivative singularity). *Let $f \in L^1(\mathbb{R};\mathbb{C})$ have compact support and constructible real and imaginary parts. There is a finite set $S \subset \mathbb{R}$ such that f is analytic off S. For every $s \in S$ and sign $\epsilon \in \{-1,1\}$, on some interval $0 < x < d_s$ one has*

$$f(s+\epsilon x) = \sum_{k=0}^{K_s} (\log x)^k \sum_{j=j_s}^{\infty} c_{s,\epsilon,j,k} x^{j/q_s}, \qquad q_s \in \mathbb{N}. \tag{200.3}$$

The inner series are convergent Laurent series in x^{1/q_s}; their negative parts are finite. After collecting equal exponents, every nonzero coefficient with $j/q_s \le -1$ vanishes. Put

$$J_{s,\epsilon}(x) = \sum_{-1<j/q_s\le 1} \sum_{k=0}^{K_s} c_{s,\epsilon,j,k} x^{j/q_s} (\log x)^k. \tag{200.4}$$

Then the difference in (200.3) *after subtracting this finite jet has, for $a = 0,1,2$, a bound*

$$\left| \frac{d^a}{dx^a}\big(f(s+\epsilon x) - J_{s,\epsilon}(x)\big) \right| \le C_{s,\epsilon} x^{\beta_s - a}(1 + |\log x|^{K_s}), \qquad \beta_s > 1, \tag{200.5}$$

after decreasing d_s. An identically zero remainder satisfies the same assertion with any $\beta_s > 1$.

Proof. The convergence used here is part of the analytic-unit representation, not a conclusion obtained by differentiating an asymptotic expansion. In [15, Definitions 2.2–2.3], a strong unit is $U \circ \phi$, where ϕ is a bounded rational-monomial map and U is analytic on an open neighborhood of its image closure. At a one-variable endpoint moved to zero, each nonconstant bounded monomial has a nonnegative exponent. Clearing denominators turns ϕ into nonnegative integral powers of $x^{1/q}$, so Taylor expansion of U converges near that endpoint. A nonzero prepared center instead gives an ordinary analytic germ there. The rational prefactor permits only a finite Laurent principal part. For a positive generator the unit is positive at zero, and its logarithm has a convergent Taylor series. These observations also follow directly from [15, Theorem 2.4] applied to the subanalytic generators.

The one-variable specialization of constructible preparation [15, Theorem 3.11] writes a germ as a finite sum of rational powers, nonnegative integral logarithmic powers, and subanalytic units. At a one-sided endpoint, a strong subanalytic unit is analytic in finitely many bounded rational monomials. Choose a common denominator for these monomials and the rational prefactors.

Expanding the analytic units gives convergent Laurent series in the corresponding root of x. Equivalently, one may first take the convergent one-variable Puiseux germs of the subanalytic generators: if a positive generator is $x^r u(x^{1/q})$ with $u(0) > 0$, its logarithm is $r \log x + \log u(x^{1/q})$. This gives exactly (200.3), not merely an undifferentiated asymptotic formula. Finitely many generators and partition endpoints suffice for all of f. Include the endpoints of its support in S.

For the integrability assertion, take the smallest exponent having a nonzero coefficient polynomial in $\log x$. Its highest nonzero log power dominates all the lower log powers; every strictly higher rational exponent is smaller by a positive power of x times a fixed logarithmic power. If that smallest exponent were at most -1, the absolute value would not be integrable at zero. This contradicts $f \in L^1$. Repeating the argument, or simply using minimality, excludes all such exponents. Equal-exponent cancellations have already been performed. The argument works also for complex coefficients.

Only finitely many terms have exponent at most one. After they are removed, each of the finitely many log coefficients is either zero or $x^\beta a(x^{1/q_s})$ with $\beta > 1$ and a analytic at zero. Differentiate this representation twice. Bounded derivatives of a on a smaller root interval give (200.5), with the least of the remaining exponents as β_s. In particular, termwise differentiation is justified by the convergent analytic representation. □

Choose disjoint neighborhoods $(s - d_s, s + d_s)$, with $0 < d_s < 1$. Let $\chi_s(x)$ be a C^2 piecewise-polynomial cutoff on $[0, \infty)$, equal to one for $x \le d_s/2$ and zero for $x \ge d_s$. Its first two derivatives vanish at the transition endpoints. Define

$$e_f(t) = \sum_{s \in S} \sum_{\epsilon = \pm 1} \mathbf{1}_{\{\epsilon(t-s)>0\}} \chi_s(\epsilon(t-s)) J_{s,\epsilon}(\epsilon(t-s)), \qquad r_f = f - e_f. \tag{200.6}$$

The cutoffs need not be analytic; their piecewise-polynomial choice keeps them subanalytic and is sufficient for two weak derivatives.

Proposition 200.5 (Exact subtraction and its derivative budget)**.** *One has $e_f \in L^1(\mathbb{R})$ and $r_f \in W^{2,1}(\mathbb{R})$, both with compact support. The representative of r_f is C^1 and has value and first derivative zero at every point of S.*

The budget for its second derivative can be recorded without suppressing its endpoint constants. Put

$$\mathcal{J}(a, k, d) = \int_0^d x^{a-1} |\log x|^k \, \mathrm{d}x = d^a \sum_{j=0}^{k} \frac{k!}{(k-j)!} \frac{(-\log d)^{k-j}}{a^{j+1}} \quad (a > 0,\ 0 < d < 1).$$

On $0 < x < d_s/2$, the contribution to $\|r_f''\|_1$ is bounded by

$$C_{s,\epsilon} \left(\mathcal{J}(\beta_s - 1, 0, d_s/2) + \mathcal{J}(\beta_s - 1, K_s, d_s/2) \right). \tag{200.7}$$

On each transition interval the contribution is bounded by the integral of $|f''|$ there plus

$$\sum_{j,k} |c_{s,\epsilon,j,k}| C_{j/q_s,k} \, d_s^{j/q_s - 1} (1 + |\log d_s|^k).$$

On the remaining compact regular intervals it is the integral of $|f''|$. Thus all constants in this finite budget are attached to actual germs and actual regular intervals.

Proof. Every extracted exponent exceeds -1, so e_f is integrable. Near s, (200.5) gives zero traces for both r_f and its first derivative on both sides, and gives an integrable second derivative since $\beta_s - 2 > -1$. Away from these points all functions are C^2, with compact support. The matching traces show by integration by parts that no point mass or derivative of a point mass remains in the second distributional derivative. This proves $r_f \in W^{2,1}$ and the asserted representative.

Integrating (200.5) for $a = 2$ gives (200.7). The identity for $\mathcal{J}$ follows by repeated integration by parts, with the boundary term at zero equal to zero. On $[d_s/2, d_s]$, two differentiations of $\chi_s(x)x^\alpha(\log x)^k$ and the scaled cutoff bounds give the stated transition estimate. The remaining regular intervals are compact subsets of analytic pieces, so their displayed integrals are finite. These are bounds for the cancellation remainder, not separate integrals of the divergent second derivatives at s. □

Remark 200.6 (Why constants and slopes are extracted). Even when the classical second derivative of a one-sided constant or linear term vanishes, its zero extension has a distributional boundary term. The constant produces a derivative of a point mass, and the slope produces a point mass. Including both exponents 0 and 1 in (200.4) gives an L^1 second derivative, stronger than merely a finite-measure second derivative. Terms such as $x^{1/2}$, $\log x$, or $x \log x$ must likewise be extracted. The selection rule concerns the prepared raw density, not the variation of an initial-coordinate observable.

200.3. **All finite-count components at once.**

Theorem 200.7 (Complete finite-count raw extraction)**.** *For each fixed n, L, R and every admissible weight, there is an exact identity of finite complex measures*

$$\mu_{n,R}^{w,L} = E_{n,R}^{w,L} + Q_{n,R}^{w,L}, \tag{200.8}$$

where E has the finite power–logarithm density (200.6) *in each nonzero lattice component, and the density $r_{n,R}^{w,L}(\ell, \cdot)$ of Q belongs to $W^{2,1}(\mathbb{R})$. In particular*

$$A_j(n, L, R, w) = \sum_\ell \|\partial_t^j r_{n,R}^{w,L}(\ell, \cdot)\|_1 < \infty \quad (j = 0, 2). \tag{200.9}$$

Its mixed Fourier transform satisfies

$$\int_{\mathbb{T}^3 \times \mathbb{R}} |\widehat{Q}| \, \mathrm{d}\omega \le 4(2\pi)^3 \sqrt{A_0 A_2}, \tag{200.10}$$

$$\int_{\mathbb{T}^3 \times \{|b| > B\}} |\widehat{Q}| \, \mathrm{d}\omega \le \frac{2(2\pi)^3}{B} A_2 \qquad (B > 0). \tag{200.11}$$

The construction includes every regular critical, singular itinerary and image-boundary contribution to the restricted measure. It does not assume that its only singularities are the retained Morse jumps.

Proof. Apply Lemma 200.2 and Proposition 200.5 to each of the finitely many lattice components. Alternatively apply them to every chart-word subdensity and then sum; that version records every word's derivative budget separately. In either construction all actual finite graph pieces are integrated in (200.2). Boundary points themselves have zero initial measure, but their effects on one-sided density limits are included in these integrals. The prepared germs of the whole density include those effects. There is no assumption that an inverse coarea map extends smoothly across such a boundary.

For a component r_ℓ, distributional integration by parts gives

$$|\widehat{r}_\ell(b)| \le \min\{\|r_\ell\|_1, |b|^{-2}\|r_\ell''\|_1\}.$$

The integral on the real line of $\min\{a, d/b^2\}$ is $4\sqrt{ad}$ for $a, d > 0$. Sum over ℓ, apply Cauchy–Schwarz, and integrate the torus of volume $(2\pi)^3$ to obtain (200.10). A zero norm is handled by continuity, or by observing that a compactly supported affine L^1 function is zero. Integration only over $|b| > B$ gives the other bound. □

The constants in (200.9) are finite for each fixed packet. Neither the preparation theorem nor compactness of I by itself bounds them uniformly in R as n, L grow. Small cutoff widths, approaching singular values, jet coefficients and the number of pieces all occur in Proposition 200.5. Their total contribution must be estimated before using (200.11) in a long-time frequency splice.

201. Count-localized inversion for the unmodified law

The collision count is one of the observed lattice coordinates. Thus a finite physical-count restriction is exact, not approximate, at every lattice point whose count lies below the cutoff. This permits the finite extraction of Section 200 to enter an identity for the original raw density on a central window. The high-count measure remains in that identity through an explicitly controlled low-frequency convolution.

Write $\mu_{n,R} = (J_{n,R})_*\nu_R^*$, and let $p_{n,R}$ denote its mixed density. The density exists by the same countable-word coarea argument used in Lemma 200.2; the nonnegative word masses sum to one. For a cutoff L, put

$$\mu_{n,R}^L = (J_{n,R})_*(\mathbf{1}_{\{N_{n,R}\le L\}}\nu_R^*), \qquad \mathsf{T}_{n,R}^L = \mu_{n,R} - \mu_{n,R}^L.$$

In this section E^L, Q^L are the exact measures constructed in Theorem 200.7 for $w = 1$, with densities e^L, r^L. We keep the dependence of these choices on n, L, R.

201.1. **An exact support property.**

Lemma 201.1 (Central labels require only a linear count cutoff)**.** *For $\ell = (k, m)$ with $m \le L$,*

$$p_{n,R}(\ell, t) = p_{n,R}^L(\ell, t) = e_{n,R}^L(\ell, t) + r_{n,R}^L(\ell, t) \quad \text{for almost every } t. \tag{201.1}$$

For $A < \infty$ define the central window

$$\mathcal{W}_{n,R,A} = \{(\ell, t) \in \mathbb{Z}^3 \times \mathbb{R} : |(\ell, t) - n\bar{G}_R| \le A\sqrt{n}\}.$$

If $\lambda > c_*^{-1} + 1$ and $L_n = \lceil \lambda n \rceil$, then (201.1) *holds throughout this window for all $n \ge \max\{1, A^2\}$, in the mixed almost-everywhere sense. Moreover, with the constants of Theorem 134.1,*

$$\|\mathsf{T}_{n,R}^L\|_{\mathrm{TV}} \le Ce^{an-cL}, \qquad \sup_\omega |\widehat{\mathsf{T}_{n,R}^L}(\omega)| \le Ce^{an-cL}. \tag{201.2}$$

Proof. The restriction $N_{n,R} \le L$ is exactly the restriction that the third lattice coordinate is at most L. The measures therefore agree on each such lattice fiber, and their densities agree almost everywhere there. The finite extraction supplies the second equality. On the displayed window the count satisfies $m \le n/c_* + A\sqrt{n} \le (c_*^{-1} + 1)n$ for $n \ge A^2$. This proves the asserted coverage. The tail has nonnegative mass equal to $\nu_R^*(N_{n,R} > L)$; use the cumulative-return tail and then the basic Fourier bound by total variation. No pointwise density bound for the high-count measure is deduced from its small mass. □

201.2. **Four terms in an exact inversion.**

Choose the cutoff χ in (144.1) also to satisfy $0 \le \chi \le 1$. Use the proved growing band, namely

$$a_n = n^{1/200}, \qquad B_n = n^{-99/200}, \qquad \chi_n(\omega) = \chi(\omega/B_n).$$

The support has physical radius $2n^{-99/200}$ and rescaled radius $2n^{1/200}$. The kernel K_n is the mixed inverse transform of χ_n, with the $(2\pi)^{-4}$ convention of (144.1).

Theorem 201.2 (An exact finite extraction on a raw central window)**.** *Take L_n as in Lemma 201.1. On $\mathcal{W}_{n,R,A}$, for all sufficiently large n,*

$$\begin{aligned} p_{n,R} = {} & K_n * \mu_{n,R} + (e_{n,R}^{L_n} - K_n * E_{n,R}^{L_n}) \\ & + \mathcal{F}^{-1}[(1 - \chi_n)\widehat{Q}_{n,R}^{L_n}] - K_n * \mathsf{T}_{n,R}^{L_n}. \end{aligned} \tag{201.3}$$

The equality holds almost everywhere. The residual inverse transform is absolutely convergent, by the proved finite extraction, and

$$n^2\|K_n * \mathsf{T}^L_{n,R}\|_\infty \le Cn^{1/50}e^{an-cL}. \tag{201.4}$$

For $L_n = \lceil \lambda n \rceil$ with $\lambda > c_*^{-1} + 1$, one also has, for every $P > 0$,

$$n^2 \sup_{\mathcal{W}_{n,R,A}} |K_n * \mathsf{T}^{L_n}_{n,R}| \le C_{A,\lambda,P} n^{-P}. \tag{201.5}$$

This last estimate does not require $c\lambda > a$. If that stronger inequality is imposed, (201.4) *additionally gives exponential smallness on the entire mixed space.*

Proof. Theorem 200.7 gives $\widehat{Q}^L \in L^1(\mathbb{T}^3 \times \mathbb{R})$, so its inverse is the continuous representative of the residual density. Splitting that inverse with χ_n gives, on the full mixed space,

$$p^L = K_n * \mu^L + (e^L - K_n * E^L) + \mathcal{F}^{-1}[(1-\chi_n)\widehat{Q}^L].$$

Now substitute $\mu^L = \mu - \mathsf{T}^L$ and use the exact fiber agreement of Lemma 201.1 on the window. Every convolution is defined because its measure is finite and K_n is bounded. This proves (201.3). Finally $\|K_n\|_\infty \le CB_n^4$ and $n^2 B_n^4 = n^{1/50}$. Combine this with (201.2) to obtain (201.4). The power $n^{1/50}$ is retained; the cutoff error is not declared to be a pointwise tail-density bound.

For the sharper central assertion, the count distance between a point of the central window and the support of T^{L_n} is at least $(\lambda - c_*^{-1} - 1)n$ when $n \ge A^2$. The Schwartz bound in (144.1), and $\|\mathsf{T}^{L_n}\|_{\mathrm{TV}} \le 1$, therefore give, for every integer $M \ge 1$,

$$n^2 \sup_{\mathcal{W}_{n,R,A}} |K_n * \mathsf{T}^{L_n}| \le C_{A,\lambda,M} n^{1/50 - 101M/200}.$$

Choose M large enough to obtain (201.5). This is decay of the proof kernel across an exactly separated count support, not decay of the original physical Fourier transform. □

Let

$$\mathfrak{g}_R(y) = \frac{1}{(2\pi)^2\sqrt{\det D_R}} \exp\Big(-\tfrac{1}{2} y^{\mathsf{T}} D_R^{-1} y\Big), \qquad y = \frac{(\ell,t) - n\bar{G}_R}{\sqrt{n}}.$$

Uniform positive definiteness was proved in Theorem 150.2. Define the actual local edge correction and finite-band residual integral by

$$\mathcal{E}^L_{n,R,A} = \operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |e^L_{n,R} - K_n * E^L_{n,R}|,$$

$$\mathcal{C}^L_{n,R}(B) = \int_{\mathbb{T}^3 \times [-B,B]} |1-\chi_n(\omega)|\,|\widehat{Q}^L_{n,R}(\omega)|\,\mathrm{d}\omega.$$

The first quantity may be infinite until the local density germs are estimated. Its definition preserves possible cancellation in the exact edge correction; separate bounds for its two terms also suffice.

Corollary 201.3 (A quantitative raw error with the finite branch sum exposed)**.** *Let $\lambda > c_*^{-1} + 1$ and $L_n = \lceil \lambda n \rceil$. For $P > 0$, $B \ge 1$, sufficiently large n, and every $R \in I$, the following extended-real inequality holds:*

$$\begin{aligned}
&\operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |n^2 p_{n,R}(\ell,t) - \mathfrak{g}_R(y)| \\
&\quad \le Cn^{-3/280}\sqrt{\log(2+n)} + Ce^{-c_0 n^{1/100}} + C_{A,\lambda,P} n^{-P} \\
&\qquad + n^2 \mathcal{E}^{L_n}_{n,R,A} + (2\pi)^{-4} n^2 \mathcal{C}^{L_n}_{n,R}(B) + \frac{n^2}{\pi B} A_2(n, L_n, R, 1).
\end{aligned} \tag{201.6}$$

Here C, c_0 are uniform in the radius; the displayed A_2 and edge terms retain their actual parameter dependence. Thus the raw central local limit follows if, for some $B = B'_n \geq 1$, the last three terms tend to zero uniformly in R for each fixed A.

Proof. In the Fourier integral for $K_n * \mu_{n,R}$, rescale $v = \sqrt{n}\,\omega$ and remove the deterministic mean phase. Theorem 145.3 bounds its difference from the cutoff Gaussian integral by $Cn^{-3/280}\sqrt{\log(2+n)}$ after multiplication by n^2. The complement of the cutoff Gaussian has $|v| \geq n^{1/200}$ and is bounded by $Ce^{-c_0 n^{1/100}}$ using uniform ellipticity. These estimates are uniform in the evaluation point, since its Fourier factor has modulus one.

Apply (201.3). Its edge term gives $n^2\mathcal{E}$, and its high-count convolution gives the third term in (201.6) by (201.5). Split the absolutely integrable residual at $|b| = B$. The part inside has precisely the stated finite-band bound. For $B \geq 1$ and sufficiently large n, $\chi_n = 0$ on $|b| > B$. Equation (200.11) bounds this part by $2(2\pi)^3 A_2/B$ before the inverse-transform factor. Since $(2\pi)^{-4}2(2\pi)^3 = 1/\pi$, the final constant is as displayed. This proves the inequality and its sufficient convergence criterion. □

201.3. **Finite bands and weighted observations.**

Proposition 201.4 (The exact cutoff and a finite-band error budget)**.** *For any frequency set Ω_n of finite volume V_n,*

$$\int_{\Omega_n} |\widehat{\mu}_{n,R} - \widehat{\mu}_{n,R}^{L_n}|\,\mathrm{d}\omega \leq CV_n e^{an - cL_n}.$$

If $V_n \leq Ce^{\kappa n}$, the choice

$$\lambda > \max\{c_*^{-1} + 1, (a + \kappa + \delta)/c\}, \qquad L_n = \lceil \lambda n \rceil,$$

gives an $O(e^{-\delta n})$ bound and still agrees exactly with the original raw density on every fixed central window for large n. For a bounded complex weight w, the same two statements hold with an extra factor $\|w\|_\infty$ in the tail estimate. When w is admissible in Definition 200.1, the extraction and the identity (201.3) *also hold for that weighted measure.*

Proof. Integrate (201.2) over Ω_n. The strict inequality in the slope leaves a positive exponential slack. The support argument is independent of the weight, and the total variation of the omitted weighted measure is at most its supremum norm times the omitted probability. Apply Theorem 200.7 to the weighted finite measure and repeat the algebra of the inversion identity. □

The last proposition is compatible with the stronger polynomial-weight truncation estimates in Proposition 134.2. It is stated here for bounded weights because that is the class used in the complete finite extraction. For a single initial or actual-return mark in the inherited BV class, the corresponding central Gaussian term has the already proved marked estimate, whenever the weight also satisfies the finite-record subanalytic condition. A finite multiple-time subanalytic indicator has the structural extraction, but is not thereby proved to have that marked central estimate. In every case the weight and its probability denominator remain unchanged.
The remaining long-time estimates. The structural finite extraction, its $W^{2,1}$ residual, the exact central support agreement and the rapidly decaying count correction are now conclusions. The quantities $\mathcal{C}_{n,R}^{L_n}(B)$, $A_2(n, L_n, R, 1)$ and $\mathcal{E}_{n,R,A}^{L_n}$ are the specific remaining estimates in (201.6). The first still contains the intervening annulus, compact nonzero torus frequencies and the roof frequencies below B; its full complement is not bounded by a one-step function defect. The second has the actual finite-count jet budget of Proposition 200.5, not a presumed polynomial growth rate. The third requires local control of singular values and coefficients on the original central windows. The same quantitative issues, with the actual indicators, remain in the exact physical conditioning application. The identity retains the original raw density throughout; no smoothing or conditioning replaces it.

202. Exponential finite-record complexity and reconstruction

The finite graph used in Lemma 197.1 has both its number of variables and its number of polynomial tests linear in the collision length. Retaining the binomial form of the component bound, rather than its fixed-dimension asymptotic estimate, gives an exponential variation bound. This removes the additional logarithm from the regularized tower budget.

202.1. **The finite binomial bound.**

Lemma 202.1 (Linear-size sign descriptions)**.** *Fix $a_0, a_1, d \geq 1$. Suppose a family of lifted graphs of length L uses at most $a_0(L+1)$ real variables and $a_1(L+1)$ polynomial sign tests of degrees at most d, and has at most a_0^{L+1} discrete alternatives. The number of connected components of any Boolean combination of those sign conditions, summed over the alternatives, is at most Ce^{CL}. The same bound holds after coordinate projection and after fixing a bounded number of coordinates or adjoining a bounded number of fixed-degree tests. The constants depend on a_0, a_1, d, not on the real coefficients.*

Proof. Use the finite bound in [14, Section 3.2, (3.3)], with zeroth Betti number and ambient variety the full space. If s is the number of tests and $k \geq 1$ the number of variables, it bounds the sum of the numbers of connected components of all realizable sign conditions by

$$d(2d-1)^{k-1} \sum_{j=0}^{k} \binom{s}{j} 4^j \leq d(2d-1)^{k-1} 5^s. \tag{202.1}$$

Here $\binom{s}{j} = 0$ for $j > s$. The right side is exponential in $k+s$, so it is Ce^{CL} in the stated range. A Boolean combination is a union of realizable sign conditions. Each of its connected components contains a connected component of one of those conditions; thus its number is no larger than their total. Summing over the exponentially many discrete alternatives preserves an exponential bound.

The image of a connected set under a coordinate projection is connected. A union of m connected sets has at most m connected components. This proves the projection assertion without using quantitative elimination of quantifiers. Fixed coordinates and additional tests only change the linear bounds by constants. A bounding-ball inequality can likewise be adjoined. In zero dimension the assertion is immediate. We use the finite inequality in (202.1), not an $s \to \infty$ expansion with constants depending on k. □

Theorem 202.2 (Exponential variation of physical finite records)**.** *There is $C_{\mathrm{g}} \geq 1$, uniform in $R \in I$, such that all the conclusions of Lemma 197.1 hold with*

$$B_L = C_{\mathrm{g}} e^{C_{\mathrm{g}} L} \tag{202.2}$$

in place of $\exp\{CL\log(2+L)\}$. In particular, for $|j| \leq L$,

$$\|\chi \circ T_R^j\|_{\mathrm{BV}} + \|h_R \circ T_R^j\|_{\mathrm{BV}} + \|\eta_R \circ T_R^j\|_{\mathrm{BV}} \leq B_L, \qquad \|g \circ T_R^j\|_{\mathrm{BV}} \leq C\|g\|_{C^1} B_L.$$

The same exponential bound holds for every binary section-membership word of length at most $2L+2$, extended by zero off its domain. The constants are independent of the moving rectangle coefficients.

Proof. Retain the first-admissible-flight graph (197.3), including its no-earlier-hit conditions, incidence signs and section-membership tests. Its degree is bounded independently of L, whereas its variable and test counts are $O(L)$. The number of label and membership alternatives is exponential in L. Fixing one initial coordinate and a superlevel of a bounded record component adds only a bounded number of variables and tests. Lemma 202.1 therefore bounds the number

of interval components of every projected slice superlevel by Ce^{CL}. A finite angular atlas and its seam contributions change only the constants.

For a bounded scalar function u on an interval, with $|u| \le M$, one-dimensional coarea gives

$$|Du| \le \int_{-M}^{M} \mathrm{Per}\{u > t\}\,\mathrm{d}t.$$

A superlevel with at most b interval components has perimeter at most $2b$, with the endpoint convention incorporated by zero extension. Thus the slice variation is at most $4Mb$. Integrating in the other initial coordinate proves both first distributional derivative bounds. The statement for binary words follows in exactly the same way by adjoining all their membership tests to the finite graph; it does not require differentiating the word's boundary pointwise.

For smooth g, cover the angular circle by the same fixed regular charts and apply the BV chain and product inequalities to the bounded coordinate record $\chi \circ T_R^j$. The trace terms at the finitely many chart seams have the same component bound. This gives the stated C^1 composition estimate. Negative iterates use the unique regular lifted fiber with the terminal state fixed, as in the original graph. All coefficient dependence enters the sign conditions, to which (202.1) applies uniformly. □

202.2. **The regularized tower with one logarithm.** Use $\mathsf{L}_{R,z,\xi}$ and $\mathcal{E}_{R,z,\xi}$ from Section 198. Set $M_f = \|f\|_\infty$ and $V_f = \|f^0\|_{\mathrm{BV}}$ for a section function. The constants $a_{\mathrm{t}}, c_{\mathrm{t}} > 0$ below are chosen so that the cumulative-return tail reads

$$\nu_R^*\{N_{m,R} > L\} \le Ce^{a_{\mathrm{t}}m - c_{\mathrm{t}}L}. \tag{202.3}$$

The one-return tail may have different positive constants.

Proposition 202.3 (Exponential regularization budget)**.** *For $L \ge 1$ and $0 < \epsilon < 1/2$, the finite-age approximant of Lemma 198.4, denoted here by Q, satisfies*

$$\|Q - \mathsf{L}_{R,z,\xi} f\|_1 \le C\{L\epsilon V_f + M_f e^{-cL}\}, \tag{202.4}$$

$$\|Q\|_\infty \le M_f, \qquad \|Q\|_{\mathrm{BV}} \le CM_f e^{CL}(1 + \epsilon^{-1} + |z|). \tag{202.5}$$

Its squared L^2 error is bounded by $2M_f$ times the right side of (202.4)*. These assertions hold for every real return phase ξ.*

Proof. The age truncation, initial smoothing, exact top convention and multiplication by $b_{R,\xi}$ are unchanged. In particular their L^1 errors are precisely those of Proposition 197.3 and Lemma 198.4. For age $j < L$, the level indicator is a binary backward membership word. Apply Theorem 202.2 to that word, the smoothed initial function evaluated at T_R^{-j}, and the accumulated phase. The product and chain inequalities bound the variation by $CM_f e^{Cj}(1 + \epsilon^{-1} + j|z|)$. Summing the ages and absorbing polynomial factors in e^{CL} gives (202.5). The age levels are disjoint, so the supremum bound is M_f, not LM_f. Multiplication by $b_{R,\xi}$ costs only its uniform BV norm. Finally both the lift and its approximant have modulus at most M_f, giving the L^2 assertion. No regularity of the full lift is presumed. □

Theorem 202.4 (A single-logarithm joint return estimate)**.** *Put*

$$\rho = (|z|^2 + \xi^2)^{1/2}, \qquad \Lambda = 1 + \log(H/\rho), \qquad H \ge 2, \quad \xi \in [-\pi, \pi].$$

There are uniform $a, b, \rho_0 > 0$ such that $0 < \rho \le \rho_0$ and $\rho^2\Lambda \le a$ imply

$$|q| = 1, \quad \|q^0\|_{\mathrm{BV}} \le H \quad \Longrightarrow \quad \|\mathcal{E}_{R,z,\xi} q\|_{L^1(\nu_R^*)} \ge b\rho^2, \tag{202.6}$$

$$\|f\|_{L^2(\nu_R^*)} = 1, \quad M_f + V_f \le H \quad \Longrightarrow \quad \|\mathcal{E}_{R,z,\xi} f\|_{L^2(\nu_R^*)} \ge b\rho^2/\Lambda. \tag{202.7}$$

In the second assertion f may vanish. The domain contains the pure spectral direction and the shifted-frequency cancellation line.

Proof. Write $\sigma = c_*\xi$ and $w = z - \sigma e_3$. On a sufficiently small fixed joint ball,

$$c_1\rho \le t = (|w|^2 + \operatorname{dist}(\sigma, 2\pi\mathbb{Z})^2)^{1/2} \le c_2\rho.$$

This follows from the invertible linear frequency change in (198.9); it does not divide by $|w|$.

For a unit section phase the exact lift has modulus one on the full collision cylinder. Choose an L^1 error $\eta = \epsilon_0\rho^2$, with $\epsilon_0 > 0$ fixed later. In Proposition 202.3, take $L \le C(1 + |\log\eta|)$ large enough for the tail and $\epsilon = \eta/(CLH)$. Circle truncation, with the same coarea threshold as in the proof of Theorem 198.5, gives a unit collision phase v with

$$\|v - \mathsf{L}_{R,z,\xi}q\|_1 \le C\eta, \qquad 1 + \log(2 + \|v\|_{\mathrm{BV}}) \le C(1 + |\log\epsilon_0|)\Lambda.$$

Indeed the logarithm in (202.5) is at most $CL + \log H + \log(1+\epsilon^{-1}) + C$, and $\log L \le C(1+|\log\eta|)$. Apply Lemma 198.2, and use the exact L^1 defect identity (198.8). The change of defect costs at most twice the approximation error. Choose ϵ_0 small enough to absorb it and then reduce the joint smallness constants. This proves (202.6).

For a normalized complex section function, set $Q_* = \mathsf{L}_{R,z,\xi}f$ and $S = \|Q_*\|_2$. The exact norm identity gives $\sqrt{c_*} \le S \le H$. If $e = \|\mathcal{E}_{R,z,\xi}f\|_2$, the collision defect of Q_*/S is at most e. Choose

$$\eta = \epsilon_0\rho^2/\Lambda, \qquad L \le C(1 + \log(H/\eta)), \qquad \epsilon = \eta^2/(CH^2L),$$

with the fixed constants large enough to make the normalized L^2 error at most η. Normalizing the approximant gives $\|v\|_2 = 1$ and

$$\|v - Q_*/S\|_2 \le 2\eta,$$
$$1 + \log(2 + \|v\|_\infty + \|v\|_{\mathrm{BV}}) + |\log t| \le C(1 + |\log\epsilon_0|)\Lambda.$$

The last inequality follows from $\log(H/\eta) = \log H + 2\log(1/\rho) + \log\Lambda + \log(1/\epsilon_0)$ and $\log\Lambda \le \Lambda$. Proposition 198.3 consequently gives

$$e + 4\eta \ge \frac{b_0c_1^2\rho^2}{C(1 + |\log\epsilon_0|)\Lambda}.$$

Since $\epsilon_0(1 + |\log\epsilon_0|) \to 0$, choose ϵ_0 first so that the approximation term is absorbable. Only then reduce a, ρ_0 to meet the collision budget. This proves (202.7) with no circular accuracy choice. □

Corollary 202.5 (Peripheral annuli and compressed resolvents)**.** *For $H \le H_0n^\zeta$, the preceding theorem applies on*

$$2n^{-99/200} \le |z| \le n^{-2/5}, \qquad |\xi| \le a_0/\sqrt{\log(2+n)}$$

for sufficiently small fixed $a_0 > 0$ and sufficiently large n. Its complex-function lower bound is $c(|z|^2 + \xi^2)/\log(2+n)$. The rescaled physical radii are still $2n^{1/200}$ and $n^{1/10}$.

For the exact section-grid compression $A_{R,h,z}$ in Section 199, let $H_h = C/h$ and $\Lambda_h = 1 + \log(H_h/\rho)$. On the joint region $0 < \rho \le \rho_0$, $\rho^2\Lambda_h \le a$,

$$\|(e^{i\xi} - A_{R,h,z})^{-1}\| \le C\Lambda_h^2/\rho^4. \tag{202.8}$$

The same bound, with twice the constant, holds when the resolvent parameter lies within $c\rho^4/\Lambda_h^2$ of $e^{i\xi}$.

Proof. On the annulus $\Lambda \le C\log(2+n)$, and $|z|^2\Lambda \to 0$ uniformly. Choosing a_0 small controls the spectral contribution to the budget. For a normalized grid vector, Lemma 199.1 gives the budget H_h. If its compressed residual is $\varepsilon \le 1$, Lemma 199.2 bounds the squared physical defect by 3ε. Theorem 202.4 therefore gives $\varepsilon \ge c\rho^4/\Lambda_h^2$. Residuals larger than one satisfy the same lower bound after changing c. This is a smallest-singular-value estimate, hence implies (202.8). Moving the scalar resolvent parameter by at most half that lower bound preserves half the singular value by the triangle inequality. No normality assumption is used. □

Proposition 202.6 (Improved finite-time consistency)**.** *The finite-time comparison of Proposition 199.5 improves to*

$$\left|\int_{Y_R^*} e^{-iz\cdot U_{n,R}}\,\mathrm{d}\nu_R^* - \langle 1, A_{R,h,z}^n 1\rangle\right| \le Cn\left\{e^{(a_{\mathrm{t}}n - c_{\mathrm{t}}L)/2} + \sqrt{h(1+|z|)e^{CL}}\right\}, \qquad L \ge n. \quad (202.9)$$

The Fourier sign and Hilbert pairing are those of that proposition; $U_{n,R} = J_{n,R} - n\bar{G}_R$ is the actual centered record.

Proof. For each $j \le n$, restrict the actual j-return phase to $\{N_{j,R} \le L\}$. Resolve this event by its binary section word up to the last collision, and use Theorem 202.2. Products of word indicators and the phase chain rule give variation at most $C(1+|z|)e^{CL}$; the number of words and all factors polynomial in L are absorbed in the exponential. The discarded part has L^2 norm at most $Ce^{(a_{\mathrm{t}}n - c_{\mathrm{t}}L)/2}$. Cellwise BV approximation on the fitted grid gives the second term inside braces. Substitute these two estimates in the exact projection power telescoping identity used in the cited proposition and sum its n terms. Neither the original law nor the matrix entries are truncated in (202.9). □

These estimates improve the reconstruction and consistency constants. They do not assert a mesh-uniform power bound for the uncompressed operator: the consistency scale and the local resolvent scale must still be reconciled, and the remaining peripheral arcs must be treated. First variation of a collision record is also distinct from the second derivative sum of the raw densities in Theorem I.

203. Several observations in a logarithmic return window

A product of observations at different actual returns is not covered by assuming a variation bound for one of its factors. We prove that such a product has a controlled approximation when its return window grows logarithmically. The final event and the record remain unchanged; the collision cutoff occurs only in the approximation used in the proof.

Let $0 \le \ell_1 < \cdots < \ell_q \le m$ and let $u_1, \dots, u_q$ be complex section functions with $\|u_j\|_\infty \le 1$ and zero extensions $u_j^0 \in \mathrm{BV}(M)$. Set

$$W_R(y) = \prod_{j=1}^{q} u_j((F_R^*)^{\ell_j} y), \qquad V = 1 + \sum_{j=1}^{q} \|u_j^0\|_{\mathrm{BV}}, \qquad \alpha_R = \int_{Y_R^*} W_R\,\mathrm{d}\nu_R^*. \quad (203.1)$$

All the data may depend on n, R. Empty products are one. No factorization of α_R into individual observation masses is asserted. For $0 \le k \le n-m$ put

$$\mathcal{C}_{n,k,R}^{W}(v) = \int_{Y_R^*} W_R((F_R^*)^k x) e^{iv\cdot U_{n,R}(x)/\sqrt{n}}\,\mathrm{d}\nu_R^*(x), \qquad U_{n,R} = J_{n,R} - n\bar{G}_R. \quad (203.2)$$

This convention uses the normalized section probability. Thus all the observations occur at their actual return states between returns k and $k+m$, including a terminal window when $k = n-m$.

203.1. **A controlled approximation to the whole window.**

Lemma 203.1 (Truncated and smoothed window weights)**.** *There is $C_{\mathrm{w}} \geq 1$, uniform in R, m, q, L, such that for integers $L \geq \max\{m, 1\}$ and $0 < \epsilon < 1/2$ there is a section function $W_{L,\epsilon}$ with*

$$\|W_{L,\epsilon}\|_{\infty} \leq 1, \qquad \|(W_{L,\epsilon})^0\|_{\mathrm{BV}} \leq C_{\mathrm{w}}(1 + q\epsilon^{-1})e^{C_{\mathrm{w}}L}, \tag{203.3}$$

$$\|W_R - W_{L,\epsilon}\|_{L^1(\nu_R^*)} \leq C\{\epsilon V + e^{a_{\mathrm{t}}m - c_{\mathrm{t}}L}\}. \tag{203.4}$$

The tail constants are those of (202.3)*.*

Proof. Let $v_j = \mathsf{S}_\epsilon u_j^0$ and define, on the actual section,

$$W_{L,\epsilon}(y) = \mathbf{1}_{\{N_{m,R}(y) \leq L\}} \prod_{j=1}^{q} v_j((F_R^*)^{\ell_j} y).$$

Extend this function by zero to M. Its supremum is at most one. Telescoping products and invariance of the actual first-return map give

$$\int_{Y_R^*} \left| \prod_j u_j \circ (F_R^*)^{\ell_j} - \prod_j v_j \circ (F_R^*)^{\ell_j} \right| \mathrm{d}\nu_R^* \leq \sum_j \|u_j - v_j\|_{L^1(\nu_R^*)} \leq C\epsilon V.$$

This use of invariance is valid before imposing the count cutoff. The cutoff changes the product by at most its event probability, which is bounded by (202.3). This proves (203.4).

To prove the variation assertion, resolve $\{N_{m,R} \leq L\}$ by its first L binary section-membership symbols. There are at most 2^L possibilities, with fixed initial membership equal to one. On each such word the collision indices of the first m actual returns are fixed integers between zero and L. Hence the product above is a product of q smooth functions $v_j \circ T_R^{t_j}$, multiplied by the word indicator and the initial section indicator. By Theorem 202.2 and $\|v_j\|_{C^1} \leq C\epsilon^{-1}$, the BV product inequality bounds its variation by $C(1 + q\epsilon^{-1})e^{CL}$ on each word. Summing the words and enlarging the exponential constant gives (203.3). All internal word jumps and the initial section trace are included. The summands have disjoint supports, which is why their number does not enter the supremum bound. Null singular sets are assigned the same convention as in the finite-record theorem. □

203.2. **A variable central band.**

Lemma 203.2 (The marked estimate on a specified subband)**.** *For $0 \leq \theta \leq 1/200$, define*

$$s_0(\theta) = 1/28 - 5\theta, \qquad s_1(\theta) = 1/14 - 4\theta.$$

For a bounded complex section function a, with zero-extension variation V_a, supremum M_a, and mass $\alpha_a = \int a \, \mathrm{d}\nu_R^*$, the marked transform under ν_R^* satisfies

$$\int_{|v| \leq 2n^{\theta}} \left| \int a((F_R^*)^k x) e^{iv \cdot U_{n,R}/\sqrt{n}} \, \mathrm{d}\nu_R^* - \alpha_a e^{-v^{\mathsf{T}} D_R v/2} \right| \mathrm{d}v$$
$$\leq C\{M_a n^{-s_0(\theta)} \sqrt{\log(2+n)} + V_a n^{-s_1(\theta)}\}. \tag{203.5}$$

The constants may be chosen on the full displayed interval of θ, and uniformly in $0 \leq k \leq n$. The physical radius is $2n^{-1/2+\theta}$, not $2n^{\theta}$.

Proof. Use the insertion $c_*^{-1}a^0$ in Lemma 146.4 and retain $\delta = \frac{1}{4}n^{-1/14}$ and a logarithmic spectral block length. Replace only the rescaled ball by $|v| \leq 2n^{\theta}$. The observable-smoothing term

integrates to $CM_a n^{5\theta-1/28}\sqrt{\log(2+n)}$ and the insertion-smoothing term to $CV_a n^{4\theta-1/14}$. The remaining polynomial margins are

$$1/14-6\theta,\quad 3/14-5\theta,\quad 1/14-7\theta,\quad 1/5-4\theta,\quad 1/5-5\theta,\quad 1/2-5\theta,\quad 1-6\theta.$$

They are strictly larger than $s_0(\theta)$ throughout $0\le\theta\le 1/200$. Their logarithmic factors are therefore absorbed uniformly; the complementary power is smaller still. The two analytic smallness conditions hold on the largest ball and hence on all the smaller balls. This proves (203.5). At $\theta=1/200$ it gives exactly the exponents $3/280$ and $9/175$ of Theorem C. Thus no inherited central estimate is replaced. □

Theorem 203.3 (Central integrals with a full return window)**.** *For the weight* (203.1)*, $n\ge 2$, $0\le k\le n-m$, $L\ge\max\{m,1\}$, $0<\epsilon<1/2$, and $0\le\theta\le 1/200$,*

$$\begin{aligned}&\int_{|v|\le 2n^\theta}\left|\mathcal{C}^W_{n,k,R}(v)-\alpha_R e^{-v^{\mathsf T}D_Rv/2}\right|\,\mathrm{d}v\\&\quad\le C\left\{n^{-s_0(\theta)}\sqrt{\log(2+n)}+(1+q\epsilon^{-1})e^{C_{\mathrm w}L}n^{-s_1(\theta)}+n^{4\theta}\big(\epsilon V+e^{a_{\mathrm t}m-c_{\mathrm t}L}\big)\right\}.\end{aligned}\tag{203.6}$$

The constant is independent of the location and the length of the window and of its observation functions. The mass α_R and the transform on the left use the full original window, with no collision cutoff and no smoothing of the observations.

Proof. Apply Lemma 203.2 to $W_{L,\epsilon}$ at return k. Lemma 203.1 controls its supremum and variation. Replacing that weight by W_R changes the transform pointwise by at most the L^1 error in (203.4); invariance allows its evaluation at $(F_R^*)^k x$. Integration over the four-dimensional ball costs $Cn^{4\theta}$ times that error. The difference of the two masses is bounded by the same error. Uniform positive definiteness gives

$$\int_{\mathbb{R}^4} e^{-v^{\mathsf T}D_Rv/2}\,\mathrm{d}v\le C,$$

so the Gaussian mass correction is no larger. This proves the result. At no step is a randomly evaluated observation replaced by the same observation at a deterministic collision time. □

203.3. **An unchanged multiple-time event.**

Corollary 203.4 (Logarithmic windows and rare-event budgets)**.** *Suppose, for fixed constants $a,b,d>0$, $\kappa\ge 0$, and $r\ge 0$,*

$$m\le a\log n,\qquad V\le Cn^{\kappa}[\log(2+n)]^r,\qquad b>a.$$

Take $L=\lceil b\log n\rceil$ and $\epsilon=n^{-d}/4$. For $0<\theta\le 1/200$ put

$$\begin{aligned}&\sigma_1=1/28-5\theta, && \sigma_2=1/14-4\theta-C_{\mathrm w}b-d,\\&\sigma_3=d-\kappa-4\theta, && \sigma_4=c_{\mathrm t}b-a_{\mathrm t}a-4\theta.\end{aligned}\tag{203.7}$$

If all four numbers are positive, the integral in (203.6) *tends to zero at a power rate, uniformly in R,k and the permitted data.*

For indicators $u_j=\mathbf{1}_{E_j}$, let

$$A_{n,k,R}=\bigcap_{j=1}^{q}\{x:(F_R^*)^{k+\ell_j}x\in E_j\},\qquad p_{n,R}=\nu_R^*(A_{n,k,R}).\tag{203.8}$$

This probability is independent of k. If $p_{n,R}\ge p_0n^{-\beta}>0$ and $\beta<\min_i\sigma_i$, then

$$\int_{|v|\le 2n^\theta}\left|\mathbb{E}_{\nu_R^*}[e^{iv\cdot U_{n,R}/\sqrt{n}}\mid A_{n,k,R}]-e^{-v^{\mathsf T}D_Rv/2}\right|\,\mathrm{d}v\longrightarrow 0\tag{203.9}$$

uniformly. The event in this conclusion is exactly (203.8)*, without an added restriction on* $N_{m,R}$*.*

Proof. Here $q \le m+1 = O(\log n)$. Substitution in (203.6) gives its four polynomial margins (203.7), multiplied only by fixed powers of $\log(2+n)$. Strict positivity absorbs those powers. For indicator weights, invariance gives $\alpha_R = p_{n,R}$, not a product of the individual probabilities. Divide the unconditional estimate by this same probability. The strict inequalities with β again absorb the logarithmic factors. No induced mixing hypothesis is used. □

The admissible region is nonempty without a relation between the complexity and return-tail constants. To exhibit one choice, enlarge C_{w} to at least one and put

$$\begin{aligned} &d = 1/56, \qquad b = 1/(56C_{\mathrm{w}}), \qquad a = \min\{b/2, c_{\mathrm{t}}b/(4a_{\mathrm{t}})\},\\ &\theta = \min\{1/448, c_{\mathrm{t}}b/32\}, \qquad \beta = \min\{1/448, c_{\mathrm{t}}b/16\}, \qquad \kappa = 0. \end{aligned} \tag{203.10}$$

For every fixed r, these constants satisfy $0 < \theta < 1/200$ and $0 < \beta < \min_i \sigma_i$. Indeed $C_{\mathrm{w}}b = d = 1/56$ and $a_{\mathrm{t}}a \le c_{\mathrm{t}}b/4$; hence the first three margins are at least $11/448$, $12/448$, and $4/448$, respectively, and the fourth is at least $5c_{\mathrm{t}}b/8$. This gives logarithmically many observations with bounded individual variation and polynomially rare exact events. The original $n^{1/200}$ band for the original single-mark class remains Theorem C; (203.7) states the cost of the enlarged class, rather than assigning it that band without proof.

203.4. **Actual moments under the same event.** Set

$$\mathcal{B} = C(1+q\epsilon^{-1})e^{C_{\mathrm{w}}L}, \qquad \mathcal{E} = \min\{1, \epsilon V + e^{a_{\mathrm{t}}m - c_{\mathrm{t}}L}\}.$$

Theorem 203.5 (Window-weighted actual moments)**.** *With the same data, uniformly in* R, k*,*

$$\left| \int W_R \circ (F_R^*)^k \, \frac{U_{n,R}}{\sqrt{n}} \, \mathrm{d}\nu_R^* \right| \le C\{n^{-1/6} + \mathcal{B}n^{-1/2} + \mathcal{E}^{1/2}\}, \tag{203.11}$$

$$\left\| \int W_R \circ (F_R^*)^k \, \frac{U_{n,R} \otimes U_{n,R}}{n} \, \mathrm{d}\nu_R^* - \alpha_R D_R \right\| \le C\{n^{-1/6} + \mathcal{B}n^{-1} + \mathcal{E}^{1/2}\}. \tag{203.12}$$

For (203.8) *with* $p_{n,R} \ge p_0 n^{-\beta}$*, the conditional mean tends to zero and the conditional covariance tends to* D_R *whenever*

$$\beta < \min\left\{ \frac{1}{6}, \frac{1}{2} - C_{\mathrm{w}}b - d, \frac{d-\kappa}{2}, \frac{c_{\mathrm{t}}b - a_{\mathrm{t}}a}{2} \right\}. \tag{203.13}$$

The explicit choice (203.10) *satisfies these inequalities as well.*

Proof. Use $c_*^{-1}(W_{L,\epsilon})^0$ in Theorem 147.6. This gives the first two terms of each bound with the approximant's exact mass. The difference between the weights has modulus at most two, so (203.4) and invariance imply

$$\|(W_R - W_{L,\epsilon}) \circ (F_R^*)^k\|_2 \le C\mathcal{E}^{1/2}.$$

The actual second and fourth moments satisfy $\|U_{n,R}/\sqrt{n}\|_2 \le C$ and $\| \, |U_{n,R}|^2/n\|_2 \le C$ by Theorem D. Cauchy–Schwarz therefore controls the two changes of weighted moment by $C\mathcal{E}^{1/2}$. The change of mass is at most $C\mathcal{E}$ and D_R is uniformly bounded. This proves (203.11)–(203.12).

For indicators, divide by the unchanged $p_{n,R}$. Under the logarithmic choices, the four margins in (203.13) give convergence of the first moment and of the second moment about the original center. Subtract the tensor square of the conditional first moment to obtain the covariance. For (203.10), these margins are at least $1/6$, $13/28$, $1/112$, and $3c_{\mathrm{t}}b/8$, all strictly larger than the displayed β. □

203.5. **The weighted raw inversion interface.** The new central estimate concerns the original window-weighted measure

$$\mu_{n,R}^{W} = (J_{n,R})_*((W_R \circ (F_R^*)^k)\nu_R^*).$$

If the observation functions also belong to the bounded finite-record subanalytic class of Theorem 200.7, that theorem supplies a finite-count extraction for this same measure. This is an additional regularity requirement: arbitrary BV observations are not thereby called subanalytic.

Here is the exact error decomposition which identifies the remaining weighted estimates. Fix $0 < \theta \le 1/200$, set $r_n = n^{-1/2+\theta}$, and choose a smooth cutoff $0 \le \chi \le 1$, equal to one on the unit ball and zero outside the ball of radius two. Let $\chi_{n,\theta}(\omega) = \chi(\omega/r_n)$ on a fixed neighborhood of zero in $\mathbb{T}^3 \times \mathbb{R}$, and let $K_{n,\theta}$ be its mixed inverse Fourier kernel. Write $\mu^{W,L} = E^{W,L} + Q^{W,L}$ for the finite extraction, $e^{W,L}$ for its extracted density, and $T^{W,L} = \mu^{W} - \mu^{W,L}$. For a central point y whose collision count does not exceed L, the same algebra as in (201.3) gives

$$\begin{aligned} p^{W}(y) = (K_{n,\theta} * \mu^{W})(y) + e^{W,L}(y) - (K_{n,\theta} * E^{W,L})(y) \\ + \mathcal{F}^{-1}[(1-\chi_{n,\theta})\widehat{Q}^{W,L}](y) - (K_{n,\theta} * T^{W,L})(y). \end{aligned} \tag{203.14}$$

The inverse transform of the residual is legitimate by Theorem I. The identity holds for the raw density representatives wherever those are defined, and hence almost everywhere.

If $L = \lceil \lambda n \rceil$, $\lambda > c_*^{-1} + 1$, every fixed central window has the required count support for large n. Since $|W_R| \le 1$, the omitted high-count measure has total variation at most one. The Poisson–Schwartz kernel estimate in Section 201, with $n^{1/200}$ replaced by n^{θ}, consequently gives

$$n^2 \sup_{y \text{ in the central window}} |(K_{n,\theta} * T^{W,L})(y)| = O(n^{-P}) \quad \text{for every } P > 0.$$

Indeed its count labels are a distance proportional to n away, $r_n n = n^{1/2+\theta}$, and the normalized kernel prefactor is $n^2 r_n^4 = n^{4\theta}$; any fixed polynomial loss is absorbed by a sufficient Schwartz order.

Theorem 203.3 bounds the normalized error in the first term of (203.14) by its right-hand side, plus $Ce^{-cn^{2\theta}}$ from the Gaussian tail. The other terms require the actual weighted local-edge correction, the complementary residual integral, and the growth of the finite second-derivative sum. For an indicator window these requirements must hold after division by the same $p_{n,R}$. Thus the central and moment inputs now include a growing multiple-time class. No weighted tail or exact physical-event replacement is inferred from them; the latter still concerns a different event from (203.8).

204. Direct estimates on finite pieces of the actual orbit

A small residual for a compressed vector and a small error in a finite-time compression are different requirements. We first record their scales. We then obtain an estimate on the uncompressed dynamics by a different argument: short-lag Gaussian estimates bound a finite Gram matrix of exact orbit vectors. The resulting estimate holds on every translate of that orbit segment. It involves no reconstruction of a long-time vector on a spatial grid.

Throughout this and the next section the Hilbert space is $\mathcal{H}_R = L^2(Y_R^*, \nu_R^*)$, with $\langle f, g\rangle_R = \int \overline{f} g \, \mathrm{d}\nu_R^*$. Write

$$\mathcal{U}_z = \mathcal{U}_{R,z}, \qquad e = \mathbf{1}, \qquad c_{j,R}(z) = \langle e, \mathcal{U}_z^j e\rangle_R = \int e^{-iz\cdot U_{j,R}} \, \mathrm{d}\nu_R^*, \quad j \ge 0. \tag{204.1}$$

Here $U_{j,R} = J_{j,R} - j\bar{G}_R$ is the centered record and $\mathcal{U}_z f = e^{-iz\cdot g_R} f \circ F_R^*$ is the unitary operator of (199.4). For negative j we use its inverse powers; in particular $c_{-j,R}(z) = \overline{c_{j,R}(z)}$. The symbol $z \in \mathbb{R}^4$ denotes physical frequency throughout.

204.1. **The two grid scales.**

Proposition 204.1 (Separation of the displayed grid budgets)**.** *Let $C_{\mathrm{g}} > 0$ be the exponential variation constant in* (202.9)*. Suppose $L_n \geq n$ and the grid contribution in that bound tends to zero, that is,*

$$n\sqrt{h_n(1+|z_n|)e^{C_{\mathrm{g}}L_n}} \longrightarrow 0. \tag{204.2}$$

If $|z_n| \geq 2n^{-99/200}$ and $\rho_n = (|z_n|^2 + \xi_n^2)^{1/2} \leq \rho_0 < 1$, then

$$\rho_n^2\{1 + \log(C_0/(h_n\rho_n))\} \longrightarrow \infty.$$

Thus these choices do not satisfy the local resolvent domain in Corollary 202.5.

Proof. Squaring (204.2) and taking logarithms gives

$$\log(1/h_n) - C_{\mathrm{g}}L_n - 2\log n - \log(1+|z_n|) \longrightarrow +\infty.$$

Consequently $1 + \log(C_0/(h_n\rho_n)) \geq cn$ for all sufficiently large n, with $c > 0$. Since $\rho_n^2 \geq 4n^{-198/200}$, the product is at least $4cn^{1/100}$. This proves the assertion. □

This statement concerns simultaneous use of the two displayed bounds, not the size of the actual approximation error for every mesh. Neither finite-dimensional theorem is contradicted. The direct argument below makes no mesh choice and proves a different long-time conclusion: average square cancellation, rather than decay at every individual return count. The fixed-return raw inversion problem is unchanged.

204.2. **A uniform estimate on the short lags.** Put

$$\gamma = \frac{1}{28} - \frac{1}{200} = \frac{43}{1400}, \qquad \alpha = \frac{200}{99}, \qquad \kappa = \alpha - 2 = \frac{2}{99}. \tag{204.3}$$

Lemma 204.2 (Pointwise Gaussian comparison on the growing band)**.** *There is C such that for $j \geq 2$ and $|z| \leq 2j^{-99/200}$,*

$$\left| c_{j,R}(z) - e^{-jz^{\mathsf{T}}D_Rz/2} \right| \leq Cj^{-\gamma}\sqrt{\log(2+j)} \tag{204.4}$$

uniformly in R. This is a pointwise estimate, not an inference from an integrated characteristic-function bound.

Proof. Set $v = -\sqrt{j}\,z$ and $\delta = j^{-1/14}/4$. Apply Lemma 145.1 at collision time $\lfloor j/c_* \rfloor$, with initial density s_R, and use the pointwise stopping estimate (145.4). The rescaled frequency obeys $|v| \leq 2j^{1/200}$. Before integration, the seven polynomial margins from smoothing, covariance replacement, amplitude, cubic remainder, stopping and integer rounding are bounded below by the entries in

$$\frac{1}{14}, \quad \frac{1}{28} - \frac{1}{200}, \quad \frac{1}{14} - \frac{2}{200}, \quad \frac{3}{14} - \frac{1}{200}, \quad \frac{1}{14} - \frac{3}{200}, \quad \frac{1}{5} - \frac{1}{200}, \quad 1 - \frac{2}{200}.$$

The second entry is the unique smallest one. It carries $\sqrt{\log(2+j)}$; all other logarithmic factors are absorbed by the strict larger margins. The complementary spectral power is exponentially small. The analytic conditions hold because $6/14 + 3/200 < 1/2$ and $2/14 + 1/200 < 1/2$. Enlarging the constant treats the finite initial range of j, since the two transforms are bounded by one. This proves (204.4) directly from the pointwise collision estimate. □

For $0 < r = |z| \leq r_0$, where r_0 is sufficiently small, choose

$$m = m(r) = \lfloor r^{-\alpha} \rfloor \geq 2. \tag{204.5}$$

Only lags shorter than m will be estimated by Gaussian comparison.

Proposition 204.3 (Summable short-lag budget)**.** *Uniformly in R and $0 < |z| = r \le r_0$,*

$$\mathcal{B}_m(R,z) := 1 + 2\sum_{j=1}^{m-1} |c_{j,R}(z)| \le C\{r^{-2} + m^{1-\gamma}\sqrt{\log(2+m)}\}, \qquad \frac{\mathcal{B}_m(R,z)}{m} \le Cr^{\kappa}. \quad (204.6)$$

Proof. For every $2 \le j < m$, the choice of m gives $rj^{99/200} \le 1$, hence the physical frequency is in the band of Lemma 204.2. Uniform ellipticity gives $e^{-jz^{\mathrm{T}}D_R z/2} \le e^{-d_0 j r^2/2}$. Summing the geometric series costs Cr^{-2}, and summing the error costs $Cm^{1-\gamma}\sqrt{\log(2+m)}$. The lag one and diagonal terms are absorbed in the first term. For small r one has $m \ge r^{-\alpha}/2$, so the normalized terms are at most

$$Cr^{\alpha-2}, \qquad Cr^{\alpha\gamma}\sqrt{1+|\log r|}.$$

Here $\alpha\gamma = 43/693$ and $\kappa = 14/693$; their difference is $29/693 > 0$. The logarithm is therefore absorbed uniformly in Cr^{κ}. No estimate on c_j for $j \ge m$ is used. □

204.3. **Finite orbit Gram matrices.**

Lemma 204.4 (Analysis and synthesis of an exact orbit segment)**.** *Let U be unitary on a complex Hilbert space, $\|e\| = 1$, and put $B_m = 1 + 2\sum_{j=1}^{m-1} |\langle e, U^j e\rangle|$. For every integer N, coefficients $a_0, \dots, a_{m-1}$ and vector b,*

$$\left\| \sum_{j=0}^{m-1} a_j U^{N+j} e \right\|^2 \le B_m \sum_{j=0}^{m-1} |a_j|^2, \quad (204.7)$$

$$\sum_{j=0}^{m-1} |\langle b, U^{N+j} e\rangle|^2 \le B_m \|b\|^2. \quad (204.8)$$

Both assertions remain true after multiplying each orbit vector by an arbitrary scalar of modulus one.

Proof. The Gram entry at (i,j) is $\langle e, U^{j-i} e\rangle$; it is independent of N. Every absolute row sum is at most B_m. For completeness, expansion of the squared norm and $2|a_i||a_j| \le |a_i|^2 + |a_j|^2$ bound that quadratic form by $B_m \sum_j |a_j|^2$. This proves (204.7). The synthesis operator from $\mathbb{C}^m$ to the Hilbert space has norm at most $\sqrt{B_m}$; its adjoint has the same norm and is the analysis map in (204.8). Multiplication by unit scalars conjugates the Gram matrix by a diagonal unitary. This proves the final assertion without assuming that the vectors are orthogonal or linearly independent. □

Theorem 204.5 (Uncompressed moving averages)**.** *Let $m = m(r)$, $0 < |z| = r \le r_0$, and $T \ge m$ be an integer. For all integers N, every $b \in \mathcal{H}_R$ and every $\xi \in \mathbb{R}$,*

$$\frac{1}{T}\sum_{j=0}^{T-1} |\langle b, \mathcal{U}_z^{N+j} e\rangle_R|^2 \le Cr^{\kappa}\|b\|_2^2, \quad (204.9)$$

$$\left\| \frac{1}{T}\sum_{j=0}^{T-1} e^{-ij\xi} \mathcal{U}_z^{N+j} e \right\|_2^2 \le Cr^{\kappa}. \quad (204.10)$$

The constants are independent of N, T, R, ξ and b. In particular these estimates apply to the original characteristic functions at arbitrarily late return counts, not to powers of a compressed operator.

Proof. Divide the T indices into complete blocks of length m and one possibly shorter block. The estimate for a shorter block follows from the same Gram argument with its row sum bounded by $\mathcal{B}_m$. There are at most $T/m + 1 \le 2T/m$ blocks. Summing (204.8) proves (204.9) with constant $2\mathcal{B}_m/m$.

For synthesis, a complete block of m unit-modulus coefficients has norm at most $\sqrt{m\mathcal{B}_m}$, and a shorter block has no larger norm. The triangle inequality across the at most $2T/m$ blocks, followed by division by T, gives squared norm at most $4\mathcal{B}_m/m$. Proposition 204.3 finishes both bounds. Modulation by $e^{-ij\xi}$ does not change any absolute Gram entry. Thus there is no restriction on the peripheral angle ξ in this theorem. □

Corollary 204.6 (One feasible scale on the full small annulus)**.** *For sufficiently large n put*

$$\mathcal{A}_n = \{z \in \mathbb{R}^4 : 2n^{-99/200} \le |z| \le n^{-2/5}\}. \tag{204.11}$$

Its rescaled radii are $2n^{1/200}$ and $n^{1/10}$. For every $z \in \mathcal{A}_n$ one has

$$m(|z|) \le 2^{-200/99} n < n/4, \qquad |z|^{\kappa} \le n^{-4/495}. \tag{204.12}$$

Consequently both conclusions of Theorem 204.5 hold for $T \ge n$ with Cr^{κ} replaced by $Cn^{-4/495}$, uniformly throughout the annulus and at every peripheral angle.

Proof. The first inequality follows by raising the lower physical radius to power $-200/99$; the exponent product is exactly one. The second follows from the upper radius and $(2/5)(2/99) = 4/495$. Thus a single explicit choice of the lag window is compatible with every averaging time $T \ge n$. No spatial mesh or long-orbit BV budget enters this choice. □

The use of the word "average" in these statements is essential. They bound a sum of squared matrix coefficients or the norm of a Cesaro average of the actual orbit. They do not bound $|c_{n,R}(z)|$ at every specified n, and they do not assert that $\|\mathcal{U}_z^n\|$ decays; that norm equals one. The next section records the weighted and peripheral consequences at precisely this level.

205. Weighted annular averages and the peripheral spectral measure

The direct orbit estimate has two consequences which require no variation bound for the final test vector: cancellation under an unchanged event, and control of the spectral measure of the constant vector on every peripheral arc. We also identify the small-frequency limit of that spectral measure. These are statements about the exact unitary return dynamics, at the level of averages or of its specified cyclic vector. They are not operator-norm power bounds.

205.1. **Weights and unchanged events.** For a complex $a \in \mathcal{H}_R$ define

$$C^a_{j,R}(z) = \int a(x) e^{-iz\cdot U_{j,R}(x)} \, \mathrm{d}\nu^*_R(x), \qquad j \ge 0. \tag{205.1}$$

The weight is not normalized. Let $\mathcal{A}_n$ be the physical annulus in (204.11).

Theorem 205.1 (Mean-square cancellation with arbitrary square-integrable weights)**.** *For n sufficiently large, $T \ge n$, $N \ge 0$, and every $z \in \mathcal{A}_n$,*

$$\frac{1}{T} \sum_{j=0}^{T-1} |C^a_{N+j,R}(z)|^2 \le Cn^{-4/495} \|a\|_2^2. \tag{205.2}$$

If A is any measurable event with $p = \nu^*_R(A) > 0$, then

$$\frac{1}{T} \sum_{j=0}^{T-1} \left| \mathbb{E}_{\nu^*_R}[e^{-iz\cdot U_{N+j,R}} \mid A] \right|^2 \le \frac{C}{p} n^{-4/495}. \tag{205.3}$$

The same event A occurs at every averaging index. The constants are uniform in R, N, T, z, a, A subject to these conditions.

Proof. Take $b = \overline{a}$ in (204.9) and use Corollary 204.6. For the conditional statement take $a = \mathbf{1}_A/p$, whose squared L^2 norm is exactly $1/p$. This proves both estimates with their actual normalizations. No smoothing, count restriction, subanalytic assumption or factorization of the event probability is used. □

For example, if $p \geq p_0 n^{-\beta}$ with $\beta < 4/495$, the average in (205.3) tends to zero uniformly. More quantitatively, for every $\eta > 0$, the fraction of indices $j < T$ at which the displayed conditional transform exceeds η in modulus is at most

$$C\eta^{-2} p^{-1} n^{-4/495}. \tag{205.4}$$

This follows by applying the elementary counting inequality to the sum of squares. The exceptional set may depend on the frequency and the event. No claim is made that one specified return count avoids it.

Corollary 205.2 (A fixed actual mark and a moving terminal mark)**.** *The bound* (205.2) *also holds for*

$$\int a((F_R^*)^k x) e^{-iz\cdot U_{N+j,R}(x)} \, \mathrm{d}\nu_R^*(x)$$

when $0 \leq k \leq N$ is fixed throughout the averaging window. It holds as well when the weight is $a((F_R^*)^{N+j} x)$, evaluated at the actual terminal return at each index.

Proof. Write $\psi_l = \mathcal{U}_z^l e$, including negative integers. The weighted-composition identity for $\mathcal{U}_z^{-k}$ is

$$\psi_{l-k}(y) = \psi_{-k}(y)\psi_l((F_R^*)^{-k} y).$$

After changing variables $y = (F_R^*)^k x$, the fixed-mark integral is $\langle \overline{a}\, \psi_{-k}, \psi_{N+j-k} \rangle_R$. Its test vector has norm $\|a\|_2$ and is independent of j. Apply (204.9).

For the terminal mark, change variables $y = (F_R^*)^{N+j} x$. The phase becomes $\overline{\psi_{-(N+j)}(y)}$. The conjugate of the resulting integral is $\langle a, \psi_{-(N+j)} \rangle_R$. Apply the same theorem to the inverse unitary, whose absolute lag correlations are unchanged. These changes of variables use the actual return map and do not replace a random observation by a deterministic collision-clock observation. □

Corollary 205.3 (A normalized annular integral)**.** *For the original law, and uniformly in $R, N \geq 0, T \geq n$,*

$$\frac{1}{|\mathcal{A}_n|} \int_{\mathcal{A}_n} \frac{1}{T} \sum_{j=0}^{T-1} |c_{N+j,R}(z)|^2 \, \mathrm{d}z \leq C n^{-4/495}. \tag{205.5}$$

The same conclusion holds for an event as in (205.3)*, with the factor p^{-1}.*

Proof. Integrate the uniform bounds in Theorem 205.1, and divide by the volume of this nonempty annulus. Finite sums and bounded integrands justify interchanging the integrals. □

The normalization in (205.5) is stated explicitly. The raw local theorem instead requires an *unnormalized* complementary integral at a specified return count, with the four-dimensional factor n^2. These requirements are not interchangeable. Indeed $|\mathcal{A}_n| \asymp n^{-8/5}$; Cauchy–Schwarz applied to the new mean-square bound would give at best $n^{2/5-2/495}$ after that raw normalization. It therefore supplies no vanishing raw error by itself. Likewise a fixed event across an averaging window is different from a separately chosen exact physical observation event for every member of that window.

205.2. **Every peripheral arc for the constant vector.** Let $\varsigma_{R,z}$ be the probability on $[-\pi,\pi)$ associated by the unitary spectral theorem with $e=\mathbf{1}$ and $\mathcal{U}_z$. Thus

$$c_{j,R}(z)=\int e^{ij\theta}\,\mathrm{d}\varsigma_{R,z}(\theta),\qquad j\in\mathbb{Z}. \tag{205.6}$$

This is a spectral measure of the return operator, not the time density of the physical record. The distinction is relevant to both of the following statements.

Theorem 205.4 (Small peripheral arcs and exterior cyclic resolvents)**.** *For $0<|z|=r\le r_0$, $m=\lfloor r^{-200/99}\rfloor$, and all $\xi\in\mathbb{R}$,*

$$\varsigma_{R,z}\{\theta:\operatorname{dist}(\theta-\xi,2\pi\mathbb{Z})\le m^{-1}\}\le Cr^{2/99}, \tag{205.7}$$

$$\left\|(1-s)(I-se^{-i\xi}\mathcal{U}_z)^{-1}e\right\|_2^2\le Cr^{2/99} \tag{205.8}$$

whenever $0<s<1$ and $(1-s)m\le 1$. All constants are uniform in R. The inverse in (205.8) *is an ordinary exterior unitary resolvent applied to one specified vector. No inverse of $e^{i\xi}-\mathcal{U}_z$ is asserted on the unit circle.*

Proof. The squared norm of $m^{-1}\sum_{j<m}e^{-ij\xi}\mathcal{U}_z^je$ is the integral against $\varsigma_{R,z}$ of

$$\left|\frac{1}{m}\sum_{j=0}^{m-1}e^{ij(\theta-\xi)}\right|^2.$$

On the arc of radius $1/m$ this kernel is bounded below by a positive absolute constant. For example, use $|\sin(mt/2)|/(m|\sin(t/2)|)\ge 2/\pi$ for $|t|\le 1/m$, with its continuous value at zero. Equation (204.7) and Proposition 204.3 bound the total integral by $Cr^{2/99}$. This proves (205.7) simultaneously for all centers, with no choice of peripheral representatives.

For the second assertion expand the resolvent in its norm-convergent Neumann series. In its block of indices $km,\dots,km+m-1$, (204.7) gives the norm bound

$$s^{km}\sqrt{\mathcal{B}_m\sum_{j=0}^{m-1}s^{2j}}\le s^{km}\sqrt{m\mathcal{B}_m}.$$

Sum the blocks and multiply by $1-s$. Since $s^m\le e^{-m(1-s)}$ and $1-e^{-x}\ge(1-e^{-1})x$ for $0\le x\le 1$, the result is at most $C\sqrt{\mathcal{B}_m/m}$. Squaring and applying (204.6) proves (205.8). The proof controls no operator norm on arbitrary input vectors. □

205.3. **A Cauchy scaling law for the exact spectral measure.** For a unit direction $u\in\mathbb{R}^4$ put

$$d_{R,u}=u^{\mathsf{T}}D_Ru,\qquad a_{R,u}=d_{R,u}/2.$$

Uniform ellipticity bounds $a_{R,u}$ above and below by positive constants. Let C_a be the Cauchy probability on $\mathbb{R}$ with density $a/[\pi(a^2+x^2)]$ and characteristic function $e^{-a|t|}$. The principal angle in (205.6) is used in the rescaling below.

Theorem 205.5 (Spectral Cauchy limit at physical frequency zero)**.** *Let $\widetilde{\varsigma}_{R,r,u}$ be the pushforward of $\varsigma_{R,ru}$ under $\theta\mapsto\theta/r^2$. Then, as $r\downarrow 0$,*

$$\sup_{R\in I,\ |u|=1}d_{\mathrm{BL}}\big(\widetilde{\varsigma}_{R,r,u},\mathsf{C}_{a_{R,u}}\big)\longrightarrow 0. \tag{205.9}$$

This describes a spectral probability. It does not claim that the physical return record has a Cauchy limit; its Gaussian limit is unchanged.

Proof. The one-return second moment is uniformly bounded by Theorem 133.3. Using $1-\cos x \le x^2/2$,

$$1 - \operatorname{Re} c_{1,R}(ru) \le Cr^2.$$

For $|\theta| \le \pi$ one has $\theta^2 \le (\pi^2/2)(1-\cos\theta)$. Therefore

$$\int |\theta| \, \mathrm{d}\varsigma_{R,ru}(\theta) \le Cr. \tag{205.10}$$

Fix $t > 0$ and put $j = \lfloor t/r^2 \rfloor$. The difference between the characteristic function of $\widetilde{\varsigma}_{R,r,u}$ at t and $c_{j,R}(ru)$ is at most Cr, by (205.10) and $|t/r^2 - j| < 1$. For sufficiently small r, $j \ge 2$, and Theorem A applied at rescaled frequency $-r\sqrt{j}\,u$ gives

$$c_{j,R}(ru) = \exp(-jr^2 d_{R,u}/2) + O_t(j^{-1/26}\sqrt{\log(2+j)}).$$

The rescaled frequencies remain in a fixed ball for fixed t. The main term tends uniformly to $e^{-td_{R,u}/2}$. Negative t uses $c_{-j} = \overline{c_j}$, and $t = 0$ is exact. This proves pointwise characteristic convergence, uniformly in the parameters, to the asserted Cauchy family.

To justify the uniform bounded-Lipschitz conclusion, take any sequence $r_l \downarrow 0$, $R_l \in I$, $|u_l| = 1$. Compactness gives a subsequence $R_l \to R_*$, $u_l \to u_*$. Continuity of D_R and the preceding characteristic estimates identify its limit as $\mathsf{C}_{a_{R_*,u_*}}$ by the characteristic-function continuity theorem. The target Cauchy laws converge as well: multiply one standard Cauchy random variable by a_{R_l,u_l}. Thus a sequence violating (205.9) cannot exist. No absolute continuity of $\varsigma_{R,z}$ for a fixed nonzero z is presumed or deduced. □

Place in the raw inversion problem. Theorems 204.5, 205.1 and 205.4 bypass the incompatible grid scales for averaged physical coefficients and for a specified cyclic resolvent vector. They cover every peripheral angle and allow square-integrable weights without a BV reconstruction hypothesis. Theorem 205.5 identifies the concentration scale of the same spectral probability. None of these statements replaces the fixed-return complementary integral in Theorem 201.2, the long-time growth of $A_2(n, L_n, R, w)$, or the local extracted-edge correction. The exact physical-event replacement remains a different assertion from the unchanged-event average proved above.

206. Finite-order collision cumulants and spectral damping

The short-lag argument of Section 204 does not estimate a prescribed later coefficient. We now return to the collision spectral splitting and prove stronger finite-order bounds before stopping. The point is that its derivatives at zero can be bounded independently of the smoothing scale, although the analytic neighborhood still shrinks with that scale. A Taylor polynomial of fixed, sufficiently high degree then gives real-frequency damping on a larger band. No analyticity of the unsmoothed twist is asserted.

All expectations in this section use the stationary collision probability ν. Put $\|u\|_{\mathcal{V}} = \|u\|_\infty + \|u\|_{\mathrm{BV}}$. Every order below is fixed before the smoothing scale, collision length, and radius are varied. Constants may depend on that order. We require no factorial bound uniform in the order.

206.1. **Connected correlations at any fixed order.**

Lemma 206.1 (Finite-product decoupling at arbitrary fixed order)**.** *For each integer $q \ge 2$ there are $C_q, \gamma_q > 0$, uniform in R, such that, for $t_1 \le \cdots \le t_q$ and $1 \le p < q$,*

$$\left| \int \prod_{j=1}^{q} u_j \circ T_R^{t_j} \, \mathrm{d}\nu - \left(\int \prod_{j \le p} u_j \circ T_R^{t_j} \, \mathrm{d}\nu \right) \left(\int \prod_{j > p} u_j \circ T_R^{t_j} \, \mathrm{d}\nu \right) \right| \le C_q e^{-\gamma_q (t_{p+1} - t_p)} \prod_{j=1}^{q} \|u_j\|_{\mathcal{V}}. \tag{206.1}$$

Negative and repeated times are allowed. The constants can be chosen simultaneously for orders at most any specified integer Q.

Proof. The proof of Lemma 147.1 works at each fixed order, and we give its scale explicitly. By multilinearity normalize all nonzero norms to one. Smooth every factor at scale ϵ. Telescoping the products and using invariance bounds the total error by $C_q \epsilon$. Translate t_1 to zero, write the expectation as the chronological word of smooth multipliers and collision powers, and replace the power across the selected gap by $\Pi_R = \nu \ell$. The new word is exactly the product of the two block expectations. The error is at most $C_q \epsilon^{-2q} \vartheta^{t_{p+1} - t_p}$: there are q smooth multipliers, all other powers are uniformly bounded, and the selected complementary power decays exponentially. For a zero gap use $I - \Pi_R$ with its uniform norm. Taking $\epsilon = \vartheta^{(t_{p+1} - t_p)/(2q+1)}/4$ proves the assertion. A finite minimum of the resulting exponents treats orders at most Q. No variation of a long pullback is estimated. □

For bounded real random variables, define the joint cumulant by

$$\operatorname{cum}(X_1, \ldots, X_q) = \sum_{\pi \in \mathcal{P}_q} (-1)^{|\pi| - 1} (|\pi| - 1)! \prod_{B \in \pi} \mathbb{E} \prod_{j \in B} X_j, \tag{206.2}$$

where $\mathcal{P}_q$ is the set of partitions of $\{1, \ldots, q\}$. This is the mixed derivative at zero of the logarithm of the joint moment generating function. In particular it is multilinear and vanishes when the variables split into two nonempty independent families. The latter assertion follows because the joint generating function factors and its logarithm is a sum of two functions in separate sets of variables. This elementary finite-order use of cumulants is distinct from an all-orders normal-approximation theorem; see also [16] for the general framework.

Proposition 206.2 (Absolute summability of fixed-order connected correlations)**.** *Let u be real and $\|u\|_{\mathcal{V}} \le B$. For each $q \ge 2$,*

$$\sum_{\mathbf{k} \in \mathbb{Z}^{q-1}} (1 + \operatorname{diam}\{0, k_2, \ldots, k_q\}) \left| \operatorname{cum}(u, u \circ T_R^{k_2}, \ldots, u \circ T_R^{k_q}) \right| \le C_{q,B}. \tag{206.3}$$

Consequently the absolutely convergent sum without the diameter factor, denoted by $\mathfrak{c}_q(R, u)$, satisfies

$$|\operatorname{cum}_q(S_{m,R} u) - m \mathfrak{c}_q(R, u)| \le C_{q,B} \qquad (m \ge 1), \qquad |\mathfrak{c}_q(R, u)| \le C_{q,B}. \tag{206.4}$$

Both statements are uniform in R and in the displayed class of u.

Proof. Order any q times and split them at a largest consecutive gap d. Compare their joint law with a law in which the entire left block and the entire right block are independent, but the two block marginals are unchanged. For every subset meeting both blocks, its moment changes by at most $C_{q,B} e^{-\gamma_q d}$ by Lemma 206.1; the gap in that subset is at least d. All other subset moments are unchanged. The finite partition polynomial (206.2), whose factors are bounded, therefore changes by at most the same order. Its value under the independent-block law is zero. Thus the original joint cumulant is bounded by $C_{q,B} e^{-\gamma_q d}$. If all times coincide, the bounded partition formula gives this estimate with $d = 0$.

For anchored times $0, k_2, \dots, k_q$, their diameter D obeys $d \ge D/(q-1)$. There are at most $(2D+1)^{q-1}$ tuples with all coordinates of absolute value at most D. Exponential decay therefore sums with the additional factor $1+D$, proving (206.3).

By multilinearity and stationarity, the exact finite sum is

$$\operatorname{cum}_q(S_{m,R}u) = \sum_{\mathbf{k}\in\mathbb{Z}^{q-1}} (m - D(\mathbf{k}))_+ \operatorname{cum}(u, u\circ T_R^{k_2}, \dots, u\circ T_R^{k_q}).$$

Indeed, after fixing the first index, the remaining indices have these relative offsets; exactly $(m-D)_+$ translates fit in the interval of m collisions. The difference between this coefficient and m has absolute value at most D. Applying the first assertion proves (206.4). This also treats repeated indices without an additional combinatorial multiplicity. □

206.2. **Uniform derivatives of the smoothed eigenvalue.** For a real unit vector $\xi \in \mathbb{R}^4$ let

$$u_{R,\delta,\xi} = \xi \cdot h_{R,\delta}, \qquad \psi_{R,\delta,\xi}(t) = \log \lambda_{R,\delta}(t\xi),$$

using the branch of the logarithm fixed in Lemma 139.1. The supremum and BV norms of $u_{R,\delta,\xi}$ are bounded uniformly, and its mean is zero.

Theorem 206.3 (A smoothing-independent finite spectral jet)**.** *For each fixed $Q \ge 3$ there is C_Q such that*

$$\begin{aligned} &\psi'(0) = 0, \qquad \psi''(0) = -\xi^{\mathsf{T}}\Gamma_{R,\delta}\xi, \\ &\psi^{(q)}(0) = i^q \mathfrak{c}_q(R, u_{R,\delta,\xi}), \qquad |\psi^{(q)}(0)| \le C_Q \quad (2 \le q \le Q). \end{aligned} \tag{206.5}$$

In particular, on $|t| \le a\delta^2$, after decreasing $a > 0$,

$$\psi(t) = -\tfrac{1}{2}t^2\xi^{\mathsf{T}}\Gamma_{R,\delta}\xi + \sum_{q=3}^{Q} \frac{\psi^{(q)}(0)}{q!}t^q + O_Q(\delta^{-2(Q+1)}|t|^{Q+1}). \tag{206.6}$$

Every constant is uniform in R, δ, ξ. The Taylor remainder still uses the shrinking complex analytic radius; it is not scale independent.

Proof. Fix R, δ, ξ temporarily. The stationary spectral pairing is

$$F_m(t) := \int e^{itS_{m,R}u_{R,\delta,\xi}}\,\mathrm{d}\nu = e^{m\psi(t)}A(t) + E_m(t), \qquad A(t) = \ell(\Pi(t\xi)\nu).$$

Here $A(0) = 1$, and the complementary term has norm at most $C\rho^m$ on the given complex neighborhood. On a smaller neighborhood, $|A(t)| \ge 1/2$ and $|e^{\psi(t)}| \ge \rho_1 > \rho$. Hence the relative error is analytic and bounded by $C(\rho/\rho_1)^m$. For large m, choose its logarithm near one. Cauchy's inequalities on a still smaller neighborhood give, for each fixed q,

$$\frac{1}{m}(\log F_m)^{(q)}(0) \longrightarrow \psi^{(q)}(0).$$

The derivatives of $\log A$ divided by m vanish. All these identification steps are made at fixed δ; their constants need not be uniform as $\delta \downarrow 0$.

Boundedness of the finite collision sum and the cumulant definition give $(\log F_m)^{(q)}(0) = i^q \operatorname{cum}_q(S_m u)$. Proposition 206.2 identifies the limit and bounds it independently of δ, R, ξ. The first derivative vanishes by centering; the second is the collision Green–Kubo variance, also identified in Lemma 139.1.

Finally the analytic logarithm is uniformly bounded on a complex disk of radius a fixed multiple of δ^2, as in the proof of that lemma. Cauchy's estimate for the Taylor tail on a concentric smaller disk is $C_Q\delta^{-2(Q+1)}|t|^{Q+1}$. This proves (206.6). Only the finitely many coefficients, not the analytic disk or an infinite cumulant series, have gained smoothing-independent bounds. □

Corollary 206.4 (Real-frequency collision damping)**.** *For each fixed $Q \ge 3$ there are $c, C, a, \delta_0 > 0$ such that $0 < \delta \le \delta_0$ and*

$$|z| \le a\delta^2, \qquad \delta^{-2(Q+1)}|z|^{Q-1} \le a \tag{206.7}$$

imply, on every local collision space,

$$|\lambda_{R,\delta}(z)| \le e^{-c|z|^2}, \qquad \|\mathcal{L}^m_{R,\delta,z}\| \le Ce^{-cm|z|^2} \quad (m \ge 0,\ z \in \mathbb{R}^4). \tag{206.8}$$

Proof. Theorem 150.2 and (138.7) give $\Gamma_{R,\delta} \ge c_1 I_4$ for all sufficiently small δ, uniformly in R. Write $z = t\xi$. The finite sum of degree at least three in (206.6) is bounded by $C_Q|t|^3$ for $|t| \le 1$. Its remainder is at most $C_Q a|t|^2$ under (206.7). Decreasing a and δ_0 therefore gives $\operatorname{Re}\psi(t) \le -c_2|t|^2$. The bounded spectral projection and the complementary power estimate then give $\|\mathcal{L}^m_z\| \le C(e^{-c_2 m|z|^2} + \rho^m)$. Decrease the constant in the exponent so that ρ^m is dominated by the same bound on the stated small real-frequency region. This also holds for $m = 0$ after increasing C. □

The estimate is a power bound for the *smoothed collision* operator on its previously constructed local distribution space. It is not an operator-norm decay assertion for the unitary twisted return Koopman operator. The next section removes smoothing and stops at the actual prescribed return, keeping those errors explicit.

207. A fixed-return inner annulus on the raw inversion scale

We apply the finite spectral jet before any averaging in the return count. The result estimates the original characteristic function at each specified count, also with an insertion at any actual return. The new band extends beyond the previously proved central ball. The farther complement and the extracted-edge estimates remain separate parts of the same raw inversion problem.

Retain M_a, V_a, α_a and $C^{[k],a}_{n,R}$ from (146.1) and (146.3). Thus a is zero outside the actual section and the transform uses $\nu|_{Y^*_R}$, not the normalized section probability. The choice $a = s_R = c_*^{-1}\mathbf{1}_{Y^*_R}$ recovers that probability. Put

$$\Phi^{[k],a}_{n,R}(z) = \int_{Y^*_R} a((F^*_R)^k x)e^{iz\cdot J_{n,R}(x)}\,\mathrm{d}\nu(x).$$

Its absolute value equals that of $C^{[k],a}_{n,R}(\sqrt{n}\,z)$. The insertion may depend on n, R, k, but is fixed as frequency varies.

207.1. **A deterministic marked estimate with damping.**

Lemma 207.1 (Damping before the two-sided stopping)**.** *Fix $Q \ge 3$. Suppose $\delta \le \delta_0$ and every frequency under consideration obeys* (206.7) *with $z = v/\sqrt{n}$. Uniformly in $0 \le k \le n$,*

$$\begin{aligned}|C^{[k],a}_{n,R}(v)| &\le CM_a\delta^{-2}e^{-c|v|^2} + C\delta V_a \\ &\quad + CM_a\left\{|v|\sqrt{\delta(1+|\log\delta|)} + (1+|v|)n^{-2/9}\right\}.\end{aligned} \tag{207.1}$$

Proof. Use the exact recentering at the marked return and then (147.8). This replaces the actual centered record by $\mathsf{H}_{m_k,m_{n-k},R}$, at cost $CM_a(1+|v|)n^{-2/9}$. Its deterministic collision length is $m_k + m_{n-k} = n/c_* + O(1)$, uniformly in the mark.

Smooth the insertion at scale δ, at L^1 cost $C\delta V_a$. The mean-zero residual $h_R - h_{R,\delta}$ has interval variance at most $Cn\delta(1+|\log\delta|)$ by (138.5). Stationarity applies to the whole interval, including its negative part. Since $|a_\delta| \le M_a$, Cauchy–Schwarz bounds the change of observable by $CM_a|v|\sqrt{\delta(1+|\log\delta|)}$.

The remaining integral is exactly

$$\ell\left(\mathcal{L}_{R,\delta,z}^{m_{n-k}} M_{a_\delta} \mathcal{L}_{R,\delta,z}^{m_k} \nu\right)$$

by (146.11). Each power now has the real-frequency damping of Corollary 206.4. The multiplier norm is at most $CM_a\delta^{-2}$. Their product is therefore bounded by $CM_a\delta^{-2}e^{-c(m_k+m_{n-k})|z|^2}$, which is the first term of (207.1). The power bound includes length zero, so the initial and terminal marks require no exceptional argument. No return-count average, induced spectral gap, or spatial compression has been used. □

207.2. **A feasible fixed-count scale.**

Theorem 207.2 (Raw-scale cancellation on a fixed-return inner annulus)**.** *For all sufficiently large n, let*

$$\mathcal{A}_n^{\mathrm{in}} = \{z \in \mathbb{R}^4 : 2n^{-99/200} \le |z| \le 2n^{-12/25}\}. \tag{207.2}$$

Its rescaled radii are $2n^{1/200}$ and $2n^{1/50}$. Uniformly in R, in $0 \le k \le n$, and in the stated insertions,

$$n^2 \int_{\mathcal{A}_n^{\mathrm{in}}} |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C\left(M_a n^{-1/200}\sqrt{\log(2+n)} + V_a n^{-13/100}\right). \tag{207.3}$$

In particular the left side tends to zero for the original probability at each prescribed n, uniformly in the physical radius.

Proof. Choose once and for all

$$Q = 19, \qquad \delta = \tfrac{1}{4}n^{-21/100}, \qquad |v| \le 2n^{1/50}, \qquad z = v/\sqrt{n}. \tag{207.4}$$

The two damping restrictions hold for large n. Indeed their powers are

$$|z|\delta^{-2} \le Cn^{-3/50}, \qquad |z|^{18}\delta^{-40} \le Cn^{-6/25}.$$

The constants, including those from $\delta = n^{-21/100}/4$, are fixed before n tends to infinity. The smoothed covariance is uniformly positive definite at this scale.

Rescaling $v = \sqrt{n}\,z$ cancels the raw factor n^2 exactly. Integrate (207.1) between the stated rescaled radii. The Gaussian damping term is bounded by $CM_a n^{21/50}e^{-cn^{1/100}}$ and hence by every fixed negative power of n times M_a. The three other integrated losses are

$$CV_a n^{-21/100+4/50}, \qquad CM_a n^{-21/200+5/50}\sqrt{\log(2+n)}, \qquad CM_a n^{-2/9+5/50}.$$

Their decay margins are respectively 13/100, 1/200, and 11/90. This proves (207.3). The four-dimensional volume factor is included in each term. Taking $a = s_R$ gives uniformly bounded M_a, V_a and unit mass. □

Corollary 207.3 (A larger integrated band at the prescribed count)**.** *Under the same conventions,*

$$\int_{|v|\le 2n^{1/50}} \left|C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf{T}} D_R v/2}\right|\,\mathrm{d}v \le C\left(M_a n^{-1/200}\sqrt{\log(2+n)} + V_a n^{-9/175}\right). \tag{207.5}$$

This is an additional wider-band estimate. The sharper rate $n^{-3/280}$ on the original $2n^{1/200}$ band in Theorem C is retained unchanged.

Proof. On the original central ball apply Theorem 146.5. On its complement up to the new radius use Theorem 207.2 and the uniformly elliptic Gaussian tail. Here $|\alpha_a| \le M_a$, $3/280 > 1/200$, and $13/100 > 9/175$. □

Proposition 207.4 (The range of the finite-order construction)**.** *Let*

$$\frac{1}{200} < \varepsilon < \frac{1}{42}, \qquad 10\varepsilon < \theta < \frac{1}{4} - \frac{\varepsilon}{2},$$

and choose a fixed integer $Q \geq 3$ with

$$(Q-1)(1/2-\varepsilon) > 2(Q+1)\theta.$$

The inner-annulus assertion holds with outer physical radius $2n^{-1/2+\varepsilon}$ and bound

$$C\left\{V_a n^{-(\theta-4\varepsilon)} + M_a n^{-(\theta/2-5\varepsilon)}\sqrt{\log(2+n)} + M_a n^{-(2/9-5\varepsilon)}\right\}. \tag{207.6}$$

The rescaled outer radius is $2n^{\varepsilon}$. Every exponent in this bound is positive, and all choices are independent of n, R, k, a.

Proof. Set $\delta = n^{-\theta}/4$. The strict upper bound on θ puts z inside the analytic disk; the inequality for Q is exactly the second damping condition, with a strictly decaying power of n. Such a fixed Q exists because $1/2 - \varepsilon > 2\theta$. Integration proceeds as above. The interval for θ is nonempty because $\varepsilon < 1/42$, and $\theta > 10\varepsilon$ makes the first two margins positive. Also $2/9 - 5\varepsilon > 0$ in this range. This is a sufficient range for the present proof, not a barrier for other methods or for the original raw theorem. □

207.3. **Unchanged marked events and raw inversion.**

Corollary 207.5 (The same marked event on the wider band)**.** *Let $E_{n,R} \subset Y_R^*$ have zero-extended indicator of variation $V_{n,R}$ and probability $p_{n,R} = \nu_R^*(E_{n,R}) > 0$. For the unchanged event $A_{n,k,R} = \{x : (F_R^*)^k x \in E_{n,R}\}$,*

$$\int_{|v|\leq 2n^{1/50}} \left|\mathbb{E}_{\nu_R^*}[e^{iv\cdot U_{n,R}/\sqrt{n}} \mid A_{n,k,R}] - e^{-v^{\mathsf{T}} D_R v/2}\right| \mathrm{d}v$$
$$\leq \frac{C}{p_{n,R}}\left(n^{-1/200}\sqrt{\log(2+n)} + V_{n,R}n^{-9/175}\right). \tag{207.7}$$

Thus convergence follows, for example, if $p_{n,R} \geq cn^{-\beta}$ and $V_{n,R} \leq Cn^{\kappa}$, where $\beta < 1/200$ and $\beta + \kappa < 9/175$.

Proof. Take $a = c_*^{-1}\mathbf{1}_{E_{n,R}}$ in Corollary 207.3. Its mass is $p_{n,R}$; invariance gives the same probability to $A_{n,k,R}$. Divide the numerator estimate by precisely this probability. No event is replaced. The factor p^{-1} here belongs to a fixed-count absolute-error estimate; it is not the squared-L^2 normalization of the averaged theorem. Its allowed events may vary with n subject to these budgets. □

For the original unweighted law, use the finite extraction and the notation of Section 201. To distinguish the new cutoff from the inherited one, put

$$\widetilde{B}_n = n^{-12/25}, \quad \widetilde{\chi}_n(\omega) = \chi(\omega/\widetilde{B}_n), \quad \widetilde{K_n} = \mathcal{F}^{-1}\widetilde{\chi}_n. \tag{207.8}$$

The support has physical radius $2n^{-12/25}$ and rescaled radius $2n^{1/50}$. Define $\widetilde{\mathcal{E}}^L_{n,R,A}$ and $\widetilde{\mathcal{C}}^L_{n,R}(B)$ by the definitions preceding Corollary 201.3, replacing K_n and χ_n by $\widetilde{K_n}$ and $\widetilde{\chi}_n$. In particular the local edge correction is recomputed with the new kernel; it is not transferred from the old kernel without an estimate.

Theorem 207.6 (Fixed-count raw inversion with the enlarged cutoff)**.** *Let $L_n = \lceil \lambda n \rceil$, $\lambda > c_*^{-1}+1$. On each central window $\mathcal{W}_{n,R,A}$, the exact identity*

$$\begin{aligned} p_{n,R} = \widetilde{K}_n * \mu_{n,R} + (e_{n,R}^{L_n} - \widetilde{K}_n * E_{n,R}^{L_n}) \\ + \mathcal{F}^{-1}[(1-\widetilde{\chi}_n)\widehat{Q}_{n,R}^{L_n}] - \widetilde{K}_n * \mathsf{T}_{n,R}^{L_n} \end{aligned} \tag{207.9}$$

holds almost everywhere for large n. For every $P > 0$ and $B \geq 1$,

$$\begin{aligned} &\operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |n^2 p_{n,R}(\ell,t) - \mathfrak{g}_R(y)| \\ &\quad \leq C n^{-1/200}\sqrt{\log(2+n)} + Ce^{-cn^{1/25}} + C_{A,\lambda,P} n^{-P} \\ &\qquad + n^2 \widetilde{\mathcal{E}}_{n,R,A}^{L_n} + (2\pi)^{-4} n^2 \widetilde{\mathcal{C}}_{n,R}^{L_n}(B) + \frac{n^2}{\pi B} A_2(n, L_n, R, 1). \end{aligned} \tag{207.10}$$

As before, this is an extended-real inequality until the displayed local-edge quantity is bounded. The remaining three raw terms retain their full radius and count dependence.

Proof. The support identity and absolutely convergent finite residual inverse used in Theorem 201.2 do not depend on the particular cutoff. Splitting with $\widetilde{\chi}_n$ gives (207.9) exactly. Corollary 207.3, with $a = s_R$, bounds the normalized inverse of the cutoff full law by the stated polynomial error. The omitted Gaussian frequencies have $|v| \geq n^{1/50}$, so their integral is at most $Ce^{-cn^{1/25}}$.

The count distance from the central window to the high-count support is at least $(\lambda - c_*^{-1} - 1)n$. The Schwartz estimate for the new kernel and the mass bound $\|\mathsf{T}^{L_n}\|_{\mathrm{TV}} \leq 1$ give, for every integer $M \geq 1$,

$$n^2 \sup_{\mathcal{W}_{n,R,A}} |\widetilde{K}_n * \mathsf{T}^{L_n}| \leq C_{A,\lambda,M} n^{2/25 - 13M/25}.$$

Choose M large enough for the desired P. The local extracted term is precisely the new $\widetilde{\mathcal{E}}$, not the old edge correction. Split the residual inverse at roof frequency B and use (200.11); its inverse-transform constant is again $1/\pi$. This proves (207.10). □

The remaining fixed-count region. The estimate (207.3) is for the actual full-law transform at the specified count, with the raw n^2 normalization and no division by annular volume. It is not deduced from Theorem K, cyclic arc mass, or the fixed-test-frequency spectral Cauchy limit. The part from the enlarged cutoff toward $n^{-2/5}$, compact nonzero lattice frequencies, and growing roof frequencies still require fixed-count control. The far-roof derivative budget and the complete local extracted-edge correction in (207.10) are unchanged analytical tasks. A first-variation estimate for collision observables does not bound those inverse-coarea derivatives. Likewise, the same-event corollary does not compare a completed-return event with an exact physical-time or lattice observation. These distinctions leave the original joint raw-density endpoint in place while moving its proved fixed-count central boundary.

208. Damped removal of smoothing and higher-moment stopping

The first-order comparison in Lemma 207.1 takes the absolute value of the entire change of observable before using spectral damping. We instead retain a finite Taylor polynomial of that change inside the chronological collision word. Every term of this polynomial still has damping over the full deterministic collision interval. Only its fourth-order remainder is estimated without damping. A second, finer smoothing scale is used for the finitely many insertions; it does not shrink the analytic domain of the twisted collision powers. We also improve the actual stopping estimate by using a fixed higher moment of the visit count.

All constants below are uniform in R. Their dependence on a fixed moment order is allowed. We neither let that order grow with n nor use an all-orders analytic expansion. Write $\|u\|_{\mathcal{V}} = \|u\|_\infty + \|u\|_{\mathrm{BV}}$ and $L_\delta = 1 + |\log \delta|$. Expectations without a subscript are with respect to the stationary collision probability ν.

208.1. **Cumulants of a small bounded-variation residual.**

Lemma 208.1 (Small-mass connected correlations)**.** *Fix $q \ge 2$ and $B < \infty$. If u is real, $\|u\|_{\mathcal{V}} \le B$, and $\|u\|_1 \le B\delta$ for $0 < \delta < 1/2$, then*

$$\sum_{\mathbf{j} \in \mathbb{Z}^{q-1}} \left| \operatorname{cum}(u, u \circ T_R^{j_2}, \dots, u \circ T_R^{j_q}) \right| \le C_{q,B} \delta L_\delta^{q-1}, \tag{208.1}$$

$$|\operatorname{cum}_q(S_{m,R} u)| \le C_{q,B} m \delta L_\delta^{q-1} \qquad (m \ge 1). \tag{208.2}$$

If in addition $\int u \, \mathrm{d}\nu = 0$, then

$$\mathbb{E}|S_{m,R} u|^4 \le C_B (m^2 \delta^2 L_\delta^2 + m \delta L_\delta^3). \tag{208.3}$$

The same fourth-moment bound holds on every deterministic integer interval, also with its orientation reversed.

Proof. In each term of the partition formula (206.2), the block containing the first factor has moment at most $B^{q-1}\|u\|_1$ in absolute value; all other block moments are bounded. Thus the joint cumulant is at most $C_{q,B}\delta$, independently of the times. The independent-block argument in Proposition 206.2 also bounds it by $C_{q,B} e^{-\gamma D}$, where $D = \operatorname{diam}\{0, j_2, \dots, j_q\}$ and $\gamma > 0$ depends on the fixed order.

There are at most $(2J+1)^{q-1}$ anchored tuples with $D \le J$. Choose J to be a sufficiently large fixed multiple of L_δ. Inside this set use the bound $C\delta$; outside it sum the exponential bound, absorbing the fixed-degree lattice counting polynomial in half of the exponential. This gives $C_{q,B}\delta L_\delta^{q-1}$. The exact anchored multiplicity $(m - D)_+$ is at most m, proving (208.2) without a boundary error independent of δ.

For a centered real variable X, the finite moment–cumulant identity is $\mathbb{E}X^4 = \operatorname{cum}_4(X) + 3\operatorname{cum}_2(X)^2$. It follows either by expanding the exponential of its cumulant generating polynomial through degree four, or directly by the partition formula. Apply (208.2) at orders two and four. Stationarity translates an arbitrary integer interval, and reversal does not change its sum. □

Lemma 208.2 (Every fixed even collision moment and its maximum)**.** *Fix an integer $p \ge 2$. For a real centered u with $\|u\|_{\mathcal{V}} \le B$ and $m \ge 1$,*

$$\mathbb{E}\left| \sum_{j=r}^{r+m-1} u \circ T_R^j \right|^{2p} + \mathbb{E} \max_{0 \le l \le m} \left| \sum_{j=r}^{r+l-1} u \circ T_R^j \right|^{2p} \le C_{p,B} m^p. \tag{208.4}$$

The location $r \in \mathbb{Z}$ is arbitrary. The estimate holds in reverse order, for the vector h_R, and under a nonnegative initial density bounded by a fixed constant.

Proof. By Proposition 206.2, every cumulant of $S_m u$ of order $2 \le j \le 2p$ is $O_{p,B}(m)$. The coefficient identity

$$\mathbb{E}X^{2p} = \sum_{\pi \in \mathcal{P}_{2p}} \prod_{B \in \pi} \operatorname{cum}_{|B|}(X)$$

is obtained by exponentiating the generating polynomial through that fixed degree. For centered X, partitions with singleton blocks vanish. Every remaining partition has at most p blocks. The moment of $S_m u$ is consequently at most $C_{p,B} m^p$. No convergence of an infinite cumulant series is used.

Enlarge m to a power of two $M < 2m$. Every prefix is a union of at most one aligned dyadic interval at each scale. At interval length $l = M2^{-j}$ the $2p$th moment of the maximum interval sum is at most $C_{p,B}(M/l)l^p$. Minkowski's inequality over scales therefore gives

$$\left\| \max_{0\le l\le M} |S_l u| \right\|_{2p} \le C_{p,B} \sum_{j=0}^{\log_2 M} (M^p 2^{-j(p-1)})^{1/(2p)} \le C_{p,B}\sqrt{M}.$$

The geometric series converges because $p > 1$. Translation and reversal use invariance, not a changed initial law. A bounded initial density is handled by domination. Summing the four component bounds proves the vector assertion. □

208.2. **Higher-moment windows for the genuine return clock.** Retain $N^{\pm}_{j,R}$, $m_j = \lfloor j/c_* \rfloor$ and $\mathsf{H}_{r,s,R}$ from Section 146. As there, the zero-return clocks and empty sums are zero.

Proposition 208.3 (Higher-moment marked stopping)**.** *For each fixed integer $p \ge 2$, all $j \ge 0$ and $b \ge 2$,*

$$\nu^*_R\{|N^{\pm}_{j,R} - m_j| > b\} \le C_p \frac{(j+b)^p}{b^{2p}}. \tag{208.5}$$

Put $s_p = p/(4p+1)$. For every $n \ge 2$, $0 \le k \le n$ and $v \in \mathbb{R}^4$, the actual marked transform obeys

$$\left| C^{[k],a}_{n,R}(v) - \int_M a(y) e^{iv\cdot \mathsf{H}_{m_k,m_{n-k},R}(y)/\sqrt{n}} \, \mathrm{d}\nu(y) \right| \le C_p M_a (1+|v|) n^{-s_p}. \tag{208.6}$$

No bound on V_a is needed for this stopping comparison.

Proof. The centered visit count $\sum_{i=1}^{L} (\eta_R - c_*) \circ T_R^{\pm i}$ has $2p$th moment at most $C_p L^p$ under ν^*_R, by Lemma 208.2 and $\nu^*_R \le c_*^{-1}\nu$. The exact visit equivalences in Lemma 146.2, followed by Markov's inequality at integer thresholds within a bounded distance of $m_j \pm b$, give (208.5). The required deviation is at least $c_* b - O(1)$ and $L \le C(j+b)$. A negative lower threshold yields an empty event; $j = 0$ is immediate. Enlarge the constant for bounded b.

Recenter at the actual marked return using (146.6). For $2 \le b \le n$, the union of the two clock exceptions has probability at most $C_p n^p / b^{2p}$. Off this union, the difference from the deterministic interval is bounded by maxima on two fixed collision windows of length $O(b)$. Lemma 208.2 bounds their second moments by $C_p b$, including windows crossing collision time zero. On the exceptional set use only the bound two for a difference of phases. Consequently the error is at most

$$C_p M_a \left(\frac{n^p}{b^{2p}} + |v| \sqrt{\frac{b}{n}} \right).$$

Taking $b = \lceil n^{(2p+1)/(4p+1)} \rceil$ gives (208.6). Finitely many small n are absorbed in the constant. The two clocks are not assumed independent, and the weight is never evaluated at a deterministic surrogate for the marked return. □

208.3. **A cubic correction retained inside the damped word.** Let $r, s \ge 0$, $m = r + s \ge 1$, and define

$$I^a_{r,s,R}(z) = \int_M a(y) e^{iz\cdot \mathsf{H}_{r,s,R}(y)} \, \mathrm{d}\nu(y).$$

The insertion need not be positive. The next result uses the coarse scale δ only for $h_{R,\delta}$ and the fine scale ϵ only for the insertion and a finite number of residual factors.

Theorem 208.4 (Damped finite-order removal of smoothing)**.** *Fix $Q \ge 3$, and suppose $\delta \le \delta_0$ and real z satisfy* (206.7)*. For every $0 < \epsilon < 1/2$,*

$$\begin{aligned} |I^a_{r,s,R}(z)| \le{}& CM_a\epsilon^{-8}(1+m|z|)^3 e^{-cm|z|^2} + C\epsilon V_a \\ &+ CM_a\epsilon(1+m|z|)^3 \\ &+ CM_a|z|^4(m^2\delta^2 L_\delta^2 + m\delta L_\delta^3). \end{aligned} \tag{208.7}$$

The constants are independent of the relative location of the mark and of the fine scale. No analytic perturbation in the fine-scale observable is asserted.

Proof. The assertion at $z = 0$ follows by increasing C, so take $z \ne 0$. Put $\xi = z/|z|$, $u = \xi \cdot (h_R - h_{R,\delta})$, $a_\epsilon = \mathsf{S}_\epsilon a$ and $u_\epsilon = \mathsf{S}_\epsilon u$. Mean-preserving smoothing gives

$$\int u \, \mathrm{d}\nu = 0, \quad \|u\|_{\mathcal{V}} \le C, \quad \|u\|_1 \le C\delta, \quad \|u - u_\epsilon\|_1 \le C\epsilon.$$

Also $\|a - a_\epsilon\|_1 \le C\epsilon V_a$ and $\|a_\epsilon\|_\infty \le M_a$. Replace the insertion first, while the full exponential still has modulus one. This costs exactly the second term of (208.7).

Translate the interval to collision times $0, \dots, m-1$. Its mark is now at time $r \in [0, m]$. With $X = |z| S_m u$, use the real-variable Taylor formula

$$e^{iX} = \sum_{j=0}^{3} \frac{(iX)^j}{j!} + \mathcal{R}_4(X), \qquad |\mathcal{R}_4(X)| \le |X|^4/24. \tag{208.8}$$

Multiplication by the coarse phase $e^{iz \cdot S_m h_{R,\delta}}$ does not change this bound. Lemma 208.1 therefore bounds the remainder integral by the last term of (208.7). This estimate is for the original residual u, not for a residual with a presumed independent law.

For $1 \le j \le 3$, expand $(S_m u)^j$ as a sum over its m^j ordered index tuples, allowing repetitions. Within each product, replace each factor $u \circ T_R^t$ by $u_\epsilon \circ T_R^t$. The other factors and the coarse phase are bounded. Telescoping and invariance bound its integrated error by $C_j M_a \epsilon$. After multiplying by $|z|^j/j!$ and summing, this costs at most $CM_a\epsilon(1+m|z|)^3$. No long pullback BV norm is used.

It remains to bound each corrected product. Write $L_z = \mathcal{L}_{R,\delta,z}$. Sort its j residual insertion times together with the mark, retaining repeated times, as $0 \le t_1 \le \cdots \le t_d \le m$, where $d = j + 1 \le 4$. Exactly one b_i is a_ϵ; the others are u_ϵ. The chronological word is

$$\ell\Big(L_z^{m-t_d} M_{b_d} L_z^{t_d - t_{d-1}} M_{b_{d-1}} \cdots L_z^{t_2 - t_1} M_{b_1} L_z^{t_1} \nu\Big). \tag{208.9}$$

For $d = 1$ this means $\ell(L_z^{m-t_1} M_{b_1} L_z^{t_1} \nu)$. The nonnegative power lengths sum to m. Smooth multiplication costs at most $CM_a\epsilon^{-2}$ for the mark and $C\epsilon^{-2}$ for each residual. Corollary 206.4 thus bounds (208.9) by $C_j M_a \epsilon^{-2(j+1)} e^{-cm|z|^2}$. Summing the finitely many degrees and all their index tuples proves the first term of (208.7).

A repeated time is a zero-length power, and a mark at 0 or m is an endpoint multiplier; both are covered by the same power bound. There is never a perturbation of L_z by u_ϵ. In particular, the fine scale does not enter the damping-domain restrictions. □

Why the new estimate has a different scale balance. The undamped observable loss in the earlier comparison was proportional to $|z|\sqrt{m\delta L_\delta}$. In (208.7), the lower-degree changes remain inside a word damped for all m collisions; its undamped residual begins with $|z|^4 m^2 \delta^2 L_\delta^2$. The polynomial cost of the much finer insertions is harmless on an annulus with a growing rescaled inner radius. This argument concerns the characteristic function itself at a deterministic collision length. The passage to a specified actual return count uses (208.6), not an average over counts.

209. A wider prescribed-count band and its raw error budget

We combine the damped removal of smoothing with higher-moment stopping. The resulting estimate reaches beyond the sufficient exponent range in Proposition 207.4. Both the raw normalization and the single actual-return insertion are retained. The unchanged-event consequence and a weighted exact inversion below use the same measures, with no normalization of a truncated law.

Use $M_a, V_a, \alpha_a, C_{n,R}^{[k],a}$ and $\Phi_{n,R}^{[k],a}$ as in Section 207. The insertion is fixed while frequency varies. It is integrated against $\nu|_{Y_R^*}$; taking $a = s_R$ gives the original probability.

209.1. **An explicit band beyond the first-order range.**

Theorem 209.1 (A wider raw-scale annulus at every prescribed return count)**.** *For every sufficiently large n, put*

$$\mathcal{A}_n^\sharp = \{z \in \mathbb{R}^4 : 2n^{-99/200} \le |z| \le 2n^{-19/42}\}. \tag{209.1}$$

The corresponding rescaled radii are $2n^{1/200}$ and $2n^{1/21}$. Uniformly in R, $0 \le k \le n$, and the stated complex insertions,

$$n^2 \int_{\mathcal{A}_n^\sharp} |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C\left(M_a n^{-5/861} + V_a n^{-59/21}\right). \tag{209.2}$$

There is no average over return counts and no division by annular volume.

Proof. Fix the orders and scales

$$Q = 19, \quad p = 10, \quad \delta = \tfrac{1}{4}n^{-1/5}, \quad \epsilon = \tfrac{1}{4}n^{-3}, \quad |v| \le 2n^{1/21}, \quad z = v/\sqrt{n}. \tag{209.3}$$

Here Q is the degree of the inherited spectral jet, $2p = 20$ is only the moment order used for the clock, and the polynomial in the residual phase has degree three. These three orders have different roles. The two analytic damping quantities satisfy

$$|z|\delta^{-2} \le Cn^{-11/210}, \qquad |z|^{18}\delta^{-40} \le Cn^{-1/7}.$$

Thus Corollary 206.4 applies for large n, uniformly in the radius. Increasing the spectral degree alone would not have improved the first-order unsmoothing loss.

Apply Proposition 208.3 with $p = 10$. The deterministic interval has length $m = m_k + m_{n-k} = n/c_* + O(1)$, so $m \asymp n$ uniformly in the mark. Use Theorem 208.4 on that interval. Rescale $v = \sqrt{n}\,z$; in four dimensions the Jacobian n^{-2} cancels the factor n^2 on the left of (209.2).

The first term of (208.7) has integral at most $CM_a n^{26} e^{-cn^{1/100}}$, after decreasing c. Indeed, $\epsilon^{-8} = O(n^{24})$ and $(1 + m|z|)^3 \le Cn^{3/2}(1 + |v|)^3$, while the lower radius is $2n^{1/200}$. This term is bounded by every fixed inverse power of n times M_a. The insertion and finite-product replacement errors have integrated bounds

$$CV_a n^{-3+4/21} = CV_a n^{-59/21}, \qquad CM_a n^{-3+3/2+7/21} = CM_a n^{-7/6}.$$

The fourth-order residual terms have bounds

$$CM_a n^{-2/5+8/21}[\log(2+n)]^2 + CM_a n^{-6/5+8/21}[\log(2+n)]^3.$$

Their power margins are $2/105$ and $86/105$. Finally the integrated stopping error is at most

$$CM_a n^{-10/41+5/21} = CM_a n^{-5/861};$$

the term without $|v|$ is smaller. Since $2/105 > 5/861$, the logarithms in the fourth-order residual terms are absorbed by the strict power slack. This proves (209.2) with the separate variation loss as stated. All orders and constants are fixed before n, R, k, a are varied. □

Corollary 209.2 (Integrated Gaussian comparison on the wider band)**.** *Under the same conventions,*

$$\int_{|v|\le 2n^{1/21}} \left| C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf T} D_R v/2} \right| \mathrm{d}v \le C\left(M_a n^{-5/861} + V_a n^{-9/175}\right). \tag{209.4}$$

The physical support radius is $2n^{-19/42}$. The sharper inherited rate on $|v| \le 2n^{1/200}$ remains a separate valid estimate.

Proof. On the original central ball apply Theorem 146.5. On its complement up to the new radius apply Theorem 209.1 and the uniformly elliptic Gaussian tail. Use $|\alpha_a| \le M_a$, $3/280 > 5/861$, and $59/21 > 9/175$. □

Proposition 209.3 (A fixed-order family of wider bands)**.** *Fix $1/200 < \varepsilon < 1/20$. Choose*

$$4\varepsilon < \theta < \frac{1}{4} - \frac{\varepsilon}{2}, \qquad \frac{p}{4p+1} > 5\varepsilon, \qquad (Q-1)(1/2-\varepsilon) > 2(Q+1)\theta, \tag{209.5}$$

with fixed integers $p \ge 2$, $Q \ge 3$. For every

$$0 < \eta < \min\{2\theta - 8\varepsilon,\ p/(4p+1) - 5\varepsilon\},$$

the annulus from $2n^{-99/200}$ to $2n^{-1/2+\varepsilon}$ satisfies

$$n^2 \int |\Phi_{n,R}^{[k],a}(z)|\, \mathrm{d}z \le C_{\varepsilon,\theta,p,Q,\eta}\left(M_a n^{-\eta} + V_a n^{-(3-4\varepsilon)}\right). \tag{209.6}$$

Its rescaled outer radius is $2n^{\varepsilon}$. Every choice in (209.5) *is feasible, and all orders remain fixed as n tends to infinity.*

Proof. Choose $\delta = n^{-\theta}/4$ and $\epsilon = n^{-3}/4$. The strict upper bound on θ leaves a nonzero analytic-disk margin, and a sufficiently large fixed Q gives the second damping margin. The interval for θ is nonempty for $\varepsilon < 1/18$, hence for the stated range. Since $p/(4p+1) \uparrow 1/4$, a fixed p with the required stopping margin exists whenever $\varepsilon < 1/20$.

The two principal integrated losses are $n^{-2\theta+8\varepsilon}(\log n)^2$ and $n^{-p/(4p+1)+5\varepsilon}$. The lower residual term has margin $1 + \theta - 8\varepsilon > 2\theta - 8\varepsilon$, because $\theta < 1$. The fine-scale replacement has margin $3/2 - 7\varepsilon$, and the insertion error has margin $3 - 4\varepsilon$. Both exceed the required positive loss margins where needed. Damping absorbs all fixed polynomial prefactors. Absorb the logarithms by the strict choice of η and repeat the raw rescaling. No constant is asserted uniform as an order grows. □

Comparison of frequency scales. The earlier sufficient range was $\varepsilon < 1/42$ (approximately 0.02381); the new one is $\varepsilon < 1/20 = 0.05$. The explicit choice $1/21$ is strictly inside the new range. Neither sufficient range reaches $1/10$, the exponent corresponding to physical radius $n^{-2/5}$. The remaining interval beyond the new cutoff is not included by a change of notation. No one of these sufficient ranges is an impossibility assertion for a different method.

209.2. **The unchanged event and a weighted raw identity.**

Corollary 209.4 (An unchanged marked-state event on the new band)**.** *Let $E_{n,R} \subset Y_R^*$ have zero-extended indicator of variation $V_{n,R}$ and probability $p_{n,R} > 0$. For the same event*

$A_{n,k,R} = \{x : (F_R^*)^k x \in E_{n,R}\}$ *in numerator and denominator,*

$$\int_{|v|\le 2n^{1/21}} \left| \mathbb{E}_{\nu_R^*}[e^{iv\cdot U_{n,R}/\sqrt{n}} \mid A_{n,k,R}] - e^{-v^{\mathsf{T}} D_R v/2} \right| \mathrm{d}v \le \frac{C}{p_{n,R}} \left(n^{-5/861} + V_{n,R} n^{-9/175} \right). \tag{209.7}$$

In particular, $p_{n,R} \ge cn^{-\beta}$ *and* $V_{n,R} \le Cn^{\kappa}$ *suffice for convergence if* $\beta < 5/861$ *and* $\beta + \kappa < 9/175$.

Proof. Set $a = c_*^{-1} \mathbf{1}_{E_{n,R}}$ in Corollary 209.2. Its insertion mass is the exact probability $p_{n,R}$, which is also the probability of $A_{n,k,R}$ by invariance. Divide by that same number. This proves the claim without a replacement of the conditioning event. □

Put

$$B_n^\sharp = n^{-19/42}, \quad \chi_n^\sharp(\omega) = \chi(\omega/B_n^\sharp), \quad K_n^\sharp = \mathcal{F}^{-1}\chi_n^\sharp. \tag{209.8}$$

The support radii are $2n^{-19/42}$ physically and $2n^{1/21}$ after rescaling. For a marked insertion define

$$w_{n,k,R}(x) = c_* a((F_R^*)^k x).$$

Then $(J_{n,R})_*(w_{n,k,R}\nu_R^*)$ is exactly $\mu_{n,R}^{[k],a}$ of (146.2), with mass α_a, not a renormalized probability.

For the next theorem require, in addition to the BV assumptions, that this weight be admissible in Definition 200.1 at the count cutoffs in use. For example a bounded insertion subanalytic on each collision chart has this property, by the finite collision graph and exact return membership. This additional requirement is not imposed on the preceding Fourier theorems. In particular arbitrary BV and finite-record subanalytic classes are not being identified.

Let $\mu^{a,L}$ be this measure restricted to $N_{n,R} \le L$, $\mathsf{T}^{a,L} = \mu^{[k],a} - \mu^{a,L}$, and let $E^{a,L}, Q^{a,L}$ be its exact finite extraction with densities $e^{a,L}, r^{a,L}$. Suppress n, k, R only in these formulas. Define anew

$$\mathcal{E}_{n,k,R,A}^{\sharp,a,L} = \operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |e^{a,L} - K_n^\sharp * E^{a,L}|,$$
$$\mathcal{C}_{n,k,R}^{\sharp,a,L}(B) = \int_{\mathbb{T}^3\times[-B,B]} |1-\chi_n^\sharp(\omega)|\,|\widehat{Q}^{a,L}(\omega)|\,\mathrm{d}\omega. \tag{209.9}$$

The edge correction is kernel-dependent, and may be infinite before its density germs are estimated.

Theorem 209.5 (Exact weighted raw error at the new cutoff)**.** *Let* $L_n = \lceil \lambda n \rceil$, $\lambda > c_*^{-1} + 1$. *For every admissible marked insertion as above, the mixed density* $p_{n,R}^{[k],a}$ *obeys, almost everywhere on* $\mathcal{W}_{n,R,A}$,

$$p_{n,R}^{[k],a} = K_n^\sharp * \mu_{n,R}^{[k],a} + (e^{a,L_n} - K_n^\sharp * E^{a,L_n}) + \mathcal{F}^{-1}[(1-\chi_n^\sharp)\widehat{Q}^{a,L_n}] - K_n^\sharp * \mathsf{T}^{a,L_n}. \tag{209.10}$$

Writing $y = ((\ell,t) - n\bar{G}_R)/\sqrt{n}$, *for* $B \ge 1$ *and* $P > 0$,

$$\begin{aligned} &\operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |n^2 p_{n,R}^{[k],a}(\ell,t) - \alpha_a \mathfrak{g}_R(y)| \\ &\quad \le C(M_a n^{-5/861} + V_a n^{-9/175}) + CM_a e^{-cn^{2/21}} + C_{A,\lambda,P} M_a n^{-P} \\ &\qquad + n^2 \mathcal{E}_{n,k,R,A}^{\sharp,a,L_n} + (2\pi)^{-4} n^2 \mathcal{C}_{n,k,R}^{\sharp,a,L_n}(B) + \frac{n^2}{\pi B} A_2(n, L_n, R, w_{n,k,R}). \end{aligned} \tag{209.11}$$

The inequality is interpreted in the extended reals until the local edge correction is bounded. The original unweighted law is the case $a = s_R$, for which $w_{n,k,R} = 1$.

Proof. The full marked measure is dominated in total variation by $c_* M_a \mu_{n,R}$, so it has a mixed density. Its restriction to a count fiber at most L_n agrees exactly with that of the full measure. Apply Theorem 200.7 to the admissible $w_{n,k,R}$; its residual Fourier inverse is absolutely convergent. Splitting this inverse by $\chi_n^\sharp$ and substituting $\mu^{a,L_n} = \mu^{[k],a} - \mathsf{T}^{a,L_n}$ proves (209.10), with the same exact support argument as Lemma 201.1.

Corollary 209.2, after rescaling in the cutoff inverse transform, bounds the first term by the first error in (209.11). The missing Gaussian frequencies satisfy $|v| \geq n^{1/21}$ and contribute at most $CM_a e^{-cn^{2/21}}$. For every integer $J \geq 1$, count-coordinate separation and the Schwartz bound on the new kernel give

$$n^2 \sup_{\mathcal{W}_{n,R,A}} |K_n^\sharp * \mathsf{T}^{a,L_n}| \leq C_{A,\lambda,J} M_a n^{4/21 - 23J/42}.$$

Here $n^2 (B_n^\sharp)^4 = n^{4/21}$ and $n B_n^\sharp = n^{23/42}$; the mass bound is $\|\mathsf{T}^{a,L_n}\|_{\mathrm{TV}} \leq c_* M_a$. Choose J for the desired P. Finally split the residual inverse at roof frequency B. The interior integral is exactly $\mathcal{C}_{n,k,R}^{\sharp,a,L_n}(B)$; the exterior estimate in (200.11), multiplied by $(2\pi)^{-4}$, gives $A_2/(\pi B)$. This proves the inequality with every weighted raw term retained. □

The remaining raw estimates. The new contribution is a fixed-count full-law estimate over a wider portion of the complement, not a function-defect or averaged-return substitute. The finite spectral jet, quartic residual estimate and higher visit moments concern bounded collision observables. They do not estimate A_2 or the local extracted density germs at a linear count cutoff. Likewise, the finite-band residual integral beyond the new cutoff still includes the farther small annulus, compact nonzero and peripheral return frequencies, and growing roof frequencies. These three raw terms must be controlled together in a compatible splice. The weighted identity specifies the common BV and subanalytic class for which both the new central theorem and the complete finite extraction apply. It does not place an arbitrary multiple-return or exact physical-time event in that class, nor compare two rare events.

210. Multiscale stopping and all fixed marked moments

A single clock window assigns the same exceptional-set cost to every larger deviation. Summing instead over the actual deviation scales recovers a fourth-root comparison. This estimate is uniform in the location of a prescribed return mark; it does not take a maximum over all marks inside the expectation. The stronger comparison will also remove the loss of central rate on the wider Fourier band.

Retain the notation of (147.4):

$$Z_{n,k,R} = \mathsf{H}_{N_{k,R}^-, N_{n-k,R}^+, R}, \qquad W_{n,k,R} = \mathsf{H}_{m_k, m_{n-k}, R}, \qquad m_j = \lfloor j/c_* \rfloor.$$

In particular, recentering the orbit at its actual marked return gives $Z_{n,k,R}(y) = U_{n,R}((F_R^*)^{-k} y)$, where $U_{n,R} = J_{n,R} - n\bar{G}_R$. All norms in the stopping argument are under the normalized section probability ν_R^*. Constants may depend on a fixed moment order, but not on R, n, k.

210.1. **Deviation shells rather than one exceptional set.**

Lemma 210.1 (Two bounds for the total clock deviation)**.** *Set*

$$D_{n,k,R} = |N_{k,R}^- - m_k| + |N_{n-k,R}^+ - m_{n-k}|.$$

For every fixed integer $p \geq 2$ there are constants $C_p, a_0, c_0 > 0$ such that, for $n \geq 2$ and $b \geq 2$,

$$\nu_R^*(D_{n,k,R} > b) \leq C_p \frac{(n+b)^p}{b^{2p}}, \tag{210.1}$$

$$\nu_R^*(D_{n,k,R} > b) \leq C_p e^{a_0 n - c_0 b}. \tag{210.2}$$

The second bound may exceed one. It will only be used beyond a fixed multiple of n.

Proof. A deviation greater than b requires one of the two clocks to deviate by more than $b/2$. Equation (208.5) and a union bound prove (210.1), with a change of constant for bounded b. This argument includes $k = 0, n$; a zero-return clock is identically zero.

Put $Q = N_{k,R}^- + N_{n-k,R}^+$. Under ν_R^*, a backward clock of j returns has the same law as a forward clock of j returns, since $N_{j,R}^-((F_R^*)^j x) = N_{j,R}^+(x)$ and the actual return map preserves that probability. The cumulative-return estimate therefore gives

$$\nu_R^*(Q > L) \leq C e^{an - cL/2} \qquad (L \geq 0)$$

by a union bound, with no independence of the two clocks. Since $D_{n,k,R} \leq Q + m_k + m_{n-k} \leq Q + c_*^{-1} n$, translating the threshold proves (210.2). For negative translated thresholds, increase a_0 and C_p so that the displayed right side is at least one. □

Theorem 210.2 (Fourth-root stopping in every fixed finite moment)**.** *For every fixed real $q \geq 1$ there is $C_q < \infty$ such that*

$$\|Z_{n,k,R}\|_q + \|W_{n,k,R}\|_q \leq C_q \sqrt{n}, \tag{210.3}$$

$$\|Z_{n,k,R} - W_{n,k,R}\|_q \leq C_q n^{1/4} \tag{210.4}$$

for all $n \geq 2$, $0 \leq k \leq n$, and $R \in I$. Consequently, for every bounded insertion a supported on the section and every real v,

$$\left| C_{n,R}^{[k],a}(v) - \int_M a(y) e^{iv \cdot W_{n,k,R}(y)/\sqrt{n}} \, \mathrm{d}\nu(y) \right| \leq C M_a |v| n^{-1/4}. \tag{210.5}$$

For $q = 2$, the proof of (210.4) *uses only the fourth collision moments, the fourth-moment clock window, and the cumulative-return tail.*

Proof. Choose a fixed integer $p \geq 2$ such that $2p > 3q/2$. The deterministic interval has length at most $c_*^{-1} n$. Its qth moment is bounded by Lemma 208.2 at a sufficiently large fixed even order, followed by domination of the initial density. For Z, split at $Q \leq An$, with Q as in the preceding proof and A a sufficiently large fixed constant. On this event, $|Z|$ is at most the sum of two deterministic collision maxima of length $\lceil An \rceil$. Their qth moments are $O_q(n^{q/2})$. On the complement, boundedness of h_R gives $|Z| \leq CQ$. Integration of the cumulative exponential tail bounds the remaining qth moment by $C_q(1+n)^q e^{-c'n}$. This proves (210.3).

We give the full multiscale argument for the difference. For each $L \geq 1$, let B_L be the sum of the four prefix or suffix maxima on the deterministic length-$\lceil L \rceil$ windows adjacent to the two endpoints $-m_k$ and m_{n-k}. All these windows are permitted to cross collision time zero. They are fixed intervals, not intervals conditioned on the clocks. By Lemma 208.2,

$$\int B_L^{2p} \, \mathrm{d}\nu_R^* \leq C_p L^p. \tag{210.6}$$

On $D_{n,k,R} \leq L$ one has the pathwise bound $|Z - W| \leq B_L$. Let $b_0 = \lceil \sqrt{n} \rceil$ and $L_j = 2^j b_0$ for $j \geq 1$. Partition the sample space into $\{D \leq b_0\}$ and the disjoint sets $\{L_j/2 < D \leq L_j\}$, $j \geq 1$. The clocks are finite almost everywhere, so these sets exhaust the section up to a null set. The first set contributes at most $C_q b_0^{q/2}$ to $\int |Z - W|^q \, \mathrm{d}\nu_R^*$.

On every other shell, Hölder's inequality gives

$$\int_{\{L_j/2<D\le L_j\}} |Z-W|^q \,\mathrm{d}\nu_R^* \le C_{p,q} L_j^{q/2} \nu_R^*(D > L_j/2)^{1-q/(2p)}. \tag{210.7}$$

The exponent is positive. Fix A large enough that the exponential bound (210.2) at threshold $L_j/2$ is at most $Ce^{-c_0 L_j/4}$ whenever $L_j > An$. For the shells $L_j \le An$, use (210.1) in (210.7). Their total is at most

$$C_{p,q,A} \sum_{L_j \le An} n^{p-q/2} L_j^{3q/2-2p} \le C_{p,q,A} n^{q/4} \sum_{j\ge 1} 2^{j(3q/2-2p)} \le C_q n^{q/4}. \tag{210.8}$$

Here $b_0 \asymp \sqrt{n}$ and $3q/2 - 2p < 0$. For the remaining shells, (210.7) is bounded by $C_{p,q} L_j^{q/2} e^{-c_q L_j}$, which is summable and contributes no more than $C_q n^{q/4}$. Taking the qth root proves (210.4). There is no logarithmic count of shells: the shell contributions form a convergent geometric series. For $q = 2$ one can take $p = 2$, in which case the moderate-shell sum is $C \sum n/L_j \le C\sqrt{n}$.

Finally, recenter at the actual mark and use $|e^{ix} - e^{iy}| \le |x - y|$. Since a vanishes outside the section, $|a|\nu \le c_* M_a \nu_R^*$ there. Equation (210.4) with $q = 1$ proves (210.5); its right side is exactly zero at $v = 0$. No martingale representation, conditional independence, or mixing of the induced map is used. □

210.2. **One anchored factor in an arbitrary fixed moment.** For an integer $d \ge 1$ and a symmetric positive semidefinite matrix D, write

$$\mathcal{G}_d(D) = \mathbb{E}\, \mathcal{Z}_D^{\otimes d}, \qquad \mathcal{Z}_D \sim N(0, D), \qquad \rho_d = \begin{cases} 1/2, & d \text{ odd}, \\ 1, & d \text{ even}. \end{cases}$$

The tensor is zero for odd d. Its norm below may be any fixed Euclidean tensor norm.

Lemma 210.3 (A fixed marked collision moment)**.** *Let a be complex and bounded-BV on the collision cylinder, with $\alpha = \int a \,\mathrm{d}\nu$. For a deterministic two-sided interval of length $m = r + s \ge 1$,*

$$\left\| \int_M a\, \mathsf{H}_{r,s,R}^{\otimes d} \,\mathrm{d}\nu - \alpha m^{d/2} \mathcal{G}_d(\Gamma_R) \right\| \le C_d(\|a\|_\infty + \|a\|_{\mathrm{BV}}) m^{d/2-\rho_d}. \tag{210.9}$$

The constant is independent of the position of zero in the interval.

Proof. First take a real unit direction ξ and put $u = \xi \cdot h_R$. Consider a joint cumulant containing $a - \alpha$ at time zero and $j \ge 1$ copies of u at integer times $i_1, \dots, i_j$. The independent-block comparison in Proposition 206.2 applies also to these different observables: the finite-order decoupling estimate is multilinear, and the partition polynomial is a finite sum of products of subset moments. After normalization of $a - \alpha$, its absolute value is bounded by

$$C_j(\|a\|_\infty + \|a\|_{\mathrm{BV}}) e^{-c_j \operatorname{diam}\{0, i_1, \dots, i_j\}}.$$

It is summable over all $\mathbb{Z}^j$. Thus the cumulant containing the single centered mark and j copies of the sum of u on any interval is bounded by $C_j(\|a\|_\infty + \|a\|_{\mathrm{BV}})$, independently of its length or position. Complex marks are handled by real and imaginary parts.

Expand the moment with one centered mark by the finite moment–cumulant partition identity. The block containing that mark must contain at least one copy of the sum. Every other nonzero block contains at least two copies, because u is centered. At most $\lfloor (d-1)/2 \rfloor$ such blocks remain. Each pure sum cumulant is $O_d(m)$ by the inherited connected-correlation theorem. Hence

$$\left| \int (a - \alpha)(\xi \cdot \mathsf{H}_{r,s,R})^d \,\mathrm{d}\nu \right| \le C_d(\|a\|_\infty + \|a\|_{\mathrm{BV}}) m^{\lfloor (d-1)/2 \rfloor}.$$

For the unweighted moment, all cumulants of order at least two are $O_d(m)$ and the second is $m\xi^{\mathsf{T}} \Gamma_R \xi + O(1)$. If d is even, the pair partitions give the Gaussian term; replacing one covariance

by its bounded correction, or using a nonpair partition, lowers the power of m by at least one. If d is odd, there are at most $(d-1)/2$ nonzero blocks and the Gaussian moment is zero. For $d=1$ the unweighted moment is exactly zero. These observations prove (210.9) against every $\xi^{\otimes d}$. Polarization in the fixed dimension four gives the symmetric tensor assertion. All cumulant orders used are fixed, at most $d+1$. □

Theorem 210.4 (All fixed Gaussian moments with an actual return mark)**.** *For every fixed integer $d \geq 1$, all $n \geq 2$ and $0 \leq k \leq n$,*

$$\left\| \int_{Y_R^*} a((F_R^*)^k x) \left(\frac{U_{n,R}(x)}{\sqrt{n}} \right)^{\otimes d} \mathrm{d}\nu(x) - \alpha_a \mathcal{G}_d(D_R) \right\| \leq C_d \left(M_a n^{-1/4} + (M_a + V_a) n^{-\rho_d} \right). \tag{210.10}$$

The convention is the restricted collision measure, with $\alpha_a = \int a \, \mathrm{d}\nu$; no positivity of a is required. In particular the actual normalized covariance and its induced Cesaro Green–Kubo expression converge to D_R at rate $O(n^{-1/4})$.

Proof. Recenter at $y = (F_R^*)^k x$. The original numerator is

$$\int_{Y_R^*} a(y) \left(\frac{Z_{n,k,R}(y)}{\sqrt{n}} \right)^{\otimes d} \mathrm{d}\nu(y).$$

For $d \geq 2$, the tensor telescoping inequality

$$\|x^{\otimes d} - y^{\otimes d}\| \leq C_d |x-y| (|x| + |y|)^{d-1}$$

and Hölder's inequality with exponents d and $d/(d-1)$ show that its change on replacing Z by W is at most $C_d M_a n^{-1/4}$, by (210.3) and (210.4). For $d=1$ use the first moment version directly. Restriction from ν to the section changes only the fixed factor c_*.

Apply Lemma 210.3 to W with $m = m_k + m_{n-k} = n/c_* + O(1)$. Division by $n^{d/2}$ gives the second error in (210.10). The Gaussian term is $\alpha_a \mathcal{G}_d((m/n)\Gamma_R)$, which differs from the stated term by $O_d(M_a/n)$ for even d and is zero for odd d. The covariance matrices are uniformly bounded. This proves the theorem.

For the unweighted assertion take $a = s_R$, so $\alpha_a = 1$ and both insertion norms are uniformly bounded. The Kac mean makes the actual first centered moment zero. At $d=2$ the theorem is therefore a covariance estimate. Expanding that exact covariance under stationarity of F_R^* gives the Cesaro expression in (147.14), with the improved error. Absolute summability of induced correlations is not asserted. □

Corollary 210.5 (Polynomial moments under the unchanged event)**.** *Let $E_{n,R} \subset Y_R^*$ have probability $p_{n,R} > 0$ and zero-extended indicator variation $V_{n,R}$. For $A_{n,k,R} = \{x : (F_R^*)^k x \in E_{n,R}\}$ and every fixed $d \geq 1$,*

$$\left\| \mathbb{E}_{\nu_R^*} \left[(U_{n,R}/\sqrt{n})^{\otimes d} \mid A_{n,k,R} \right] - \mathcal{G}_d(D_R) \right\| \leq \frac{C_d}{p_{n,R}} \left(n^{-1/4} + (1 + V_{n,R}) n^{-\rho_d} \right). \tag{210.11}$$

For $p_{n,R} \geq c n^{-\beta}$ and $V_{n,R} \leq C n^{\kappa}$, $\beta < 1/4$ and $\beta + \kappa < \rho_d$ suffice. Every fixed polynomial moment and the conditional covariance converge under $\beta < 1/4$ and $\beta + \kappa < 1/2$.

Proof. Take $a = c_*^{-1} \mathbf{1}_{E_{n,R}}$ in Theorem 210.4. Its mass equals $p_{n,R}$, also the probability of $A_{n,k,R}$ by invariance. Divide by this unchanged denominator. The conditional first moment tends to zero under the stronger stated conditions, so subtracting its tensor square from the conditional

second moment proves the covariance assertion. A fixed polynomial is a finite linear combination of the displayed tensor coordinates. □

Scope of the moment improvement. The fourth-root rate is obtained for the actual unbounded record at one prescribed count and mark. It is not a maximal-in-k theorem, a statement about growing moment orders, or an event-replacement estimate. In particular it neither assumes induced mixing nor treats a first-variation bound as a bound for inverse-coarea second derivatives.

211. A wider fixed-count band with the original central rate

The multiscale stopping estimate can be combined with damped cubic unsmoothing without altering the actual return mark. We choose the outer radius so that the integrated stopping error has the same power as the original marked central estimate. Thus the wider band no longer requires a slower probability-decay budget for a single marked event.

Set

$$\varepsilon_\natural = \frac{67}{1400}, \qquad V_n^\natural = n^{67/1400}, \qquad B_n^\natural = n^{-633/1400}. \tag{211.1}$$

The smooth Fourier support radii are $2V_n^\natural$ after rescaling and $2B_n^\natural$ in physical frequency. The preceding radius $n^{1/21}$ is strictly smaller than $V_n^\natural$. All statements below are at the prescribed return count n, not averaged in n.

Theorem 211.1 (Rate-preserving prescribed-count annulus)**.** *Uniformly in $R \in I$, $0 \le k \le n$ and the marked bounded-BV insertion a, for all sufficiently large n,*

$$n^2 \int_{2n^{-99/200} \le |z| \le 2B_n^\natural} |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C\left(M_a n^{-3/280} + V_a n^{-983/350}\right), \tag{211.2}$$

$$\int_{|v| \le 2V_n^\natural} |C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf{T}} D_R v/2}|\,\mathrm{d}v \le C\left(M_a n^{-3/280}\sqrt{\log(2+n)} + V_a n^{-9/175}\right). \tag{211.3}$$

The original probability corresponds to $a = s_R$. The transform with a general insertion is still integrated against $\nu|_{Y_R^*}$, with mass $\alpha_a = \int a\,\mathrm{d}\nu$.

Proof. Recenter at the actual mark. Equation (210.5) compares the original transform with $I^a_{m_k, m_{n-k}, R}(v/\sqrt{n})$ at cost $CM_a|v|n^{-1/4}$. The deterministic length satisfies $m_k + m_{n-k} = n/c_* + O(1)$. Apply Theorem 208.4 with

$$Q = 19, \qquad \delta = \tfrac{1}{4}n^{-1/5}, \qquad \epsilon = \tfrac{1}{4}n^{-3}, \qquad |v| \le 2n^{67/1400}.$$

The coarse scale determines the collision perturbation disk, while only the insertions use the fine scale. For $z = v/\sqrt{n}$, the two required quantities satisfy

$$\begin{aligned} |z|\delta^{-2} &= O(n^{-73/1400}), \\ |z|^{18}\delta^{-40} &= O(n^{-97/700}). \end{aligned} \tag{211.4}$$

Hence the damping theorem applies for all sufficiently large n. On the annulus $2n^{1/200} \le |v| \le 2V_n^\natural$, the damped polynomial term is smaller than every fixed inverse power of n: its prefactor is polynomial and its exponential is at most $e^{-cn^{1/100}}$.

We display every nondamping integrated exponent. Four-dimensional volume and the largest frequency factor give respectively

$$\begin{aligned}
\epsilon V_a (V_n^\natural)^4 &\le C V_a n^{-983/350},\\
M_a \epsilon (1+\sqrt{n} V_n^\natural)^3 (V_n^\natural)^4 &\le C M_a n^{-233/200},\\
M_a \delta^2 L_\delta^2 (V_n^\natural)^8 &\le C M_a n^{-3/175} [\log(2+n)]^2,\\
M_a n^{-1} \delta L_\delta^3 (V_n^\natural)^8 &\le C M_a n^{-143/175} [\log(2+n)]^3,\\
M_a n^{-1/4} (V_n^\natural)^5 &= M_a n^{-3/280}.
\end{aligned} \tag{211.5}$$

The quartic residual exponent exceeds $3/280$ by $9/1400$, so its logarithms are absorbed by this strict power margin. Every other supremum-norm error also decays faster than the last term. Since $v = \sqrt{n} z$ has Jacobian $\mathrm{d}v = n^2 \, \mathrm{d}z$ in four dimensions, these estimates prove (211.2) with the raw factor n^2 and no division by annular volume.

On $|v| \le 2n^{1/200}$, use the sharper inherited Theorem 146.5. On the rest of the new ball, use (211.2) and the uniformly elliptic Gaussian tail. Since $983/350 > 9/175$, their sum gives (211.3). This does not replace the old small-ball theorem by a weaker one. □

Remark 211.2 (The choice of radius). The equality

$$\frac{1}{4} - 5\varepsilon_\natural = \frac{3}{280}$$

selects the radius for which this stopping estimate retains the original central power. It is a balance of the displayed error bounds, not an assertion of optimality or an obstruction to another method. With a smaller positive rate, the preceding construction still reaches every fixed $1/200 < \varepsilon < 1/20$. Neither statement covers the rescaled endpoint $n^{1/10}$, corresponding to physical frequency $n^{-2/5}$.

Corollary 211.3 (The original rare-event budget on the wider band)**.** *For the unchanged marked-state event $A_{n,k,R}$, probability $p_{n,R}$, and indicator variation $V_{n,R}$ of Corollary 210.5,*

$$\begin{aligned}
&\int_{|v| \le 2V_n^\natural} \left| \mathbb{E}_{\nu_R^*} [e^{iv \cdot U_{n,R}/\sqrt{n}} \mid A_{n,k,R}] - e^{-v^{\mathsf{T}} D_R v/2} \right| \mathrm{d}v \\
&\qquad \le \frac{C}{p_{n,R}} \left(n^{-3/280} \sqrt{\log(2+n)} + V_{n,R} n^{-9/175} \right).
\end{aligned} \tag{211.6}$$

Thus $p_{n,R} \ge c n^{-\beta}$ and $V_{n,R} \le C n^{\kappa}$ suffice when $\beta < 3/280$ and $\beta + \kappa < 9/175$. Under these conditions, every fixed conditional polynomial moment also has the Gaussian limit.

Proof. Use $a = c_*^{-1} \mathbf{1}_{E_{n,R}}$ in (211.3). Its mass is the exact denominator $p_{n,R}$. Division proves the bound without changing the event. The stated inequalities also imply the moment conditions in Corollary 210.5. □

211.1. **The raw identity with the new kernel.** Define

$$\chi_n^\natural(\omega) = \chi(\omega / B_n^\natural), \qquad K_n^\natural = \mathcal{F}^{-1} \chi_n^\natural.$$

For a marked insertion additionally satisfying the finite-record admissibility in Definition 200.1, use the same weight $w_{n,k,R} = c_* a \circ (F_R^*)^k$ and the same exact finite extraction $\mu^{a,L} = E^{a,L} + Q^{a,L}$ as in Theorem 209.5. Set, with the new kernel throughout,

$$\begin{aligned}
\mathcal{E}_{n,k,R,A}^{\natural,a,L} &= \operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |e^{a,L} - K_n^\natural * E^{a,L}|,\\
\mathcal{C}_{n,k,R}^{\natural,a,L}(B) &= \int_{\mathbb{T}^3 \times [-B,B]} |1 - \chi_n^\natural(\omega)| \, |\widehat{Q}^{a,L}(\omega)| \, \mathrm{d}\omega.
\end{aligned} \tag{211.7}$$

In particular the edge correction is not copied from a different cutoff. It may still be infinite before its germs are estimated.

Corollary 211.4 (Exact weighted raw budget with the retained central rate)**.** *Let* $L_n = \lceil \lambda n \rceil$ *with* $\lambda > c_*^{-1} + 1$*. For* $B \geq 1$*,* $P > 0$*, and* $y = ((\ell, t) - n\bar{G}_R)/\sqrt{n}$*,*

$$\begin{aligned}
&\operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |n^2 p_{n,R}^{[k],a}(\ell, t) - \alpha_a \mathfrak{g}_R(y)| \\
&\quad \leq C\left(M_a n^{-3/280}\sqrt{\log(2+n)} + V_a n^{-9/175}\right) + CM_a e^{-cn^{67/700}} + C_{A,\lambda,P} M_a n^{-P} \\
&\qquad + n^2 \mathcal{E}_{n,k,R,A}^{\natural,a,L_n} + (2\pi)^{-4} n^2 \mathcal{C}_{n,k,R}^{\natural,a,L_n}(B) + \frac{n^2}{\pi B} A_2(n, L_n, R, w_{n,k,R}).
\end{aligned} \tag{211.8}$$

The inequality has the extended-real interpretation when needed. Both cutoff radii are as in (211.1)*.*

Proof. On the central count labels the high-count tail has no density contribution. The same exact support and absolute-residual-inversion argument gives

$$\begin{aligned}
p_{n,R}^{[k],a} = K_n^{\natural} * \mu_{n,R}^{[k],a} + (e^{a,L_n} - K_n^{\natural} * E^{a,L_n}) \\
+ \mathcal{F}^{-1}[(1 - \chi_n^{\natural})\widehat{Q}^{a,L_n}] - K_n^{\natural} * \mathsf{T}^{a,L_n}.
\end{aligned}$$

Use (211.3) for the first term. The missing Gaussian frequencies have $|v| \geq V_n^{\natural}$ and give the stated exponential tail. Count separation is unchanged, but the kernel scale is new. Its Schwartz estimate is

$$n^2 \sup_{\mathcal{W}_{n,R,A}} |K_n^{\natural} * \mathsf{T}^{a,L_n}| \leq C_{A,\lambda,J} M_a n^{67/350 - 767J/1400}.$$

Indeed $n^2 (B_n^{\natural})^4 = n^{67/350}$ and $nB_n^{\natural} = n^{767/1400}$. Choose J for the prescribed power P. Splitting the residual inverse at roof frequency B gives exactly the last two terms, by the inherited absolute second-derivative far-tail estimate. The local edge term stays in its combined form. □

What remains beyond this band. The new fixed-count estimate improves the physical transform, rather than an average or a compressed operator. The outer region beginning at $2n^{-633/1400}$, compact nonzero and peripheral return frequencies, and growing roof frequencies still enter the complementary residual integral. The long-time bound for A_2 at $L_n \asymp n$ and the local edge correction remain distinct terms in (211.8). A relative comparison with an exact physical-time event is not performed. The new all-moment and same-event conclusions preserve the actual indicator and its exact probability; they are not a substitute for those raw estimates.

212. Damped removal of smoothing at arbitrary fixed order

The cubic expansion is sufficient for the isotropic band of Section 211. Different widths in the four physical frequency coordinates require a separate balance. We first prove a finite-order version in which the residual degree, the spectral Taylor degree, and the two smoothing scales are independent. The expansion order is fixed before the collision length varies. No estimate uniform in that order, and no analytic unsmoothed twist, is used.

212.1. **Small-mass moments and heterogeneous cumulants.** Write $L_\delta = 1 + |\log \delta|$ and use the stationary probability ν in this subsection.

Lemma 212.1 (Every fixed even small-mass moment)**.** *Fix an integer $P \ge 1$ and $B < \infty$. Suppose that u is real, centered, $\|u\|_{\mathcal{V}} \le B$, and $\|u\|_1 \le B\delta$, with $0 < \delta < 1/2$. Then*

$$\int |S_{m,R}u|^{2P} \, \mathrm{d}\nu \le C_{P,B} \sum_{b=1}^{P} (m\delta)^b L_\delta^{2P-b} \qquad (m \ge 1). \tag{212.1}$$

The estimate holds on any translated or reversed deterministic interval. Its constant is uniform in R, u, δ, m within the stated class, but may depend on P.

Proof. Lemma 208.1 gives, simultaneously for the finitely many orders $2 \le j \le 2P$,

$$|\operatorname{cum}_j(S_{m,R}u)| \le C_{P,B} m\delta L_\delta^{j-1}.$$

Expand the $2P$th moment by the finite partition formula. Centering removes singleton blocks. A remaining partition with b blocks has $1 \le b \le P$, and its absolute contribution is at most

$$C_{P,B}(m\delta)^b L_\delta^{\sum_{A\in\pi}(|A|-1)} = C_{P,B}(m\delta)^b L_\delta^{2P-b}.$$

There are finitely many such partitions at the fixed order. Absorb their numbers in $C_{P,B}$ and sum over b. The even moment is the absolute even moment. Invariance translates an interval, and reversal only changes the order of summation. This is an exact finite moment–cumulant calculation, not a truncation of a convergent infinite cumulant series. □

Lemma 212.2 (One heterogeneous cumulant comparison)**.** *For each fixed $q \ge 2$, and bounded-BV collision observables $u_1, \dots, u_q$, the joint cumulant at times $t_1, \dots, t_q$ satisfies*

$$|\operatorname{cum}(u_1 \circ T_R^{t_1}, \dots, u_q \circ T_R^{t_q})| \le C_q e^{-c_q \operatorname{diam}\{t_1,\dots,t_q\}} \prod_{j=1}^{q} \|u_j\|_{\mathcal{V}}. \tag{212.2}$$

The same bound with a fixed factor $1 + \operatorname{diam}$ is summable over all times relative to one anchored time.

Proof. Normalize the nonzero norms to one and order the times, preserving the identity of each observable. Split at a largest consecutive gap d. Under a comparison law, keep the entire left and right marginal laws unchanged but make them independent. If m_A denotes the moment of the variables in a nonempty subset A, its change satisfies $|m_A - m_A^\circ| \le C_q e^{-\gamma_q d}$ whenever A meets both sides, by Lemma 206.1; otherwise its change is zero. For a partition $\pi = \{A_1, \dots, A_b\}$, write explicitly

$$\prod_{j=1}^{b} m_{A_j} - \prod_{j=1}^{b} m_{A_j}^\circ = \sum_{i=1}^{b} (m_{A_i} - m_{A_i}^\circ) \prod_{j<i} m_{A_j} \prod_{j>i} m_{A_j}^\circ.$$

Every factor is bounded. Substitute this identity into (206.2). The comparison cumulant vanishes by independent-block cancellation, so the original one is $O_q(e^{-\gamma_q d})$. The diameter is at most $(q-1)d$; coincident times are treated by the bounded partition formula. Restoring the norms proves the bound and exponential summability.

For example, take four centered, possibly different variables A, B, C, D in chronological order, with the chosen split between B and C. The fourth cumulant is

$$\mathbb{E}(ABCD) - \mathbb{E}(AB)\mathbb{E}(CD) - \mathbb{E}(AC)\mathbb{E}(BD) - \mathbb{E}(AD)\mathbb{E}(BC).$$

Under the comparison law, the first two terms agree and the last two vanish. Each difference is controlled by a cross-gap subset moment, with the unchanged same-side moments bounded. This includes $A = a - \alpha$ and three collision factors, or a split on either side of that mark. No identical-distribution assumption is involved. For the moment theorem of Section 210, the mark block still consumes at least one collision factor; the remaining nonzero blocks consume at least

two each. Thus the parity rates there remain $\rho_d = 1/2$ for odd d and $\rho_d = 1$ for even d. The constants in this comparison are fixed-order constants. □

212.2. **A corrected chronological word of fixed arbitrary degree.** Use $I^a_{r,s,R}(z)$ from Section 208, with $m = r + s \geq 1$. Its mark is a bounded-BV function on the collision cylinder; it need not be a probability density.

Theorem 212.3 (High-order damped unsmoothing)**.** *Fix integers $P \geq 1$ and $Q \geq 3$. Suppose $0 < \delta \leq \delta_0(Q)$ and real z satisfy* (206.7) *for the spectral degree Q. For any fine scale $0 < \epsilon < 1/2$,*

$$\begin{aligned} |I^a_{r,s,R}(z)| \leq{}& C_{P,Q} M_a \epsilon^{-4P} (1 + m|z|)^{2P-1} e^{-cm|z|^2} + C\epsilon V_a \\ &+ C_P M_a \epsilon (1 + m|z|)^{2P-1} \\ &+ C_P M_a |z|^{2P} \sum_{b=1}^{P} (m\delta)^b L_\delta^{2P-b}. \end{aligned} \tag{212.3}$$

The constants are independent of the mark location $0 \leq r \leq m$, $R, m, \delta, \epsilon, z$ in this domain. The residual Taylor degree is $2P - 1$, not Q. The fine scale enters only the multipliers and not the analytic damping domain.

Proof. For $z \neq 0$ put $u = (z/|z|) \cdot (h_R - h_{R,\delta})$. It is real, centered, uniformly bounded-BV and has L^1 norm at most $C\delta$. The case $z = 0$ follows by increasing the constant. First replace a by $a_\epsilon = \mathsf{S}_\epsilon a$, while the full phase has modulus one. This costs $C\epsilon V_a$. Translate the interval so that it begins at zero and the mark is at time r. The exact real-variable Taylor formula gives

$$e^{iX} = \sum_{j=0}^{2P-1} \frac{(iX)^j}{j!} + \mathcal{R}_{2P}(X), \qquad |\mathcal{R}_{2P}(X)| \leq \frac{|X|^{2P}}{(2P)!}, \qquad X = |z| S_m u.$$

The coarse phase has modulus one, and Lemma 212.1 bounds the remainder integral by the last line of (212.3). In particular the term with $b = 1$ is retained even when $m\delta$ is small; only the term with $b = P$ would not be a uniform bound.

For degree $j \leq 2P - 1$, expand the jth power of the residual sum as the sum over its m^j time tuples, including repetitions. In each product replace every u by $u_\epsilon = \mathsf{S}_\epsilon u$. Telescoping, invariance and the uniform supremum bounds give an integrated cost at most $C_j M_a \epsilon$ per tuple. Summation over the fixed degrees gives the third term of (212.3).

Sort the residual times together with the mark as $0 \leq t_1 \leq \cdots \leq t_d \leq m$, $d = j + 1 \leq 2P$. The corrected integral is exactly

$$\ell\big(L_z^{m-t_d} M_{b_d} L_z^{t_d - t_{d-1}} \cdots M_{b_2} L_z^{t_2 - t_1} M_{b_1} L_z^{t_1} \nu\big), \qquad L_z = \mathcal{L}_{R,\delta,z}.$$

One multiplier is a_ϵ and the other j multipliers are u_ϵ. Each costs at most its fixed supremum bound times $C\epsilon^{-2}$. The power lengths sum to exactly m, including zero-length powers for repeated times and endpoint marks. Corollary 206.4 bounds the word by $C_{j,Q} M_a \epsilon^{-2(j+1)} e^{-cm|z|^2}$. The product of its finitely many power-bound constants is absorbed in $C_{j,Q}$; no count-dependent number of factors is present. Summing the tuples and degrees proves the first term of (212.3). This proves the assertion without perturbing the operator by the fine-scale residual. □

Remark 212.4 (Fixed orders and the retained cubic theorem). At $P = 2$, the last sum is $m^2\delta^2 L_\delta^2 + m\delta L_\delta^3$, and the first multiplier loss is ϵ^{-8}. Thus Theorem 208.4 is recovered exactly. Higher fixed P gives a different frequency budget; it does not assert that P or Q may grow with n. The original cubic theorem and the sharper small-ball estimate remain separate statements.

213. Anisotropic Fourier bands at the prescribed return count

The four coordinates of the raw record have different roles in the final inversion: three are discrete and the fourth is flight time. The same prescribed-count argument permits different Fourier widths in these coordinates. We prove a family of such estimates, retaining the four-dimensional Lebesgue factor in frequency and the original return count and mark. It gives new portions of the farther annulus; no average over counts, directions, or frequency volume is taken.

213.1. **A feasible region of four width exponents.** Fix $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3, \theta_4)$ and set

$$\theta_* = \max_i \theta_i, \quad \Theta = \sum_{i=1}^{4} \theta_i, \quad B_{n,i} = n^{-1/2+\theta_i}, \quad \mathcal{B}_n = \prod_{i=1}^{4} [-2n^{\theta_i}, 2n^{\theta_i}]. \tag{213.1}$$

Thus the physical box is $n^{-1/2}\mathcal{B}_n$ with half-widths $2B_{n,i}$. The coordinates are ordered as (K_1, K_2, N, T) and their dual frequencies as (u_1, u_2, s, b). For all large n the first three physical intervals lie in the chosen local chart of $\mathbb{T}^3$.

Theorem 213.1 (Prescribed-count cancellation with separate widths)**.** *Suppose*

$$\min_i \theta_i \geq \frac{1}{200}, \qquad \theta_* < \frac{1}{10}, \qquad \Theta + \theta_* < \frac{1}{4}. \tag{213.2}$$

Fix $0 < \eta \leq 1/4 - \Theta - \theta_*$. There is a number $\sigma > 0$ such that, uniformly in $R \in I$, $0 \leq k \leq n$ and bounded-BV section insertions a, for all sufficiently large n,

$$n^2 \int_{\substack{z \in n^{-1/2}\mathcal{B}_n \\ |z| \geq 2n^{-99/200}}} |\Phi_{n,R}^{[k],a}(z)| \, \mathrm{d}z \leq C \left(M_a n^{-\eta} + V_a n^{-\sigma} \right). \tag{213.3}$$

One may choose σ larger than any prescribed fixed number by reducing only the fine multiplier scale. If $\eta \geq 3/280$ and $\sigma \geq 9/175$, then

$$\int_{\mathcal{B}_n} |C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf{T}} D_R v/2}| \, \mathrm{d}v \leq C \left(M_a n^{-3/280} \sqrt{\log(2+n)} + V_a n^{-9/175} \right). \tag{213.4}$$

The measure convention is $\nu|_{Y_R^*}$ and $\alpha_a = \int a \, \mathrm{d}\nu$. The original probability is $a = s_R$. All exponent choices and expansion orders are fixed independently of n, R, k, a.

Proof. Here are choices with strict analytic and residual margins. Select

$$2\theta_* < \alpha < \frac{1}{4} - \frac{\theta_*}{2}, \quad P(\alpha - 2\theta_*) > \Theta + \eta, \quad (Q-1)(1/2 - \theta_*) > 2\alpha(Q+1), \tag{213.5}$$

where $P \geq 1$ and $Q \geq 3$ are integers. The interval for α is nonempty because $\theta_* < 1/10$. Once α is chosen, choose P and Q sufficiently large; both displayed strict inequalities have positive coefficients at large order. Choose a fixed β so that

$$\beta - \Theta - (2P-1)(1/2 + \theta_*) > \eta. \tag{213.6}$$

Set $\delta = n^{-\alpha}/4$ and $\epsilon = n^{-\beta}/4$. There is no relation $Q = 2P - 1$.

Recenter the orbit at its prescribed actual-return mark and use (210.5). Its integrated cost is

$$CM_a n^{-1/4} \int_{\mathcal{B}_n} |v| \, \mathrm{d}v \leq CM_a n^{-1/4+\Theta+\theta_*} \leq CM_a n^{-\eta}. \tag{213.7}$$

The deterministic interval has length $m = n/c_* + O(1)$ uniformly in k. For $z = v/\sqrt{n}$ in the box, $|z| \leq 4n^{-1/2+\theta_*}$. The two coarse damping quantities are therefore bounded by constants times

$$n^{-(1/2-\theta_*-2\alpha)}, \qquad n^{-[(Q-1)(1/2-\theta_*)-2\alpha(Q+1)]}.$$

Both powers decay by (213.5). Apply Theorem 212.3. On $|v| \ge 2n^{1/200}$ its first term, even after integration, is smaller than every fixed inverse power of n: the prefactor is a fixed polynomial, and $m|z|^2 \ge cn^{1/100}$.

The fine insertion error integrates to $CV_a n^{-(\beta-\Theta)}$. The fine residual replacement is at most

$$CM_a n^{-[\beta-\Theta-(2P-1)(1/2+\theta_*)]}.$$

For each of the P residual moment terms, the integrated bound is

$$CM_a n^{\Theta+2P\theta_*-P+(1-\alpha)b}[\log(2+n)]^{2P-b}, \qquad 1 \le b \le P. \tag{213.8}$$

Since $\alpha < 1$, the largest exponent occurs at $b = P$ and is $\Theta - P(\alpha - 2\theta_*) < -\eta$. Each logarithmic power is fixed and absorbed by the strict margin. This treats also the connected terms with fewer than P blocks; they have additional powers of $n^{-(1-\alpha)}$. Put $\sigma = \beta - \Theta$. The change of variables satisfies $\mathrm{d}v = n^2\,\mathrm{d}z$, exactly, so the bounds prove (213.3) with the raw factor n^2. Increasing β increases σ without changing the coarse damping domain or the polynomial-versus-exponential argument.

The ball $|v| \le 2n^{1/200}$ is contained in $\mathcal{B}_n$. Use Theorem 146.5 there. On the rest use (213.3) and the uniformly elliptic Gaussian tail. Under the stated inequalities on η, σ, this gives (213.4). □

Remark 213.2 (Quantifiers and frequency geometry). The inequalities in (213.2) describe a sufficient region for this construction, not an optimality theorem. They allow a single width arbitrarily close to rescaled exponent $1/10$, while the other widths remain smaller and the sum budget is respected. They do not give the full isotropic ball with that exponent. All constants and the threshold in n may depend on the fixed vector $\boldsymbol{\theta}$ and the fixed orders. No limit of growing Taylor orders is taken. The scalar covariance and induced Cesaro conclusions of earlier sections are unchanged.

213.2. **A concrete extension in the flight-time frequency.** Choose, in the physical coordinate order above,

$$\boldsymbol{\theta} = \left(\frac{1}{100}, \frac{1}{100}, \frac{1}{100}, \frac{9}{100}\right), \quad \alpha = \frac{19}{100}, \quad P = 16, \quad Q = 29, \quad \beta = 20. \tag{213.9}$$

Write $\mathcal{B}_n^{\mathrm{roof}}$ for the resulting rescaled box. Its physical support is

$$|u_1|, |u_2|, |s| \le 2n^{-49/100}, \qquad |b| \le 2n^{-41/100}. \tag{213.10}$$

The rescaled half-widths are $2n^{1/100}$ and $2n^{9/100}$.

Corollary 213.3 (A fixed-count roof-direction extension)**.** *For the same radius, count, mark and insertion uniformity,*

$$n^2 \int_{\substack{z \in n^{-1/2}\mathcal{B}_n^{\mathrm{roof}} \\ |z| \ge 2n^{-99/200}}} |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C\left(M_a n^{-1/40} + V_a n^{-497/25}\right). \tag{213.11}$$

The marked central comparison on $\mathcal{B}_n^{\mathrm{roof}}$ has the original error on the right of (213.4)*. The same central error, with a changed constant, holds on the union*

$$\mathcal{U}_n = \mathcal{B}_n^{\mathrm{roof}} \cup \{v : |v| \le 2n^{67/1400}\}. \tag{213.12}$$

Thus the old isotropic band is retained and genuinely new frequencies beyond it are covered.

Proof. Here $\Theta = 3/25$ and $\theta_* = 9/100$. The distinct error budgets in Theorem 213.1 are

$$\begin{aligned} 1/2 - \theta_* - 2\alpha &= 3/100, & (Q-1)(1/2-\theta_*) - 2\alpha(Q+1) &= 2/25, \\ \beta - \Theta &= 497/25, & \beta - \Theta - (2P-1)(1/2+\theta_*) &= 159/100, \\ P(\alpha - 2\theta_*) - \Theta &= 1/25, & 1/4 - \Theta - \theta_* &= 1/25. \end{aligned} \tag{213.13}$$

The residual with $b = P$ has the factor $n^{-1/25}[\log(2+n)]^{16}$; every other residual term has a strictly faster power, with its own fixed logarithm. They are all $O(n^{-1/40})$. The residual Taylor degree is 31 and the spectral degree is 29; the two numbers refer to different expansions. The theorem proves (213.11), and $1/40 > 3/280$ gives the central estimate. For the union, add the nonnegative integrals over the two sets and use Theorem 211.1 for the isotropic ball. The new roof exponent satisfies $9/100 > 67/1400$, so the difference of the two regions is nonempty for all sufficiently large n. □

Corollary 213.4 (The unchanged marked-state event on the union)**.** *Let $A_{n,k,R}$ be the same single marked-state event as in Corollary 211.3, with probability $p_{n,R} > 0$ and zero-extension variation $V_{n,R}$. Then*

$$\int_{\mathcal{U}_n} \left| \mathbb{E}_{\nu_R^*}[e^{iv\cdot U_{n,R}/\sqrt{n}} \mid A_{n,k,R}] - e^{-v^{\mathsf{T}} D_R v/2} \right| \mathrm{d}v \\ \le \frac{C}{p_{n,R}} \left(n^{-3/280}\sqrt{\log(2+n)} + V_{n,R} n^{-9/175} \right). \tag{213.14}$$

The sufficient conditions $p_{n,R} \ge cn^{-\beta_0}$, $V_{n,R} \le Cn^{\kappa_0}$ remain $\beta_0 < 3/280$ and $\beta_0 + \kappa_0 < 9/175$.

Proof. Insert $a = c_*^{-1}\mathbf{1}_{E_{n,R}}$, whose mass is exactly $p_{n,R}$, and divide the union estimate by that same probability. The event, its lattice coordinates, and its denominator are not replaced. The all-moment statements retain their separate conditions with $\rho_d = 1/2$ in odd degree and $\rho_d = 1$ in even degree. □

213.3. **Raw inversion with a coordinatewise kernel.** Fix a real even $\chi_0 \in C_c^\infty(\mathbb{R})$ with $0 \le \chi_0 \le 1$, $\chi_0 = 1$ on $[-1,1]$ and support in $[-2,2]$. With the widths in (213.9), set

$$\chi_n^{\mathrm{box}}(\omega) = \prod_{i=1}^{4} \chi_0(\omega_i/B_{n,i}), \qquad K_n^{\mathrm{box}} = \mathcal{F}^{-1}\chi_n^{\mathrm{box}}. \tag{213.15}$$

The inverse is on $\mathbb{Z}^3 \times \mathbb{R}$ with the inherited Fourier normalization. This kernel uses the box alone; the transform estimate on the larger union does not require silently identifying two kernels.

For a marked insertion which also satisfies the finite-record admissibility of Definition 200.1, put $w_{n,k,R} = c_* a \circ (F_R^*)^k$ and use the exact extraction $\mu^{a,L} = E^{a,L} + Q^{a,L}$. Define the active-kernel terms by

$$\mathcal{E}_{n,k,R,A}^{\mathrm{box},a,L} = \operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |e^{a,L} - K_n^{\mathrm{box}} * E^{a,L}|,$$
$$\mathcal{C}_{n,k,R}^{\mathrm{box},a,L}(B) = \int_{\mathbb{T}^3\times[-B,B]} |1 - \chi_n^{\mathrm{box}}(\omega)||\widehat{Q}^{a,L}(\omega)|\,\mathrm{d}\omega. \tag{213.16}$$

The local edge quantity is allowed to be infinite before its germs are bounded. Bounded variation alone does not imply finite-record admissibility or a uniform density-derivative estimate.

Corollary 213.5 (Exact marked raw budget at separate coordinate scales)**.** *Let $L_n = \lceil \lambda n \rceil$ with $\lambda > c_*^{-1} + 1$. For $B \ge 1$, $P_0 > 0$, and $y = ((\ell,t) - n\bar{G}_R)/\sqrt{n}$, one has*

$$\operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |n^2 p_{n,R}^{[k],a}(\ell,t) - \alpha_a \mathfrak{g}_R(y)| \\ \le C\left(M_a n^{-3/280}\sqrt{\log(2+n)} + V_a n^{-9/175} \right) + CM_a e^{-cn^{1/50}} + C_{A,\lambda,P_0} M_a n^{-P_0} \\ + n^2 \mathcal{E}_{n,k,R,A}^{\mathrm{box},a,L_n} + (2\pi)^{-4} n^2 \mathcal{C}_{n,k,R}^{\mathrm{box},a,L_n}(B) + \frac{n^2}{\pi B} A_2(n, L_n, R, w_{n,k,R}). \tag{213.17}$$

This is an extended-real inequality where necessary, with physical cutoffs (213.10) *and rescaled cutoffs* $2n^{1/100}$, $2n^{9/100}$.

Proof. For all sufficiently large n, the lattice-frequency support lies strictly inside the fundamental torus cube. Hence Fourier inversion and one-dimensional rescaling give the product-kernel estimate

$$|K_n^{\mathrm{box}}(x)| \le C_J \prod_{i=1}^{4} B_{n,i} \prod_{i=1}^{4} (1 + B_{n,i}|x_i|)^{-J} \qquad (x \in \mathbb{Z}^3 \times \mathbb{R}). \tag{213.18}$$

It follows directly from the Schwartz decay of the inverse of χ_0; no lattice coordinate is approximated by a continuous variable.

Let T^{a,L_n} be the high-count part of the marked measure. On the actual central count labels it has no density contribution, and the count coordinate of its support is separated from these labels by at least $c_{A,\lambda} n$ for large n. Its total variation is at most $c_* M_a$. Since the third coordinate is the collision count, (213.18) gives

$$n^2 \sup_{\mathcal{W}_{n,R,A}} |K_n^{\mathrm{box}} * \mathsf{T}^{a,L_n}| \le C_{A,\lambda,J} M_a n^{3/25 - 51J/100}. \tag{213.19}$$

Here $n^2 \prod_i B_{n,i} = n^{3/25}$ and $n B_{n,3} = n^{51/100}$. Select the fixed Schwartz order J so that the bound is $O(M_a n^{-P_0})$.

The exact finite-count decomposition and support identity now give

$$\begin{aligned} p_{n,R}^{[k],a} = {} & K_n^{\mathrm{box}} * \mu_{n,R}^{[k],a} + (e^{a,L_n} - K_n^{\mathrm{box}} * E^{a,L_n}) \\ & + \mathcal{F}^{-1}[(1 - \chi_n^{\mathrm{box}}) \widehat{Q}^{a,L_n}] - K_n^{\mathrm{box}} * \mathsf{T}^{a,L_n}. \end{aligned}$$

For the first term use the central comparison on the box, whose cutoff is between zero and one. Missing Gaussian frequencies have at least one $|v_i| \ge n^{\theta_i}$ and hence $|v| \ge n^{1/100}$. Uniform ellipticity bounds their integral by $Ce^{-cn^{1/50}}$. The finite-band residual inverse and its far-roof part are bounded exactly as in Corollary 211.4, giving the last two terms of (213.17). The local combined edge correction is the one in (213.16), not a correction for either preceding isotropic kernel. This proves the inequality. □

Position in the full raw argument. The new theorem enlarges the proved prescribed-count domain in a specific, four-dimensionally integrated way, including flight-time frequencies of size $n^{-41/100}$. The previously proved isotropic ball is preserved by (213.12). Neither set contains the full ball of physical radius $n^{-2/5}$: the transverse widths, compact nonzero torus frequencies, full peripheral regimes and growing roof ranges require additional estimates. The terms involving $\mathcal{E}^{\mathrm{box}}$, $\mathcal{C}^{\mathrm{box}}$ and the true long-time A_2 in (213.17) are not bounded by the high-order collision moment lemma. Weighted denominator asymptotics and relative physical-event replacement remain separate steps in the same raw mixed-density problem.

214. Shape-adaptive Fourier cancellation at a prescribed count

The preceding box theorem assigns one fixed exponent to each coordinate. Here the frequency region may depend on n without being a box. A weighted-volume bound replaces the sum of its four width exponents. We then construct a single nonrectangular region reaching all four coordinate directions. The logarithmic factor in its definition pays for the entire dyadic sum; it does not enter the final central rate.

All integrals in this section are in the rescaled variable $v = \sqrt{n} z$. The coordinate order remains (K_1, K_2, N, T), with dual physical coordinates (u_1, u_2, s, b). Put

$$\tau_0 = \frac{1}{200}, \qquad \theta_0 = \frac{9}{100}, \qquad s_0 = \frac{67}{280}, \qquad d_0' = s_0 - \tau_0 = \frac{41}{175}, \qquad \eta_0' = \frac{1}{4} - s_0 = \frac{3}{280}. \tag{214.1}$$

The symbols d_0' and η_0' in this section are exponents, not covariance lower bounds or phase-separation constants.

214.1. **A measurable-region estimate with fixed expansion orders.**

Theorem 214.1 (A weighted-volume criterion)**.** *Fix $C_0 < \infty$. For each n, let $\Omega_n \subset \mathbb{R}^4$ be any measurable set such that*

$$\sup_{v\in\Omega_n} |v| \le C_0 n^{\theta_0}, \qquad |\Omega_n| \le C_0 n^{d_0'}, \qquad \int_{\Omega_n} |v|\,\mathrm{d}v \le C_0 n^{s_0}. \tag{214.2}$$

The supremum may be interpreted essentially. Uniformly in these sets, $R \in I$, the prescribed count n, the mark $0 \le k \le n$, and bounded-BV section insertions a, for all sufficiently large n,

$$n^2 \int_{\substack{z\in n^{-1/2}\Omega_n \\ |z|\ge 2n^{-99/200}}} |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C_{C_0}\left(M_a n^{-3/280} + V_a n^{-5559/175}\right). \tag{214.3}$$

No regularity of $\partial\Omega_n$, and no fixed choice of width exponents for Ω_n, is required.

Proof. Use the exact marked recentering and (210.5). Its integral is at most $CM_a n^{-1/4}\int_{\Omega_n}|v|\,\mathrm{d}v \le CM_a n^{-3/280}$. The deterministic collision length is $m = m_k + m_{n-k} = n/c_* + O(1)$ uniformly in k.

Apply Theorem 212.3 with the fixed choices

$$P = 26, \qquad Q = 29, \qquad \delta = \tfrac{1}{4}n^{-19/100}, \qquad \epsilon = \tfrac{1}{4}n^{-32}. \tag{214.4}$$

The residual Taylor degree is 51 and the spectral degree is 29. For $z = v/\sqrt{n}$ the two analytic margins are $3/100$ and $2/25$, exactly as for the roof box. The harmless fixed constant C_0 is absorbed by increasing the threshold in n. On $|v| \ge 2n^{1/200}$, the damped term is at most a fixed polynomial in n times $e^{-cn^{1/100}}$, including its 52 multiplier factors. Its integral is smaller than every fixed inverse power of n.

The fine insertion loss has exponent $32 - d_0' = 5559/175$. The fine residual replacement has margin

$$32 - d_0' - 51(1/2 + \theta_0) = \frac{1173}{700}. \tag{214.5}$$

For the residual term with b nonsingleton blocks, $1 \le b \le 26$, (212.3) and the volume bound give

$$CM_a n^{d_0'+52\theta_0-26+(81/100)b}[\log(2+n)]^{52-b}.$$

The slowest term has $b = 26$ and decay exponent

$$26(19/100 - 2\theta_0) - d_0' = \frac{9}{350} = \frac{3}{280} + \frac{3}{200}. \tag{214.6}$$

Every smaller block count improves the power by $81/100$. Thus all fixed logarithms are absorbed by the strict margin $3/200$. The $b = 1$ contribution is included in this calculation. Finally $\mathrm{d}v = n^2\,\mathrm{d}z$ and the centering phase has modulus one. This proves (214.3) with constants independent of the shape of Ω_n. Neither expansion order grows with n. □

214.2. **A summable family of coordinate scales.** For $\mathbf{j} = (j_1, j_2, j_3, j_4) \in \mathbb{Z}^4_{\ge 0}$, write $S(\mathbf{j}) = \sum_i j_i$ and $J(\mathbf{j}) = \max_i j_i$. The following elementary counting fact will be used both for frequency volume and for the physical convolution kernel.

Lemma 214.2 (The weighted dyadic sum)**.** *For each integer $K \ge 0$,*

$$\sum_{S(\mathbf{j})+J(\mathbf{j})\le K} 2^{S(\mathbf{j})+J(\mathbf{j})} \le C2^K(K+1)^3. \tag{214.7}$$

Consequently, if $a > 0$, $T \ge a^5$ and $U \ge a$, the finite set

$$\mathcal{D}(a,T,U) = \{\mathbf{j} : a^5 2^{S(\mathbf{j})+J(\mathbf{j})} \le T,\ a2^{J(\mathbf{j})} \le U\}$$

satisfies, with $\lambda_i = a2^{j_i}$ and $\lambda_* = \max_i \lambda_i$,

$$\sum_{\mathbf{j}\in\mathcal{D}} \prod_i \lambda_i\lambda_* \le CT\,[1+\log(T/a^5)]^3,$$
$$\sum_{\mathbf{j}\in\mathcal{D}} \prod_i \lambda_i \le C(T/a)\,[1+\log(T/a^5)]^3. \tag{214.8}$$

Proof. Fix $t = S + J$ and choose one coordinate attaining the maximum J. For this choice the other three coordinates sum to $t - 2J$. Ignoring their upper bound by J only increases their number, which is at most $(t+2)^2$. There are at most $t+1$ choices for J and four choices of its attaining coordinate. Thus the number of tuples with $S + J = t$ is at most $C(t+1)^3$. Overcounting tied maxima is harmless. Sum $2^t C(t+1)^3$ for $0 \le t \le K$ and use $\sum_{t=0}^{K} 2^t \le 2^{K+1}$. This proves the first assertion. Take $K = \lfloor \log_2(T/a^5) \rfloor$ and multiply by a^5. The bound on U only removes terms. The second scale sum follows from $\lambda_* \ge a$. □

Set

$$a_n = 2n^{1/200}, \quad U_n = n^{9/100}, \quad T_n = \frac{n^{67/280}}{[\log(2+n)]^3}, \quad \mathcal{D}_n = \mathcal{D}(a_n, T_n, U_n). \tag{214.9}$$

All assertions concerning these sets are for sufficiently large n, so $T_n \ge a_n^5$ and $U_n \ge a_n$. Define the closed union of boxes

$$\mathcal{H}_n = \bigcup_{\mathbf{j}\in\mathcal{D}_n} \prod_{i=1}^{4} [-2\lambda_i, 2\lambda_i], \qquad \lambda_i = a_n 2^{j_i}. \tag{214.10}$$

This is a nonrectangular frequency region. A useful description uses $x_i(v) = \max\{a_n, |v_i|\}$ and $x_*(v) = \max_i x_i(v)$. It contains

$$\left\{ v : x_*(v) \le U_n/2, \quad x_*(v) \prod_i x_i(v) \le T_n/32 \right\}, \tag{214.11}$$

and is contained in the same product region with thresholds $2U_n$ and $32T_n$. These constants also allow a smooth cutoff below.

Lemma 214.3 (Size and interior of the nonrectangular region)**.** *The set $\mathcal{H}_n$ satisfies* (214.2) *with one fixed constant. It contains $\{|v| \le 2n^{1/200}\}$. The inclusions just stated hold for all sufficiently large n.*

Proof. Every box has volume $256 \prod_i \lambda_i$ and $|v| \le 4\lambda_*$ on it. The two scale sums in (214.8) bound its union volume and first radial moment. Since $1 + \log(T_n/a_n^5) \le C\log(2+n)$, the factor $[\log(2+n)]^{-3}$ in T_n cancels this entire counting loss. This gives $|\mathcal{H}_n| \le Cn^{41/175}$ and $\int_{\mathcal{H}_n} |v|\,\mathrm{d}v \le Cn^{67/280}$. Its radius is at most $4U_n$. The zero tuple supplies the central ball.

For the inner inclusion, select scales with $x_i \le \lambda_i \le 2x_i$, taking $j_i = 0$ when $x_i = a_n$. They obey $\lambda_* \prod_i \lambda_i \le 32 x_* \prod_i x_i \le T_n$ and $\lambda_* \le 2x_* \le U_n$. This gives an admissible box containing v. Conversely, membership in a box implies $x_i \le 2\lambda_i$, because $\lambda_i \ge a_n$, and hence the stated outer bounds. □

Theorem 214.4 (One retained-rate region in all four directions)**.** *The physical region is $n^{-1/2}\mathcal{H}_n$. On its part outside $|z| \le 2n^{-99/200}$,*

$$n^2 \int |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C\left(M_a n^{-3/280} + V_a n^{-5559/175}\right). \tag{214.12}$$

Let

$$\mathcal{U}_n^\sharp = \mathcal{H}_n \cup \mathcal{B}_n^{\mathrm{roof}} \cup \{|v| \le 2n^{67/1400}\}. \tag{214.13}$$

Then

$$\int_{\mathcal{U}_n^\sharp} |C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf{T}} D_R v/2}|\,\mathrm{d}v \le C\left(M_a n^{-3/280}\sqrt{\log(2+n)} + V_a n^{-9/175}\right). \tag{214.14}$$

Both estimates are uniform in R, n, k, a in the original marked class. Every previously proved isotropic and roof-box frequency is retained.

Proof. Theorem 214.1 and Lemma 214.3 give the first assertion. On the central ball use Theorem 146.5. On its complement the uniformly positive covariance gives an exponentially small Gaussian integral, while the first assertion gives the physical transform integral. Finally add the inherited roof-box and isotropic bounds. Their overlap can only increase the upper bound. No division by the volume of any region is made. □

Corollary 214.5 (New mixed directions and the unchanged event)**.** *For a fixed exponent vector $\boldsymbol{\rho}$ with $\rho_i \ge 0$, set $\widetilde{\rho}_i = \max\{1/200, \rho_i\}$ and $\widetilde{\rho}_* = \max_i \widetilde{\rho}_i$. If*

$$\widetilde{\rho}_* < 9/100, \qquad \sum_i \widetilde{\rho}_i + \widetilde{\rho}_* < 67/280, \tag{214.15}$$

then every fixed multiple of the box $|v_i| \le n^{\rho_i}$ is contained in $\mathcal{H}_n$ for all sufficiently large n. Also $|v_i| \le U_n/4$ on any one coordinate axis is contained in $\mathcal{H}_n$. Thus each physical coordinate has a new axial reach of order $n^{-41/100}$, not only the roof coordinate.

For the unchanged marked-state event with probability $p_{n,R} > 0$ and variation $V_{n,R}$, the right side of (214.14) *becomes*

$$\frac{C}{p_{n,R}}\left(n^{-3/280}\sqrt{\log(2+n)} + V_{n,R} n^{-9/175}\right).$$

Its probability and variation conditions are unchanged.

Proof. Apply the inner description (214.11). The strict power margins absorb both fixed constants and $[\log(2+n)]^3$. On an axis the product exponent is $2(9/100) + 3/200 = 39/200 < 67/280$, so the same argument works at the asserted constant multiple of U_n. As a mixed example, $\boldsymbol{\rho} = (1/25, 1/25, 1/25, 1/20)$ has sum plus maximum $11/50 < 67/280$ and extends beyond the old isotropic exponent. The event statement follows by taking $a = c_*^{-1}\mathbf{1}_E$ and dividing by the exact probability, as in Corollary 213.4. □

Extent of the result. This is not the isotropic ball of rescaled radius $n^{1/10}$. For example, a vector with four coordinates of order $n^{3/50}$ is outside the product budget and outside both retained regions for large n. The expansion orders are fixed. The new result controls a single explicit nonrectangular domain, rather than a collection of estimates whose constants might depend on n-varying exponents. The next section connects this domain to the already used raw extraction without imposing a new, unrelated edge hypothesis.

215. A coherent raw correction under changes of cutoff

Increasing the low-frequency region changes both the extracted-edge convolution and the residual inverse transform. Estimating their changes separately loses an exact cancellation. We retain that cancellation and prove a quantitative comparison with the existing box and isotropic cutoffs, for the same finite extraction and the same physical weight. This closes the change-of-cutoff contribution; it does not assume smallness of the remaining common raw correction.

215.1. **A smooth cutoff for the nonrectangular region.** Choose the function χ_0 in Section 213 also nonincreasing on $[0,\infty)$. Define

$$\varphi_0(t)=\chi_0(t),\qquad \varphi_j(t)=\chi_0(t/2^j)-\chi_0(t/2^{j-1})\quad (j\ge 1).$$

These functions are nonnegative and sum to one on $\mathbb{R}$. With $\mathcal{D}_n, a_n$ from (214.9), put

$$\Xi_n(v)=\sum_{\mathbf{j}\in\mathcal{D}_n}\prod_{i=1}^{4}\varphi_{j_i}(v_i/a_n),\qquad \chi_n^\sharp(\omega)=\Xi_n(\sqrt{n}\,\omega),\qquad K_n^\sharp=\mathcal{F}^{-1}\chi_n^\sharp. \tag{215.1}$$

For large n, the first three physical supports are strictly inside the chosen torus chart; the expression is extended by zero there. Its largest coordinate support is $2n^{-41/100}$ and its central plateau contains the physical ball $|\omega|\le 2n^{-99/200}$. The corresponding rescaled scales are $2n^{9/100}$ and $2n^{1/200}$.

Lemma 215.1 (Cutoff plateau and count-coordinate kernel tail)**.** *The cutoff satisfies $0\le\Xi_n\le 1$, is supported in $\mathcal{H}_n$, and is one on the central ball and on the inner product region* (214.11)*. For every fixed integer $J\ge 0$,*

$$|K_n^\sharp(x)|\le C_J\sum_{\mathbf{j}\in\mathcal{D}_n}\prod_{i=1}^{4}B_{\mathbf{j},i}(1+B_{\mathbf{j},i}|x_i|)^{-J},\qquad B_{\mathbf{j},i}=\lambda_i/\sqrt{n},\quad x\in\mathbb{Z}^3\times\mathbb{R}. \tag{215.2}$$

In particular, for $L_n=\lceil\lambda n\rceil$ with $\lambda>c_*^{-1}+1$, and the high-count part T^{a,L_n} of the original marked law,

$$n^2\sup_{\mathcal{W}_{n,R,A}}|K_n^\sharp*\mathsf{T}^{a,L_n}|\le C_{A,\lambda,J}M_a n^{41/175-101J/200}. \tag{215.3}$$

The third coordinate, not the largest roof scale, supplies this decay.

Proof. The nonnegative full product partition sums to one, so a subsum lies between zero and one. Each summand is supported in its box from (214.10). If a summand is nonzero at v, then $\lambda_i\le 2\max\{a_n,|v_i|\}$ for every i; this is immediate for $j_i=0$, and follows from $|v_i|>\lambda_i/2$ for $j_i>0$. At a point of the inner product region all nonzero summands therefore belong to $\mathcal{D}_n$, which proves the plateau. On the central ball only the zero tuple can contribute and its value is one.

For $j\ge 1$, write $\varphi_j(v/a_n)$ as $\chi_0(v/\lambda_j)-\chi_0(2v/\lambda_j)$. Thus the inverse of every factor is a dilation of one of two fixed Schwartz functions. The mixed inverse transform, with the three integer output coordinates left discrete, gives (215.2). Its constants are uniform over all dyadic indices. The sum is finite at every n.

The total variation of the high-count marked measure is at most c_*M_a. On a central window its third-coordinate support is separated by at least $c_{A,\lambda}n$, by Lemma 201.1. Since every $B_{\mathbf{j},3}\ge a_n/\sqrt{n}=2n^{-99/200}$, each count factor has decay at most $Cn^{-101J/200}$. Multiply the remaining scale sum by n^2 and use (214.8); the result is $Cn^{41/175}$. This proves (215.3). Arbitrary inverse powers follow by choosing a fixed J large enough. □

215.2. **The exact signed correction and its cancellation.** Require now the additional finite-record admissibility of the weight $w_{n,k,R}=c_*a\circ(F_R^*)^k$ from Definition 200.1. Fix one extraction

$$\mu^{a,L}=E^{a,L}+Q^{a,L},\qquad p^{a,L}=e^{a,L}+r^{a,L},\qquad \widehat{Q}^{a,L}\in L^1(\mathbb{T}^3\times\mathbb{R}).$$

All cutoffs in this subsection use precisely these same measures. For a real smooth compactly supported frequency cutoff χ, with kernel $K_\chi=\mathcal{F}^{-1}\chi$, define

$$\mathcal{R}_\chi^{a,L}=e^{a,L}-K_\chi*E^{a,L}+\mathcal{F}^{-1}[(1-\chi)\widehat{Q}^{a,L}]. \tag{215.4}$$

This is a signed or complex function defined almost everywhere; it is not the sum of the absolute edge and residual bounds. Its essential supremum may be infinite. The residual inverse is absolutely convergent, and the other convolutions are well defined for finite measures.

Proposition 215.2 (Exact transport of a raw correction)**.** *For two such cutoffs χ, ψ, almost everywhere on the mixed space,*

$$\begin{aligned} \mathcal{R}_{\chi}^{a,L} &= p^{a,L} - K_{\chi} * \mu^{a,L}, \\ \mathcal{R}_{\chi}^{a,L} - \mathcal{R}_{\psi}^{a,L} &= (K_{\psi} - K_{\chi}) * \mu^{a,L}. \end{aligned} \tag{215.5}$$

In particular their difference has a bounded continuous representative, and

$$\|\mathcal{R}_{\chi}^{a,L} - \mathcal{R}_{\psi}^{a,L}\|_{\infty} \le (2\pi)^{-4} \int |\chi - \psi|\, |\widehat{\mu}^{a,L}|\, \mathrm{d}\omega. \tag{215.6}$$

The identities hold for any exact finite extraction with the stated integrable residual transform. They introduce no estimate for A_2.

Proof. Absolute residual inversion gives $\mathcal{F}^{-1}[(1-\chi)\widehat{Q}] = r - K_{\chi} * Q$. Substitute this in (215.4) and use $E + Q = \mu^{a,L}$. Subtract the two identities. In frequency form the edge change is $(\psi - \chi)\widehat{E}$ and the residual change is $(\psi - \chi)\widehat{Q}$; their sum is $(\psi - \chi)\widehat{\mu}^{a,L}$. This is the cancellation that is lost by taking absolute values of the two changes first. The last integrand is integrable, since the cutoff difference has compact support and the measure transform is bounded. Mixed Fourier inversion therefore proves the norm bound and continuity. The equalities are almost-everywhere equalities of actual densities, not a subtraction of two extended-real essential suprema. □

215.3. **Comparison with the already used cutoffs.** Retain $\chi_n^{\mathrm{box}}, K_n^{\mathrm{box}}$ from (213.15) and $\chi_n^{\natural}, K_n^{\natural}$ from Section 211. Each is one on the original central ball for sufficiently large n. Define

$$\Delta_n(a) = M_a n^{-3/280} + V_a n^{-983/350}. \tag{215.7}$$

This exponent for the variation is the slowest of the three complementary-band estimates, not the weaker central-ball exponent.

Theorem 215.3 (A vanishing complete change-of-cutoff contribution)**.** *For χ, ψ any two of $\chi_n^{\sharp}, \chi_n^{\mathrm{box}}, \chi_n^{\natural}$, $L_n = \lceil \lambda n \rceil$, $\lambda > c_*^{-1} + 1$, and fixed $A, P_0 < \infty$ with $P_0 > 0$,*

$$n^2 \operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |\mathcal{R}_{\chi}^{a,L_n} - \mathcal{R}_{\psi}^{a,L_n}| \le C\Delta_n(a) + C_{A,\lambda,P_0} M_a n^{-P_0}. \tag{215.8}$$

The constant is uniform in R, n, k, a in the common weight class. If in addition $c\lambda > a$ for the constants in (201.2)*, the comparison is global, with an exponentially small tail in place of the last term.*

Proof. The cutoff difference vanishes on $|z| \le 2n^{-99/200}$ and is supported in $n^{-1/2}\mathcal{U}_n^{\sharp}$. It has modulus at most one. Equations (214.12), (213.11), and (211.2) therefore imply

$$(2\pi)^{-4} n^2 \int |\chi - \psi|\, |\widehat{\mu}_{n,R}^{[k],a}|\, \mathrm{d}z \le C\Delta_n(a).$$

Here $1/40 > 3/280$, and the two other variation exponents are larger than $983/350$. This is an estimate for the full law at the prescribed return count, not for an averaged coefficient.

In (215.5), write $\mu^{a,L_n} = \mu_{n,R}^{[k],a} - \mathsf{T}^{a,L_n}$. The full-law part has the global bound just proved. The remaining convolution is bounded on the central window by the two relevant count-separation bounds: (215.3), (213.19), or the isotropic bound in the proof of Corollary 211.4. Choose their fixed Schwartz orders for P_0. This proves the first assertion without assuming $c\lambda > a$. With that stronger condition the tail total variation is exponentially small, and all three kernel suprema are polynomially bounded. Their products are exponentially small globally. □

Corollary 215.4 (One common local raw remainder)**.** *Let $\mathcal{A}_n$ be a class of common admissible weights and marks for which $\sup_{a\in\mathcal{A}_n}\Delta_n(a)\to 0$ and M_a has at most fixed polynomial growth. Then*

$$\sup_{R,k,a\in\mathcal{A}_n} n^2\|\mathcal{R}_\chi^{a,L_n}\|_{L^\infty(\mathcal{W}_{n,R,A})}\longrightarrow 0$$

holds for one of the three cutoffs if and only if it holds for all three. Finite values of the norms likewise transfer, and the difference of any two finite norms is at most the bound in (215.8)*. For a fixed bounded-BV weight class the rate is $O(n^{-3/280})$.*

Proof. Apply the triangle inequality to the almost-everywhere identity in Theorem 215.3 in both directions, choosing P_0 larger than the polynomial growth exponent of M_a. If one norm is finite, the bounded difference makes the other finite. No difference of infinite norms is formed. The uniform convergence statement follows. □

The corollary does not transfer smallness of the isolated edge term or of the absolute complementary integral. Cancellation between them is retained in $\mathcal{R}$. In particular a small value of their signed sum cannot be used to bound either of the two absolute quantities. This distinction permits one fixed raw problem to be used throughout the expansion of the known Fourier region.

215.4. **The exact raw budget at the new cutoff.** For the new kernel define, immediately before use,

$$\mathcal{E}^{\sharp,a,L}_{n,k,R,A}=\operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}}|e^{a,L}-K_n^\sharp * E^{a,L}|,$$

$$\mathcal{C}^{\sharp,a,L}_{n,k,R}(B)=\int_{\mathbb{T}^3\times[-B,B]}|1-\chi_n^\sharp|\,|\widehat{Q}^{a,L}|\,\mathrm{d}\omega. \tag{215.9}$$

The first quantity is allowed to be infinite. The exact identity on the central count labels is

$$p_{n,R}^{[k],a}=K_n^\sharp * \mu_{n,R}^{[k],a}+\mathcal{R}^{a,L_n}_{\chi_n^\sharp}-K_n^\sharp * \mathsf{T}^{a,L_n}. \tag{215.10}$$

It follows from central count support and the first identity in Proposition 215.2.

Corollary 215.5 (Raw inversion with the coherent cutoff)**.** *For $B\ge 1$, $P_0>0$, and $y=((\ell,t)-n\bar{G}_R)/\sqrt{n}$,*

$$\begin{aligned}&\operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}}|n^2p_{n,R}^{[k],a}(\ell,t)-\alpha_a\mathfrak{g}_R(y)|\\&\quad\le C\left(M_an^{-3/280}\sqrt{\log(2+n)}+V_an^{-9/175}\right)+CM_ae^{-cn^{1/100}}+C_{A,\lambda,P_0}M_an^{-P_0}\\&\qquad+n^2\mathcal{E}^{\sharp,a,L_n}_{n,k,R,A}+(2\pi)^{-4}n^2\mathcal{C}^{\sharp,a,L_n}_{n,k,R}(B)+\frac{n^2}{\pi B}A_2(n,L_n,R,w_{n,k,R}).\end{aligned} \tag{215.11}$$

The inequality has its extended-real meaning if necessary.

Proof. Apply (214.14) to the first term of (215.10), using $0\le\Xi_n\le 1$. The omitted Gaussian frequencies are outside the plateau containing $|v|\le 2n^{1/200}$; uniform ellipticity gives the exponential term. Use (215.3) for the high-count convolution. Bound the two parts of the coherent correction separately only now: its edge term is the one in (215.9). Splitting the absolutely convergent residual inverse at $|b|=B$ and using (200.11) gives the last two terms. This keeps the true derivative sum at $L_n\asymp n$ unchanged. □

The remaining absolute and conditional estimates. Theorem 215.3 is an unconditional estimate for the entire signed change of the raw correction, and Corollary 215.4 makes that comparison independent of which of the three cutoffs is used. It supplies neither an absolute value for the common correction nor a bound on the genuine long-time A_2. Frequencies outside the retained union, the common local raw remainder, weighted raw denominators and relative physical event replacement still require estimates for the same original law. The last comparison concerns the same indicator on both sides. No physical observation event has been substituted.

216. Unsmoothed windows and an averaged common raw remainder

The transport identity compares two raw corrections but does not bound one of them. Positivity gives a different estimate: it bounds the integral of the common correction over an actual observation window. The window is not convolved with noise. Its side lengths must exceed the resolution of a proved Fourier region. We keep this restriction explicit, in particular in the three integer coordinates.

Put $X_{n,R} = U_{n,R}/\sqrt{n}$, $U_{n,R} = J_{n,R} - n\bar{G}_R$, and set

$$e = \frac{3}{280}, \quad v_0 = \frac{9}{175}, \qquad \mathcal{E}_n(a) = M_a n^{-e}\sqrt{\log(2+n)} + V_a n^{-v_0}. \tag{216.1}$$

The centered pushforward of the marked measure in (146.2) is denoted by $\lambda_{n,R}^{[k],a}$. Its mass is α_a. Theorem 214.4 gives the integrated characteristic error $C\mathcal{E}_n(a)$ on $\mathcal{U}_n^\sharp$. Frequencies here are rescaled; a frequency v corresponds to the physical frequency $v/\sqrt{n}$.

216.1. **Order-preserving interval approximation.** We give the approximation argument, including its values at endpoints. These values matter because three coordinates of the record are discrete. The construction is the classical Beurling–Selberg interval construction; see [17, Introduction]. Only the elementary facts proved below are used, not an extremality assertion.

Lemma 216.1 (Bandlimited envelopes, including endpoints)**.** *For an interval I of length $l > 0$ and an angular bandwidth $b > 0$, there are real continuous integrable functions L_I, U_I with Fourier support in $[-b, b]$ such that*

$$L_I \le \mathbf{1}_I \le U_I, \quad 0 \le U_I \le 3, \quad \int U_I = l + \frac{2\pi}{b}, \quad \int L_I = l - \frac{2\pi}{b}. \tag{216.2}$$

Either convention at either endpoint is allowed in the same inequalities. The difference $E_I = U_I - L_I$ is nonnegative and has integral $4\pi/b$. The Fourier convention in this statement is $\widehat{f}(v) = \int e^{-ivx} f(x)\,\mathrm{d}x$.

Proof. Use the continuous extensions of

$$K(x) = \left(\frac{\sin \pi x}{\pi x}\right)^2, \qquad H(x) = \sum_{m\in\mathbb{Z}} \operatorname{sgn}(m) K(x-m) + 2xK(x), \quad \operatorname{sgn}(0) = 0.$$

The identity $\sum_{m\in\mathbb{Z}} K(x-m) = 1$ follows from the partial-fraction identity for $\pi^2/\sin^2 \pi x$; the limiting integer cases agree. For $x > 0$ not an integer,

$$1 - H(x) = K(x) + 2\left(\frac{\sin \pi x}{\pi}\right)^2 \left(\sum_{j\ge 1} \frac{1}{(x+j)^2} - \frac{1}{x}\right).$$

The decreasing-integrand comparisons give $1/(x+1) \le \sum_{j\ge1}(x+j)^{-2} \le 1/x$. Since $1/(x+1) \ge 1/x - 1/x^2$, it follows that $|1 - H(x)| \le K(x)$. Oddness proves $|\operatorname{sgn}(x) - H(x)| \le K(x)$ everywhere. Moreover $H - \operatorname{sgn}$ is an odd integrable function.

In the Fourier convention $e^{-2\pi i\xi x}$, K has transform $(1-|\xi|)_+$: write the sinc as the inverse transform of an interval and take its square. Each translate of K, and xK as a tempered distribution, has support in $[-1,1]$ after Fourier transformation. The partial sums defining H converge as tempered distributions, because their absolute values are bounded by $\sum_m K(x-m) = 1$. Thus H has the same distributional Fourier support.

Write the interval endpoints as $c < d$, and put $\delta = b/(2\pi)$. Set

$$U_I(x) = \tfrac{1}{2}\{H(\delta(x-c)) - H(\delta(x-d)) + K(\delta(x-c)) + K(\delta(x-d))\},$$

and change the two plus signs before K to minus signs to obtain L_I. The sign comparison proves the bounds off the endpoints. At $x = c$, $U_I(c) = (H(\delta(d-c)) + 1 + K(\delta(d-c)))/2 \geq 1$ and $L_I(c) = (H(\delta(d-c)) - 1 - K(\delta(d-c)))/2 \leq 0$; the other endpoint is the same. This proves both endpoint conventions. The pointwise errors are bounded by the sum of the two K terms, so $0 \leq U_I \leq 3$. The odd integrable error of H has integral zero, and $\int K = 1$. Hence the stated integrals follow. The support statement follows by translation and dilation of the compactly supported distributions above; the final functions are integrable, so their Fourier transforms are continuous. In particular Fourier inversion and integration against a finite measure are valid. □

For $B = \prod_{i=1}^d I_i$, where $d = 3$ or 4, write $|B| = \prod_i l_i$ and $\mathcal{Q}(\mathbf{b}) = \prod_i [-b_i, b_i]$. If $b_i l_i \geq 1$, tensorization gives envelopes with support in $\mathcal{Q}(\mathbf{b})$ and

$$L_B \leq \mathbf{1}_B \leq U_B, \quad \|L_B\|_1 + \|U_B\|_1 \leq C_d|B|, \quad \int (U_B - \mathbf{1}_B) + \int (\mathbf{1}_B - L_B) \leq C_d|B| \sum_i (b_i l_i)^{-1}. \tag{216.3}$$

Here the minorant is not the product of the minorants, which can have the wrong sign. Precisely, set

$$U_B = \prod_i U_{I_i}, \qquad L_B = U_B - \sum_i E_{I_i} \prod_{j \neq i} U_{I_j}. \tag{216.4}$$

To verify the lower bound, telescope $\prod_i U_{I_i} - \prod_i \mathbf{1}_{I_i}$ and use $0 \leq U_{I_i} - \mathbf{1}_{I_i} \leq E_{I_i}$ and $\mathbf{1}_{I_j} \leq U_{I_j}$. Integration of these nonnegative error bounds proves (216.3). Products here involve different variables, so they do not increase a coordinate bandwidth.

216.2. **An actual box denominator, without a smoothing event.** Let $\mathbf{h} = (h_1, \dots, h_4)$, $0 < h_i \leq 1$, and define

$$B(\xi, \mathbf{h}) = \prod_i [\xi_i - h_i, \xi_i + h_i], \quad V(\mathbf{h}) = \prod_i 2h_i, \quad \mathcal{B}_{n,R}(\xi, \mathbf{h}) = n\bar{G}_R + \sqrt{n}B(\xi, \mathbf{h}). \tag{216.5}$$

In the last expression the first three coordinates are intersected with $\mathbb{Z}^3$. Write $\mathfrak{m}$ for counting measure times Lebesgue measure on the mixed record space, and put

$$\mathfrak{G}_{n,R}(\mathcal{B}) = n^{-2} \int_{\mathcal{B}} g_{D_R}\left(\frac{x - n\bar{G}_R}{\sqrt{n}}\right) \mathrm{d}\mathfrak{m}(x). \tag{216.6}$$

The Gaussian density has the standard probability normalization.

Theorem 216.2 (Unsmoothed window probabilities at a prescribed count)**.** *Fix $A < \infty$. Suppose $|\xi| \leq A$, $b_i h_i \geq 1$, $0 < b_i \leq \sqrt{n}/2$, and $\mathcal{Q}(\mathbf{b}) \subset \mathcal{U}_n^\sharp$. Then, uniformly in R, n, k, a and these boxes,*

$$\left|\mu_{n,R}^{[k],a}(\mathcal{B}) - \alpha_a \mathfrak{G}_{n,R}(\mathcal{B})\right| \leq C_A V(\mathbf{h}) \left(\mathcal{E}_n(a) + M_a \sum_i (b_i h_i)^{-1}\right). \tag{216.7}$$

For $a \geq 0$, the factor M_a in the second term may be replaced by α_a. If $\alpha_a > 0$, $a \geq 0$, and

$$\epsilon_n(a, \mathbf{b}, \mathbf{h}) = \mathcal{E}_n(a)/\alpha_a + \sum_i (b_i h_i)^{-1} \longrightarrow 0, \tag{216.8}$$

then the ratio of the probability to its displayed Gaussian mass converges to one uniformly on these central windows.

Proof. First suppose $a \geq 0$, so $\lambda_{n,R}^{[k],a}$ is a positive measure. Integrate the two envelopes against it. Their transforms are supported in $\mathcal{Q}(\mathbf{b})$ and bounded by $CV(\mathbf{h})$, by (216.3). Fourier inversion and the inherited integrated characteristic estimate give

$$\left| \int F \, \mathrm{d}\lambda_{n,R}^{[k],a} - \alpha_a \int F g_{D_R} \, \mathrm{d}x \right| \leq CV(\mathbf{h}) \mathcal{E}_n(a), \qquad F = L_B, U_B.$$

This integrates an entire test function against the actual discrete coordinates; no density with respect to four-dimensional Lebesgue measure is assigned to that law. Uniform ellipticity bounds $\|g_{D_R}\|_\infty$. The envelope excess then proves the same estimate with $\int_B g_{D_R}$ as Gaussian mass, with an additional $C\alpha_a V(\mathbf{h}) \sum_i (b_i h_i)^{-1}$.

We now compare that continuous Gaussian mass with the mixed mass in (216.6). The grid spacing is $n^{-1/2}$ in precisely the first three standardized coordinates. In one coordinate, assign to each included grid point its interval of that length. The symmetric difference from the target interval has length at most $2n^{-1/2}$. On full cells use the uniform first-derivative bound of the Gaussian. Iterating in the three grid coordinates gives

$$\left| \mathfrak{G}_{n,R}(\mathcal{B}) - \int_B g_{D_R} \, \mathrm{d}x \right| \leq C_A V(\mathbf{h}) \sum_{i=1}^{3} (\sqrt{n} h_i)^{-1}. \tag{216.9}$$

The conditions imply $\sqrt{n} h_i \geq 2$, so all intermediate grid volumes are comparable to the interval lengths. Since $b_i \leq \sqrt{n}/2$, the grid error is absorbed by the stated budget. This explicitly pays for endpoints and the lattice convention.

For complex a, decompose its real and imaginary parts into positive and negative parts. Their supremum and variation norms have sum at most a fixed multiple of $M_a + V_a$, by contraction of BV under positive part. Their masses have sum at most CM_a. Add the positive-measure estimates to obtain (216.7). For the last assertion, uniform ellipticity and $|\xi| \leq A$, $h_i \leq 1$ give $\mathfrak{G}_{n,R}(\mathcal{B}) \geq c_A V(\mathbf{h})$ for all sufficiently large n under the vanishing budget. Divide the positive estimate by that lower bound times α_a. □

Corollary 216.3 (A concrete rare-window law)**.** *Take $a = s_R$, so $\alpha_a = 1$ and the original section probability is recovered. Set $\rho = 67/1400$ and $h_i = n^{-1/25}$. Uniformly for $|\xi| \leq A$,*

$$\nu_R^*\{X_{n,R} \in B(\xi, \mathbf{h})\} = 16 n^{-4/25} g_{D_R}(\xi) \left(1 + O_A(n^{-11/1400} \sqrt{\log(2+n)})\right). \tag{216.10}$$

The event uses the original record, integer intervals in its first three coordinates, and a time interval in its fourth. Each physical half-width is $n^{23/50}$. No convolution is made in its definition.

Proof. Choose $b_i = n^\rho/4$. The product frequency box lies in the retained isotropic ball of radius $2n^\rho$. Here $\rho - 1/25 = 11/1400 > 0$, $e = 15/1400$, and $v_0 > e$. Theorem 216.2 and the Gaussian first-derivative bound give the conclusion, since replacing the Gaussian mass by its value at the center times volume costs relative $O(n^{-1/25})$. The grid error is smaller. The Fourier supports are rescaled b_i and physical $b_i/\sqrt{n} = n^{-633/1400}/4$. □

216.3. **A size bound for the common signed correction.**

Theorem 216.4 (The common correction averaged over an actual window)**.** *In addition to the hypotheses of Theorem 216.2, suppose the original marked weight is finite-record admissible, and*

take the same finite extraction and $L_n = \lceil \lambda n \rceil$, $\lambda > c_*^{-1} + 1$, *as in Section 215. For any of the three cutoffs* $\chi_n^\sharp, \chi_n^{\mathrm{box}}, \chi_n^\natural$, *and any fixed* $P > 0$,

$$\frac{n^2}{\mathfrak{m}(\mathcal{B})} \left| \int_{\mathcal{B}} \mathcal{R}_\chi^{a,L_n} \, \mathrm{d}\mathfrak{m} \right| \le C_A \left(\mathcal{E}_n(a) + M_a \sum_i (b_i h_i)^{-1} + M_a n^{-P} \right). \tag{216.11}$$

Thus for the concrete windows in (216.10) *the common correction itself, rather than only a change between two cutoffs, has a vanishing normalized signed window integral. The absolute value is outside the integral.*

Proof. Every cutoff is between zero and one, equals one on the original central ball, and is supported in the retained Fourier union. Mixed inversion and the integrated characteristic estimate imply

$$\sup_x \left| n^2 (K_\chi * \mu_{n,R}^{[k],a})(x) - \alpha_a g_{D_R}((x - n\bar{G}_R)/\sqrt{n}) \right| \le C\mathcal{E}_n(a) + C_P M_a n^{-P}. \tag{216.12}$$

The omitted Gaussian frequencies are bounded by uniform ellipticity; the first three physical supports are inside one torus chart. On the central box, $N \le L_n$ for all sufficiently large n, so $\mu^{a,L_n}(\mathcal{B}) = \mu_{n,R}^{[k],a}(\mathcal{B})$ exactly. The exact identity (215.5) gives

$$\int_{\mathcal{B}} \mathcal{R}_\chi^{a,L_n} = \mu_{n,R}^{[k],a}(\mathcal{B}) - \int_{\mathcal{B}} K_\chi * \mu_{n,R}^{[k],a} + \int_{\mathcal{B}} K_\chi * \mathsf{T}^{a,L_n}.$$

All integrals use $\mathfrak{m}$. Apply the preceding theorem to the first term, (216.12) to the second, and count-coordinate kernel separation to the last. The third-coordinate support gap is of order n, and each of the three kernels gives any fixed inverse power on that gap. Finally $\mathfrak{m}(\mathcal{B}) \asymp n^2 V(\mathbf{h})$ by the same grid count used in (216.9). This proves the estimate. No derivative norm of the extracted density was used. □

Resolution and the raw endpoint. The signed estimate (216.11) is a bound for one common correction tested on unsmoothed windows, not a bound for its essential supremum or for its absolute integral. The resolution condition $b_i h_i \to \infty$ cannot hold for a singleton lattice coordinate when $b_i = o(\sqrt{n})$. It therefore does not establish a fixed-label raw denominator or the microscopic local limit theorem. The complete complementary integral, the common correction in the required pointwise norm, and the long-time raw preparation budget remain the requirements for that same endpoint. Positivity has here supplied actual probability and signed-average estimates, not an unproved inference from cutoff transport.

217. Relative comparison at an exact physical observation time

The previous section gives a denominator for an unchanged, unsmoothed window. We now compare two different events relative to such a denominator. The time of physical observation is deterministic. The initial law in this section is ν_R^* at a section collision, not the stationary suspension law. The window widths will be specified; we do not replace a singleton lattice event by a larger one in the raw theorem.

217.1. **A three-dimensional denominator from the four-dimensional band.** Put

$$\rho = \frac{67}{1400}, \qquad \sigma = \frac{1}{25}, \qquad \epsilon_n = n^{-11/1400} \sqrt{\log(2+n)}. \tag{217.1}$$

For a positive definite $d \times d$ matrix C, g_C denotes its normalized d-dimensional Gaussian density.

Lemma 217.1 (Projection of the unsmoothed window theorem)**.** *Let* $P_{n,R} : \mathbb{R}^4 \to \mathbb{R}^3$ *be any linear map which extends to an invertible* 4×4 *matrix* $S_{n,R}$ *with* $\|S_{n,R}\| + \|S_{n,R}^{-1}\| \le C_0$. *Under*

ν_R^*, *put* $Y = P_{n,R}U_{n,R}/\sqrt{n}$ *and* $\Sigma_{n,R} = P_{n,R}D_RP_{n,R}^{\mathsf{T}}$. *For* $|\xi| \le A$ *and half-widths satisfying* $c_0 n^{-\sigma} \le h_i \le C_0 n^{-\sigma}$, $1 \le i \le 3$,

$$\nu_R^*\{Y \in B_3(\xi, \mathbf{h})\} = \int_{B_3(\xi,\mathbf{h})} g_{\Sigma_{n,R}}(y)\,\mathrm{d}y + O_{A,c_0,C_0}\left(\prod_{i=1}^{3}(2h_i)\epsilon_n\right). \tag{217.2}$$

The event is the actual projection of the record. No restriction of an $L^1(\mathbb{R}^4)$ *Fourier estimate to a measure-zero frequency plane is used.*

Proof. Write $Y' = S_{n,R}U_{n,R}/\sqrt{n}$ and truncate its fourth coordinate to $[-T, T]$, where $T = n^{1/400}$. The characteristic error for Y' on the cube $|v_i| \le b$, $b = c(C_0)n^{\rho}$, has integral at most $C\mathcal{E}_n(s_R)$. Indeed $S_{n,R}^{\mathsf{T}}$ maps this cube into the retained isotropic ball $|v| \le 2n^{\rho}$, and its determinant and inverse determinant are bounded. This is a four-dimensional change of variables, not a trace estimate.

Apply the product envelopes (216.4) to $B_3(\xi, \mathbf{h}) \times [-T, T]$. Their L^1 norms are $O(TV_3)$, where $V_3 = \prod_{i=1}^{3}(2h_i)$. Fourier comparison thus costs $CTV_3\mathcal{E}_n(s_R)$. We keep a sharper bound for the Gaussian envelope error. Uniform ellipticity of $S_{n,R}D_RS_{n,R}^{\mathsf{T}}$ gives a joint density bounded by $Ce^{-c|y|^2}$. For a term containing an error factor E_{I_i} with $i \le 3$, integrate that factor and the other two output factors in L^1; integrate the fourth majorant against $e^{-cy_4^2}$, using $0 \le U_{[-T,T]} \le 3$. This is at most $CV_3/(bh_i)$, independently of T. The fourth-coordinate error term has integral at most CV_3/b. The same bounds apply to both envelopes by (216.4).

The fourth coordinate of Y' has bounded sixty-fourth moment, uniformly in $n, R, S_{n,R}$, by (210.3). Removing its truncation costs at most CT^{-64}. For the Gaussian, the corresponding cost on the output box is at most $CV_3e^{-cT^2}$. Consequently the absolute probability error divided by V_3 is at most

$$C\left[T\mathcal{E}_n(s_R) + \sum_{i=1}^{3}(bh_i)^{-1} + b^{-1} + T^{-64}/V_3 + e^{-cT^2}\right]. \tag{217.3}$$

The four relevant power margins are

$$e - \frac{1}{400} = \frac{23}{2800}, \qquad \rho - \sigma = \frac{11}{1400}, \qquad \frac{64}{400} - 3\sigma = \frac{1}{25}, \qquad \rho > \frac{11}{1400}.$$

Also $v_0 - 1/400 > 23/2800$, and the norms of s_R are uniform. Each term in (217.3) is therefore $O(\epsilon_n)$. This proves the lemma, including the discrete endpoint conventions inherited from the envelopes. □

217.2. **Maximal actual-return increments and inverse clocks.**

Lemma 217.2 (A deterministic return-window maximal estimate)**.** *For every fixed real* $p > 2$, *every integer* $q \ge 1$, *and* $n \ge q$,

$$\int \max_{|j|\le q} |U_{n+j,R} - U_{n,R}|^p \,\mathrm{d}\nu_R^* \le C_p q^{p/2}. \tag{217.4}$$

Proof. Every centered return increment over a deterministic interval of length l has pth moment at most $C_p l^{p/2}$, by invariance of F_R^* and (210.3). Enlarge q to a power of two $Q < 2q$. At scale l, the maximum over its Q/l aligned intervals has L^p norm at most $C_pQ^{1/p}l^{1/2-1/p}$. A prefix is a sum of at most one interval per scale. Minkowski's inequality and $p > 2$ give a convergent geometric sum bounded by $C_p\sqrt{Q}$. Translation by the deterministic index n is justified by return-map invariance. The backward part uses suffixes of the length-q interval ending at n, and has the same bound. Adding the two parts proves the assertion. The maximum is over a short deterministic return window, not over all marks in a collision Fourier theorem. □

Start the physical orbit at $x \in Y_R^*$, with physical time zero at that outgoing collision. Let $W_R^Y(t)$ be its lattice displacement and $C_R^Y(t)$ its completed collision count, using the physical convention of Section 137. Put

$$V_R^Y(t) = \left(W_R^Y(t), C_R^Y(t) - t/\bar{\tau}_R\right), \quad B_R(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3 - x_4/\bar{\tau}_R). \tag{217.5}$$

The chosen displacement convention can differ from the sum of completed flight labels by one uniformly bounded cell-position term; that term is retained in the estimates below. Define the actual completed-return clock and its deterministic mean index by

$$M_R(t) = \max\{m \geq 0 : T_{m,R} \leq t\}, \qquad n_R(t) = \lfloor c_* t/\bar{\tau}_R \rfloor. \tag{217.6}$$

The positive lower free-flight length makes the maximum finite and bounds it deterministically by $C(1+t)$.

Proposition 217.3 (A coupling at the deterministic physical time)**.** *For every fixed $P > 0$, there is C_P such that, uniformly in R and all sufficiently large t,*

$$\nu_R^*\left\{|V_R^Y(t) - B_R U_{n_R(t),R}| > C_P t^{3/8}\right\} \leq C_P t^{-P}. \tag{217.7}$$

The same bound holds for the difference $B_R U_{M_R(t),R} - B_R U_{n_R(t),R}$.

Proof. Put $n = n_R(t)$, $q = \lceil t^{5/8} \rceil$; for large t, $q < n$ and $n \asymp t$ uniformly. The return flight-time mean is $\bar{\tau}_R/c_*$. The equivalences for the increasing times T_m turn $|M_R(t) - n| > q$ into a deviation of T_{n+q} or T_{n-q} from its mean of size at least cq. The rounding error is uniformly bounded. For every fixed p, (210.3) and Markov's inequality therefore give

$$\nu_R^*\{|M_R(t) - n| > q\} \leq C_p t^{p/2} q^{-p} \leq C_p t^{-p/8}.$$

Off this event, (217.4) bounds the probability that $|U_{M_R(t)} - U_n| > t^{3/8}$ by $C_p q^{p/2} t^{-3p/8} \leq C_p t^{-p/16}$. The operator norm of B_R is uniform, and $B_R \bar{G}_R = 0$.

At the last completed section return, the vector is $B_R J_{M_R(t)} = B_R U_{M_R(t)}$. Between this return and time t, the change in displacement, collision count, and elapsed time is bounded by $C r_R^* \circ (F_R^*)^{M_R(t)} + C$. Uniformly bounded single flights and labels justify this assertion even when the next return has not yet occurred. Since $M_R(t) \leq C(1+t)$, invariance and the exponential one-return tail give

$$\nu_R^*\left\{\max_{0 \leq j \leq C(1+t)} r_R^* \circ (F_R^*)^j > t^{3/8}\right\} \leq C(1+t) e^{-c t^{3/8}}.$$

This is a union bound, not a conditional independence statement. Choose a fixed $p > \max\{2, 16P\}$ and combine the three events. The count of moment factors is fixed for each P. This proves both assertions with the actual physical observation time left unchanged. □

217.3. **A relative event theorem with its actual denominator.** Recall the uniformly positive physical covariance

$$\mathcal{V}_R = \bar{\tau}_R^{-1} B_R \Gamma_R B_R^{\mathsf{T}} = (c_*/\bar{\tau}_R) B_R D_R B_R^{\mathsf{T}}.$$

For $h = t^{-1/25}$ and $|\xi| \leq A$, let $B_t = B_3(\xi, (h,h,h))$. On the same initial probability space (Y_R^*, ν_R^*) define

$$\begin{aligned} A_t^{\mathrm{ret}} &= \{t^{-1/2} B_R U_{n_R(t),R} \in B_t\}, \\ A_t^{\mathrm{last}} &= \{t^{-1/2} B_R U_{M_R(t),R} \in B_t\}, \\ A_t^{\mathrm{phys}} &= \{t^{-1/2} V_R^Y(t) \in B_t\}. \end{aligned} \tag{217.8}$$

All three are exact events. The third is measured at deterministic physical time t; it uses integer lattice and collision-count intervals of physical half-width $t^{23/50}$.

Theorem 217.4 (An unsmoothed physical denominator and relative replacement)**.** *Uniformly for $R \in I$, $|\xi| \le A$, and large t, all three events in* (217.8) *have probability*

$$8t^{-3/25} g_{\mathcal{V}_R}(\xi) \left(1 + O_A(t^{-11/1400}\sqrt{\log(2+t)})\right). \tag{217.9}$$

For either $A_t = A_t^{\mathrm{last}}$ or A_t^{phys},

$$\frac{\nu_R^*(A_t \triangle A_t^{\mathrm{ret}})}{\nu_R^*(A_t^{\mathrm{ret}})} \le C_A t^{-11/1400}\sqrt{\log(2+t)}. \tag{217.10}$$

The total variation distance, defined as the supremum over measurable sets, between the two conditional initial laws is bounded by the same right-hand side with a changed constant. Consequently the comparison holds for every common bounded measurable path statistic, with a factor twice its supremum norm for the difference of expectations.

Proof. Set $n = n_R(t)$ and take $P_{n,R} = \sqrt{n/t}\, B_R$ in Lemma 217.1. Adjoin the fourth row $(0,0,0,1)$ to obtain $S_{n,R}$. The ratio n/t is bounded above and away from zero uniformly, as are $\bar{\tau}_R$ and its reciprocal; thus the matrix and its inverse satisfy the lemma's bounds. Its projected covariance is $(n/t)B_R D_R B_R^{\mathsf{T}}$, which differs from $\mathcal{V}_R$ by $O(t^{-1})$. Uniform ellipticity and the Gaussian formula make the density error $O_A(t^{-1})$ on central boxes. Replacing the box Gaussian mass by its center value costs relative $O_A(h)$. The lemma proves (217.9) for A_t^{ret}, including a lower bound $c_A t^{-3/25}$.

For the event comparison, use Proposition 217.3 with $P = 2$. Outside a set of probability $O(t^{-2})$, the standardized vector changes by at most $d_t = Ct^{-1/8}$. Hence the symmetric difference is contained in that exceptional set and the event that the return vector lies in $B_t^+ \setminus B_t^-$, where the half-widths are $h + d_t$ and $h - d_t$. For large t they lie between $h/2$ and $2h$. Apply the projected-window lemma separately to these two boxes and subtract their probabilities. The Gaussian boundary volume is at most $C_A h^3 d_t/h$, with

$$d_t/h = O(t^{-17/200}).$$

The two Fourier-envelope errors have total at most $C_A h^3 t^{-11/1400}\sqrt{\log(2+t)}$. Thus the symmetric-difference probability divided by the proved denominator is at most

$$C_A\left[t^{-17/200} + t^{-11/1400}\sqrt{\log(2+t)} + t^{-2+3/25}\right].$$

The first and third terms have strictly faster power decay, so this proves (217.10). This argument subtracts masses of enlarged and contracted boxes; it does not require a separate Fourier estimate for a thinner unresolved boundary slab. The difference of event probabilities is bounded by their symmetric difference, proving (217.9) for the other events as well.

Finally, if $p = \nu_R^*(A_t^{\mathrm{ret}})$ and the symmetric-difference mass is $\delta \le p/2$, then $\nu_R^*(A_t) \ge p/2$. Writing the two normalized indicator densities shows directly that the conditional total variation distance is at most $2\delta/p$. Integration of any common bounded statistic proves the final assertion. No event was silently substituted in a numerator or a denominator. □

The still required microscopic comparison. Theorem 217.4 proves a relative comparison for different events and a nonzero denominator at deterministic physical time, but at the displayed growing physical window widths and under a section-start law. The original raw application also requires fixed lattice labels, its own exact time constraint, and its stated initial law. Those are not the events of (217.8). In particular neither this theorem nor the signed average estimate in the preceding section supplies the full pointwise common remainder, the long-time density jet

budget, or the full weighted fixed-count Fourier complement. The same raw mixed-density target and all its existing mechanisms are retained. The new results identify precisely a range in which actual denominators and relative event errors can already be proved, without presupposing the microscopic theorem.

218. A wider collision band and projected window probabilities

A prescribed collision interval does not incur a return-clock stopping error. We exploit this distinction before passing to physical time. It permits a full isotropic collision band with rescaled radius $m^{9/100}$, without asserting such a band at the prescribed return count. The initial collision density may be nonstationary, provided that its supremum and initial-coordinate variation are bounded. This includes the length-biased collision marginal of equilibrium flow.

Write $S_{m,R}h_R = \sum_{j=0}^{m-1} h_R \circ T_R^j$ and put

$$e_0 = \frac{3}{280}, \qquad \mathfrak{e}_m = m^{-e_0}\sqrt{\log(2+m)}.$$

All initial densities in this section are relative to the collision probability ν, not to ν_R^*.

218.1. **Eliminating the stopping loss on a fixed collision interval.**

Theorem 218.1 (An isotropic collision band with bounded-BV initial data)**.** *Fix $B, C_0 < \infty$. For every complex collision function a with $\|a\|_\infty + \|a\|_{\mathrm{BV}} \le B$, and $\alpha_a = \int a \,\mathrm{d}\nu$, one has*

$$\int_{|v|\le C_0 m^{9/100}} \left| \int a e^{iv\cdot S_{m,R}h_R/\sqrt{m}} \,\mathrm{d}\nu - \alpha_a e^{-v^{\mathsf{T}}\Gamma_R v/2} \right| \mathrm{d}v \le C_{B,C_0}\mathfrak{e}_m. \tag{218.1}$$

The bound is uniform in $R \in I$ and $m \ge 2$. The physical frequencies have radius $C_0 m^{-41/100}$. There is no stopping time in the transform.

Proof. On $|v| \le 2m^{1/200}$, integrate Lemma 145.1 with $\delta = m^{-1/14}/4$. The six resulting terms, with $U = m^{1/200}$ and $L_\delta = 1 + |\log\delta|$, are

$$C_B\left(\delta U^4 + \sqrt{\delta L_\delta}\,U^5 + \delta L_\delta U^6 + m^{-1/2}\delta^{-4}U^5 + m^{-1/2}\delta^{-6}U^7 + \delta^{-2}\rho^m U^4\right).$$

The slowest exponent is $1/28 - 5/200 = 3/280$. The cubic exponent $1/2 - 6/14 - 7/200 = 51/1400$ and all other powers have a strict margin over $3/280$. Thus this integral is $O_B(\mathfrak{e}_m)$. No return-clock comparison is used in this calculation.

For the annulus $2m^{1/200} \le |v| \le C_0 m^{9/100}$ use Theorem 212.3, with a mark at the initial collision, and the fixed choices

$$P = 40, \quad Q = 29, \quad \delta = m^{-19/100}/4, \quad \epsilon = m^{-50}/4. \tag{218.2}$$

The residual Taylor degree is 79; the spectral Taylor degree is 29. Neither varies with m. For $z = v/\sqrt{m}$, the two analytic margins in (206.7) are

$$\frac{1}{2} - \frac{9}{100} - 2\frac{19}{100} = \frac{3}{100}, \qquad 28\left(\frac{1}{2} - \frac{9}{100}\right) - 2\frac{19}{100}30 = \frac{2}{25}.$$

Fixed multiplicative constants C_0 are absorbed by taking m sufficiently large. The volume is $O_{C_0}(m^{9/25})$. The fine insertion and fine residual errors have decay exponents

$$50 - \frac{9}{25} = \frac{1241}{25}, \qquad 50 - \frac{9}{25} - 79\left(\frac{1}{2} + \frac{9}{100}\right) = \frac{303}{100}.$$

For the residual moment term indexed by $1 \le b \le 40$, the power before taking its negative decay sign is

$$\frac{9}{25} - 40 + 80\frac{9}{100} + \frac{81}{100}b. \tag{218.3}$$

Its largest value, at $b = 40$, is $-1/25$. Every smaller b improves this by $81/100$ per removed block. In particular the connected $b = 1$ term is included. The fixed logarithmic powers are absorbed by $1/25 - 3/280 = 41/1400 > 0$.

The damped word has a fixed polynomial prefactor, possibly large, and $m|z|^2 \geq 4m^{1/100}$. It is smaller than every fixed inverse power after integration. The omitted Gaussian contribution has the same property by the already proved uniform ellipticity of $\Gamma_R = c_* D_R$. This proves the annular estimate and then (218.1). Increasing the constant covers the finitely many remaining m. The argument enlarges a collision-clock band, not the known prescribed-return-count Fourier region. □

218.2. **Projection without taking a trace of a Fourier estimate.** For a positive definite matrix C, write g_C for its normalized Gaussian density. Put

$$\sigma_1 = \frac{2}{25}, \qquad \kappa_1 = \frac{23}{2800}, \qquad \varepsilon_m = m^{-\kappa_1}\sqrt{\log(2+m)}. \tag{218.4}$$

Theorem 218.2 (Projected collision windows under bounded initial densities)**.** *Let $\sigma_R \geq 0$ satisfy $\int \sigma_R \, \mathrm{d}\nu = 1$ and $\|\sigma_R\|_\infty + \|\sigma_R\|_{\mathrm{BV}} \leq B$. Let $P_{m,R} : \mathbb{R}^4 \to \mathbb{R}^3$ extend to a matrix $\mathsf{A}_{m,R}$ with $\|\mathsf{A}_{m,R}\| + \|\mathsf{A}_{m,R}^{-1}\| \leq C_0$. Set*

$$Y = \frac{1}{\sqrt{m}} P_{m,R} \sum_{j=0}^{m-1} h_R \circ T_R^j.$$

For $|\xi| \leq A$ and $c_0 m^{-\sigma_1} \leq h_i \leq C_0 m^{-\sigma_1}$,

$$\int \sigma_R \mathbf{1}_{\{Y \in B_3(\xi,\mathbf{h})\}} \, \mathrm{d}\nu = \int_{B_3(\xi,\mathbf{h})} g_{P_{m,R}\Gamma_R P_{m,R}^{\mathsf{T}}}(y) \, \mathrm{d}y + O_{A,B,c_0,C_0}\left(V_3 \varepsilon_m\right), \tag{218.5}$$

where $V_3 = \prod_{i=1}^3 (2h_i)$. Endpoints in the exact event may be included or excluded. No smoothing is added to that event.

Proof. Write $\mathsf{A} = \mathsf{A}_{m,R}$. Use the full four-dimensional variable $Y' = \mathsf{A}(\sum_{j<m} h_R \circ T_R^j)/\sqrt{m}$. Truncate its fourth coordinate to $[-T, T]$, with $T = m^{1/400}$. Choose the same bandwidth $b = c(C_0) m^{9/100}$ in all four output coordinates. The linear change of frequency $v \mapsto \mathsf{A}^{\mathsf{T}} v$ maps this cube into the band of Theorem 218.1, with bounded determinant factors. The characteristic error integrated on the cube is $O(\mathfrak{e}_m)$.

Apply the endpoint-safe product envelopes (216.4) to $B_3(\xi, \mathbf{h}) \times [-T, T]$. Their L^1 norms are $O(TV_3)$, so Fourier comparison costs $CTV_3\mathfrak{e}_m$. To estimate the Gaussian envelope error, use a joint density bounded by $C\exp(-c|y|^2)$, uniformly in all parameters. In an error term for one of the first three coordinates, integrate that error factor and the other two output majorants in L^1, but integrate the fourth majorant against the Gaussian decay. Since it is at most three, this contributes $CV_3/(bh_i)$, without a factor T. The error factor for the fourth coordinate contributes CV_3/b.

Lemma 208.2 at order 128, followed by $\sigma_R \leq B$, gives a uniform 128th moment for Y_4'. Its truncation error is at most CT^{-128}. The Gaussian tail costs at most $CV_3 e^{-cT^2}$. Thus the absolute error divided by V_3 is bounded by

$$C\left(T\mathfrak{e}_m + \sum_{i=1}^3 (bh_i)^{-1} + b^{-1} + T^{-128}/V_3 + e^{-cT^2}\right). \tag{218.6}$$

The relevant exponents are

$$\frac{3}{280} - \frac{1}{400} = \frac{23}{2800}, \quad \frac{9}{100} - \frac{2}{25} = \frac{1}{100}, \quad \frac{128}{400} - 3\frac{2}{25} = \frac{2}{25}.$$

They give (218.5). Fourier comparison has used a genuine four-dimensional integral and then removed a coordinate by a moment bound; it has not restricted an $L^1(\mathbb{R}^4)$ estimate to a plane. Positivity and the endpoint envelope inequalities apply directly even when output coordinates have atoms. □

219. Equilibrium physical windows and relative conditional laws

A typical physical observation does not start at a section collision. The section state seen from equilibrium is biased by the duration of its return excursion. We retain this bias exactly. A wider collision-clock Fourier band supplies the denominator, while the return-clock moment estimates compare the same stationary physical event with its completed-return descriptions. This avoids assuming bounded variation of a length-biased entrance distribution.

219.1. **Two descriptions of one equilibrium probability.** Write $\Theta_R(y) = \tau_R^*(y)$, $r_R^*(y) = r(y)$, and $\bar\Theta_R = \int \Theta_R \,\mathrm{d}\nu_R^* = \bar\tau_R/c_*$. By Proposition 131.2, equilibrium flow is represented on

$$\mathcal{X}_R = \{(y,s) : y \in Y_R^*,\ 0 \le s < \Theta_R(y)\}, \qquad \mathrm{d}\mathbb{P}_R(y,s) = \bar\Theta_R^{-1}\,\mathrm{d}\nu_R^*(y)\,\mathrm{d}s. \tag{219.1}$$

The physical observation starts at the flow state reached after time s from y. For $0 \le j \le r(y)$ put $\tau_{j,R}^{\mathrm{sum}}(y) = \sum_{i<j} \tau_R(T_R^i y)$. There is a unique $0 \le j < r(y)$ such that $\tau_{j,R}^{\mathrm{sum}} \le s < \tau_{j+1,R}^{\mathrm{sum}}$, apart from null endpoint conventions. Define the current outgoing collision and its elapsed flight age by

$$x = T_R^j y, \qquad u = s - \tau_{j,R}^{\mathrm{sum}}(y).$$

Lemma 219.1 (The two length biases and exceptional-set transfer)**.** *Under $\mathbb{P}_R$, the marginal law of y is $(\Theta_R/\bar\Theta_R)\nu_R^*$, whereas the law of (x,u) is*

$$\bar\tau_R^{-1}\mathbf{1}_{\{0\le u<\tau_R(x)\}}\,\mathrm{d}\nu(x)\,\mathrm{d}u. \tag{219.2}$$

In particular the outgoing collision marginal is $\sigma_R\nu$, where $\sigma_R = \tau_R/\bar\tau_R$ is a probability density with uniformly bounded supremum and BV norms. There are uniform $C, c > 0$ such that, for measurable $E \subset Y_R^*$ and $L \ge 0$,

$$\mathbb{P}_R\{y \in E\} \le C\nu_R^*(E)^{1/2}, \qquad \mathbb{P}_R\{\Theta_R(y) > L\} \le Ce^{-cL}. \tag{219.3}$$

Proof. Integrating s in (219.1) gives the first marginal. For an integrable test F of (x,u), partition each excursion into its actual collision flights. The tower identity gives

$$\begin{aligned}\int F(x,u)\,\mathrm{d}\mathbb{P}_R &= \frac{1}{\bar\Theta_R}\int_{Y_R^*}\sum_{j<r(y)}\int_0^{\tau_R(T_R^j y)} F(T_R^j y,u)\,\mathrm{d}u\,\mathrm{d}\nu_R^*(y)\\ &= \frac{1}{c_*\bar\Theta_R}\int_M\int_0^{\tau_R(x)} F(x,u)\,\mathrm{d}u\,\mathrm{d}\nu(x).\end{aligned}$$

Since $c_*\bar\Theta_R = \bar\tau_R$, this proves the second marginal. Lemma 138.1 and the positive lower bound for $\bar\tau_R$ give the density bounds. No assertion that $\Theta_R/\bar\Theta_R$ is bounded-BV is needed or made.

The return roof has a uniform exponential moment by Theorem 133.3. In particular $\|\Theta_R\|_{L^2(\nu_R^*)}$ is uniform. Cauchy–Schwarz proves the first estimate in (219.3). Integrating the exponential roof tail after multiplication by Θ_R, and reducing the tail exponent, proves the second. The positive normalizing mean is bounded away from zero. This treats the physical-time length bias, not merely the collision-count bias by $r(y)$. □

219.2. **Coupling collision, return and physical observations.** Let $V_R(t) = (W_R(t), C_R(t) - t/\bar{\tau}_R)$ be the actual physical increment during the interval of length t starting at (y, s). The lattice convention is the one used throughout the article; any bounded cell-position term is included in the following error bounds. Retain B_R from (217.5). Set

$$\begin{aligned} m_R(t) &= \lfloor t/\bar{\tau}_R \rfloor, & n_R(t) &= \lfloor c_* t/\bar{\tau}_R \rfloor, \\ L_R(t; y, s) &= \max\{l \geq 0 : T_{l,R}(y) \leq t + s\}, \\ Z_t^{\mathrm{coll}} &= t^{-1/2} B_R \sum_{i < m_R(t)} h_R(T_R^i x), & Z_t^{\mathrm{ret}} &= t^{-1/2} B_R U_{n_R(t),R}(y), \\ Z_t^{\mathrm{last}} &= t^{-1/2} B_R U_{L_R(t;y,s),R}(y), & Z_t^{\mathrm{phys}} &= t^{-1/2} V_R(t). \end{aligned} \tag{219.4}$$

These are random variables on the same probability space $(\mathcal{X}_R, \mathbb{P}_R)$. The return reference starts at the actual section origin y preceding the stationary observation, not at a newly sampled independent section state.

Proposition 219.2 (Uniform stationary observation coupling)**.** *For each fixed $P > 0$ there is C_P such that, for all large t,*

$$\mathbb{P}_R \left\{ \max_{j \in \{\mathrm{ret}, \mathrm{last}, \mathrm{phys}\}} |Z_t^j - Z_t^{\mathrm{coll}}| > C_P t^{-1/8} \right\} \leq C_P t^{-P} \qquad (R \in I). \tag{219.5}$$

Proof. First compare the collision reference with the physical vector. The age u is bounded by the uniform one-flight horizon. If $\ell(t)$ is the completed-flight index measured from x, then its threshold differs from t by at most that horizon. The flight roof is bounded-BV and has mean $\bar{\tau}_R$. Fixed collision moments and the monotone inverse-clock equivalences therefore give, with $m = m_R(t)$, $q = \lceil t^{5/8} \rceil$, and each fixed p,

$$\mathbb{P}_R\{|\ell(t) - m| > q\} \leq C_p t^{p/2} q^{-p} \leq C_p t^{-p/8}.$$

Indeed the outgoing collision density is bounded by Lemma 219.1, so stationary collision moment estimates apply by domination; independence of u and x is not used. Off this event the centered collision-sum difference lies in the deterministic length-q windows adjacent to m. Lemma 208.2, at a sufficiently large fixed even order, bounds the probability that its size exceeds $t^{3/8}$ by $C_p q^{p/2} t^{-3p/8} \leq C_p t^{-p/16}$. The initial and final partial flights contribute $O(1)$ after applying B_R, because $B_R h_R = (\kappa_R, 1 - \tau_R/\bar{\tau}_R)$. Choose $p > 16P$. This proves the physical-to-collision comparison.

For the return reference, apply Proposition 217.3 under ν_R^*, with its exceptional probability exponent chosen larger than $2P$. The first estimate in (219.3) transfers this exceptional set to $\mathbb{P}_R$ with probability $O(t^{-P})$. The actual interval from time s to $s + t$ differs from the section-start interval from zero to t by two end intervals of length at most $s \leq \Theta_R(y)$. The positive minimum free-flight length and bounded one-flight displacement give, deterministically,

$$|V_R(t) - V_R^Y(t; y)| \leq C(1 + \Theta_R(y)).$$

This includes collision-count centering and cell-position conventions. The exponential tail in (219.3) makes this error at most $Ct^{3/8}$ outside a smaller exceptional set. Thus the return reference and the physical vector obey the required comparison.

For the last-return reference, its difference from the physical vector is bounded before normalization by

$$C(1 + \Theta_R(y) + r_R^*((F_R^*)^{L_R(t;y,s)} y)).$$

The initial partial return contributes the roof term and the final partial return contributes the last term. On $\Theta_R \leq t$, the positive minimum flight length gives $L_R(t; y, s) \leq C(1 + t)$. Under ν_R^*

the union of the events $r_R^* \circ (F_R^*)^j > t^{3/8}$ over these deterministic indices has probability at most $C(1+t)e^{-ct^{3/8}}$, by return invariance. The first bound in (219.3) transfers this union to $\mathbb{P}_R$; the event $\Theta_R > t$ has exponential probability as well. This proves the last-return comparison. A triangle inequality and union bound now give (219.5) for all three variables against the common collision reference. All moment orders are fixed for a chosen P. □

219.3. **The stationary denominator and relative replacement.** Let $h = t^{-2/25}$ and $B_t = B_3(\xi, (h,h,h))$, $|\xi| \le A$. For each $j \in \{\mathrm{coll}, \mathrm{ret}, \mathrm{last}, \mathrm{phys}\}$ define

$$A_t^j = \{Z_t^j \in B_t\}. \tag{219.6}$$

The physical event is an observation at the original deterministic time t, with exact integer displacement and collision-count intervals of physical half-width $t^{21/50}$.

Theorem 219.3 (Stationary physical windows and relative conditional laws)**.** *Uniformly for $R \in I$, $|\xi| \le A$, and large t, each event in* (219.6) *satisfies*

$$\mathbb{P}_R(A_t^j) = 8t^{-6/25} g_{\mathcal{V}_R}(\xi)\left(1 + O_A(t^{-23/2800}\sqrt{\log(2+t)})\right). \tag{219.7}$$

For $j = \mathrm{coll}, \mathrm{last}, \mathrm{phys}$,

$$\frac{\mathbb{P}_R(A_t^j \triangle A_t^{\mathrm{ret}})}{\mathbb{P}_R(A_t^{\mathrm{ret}})} \le C_A t^{-23/2800}\sqrt{\log(2+t)}. \tag{219.8}$$

The conditional laws $\mathbb{P}_R(\cdot \mid A_t^j)$ and $\mathbb{P}_R(\cdot \mid A_t^{\mathrm{ret}})$ on the full stationary initial-state space have total variation distance bounded by the same right side with a changed constant. Thus the comparison holds for every common bounded measurable statistic of the entire orbit.

Proof. Take $m = m_R(t)$ and the initial density $\sigma_R = \tau_R/\bar{\tau}_R$. Apply Theorem 218.2 with $P_{m,R} = \sqrt{m/t}\, B_R$, augmenting by the fourth row $(0,0,0,1)$. The ratios m/t, $\bar{\tau}_R$ and their reciprocals have uniform bounds for large t, so the augmenting matrix and its inverse do also. The covariance is

$$(m/t) B_R \Gamma_R B_R^{\mathsf{T}} = \mathcal{V}_R + O(t^{-1}).$$

The window half-width $t^{-2/25}$ is comparable to $m^{-2/25}$. Uniform ellipticity controls the change in the Gaussian density from this $O(t^{-1})$ perturbation. Replacing its integral on the window by $8h^3$ times its central value costs relative $O_A(h)$. This proves (219.7) for A_t^{coll} and gives a lower bound $c_A h^3$.

Choose $P = 2$ in Proposition 219.2 and put $d_t = C_2 t^{-1/8}$. If A_t^j and A_t^{coll} differ outside the exceptional set, then Z_t^{coll} belongs to the shell between boxes of half-widths $h + d_t$ and $h - d_t$ about ξ. For large t those widths are between $h/2$ and $2h$. The projected collision-window theorem applies to both boxes with the same constants. Subtract their probabilities before taking absolute values. The Gaussian shell mass is at most $C_A h^2 d_t$; the total envelope comparison error is at most $C_A h^3 \varepsilon_t$. Consequently

$$\frac{\mathbb{P}_R(A_t^j \triangle A_t^{\mathrm{coll}})}{\mathbb{P}_R(A_t^{\mathrm{coll}})} \le C_A\left(t^{-9/200} + \varepsilon_t + t^{-2+6/25}\right) \le C_A \varepsilon_t. \tag{219.9}$$

This also proves the denominator asymptotic for each j. The symmetric-difference triangle inequality and the resulting comparison of the denominators give (219.8). No Fourier estimate of an unresolved thin boundary slab is assumed.

For measurable events A, B with $p = \mathbb{P}_R(A) > 0$ and $\delta = \mathbb{P}_R(A \triangle B) \le p/2$, direct subtraction of $\mathbf{1}_A/p$ and $\mathbf{1}_B/\mathbb{P}_R(B)$ gives $d_{\mathrm{TV}}(\mathbb{P}_R(\cdot \mid A), \mathbb{P}_R(\cdot \mid B)) \le 2\delta/p$. Apply this to the proved event comparison. The expectation difference for a common bounded statistic is at most twice its

supremum norm times that distance. This is a comparison of conditional measures, not a claim of a conditional invariance principle or of independence from the event. □

Corollary 219.4 (Additional bounded path selection)**.** *Let $w_t \ge 0$ be any measurable function on $\mathcal{X}_R$, possibly depending on the whole path, with $\|w_t\|_\infty \le B$ and*

$$\int_{A_t^{\mathrm{ret}}} w_t \,\mathrm{d}\mathbb{P}_R \ge a_t \mathbb{P}_R(A_t^{\mathrm{ret}}), \qquad a_t > 0.$$

If $B\varepsilon_t/a_t \to 0$, the conditional laws with densities proportional to $w_t \mathbf{1}_{A_t^j}$ and $w_t \mathbf{1}_{A_t^{\mathrm{ret}}}$ have total variation distance $O_A(B\varepsilon_t/a_t)$.

Proof. Their unnormalized difference has L^1 norm at most $B\mathbb{P}_R(A_t^j \triangle A_t^{\mathrm{ret}})$. Divide by the stated lower bound for the weighted return denominator, then use the same normalization inequality as in the preceding proof. The lower bound involving a_t is an explicit hypothesis, not a weighted raw denominator theorem inferred from the unweighted one. □

Relation to the original raw problem. Theorems 218.1 and 219.3 supply a collision-clock Fourier estimate and a relative physical-event theorem under the actual equilibrium law. They are not statements under the unbiased section-start law. The physical half-width $t^{21/50}$ is smaller than the section-start half-width $t^{23/50}$ above, but still grows: it is not a singleton lattice label. The probability $t^{-6/25}$ and the relative error in (219.8) refer to these exact windows and to no substituted event. The bounded path-selection corollary states its own additional denominator requirement.

The full prescribed-return-count complement, the pointwise common signed correction, the long-time inverse-coarea derivative budget, and the original microscopic conditioning theorem remain distinct requirements of the same raw mixed-density problem. In particular removing the collision stopping loss has not removed the return-count stopping loss from Theorems P and Q. The common raw correction is still bounded only in the signed window norm proved in Theorem 216.4, unless its stronger pointwise norm is established separately.

220. Joint collision blocks under a shrinking endpoint window

The comparison of two conditioned initial measures does not identify their common limit. We now retain finitely many path frequencies while integrating the endpoint frequency. The estimates are uniform in the positions of the block boundaries, including coincident boundaries. This uniformity will be used for conditional tightness, not only for finite-dimensional convergence.

Write $S_m h = S_{m,R} h_R$, $e_0 = 3/280$, and $\mathfrak{e}_m = m^{-e_0}\sqrt{\log(2+m)}$, as in Section 218. All collision expectations in this section are under $\sigma\nu$, where

$$\sigma \ge 0, \qquad \int \sigma \,\mathrm{d}\nu = 1, \qquad \|\sigma\|_\infty + \|\sigma\|_{\mathrm{BV}} \le B. \tag{220.1}$$

The constants may depend on the fixed number of blocks, but not on any of their lengths.

220.1. **A uniform joint Fourier estimate.** Fix an integer $q \ge 1$ and integers $0 = j_0 \le j_1 \le \cdots \le j_q = m$. Put $m_l = j_l - j_{l-1}$, $d_l = m_l/m$, and $H_l = (S_{j_l} h - S_{j_{l-1}} h)/\sqrt{m}$. For $u_1, \dots, u_q \in \mathbb{R}^4$ define

$$\mathcal{C}_m(v; \mathbf{u}) = \int \sigma \exp\left(i \sum_{l=1}^{q} (v + u_l) \cdot H_l \right) \mathrm{d}\nu,$$

$$\mathcal{G}_m(v; \mathbf{u}) = \exp\left(-\frac{1}{2} \sum_{l=1}^{q} d_l (v + u_l)^{\mathsf{T}} \Gamma_R (v + u_l) \right). \tag{220.2}$$

An interval of length zero contributes zero to both expressions.

Theorem 220.1 (The endpoint band with joint block frequencies)**.** *For fixed $q, B, C_0 < \infty$, uniformly in R, the subdivision, and $\max_l |u_l| \le m^{1/200}$,*

$$\int_{|v|\le C_0 m^{9/100}} |\mathcal{C}_m(v;\mathbf{u}) - \mathcal{G}_m(v;\mathbf{u})|\,\mathrm{d}v \le C_{q,B,C_0}\mathfrak{e}_m. \tag{220.3}$$

Here v and u_l are rescaled frequencies. Each actual collision frequency is $(v+u_l)/\sqrt{m}$.

Proof. On $|v| \le 2m^{1/200}$ take $\delta = m^{-1/14}/4$ and smooth σ and h_R. Put $z_l = (v+u_l)/\sqrt{m}$. In chronological order the smoothed pairing is

$$\ell\left(L_{z_q}^{m_q}\cdots L_{z_1}^{m_1}(\mathsf{S}_\delta\sigma)\nu\right), \qquad L_z = \mathcal{L}_{R,\delta,z}.$$

Use the decomposition in Lemma 139.1. For a zero power its complementary part means $I-\Pi(z)$. The complementary bound $C\rho^l$ and its annihilation of $\Pi(z)$ then hold also at $l=0$.

We explain why a short block is harmless. In the expansion of this fixed product, the term with only complementary factors is at most $C_{q,B}\delta^{-2}\rho^m$. Every other nonprincipal term has two adjacent factors of different types. At such a boundary, for example,

$$N(z)^l\Pi(w) = N(z)^l(\Pi(w)-\Pi(z)).$$

The reversed product is handled in the same way. Analytic projection bounds give $\|\Pi(w)-\Pi(z)\| \le C\delta^{-2}|w-z|$ on this ball. All remaining powers are uniformly bounded: their eigenvalue factors have modulus at most $\exp(Cm\delta^{-6}\max_l|z_l|^3) \le C$, and the complementary powers have their uniform bounds. Thus the sum of the mixed terms, including the initial density norm, is at most $C_{q,B}\delta^{-4}(|v|+U)/\sqrt{m}$, where $U = \max_l|u_l|$. This argument uses no positive lower bound for m_l/m.

The product of principal factors has amplitude $1 + O_{q,B}(\delta^{-4}(|v|+U)/\sqrt{m})$. The eigenvalue product has the Gaussian logarithm with $\Gamma_{R,\delta}$, up to a remainder bounded by $C_q\delta^{-6}m^{-1/2}(|v|+U)^3$. Changing the density costs $C_B\delta$. For the observable change, the small-mass variance on each deterministic interval and Minkowski's inequality give

$$\int\left|\sum_l z_l\cdot(S_{j_l}-S_{j_{l-1}})(h_R-h_{R,\delta})\right|(\mathsf{S}_\delta\sigma)\,\mathrm{d}\nu \le C_{q,B}(|v|+U)\sqrt{\delta(1+|\log\delta|)}.$$

Changing the covariance costs $C_q\delta(1+|\log\delta|)(|v|+U)^2$. Since $U \le m^{1/200}$, integration of these bounds has exactly the six power budgets in the proof of Theorem 218.1. The slowest is $1/28 - 5/200 = 3/280$. This proves the central estimate, uniformly in the permitted growing block frequencies.

On $2m^{1/200} \le |v| \le C_0 m^{9/100}$ use the fixed orders and scales

$$P = 40, \quad Q = 29, \quad \delta = m^{-19/100}/4, \quad \epsilon = m^{-50}/4.$$

Here $|v+u_l| \ge |v|/2$ and $|v+u_l| \le 3|v|/2$. We give the multiblock version of the argument in Theorem 212.3 to account for the additional frequencies. Let $r_\delta = h_R - h_{R,\delta}$ and $Z = \max_l|z_l|$. For $Z > 0$, write the residual phase as

$$X = \sum_l z_l\cdot(S_{j_l}-S_{j_{l-1}})r_\delta.$$

The all-small-mass moment bound, on each block, followed by Minkowski's inequality gives

$$\int |X|^{2P}\,\mathrm{d}\nu \le C_{q,P}Z^{2P}\sum_{b=1}^{P}(m\delta)^b(1+|\log\delta|)^{2P-b}.$$

Domination gives the same bound with the smoothed initial density. Expand e^{iX} through degree 79. Smooth its residual factors and the initial density at scale ϵ. Each corrected monomial is a

chronological word with at most 80 smooth multipliers and the $q-1$ additional block boundaries. The power lengths in block l sum to m_l, so its damping is

$$\exp\left(-c\sum_l m_l|z_l|^2\right)\le e^{-c'|v|^2}.$$

The number of factors is bounded in terms of q and the fixed order, not in terms of m. The fine insertion, fine residual, and residual moment errors are consequently bounded exactly as in (212.3), with $|z|$ replaced by Z and a changed fixed constant. The two analytic margins are $3/100$ and $2/25$. The integrated fine errors have decay exponents $1241/25$ and $303/100$. The largest residual-moment power is $-1/25$, attained at $b=40$; every smaller block number improves it by $81/100$. Thus the margin $41/1400$ pays all fixed logarithms. The Gaussian in (220.2) has the same annular damping. Combining the two regions proves the theorem for large m; enlarging the constant covers the finitely many remaining m. □

220.2. **A shrinking endpoint with oscillatory path weights.** Let $P_m:\mathbb{R}^4\to\mathbb{R}^3$ be the first three rows of a matrix A_m satisfying $\|\mathsf{A}_m\|+\|\mathsf{A}_m^{-1}\|\le C_0$. Put

$$C_m=P_m\Gamma_R P_m^{\mathsf{T}},\qquad Y_m=P_mS_mh/\sqrt{m},\qquad E_m=\{Y_m\in B_3(\xi,\mathbf{h})\},$$

where $|\xi|\le A$ and $c_0m^{-2/25}\le h_i\le C_0m^{-2/25}$. These assumptions make C_m uniformly positive definite. Write $V_3=\prod_i 2h_i$ and

$$\varepsilon_m=m^{-23/2800}\sqrt{\log(2+m)}.$$

For $\theta_1,\dots,\theta_q\in\mathbb{R}^3$ define

$$\mathcal{B}_m(\boldsymbol{\theta};y)=\exp\Big(i\sum_l d_l\theta_l\cdot y-\frac12\sum_{l,r}(d_l\mathbf{1}_{l=r}-d_ld_r)\theta_l^{\mathsf{T}}C_m\theta_r\Big).\tag{220.4}$$

This is the characteristic function of Gaussian increments of lengths d_l, conditioned on their sum being y.

Proposition 220.2 (Uniform conditional block characteristic functions)**.** *There is $c(C_0)>0$ such that $\max_l|\theta_l|\le c(C_0)m^{1/200}$ implies*

$$\left|\mathbb{E}_{\sigma\nu}\left[\exp\left(i\sum_l\theta_l\cdot P_mH_l\right)\middle|E_m\right]-\mathcal{B}_m(\boldsymbol{\theta};\xi)\right|\le C_{q,A,B,c_0,C_0}\varepsilon_m.\tag{220.5}$$

The bound is uniform over all subdivisions, including zero-length blocks. It concerns the original endpoint indicator, with either choice of inclusion of its discrete endpoints.

Proof. First retain the Gaussian integral over the endpoint box rather than its center. Augment Y_m by the fourth row of A_m and truncate that fourth coordinate to $[-T,T]$, $T=m^{1/400}$. Use the four-dimensional endpoint envelopes of (216.4) at bandwidth $b=c_1(C_0)m^{9/100}$. After the change of frequency $v\mapsto\mathsf{A}_m^{\mathsf{T}}v$, the oscillatory comparison is Theorem 220.1, with $u_l=P_m^{\mathsf{T}}\theta_l$. Its error for either envelope is $CTV_3\mathfrak{e}_m$.

Positivity cannot sandwich an oscillatory numerator. Instead let $W=\exp(i\sum_l\theta_l\cdot P_mH_l)$, so $|W|=1$, and use

$$|\mathbb{E}[W(\mathbf{1}_{B_3\times[-T,T]}-U)]|\le\mathbb{E}(U-\mathbf{1}_{B_3\times[-T,T]})\le\mathbb{E}(U-L).$$

The latter is bounded by the unweighted Fourier comparison plus the Gaussian envelope excess. The Gaussian weighted endpoint measure also has total variation bounded by its unweighted Gaussian marginal, because its conditional characteristic function has modulus at most one. Thus the same replacement is valid on the Gaussian side. Integrating the fourth Gaussian coordinate

against its decay, as in (218.6), gives an error $CV_3[\sum_i (bh_i)^{-1} + b^{-1}]$, without a factor T. The 128th collision moment bounds the omitted actual tail by CT^{-128}; $|W| = 1$ makes this estimate unchanged. The Gaussian tail is at most $CV_3 e^{-cT^2}$.

It follows that the numerator differs from

$$\int_{B_3(\xi,\mathbf{h})} g_{C_m}(y)\mathcal{B}_m(\boldsymbol{\theta}; y)\,\mathrm{d}y$$

by at most $CV_3\varepsilon_m$. The three exponent margins are 23/2800, 1/100 and 2/25, exactly as in the unweighted projection. The Gaussian formula follows by subtracting d_l times the total from independent centered increments of covariance $d_l C_m$. These differences have covariance $(d_l \mathbf{1}_{l=r} - d_l d_r)C_m$ and are independent of the total.

Theorem 218.2 gives $\mathbb{P}_{\sigma\nu}(E_m) = V_3 g_{C_m}(\xi)(1 + O(\varepsilon_m))$, including a lower bound $c_A V_3$. On the box the change in (220.4) from its value at ξ is at most $Cm^{-2/25}\max_l |\theta_l|$; the nonoscillatory Gaussian density changes by $O_A(m^{-2/25})$. The first error is $O(m^{-3/40})$, hence smaller than ε_m even for the permitted growing frequencies. Divide by the exact denominator. This proves the proposition, without taking a trace of a four-dimensional L^1 Fourier bound. □

Uniformity needed below. The number of blocks in these statements is fixed. Their boundaries may nevertheless depend on m and may approach each other or an endpoint. The permitted growth of the test frequencies and this boundary uniformity are both needed to prove tightness after division by a probability of order $m^{-6/25}$. Finite-dimensional convergence alone would not justify such a division in a path-space assertion.

221. Conditioned physical paths and their Gaussian bridges

We identify the common conditional law of Theorem 219.3. The conditioning remains an exact observation at deterministic physical time, with the same shrinking standardized endpoint box. The result is a functional limit, not merely a comparison of two conditional measures. Its proof pays for the rare denominator in the tightness and clock estimates.

For a positive definite 3×3 matrix C and $\xi \in \mathbb{R}^3$, let

$$\mathbb{B}_{C,\xi}(s) = s\xi + C^{1/2}(W(s) - sW(1)), \qquad 0 \le s \le 1, \tag{221.1}$$

where W is standard three-dimensional Brownian motion. This law on $\mathcal{C} = C([0,1], \mathbb{R}^3)$ has mean $s\xi$ and covariance $(\min\{s,t\} - st)C$. We use the supremum norm on $\mathcal{C}$ and the bounded-Lipschitz distance on probability laws.

221.1. **Tightness after conditioning on the endpoint.** Under the hypotheses of Proposition 220.2, let X_m be the polygonal interpolation with

$$X_m(j/m) = P_m S_j h/\sqrt{m}, \qquad 0 \le j \le m.$$

Retain E_m, C_m and ε_m from that proposition.

Lemma 221.1 (A trigonometric tail test)**.** *For each positive integer p there are $c_p, C_p > 0$ such that*

$$c_p \mathbf{1}_{\{|x| \ge 1\}} \le F_p(x) := \int_0^1 (1 - \cos(ux))^p\,\mathrm{d}u \le C_p |x|^{2p}.$$

The integrand is a trigonometric polynomial of degree p in ux.

Proof. The upper bound follows from $1 - \cos v \le v^2/2$. The function F_p is continuous, even, and strictly positive away from zero. On $1 \le |x| \le 4\pi$ it has a positive minimum. For $|x| \ge 4\pi$, write it as the average of the nonnegative, 2π-periodic function $(1 - \cos v)^p$ over an interval of length $|x|$. The complete periods occupy at least half that length and have a positive mean. This gives the lower bound. Expanding the power gives the last assertion. □

Theorem 221.2 (A collision bridge under an unsmoothed endpoint window)**.** *Uniformly in R, the initial densities* (220.1)*, the matrices and windows in Proposition 220.2,*

$$d_{\mathrm{BL}}\left(\mathrm{Law}_{\sigma\nu}(X_m \mid E_m), \mathrm{Law}(\mathbb{B}_{C_m,\xi})\right) \longrightarrow 0. \tag{221.2}$$

No independence of the actual collision increments is assumed.

Proof. For a fixed finite list of times, use (220.5) with bounded test frequencies and the corresponding consecutive blocks. Rounding times to collision indices and polygonal interpolation change a fixed finite list by $O(m^{-1/2})$, since h_R and P_m are bounded. The displayed Gaussian increment formula proves the finite-dimensional bridge limit, uniformly in the parameters.

For tightness take any collision-grid increment of length $\Delta = (k-j)/m$. Apply Lemma 221.1 with $p = 4$ to each of its three coordinates divided by $a/\sqrt{3}$. The characteristic functions involved have three consecutive blocks and test frequencies at most $4\sqrt{3}/a$. Therefore, whenever $4\sqrt{3}/a \le c(C_0)m^{1/200}$,

$$\mathbb{P}_{\sigma\nu}\{|X_m(k/m) - X_m(j/m)| > a \mid E_m\} \le C_A a^{-8}\Delta^4 + C_A\varepsilon_m. \tag{221.3}$$

Indeed the corresponding Gaussian bridge increment has mean $\Delta\xi$, covariance $\Delta(1-\Delta)C_m$, and eighth moment at most $C_A\Delta^4$. The finite trigonometric coefficient sum is fixed, so its Fourier comparison costs $C_A\varepsilon_m$, not a negative power of the endpoint probability.

Set

$$\eta = \frac{1}{200}, \qquad \alpha = \frac{1}{8}, \qquad J_m = \lfloor \eta \log_2 m \rfloor. \tag{221.4}$$

Use the nested grids with vertices $\lfloor ml2^{-j}\rfloor/m$, $0 \le l \le 2^j$, for $j \le J_m$. Each adjacent increment has length at most 2^{1-j}. Fix $\epsilon > 0$ and assign threshold $a_j = \epsilon 2^{-\alpha j}$. For all sufficiently large m, all frequencies in (221.3) are permitted, because their largest order is $C_\epsilon m^{\eta\alpha} = C_\epsilon m^{1/1600}$, strictly below $m^{1/200}$. A union bound over levels $j_0 \le j \le J_m$ and their adjacent increments gives

$$C_A\epsilon^{-8}\sum_{j\ge j_0} 2^{-2j} + C_A 2^{J_m}\varepsilon_m \le C_A\epsilon^{-8}2^{-2j_0} + C_A m^{-9/2800}\sqrt{\log(2+m)}. \tag{221.5}$$

Here $4(1-2\alpha) - 1 = 2$ and $23/2800 - 1/200 = 9/2800$. On the complementary event, dyadic chaining bounds the modulus of the polygon through the finest grid vertices by a constant times $\epsilon 2^{-\alpha j_0}$ on intervals of length 2^{-j_0}: pass from each vertex to its ancestor at level j_0 and sum at most two increments per finer level. Adjacent ancestors add at most a fixed number of level-j_0 increments. Rounding the nested grids only changes these constants.

It remains to control the motion inside the finest cells. This is where ordinary tightness without a rare-event budget would be insufficient. Let $\widetilde{X}_m$ be the polygon through those vertices. Lemma 208.2, at the fixed order 128, gives on each cell, under the bounded density σ,

$$\mathbb{E}\sup_{s \text{ in the cell}} |X_m(s) - \widetilde{X}_m(s)|^{128} \le C 2^{-64J_m}.$$

The bound follows from deterministic-window partial sums and their endpoint increment; the largest cell has $O(m2^{-J_m})$ collisions. There are 2^{J_m} cells. Since $\mathbb{P}_{\sigma\nu}(E_m) \ge c_A m^{-6/25}$,

$$\mathbb{P}_{\sigma\nu}\{\|X_m - \widetilde{X}_m\|_\infty > \epsilon \mid E_m\} \le C_A\epsilon^{-128}m^{6/25}2^{-63J_m} \le C_A\epsilon^{-128}m^{-3/40}. \tag{221.6}$$

The strict exponent is $63/200 - 6/25 = 3/40$. First let $m \to \infty$ in (221.5) and (221.6), and then let $j_0 \to \infty$. They prove conditional tightness in $\mathcal{C}$.

Finally suppose uniformity in (221.2) failed. Along a violating sequence select a subsequence for which C_m and ξ converge; these lie in compact sets of uniformly positive matrices and bounded vectors. Tightness and the uniform finite-dimensional estimate identify every subsequential law as the bridge with those limiting parameters. The comparison bridge laws also converge: couple

them by the same W in (221.1) and use continuity of the positive matrix square root. This contradicts the violating bounded-Lipschitz distance and completes the proof. □

221.2. **The whole physical clock on the same stationary space.** Use the stationary suspension $(\mathcal{X}_R, \mathbb{P}_R)$ and the four exact endpoint events A_t^j of (219.6), with half-width $t^{-2/25}$. Let $\mathsf{X}_{t,R}$ be the continuous polygon through the values

$$t^{-1/2}V_R(k), \qquad k = 0, 1, \dots, \lfloor t \rfloor, \quad \text{and } t^{-1/2}V_R(t),$$

at times k/t and 1, where $V_R(u) = (W_R(u), C_R(u) - u/\bar{\tau}_R)$. The last point is included only once when t is an integer. Its distance from the original physical step path is $O(t^{-1/2})$: the minimum flight length is positive and a unit physical time interval has bounded displacement and bounded collision count.

Let $m = m_R(t)$ and let $\mathsf{Y}_{t,R}$ be the collision polygon with values $t^{-1/2}B_R S_j h_R(x)$ at times j/m. The state x is the actual outgoing collision preceding the stationary observation, not an independently chosen initial state.

Lemma 221.3 (A uniform pathwise physical-to-collision comparison)**.** *For every fixed $P > 0$,*

$$\mathbb{P}_R\{\|\mathsf{X}_{t,R} - \mathsf{Y}_{t,R}\|_\infty > C_P t^{-1/8}\} \le C_P t^{-P}. \tag{221.7}$$

Consequently the same exceptional probability under any of the four endpoint conditionings is at most $C_{A,P}t^{-P+6/25}$.

Proof. Use Proposition 219.2 at each integer observation length $k \ge t^{1/4}$ and at t. In unnormalized variables its physical-to-collision error is $C_Q k^{3/8}$ outside probability $C_Q k^{-Q}$. Summing over at most $t + 2$ observation lengths gives $C_Q t^{1-Q/4}$. Take $Q > 4(P + 1)$. Every retained error is at most $C_Q t^{3/8}$.

For $k < t^{1/4}$ both the physical vector and its deterministic collision reference have size at most $C(1 + k)$: use bounded speed and cell-position terms, the positive minimum flight length, and boundedness of h_R. Their difference is therefore at most $Ct^{1/4}$, which is smaller than the same threshold. At the physical grid point k/t the index $\lfloor mk/t \rfloor$ differs from $m_R(k)$ by at most two. This changes the collision sum by a bounded amount. Between adjacent physical grid points, the physical interpolation and the collision polygon each change by a bounded amount before normalization. Division by $\sqrt{t}$ proves (221.7). No union over a continuum of clocks has been taken. The lower bounds in (219.7) prove the last assertion. □

Theorem 221.4 (Stationary physical bridges in exact endpoint windows)**.** *For every $A < \infty$, uniformly for $R \in I$, $|\xi| \le A$, and $j \in \{\text{coll}, \text{ret}, \text{last}, \text{phys}\}$,*

$$d_{\mathrm{BL}}\left(\mathrm{Law}_{\mathbb{P}_R}(\mathsf{X}_{t,R} \mid A_t^j), \mathrm{Law}(\mathbb{B}_{\mathcal{V}_R,\xi})\right) \longrightarrow 0. \tag{221.8}$$

In particular the actual physical observation conditions the entire physical displacement and collision-fluctuation path to a Gaussian bridge with terminal value ξ.

Proof. For A_t^{coll} apply Theorem 221.2 under the exact collision marginal $\sigma_R\nu$, $\sigma_R = \tau_R/\bar{\tau}_R$, with $P_m = \sqrt{m/t}\, B_R$. This matrix has the uniformly bounded augmentation used in Theorem 219.3. Its covariance is $C_m = (m/t)B_R\Gamma_R B_R^{\mathsf{T}} = \mathcal{V}_R + O(t^{-1})$, and its endpoint window widths are comparable to $m^{-2/25}$. The resulting polygon is exactly $\mathsf{Y}_{t,R}$. Lemma 221.3, with $P = 2$, replaces it by $\mathsf{X}_{t,R}$ under this conditioning, since the conditional exceptional probability is $O(t^{-2+6/25})$. Continuity of Gaussian bridge laws in their positive covariance gives the stated limit.

Theorem 219.3 and the triangle inequality bound the other three conditional initial laws against the collision-conditioned law in total variation by

$$C_A t^{-23/2800}\sqrt{\log(2 + t)}.$$

Pushforward by the same physical path cannot increase this distance. This proves all four assertions. The return-conditioned statistic remains the actual physical path on the original stationary suspension; neither its initial section point nor its elapsed age is resampled. □

221.3. **Actual denominators for a class of path selectors.**

Corollary 221.5 (Path-weighted window denominators)**.** *Let $F : \mathcal{C} \to [0, \infty)$ be fixed, bounded and continuous. Uniformly on the parameter sets of the preceding theorem,*

$$\mathbb{E}_R[F(\mathsf{X}_{t,R})\mathbf{1}_{A_t^j}] = 8t^{-6/25} g_{\mathcal{V}_R}(\xi) \left(\mathbb{E}F(\mathbb{B}_{\mathcal{V}_R,\xi}) + o(1)\right). \tag{221.9}$$

Whenever the bridge expectation has a positive uniform lower bound, the conditional path law with weight $F(\mathsf{X}_{t,R})\mathbf{1}_{A_t^j}$ converges to the bridge law weighted by F. The relative comparison between any two of these weighted endpoint events retains the rate $O_A(t^{-23/2800}\sqrt{\log(2+t)})$, with a constant depending on the stated lower bound and $\|F\|_\infty$.

Proof. Uniform weak convergence and uniform tightness give convergence of expectations of every fixed bounded continuous function: on a common compact path set approximate it uniformly by bounded Lipschitz functions, and use boundedness outside that set. Multiply this conclusion by (219.7). For the reweighted law apply it also to FG for a bounded continuous path test G, and divide by the positive denominator. For the comparison, the difference of the two unnormalized densities is supported on the original symmetric difference and is at most $\|F\|_\infty$ there. Normalization costs at most twice its integral divided by either weighted mass. The relative event theorem and (221.9) give the claimed rate. □

Corollary 221.6 (A concrete nonsingle-time indicator class)**.** *Fix $0 < s_1 < \cdots < s_r < 1$ and bounded boxes $D_1, \dots, D_r \subset \mathbb{R}^3$ with nonempty interiors. Define the exact path event*

$$D_t = \bigcap_{l=1}^{r} \{t^{-1/2} V_R(t s_l) \in D_l\}.$$

Then, uniformly for $R \in I$, $|\xi| \le A$, and all four endpoint events,

$$\mathbb{P}_R(D_t \cap A_t^j) = 8t^{-6/25} g_{\mathcal{V}_R}(\xi) \mathfrak{b}_R(\xi; \mathbf{s}, \mathbf{D})(1 + o(1)), \tag{221.10}$$

where

$$\mathfrak{b}_R(\xi; \mathbf{s}, \mathbf{D}) = \int_{\prod_l D_l} g_{\Sigma_R}((y_l - s_l\xi)_{l=1}^r) \prod_l \mathrm{d}y_l, \quad (\Sigma_R)_{lk} = (\min\{s_l, s_k\} - s_l s_k)\mathcal{V}_R.$$

The factor $\mathfrak{b}_R$ has a positive uniform lower bound. The conditional physical path converges to the bridge restricted to these cylinder boxes. The conditional initial laws for $D_t \cap A_t^j$ and $D_t \cap A_t^{j'}$ are within $O_A(t^{-23/2800}\sqrt{\log(2+t)})$ in total variation.

Proof. For distinct interior times the scalar bridge covariance matrix is positive definite. This follows, for example, by expressing its values as independent Gaussian increments conditioned on their total: no nonzero linear combination of the interior partial sums is a multiple of that total. Uniform ellipticity of $\mathcal{V}_R$ gives positive definiteness of Σ_R. The Gaussian density is uniformly positive on any fixed compact subbox of $\prod_l \operatorname{int} D_l$, for $R \in I$ and $|\xi| \le A$. This proves the lower bound. The cylinder boundary has zero measure under every limiting bridge. The actual observations differ from the interpolated values by $O(t^{-1/2})$ uniformly. Enlarging and contracting the fixed cylinder boxes therefore shows that the path convergence applies to the exact physical indicator and to that indicator times a bounded continuous path test. Uniformity follows by the same compact-parameter subsequence argument; every subsequential Gaussian has this boundary-null property. These observations prove the denominator and the restricted bridge

limit. Finally $(D_t \cap A_t^j) \triangle (D_t \cap A_t^{j'}) \subset A_t^j \triangle A_t^{j'}$; divide the inherited relative bound by the newly proved positive cylinder factor. □

Relation to the raw conditioning problem. The result supplies an actual conditional functional limit and explicit weighted denominators for fixed interior cylinder observations, without assuming their long pullbacks have controlled variation. The endpoint width remains $t^{21/50}$ in physical coordinates. Fixed lattice labels, microscopic time windows, the pointwise common signed raw correction, and the complete fixed-return Fourier complement still require their own estimates. No restriction on those original target events has been silently imposed: the theorem above concerns the displayed window class and leaves the original raw mixed-density endpoint intact.

222. A quantitative geometric extraction on every finite itinerary

The power–logarithm extraction gives a residual at each finite packet, but does not bound its derivative norm as the collision cutoff grows. We construct a different exact extraction. Its removed measure has a quantified total variation, and its residual has a uniform, explicit second-derivative budget. The construction takes place before coarea, on the physical collision cylinder. In particular it neither estimates individual prepared jets nor assumes separated critical values.

Write $L_m = S_{m,R}\tau_R$, and use (α, p) throughout for derivatives and area. All constants in this section are uniform in $R \in I$. Put

$$R_- = 9/20, \qquad R_+ = 47/100, \qquad \ell_0 = 3/50, \qquad H_0 = 1 + R_+/\ell_0 = 53/6.$$

The integer m is a number of physical collisions, not returns.

222.1. **A direction transverse to every regular roof level.** Lemma 224.1 gives a self-contained derivation of the coordinate differential and action used below, with both signs and the determinant in invariant coordinates.

Lemma 222.1 (A uniform nonstationary direction)**.** *On a regular m-collision branch, $m \geq 1$, write*

$$DT_R^m = \begin{pmatrix} A_m & B_m \\ C_m & D_m \end{pmatrix}.$$

Then $B_m \neq 0$ and $0 < A_m/B_m \leq H_0$. The total roof satisfies

$$|\nabla L_m| \geq \frac{R_-}{\sqrt{1+H_0^2}}|p|. \tag{222.1}$$

On $p \neq 0$ the vector field

$$X_m = -\frac{1}{Rp}\left(\partial_\alpha - \frac{A_m}{B_m}\partial_p\right) \quad \textit{satisfies} \quad X_m L_m = 1. \tag{222.2}$$

No incidence bound at the interior collisions is required for these three assertions.

Proof. Let $c = \sqrt{1-p^2}$, $c' = \sqrt{1-(p')^2}$ and $v = \tau_R/R$ for one regular flight. In the coordinates used here its differential is

$$DT_R = -\begin{pmatrix} (v+c)/c' & v/(cc') \\ v+c+c' & (v+c')/c \end{pmatrix} =: -P. \tag{222.3}$$

One obtains this formula by differentiating the straight flight and its reflection: in arc length and outgoing angle the four entries, before multiplication by $-1/c'$, are $\tau_R/R+c$, τ_R, $\tau_R/R^2+(c+c')/R$, and $\tau_R/R + c'$. Change variables by $r = R\alpha$, $p = \sin\varphi$. This is also the fixed-table specialization of the differential in [18, Section 3.1]. Its determinant in (α, p) is one.

All entries of P are positive. The ratios of the first to the second entry in either row are

$$c+\frac{c^2}{v},\qquad c+\frac{c^2}{v+c'},$$

respectively, and are at most H_0. The first row of the positive product $P_{m-1}\cdots P_0$ is a positive linear combination of the two rows of P_0. Its entry ratio is consequently a weighted average of these two ratios, with positive weights. This proves the assertion about A_m/B_m, including $m=1$.

The exact action identity of Lemma 149.2 gives

$$\partial_\alpha L_m = R(p_m A_m - p),\qquad \partial_p L_m = R p_m B_m.$$

Subtracting A_m/B_m times the second identity from the first gives $-Rp$. Cauchy–Schwarz proves (222.1); the same calculation proves (222.2). □

222.2. **A one-collision regularity cutoff.**

Lemma 222.2 (Polynomial sublevel sets in a rectangle)**.** *Fix $D\ge 1$ and $c_0>0$. A real polynomial P in two variables, of degree at most D in each variable and with maximum coefficient modulus at least c_0, satisfies*

$$|\{(x,y)\in[-1,1]^2 : |P(x,y)|\le h\}| \le C_{D,c_0} h^{1/(2D)},\qquad 0<h<1. \tag{222.4}$$

An upper bound on the other coefficients is not needed.

Proof. For a one-variable polynomial of degree at most D, a sublevel set of length $s>0$ contains $D+1$ points whose pairwise distances are at least $c_D s$. For example choose successive quantiles of its restricted Lebesgue measure, discarding a small part at the endpoints. Lagrange interpolation on these points bounds every coefficient by $C_D h s^{-D}$ when the polynomial is at most h there. It follows that a coefficient of modulus at least $b>0$ gives the one-variable bound $C_D(h/b)^{1/D}$; when this exceeds two the bound is automatic.

Write $P(x,y)=\sum_j a_j(y)x^j$ and choose a_j with a coefficient of modulus at least c_0. The set where $|a_j(y)|\le h^{1/2}$ has length at most $C_{D,c_0}h^{1/(2D)}$. On its complement, apply the one-variable bound to $P(\cdot,y)$ with $b=h^{1/2}$. Integrating over y proves (222.4). Constant polynomials cause no difficulty in either step. □

Fix an even $\theta\in C^\infty(\mathbb{R};[0,1])$, zero on $[-1,1]$ and one outside $(-2,2)$. A finite, parameter-independent set $\mathcal{D}$ of nonzero lattice centers contains every possible next collision center, by the uniform finite horizon. For $d\in\mathcal{D}$ put

$$\Delta_{d,R}(\alpha,p) = R^2 - \big((d-Rn_\alpha)\times(\sqrt{1-p^2}\,n_\alpha + p t_\alpha)\big)^2. \tag{222.5}$$

The cross denotes the scalar planar cross product. We use all candidate discriminants, including candidates that are not selected by the actual flight. This prevents a discontinuity from being missed on the side where a tangent disk ceases to be the first collision. Lemma 224.2 specifies the finite candidate set, four bounded charts and uniform coefficient minimum, and proves that every label transition is covered.

Lemma 222.3 (Uniform regularity cutoffs)**.** *There are $\varepsilon_0>0$ and smooth functions $\chi_{R,\varepsilon}: M\to[0,1]$, $0<\varepsilon\le\varepsilon_0$, vanishing on neighborhoods of the one-collision singularities, grazing endpoints and every boundary of Y_R^*, such that, with $\kappa_0=1/16$,*

$$\nu\{\chi_{R,\varepsilon}\ne 1\}\le C\varepsilon^{\kappa_0}. \tag{222.6}$$

On the support of this cutoff and its first three derivatives, every selected one-collision branch and roof, as well as the cutoff itself, have $C^3(\alpha,p)$ bounds at most $(C/\varepsilon)^C$. The actual section

membership and lattice label are locally constant where needed. Values at the excluded singularities are fixed to zero.

Proof. Let $\mathcal{A}_R$ contain the angular endpoints and $\mathcal{B}_R$ the p-endpoints of all the finitely many rectangles defining Y_R^*. Set $c(p) = \sqrt{1-p^2}$ and take the product

$$\theta(c(p)/\varepsilon) \prod_{d\in\mathcal{D}} \theta(\Delta_{d,R}/\varepsilon) \prod_{a\in\mathcal{A}_R} \theta(\sin(\alpha-a)/\varepsilon) \prod_{b\in\mathcal{B}_R} \theta((p-b)/\varepsilon). \tag{222.7}$$

The extra angular zeros at antipodal points only enlarge the removed set by finitely many strips. All factors are globally defined on the angular circle. The first factor makes the product identically zero near $p = \pm 1$, so its zero extension is smooth at these endpoints.

Here is an explicit uniform sublevel estimate for each discriminant. On a bounded rational chart for α, write t for its half-angle coordinate, and put $s = \tan(\varphi/2)$, so that $|s| \le 1$, $c = (1-s^2)/(1+s^2)$ and $p = 2s/(1+s^2)$. The cross product in (222.5) has denominator $(1+t^2)(1+s^2)$ and numerator of total degree at most four. Thus $\Delta_{d,R}$ has a bounded positive denominator and a numerator of total degree at most eight. That numerator is never the zero polynomial: at any fixed initial boundary point, $|d - Rn_\alpha| > R$, and its cross product with a varying unit direction cannot have constant square R^2 on an interval of directions. Compactness of I and finiteness of $\mathcal{D}$ and the charts give a positive lower bound on the maximum coefficient modulus. Apply Lemma 222.2 with $D = 8$. The coordinate Jacobian of ν in these bounded (t, s) charts is bounded above. This proves the bound $C\varepsilon^{1/16}$ for the discriminant factors. The remaining factors remove finitely many strips of measure $O(\varepsilon)$, or $O(\varepsilon^2)$ at grazing. A union bound proves (222.6).

Every possible change of first-hit label occurs at a candidate tangency: a positive entry root cannot cross zero because distinct disks are separated, and two distinct disks cannot have the same entry point at the same positive time. Those tangencies are among the zero sets in (222.5). On a selected branch with $\chi_{R,\varepsilon} > 0$, its entry root is

$$(d - Rn_\alpha)\cdot v - \sqrt{\Delta_{d,R}}, \qquad \Delta_{d,R} > \varepsilon,$$

with $c(p) > \varepsilon$. Differentiating this formula and the Euclidean reflection formula through order three gives a fixed polynomial loss in ε^{-1}. The outgoing point lies on a circle of radius at least R_-, so local angular coordinates add only fixed losses. Changing from φ to p costs further fixed powers of $c(p)^{-1}$. Differentiating the factors of (222.7) has the same property. The estimate also holds on their derivative supports, because a smooth factor which vanishes on an interval is flat at its endpoints. Section membership can change only on the explicitly removed rectangle boundaries. □

For $L \ge 1$ define, on regular orbits and then by zero extension,

$$\Gamma_{L,R,\varepsilon}(x) = \theta(p(x)/\varepsilon) \prod_{j=0}^{L} \chi_{R,\varepsilon}(T_R^j x). \tag{222.8}$$

The first factor removes a thin normal strip at the initial state. This one strip, rather than neighborhoods chosen separately at every critical value, suffices for (222.1).

Lemma 222.4 (Whole-itinerary bounds)**.** *The function in* (222.8) *is globally C^3 after zero extension across its excluded boundaries, and*

$$\int (1 - \Gamma_{L,R,\varepsilon})\, \mathrm{d}\nu_R^* \le C(L+1)\varepsilon^{1/16}, \tag{222.9}$$

$$\|\Gamma_{L,R,\varepsilon}\|_{C^3} \le \exp\{C(L+1)\log(C/\varepsilon)\}. \tag{222.10}$$

On its derivative supports, for $1 \le m \le L$, the field X_m has C^2 norm bounded by the right side of (222.10)*, after changing C.*

Proof. Use $1 - \prod b_j \le \sum(1 - b_j)$ for $0 \le b_j \le 1$, invariance of ν, and $\nu_R^* \le c_*^{-1}\nu$. The initial normal strip has measure $O(\varepsilon)$, and each of the other $L+1$ terms is bounded by (222.6). This proves (222.9) without counting itinerary cells.

On a nonzero derivative support all earlier branches are regular and all the one-step bounds of Lemma 222.3 apply. The chain rule through fixed order three, iterated j times, gives at most $(C/\varepsilon)^{C(j+1)}$. The product rule for $L+2$ factors contributes at most a fixed power of $L+2$, absorbed by $\exp(C(L+1))$. At any boundary of a finite itinerary, an earlier cutoff factor vanishes on a neighborhood on both sides. At a section boundary the corresponding factor at that time does the same. Consequently these piecewise derivatives extend by zero and give the global estimate.

Every entry of the positive matrix in (222.3) is at least $v_0 = \ell_0/R_+ = 6/47$. In particular $|B_m| \ge v_0^m$. The quotient rule, the C^3 bounds for T_R^m, and this lower bound give a C^2 bound for A_m/B_m of the same exponential form. On the relevant supports $|p| \ge \varepsilon$. Formula (222.2) therefore gives the assertion about X_m. □

222.3. **An exact residual for bounded-variation return marks.** Fix $q \ge 0$, indices $0 \le k_1, \dots, k_q \le n$, and complex section functions $a_1, \dots, a_q$ with bounded supremum and BV zero extensions a_i^0 to M. The number q is fixed; repeated marks are allowed. Put

$$\begin{aligned} w(x) &= \prod_{i=1}^{q} a_i((F_R^*)^{k_i}x), \\ M &= \prod_{i=1}^{q} \|a_i\|_\infty, \qquad V = \sum_{i=1}^{q} \|a_i^0\|_{\mathrm{BV}} \prod_{j\ne i} \|a_j\|_\infty. \end{aligned} \tag{222.11}$$

For $q = 0$ these conventions give $w = 1$, $M = 1$ and $V = 0$. All measures below use the normalized section law ν_R^* and retain the exact weight w. For the earlier unnormalized single-mark convention, choose $w = c_* a \circ (F_R^*)^k$.

Theorem 222.5 (All-itinerary geometric extraction)**.** *For $1 \le n \le L$ and $0 < \varepsilon \le \varepsilon_0$, there is an exact decomposition of the actual finite-count measure*

$$\mu_{n,R}^{w,L} = E_{n,R,\varepsilon}^{w,L} + Q_{n,R,\varepsilon}^{w,L} \tag{222.12}$$

with compactly supported mixed densities e_ε and q_ε, respectively. The residual satisfies

$$\begin{aligned} \|Q_{n,R,\varepsilon}^{w,L}\|_{\mathrm{TV}} &\le M, \\ \|E_{n,R,\varepsilon}^{w,L}\|_{\mathrm{TV}} &\le C_q\{\varepsilon V + M(L+1)\varepsilon^{1/16}\}, \end{aligned} \tag{222.13}$$

$$\mathcal{A}_{2,\varepsilon}(n,L,R,w) := \sum_{\ell\in\mathbb{Z}^3} \|\partial_t^2 q_\varepsilon(\ell,\cdot)\|_1 \le C_q M \exp\{C_q(L+1)\log(C_q/\varepsilon)\}. \tag{222.14}$$

Each residual component belongs to $W^{2,1}(\mathbb{R})$. The constants do not depend on n, the marks or the radius. When $w = 1$, both terms of (222.12) *are positive measures. No subanalytic assumption on the marked functions is imposed.*

Proof. Set $a_{i,\varepsilon} = \mathsf{S}_{\varepsilon/8} a_i^0$, using the mean-preserving smoothing of Lemma 138.2, and let w_ε be the corresponding product at the same actual marks. Differentiating the same smoothing kernel three times gives $\|a_{i,\varepsilon}\|_{C^3} \le C\varepsilon^{-3}\|a_i\|_\infty$; composition with $p = \sin\varphi$ is not needed for this flat-coordinate bound. Define

$$Q_{n,R,\varepsilon}^{w,L} = (J_{n,R})_*(\mathbf{1}_{\{N_{n,R}\le L\}}\Gamma_{L,R,\varepsilon} w_\varepsilon \nu_R^*), \qquad E_{n,R,\varepsilon}^{w,L} = \mu_{n,R}^{w,L} - Q_{n,R,\varepsilon}^{w,L}. \tag{222.15}$$

An empty product is not smoothed. Supremum contraction gives $|w_\varepsilon| \le M$. Telescoping the products and invariance of F_R^* give

$$\int |w - w_\varepsilon| \, \mathrm{d}\nu_R^* \le c_*^{-1} \sum_i \|a_i^0 - a_{i,\varepsilon}\|_{L^1(\nu)} \prod_{j \ne i} \|a_j\|_\infty \le C_q \varepsilon V.$$

Combining this with (222.9) proves the two variation bounds. When $w = 1$ the residual source weight is exactly $\Gamma_{L,R,\varepsilon} \mathbf{1}_{\{N_n \le L\}}$, which lies between zero and the source weight of $\mu_{n,R}^{1,L}$. This proves positivity.

It remains to establish the derivative estimate including the sum over all components. Partition the initial section by its finite regular collision itinerary through L, all its section decisions, and the value $m = N_{n,R} \le L$. There are finitely many pieces, but no estimate of their number will be used. On one such piece D, every mark is $T_R^{j_i} x$ for a fixed $j_i \le m$, the lattice label $\ell = (\sum_{j<m} \kappa_R \circ T_R^j, m)$ is constant, and the time coordinate is $L_m(x)$. The density on the initial cylinder is

$$b_D = (4\pi c_*)^{-1} \mathbf{1}_D \Gamma_{L,R,\varepsilon} w_\varepsilon.$$

Lemma 224.3 supplies the formal partition and the earliest-forbidden-time argument. Its zero extension across the boundaries of D is C^2: all changes of itinerary or section decision have already been removed on both sides by (222.8). Fixed-order chain and product rules, smoothing in C^3, and Lemma 222.4 give on D

$$|b_D| + |\operatorname{div}(b_D X_m)| + |\operatorname{div}(X_m \operatorname{div}(b_D X_m))| \le C_q M \exp\{C_q(L+1)\log(C_q/\varepsilon)\}. \tag{222.16}$$

All vector-field products are extended by zero outside their regular pieces; the orbit cutoffs vanish on neighborhoods of every excluded boundary. A further smooth cutoff in p can remove the factor $1/p$ away from the supports. No extension of the billiard map through a singularity is used. Angular integration is periodic, and all source functions vanish near $p = \pm 1$. Hence there are no coordinate boundary terms.

For a test function $\phi \in C_c^\infty(\mathbb{R})$, $X_m(\phi \circ L_m) = \phi' \circ L_m$. Integration by parts therefore identifies the first and second distributional derivatives of the pushforward density as the pushforwards of

$$\operatorname{div}(b_D X_m), \qquad \operatorname{div}(X_m \operatorname{div}(b_D X_m)), \tag{222.17}$$

respectively. The signs are positive in both distributional identities. On these supports L_m is a submersion by (222.1), so coarea gives L^1 densities for all three pushforwards. Their norms are at most the integrals of the absolute source functions.

The interiors of the pieces D are disjoint. Summing these source integrals, even after summing over all output labels, pays only the fixed area of the collision cylinder. It does not multiply (222.16) by an itinerary count. This proves (222.14) and the $W^{2,1}$ assertion. The actual finite-count measure has a density for bounded measurable weights by coarea on its regular branches; alternatively (222.1) excludes a zero gradient outside the null initial line $p = 0$. Subtraction gives the density of E. All time supports lie in $[0, L\tau_{\max}]$, and all lattice supports are finite. □

Remark 222.6 (What is extracted). The measure E includes the discarded parts of every regular critical word, every grazing and first-hit transition, every section-decision boundary, and all their iterated preimages. Any singular output image created by those parts remains in E. There is no unlisted branch in (222.12). This extraction is not the power–logarithm jet extraction of Section 200; its new derivative bound does not bound the old jet residual. In particular, small total variation of E does not imply a small essential supremum of e_ε.

223. Finite-band inversion at fixed lattice labels

The geometric extraction has two uses at the original microscopic resolution. First, it supplies a uniform far-roof estimate for a new exact residual at a linear collision cutoff. Second, it reduces exact fixed-label interval probabilities to a finite Fourier integral of the *full actual law*, with an arbitrarily small polynomial error. Neither conclusion evaluates that finite integral asymptotically.

Let $\mu^w_{n,R} = (J_{n,R})_*(w\nu^*_R)$ and retain the weights (222.11). The uncentered transform is

$$\widehat{\mu}^w_{n,R}(u,b) = \int e^{i(u\cdot\ell+bt)}\,\mathrm{d}\mu^w_{n,R}(\ell,t), \qquad (u,b)\in[-\pi,\pi]^3\times\mathbb{R}.$$

Thus u_3 is the collision-count frequency and b is roof frequency. For a time interval $I=[s,s+h]$, $h\ge 0$, put

$$H_I(b) = \int_I e^{-ibt}\,\mathrm{d}t, \qquad H_I(0)=h, \qquad |H_I(b)|\le \min\{h, 2/|b|\}. \tag{223.1}$$

All endpoint conventions give the same probability, because the roof components of the law are absolutely continuous.

Proposition 223.1 (Far-roof bounds for the geometric residual)**.** *For the exact residual of Theorem 222.5, write $\mathcal{A}=\mathcal{A}_{2,\varepsilon}(n,L,R,w)$. For $B>0$,*

$$\frac{1}{(2\pi)^4}\int_{[-\pi,\pi]^3}\int_{|b|>B}|\widehat{Q}_\varepsilon(u,b)|\,\mathrm{d}b\,\mathrm{d}u \le \frac{\mathcal{A}}{\pi B}, \tag{223.2}$$

$$\frac{1}{(2\pi)^4}\int_{[-\pi,\pi]^3}\int_{|b|>B}|H_I(b)\widehat{Q}_\varepsilon(u,b)|\,\mathrm{d}b\,\mathrm{d}u \le \frac{\mathcal{A}}{\pi B^2}. \tag{223.3}$$

These estimates include the sum over all lattice labels in $\mathcal{A}$.

Proof. Distributional integration by parts twice on each compactly supported $W^{2,1}$ component gives

$$|\widehat{Q}_\varepsilon(u,b)|\le \mathcal{A}/|b|^2 \qquad (b\neq 0).$$

For $|b|\le 1$ its transform is bounded by M, and for $|b|>1$ it is bounded by $\mathcal{A}/|b|^2$. Thus the transform is integrable on the full product of the frequency torus and roof line, and absolute inversion is legitimate. The torus volume is $(2\pi)^3$. Integrating the displayed bound over both roof tails gives (223.2). Multiply it by $2/|b|$ before integrating to obtain (223.3). The angular Fourier convention is responsible for the factors π^{-1}. □

Define the finite integral of the actual law by

$$\mathcal{I}^w_{n,R,B}(\ell,I) = \frac{1}{(2\pi)^4}\int_{[-\pi,\pi]^3}\int_{-B}^{B} e^{-iu\cdot\ell}H_I(b)\widehat{\mu}^w_{n,R}(u,b)\,\mathrm{d}b\,\mathrm{d}u. \tag{223.4}$$

It is well defined for every finite B. There is no cutoff or conditioning inside $\mu^w_{n,R}$ in this definition.

Theorem 223.2 (Microscopic finite-band reduction)**.** *There are $a,c>0$, as in Theorem 134.1, such that, for $n\le L$, $0<\varepsilon\le\varepsilon_0$, $B\ge 1$, every $\ell\in\mathbb{Z}^3$ and every finite interval I of length h,*

$$\begin{aligned}&\left|\mu^w_{n,R}(\{\ell\}\times I) - \mathcal{I}^w_{n,R,B}(\ell,I)\right| \\ &\quad\le C_q\left\{Me^{an-cL}+\varepsilon V + M(L+1)\varepsilon^{1/16}\right\}\left[1+\log(1+Bh)\right] \\ &\qquad + \frac{C_qM\exp\{C_q(L+1)\log(C_q/\varepsilon)\}}{\pi B^2}.\end{aligned} \tag{223.5}$$

In particular no lattice coordinate is averaged, and the time interval need not grow with n.

Proof. Let $Q = Q^{w,L}_{n,R,\varepsilon}$. The cumulative-return tail and (222.13) give

$$\|\mu^w_{n,R} - Q\|_{\mathrm{TV}} \le \delta := C_q\{Me^{an-cL} + \varepsilon V + M(L+1)\varepsilon^{1/16}\}. \tag{223.6}$$

The direct change of the interval probability costs at most δ. The corresponding change inside the finite integral is at most

$$\frac{\delta}{2\pi}\int_{-B}^{B} \min\{h, 2/|b|\}\,\mathrm{d}b \le C\delta[1+\log(1+Bh)].$$

This logarithmic cost is essential. Bounding H_I by its supremum on the whole band would pay a factor of B and lose the conclusion below. The remaining difference, for Q, is its absolutely invertible roof tail, bounded by (223.3). Add the three errors and use (222.14). □

Corollary 223.3 (A uniform long-time budget)**.** *Fix q, $P > 0$, $H < \infty$, and nonnegative exponents β, γ. Suppose*

$$M \le M_0 n^\beta, \qquad V \le V_0 n^\gamma.$$

There are a slope $\lambda > a/c$, a fixed $d > 0$, and explicit choices $L_n = \lceil \lambda n \rceil$, $\varepsilon_n = \varepsilon_0 n^{-d}$, and $B_n \ge 1$ with

$$\log B_n \le Cn\log(2+n) \tag{223.7}$$

such that, uniformly in R, the marks, all $\ell \in \mathbb{Z}^3$ and all intervals of length at most H,

$$n^2\left|\mu^w_{n,R}(\{\ell\}\times I) - \mathcal{I}^w_{n,R,B_n}(\ell, I)\right| \le C_P n^{-P}. \tag{223.8}$$

For the same choices the total variation in (223.6) *is $O(n^{-P-4})$, and the residual has a second-derivative bound $\mathcal{A}_{2,\varepsilon_n} \le A_n$ with $\log A_n \le Cn\log(2+n)$. Its pointwise far-roof inverse, multiplied by n^2, is at most $C_P n^{-P-2}$.*

Proof. Choose $\lambda > \max\{1, a/c\} + 1$ and

$$d > \max\{16(P+5+\beta),\ P+4+\gamma\}.$$

The two geometric errors in (223.6) are then $O(n^{-P-4})$. The truncated high-count error is exponentially small and hence satisfies the same bound, with a changed constant. Set

$$\begin{aligned} A_n &= 1 + C_q M_0 n^\beta \exp\{C_q(L_n+1)\log(C_q/\varepsilon_n)\},\\ B_n &= n^{P+4} A_n. \end{aligned} \tag{223.9}$$

These are numerical expressions in the constants already proved; no unknown value of a density derivative occurs in the choice of B_n. Both logarithms satisfy (223.7). For $h \le H$, the logarithmic factor in (223.5) is at most $Cn\log(2+n)$. Its contribution after multiplication by n^2 is $O(n^{-P-1}\log(2+n))$, bounded by $C_P n^{-P}$. The last term is smaller. Finally (223.2) and $\mathcal{A}_{2,\varepsilon_n} \le A_n$ give the claimed pointwise tail. □

Remark 223.4 (Microscopic resolution and the remaining finite integral)*.* The three lattice coordinates in (223.8) are singletons. Its time interval may have any fixed positive length, or may shrink; the stated error is absolute, not relative to an unproved denominator. Both the observable and its law are the original ones. Thus the reduction is not a mesoscopic replacement of the raw event. The price is a roof bandwidth whose logarithm can grow as $n\log n$. The preceding collision and prescribed-return central bands do not reach this bandwidth. No estimate of the finite integral (223.4) on its remaining complement is supplied by this corollary.

223.1. **The common pointwise correction after the new extraction.** The new extraction also makes one precise alternative to the old jet budget available in the pointwise inversion identity. Let χ be one of the retained smooth cutoffs and K_χ its mixed inverse kernel. For the same weight and finite count packet, let

$$\mathcal{R}_\chi^{w,L} = p_{n,R}^{w,L} - K_\chi * \mu_{n,R}^{w,L}.$$

This is the intrinsic signed correction in (215.4); the definition does not depend on the chosen extraction. Substituting (222.12) gives the almost-everywhere identity

$$\begin{aligned}\mathcal{R}_\chi^{w,L} &= e_\varepsilon - K_\chi * E_\varepsilon \\ &\quad + \frac{1}{(2\pi)^4}\int_{[-\pi,\pi]^3}\int_{-B}^{B} e^{-i(u\cdot\ell+bt)}(1-\chi(u,b))\widehat{Q}_\varepsilon(u,b)\,\mathrm{d}b\,\mathrm{d}u + \mathcal{T}_{\varepsilon,B}(\ell,t),\end{aligned} \tag{223.10}$$

provided B contains the roof support of χ. Here

$$\|K_\chi * E_\varepsilon\|_\infty \le \|K_\chi\|_\infty \|E_\varepsilon\|_{\mathrm{TV}}, \qquad \|\mathcal{T}_{\varepsilon,B}\|_\infty \le \frac{\mathcal{A}_{2,\varepsilon}}{\pi B}. \tag{223.11}$$

The identities are formed before taking any essential supremum. They do not subtract extended-real norm values.

For clarity, the terms in the final raw formula have the following status in this extraction.

Term	Estimate and scope
Known central convolution	The inherited central and fixed-count bands are unchanged, with their original weight classes.
$K_\chi * E_\varepsilon$	The total-variation estimate above and the known kernel supremum give a quantitative pointwise bound.
$\mathcal{T}_{\varepsilon,B}$	The explicit derivative budget gives a pointwise far-roof bound, including at linear L.
e_ε	Its L^1 norm is small; no small essential supremum is proved. It is positive for the original unweighted law.
Finite complementary integral	Balanced, compact nonzero, peripheral and growing-roof regimes outside the proved domains still require a compatible bound.

In particular, the geometrically removed density can be tall on a very small set. The elementary estimate

$$m\{|e_\varepsilon| > \eta\} \le \|E_\varepsilon\|_{\mathrm{TV}}/\eta, \qquad \eta > 0,$$

uses the mixed counting–Lebesgue measure m, and cannot be upgraded to an essential-supremum estimate. The two uncontrolled terms in (223.10) may still need to be estimated together with their sign. The coherent transport identities remain available for that purpose.

For a single marked BV weight the geometric extraction and the inherited Fourier estimates now apply to the same law without a further subanalytic-weight assumption for the extraction. For a fixed product of several return-state marks, the new extraction and (223.8) hold, but the inherited single-mark central theorem is not thereby a multiple-mark theorem. The stationary path cylinders of Theorem U and the original microscopic physical-time event are different weight classes. Their raw denominator and relative microscopic event replacement are not consequences of (223.8).

224. Source boundaries and the stationary geometric extraction

We give the coordinate and boundary arguments for the geometric extraction in one place. We then establish the corresponding positive extraction under the stationary suspension law. The latter is not the section law: its entrance marginal has an unbounded roof bias.

224.1. **The differential and the action in invariant coordinates.**

Lemma 224.1 (Collision differential and action)**.** *On a regular flight from Rn_α to $d + Rn_{\alpha'}$, set $c = \sqrt{1-p^2}$, $c' = \sqrt{1-(p')^2}$ and $v_0 = \tau_R/R$. In the coordinates (α, p) the derivative is*

$$DT_R = -\begin{pmatrix} (v_0+c)/c' & v_0/(cc') \\ v_0+c+c' & (v_0+c')/c \end{pmatrix}, \qquad \det DT_R = 1,$$

and

$$d\tau_R = Rp'\,d\alpha' - Rp\,d\alpha. \tag{224.1}$$

These formulas use the outgoing value of p' after reflection.

Proof. Write $\mathcal{J}$ for counterclockwise rotation by $\pi/2$, and $V = cn_\alpha + pt_\alpha$ for the unit flight velocity. If β is its direction angle on a local chart, then

$$d\beta = d\alpha + dp/c, \qquad dV = \mathcal{J}V\,d\beta.$$

At arrival, $V = -c'n_{\alpha'} + p't_{\alpha'}$: reflection preserves the tangential component and reverses the normal component. Thus $d\beta = d\alpha' - dp'/c'$. Differentiating the flight equation and taking its scalar product with $\mathcal{J}V$ gives

$$-Rc'\,d\alpha' = Rc\,d\alpha + \tau_R\,d\beta,$$

since $\mathcal{J}V \cdot t_\alpha = c$ and $\mathcal{J}V \cdot t_{\alpha'} = -c'$. Substitution gives the first row of the displayed matrix. The identity $dp' = c'(d\alpha' - d\beta)$ gives its second row. Its determinant is

$$\frac{(v_0+c)(v_0+c') - v_0(v_0+c+c')}{cc'} = 1.$$

Taking instead the scalar product of the differentiated flight equation with V gives (224.1). The lattice center is constant on this branch. Summing the action formula gives the two derivatives of L_m used in Lemma 222.1. This derivation also specifies the sign and coordinate conversion in the fixed-table specialization of [18, Section 3.1]. □

224.2. **A complete first-hit guard.** Let $\tau_{\max}$ be a uniform horizon from the earlier geometric estimates, and choose once and for all

$$\mathcal{D} = \{d \in \Lambda \setminus \{0\} : |d| \le \tau_{\max} + 2R_+ + 1\}. \tag{224.2}$$

Here Λ is the physical triangular lattice, not a rescaled collision coordinate lattice.

Lemma 224.2 (First-hit transitions and uniform chart coefficients)**.** *The set* (224.2) *is finite and contains every next collision center for all $R \in I$. Away from initial grazing and the zero sets of all $\Delta_{d,R}$, the first-hit label is locally constant and the selected positive entry root is smooth. In four bounded angular half-angle charts, the numerator of each discriminant has degree at most eight, its positive denominator lies in $[1, 16]$, and the maximum modulus of its coefficients is bounded below uniformly in R, the chart and $d \in \mathcal{D}$.*

Proof. A flight of length at most $\tau_{\max}$ from one boundary point to the next gives $|d| \le \tau_{\max} + 2R_+$, proving the first assertion. For a candidate let $a = d - Rn_\alpha$, $A = a \cdot V$ and $\Delta = R^2 - |a|^2 + A^2$. Since $|d| \ge 1$ and $R \le 47/100$,

$$|a| \ge 1 - R > R, \qquad A^2 - \Delta = |a|^2 - R^2 > 0.$$

On $\Delta > 0$ the entry root is $A - \sqrt{\Delta}$. It is positive exactly when $A > 0$. Its sign cannot change while $\Delta > 0$, and a positive root cannot pass through zero. Two distinct positive entry roots cannot be equal: the corresponding point on the ray would lie on two disjoint closed disks. Consequently the ordering of all admissible entry roots persists locally until a candidate discriminant vanishes. The margin in (224.2) prevents a center from entering through the edge of the candidate list. The outgoing half-ray cannot reenter the initial convex disk. This accounts

for all first-hit transitions, including the side of a tangency on which the candidate is no longer hit.

Take chart centers $\alpha_j = j\pi/2$, $0 \le j \le 3$, and use $\alpha = \alpha_j + 2\arctan t$ on $|t| < 1$; their open images cover the angular circle. Bounds may be taken on the closed chart intervals. Let $s = \tan(\varphi/2) \in [-1, 1]$, where $p = \sin\varphi$. Then

$$p = \frac{2s}{1+s^2}, \qquad c = \frac{1-s^2}{1+s^2}, \qquad a \times V = c(d \times n_\alpha) + p(d \times t_\alpha - R).$$

The last expression has denominator $(1+t^2)(1+s^2)$ and a numerator of total degree at most four. Thus the discriminant has denominator $(1+t^2)^2(1+s^2)^2 \in [1, 16]$ and a numerator of total degree at most eight. The coordinate Jacobian of ν is bounded above by $1/\pi$, since $|d\alpha/dt| \le 2$ and $|dp/ds| \le 2$.

For each fixed d, j, R, the numerator is not identically zero. Fix an initial point in the chart. Its vector a has length greater than R; the square of its cross product with directions in an open angular interval cannot be the constant R^2. The maximum modulus of the finitely many polynomial coefficients is a continuous, strictly positive function of R. Its minimum on I is positive. Taking the minimum over the finite candidate and chart sets proves the uniform coefficient bound. This justifies the uniform use of Lemma 222.2, including near tangency. □

Lemma 224.3 (Partition and zero extension)**.** *Fix $n \le L$ and $\varepsilon > 0$. Up to a ν-null set, the source $Y_R^* \cap \{N_{n,R} \le L\}$ is the union of finitely many disjoint open pieces D relative to the collision cylinder. On each piece the center itinerary, section decisions, mark collision indices, output label and $m = N_{n,R}$ are fixed. The source functions b_D in the proof of Theorem 222.5, and the vector-field products used there, extend by zero with the regularity needed for both integrations by parts. In particular there are no distributions supported on itinerary or section boundaries in those identities.*

Proof. At each collision record the first-hit center in the finite set (224.2) and the section membership decision. There are finitely many such records through L collisions. Remove all encountered tangencies, grazing points and section boundaries. Their union is null: the one-step discriminant zero sets are null by Lemma 224.2 and the polynomial sublevel bound, section sides are null, and the collision map preserves ν. The regular decision sets are open. They need not be connected for the argument; all their components with one record are treated together. On each such set every return index and output label is fixed. No artificial angular seam is introduced in this partition.

Here is the boundary extension explicitly. At a boundary point choose the first singular collision or the first section decision at which the record can change. All preceding regular branches persist on a neighborhood. At a first-hit tangency the input discriminant of the preceding flight is zero. At an initial grazing point the initial guard is zero; grazing produced by a flight is already its tangency. At a section decision the corresponding rectangle-side guard is zero. In each case, at the earliest such input time j, the previous maps are continuous and $\chi_{R,\varepsilon} \circ T_R^j$ is identically zero on a two-sided neighborhood of the boundary point. The factors at later, possibly undefined, collision states need not be evaluated: the product is defined to be zero throughout that neighborhood.

At a boundary of a cutoff transition inside a regular decision set, ordinary smoothness and the flat endpoint derivatives of θ give the same matching of derivatives. The initial factor $\theta(p/\varepsilon)$ removes the pole of X_m. Hence b_D is C^2 after zero extension, $b_D X_m$ is C^2, and $X_m \operatorname{div}(b_D X_m)$ is C^1. Their derivatives have the bounds of (222.16). For fixed L and ε their supports are separated from the excluded boundaries; no uniform lower bound for that separation is used. The cylinder is periodic in α, and the supports avoid $p = \pm 1$. The distributional integration-by-parts formulas

therefore have no boundary terms. Since the pieces have disjoint interiors, their absolute source integrals sum over a subset of the initial cylinder, not over multiple copies of it. □

224.3. **The preceding section under the equilibrium law.** Write $r = r_R^*$ and $\tau^* = \tau_R^*$. On the return suspension put

$$\Omega_R = \{(y,s) : y \in Y_R^*,\ 0 \le s < \tau^*(y)\}, \qquad d\mathbb{P}_R^{\mathrm{eq}} = \frac{c_*}{\bar\tau_R}\, d\nu_R^*(y)\, ds. \tag{224.3}$$

Its entrance marginal is $\rho_R \nu_R^*$, where $\rho_R = c_* \tau^*/\bar\tau_R$. This is the stationary normalization of Lemma 219.1. The same y is used for $J_{n,R}(y)$ and for every physical trajectory observable on Ω_R.

Proposition 224.4 (Positive extraction with the actual entrance bias)**.** *Let $\mu_{n,R}^{\mathrm{eq}} = (J_{n,R})_*(\rho_R \nu_R^*)$. There is a positive measure $Q \le \mu_{n,R}^{\mathrm{eq}}$ with a compactly supported mixed density q whose roof components belong to $W^{2,1}(\mathbb{R})$, and constants independent of R, n, L, ε such that*

$$\delta_{\mathrm{eq}} := \|\mu_{n,R}^{\mathrm{eq}} - Q\|_{\mathrm{TV}} \le C\left(e^{(an-cL)/2} + (L+1)^{1/2}\varepsilon^{1/32}\right), \tag{224.4}$$

$$\sum_{\ell \in \mathbb{Z}^3} \|\partial_t^2 q(\ell,\cdot)\|_1 \le C \exp\{C(L+1)\log(C/\varepsilon)\} =: A_2. \tag{224.5}$$

In particular $\|Q\|_{\mathrm{TV}} \le 1$. For the section probability the same conclusions hold with the stronger removed-mass estimate $C(e^{an-cL} + (L+1)\varepsilon^{1/16})$.

Proof. Define

$$Q = (J_{n,R})_*(\rho_R \mathbf{1}_{\{N_{n,R} \le L\}} \Gamma_{L,R,\varepsilon} \nu_R^*).$$

All factors are nonnegative and the last two are at most one. The one-return exponential moment and the positive lower bound for $\bar\tau_R$ give $\sup_R \|\rho_R\|_{L^2(\nu_R^*)} < \infty$. Cauchy–Schwarz, the cumulative return tail and (222.9) consequently give

$$\int \rho_R \mathbf{1}_{\{N_n > L\}}\, d\nu_R^* \le C e^{(an-cL)/2}, \qquad \int \rho_R (1 - \Gamma_{L,R,\varepsilon})\, d\nu_R^* \le C\big((L+1)\varepsilon^{1/16}\big)^{1/2}.$$

These imply (224.4).

On a source piece of Lemma 224.3, its first return collision time j_1 is fixed and at most L. There $\tau^* = \sum_{j<j_1} \tau_R \circ T_R^j$. The cutoff estimates and chain rule bound this function and its first three derivatives, on the required supports, by $C\exp\{C(L+1)\log(C/\varepsilon)\}$. Multiplying b_D by ρ_R therefore preserves the divergence budget after increasing C. The earliest-decision argument gives the same zero extensions. The two source integrations by parts and disjoint-source summation prove (224.5). The argument does not assume that the untruncated entrance bias is BV. The section-law assertion is the unweighted case of Theorem 222.5 and its cumulative-tail extension. □

225. Positive finite-band inversion and microscopic conditioning

A sharp truncation of a Fourier integral need not define a positive measure. For conditioning it is useful to retain positivity before normalization. We construct a finite-band likelihood on the original probability space and control it uniformly against every bounded trajectory statistic. Only the unweighted law is regularized; the trajectory statistic is never differentiated or smoothed.

225.1. **A positive kernel and a source-to-density estimate.** Use the real-line transform $\widehat{f}(b) = \int e^{ibt} f(t)\,dt$. Define, with the continuous value at zero,

$$k(t) = \frac{3}{8\pi}\left(\frac{\sin(t/4)}{t/4}\right)^4, \qquad k_B(t) = Bk(Bt), \qquad B > 0. \tag{225.1}$$

Lemma 225.1 (Kernel normalization and moments)**.** *The kernel k is even and nonnegative, $\int k = 1$, and*

$$\widehat{k}(b) = \eta(b) = \begin{cases} 1 - 6b^2 + 6|b|^3, & |b| \le 1/2, \\ 2(1-|b|)^3, & 1/2 \le |b| \le 1, \\ 0, & |b| \ge 1. \end{cases} \tag{225.2}$$

Moreover $\int tk(t)\,dt = 0$, $\int t^2 k(t)\,dt = 12$, and $\kappa_1 := \int |t|k(t)\,dt \le \sqrt{12}$.

Proof. The identity $\sin(t/4)/(t/4) = 2\int_{-1/4}^{1/4} e^{iut}\,du$ expresses the fourth power by the fourfold convolution of the interval indicator. Evaluating that elementary convolution gives (225.2), including the normalization $\eta(0) = 1$. The kernel decays as $O((1+|t|)^{-4})$ and hence has a finite second moment. Differentiating its transform twice at zero therefore gives $\int t^2 k(t)\,dt = -\eta''(0) = 12$. Evenness gives the first signed moment; Cauchy–Schwarz gives the absolute first-moment bound. □

Lemma 225.2 (An all-label interpolation inequality)**.** *Let $q_\ell \in W^{2,1}(\mathbb{R})$ have compact support, and suppose $A_j = \sum_\ell \|q_\ell^{(j)}\|_1 < \infty$ for $j = 0, 2$. Then*

$$A_1 \le 2\sqrt{A_0 A_2}, \qquad \sum_\ell \|q_\ell\|_\infty \le A_1/2, \qquad \sum_\ell \|q_\ell - k_B * q_\ell\|_1 \le 6A_2/B^2. \tag{225.3}$$

The same statements hold for real or complex functions.

Proof. Taylor's formula in $W^{2,1}$, summed over labels, gives for every $h > 0$

$$A_1 \le 2A_0/h + hA_2/2.$$

Taking $h = 2\sqrt{A_0/A_2}$ proves the first inequality when both quantities are positive. If either vanishes the claim follows by letting h tend to zero or infinity. Each component tends to zero at both ends of the line. Integrating its derivative from either end bounds its supremum by half its total variation, proving the second inequality. Finally

$$\|q_\ell(\cdot - y) - q_\ell + yq_\ell'\|_1 \le y^2 \|q_\ell''\|_1/2.$$

Integrate against $k_B(y)$, use its zero signed first moment and second moment $12/B^2$, and sum. All sums and integrals are justified by the nonnegative bounds just displayed. □

Here and below $\|\cdot\|_{\mathrm{TV}}$ is the total variation norm of a signed measure, equivalently the supremum of its integrals against measurable complex functions of modulus at most one. Thus it is the L^1 norm for densities, without a factor of one half.

Theorem 225.3 (Positive spectral approximation of a mixed law)**.** *Let $Z = (\mathsf{L}, \mathsf{T})$ on a probability space $(\Omega, \mathbb{P})$ take values in $\mathbb{Z}^3 \times \mathbb{R}$. Suppose its law μ admits a positive submeasure $Q \le \mu$ with compactly supported roof densities $q_\ell \in W^{2,1}(\mathbb{R})$. Write*

$$\delta = \|\mu - Q\|_{\mathrm{TV}}, \qquad A_2 = \sum_\ell \|q_\ell''\|_1 < \infty.$$

Then

$$p_B(\ell, t) = \frac{1}{(2\pi)^4} \int_{[-\pi,\pi]^3} \int_{-B}^{B} e^{-i(u\cdot\ell + bt)} \eta(b/B)\widehat{\mu}(u, b)\,db\,du \tag{225.4}$$

is a nonnegative density of total mass one. It smooths only the roof coordinate and preserves all three lattice coordinates exactly. The corresponding probability μ_B satisfies

$$\|\mu - \mu_B\|_{\mathrm{TV}} \le 2\delta + 6A_2/B^2. \tag{225.5}$$

If μ has density p, this is the global mixed L^1 estimate for $p - p_B$, including the sum over all labels.

Proof. Fourier inversion for k_B and torus orthogonality give

$$p_B(\ell, t) = \mathbb{E}\big[\mathbf{1}_{\{\mathsf{L}=\ell\}} k_B(t - \mathsf{T})\big].$$

The frequency integral is absolutely convergent, since the band is finite and $|\widehat{\mu}| \le 1$. This identity proves positivity, normalization and exact preservation of the lattice marginal. Write $\mu = Q + E$ with $E \ge 0$ and $\|E\|_{\mathrm{TV}} = \delta$. Roof convolution is a contraction on finite measures, so $\|E - k_B * E\|_{\mathrm{TV}} \le 2\delta$. The last estimate of Lemma 225.2 bounds the remaining difference $Q - k_B * Q$, proving (225.5). □

225.2. **A microscopic event as a finite-band likelihood.** Consider an event $A = \{Z \in \mathcal{A}\}$ with fibres $\mathcal{A}_\ell \subset \mathbb{R}$, each a union of at most J bounded intervals. Empty fibres are permitted. Endpoint conventions do not matter for the billiard laws considered below. Define on the original probability space

$$\lambda_{A,B} = (k_B * \mathbf{1}_{\mathcal{A}_{\mathsf{L}}})(\mathsf{T}). \tag{225.6}$$

In particular $0 \le \lambda_{A,B} \le 1$. This is a weight on the same initial trajectory, not a newly sampled section point or a change to its physical evolution.

Theorem 225.4 (Uniform weighted inversion on the original space)**.** *Under the hypotheses of Theorem 225.3, let*

$$E_B(J) = \delta + 2\kappa_1 J\sqrt{A_2}/B. \tag{225.7}$$

Then

$$\|\mathbf{1}_A - \lambda_{A,B}\|_{L^1(\mathbb{P})} \le E_B(J). \tag{225.8}$$

For every bounded measurable trajectory statistic W, including a statistic depending on n or B,

$$|\mathbb{E}[W\mathbf{1}_A] - \mathbb{E}[W\lambda_{A,B}]| \le \|W\|_\infty E_B(J). \tag{225.9}$$

No BV, continuity or finite-record hypothesis on W is required. For $\mathcal{A} = \{\ell\} \times I$, the second expectation is exactly

$$\frac{1}{(2\pi)^4} \int_{[-\pi,\pi]^3} \int_{-B}^{B} e^{-iu\cdot\ell} H_I(b)\eta(b/B)\mathbb{E}\big[We^{i(u\cdot\mathsf{L}+b\mathsf{T})}\big]\, db\, du. \tag{225.10}$$

The bound is independent of the length of I. Finite unions of fibres have the corresponding finite sum of these formulas.

Proof. For a set I that is a union of at most J intervals,

$$\int_{\mathbb{R}} |\mathbf{1}_I(t) - \mathbf{1}_I(t - y)|\, dt \le 2J|y|.$$

Positivity and unit mass of k_B imply pointwise

$$|\mathbf{1}_I(t) - (k_B * \mathbf{1}_I)(t)| \le \int k_B(y)|\mathbf{1}_I(t) - \mathbf{1}_I(t - y)|\, dy.$$

On the positive discarded measure $E = \mu - Q$ the integrand is at most one, and so costs at most δ. On Q, bound each component by $\|q_\ell\|_\infty$ and apply the preceding translation estimate. Summation and Lemma 225.2 give

$$\int |\mathbf{1}_{\mathcal{A}} - k_B * \mathbf{1}_{\mathcal{A}}|\, dQ \le \frac{2\kappa_1 J}{B} \sum_{\ell} \|q_\ell\|_\infty \le \frac{\kappa_1 J A_1}{B} \le \frac{2\kappa_1 J \sqrt{A_2}}{B},$$

since $A_0 = \|Q\|_{\mathrm{TV}} \le 1$. This proves (225.8). It also handles countably many nonempty fibres by summation of nonnegative bounds.

Multiplication by W proves (225.9); taking the supremum over $|W| \le 1$ gives precisely the total variation norm of $\mathbf{1}_A \mathbb{P} - \lambda_{A,B} \mathbb{P}$. Fourier inversion for k_B, followed by integration over the bounded interval I and torus orthogonality, proves (225.10). Fubini applies because the frequency band and interval are finite and W is bounded. No derivative of the weighted transform is used. In particular the proof does not claim a global L^1 convolution approximation uniform in arbitrary oscillating weights W; it proves the stated interval-event operator estimate instead. □

Corollary 225.5 (Relative conditioning and selected denominators)**.** *Let $p_A = \mathbb{P}(A)$ and $p_{A,B} = \mathbb{E}\lambda_{A,B}$. If $p_A > E_B(J)$, the probability $\Pi_{A,B} = p_{A,B}^{-1} \lambda_{A,B} \mathbb{P}$ is defined and*

$$\|\mathbb{P}(\cdot \mid A) - \Pi_{A,B}\|_{\mathrm{TV}} \le 2E_B(J)/p_A. \tag{225.11}$$

Alternatively, $p_{A,B} > E_B(J)$ proves the rigorous denominator bound $p_A \ge p_{A,B} - E_B(J) > 0$ and the upper bound $2E_B(J)/(p_{A,B} - E_B(J))$ in (225.11)*.*

For every measurable selector $D \subset \Omega$, the same statements hold with $A \cap D$ and the likelihood $\mathbf{1}_D \lambda_{A,B}$, using the identical error budget $E_B(J)$. More generally they hold for a nonnegative path weight bounded by one, with its corresponding weighted denominator. No lower bound for that denominator is assumed without being stated.

Proof. Equation (225.8) bounds both the variation of the unnormalized measures and the difference of their masses. For two nonnegative densities f, g of masses $a, b > 0$,

$$\|f/a - g/b\|_1 \le \|f - g\|_1/a + |a - b|/a.$$

Apply this with $f = \mathbf{1}_A$ and $g = \lambda_{A,B}$. Positivity of the needed masses follows from their difference being at most $E_B(J)$. Multiplication by $\mathbf{1}_D$, or by a nonnegative weight at most one, can only decrease the unnormalized L^1 error. This proves every assertion with no regularity assumption on the selector. □

225.3. **Uniform long-time budgets for the actual billiard laws.** Let $\mathbb{P}_R^{\mathrm{sec}} = \nu_R^*$ and let $\mathbb{P}_R^{\mathrm{eq}}$ be (224.3). In the first case set $Z = J_{n,R}(y)$; in the second use the same record from the actual preceding section origin y, retaining the elapsed age s for all path selectors. Denote the two output densities by $p_{n,R}^{\mathrm{j}}$, $\mathrm{j} \in \{\mathrm{sec}, \mathrm{eq}\}$.

Theorem 225.6 (Microscopic positive inversion with arbitrary selectors)**.** *Fix $P > 0$, $J_0 < \infty$ and $\kappa \ge 0$. There are choices $L_n = \lceil \lambda n \rceil$, $\varepsilon_n = \varepsilon_0 n^{-d}$ and $B_n \ge 1$, uniform in R, for which*

$$\log B_n \le Cn \log(2 + n), \tag{225.12}$$

and the following assertions hold for both initial laws. Let p_{n,R,B_n}^{j} be the positive finite-band density (225.4)*. Then*

$$n^2 \|p_{n,R}^{\mathrm{j}} - p_{n,R,B_n}^{\mathrm{j}}\|_{L^1(\mathbb{Z}^3 \times \mathbb{R})} \le C_P n^{-P}, \tag{225.13}$$

$$n^2 \sup_{\substack{\mathcal{A}:\ J \le J_0 n^\kappa \\ |W| \le 1}} \left| \mathbb{E}_R^{\mathrm{j}}[W \mathbf{1}_{\{J_{n,R} \in \mathcal{A}\}}] - \mathbb{E}_R^{\mathrm{j}}[W \lambda_{A,B_n}] \right| \le C_P n^{-P}. \tag{225.14}$$

The supremum is over the interval-fibre events defined above and all bounded measurable statistics on the respective original probability space. In particular singleton lattice labels and arbitrarily shrinking roof intervals are allowed. Their Fourier representation is (225.10)*, with* $\widehat{\mu}^{W}$ *computed under the same initial law.*

Proof. Choose $\lambda > \max\{1, a/c\} + 1$ and $d > 32(P+5)$. Proposition 224.4 gives, for both laws, positive extractions with

$$\delta_n \le C_P n^{-P-4}, \qquad A_{2,n} \le A_n := 1 + C\exp\{C(L_n+1)\log(C/\varepsilon_n)\}.$$

Indeed the slower stationary geometric mass is bounded by $Cn^{1/2-d/32}$, and its high-count term is exponentially small. Increase the common C to cover both derivative budgets and all initial values of $n \ge 2$. Set

$$B_n = n^{P+\kappa+4}\sqrt{A_n}. \tag{225.15}$$

This proves (225.12). Also

$$2\delta_n + 6A_n/B_n^2 \le C_P n^{-P-4}, \qquad \delta_n + 2\kappa_1 J_0 n^{\kappa}\sqrt{A_n}/B_n \le C_P n^{-P-4}.$$

Apply Theorems 225.3 and 225.4, then multiply by n^2. The stronger intermediate powers absorb constants and rounding. All bandwidths here are finite; their logarithms, not the bandwidths themselves, have order $n\log n$. □

Location of the remaining local theorem. The positive likelihood gives a finite-band representation of microscopic conditioning on the original probability space, including arbitrary bounded path selectors and the true stationary entrance bias. It does not evaluate that finite integral by a Gaussian. Its relative conclusions use the denominator bounds stated in Corollary 225.5. Likewise (225.13) is a global L^1 approximation, not an essential-supremum estimate on a prescribed central window. The full microscopic conclusion remains governed by (223.10): the finite complement and the pointwise common correction must still be estimated, followed by the original physical-event comparison at that denominator scale.

226. Exact stationary endpoint inversion by flight-age integration

The return record from the preceding section and the observation at a fixed physical time are different microscopic events. In this section we invert the latter directly. Integrating the actual initial flight age produces an overlap of two intervals. This overlap has two distributional derivatives with a uniform measure bound, independently of the number of intervening collisions. The initial and terminal cell corrections are retained exactly, not bounded and discarded.

Throughout this section $m \ge 1$ is a prescribed *physical collision count*, not a return count. Put

$$S_{m,R} = \sum_{j=0}^{m-1} \tau_R \circ T_R^j, \qquad K^{\mathrm{c}}_{m,R} = \sum_{j=0}^{m-1} \kappa_R \circ T_R^j, \qquad 0 < \tau_- \le \tau_R \le \tau_+.$$

Here κ_R is the physical next-disk label and τ_-, τ_+ are uniform finite-horizon bounds. The equilibrium probability in current-collision coordinates is, by Lemma 219.1,

$$d\mathbb{P}_R(x,a) = \bar{\tau}_R^{-1} d\nu(x)\, da, \qquad 0 \le a < \tau_R(x). \tag{226.1}$$

This is the same stationary trajectory as in the return-suspension coordinates. No new initial point is sampled.

226.1. **Cell labels during a physical flight.** Fix the fundamental-cell convention used for $W_R(t)$. We take a bounded convex polygonal fundamental cell $\mathcal{P}$; in particular one may use a fixed half-open lattice parallelogram. Let $\mathcal{L} : \mathbb{R}^2 \to \mathbb{R}^2$ send the standard integer basis to the triangular lattice basis. For the lifted flight beginning at the disk with center zero define

$$d_R(x, a) = d \quad \Longleftrightarrow \quad Rn_\alpha + av(x) \in \mathcal{P} + \mathcal{L}d, \qquad d \in \mathbb{Z}^2, \tag{226.2}$$

with the fixed half-open boundary convention. The same cell convention is used at the two observation times.

Lemma 226.1 (A uniform finite flight partition)**.** *There is a fixed finite set $\mathcal{D} \subset \mathbb{Z}^2$ such that, for almost every regular x and every R, the sets*

$$I_{R,d}(x) = \{a \in [0, \tau_R(x)) : d_R(x, a) = d\}, \qquad d \in \mathcal{D},$$

are intervals or empty and partition the flight up to endpoints. Write $J_* = |\mathcal{D}|$. Both J_* and $\max_{d\in\mathcal{D}} |d|$ are uniform in R.

If $C_R(t) = m$ for a trajectory starting at (x, a), its terminal collision state is $y = T_R^m x$, its terminal age is $v = t + a - S_{m,R}(x)$, and exactly

$$0 \le v < \tau_R(y), \qquad W_R(t) = K^{\mathrm{c}}_{m,R}(x) + d_R(y, v) - d_R(x, a). \tag{226.3}$$

Proof. The segment in (226.2) is contained in the fixed ball of radius $47/100 + \tau_+$. Only finitely many translates of the bounded fundamental cell meet that ball; take their labels as $\mathcal{D}$. Convexity makes the intersection of each cell with a straight flight an interval. Flights lying along a cell edge form a null set and may be assigned the fixed boundary convention. This proves the partition and uniformity without differentiating a cell crossing time.

The jth future collision occurs at $S_{j,R}(x) - a$. These times are strictly increasing, so the terminal age condition is equivalent to $C_R(t) = m$, apart from suspension-null endpoints. The terminal lifted position is $\mathcal{L}K^{\mathrm{c}}_{m,R}(x) + Rn_{\alpha_y} + vv(y)$. Subtracting the initial cell label from its cell label gives (226.3). In particular the last two terms cannot be omitted at a singleton lattice label. □

For intervals $I, J \subset \mathbb{R}$ put

$$H_{I,J}(s) = \int_{\mathbb{R}} \mathbf{1}_I(a)\mathbf{1}_J(s + a)\, da = (\mathbf{1}_{-I} * \mathbf{1}_J)(s). \tag{226.4}$$

The value does not depend on choices at interval endpoints. Define $p_{m,R}(k, t) = \mathbb{P}_R\{W_R(t) = k, C_R(t) = m\}$ for $t \ge 0$ and extend it by zero to $t < 0$.

Theorem 226.2 (Exact overlap formula and uniform physical-time regularity)**.** *For all $m \ge 1$ the profiles $p_{m,R}(k, \cdot)$ have continuous, compactly supported representatives given by*

$$p_{m,R}(k, t) = \frac{1}{\bar\tau_R} \int_M \sum_{d,e\in\mathcal{D}} \mathbf{1}_{\{K^{\mathrm{c}}_{m,R}(x)+e-d=k\}} \cdot H_{I_{R,d}(x), I_{R,e}(T_R^m x)}(t - S_{m,R}(x))\, d\nu(x). \tag{226.5}$$

There are constants $A_0, A_1, A_2, L_0 < \infty$, independent of m, R, for which

$$\sum_k \|p_{m,R}(k, \cdot)\|_1 \le A_0, \qquad \sum_k \|\partial_t p_{m,R}(k, \cdot)\|_1 \le A_1, \tag{226.6}$$

$$\sum_k \|D_t^2 p_{m,R}(k, \cdot)\|_{\mathrm{TV}} \le A_2, \qquad \sum_k |p_{m,R}(k, t) - p_{m,R}(k, s)| \le L_0|t - s|. \tag{226.7}$$

One may take, writing $\bar\tau_- = \min_R \bar\tau_R$,

$$A_0 = \tau_+^2/\bar\tau_-, \quad A_1 = 2J_*\tau_+/\bar\tau_-, \quad A_2 = 4J_*^2/\bar\tau_-, \quad L_0 = 2J_*/\bar\tau_-.$$

Here $D_t^2 p$ is a finite signed measure; an L^1 density for this second derivative is neither needed nor asserted. The profiles are probabilities at fixed observation times, not a probability density of total mass one on $\mathbb{Z}^2 \times \mathbb{R}$.

Proof. Insert (226.3) into (226.1) and integrate the initial age. This is exactly (226.5). Since $S_m \geq \tau_R(x)$, its integrand vanishes for $t < 0$; it is supported in $0 \leq t \leq (m+1)\tau_+$. There are only finitely many lattice labels for each m, by finite horizon.

For an interval, $D\mathbf{1}_I$ is the difference of its endpoint masses, with total variation at most two. Consequently

$$\|H_{I,J}\|_1 = |I||J|, \qquad \|H'_{I,J}\|_1 \leq 2|I|, \qquad \|D^2 H_{I,J}\|_{\mathrm{TV}} \leq 4.$$

Apply these inequalities before integrating over x. Summing over k counts each pair (d, e) exactly once. The interval partitions give $\sum_d |I_{R,d}(x)| = \tau_R(x)$ and the analogous terminal identity. They yield all three bounds A_0, A_1, A_2. Differentiation is justified in the distributional sense by testing against compactly supported smooth functions and using these finite absolute bounds.

For the Lipschitz assertion it is sharper to work on the source. Denote the integrand event indicator at time t by $a_k(t; x, a)$. The unique initial interval and the final partition imply

$$\sum_k |a_k(t; x, a) - a_k(s; x, a)| \leq \sum_{d,e} \mathbf{1}_{I_{R,d}(x)}(a) \big|\mathbf{1}_{I_{R,e}(T^m x)}(t + a - S_m) - \mathbf{1}_{I_{R,e}(T^m x)}(s + a - S_m)\big|.$$

Sum first over d, integrate in a, and enlarge its domain to the real line. Translation of each final interval costs at most $2|t - s|$. Integrating over ν proves the last bound with L_0. It also proves continuity of the formula, including at $t = 0$. □

226.2. **An absolutely convergent physical inversion.** Use $\widehat{f}(b) = \int e^{ibt} f(t)\, dt$. For $u \in \mathbb{T}^2$ and $b \in \mathbb{R}$ define the single-flight amplitudes

$$A_R^-(x; u, b) = \int_0^{\tau_R(x)} e^{-i(u \cdot d_R(x,a) + ba)}\, da,$$

$$A_R^+(x; u, b) = \int_0^{\tau_R(x)} e^{i(u \cdot d_R(x,a) + ba)}\, da, \tag{226.8}$$

$$\mathcal{F}_{m,R}(u, b) = \int_M e^{i(u \cdot K^{\mathrm{c}}_{m,R} + bS_{m,R})} A_R^-(x; u, b) A_R^+(T_R^m x; u, b)\, d\nu(x). \tag{226.9}$$

The torus is represented by $[-\pi, \pi]^2$. For real frequencies the two single-flight amplitudes are complex conjugates at the same state.

Theorem 226.3 (Uniform pointwise far-frequency bound)**.** *The probability of the original singleton physical event satisfies*

$$p_{m,R}(k, t) = \frac{1}{(2\pi)^3 \bar{\tau}_R} \int_{[-\pi,\pi]^2} \int_{\mathbb{R}} e^{-i(u \cdot k + bt)} \mathcal{F}_{m,R}(u, b)\, db\, du. \tag{226.10}$$

The integral is absolutely convergent. If $p^{\mathrm{sharp}}_{m,R,B}$ denotes the same inverse restricted to $|b| \leq B$, then

$$\sum_k \|p_{m,R}(k, \cdot) - p^{\mathrm{sharp}}_{m,R,B}(k, \cdot)\|_\infty \leq A_2/(\pi B), \qquad B > 0. \tag{226.11}$$

In particular $B_m = m^{P+3/2}$ gives error $O(m^{-P})$ after multiplication by the physical local scale $m^{3/2}$.

Proof. Each amplitude is a sum of at most J_* interval integrals, so

$$|A_R^\pm(x; u, b)| \leq \min\{\tau_+, 2J_*/|b|\}. \tag{226.12}$$

The product in (226.9) is therefore bounded by $\min\{\tau_+^2, 4J_*^2/b^2\}$, uniformly in m, R, u. Fourier transformation of (226.4) shows that $\mathcal{F}_{m,R}/\bar{\tau}_R$ is exactly the Fourier transform of the finite family of time profiles. Fourier inversion and torus orthogonality prove (226.10); continuity from Theorem 226.2 makes the identity pointwise. No dynamical decay inside the finite band is invoked.

For the stronger summed bound, integration by parts in the sense of distributions gives, for $b \neq 0$,

$$\sum_k |\widehat{p}_{m,R}(k,b)| \le |b|^{-2} \sum_k \|D_t^2 p_{m,R}(k,\cdot)\|_{\mathrm{TV}} \le A_2/b^2.$$

Integrating over $|b| > B$ with the one-dimensional inverse factor $1/(2\pi)$ proves (226.11). The factor from the lattice inversion has already been accounted for in each Fourier coefficient. This avoids a factor equal to the number of spatial labels. □

226.3. **A positive likelihood on the same physical trajectory.** Let k_B and η be the positive kernel and its transform from Lemma 225.1. For a set $\mathcal{K} \subset \mathbb{Z}^2$ define, on the original stationary source,

$$A_{m,\mathcal{K}}(t) = \{C_R(t) = m, W_R(t) \in \mathcal{K}\}, \qquad \lambda_{m,\mathcal{K},B}(t;\omega) = \int_{\mathbb{R}} k_B(h) \mathbf{1}_{A_{m,\mathcal{K}}(t-h)}(\omega)\, dh. \quad (226.13)$$

Negative-time events in this formula are zero. These are numerical weights on one trajectory. They do not change its physical dynamics, its age, its lattice convention or the target event.

Theorem 226.4 (Polynomial-band reconstruction of the original event)**.** *For every $m \ge 1$, R, $t \ge 0$, $B > 0$ and spatial set $\mathcal{K}$,*

$$0 \le \lambda_{m,\mathcal{K},B} \le 1, \qquad \|\mathbf{1}_{A_{m,\mathcal{K}}(t)} - \lambda_{m,\mathcal{K},B}(t)\|_{L^1(\mathbb{P}_R)} \le L_0\kappa_1/B =: E_B^{\mathrm{phys}}. \quad (226.14)$$

Consequently, for every bounded measurable trajectory statistic W,

$$|\mathbb{E}_R[W\mathbf{1}_{A_{m,\mathcal{K}}(t)}] - \mathbb{E}_R[W\lambda_{m,\mathcal{K},B}(t)]| \le \|W\|_\infty E_B^{\mathrm{phys}}. \quad (226.15)$$

Uniformity permits W to depend on m, t, B. It follows by source contraction, not a spectral estimate for the weighted transform.

For $\mathcal{K} = \{k\}$ the second expectation has the finite formula

$$\frac{1}{(2\pi)^3\bar{\tau}_R} \int_{[-\pi,\pi]^2} \int_{-B}^{B} e^{-i(u\cdot k + bt)} \eta(b/B) \cdot \int_M e^{i(u\cdot K^{\mathrm{c}}_{m,R} + bS_{m,R})} A^-_{R,W}(x;u,b) A^+_R(T_R^m x;u,b)\, d\nu(x)\, db\, du, \quad (226.16)$$

where

$$A^-_{R,W}(x;u,b) = \int_0^{\tau_R(x)} W(x,a) e^{-i(u\cdot d_R(x,a) + ba)}\, da.$$

The statistic W is held fixed while taking the observation-time transform. A fixed bounded target-time interval I may also be integrated: its absolute weighted error is at most $|I|\|W\|_\infty E_B^{\mathrm{phys}}$.

Proof. The source argument in the proof of Theorem 226.2, before averaging the event indicators, shows the stronger estimate

$$\sum_k \mathbb{E}_R|\mathbf{1}_{A_{m,\{k\}}(t)} - \mathbf{1}_{A_{m,\{k\}}(s)}| \le L_0|t-s|$$

for all real t, s, with the zero extension already established. A spatial set can only reduce the corresponding bound. Average it against $k_B(t-s)$ and use positivity and $\int |h| k_B(h)\, dh = \kappa_1/B$. This proves (226.14). Multiplication by W and then integration prove (226.15). In particular

this is total variation of unnormalized measures on the original stationary source, not merely a difference of scalar probabilities.

For fixed (x,a) and m, the time event is supported in an interval of length at most τ_+. Its time Fourier transform is obtained by the substitution $t = S_m + v - a$. Integrate v over the final cell intervals and a over the initial ones. The factor $W(x,a)$ remains inside the latter integral and gives $A^-_{R,W}$. Invert the positive kernel and the two lattice coordinates to obtain (226.16). The band is finite and all age integrals and the statistic are bounded, so Fubini is legitimate. Finally integrate the established error over I. □

Corollary 226.5 (Denominator-sensitive physical posteriors)**.** *Let $A = A_{m,\mathcal{K}}(t)$, $p_A = \mathbb{P}_R(A)$ and $p_{A,B} = \mathbb{E}_R\lambda_{m,\mathcal{K},B}(t)$. If $p_{A,B} > E_B^{\mathrm{phys}}$, then*

$$p_A \ge p_{A,B} - E_B^{\mathrm{phys}} > 0, \qquad \left\| \mathbb{P}_R(\cdot \mid A) - \frac{\lambda_{m,\mathcal{K},B}(t)}{p_{A,B}} \mathbb{P}_R \right\|_{\mathrm{TV}} \le \frac{2E_B^{\mathrm{phys}}}{p_{A,B} - E_B^{\mathrm{phys}}}.$$

If instead a lower bound for $p_A > E_B^{\mathrm{phys}}$ is known, the right side may be replaced by $2E_B^{\mathrm{phys}}/p_A$. These assertions hold also after multiplication by any fixed nonnegative selector at most one, using its own denominator. For $p_A \ge cm^{-3/2}$ and $B_m = m^{P+3/2}$, the relative error is $O(m^{-P})$.

Proof. Apply the normalization inequality in Corollary 225.5 to (226.14). Multiplication by a selector contracts the source L^1 error. The last assertion is division by the explicitly stated denominator bound, not a proof of that bound. □

Which tail has been closed. Equations (226.11) and (226.14) use a polynomial bandwidth for the *original stationary observation* at a fixed collision count. They require no removal of a small-mass source set, no long-itinerary coarea derivative bound, and no comparison with an event at the preceding section origin. The time profiles have uniform distributional curvature because two physical flight ages have been integrated. The return density in Theorem W has a different output and does not inherit this estimate. In particular a bound for these profiles is not a pointwise estimate for the four-coordinate return density.

227. The Gaussian central term for an exact physical singleton

We now evaluate the central part of (226.10). This calculation takes place in the actual three frequency variables (u_1, u_2, b) of a prescribed physical collision count. It is not a trace of a four-dimensional integrated estimate. Together with the preceding uniform time tail it reduces the original stationary singleton probability to one finite, signed middle-frequency integral at a polynomial bandwidth.

Define the bounded centered collision record and its covariance by

$$f_R^{\mathrm{c}} = (\kappa_{R,1}, \kappa_{R,2}, \tau_R - \bar\tau_R) = P_R h_R, \qquad P_R(w_1, w_2, a, b) = (w_1, w_2, b - \bar\tau_R a), \qquad \Sigma_R = P_R \Gamma_R P_R^{\mathsf{T}}. \tag{227.1}$$

The map P_R has full row rank with uniformly bounded coefficients. Thus Σ_R is uniformly positive definite by Theorem E. Its Green–Kubo formula and its smoothed version follow by applying P_R to the collision formulas; no new correlation theorem is needed.

227.1. **A three-dimensional endpoint-weighted central estimate.**

Lemma 227.1 (Two roof weights at the collision endpoints)**.** *For $m \ge 2$, uniformly in R,*

$$\int_{|v|\le 2m^{1/200}} \left| \int_M \tau_R(x)\tau_R(T_R^m x) e^{iv\cdot\sum_{j<m} f_R^{\mathrm{c}}(T_R^j x)/\sqrt{m}}\, d\nu(x) - \bar\tau_R^2 e^{-v^{\mathsf{T}}\Sigma_R v/2} \right| dv \le Cm^{-11/700}\sqrt{\log(2+m)}. \tag{227.2}$$

The integration is with respect to three-dimensional Lebesgue measure.

Proof. Put $h_\delta = \mathsf{S}_\delta h_R$ and $\tau_\delta = \mathsf{S}_\delta \tau_R$, and let $z = P_R^{\mathsf{T}} v/\sqrt{m}$. The collision transfer convention gives exactly the smoothed pairing

$$\ell_j(M_{\tau_\delta}\mathcal{L}^m_{R,\delta,z}(\tau_\delta \nu)).$$

The initial vector and terminal functional each have norm $O(\delta^{-2})$. The spectral splitting of Lemma 139.1 therefore has complementary error $O(\delta^{-4}\rho^m)$. Its principal amplitude at $z = 0$ is $\bar\tau_R^2$, because smoothing preserves the mean. The difference of that amplitude from its value at zero is bounded by $C\delta^{-6}|v|/\sqrt{m}$: two endpoint norms and one projection difference have all been paid.

Write $\ell_\delta = 1 + |\log \delta|$. The same spectral lemma gives

$$m \log \lambda_{R,\delta}(z) = -\tfrac{1}{2} v^{\mathsf{T}} P_R \Gamma_{R,\delta} P_R^{\mathsf{T}} v + O(\delta^{-6}|v|^3/\sqrt{m}).$$

When the displayed cubic remainder and $|z|\delta^{-2}$ are small, positive semidefiniteness of the smoothed covariance bounds the principal power by a fixed constant. The covariance replacement costs $C|v|^2\delta\ell_\delta$.

Replacing either endpoint τ_δ by τ_R costs $C\delta$, by invariance, the L^1 smoothing error and the uniformly bounded other factors. The product of the two smoothed roofs is bounded pointwise independently of m, δ. The small-BV variance bound (138.5), applied to each component of $P_R(h_R - h_\delta)$, consequently bounds the unsmoothing of the phase by $C|v|\sqrt{\delta\ell_\delta}$. The resulting pointwise error is

$$C\left[\delta + |v|\sqrt{\delta\ell_\delta} + |v|^2\delta\ell_\delta + m^{-1/2}\delta^{-6}(|v| + |v|^3) + \delta^{-4}\rho^m\right].$$

No long pullback of an endpoint weight is estimated in BV.

Take $\delta = m^{-1/14}/4$ and integrate on the three-dimensional ball of radius $2m^{1/200}$. In the order just displayed, the five polynomial decay exponents are

$$\frac{79}{1400}, \qquad \frac{11}{700}, \qquad \frac{13}{280}, \qquad \frac{9}{175}, \qquad \frac{29}{700}.$$

The second term is the largest; the extra logarithm in the covariance term is absorbed by its strict power margin. The spectral smallness requirements follow from

$$\frac{1}{2} - \frac{1}{200} - \frac{1}{7} > 0, \qquad \frac{1}{2} - \frac{3}{200} - \frac{3}{7} > 0.$$

The complementary term is exponentially small. Increasing C for finitely many initial m proves (227.2). □

Set $r_m = 2m^{-99/200}$, so that $\sqrt{m}\, r_m = 2m^{1/200}$. In the neighborhood $|(u,b)| \le r_m$, the bounded cell labels and ages in Lemma 226.1 give

$$|A_R^{\pm}(x; u, b) - \tau_R(x)| \le C|(u,b)|. \tag{227.3}$$

This estimate uses the exact cell labels. It does not require the amplitudes themselves to have a uniform BV norm at every frequency. For m sufficiently large, this ball is inside the chosen torus chart. Define its inverse contribution by

$$G^{\mathrm{cen}}_{m,R}(k,t) = \frac{1}{(2\pi)^3 \bar\tau_R} \int_{|(u,b)|\le r_m} e^{-i(u\cdot k + bt)} \mathcal{F}_{m,R}(u,b)\, du\, db. \tag{227.4}$$

Theorem 227.2 (Evaluated Gaussian central term)**.** *Writing*

$$\zeta_{m,R}(k,t) = \left(\frac{k}{\sqrt{m}}, \frac{t - m\bar\tau_R}{\sqrt{m}}\right),$$

one has, uniformly in R, k, t and all sufficiently large m,

$$\left|m^{3/2}G^{\mathrm{cen}}_{m,R}(k,t) - \bar\tau_R g_{\Sigma_R}(\zeta_{m,R}(k,t))\right| \le Cm^{-11/700}\sqrt{\log(2+m)}. \tag{227.5}$$

Here g_Σ is the normalized Gaussian density on $\mathbb{R}^3$.

Proof. Remove $e^{ibm\bar\tau_R}$ from (226.9). Substitution $(u,b) = v/\sqrt{m}$ in (227.4) gives an inverse on the actual three-dimensional ball. By (227.3), replacement of the two amplitudes by the endpoint roofs costs at most

$$Cm^{-1/2}\int_{|v|\le 2m^{1/200}} |v|\,dv \le Cm^{-12/25}$$

after multiplication by $m^{3/2}$. Apply Lemma 227.1 to the resulting transform. The error in an inverse is bounded by its integrated transform error, uniformly in the target. Uniform ellipticity gives a Gaussian tail $Ce^{-cm^{1/100}}$ outside this ball. The full Gaussian inverse is $g_{\Sigma_R}(\zeta)$ with factor $\bar\tau_R^2/\bar\tau_R$. This proves the normalization and the asserted bound. □

227.2. **One finite signed complement for the original physical event.** For $B \ge 1$ and sufficiently large m put

$$\mathfrak{M}_{m,R,B}(k,t) = \frac{1}{(2\pi)^3\bar\tau_R}\int_{\substack{u\in[-\pi,\pi]^2,\ |b|\le B\\ |(u,b)|>r_m}} e^{-i(u\cdot k+bt)}\mathcal{F}_{m,R}(u,b)\,db\,du. \tag{227.6}$$

This is a signed inverse evaluated at the target, not an integral of an absolute value. Its domain contains all noncentral spatial torus frequencies and all remaining time frequencies below B.

Theorem 227.3 (A pointwise Gaussian-plus-complement identity)**.** *For every $B \ge 1$ and all sufficiently large m,*

$$\sup_{R,k,t}\left|m^{3/2}p_{m,R}(k,t) - \bar\tau_R g_{\Sigma_R}(\zeta_{m,R}(k,t)) - m^{3/2}\mathfrak{M}_{m,R,B}(k,t)\right|$$
$$\le C\left(m^{-11/700}\sqrt{\log(2+m)} + \frac{m^{3/2}}{B}\right). \tag{227.7}$$

There is no discarded-density term in this identity. For $B_m = m^{P+3/2}$, its last term is $O(m^{-P})$.

More intrinsically, put

$$\xi_{t,m,R}(k) = \left(\frac{k}{\sqrt{t}}, \frac{m - t/\bar\tau_R}{\sqrt{t}}\right).$$

For every fixed $M < \infty$, uniformly in R and k, m, t satisfying $t \to \infty$ and $|\xi_{t,m,R}(k)| \le M$,

$$t^{3/2}p_{m,R}(k,t) = g_{\mathcal{V}_R}(\xi_{t,m,R}(k)) + t^{3/2}\mathfrak{M}_{m,R,B_m}(k,t)$$
$$+ O_M\left(t^{-11/700}\sqrt{\log(2+t)} + t^{-P}\right). \tag{227.8}$$

Thus the Gaussian denominator asymptotic for these original physical singletons is equivalent to the vanishing of the displayed normalized signed complement on each such central set.

Proof. Split the absolutely convergent integral in (226.10) into its central ball, the finite middle region (227.6), and $|b| > B$. Theorem 227.2 evaluates the first part and Theorem 226.3 bounds the third. This gives (227.7) for the actual event, pointwise in observation time. No essential supremum of a discarded return density has entered the argument.

Let $D_R^{\mathrm{clk}} = \mathrm{diag}(1, 1, -1/\bar\tau_R)$. The physical covariance defined earlier satisfies exactly

$$\mathcal{V}_R = \bar\tau_R^{-1}D_R^{\mathrm{clk}}\Sigma_R(D_R^{\mathrm{clk}})^{\mathsf{T}}, \qquad \det\mathcal{V}_R = \det\Sigma_R/\bar\tau_R^5. \tag{227.9}$$

Indeed $D_R^{\mathrm{clk}}P_R$ is the map B_R of the physical-time functional theorem. If $|\xi| \le M$, then $m = t/\bar{\tau}_R + O_M(\sqrt{t})$ uniformly. Set $A_R = \sqrt{\bar{\tau}_R}\,\mathrm{diag}(1,1,-\bar{\tau}_R)$. Its absolute determinant is $\bar{\tau}_R^{5/2}$ and $A_R\mathcal{V}_R A_R^{\mathsf{T}} = \Sigma_R$. Also

$$\zeta_{m,R} = A_R\xi + O_M(t^{-1/2}), \qquad \bar{\tau}_R(t/m)^{3/2} = \bar{\tau}_R^{5/2} + O_M(t^{-1/2}).$$

Uniform Gaussian derivative bounds on compact sets therefore show

$$\bar{\tau}_R(t/m)^{3/2} g_{\Sigma_R}(\zeta_{m,R}) = g_{\mathcal{V}_R}(\xi) + O_M(t^{-1/2}).$$

Multiplying (227.7) by $(t/m)^{3/2}$ proves (227.8). Since the remaining explicit error tends to zero, subtraction proves both directions of the asserted equivalence. A signed complement may vanish even if its absolute transform integral is not small. □

Corollary 227.4 (What a complement estimate would imply)**.** *If the normalized middle term in* (227.8) *tends to zero uniformly on a fixed central set, then there*

$$\mathbb{P}_R\{W_R(t) = k, C_R(t) = m\} = t^{-3/2} g_{\mathcal{V}_R}(\xi_{t,m,R}(k))(1 + o(1)),$$

and this denominator is uniformly bounded below by $c_M t^{-3/2}$. For any $P > 0$, Theorem 226.4 with $B_m = m^{P+3/2}$ then approximates its conditional initial law by the positive finite-band likelihood with total variation error $O_M(t^{-P})$. A selected event needs its own selected denominator; the unselected Gaussian formula does not establish it.

Proof. Uniform ellipticity makes the Gaussian density uniformly positive on the compact central set. Use Theorem 227.3 and divide the source error in Corollary 226.5 by this lower bound. The final sentence specifies the normalization required after multiplication by a selector. □

227.3. **Relation to the return-density and suspension theories.** The cell-displacement local theorem of [2] and the suspension mixing-local-limit framework of [3, Definition 2.2 and Section 2.4] are natural comparisons. In the latter framework a local theorem on the base is an input to the suspension argument. We have not imported that input for the moving-section four-coordinate record. The present calculation instead keeps one prescribed physical collision count and the complete cell corrections, integrates the true flight ages, and obtains a uniform pointwise far-time-frequency estimate directly. Its two single-flight amplitudes and their inverse-square product are the mechanism behind the polynomial bandwidth. They do not by themselves evaluate the middle-frequency collision correlation.

For reference the three norms and the completed steps can be recorded without identifying their endpoints:

Object	Proved estimate	Remaining local task
Preceding-section return law	Theorem W: global mixed L^1 and source-TV approximation; B_n may be $\exp(O(n\log n))$	The common pointwise return correction and its finite complement.
Actual physical singleton	Theorem X: a pointwise $O(B^{-1})$ tail, a Gaussian central term and a polynomial-band exact ledger	The normalized signed middle term in (227.8).
Physical weighted event	Absolute source-TV error E_B^{phys} for every bounded selector	Its evaluated weighted formula and positive selected denominator before relative conditioning.

The local physical remainder is now one explicitly specified integral, with no discarded-density or event-replacement error. This is a direct analysis of the original physical event, not a proof that it coincides with a preceding-section event. The full four-coordinate return LLT and the microscopic

Gaussian denominator remain the stated targets; neither is relabeled as a consequence of the new tail estimate.

228. Conditioning, clocks, and geometric prediction

Proposition 228.1 (Central density conditioning)**.** *Suppose an unweighted and an f-weighted joint density satisfy*

$$n^2 p_n(z) = G_n(z) + r_n(z), \qquad n^2 p_n^f(z) = a_f G_n(z) + r_n^f(z)$$

on a common central window, with $G_n \geq g_0 > 0$ and $|r_n|, |r_n^f| \leq \varepsilon_n < g_0$. Then their density ratio differs from a_f by at most

$$\frac{(1+|a_f|)\varepsilon_n}{g_0 - \varepsilon_n}. \tag{228.1}$$

The same bound holds after conditioning on fixed lattice coordinates and any positive-length roof interval contained in that window, including intervals whose lengths shrink with n.

Proof. Subtract a_f times the first identity from the second and divide by $G_n + r_n \geq g_0 - \varepsilon_n$. For an interval, integrate both identities first; the errors are at most its length times ε_n and the denominator is at least its length times $g_0 - \varepsilon_n$. The length cancels. □

At an exact roof value this is a density-defined version of a regular conditional expectation for almost every value with positive density. It is not conditioning on a positive-probability singleton. It supplies no estimate outside the common central window and no functional path convergence by itself. These are the precise limitations of this interface to stopped-path large deviations.

Proposition 228.2 (A conditional physical-clock transfer)**.** *Let $(X_j, \tau_j) \in \mathbb{R}^d \times (0,\infty)$ be stationary dynamical observations with means (a,μ), $\mu > 0$, and a joint functional CLT with covariance Σ. Assume the uniform law of large numbers for their clock and that initial and terminal residual records on bounded rescaled time intervals are $o_P(\sqrt{t})$. Then the reward accumulated up to physical time t, centered by $(a/\mu)t$, has limiting covariance per unit time*

$$\frac{1}{\mu} Q \Sigma Q^T, \qquad Q = (I_d, -a/\mu). \tag{228.2}$$

For $X = (\kappa_1, \kappa_2, r)$ this transfers induced displacement and collision counts, not the number of experimental bits.

Proof. The clock law gives its inverse index $N(t)/t \to 1/\mu$. The centered functional CLT evaluated at this inverse time and the identity $S_{N(t)}\tau = t + o_P(\sqrt{t})$ give

$$S_{N(t)}X - (a/\mu)t = (S_{N(t)}X - N(t)a) - (a/\mu)(S_{N(t)}\tau - N(t)\mu) + o_P(\sqrt{t}).$$

The continuous limit permits the random-time substitution. Projection by Q and evaluation at time $1/\mu$ prove (228.2). The assumed residual estimate includes unfinished induced returns, which are not bounded merely because individual flights are bounded. □

Lemma 228.3 (Covariance continuity from a finite-time comparison)**.** *Suppose a centered correlation matrix satisfies $\|C_j(R)\| \leq Ce^{-cj}$ and $\|C_j(R) - C_j(R')\| \leq C\delta^\alpha e^{aj}$, where $\delta = |R - R'|$, $a, c, \alpha > 0$. Then the absolutely convergent Green–Kubo covariance satisfies*

$$\|\Sigma_R - \Sigma_{R'}\| \leq C' \delta^{\alpha c/(a+c)}.$$

If $a = 0$, the bound is instead $C'\delta^\alpha(1 + |\log \delta|)$.

Proof. Split the defining correlation series at $J = \lfloor \alpha \log(1/\delta)/(a+c) \rfloor$. For $a > 0$, the finite sum is bounded by $C\delta^{\alpha} e^{aJ}$ and the remaining sum by Ce^{-cJ}. Both have the stated order. For $a = 0$, sum the constant comparison bound over $J+1$ terms and use the exponential tail. Symmetrizing the correlations only changes the constant. □

The hypotheses of this lemma must concern the actual induced observables and their varying domains. Spectral continuity for a different observable norm is not a substitute for this comparison.

Proposition 228.4 (Geometric error and the central prediction scale)**.** *Assume* (143.2) *with error* ϵ_n*, uniformly elliptic covariances, and estimates* $(\widehat{m}, \widehat{\Sigma})$ *satisfying* $|\widehat{m} - m_R| + \|\widehat{\Sigma} - \Sigma_R\| \le \omega(\nu)$*. If* $\sqrt{n}\,\omega(\nu) \le 1$*, replacing the Gaussian parameters in* (143.2) *changes its right side on a fixed central window by at most* $C(1+\sqrt{n})\omega(\nu)$*. The total prediction error is bounded by this quantity plus* ϵ_n*.*

Proof. The argument of the Gaussian moves by $\sqrt{n}(\widehat{m} - m_R)$. On a slightly enlarged compact window the Gaussian and its derivatives in its argument and covariance are bounded uniformly under ellipticity. The mean value theorem gives the bound. Adding the LLT error finishes the proof. □

There is also a direct equilibrium conclusion that needs no local-limit hypothesis. The mean collision rate is

$$\lambda(R) = \frac{2R}{\sqrt{3}/2 - \pi R^2}.$$

Its derivative is $2/A_R + 4\pi R^2/A_R^2$, where $A_R = \sqrt{3}/2 - \pi R^2$. On I, $A_R > 17/100$, using $\sqrt{3} > 173/100$ and $\pi < 22/7$, and direct substitution gives $0 < \lambda'(R) < 110$. Therefore a radius estimate with error at most ν gives collision-rate error at most 110ν. On the circular submodel the average support function identifies the radius, independently of translation. The finite geometric conclusion of [1] therefore transfers to this rate with its geometric observation cost, without recovering the entire unknown launch law.

229. A logarithmic unfinished-block bound under conditioning

The stationary law of an induced block is length-biased. We use this law for the initial block and the exact visit intensity for subsequent blocks. These two ingredients avoid applying a launch-law union bound to a stationary initial state.

Write $\overline{\tau}_R^* = \bar{\tau}_R/c_*$ and represent the flow as the suspension over (Y_R^*, F_R^*, ν_R^*) with roof τ_R^*. Let t_j be its two-sided visit times and x_j their outgoing section states. The minimum time between consecutive visits is at least $3/50$.

Lemma 229.1 (Marked visit intensity)**.** *For every nonnegative measurable* f *on* Y_R^* *and every* $t \ge 0$*,*

$$\mathbb{E}_{\mu_R} \sum_{t_j \in (0,t]} f(x_j) = \frac{t}{\overline{\tau}_R^*} \int f \, \mathrm{d}\nu_R^*. \tag{229.1}$$

Proof. First take bounded f. Stationarity makes the locally finite measure $A \mapsto \mathbb{E}_{\mu_R} \sum_{t_j \in A} f(x_j)$ translation-invariant on $\mathbb{R}$, hence equal to i_f times Lebesgue measure. Local finiteness follows from the positive lower roof bound. Choose $0 < \delta < 3/50$. At a stationary time, being in the first δ units of an induced roof is the same event as a visit in the preceding interval of length δ, apart from null endpoints. There is at most one such visit, and its mark is the current base state. Integration in the suspension therefore gives

$$\delta i_f = \frac{1}{\overline{\tau}_R^*} \int_{Y_R^*} \int_0^{\delta} f(x) \, \mathrm{d}s \, \mathrm{d}\nu_R^*(x) = \frac{\delta}{\overline{\tau}_R^*} \int f \, \mathrm{d}\nu_R^*.$$

This determines i_f and proves the formula for bounded f. Monotone convergence extends it to arbitrary nonnegative marks, including the case of an infinite integral. No renewal independence is involved. □

Let $\mathcal{M}_R(t)$ be the maximum of Q_R over all complete induced blocks whose time intervals meet $[0,t]$, including the block containing time zero. Endpoint conventions at a visit are immaterial for μ_R. The marked size of any unfinished piece is bounded by a fixed multiple of $1+\mathcal{M}_R(t)$.

Theorem 229.2 (Uniform exponential maximum and conditioned residual)**.** *There are constants $a_2, C_2 > 0$ such that, for every $R \in I$, $t \geq 0$ and $q \geq 0$,*

$$\mu_R\{\mathcal{M}_R(t) > q\} \leq C_2(1+t)e^{-a_2 q}. \tag{229.2}$$

Consequently, for any measurable event $B_{R,t}$ of probability at least $d_t > 0$,

$$\mu_R\{\mathcal{M}_R(t) > q \mid B_{R,t}\} \leq \min\{1, C_2(1+t)e^{-a_2 q}/d_t\}. \tag{229.3}$$

In particular, if $d_t \geq d(1+t)^{-A}$ with $d > 0$ and $A \geq 0$, then $\mathcal{M}_R(t) = O_P(\log(2+t))$ under these conditional laws, uniformly in R and in the events. The maximum marked unfinished return during $[0,t]$ is then $o_P(\sqrt{t})$ under the same laws. More generally the last assertion holds if $\log((1+t)/d_t) = o(\sqrt{t})$.

Proof. The initial containing block has law $\tau_R^* \, \mathrm{d}\nu_R^*/\overline{\tau}_R^*$. If the maximum exceeds q, either that block has $Q_R > q$ or a visit in $(0,t]$ starts a block with $Q_R > q$. The union bound and Lemma 229.1 give

$$\mu_R\{\mathcal{M}_R(t) > q\} \leq \frac{1}{\overline{\tau}_R^*} \left(\int \tau_R^* \mathbf{1}_{\{Q_R > q\}} \, \mathrm{d}\nu_R^* + t \, \nu_R^*(Q_R > q) \right). \tag{229.4}$$

The denominator is bounded below on I. Since $\tau_R^* \leq Q_R$, the exponential moment in (133.5) bounds the first integral by $Ce^{-a_1 q/2}$ and the probability by $Ce^{-a_1 q}$. For example, $Q \leq (2/a_1)e^{a_1 Q/2}$ and $e^{a_1 Q/2}\mathbf{1}_{Q>q} \leq e^{a_1 Q}e^{-a_1 q/2}$ suffice for the first bound. This proves (229.2).

Dividing the unconditional exceptional probability by $\mu_R(B_{R,t})$ proves (229.3). With a polynomial lower bound on d_t, setting $q = K\log(2+t)$ and taking $a_2 K > A+1$ makes the right side tend to zero. With the more general hypothesis, set $q = \epsilon\sqrt{t}$ for any $\epsilon > 0$; the logarithm of the displayed upper bound tends to minus infinity. The deterministic bound of each marked piece by $C(1+Q_R)$ proves the assertions for all unfinished pieces at once. The event may depend on the whole orbit and on an estimator from that orbit; no independence is assumed. □

For a stationary starting point (x,s) put $V_R(v) = \max\{j : T_{j,R}(x) \leq s+v\}$. Define the two path records

$$Z_R(v) = (V_R(v), C_R(v), W_R(v)), \qquad Z_R^\circ(v) = (V_R(v), N_{V_R(v),R}(x), K_{V_R(v),R}(x)).$$

The second record counts full blocks from the beginning of the initial containing roof, not from the observation time. It is used solely as an exact intermediate record for comparison.

Corollary 229.3 (A conditional path-comparison budget)**.** *There is C_3 such that, pathwise outside the collision singularities,*

$$\sup_{0 \leq v \leq t} |Z_R(v) - Z_R^\circ(v)| \leq C_3(1+\mathcal{M}_R(t)). \tag{229.5}$$

Let $\widehat{Z}_R$ and $\widehat{Z}_R^\circ$ be the corresponding paths on $[0,1]$, divided by $\sqrt{t}$ and with any identical deterministic centering. For $|F| \leq 1$ Lipschitz with constant L_F for the uniform path metric and $\mu_R(B_{R,t}) \geq d_t > 0$,

$$\left| \mathbb{E}_{\mu_R}[F(\widehat{Z}_R) \mid B_{R,t}] - \mathbb{E}_{\mu_R}[F(\widehat{Z}_R^\circ) \mid B_{R,t}] \right| \leq \frac{C_3 L_F(1+q)}{\sqrt{t}} + \frac{2C_2(1+t)e^{-a_2 q}}{d_t}. \tag{229.6}$$

For $t \geq 2$ choose

$$q = a_2^{-1} \log\left(\frac{\max\{1, 2C_2\}(1+t)\sqrt{t}}{d_t}\right). \tag{229.7}$$

The second term is at most $t^{-1/2}$; for polynomially bounded conditioning probabilities the total error is $O((1+L_F)\log t/\sqrt{t})$.

Proof. The collision counts differ by the initial prefix and the final incomplete block. Each is bounded by the full block's collision count. The same holds for the lattice displacement after multiplying by the uniform bound on a single-flight lattice step and adding a fixed fundamental-cell boundary constant. The visit counts agree exactly by construction. These estimates are simultaneous for all $v \in [0,t]$, and prove (229.5). On $\{\mathcal{M}_R(t) \leq q\}$ the Lipschitz bound gives the first term in (229.6). On its complement the difference of two values of F is at most two. Apply (229.3). The stated choice of q and substitution prove the final assertions. □

Remark 229.4 (Which conditional interface has been closed). The square-root unfinished-block requirement is now a theorem for the actual mechanical family, even after any conditioning of polynomially small positive probability and over the whole observation interval. A functional central limit theorem for completed records is still a separate premise in Proposition 228.2. For a local limit, the conditioning event on the two sides of (229.6) is the same event. Replacing an exact lattice observation by the completed-block observation changes that event and needs a further boundary estimate. If two events B, B' have probabilities at least d, then for $|f| \leq 1$ direct subtraction of the ratios gives

$$|\mathbb{E}[f \mid B] - \mathbb{E}[f \mid B']| \leq 2\mu_R(B \triangle B')/d.$$

Thus an $O(\log t)$ error in an integer record is not discarded when conditioning on one lattice value. The weighted raw-density estimates of Proposition 228.1 remain the requisite local input. A zero-probability singleton in the continuous coordinate is not an event to which (229.3) applies.

230. Induced covariance and raw-summation criteria

We record the exact analytical implications needed in the low-frequency and outer-frequency parts of the original density theorem. These criteria distinguish proved finite-record facts from the two infinite sums that still require estimates. The collision matrix Γ_R and the actual Gaussian covariance D_R have already been established in Theorems 138.4 and 139.5. The following induced Green–Kubo criterion is stronger: neither its infinite sum nor its periodic-evaluable representative is presumed by those theorems. The direct argument of Theorem 150.2 now proves positive definiteness of D_R without either of these additional premises; the criterion below is retained for the stronger, absolutely convergent induced Green–Kubo representation.

Proposition 230.1 (A covariance closure criterion)**.** *Put $g_R = G_R - \bar{G}_R$. Suppose that*

$$\lim_{m\to\infty} \sup_R \sum_{k>m} \left\| \int g_R \otimes (g_R \circ (F_R^*)^k)\, \mathrm{d}\nu_R^* \right\| = 0. \tag{230.1}$$

Suppose also that zero asymptotic variance of $v \cdot g_R$ implies a coboundary whose transfer function has a representative for which the coboundary identity telescopes on the five real-rank periods of Corollary 128.5. Then the Green–Kubo matrices

$$\Sigma_R = \int g_R \otimes g_R \,\mathrm{d}\nu_R^* + \sum_{k\geq 1} \int \left(g_R \otimes (g_R \circ F^k) + (g_R \circ F^k) \otimes g_R\right) \mathrm{d}\nu_R^*$$

are continuous and satisfy $cI_4 \leq \Sigma_R \leq CI_4$ on I for some $c, C > 0$.

Proof. Corollary 133.4 makes every fixed-lag matrix continuous, including the zero-lag term. The uniformly summable tail in (230.1) therefore gives continuity of the series. The usual expansion of the covariance of $\sum_{j<n} g_R \circ F^j$ is a finite identity; dividing it by n and using absolute summability identifies its limit with Σ_R. Thus Σ_R is positive semidefinite and its directional quadratic form is the asymptotic variance.

If $v^T \Sigma_R v = 0$, the stipulated coboundary identity gives on each of the five selected periods

$$v_1 K_1 + v_2 K_2 + v_3 N + v_4 T - h(v \cdot \bar{G}_R) = 0.$$

The augmented vector $(v_1, v_2, v_3, -v \cdot \bar{G}_R, v_4)$ annihilates their five-dimensional record matrix, whose determinant is nonzero by Corollary 128.5. Hence $v = 0$. Each matrix is positive definite. Continuity and compactness of I give uniform lower and upper eigenvalue bounds. The periodic representative premise is used explicitly here; a merely measurable identity is insufficient. □

Lemma 230.2 (An absolute raw-residual summation criterion)**.** *Let $\mathcal{W}$ be countable. For $w \in \mathcal{W}$ assign a lattice label $k_w \in \mathbb{Z}^3$ and a function $q_w \in L^1(\mathbb{R})$ whose second distributional derivative is a finite signed measure. Suppose*

$$A_0 = \sum_w \|q_w\|_1 < \infty, \qquad A_2 = \sum_w \|D^2 q_w\|_{\mathrm{TV}} < \infty.$$

Then $H(u,b) = \sum_w e^{iu\cdot k_w} \widehat{q}_w(b)$ belongs to $L^1(\mathbb{T}^3 \times \mathbb{R})$ and, for $B \geq 1$,

$$\int_{\mathbb{T}^3 \times \mathbb{R}} |H| \leq 2(2\pi)^3 (A_0 + A_2), \tag{230.2}$$

$$\int_{\mathbb{T}^3 \times \{|b|>B\}} |H| \leq 2(2\pi)^3 A_2 / B.$$

The same estimates hold uniformly for a parameter family whenever the corresponding sums are uniformly bounded.

Proof. Distributional differentiation and the stated Fourier convention give $-b^2 \widehat{q}_w(b) = \widehat{D^2 q_w}(b)$. Thus

$$|\widehat{q}_w(b)| \leq \min\{\|q_w\|_1, |b|^{-2} \|D^2 q_w\|_{\mathrm{TV}}\} \quad (b \neq 0).$$

Integrate the first estimate on $|b| \leq 1$ and the second on $|b| > 1$, and then sum by Tonelli. The torus has volume $(2\pi)^3$. The series itself converges uniformly as a transform of the absolutely summable L^1 densities. Integrating only $|b| > B$ gives the second bound. All lattice multiplicities are included in the two sums; no bound on the number of words has been suppressed. □

To use this lemma, the residuals must be those of a decomposition of the actual full measure, including singular itinerary boundaries. A jump left in q_w contributes a derivative of a point mass to $D^2 q_w$ and does not satisfy the hypothesis. The individual jump estimates of Proposition 142.3 do not prove the sum $A_2 < \infty$. Nor does the scalar tail (133.5) bound inverse Jacobian factors or the variation of a density near a critical value. If A_2 grows with n, its actual growth must be used in the frequency splice rather than treated as a fixed constant.

There is nevertheless a useful exact truncation budget. For an insertion $|w| \leq M_0$, deleting the physical words with $N_{n,R} > L$ changes its record measure, and its raw transform in uniform norm, by at most

$$M_0 C_1 e^{-a_1 L/n}. \tag{230.3}$$

Indeed total variation contracts under pushforward, and (133.6) bounds the discarded mass. On a frequency band of volume V the integrated error is at most $V M_0 C_1 e^{-a_1 L/n}$. Thus the sufficient physical-count budget for a prescribed integrated error ε is

$$L \geq \frac{n}{a_1} \max\{0, \log(V M_0 C_1 / \varepsilon)\}.$$

This controls a finite-band error, not a pointwise density error or an infinite frequency band. It is compatible with retaining the original raw datum and with the separate all-branch derivative contract above.

231. The joint local-limit conclusion and its realization

The endpoint of this article remains the parameter-uniform raw mixed density local limit (143.2) for the physical record $(K_{n,R}, N_{n,R}, T_{n,R})$, with the weighted positive-denominator interface of Proposition 228.1. Neither the return section nor the recorded observables are replaced by independent flights, a smoothed observation, or an unrelated geometric problem.

The change from Y_R to Y_R^* is explicit, and the periodic records are those of its actual first return. The joint periodic annihilator is trivial. The adjacent-increment comparison yields a uniform positive period discrepancy with logarithmic collision length, and Theorem 141.1 gives its positive-measure consequence for controlled regular phases. The phase-reconstruction and Fredholm hypotheses of Theorem 141.2 state precisely how these estimates would give an induced resolvent bound. Their verification on the actual induced strong/weak spaces is still required. No division by an uncontrolled spectral modulus is implicit.

The collision-space spectral input has a different, now completed use: Theorem 133.3 proves uniform exponential moments of the genuine return block. It strengthens finite-record stability to all finite moments. The common-chart construction in Proposition 136.2 gives an explicit initial-state remainder for the moving-domain density comparison. The bounded compensation of Section 138 then proves a uniformly summable collision Green–Kubo formula. The exact stopped identity gives the actual Gaussian covariance $D_R = c_*^{-1}\Gamma_R$, its continuity, and its L^2 cohomological kernel. These conclusions no longer depend on an assumed induced spectral expansion. The stronger induced-correlation criterion of Proposition 230.1 remains available, but is not used to obtain these Gaussian laws. The actual normalized second moments now converge by Theorem 147.6, and Theorem 150.2 proves uniform positive definiteness. Its measurable phase argument removes the section term before using symplectic area and finite physical covers; periodic evaluation of the transfer function is not required.

At the physical-clock level, the first-order functional law and its geometric rate comparison remain valid. The stronger maximum estimate in Theorem 229.2 now controls all unfinished returns on a growing interval, at logarithmic scale, even after conditioning of polynomially small positive probability. Corollary 229.3 supplies the corresponding explicit square-root path error. Thus the unfinished-block premise no longer needs to be assumed in that range of conditioning probabilities. Theorem 140.3 now supplies the unconditioned functional Gaussian theorem for the completed physical return record; Corollary 140.4 gives unconditioned physical-time fluctuations as well. Neither an unconditioned functional limit nor a path comparison under one event justifies changing an exact lattice conditioning event. That boundary comparison is a separate weighted local-density problem, as recorded in Remark 229.4.

For the raw inversion itself, the exact edge subtraction and the local edge criterion retain every correction term. The all-branch residual must include central critical words and singular itinerary boundaries. Lemma 230.2 specifies an absolute derivative-norm criterion for that sum; no uncontrolled word multiplicity is hidden in it. The required uniform sum has not been established here. Its growth constants, together with the actual induced operator estimates, must satisfy the strict splice inequalities (143.3) or directly prove (144.2). The weighted insertion class needs the same verification. A positive conditioning denominator then follows from the raw LLT on central windows by Proposition 228.1.

The resulting dependency chain is exact: the geometry and periodic coercivity, the genuine exponential return tail, the common finite-record charts, and the conditioned unfinished-block

estimate are proved; the continuous collision Green–Kubo formula, actual Gaussian laws, functional limits, and measurable zero-variance cohomology are also proved. The marked moment laws and full joint nondegeneracy are proved as well. Quantitative induced phase reconstruction or a direct complete complementary-frequency estimate, together with the global raw residual and weighted bounds, remains necessary to instantiate the full density theorem. No complete unconditional raw LLT or full A3 density input is asserted before those premises are verified. The collision spectral theory is used at its stated smooth-multiplier scope for killing and for the compensated Gaussian limit. No spectral theorem about the unbounded induced observables is imported.

Acknowledgment and reproducibility. AI-assisted analysis contributed to this manuscript. The supplementary source audit fixes the mathematical baseline, the referee report and all imported source files; the response describes the revisions to each numbered request. Finite computations check algebra and explicit orbit margins and do not replace the analytical proofs. Independent human specialist review is not claimed by this manuscript.

Appendix A. Uniform variation and the collision-space embeddings

We give the geometric and norm details used in Section 138. Variation is always taken in initial collision coordinates. The statements below do not concern variation of a density after a many-collision pushforward.

Lemma A.1 (A finite set of possible next centers)**.** *There is a finite subset $\mathcal{D}$ of the triangular lattice such that every regular next collision, for every $R \in I$, is with a disk centered at a member of $\mathcal{D}$. The cardinality and the set may be chosen independently of the initial collision state.*

Proof. Let H be a uniform upper bound for the free flight, as supplied by the finite-horizon geometry in Lemma 133.1. A flight from $q = Rn_\alpha$ to the disk centered at d ends at $q' = d + Rn'$ and has length at most H. Hence $|d| \le |q| + |q' - q| + |q' - d| \le H + 2R_{\max}$. The intersection of the lattice with this fixed ball is finite. For example the open disks of radius $1/2$ about its lattice points are disjoint, and lie in a ball of radius $H + 2R_{\max} + 1/2$; area comparison gives a bound of $4(H + 2R_{\max} + 1/2)^2$ for their number. Removing the initial disk or inadmissible centers only decreases this bound. □

Proposition A.2 (Uniform semialgebraic slicing)**.** *+Let Λ be a semialgebraic parameter set and $Q = (a, b) \times (c, d)$ a bounded rectangle. Suppose $u : \Lambda \times Q \to \mathbb{R}$ is semialgebraic and $|u| \le M$. There is an integer N, depending on the family but not on $\lambda \in \Lambda$, such that $u_\lambda \in \mathrm{BV}(Q)$ and*

$$|D_1 u_\lambda|(Q) \le 4NM(d - c), \qquad |D_2 u_\lambda|(Q) \le 4NM(b - a). \tag{A.1}$$

The zero extension outside Q has bounded variation as well, with an additional bound $2M[(b-a)+(d-c)]$ for the sum of its two variations. The assertions depend only on the almost-everywhere class of the functions, not on chosen values on lower-dimensional sets.

Proof. Apply differentiable cell decomposition to the graph of u, ordering (λ, y) before x, and refine the resulting cells by the sign of the derivative in x wherever that derivative is defined. Uniform finiteness and one-dimensional monotonicity give a single bound N for the number of monotone intervals and exceptional points in every fiber. These are the parameterized cell and monotonicity theorems of [9, Sections 2.1–2.2 and 6.2]. On a monotone interval the variation is at most $2M$. Jumps between intervals are also at most $2M$. Isolated assigned values are immaterial for distributional variation. Thus the variation of the almost-everywhere one-dimensional fiber is bounded by $4NM$, after enlarging N if necessary.

For $\phi \in C_c^1(Q)$, Fubini's theorem and the one-dimensional integration-by-parts inequality now give

$$\left| \int_Q u_\lambda \, \partial_x \phi \, \mathrm{d}x \, \mathrm{d}y \right| \le 4NM(d-c)\|\phi\|_\infty.$$

This bounded functional defines a finite distributional derivative measure, proving the first estimate. Reorder the parameters as (λ, x) before y and repeat the argument for the second. Taking the larger of the two finite fiber bounds gives the stated single N. Extending each fiber by zero adds at most two endpoint jumps, each of size at most M. The same Fubini argument proves the claimed zero-extension bound. In particular no bound on a derivative in the parameter is needed. □

Proposition A.3 (Angular chart gluing for the physical record)**.** *The one-collision record f_R has a uniformly bounded supremum norm and flat-coordinate variation on M. Adding any fixed finite union of moving rectangular section indicators with uniformly bounded perimeters preserves these bounds. This proves the variation assertions for h_R and s_R in Lemma 138.1.*

Proof. Choose a fixed finite angular atlas on which $t = \tan((\alpha - \alpha_0)/2)$ lies in a bounded interval, and choose a smooth partition of unity with supports strictly inside these charts. The chart maps and their inverses have bounded first derivatives. The components of n_α and t_α are rational in t, and $\sqrt{1-p^2}n_\alpha + pt_\alpha$ is semialgebraic in (t,p). Lemma A.1 reduces the next collision to finitely many candidate roots. The conditions for a root to be real, positive, and the first admissible entry, and all comparisons between two such roots, are semialgebraic. Therefore its minimum and the finite lattice label define bounded semialgebraic families in (R,t,p). Ties and singular trajectories can be assigned a fixed bounded value. They do not change the almost-everywhere record.

Proposition A.2 gives a uniform variation bound on each chart. Multiplication by a fixed smooth partition function bounds the variation by its supremum times the original variation plus its derivative supremum times the original L^1 norm. A bounded smooth change of the angular variable preserves a variation bound with a constant depending only on the chart and its inverse. For completeness, this follows for smooth functions by the chain rule and change of variables, and for a bounded-variation function by convolution on a slightly smaller chart followed by lower semicontinuity of distributional variation. The cutoff has support in that smaller chart. There is therefore no unaccounted jump on the chart boundary. The sum of the finitely many localized functions gives the asserted bound on the cylinder. At $p = \pm 1$, the fiber proof already includes finite one-sided traces; no inverse transformation from p to the angle of incidence is used.

A rectangle indicator has variation equal to its coordinate perimeter, up to the fixed normalization of ν. The variation of the indicator of a finite union is bounded by the sum of these perimeters, including rectangles represented across the angular seam. Its moving center does not change that estimate. The bounded coefficients in (138.1) finish the assertion for h_R and s_R. □

Lemma A.4 (Smooth densities and test pairings in the local spaces)**.** *On each local space $(\mathcal{B}_j, \mathcal{B}_{w,j})$ used in Lemma 133.1, smooth functions a, g, b satisfy, uniformly for the radii in that neighborhood,*

$$\|a\nu\|_{\mathcal{B}_j} + |a\nu|_{w,j} \le C\|a\|_{C^1}, \tag{A.2}$$

$$\|M_g V\|_{\mathcal{B}_j} \le C\|g\|_{C^2}\|V\|_{\mathcal{B}_j}, \qquad |\ell_j(M_b V)| \le C\|b\|_{C^2}\|V\|_{\mathcal{B}_j}. \tag{A.3}$$

The function norms are in the collision coordinates used to define the local spaces. In particular, for the mean-preserving smoothing,

$$\|(\mathsf{S}_\delta a)\nu\|_{\mathcal{B}_j} \le C\delta^{-2}\|a\|_\infty. \tag{A.4}$$

Here a is a density relative to ν, not relative to unweighted area in the incidence-angle coordinates.

Proof. We record the norm identification explicitly. In [4, Sections 3.2–3.3, (3.11)–(3.13)], a smooth function is identified with its density times the invariant reference measure $\mu = c\cos\varphi\,\mathrm{d}r\,\mathrm{d}\varphi$. Under the local boundary reparametrization $r = R_0\alpha$, this probability is precisely ν. Thus the smooth function representing $a\nu$ is a; an additional factor of $\cos\varphi$ is not inserted into that function.

For the weak norm, each stable-curve integral is bounded by $C\|a\|_\infty$ because the stable curves have uniformly bounded length and the test functions have bounded supremum. For the strong stable norm, tests on a curve W have supremum at most $|W|^{-\alpha}$. The integral is therefore at most $\|a\|_\infty|W|^{1-\alpha}$, uniformly bounded since $\alpha < 1$. For the strong unstable norm, take two matched graph curves at distance at most ϵ in the source metric. Their unmatched parameter intervals have length $O(\epsilon)$. On their common interval the graph coordinates and arclength factors differ by $O(\epsilon)$, and the matched tests differ in supremum by at most ϵ. The difference of the two integrals is consequently bounded by $C\|a\|_{C^1}\epsilon$. Dividing by ϵ^β is harmless, since the source exponent satisfies $\beta < 1$. These three estimates prove (A.2), with constants uniform in the finite local cover and its fixed charts.

For a multiplier, stable and weak tests are simply multiplied by the restriction of g to their curves. In the matched-curve term, the two restrictions differ in C^q by at most $C\|g\|_{C^2}\epsilon^{1-q}$, by interpolation of their supremum and Lipschitz bounds. The admissible source exponents have $\beta < 1 - q$. Splitting into matched tests and this difference gives the first bound in (A.3); it is the multiplier estimate already used in Lemma 133.1. Applying the bounded mass functional proves the second. Finally (138.3) bounds the smoothed C^2 norm by $C\delta^{-2}\|a\|_\infty$. Composition with $p = \sin\varphi$ uses only bounded derivatives of this forward map. This proves (A.4). □

The estimates above concern smooth functions on the local collision spaces. They do not make the hard section projection a bounded multiplier on those spaces, nor do they realize an unbounded induced twist. The use of that projection in the compensation argument remains entirely on the level of ordinary functions and smooth approximations.

Appendix B. Chronological spectral products and coarse interpolation

This appendix supplies explicit product and interpolation estimates for Section 140. It also separates the treatment of short increments from the long-block spectral argument.

Lemma B.1 (Products of differently twisted collision powers)**.** *Fix an integer $q \ge 1$. Use one local collision space, a common smoothing scale δ, and real vectors $z_1, \dots, z_q$ with $|z_j| \le c\delta^2$. Let $m_j \ge 1$ and assume*

$$\delta^{-6}\sum_{j=1}^{q} m_j|z_j|^3 \le c.$$

Write $L_j = \mathcal{L}_{R,\delta,z_j}$ and let a be a complex initial insertion with bounded supremum and variation norms. Then

$$\left|\ell_j(L_q^{m_q}\cdots L_1^{m_1}(\mathsf{S}_\delta a)\nu) - \left(\int a\,\mathrm{d}\nu\right)\exp\left(-\tfrac{1}{2}\sum_{j=1}^{q} m_j z_j^{\mathsf{T}}\Gamma_{R,\delta}z_j\right)\right|$$
$$\le C_q\left\{\delta^{-4}\sum_{j=1}^{q}|z_j| + \delta^{-6}\sum_{j=1}^{q} m_j|z_j|^3 + \delta^{-2}\sum_{j=1}^{q}\rho^{m_j}\right\}. \tag{B.1}$$

The index on ℓ_j names the chosen local space and is fixed in the formula. The powers are in physical chronological order: the earliest increment acts first, on the right.

Proof. Write $L_j^{m_j} = \lambda_j^{m_j}\Pi_j + E_j$. Lemma 139.1 gives $\|\Pi_j\| \le C$, $\|E_j\| \le C\rho^{m_j}$ and $\|\Pi_j - \Pi_R\| \le C\delta^{-2}|z_j|$. The real-frequency quadratic terms in $m_j \log \lambda_j$ are nonpositive. The assumption on the sum of the cubic remainders bounds each principal scalar power and every product of these powers by a fixed constant. Expand the product into its 2^q terms. Every term with a complementary factor is bounded, after evaluation, by a constant times $\delta^{-2}\rho^{m_j}$ for at least one of its complementary indices. The factor δ^{-2} comes only from the initial vector, through (A.4); the other projectors and powers have uniform bounds. Summing the finitely many terms gives the last term of (B.1).

For the all-principal term, telescope the product of projectors about Π_R, retaining their order. Since $\Pi_R^2 = \Pi_R = \nu\ell_j$,

$$\|\Pi_q \cdots \Pi_1 - \Pi_R\| \le C_q\delta^{-2}\sum_j |z_j|.$$

Evaluation on the initial vector gives the first error in (B.1). Its zero-frequency amplitude is $\int a \,\mathrm{d}\nu$. The logarithmic expansion and $|e^w - 1| \le |w|e^{|w|}$ give the second error. The principal scalar powers remain bounded during this last comparison. This proves the claim without introducing a factor δ^{-2q}. □

Corollary B.2 (Finite-dimensional scales, zero blocks, and short blocks)**.** *For fixed distinct times $0 = t_0 < t_1 < \cdots < t_q$ and fixed vectors v_j, put $m_j = \lfloor nt_j \rfloor - \lfloor nt_{j-1} \rfloor$, $z_j = v_j/\sqrt{n}$, and $\delta_n = \frac{1}{4}n^{-1/32}$. The spectral product error is bounded by*

$$C_q\left\{\delta_n^{-4}n^{-1/2} + \delta_n^{-6}n^{-1/2} + \delta_n^{-2}\sum_j \rho^{m_j}\right\} = o(1).$$

Repeated times give zero blocks, which may be omitted exactly. A fixed number of additional deterministic increments of lengths $m_j = o(n)$ may instead be removed from a real characteristic exponent at a cost $C\sum_j |v_j|\sqrt{m_j/n}$ for the original collision observable. The estimates are uniform for initial probability densities with uniformly bounded supremum and variation norms. The short-increment estimate uses only the supremum bound.

Proof. Distinct fixed times make all m_j positive multiples of n for large n. Substitute the scales in (B.1). Smoothing the initial density costs $O(\delta_n)$; smoothing the covariance costs $O(\delta_n\ell_{\delta_n})$ by (138.7). The residual-path bound (140.2) removes smoothing from the original finite-dimensional process. Integer parts in the covariance time coefficients cost $O(n^{-1})$. This gives precisely the independent Gaussian increments in Theorem 140.2.

When an interval has length zero its operator is the identity, not a principal spectral projector, so it is omitted before this expansion. For a deterministic short interval, uniform collision correlation summability bounds the second moment of its original centered sum by Cm_j, from every deterministic starting index under a bounded initial density. The difference of the two characteristic exponents, with the other increments left at their original times, is therefore bounded in L^1 by $C|v_j|\sqrt{m_j/n}$. The elementary exponential difference bound proves the last assertion. This argument does not use a complementary spectral power for a short block, and does not move the other increments in physical time. □

Lemma B.3 (The complex bound before the fourth derivative)**.** *For one real component u of $h_{R,\delta}$, the stationary pairing has, on a complex disk $|z| < c\delta^2$, a decomposition*

$$\int e^{izS_{m,R}u}\,\mathrm{d}\nu = e^{m\psi(z)}A(z) + E_m(z)$$

with $|A(z)| + |\psi(z)| \le C$ and $|E_m(z)| \le C\rho^m$, uniformly on a slightly smaller such disk. Consequently $|E_m^{(4)}(0)| \le C\delta^{-8}\rho^m$.

Proof. The two fixed resolvent contours in Lemma 139.1 give uniformly bounded spectral projections and complementary powers throughout the complex ball, not just on its real diameter. Both the vector ν and the mass functional have bounded norms. Evaluating the complementary power between them gives the displayed complex bound for E_m. The eigenvalue lies in a fixed disk about one not meeting zero, so its chosen logarithm and the principal amplitude are uniformly bounded. Cauchy's estimate on a circle of radius proportional to δ^2 now yields the fourth-derivative estimate. There is no need to bound the principal power uniformly for complex z and arbitrary m. Its derivative is calculated by the product rule, as in Lemma 140.1. □

Proposition B.4 (Coarse polygonal interpolation)**.** *Let (u_k) be bounded real-vector random variables, $|u_k| \le M$. Let $L \ge 1$ be an integer and suppose, under the probability in question, that every deterministic interval of $m \ge L$ indices has*

$$\mathbb{E}\left|\sum_{k=r}^{r+m-1} u_k\right|^4 \le Cm^2.$$

At times jL/n, interpolate the sums normalized by $\sqrt{n}$ to obtain Z_n. Let X_n be the polygonal interpolation using every index. Then, with constants independent of L, n,

$$\mathbb{E}|Z_n(t) - Z_n(s)|^4 \le C'|t-s|^2, \qquad \|Z_n - X_n\|_\infty \le 2ML/\sqrt{n}. \tag{B.2}$$

A final incomplete coarse interval is defined by extending one full block beyond the observation horizon and restricting the interpolation.

Proof. Write $\Delta = L/n$. For a segment of duration $h \le \Delta$ inside one block, linear interpolation multiplies the full-block increment by h/Δ. Its fourth moment is at most $C\Delta^2(h/\Delta)^4 \le Ch^2$. For an interval spanning two adjacent blocks, apply this estimate separately to the durations h_1, h_2 and use the L^4 triangle inequality. Since $(\sqrt{h_1} + \sqrt{h_2})^4 \le 4(h_1 + h_2)^2$, the same form holds.

For a general interval, split it into its first fractional block, its complete-block middle interval, and its last fractional block. The middle interval, when nonempty, contains at least L indices, so the hypothesis gives the required fourth moment for its total duration. The L^4 triangle inequality and $(\sqrt{h_1} + \sqrt{h_2} + \sqrt{h_3})^4 \le 9(h_1 + h_2 + h_3)^2$ prove the first bound with one fixed constant. Empty pieces are simply omitted. Within one coarse block both interpolations remain within $ML/\sqrt{n}$ of the sum at the left endpoint, proving the deterministic second bound. The endpoint convention uses a full block, so these same estimates apply on a fixed observation interval without a new constant. □

Applying this proposition with $u_k = h_{R,\delta_n} \circ T_R^k$ and $L_n = \lceil \delta_n^{-8} \rceil$ gives the interpolation estimates in Theorem 140.2. Indeed (140.1) is bounded by Cm^2 for $m \ge L_n$. Stationarity gives the interval estimates under ν, and bounded initial densities give them by domination. The fourth-moment constants are uniform. Together with the residual maximal estimate this proves tightness of the original process, not just of its coarse approximation.

Lemma B.5 (Continuity of the covariance square root)**.** *On every bounded set of positive semidefinite matrices, the map $C \mapsto C^{1/2}$ is continuous in operator norm. Consequently $R \mapsto \mathrm{Law}(\mathcal{W}_{\Gamma_R})$ is continuous on continuous path space, including at singular matrices.*

Proof. Uniformly approximate the scalar square-root function on a bounded interval containing all the spectra by polynomials. The spectral theorem bounds the matrix approximation by the same uniform scalar error, and each polynomial in the matrix is norm-continuous. This proves

the first claim. On one probability space take a standard four-dimensional Brownian motion W and set $\mathcal{W}_C = C^{1/2}W$. On any compact time interval its sample paths are bounded almost surely. Norm continuity of the square root therefore gives almost-sure uniform convergence of the coupled paths and hence convergence of their laws. $\square$

Appendix C. Locations of the collision norm definitions

The following map fixes the source conventions in Lemma A.4. Locations refer to the 45-page preprint of [4]; the last column records the deduction used here, not an additional theorem attributed to that source.

Object	Source location	Use in this article
Relative density	Section 3.2; Section 3.3, footnote 7	A smooth a represents $a\nu$, without another cosine factor.
Weak norm	Equation (3.11)	Bounded stable-curve integrals.
Stable norm	Equation (3.12)	The test normalization gives $\|a\|_\infty \lvert W\rvert^{1-\alpha}$.
Unstable norm	Equation (3.13)	Matched curves give $C\|a\|_{C^1}\epsilon^{1-\beta}$.
Exponent restrictions	Section 3.3, before (3.12)	$\beta \le p-q$, $p \le 1/3$, hence $\beta < 1-q$.
Mass functional	Lemma 3.4, with test 1	$\lvert\ell(V)\rvert \le C\lvert V\rvert_w$.
Complementary powers	Theorem 2.1(3)	Local uniform spectral power bound.

Smooth multiplication is proved in Lemma 133.1 from these definitions. The changing boundary length is handled by Remark 2.7(b) of the source. The table concerns collision-space distributions. It neither embeds the hard induced twist nor estimates second derivatives of raw densities.

Appendix D. Dependencies and the remaining raw estimates

This guide distinguishes the completed input theorems, the two exact residual constructions and the remaining pointwise local theorem. The covariance is D_R throughout; the physical projection is $\mathcal{V}_R$. All finite-frequency reconstructions use the original record, but are not Gaussian evaluations of their finite integrals.

Stage	Established result	Interface still required
Collision and return statistics	Gaussian and functional laws, fixed moments, actual covariance and phase-defect results	Their specified frequency and regularity budgets must be respected.
Prescribed-count Fourier bands	Theorems N–P: isotropic, coordinate and shape-adaptive regions	The full fixed-count complement is not covered by those regions or by averaged-orbit bounds.
Coherent raw correction	Theorem Q: exact and quantitative transport between cutoffs; Theorem R: signed window control	The common pointwise correction itself still needs a local bound.
Power–logarithm residual	Exact finite-record preparation and $W^{2,1}$ residual	Its original long-time jet budget is not supplied by the different geometric residual.
Geometric residual	Theorem V: all-label second-derivative budget at linear collision cutoff; Proposition 224.4: actual entrance-bias extension	Removed mass is small, but its density is not known to be uniformly small on central windows.
Positive reconstruction	Theorem W: mixed L^1 approximation and source-TV microscopic event approximation, uniform in bounded selectors	Evaluate the remaining finite integral; a positive likelihood alone does not provide a Gaussian denominator.
Physical conditioning	Theorems S–U: mesoscopic denominators, relative comparison and Gaussian bridges	The original microscopic physical-event comparison requires its own raw-scale denominator.

The exact pointwise responsibility is recorded in (223.10). The far-roof inverse and the convolution of the discarded measure have quantitative bounds. Its remaining signed terms are the discarded density and the finite complementary inverse. Their joint local essential supremum, or an equivalent complete raw inversion estimate, has not been derived from the total-variation reconstruction. The geometric bandwidth satisfies $\log B_n = O(n \log n)$; it is not merely of size $n \log n$.

For weights, three statements must not be interchanged. Finite-product BV return marks admit the geometric extraction. The inherited central Fourier theorems have their own insertion classes. The positive event formula in Theorem W permits every bounded measurable trajectory selector because its error is measured on the original source, not because all such selectors are smooth multipliers. Its normalized bound contains the actual selected denominator, or the explicit finite-band lower-bound test of Corollary 225.5.

The section law and the stationary entrance law are also different. The latter uses the original roof bias and elapsed age in (224.3). It is not replaced by the bounded current-collision marginal. Finally, collision correlation summability and induced Cesaro covariance convergence remain distinct. The norm imports are in Appendix C; the new boundary and positivity arguments are in Sections 224 and 225.

Appendix E. Complementary statements for the actual return record

The statements below collect the return-record conclusions and the intermediate inversion formulas used in the article. Their hypotheses and proofs are unchanged. Theorem 1 now evaluates the complete stationary physical probability in Theorem X; the four-coordinate return-density problem remains a distinct endpoint. The letters A–X retain the cross-reference convention used in the geometric and probabilistic sections.

The physical record. Let T_R be the collision map, Y_R^* the section constructed below, and F_R^* its actual first-return map. Write $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ for the four-coordinate record of n returns, with the two coordinates of $K_{n,R}$ displayed separately. Its law is taken under the normalized collision measure ν_R^*. The following synopsis records the unconditional probabilistic conclusions. The geometry, normalization, and limiting matrices are specified intrinsically in the cited theorems.

Theorem A (Gaussian laws of the physical record)**.** *For $R \in I = [0.45, 0.47]$ set*

$$c_* = \frac{91}{10000\pi}, \qquad \bar\tau_R = \frac{\sqrt{3}/2 - \pi R^2}{2R}, \qquad \bar G_R = (0, 0, c_*^{-1}, \bar\tau_R/c_*).$$

There is a continuous positive semidefinite matrix D_R such that, for every $K < \infty$ and $n \ge 2$,

$$\sup_{R\in I,\ |t|\le K} \left| \int e^{it\cdot(J_{n,R} - n\bar G_R)/\sqrt{n}}\, \mathrm{d}\nu_R^* - e^{-t^{\mathsf T} D_R t/2} \right| \le C_K n^{-1/26}\sqrt{\log(2+n)}.$$

The polygonal centered return processes converge, uniformly in R, to Brownian motion with covariance D_R. The stationary physical processes of lattice displacement and centered collision count also satisfy a parameter-uniform functional Gaussian limit.

More precisely, $D_R = c_*^{-1}\Gamma_R$, where Γ_R is the absolutely convergent collision Green–Kubo matrix of $h_R = f_R - \bar G_R \mathbf{1}_{Y_R^*}$. Its kernel consists exactly of directions v for which $v \cdot (G_R - \bar G_R)$ is an L^2 coboundary for F_R^*. No nondegeneracy assumption is used in these conclusions.

Proof. Theorems 138.4, 139.5, 139.6, and 140.3, together with Corollary 140.4, prove the assertions. □

Theorem B (An integrated central band)**.** *For the same actual return record and the matrix D_R in Theorem A,*

$$\sup_{R\in I} \int_{|v|\le 2n^{1/200}} \left| \int e^{iv\cdot(J_{n,R} - n\bar G_R)/\sqrt{n}}\, \mathrm{d}\nu_R^* - e^{-v^{\mathsf T} D_R v/2} \right| \mathrm{d}v \le C n^{-3/280}\sqrt{\log(2+n)}.$$

The variable v is rescaled: the corresponding physical frequencies have size at most $2n^{-1/2+1/200}$. The same estimate holds for complex initial insertions whose zero extensions to the collision cylinder have uniformly bounded supremum and variation norms, with the Gaussian multiplied by the integral of the insertion.

Proof. Apply Theorem 145.3; the original probability is the admissible insertion $a_R = s_R$. □

Complex insertion convention. In Theorem B the insertion is a density with respect to the collision probability ν. Its amplitude $\alpha_R = \int a_R \,\mathrm{d}\nu$ may be complex; the associated pushforward is a finite complex measure, not in general a probability. The original section probability corresponds to $a_R = s_R = c_*^{-1}\mathbf{1}_{Y_R^*}$.

Theorem C (A mark at any actual return)**.** *Let a be a complex function on the collision cylinder, zero outside Y_R^*, and put $M_a = \|a\|_\infty$, $V_a = \|a\|_{\mathrm{BV}}$, and $\alpha_a = \int a \,\mathrm{d}\nu$. For $0 \le k \le n$ set*

$$C_{n,R}^{[k],a}(v) = \int_{Y_R^*} a((F_R^*)^k x) e^{iv\cdot(J_{n,R}(x) - n\bar G_R)/\sqrt{n}}\, \mathrm{d}\nu(x).$$

There is a constant C, independent of R, n, k, a, such that, for $n \geq 2$,

$$\int_{|v|\leq 2n^{1/200}} \left| C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf{T}} D_R v/2} \right| \mathrm{d}v \leq C \left(M_a n^{-3/280} \sqrt{\log(2+n)} + V_a n^{-9/175} \right).$$

Thus the same central band accommodates initial, terminal, and one intermediate return-state insertion, including a mark whose index varies with n. The displayed transform is integrated against the unnormalized restricted collision measure $\nu|_{Y_R^*}$; division by c_* gives the corresponding transform under the section probability ν_R^*. No positivity of a or nonsingularity of D_R is assumed.

Proof. This is Theorem 146.5. □

Theorem D (Actual second moments and a marked quadratic law)**.** *Put $U_{n,R} = J_{n,R} - n\bar{G}_R$. Uniformly for $R \in I$,*

$$\int |U_{n,R}|^4 \, \mathrm{d}\nu_R^* \leq Cn^2, \qquad \left\| \frac{1}{n} \int U_{n,R} \otimes U_{n,R} \, \mathrm{d}\nu_R^* - D_R \right\| \leq Cn^{-1/6}.$$

The count component satisfies $e_3^{\mathsf{T}} D_R e_3 \geq d_N > 0$ uniformly in R, where $e_3 = (0,0,1,0)$. For the complex insertion and normalization of Theorem C,

$$\left\| \int_{Y_R^*} a((F_R^*)^k x) \frac{U_{n,R}(x) \otimes U_{n,R}(x)}{n} \, \mathrm{d}\nu(x) - \alpha_a D_R \right\| \leq C \left(M_a n^{-1/6} + (M_a + V_a) n^{-1} \right), \qquad 0 \leq k \leq n.$$

The matrix D_R is also the uniform limit of the induced Green–Kubo formula with Cesaro weights. No induced mixing or absolute convergence of its correlation series is assumed.

Proof. Apply Theorems 147.4 and 147.6, and Corollaries 147.7 and 148.1. □

Theorem E (Uniform nondegeneracy of the joint record)**.** *The covariance of the same actual four-coordinate return record satisfies*

$$d_0 I_4 \leq D_R \leq D_0 I_4 \qquad (R \in I)$$

for constants $0 < d_0 \leq D_0 < \infty$. Thus every nonzero real joint direction has positive Gaussian variance. For all sufficiently large n, the normalized actual covariance in Theorem D is at least $(d_0/2) I_4$, uniformly in R. The three-dimensional physical-time fluctuation covariance is uniformly positive definite as well.

Proof. Apply Theorem 150.2 and Corollary 150.4. □

In every subsequent use of the matrix from Theorems A–D, the stronger uniform positive definiteness of Theorem E is available. The positive semidefinite formulation of the earlier statements records their weaker logical input, not a remaining covariance hypothesis.

Theorem F (Physical phase arithmetic and quantitative defects)**.** *Let $\omega = (u, s, t) \in \mathbb{T}^2 \times \mathbb{T} \times \mathbb{R}$ and $\phi_{R,\omega} = u \cdot \kappa_R + s + t\tau_R$ on the collision cylinder. A measurable circle-valued solution of $q \circ T_R = e^{i\phi_{R,\omega}} q$ exists exactly when $\omega = 0$; in that case it is constant almost everywhere.*

For each fixed R and compact nonzero frequency set $\mathcal{K}$, there is $c_{R,\mathcal{K}} > 0$ such that, for $H \geq 2$,

$$\inf_{\omega \in \mathcal{K}} \|q \circ T_R - e^{i\phi_{R,\omega}} q\|_1 \geq \frac{c_{R,\mathcal{K}}}{1 + \log H} \quad (|q| = 1, \ \|q\|_{\mathrm{BV}} \leq H).$$

For complex f with $\|f\|_2 = 1$ and $\|f\|_\infty + \|f\|_{\mathrm{BV}} \leq H$, the corresponding bound is

$$\inf_{\omega \in \mathcal{K}} \|f \circ T_R - e^{i\phi_{R,\omega}} f\|_2 \geq \frac{c_{R,\mathcal{K}}}{(1 + \log H)^2}.$$

No lower bound on $|f|$ is assumed. For fixed H and $\mathcal{K}$, the second infimum is also positive uniformly over $R \in I$. The latter assertion does not specify a uniform logarithmic rate in H.

Proof. Theorem 194.2 proves the exact assertion. Theorems 195.4, 195.6 and 195.8 prove the defect bounds. □

Theorem G (Near-origin defects for the actual return record)**.** *Set* $g_R = G_R - \bar{G}_R$ *and*

$$\Lambda = 1 + \log(H/r), \qquad K = \Lambda \log(2 + \Lambda), \qquad H \geq 2, \quad 0 < r < 1/2.$$

There are $a, c, r_* > 0$*, independent of* R*, such that* $r \leq r_*$ *and* $rK \leq a$ *imply the following conclusions for* $|z| = r$*. Every unit-modulus section function* q *whose zero extension has variation at most* H *satisfies*

$$\|q \circ F_R^* - e^{iz \cdot g_R} q\|_{L^1(\nu_R^*)} \geq cr^2.$$

Every complex section function f *with* $\|f\|_{L^2(\nu_R^*)} = 1$ *and* $\|f\|_\infty + \|f^0\|_{\mathrm{BV}} \leq H$ *satisfies*

$$\|f \circ F_R^* - e^{iz \cdot g_R} f\|_{L^2(\nu_R^*)} \geq cr^2/K.$$

The function may vanish. With $H \leq H_0 n^\zeta$*, these bounds apply uniformly on the physical annulus* $2n^{-99/200} \leq |z| \leq n^{-2/5}$ *for all sufficiently large* n*. The corresponding rescaled radii are* $2n^{1/200}$ *and* $n^{1/10}$*.*

Proof. Apply Theorems 197.4 and 197.5, and Corollary 197.6. □

The spectral parameter in Theorem G.. The displayed defect in Theorem G has unit spectral parameter for the centered return multiplier. A constant per return is not the same as a constant per collision. The next theorem treats both the physical frequency z and the return spectral phase ξ, with $\lambda = e^{i\xi}$, and strengthens the regularity constraint.

Theorem H (A joint peripheral neighborhood)**.** *For* $\xi \in [-\pi, \pi]$*, set*

$$\rho = (|z|^2 + \xi^2)^{1/2}, \quad \Lambda = 1 + \log(H/\rho), \quad K = \Lambda \log(2 + \Lambda).$$

There are $a, b, \rho_0 > 0$*, independent of* R*, such that* $0 < \rho \leq \rho_0$ *and* $\rho^2 K \leq a$ *imply*

$$\|q \circ F_R^* - e^{i(\xi + z \cdot g_R)} q\|_{L^1(\nu_R^*)} \geq b\rho^2$$

for every unit section phase with $\|q^0\|_{\mathrm{BV}} \leq H$*, and*

$$\|f \circ F_R^* - e^{i(\xi + z \cdot g_R)} f\|_{L^2(\nu_R^*)} \geq b\rho^2/K$$

for every normalized complex section function with $\|f\|_\infty + \|f^0\|_{\mathrm{BV}} \leq H$*. Zeros are allowed. These estimates include* $z = 0$*,* $\xi \neq 0$*. They are local in the joint parameters, not estimates over the full peripheral circle.*

For the physical annulus $2n^{-99/200} \leq |z| \leq n^{-2/5}$*,* $|\xi| \leq n^{-\gamma}$*, and any* $0 < \beta < \min\{4/5, 2\gamma\}$*, they apply to regularity budgets* $H \leq H_0 \exp(n^\beta)$ *for all sufficiently large* n*. The rescaled annulus is* $2n^{1/200} \leq |\sqrt{n}\, z| \leq n^{1/10}$*.*

Proof. Apply Theorem 198.5 and Corollary 198.6. □

Theorem 199.3 converts these physical defects into a lower singular-value bound for finite-rank orthogonal compressions of the actual return operator. The reconstruction has supremum and variation budget C/h at mesh h; no positive lower modulus is assumed. Proposition 199.5 separately bounds the difference between the compressed powers and the unmodified characteristic function. A mesh-uniform spectral limit and the complete complementary-frequency integral are not inferred from that finite-time comparison.

Theorem I (Finite-count extraction of the raw density)**.** *Fix $R \in I$, $n \geq 1$ and a collision-count cutoff L. For every bounded finite-record subanalytic weight in Definition 200.1, the restricted actual law*

$$\mu_{n,R}^{w,L} = (J_{n,R})_*(w\mathbf{1}_{\{N_{n,R}\leq L\}}\nu_R^*)$$

has an exact decomposition $\mu_{n,R}^{w,L} = E^L + Q^L$. The extracted density is a finite sum of cutoff one-sided terms $x^\alpha(\log x)^k$, with $\alpha \in \mathbb{Q}$, $-1 < \alpha \leq 1$, and k a nonnegative integer. It includes all singular boundary contributions to the finite-count law. Each lattice component of the residual density r^L belongs to $W^{2,1}(\mathbb{R})$. In particular, if $A_2 = \sum_\ell \|\partial_t^2 r^L(\ell,\cdot)\|_1$, then

$$\widehat{Q}^L \in L^1(\mathbb{T}^3 \times \mathbb{R}), \qquad \int_{\mathbb{T}^3\times\{|b|>B\}} |\widehat{Q}^L|\,\mathrm{d}\omega \leq 2(2\pi)^3 A_2/B.$$

Here A_2 is finite at each fixed (n, L, R, w); no uniform long-time bound for this quantity is part of the assertion.

For the original unweighted law, choose $L_n = \lceil \lambda n \rceil$ with $\lambda > c_*^{-1} + 1$. Its density agrees exactly with the restricted density on every fixed central window for large n. The finite extraction gives the raw inversion identity (201.3) *on that window. The high-count measure enters only through a low-frequency convolution, whose normalized central contribution is $O(n^{-P})$ for every $P > 0$. The physical cutoff is $2n^{-99/200}$, or $2n^{1/200}$ after rescaling.*

Proof. Apply Theorem 200.7, Lemma 201.1, and Theorem 201.2. □

Theorem I supplies the complete finite-count residual, rather than only an initial-coordinate variation bound. Corollary 201.3 displays the remaining uniform residual-growth, complementary-integral and local-edge estimates for the original raw density.

Theorem J (Exponential reconstruction and multiple observations)**.** *The finite-record variation budget may be taken as Ce^{CL}. Consequently, with $\rho = (|z|^2 + \xi^2)^{1/2}$ and $\Lambda = 1 + \log(H/\rho)$, the joint return spectral estimates of Theorem H strengthen to*

$$\|\mathcal{E}_{R,z,\xi} q\|_1 \geq b\rho^2, \qquad \|\mathcal{E}_{R,z,\xi} f\|_2 \geq b\rho^2/\Lambda$$

whenever $0 < \rho \leq \rho_0$ and $\rho^2\Lambda \leq a$. Here q is a unit section phase of zero-extension variation at most H, while f is a normalized complex section function with supremum plus zero-extension variation at most H. The constants are uniform in R, and no positive lower modulus of f is required.

There are constants $a_*, \theta_*, \beta_* > 0$ for the original family with the following property. Let $m_n \leq a_* \log n$ and choose any $0 \leq \ell_1 < \cdots < \ell_q \leq m_n$. Let $E_j \subset Y_R^*$ be observation sets whose zero-extended indicators have total variation at most a fixed power of $\log(2+n)$. If the unchanged event

$$A_{n,k,R} = \bigcap_{j=1}^{q} \{x : (F_R^*)^{k+\ell_j} x \in E_j\}, \qquad 0 \leq k \leq n - m_n,$$

has probability at least $p_0 n^{-\beta_*}$, then, for $U_{n,R} = J_{n,R} - n\bar{G}_R$,

$$\sup_{R,k} \int_{|v|\leq 2n^{\theta_*}} \left| \mathbb{E}[e^{iv\cdot U_{n,R}/\sqrt{n}} \mid A_{n,k,R}] - e^{-v^{\mathsf{T}} D_R v/2} \right| \mathrm{d}v \longrightarrow 0.$$

Its conditional normalized mean tends to zero and its conditional covariance tends to D_R, uniformly. All expectations use ν_R^*. The physical cutoff is $2n^{-1/2+\theta_*}$. Explicit constants and more general probability, variation and frequency tradeoffs are given in (203.7) *and* (203.10).

Proof. Theorems 202.2 and 202.4 prove the first part. Corollary 203.4 and Theorem 203.5, with (203.10), prove the remaining assertions. □

The single-logarithm estimate is local in the joint physical and return spectral parameters; its exact lifting identity remains valid at every return phase. The growing observation window is a new class, not an assumed BV bound on a long pullback. Its smaller positive central band is additional to the unchanged $n^{1/200}$ band of Theorem C. Neither statement replaces the complementary-integral or local-edge estimates for the raw density. The original multiple-time event is never exchanged for an event with a collision cutoff.

Theorem K (Direct annular averages for the uncompressed record)**.** *Let $\mathcal{U}_{R,z} f = e^{-iz\cdot g_R} f \circ F_R^*$ on $L^2(\nu_R^*)$. For all sufficiently large n, all $T \geq n$, $N \geq 0$, $a \in L^2(\nu_R^*)$, and physical frequencies*

$$2n^{-99/200} \leq |z| \leq n^{-2/5}, \qquad 2n^{1/200} \leq |\sqrt{n}\, z| \leq n^{1/10},$$

one has, uniformly in R,

$$\frac{1}{T}\sum_{j=0}^{T-1} \left| \int a e^{-iz\cdot U_{N+j,R}} \, \mathrm{d}\nu_R^* \right|^2 \leq C n^{-4/495} \|a\|_2^2,$$

$$\sup_{\xi\in\mathbb{R}} \left\| \frac{1}{T}\sum_{j=0}^{T-1} e^{-ij\xi} \mathcal{U}_{R,z}^{N+j} \mathbf{1} \right\|_2^2 \leq C n^{-4/495}.$$

For an unchanged event A of probability $p > 0$, the average of the squared conditional characteristic functions is at most $Cp^{-1}n^{-4/495}$. No BV condition on $\mathbf{1}_A$ is needed.

If $\varsigma_{R,ru}$ is the spectral probability of $\mathbf{1}$ for $\mathcal{U}_{R,ru}$, where $|u| = 1$, its principal angle divided by r^2 converges in law, uniformly in R, u, to the Cauchy law of scale $u^{\mathsf{T}} D_R u/2$.

Proof. Theorems 204.5 and 205.1, with Corollary 204.6, prove the averaged bounds. Theorem 205.5 proves the spectral conclusion. □

The prescribed-count target. Theorem K concerns averages in the return count. The raw local theorem instead requires, at the prescribed n, an estimate of the form $n^2 \int |1-\chi_n|\, |H_{n,R}| \, \mathrm{d}\omega = o(1)$ for the appropriate edge-subtracted residual, together with its local extracted-edge term. The averaged bound does not imply this estimate: its existing annular normalization gives only $n^{2/5-2/495}$ after Cauchy–Schwarz and the raw factor n^2.

The common lag scale is $m(|z|) = \lfloor |z|^{-200/99} \rfloor < n/4$. It requires no spatial mesh, and no BV reconstruction of a time-n vector. The estimates are time averages or cyclic spectral statements, not decay at each specified return count. The physical record still has the Gaussian law in Theorem A; the Cauchy law here concerns a different, explicitly identified spectral probability. The Cauchy characteristic convergence is for fixed rescaled spectral test frequency. It supplies no uniform estimate at $t = n|z|^2$ when that quantity grows across the physical annulus.

Theorem L (Fixed-return cancellation and a larger central band)**.** *For the insertion and measure conventions of Theorem C, let $\Phi_{n,R}^{[k],a}(z) = \int_{Y_R^*} a((F_R^*)^k x) e^{iz\cdot J_{n,R}(x)} \, \mathrm{d}\nu(x)$. Uniformly for $R \in I$ and $0 \leq k \leq n$,*

$$n^2 \int_{2n^{-99/200} \leq |z| \leq 2n^{-12/25}} |\Phi_{n,R}^{[k],a}(z)| \, \mathrm{d}z \leq C \left(M_a n^{-1/200} \sqrt{\log(2+n)} + V_a n^{-13/100} \right).$$

This holds at every prescribed sufficiently large return count, not only on average. Moreover

$$\int_{|v|\le 2n^{1/50}} |C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf T} D_R v/2}|\,\mathrm{d}v \le C\left(M_a n^{-1/200}\sqrt{\log(2+n)} + V_a n^{-9/175}\right).$$

The new physical support radius is $2n^{-12/25}$; the old, sharper estimate on $2n^{-99/200}$ remains unchanged. The unweighted law is $a = s_R$. Theorem 207.6 places the enlarged cutoff in the exact raw-density decomposition.

Proof. Apply Theorem 207.2 and Corollary 207.3. □

Theorem M (Damped unsmoothing and a wider fixed-return band)**.** *For the insertion, measure and frequency conventions of Theorems C and L,*

$$n^2 \int_{2n^{-99/200}\le|z|\le 2n^{-19/42}} |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C\left(M_a n^{-5/861} + V_a n^{-59/21}\right),$$
$$\int_{|v|\le 2n^{1/21}} \left|C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf T} D_R v/2}\right|\,\mathrm{d}v \le C\left(M_a n^{-5/861} + V_a n^{-9/175}\right).$$

Both estimates hold at every sufficiently large prescribed return count, uniformly in R, $0 \le k \le n$ and the stated bounded-BV insertions. The physical support radius is $2n^{-19/42}$ and its rescaled radius is $2n^{1/21}$. More generally, a fixed-order construction reaches every rescaled exponent $1/200 < \varepsilon < 1/20$ with a positive decay rate. The prior central and inner-annulus theorems remain valid on their bands.

Proof. Theorems 208.4 and 209.1, Corollary 209.2, and Proposition 209.3 prove the assertions. □

The new weighted raw identity in Theorem 209.5 uses this larger cutoff and recomputes its local edge correction. Its finite-band residual, long-time derivative budget and exact-event comparison are not replaced by the new marked characteristic estimate.

Theorem N (Multiscale stopping and a rate-preserving band)**.** *For the actual return record and insertion convention of Theorems C and D, let $\mathcal{G}_d(D)$ denote the degree-d moment tensor of $N(0,D)$. For every fixed integer $d \ge 1$, put $\rho_d = 1/2$ for odd d and $\rho_d = 1$ for even d. Then*

$$\left\| \int_{Y_R^*} a((F_R^*)^k x) \left(\frac{U_{n,R}(x)}{\sqrt{n}}\right)^{\otimes d} \mathrm{d}\nu(x) - \alpha_a \mathcal{G}_d(D_R) \right\| \le C_d\left(M_a n^{-1/4} + (M_a + V_a) n^{-\rho_d}\right).$$

In particular, the actual normalized covariance and its induced Cesaro formula converge to D_R at rate $O(n^{-1/4})$. Moreover,

$$n^2 \int_{2n^{-99/200}\le|z|\le 2n^{-633/1400}} |\Phi_{n,R}^{[k],a}(z)|\,\mathrm{d}z \le C\left(M_a n^{-3/280} + V_a n^{-983/350}\right),$$
$$\int_{|v|\le 2n^{67/1400}} |C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf T} D_R v/2}|\,\mathrm{d}v \le C\left(M_a n^{-3/280}\sqrt{\log(2+n)} + V_a n^{-9/175}\right).$$

The estimates are uniform in R, the prescribed sufficiently large return count n, and one arbitrary mark $0 \le k \le n$. The two support radii are $2n^{-633/1400}$ physically and $2n^{67/1400}$ after rescaling. No return-count averaging or normalization by annular volume is used. The same-event consequence retains the original probability and variation budget of Theorem C on this larger band.

Proof. Apply Theorems 210.2, 210.4, and 211.1, together with Corollary 211.3. □

Theorem O (Separate coordinate widths at the prescribed count)**.** *Let*

$$\mathcal{B}_n^{\mathrm{roof}} = [-2n^{1/100}, 2n^{1/100}]^3 \times [-2n^{9/100}, 2n^{9/100}].$$

For the same insertion and measure convention as Theorem C,

$$n^2 \int_{\substack{\sqrt{n}z \in \mathcal{B}_n^{\mathrm{roof}} \\ |z| \geq 2n^{-99/200}}} |\Phi_{n,R}^{[k],a}(z)| \, \mathrm{d}z \leq C \left(M_a n^{-1/40} + V_a n^{-497/25} \right).$$

The estimate is uniform in R, one prescribed mark $0 \leq k \leq n$, and all sufficiently large return counts n. On the union $\mathcal{U}_n = \mathcal{B}_n^{\mathrm{roof}} \cup \{|v| \leq 2n^{67/1400}\}$,

$$\int_{\mathcal{U}_n} |C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf{T}} D_R v/2}| \, \mathrm{d}v \leq C \left(M_a n^{-3/280} \sqrt{\log(2+n)} + V_a n^{-9/175} \right).$$

The new physical widths are $2n^{-49/100}$ in the three discrete coordinates and $2n^{-41/100}$ in flight time. They extend the proved region without asserting cancellation on the whole isotropic annulus. More generally separate rescaled exponents $\theta_i \geq 1/200$ are admissible when $\max_i \theta_i < 1/10$ and $\sum_i \theta_i + \max_i \theta_i < 1/4$, with the rates specified in Theorem 213.1.

Proof. Apply Theorem 212.3, Theorem 213.1, and Corollary 213.3. □

Theorem P (A nonrectangular region at the original central rate)**.** *Let $\mathcal{H}_n$ be the dyadic region defined in* (214.9)–(214.10)*. Its base scale is $2n^{1/200}$, its maximal coordinate scale is $n^{9/100}$, and its product-times-maximum budget is $n^{67/280}/[\log(2+n)]^3$. Put*

$$\mathcal{U}_n^\sharp = \mathcal{H}_n \cup \mathcal{B}_n^{\mathrm{roof}} \cup \{|v| \leq 2n^{67/1400}\}.$$

For the actual prescribed count, mark and measure convention of Theorem C,

$$\int_{\mathcal{U}_n^\sharp} |C_{n,R}^{[k],a}(v) - \alpha_a e^{-v^{\mathsf{T}} D_R v/2}| \, \mathrm{d}v \leq C \left(M_a n^{-3/280} \sqrt{\log(2+n)} + V_a n^{-9/175} \right).$$

The physical region is $n^{-1/2}\mathcal{U}_n^\sharp$. It retains both the old isotropic ball and the roof box, and reaches physical scale $n^{-41/100}$ on every coordinate axis. It is not an isotropic ball of radius $n^{-2/5}$. Constants are uniform in R, n, k, a; the residual order $P = 26$ and spectral degree $Q = 29$ are fixed.

Proof. Use Theorem 214.4 and Corollary 214.5. □

Theorem Q (Coherent transport of the raw correction)**.** *For an additionally finite-record admissible marked weight, fix one exact finite-count extraction $\mu^{a,L} = E^{a,L} + Q^{a,L}$. For a cutoff χ with kernel K_χ, put*

$$\mathcal{R}_\chi^{a,L} = e^{a,L} - K_\chi * E^{a,L} + \mathcal{F}^{-1}[(1-\chi)\widehat{Q}^{a,L}].$$

For any two of the new dyadic, preceding box and preceding isotropic cutoffs, and $L_n = \lceil \lambda n \rceil$ with $\lambda > c_*^{-1} + 1$, on each fixed central window,

$$n^2 \operatorname*{ess\,sup}_{\mathcal{W}_{n,R,A}} |\mathcal{R}_\chi^{a,L_n} - \mathcal{R}_\psi^{a,L_n}| \leq C \left(M_a n^{-3/280} + V_a n^{-983/350} \right) + C_{A,\lambda,P_0} M_a n^{-P_0}$$

for every fixed $P_0 > 0$. The difference is a bounded continuous function even when an individual correction has infinite essential supremum. This compares the entire signed correction, not either of its two absolute bounds separately.

Proof. Proposition 215.2 and Theorem 215.3 prove the assertion. □

Theorem R (Unsmoothed windows and the common signed correction)**.** *Let* $U_{n,R} = J_{n,R} - n\bar{G}_R$, $\rho = 67/1400$, *and* $B_n(\xi) = \prod_{i=1}^4 [\xi_i - n^{-1/25}, \xi_i + n^{-1/25}]$. *Uniformly on* $|\xi| \le A$ *and* $R \in I$,

$$\nu_R^*\{U_{n,R}/\sqrt{n} \in B_n(\xi)\} = 16n^{-4/25} g_{D_R}(\xi) \left(1 + O_A(n^{-11/1400}\sqrt{\log(2+n)})\right).$$

The event uses the unmodified record and actual integer intervals. More generally, a rectangular frequency box contained in the retained region gives the window estimates in Theorem 216.2, including positive marked weights with their exact mass.

For a finite-record admissible complex mark, the same physical windows $\mathcal{B} = n\bar{G}_R + \sqrt{n}B_n(\xi)$ *satisfy*

$$\frac{n^2}{\mathfrak{m}(\mathcal{B})} \left| \int_{\mathcal{B}} \mathcal{R}_\chi^{a,L_n} \, \mathrm{d}\mathfrak{m} \right| \le C_A \left(M_a n^{-11/1400} \sqrt{\log(2+n)} + V_a n^{-9/175} \right)$$

for each of the three retained cutoffs and $L_n = \lceil \lambda n \rceil$, $\lambda > c_*^{-1} + 1$. *Here* $\mathfrak{m}$ *is counting measure times Lebesgue measure. The absolute value is outside the integral; the assertion is not an essential-supremum bound.*

Proof. Apply Corollary 216.3 and Theorem 216.4. □

Theorem S (Relative windows at deterministic physical time)**.** *Start the orbit at a section collision with law* ν_R^*. *Let* $V_R^Y(t) = (W_R^Y(t), C_R^Y(t) - t/\bar{\tau}_R)$ *and* $n_R(t) = \lfloor c_* t/\bar{\tau}_R \rfloor$. *For* $B_t = \prod_{i=1}^3 [\xi_i - t^{-1/25}, \xi_i + t^{-1/25}]$, *define*

$$A_t^{\mathrm{phys}} = \{V_R^Y(t)/\sqrt{t} \in B_t\}, \qquad A_t^{\mathrm{ret}} = \{B_R U_{n_R(t),R}/\sqrt{t} \in B_t\},$$

where $B_R(x) = (x_1, x_2, x_3 - x_4/\bar{\tau}_R)$. *Uniformly for* $|\xi| \le A$ *and* $R \in I$, *both probabilities equal*

$$8t^{-3/25} g_{\mathcal{V}_R}(\xi) \left(1 + O_A(t^{-11/1400}\sqrt{\log(2+t)})\right),$$

and

$$\frac{\nu_R^*(A_t^{\mathrm{phys}} \triangle A_t^{\mathrm{ret}})}{\nu_R^*(A_t^{\mathrm{ret}})} \le C_A t^{-11/1400} \sqrt{\log(2+t)}.$$

The conditional initial laws obey the same total variation bound. These are exact physical window events of half-width $t^{23/50}$, *not singleton lattice events or events under a different initial law.*

Proof. This is Theorem 217.4. □

Theorem T (Stationary physical observations and conditional laws)**.** *Represent equilibrium flow by the actual return suspension*

$$\mathrm{d}\mathbb{P}_R(y,s) = \frac{c_*}{\bar{\tau}_R} \, \mathrm{d}\nu_R^*(y) \, \mathrm{d}s, \qquad 0 \le s < \tau_R^*(y).$$

Let A_t^{phys} *be the event that the stationary physical increment* $(W_R(t), C_R(t) - t/\bar{\tau}_R)/\sqrt{t}$ *lies in the three-dimensional box of center* ξ *and half-width* $t^{-2/25}$. *For* $n_R(t) = \lfloor c_* t/\bar{\tau}_R \rfloor$, *let* A_t^{ret} *be the event that* $B_R U_{n_R(t),R}(y)/\sqrt{t}$ *lies in the same box, where* $B_R(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3 - x_4/\bar{\tau}_R)$. *Then, uniformly in* R *and bounded* ξ,

$$\mathbb{P}_R(A_t^{\mathrm{phys}}) = 8t^{-6/25} g_{\mathcal{V}_R}(\xi) \left(1 + O(t^{-23/2800}\sqrt{\log(2+t)})\right),$$

$$\frac{\mathbb{P}_R(A_t^{\mathrm{phys}} \triangle A_t^{\mathrm{ret}})}{\mathbb{P}_R(A_t^{\mathrm{ret}})} \le C t^{-23/2800} \sqrt{\log(2+t)}.$$

The return event has the same denominator asymptotic. The same comparison holds with the actual last-completed-return event, and for the conditional laws of any common bounded path statistic. The three physical coordinate half-widths are $t^{21/50}$. The return origin y has its equilibrium roof-length bias; it is not drawn from the unbiased section probability.

Proof. Theorems 218.1 and 218.2 provide the collision-clock window estimate. Lemma 219.1 and Proposition 219.2 give the normalization and coupling. Theorem 219.3 proves the displayed assertions and their conditional-law consequence. □

Theorem U (Physical paths conditioned on stationary windows)**.** *Let $\mathsf{X}_{t,R}$ be a continuous interpolation of*

$$s \longmapsto t^{-1/2}(W_R(ts), C_R(ts) - ts/\bar{\tau}_R), \qquad 0 \le s \le 1,$$

whose supremum distance from the physical path is $O(t^{-1/2})$. With the stationary initial law and the exact endpoint windows of Theorem T, uniformly in R and bounded ξ,

$$\mathrm{Law}(\mathsf{X}_{t,R} \mid A_t^{\mathrm{phys}}) \implies \mathrm{Law}\big(s\xi + \mathcal{V}_R^{1/2}(W(s) - sW(1))\big) \quad \text{on } C([0,1],\mathbb{R}^3).$$

Here W is standard three-dimensional Brownian motion. Uniformity is in bounded-Lipschitz distance. The same limit holds under the prescribed-return, last-completed-return and collision reference events on the same stationary space. For fixed interior times and bounded boxes of nonempty interior, imposing those additional path observations multiplies the endpoint denominator by their explicitly positive Gaussian bridge probability; the resulting conditional path limit is the bridge restricted to those boxes. The physical endpoint half-width remains $t^{21/50}$.

Proof. Theorem 220.1 and Proposition 220.2 give joint endpoint-window comparisons uniform in the block boundaries. Theorem 221.2 proves conditional tightness. Lemma 221.3, Theorem 221.4 and Corollary 221.6 give the physical conclusions. □

Microscopic reconstruction and conditional-measure reductions. Theorems V and W below are unconditional approximation theorems at the original lattice resolution. They are separated from the Gaussian asymptotic conclusions above: the remaining finite integral is not identified with a Gaussian main term.

Theorem V (Quantitative extraction and microscopic inversion)**.** *Use the normalized section probability and the exact finite-count measure $\mu_{n,R}^{w,L}$. For a fixed product of bounded-BV return-state marks, let M, V be the budgets in* (222.11)*. There is an exact decomposition $\mu_{n,R}^{w,L} = E_\varepsilon + Q_\varepsilon$ whose residual density q_ε satisfies, uniformly in $R, n \le L$ and the mark locations,*

$$\|E_\varepsilon\|_{\mathrm{TV}} \le C\{\varepsilon V + M(L+1)\varepsilon^{1/16}\},$$
$$\sum_{\ell \in \mathbb{Z}^3} \|\partial_t^2 q_\varepsilon(\ell, \cdot)\|_1 \le CM \exp\{C(L+1)\log(C/\varepsilon)\}.$$

For $w = 1$ both measures are positive. For polynomial M, V, any fixed $P > 0$ and any fixed upper bound on $|I|$, a roof cutoff with $\log B_n = O(n \log n)$ gives

$$\sup_{R,\ell,I} n^2 \left| \mu_{n,R}^w(\{\ell\} \times I) - \mathcal{I}_{n,R,B_n}^w(\ell, I) \right| \le C_P n^{-P}.$$

Here $\mathcal{I}$ is the finite Fourier integral of the full, unmodified actual law in (223.4)*. No lattice coordinate is averaged, and I need not grow. This is a reduction of the microscopic probability, not its Gaussian asymptotic or a pointwise bound for the removed density.*

Proof. Theorem 222.5 proves the extraction. Proposition 223.1, Theorem 223.2 and Corollary 223.3 give the inversion. □

Theorem W (Positive microscopic reconstruction on the original space)**.** *Use either the normalized section law or the stationary return suspension law* (224.3)*. In the latter case the record starts at the actual preceding section origin and all trajectory statistics retain the original elapsed age. For every $P > 0$ there is a roof bandwidth B_n with $\log B_n = O(n \log n)$ and a nonnegative mixed density p_{n,R,B_n} of mass one such that*

$$n^2 \|p_{n,R} - p_{n,R,B_n}\|_{L^1(\mathbb{Z}^3 \times \mathbb{R})} \le C_P n^{-P}$$

uniformly in R. The density is the finite integral (225.4)*; no lattice coordinate is smoothed. For every microscopic event $A = \{J_{n,R} \in \{\ell\} \times I\}$, there is a likelihood $0 \le \lambda_{A,B_n} \le 1$ on the same probability space with*

$$n^2 \sup_{|W| \le 1} |\mathbb{E}[W \mathbf{1}_A] - \mathbb{E}[W \lambda_{A,B_n}]| \le C_P n^{-P}.$$

Here I is any bounded interval, including a shrinking interval, and W is any bounded measurable trajectory statistic. The second expectation has the finite Fourier formula (225.10)*. More generally, a polynomial number of intervals per lattice fibre is permitted. Normalization gives a conditional total-variation bound after division by the same actual denominator, as stated in Corollary 225.5.*

Proof. Proposition 224.4 supplies the positive residual for both initial laws. Apply Theorem 225.6 and Corollary 225.5. □

Theorem X (Exact inversion and Gaussian-plus-complement reduction)**.** *Fix the same convex polygonal fundamental cell at both physical observation times. For a prescribed physical collision count $m \ge 1$, let*

$$p_{m,R}(k,t) = \mathbb{P}_R\{W_R(t) = k,\ C_R(t) = m\}.$$

The initial and terminal within-flight cell corrections enter the exact three-frequency inverse (226.10)*. Uniformly in m, R, its time profiles satisfy*

$$\sum_k \|D_t^2 p_{m,R}(k,\cdot)\|_{\mathrm{TV}} \le C, \qquad \sum_k \|p_{m,R}(k,\cdot) - p^{\mathrm{sharp}}_{m,R,B}(k,\cdot)\|_\infty \le C/B.$$

The derivative is a signed measure. With Σ_R and $\zeta_{m,R}$ defined in Section 227, the central physical radius $2m^{-99/200}$, or $2m^{1/200}$ after rescaling, gives

$$\begin{aligned} m^{3/2} p_{m,R}(k,t) = \bar{\tau}_R g_{\Sigma_R}(\zeta_{m,R}(k,t)) + m^{3/2} \mathfrak{M}_{m,R,B}(k,t) \\ + O\left(m^{-11/700} \sqrt{\log(2+m)} + m^{3/2}/B\right), \end{aligned}$$

where $\mathfrak{M}_{m,R,B}$ is the explicitly specified signed finite middle-frequency inverse. Thus $B_m = m^{P+3/2}$ gives a pointwise far-frequency error $O(m^{-P})$ on the normalized physical scale, without a discarded-density term.

For the same original physical event there is a positive likelihood $0 \le \lambda_{m,k,B} \le 1$ whose unnormalized source total-variation error is at most C/B, uniformly against all bounded measurable path selectors. A relative posterior estimate requires its actual selected denominator. The microscopic Gaussian denominator requires the signed middle term to vanish; it is not asserted by the tail bound alone.

Proof. Theorems 226.2, 226.3, 226.4 and 227.3 prove the statements. □

Appendix F. Retained leading statements and their proof routes

The following statements reproduce the preceding leading theorem sequence. Their detailed proofs remain in the unchanged core sections. All references use labels rather than the historical order of the front matter.

Theorem 1 (The stationary microscopic local law)**.** *There is a continuous uniformly positive definite matrix $\mathcal{V}_R$ such that, for every $M < \infty$,*

$$\sup_{\substack{R\in I,\ k\in\mathbb{Z}^2,\ m\in\mathbb{N}\\ |\xi_{t,R}(k,m)|\le M}} \left| t^{3/2}\mathbb{P}_R\{W_R(t)=k, C_R(t)=m\} - g_{\mathcal{V}_R}(\xi_{t,R}(k,m))\right| \longrightarrow 0, \tag{F.1}$$

where $I=[0.45,0.47]$ and $\xi_{t,R}(k,m)=(k/\sqrt{t},(m-t/\bar\tau_R)/\sqrt{t})$. The covariance is that of the stationary physical functional limit. For bounded regular initial and terminal flight-state functions χ_R,ψ_R, the same formula holds with the probability replaced by its $\chi_R\psi_R$-weighted expectation and the Gaussian multiplied by $\mu_R(\chi_R)\mu_R(\psi_R)$.

For every fixed $P>0$, the entire signed complement in (227.6)*, with $B_m=m^{P+3/2}$ and physical central radius $2m^{-99/200}$, satisfies $t^{3/2}\mathfrak{M}_{m,R,B_m}(k,t)=o(1)$ on these central sets. The positive likelihood for the same microscopic event then approximates its normalized conditional measure on the original trajectory space in total variation with error $O(m^{-P})$. This comparison is uniform against all bounded measurable trajectory tests.*

Proof. Theorems 124.2 and 124.4, followed by Corollaries 124.3 and 124.5, prove the assertions. The regular endpoint class is specified before Theorem 124.4. □

Theorem 2 (An action principle beyond circular tables)**.** *Place one oriented ellipse at every point of the same triangular lattice, with semiaxes $a,b\in[0.45,0.47]$ and orientation $\theta\in\mathbb{R}/2\pi\mathbb{Z}$. The stationary local law* (F.1)*, its regular endpoint-selected form and its normalized positive-likelihood comparison hold uniformly in (a,b,θ). The mean roof is*

$$\bar\tau_{a,b,\theta}=\frac{\pi(\sqrt{3}/2-\pi ab)}{L_{a,b}},$$

where $L_{a,b}$ is the perimeter. The limiting physical covariance is continuous and uniformly positive definite, including at $a=b$.

More generally, these conclusions hold for compact families satisfying the separated geometry, local-space, action, finite-cover mixing, parameter continuity, preceding-flight and regular observation conditions in Section 120. For these families the physical phase equation $q\circ T=e^{i(u\cdot\kappa+s+b_0\tau)}q$ has a measurable circle-valued solution only when $b_0=0$, $u\in(2\pi\mathbb{Z})^2$ and $s\in2\pi\mathbb{Z}$. Neither a collision rotation nor a quantitative bound on growing-band spectral constants is assumed.

Proof. Theorem 119.2, Theorem 120.1, Lemma 120.2 and Corollary 120.4 give the assertions. The original circular theorem is the restriction $a=b=R$; the signed middle identity in its statement retains its previously specified circular record and cutoff. □

Theorem 3 (Local constraints on the actual return record)**.** *For the original circular family let $c=\nu(Y_R^*)=91/(10000\pi)$. There are continuous $\beta_R\in\mathbb{R}^{1\times3}$ and $\sigma^2_{A,R}>0$, with uniform positive variance bounds, such that, for every fixed bounded interval J of positive length and fixed $a<b$, uniformly for $R\in I$ and $|z|\le M$,*

$$\begin{aligned} m^{3/2}\sum_{\substack{1\le n\le m\\ a\le(n-mc)/\sqrt{m}\le b}} &\nu_R^*\{K_{n,R}=k,\ N_{n,R}=m,\ T_{n,R}-t\in J\}\\ &- c|J|g_{\Sigma_R}(z)\int_a^b g_{\sigma^2_{A,R}}(y-\beta_R z)\,dy\longrightarrow 0, \qquad z=\left(\frac{k}{\sqrt{m}},\frac{t-m\bar\tau_R}{\sqrt{m}}\right). \end{aligned} \tag{F.2}$$

Here Σ_R is the collision displacement–roof covariance. The Schur data $\beta_R, \sigma^2_{A,R}$ are those of the joint covariance after adjoining the true section occupation. The summands are probabilities of the original return records; neither the section nor the event is replaced. The probability sum is bounded below by $c_{M,a,b,J} m^{-3/2}$ for all sufficiently large m.

Proof. Lemma 151.1 realizes the exact occupation. Theorem 151.3 proves its mixed local–central law. Proposition 152.1 and Theorem 152.2 give the statement; equation (152.5) identifies its amplitude with the Gaussian normalization of the original induced covariance D_R. □

Theorem 4 (Fine resolution of the actual return index)**.** *For the original circular family and every bounded interval J,*

$$\nu_R^*\{K_{n,R} = k, N_{n,R} = m, T_{n,R} - t \in J\} \le C(1 + |J|)m^{-2} \qquad (m \ge n \ge 1),$$

uniformly in the radius and all targets. Let J now have fixed positive length. Let $\mathcal{B}_m$ consist of H_m consecutive actual return indices, with center ℓ_m, where $H_m \to \infty$ and $H_m/\sqrt{m} \to 0$. Put

$$Z_{m,R} = \left(\frac{k}{\sqrt{m}}, \frac{t - m\bar{\tau}_R}{\sqrt{m}}, \frac{\ell_m - mc}{\sqrt{m}}\right), \qquad \Omega_R = cL_R D_R L_R^{\mathsf{T}},$$

with L_R specified in (151.7)*. Then*

$$\frac{m^2}{H_m} \sum_{n \in \mathcal{B}_m} \nu_R^*\{K_{n,R} = k, N_{n,R} = m, T_{n,R} - t \in J\} = c|J| g_{\Omega_R}(Z_{m,R}) + o(1), \tag{F.3}$$

uniformly on central compact sets and over widths between any $h_m \to \infty$ and $u_m\sqrt{m}$ with $u_m \to 0$. The resulting denominator is at least $c_{M,J} H_m m^{-2}$. No lower polynomial growth restriction is imposed on H_m. The upper bound applies to one index; the Gaussian asymptotic in this statement uses a diverging block, and the roof interval is not replaced by a pointwise density. The convergence is qualitative, with no asserted polynomial rate for the $o(1)$ term.

Proof. Theorem 154.1 gives the assertions. Corollary 154.2 writes the same limit in the original D_R normalization. The explicit two-band budget is (153.12); Proposition 154.4 isolates the remaining single-index occupation-torus inverse. □

Theorem 5 (An exact return index with its arithmetic coefficient)**.** *For the original circular table and the unchanged section Y_R^*, all measurable occupation resonances have zero roof coordinate. Their group is finite cyclic, of order $d_R \le D$ uniformly in R. At each fixed R, let $a_R^{\mathrm{ar}} \in (\mathbb{Z}/d_R\mathbb{Z})^2$, $h_R^{\mathrm{ar}} \in \mathbb{Z}/d_R\mathbb{Z}$ and $w_{R,l} \ge 0$ be the residue data in* (159.2)*, with $\sum_l w_{R,l} = c$. Put*

$$\mathfrak{a}_R(k,n,m) = \frac{d_R}{c^2} \sum_{l \bmod d_R} w_{R,l} w_{R,l + a_R^{\mathrm{ar}} \cdot k + n + h_R^{\mathrm{ar}} m \cdot}$$

For every fixed bounded interval J of positive length and each fixed radius,

$$n^2 \nu_R^*\{K_{n,R} = k, N_{n,R} = m, T_{n,R} - t \in J\} = |J| \mathfrak{a}_R(k,n,m) g_{D_R}(V_n) + o_R(1), \tag{F.4}$$

where $V_n = n^{-1/2}(k_1, k_2, m - n/c, t - n\bar{\tau}_R/c)$ and the error is uniform over its compact sets. The coefficient is nonnegative, its average over d_R consecutive return indices is one, and a zero coefficient makes the exact event null. There is a positive microscopic denominator on each positive residue class.

For every fixed roof band, the complete fixed-count inverse also has a finite spectral reduction uniform over all $R \in I$, as stated in (159.1)*. This uniform formula retains near-resonant branches through parameter changes. No unmodulated radius-uniform singleton law or pointwise roof-density statement is included in* (F.4)*.*

Proof. Apply Theorems 159.1, 158.1 and 159.2. □

Theorem 6 (Uniform transitions and the exact-event return bridge)**.** *Let J be a fixed bounded interval of positive length and put*

$$E = \{K_{n,R} = k, N_{n,R} = m, T_{n,R} - t \in J\}, \qquad Z = m^{-1/2}(k_1, k_2, t - m\bar{\tau}_R, n - mc).$$

The probability of this exact return event satisfies

$$m^2 \nu_R^*(E) = |J| \mathcal{L}_{m,R}(k_1, k_2, t, n) + o(1)$$

uniformly in R and the targets, where $\mathcal{L}_{m,R}$ is the finite Gaussian expression in (160.5)*. It retains the damping and drift of branches approaching the unit circle; at fixed R it reduces to the arithmetic Gaussian in Theorem 5.*

Let $\mathcal{Y}_{n,R}$ be the polygonal path with values $n^{-1/2}(J_{l,R} - (l/n)J_{n,R})$ at l/n. For every $M, \epsilon > 0$, uniformly on $|Z| \le M$ and $m^2 \nu_R^*(E) \ge \epsilon$,

$$d_{\mathrm{BL}}\left(\mathrm{Law}(\mathcal{Y}_{n,R} \mid E), \mathrm{Law}(\mathbb{B}_{D_R})\right) \longrightarrow 0.$$

Here $\mathbb{B}_{D_R}$ is the centered continuous Gaussian bridge with covariance $(s \wedge t - st)D_R$. At each fixed radius the mass condition holds on every positive arithmetic residue class. There is also a fixed finite integer L, independent of m and R, such that

$$\frac{m^2}{L} \sum_{j=0}^{L-1} \nu_R^*\{K_{n+j,R} = k, N_{n+j,R} = m, T_{n+j,R} - t \in J\} = c|J| g_{\Omega_R}(Z) + o(1)$$

uniformly on central compact sets. This last packet assertion does not replace the single-index event used in the bridge statement.

Proof. Apply Theorems 160.3, 161.5 and 162.2. Theorem 162.1 characterizes the additional condition for the radius-uniform unmodulated single-index interval law. □

Arithmetic and cancellation at an exact index. The new theorem estimates the full occupation torus at a prescribed collision count. Section 157 refines each regular inverse curve by at most a linear number of occupation cuts. The resulting polynomial losses are absorbed in the three collision norm inequalities. Section 158 proves that every occupation holonomy differs from its roof holonomy by an integer; countability then removes nonzero roof resonances. It also calculates the true section-to-section coefficient at every remaining scalar eigenvalue. Section 159 performs the fixed-count inverse and sums those residues by finite Fourier orthogonality.

The earlier source spectral and Abel identities in Section 156 remain exact. They are not used to infer decay of a fixed coefficient. The signed complement from (154.7) is now evaluated at fixed R: its main term is proportional to $\mathfrak{a}_R - 1$, not automatically zero. A fixed interval is not a pointwise roof density. Section 160 now evaluates arithmetic transitions uniformly by retaining their damped spectral branches. Sections 161 and 162 give the exact-event path bridge and a finite-packet consequence. Triviality of the section residues and the common pointwise raw correction remain separate requirements for the original full raw theorem.

Theorem 7 (Regular critical packets and arithmetic endpoint laws)**.** *For the original circular family and the actual section, every finite collision cutoff $L \ge n$ admits the exact mixed-density decomposition*

$$p_{n,R} = e_{n,R}^{L,\varepsilon} + q_{n,R}^{L,\varepsilon} + r_{n,R}^{L,\varepsilon}, \qquad r \ge 0.$$

The finite edge sum e retains every guarded interior critical word; no initial normal strip is discarded. Each edge has the explicit form $\mathbf{1}_{s \ge 0} e^{-s}\{a_0 + (a_0 + a_1)s\}$, where a_0, a_1 are the intrinsic

right roof jets. Its coefficients stabilize on each word with positive guard margins as the guards are removed. Uniformly in R,

$$\|r\|_{L^1(\mathbb{Z}^3\times\mathbb{R})} \le C\{e^{an-c_0L} + (L+1)\varepsilon^{1/16}\},$$
$$\sum_z (|a_{0,z}| + |a_{1,z}|) + \sum_\ell \|q(\ell,\cdot)\|_{W^{2,1}} \le \exp\{C(L+1)\log(C/\varepsilon)\}.$$

The residual inverse truncated at roof frequency B has pointwise error at most the last bound divided by πB. The uncut source has in addition the nonnegative pointwise error $r(\ell,t)$; its small mass alone is not a uniform pointwise estimate.

At each fixed radius, let w_l be the actual section phase masses and $r_0 = r_R(k,n,m)$ its arithmetic residue. Under the unchanged central exact-index event with a fixed positive roof interval and $\sum_l w_l w_{l+r_0} > 0$, the endpoint phase pair has limiting masses

$$\pi_{R,r_0}(l,l') = \mathbf{1}_{\{l'=l+r_0\}} \frac{w_l w_{l+r_0}}{\sum_h w_h w_{h+r_0}}.$$

These phases are asymptotically independent of the actual pinned return Gaussian bridge. No regularity of the measurable phase indicators as strong multipliers is assumed.

Proof. Theorem 163.3, Corollary 163.4, Theorem 164.2 and Corollary 164.3 give the assertions. □

Theorem 8 (Deconcentration of regular critical edges)**.** *For $0 < \varepsilon < 1/10$ let $\mathcal{J}^{\varepsilon}_{n,k,m,R}$ be the positive measure of intrinsic critical jump coefficients in* (170.5)*, using the exponentially relaxed margins of Section 169. There are $C, h_* > 0$, independent of ε, such that for every fixed $0 < h < h_*\varepsilon^6$,*

$$\limsup_{m\to\infty} \sup_{R,n,k,t} m^2 \mathcal{J}^{\varepsilon}_{n,k,m,R}([t-h,t+h]) \le Ch.$$

The sum at any single critical value is therefore $o(m^{-2})$, uniformly in the exact labels and radius for each fixed ε. For every sequence $r_m \downarrow 0$, the original source restricted to the physical critical collars of roof excess less than r_m has density $p^{\varepsilon,\mathrm{crit}}$ satisfying

$$\sup_{R,n,k} m^2 \|p^{\varepsilon,\mathrm{crit}}_{n,k,m,R}\|_\infty \longrightarrow 0.$$

The same conclusion holds for arbitrary bounded measurable insertions on this restricted source. Its contribution to $p - K_B * p$ is pointwise $o(m^{-2})$, uniformly for all $B > 0$. No assertion about the full remaining signed correction or about a prescribed shrinking margin $\varepsilon = \varepsilon_m$ is included.

Proof. Apply Theorems 170.2 and 170.3, followed by Corollary 170.4. □

Arithmetic and the pointwise return target. For the exact return index we retain the factor $\mathfrak{a}_R$ at fixed radius and the transition kernel $\mathcal{L}_{m,R}$ uniformly in the radius. The unmodulated specialization requires the zero-residue criterion; it is not imposed by notation. Theorem 143.1 remains the inherited raw inversion criterion, not a statement that its remaining pointwise hypotheses have been verified. Theorem 7 advances that same raw problem by retaining, rather than removing, the regular critical source. It does not replace the pointwise task by an existential shrinking interval.

Theorem 9 (Protected raw inversion and margin localization)**.** *Let $\Xi^{\varepsilon}_{m,R}$ be the graded source guard in* (171.2)*, equal to one when all the graded margins exceed twice their thresholds. For the original exact labels put*

$$f^{\varepsilon}(t)\,dt = (T_{n,R})_*(\mathbf{1}_{\{K_{n,R}=k,\, N_{n,R}=m\}} \Xi^{\varepsilon}_{m,R}\, \nu^*_R), \qquad b^{\varepsilon} = p_{n,R}(k,m,\cdot) - f^{\varepsilon} \ge 0.$$

For every fixed $\varepsilon > 0$ and $B \geq C_2\varepsilon^{-12}$,

$$\limsup_{m\to\infty} \sup_{R,n,k} m^2 \|f^\varepsilon - K_B * f^\varepsilon\|_\infty \leq CB^{-1/2}. \tag{F.5}$$

The same correction tends to zero for every $B_m \to \infty$ with ε fixed, without an asserted rate in m. Regular endpoint insertions satisfy the corresponding estimate with their stated supremum and endpoint-gradient budgets. The positive remainder obeys

$$\sup_{m,R} \sum_{n=1}^{m} \sum_{k} \|b^\varepsilon\|_1 \leq C\varepsilon.$$

With $\varepsilon(B) = AB^{-1/12}$ for a fixed sufficiently large A, the full raw arithmetic local law with main term $\mathcal{L}_{m,R}$ is equivalent to vanishing of the normalized signed correction $b^{\varepsilon(B)} - K_B * b^{\varepsilon(B)}$ on each central compact set, first in collision limsup and then as $B \to \infty$. The mass bound alone does not supply this last estimate.

Proof. Apply Theorems 171.4, 172.2 and 172.5, and Proposition 172.4. □

Theorem 10 (Pointwise removal of endpoint decision boundaries)**.** *Fix an endpoint depth $J \geq 0$ and a margin $\varepsilon > 0$. Let $G^{\varepsilon,J}_{m,R}$ be the original graded protection with the section factors at $0, \dots, J, m-J, \dots, m$ omitted. Keep every incidence and clearance factor and every other section factor. Let $E^{\varepsilon,J}_{m,R}$ be the product of the omitted factors, and for the original exact labels define*

$$d^a(t)\,dt = (T_{n,R})_*(\mathbf{1}_{\{K_{n,R}=k, N_{n,R}=m\}} a G^{\varepsilon,J}_{m,R}(1 - E^{\varepsilon,J}_{m,R})\nu^*_R).$$

Uniformly for all measurable insertions with $|a| \leq M$,

$$\limsup_{m\to\infty} \sup_{R,n,k,a} m^2 \|d^a\|_\infty \leq CM(J+1)\varepsilon.$$

The same bound holds for $d^a - K_B * d^a$, uniformly over every $B > 0$. All original integer labels and the roof observable on the left are unchanged. The finite occupation enlargement occurs only in a positive upper comparison, not as an average replacing the target coefficient.

In particular, with $\varepsilon(B) = AB^{-1/12}$ and $J(B) = \lfloor B^{1/24} \rfloor$, the full unweighted correction differs from the signed correction of $1 - G^{\varepsilon(B),J(B)}$ by at most $CB^{-1/24}$ at the normalized collision-limsup scale. This residual splits by its first bad margin into incidence, clearance and interior section-decision sources. Its pointwise correction remains to be estimated, with the arithmetic main term $\mathcal{L}_{m,R}$ retained.

Proof. Apply Theorem 173.6, Proposition 174.1 and Theorem 174.2. □

Theorem 11 (Every section-decision depth)**.** *Let $H^\eta_{m,R}$ contain all graded incidence and clearance factors, and let $D^\varepsilon_{m,R}$ contain all graded section-decision factors, as in* (176.1)*. For the original exact event $\Pi_{n,k,m,R}$ and any bounded measurable source insertion $|w| \leq M$, define*

$$d^{\eta,\varepsilon,w}_{n,k,m,R}(t)\,dt = (T_{n,R})_*(\mathbf{1}_{\Pi_{n,k,m,R}} w H^\eta_{m,R}(1 - D^\varepsilon_{m,R})\nu^*_R).$$

For fixed positive η, ε sufficiently small,

$$\limsup_{m\to\infty} \sup_{R,n,k,\,|w|\leq M} m^2 \|d^{\eta,\varepsilon,w}_{n,k,m,R}\|_\infty \leq CM\varepsilon^{1/16}.$$

The same bound holds for $d - K_B * d$, uniformly over every $B > 0$. For the positive depth telescope of (176.2)*, its tail beyond depth $J \geq -1$ satisfies the finite-count bound*

$$m^2 \|d^{\eta,\varepsilon,w}_{>J}\|_\infty \leq CM\varepsilon^{1/16}(J+2)(47/53)^{(J+1)/64} + C_{\eta,\varepsilon} M m^4 \rho^{m/2}_{\eta,\varepsilon}, \qquad \rho_{\eta,\varepsilon} < 1.$$

Consequently an increasing decision depth cannot escape this estimate at fixed physical margins. With $\varepsilon(B) = AB^{-1/12}$ fixed before the collision limit, the complete signed correction differs from the incidence–clearance correction by $O(B^{-1/192})$ in normalized essential supremum. The exact labels and arithmetic transition main term remain unchanged.

Proof. Apply Theorem 176.4, Corollary 176.5, and Theorem 177.1. □

Mechanism of the microscopic estimate. The exact age-overlap formula (226.5) retains the initial and terminal cell offsets. Its two interval endpoints give a pointwise far-frequency error $O(B^{-1})$, but that estimate alone says nothing about intermediate frequencies. Theorem 122.4 supplies the missing bounded-band dynamical mechanism for the unsmoothed collision triple $(\kappa_1, \kappa_2, \tau - \bar{\tau})$. It realizes the twist on the image side of the collision operator. The exact action formula controls the accumulated phase on stable curves and on the matched curves needed for the unstable norm; physical measurable phase rigidity excludes every nonzero peripheral frequency.

Upper and lower Fourier envelopes, applied to positive source measures, give a joint collision local limit with both endpoint states. The order of limits is a fixed frequency band, the collision count tending to infinity, and finally the band tending to infinity. The original physical overlap is a legitimate test of the resulting local measure. This evaluates the complete probability before the Gaussian-plus-complement identity is used to deduce the signed middle estimate. It requires neither a bound on growing-band operator constants nor an absolute estimate for the modulus of that middle transform.

Actual returns and conditioning. The physical local theorem does not replace the four-coordinate raw return problem by a different record. All the return geometry, nondegeneracy, Gaussian, marked, path and raw-inversion results remain in the body. Their complementary formulations are collected in Appendix E, with the letters A–X used in the cross-references. The current theorem provides a direct route to microscopic physical conditioning, using the original event and a proved denominator; it needs no return-to-physical event replacement. Its arbitrary-test total-variation comparison is not an assertion that every selected path event has a positive Gaussian amplitude. The Gaussian amplitude is evaluated here for the stated regular endpoint class. The general weighted return-density theorem remains separate from the exact-index, fixed-interval conditional return bridge proved in Theorem 6. A pointwise roof-conditioned path law is not asserted.

Relation to previous local limit methods. The theorem-level comparison in Section 116 distinguishes existing fixed-table local limits from the compact-family action and source estimates here. The collision theory of [8, 4] and the image-side displacement multiplier of [12] provide the anisotropic norm framework. Our compact-frequency argument includes the physical roof by verifying its action bounds in those norms; a spatial local limit theorem is not cited as a theorem for the three-coordinate record. The endpoint form of a mixing local limit is also natural in the flow theory of [3]. Here uniformity in the moving family, the actual stationary age, and the two cell offsets are treated together. The original four-coordinate return-density analysis remains tied to the periodic arithmetic and extracted critical edges developed below.

Appendix G. Inputs to the pointwise lower law

The shortest new argument uses the following four inputs from the retained actual-return theory. The source partition is (181.4); the protected correction is Theorem 172.2; the complete decision correction is Corollary 176.5; and the physical local-variation estimate is Theorem 179.1. The fixed-band main term is the full-source formula in the proof of Theorem 166.3. None of these inputs is a pointwise bound on the unprotected physical remainder.

For the inherited physical multiplier the following table fixes the hypothesis-to-source correspondence. External lemma numbering follows arXiv:1902.06850v1, as specified in [12].

Requirement	Location and convention in this article
Finite partition complexity	Lemma 178.1: a fixed candidate-center set and bounded trigonometric degree on each homogeneity strip.
Stable transversality	Equation (178.4): clearance levels have positive slope, stable graphs negative slope.
Boundary neighborhoods	The same lemma: an interval of width s has length $O(s)$ on a stable graph; coalescing boundaries introduce no inverse width.
Grazing thresholds	Two horizontal angle cuts; the original homogeneity strips remain in force. No inverse change from p is used at grazing.
Multiplier exponents	Equation (117.4) and [12, Lemma 3.3(b)]; the indicator is constant on each partition cell.
Weak gain and physical product	Proposition 178.2: stable-length exponent $1/8$, with a uniform strong bound, not a small strong bound.

The preceding incident disk does not thicken the clearance envelope: its physical incoming line intersects that disk and therefore has line distance at most R, while the envelope requires at least R. Their common boundary is the tangency condition. Clearance of flight j remains marked at collision $j+1$, and the occupation sum is $\sum_{a=0}^{m-1} \eta_R \circ T_R^a$. The source factor $1/c$ is paid when dropping the section restrictions in the positive upper comparison, before the new argument begins.

These are identified continuum inputs, not claims of independent specialist verification. In particular the matching geometry, the strong-completion multiplication, the moving-peak interpolation and the all-depth finite-count remainder still carry the audit scope described in the relevant proofs.

Comparison of theorem statements. The cell-index mixing local theorem with endpoint observables and scatterer perturbations in [12], and the abstract suspension local theorem in [3], remain the relevant comparison points. The new conclusion here is neither a differentiation of their interval laws nor an assertion that a singleton lattice event is a new topology. It is the lower inequality for the actual four-coordinate return density, obtained from the positive source partition above. The abstract real-analysis proposition 181.2 makes this inference reusable, but does not by itself verify its dynamical assumptions for another family.

Appendix H. Retained local, endpoint and pointwise statements

The original record and its density. Let T_R be the physical collision map and F_R^* its actual first return to Y_R^*. The collision probability is ν and the return probability is $\nu_R^* = c^{-1}\nu|_{Y_R^*}$, where $c = 91/(10000\pi)$. For n genuine returns write $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$. Define $p_{n,R}$ by

$$\nu_R^*\{K_{n,R} = k, N_{n,R} = m, T_{n,R} \in du\} = p_{n,R}(k, m, u)\, du.$$

Its existence on the complete, unprotected source is Theorem 167.3. The counting coordinates are $k \in \mathbb{Z}^2$ and $m \geq n$; only the roof coordinate u is continuous.

Theorem 12 (An explicit weak endpoint for the original arithmetic density)**.** *Let $\mathcal{L}_{m,R}$ be the evaluated finite transition kernel in* (160.5)*. For every fixed $h > 0$,*

$$\lim_{m\to\infty} \sup_{R,\, 1\leq n\leq m,\, k,t} \frac{1}{h}\int_t^{t+h} |m^2 p_{n,R}(k, m, u) - \mathcal{L}_{m,R}(k_1, k_2, u, n)|\, du = 0.$$

The original exact return index is retained. At fixed radius and on central compact sets the main term is $c\mathfrak{a}_R(k, n, m)g_{\Omega_R}$ in the stated collision normalization. There is no zero-residue assumption.

Put $p_c = 145/144$. The normalized original density $P = m^2 p_{n,R}$ satisfies

$$\sup_{R,n,k,m} \sup_{L>0} L^{p_c}|\{u \in [t, t + h] : P(u) > L\}| \leq C(1 + h).$$

For every $1 < q < p_c$,

$$\sup_{R,n,k,t} \frac{1}{h}\int_t^{t+h} |m^2 p_{n,R}(k, m, u) - \mathcal{L}_{m,R}(k_1, k_2, u, n)|^q\, du \to 0.$$

For $\beta > 1$, the same assertion holds with the qth power replaced by the convex function Φ_β in (186.3)*, which is comparable at infinity to $x^{p_c}/[\log(e + x)]^\beta$. This is a weak endpoint and an endpoint-modular law, not a strong L^{p_c} assertion.*

On fixed windows with pointwise arithmetic reference floor $\mathcal{L}_{m,R} \geq d > 0$, the unchanged conditional roof likelihood converges to one in every L^q of the reference law, $1 < q < p_c$, and in each of these endpoint modulars. Forward relative entropy and forward Rényi divergences of orders in $(1, p_c)$ tend to zero. No collision-count rate or forward essential likelihood estimate is claimed.

There is also a resolution-uniform localization. For each large fixed B, let $O_{B,m}$ be the positive physical maximal set of (187.12)*. Its length on the roof line is at most $C(1 + h)B^{-1/384}$, and its source mass on the displayed window has collision limsup at most the same bound. At every anchor outside $O_{B,m}$, all subinterval averages of the scalar and collision-path raw errors are bounded by $r_{B,m}$, with $\limsup_m r_{B,m} \leq CB^{-1/384}$. The subintervals have no minimum length. The same-roof statement holds almost everywhere on that complement; it does not assert height control inside $O_{B,m}$.*

The same limit is uniform over all measurable roof selectors of modulus at most one supported in the displayed interval. On bounded central regions at a fixed return index, it gives total-variation approximation of the complete mixed measure. If the positive-part transition mass of a roof interval is at least dh, $d > 0$, the exact conditional roof law is approximated in total variation by its normalized positive-part transition density.

Proof. Apply Theorem 180.1 and Corollaries 180.2, 180.3 and 180.4, followed by Theorems 183.4, 184.2 and 184.3, followed by Corollary 185.5 and Theorems 186.4 and 186.5, together with Corollary 187.4 and Theorem 188.1. □

Theorem 13 (Pointwise arithmetic lower law and positive concentration)**.** *Write* $P(u) = m^2 p_{n,R}(k,m,u)$ *and* $G(u) = \mathcal{L}_{m,R}(k_1,k_2,u,n)$*. Then*

$$\sup_{R,1\le n\le m,k} \operatorname*{ess\,sup}_u (G(u)-P(u))_+ \longrightarrow 0.$$

More precisely, at each sufficiently large fixed band B*, let* $\mathfrak{b}_B = m^2 b^{A_0 B^{-1/12},1}_{n,k,m,R} \ge 0$ *be the original positive physical remainder. Then*

$$\limsup_{m\to\infty} \sup_{R,n,k} \|P - G - \mathfrak{b}_B\|_\infty \le C B^{-1/192}.$$

The full two-sided pointwise theorem on central compact sets is equivalent to the vanishing of the positive height of $\mathfrak{b}_B$ *in the ordered limit* $m\to\infty$*, then* $B\to\infty$*. This condition is also equivalent to the separate vanishing of its positive incidence and clearance components.*

Where the transition kernel is bounded below by $d>0$*, the original roof density is eventually at least* $d/(2m^2)$ *almost everywhere. On fixed roof windows with that lower bound, the exact conditional roof law dominates* $(1-o(1))$ *times the normalized arithmetic reference as a positive measure, uniformly in all targets.*

Proof. Theorem 181.4, Corollaries 181.5 and 182.1, and Theorem 182.2 give the assertions. □

Theorem 14 (Path-valued raw limits and same-roof conditional bridges)**.** *Let* $\mathsf{Q}^{n,k,u}_{m,R}$ *be the conditional law of the pinned collision path given the original exact discrete labels and its own roof value* u*, and let* W_R *be the Gaussian bridge law with covariance* Ω_R*. The norm* BL^* *is the bounded-Lipschitz dual norm on continuous path space. For every fixed* $h>0$*,*

$$\sup_{R,n,k,t} \frac{1}{h}\int_t^{t+h} \left\| m^2 p_{n,R}(k,m,u)\mathsf{Q}^{n,k,u}_{m,R} - \mathcal{L}_{m,R}(k_1,k_2,u,n)\mathsf{W}_R \right\|_{\mathrm{BL}^*} du \longrightarrow 0.$$

At every sufficiently large fixed band B*, the complete path-valued error differs, in essential supremum of this norm, from the original positive physical path measure by at most* $CB^{-1/192}$ *after the collision limsup. The same physical measure works simultaneously for all unit bounded-Lipschitz path tests.*

On fixed central target sets, the displayed local limit also holds for the actual return path and the Gaussian bridge of covariance D_R*. Whenever the positive-part arithmetic reference mass on the roof interval is at least* dh*,* $d>0$*, the conditional bridge kernel at* u *converges in mean bounded-Lipschitz distance, uniformly, under the original conditional roof law. In particular it converges uniformly outside sets of roof values whose conditional probabilities tend to zero. These kernels fix the same exact event and the same roof value; no pointwise conditioning event is replaced by a window event.*

For each fixed central window and some $a>0$*, the unnormalized exact source also satisfies*

$$\frac{m^2}{h}\int_E e^{a(\|\mathcal{B}_{m,R}\|_\infty + \|\mathcal{Y}_{n,R}\|_\infty)} d\nu^*_R \le C.$$

Consequently the path-valued local laws hold for polynomial-growth Lipschitz tests, and for every finite $p\ge 1$*,*

$$\int W_p(\mathsf{Q}^u,\mathsf{W}_R)^p \, d\pi_m(u) + \int W_p(\mathsf{R}^u,\mathsf{V}_R)^p \, d\pi_m(u) \longrightarrow 0.$$

Here π_m *is the unchanged conditional roof law and* W_p *uses the unbounded path supremum metric. Conditional covariance kernels and path-maximum moments converge in the corresponding roof mean. The same unbounded-cost assertions under the reference roof law additionally require a pointwise positive reference floor on the window.*

The scalar positive-height criterion of Theorem 13 also suffices for the uniform essential-supremum same-roof bridge on positive-reference central sets. This implication does not assert that the remaining height criterion has been proved.

Proof. Theorem 190.3, Lemma 191.1, Theorem 191.2, Corollary 191.3 and Proposition 191.4 give the bounded-test assertions. Theorem 192.3, Theorem 193.2, Theorem 193.4 and Corollary 193.5 prove the strengthened moment and transport conclusions. □

A direct route to the path numerator. The new route requires four interfaces, not a new inducing scheme. The endpoint influence estimate in Lemma 171.1 bounds the whole path map in Lemma 190.1. The chronological peak calculation and unnormalized fourth moments in Section 161 yield fixed-band inversion uniformly over path tests. The protected correction, all-depth decision estimate and physical local variation then prove Theorem 190.3. Section 191 disintegrates this actual numerator and transfers the pinned clock. The norm integrates an absolute path-dual error in the roof coordinate; it is neither path-space total variation nor a two-sided scalar essential-height theorem.

A cap-free route to the explicit endpoint. The new scalar route uses only three finite inputs: the local all-margin mass bound of Lemma 183.1, the polynomial protected height of Proposition 183.2, and the exact global strip mass of Lemma 185.1. At large m relative to $\log(1/\varepsilon)$, the local estimate absorbs its exponential error. At smaller m, the global mass is stronger. Lemma 185.2 therefore gives $C\varepsilon^{1/16}$ with no remaining finite-count error. The positive-source split then gives a density-weighted tail $CL^{-1/144}$ and a weak $L^{145/144}$ estimate. The full exponential height cap is no longer an input to this route, and the exponent does not depend on a ratio of spectral decay to geometric height growth. All prior proofs, including the earlier cap-based argument, are retained.

Section 186 proves, for each physical component,

$$\sup_{R,n,k,t,m}\int_t^{t+h}|m^2 b^{\varepsilon,j,w}_{n,k,m,R}(u)|^q\,du\le C_qM^q(1+h)\varepsilon^{9(p_c-q)},\quad 1<q<p_c.$$

The same estimate holds for its complete smoothing correction at every bandwidth. These finite inequalities allow a count-dependent protection scale, but do not use a count-dependent spectral band. The scalar and path-valued endpoint laws and the original roof likelihood follow from the same distribution-tail argument. The arithmetic factor and finite transition kernel are permanent parts of the statements, not terms assumed to cancel.

From local variation to a one-sided pointwise theorem. The new argument uses the actual positive source partition, not the invalid implication from local L^1 convergence to uniform convergence. Section 181 first estimates the convolution of the physical remainder in essential supremum, with the reconstruction band fixed. The protected and complete section-decision corrections already have pointwise bounds. Subtracting these quantities leaves a positive physical source, which proves the lower law and the sharpened positive-height criterion. Section 182 then obtains exact conditional minorization. Positive spikes and the zero-residue specialization do not follow from those statements. The new path numerator is proved separately in Section 190; its same-roof convergence in mean is distinguished from the uniform pointwise conclusion which still requires the scalar height criterion.

The physical remainder and the ordered estimate. For a locally integrable density set $\|f\|^{\mathrm{loc}}_{1,h}=\sup_t\int_t^{t+h}|f|$. The actual physical remainder $b^{\varepsilon,w}$ contains all histories with an incidence or clearance defect, whether or not they also carry section-decision defects. For arbitrary $|w|\le M$, Theorem 179.1 proves

$$\sup_{R,n,k}m^2\|b^{\varepsilon,w}_{n,k,m,R}\|^{\mathrm{loc}}_{1,h}\le CM\varepsilon^{1/16}(h+B_*^{-1})+C_{B_*}M(1+B_*h)m^3\rho_{B_*}^{m/2}.$$

The auxiliary band B_* is fixed, but the defect width and mark may vary with m. The full correction, with no guard on the original density, therefore satisfies

$$\limsup_{m\to\infty} \sup_{R,n,k} \frac{m^2}{h} \|p_{n,R}(k,m,\cdot) - K_B * p_{n,R}(k,m,\cdot)\|_{1,h}^{\mathrm{loc}} \le CB^{-1/192}.$$

Here B is fixed before the collision limsup and $\varepsilon(B) = AB^{-1/12}$. The auxiliary band is enlarged only after that limsup. No rate in m or growing-band spectral estimate is asserted.

A short proof route. Section 178 proves the new geometric multiplier: clearance is tested at the next collision, where its boundaries have positive slope and are transverse to stable curves. It combines this with the previously proved moving-peak interpolation. Section 179 sums all physical marks with a uniform finite-count remainder and estimates their signed correction in the local density norm. Section 180 combines this with the already proved protected and section-decision corrections and the fixed-band arithmetic kernel. The earlier leading statements are compiled in Appendix F; all earlier core proofs remain.

Pointwise raw inversion remains the original target. Local total variation is stronger than testing a fixed roof interval: the test may be chosen measurably after the collision count. It is not an essential-supremum statement. The same original pointwise criterion (177.5) still requires control of possible narrow physical-boundary spikes. We have neither changed that criterion nor replaced its arithmetic main term. The scalar conditional result of Theorem 12 is for the unchanged event with $T_{n,R}$ in a fixed interval. Theorem 14 estimates the disintegrated path kernels at their own roof values in conditional mean; it does not assert convergence at every prescribed roof value.

Relation to previous local limit methods. The collision-space framework and piecewise multiplier criterion are those of [8, 4, 12]; the comparisons with collision and suspension local limits remain in Section 116. The new conclusion is not obtained by differentiating their interval statements. It uses the physical thin-layer estimate and previously proved protected regularity to control an absolute density norm on the full source. No claim of priority for general mixing local limits is made.

Appendix I. Retained complete-record theorem and its proof route

Record, source and arithmetic. Let T_R be the collision map of the triangular Lorentz table, $R \in [0.45, 0.47]$, and F_R^* its actual first return to the specified section Y_R^*. Put $c = \nu(Y_R^*) = 91/(10000\pi)$ and $\nu_R^* = c^{-1}\nu|_{Y_R^*}$. At a fixed return index n the original record $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ has the complete density

$$\nu_R^*\{K_{n,R} = k, N_{n,R} = m, T_{n,R} \in du\} = p_{n,R}(k, m, u)\, du.$$

The measure is counting in $(k, m) \in \mathbb{Z}^2 \times \{n, n+1, \ldots\}$ and Lebesgue in $u > 0$. The finite Gaussian peak sum $\mathcal{L}_{m,R}(k_1, k_2, u, n)$ is defined in (160.5). It retains every occupation residue and every spectrally damped branch needed for parameter-uniformity. Define $\gamma_{n,R} = m^{-2}\mathcal{L}_{m,R}$ and let $\mathsf{A}_{n,R}$ be its normalized positive part on this whole mixed space, as in (114.8).

Theorem 15 (The full arithmetic record and its coupled bridges)**.** *Uniformly in R, as the actual return index $n \to \infty$,*

$$\sum_{m=n}^{\infty} \sum_{k \in \mathbb{Z}^2} \int_0^{\infty} |p_{n,R}(k,m,u) - m^{-2} \mathcal{L}_{m,R}(k_1,k_2,u,n)|\, du \to 0.$$

The total mass of $(\gamma_{n,R})_+$ tends to one. Consequently $d_{\mathrm{TV}}(\mathrm{Law}_{\nu_R^*}(J_{n,R}), \mathsf{A}_{n,R}) \to 0$. Here probability total variation is one half of variation mass. No average of return indices is taken.

Let $\mathcal{B}_{N_{n,R},R}$ and $\mathcal{Y}_{n,R}$ be the collision and actual-return paths pinned to their own terminal record. Put $A_R = c^{-1/2} L_R^{-1}$, where L_R is the covariance change in (151.7)*, and set*

$$\mathsf{H}_R = \mathrm{Law}(\mathbb{B}_{\Omega_R}, A_R \mathbb{B}_{\Omega_R}).$$

Then

$$\left\| \mathrm{Law}_{\nu_R^*}(J_{n,R}, \mathcal{B}_{N_{n,R},R}, \mathcal{Y}_{n,R}) - \mathsf{A}_{n,R} \otimes \mathsf{H}_R \right\|_{\mathrm{obs,BL}} \to 0$$

uniformly. The norm allows every bounded measurable test of the record, with a unit bounded-Lipschitz test of the two paths chosen measurably from that record. It is not total variation in the path coordinates. The two Gaussian bridges have covariances Ω_R, D_R and are coupled by A_R, not independently sampled.

If Δ_n denotes the uniform error in the last display, applying the same arbitrary observation kernel to both records cannot increase it. Conditioning on the full output event of reference probability at least $r_n > \Delta_n$ gives output total-variation error at most $\Delta_n/(r_n - \Delta_n)$ and joint path-dual error at most $2\Delta_n/(r_n - \Delta_n)$. In particular these errors vanish when $\Delta_n/r_n \to 0$. No polynomial return-count rate is asserted.

Proof. Theorems 114.4, 115.2 and 115.3 prove the assertions. □

The new spectral estimate. For the retained peak data in (160.1), we prove

$$|\mu_j(R) - \mu_R|^2 \le C \kappa_j(R), \qquad |\mathcal{L}_{m,R}(y)| \le C e^{-a|y - m\mu_R|^2/m}.$$

The first inequality is stronger than continuity at a resonance. Its proof dominates a branch with a nonzero smooth physical pairing by the positive exponential moment at an imaginary tilt, then optimizes that tilt. It does not require a nonzero section residue. The second inequality prevents weakly damped peaks from carrying mass to remote centers as the radius varies. It justifies summation of all collision and displacement labels and integration over the full roof line. The original return fourth moment controls the source tail.

Arithmetic is part of the theorem. At each fixed radius the normalized reference may equivalently use

$$n^{-2} \mathfrak{a}_R(k,n,m) g_{D_R}\left(\frac{k_1, k_2, m - n/c, u - n\bar{\tau}_R/c}{\sqrt{n}} \right),$$

by Corollary 114.5. Zero residue classes are retained. The uniform-in-radius theorem uses the full transition kernel instead of assuming that arithmetic types are constant. An unmodulated Gaussian specialization is not asserted without verification of the concrete section residue criterion.

The unrestricted pointwise target. The raw object remains this same four-coordinate return record. The positive identity of Theorem 181.4 reduces its unrestricted two-sided pointwise theorem to the essential heights of both the incidence and clearance sources at the original exact labels. Those requirements are not consequences of the total-variation result above. The local weak $L^{145/144}$ law, positive lower law and common all-resolution exceptional sets remain stated and proved, together with their exact quantifiers, in Appendix H and the cited core sections. Every inherited mathematical module remains compiled.

Proof route and comparison. Sections 114 and 115 give the new route first. They explicitly identify the inherited occupation spectrum, local raw variation and joint clock identity used at each step. The later parts retain the complete geometric and raw inversion proofs. The source analysis goes beyond a fixed interval limit because its roof tests are allowed to depend measurably on all exact labels. The new passage from local to global needs an arithmetic reference-tail theorem, not merely tightness of the true process. Positive-pressure domination, Gaussian completion of the square and postselection normalization are not claimed as new general methods. Their use here is to keep moving occupation peaks, the complete raw record and the two coupled clocks in one statement. The theorem-level comparisons with [12, 3] remain in Section 116; neither cited interval theorem is differentiated to obtain a density value.

Appendix J. Retained revision-65 abstract and leading statements

This appendix retains the preceding abstract and leading statements. Their theorem labels, hypotheses, conclusions and proof references are unchanged. The new count-uniform contact and physical-trace results are stated at the beginning of the present article.

The preceding abstract. We study the joint displacement, collision count and flight time of actual returns in a triangular periodic Lorentz gas. A visible spectral branch is dominated at imaginary frequency by the positive pressure of the original occupation record. At an exact resonance their quadratic terms agree. We prove that this flat contact makes the squared drift negligible compared with the damping. Consequently the entire arithmetic transition kernel has one Gaussian shape, centered at the physical mean and with the physical covariance, uniformly through changes of arithmetic type. Its finite phases and damped residues are retained.

The complete fixed-return law is therefore approximated in total variation on its untruncated mixed record space by a damped arithmetic factor times the physical return Gaussian. The same reference applies to the graph-coupled collision and actual-return bridges, and to record-only observations with their full selection events. The pressure principle is stated in arbitrary dimension with separate physical hypotheses and is also verified in a deterministic singular hyperbolic baker map. We additionally verify the pressure hypotheses on a correlated Markov baker map with a continuous deterministic roof. Its exact stable coarea formula yields a pointwise roof local limit and positive boundary-source height bounds uniformly throughout a nonzero perturbation interval. Keeping two integer counts before scalar projection resolves the arithmetic transition. For fixed endpoint witnesses the critical density profile has exactly the damping predicted by pressure contact. A separate positive coarea criterion states the exact-label inputs needed for source heights. These independent realizations do not replace the Lorentz record.

For the original Lorentz source, the endpoint contact determinant also cancels a product of inverse incidences at every finite count. This yields an incidence-source height gain. A two-coordinate physical coarea formula bounds transverse clearance sources and retains their rank-deficient part as a positive remainder. The count constants in these estimates are exponential and are not identified with central-scale bounds. Finite-type coarea further controls clearance tangent to a regular roof fiber, including second-order folds. On an incidence-capped physical graph, the remaining seam/rank caustic has finite fibers in parameter–roof space. We localize the original positive clearance source near that caustic, with its count-dependent constants retained. A buffered algebraic caustic gives explicit separation and tube moduli in both collision count and incidence cap. Fixed-block real quantifier elimination supplies these finite-count bounds.

For normal-to-normal physical words we identify the intrinsic quadratic coefficient with a reversible square-root determinant. Direct angular coarea gives the density inside roof-critical clearance caustics, including multiple seams and exact section decisions, with a width-uniform finite-jet error.

The unrestricted pointwise raw-density target remains distinct from these integrated conclusions. It still requires essential-height control of the two positive physical boundary sources. No source mass estimate is identified with that height control.

The preceding leading statements.
The original record. Let $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ record n actual returns of the triangular Lorentz collision map to Y_R^*, under $\nu_R^* = c^{-1}\nu|_{Y_R^*}$, where $c = 91/(10000\pi)$. The density $p_{n,R}(k,m,u)$ is with respect to counting measure in $(k,m) \in \mathbb{Z}^2 \times \{n, n+1, \dots\}$ and Lebesgue measure in $u > 0$. No collision or return is removed. Write

$$V_{n,R}(k,m,u) = \frac{(k_1, k_2, m - n/c, u - n\bar{\tau}_R/c)}{\sqrt{n}}.$$

The physical return covariance is D_R; the collision occupation covariance is $\Omega_R = cL_R D_R L_R^{\mathsf{T}}$, with $\det L_R = c$. The finite peak data in (160.1) retain the exact occupation arithmetic.

Theorem 16 (Flat contact and the common Gaussian shape)**.** *For every retained peak put $v_j = \mu_j - \mu_R$ and $h_j = \|\Omega_R - \operatorname{Re} H_j\|$. Uniformly near zero damping,*

$$|v_j| \le C\{\sqrt{\kappa_j h_j} + \kappa_j^{2/3}\}, \qquad \sup_{0<\kappa_j\le\delta} \frac{|v_j|^2}{\kappa_j} \longrightarrow 0 \quad (\delta \downarrow 0).$$

Thus $\sqrt{m}\, v_j \to 0$ uniformly on each set $m\kappa_j \le L < \infty$. For the complete transition kernel, set

$$\Phi_{m,R}(y) = c^{-1} \operatorname{Re} \sum_j e^{-ia_j(R)\cdot y} z_j(R)^m B_j(R).$$

There are $a > 0$ and $\theta_m \to 0$ such that

$$\sup_{R,y} e^{a|y-m\mu_R|^2/m} \left| \mathcal{L}_{m,R}(y) - \Phi_{m,R}(y) g_{\Omega_R}\left(\frac{y - m\mu_R}{\sqrt{m}}\right)\right| \le \theta_m.$$

All damping, phases, residues and partition amplitudes are retained. The covariance and center in the Gaussian are the physical ones, also through changes of arithmetic type.

Proof. Apply Theorem 107.1, Corollary 107.3 and Theorem 108.1. □

Theorem 17 (The complete arithmetic Gaussian return law)**.** *Set*

$$\alpha_{m,R}(k,n,u) = c^{-1}\Phi_{m,R}(k_1,k_2,u,n), \qquad q^\circ_{n,R}(k,m,u) = n^{-2}\alpha_{m,R}(k,n,u) g_{D_R}(V_{n,R}(k,m,u)).$$

Uniformly in $R \in [0.45, 0.47]$, as the fixed return index $n \to \infty$,

$$\sum_{m=n}^{\infty} \sum_{k\in\mathbb{Z}^2} \int_0^\infty |p_{n,R}(k,m,u) - q^\circ_{n,R}(k,m,u)|\, du \to 0.$$

The signed reference has vanishing negative mass and its positive part has mass tending to one. Its normalized positive part $\mathsf{A}^\circ_{n,R}$ approximates the original record law in probability total variation. At fixed radius the damped factor may be replaced by the exact finite residue $\mathfrak{a}_R(k,n,m)$, including zero classes.

For the actual collision and return bridges from the same trajectory,

$$\left\| \operatorname{Law}_{\nu_R^*}(J_{n,R}, \mathcal{B}_{N_{n,R},R}, \mathcal{Y}_{n,R}) - \mathsf{A}^\circ_{n,R} \otimes \operatorname{Law}(\mathbb{B}_{\Omega_R}, c^{-1/2} L_R^{-1} \mathbb{B}_{\Omega_R}) \right\|_{\text{obs,BL}} \to 0.$$

The norm is variation in the observation and bounded-Lipschitz dual in the two paths, not path-space total variation. The record becomes independent of this coupled bridge pair; its two paths are not independently sampled. A record-only observation kernel contracts the error. If that error is Δ°_n, selection on its complete output event of reference mass at least $r_n > \Delta^\circ_n$ costs at most $\Delta^\circ_n/(r_n - \Delta^\circ_n)$ in probability TV and twice that amount in the joint path-dual norm.

Proof. Theorem 108.3 and Corollary 108.4 prove these statements. Probability TV is one half of variation mass. □

Why the centers can be removed. The positive pressure dominates a visible centered branch at an imaginary tilt. Their difference is nonnegative, has value κ_j and gradient v_j, and its Hessian at zero is $\Omega_R - \operatorname{Re} H_j$. At zero damping this Hessian vanishes. Retaining it in Taylor's formula gives the first theorem; a bound by an unspecified quadratic constant gives only $|v_j|^2 \le C\kappa_j$. The new proof also gives an explicit modulus in the covariance mismatch, but does not assert a polynomial rate in the collision count. It uses no derivative in the table parameter and no spectral estimate at a growing Fourier band.

A reusable statement and a second realization. Section 107 separates positive physical moments, branch visibility and second-order contact as hypotheses of a scalar theorem in arbitrary dimension. They are verified for the original occupation spectrum and, independently, for a deterministic three-strip baker map. In the latter example a peak has $\kappa_\epsilon = \pi^2\epsilon^2/9 + O(\epsilon^3)$ but $v_\epsilon = \pi^2\epsilon^3/18 + O(\epsilon^4)$. This is a second singular hyperbolic realization of the spectral principle, not a substitute for the Lorentz density theorem. The baker observable has finite support; no continuous density theorem is asserted for it.

A correlated roof and a direct height calculation. Sections 109 and 110 verify the pressure/contact inputs in a singular Markov baker map whose state correlations do not vanish. Its positive roof is a finite-state reward plus the actual stable endpoint difference, not an added random smoothing. For its unperturbed roof density p_m^{M}, with $\bar\tau = 24/7$ and $\sigma^2 = 204/343$, we prove

$$\operatorname*{ess\,sup}_{t\in\mathbb{R}} \left|\sqrt{m}\, p_m^{\mathrm{M}}(t) - g_{\sigma^2}((t - m\bar\tau)/\sqrt{m})\right| \le C \log(2+m)/\sqrt{m}.$$

The three stated boundary sources have normalized heights bounded by $C\{s + (3/4)^{\lfloor m/2\rfloor}\}$ at zero perturbation. Those results remain in Section 110. The following theorem places density inversion, source heights and the moving peak on the same perturbed family. It uses the original two integer counts before projection, not a presumed common scalar lattice span.

Theorem 18 (Pointwise inversion through a perturbed arithmetic family)**.** *On the Markov baker with transition matrix* $P = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/3 & 2/3 \end{smallmatrix}\right)$ *and invariant state weights* $\pi = (4/7, 3/7)$*, retain the positive roof* $\tau_\epsilon = 3 + I_1 + \epsilon I_0 I_1 + y_1 - y_0$*. For every fixed* $0 < \epsilon_0 < 1/4$*, uniformly for* $|\epsilon| \le \epsilon_0$*,*

$$\operatorname*{ess\,sup}_{t\in\mathbb{R}} \left|\sqrt{m}\, p_{m,\epsilon}(t) - g_{\sigma_\epsilon^2}((t - m\bar\tau_\epsilon)/\sqrt{m})\right| \le C \frac{\log(2+m)^{3/2}}{\sqrt{m}},$$

where $\bar\tau_\epsilon = (24 + 2\epsilon)/7$ *and* $\sigma_\epsilon^2 = (204 + 384\epsilon + 198\epsilon^2)/343$*. For the three original boundary sources,*

$$\sqrt{m}\left(\|p^{\mathrm{vert}}_{m,\epsilon,s}\|_\infty + \|p^{\mathrm{stab}}_{m,\epsilon,s}\|_\infty + \|p^{\mathrm{age}}_{m,\epsilon,s}\|_\infty\right) \le C\{s + (3/4)^{L_m}\}, \quad L_m = \min(\lfloor m/2\rfloor, \lfloor\sqrt{m}\rfloor).$$

Lipschitz endpoint weights retain the explicit arithmetic profile (112.8)*. On the critical scale* $\epsilon = c_m/\sqrt{m}$*, bounded* c_m*, the fixed witness pair* $e^{2\pi i y_0}, e^{-2\pi i y_m}$ *has leading density*

$$g_{204/343}(z) e^{-2\pi i (r_\epsilon - (16/17) c_m z)} e^{-12\pi^2 c_m^2/119}, \quad z = \frac{t - m\bar\tau_\epsilon}{\sqrt{m}}, \quad r_\epsilon = t - 3m - 2m\epsilon/7,$$

with the same normalized error order. In particular the damping coefficient agrees with the moving pressure peak on this actual source.

Proof. Apply Theorems 112.3, 113.1 and 113.2. □

Finite-state Markov additive density theory and compact-family versions already exist [20, 21]. The new conclusion in this proof chain concerns exact coarea through the scalar arithmetic transition, positive source heights and the visible critical endpoint profile. It does not assert a Lorentz height bound by analogy.

Density estimates on the original physical boundary source. For a regular m-collision orbit set $\mathcal{I}_{m,R} = \prod_{j=0}^{m} \cos(\varphi \circ T_R^j)^{-1}$. Theorem 97.3 proves that this unbounded insertion has a roof density at most CA^m, for fixed $C > 0$, $A > 1$. The same determinant controls all contact incidences simultaneously. For the original first-incidence source, and for the part of the original clearance source on which $|\det D_{(\alpha,p)}(L_{m,R}, Z_d \circ T_R^{j+1})| \ge \eta$, Corollary 97.4 and Theorem 98.2 give

$$\operatorname*{ess\,sup}_u \sum_{n=1}^{m} \sum_k b^{\varepsilon,\mathrm{inc},1}_{n,k,m,R}(u) \le CA^m\varepsilon, \qquad \operatorname*{ess\,sup}_u \sum_{n=1}^{m} \sum_k b^{\varepsilon,\mathrm{tr},\eta,1}_{n,k,m,R}(u) \le CA^m\varepsilon/\eta.$$

These estimates allow count-dependent thresholds and bounded source insertions. They are derived on the true circular collision geometry, not on a Markov replacement. They do not bound the rank-deficient clearance source or remove the exponential count factor.

The unrestricted raw target and the proof route. The paper retains its four-coordinate record and unrestricted two-sided pointwise target. The positive incidence and clearance sources in Corollary 181.5 still require essential-height bounds at the exact labels. Removing the reference's moving centers does not remove those physical sources. The new reference positivity statement likewise does not give a pointwise upper bound for the true density. No unrestricted same-roof bridge or forward essential likelihood conclusion is drawn from the integrated theorem.

The direct physical source calculation is in Sections 97 and 98. Its inputs are the endpoint action and collision-graph complexity of Section 167, the exact contact determinant of Section 169, and the next-collision clearance geometry of Section 178. Thus its short proof route does not require the independent Markov realization. The positive exact-label criterion is in Section 111. Its perturbed verification and critical density profile are in Sections 112 and 113; the required two-count Fourier and positive cylinder arguments are proved there. The scalar root, finite-witness and complex Gaussian details are given first in Section 106. The canonical Lorentz route then runs through Sections 107 and 108, followed by the full-record tail and joint clock arguments of Sections 114 and 115. They state their inherited inputs locally. The entire earlier proof chain and all earlier statements remain compiled; the previous front matter is in Appendix I. The comparison with billiard endpoint and suspension local limits [12, 3] concerns the precise observable and topology, not a claim that pressure domination or Gaussian differentiation is new. The additional conclusion here is the uniform common Gaussian shape of a visible moving resonance family, proved before summing the original exact-return law.

Finite-type clearance and its physical caustic. The first-clearance source is further resolved on the actual roof fibers in Sections 99 and 100. If a coordinate derivative of the roof is at least a and the rth clearance derivative along its level curve is at least b, its exact-label essential height is at most $C_r A_r^m a^{-1}(\varepsilon/b)^{1/r}$. In particular the second-order estimate includes fold contacts with vanishing two-coordinate Jacobian. The complete complementary source remains positive.

For fixed m and incidence cap $\chi > 0$, the physical clearance caustic in (R, t) has at most CA^m points in each fixed-radius fiber. A compact semialgebraic separation inequality bounds the original source away from its neighborhood and gives an exact raw-error formula retaining only the caustic source and explicit finite-count errors. The neighborhood also has a power measure bound, which combines with the inherited weak-integrability estimate. These are geometric reductions for the Lorentz source, not a transfer of the Markov model theorem. Their count-dependent constants do not establish the two ordered central-height limits.

Effective moduli on the physical graph. Sections 101 and 102 quantify a buffered version of this localization. For one fixed C_0, put

$$E_m = \left\lceil 2^{C_0(m+1)^{16}} \right\rceil, \qquad K_m = 2^{E_m}.$$

The source cap is χ, while the critical-value reference uses $\chi/2$. Theorems 101.3 and 101.4 give

$$(\chi d)^{E_m} \le K_m\big(\big||Z| - R\big| + |\mathcal{J}|\big), \qquad \sup_R |\text{vertical } h\text{-tube}| \le K_m \chi^{-1} h^{1/E_m}.$$

The reference is an explicitly enlarged weak-first-hit critical graph, not a replacement source. The original first-clearance source is restricted positively to its neighborhood; its complement has the height bound in Theorem 102.1. The cap buffer makes the compactness argument valid while the cap varies. Standard effective quantifier elimination supplies the algebraic bounds, with its integer coefficient control. The new application is the fixed-format physical graph and the exact

positive source formula, not a new quantifier-elimination theorem. The very large count budgets and the remaining tube height are kept distinct from the ordered pointwise endpoint.
Intrinsic weights and density inside the caustic. The physical caustic is not used solely as an exceptional roof set. Sections 103–105 calculate its roof-critical contribution locally on the same collision source. They retain normal endpoints, positive selected incidences, all competing-hit restrictions and the original section decisions. A boundary branch linearization is distinguished from the derivative of the billiard map at a singular state.

Theorem 19 (Reversible weights and physical caustic profiles)**.** *For a normal-to-normal m-flight word, let M_z be its canonical half-word derivative, $S = \operatorname{diag}(1,-1)$ and $P_z = SM_z^{-1}SM_z$. The full local quadratic coefficient for ν/c_* is*

$$J_z = \frac{1}{Rc_*\sqrt{|\det(I-P_z)|}} = \frac{1}{Rc_*}\frac{\lambda_z^{-1/2}}{1-\lambda_z^{-1}}, \qquad \lambda_z > 1.$$

At a regular word $P_z = DT_R^{2m}(z)$; at a competing-hit boundary it denotes the physical-side branch expression only. The original labels remain those of the first m flights.

At a simple clearance seam through a roof minimum t_z, the original local strip $0 < g < s$ has density

$$b_{z,s}(t_z+h) = \frac{J_z}{\pi}\operatorname{arcsinmin}\left\{1, \frac{s}{\kappa_z\sqrt{2h}}\right\} + O_z(J_z h^{1/4}), \qquad h > 0,$$

uniformly in the width-to-roof ratio. Here $\kappa_z^2 = dg(z)(\nabla^2 F(z))^{-1}dg(z)^{\mathsf{T}} > 0$. In particular its physical one-sided coefficient is $J_z/2$. For any fixed word with positive selected incidences, multiple clearance seams and section-boundary coincidences are allowed in the affine angular profile of Section 105. For all sufficiently small widths relative only to the local incidence cap and the fixed near-clearance scale, it satisfies

$$\sum_{n,k}\left|b_{z,n,k}^{\varepsilon,\mathrm{clr}}(t_z+h) - \frac{J_z}{2\pi}\mathcal{A}_{z,n,k}(\varepsilon,h)\right| \le C_z J_z h^{1/4}, \qquad 0 \le \sum_{n,k}\mathcal{A}_{z,n,k} \le 2\pi.$$

All constants and neighborhoods in these profile assertions are local finite-word data. No ordered long-count height conclusion is included in their statements.

Proof. Theorem 103.2 and Proposition 103.4 give the coefficient. Theorems 104.2 and 105.3 give the two local profiles. □

The determinant relation is an explicit two-dimensional application of the action–monodromy principle, not a replacement for the classical Hill theory [24]. The density calculation supplies the original positive source fraction and its exact-label sum. Its affine offsets avoid presuming that the fixed reconstruction width is smaller than every nonzero clearance margin. The ordered trace transfer in Corollary 104.5 states the further concentration input explicitly. Neither that input nor uniform coverage of all long words is supplied by a finite-count profile, and the full pointwise arithmetic target remains unchanged.

Appendix K. The retained revision-66 leading statements

The following abstract and introduction are retained verbatim as the preceding stage of the argument. The active statements are at the beginning of the present article.

Retained abstract. We study the joint displacement, collision count and flight time of actual returns in a triangular periodic Lorentz gas. The complete arithmetic return law, its common Gaussian shape through changes of arithmetic type and its coupled collision–return bridges are retained. We develop the geometry needed for pointwise inversion on the same physical source.

For normal-to-normal collision words we prove a count-uniform selected-contact continuation under an exponentially tapered incidence bound. Competing-hit and section boundaries may cross this chart. An incidence-normalized tridiagonal inverse gives uniform contact jets, Morse radii and relative source distortion. Keeping the nonlinear physical decisions exactly, we obtain a multiplicative angular formula for the original first-clearance density. Its coefficient is the reversible square-root determinant of the original half-word.

A positive comparison with the two-endpoint local law gives an ordered central-scale bound for the resulting finite-width physical collar trace. This comparison permits coincident critical values and changing exact labels, and uses no word-count multiplier. The zero-width observable trace has the additional physical angular fraction. An exact positive inequality separates it from the finite-width trace by a specified angular-loss term. That term, the complete incidence height and the positive source outside the uniform charts are not estimated here. They remain distinct inputs to the unrestricted pointwise arithmetic raw-density theorem; no order of limits is interchanged.

The record and the pointwise problem. Let $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ be the record of n actual returns of the circular triangular Lorentz gas to the original section Y_R^*, under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, where $c_* = 91/(10000\pi)$ and $R \in [9/20, 47/100]$. Its density $p_{n,R}(k, m, t)$ uses counting measure in $(k, m) \in \mathbb{Z}^2 \times \{n, n+1, \dots\}$ and Lebesgue measure in the roof t. Neither a collision nor a return is removed. The joint occupation convention counts $0, \dots, m-1$; terminal membership is checked separately at collision m.

The arithmetic transition kernel $\mathcal{L}_{m,R}$ retains its finite phases and damped residues. The complete record theorem (Theorem 17) concerns the untruncated law in variation, with coupled path conclusions in the stated path-dual norm. For each fixed M, put $Z_{m,R} = (k_1, k_2, t - m\bar{\tau}_R, n - mc_*)/\sqrt{m}$. The stronger pointwise target, as $m \to \infty$, is

$$\sup_{R,n,k\ t:\ |Z_{m,R}|\le M} \operatorname*{ess\,sup} \left| m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n) \right| \longrightarrow 0.$$

The positive raw-error identity requires the complete ordered incidence and clearance heights for this target. The present geometric results apply directly to those original sources; they do not replace the pointwise problem by a different observable.

A chart may cross a physical boundary. Put $q = 47/53$, $d_j = \min(j, m-j)$ and $e_j = q^{d_j/4}$. A normal-to-normal word z in the physical-side closure is χ-incident when its selected incidence cosines satisfy $c_j(z) \ge \chi e_j$, $0 < \chi \le 1/10$. No lower bound is imposed on its clearance or section margins. Its selected contact equations can therefore continue beyond the physical word. Every such nonphysical point will have zero source weight. This distinction allows uniform coordinates without replacing the physical orbit space by an overcover.

Theorem 20 (Uniform charts and the physical angular density)**.** *There are constants $r_0, C > 0$, independent of the collision count, such that every χ-incident word has a Morse chart on $|y| < r_\chi = r_0\chi^3$ with*

$$F_z = t_z + |y|^2/2, \qquad \left| \log \frac{2\pi\widetilde{w}_z(y)}{J_z} \right| \le K_\chi |y|, \qquad K_\chi = C\chi^{-2}.$$

Here $\widetilde{w}_z$ is the original selected-coordinate density before imposing physical restrictions, and

$$J_z = \frac{1}{Rc_*\sqrt{|\det(I - SM_z^{-1}SM_z)|}}, \qquad S = \operatorname{diag}(1,-1).$$

The matrix M_z is the original canonical half-word derivative, or its physical-side branch value at a seam. There is no primitive-period quotient, and no label is read from the doubled word.

For $0 < \varepsilon \le \chi/4$ let $A^{\varepsilon}_{z,n,k}(h)$ be the angular integral of the exact physical word indicator, exact return label and unchanged first-clearance multiplier on $|y| = \sqrt{2h}$, as in (95.4)*. Then $\sum_{n,k} A^{\varepsilon}_{z,n,k} \le 2\pi$ and the original restricted density satisfies, almost everywhere,*

$$e^{-K_\chi\sqrt{2h}}\frac{J_z}{2\pi}A^{\varepsilon}_{z,n,k}(h) \le b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z+h) \le e^{K_\chi\sqrt{2h}}\frac{J_z}{2\pi}A^{\varepsilon}_{z,n,k}(h), \quad 0 < h < r_\chi^2/2.$$

The constants require no boundary-gradient lower bound or angle between boundary gradients. The nonlinear angular sets themselves are retained, not replaced by a uniformly evaluated affine profile.

Proof. The weighted inverse in Lemma 94.1 gives Theorems 94.2 and 94.3. Theorem 103.2 identifies the coefficient. Theorem 95.2 applies positive polar coarea with the exact source indicators. □

Theorem 21 (The ordered physical collar trace)**.** *For a family of these charts, each counted once, define*

$$\mathcal{Q}_{m,s;n,k,R} = \sum_z \left(\frac{1}{s}\int_0^s b^{\varepsilon,\mathrm{clr}}_{z,n,k}(t_z+u)\,du\right)\delta_{t_z}.$$

For each fixed χ, ε, $0 < s < r_\chi^2/2$ and $h > 0$,

$$\limsup_{m\to\infty} \sup_{R,n,k,t} m^2\mathcal{Q}_{m,s;n,k,R}([t-h,t+h]) \le C(2h+s).$$

The constant is independent of the permitted widths. The estimate keeps the two exact integer labels and permits coincident critical values and section-boundary coincidences. In particular the expression with $h = s$ tends to zero when $s \downarrow 0$ after the collision limit.

Proof. Every contributing physical collar lies in the same exact-label event, in a roof interval of length $2h + s$, and in two normal endpoint strips of width $C\sqrt{s}$. The collars are disjoint. Their positive comparison with Lemma 170.1 gives Theorem 96.1. □

The zero-width distinction. At a clearance seam the fixed-count limiting coefficient is $\beta_{z,n,k} = J_z\theta_{z,n,k}$, where $\theta_{z,n,k}$ is the physical and exact-label angular fraction. The resulting atomic trace $\mathcal{B}_m$ satisfies

$$\mathcal{B}_m(I) \le e^{K_\chi\sqrt{2s}}\mathcal{Q}_{m,s}(I) + \mathcal{D}_{m,s}(I),$$

where $\mathcal{D}$ is the positive loss of the actual angular set relative to its tangent angular fraction, defined in (96.7). Proposition 96.3 states the additional uniform estimate needed to pass to $\mathcal{B}_m$. That estimate is not supplied by bounded contact jets. Nor does $\mathcal{B}_m$ equal the larger trace of all full coefficients J_z. The complete first-incidence source and the positive clearance source outside the charts remain in (95.8) and in the original raw-error identity. In particular the two preceding theorems do not assert the unrestricted pointwise local limit or its same-roof conditional laws.

Proof route and relation to the preceding article. The new proof route is Sections 94–96. It uses the contact matrix and source normalization of Sections 169 and 103, and the inherited two-endpoint local law in Section 170. No periodic-orbit counting theorem is substituted for that positive comparison. The square-root weight retains the action–monodromy relation discussed in [24]; the new ingredients are the incidence-normalized weighted inverse, uniform selected-contact jets across physical seams, and the exact physical collar trace.

All preceding core proofs, the arithmetic return law, the independent model realizations and their scopes remain in the article. The previous abstract and leading statements are compiled verbatim in Appendix J; the earlier appendices also remain. The effective finite-count budgets of Section 102 are retained but not used in the new uniform argument. Their threshold $4K_m\varepsilon \le (\chi h)^{E_m}$ fails eventually for fixed $\varepsilon > 0$ and $\chi, h \le 1$. No choice of a growing reconstruction band is made here.

Appendix L. Retained leading statements of revision 67

This appendix preserves the preceding abstract and introduction verbatim. Its historical wording records the preceding stage of the argument.

The preceding abstract. We study displacement, collision count and flight time of actual returns in a triangular periodic Lorentz gas, retaining the complete arithmetic record law and the original pointwise inversion problem. Count-uniform selected-contact coordinates and exact nonlinear physical masks give a positive finite-width collar trace at normal roof-critical words. We pass from this trace to zero-width physical coefficients on explicit finite-jet strata. The decisive estimate measures angular error relative to the physical coefficient, so that it can be absorbed before taking the collision limit; no concentration of the larger reversible trace is used. The resulting critical-value bound has a constant independent of the stratum cutoff and gives an ordered essential-height estimate for the corresponding original-source caustic collars. An explicit weighted tail of the angular condition number controls the extension to all seam charts. That tail and the complete incidence and positive complementary clearance heights remain separate requirements for the unrestricted pointwise arithmetic law. Every collision, exact return label and arithmetic zero class is retained.

The preceding introduction.

The record and the pointwise problem. For the original section Y_R^* write $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$, under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, where $c_* = 91/(10000\pi)$ and $R \in [9/20, 47/100]$. The density $p_{n,R}(k,m,t)$ uses counting measure in (k,m) and Lebesgue measure in t. Occupation counts collisions $0,\dots,m-1$; terminal section membership at m is a separate condition. The complete arithmetic record theorem and coupled path laws retain their scope (Theorem 17). Put $Z_{m,R} = (k_1, k_2, t - m\bar{\tau}_R, n - mc_*)/\sqrt{m}$. For each fixed M the stronger target remains

$$\sup_{R,n,k}\ \operatorname*{ess\,sup}_{t:\,|Z_{m,R}|\le M} \left| m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n) \right| \longrightarrow 0.$$

The finite arithmetic transition kernel $\mathcal{L}_{m,R}$ is not replaced by an unmodulated Gaussian. We develop a new density-height estimate on its original positive boundary source.

Physical coefficients at a normal critical word. Fix χ, ε with $0 < \chi \le 1/10$ and $0 < \varepsilon \le \chi/4$. The selected incidence cosines of a word satisfy $c_j \ge \chi(47/53)^{\min(j,m-j)/4}$. Its Morse chart has $F_z = t_z + |y|^2/2$, radius $r_\chi = r_0\chi^3$ and relative density distortion $K_\chi = C\chi^{-2}$. Put $h_\chi = r_\chi^2/2$. The chart is a coordinate continuation, not an assertion of physical continuation across a first-hit seam. Every actual decision is kept as an exact mask. With $S = \operatorname{diag}(1,-1)$ the coefficient is

$$J_z = \frac{1}{Rc_*\sqrt{|\det(I - SM_z^{-1}SM_z)|}}.$$

Here M_z is the canonical derivative of the original half-word, or its physical-side branch value. There is no primitive-period quotient. At a clearance seam the zero-width physical coefficient is $\beta_{z,n,k} = J_z\theta_{z,n,k}$, where θ is the angular fraction assigned by the original exact-label rule. In particular this coefficient need not equal J_z.

The geometric cutoff. In a sufficiently small saturated disk, take every scalar physical and section margin f, with $b_f = f(0)$, $A_f = |Df(0)|$ and $L_f = \frac{1}{2}\|D^2 f\|_\infty$. Let ρ_z be the explicit inactive-sign radius in (91.1), including the radius on which this complete Boolean description is valid. For $\theta_{z,n,k} > 0$ put

$$\mathfrak{a}_{z,n,k} = \max\left\{ \rho_z^{-1}, \frac{1}{\theta_{z,n,k}} \sum_{b_f = 0} \frac{L_f}{A_f} \right\}.$$

At every fixed word this is finite; no all-count bound is assumed. The finite-jet stratum consists of pairs $(z,(n,k))$ with $\mathfrak{a}_{z,n,k} \le K$. It can contain simultaneous seams and coincident or arbitrarily close tangent directions.

Theorem 22 (Zero-width physical trace and caustic height)**.** *Let $\mathcal{B}^K_{m;n,k,R}$ sum the coefficients $\beta_{z,n,k}$ at t_z on the finite-jet stratum, and let $\mathcal{Q}^K_{m,s;n,k,R}$ be the corresponding averages of the actual physical source over $0 < F_z - t_z < s$. For fixed K, χ, ε and $K\sqrt{2s} < 1$, $0 < s < h_\chi$, one has, for every $h > 0$,*

$$\mathcal{B}^K_m(I) \le \frac{e^{K_\chi \sqrt{2s}}}{1 - K\sqrt{2s}} \mathcal{Q}^K_{m,s}(I), \qquad \limsup_m \sup_{R,n,k,t} m^2 \mathcal{B}^K_m([t-h, t+h]) \le 2Ch.$$

The constant C is independent of K. If $b^{K,H}$ is the same original source restricted to $0 < F_z - t_z < H$, then

$$\limsup_m \sup_{R,n,k} m^2 \| b^{K,H}_{m;n,k,R} \|_\infty \le CH e^{K_\chi \sqrt{2H}} (1 + K\sqrt{2H}), \quad K\sqrt{2H} < 1, \quad H < h_\chi.$$

All widths are fixed before the collision limit. No word-count factor, label averaging or full reversible-trace bound occurs.

Proof. A homogeneous sign estimate gives a relative angular error at most $K\sqrt{2u}$ (Theorem 91.2). Positivity then gives Theorem 92.1 from the inherited two-endpoint collar law. Corollary 92.2 converts the critical-value bound to essential height using the same nonlinear profile, not differentiation of an interval local limit. □

Theorem 23 (The positive cost of exhausting the strata)**.** *Let $E(K)$ be the collision-limsup unit-window height of the atomic physical coefficients with $\mathfrak{a}_{z,n,k} > K$, as in* (93.2)*. The complete seam-chart angular loss of* (96.7)*, for fixed K, χ, ε and $0 < s < h_\chi$ with $K\sqrt{2s} < 1$, satisfies*

$$\limsup_m \sup_{R,n,k,t} m^2 \mathcal{D}_{m,s}([t-1, t+1]) \le \tfrac{4}{3} CK\sqrt{2s} + E(K).$$

Thus $E(K) \to 0$ is sufficient for the full seam-chart angular-loss limit and the corresponding zero-width concentration theorem. The bound on $E(K)$ is an additional physical input, not a conclusion of uniform selected-contact jets.

Proof. This is Proposition 93.1: the lost angular fraction in a bad stratum is at most its entire physical tangent fraction. The good part is estimated before taking either outer limit. □

The complete source and the proof route. The three new sections below prove these statements from the primitive margin budgets, the exact nonlinear profile and the positive two-strip comparison. The density $b^{K,H}$ is a restriction, and the complete clearance source is $b^{K,H} + r^{\varepsilon,\chi,K,H}$ with $r \ge 0$. The latter retains selected grazing, failed incidence taper, noncritical rank components, non-seam centers, zero-angular-fraction labels, bad strata and omitted annuli. Neither its essential height nor the complete first-incidence height is inferred from the new theorem. The positive raw-error identity and the unrestricted same-roof consequences retain exactly their prior logical requirements.

The proof spine is Sections 91–93, with the geometric and dynamical inputs in Sections 94–96. The comparison with local-limit literature is made at the end of Section 93. All preceding core proofs and appendices remain compiled. The preceding leading statements are retained in Appendix K; historical wording there describes the earlier stage of the argument.

Appendix M. Retained revision-68 opening statements

The abstract and introduction of the preceding revision follow. They record the scope of its guarded-persistence theorem; the absolute-guard and finite-scale extensions are proved in the opening sections of the present article.

Revision-68 abstract.

> We study the joint displacement, collision count and flight time of actual returns in a triangular periodic Lorentz gas, with the complete arithmetic record and the original pointwise inversion problem retained. At a normal roof-critical word, the exact physical and section decisions have analytic angular germs. We prove that every nonzero germ has a positive leading radial coefficient, including labels with zero tangent fraction and arbitrarily high finite order of birth. A relative estimate retaining the original clearance guard converts this description into radial persistence. One positive comparison with an outer annulus then gives an ordered essential-height bound for the original nonlinear caustic collars on explicit persistence strata. Its constant is independent of the birth order; no full reversible-trace estimate or zero-width coefficient for a born sector is used. Non-seam centers with positive center guard are included. The complete incidence source and the complementary clearance height remain explicit in the positive identity governing the unrestricted pointwise arithmetic theorem.

The physical record and the pointwise theorem. Let $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ be the actual return record to Y_R^* under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, where $c_* = 91/(10000\pi)$ and $R \in [9/20, 47/100]$. The density $p_{n,R}(k,m,t)$ is with respect to counting measure in (k,m) and Lebesgue measure in t. Occupation counts collisions $0,\dots,m-1$, with terminal section membership at m imposed separately. The complete arithmetic record theorem and coupled path laws are retained (Theorem 17). With $Z_{m,R} = (k_1, k_2, t - m\bar{\tau}_R, n - mc_*)/\sqrt{m}$, the original unrestricted pointwise target is, for every fixed M,

$$\sup_{R,n,k} \operatorname*{ess\,sup}_{t:\,|Z_{m,R}|\le M} \left| m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n) \right| \longrightarrow 0. \tag{M.1}$$

The finite arithmetic transition kernel and its zero classes are not replaced by an unmodulated Gaussian. We address a nonlinear source which a zero-width tangent trace cannot see.

Nonlinear angular birth. Fix $0 < \chi \le 1/10$ and $0 < \varepsilon \le \chi/4$. At a normal critical word satisfying $c_j \ge \chi(47/53)^{\min(j,m-j)/4}$, the selected-contact Morse chart has $F_z = t_z + |y|^2/2$, radius $r_\chi = r_0\chi^3$, and relative density constant $K_\chi = C\chi^{-2}$. All physical first-hit and section tests remain exact masks on that continuation. For $\lambda = (n,k)$ let $B_{z,\lambda}$ be their one-hot output and put $\ell_{z,\lambda}(r) = (2\pi)^{-1}\int B_{z,\lambda}(r\omega_\phi)\,d\phi$. Its value at zero is the tangent fraction $\theta_{z,\lambda}$. The unchanged normalization is

$$J_z = \frac{1}{Rc_*\sqrt{|\det(I - SM_z^{-1}SM_z)|}}, \qquad S = \operatorname{diag}(1,-1),$$

where M_z is the canonical derivative of the original half-word, or its physical-side branch value. There is no primitive-period quotient.

Theorem 24 (The full angular germ)**.** *At every fixed such word, each exact-label angular fraction is either identically zero on a smaller disk or has the form*

$$\ell_{z,\lambda}(r) = \eta_{z,\lambda} r^{d_{z,\lambda}}(1 + O_z(r)), \qquad \eta_{z,\lambda} > 0, \quad d_{z,\lambda} \in \{0,1,2,\dots\}.$$

If $\theta_{z,\lambda} > 0$, its order is zero and its leading coefficient is $\theta_{z,\lambda}$. Every nonzero nonlinear germ with zero tangent fraction has a positive finite order and is retained. No uniform bound on that order or on the analytic radius is asserted.

Proof. Every active analytic scalar margin has two simple angular roots. Their analytic germs have a fixed cyclic order at sufficiently small positive radius, with coincident germs kept as simultaneous tests. The exact angular fraction is a finite sum of analytic arc lengths. The complete argument is Theorem 88.3. □

A physical persistence cutoff. Let $g_z = D_z^{\varepsilon}(0)$ be the original clearance guard at the center. For $g_z > 0$ and a nonzero angular germ, a recorded analytic radius and relative Taylor remainder, together with $\mathrm{Lip}(D_z^{\varepsilon})/g_z$, define the finite number $\mathfrak{p}_{z,\lambda}$ in (89.3). On its stratum $\mathfrak{p}_{z,\lambda} \le K$, with $r = \sqrt{2u} < K^{-1}$, the actual density satisfies

$$e^{-K\chi r}(1 - Kr)\gamma_{z,\lambda} r^{d_{z,\lambda}} \le b_{z,\lambda}^{\varepsilon,\mathrm{clr}}(t_z + u) \le e^{K\chi r}(1 + Kr)\gamma_{z,\lambda} r^{d_{z,\lambda}}, \qquad \gamma_{z,\lambda} = J_z g_z \eta_{z,\lambda}.$$

At a seam $g_z = 1$. Non-seam centers with positive g_z are also admitted. For positive birth order, γ is not a zero-width atom.

Theorem 25 (Nonlinear collar height, uniformly in the birth order)**.** *Let $b^{\mathrm{p},K,H}$ be the original exact-label clearance source on the persistence stratum, restricted to $0 < F_z - t_z < H$. For fixed K, χ, ε, H with $2H < h_\chi = r_\chi^2/2$ and $2K\sqrt{H} < 1$,*

$$\limsup_{m\to\infty} \sup_{R,n,k} m^2 \|b_{m;n,k,R}^{\mathrm{p},K,H}\|_\infty \le 6CHe^{(2+\sqrt{2})K\chi\sqrt{H}} \frac{1 + \sqrt{2}K\sqrt{H}}{1 - 2K\sqrt{H}}.$$

The constant is independent of all finite birth orders and of the number of words, labels and coincident critical values. This ordered height tends to zero as $H \downarrow 0$ at fixed K, χ, ε.

Proof. The value at any inner roof level is bounded by a fixed multiple of the original source averaged over the outer annulus $H < u < 2H$. The factor is independent of the birth order because r^d is nondecreasing for every $d \ge 0$. Positivity compares that annular average with the inherited collar trace at width $2H$. Contributing critical centers lie in $[t - H, t]$, so its local bound is $3CH$. Proposition 89.2 and Theorem 90.1 give the constants and the common coarea representative. All scales are fixed before the collision limit. □

The complete source and the proof route. The source in the second theorem includes nonlinear zero-tangent sectors of every finite order on the stated strata. The exact partition is nevertheless $b^{\varepsilon,\mathrm{clr}} = b^{\mathrm{p},K,H} + r_{68}^{\varepsilon,\chi,K,H}$ with $r_{68} \ge 0$. Large persistence numbers, zero-center-guard births, omitted annuli, failed incidence taper, selected grazing and all noncritical or uncovered components remain in this complement. Its height and the complete first-incidence height are still required in (90.10) to obtain (M.1). Fixed-word analytic exhaustion is not a collision-limsup tail theorem. In particular the original physical angular tail $E(K)$ is not declared controlled.

The new proof is Sections 88–90. Its load-bearing inherited inputs are the complete scalar decisions, selected-contact Morse chart, relative physical density and positive fixed-width collar law. The earlier zero-width stratum theorem remains in Sections 91–93; it gives the sharper critical-value statement for its original order-zero coefficient. Every preceding core proof and appendix is compiled below, and the preceding abstract and introduction are retained verbatim in Appendix L. The present argument uses neither density differentiation of a local limit nor a uniform bound on the analytic birth order.

Appendix N. The preceding opening statements

The preceding abstract and introduction are retained verbatim below. Their scoped theorems and the full pointwise target keep their original mathematical content.

Preceding abstract. We study the joint displacement, collision count and flight time of actual returns in a triangular periodic Lorentz gas, retaining the complete arithmetic record and the original pointwise inversion problem. We construct a positive comparison trace on the unchanged physical itineraries and exact labels. Absolute control of the clearance guard then yields an ordered caustic-height theorem on angular-persistence strata, including zero center guards and all finite angular birth orders. The zero-center contribution has an additional square-root gain in the collar width; infinitely flat guards require no analytic classification. A second comparison uses finite-scale angular occupation and applies beyond the certified germ disk, including labels born only at positive radius. We give its physically weighted tail transfer and an exact disjoint source decomposition. The complete incidence height, the radial tail and the exterior clearance height remain explicit in the positive identity for the unrestricted pointwise arithmetic theorem.

Preceding introduction.
The physical record. For $R \in [9/20, 47/100]$ let $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ be the actual return record to Y_R^* under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, with $c_* = 91/(10000\pi)$. The density $p_{n,R}(k,m,t)$ uses counting measure in (k,m) and Lebesgue measure in t. Occupation counts collisions $0,\dots,m-1$; terminal section membership is imposed at collision m. The original pointwise problem is

$$\sup_{R,n,k}\ \operatorname*{ess\,sup}_{t:\,|Z_{m,R}|\le M}\left|m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)\right| \longrightarrow 0, \tag{N.1}$$

where $Z_{m,R} = (k_1, k_2, t - m\bar{\tau}_R, n - mc_*)/\sqrt{m}$. The arithmetic transition kernel in $\mathcal{L}_{m,R}$, including its zero classes, is retained. Our argument concerns the positive physical sources which enter this pointwise inversion, not a replacement process or an averaged return label.
The comparison source. At a selected normal critical word write $F_z = t_z + |y|^2/2$, $h_\chi = r_\chi^2/2$, and $K_\chi = C\chi^{-2}$. Let $B_{z,\lambda}$ be the exact physical indicator with label $\lambda = (n,k)$, and let D_z^ε be the unchanged first-clearance multiplier. Its center value g_z may vanish; its Lipschitz constant is at most $L_g = C_g\varepsilon^{-1}$. The source with this multiplier omitted, but every physical and label test retained, has density $\widehat{b}_{z,\lambda}$. Pointwise domination gives

$$b_{z,\lambda}^{\varepsilon,\mathrm{clr}}(t_z+u) \le \min\{1, g_z + L_g\sqrt{2u}\}\,\widehat{b}_{z,\lambda}(t_z+u).$$

Its positive collar trace satisfies the same ordered two-endpoint bound as the guarded trace, by a direct comparison with the original exact-return event.

Theorem 26 (Zero-center and all-order collar height)**.** *Let $b^{\mathrm{a,K,H};\le\delta}$ be the original clearance source on the angular-persistence stratum $\mathfrak{p}_{z,\lambda}^{\mathrm{a}} \le K$, with $g_z \le \delta$, restricted to $0 < F_z - t_z < H$. For fixed $\chi, \varepsilon, K, H, \delta$ with $2H < h_\chi$ and $2K\sqrt{H} < 1$,*

$$\limsup_m \sup_{R,\lambda} m^2 \|b_{m;\lambda,R}^{\mathrm{a,K,H};\le\delta}\|_\infty \le 6CHF_{\chi,K}(H)\min\{1, \delta + L_g\sqrt{2H}\},$$

where

$$F_{\chi,K}(H) = e^{(2+\sqrt{2})K_\chi\sqrt{H}}\frac{1+\sqrt{2}K\sqrt{H}}{1-2K\sqrt{H}}.$$

All finite angular birth orders are included with the same constant. There is no positive-center-guard assumption. At $\delta = 0$ the bound is $6\sqrt{2}CC_g\varepsilon^{-1}H^{3/2}F_{\chi,K}(H)$.

Proof. The angular certificate alone compares every inner level of the unguarded physical density to one positive outer-annulus average. Absolute guard domination is applied on the inner source, rather than used in the denominator of the comparison. The contributing critical centers lie in $[t-H,t]$. The positive comparison trace at width $2H$ has ordered bound $3CH$. The complete proof is Theorem 86.2. □

Beyond an angular germ. For fixed $0<S<h_\chi$ define

$$V_z(S)=\frac{\operatorname{ess\,sup}_{0<u<S}\ell_{z,\lambda}(\sqrt{2u})}{S^{-1}\int_0^S\ell_{z,\lambda}(\sqrt{2u})\,du},$$

with value zero when the denominator vanishes. This is a finite-scale geometric ratio, not either of the earlier angular condition numbers. It is finite at each fixed chart and label. Its tail must nevertheless be controlled after the parameter and label suprema and the collision limsup; fixed-count finiteness does not establish that control.

Theorem 27 (Complete fixed-radius comparison)**.** *The original clearance source on $0<F_z-t_z<S$ and $V_z(S)\le T$ satisfies*

$$\mathcal{H}(b^{(0,S),S,V\le T})\le 2CTSe^{2K_\chi\sqrt{2S}}.$$

This includes portions beyond any angular-germ certificate, zero center guards, and labels absent near the center but present at positive radius. For $\alpha>0$, the remaining $V_z(S)>T$ source satisfies

$$\mathcal{H}(b^{(0,S),S,V>T})\le e^{2K_\chi\sqrt{2S}}T^{-\alpha}\mathcal{M}_\alpha(S),$$

where $\mathcal{M}_\alpha(S)$ is the exact physical collar-trace moment in (87.6)*. This implication requires a finite moment bound; such a bound is not supplied by finiteness of individual ratios.*

Proof. The physical density distortion compares the radial supremum to its positive collar average with cost $e^{2K_\chi\sqrt{2S}}V_z(S)$. At roof t, centers lie in $[t-S,t]$. The ordered comparison trace bounds this interval by $2CS$. The tail follows from $V\mathbf{1}_{\{V>T\}}\le T^{-\alpha}V^{1+\alpha}$. See Theorem 87.2 and Proposition 87.3. □

The complete endpoint. The two theorems enlarge the controlled part of the original Lorentz source. They are combined without overlapping source mass in (87.8). The complete first-incidence term, the high-ratio radial term and the exterior clearance term remain in (87.9). The exterior includes failed taper, selected grazing, noncritical rank, uncovered points and omitted outer annuli. No full-source height, unrestricted same-roof bridge or essential-likelihood theorem is inferred from a partial source estimate. The target (N.1) and its arithmetic factor are unchanged.

The new proofs occupy Sections 85–87. Their shortest inherited route is the exact physical chart and guard, the two-endpoint local bound, and the analytic angular classification for the first theorem only. All earlier mathematical modules remain compiled below. The preceding opening statements are retained in Appendix M; source and validation records are kept outside the principal argument.

Appendix O. The preceding opening statements

The preceding abstract and introduction are retained verbatim below. Their scoped theorems and the original pointwise target are unchanged.

Preceding abstract. We study exact displacement, collision count and flight time for actual returns in a triangular periodic Lorentz gas. A positive finite-scale angular-overlap construction separates radial growth from angular loss on the original physical source. Its height is controlled by one physical collar trace, without a radial-ratio cutoff or a moment hypothesis. For angular occupation with no loss to an outer annulus, this gives the complete inner-collar estimate, including zero-center guards and later positive-radius births. For each fixed word we identify the signed radial current of the exact physical angular set and retain its jump contributions. On the whole selected Morse disk the overlap part has ordered height $O(\chi^6)$. We insert this bound into the original pointwise arithmetic error identity, retaining explicitly the angular loss, the uncovered physical source and the first-incidence source.

Preceding introduction.

The original local problem. For $R \in [9/20, 47/100]$ the actual return record is $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, with $c_* = 91/(10000\pi)$. Its density $p_{n,R}(k,m,t)$ uses counting measure in (k,m) and Lebesgue measure in t. Occupation counts collisions $0,\dots,m-1$; terminal membership is imposed at collision m. The pointwise problem throughout this article is

$$\sup_{R,n,k}\ \operatorname*{ess\,sup}_{t:\,|Z_{m,R}|\le M} \Big| m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n) \Big| \longrightarrow 0, \tag{O.1}$$

where $Z_{m,R} = (k_1, k_2, t - m\bar{\tau}_R, n - mc_*)/\sqrt{m}$. The finite arithmetic transition kernel in $\mathcal{L}_{m,R}$, including its zero classes, is unchanged. The results below concern the positive sources entering this exact pointwise problem.

Angular overlap. At a selected critical chart write $F_z = t_z + |y|^2/2$, $r_\chi = r_0\chi^3$, $h_\chi = r_\chi^2/2$, and $K_\chi = C\chi^{-2}$. For the complete physical indicator with exact label $\lambda = (n,k)$, put

$$L_z(u) = \frac{1}{2\pi}\int_0^{2\pi} B_{z,\lambda}(\sqrt{2u}\,\omega_\phi)\,d\phi.$$

The guard is still D_z^ε, with center $g_z \in [0,1]$ and Lipschitz constant $C_g\varepsilon^{-1}$. Given a fixed receiving interval $J \subset (0, h_\chi)$, form

$$a_{z,J}(u) = \frac{1}{|J|}\int_J \min\{1, L_z(v)/L_z(u)\}\,dv, \qquad \Delta_{z,J}(u) = \frac{1}{|J|}\int_J (L_z(u) - L_z(v))_+\,dv.$$

The first expression is set to one when $L_z(u) = 0$; the corresponding source is then zero. Multiplication of the original clearance source by $a_{z,J}$ and $1 - a_{z,J}$ gives complementary positive weights. No point is transported to a nonphysical continuation, and no section decision is changed.

Theorem 28 (Full inner collar with no angular loss)**.** *Fix $0 < \varepsilon \le \chi/4$, $2H < h_\chi$ and $0 \le \delta \le 1$. On any selected subfamily with $g_z \le \delta$ and $L_z(u) \le L_z(v)$ for almost every $(u, v) \in (0, H) \times (H, 2H)$, the entire original clearance source on $0 < F_z - t_z < H$ satisfies*

$$\mathcal{H}(b) \le 6CHe^{(2+\sqrt{2})K_\chi\sqrt{H}} \min\{1, \delta + C_g\varepsilon^{-1}\sqrt{2H}\}.$$

Here $\mathcal{H}$ is the ordered m^2 essential height. The bound requires no analytic-germ radius, birth-order bound or radial moment. At zero center guard its order is $\varepsilon^{-1}H^{3/2}$.

Proof. The loss $\Delta_{z,(H,2H)}$ vanishes. The scalar overlap is bounded by the outer angular average, which the relative physical Jacobian compares with the unchanged positive collar trace. The contributing centers occupy an interval of length H. This gives the displayed constant by Corollary 82.3. □

Loss and physical boundaries. The no-loss hypothesis is not imposed on all physical words. For an arbitrary angular profile the same comparison controls its overlap part, while the remaining positive density is bounded by its exact $\Delta_{z,J}$. Theorem 83.1 proves, at every fixed word and on every compact radial interval, that L_z is of bounded variation and

$$DL_z = (2\pi)^{-1}\pi_{\#}D_u\mathbf{1}_{E_{z,\lambda}}, \qquad E_{z,\lambda} = \{(u, \phi) : B_{z,\lambda}(\sqrt{2u}\omega_\phi) = 1\}.$$

Its negative part bounds loss to an outer collar. Radial deaths are retained as measure atoms, rather than omitted from a classical derivative. Fixed-word finiteness does not supply an ordered weighted tail estimate.

Theorem 29 (The complete selected disk)**.** *With $I = (0, h_\chi)$ and $J = (0, h_\chi/2)$, the original clearance source has an exact positive weight decomposition*

$$b^{\varepsilon,\mathrm{clr}} = s_{70} + d_{70} + e_{70}, \qquad \mathcal{H}(s_{70}) \le \frac{3}{2}Ch_\chi e^{(1+1/\sqrt{2})K_\chi r_\chi} \le C'\chi^6.$$

Here s_{70} is the overlap part on every selected physical disk, d_{70} is its complementary angular-loss part, and e_{70} is the original source outside those disks. This allocation includes all outer annuli of the selected disks, not merely their analytic-germ neighborhoods.

Proof. The complete interval comparison, with receiving width $h_\chi/2$, has center-interval plus collar length $3h_\chi/2$. Sum the complementary weights before adding the original outside event. The full argument is Theorem 84.1. □

For $\varepsilon(B) = A_0B^{-1/12}$ and $\chi(B) = \sqrt{\varepsilon(B)}$, the controlled overlap height is $O(B^{-1/4})$. These geometric choices are fixed before the collision limsup. The complete endpoint estimate is (84.3); it retains the heights of d_{70}, e_{70} and the complete first-incidence source. Their vanishing, and hence (O.1) and its unrestricted same-roof consequences, are not asserted by the overlap theorem. The new argument isolates angular loss rather than treating a large radial ratio itself as an obstruction.

The principal proof is in Sections 82–84. Its quantitative input is the unchanged physical collar trace of Theorem 85.2, proved on the same exact record. The signed-current identification additionally uses the analytic physical decisions at a fixed word. All preceding mathematical modules are retained below; the preceding opening statements appear in Appendix N.

Appendix P. The preceding opening

Previous abstract. We study exact displacement, collision count and flight time for actual returns in a triangular periodic Lorentz gas. A common physical collar recovers angular loss by combining receiving surpluses before taking a positive deficit. The construction increases the previously controlled source, preserves its $O(\chi^6)$ ordered height, and reduces the residual without changing any physical or integer label. We express the new residual through a signed two-sided radial current, including late births, atomic deaths and cancellation at a common roof. This gives the complete selected-source height on a receiving- reserve class without a wordwise variation or monotonicity bound. The original arithmetic raw-error identity then retains only the net current excess, the outside source and complete first incidence.

The exact local problem. For $R \in [9/20, 47/100]$ consider the actual return record $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, where $c_* = 91/(10000\pi)$. Write $p_{n,R}(k,m,t)$ for its density relative to counting measure in (k,m) and Lebesgue measure in t. Occupation counts collisions $0,\dots,m-1$; terminal membership is imposed at collision m. The pointwise target remains

$$\sup_{R,n,k} \operatorname*{ess\,sup}_{t:\,|Z_{m,R}|\le M} \left| m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n) \right| \longrightarrow 0, \tag{P.1}$$

where $Z_{m,R} = (k_1, k_2, t - m\bar{\tau}_R, n - mc_*)/\sqrt{m}$. The finite arithmetic transition kernel in $\mathcal{L}_{m,R}$, including its zero classes, is retained throughout. The estimates below concern the complete positive sources in this same problem.

A common receiving capacity. For a selected critical chart use $F_z = t_z + |y|^2/2$, $r_\chi = r_0\chi^3$, $h_\chi = r_\chi^2/2$ and $K_\chi = C\chi^{-2}$. Its angular fraction $L_{z,\lambda}(u)$ includes every physical and section test and the exact label $\lambda = (n,k)$. Set $h = h_\chi$, $j = h/2$ and, at a common roof t, sum only over $0 < u_z = t - t_z < h$:

$$\mathsf{X}_\lambda(t) = \sum_z J_z L_{z,\lambda}(u_z), \qquad \mathsf{Y}_\lambda(t) = \sum_z \frac{J_z}{j} \int_0^j L_{z,\lambda}(v)\, dv.$$

Both sums use the same active charts, the same collision count and the same exact label. The physical coefficient J_z already contains the section normalization. The old angular overlap defines the controlled source s_{70} and its positive loss d_{70} on these complete disks. We keep s_{70} and recover a bounded measurable fraction of d_{70} using the receiving capacity not yet used by the old overlap.

Theorem 30 (Recovery on the original physical source)**.** *Fix $0 < \varepsilon \le \chi/4$, $0 < \chi \le 1/10$ and $K \ge 1$ before the collision limit. There are complementary physical weights $s_{71,K}, d_{71,K}$ such that*

$$b^{\varepsilon,\mathrm{clr}} = s_{71,K} + d_{71,K} + e_{70}, \qquad s_{71,K} \ge s_{70}, \quad 0 \le d_{71,K} \le d_{70},$$

where e_{70} is exactly the original outside source. Moreover

$$\mathcal{H}(s_{71,K}) \le \tfrac{3}{2} CKh_\chi e^{(1+1/\sqrt{2})K_\chi r_\chi} \le C' K\chi^6,$$
$$d_{71,K}(t) \le e^{K_\chi r_\chi}(\mathsf{X}_\lambda(t) - K\mathsf{Y}_\lambda(t))_+.$$

All selected-disk annuli are included. No physical trajectories are coupled and no capacity is borrowed from a different exact label.

Proof. The capacity identities of Lemma 80.1 recover the maximal common fraction allowed by the unused receiving capacity. Theorem 80.2 compares the resulting source with one positive physical collar trace after summing all active charts. The center interval and collar width add to $3h_\chi/2$. □

Signed radial balance. Let $\mu_{z,\lambda} = DL_{z,\lambda}$ be the physical radial current. It is a signed measure at each fixed word; it retains jump atoms and the actual Boolean boundary orientation. With $F_j(s) = \min(s/j, 1)$, set

$$k_j(u,s) = F_j(s)\mathbf{1}_{s<u} - (1 - F_j(s))\mathbf{1}_{s>u}.$$

Theorem 81.2 proves the exact identity

$$\mathsf{X}_\lambda(t) - \mathsf{Y}_\lambda(t) = \sum_{z:\,0<t-t_z<h} J_z \int_{(0,h)} k_j(t - t_z, s)\, d\mu_{z,\lambda}(s). \tag{P.2}$$

An inward comparison pays positive births below the source roof as well as negative deaths above it. Their signed contributions are added with the physical weights and common-roof translations before a positive part is taken. Fixed-word bounded variation is not an ordered bound for that sum.

Theorem 31 (The complete selected source with a reserve)**.** *Suppose* $\mathsf{X}_\lambda(t) \le K\mathsf{Y}_\lambda(t)$ *for almost every roof, uniformly in* R, λ *at all sufficiently large counts. Then the entire selected-chart source, including its old angular-loss component, has ordered height at most* $C'K\chi^6$*. In particular a nonpositive signed current in* (P.2) *suffices with* $K = 1$*. Individual profiles may have arbitrarily many births and deaths; no wordwise variation bound or individual no-loss condition is required.*

Proof. The residual in Theorem 30 is zero. Apply the same ordered estimate to the complete selected source; see Corollary 80.3. □

This condition is not asserted for every Lorentz record. It identifies a larger complete-source class than a chartwise no-loss condition, while retaining the original model and endpoint. For general records write $\mathcal{N}_\chi(K)$ for the ordered essential height of $(\mathsf{X} - K\mathsf{Y})_+$. The legal fixed-band choices

$$\varepsilon(B) = A_0 B^{-1/12}, \qquad \chi(B) = \sqrt{\varepsilon(B)}$$

give the complete estimate

$$\begin{aligned}\mathcal{E}_M \le{}& C_M B^{-1/192} + C'K(B)B^{-1/4} + C''\mathcal{N}_{\chi(B)}(K(B)) \\ &+ \mathcal{H}(b^{\varepsilon(B),\mathrm{inc}}) + \mathcal{H}(e_{70}).\end{aligned} \tag{P.3}$$

The ordered vanishing of the last three heights remains to be proved on the unrestricted physical family. Neither source mass, finite-word perimeter nor a collision-count-dependent band is substituted for these estimates. In particular the unrestricted same-roof and essential-likelihood conclusions are not inferred from the recovery construction alone.

The principal new argument is contained in Sections 80–81. Its quantitative input is the unchanged physical comparison collar of Theorem 85.2; its geometric current input is Theorem 83.1. The complete preceding mathematical development follows, and the preceding opening is retained in Appendix O.

Appendix Q. The retained opening of revision 72

The following opening is retained as a historical statement of the preceding route. Its finite-measure qualification is superseded by Theorem 74.3; its collision-uniform qualification is not. In its source notation, e_{70} means first clearance outside the selected disks, with incidence separate. Preceding abstract. We study exact displacement, collision count and flight time for actual returns in a triangular periodic Lorentz gas. We construct the global distributional current of the complete physical boundary source, including first incidence, competing-hit clearance and all states outside the selected critical disks. Its chart representation retains the entrance and exit atoms of every moving window. An exact compact primitive identifies the discrepancy between the original source and a fixed-width average. Keeping the previously controlled source, we recover a portion of every remaining physical defect with ordered height $O(\chi^6 + K\varepsilon^{1/16})$. A directed local-flux bound then gives an explicit sufficient estimate for the unrestricted pointwise arithmetic law. The quantitative decay of the complete current excess remains to be established; neither a window law nor a finite-count variation bound is substituted for that estimate.

The exact record. For $R \in [9/20, 47/100]$ consider the actual return record $J_{n,R} = (K_{n,R}, N_{n,R}, T_{n,R})$ under $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, where $c_* = 91/(10000\pi)$. Occupation counts collisions $0, \dots, m-1$; terminal membership is imposed at collision m. The density $p_{n,R}(k, m, t)$ uses counting measure in (k, m) and Lebesgue measure in t. Our pointwise problem is unchanged:

$$\sup_{R,n,k} \operatorname*{ess\,sup}_{t:\,|Z_{m,R}|\le M} \left| m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n) \right| \longrightarrow 0, \tag{Q.1}$$

where $Z_{m,R} = (k_1, k_2, t - m\bar{\tau}_R, n - mc_*)/\sqrt{m}$. The finite arithmetic transition kernel in $\mathcal{L}_{m,R}$, including its zero classes, is retained. No exact label is replaced by a block or by an average over return indices.

A current on the complete source. At fixed m, R, n, k, ε, let $b = b^{\varepsilon}_{n,k,m,R}$ be the density of the original first-physical-defect source. Its complete positive decomposition is

$$b = s_{71,1} + d_{71,1} + b^{\varepsilon,\mathrm{inc}} + e_{70}.$$

Here $s_{71,1}$ is the previously recovered selected-disk source, and e_{70} includes every original physical state outside those disks. We define its distributional current directly on the original collision space by

$$\langle \mathfrak{D}^{\varepsilon}, \phi \rangle = -\int_{\Pi_{n,k,m,R}} (1 - H^{\varepsilon}_{m,R}) \phi'(L_{m,R})\, d\nu_R^*.$$

It is Db, independently of a chart partition. Theorem 77.1 gives the missing entrance and exit atoms when a moving selected-chart window is differentiated. Theorem 78.1 includes the incidence and outside pieces, and Proposition 78.2 identifies the oriented coarea flux on regular physical patches. No finite-measure or collision-uniform variation bound is built into this definition.

Theorem 32 (Recovery on the whole physical source)**.** *Fix $0 < \varepsilon \le \chi/4$, $0 < \chi \le 1/10$, $\delta > 0$ and $K \ge 1$ before the collision limit. Write $A_\delta b(t) = (2\delta)^{-1} \int_{t-\delta}^{t+\delta} b(v)\, dv$. There are complementary weights on the original physical source, with densities $s_{72,K}, r_{72,K}$, such that*

$$\begin{aligned} b &= s_{72,K} + r_{72,K}, \\ s_{72,K} \ge s_{71,1}, &\qquad \mathcal{H}(s_{72,K}) \le C\chi^6 + CK\varepsilon^{1/16}, \\ 0 \le r_{72,K} &\le [b - KA_\delta b]_+. \end{aligned}$$

The residual is a displayed nonnegative weighting of all three original remainders $d_{71,1}$, $b^{\varepsilon,\mathrm{inc}}$ and e_{70}. All original states, guards and exact labels remain accounted for.

Proof. Lemma 79.1 allocates only unused capacity and preserves the old controlled source. The original fixed-width local-variation bound estimates $A_\delta b$ in the legally ordered limit. Theorem 79.2 then proves the height bound without a word or witness count. □

The quantitative remaining estimate. The compact odd kernel κ_δ of (78.5) satisfies $D\kappa_\delta = \delta_0 - (2\delta)^{-1}\mathbf{1}_{(-\delta,\delta)}dt$. Consequently

$$b - A_\delta b = \kappa_\delta * \mathfrak{D}^\varepsilon$$

as distributions and almost everywhere as densities. This identity holds even when the current is not a measure. Define

$$\mathcal{U}_{\varepsilon,\delta}(K) = \limsup_m \sup_{R,n,k} m^2 \|[b - KA_\delta b]_+\|_\infty.$$

It uses the complete physically weighted current at a common roof, not wordwise absolute variations. With the fixed-band choices $\varepsilon(B) = A_0 B^{-1/12}$ and $\chi(B) = \sqrt{\varepsilon(B)}$, Theorem 79.3 gives

$$\begin{aligned}\mathcal{E}_M \le C_M B^{-1/192} + CB^{-1/4} + CK(B)B^{-1/192} \\ + \mathcal{U}_{\varepsilon(B),\delta(B)}(K(B)).\end{aligned} \tag{Q.2}$$

Thus $K(B)B^{-1/192} \to 0$ and vanishing of the last term suffice for (Q.1). For a complete current which is a finite measure, Proposition 79.4 bounds that last term by half its directed local variation. In particular a bound $\mathcal{V}_\varepsilon(\delta) \le A\varepsilon^{-q}\delta^\beta$ would give the explicit rate in (79.14). These current bounds are not proved on the unrestricted Lorentz family. The current construction and recovery theorem therefore do not by themselves establish the unrestricted pointwise law or its same-roof conditional consequences.

The new argument is given in Sections 77–79. Its principal quantitative inputs are the unchanged full-source local-variation theorem, Theorem 179.1, and the selected-source estimate, Theorem 80.2. All preceding mathematical inputs are retained below. The exact preceding opening is retained in Appendix P.

References

[1] Q. Qi, *Scalar collision laws and recognition of periodic dispersing billiards*, A2 v43, primary article and technical supplement, theta-theory repository, 2026.

[2] D. Szász and T. Varjú, *Local limit theorem for the Lorentz process and its recurrence in the plane*, Ergodic Theory Dynam. Systems **24** (2004), 257–278.

[3] D. Dolgopyat and P. Nándori, *On mixing and the local central limit theorem for hyperbolic flows*, Ergodic Theory Dynam. Systems **40** (2020), 142–174; arXiv:1710.08568v1, Definition 2.2 and Proposition 6.3.

[4] M. F. Demers and H.-K. Zhang, *A functional analytic approach to perturbations of the Lorentz gas*, Comm. Math. Phys. **324** (2013), 767–830; arXiv:1210.1261.

[5] V. Baladi, M. F. Demers and C. Liverani, *Exponential decay of correlations for finite horizon Sinai billiard flows*, Invent. Math. **211** (2018), 39–177; arXiv:1506.02836v3, including the correction note.

[6] L.-S. Young, *Statistical properties of dynamical systems with some hyperbolicity*, Ann. of Math. (2) **147** (1998), 585–650; author preprint ESI 445 (1997), Section 8, Theorem 6.

[7] H. Bruin, M. Holland and M. Nicol, *Livšic regularity for Markov systems*, Ergodic Theory Dynam. Systems **25** (2005), 1739–1765; doi:10.1017/S0143385705000179; arXiv:math/0503690.

[8] M. F. Demers and H.-K. Zhang, *Spectral analysis of the transfer operator for the Lorentz gas*, J. Mod. Dyn. **5** (2011), 665–709; doi:10.3934/jmd.2011.5.665.

[9] M. Coste, *An introduction to o-minimal geometry*, lecture notes, November 1999; Dottorato di Ricerca in Matematica, Università di Pisa, Istituti Editoriali e Poligrafici Internazionali, 2000.

[10] S. Gouëzel, *Sharp polynomial estimates for the decay of correlations*, Israel J. Math. **139** (2004), 29–65; arXiv:math/0202147.

[11] I. Melbourne and M. Nicol, *Almost sure invariance principle for nonuniformly hyperbolic systems*, Comm. Math. Phys. **260** (2005), 131–146; arXiv:math/0503693.

[12] M. F. Demers, F. Pène and H.-K. Zhang, *Local limit theorem for randomly deforming billiards*, Comm. Math. Phys. **375** (2020), 2281–2334; doi:10.1007/s00220-019-03670-7. Lemma numbering here follows arXiv:1902.06850v1.

[13] L. Ambrosio, N. Fusco and D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford Mathematical Monographs, Clarendon Press, Oxford, 2000.

[14] S. Basu, R. Pollack and M.-F. Roy, *An asymptotically tight bound on the number of semi-algebraically connected components of realizable sign conditions*, Combinatorica **29** (2009), 523–546; arXiv:math/0603256v3, Section 3.2, equation (3.3).

[15] R. Cluckers and D. J. Miller, *Stability under integration of sums of products of real globally subanalytic functions and their logarithms*, Duke Math. J. **156** (2011), no. 2, 311–348; arXiv:0911.4373v1, Theorems 1.3 and 3.11.

[16] H. Döring, S. Jansen and K. Schubert, *The method of cumulants for the normal approximation*, Probab. Surv. **19** (2022), 185–270; doi:10.1214/22-PS7.

[17] E. Carneiro and J. D. Vaaler, *Some extremal functions in Fourier analysis, II*, Trans. Amer. Math. Soc. **362** (2010), 5803–5843; arXiv:0809.4050, Introduction.

[18] M. Stenlund, L.-S. Young and H. Zhang, *Dispersing billiards with moving scatterers*, Comm. Math. Phys. **322** (2013), 909–955; arXiv:1210.0011v4, Section 3.1.

[19] C. Villani, *Optimal Transport: Old and New*, Grundlehren der mathematischen Wissenschaften **338**, Springer, Berlin, 2009; Chapters 5–6; doi:10.1007/978-3-540-71050-9.

[20] L. Hervé and J. Ledoux, *A local limit theorem for densities of the additive component of a finite Markov Additive Process*, Statist. Probab. Lett. **83** (2013), 2119–2128; arXiv:1306.5353; doi:10.1016/j.spl.2013.05.032.

[21] L. Hervé and J. Ledoux, *Additional material on local limit theorem for finite Additive Markov Processes [HL13]*, arXiv:1305.5644 (2013), Section 6.

[22] M. Coste, *An introduction to semialgebraic geometry*, Institut de Recherche Mathématique de Rennes, October 2002, Sections 2.2–2.3 and Theorem 2.12.

[23] S. Basu and A. Mohammad-Nezhad, *Improved effective Łojasiewicz inequality and applications*, Forum Math. Sigma **12** (2024), e115. DOI: 10.1017/fms.2024.66. The quantifier-elimination statement and numbering used here are Theorem 4 of the pinned arXiv:2211.10034v2 version.

[24] S. Bolotin and D. Treschev, *Hill's formula*, Russian Math. Surveys **65** (2010), no. 2; doi:10.1070/RM2010v065n02ABEH004671. The discrete and reversible comparison is with the original version arXiv:1006.1532v1. The two-dimensional identity used here is proved in full in Section 103.

[25] V. Baladi and M. F. Demers, *Thermodynamic formalism for dispersing billiards*, J. Mod. Dyn. **18** (2022), 415–493; arXiv:2009.10936v3, Section 1.1, equations (2.4), (2.11), and the proof of Lemma 3.8.

[26] M. F. Demers and H.-K. Zhang, *A functional analytic approach to perturbations of the Lorentz gas*, arXiv:1210.1261v1 (2012), Theorem 2.5, Remark 2.7 and Lemma 3.2(b); doi:10.1007/s00220-013-1820-0.